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Preface to the Fourth Edition

Common Admission Test (CAT) is known to usher surprises by bringing about changes in its format. CAT 2015 was another such test, which reverted to the 3-section format exam with sectional time-limit from the erstwhile 2-section exam. CAT 2015 also introduced, for the first time, questions in Quantitative Aptitude and Verbal Ability that are not option based. Although the focus areas in CAT 2015 proportionately remained the same as that in the previous years, the test's new format has been set forth with significant changes such as clubbing Data Interpretation with Logical Reasoning once again. Hence, we felt the need to bring out this new edition of *Quantitative Aptitude for the CAT*.

This new edition has incorporated major modifications that will make it more comprehensive and beneficial for all users of the book. The nature and extent of these modifications can be encapsulated as follows:

(a) Chapters have been re-organized with emphasis on QA section in keeping with the current pattern of CAT.
(b) A new chapter on Simple Interest and Compound Interest has been added.
(c) Complete solutions have been added to chapters on Permutation and Combination, and Probability.
(d) A special section catering to Campus Recruitment Tests has been appended at the end.
(e) Model test papers (based upon previous years' examinations) for XAT, IIFT, and SNAP have been included for the benefit of students who intend to appear for exams of other B-schools.
(f) A CD designed to simulate the same GUI experience as that of CAT 2015 has been included with the book. The CD contains five Section Tests based on the pattern of CAT 2015 (both options based and non-options based) to provide students with a hands-on experience and enhance their confidence.

I am sure that the changes made to this new edition will help students to derive more from this book. In case you have any question, you can connect with me on Facebook or Quora.

For any feedback or clarification, I can be reached at nsinha.alexander@gmail.com

Happy Learning!

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Preface

The fact that there was no surprise element in CAT 2010 with respect to CAT 2009 was itself a surprise. Probably IIMs/ CAT committee wanted to regain the ground that they might have lost during CAT 2009 online test glitches.

So everybody was expecting some changes in CAT pattern in 2011, and CAT committee obliged. The CAT changed again in 2011—with the introduction of (a) 2-section format instead of 3-section format and (b) sectional time limit. This is a welcome change as now the CAT is similar to other global exams like the GMAT and GRE. They also comprise only two sections and have sectional time limit.

This book covers everything that you need to prepare for the Quantitative Aptitude and Data Interpretation Section as per CAT 2011 pattern. Before we move ahead, let us see the cut-off percentile of different IIMs for CAT 2011 (for general category):

S. No	IIM's	Overall score	QA/DI	VA/LR
1	IIM-A	99	94	94
2	IIM-B	90	80	90
3	IIM-C	99.55	94.24	93.73
4	IIM-L	90	85	85
5	IIM-I	90	85	85
6	IIM Trichy	80	70	70
7	IIM Kashipur	75	70	70
8	IIM Udaipur	80	70	70
9	IIM Ranchi	90	70	70
10	IIM Shillong	***	65	65
11	IIM Kozhikode	85	55	55

*** Not given.

This book is divided into four parts: Part 1—Quantitative Aptitude; Part 2—CAT Papers; Part 3—Other MBA Entrance Papers; and Part 4—Section Tests. First part is further distributed among modules to facilitate the learning of students. Once a student is through with chapters, there are benchmarking tests with calibrated percentile. Moving on further, there are CAT papers to know the level of your preparedness. Finally there are three section tests to help you assess your preparation level.

Further, to facilitate the online testing, a CD containing Section Tests and Full Length Tests is appended.

Your comments and suggestions would be very useful in improving subsequent editions of this book. Please mail me your suggestions at: nsinha.alexander@gmail.com

Nishit K. Sinha

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About the Author

Nishit K. Sinha, an IIM Lucknow alumnus, has been training students for the CAT and other B-school entrance examinations for more than a decade. During this period, he has successfully trained more than 10,000 students of varying backgrounds to clear various MBA entrance examinations. To best analyse the pattern of all the major B-school entrance tests, as well as to remain updated on their pattern, he sits for examinations such as the CAT and XAT every year.

Currently, he is associated with Graphic Era Hill University, Dehradun.

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Acknowledgements

This book bears the imprint of many people—my colleagues, my students, and my teachers who have had a significant impact on my thought process and have generously extended help whenever I needed.

Special mention of thanks to Prof. (Dr) Kamal Ghanshala and Prof. (Dr) Sanjay Jasola for providing motivation and guidance to keep myself updated and bring out this new edition.

I would like to thank my teachers Mr Anoop Singhania, Mr Vinay Singh, Mr M.K. Alam Bhutto, Mr Jairam Singh, and Mr Arun Sharma.

My special thanks to my brothers Ravi Shankar Prasad, Sharat Chandra Mayank, Amit Kumar, and Vinit Kumar.

I extend my heartfelt thanks to my colleagues Mr P.A. Anand, Mr Narendra Bisht, Ms Divya Paul, Ms Nishu Chawla, Ms Aanchal, and Mr Anurag Chauhan. This book would not have been possible without the contribution from all my students, past and present, who have helped me improve the content and the presentation of the book, and its new edition.

I would like to thank Ms Sharel Simon and Ms G. Sharmilee for giving the book the final shape. Thanks to Mr Vikas Sharma and Mr H. Nagaraja for ensuring that I get the timely and accurate feedback of the users.

Thanks would be a small word for my wife, who took care of family and home, giving me enough time to complete this project. Love to my son who gave me company during late nights. Sudhir, my man-Friday, who took care of my small necessities, your contribution is noteworthy.

I may have forgotten some names here. I wish to express my gratitude to all who have contributed in the making of this book.

Nishit K. Sinha

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CAT Demystified

CAT stands for the Common Admission Test. It is a test conducted by IIMs for admission into several programs offered by them. Besides IIMs, there are a good number of colleges which accept CAT score in their first round of selection process. As of now, there are 19 IIMs offering PGP at the following places: Ahmedabad, Bangalore, Calcutta, Lucknow, Indore, Kozhikode, Shillong, Ranchi, Rohtak, Raipur, Udaipur, Trichy, Kashipur, Bodhgaya, Nagpur, Sambalpur, Sirmaur, Amritsar, and Vishakhapatnam.

History of the CAT

Almost for the past three decades, since the CAT has been started, it has changed its colours many a time in terms of number of questions, sections asked, and orientation of those questions. Here, we will discuss the pattern of CAT 2000 onwards.

Chart 1

	Number of sections	Total number of questions	Total marks	Time allowed
CAT 2000	3	165	N.A.	120 minutes
CAT 2001	3	165	N.A.	120 minutes
CAT 2002	3	150	N.A.	120 minutes
CAT 2003	3	150	N.A.	120 minutes
CAT 2004	3	123	150	120 minutes
CAT 2005	3	90	150	120 minutes
CAT 2006	3	75	300	150 minutes
CAT 2007	3	75	300	150 minutes
CAT 2008	3	90	360	150 minutes
CAT 2009	3	60	450 (scaled score)	135 minutes
CAT 2010	3	60	450 (scaled score)	135 minutes
CAT 2011	2	60	450 (scaled score)	140 minutes
CAT 2012	2	60	450 (scaled score)	140 minutes
CAT 2013	2	60	450 (scaled score)	140 minutes
CAT 2014	2	100	300	170 minutes
CAT 2015	3	100	300	180 minutes

The CAT online examinations

The CAT examinations held from 2011 to 2014 had two sections: (a) Quantitative Ability and Data Interpretation (b) Verbal Ability and Logical Reasoning with 30 questions in each section. It also have sectional time limit of 70 minutes

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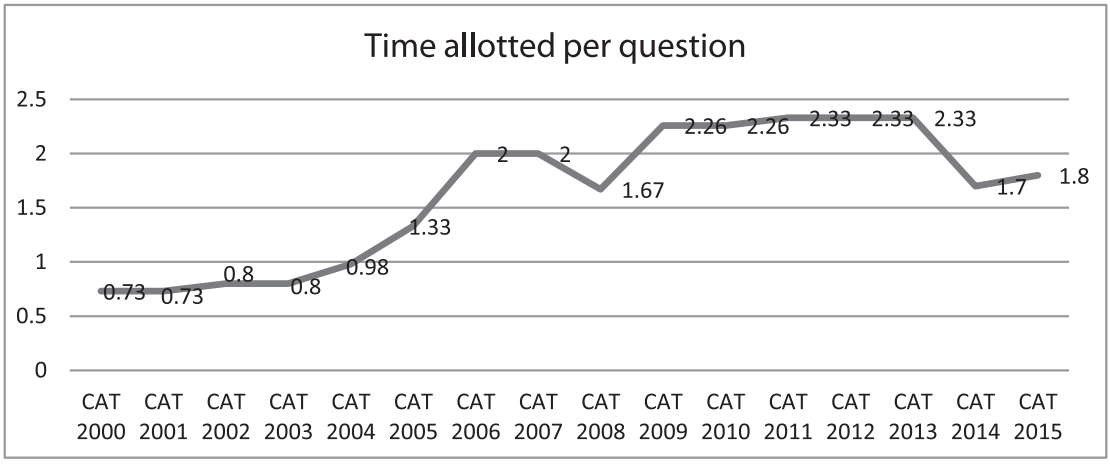
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for each section. CAT 2015 had 100 questions: 34 questions in QA, 34 questions in VA/RC, and 32 in LR/DI. Each section had a sectional time limit of one hour.

Before CAT 2004, the CAT did not mention how many marks one question carried? Marks carried per question were announced for the first time in CAT 2004.

Quite obvious from the above table that time allotted per question has risen sharply from CAT 2000 to CAT 2011. One possible conclusion drawn from here is that the CAT is focussing more on accuracy than speed, and secondly, it expects students to gain a certain level of competence across all the areas in a particular section. With the number of questions going down and time going up, students do not have much choice of questions to choose from.

Chart 2



Sectional Breakups and Getting an IIM Call

One thing that has remained constant during this period of CAT 2000 to CAT 2010 is the number of sections and the way these sections have been joined—Quantitative Aptitude (QA), Logical Reasoning and Data Interpretation (LR/DI), and English Usage/Reading Comprehension (EU/RC). Though CAT 2011 changed it all:

Chart 3

Year	QA	LR/DI	EU/RC	Total number of questions
CAT 2000	55	55	55	165
CAT 2001	50	50	50	150
CAT 2002	50	50	50	150
CAT 2003	50	50	50	150
CAT 2004	35	38	50	123
CAT 2005	30	30	30	90
CAT 2006	25	25	25	75
CAT 2007	25	25	25	75
CAT 2008	25	25	40	90
CAT 2009	20	20	20	60
CAT 2010	20	20	20	60
CAT 2011	30 (QA + DI)		30 (Verbal + LR)	60
CAT 2012	30 (QA + DI)		30 (Verbal + LR)	60

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Year	QA	LR/DI	EU/RC	Total number of questions
CAT 2013	30 (QA + DI)		30 (Verbal + LR)	60
CAT 2014	50 (QA + DI)		50 (Verbal + LR)	100
CAT 2015	34	32	34	100

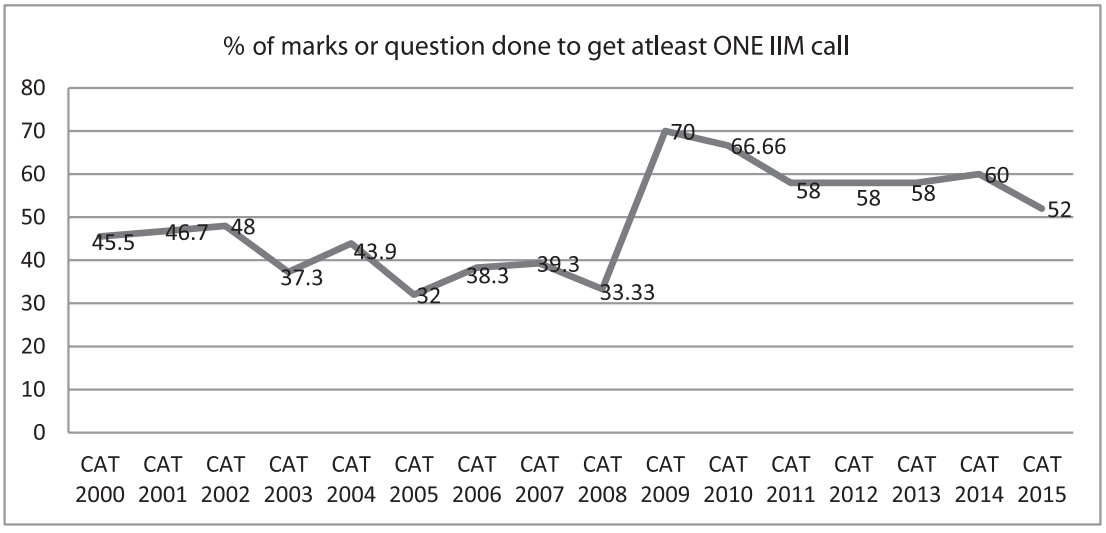
However, past CAT trends show that a student is required to get around 70% marks to get at least one IIM call (with clearing the sectional cut-off). The following table and bar chart give us some clarity regarding the same:

Chart 4

Year	Total marks or questions	Marks/Qs required to get at least one IIM call
CAT 2000	165	75
CAT 2001	150	70
CAT 2002	150	72
CAT 2003	150	56
CAT 2004	123	54
CAT 2005	150	48
CAT 2006	300	115
CAT 2007	300	118
CAT 2008	360	120
CAT 2009	60	42
CAT 2010	60	40
CAT 2011	60	35
CAT 2012	60	35
CAT 2013	60	35
CAT 2014	100	60
CAT 2015	100	52

Following line chart gives questions solved or marks required as a percentage of total marks or total questions (as applicable):

Chart 5



Note: Above calculation is based upon the data collected from the students who got IIM calls in that particular year.

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So, to get at least one IIM call in CAT 2008, a student was required to get 33.33% marks out of the total with clearing the cut-off across the sections. Although in the online format of the CAT (since 2009), percentage questions to be done to get at least one IIM call has gone up, it is primarily because the exam is perceived to be easier in its totality than pen and paper-based exams. A student might have got 10 easy questions out of 20 questions in a section.

If we convert the requirement of marks to be obtained from the above line chart into questions to be done, we get the following table:

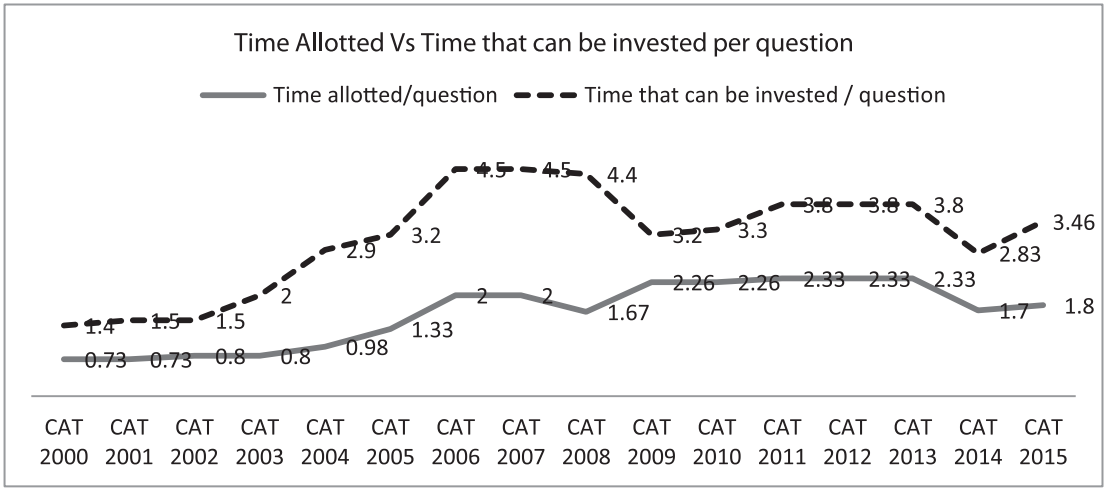
Chart 6

Year	Number of questions to be solved	Time allotted	Time invested per question
CAT 2000	85	120 minutes	1.4
CAT 2001	80	120 minutes	1.5
CAT 2002	81	120 minutes	1.5
CAT 2003	60	120 minutes	2.0
CAT 2004	42	120 minutes	2.9
CAT 2005	38	120 minutes	3.2
CAT 2006	33	150 minutes	4.5
CAT 2007	33	150 minutes	4.5
CAT 2008	34	150 minutes	4.4
CAT 2009	42	135 minutes	3.2
CAT 2010	40	135 minutes	3.3
CAT 2011	35	135 minutes	3.8
CAT 2012	35	135 minutes	3.8
CAT 2013	35	135 minutes	3.8
CAT 2014	60	170 minutes	2.83
CAT 2015	52	180 minutes	3.46

Chart 6—For CAT 2000 to CAT 2008, at 90% accuracy with 1/4 negative marking, these are the approximate number of questions to be done. For CAT 2009 to CAT 2011, net of these many questions is to be done.

To summarize this whole discussion till now, we will compare Chart 2 with Chart 6 and present them in a unified line chart given below (Chart 7).

Chart 7



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This line brings to us an important information—for CAT 2006, CAT 2007, or CAT 2008, even if a student has taken approximately 4.5 minutes to solve a question with 90% accuracy, s/he has got enough marks to get at least one IIM call (provided s/he clears the sectional cut-off too). For online formats for CAT, it is around 3.3 minutes per question. For CAT 2015, it was around 3.5 minutes.

So, it's more about accuracy now than speed.

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How to Prepare for QA

Purpose of QA Being Included in the CAT

Have you ever thought why Quantitative Aptitude (QA) questions are asked in the CAT? Is it to check your mathematical ability with questions from Algebra, or Permutation and Combination? If that were the case, they would have asked questions also from Group Theory and Applied Mathematics, which they don't. The questions asked in Quantitative Aptitude help to test the student's reasoning ability based on mathematical skills. They also test the student's comprehension of basic mathematical concepts. And even if somebody has not studied mathematics after class 10, they can solve these problems with a little practice. What is required is an understanding of basic concepts. It is, therefore, clear that through the test of QA, the examiners intend to check functional ability pertaining to basic mathematical operations.

Five-point Action Plan for QA

1. Address your math-phobia first You will not be in the best state of mind to crack the CAT if past failures or inabilities in mathematics haunt you. As it is said, it's okay to lose a couple of battles. Keep your eyes set on the war!
2. Make a roadmap 'Divide and Win' is the key. Your preparation should be divided into topics and every topic should be further divided into sub-topics. Once you have a topic-centric roadmap of the entire QA syllabus, carefully distribute the time you want to devote to each section on a daily basis.
3. Strengthen your fundamentals Till you have understood the concepts and their various applications thoroughly, do not start solving the problems. Let this process take some time. Now, before you start solving questions, go over the fundamental concepts once again and see which concept should be applied to get the right answer.
4. Adopt a systematic practice technique Don't jump to solving problems immediately. Make an effort to understand the basic theories behind the mathematical concepts, however trivial they may seem. And only then solve the problems.
5. Focus on weaknesses If students are weak in a particular area (say, geometry or permutations and combinations), they tend to neglect these questions and leave the solutions to choice. This is not only unproductive but also damaging. Identify the areas with which you are not comfortable. Numbers, geometry and algebra account for a major portion of the questions in the CAT; hence, they deserve due attention.

Consistency with Perseverance is the Key

If you do not find ways to solve a problem on the first day, sleep on it and tackle it the next day. Sometimes even the simplest of answers elude you and lead to frustration. Even the best of mathematicians face this problem; therefore, leave it for another day. And if you are unable to solve this problem even on the second day, try it on the third day or the third time. If you are still not able to solve it, check the solution provided at the end of the book or take help. The bottom line is that only consistent efforts and practice can bring positive results. If you want to go through this book in 10 days' time, you can do that also. And you will be able to learn the fundamentals, but remember it will be only short-term. You will be required to go through the same chore again after a few days and repeat this exercise.

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How to Use This Book

Let me share with you what my feelings were before I started working on this book. I thought of the kind of book I would prefer if I had to learn the theories of public administration when I do not have any background of this subject. I realized I would want a book which is strong in fundamental concepts but lucid in language; I would need a book which is self-explanatory but not verbose; I would need a book which is all-encompassing but not irrelevant; I would need a book with a good number of practice questions but no repetitive questions (if I cannot learn a concept with 200 relevant questions, chances are low that I will be able to learn that concept with 250 questions having the same level of relevance); I would need a book which can help me judge my progress from time to time. And this is what I have sincerely tried to provide in this book.

How to Go Ahead with This Book

Step 1 Go through the learning objectives and remain focused on them. After completing the chapter, check if these objectives have been fulfilled.

Step 2 Start with the concepts. Before moving on to the next concept, go through the worked-out examples related to that concept. Move ahead only when you have internalized them. Sometimes this might appear to be drudgery, but you must do it.

Step 3 Do the warm-up exercise (in QA part only). It is a precursor to the problems. Most of the questions in the warm-up exercise will help you check only your understanding of the concept and not the application of it. Ideally, you should not give more than 2 minutes to any question in the warm-up exercise. If you get less than 75 per cent questions correct, go over the concepts for which you got the answers wrong.

Step 4 Do the foundation exercise. Solve the problems in this set without any time constraint. This level tests your comprehension of the concept, and is a precursor to the application-based questions. Try to solve all the questions in one sitting, whatever time it takes (one or two breaks of 10 minutes can be taken). If you don't get the correct answers, attempt the same question once again in the next sitting (preferably the next day). This will help you to develop:

- A thorough understanding of the concepts
- An experience of the problems being asked at the basic level
- Confidence building

Step 5 Next is the moderate exercise. This level tests your ability to apply a particular concept, or a combination of concepts, to a problem. You might find that concepts of geometry are being used in problems on time, speed and distance. This will help you to develop:

- The ability to identify easy and difficult questions
- Mental images of the problems
- A logical connection between concepts and their application.

Step 6 Do not go for the advanced level immediately. Relax. Do not solve any QA question for one day. The following day, take the benchmarking test. Your performance in this test can tell you a lot about your progress. Ideally, you should aim for 85 percentile or more.

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Step 7 After you have done all the moderate exercises and benchmarking tests, go over the topics again. Only then attempt the advanced questions. Most of the questions here are above the regular CAT level. The idea is to prepare you for a higher level. If you excel at this level, CAT will be a cakewalk for you.

Step 8 Take the Review Tests. Target anything above 98 percentile. If you get it consistently in all the RTs, you are prepared for the CAT. If not, repeat the process from Step 5. If in any one of the RTs, you get less than 85 percentile, repeat the process from Step 4.

All the best!

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PART

1

QUANTITATIVE APTITUDE

MODULE 1: Numbers and their Properties

MODULE 2: Arithmetic

MODULE 3: X+2 Maths

MODULE 4: Measurement

MODULE 5: Miscellaneous

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MODULE 1

NUMBERS AND THEIR PROPERTIES

- Vedic Mathematics
- Number System

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CHAPTER 1

Vedic Mathematics

LEARNING OBJECTIVES

After completion of this chapter, the reader should be able to understand:

- How to do faster calculations?
- Multiplications
- Squares
- Cubes
- Properties of squares and cubes

INTRODUCTION TO THE TOPIC

Vedic mathematics is the ancient system of mathematics drawn from the Vedas. The Vedas are ancient texts that encompass a broad spectrum of knowledge, covering all aspects of life. These include the sutras (verses) pertaining to mathematics. In the early 20th century, Swami Shri Bharati Krishna Tirthaji Maharaja claimed to have rediscovered a collection of 16 ancient mathematical sutras from the Vedas and published it in a book titled *Vedic Mathematics*. Historians, however, do not agree on whether or not these were truly a part of the Vedic tradition. If these sutras dated back to the Vedic era, then they would be a part of an oral rather than a written tradition. In spite of these controversies, they are a novel and useful approach to computation; they are flexible in application and easy to remember. They can often be applied in the algebraic contexts and in simple arithmetic as well.

TYPES OF CALCULATIONS

The different types of calculations that form the basis of mathematics are:

1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Ratio comparison
6. Percentage calculations

When we talk about the techniques of calculations, addition and subtraction can simply not have any short-cuts. Since addition and subtraction are the basic units, we can at best only approximate the values.

In case of multiplication, the techniques of Vedic maths can be used.

Ratio comparison techniques are discussed in the chapter on ratio, proportion, and variation, and the percentage calculations in the chapter on percentage.

VEDIC MATH TECHNIQUES IN MULTIPLICATION

There are several techniques of multiplication. We will discuss them one by one.

Method 1: Base Method

In this method, one number is used as a base; for example, 10, 50, 100, etc. The number that is closer to both the numbers should be taken as the base.

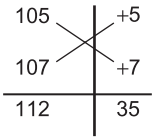
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Example 1 105×107

Solution In this case, both the numbers are close to 100, so 100 is taken as the base. We will now find the deficit/surplus from the base.

Base = 100, Surplus = 5 and 7



The right part (after slash) ⇒ this is the product of the surplus. Since the base = 100 and the surpluses are 5 and 7, the product would be $5 \times 7 = 35$.

The left part (before slash) ⇒ It could be either of the numbers plus the surplus of the other multiplicand. Hence, the left part would be either (105 + 7) or (107 + 5) = 112 (both will always be the same), i.e., 112.

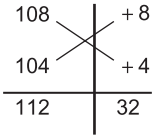
The left part would be equivalent to the number × 100. In this case, $112 \times 100 = 11{,}200$

Now, we add both the right part and the left part = 11,200 + 35 = 11,235

Hence, the result of the multiplication would be 11,235.

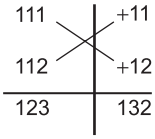
Example 2 108×104

Solution



Example 3 111×112

Solution

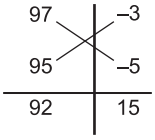


Here, it is $11 \times 12 = 132$. But it can have only two digits. Thus, 1 will be carried over to the left part and the right part will be only 32. Left part will be either 111 + 12 + 1 (1 for the carry over) or (112 + 11 + 1), i.e., 124. So, the result will be 12,432.

For 102×104, the answer will be 10,608. Please note that the right part will be 08 and not simply 8.

Example 4 97×95

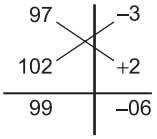
Solution



Base = 100, Deficit = 97 − 100 = −3 and 95 − 100 = −5

Example 5 97×102

Solution



97×102

Base = 100, Deficit = 97 − 100 = −3,

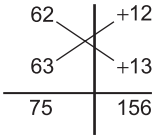
Surplus = 102 − 100 = 2

The right part will now be $(-3) \times 2$, i.e., −06. To take care of the negative, we will borrow 1 from the left part, which is equivalent to borrowing 100 (because we are borrowing from the hundred digits of the left part). Therefore, this part will be 100 − 06 = 94.

So, the answer = 9894

Example 6 62×63

Solution



We will assume here the base as 50 owing to the fact that the numbers are close to 50.

Base = 50, Surplus = 62 − 50 = 12,

Surplus = 63 − 50 = 13

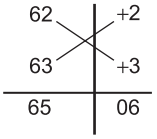
The left-hand side = 156 and the right-hand side = 75. Since the base is assumed to be equal to 50, so the value of the right-hand side = $75 \times 50 = 3750$. Besides, only two digits can be there on the right-hand side, so 1(100) is transferred to the left-hand side leaving 56 only on the left-hand side.

So, the value on the right-hand side = 3750 + 100 = 3850

Value on the left-hand side = 56

Net value = 3850 + 56 = 3906

Let us do the same multiplication by assuming 60 as the base.



Base = 60, Surplus = 62 − 60 = 2, Surplus = 63 − 60 = 3

Since the base is assumed to be equal to 60, the value of the right-hand side = $65 \times 60 = 130 \times 30 = 3900$

So, net value = 3906

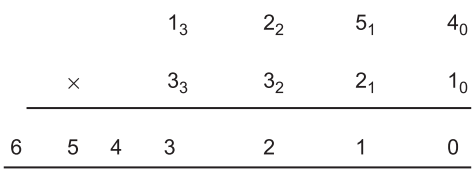
Method 2: Place Value Method

In this method of multiplication, every digit is assigned a place value and the multiplication is done by equating the place values of multiplicands with the place value of the product.

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Let us see this with some examples:

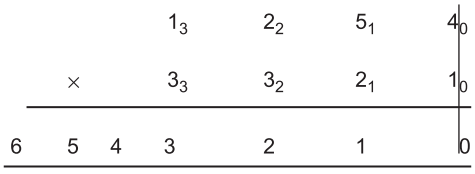


Conventionally, the unit digit is assigned a place value 0, the tens place digit is assigned a place value 1, the hundreds place digit is assigned a place value 2, the thousands place digits is assigned a place value 3 and so on.

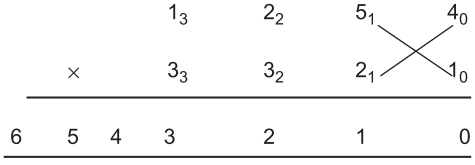
This multiplication is a two-step process.

Step 1 Add the place values of the digits of the numbers given (1254 × 3321) to obtain the place value of the digits of the product.

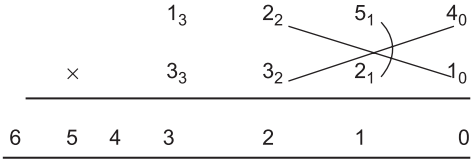
For example, using the place values of the multiplicands, i.e., using 0, 1, 2, and 3 of the number 1254 and the same place values 0, 1, 2, and 3 of the another multiplicand 3321, we can get 0 place value in the product in just one way, i.e., adding 0 and 0.



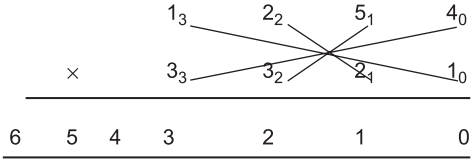
Place value 1 in the product can be obtained in two ways.



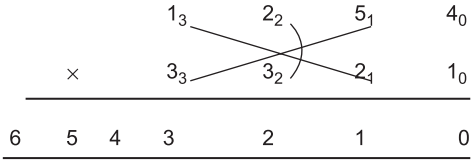
Place value 2 can be obtained in three ways.



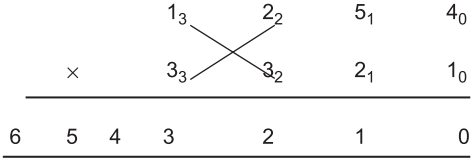
Place value 3 can be obtained in four ways.



Place value 4 can be obtained in three ways.



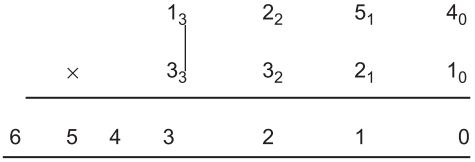
Place value 5 can be obtained in two ways.

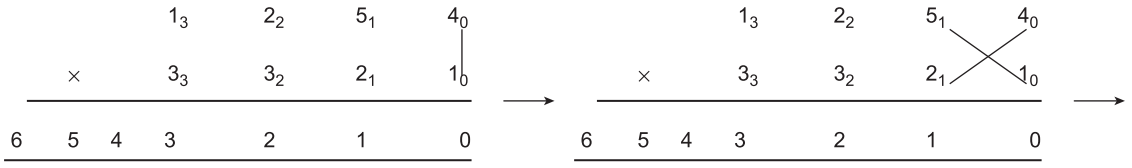


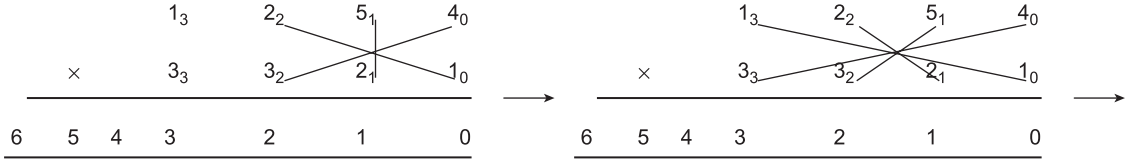
Place value 6 can be obtained in one way.

And this is the maximum place value that can be obtained.

Step 2 Multiply the corresponding numbers one by one.







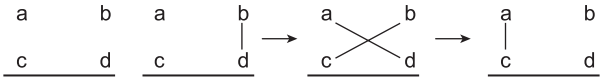
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In this manner, we can find the product = 41,64,534

This method is most useful in case of the multiplications of 2 digits × 2 digits or 2 digits × 3 digits or 3 digits × 3 digits multiplication.

Example $ab \times cd$



Similarly, we can have a proper mechanism of multiplication of 2 digits × 3 digits or 3 digits × 3 digits using the place value method.

Method 3: Units Digit Method

This method of multiplication uses the sum of the unit's digit, provided all the other digits on the left-hand side of the unit digit are the same.

Example 7 75×75

Solution

1.0 + 7	5
7	5
56	25

The sum of the units digit = 10, so we add 1.0 in one of the digits on the left-hand side.

Example 8 62×63

Solution

0.5 + 6	3
6	2
39	06

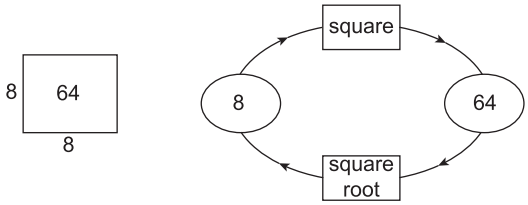
The sum of the units digit = 5, so we add 0.5 in one of the digits on the left-hand side.

SQUARING

A **square number**, also called a **perfect square**, is an integer that can be written as the square of some other integer. In other words, a number whose square root is an integer is known as the square number of a perfect square.

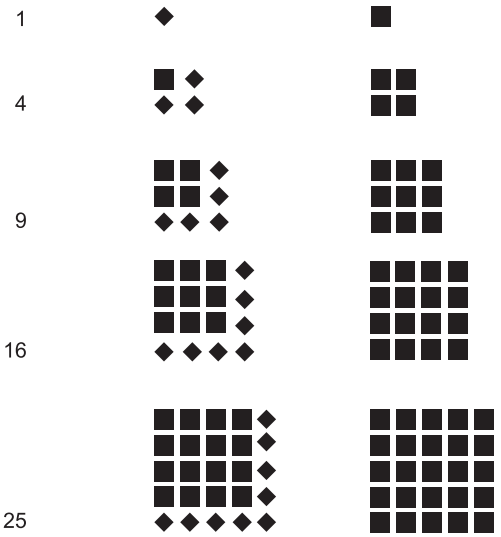
For example, 9 is a square number since it can be written as 3×3.

This can be seen through the following flow-chart also.



Properties of a Square Number

1. The number N is a square number if it can be arranged as N points in a square.



Therefore, it can be deduced that the formula for the nth square number is n^2. This is also equal to the sum of the first n odd numbers $n^2 = \sum_{k=1}^{n}(2k-1)$ as can be seen in the above figure, where a square results from the previous one by adding an odd number of points (marked as '◆'). For example, $5^2 = 25 = 1 + 3 + 5 + 7 + 9$.

It should be noted that the square of any number can be represented as the sum $1 + 1 + 2 + 2 + \ldots + n - 1 + n - 1 + n$. For instance, the square of 4 or 4^2 is equal to $1+1+2+2+3+3+4=16$. This is the result of adding a column and row of thickness 1 to the square graph of three. This can be useful for finding the square of a big number quickly. For instance, the square of $52 = 50^2 + 50 + 51 + 51 + 52 = 2500 + 204 = 2704$.

2. A square number can only end with digits 00, 1, 4, 6, 9, or 25 in base 10, as follows:
3. If the last digit of a number is 0, then its square ends in 00 and the preceding digits must also form a square.
4. If the last digit of a number is 1 or 9, then its square ends in 1 and the number formed by its preceding digits must be divisible by four.
5. If the last digit of a number is 2 or 8, then its square ends in 4 and the preceding digit must be even.
6. If the last digit of a number is 3 or 7, then its square ends in 9 and the number formed by its preceding digits must be divisible by four.
7. If the last digit of a number is 4 or 6, then its square ends in 6 and the preceding digit must be odd.

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8. If the last digit of a number is 5, then its square ends in 25 and the preceding digits (other than 25) must be 0, 2, 06, or 56.
9. A square number cannot be a perfect number. (If the sum of all the factors of a number excluding the number itself is equal to the number, then the number is known to be a perfect number.)
10. The digital sum of any perfect square can be only 0, 1, 4, 9, and 7. (Digital sum of any number is obtained by adding the digits of the number until we get a single digit. Digital sum of 385 = 3 + 8 + 5 = 1 + 6 = 7.)

An easy way to find the squares is to find two numbers that have a mean of it. This can be seen through the following example:

To find the square of 21, take 20 and 22, then multiply the two numbers together and add the square of the distance from the mean: $22 \times 20 = 440 + 1^2 = 441$. Here, we have used the following formula $(x - y)(x + y) = x^2 - y^2$ known as the difference of two squares. Thus,

$$(21 - 1)(21 + 1) = 21^2 - 1^2 = 440.$$

Odd and Even Square Numbers

Squares of even numbers are even, as $(2n)^2 = 4n^2$.

Squares of odd numbers are odd, as $(2n + 1)^2 = 4(n^2 + n) + 1$.

Hence, we can infer that the square roots of even square numbers are even and square roots of odd square numbers are odd.

Methods of Squaring

Like multiplication, there are several methods for squaring also. Let us see the methods one by one.

Method 1: Base 10 Method

Understand it by taking few examples:

- Let us find out the square of 9. Since 9 is 1 less than 10, decrease it still further to 8. This is the left side of our answer.
- On the right-hand side, put the square of the deficiency that is 1^2. Hence, the answer is 81.
- Similarly, $8^2 = 64$, $7^2 = 49$
- For numbers above 10, instead of looking at the deficit we look at the surplus. For example,
  $11^2 = (11 + 1); 10 + 1^2 = 121$
  $12^2 = (12 + 2); 10 + 2^2 = 144$
  $14^2 = (14 + 4); 10 + 4^2 = 18; 10 + 16 = 196$
  and so on

This is based on the identities $(a + b)(a - b) = a^2 - b^2$ and $(a + b)^2 = a^2 + 2ab + b^2$.

We can use this method to find the squares of any number, but after a certain stage, this method loses its efficiency.

Method 2: Base 50n Method here, (n is any natural number)

This method is nothing but the application of $(a + b)^2 = a^2 + 2ab + b^2$.

This can be seen in the following examples:

Example 9 Find the square of 62.

Solution Because this number is close to 50, we will assume 50 as the base.

$$(62)^2 = (50 + 12)^2 = (50)^2 + 2 \times 50 \times 12 + (12)^2$$
$$= 2500 + 1200 + 144$$

To make it self-explanatory, a special method of writing is used.

$(62)^2$ = [100's in (Base)]^2 + Surplus | Surplus^2
= 25 + 12 | 144 = 38 | 44 [Number before the bar on its left-hand side is number of hundreds and on its right-hand side is the last two digits of the number.]

$(68)^2 = 25 + 18 \mid 324 = 46 \mid 24$

$(76)^2 = 25 + 26 \mid 676 = 57 \mid 76$

$(42)^2 = 25 - 8 \mid 64 = 17 \mid 24$ $[(a - b)^2 = a^2 - 2ab + b^2]$

Example 10 Find the square of 112.

Solution Since this number is closer to 100, we will take 100 as the base.

$$(112)^2 = (100 + 12)^2 = (100)^2 + 2 \times 100 \times 12 + (12)^2 =$$
$$10{,}000 + 2 \times 1200 + 144$$

$(112)^2$ = [100's in (Base)]^2 + 2 × Surplus | Surplus^2

$= 100 + 2 \times 12 \mid 12^2 = 125 \mid 44$

Alternatively, we can multiply it directly using base value method.

Had this been 162, we would have multiplied 3 in surplus before adding it into [100's in (Base)]^2 because assumed base here is 150.

$(162)^2$ = [100's in (Base)]^2 + 3 × Surplus | Surplus^2

$= 225 + 3 \times 12 \mid 12^2 = 262 \mid 44$

Method 3: 10^n Method

This method is applied when the number is close to 10^n.

With base as 10^n, find the surplus or deficit (×).

Again, the answer can be arrived at in two parts.

$(B + 2x) \mid x^2$

The right-hand part will consist of n digits. Add leading zeros or carry forward the extra to satisfy this condition.

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$108^2 = (100 + 2 \times 8) \mid 8^2 = 116 \mid 64 = 11{,}664$

$102^2 = (100 + 2 \times 2) \mid 2^2 = 104 \mid 04 = 10{,}404$

$93^2 = (100 - 2 \times 7) \mid (-7)^2 = 86 \mid 49 \Rightarrow 8649$

$1006^2 = (1000 + 2 \times 6) \mid 6^2 = 10 \mid 12 \mid 036 = 10{,}12{,}036$

The right-hand part will consist of 2 digits. Add leading zeros or carry forward the extra to satisfy this condition.

$63^2 = (25 + 13) \mid 13^2 = 38 \mid 169 = 3969$

$38^2 = (25 - 12) + (-12)^2 = 13 \mid 144 = 1444$

Square Mirrors

$14^2 + 87^2 = 41^2 + 78^2$

$15^2 + 75^2 = 51^2 + 57^2$

$17^2 + 84^2 = 71^2 + 48^2$

$26^2 + 97^2 = 62^2 + 79^2$

$27^2 + 96^2 = 72^2 + 69^2$

Some Special Cases

1. Numbers ending with 5

If a number is in the form of $n5$, the square of it is $n(n + 1) \mid 25$

Example $45^2 = 4 \times 5 \mid 25 = 2025$

$135^2 = 13 \times 14 \mid 25 = 18{,}225$

This is nothing but the application of the multiplication method using the sum of unit's digits.

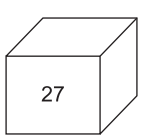
We can use this method to find out the squares fractions also like $1\frac{1}{2}, 2\frac{1}{2}, 3\frac{1}{2}$, etc.

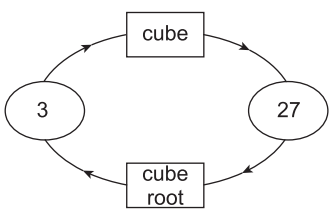
Process: Multiply the integral portion by the next higher integer and add $\frac{1}{4}$.

For example, $\left(6\frac{1}{2}\right) = 6 \times 7 + \frac{1}{4} = 42\frac{1}{4}$

CUBING

A number whose cube root is an integer is called a perfect cube.





Properties of a Cube

1. The sum of the cubes of any number of consecutive integers starting with 1 is the square of some integer. (e.g., $1^3 + 2^3 = 9 = 3^2$, $1^3 + 2^3 + 3^3 = 36 = 6^2$, etc.)
2. Unit digit of any cube can be any digit from 0 to 9.

Methods of Cubing

We can find the cube of any number close to a power of 10 say 10^n with base = 10^n by finding the surplus or the deficit (x). The answer will be obtained in three parts. $B + 3x \mid 3 \cdot x^2 \mid x^3$

The left two parts will have n digits.

104^3

Base B = 100 and surplus = $x = 4$

$(100 + 3 \times 4) \mid 3 \times 4^2 \mid 4^3 = 112 \mid 48 \mid 64 = 11{,}24{,}864$

109^3

Base B = 100 and $x = 9$

$(100 + 3 \times 9) \mid 3 \times 9^2 \mid 9^3 = 127 \mid 243 \mid 729 = 12{,}95{,}029$

98^3

Base B = 100 and $x = -2$

$(100 - 3 \times 2) \mid 3 \times (-2)^2 \mid (-2)^3 = 94 \mid 12 \mid -8 = 94 \mid 11 \mid 100 - 8 = 941{,}192$

VEDIC MATHS TECHNIQUES IN ALGEBRA

1. If one is in ratio, the other one is zero

This formula is often used to solve simple simultaneous equations that may involve big numbers. However, these equations in special cases can be visually solved because of a certain ratio between the co-efficients. Consider the following example:

$6x + 7y = 8$

$19x + 14y = 16$

Here, the ratio of co-efficients of y is the same as that of the constant terms. Therefore, the 'other' is zero, i.e., $x = 0$. Hence, the solution of the equations is $x = 0$ and $y = 8/7$.

Alternatively,

$19x + 14y = 16$ is equivalent to $(19/2)x + 7y = 8$.

Therefore, x has to be zero and no ratio is needed; just divide by 2!

Note that it would not work if both had been 'in ratio':

$6x + 7y = 8$

$12x + 14y = 16$

This formula is easily applicable to more general cases with any number of variables. For instance,

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$$ax + by + cz = a$$

$$bx + cy + az = b$$

$$cx + ay + bz = c$$

which yields $x = 1$, $y = 0$, and $z = 0$

2. When samuccaya is the same, then that samuccaya is zero

Consider the following symbols: N_1 – Numerator 1, N_2 – Numerator 2, D_1 – Denominator 1, D_2 – Denominator 2 and so on.

This formula is useful for solving equations that can be solved visually. The word 'samuccaya' has various meanings in different applications. For instance, it may mean a term, which occurs as a common factor in all the terms concerned. For example, an equation '$12x + 3x = 4x + 5x$'. Since 'x' occurs as a common factor in all the terms, therefore, $x = 0$ is the solution. Alternatively, samuccaya is the product of independent terms. For instance, in $(x + 7)(x + 9) = (x + 3)(x + 21)$, the samuccaya is $7 \times 9 = 3 \times 21$; therefore, $x = 0$ is the solution. It is also the sum of the denominators of two fractions having the same numerical numerator, for example:

$1/(2x - 1) + 1/(3x - 1) = 0$ means $5x - 2 = 0$

The more commonly used meaning is 'combination' or total. For instance, if the sum of the numerators and the sum of denominators are the same, then that sum is zero. Therefore,

$$\frac{2x+9}{2x+7} = \frac{2x+7}{2x+9}$$

Therefore, $4x + 16 = 0$ or $x = -4$

This meaning ('total') can also be applied in solving the quadratic equations. The total meaning not only imply sum but also subtraction. For instance, when given $N_1D_1 = N_2/D_2$, if $N_1 + N_2 = D_1 + D_2$ (as shown earlier), then this sum is zero. Mental cross multiplication reveals that the resulting equation is quadratic (the co-efficients of x^2 are different on the two sides). So, if $N_1 - D_1 = N_2 - D_2$, then that samuccaya is also zero. This yields the other root of a quadratic equation.

The interpretation of 'total' is also applied in multi-term RHS and LHS. For instance, consider

$$\frac{1}{x-7} + \frac{1}{x-9} = \frac{1}{x-6} + \frac{1}{x-10}$$

Here, $D_1 + D_2 = D_3 + D_4 = 2x - 16$. Therefore, $x = 8$.

There are several other cases where samuccaya can be applied with great versatility. For instance, 'apparently cubic' or 'biquadratic' equations can be easily solved as shown below:

$$(x - 3)^2 + (x - 9)^3 = 2(x - 6)^3$$

Note that $x - 3 + x - 9 = 2(x - 6)$. Therefore, $(x - 6) = 0$ or $x = 6$.

Consider

$$\frac{(x+3)^3}{(x+5)^3} = \frac{x+1}{x+7}$$

Observe: $N_1 + D_1 = N_2 + D_2 = 2x + 8$

Therefore, $x = -4$

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CHAPTER

2

Number System*

LEARNING OBJECTIVES

After completion of this chapter, the reader should be able to understand:

- Numbers and their different types
- Definitions and properties of these numbers
- Concepts related to these numbers
- Different types of questions that are covered in the CAT
- Methods for solving these questions

INTRODUCTION

Here, we will be discussing the relative importance of various concepts in the number system with respect to CAT preparation as it has been one of the important topics in QA historically. From the past 15 years, CAT paper consisted of questions (almost 20%) from the number system. However, in this chapter, logic has an important role to play than the numbers. In other words, we can say that the logical processes take precedence over calculations in finding solution to exceptionally complex mathematical problems in number system. Students are expected to have a clear understanding of the definitions as well as concepts and develop a keen insight on numbers and their properties. Apart from these skills, they should try to maximize the potential of learning and solving every question.

The questions are asked based on the following two approaches:

1. ***Definitions and properties of numbers:*** In this section, questions will be based upon the definitions of different kinds of numbers. Apart from this, questions can be asked from some of the very basic calculations, formula, or properties of numbers.
2. ***Concepts:*** Some of the concepts on which questions are being asked are as follows:
   (a) LCM and HCF
   (b) Divisibility rules (for base 10)
   (c) Number of divisors
   (d) Number of exponents
   (e) Remainders
   (f) Base system
   (g) Units digit
   (h) Tens digit
   (i) Pigeonhole principle

CLASSIFICATION OF NUMBERS/ INTEGERS

Natural Numbers

Natural numbers are counting numbers, that is, the numbers that we use to count any number of things. For example, 1, 2, 3, ….

The lowest natural number is 1.

Whole Numbers

When zero is included in the list of natural numbers, then they are known as whole numbers. For example, 0, 1, 2, …

The lowest whole number is 0.

Integers

Integers are whole numbers, negative of whole numbers, and zero. For example, 43, 434, 235, 28, 2, 0, –28, and –3030

*To read more about Number System, you may use the book "Number System For CAT" by Nishit Sinha.

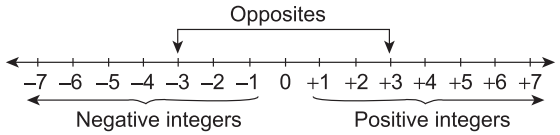
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are integers; however, numbers such as 1/2, 4.00032, 2.5, Pi, and –9.90 are not whole numbers.

Number Line

The number line is used to represent the set of real numbers. The following is the brief representation of the number line:



Properties of Number Line

1. The number line goes till infinity in both directions. This is indicated by the arrows.
2. Integers greater than zero are called positive integers. These numbers are to the right of zero on the number line.
3. Integers less than zero are called negative integers. These numbers are to the left of zero on the number line.
4. The integer zero is neutral. It is neither positive nor negative.
5. The sign of an integer is either positive (+) or negative (–), except zero, which has no sign.
6. Two integers are opposites if each of them is at the same distance from zero, but on opposite sides of the number line. One will have a positive sign, and the other will have a negative sign. In the abovementioned number line, +3 and –3 are labelled as opposites. In other words, the whole negative number scale looks like a mirror image of the positive number scale, with a number like –15 being the same distance away from 0 as for number 15.
7. The number halfway between –1 and –2 is –1.5 and the number half way between 1 and 2 is 1.5.
8. We represent positive numbers without using a positive sign. For example, we would write 29.1 instead of +29.1. However, when we talk of negative numbers, the sign must be present.

Prime Numbers and Composite Numbers

Prime Numbers

Among natural numbers, we can distinguish prime numbers and composite numbers.

All the numbers that are divisible by 1 and itself only are known as prime numbers.

As mentioned earlier, primes can be natural numbers only. In other words, we can say that all the numbers that have only two factors are known as prime numbers. Prime numbers can also be seen as the building blocks. Further, we combine two or more than two or same or distinct prime numbers to create numbers larger than these prime numbers,

For example, $3 \times 2 = 6$

The following is the list of all prime numbers that are less than 1000.

2	3	5	7	11	13	17	19	23	29	31	37	41	43	47	53	59	61
67	71	73	79	83	89	97	101	103	107	109	113	127	131	137	139	149	151
157	163	167	173	179	181	191	193	197	199	211	223	227	229	233	239	241	251
257	263	269	271	277	281	283	293	307	311	313	317	331	337	347	349	353	359
367	373	379	383	389	397	401	409	419	421	431	433	439	443	449	457	461	463
467	479	487	491	499	503	509	521	523	541	547	557	563	569	571	577	587	593
599	601	607	613	617	619	631	641	643	647	653	659	661	673	677	683	691	701
709	719	727	733	739	743	751	757	761	769	773	787	797	809	811	821	823	827
829	839	853	857	859	863	877	881	883	887	907	911	919	929	937	941	947	953
967	971	977	983	991	997												

Using the following table, we can find number of prime numbers between every 100 numbers.

Numbers from-to	1–100	101–200	201–300	301–400	401–500	501–600	601–700	701–800	801–900	901–1000
Number of primes	25	21	16	16	17	14	16	14	15	14

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Today, the largest known prime number is 78,16,230-digit prime number $2^{259,64,951} - 1$. It was found in early 2005; but, how big have the 'largest known primes' been historically?, and when might we see the first billion-digit prime number?

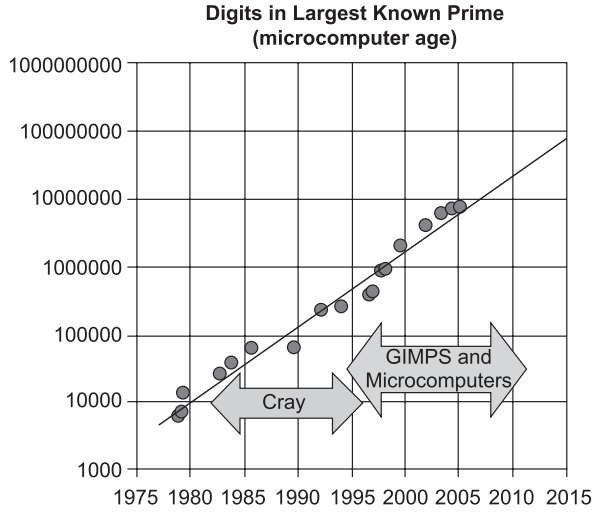
The following are the records that were used before the invention of electronic computers.

Number	Digits	Year	Prover	Method
$2^{17} - 1$	6	1588	Cataldi	Trial division
$2^{19} - 1$	6	1588	Cataldi	Trial division
$2^{31} - 1$	10	1772	Euler	Trial division
$(2^{59} - 1)/179,951$	13	1867	Landry	Trial division
$2^{127} - 1$	39	1876	Lucas	Lucas sequences
$(2^{148} + 1)/17$	44	1951	Ferrier	Proth's theorem

Until 1951, the prime number found by Lucas in 1876 was accepted as the largest prime number. In 1951, Ferrier used a mechanical desk calculator and techniques that are based on partial inverses of Fermat's little theorem (see the pages on remainder theorem). Using these techniques, he slightly improved this record by finding a 44-digit prime number.

Thus, in 1951, Ferrier found the prime $(2^{148} + 1)/17 = 209889366574405864861512642566102225938639 21$.

When will we have a one billion-digit prime number?



However, this record was very short-lived. In the same year (i.e., 1951), the advent of electronic computers helped human being in finding a bigger prime number than the earlier one.

In 1951, Miller and Wheeler began the electronic computing age by finding several primes and they had a new 79-digit prime number: $2^{127} - 1$. In addition, we know that this was the computer age and everybody was working hard to find the primes with the help of computers. Therefore, records were broken with a never-before pace.

Can we have a single formula representing all the prime numbers?

Until now, all the attempts done in this regard have proved to be fruitless. This is because there is no symmetricity between the differences among the prime numbers. Sometimes, two consecutive prime numbers differ by 2, 4, 10,000 or more. Therefore, there is no standard formula that can represent the prime numbers.

However, there are some standard notations that give us limited number of prime numbers: $N^2 + N + 41$. For all the values of N from -39 to $+39$, this expression gives us a prime number. Another similar example is $N^2 + N + 17$.

It is necessary to remember that all the prime numbers (>3) are of the form $6n \pm 1$ form (where n is any natural number); that is, all the prime numbers (>3) when divided by 6 give either 1 or 5 as the remainder.

Note: Here, it is important to know that if a number gives a remainder of 1 or 5 when divided by 6, it is not necessarily a prime number. For example, 25 when divided by 6 gives remainder = 1; however, 25 is not a prime number.

Composite Numbers

A number is composite if it is the product of two or more than two distinct or same prime numbers. For example, 4, 6, 8,....

$4 = 2^2$

$6 = 2^1 \times 3^1$

The lowest composite number is 4.

All the composite numbers will have at least 3 factors.

Even and Odd Numbers

Let us assume N as an integer. If there exists an integer P such that $N = 2P + 1$, then ***N* is an odd number**. If there exists an integer P such that $N = 2P$, then ***N* is an even number**.

In simple language, even numbers are those integers that are divisible by 2 and odd numbers are those integers that are not divisible by 2. Even and odd numbers can be positive as well as negative also.

In other words, if x is an integer (even or not), then $2x$ will be an even integer; this is because it is a multiple of 2. Further, x raised to any positive integer power will be an even number, and therefore, x^2, x^3, x^4, etc., will be even numbers.

Any integer that is not a multiple of 2 is called an *odd* number. For instance, -1, 3, 6883, and -8147 are all odd numbers. Any odd number raised to a positive integer power

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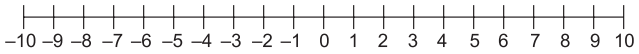
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will also be an odd number, and therefore, if x is an odd number, then x^2, x^3, x^4, etc., will be odd numbers.

The concept of even and odd numbers are most easily understood in the binary base. The abovementioned definition simply states that even numbers end with a 0 and odd numbers end with a 1.

Comparing Integers

We can compare two different integers by looking at their positions on the number line. For any two different places on the number line, the integer on the right-hand side is greater than the integer on the left-hand side. Note that every positive integer is greater than any negative integer.



For example, $9 > 4$, $6 > -9$, $-2 > -8$, and $0 > -5$, $-2 < 1$, $8 < 11$, $-7 < -5$, and $-10 < 0$

Remember

1. 1 is neither prime nor composite.
2. 0 is neither positive nor negative.

Example 1 Two of a, b, c, and d are even and two are odd, not necessarily in order. Which of the following is definitely even?

(a) $a + b + c - 2d$ (b) $a + 2b - c$
(c) $a + b - c + d$ (d) $2a + b + c - d$

Solution Since we do not know which two are even and two are odd, we will have to do trial-and-error method to solve this problem using the given options.

In option (a), if a and b are even, and c and d are odd, then this will lead us to odd number.

In option (b), if a and b are even, and c is odd, then this will lead us to odd number.

In option (d), if a and b are odd, and c and d are even, then this will lead us to odd number.

In option (c), whatever is the value of a, b, c, and d, it is always going to be an even number.

Thus, we can say that any type of calculation done with two even and two odd numbers will always result in an even number. Hence, the answer is option (c).

Example 2 If N, $(N + 2)$, and $(N + 4)$ are prime numbers, then the number of possible solutions for N is/are **(*CAT 2003*)**

(a) 1 (b) 2
(c) 3 (d) None of these

Solution There is only one triplet of prime numbers where difference between any two prime number is 2, that is, 3, 5, and 7. Therefore, $N = 3$ is the only solution. Hence, answer is (a).

Proof We know that the prime numbers are of the form $6M \pm 1$ (except 2 and 3). Now, if N is of the format $(6M + 1)$, then $(N + 2)$ will be of $(6M + 3)$ format and $(N + 4)$ will be of $(6M + 5)$ format. From these three numbers, since $(N + 2)$ is of $(6M + 3)$ format, it will be divisible by 3.

Similarly, if N is of the format $(6M - 1)$, then $(N + 2)$ will be of $(6M + 1)$ format and $(N + 4)$ will be of $(6M + 3)$ format. From these three numbers, since $(N + 4)$ is of $(6M + 3)$ format, it will be divisible by 3.

In both the cases, we find that one number that is given three numbers and it is divisible by 3. In the abovementioned example, (3, 5, and 7), one of the given three numbers is divisible by 3.

Example 3 Let x and y be the positive integers such that x is prime and y is composite. Then, which of the following is true? **(*CAT 2003*)**

(a) $y - x$ cannot be an even integer.
(b) xy cannot be an even integer $x + y$
(c) $\frac{x+y}{x}$ cannot be an even integer.
(d) None of these

Solution Eliminating the options,

To eliminate option (a): If $y = 4$ and $x = 2$, then $y - x$ can be even.

To eliminate option (b): If $y = 4$ and $x = 2$, then yx can be even.

To eliminate option (c): If $y = 6$ and $x = 2$, then it can also be even.

Therefore, answer is option (d).

QUESTIONS BASED UPON CONCEPTS

LCM

A common multiple is a number that is a multiple of two or more than two numbers. The common multiples of 3 and 4 are 12, 24, ...

The least common multiple (LCM) of two numbers is the smallest positive number that is a multiple of both.

Multiples of 3 — 3, 6, 9, 12, 15, 18, 21, 24, ...

Multiples of 4 — 4, 8, 12, 16, 20, 24, 28, ...

Therefore, LCM of 3 and 4 will be 12, which is the lowest common multiple of 3 and 4.

First of all, the basic question which lies is—for what kind of numbers, we can use LCM?

Let us explain it through an example: LCM of 10, 20, and 25 is 100. It means that 100 is the lowest number, which is divisible by all these three numbers.

Since (–100) is lower than 100 and divisible by each of 10, 20, and 25, can the LCM be (–100)? or can it be 0?

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Further, what will be the LCM of (–10) and 20? Will it be (–20) or (–200) or (–2000) or smallest of all the numbers, that is, <–∞?

Answer to all these questions is very simple: LCM is a concept defined only for positive numbers, whether the number is an integer or a fraction. In other words, **LCM is not defined for negative numbers or zero**.

Now, we will define a different method for finding the LCM of two or more than two positive integers.

Process to Find LCM

Step 1 Factorize all the numbers into their prime factors.
Step 2 Collect all the distinct factors.
Step 3 Raise each factor to its maximum available power and multiply.

Example 4 LCM of 10, 20, 25.

Solution

Step 1 $10 = 2^1 \times 5^1$
$20 = 2^2 \times 5^1$
$25 = 5^2$
Step 2 2, 5
Step 3 $2^2 \times 5^2 = 100$

One of the principal advantage of using this method is that we can find the LCM of any number of numbers in a straight line without using the conventional method. The following explains this using the previous example:

The LCM of 10 and 20 = 20, and LCM of 20 and 25 = 100 (For this, we have to know which factor of 25 is not present in 20; then, we need to multiply it by this factor. Therefore, 25 is having 5^2 and 20 is having 5^1 only, and hence, we will multiply 20 by 5.)

Example 5 LCM of 35, 45, 55.

Solution First, let us determine the LCM of 35 and 45.

Now, $35 = 5^1 \times 7^1$ and $45 = 3^2 \times 5^1$.

Therefore, it can be observed here that 35 is not having 3^2 in it, and hence, we will multiply 35 by 3^2.

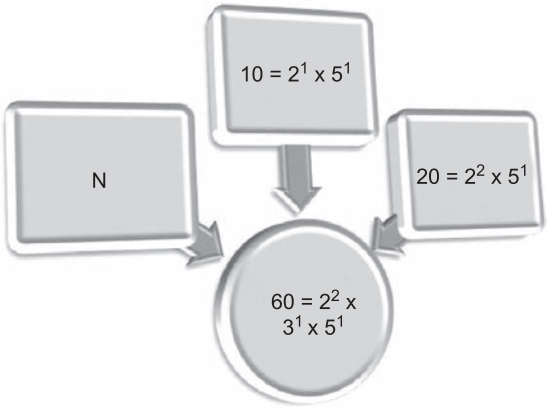
Thus, LCM of 35 and 45 = 35×3^2 (Further, you can start with 45 to find the missing factors of 35 in 45.)

Now, we will find the LCM of 35×3^2 and $55 = 5^1 \times 11^1$
$55 = 5^1 \times 11^1$

Now, 11^1 is not present in 35×3^2. Therefore, we will multiply 35×3^2, and finally, LCM = $35 \times 3^2 \times 11^1 = 3465$.

Example 6 LCM of three natural numbers 10, 20, and $N = 60$. How many values of N are possible?

Solution We have already discussed that to generate the LCM, we must multiply the prime numbers with the highest available power. Therefore, let us start with factorizing the number.



$2^2 \times 5^1$ is already present in 20; however, 3 is not present in either 10 or 20. Therefore, we can conclude that 3^1 has to be from N. This is the minimum value of $N = 3$. Second, we can also say that N may contain powers of 2 and 5 as long as the maximum power of 2 = 2 and maximum power of 5 = 1 (as in $2^2 \times 5^1$).

Therefore, the total different values of $N = (3^1 \times 2^0 \times 5^0)$, $(3^1 \times 2^1 \times 5^0)$, $(3^1 \times 2^2 \times 5^0)$, $(3^1 \times 2^0 \times 5^1)$, $(3^1 \times 2^1 \times 5^1)$, $(3^1 \times 2^2 \times 5^1) = 3, 6, 12, 15, 30, 60 = 6$ values

Highest Common Factor (HCF)

The factors that are positive integral values of a number and can divide that number is called HCF. HCF, which is also known as Greatest Common Divisor (GCD), is the highest value that can divide the given numbers.

Factors of 20 — 1, 2, 4, 5, 10, 20.
Factors of 30 — 1, 2, 3, 5, 6, 10, 15, 30.
Therefore, 10 will be the HCF of 20 and 30.

Process to Find HCF

Step 1 Factorize all the numbers into their prime factors.
Step 2 Collect all the common factors.
Step 3 Raise each factor to its minimum available power and multiply.

Example 7 HCF of 100, 200, and 250

Solution

Step 1 $100 = 2^2 \times 5^2$
$200 = 2^3 \times 5^2$
$250 = 5^3 \times 2^1$
Step 2 2, 5
Step 3 $2^1 \times 5^2 = 50$

Alternatively, to find HCF of numbers such as 100, 200, and 250, we have to observe the common quantity that can be taken from these numbers. To do this, we can write these

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numbers as $(100x + 200y + 250z)$, and now, it can be very easily observed that we can take 50 as the common number from the given numbers.

The LCM and HCF can be summarized as follows: it is very essential to understand the mechanism of determining LCM and HCF. These two concepts can be understood easily by the following example:

Example 8 Find the LCM and HCF of 16, 12, 24.

Solution

No.	Multiples	Factors
16	16, 32, 48, 64, 80, 96, 112, 128, ...	1, 2, 4, 8, 16
12	12, 24, 36, 48, 60, 72, 84, 96, 108, ...	1, 2, 3, 4, 6, 12
24	24, 48, 72, 96, 120, 144, 168, 192, ...	1, 2, 3, 4, 6, 8, 12, 24

Common Multiple	Common Factor
48	1, 2, 3, 4
Lowest common multiple	Highest common factor
48	4

The standard formulae are as follows:

1. LCM × HCF = product of two numbers.
   This formula can be applied only in the case of two numbers. However, if the numbers are relatively prime to each other (i.e., HCF of numbers = 1), then this formula can be applied for any number of numbers.
2. LCM of fractions = LCM of numerator of all the fractions/HCF of denominator of fractions.
3. HCF of fractions = HCF of numerator of all the fractions/LCM of denominator of fractions.
4. HCF of (sum of two numbers and their LCM) = HCF of numbers.

Example 9 HCF of two natural numbers A and B is 120 and their product is 10,000. How many sets of values of A and B is/are possible?

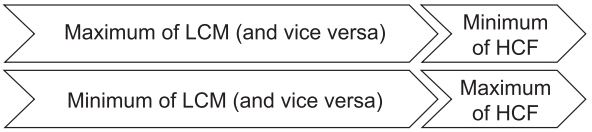
Solution HCF $(A, B) = 120 \Rightarrow 120$ is a common factor of both the numbers (120 being the HCF). Hence, 120 is present in both the numbers. Therefore, the minimum product of A and $B = 120 \times 120 = 14{,}400$. Hence, no set of A and B are possible for satisfying the conditions.

Maxima and Minima in LCM/HCF

If the product of two numbers is given, and none of LCM or HCF is given, then this gives rise to the case of maxima and minima.

Primarily, we use the formula LCM × HCF = product of two numbers. Although this formula only provides the basic framework, we need to visualize the situation to solve these questions.

By using the formula, LCM × HCF = product of two number, we can say that since RHS is constant, LHS will be inversely proportional to HCF (subject to the values being natural numbers).



Example 10 Product of two natural numbers = 144. What is the (a) largest possible and (b) smallest possible HCF of these two natural numbers?

Solution Let us first factorize $144 = 12 \times 12$
$= (2^2 \times 3) \times (2^2 \times 3)$

The largest possible HCF occurs when LCM = HCF and numbers are equal.

We already know that product of two natural numbers = LCM × HCF.

Since numbers have to be equal, each of the numbers = 12, and the largest possible HCF = 12.

Therefore, the smallest possible HCF has to be equal to 1 (possible set of numbers = 144, 1).

Example 11 Product of two natural numbers = 144. How many different values of LCM are possible for these two natural numbers?

Solution We have already seen in the abovementioned question that the largest possible value of HCF = 12 and consequently, small values of LCM = 12.

Let us see the different values of HCF and corresponding LCM.

Therefore, total different values of LCM = 6

HCF = 12	HCF = 6	HCF = 4	HCF = 3	HCF = 2	HCF = 1
LCM = 12	LCM = 24	LCM = 36	LCM = 48	LCM = 72	LCM = 144

The following gives some questions based on the standard application of LCM and HCF:

Case I Time and Work

Example 12 Tatto can do a work in 10 days and Tappo can do the same work in 12 days. How many days will it take if both start working together?

Solution Let us assume total work = LCM of (10, 12) units = 60 units. Now, 60 units of work is being done by Tatto in 10 days and Tatto is doing 6 units of work per day; similarly, Tappo is doing 5 units of work per day. Hence, they are doing 11 units of work in one day together.

Thus, they will take $\frac{60}{11} = 5\frac{5}{11}$ days to complete the work.

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Case II Time, speed, and distance — circular motion

Example 13 The speed of *A* is 15 m/s and speed of *B* is 20 m/s. They are running around a circular track of length 1000 m in the same direction. Let us find after how much time, will they meet at the starting point if they start running at the same time.

Solution Time taken by *A* and *B* in taking one circle are 66.66 s and 50 s, respectively. Therefore, LCM (66.66, 50) = 20s.

Case III Number system— tolling the bell

Example 14 There are two bells in a temple. Both the bells toll at a regular interval of 66.66 s and 50 s, respectively. After how much time, will they toll together for the first time?

Solution Time taken by Bell 1 and Bell 2 to toll is 66.66 s and 50 s. Therefore, LCM (66.66, 50) = 200 s.

Here, it can be observed that the mathematical interpretation of both the questions are same, only the language has been changed.

Case IV Number System— Number of Rows

Example 15 There are 24 peaches, 36 apricots, and 60 bananas and they have to be arranged in several rows in such a way that every row contains same number of fruits of one type. What is the minimum number of rows required for this arrangement?

Solution We can arrange one fruit in one row, and still in (24 + 36 + 60) = 120 rows, we can arrange all the fruits. Further, even we can arrange two fruits in one row and can arrange all the fruits in 60 rows. However, for the rows to be minimum, the number of fruits should be maximum in one row.

HCF of 24, 36, 60 = 12, and therefore, 12 fruits should be there in one row.

Hence, the number of rows = 10

Case V Number System— finding remainder

Example 16 Find the lowest three-digit number that when divided by 4 and 5 gives 3 as the remainder.

Solution Let us assume that there is no remainder. Therefore, the number has to be a multiple of LCM of 4 and 5. Now, LCM (4, 5) = 20

However, there is a remainder of 3 when divided by 4 and 5. Therefore, the number will be in the form of (207*N* + 3).

Hence, numbers are 23, 43, 63, 83, 103, and so on.

Thus, the three-digit number is 103.

Divisibility Rules (For Decimal System)

Divisibility rules are quite imperative. This is because using this, we can infer if a particular number is divisible by other number or not, without actually dividing it.

Divisibility rules of numbers are specific to that particular number only. It simply means that divisibility rules of different numbers will be different. We shall now see a list of divisibility rules for some of the natural numbers.

Divisibility Rules

For 2 If units digit of any number is 0, 2, 4, 6 or 8, then that number will be divisible by 2.

For 3 If sum total of all the digits of any number is divisible by 3, then the number will be divisible by 3 (e.g., 123, 456, etc.)

Example 17 How many values of *A* are possible if 3245684 *A* is divisible by 3?

Solution Sum total of the number = 32 + *A*

For this number to be divisible by 3, *A* can take three values namely 1 or 4 or 7. (No other values are possible since *A* is the units digit of the number.)

For 4 If the last two digit of a number is divisible by 4, then that number will be divisible by 4 (e.g., 3796, 248, 1256, etc.)

For 5 If the last digit of the number is 5 or 0, then that number will be divisible by 5.

For 6 If the last digit of the number is divisible by two and sum total of all the digits of number is divisible by 3, then that number will be divisible by 6.

For 7 The integer is divisible by 7 if and only if the difference of the number of its thousands and the remainder of its divisible by 1000 is divisible by 7.

Example: Let us take the number 795. The units digit is 5 and when it is doubled, we get 10. The remaining part of the number (i.e., the tens) is 79. If 10 is subtracted from 79, we get 69. Since this result is not divisible by 7, the original number 695 is also not divisible by 7.

For 8 If the last 3 digits of number is divisible by 8, then the number itself will be divisible by 8 (e.g., 128, 34568, 76232, etc).

For 9 If the sum of digits of the number is divisible by 9, then the number will be divisible by 9 (e.g., 1,298,35,782).

1 + 2 + 9 + 8 + 3 + 5 + 7 + 8 + 2 = 45. Since 45 is divisible by 9, number will be divisible by 9.

If units digit of any number is 0, 2, 4, 6, or 8, then that number will be divisible by 2.

Example 18 How many pairs of *A* and *B* are possible in number 89765*A*4*B* if it is divisible by 9, given that the last digit of number is even?

Solution Sum of the digits of number is 8 + 9 + 7 + 6 + 5 + *A* + 4 + *B* = 39 + *A* + *B*.

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Therefore, $(A + B)$ should be 6 or 15. Next value should be 24; since A and B are digits, so it cannot be more than 18. Possible pairs of A and B are as follows:

A	B
0	6
1	5
2	4
3	3
4	2
5	1
6	0
7	8
8	7
9	6
6	9

Since B is even, six possible set of values of A and B are there.

For 11 A number is divisible by 11, if the difference between the sum of the digits at even places and the sum of the digits at odd places is divisible by 11 (zero is divisible by 11).

For example**,** 65,95,149 is divisible by 11 as the difference of $6 + 9 + 1 + 9 = 25$ and $5 + 5 + 4 = 14$ is 11.

For 12 If the number is divisible by 3 and 4, then the number will be divisible by 12 (e.g., 144, 348).

For 13 $(A + 4B)$, where B is the units place digit and A is all the remaining digits.

For example, let us check the divisibility of 1404 by 13. Here, $A = 140$ and $B = 4$, then $A + 4B = 140 + 4 \times 4 = 156$. This 156 is divisible by 13, and therefore, 1404 will be divisible by 13.

For 14 If the number is divisible by both 2 and 7, then the number will be divisible by 14.

For 15 A number is divisible by 15, if the sum of the digits is divisible by 3 and units digit of the number is 0 or 5.

For example**,** 225, 450, 375, etc.

For 16 A number is divisible by 16, if the number formed by the last 4 digits of the given number is divisible by 16.

For example, 125,78,320 is divisible by 16, since the last 4 digits of the number 8320 is divisible by 16.

For 17 $(A - 5B)$ where B is the unit's place digit and A is all the remaining digits.

For 18 Number should be divisible by both 9 and 2.

For 19 $(A + 2B)$ where B is the unit's place digit and A is all the remaining digits.

If the sum of the number of tens in the number and twice the units digit is divisible by 19, then the number is divisible by 19.

For example, let us take 665. The units digit is 5 and when it is doubled, we get 10. The remaining part of the number is 66. If 10 (which is the units digit doubled) is added to 66, we get 76. Since this result (76) is divisible by 19, it means the original number 665 is also divisible by 19.

For 20 Number should be divisible by 4 and 5.

The process to find the divisibility rule for prime numbers is simple; however, it is difficult to express in words. Let us discuss it in the following.

We are creating the divisibility rule for P, a prime number.

Step 1 Find the multiple of P, closest to any multiple of 10. (This will be essentially of the form $(10K + 1)$ or $(10K - 1)$.)

Step 2 If it is $(10K - 1)$, then the divisibility rule will be $(A + KB)$, and if it is $(10K + 1)$, then the divisibility rule will be $(A - KB)$, where B is the unit's place digit and A is all the remaining digits.

For example, let us find the divisibility rule of 23: Lowest multiple of 23, which is closest to any multiple of 10 = 69 = $(7 \times 10 - 1)$

Therefore, rule is $(A + 7B)$.

Number of Divisors

If one integer can be divided by another integer an exact number of times, then the first number is said to be a multiple of the second, and the second number is said to be a factor of the first.

For example, 48 is a **multiple** of 6 because it can divide 48 an exact number of times (in this case, it is 8 times). In other words, if you have 48 apples, we can distribute them among 6 persons equally without splitting any apple.

Similarly, 6 is **a factor** of 48. On the other hand, 48 is not a multiple of 5, because 5 cannot divide 48 an exact number of times. Therefore, 5 is not a factor of 48.

When we talk about number of divisors of any number, we are talking about positive integral divisor of that number.

For example, it can be observed that 20 has six divisors, namely 1, 2, 4, 5, 10, and 20.

Formation of Divisors

$$20 = 2^2 \times 5^1$$

Now, let us assume that 20 will be divisible by which numbers:

$$\frac{2^2 \times 5^1}{7}$$ Yes/No

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$\frac{2^2 \times 5^1}{2^1}$ Yes/No

$\frac{2^2 \times 5^1}{2^3}$ Yes/No

$\frac{2^2 \times 5^1}{2^1 \times 5^1}$ Yes/No

Answer to the abovementioned posers can be given in the following order—No, Yes, No, Yes.

We can observe that the denominator should have powers of only 2 and 5—powers of 2 should be from 0 to 2 and powers of 5 should be 0 to 1.

$$\frac{2^2 \times 5^1}{2^{0-2} \times 5^{0-1}}$$

Hence, we will take three powers of 2, that is, 2^0, 2^1, and 2^2 and two powers of 5, that is, 5^0 and 5^1.

Divisors will come from all the possible arrangements of powers of 2 and 5.

$$2^0 \times 5^0 = 1$$

$$2^0 \times 5^1 = 5$$

$$2^1 \times 5^0 = 2$$

$$2^1 \times 5^1 = 10$$

$$2^2 \times 5^0 = 4$$

$$2^2 \times 5^1 = 20$$

By summarizing these calculations, following formula can be derived:

If N is any number that can be factorized like $N = a^p \times b^q \times c^r \times ...$, where a, b, and c are prime numbers, then the number of divisors = $(p + 1)(q + 1)(r + 1)$

Example 19 Find the number of divisors of $N = 420$.

Solution $N = 420 = 2^2 \times 3^1 \times 7^1 \times 5^1$

Therefore, the number of divisors = $(2 + 1)(1 + 1)(1 + 1)(1 + 1) = 24$.

Example 20 Find the total number of even and prime divisors of $N = 420$.

Solution $N = 420 = 2^2 \times 3^1 \times 7^1 \times 5^1$

Odd divisors will come only if we take zero power of 2 (since any number multiplied by any power (≥1) of 2 will give us an even number). Odd divisors will come if we take $N_1 = 2^0 \times 3^1 \times 7^1 \times 5^1$

Number of odd divisors = $(0 + 1)(1 + 1)(1 + 1)(1 + 1) = 8$

Therefore, total number of even divisors = total number of divisors − number of odd divisors = 24 − 8 = 16

Alternatively, we can also find the number of even divisors of N as 420 (in general, for any number).

$$420 = 2^2 \times 3^1 \times 7^1 \times 5^1$$

To obtain the factors of 420, which are even, we will not consider 2^0, since $2^0 = 1$

Therefore, number of even divisors of 420 = (2) (1 + 1) (1 + 1) (1 + 1) = 16.

(We are not adding 1 in the power of 2, since we are not taking 2^0, that is, we are not taking one power of 2.)

Prime divisor = 4 (namely 2, 3, 5, and 7 only)

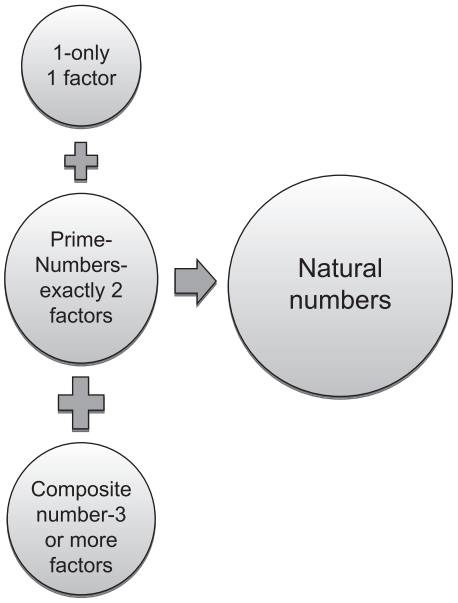
Example 21 $N = 2^7 \times 3^5 \times 5^6 \times 7^8$. How many factors of N are divisible by 50 but not by 100?

Solution All the factors that are divisible by 50 but not divisible by 100 will have at least two powers of 5, and one power of 2. Further, its format will be $2^1 \times 5^{2+y}$.

Therefore, the number of divisors = $1 \times 6 \times 5 \times 9 = 270$.

The following discusses the determination of prime factors and composite factors:

We know that natural number line (starting from 1, 2, 3, …) can be classified on the basis of number of factors to the natural number.



From the given graphics, we conclude the following:

(i) On the basis of number of factors, natural number line can be categorized into three parts: (a) 1, (b) prime number, and (c) composite factors.

(ii) Lowest composite number = 4.

The essence of the whole discussion lies in the fact that the total number of factors of any natural number = 1 (number 1 is a factor of all the natural numbers) + prime factors + composite factors.

Therefore, once we complete the prime factorization for finding the number of prime factors, we just need to

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count the number of prime factors. To calculate the number of composite factors, we will subtract the number of prime factors and 1 from the total number of factors.

Example 22 Find the number of prime factors and composite factors of $N = 420$.

Solution $420 = 2^2 \times 3^1 \times 5^1 \times 7^1$

Number of prime factors = 4 (namely 2, 3, 5, 7).

Total number of factors = $(2 + 1)(1 + 1)(1 + 1)(1 + 1) = 3 \times 2 \times 2 \times 2 = 24$

Therefore, the total number of composite factors = total number of factors – prime factors – 1 = 24 – 4 – 1 = 19.

The following discusses the determination of factors that are perfect squares or cubes or higher power:

A number will be perfect square only if all the prime factors of this number will have even powers. Therefore, a number of the format 2^x will be a perfect square only if $x = 0, 2, 4, 6, 8$, etc.

Similarly, a number will be perfect cube only if all the prime factors of this number will have powers divisible by 3. Therefore, a number of the format 2^y will be a cube only if $x = 0, 3, 6, 9$, etc.

Example 23 How many factors of the number $N = 720$ will be (a) perfect square, (b) cube, and (c) both a perfect square and cube?

Solution $N = 720 = 2^4 \times 3^2 \times 5^1$

(a) For a factor of $N = 720$ to be a perfect square, it should have only the following powers of its prime factors:

Powers of 2	Powers of 3	Powers of 5
2^0	3^0	5^0
2^2	3^2	
2^4		

Number of powers of 2 = 3
Number of powers of 3 = 2
Number of powers of 5 = 1
Hence, the total number of factors of $N = 720$ that are perfect square = $3 \times 2 \times 1 = 6$

(b) For a factor of $N = 720$ to be a cube, it should have only the following powers of its prime factors:

Powers of 2	Powers of 3	Powers of 5
2^0	3^0	5^0
2^3		

Number of powers of 2 = 2
Number of powers of 3 = 1
Number of powers of 5 = 1
Hence, the total number of factors of $N = 720$ that are cubes = $2 \times 1 \times 1 = 2$.

(c) For a factor of $N = 720$ to be both a cube and a square, it should have only the following powers of its prime factors:

Powers of 2	Powers of 3	Powers of 5
2^0	3^0	5^0

Number of powers of 2 = 1
Number of powers of 3 = 1
Number of powers of 5 = 1
Hence, the total number of factors of $N = 720$ that are cubes = $1 \times 1 \times 1 = 1$

The condition for two divisors of any number *N* to be co-prime to each other can be explained as follows:

Two numbers are said to be co-prime to each other if their HCF = 1. This can happen only if none of the factors of the first number (other than 1) is present in the second number and vice versa.

Let us see it for $N = 12$

Total number of factors of 12 = 6 (namely 1, 2, 3, 4, 6, 12). Now, if we have to find set of factors of this number that are co-prime to each other, we can start with 1.

Number of factors that are co-prime to 1 = 5 (namely, 2, 3, 4, 6, 12).

Next, the number of factors that are co-prime to 2 = 1 (namely 3)

Therefore, the total number of set of factors of 12 that are co-prime to each other = 6

Thus, we can induce that if we have to find the set of factors that are co-prime to each other for $N = a^p \times b^q$, it will be equal to $[(p + 1)(q + 1) - 1 + pq]$.

If there are three prime factors of the number, that is, $N = a^p \times b^q \times c^r$, then set of co-prime factors can be given by $[(p + 1)(q + 1)(r + 1) - 1 + pq + qr + pr + 3pqr]$

Alternatively, we can find the set of co-prime factors of this number by pairing up it first, and then finding the third factor.

Example 24 Find the set of co-prime factors of the number $N = 720$.

Solution $720 = 2^4 \times 3^2 \times 5^1$

Using the formula for three prime factors $[(p + 1)(q + 1)(r + 1) - 1 + pq + qr + pr + 3pqr]$,

we get $[(4 + 1)(2 + 1)(1 + 1) - 1 + (4 \times 2) + (2 \times 1) + (4 \times 1) + (3 \times 4 \times 2 \times 1)] = 67$

Alternatively, let us find the first for $2^4 \times 3^2 = [(4 + 1)(2 + 1) - 1 + (4 \times 2)] = 22$

Now, $p^{22} \times 5^1$ will give us $[(22 + 1)(1 + 1) - 1 + 22 \times 1] = 67$.

Sum of Divisors

We can find the sum of divisors similar to the number of divisors of any number. If N is any number that can be

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factorized like $N = a^p \times b^q \times c^r x$, where a, b, and c are prime numbers, then

$$\text{Sum of the divisors} = \frac{(a^{p+1}-1)(b^{q+1}-1)(c^{r+1}-1)}{(a-1)(b-1)(c-1)}$$

Remainders

Dividend = quotient × divisor + remainder

The basic framework of remainder are as follows:

1. If N is a number divisible by 7, it can be written as $7K = N$, where K is the quotient.
2. When N is divided by 7, remainder obtained is 3. Therefore, it can be written as $7K + 3 = N$, where K is the quotient.
3. When N is divided by 7, remainder obtained is 3 and it is equivalent of saying remainder obtained is (–4) when divided by 7. It can be understood that when N is divided by 7, remainder obtained is $3 = N$ is 3 more than a multiple of 7 => Therefore, N is 4 short of another multiple of 7. Therefore, remainder obtained = –4.
4. When N is divided by 8, different remainders can be obtained. They are 0, 1, 2, 3, 4, 5, 6, 7 (8 different remainders). Similarly, when it is divided by 5, remainders 0, 1, 2, 3, 4 (5 different remainders) are obtained.

Basics of Remainder

1. If any positive number A is divided by any other positive number B and if $B > A$, then the remainder will be A itself. In other words, if the numerator is smaller than the denominator, then the numerator is the remainder. For example,

   Remainder of $\frac{5}{12} = 5$

   Remainder of $\frac{21}{45} = 21$

2. Remainder should always be calculated in its actual form, that is, we cannot reduce the fraction to its lower ratio. For example,

   Remainder of $\frac{1}{2} = 1$

   Remainder of $\frac{2}{4} = 2$

   Remainder of $\frac{3}{6} = 3$

   It can be observed that despite all the fractions being equal, remainders are different in each case.

Example 25 What is the remainder when 5×10^5 is divided by 6×10^6?

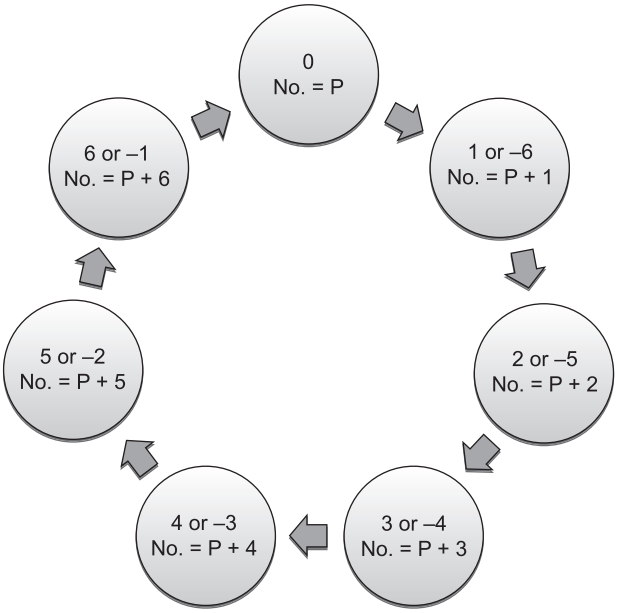
Solution As we know that we cannot reduce the fractions to its lower terms and numerator is less than denominator, the remainder obtained will be equal to 5×10^5.

3. The concept of negative remainder—as obvious from the name, this remainder implies that something has been left or something remains there. Therefore, remainder can simply never be negative. Its minimum value can be zero only and non-negative. For example, What is the remainder when –50 is divided by 7?

   Solution is $\frac{-50}{7} = \frac{-56+6}{7}$; this gives a remainder of 6

However, when we divide –50 by 7, we get –1 as the remainder. Now, since remainder has to be non-negative, we add 7 (quotient) to it that makes the final remainder as $(-1 + 7) = 6$. It can be explained in the following figure.

Let us assume that when P is divided by 7, remainder obtained as 0.



Therefore, when $(P + 1)$ will be divided by 7, remainder obtained will be either 1 or –6. Similarly, when $(P + 2)$ is divided by 7, the remainder obtained will be 2 or –5, and so on.

Now, there are two methods to find the remainder of any expression:

1. **Cyclicity method**
   for every expression of the remainder, there comes attached a specific cyclicity of remainders.

Example 26 What is the remainder when 4^{1000} is divided by 7?

Solution To find the cyclicity, we keep finding the remainders until some remainder repeats itself. It can be understood with the following example:

Number/7—	4^1	4^2	4^3	4^4	4^5	4^6	4^7	4^8
Remainder—	4	2	1	4	2	1	4	2

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Now, 4^4 gives us the same remainder as 4^1; therefore, the cyclicity is of 3 (this is because remainders start repeating themselves after 4^3).

Thus, any power of 3 or a multiple of 3 will give a remainder of 1, and hence, 4^{999} will give 1 as the remainder. Final remainder = 4.

Example 27 What is the remainder when 4^{96} is divided by 6?

Solution Let us find the cyclicity.

Number/6—4^1	4^2	4^3	4^4	4^5	4^6	4^7	4^8
Remainder—4	4	4	4	4	4	4	4

In all cases, the remainder is 4, and therefore, the final remainder will be 4. Actually, it is not required to find remainders till 4^8 or even 4^3. 4^2 itself gives us a remainder of 4 when divided by 6, which is same as the remainder obtained when 4^1 is divided by 6. Therefore, the length of cycle = 1.

Hence, final remainder = 4. Further, it can be observed here that if we write $4^{100}/6 = 2^{200}/6 = 2^{199}/3$, then remainder obtained will be 2, which is not the right answer (as given in the CAT brochure of next year, i.e., CAT 2004.)

2. Remainder Theorem Method

The product of any two or more than two natural numbers has the same remainder when divided by any natural number as the product of their remainders.

Let us understand this through an example:

Example 28 Remainder $\frac{12\times13}{7}$ = Remainder $\frac{156}{7}$ = 2

Solution The conventional way of doing this is Product →→→Remainder

Using the theorem method, we get

Remainder→→→

Product →→→ Remainder

Therefore, first, we will find the remainders of each individual number, and then, we will multiply these individual remainders to find the final remainder.

Remainder 12/7 = 5

Remainder 13/7 = 6

Remainder $\frac{12\times13}{7}$ = (5 × 6)/7 = Remainder 30/7 = 2

Example 29 What is the remainder obtained when (1421 × 1423 × 1425) is divided 12?

Solution Remainder of 1421/12 = 5

Remainder of 1423/12 = 7

Remainder of 1425/12 = 9

Remainder (1421 × 1423 × 1425)/12 = Remainder (5 × 7 × 9)/12 = Remainder (5 × 63)/12 = Remainder (5 × 3)/12 = 3

Successive Division

Let us assume that N is any number that is divided successively by 3 and 5. Here, we mean to say that at first, we divide N by 3, and then, the quotient obtained is divided by 5.

For example, let us consider the case where 50 is divided by 5 and 3 successively.

50 divided by 5 gives 10 as the quotient. Now, we will divide 10 by 3. Finally, it gives a quotient of 3 and remainder of 1.

Example 30 When a number N is divided successively by 3 and 5, remainder obtained are 1 and 2, respectively. What is the remainder when N is divided by 15?

Solution It can be seen that we are required to calculate it from back-end.

The family of numbers which when divided by 5 gives remainder as $2 = 5S + 2$

Therefore, $N = 3(5S + 2) + 1 = 15S + 7$

Now, if N is divided by 15, remainder = 7

Fermat's Remainder Theorem

Let P be a prime number and N be a number non-divisible by P. Then, remainder obtained when A^{P-1} is divided by P is 1.

(The remainder obtained when $\frac{A^{p-1}}{P} = 1$, if HCF $(A, P) = 1$.)

Example 31 What is the remainder when 2^{100} is divided by 101?

Solution Since it satisfies the Fermat's theorem format, remainder = 1.

Derivations

1. $\frac{(A+1)^N}{A}$ will always give 1 as the remainder (for all natural values of A and N).

Example 32 What is the remainder when 9^{100} is divided by 8?

Solution For $A = 8$, it satisfies the abovementioned condition. Therefore, remainder = 1.

Alternatively, we can apply either of cyclicity or theorem method to find the remainder.

2. $\frac{(A)^N}{A+1}$ when N is even, remainder is 1, and when N is odd, then remainder is A (for all natural values of A and N).

Example 33 What is the remainder when 2^{10} is divided by 3?

Solution Since N is even, remainder = 1

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3. i. $(a^n + b^n)$ is divisible by $(a + b)$, if n is odd.

The extension of the abovementioned formula $(a^n + b^n + c^n)$ is divisible by $(a + b + c)$, if n is odd and a, b, and c are in arithmetic progression.

Example 34 What is the remainder obtained when $\frac{7^7 + 10^7 + 13^7 + 16^7}{46}$?

Solution It can be seen that 7, 10, 13, and 16 are in arithmetic progression and power n is odd. Further, denominator = 7 + 10 + 13 + 16 = 46. Hence, it will be divisible. Therefore, remainder obtained = 0.

Similarly, the abovementioned situation can be extended for any number of terms.

- $(a^n - b^n)$ is divisible by $(a + b)$, if n is even.
- $(a^n - b^n)$ is divisible by $(a - b)$, if n is even

Example 35 What is the remainder when $(15^{23} + 23^{23})$ is divided by 19? ***(CAT 2004)***

Solution It can be observed that $(15^{23} + 23^{23})$ is divisible by 38, and therefore, it will be divisible by 19 also. Hence, remainder = 0.

Alternatively, this problem can be done either by cyclicity method or theorem method.

Example 36 What is the remainder when $(16^3 + 17^3 + 18^3 + 19^3)$ is divided by 70? ***(CAT 2005)***

Solution We know that this is a basic multiplication and division question. However, using the abovementioned approach makes it very simple.

We know that $(a^n + b^n)$ is divisible by $(a + b)$, if n is odd. From this, we can say that $(a^n + b^n + c^n)$ is divisible by $(a + b + c)$, if n is odd, and similarly, $(a^n + b^n + c^n + d^n)$ is divisible by $(a + b + c + d)$. Now, (16 + 17 + 18 + 19) = 70, and therefore, remainder is zero.

The following table lists out different types of problems:

	Problem	Solution
1	Find the greatest number that will exactly divide *a*, *b*, and *c*	Required number = HCF of *a*, *b*, and *c*
2	Find the greatest number that will divide *x*, *y*, and *z* leaving remainders *a*, *b*, and *c*, respectively.	Required number (greatest divisor) = HCF of $(x - a)$, $(y - b)$ and $(z - c)$.
3	Find the least number that is exactly divisible by *a*, *b*, and *c*.	Required number = LCM of *a*, *b*, and *c*
4	Find the least number that when divided by *x*, *y*, and *z* leaves the remainders *a*, *b*, and *c*, respectively, and $(x - a) = (y - b) = (z - c) = N$	Required number = LCM of (*x*, *y*, and *z*) – *N*
5	Find the least number that when divided by *x*, *y*, and *z* leaves the same remainder *r* in each case.	Required number = (LCM of *x*, *y*, and *z*) + *r*

Units Digit

As discussed earlier, cyclicity exists for units digit of the numbers also. (However, it is necessary to remember that there is no relation between the cyclicity of remainders and the units digit.) Let us consider a simple example: $-2^5 = 32$. Here, we know that units digit of 2^5 is 2. However, problem occurs when we start taking large numbers like $25{,}678^{2345}$, and so on. To find the units digit of these numbers, we have some standard results, which we use as formula.

(Any even number)^{4n} = …6

It means that any even number raised to any power, which is a multiple of 4, will give 6 as the units digit.

(Any odd number)^{4n} = …1

It means that any odd number raised to any power, which is a multiple of 4, will give 1 as the units digit.

Exception: 0, 1, 5, 6 [These are independent of power, and units digit will be the same.]

Example 37 Find the units digit of $25{,}678^{2345} \times 3485^{4857}$.

Solution Units digit of $25{,}678^{2345}$ = units digit of 8^{45}

(To find the units digit, we need to have units digits only. Similarly, to find tens digit, we need to have the tens and units digit only. In the present case, we are considering only last two digits of the power because divisibility rule of 4 needs only the last two digits of the number.)

$$8^{45} = 8^{44+1} = 8^{44} \times 8^1 = (\ldots 6) \times 8 = \ldots 8$$

Example 38 What is the units digit of ${}_{32}32^{32}$?

Solution 32 is an even number that is having a power of the form 4^n. Therefore, it will give 6 as the units digit.

Example 39 When 3^{32} is divided by 50, it gives a number of the format (*asdf*…•*xy*) (*xy* being the last two digits after decimal). Find *y*.

Solution It can be observed that units digit of $3^{32} = 1$. Now, any number having 1 as the units digit will always give 2 at the units place when divided by 50. Therefore, the answer is 2.

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Example 40 What is the last non-zero digit of the number 30^{2720}?

Solution $30^{2720} = [30^4]^{680} = ...1$

Units digit can also be determined by cyclicity method. It can be seen that

Units digit of $2^1 = 2$
Units digit of $2^2 = 4$
Units digit of $2^3 = 8$
Units digit of $2^4 = 6$
Units digit of $2^5 = 2$

Therefore, it can be inferred that units digit of 2^1 = units digit of 2^5 = units digit of 2^9.

Hence, the cyclicity of 2 = 4, that is, every fourth power of 2 will give same units digit.

Similarly, cyclicity of 3 = 4

Cyclicity of 4 = 2
Cyclicity of 7 = 4
Cyclicity of 8 = 4
Cyclicity of 9 = 2

Cyclicity of 0 or Cyclicity of 1 or Cyclicity of 5 or Cyclicity of 6 = 1

Tens Digit

Method 1: Cyclicity Method

Digits	Cyclicity
2, 3, 8	20
4, 9	10
5	1
6	5
7	4

Example 41 What is the tens place digit of 12^{42}?

Solution For this, we need to break 12^{42} first by using binomial theorem as $(10 + 2)^{42}$. Obviously, this expression will have 43 terms, and out of these 43 terms, the first 41 terms will have both of their tens and units place digit as 0.

The last two terms will be ${}^{42}C_{41} \times 10^1 \times 2^{41} + {}^{42}C_{42} \times 10^0 \times 2^{42}$. Now, we will find the tens place digit of all these terms individually.

Tens digit of ${}^{42}C_{41} \times 10^1 \times 2^{41} = 42 \times 10 \times (02)$ [Cyclicity of 2 is 20; 2^{41} will have same tens digits as 2^1] = 840, and therefore, 40 are the last two digits.

Similarly, ${}^{42}C_{42} \times 10^0 \times 2^{42} = 1 \times 1 \times 04 = 04$. Finally, the last two digits are (40 + 04) = 44, and therefore, 4 is the tens place digit.

Note: $(25)^n$ and $(76)^n$ will always give 25 and 76 as the last two digits for any natural number value of n.

Method 2: Generalization Method

(i) (Any even number)^{20N} will give 76 as its last two digits (where N is any natural number).
However, if units digit = 0, then it will give '00' as the last two digits.

(ii) (Any odd number)^{20N} will give 01 as its last two digits (where N is any natural number). However, if units digit = 5, then it will give '25' as the last two digits.
Let us solve the previous worked-out example once again using this method.

Example 42 What is the tens place digit of 12^{42}?

Solution Using generalization (i),

we get $12^{20} = \ldots 76$ (76 as last two digits)

$12^{20} \times 12^{20} = 12^{40}$
$= (\ldots 76)$
$\times (\ldots 76)$
$= (\ldots 76)$

$12^{42} = 12^{40} \times 12^2 = (\ldots 76) \times (144)$

Since we are required to calculate the last two digits, we will focus only on the last two digits of both the numbers.

$(\ldots 76) \times (44) = 3344$. Hence, 44 is the last two digits of 12^{42}.

Note: we are not certain if 3 is at 100s place of this number.

Example 43 Find the tens place digit of 784^{1000}.

Solution Tens place digit of 784^{1000} = Tens place digit of 841,000

As discussed earlier, (any even number)^{20N} will give 76 as the last two digits.

$84^{1000} = (84)^{20 \times 50} = (84)^{20N}$. This will have 76 as the last two digits.

Number of Exponents

Let us take a simple number 10^5

This is read as 10 to the power 5, or we can say that the exponent of 10 is 5.

In simple terms, exponents are also known as power.

Example 44 What is the maximum value of s if $N = (35 \times 45 \times 55 \times 60 \times 124 \times 75)$ is divisible by 5^x?

Solution If we factorize $N = (35 \times 45 \times 55 \times 60 \times 124 \times 75)$, then we can observe that 5 appears 6 times, it means N is divisible by 5^6.

Thus, the maximum value of $x = 6$

The exponent of any prime number P in $n!$ is given as

$= \left[\frac{n}{p}\right] + \left[\frac{n}{p^2}\right] + \left[\frac{n}{p^3}\right] + \ldots\ldots \left[\frac{n}{p^x}\right]$, where $n \geq p^x$ and [.] denotes the greatest integer value, that is, we have to consider only the integral value

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Let us find the exponent of 5 in 1000! $= \frac{1000}{5} + \frac{1000}{5^2} + \frac{1000}{5^3} + \frac{1000}{5^4} = 200 + 40 + 8 + 1 = 249$

Example 45 What is the highest power of 5 that can divide $N = (22! + 17{,}894!)$?

Solution The number of times this number is divisible by 5 is same as the number of zeroes at the end of this number. Here, 22! have 4 zeroes at its end, and therefore, N will also be having only four zeroes at its end. Hence, the highest power of 5 that can divide N is 4.

The process to find the exponent of any composite number in *n*! is given as

We have got three different kinds of composite numbers:

1. Product of two or more than two prime numbers with unit power of all the prime numbers.
   For example, $15(5 \times 3)$, $30(2 \times 3 \times 5)$, etc.
2. (Any prime number)^n = where $n > 1$. For example, $4(2^2)$, $27(3^3)$.
3. Product of two or more than two prime numbers with power of any one prime number more than 1.
   For example, $12(2^2 \times 3)$, $72(2^3 \times 3^2)$, etc.

Let us find the exponents of the abovementioned composite numbers:

1. Let us find the exponent of 15 in 100! 15 is the product of two distinct prime numbers 5 and 3. To find the exponents of 15, we need to find the exponents of 5 and 3 individually.
   Therefore, we will apply the same formula for finding the exponents for any prime number in both of these cases individually, and minimum of those two will be the solution.
   $100/5^x = [100/5] + [100/5^2] = 20 + 4 = 24$
   $100/3^x = [100/3] + [100/3^2] + [100/3^3] + [100/3^4]$
   $= 33 + 11 + 3 + 1 = 48$
   Obviously, 24 is the answer.
2. Let us find the exponent of 25 in 100!
   Similarly, we can find solution for the third category numbers also $(25 = 5^2)$
   In this case, we will first find the exponents of 5 and then divide it by 2 (actually the power) to find the exponents of 25.
   $100/5^x = [100/5] + [100/5^2] = 20 + 4 = 24$; $100/25^x = 24/2 = 12$
3. Similarly, we can find the solution for the third category numbers also.

Base System

In our decimal system of writing the numbers, we use 10 digits (0–9). In this system, the largest number of single digit is 9, and the moment we have to form a number larger than this number, we take the two-digit numbers starting from 10. Similarly, the largest number of two digits is 99, and after this, we have 100 (which is a three-digit number). Further, it is very plain and simple.

Now, let us assume a system of writing where we use only 6 digits (0–5). The largest single-digit number in this system will be 5 and next to this will be 10. Similarly, the largest two-digit number will be 55 and next is 100.

This whole procedure can be summed up in the following table:

$(0–9)_{10}$	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
$(0–8)_9$	0	1	2	3	4	5	6	7	8	10	11	12	13	14	15	16	17	18
$(0–7)_8$	0	1	2	3	4	5	6	7	10	11	12	13	14	15	16	17	20	21
$(0–6)_7$	0	1	2	3	4	5	6	10	11	12	13	14	15	16	20	21	22	23
$(0–5)_6$	0	1	2	3	4	5	10	11	12	13	14	15	20	21	22	23	24	25
$(0–3)_4$	0	1	2	3	10	11	12	13	20	21	22	23	30	31	32	33	100	101
$(0–2)_3$	0	1	2	10	11	12	20	21	22	100	101	102	110	111	112	120	121	122

Questions from this concept are asked in three different ways:

1. $(\text{Base})_{10}$ to any other base and vice versa
2. (Base) to (Base) and vice versa; none of x and y being equal to 10, but x and y will be given.
3. $(\text{Base})_x$ to $(\text{Base})_y$, the value of x and y will not be given.

1. $(Base)_{10}$ *to any other base and vice versa*

Method 1

Let us consider (74).

$(74)_{10} = 7 \times 10^1 + 4 \times 10^0$, since the base is 10.

Now, if we have to convert this number to 9 base, then we will try to write it in terms of powers of 9.

$(74)_{10} = 8 \times 9^1 + 2 \times 9^0 = (82)_9$
$(74)_{10} = 1 \times 8^2 + 1 \times 8^1 + 2 \times 8^0 = (112)_8$
$(74)_{10} = 1 \times 7^2 + 3 \times 7^1 + 4 \times 7^0 = (134)_7$
$(74)_{10} = 2 \times 6^2 + 0 \times 6^1 + 2 \times 6^0 = (202)_6$

While converting the numbers from decimal system to any other system of writing the numbers, we should be concerned with the following two rules:

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- Take maximum possible power of the base and then keep writing rest of the number with the help of lesser power of base (as illustrated in the earlier example).
- Once we have used (base)^n, where n is the maximum power, then we will be required to write the coefficient of all the powers of base from 0 to $(n-1)$ as in the case of $(74)_{10} = (202)_6$.

Now, let us assume that we have to convert $(356)_7$ in the base of 10. $(356)_7 = 3\times7^2 + 5\times7^1 + 6\times7^0 = (188)_{10}$

Method 2

Let us convert $(74)_{10}$ to the base of $(\ )_9$

Base	74	Reminder
9	8	2 ↑
	→	

Therefore, $(74)_{10} = (82)_9$

Let us convert $(74)_{10}$ to the base of $(\ )_8$

Base	74	Reminder
9	9	2 ↑
8	1	1
	→	

Therefore, $(74)_{10} = (112)_8$.

Let us convert $(74)_{10}$ to the base of $(\ )_7$

Base	74	Reminder
7	10	4 ↑
7	1	3
	1 →	

Therefore, $(74)_{10} = (134)_7$.

Let us convert $(74)_{10}$ to the base of $(\ )_6$

Base	74	Reminder
6	12	2 ↑
6	2	0
	↑ →	
	Quotient	

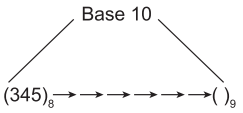
Therefore, $(74)_{10} = (202)_6$.

Task for students

Convert $(123)_{10}$ into base 9, base 8, base 7, base 15, base 20. Answers are given at the end of topic.

(Base)$_x$ to (Base)$_y$ and vice versa; none of x and y being equal to 10 but x and y will be given.

Converting $(345)_8$ to the base of $(\ )_9$:



We will do this problem with the help of creating a bridge of base 10 between base 8 and base 7.

Step 1 Convert $(345)_8$ to base 10.

$345 = 3\times8^2 + 4\times8^1 + 5\times8^0 = (229)_{10}$

Step 2 Now, convert this number in base 10 to base 9.

$(229)_{10} = 2\times9^2 + 7\times9^1 + 4\times9^0 = (274)_9$

However, if new base is a power of old base and vice versa, then it can be converted directly to the new base, that is, it is not necessary to go to base 10 for these types of conversions.

Let us convert (74) to the base of $(\ )_8$:

Converting $(101110010)_2$ to Octal $(\ )_8$ system:

At first, we will club three digits of binary number into a single block, and then, we will write the decimal equivalent of each group (left to right).

Therefore, $(101110010)_2$ is now $(101)_2(110)_2(010)_2$.

Now, $(101)_2 = 1\times2^2 + 0 + 1\times2^0 = 5$

$(110)_2 = 1\times2^2 + 1\times2^1 + 0\times2^0 = 6$

$(010)_2 = 0\times2^2 + 1\times2^1 + 0\times2^0 = 2.$

Thus, $(101110010)_2 = (562)_8$

Converting $(101110010)_2$ to hexa-decimal $(\ )_{16}$ system:

At first, we will club four digits of binary number into a single block, and then, we will write the decimal equivalent of each group (left to right).

Therefore, $(101110010)_2$ is now $(0001)_2(0111)_2(0010)_2$.

Now, we have the following:

Decimal equivalent of $(0001)_2 = 1$

Decimal equivalent of $(0111)_2 = 7$

Decimal equivalent of $(0010)_2 = 2$

$(101110010)_2 = (172)_{16}$

(Base)$_x$ to (Base)$_y$, value of x and y will not be given.

Normally, in these types of questions, some calculation is given in some unknown system of writing numbers. On the basis of that calculation, we will be required to solve questions.

Example 46 In a system of writing of N digits, we have $4\times6 = 30$ and $5\times6 = 36$. What will be the value of $N = 3\times4\times5$ in the same system of writing?

Solution Let us assume that there are N digits in this system of writing.

$$(30)_N = 3\times N + 0\times N^0 = 24$$
$$\Rightarrow \quad 3N = 24$$
$$\Rightarrow \quad N = 8$$

Therefore, this system of writing has 8 digits.

In this system, $3\times4\times5 = 60$ will be written as 74 ($60 = 7\times8^1 + 4\times8^0$).

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Alternatively, since this system is having 6 as one of its digits, minimum value of N will be 7. Again, 24 is written as 30 in this system, then N is less than 10. Now, use trial-and error method for $N = 7$ or 8 or 9 to find N in $24 = (30)_N$.

Decimal Calculation

So far, we have seen the calculations involving natural numbers only. Now, let us work with decimals.

Let us see the process of converting decimal system numbers to any other system:

Let us assume that (12.725) is a number in decimal system, which is required to be converted into octal system (8 digits).

We will first convert 12 into octal system, i.e., $(12)_{10} = (14)_8$

Now, to convert (0.725) into $(\ )_8$, we will apply the following method:

$0.725 \times 8 = \mathbf{5}.8$ eliminate the integral part from here.

$0.8 \times 8 = \mathbf{6}.4$ eliminate the integral part from here.

$0.4 \times 8 = \mathbf{3}.2$ eliminate the integral part from here.

$0.2 \times 8 = \mathbf{1}.6$ eliminate the integral part from here.

Further, we keep doing this until we get decimal part as zero, that is, the product should be an integer.

$(0.725)10 = (0.5632\ldots)_8$

Thus, $(12.725) = (14.5632\ldots)_8$

Let us discuss the process of converting any other system numbers to decimal system:

Now, let us assume that if $(15.453)_7$ is to be converted into decimal system, then the process is as follows:

We will first convert $(15)_7$ into decimal system.

$(15)_7 = 1 \times 7^1 + 5 \times 7^0 = (12)_{10}$

Let us discuss the basic algebraic calculations involving the base systems:

Addition

$$\begin{array}{r} 325_7 \\ +\,456_7 \\ \hline \end{array}$$

Start with the units place digit, $5 + 6 = 11$, which is 14_7. Thus, units digit is 4 and carry over is 1.

Next is tens place digit, $2 + 5 + 1$ (carry over) $= 8$, which is 11_7. Hence, tens digit is 1 and carry over is again 1.

Next is $3 + 4 + 1$ (carry over) $= 8$, which is 11_7.

$$\begin{array}{r} 325_7 \\ +\,456_7 \\ \hline 1114_7 \end{array}$$

Subtraction

$$456_8 - 367_8$$

Let us start with the units digit; since 6 is smaller than 7, we will borrow 1 from the tens place digit. Therefore, now, it is 14 (when the base is 10, we get 10; however, here, the base is 8, and hence, we will get 8.). When 7 subtracted from 14, we have 7, which is the units digit.

Next, tens digit is now 4 and we have to subtract 5 from it. We will again borrow 1 from hundreds place digit. Now, it is 12, and $12 - 6 = 6$, which is the tens place digit.

Now, hundreds place digit is $3(4 - 1)$, and $(3 - 3) = 0$.

$$\begin{array}{r} 456_8 \\ -367_8 \\ \hline 67_8 \end{array}$$

Note: Another method of calculation is (i) converting these values (in whatever base) into decimal system, (ii) performing the actual calculation in decimal system itself, and (iii) converting the numbers into the required or given system.

The following are some of the standard system of writing:

Decimal system
Digits used—0, 1, 2, 3, 4, 5, 6, 7, 8, 9.
Total digits used = 10 digits
Hexa-decimal system
Digits used—0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F. Total digits used = 16
Octal system
Digits used—0, 1, 2, 3, 4, 5, 6, 7.
Total digits used = 8
Binary system
Digits used—0, 1
Total digits used—2

The divisibility rules for systems other than decimal system are as follows:

Here, we emphasize that different number systems are just different ways to write numbers. Thus, the divisibility of one number by another does not depend on the particular system in which they are written.

At the same time, in each system, there are some techniques to determine divisibility by certain specific numbers. These are the divisibility tests.

Now, let us investigate the other, less trivial divisibility tests. Perhaps, the most well-known of these are the tests for divisibility by 3 and 9. We will try to generalize these tests for any number base system. Is $1,23,45,65,64,231_7$ divisible by 6?

We know the divisibility rule for 9—sum of digits of the number should be divisible by 9.

Sum of digits of this number is 42.

Now, we can answer this question easily: since the sum of the digits (which is 42_{10}) is divisible by 6, the number itself is divisible by 6.

In general, the sum of the digits of a number written in the base n system is divisible by $(n - 1)$ if and only if the number itself is divisible by $(n - 1)$.

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Therefore, divisibility rule for 4 in a base system of 5—sum of digits of the number should be divisible by 4. For example, 31_5 is divisible by 4.

Similarly, if we have to find the divisibility rule of 12 in the base of 11, it will be nothing but same as the divisibility rule of 11 in the base of 10. Generalizing the whole concept, we can say that the divisibility rule of any natural number N in the base of $(N-1)$ will be same as divisibility rule of 11 on base 10.

Pigeonhole Principle

Despite not being very much in vogue with respect to the CAT preparation (only a few questions have been asked from this concept so far in CAT), the importance of this topic lies in the fact that this concept is purely logical.

General Statement of Pigeonhole Principle

If we assume $(N+1)$ or more pigeons in N holes (nests), then at least one hole will be there, which will have 2 or more pigeons.

Example 47 What is the minimum number of people in any group of five people who have an identical number of friends within the group, provided if A is friend of B, then B is also friend of A?

Solution Since there are five persons in the group, the possible number of friends is 0, 1, 2, 3, 4. It seems here that everybody is having different number of friends, and hence, the answer is zero. However, anybody having four friends ensures that nobody is having 0 friends. Thus, at least two persons must have same number of friends.

Practice Exercises

Warm Up

Q.1 Which of the following is the smallest?
(a) $5^{1/2}$ (b) $6^{1/3}$ (c) $8^{1/4}$ (d) $12^{1/6}$

Q.2 A number N is divisible by 6 but not divisible by 4. Which of the following will not be an integer?
(a) $N/3$ (b) $N/2$ (c) $N/6$ (d) $N/12$

Q.3 If a, b, and c are consecutive positive integers, then the largest number that always divides $(a^2+b^2+c^2)$
(a) 14 (b) 55
(c) 3 (d) None of these

Q.4 $\dfrac{(3.134)^3+(1.866)^3}{(3.134)^2-3.134\times1.866+(1.866)^2}=?$
(a) 25 (b) 2.68 (c) 1.038 (d) 5

Q.5 If n^2 is a perfect cube, then which of the following statements is always true?
(a) n is odd.
(b) n is even.
(c) n^3 is a perfect square.
(d) n is a perfect cube.

Q.6 If $(5x+11y)$ is a prime number for natural number values of x and y, then what is the minimum value of $(x+y)$?
(a) 2 (b) 3 (c) 4 (d) 5

Q.7 For what values of x is (25^x+1) divisible by 13?
(a) All real values of x
(b) Odd natural values of x
(c) Even values of x
(d) All the integral values of x

Q.8 Which of the following numbers lies between 5/6 and 6/7?
(a) 71/84 (b) 31/42
(c) 129/168 (d) 157/339

Q.9 By multiplying with which of the following numbers, does the product of $8\times9\times10\times11\times12$ become a perfect square?
(a) 55 (b) 11 (c) 165 (d) 310

Q.10 What is the difference between the sum of the cubes and that of squares of the first 10 natural numbers?
(a) 5280 (b) 2640 (c) 3820 (d) 4130

Q.11 If $3-9+15-21+\ldots$ up to 19 terms $=x$ then x is a/an
(a) odd number (b) even number
(c) prime number (d) irrational number

Q.12 What is the units digit of $21^3\times21^2\times34^7\times46^8\times77^8$?
(a) 4 (b) 8 (c) 6 (d) 2

Q.13 If the units digit in the product $(47n\times729\times345\times343)$ is 5, what is the maximum number of values that n may take?
(a) 9 (b) 3 (c) 7 (d) 5

Q.14 In how many ways, can 846 be resolved into two factors?
(a) 9 (b) 11
(c) 6 (d) None of these

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Q.15 If a number is divided by 15, it leaves a remainder of 7. If thrice the number is divided by 5, then what is the remainder?
(a) 5 (b) 6 (c) 7 (d) 1

Q.16 A number when divided by 391 gives a remainder of 49. Find the remainder when it is divided by 39.
(a) 10 (b) 9
(c) 11 (d) Cannot be determined

Q.17 p and q are two prime numbers such that $p < q < 50$. In how many cases, would $(q + p)$ be also a prime number?
(a) 5 (b) 6
(c) 7 (d) None of these

Q.18 How many distinct factors of 1600 are perfect cubes?
(a) 3 (b) 4 (c) 6 (d) 2

Q.19 The LCM of 96,144 and N is 576. If their HCF is 48, then which of the following can be one of the values of N?
(a) 168 (b) 192 (c) 144 (d) 244

Q.20 If p and q are consecutive natural numbers (in increasing order), then which of the following is true?
(a) $q^2 < p$ (b) $2p > p^2$
(c) $(q + 1)^2 > p^2$ (d) $(p + 2)^3 < q^3$

Q.21 $(17^{21} + 19^{21})$ is not divisible by
(a) 36 (b) 8 (c) 9 (d) 18

Q.22 Which of the following will divide $11^{12296} - 1$?
(a) 11 and 12 (b) 11 and 10
(c) 10 and 12 (d) 11 only

Q.23. If a, b, c, and d are consecutive odd numbers, then $(a^2 + b^2 + c^2 + d^2)$ is always divisible by
(a) 5 (b) 7 (c) 3 (d) 4

Q.24. Four bells toll at intervals of 14, 21, and 42 min, respectively. If they toll together at 11:22 am, when will they toll together for the first time after that?
(a) 11:56 am (b) 12:04 pm
(c) 12:06 pm (d) 11:48 am

Q.25 When x is divided by 6, remainder obtained is 3. Find the remainder when $(x^4 + x^3 + x^2 + x + 1)$ is divided by 6.
(a) 3 (b) 4 (c) 1 (d) 5

Q.26 I have 7^7 sweets and I want to distribute them equally among 2^4 students. After each of the student got maximum integral sweets, how many sweets are left with me?
(a) 8 (b) 5
(c) 1 (d) None of these

Q.27 When I distribute some chocolates to my 40 students, three chocolates will be left. If I distribute the same number of chocolates to my students and my colleague Manoj Dawrani, seven chocolates are left. Find the minimum number of chocolates I have.
(a) 1443 (b) 1476
(c) 1480 (d) None of these

Q.28 The LCM of two numbers is 40 times of their HCF. The sum of the LCM and HCF is 1476. If one of the numbers is 288, find the other numbers.
(a) 169 (b) 180 (c) 240 (d) 260

Q.29 1010101...94 digits is a 94-digit number. What will be the remainder obtained when this number is divided by 375?
(a) 10 (b) 320
(c) 260 (d) None of these

Q.30 Chandrabhal adds first N natural numbers and finds the sum to be 1850. However, actually one number was added twice by mistake. Find the difference between N and that number.
(a) 40 (b) 33 (c) 60 (d) 17

Q.31 When I distribute a packet of chocolates to 7 students, I am left with 4 chocolates. When I distribute the same packet of chocolates to 11 students, I am left with 6 chocolates. How many chocolates will be left with me if I distribute the same packet of chocolates among 13 students (a packet of chocolate contains total number of N chocolates, where $1000 < N < 1050$)?
(a) 2 (b) 0 (c) 6 (d) 7

Q.32 How many prime numbers are there between 80 and 105?
(a) 3 (b) 4 (c) 5 (d) 8

Q.33 If x and y are consecutive natural numbers in an increasing order, then which of the following is always true?
(a) $x^y > y^x$ (b) $y^x > x^y$
(c) $x^x > y^y$ (d) $y^y > x^x$

Q.34 What is the remainder when 5^{79} is divided by 7?
(a) 1 (b) 0 (c) 5 (d) 4

Foundation

Q.1 The LCM of two natural numbers is 590 and their HCF is 59. How many sets of values are possible?
(a) 1 (b) 2
(c) 5 (d) 10

Q.2 MUL has a waiting list of 5005 applicants. The list shows that there are at least 5 males between any two females. The largest number of females in the list could be:
(a) 920 (b) 835 (c) 721 (d) 1005

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Q.3 HCF of two numbers A and B is 24. HCF of two other numbers C and D is 36. What will be the HCF of A, B, C, and D?
(a) 12 (b) 24 (c) 36 (d) 6

Q.4 How many zeroes will be there at the end of $25 \times 35 \times 40 \times 50 \times 60 \times 65$?
(a) 6 (b) 8 (c) 5 (d) 7

Q.5 What is the units digit of $576{,}847 \times 564{,}068 \times 96{,}467 \times 458{,}576$?
(a) 2 (b) 4 (c) 6 (d) 8

Q.6 What is the units digit of $1! + 2! + 3! + 99! + 100!$?
(a) 3 (b) 1 (c) 5 (d) 6

Q.7 How many divisors will be there of the number 1020?
(a) 12 (b) 20 (c) 24 (d) 36

Q.8 In Q.7, what is the difference between the number of even divisors and number of prime divisors?
(a) 13 (b) 12
(c) 11 (d) None of these

Q.9 $N = 7!^3$. How many factors of N are multiples of 10?
(a) 736 (b) 1008
(c) 1352 (d) 894

Q.10 A number N has odd number of divisors. Which of the following is true about N?
(a) All the divisors of this number will be odd.
(b) There will be at least $(N - 11)$ prime divisors.
(c) N will be a perfect square.
(d) At least one divisor of the number should be odd.

Q.11 How many zeroes will be there at the end of the expression $N = 2 \times 4 \times 6 \times 8 \times \ldots \times 100$?
(a) 10 (b) 12
(c) 14 (d) None of these

Q.12. How many zeroes will be there at the end of the expression $N = 10 \times 20 \times 30 \ldots \times 1000$?
(a) 1280 (b) 1300
(c) 1320 (d) None of these

Q.13 How many zeroes will be there at the end of the expression $N = 7 \times 14 \times 21 \times \ldots \times 777$?
(a) 24 (b) 25
(c) 26 (d) None of these

Q.14 The number from 1 to 33 are written side by side as follows: 123,456... 33. What is the remainder when this number is divided by 9?
(a) 0 (b) 1 (c) 3 (d) 6

Q.15 The number 444,444 ... (999 times) is definitely divisible by:
(a) 22 (b) 44
(c) 222 (d) All of these

Q.16 Find the units digit of $N = {}_{17}27!^{37!}$
(a) 1 (b) 3 (c) 7 (d) 9

Q.17 How many divisors of $N = 420$ will be of the form $(4n + 1)$, where n is a whole number?
(a) 3 (b) 4 (c) 5 (d) 8

Q.18 $N = 2^3 \times 5^3 \times 7^2$. How many sets of two factors of N are co-prime?
(a) 72 (b) 64
(c) 36 (d) None of these

Q.19 What is the units digit of $2^{3^{4^5}}$?
(a) 2 (b) 4 (c) 8 (d) 6

Q.20 How many zeroes will be there at the end of $1003 \times 1001 \times 999 \times \ldots \times 123$?
(a) 224 (b) 217
(c) 0 (d) None of these

Q.21 How many zeroes will be there at the end of $36!^{36!}$?
(a) $7 \times 6!$ (b) $8 \times 6!$
(c) $7 \times 36!$ (d) $8 \times 36!$

Q.22 The number formed by writing any digit 6 times (e.g., 111,111, 444,444, etc.) is always divisible by:
(i) 7 (ii) 11 (iii) 13
(a) (i) and (ii) (b) (ii) and (iii)
(c) (i) and (iii) (d) (i), (ii) and (iii)

Q.23 What is the maximum value of HCF of $[n^2 + 17]$ and $[(n + 1)^2 + 17]$?
(a) 69 (b) 85
(c) 170 (d) None of these

Q.24 What is the number of pairs of values of (x, y), which will satisfy $2x - 5y = 1$, where $x < 200$, and x and y are positive integers?
(a) 38 (b) 39 (c) 40 (d) 41

Q.25 $N = 2^3 \times 5^3$. How many sets of two distinct factors of N are co-prime to each other?
(a) 12 (b) 24 (c) 23 (d) 11

Q.26 What is the sum of digits of the least multiple of 13, which when divided by 6, 8, and 12 leave 5, 7, and 11 as the remainder?
(a) 5 (b) 6 (c) 7 (d) 8

Q.27 What is the units digit of $7^{11^{22^{33}}}$?
(a) 1 (b) 3 (c) 7 (d) 9

Q.28 What is the remainder when (1! + 2! + 3! + ... 1000!) is divided by 5?
(a) 1 (b) 2 (c) 3 (d) 4

Q.29 If $A = 3^{150} \times 5^{76} \times 7^{140}$, $B = 3^{148} \times 5^{76} \times 7^{141}$, $C = 3^{148} \times 5^{80} \times 7^{139}$, and $D = 3^{151} \times 5^{80} \times 7^{142}$, then the order of A, B, C, and D from largest to smallest is:
(a) DACB (b) CDBA
(c) CDAB (d) DCAB

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Q.30 The HCF of 0.3, 0.15, 0.225, 0.0003 is:
(a) 0.0003 (b) 0.3 (c) 0.15 (d) 0.0015

Q.31 How many numbers between 1 and 250 are divisible by 5 but not by 9?
(a) 98 (b) 97
(c) 101 (d) None of these

Q.32 A and B are two distinct digits. If the sum of the two-digit numbers formed by using both the digits is a perfect square, what is the value of $(A + B)$?
(a) 9 (b) 11 (c) 13 (d) 17

Q.33 A number $N = 897324P64Q$ is divisible by both 8 and 9. Which of the following is the value of $(P + Q)$?
i. 2 ii. 11 iii. 9
(a) Either i or ii (b) Either ii or iii
(c) Either i or ii or iii (d) None of these

Direction for Questions 34 and 35: Read the following passage below and solve the questions based on it.

A = Set of first N positive numbers. There are 16 numbers in A that are divisible by both X and Y. There are 50 numbers in A that are divisible by X but not by Y and 34 numbers in A divisible Y but not by X.

Q.34 How many numbers in A are divisible by any of the two numbers?
(a) 100 (b) 50
(c) 200 (d) None of these

Q.35 How many numbers in N are divisible by X?
(a) 42 (b) 56
(c) 66 (d) None of these

Q.36 Nitin had forgotten his 6-digit bank account number but only remembered that it was of the form $X515X0$ and was divisible by 36. What was the value of X?
(a) 4 (b) 7 (c) 8 (d) 9

Q.37 Students from the Delhi Public School are writing their exams in Kendriya Vidyalaya. There are 60 students writing their Hindi exams, 72 students writing French exam, and 96 students writing their English exam. The authorities of the Kendriya Vidyalaya have to make arrangements such that each classroom contains equal number of students. What is the minimum number of classrooms required to accommodate all students of Delhi Pubic School?
(a) 19 (b) 38 (c) 13 (d) 6

Q.38 In the Jyotirmayi school, all classes started at 9:00 am. The school has three sections: primary, middle, and secondary. Each class for the primary section lasts for 30 min, for the middle section for 45 min, and for the secondary section for 30 min. A lunch break has to be given for the entire school when each of three sections have just finished a respective class and are free. What is the earliest time for the lunch break?
(a) 11:00 am (b) 10:30 am
(c) 12:00 pm (d) 12:30 pm

Q.39 In the firing range, four shooters are firing at their respective targets. The first, the second, the third, and the fourth shooter hit the target once every 5 s, 6 s, 7 s, and 8 s, respectively. If all of them hit their target at 10:00 am, when will they hit their target together again?
(a) 10:14 am (b) 10:28 am
(c) 10:30 am (d) 10:31 am

Q.40 Two friends Harry and Jayesh were discussing about two numbers. They found the two numbers to be such that one was twice the other. However, both had the same number of prime factors, while the larger one had 4 more factors than the smaller one. What are the numbers?
(a) 40, 80 (b) 20, 40
(c) 30, 60 (d) 50, 100

Q.41 To celebrate their victory in the World Cup, the Sri Lankans distributed sweets. If the sweets were distributed among 11 players, 2 sweets were left. When the sweets were distributed to 11 players, 3 extra players, and 1 coach, even then 2 sweets were left. What is the minimum number of sweets in the box?
(a) 167 (b) 334 (c) 332 (d) 165

Q.42 The first 20 natural numbers from 1 to 20 are written next to each other to form a 31-digit number $N = 12345678910111213141516171819 20$. What is the remainder when this number is divided by16?
(a) 0 (b) 4 (c) 7 (d) 9

Q.43 Two friends Kanti and Sridhar were trying to find the HCF of 50 distinct numbers. If they were finding the HCF of two numbers at a time, how many times this operation should be repeated to find the HCF of 50 numbers?
(a) 20 (b) 25 (c) 49 (d) 50

Q.44 How many zeroes will be there at the end of $N = 18! + 19!$?
(a) 3 (b) 4
(c) 5 (d) Cannot be determined

Q.45 Manish was dividing two numbers by a certain divisor and obtained remainders as 437 and 298, respectively. When he divides the sum of the two numbers by the same divisor, the remainder is 236. What is the divisor?
(a) 499 (b) 735
(c) 971 (d) None of these

Q.46 I purchased a ticket for the football match between France and Italy in the World Cup. The number on the ticket was a 5-digit perfect square such that the first and

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the last digit were the same and the second and fourth digit were the same. If the third digit was 3, then what was the ticket number?
(a) 24,342 (b) 12,321
(c) 21,312 (d) None of these

Q.47 How many integers N in the set of integers {1, 2, 3,..., 100} are there such that $N^2 + N^3$ is a perfect square?
(a) 5 (b) 7 (c) 9 (d) 11

Q.48 In a birthday party, all the children were given candy bars. All the children got three candy bars each, except the child sitting at the end who got only 2 candy bars. If each child had been given only 2 candy bars, there would have been 8 candy bars remaining. How many children were there and how many candy bars were distributed?
(a) 9, 26 (b) 6, 26 (c) 9, 18 (d) 6, 15

Q.49 A natural number N satisfies following conditions.
(A) Number is having all the 9s.
(B) It is divisible by 13.
How many digits are there in N?
(a) 5 (b) 6 (c) 7 (d) 8

Q.50 What is the minimum number of identical square tiles required to cover a floor of dimension 3.78 m and 4.8 m?
(a) 2520 (b) 3780
(c) 5040 (d) 6480

Q.51 What is the smallest five-digit number which when divided by 7, 11, and 21 leave a remainder of 3 in each case?
(a) 10,019 (b) 10,001
(c) 10,111 (d) 10,167

Q.52 A milkman has 3 jars containing 57 L, 129 L, and 177 L of pure milk, respectively. A measuring can leaves the same amount of milk unmeasured in each jar after a different number of exact measurements of milk in each jar. What is the volume of largest such can?
(a) 12 L (b) 16 L (c) 24 L (d) 48 L

Q.53 A boy was carrying a basket of eggs. He fell down and some of the eggs were broken. The boy has 10 eggs left with him. When asked by his mother, how many eggs were broken, the boy could not recall. However, he recalled that when 1 egg was left, he counted the total number of eggs as 3. When counted 4 at a time, 1 egg was left and when counted 5 at a time, no egg was left. How many eggs were broken?
(a) 15 (b) 25 (c) 30 (d) 35

Q.54 How many prime numbers are there which, when divided by another prime number, gives a quotient which is same as the remainder?
(a) 0 (b) 1
(c) 2 (d) More than 2

Q.55 Let A, B, and C be digits such that $(100A + 10B + C)(A + B + C) = 2005$. What is the value of A?
(a) 4 (b) 2 (c) 3 (d) 1

Q.56 Find the product of all the factors of 3^{16}.
(a) 3^{33} (b) 3^{68} (c) 3^{136} (d) 3^{128}

Q.57 What is the remainder when 90^{91} is divided by 13?
(a) 0 (b) 7 (c) 12 (d) 1

Q.58 Find the remainder when the product of 10 consecutive natural numbers starting from 8641 is divided by 8640.
(a) 1 (b) 55 (c) 10 (d) 0

Q.59 A faulty car odometer proceeds from digit 3 to digit 5, always skipping digit 4, regardless of position. If the odometer now reads 002,005, how many miles has the car actually travelled?
(a) 1404 (b) 1462 (c) 1604 (d) 1605

Q.60 How many numbers are there between 400 and 600 in which 8 occurs only once?
(a) 36 (b) 18 (c) 19 (d) 38

Q.61 If $n^2 = 123.45654321$, which of the following is the exact value of n?
(a) 11.1001 (b) 11.1101 (c) 11.1111 (d) 11.1011

Q.62 A mule said to a horse, 'If I take one sack off your back, my load will be double of yours and if you take one off my back, our loads will be the same.' How many sacks in all were they carrying?
(a) 5 (b) 7 (c) 12 (d) 14

Q.63 Divide 45 into 4 parts such that if the first is increased by 2, the second is decreased by 2, the third multiplied by 2, and the fourth divided by 2, the result is the same.
(a) 20, 8, 5, 12
(b) 12, 5, 20, 8
(c) 5, 8, 12, 20
(d) 8, 12, 5, 20

Q.64 Find the remainder when $3x^2 - x^6 + 31x^4 + 21x + 5$ is divided by $(x + 2)$.
(a) 10 (b) 12
(c) 11 (d) None of these

Q.65 Four prime numbers are in ascending order of their magnitudes. The product of the first three is 385 and that of the last three is 1001. The largest given prime number is:
(a) 11 (b) 13 (c) 17 (d) 19

Q.66 What is the remainder when 4^{44} is divided by 15?
(a) 1 (b) 2 (c) 3 (d) 4

Q.67 LCM of two integers P and Q is 211. What is the HCF of P and Q?
(a) 37 (b) 1
(c) 3 (d) Cannot be determined

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Q.68 How many times does the digit 6 appear when we count from 11 to 400?
(a) 34 (b) 74 (c) 39 (d) 79

Q.69 In Q.68, how many numbers will be having 8 as its digit?
(a) 74 (b) 75 (c) 76 (d) 77

Q.70 S is a number formed by writing 8 for 88 times. What will be the remainder of this number when divided by 7?
(a) 4 (b) 5 (c) 8 (d) 1

Q.71 We are writing all the multiples of 3 from 111 to 324. How many times will we write digit 3?
(a) 18 (b) 19 (c) 21 (d) 22

Q.72 What is the remainder when $7 + 77 + 777 + 7777 + \ldots$ (till 100 terms) is divided by 8?
(a) 0 (b) 2 (c) 4 (d) 6

Q.73 A number has exactly 15 composite factors. What can be the maximum number of prime factors of this number?
(a) 2 (b) 3 (c) 4 (d) 5

Q.74 $N = 204 \times 221 \times 238 \times 255 \times \ldots \times 850$. How many consecutive zeroes will be there at the end of this number N?
(a) 8 (b) 10 (c) 11 (d) 12

Q.75 1st 126 natural numbers are put side by side in the ascending order to create a large number N = 123456…125126. What will be the remainder when N is divided by 5625?
(a) 5126 (b) 26 (c) 126 (d) 156

Q.76 When a number S is divided by 3, 4, and 7 successively, remainders obtained are 2, 1, and 4 respectively. What will be the remainder when the same number is divided by 84?
(a) 43 (b) 53 (c) 63 (d) 73

Q.77 What is the remainder when $1714 \times 1715 \times 1717$ is divided by 12?
(a) 3 (b) 8 (c) 2 (d) 9

Q.78 $N^2 = 12345678987654321$. Find N.
(a) 101010101 (b) 11111
(c) 111111111 (d) 1000000001

Q.79 If a, b, c, and d are distinct integers in the range 10–15 (both inclusive), find the greatest value of $(a + b)(c + d)$ is.
(a) 750 (b) 731 (c) 700 (d) 729

Q.80 The smallest natural number which is a perfect square and is of the form $abbb$ lies in between:
(a) 1000 to 2000 (b) 2000 to 3000
(c) 3000 to 4000 (d) 4000 to 5000

Moderate

Q.1. How many number of zeroes will be there at the end of 12! expressed in base 6?
(a) 4 (b) 5 (c) 6 (d) 7

Q.2 Find the remainder when $2222^{5555} + 5555^{2222}$ is divided by 7.
(a) 1 (b) 3 (c) 0 (d) 5

Q.3 LCM of first 100 natural numbers is N. What is the LCM of first 105 natural numbers?
(a) $5! \times N$ (b) 10,403 N (c) 105N/103 (d) 4N

Q.4 How many divisors of 10^5 end with a zero?
(a) 1 (b) 3 (c) 9 (d) 16

Q.5 Following expression holds true if we replace some of '+' signs by '×' signs.
$1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10 = 100$
How many '+' signs are needed to be replaced by '×'?
(a) 2 (b) 3 (c) 4 (d) 1

Q.6 In a particular country, all the numbers are expressed with the help of three alphabets a, b, and c.
15 is written as abc.
6 is written as bc.
60 is written as $bcbc$.
How would one write 17 in that country?
(a) abb (b) bab (c) baa (d) aba

Q.7 When a certain two-digit number is added to another two digit number having the same digits in reverse order, the sum is a perfect square. How many such two-digit numbers are there?
(a) 4 (b) 6 (c) 8 (d) 10

Q.8 What is the remainder when ${}_{32}32^{32}$ is divided by 7?
(a) 2 (b) 3 (c) 4 (d) 6

Q.9 N is a 1001 digit number consisting of 1001 sevens. What is the remainder when N is divided by 1001?
(a) 7 (b) 700
(c) 777 (d) None of these

Q.10 Find four positive numbers such that the sum of the first, third, and fourth exceeds the second by 8; the sum of the squares of the first and second exceed the sum of the squares of the third and fourth by 36; the sum of the products of the first and second, and of the third and fourth is 42; the cube of the first is equal to the sum of the cubes of the second, third, and fourth.

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(a) 2, 1, 9, 3 (b) 2, 4, 6, 8
(c) 6, 5, 4, 3 (d) None of these

Q.11 Digital sum of a number is obtained by adding all the digits of a number until a single digit is obtained. Find the digital sum of 19^{100}.
(a) 1 (b) 4 (c) 7 (d) 9

Q.12 Find the HCF of $(2^{100} - 1)$ and $(2^{1}1)$.
(a) $2^{10} - 1$ (b) $2^{20} - 1$
(c) 1 (d) None of these

Q.13 Let S be the set of positive integers n for which $1/n$ has the repeating decimal representation $0.\overline{ab} = 0.ababab\ldots$, with a and b as different digits. What is the sum of the elements of S?
(a) 11 (b) 44 (c) 110 (d) 143

Q.14 An intelligence agency forms a code of two distinct digits selected from 0, 1,..., 9 such that the first digit of the code is non-zero. However, the code, handwritten on a slip, can potentially create confusion when read upside down, for example, code 91 may appear as 16. How many codes are there for which no such confusion can arise?
(a) 80 (b) 63
(c) 71 (d) None of these

Q.15 If p, $(p + 2)$, and $(p + 4)$ are prime numbers, then the number of possible solutions for p is:
(a) 0 (b) 1
(c) 2 (d) None of these

Q.16 Suppose N is an integer such that the sum of the digits of N is 2, and $10^9 < N < 10^{10}$. How many values of N are possible?
(a) 11 (b) 10 (c) 9 (d) 8

Q.17 Ten students solved a total of 35 questions in a Maths Olympiad. Each question was solved by exactly one student. There is at least one student who solved exactly one problem, at least one student who solved exactly two problems, and at least one student who solved exactly three problems. What is the minimum number of students who has/have solved at least five problems?
(a) 1 (b) 2
(c) 3 (d) None of these

Q.18 N has 37 zeroes at its end. How many values of N is/are possible?
(a) 0 (b) 1 (c) 5 (d) Infinite

Q.19 In the abovementioned question, how many values of N will be even?
(a) 0 (b) 2 (c) 3 (d) Infinite

Q.20 $N!$ is having 30 zeroes at its end. How many values of N is/are possible?
(a) 0 (b) 1 (c) 5 (d) Infinite

Q.21 What is the remainder when $(1^1 + 2^2 + 3^3 + \ldots + 100^{100})$ is divided by 4?
(a) 0 (b) 1 (c) 2 (d) 3

Q.22 A three-digit number in which all the three-digit numbers are odd is such that if the cubes of the digit are added, the sum would be equal to the number itself. If one of the digit is 7, find the number.
(a) 171 (b) 371 (c) 575 (d) 775

Q.23 A teacher said that there were 100 students in his class, 24 of whom were boys and 32 were girls. Which base system did the teacher use in this statement?
(a) 9 (b) 5 (c) 6 (d) 8

Q.24 What is the remainder when 3^{450} is divided by 108?
(a) 3 (b) 1 (c) 27 (d) 81

Q.25 P is a natural number. $2P$ has 28 divisors and $3P$ has 30 divisors. How many divisors of $6P$ will be there?
(a) 35 (b) 40 (c) 45 (d) 48

Q.26 pqr is a three-digit natural number such that $pqr = p^3 + q^3 + r^3$. What is the value of r?
(a) 0 (b) 1
(c) 3 (d) Cannot be determined

Q.27 There are two three-digit numbers. When one number is divided by another number, quotient obtained is 6 and remainder is 0. The sum total of both the numbers is a multiple of 504. What is the difference between the numbers?
(a) 720 (b) 360 (c) 120 (d) 420

Q.28 LCM of 12^{24}, 16^{18}, and N is 24^{24}. The number of all the possible values of $N = S$. What is the value of S?
(a) 25 (b) 1800
(c) 1825 (d) None of these

Q.29 Each of P, Q, R, and S equals either 0 or 1. It is given that
If $Q = 0$, then $R = 1$;
If $R = 0$, then $P = S$;
If $S = 0$, then $P = 1$.
Let us assume that $R = 0$, find the value of $(P + Q + R + S)$?
(a) 0 (b) 1 (c) 2 (d) 3

Q.30 When asked about his date of birth in 1996, Mayank replied that 'last two digits of my birth year stands for my age.' When Siddharth was asked about his age, he also replied the same. However, Siddharth is older to Mayank. What is the difference in their age?
(a) 46 (b) 50
(c) 0 (d) Cannot be determined

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Direction for Questions 31 to 35: Read the following passage and solve the questions based on it.

There is a prison with 100 cells inside it. Cells are numbered from 1 to 100 and every cell is occupied by one prisoner only. One day jailer decides to release some of the prisoners, and for this, he defines an algorithm of 100 steps, which are as follows:

Step 1 Reverse the position of all the cells that are divisible by 1.

Step 2 Reverse the position of all the cells that are divisible by 2.

Step 3 Reverse the position of all the cells that are divisible by 3.

.
.
.

Step 99 Reverse the position of all the cells that are divisible by 99.

Step 100 Reverse the position of all the cells that are divisible by 100.

Initially, all the cells are closed. After executing all these steps, prisoners of all the cells that remain open are released.

Q.31 How many prisoners are released?
(a) 25 (b) 10
(c) 5 (d) None of these

Q.32 Which of the following cell number will be open at the end?
(a) Cell number 56 (b) Cell number 64
(c) Cell number 72 (d) Cell number 84

Q.33 Which of the following is true about the family of cell numbers N that will be open at the end?
(a) All the elements of N will be having only two factors including the number itself.
(b) All the elements of N will be having odd number of factors excluding the number itself.
(c) All the elements of N will be having odd number of factors including the number itself.
(d) None of these

Q.34 If number of prisoners in each cell is equal to the cell number, then how many prisoners will be released?
(a) 285 (b) 385
(c) 485 (d) None of these

Q.35 There are many islands on earth for which we do not have much information, or sometime we do not have any information about those islands. On such an island, which uses a system of S digits ($S \neq 10$) to write the numbers, selling price of a goat is ₹1143. Parul bought a goat and paid the shopkeeper ₹1150. Now, the shopkeeper returned Parul ₹5. How many digits are used to write the numbers in that system?
(a) 9 (b) 8 (c) 7 (d) 6

Q.36 What is the value of N in the following expression $345_6 + 632_7 + 487_9 = (N)_5$?
(a) 11,412 (b) 11,214
(c) 10,412 (d) 21,412

Q.37 In a certain system of writing of numbers, we write $(12\Phi)_4 = (124)_\Phi$, where Φ is the units digit in the first place and number of digits in that system in the second place.
How many values of Φ are possible?
(a) 0 (b) 1 (c) 3 (d) Infinite

Q.38 Digital sum of a number is obtained by adding all the digits of a number until a single digit is obtained. What is the digital sum of 4444^{4444}?
(a) 2 (b) 6 (c) 7 (d) 3

Direction for Questions 39 to 41: Read the following passage and solve the questions based on it.

$N = A^2 \times B^3 \times C^4$, where A, B, and C are prime numbers.

Q.39 How many factors of N are perfect squares?
(a) 6 (b) 12
(c) 8 (d) None of these

Q.40 How many factors of N are cube of any natural number?
(a) 2 (b) 3 (c) 4 (d) 6

Q.41 How many factors of N will be having exactly 3 factors?
(a) 2 (b) 3
(c) 4 (d) None of these

Direction for Questions 42 and 43: Read the following passage and solve the questions based on it.

M and N are two natural numbers such that $M + N = 949$. LCM of M and N is 2628.

Q.42 What is the HCF of M and N?
(a) 23 (b) 73
(c) 69 (d) None of these

Q.43 When M and N are divided by P, remainder obtained is same. How many values of P are possible?
(a) 0 (b) 1 (c) 2 (d) 3

Q.44 What is the difference between the largest four-digit number and the smallest four-digit number, when written in hexadecimal system?
(a) FFFF (b) EFFF (c) EEEE (d) EFEF

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Q.45 A 10-digit number is said to be interesting if its digits are all distinct and it is a multiple of 11,111. How many interesting integers are there?
(a) 1001 (b) 101
(c) 10,001 (d) None of these

Q.46 Sum of a three-digit number and its mirror image is a multiple of 111. What is this sum (provided 6 is one of the digits of this three-digit number)?
(a) 999 (b) 666 (c) 888 (d) 777

Q.47 Consider the following statements:
(i) If $(2)_N \times (4)_N = (8)_N$, where $(X)_N$ is a number written in a system of writing having N digits, then N can have infinite values.
(ii) If $(4)_N \times (5)_N = (24)_N$, where $(X)_N$ is a number written in a system of writing having N digits, then N will have more than 1, but finite values.
(iii) If $(5)_N \times (6)_N = (3A)_N$, where $(X)_N$ is a number written in a system of writing having N digits and A is the units digit of this number $(3A)$ written in a system of writing the numbers having N digits, then A can have four values.
How many of the abovementioned statements are true?
(a) 1 (b) 2
(c) 3 (d) None of these

Q.48 How many numbers between 1 and 250 can be represented in the form x^y, where $y > x > 1$?
(a) 6 (b) 7 (c) 8 (d) 9

Q.49 What will be the tens place digit in the following expression:
$1! + 2! + 3! + ... + 70!$?
(a) 1 (b) 3 (c) 5 (d) 6

Q.50 A two-digit number is K times the sum of its digits. When these numbers are interchanged, it equals N multiplied by the sum of digits. Which of the following is equal to N?
(a) $K - 11$ (b) $99 - K$
(c) $11 - K$ (d) $K - 99$

Q.51 What is the remainder when ${}_{32}33^{34}$ is divided by 7?
(a) 4 (b) 2 (c) 3 (d) 1

Q.52 1s are given 100 times, 2s are given 100 times, and 3s are given 100 times. Now, numbers are made by arranging these 300 digits in all possible ways. How many of these numbers will be perfect squares?
(a) 10101 (b) 1001001
(c) 100010001 (d) None of these

Q.53 Which of the following number is the greatest number dividing a family of numbers $(N^5 - N)$, where N is any natural number?
(a) 0 (b) 15 (c) 30 (d) 120

Q.54 When Rahul Ghosh distributes ₹5 to some of his employees, he is left with ₹1. When he distributes ₹6 to some of his employees, he is left with ₹5. What is the sum of digits of the minimum possible sum with Rahul Ghosh if he is having more than ₹100?
(a) 1 (b) 2 (c) 3 (d) 4

Q.55 The integers 3441 and 32,506 when divided by a three-digit integer N leave the same remainder. What can be the value of N?
(a) 289 (b) 307 (c) 317 (d) 319

Q.56 In the abovementioned question, how many values of N is/are possible?
(a) 2 (b) 1
(c) 4 (d) None of these

Q.57 What is the remainder when 77,777... up to 56 digits is divided by 19?
(a) 1 (b) 7 (c) 9 (d) 13

Q.58 There are three consecutives odd integers such that the product is a prime number. Find the largest of the prime number.
(a) 1 (b) 3 (c) 5 (d) 7

Q.59 A square floor of the dimensions 72 cm × 72 cm has to be laid with rectangular tiles whose length and breadth are in the ratio 3:2. What is the difference between the maximum number of tiles and minimum numbers of tiles, given that the length and the breath are integers?
(a) 858 (b) 864 (c) 868 (d) 872

Q.60 A man took a five-digit number ending in 9 and raised it to an even power greater than 50. He then multiplied it with 17 raised to a multiple of 4. What is the last digit of the resulting number?
(a) 1 (b) 9 (c) 3 (d) 7

Q.61 Rahul took an odd number of prime numbers and added them. He found the result to be even. Which of the following could be concluded?
(a) The numbers include an even number.
(b) The numbers do not include an even number.
(c) There are even number of even numbers.
(d) None of these.

Q.62 When asked as to how many guavas Lovely has, she replies, 'if you add three quarters of the number of guavas that I have to three quarters of one guava, you will get the numbers of guavas I have'. How many guavas does she have?
(a) 2 (b) 5
(c) 7 (d) None of these

Direction for Questions 63 to 65: Read the following passage and solve the questions based on it.

Rohan purchased some pens, pencils, and erasers for his young brothers and sisters for the ensuing examinations.

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He had to buy at least 11 pieces of each items in a manner that the number of pens purchased be more than the number of pencils, which is more than the number of erasers. He purchased a total of 38 pieces.

Q.63 How many erasers did Rohan purchase?
(a) 11 (b) 10
(c) 8 (d) Cannot be determined

Q.64 If each eraser costs ₹3, each pencil ₹2, and each pen ₹10, what is the maximum amount that Rohan could have spent?
(a) 207 (b) 255 (c) 288 (d) 300

Q.65 If the number of pencils cannot be equally divided among his 4 brothers and sisters, how many pens did he purchase?
(a) 11 (b) 12 (c) 13 (d) 14

Q.66 What is the remainder when [100! + 1] is divided by 101?
(a) 100 (b) 1 (c) 0 (d) 2

Q.67 A is a set of first 10 natural numbers. B is a subset of A such that B consists of exactly 2 numbers, both co-primes to each other. Find the maximum number of such possible subsets.
(a) 29 (b) 28 (c) 33 (d) 31

Q.68 When written in the lowest term, rational numbers A and B have denominators 60 and 70, respectively. Now, A and B have been added. What is the smallest possible denominator of $(A + B)$?
(a) 72 (b) 84 (c) 105 (d) 148

Direction for Questions 69 and 70: Read the passage below and solve the questions based on it.

The multiplication of two numbers is shown below.

$$\begin{array}{r} AD4 \\ \times E \\ \hline A206 \end{array}$$

where A, D, and E are all distinct digits.

Q.69 The value of D is:
(a) 3 (b) 9 (c) 2 (d) 7

Q.70 The value of $(A + E)$ is:
(a) 6 (b) 9 (c) 11 (d) 10

Q.71 P, Q, R, S, and T are five prime numbers, where $P < Q < R < S < T$. It is also given that $P + Q + R + S + T = 482$. What is the value of P^5?
(a) 243 (b) 32
(c) 16,807 (d) More than one value

Q.72 If n is a natural number greater than 1, then $1075^n - 1075^{n-1}$ is not divisible by:
(a) 15 (b) 23 (c) 43 (d) 179

Direction for Questions 73 and 74: Read the following passage and solve the questions based on it.

(a, b) is an ordered pair such that a and b are two-digit numbers and a is the number obtained by interchanging the digits of b.

Q.73 How many ordered pairs are there such that $(a + b)$ is a perfect square?
(a) 4 (b) 8
(c) 3 (d) None of these

Q.74 How many ordered pairs are there such that $(a - b)$ is a perfect square greater than zero?
(a) 3 (b) 5
(c) 13 (d) None of these

Q.75 N is an even number 8 when written in the decimal system. If S is the sum of the digits of N, when it is written in base 7, we can surely say that S would always be:
(a) Even (b) Odd
(c) Composite (d) Both (a) and (c)

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Q.1 Find the last non zero digit of 96!.
(a) 2 (b) 4 (c) 6 (d) 8

Q.2 What will be remainder when 1212121212... 300 times is being divided by 99?
(a) 18 (b) 81 (c) 54 (d) 36

Q.3 Two players A and B are playing a game of putting '+' and '–' signs in between any two integers written from 1 to 100. A starts the game by placing a + sign anywhere between any two integers. Once all the signs have been placed, the result is calculated. If it is even, then A wins and if it is odd, then B wins, provided they are placing signs by taking turns one by one and either of them can place any sign anywhere between any two integers. Who will win at the end?
(a) A (b) B
(c) Either A or B (d) Cannot be determined

Q.4 What is the remainder when 2^{100} is divided by 101?
(a) 1 (b) 100
(c) 99 (d) None of these

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Q.5 If $S = \{X_1, X_2, X_3, \ldots X_n\}$, where $n > 0$, $X_1 = 1$ is a set of positive numbers where $X = X_{n-1} + 2^n - 1$. Which of the following sets of numbers does S resemble?
(a) Fibonacci (b) Cube
(c) Square (d) Prime

Q.6 A three-digit number ABC is a perfect square and the number of factors of this number is also a perfect square. If $(A + B + C)$ is also a perfect square, then what is the number of factors of the six-digit number $ABCABC$?
(a) 32 (b) 52
(c) 72 (d) Cannot be determined

Q.7 How many divisors of 10^5 will have at least one zero at its end?
(a) 9 (b) 12 (c) 15 (d) 25

Q.8 Let $V_1, V_2, V_3, \ldots, V_{100}$ be hundred positive integers such that $V_i + V_{i+1} + V_{i+2} + V_{i+3} = K$, where K is a constant and $i = 1, 2, 3, \ldots, 97$. If $V_3 = 9$, then what is the value of V_{99}?
(a) 9 (b) $(K - 9)$
(c) $(K/2 - 9)$ (d) Cannot be determined

Q.9 In the abovementioned question, if $V_5 = 7$, then what is the value of V_{90}?
(a) 7 (b) $(K - 7)$
(c) $(K - 7)/2$ (d) Cannot be determined

Q.10 What is the largest integer that is a divisor of $(n + 1)(n + 3)(n + 5)(n + 7)(n + 9)$ for all positive even integers n?
(a) 3 (b) 5 (c) 11 (d) 15

Q.11 If K is any natural number, such that $100 < K < 200$, how many values of K exist such that $K!$ has 'z' zeroes at its end and $(K + 2)!$ has '$z + 2$' zeroes at its end?
(a) 2 (b) 4
(c) 6 (d) None of these

Q.12 Tatto bought a notebook containing 96 leaves and numbered them which came to 192 pages. Tappo tore out the latter 25 leaves of the notebook and added the 50 numbers she found on those pages. Which of the following is not true?
(a) She could have found the sum of pages as 1990.
(b) She could have found sum of pages as 1275.
(c) She could have got sum of pages as 1375.
(d) None of these

Direction for Questions 13 to 15: Read the following passage and solve the questions based on it.

There are 50 integers $a1, a2, a3, \ldots, a50$; not all of them are necessarily different. Let the greatest integer of these integers be referred to as G and smallest integer be referred to as L. The integers $(a1–a24)$ form a sequence and the rest form a sequence $S2$. Each member of $S1$ is less than or equal to each member of $S2$.

Q.13 All values in $S1$ are changed in sign, while those in $S2$ remain unchanged. Which of the following statements is true?
(a) Every member of $S1$ is greater than or equal to every member of $S2$.
(b) G is in $S1$.
(c) If all the numbers originally in $S1$ and $S2$ had the same sign, then after the change of sign, the largest number of $S1$ and $S2$ is in $S1$.
(d) None of these

Q.14 Elements of $S1$ are in ascending order and those of $S2$ are in descending order. If $a24$ and $a25$ are interchanged, then which of the following is true?
(a) $S1$ continues to be in ascending order.
(b) $S2$ continues to be in descending order.
(c) Both (a) and (b)
(d) Cannot be determined

Q.15 Every element of $S1$ is made greater than or equal to every element of $S2$ by adding an integer x to each element of $S1$. Then, x cannot be less than:
(a) 2^{10}
(b) The smallest value of $S2$
(c) The largest value of $S2$
(d) $(G - L)$

Q.16 Twenty-five boxes of sweets are delivered to Mr Roy's home. Mr Roy had ordered three different types of sweets. What is the minimum number of boxes of sweets that are having sweets of same type?
(a) 1 (b) 8
(c) 9 (d) Cannot be determined

Q.17 A warehouse contains 200 shoes of size 8, 200 shoes of size 9, and 200 shoes of size 10. Of these 600 shoes, there are 300 left shoes and 300 right shoes. What is the minimum number of usable shoes?
(a) 50 (b) 100
(c) 200 (d) None of these

Q.18 A teacher was doing some calculation exercise on the blackboard. When the teacher went out, a naughty student Chunmun erased some of the numbers written on the blackboard. Now, it appeared like this

$$\begin{array}{r} 23_5_ \\ +1_642 \\ \hline 42423 \end{array}$$

When teacher entered the room, he realized that still this calculation was right, but in some other system of writing (i.e., not 10). How many digits are there in that system?
(a) 11 (b) 9
(c) 7 (d) 8

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Q.19 Tatto, Tappo, and Bubbly were solving problems from a problem book. Each solved exactly 60 problems, but they solved only 100 problems altogether. Any problem is known as 'easy' if it was solved by all of them, and 'difficult' if it was solved by only one of them. What is the difference between the number of 'difficult' problems and number of 'easy' problems?
(a) 10 (b) 20 (c) 30 (d) 40

Q.20 LCM of two numbers A and $B = P^x \times Q^y$, where P and Q are prime numbers and x and y are positive whole numbers. How many set of values are possible for A and B?
(a) $xy\,(x+y)$ (b) $xy\,(x-y)$
(c) $x^2y^2\,(x+y)$ (d) None of these

Q.21 When 7179 and 9699 are divided by another natural number N, remainder obtained is same. How many values of N will be ending with one or more than one zeroes?
(a) 24 (b) 124
(c) 46 (d) None of these

Q.22 There exists a five-digit number N with distinct and non-zero digits such that it equals the sum of all distinct three-digit numbers whose digits are all different and are all digits of N. Then, the sum of the digits of N is necessarily a:
(a) Perfect square (b) Cube
(c) Even (d) None of these

Q.23 Starting with 1, positive integers are written one after the other. What is the 400,000th digit that will be written?
(a) 3 (b) 6
(c) 8 (d) None of these

Q.24 Which of the following would always divide a six-digit number of the form *ababab*?
(a) 10,101 (b) 11,111
(c) 10,001 (d) None of these

Q.25 If the number system of a particular country, 25 means 5 tens and 2 units, 467 means 7 hundreds, 6 tens, and 4 units. Then, find the value of 173×425?
(a) 4,04,491 (b) 7,35,255
(c) 6,22,744 (d) 5,25,376

Q.26 Let A be the set of integers N such that
(i) $100 \le N \le 500$
(ii) N is even
(iii) N is divisible by either 2 or 3 or 4 but not by 7.
How many elements are there in set A?
(a) 171 (b) 172 (c) 170 (d) 173

Q.27 Three distinct prime numbers, which are less than 10, are taken and all the numbers that can be formed by arranging all the digits are taken. Now, difference between the largest and the smallest number formed is equal to 495. It is also given that the sum of the digits is more than 13. What is the product of the numbers?
(a) 30 (b) 70 (c) 105 (d) 315

Q.28 What is the remainder when ${}_{55}56^{57}$ is divided by 17?
(a) 1 (b) 4 (c) 13 (d) 17

Q.29 What is the remainder when ${}_{20}51^{97}$ is divided by 17?
(a) 10 (b) 9 (c) 15 (d) 7

Q.30 $[111 \ldots 111(200 \text{ digits}) - 222\ldots22(100 \text{ digits})]^{1/2}$ is equal to
(a) 1313... 1313(100 digits)
(b) 2121...2121(100 digits)
(c) 1111...1111(100 digits)
(d) 3333...333(100 digits)

Q.31 N is a number which when divided by 10 gives 9 as the remainder,
when divided by 9 gives 8 as the remainder,
when divided by 8 gives 7 as the remainder,
when divided by 7 gives 6 as the remainder,
when divided by 6 gives 5 as the remainder,
when divided by 5 gives 4 as the remainder,
when divided by 4 gives 3 as the remainder,
when divided by 3 gives 2 as the remainder,
when divided by 2 gives 1 as the remainder.
What is N?
(a) 2519 (b) 841
(c) 839 (d) 2521

Q.32 What is the remainder when $(10^3 + 9^3)^{752}$ is divided by 12^3?
(a) 1 (b) 729 (c) 752 (d) 1000

Q.33 $f(x, x) = x^x$ and $f(f(x, x)) = {}_x x^x$ and so on. What is the value of $f(f(f(f(7,7))))$ when divided by 5?
(a) 1 (b) 2 (c) 3 (d) 4

Q.34 The students of class 10th of Morgan High School took a test, which had a maximum of 50 marks. The teacher misplaced the text notebooks of two of the students—Robin and Garry; however, she remembered that Garry had scored something between 10 and 15 and Robin something between 32 and 40. She also remembered that the product of the marks obtained by the two students was also equal to 10 times the marks obtained by two of them. How many marks did Garry scored?
(a) 11 (b) 12 (c) 13 (d) 14

Q.35 The History teacher was referring to a year in the 19th century. Rohan found an easy way to remember the year. He found that the number, when viewed in a mirror, increased 4.5 times. Which year was the teacher referring to?
(a) 1,801 (b) 1,810
(c) 1,818 (d) More than one value

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Q.36 Srini wrote his class 10th board examination this year. When the result were out, he searched for his hall ticket to see his roll number, but could not trace it. He could remember only the first three digits of the six-digit number as 267. However, his father remembered that the number was divisible by 11. His mother gave the information that the number was also divisible by 13. They tried to recollect the number when all of a sudden Srini told that the number was a multiple of 7. What was the units digits of the number?

(a) 5 (b) 7
(c) 2 (d) Cannot be determined

Q.37 Prof. Mathur and Prof. Singh attended the All India Historian's meet last week. Prof. Mathur told Prof. Singh, 'I found out that your teaching experience is twice that of mine'. Prof. Singh replied in the affirmative. Prof'. Mathur continued, 'But last time when both of us came for the same meet, I remember that your teaching experience was thrice that of mine'. 'That was 2 years ago.' Prof. Singh said. How many years has Prof. Singh been working?

(a) 8 (b) 10 (c) 12 (d) 16

Direction for Questions 38 and 39: Read the passage below and solve the questions based on it.

ABCDEF is a six-digit number with distinct digits. Further, the number is divisible by 11 and the sum of its digits is 24. Further, $A > C > E$ and $B > D > F$.

Q.38 The sum $(A + C + E)$ is equal to:

(a) 12 (b) 6
(c) 8 (d) Cannot be determined

Q.39 $(A + B)$ is always:

(a) 10 (b) 9
(c) 6 (d) Cannot be determined

Q.40 Raju had to divide 1080 by N, a two-digit number. Instead, he performed the division using M, which is obtained by reversing the digits of N, and ended up with a quotient that was 25 less than what he should have obtained otherwise. If 1080 is exactly divisible both by N and M, find the sum of the digits of N.

(a) 6 (b) 8
(c) 9 (d) None of these

Q.41 Let $S = \{1, 2, 3,... n\}$ be a set of N natural numbers. Let T be a subset of S such that the sum of any three elements of T is not less than N. Find the maximum number of elements in any such subset T for $N = 40$?

(a) 26 (b) 27
(c) 28 (d) None of these

Q.42 The last digit of the LCM of $(3^{2003} - 1)$ and $(3^{2003} + 1)$ is:

(a) 8 (b) 2 (c) 4 (d) 6

Q.43 a, b, and c are positive integers such that $(a + b + c) = 2003$. Let $E = (-1)^a + (-1)^b + (-1)^c$. Find the number of possible values of E.

(a) 2004 (b) 3 (c) 1003 (d) 2

Q.44 Ajay took a four-digit number in base 5 notation. He subtracted the sum of the digits of the numbers from the number. From the result, he struck off one of the digits. The remaining three digits were 1, 0, and 2. Then, the digit struck off by Ajay was:

(a) 2 (b) 1
(c) 4 (d) Cannot be determined

Direction for Questions 45 and 46: Read the following passage and solve the questions based on it.

N is a single digit integer satisfying the following two conditions.

(i) N is non-zero.
(ii) N is the right most digit of the number $(n!)^4$, where n is a natural number greater than 1.

Q.45 What is the number of possible values of V?

(a) 1 (b) 2
(c) 0 (d) None of these

Q.46 If condition (a) is relaxed, the number of possible values of V is:

(a) 1 (b) 2
(c) 0 (d) More than 2

Q.47 A teacher wrote a number on the blackboard and the following observations were made by the students.
The number is a four-digit number.
The sum of the digits equals the product of the digits.
The number is divisible by the sum of the digits.
The sum of the digits of the number is:

(a) 8 (b) 10 (c) 12 (d) 14

Q.48 N of odd numbers are taken. The product of these odd numbers is of the form $(4n + 1)$, where n is any natural number. Which of the following is true regarding the number of numbers?

(a) There must have been an odd number of numbers of the form $(4n + 1)$.
(b) There must have been an even number of numbers of the form $(4n + 1)$.
(c) There must have been an even number of numbers of the form $(4n + 3)$.
(d) None of these

Q.49 16 students were writing a test in a class. Rahul made 14 mistakes in the paper, which was the highest number of mistakes made by any student. Which of the following statements is definitely true?

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(a) At least two students made the same number of mistakes.
(b) Exactly two students made the same number of mistakes.
(c) At most two students made the same number of mistakes.
(d) All students made different number of mistakes.

Q.50 The sum of the factorials of the three digits of a three-digit number is equal to the three-digit number formed by these three digits taken in the same order. Which of the following is true of the number of such three-digit numbers, if no digit occurs more than once?
(a) No such number exists.
(b) Exactly one such number exists.
(c) There is more than one such number, but they are finite in number.
(d) There are infinite such numbers.

Direction for Questions 51 and 52: Read the following passage and solve the questions based on it.

The inhabitants of planet Romeo, a remote planet in the Milky Way, use a number system that is similar to the decimal system used here on the earth, except that it has 11 distinct digits instead of the usual ten digits. The extra digit is an alien digit called x, which is inserted between the digits 7 and 8. Now, $8 - 7 = 2$ and x is the digit, which is equidistance from 7 and 8, that is, $x - 7 = 8 - x = 1$. All the algebraic signs and operations carry the same meaning as in the usual sense.

Q.51 Romeo, an inhabitant of planet Romeo, evaluated an expression and arrived at an answer of '6x'. Then, which of the following can be the expression that Romeo evaluated?
(a) $x^2 + x - 1$ (b) $x^2 + x + 1$
(c) $x^2 + x + 2$ (d) None of these

Q.52 What is the decimal equivalent of the two-digit number '9x'?
(a) 98 (b) 107
(c) 118 (d) None of these

Q.53 If P is the product of four consecutive positive integers, then which of the following statements is not true?
(a) P is perfectly divisible by 24.
(b) P is not a perfect square.
(c) $(P + 1)$ is a perfect square.
(d) $(P - 2)$ is a perfect square.

Q.54 Let S be a two-digit number such that both S and S^2 end with the same digit and none of the digits in S equals zero. When the digits of S are written in the reverse order, the square of the new number so obtained has the last digit as 6 and is less than 3000. How many values of S are possible?
(a) 3 (b) 4 (c) 5 (d) 6

Q.55 How many different four digit numbers are there in the octal (base 8) system when expressed in that system?
(a) 3584 (b) 2058 (c) 6000 (d) 7000

Q.56 How many numbers less than or equal to 500 are there, each of which is the product of more than three distinct prime numbers?
(a) 3 (b) 4 (c) 5 (d) 6

Q.57 How many numbers below 100 can be expressed as a difference of two perfect squares in only one way?
(a) 15 (b) 25 (c) 35 (d) 45

Q.58 The LCM of the two quotients obtained when two numbers a and b are divided by their HCF is 77. LCM of another pair of quotients, when other two numbers c and d have the same HCF as that of a and b, and when divided by it is 221. Which of the following could be the ratio of the sum of a and b to that of the difference between c and d?
(a) 4:17 (b) 11:4 (c) 2:9 (d) 39:2

Q.59 Let p, q, and s be integers such that $p^2 = sq^2$. Then, it follows that
(a) p is an even number.
(b) if s divides p, then s is a perfect square.
(c) s divides p.
(d) q^2 divides p.

Q.60 Let N be a positive integer not equal to 1. Then, none of the numbers 2, 3,...., N is a divisor of $(N! - 1)$. Thus, we can conclude that
(a) $(N! - 1)$ is a prime number.
(b) at least, one of the numbers $(N + 1)$, $(N + 2)$,..., $(N! - 2)$ is a divisor of $(N! - 1)$.
(c) the smallest number between N and $N!$, which is a divisor of $(N! + 1)$, is a prime number.
(d) none of the foregoing statement is necessarily correct.

True/False

Q.1 A natural number M is divided by another natural number N and the remainder obtained = $R \neq 0$ and $M > N$. It is always possible to find the exact remainder when M is divided by P (where P is a factor of N).
State whether True or False.

Q.2 A natural number M is divided by another natural number N and the remainder obtained = $R \neq 0$ and $M > N$. It is always possible to find the exact remainder when M is divided by P (where P is a multiple of N).
State whether True or False.

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Q.3 A, B, and C are three natural numbers. If we know the LCM of A, B, and LCM of B, C separately, then it is always possible to find the LCM of A, B, C.
State whether True or False.

Q.4 A, B, C are three natural numbers. If we know the HCF of A, B and HCF of B, C separately, then it is always possible to find the HCF of A, B, C.
State whether True or False

Q.5 A, B, C, D are four natural numbers. If we know the LCM of A, B and LCM of C, D separately, then it is always possible to find the LCM of A, B, C, D.
State whether True or False.

Q.6 A, B, C, D are four natural numbers. If we know the HCF of A, B and HCF of C, D separately, then it is always possible to find the HCF of A, B, C, D.
State whether True or False.

Q.7 If we know the total number of odd factors of a number, then we can always find the total number of factors of that number.
State whether True or False.

Q.8 If we know the total number of even factors of a number, then we can always find the total number of factors of that number.
State whether True or False.

Q.9 If a number is odd, then it cannot have total number of factors as an even number.
State whether True or False.

Answers

Warm Up

1. (d)	**2.** (d)	**3.** (d)	**4.** (d)	**5.** (d)	**6.** (d)	**7.** (b)	**8.** (a)	**9.** (c)	**10.** (b)
11. (a)	**12.** (a)	**13.** (d)	**14.** (c)	**15.** (d)	**16.** (d)	**17.** (b)	**18.** (a)	**19.** (b)	**20.** (c)
21. (b)	**22.** (c)	**23.** (d)	**24.** (b)	**25.** (c)	**26.** (d)	**27.** (d)	**28.** (b)	**29.** (d)	**30.** (a)
31. (b)	**32.** (c)	**33.** (d)	**34.** (c)						

Foundation

1. (b)	**2.** (b)	**3.** (a)	**4.** (a)	**5.** (a)	**6.** (a)	**7.** (c)	**8.** (b)	**9.** (b)	**10.** (c)
11. (b)	**12.** (d)	**13.** (c)	**14.** (c)	**15.** (c)	**16.** (a)	**17.** (b)	**18.** (d)	**19.** (a)	**20.** (c)
21. (d)	**22.** (d)	**23.** (a)	**24.** (c)	**25.** (b)	**26.** (d)	**27.** (c)	**28.** (c)	**29.** (d)	**30.** (a)
31. (b)	**32.** (b)	**33.** (a)	**34.** (a)	**35.** (c)	**36.** (c)	**37.** (a)	**38.** (b)	**39.** (a)	**40.** (c)
41. (a)	**42.** (a)	**43.** (c)	**44.** (b)	**45.** (a)	**46.** (b)	**47.** (c)	**48.** (a)	**49.** (b)	**50.** (c)
51. (d)	**52.** (c)	**53.** (a)	**54.** (b)	**55.** (a)	**56.** (c)	**57.** (c)	**58.** (d)	**59.** (b)	**60.** (a)
61. (c)	**62.** (c)	**63.** (d)	**64.** (d)	**65.** (b)	**66.** (a)	**67.** (b)	**68.** (d)	**69.** (b)	**70.** (b)
71. (d)	**72.** (d)	**73.** (a)	**74.** (b)	**75.** (c)	**76.** (b)	**77.** (c)	**78.** (c)	**79.** (d)	**80.** (a)

Moderate

1. (b)	**2.** (c)	**3.** (b)	**4.** (c)	**5.** (b)	**6.** (a)	**7.** (c)	**8.** (c)	**9.** (b)	**10.** (c)
11. (a)	**12.** (b)	**13.** (d)	**14.** (c)	**15.** (b)	**16.** (b)	**17.** (a)	**18.** (c)	**19.** (c)	**20.** (a)
21. (a)	**22.** (b)	**23.** (c)	**24.** (d)	**25.** (a)	**26.** (d)	**27.** (a)	**28.** (c)	**29.** (d)	**30.** (b)
31. (b)	**32.** (b)	**33.** (c)	**34.** (b)	**35.** (b)	**36.** (a)	**37.** (a)	**38.** (c)	**39.** (b)	**40.** (c)
41. (d)	**42.** (b)	**43.** (c)	**44.** (b)	**45.** (d)	**46.** (c)	**47.** (a)	**48.** (b)	**49.** (a)	**50.** (c)
51. (d)	**52.** (d)	**53.** (c)	**54.** (b)	**55.** (b)	**56.** (c)	**57.** (a)	**58.** (b)	**59.** (a)	**60.** (a)
61. (a)	**62.** (d)	**63.** (a)	**64.** (a)	**65.** (c)	**66.** (c)	**67.** (d)	**68.** (b)	**69.** (a)	**70.** (d)
71. (b)	**72.** (b)	**73.** (b)	**74.** (c)	**75.** (d)					

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Advanced

1. (d)	**2.** (a)	**3.** (a)	**4.** (a)	**5.** (c)	**6.** (d)	**7.** (d)	**8.** (a)	**9.** (d)	**10.** (d)
11. (c)	**12.** (a)	**13.** (d)	**14.** (a)	**15.** (d)	**16.** (c)	**17.** (b)	**18.** (c)	**19.** (b)	**20.** (d)
21. (c)	**22.** (a)	**23.** (d)	**24.** (a)	**25.** (a)	**26.** (b)	**27.** (b)	**28.** (a)	**29.** (a)	**30.** (d)
31. (a)	**32.** (a)	**33.** (c)	**34.** (d)	**35.** (c)	**36.** (b)	**37.** (a)	**38.** (a)	**39.** (c)	**40.** (c)
41. (d)	**42.** (c)	**43.** (d)	**44.** (b)	**45.** (a)	**46.** (b)	**47.** (a)	**48.** (c)	**49.** (a)	**50.** (b)
51. (c)	**52.** (c)	**53.** (d)	**54.** (a)	**55.** (d)	**56.** (b)	**57.** (c)	**58.** (d)	**59.** (b)	**60.** (d)

True/False

1. True
To find the remainder when M divided by P, we simply need to divide R by P.

2. False
If we divide M by P, we would get a range of remainders in terms of R and not the exact value of remainder in all the cases.

3. True
LCM, by its meaning, is the lowest number divisible by all the numbers constituting it. Final LCM will be the LCM of the pairs of numbers.

4. True
HCF, by its meaning, is the highest number that can divide the numbers constituting it. Final HCF will be the HCF of the pairs of numbers.

5. True
LCM, by its meaning, is the lowest number divisible by all the numbers constituting it. Final LCM will be the LCM of the pairs of numbers.

6. True
HCF, by its meaning, is the highest number that can divide the numbers constituting it. Final HCF will be the HCF of the pairs of numbers.

7. False

8. False

9. False
Total number of factors do not have any relationship with the number being odd or even. For example, all the perfect squares (irrespective of being odd or even) have total number of factors = odd number.

Hints and Solutions

Warm Up

1. Numbers are $5^{1/2}$, $6^{1/3}$, $8^{1/4}$, and $12^{1/6}$.

To solve such questions, we raise each number to a common power so that the powers of the numbers are natural numbers.

In this case, raise each number to the power 12 (LCM of 2, 3, 4, and 6).

Therefore, numbers obtained = $(5^{1/2})^{12}$, $(6^{1/3})^{12}$, $(8^{1/4})^{12}$, and $(12^{1/6})^{12} = 5^6$, 6^4, 8^3, and 12^2

Now, the smallest number in these numbers is 12^2. Therefore, the smallest number is $12^{1/6}$.

2. $\frac{N}{12} = \frac{N}{3} \times \frac{N}{4}$.

However, it is given that $\frac{N}{4}$ is not an integer, and therefore, $\frac{N}{12}$ will not be an integer.

3. Let the numbers are $(x-1)$, x, and $(x+1)$.

Then, $(x-1)^2 + x^2 + (x+1)^2 = (3x^2 + 2)$

When $x = 2$, then $3x^2 + 2 = 14$

Further, when $x = 3$, then $3x^2 + 2 = 29$

Therefore, the largest number that will always divide $(a^2 + b^2 + c^2) = 1$

4. Let $3.134 = a$ and $1.866 = b$

Then,

$$\frac{a^3+b^3}{a^2-ab+b^2} = \frac{(a+b)^3 - 3ab(a+b)}{a^2-ab+b^2}$$
$$= \frac{(a+b)\left[a^2-ab+b^2\right]}{a^2-ab+b^2}$$
$$= (a+b)$$

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Therefore,

$$\frac{(3.134)^3+(1.866)^3}{(3.134)^2-3.134\times1.866+(1.866)^2}$$

$= 3.134 + 1.866 = 5$

5. If n is a perfect cube, then n^2 will also be a perfect cube. Therefore, the answer is option (d).

6. For $5x + 11y = 31$

The values of x and y will be 4 and 1, respectively, which are the minimum values of x and y. Then, $x + y = 5$

7. $\frac{25^x+1}{13}=\frac{(-1)^x}{13}\frac{1}{13}$

Hence, for odd natural values of x, $25^x + 1$ will be divisible.

8. Solve it through actual calculation. The number is 71/84.

9. Let the number be x.

Hence, $x \times 8 \times 9 \times 10 \times 11 \times 12 = x \times 2^6 \times 3^2 \times 3 \times 5 \times 11$

Here, we can say that for being a perfect square, x should be $3 \times 5 \times 11 = 165$.

10. General term would be $n^3 - n^2 = n^2(n - 1)$

Therefore, summation would be $= 0 + 4 + 18 + 48 + 100 + 180 + 294 + 448 + 648 + 900 = 2640$

11. $3 - 9 + 15 - 21 +$ _____19 terms

$(3 \times 1) - (3 \times 3) + (3 \times 5) - (3 \times 7) + (3 \times 9)$_____ 19 terms

Here, we can say that every term of this series will be an odd number.

Hence, odd − odd = even number.

Therefore, we can say that till the 18th term, they all will become even numbers and 19th term is an odd number.

12. Units digit of $21^3 \times 21^2 \times 34^7 \times 46^8 \times 77^8 = 1 \times 1 \times 4 \times 6 \times 1 = 24$

Therefore, units digit = 4

13. $47n \times 729 \times 345 \times 343 = 47n \times 862,66,215$

It is given that the units digit of $47n \times 862,66,215$ or $n \times 5$ is 5.

Therefore, the values of n are all odd digits. Hence, option (d) is the answer.

14. Total number of factors of $846 = 2 \times 3^2 \times 47$ are

$(1 + 1)(2 + 1)(1 + 1) = 2 \times 3 \times 2 = 12$

Therefore, total sets $= \frac{12}{2} = 6$

15. Let the original number is $x + 7$. Hence, thrice the number $= 3(x + 7) = 3x + 21$

It is given that x is divisible by 15, then $3x$ will also be divisible by 15 or by 5.

Therefore, remainder obtained when $(3x + 21)$ divided by 5 = remainder obtained when 21 divided by 5 = 1

16. The first number is 49, next number is (391 + 49), next number = $(2 \times 391 + 49)$, etc.

Since there are many numbers, the answer cannot be determined.

17. Do it from actual calculation.

The values of $P + Q = 5, 7, 13, 19, 31, 43$. Hence, the answer is 6.

18. Prime factors of $1600 = 2^6 \times 5^2$

Hence, for a perfect cube, we can take the values of $2 = 2^0$, 2^3 and 2^6 and the value of 5 is 5^0.

Therefore, the number of perfect cube factors $= 3 \times 1 = 3$

19. LCM of 96, 144 and $N = 576$

Or, LCM of $(2^5 \times 3, 2^4 \times 3^2$ and $N) = 2^6 \times 3^2$.

From here, N should be $2^6 \times 3^2$ or $2^6 \times 3$ or 2^6

However, it is given that HCF is $48 = 2^4 \times 3$.

Hence, $N = 2^6 \times 3$

20. It is given that p and q are consecutive natural numbers, such that $p < q$.

Hence, option (a) is incorrect for every possible value of p and q.

Option (b) is incorrect for p equals to 1 and q equals to 2.

Option (d) is incorrect for every possible value of p and q.

Therefore, the answer is option (c).

21. We know that $(a^n + b^n)$ is divisible by $(a + b)$ if n is an odd number.

It means $(17^{21} + 19^{21})$ is divisible by 36 and all the factors of 36.

Therefore, the answer is 8 because 8 is not a factor of 36.

22. $11^{12,296} - 1$ is divisible by 10 and 12. Because

$$\frac{11^{12,296}-1}{10}=\frac{(1)^{12,296}-1}{10}=\frac{1-1}{10}=0$$

$$\frac{11^{122,296}-1}{12}=\frac{(-1)^{12,296}-1}{12}=\frac{1-1}{12}=0$$

23. **Method 1**

Assume that the numbers are $(2a - 3)$, $(2a - 1)$, $(2a + 1)$, and $(2a + 3)$.

Given $(2a - 3)^2 + (2a - 1)^2 + (2a + 1)^2 + (2a + 3)^2$ $4a^2 - 12a + 9 + 4a^2 - 4a + 1 + 4a^2 + 1 + 4a + 4a^2 + 9 + 12a$

$= 16a^2 + 20 = 4(4a^2 + 5)$

Method 2

Assume numbers to be 3, 5, 7, and 9.

Therefore, $a^2 + b^2 + c^2 + d^2$

$= 3^2 + 5^2 + 7^2 + 9^2$

$= 9 + 25 + 49 + 81 = 164$

This is divisible by 4 (maximum value).

Hence, option (d) is the answer.

24. LCM of 14, 21, and 42 is 42.

It means that after every 42 minutes, all bells will toll together.

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Then, after 11:22 am, they will toll at 11:22 + 42 = 11:64 – 12:04 pm

25. $\frac{x^4+x^3+x^2+x+1}{6}=\frac{(3)^4+(3)^3+(2)^2+(2)+1}{6}$

$\left[\text{since } \frac{x}{6}=\frac{3}{6}\right]=\frac{121}{6}=1$

26. The question is asking about the remainder when we divide 7^7 by 2^4. Remainder is 7.

27. Let the number be $(41K + 7)$. Now, divide $(41K + 7)$ by 40.

$\frac{41K+7}{40}=\frac{40K+K+7}{40}=\frac{K+7}{40}$

Now, substitute the value of K for which $(x + 7)$ will give a remainder of 3, which is $K = 36$.

Therefore, the original number = $(41K + 7) = 41 \times 36 + 7 = 1479$

28. Let the HCF be x.

Then, LCM + HCF = 1476

$40x + x = 1476$ or $x = 36$

Therefore, HCF = 36 and LCM = $40x = 1440$

We know that product of numbers = LCM × HCF. Now, you can solve the equation. Answer is 180.

29. 101,010 94 ... digits can be written as:

$\frac{101,010...1,000,000\ (94\ \text{digits})+1010}{125\times 3}$

$\frac{101,010...100,000}{125\times 3}+\frac{1010}{375}$

Remainder obtained when $\frac{101,010...100,000}{125\times 3}=0$

Remainder obtained when $\frac{1010}{375}=260$

Therefore, net remainder = 260

Hence, option (d) is the answer.

30. Sum of 1st 60 numbers $=\frac{60\times 61}{2}=1830$

Therefore, the number that has been added twice

= 1850 – 1830 = 20

Hence, $N - 20 = 60 - 20 = 40$.

31. Let the number is $11x + 6$.

Divide $(11x + 6)$ by 7

$\frac{11x+6}{7}=\frac{7x+4x+6}{7}=\frac{4x+6}{7}$

Now, substitute the value of x in $(4x + 6)$, so that the remainder will be 4, which is $x = 3$.

Therefore, the value of $11x + 6 = 39$. Now, the remainder 39 when divided by 13 is zero.

It remains same for every number that satisfy the given condition.

32. Count the number by actual counting method. The numbers are 83, 89, 87, 101, 103.

33. Solve the question by taking different value of x and y. For option (b), $x = 2$ and $y = 3$

$y^x = 3^2 = 9$ and $x^y = 2^3 = 8$

34. Using Fermat's theorem

$$\frac{5^{79}}{7}=\frac{5^{78}\times 5}{7}=\frac{5^{6[13]}\times 5}{7}=\frac{15}{7}=5$$

FOUNDATION

1. It is given that LCM = $590 = 59 \times 2^2 \times 5$ and HCF = 59

Therefore, numbers can be assumed as $59a$ and $59b$.

We know that the product of two numbers = LCM × HCF

Therefore, $59a \times 59b = 590 \times 59$

Hence, $ab = 10$ => Sets possible for a and b = (10,1) and (5, 2).

From here, the sets of value of a and b are

(i) 59×2 and 59×5

(ii) $59 \times 2 \times 5$ and 59

2. Let the first applicant be female. The remaining applicants = 5005 – 1 = 5004

For maximum female applicants, for every six applicants, there should be a female.

Therefore, number of females = 1 + (5004/6) = 1 + 834 = 835

3. HCF of A and $B = 24 = 2^3 \times 3$ and HCF of C and $D = 36 = 2^2 \times 3^2$

Then, HCF of A, B, C, and D = HCF of 24 and $36 = 2^2 \times 3 = 12$

4. $25 \times 35 \times 40 \times 50 \times 60 \times 65 = (5)^2 \times (5 \times 7) \times (5 \times 8) \times (5^2 \times 2) \times (5 \times 12) \times (5 \times 13)$

$= 5^8 \times 2^6 \times 3 \times 7 \times 13$.

There are eight 5s and six 2s.

Number of zeroes = number of sets of 2 and 5 = minimum of (number of 2s and number of 5s) = 6

5. Units digit of 576,847 × 564,068 × 96,467 × 458,576 = units digit of $7 \times 8 \times 7 \times 6 = 56 \times 42 = 6 \times 2 = 12 = 2$

6. Units digit of 1! + 2! + 3! + 4! + 5! + 6! …

= 1 + 2 + 6 + 24 + 120 + 0 …= 3

Note: We know that units digit of 5! or for all the numbers greater than 5! is zero.

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7. Factors of 1020 will divide 1020 properly. Therefore, factors of $1020 = 2^2 \times 3 \times 5 \times 7$
$= (2 + 1)(1 + 1)(1 + 1)(1 + 1) = 24$

8. Number of prime divisors or factors = 4 (namely: 2, 3, 5, and 7)
Number of even factors $= 2 \times 2 \times 2 \times 2 = 16$. Therefore, required factors $= 16 - 4 = 12$

9. Prime factorization of $(7!)^3 = (2^4 \times 3^2 \times 5 \times 7)^3 = 2^{12} \times 3^6 \times 5^3 \times 7^3$

Now, for a multiple of 10, there should be at least one 5 and at least one 2 present in the number. Therefore, the number can be like $= 2^{1-12} \times 3^{0-6} \times 5^{1-3} \times 7^{1-3}$. Hence, number of factors $= 12 \times 7 \times 3 \times 4 = 1008$

10. If a number has odd number of divisors, then it is a perfect square.

11. $N = 2 \times 4 \times 6 \times 8 \times \ldots 100$

Count the number of 5s in N, which is 12. Therefore, number of zeroes are 12.

12. $N = 10 \times 20 \times 30 \ldots \times 1000$

There is one 5 in the multiple of 10.

There are two 5s in the multiple of 25 and three 5s in the multiple of 125.

Now, count the multiple of 5s in the expression, which are $(100 + 20 + 4) = 124$

13. $N = 7 \times 14 \times 21 \times \ldots \times 777$

Method 1

In this expression, every fifth term is a multiple of 5.

Now, there are 111 terms in the expression.

Therefore, number of 5s $= (111/5) + (111/25) = 22 + 4 = 26$

Method 2

$N = 7 \times 14 \times 21 \times \ldots \times 777 = (7 \times 1) \times (7 \times 2) \times (7 \times 3) \ldots \times (7 \times 111) = 7^{111} \times (1 \times 2 \times 3 \times \ldots \times 111) = 7^{111} \times 111!$

Number of zeroes in $111! = (111/5) + (111/5^2)$
$= 22 + 4 = 26$

14. If the sum of digits is divisible by 9, then the number will also be divisible by 9.

Therefore, the sum of 1 to $33 = \dfrac{33 \times 34}{2} = 561$

Now, the remainder, when 561 is divided by 9 = 3

15. Since there are 999 terms in the number, then it is divisible by 222.

Because every term will be divisible by 222, and therefore, all 999 terms will also be divisible by 222.

16. $7^1 = 7$, $7^2 = 9$, $7^3 = 3$ and $7^4 = 1$

Therefore, the cycle of 7 is 4, and $27!^{371!}$ is divisible by 4. Therefore, units digit is 1.

17. $N = 420 = 2^2 \times 3 \times 5 \times 7$

Odd factors in $N = 1, 3, 5, 7, 15, 21, 35, 105$

Now, $(4n + 1)$ format $\rightarrow$ Remainder obtained when divided by 4 is 1.

Therefore, $(4n + 1)$ format number = 1, 5, 21, 105

18. $N = 2^3 + 5^3 \times 7^2$

First number of sets of co-prime factors in $2^3 \times 5^3 = (x + 1)(y + 1) + xy = 16 + 9 = 25$. Now, number of sets of co-prime factors in $2^3 \times 5^3 \times 7^2 = A^{25} \times 7^2 = (x + 1)(y + 1) + xy = 26 \times 3 + 50 = 128$

19. We know that $2^1 = 2$, $2^2 = 4$, $2^3 = 8$, $2^4 = 16$ and $2^5 = 32$

Therefore, cycle of 2 is 4.

Now, $\dfrac{3^{4^5}}{4} = \dfrac{(-1)^{4^5}}{4} = \dfrac{1}{4}$

Therefore, remainder is 1, so units digit $= 2^1 = 2$

20. Since all the numbers in the expression are odd. Therefore, the product of all odd numbers would also be odd. Hence, number of zeros is zero.

21. Number of 5s in $36! = \left[\dfrac{36}{5}\right] + \left[\dfrac{36}{25}\right] = 7 + 1 = 8$

Therefore, zeros in $(36!)^{36!} = 8 \times 36!$

22. See the divisibility rule of 7, 11, and 13. These types of number will always divisible by 3, 7, 11, 13, and 37.

24. It is given that $2x - 5y = 1$

Smallest positive value of x is 3, when y is 1;

Next sets are (8, 3), (13, 5), and so on.

Now, it is clear that in every five consecutive numbers, there is a value of x, which satisfy $2x - 5y = 1$

Then, number of values of $x = 1 + \dfrac{200 - 2}{5} = 1 + 39 = 40$

25. For $N = 2^3 \times 5^3$

The number of sets of factors co-prime to each other $= (x + 1)(y + 1) + xy = (3 + 1)(3 + 1) + 3 \times 3 = 25$. However, for co-prime set (1, 1), factors are not distinct. Therefore, number of sets $= 25 - 1 = 24$

26. Do this question by actual calculation, and the number is 143.

Therefore, sum of digits $= 1 + 4 + 3 = 8$

27. We know that $7^1 = 7$, $7^2 = 9$, $7^3 = 3$, and $7^4 = 1$. Therefore, the cycle of 7 is four.

Now, divide $11^{22^{33}}$ by 4.

$$\frac{11^{22^{33}}}{4} = \frac{(-1)^{22^{33}}}{4} = \frac{1}{4}$$

Remainder obtained is 1 and therefore, units digit $= 7^1 = 7$

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28. We know that 5! or greater than 5! will be divisible by 5. Therefore, when (1! + 2! + 3!…. 1000!) is divisible by 5 equals to when (1! + 2! + 3! + 4!) is divided by 5, we get the remainder as

$$\frac{1!+2!+3!+4!}{5}=\frac{33}{5}=\frac{3}{5}$$

Hence, remainder obtained = 3

29. Let $x = 3^{148}$, $y = 5^{76}$, and $z = 7^{139}$

Then, $A = x \times 3^2\, xy \times z \times 7 = 63xyz$

$B = x \times y \times z \times 7^2 = 49xyz$

$C = x \times y \times 5^4 \times z = 625xyz$

$D = x \times 3^3 \times y \times 5^4 \times z \times 7^3 = 5788125xyz$

Therefore, the order of A, B, C, and $D = DCAB$

30. The smallest number is 0.0003 and it will also divide all the other numbers properly.

Therefore, HCF = 0.0003

31. Numbers divided by 5, but not by 9 $= \left[\frac{550}{5}-\frac{550}{5\times 9}\right] =$ 110 − 12 = 98

32. It is given that $(AB + BA)$ = perfect square

$(10A + B) + (10B + A)$ = perfect square

$11(A + B)$ = perfect square.

For being a perfect square, $(A + B)$ should be 11.

33. $N = 897324P64Q$

For N divisible by 8, last three digits should be divisible by 8.

For N divisible by 9, sum of digits should be divisible by 9.

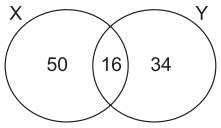
However, $64Q$ is divisible by 8 when Q equals 0 and 8.

Now, if $Q = 0$, then P should be 2,

and if $Q = 8$, then P should be 3.

Then, $(P + Q) = 2$ and 11

Answers to Q.34 to 35:



34. Number in A is divisible by any of the two numbers = 50 + 16 + 34 = 100

35. Numbers are divisible by $X = 50 + 16 = 66$

36. Divisibility rule of 9 is that the sum of all digits should be divisible by 9. Therefore,

$$\frac{x+5+1+5+x+0}{9}=\frac{11+2x}{9}$$

From here, x should be 8.

Therefore, the number is 851,580, which is also divisible by 4.

37. For minimum number of classrooms, the maximum number of students should be in a classroom.

This can be obtained by calculating the HCF of 60, 72, and 96 = 12

It means that every classroom should contain 12 students. Hence, number of classroom $= \frac{60}{12}+\frac{72}{12}+\frac{96}{12}=19$

38. This question is about the LCM of 30 min, 45 min, and 30 min.

Therefore, LCM = 90 min = 1 h 30 min

Hence, the earliest time for the lunch break = 9 am + 1 h 30 min = 10:30 am

39. This question is about the LCM of 5s, 6s, 7s, and 8s.

Then, LCM of 5s, 6s, 7s, and 8s = 840 sec = 14 min

Hence, the time obtained when they hit target together is = 10:14 am.

40. Go through the options.

The answer is option (c).

Because the number of factors of 30 $(2 \times 3 \times 5) = 8$

and the number of factors of 60 $(2^2 \times 3 \times 5) = 12$

41. This question is about a number which when divided by 11 gives remainder 2 and when divided by 15 gives remainder 2 again.

Now, find the number from actual calculation and the number is 167.

42. If the last four digits of a number is divisible by 16, then the number will also be divisible by 16.

43. For n numbers, the operation should be repeated for $(x - 1)$ times, and therefore, for 50 numbers, the operation should be repeated for $(50 - 1) = 49$ times

44. The number of 5s in 18! $= \frac{18}{5} = 3$ and in 19! $= \frac{19}{5} = 3$

Therefore, number zeroes in 18! is 3 and in 19! is 3. Hence, number of zeroes in 18! + 19! = 3 zeroes

45. Let the number be x.

It is given that if we divide the sum of two numbers, then the remainder is 236.

Hence, it means when we divide (437 + 298) by x, then the remainder is 236.

From here, the number x should be 499.

46. For being a perfect square, the last digit of the number should be 1, 4, 5, 6, and 9, and the digital sum of the number should be 1, 4, 9, and 7.

47. $N^2 + N^3 = N^2\,(N + 1)$

For $(N^2 + N^3)$ to be a perfect square, $(N + 1)$ should be a perfect square.

Further, we know that there are 10 perfect squares till 100. However, we cannot take $(N + 1) = (1 \rightarrow N) = 0$

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Therefore, there are 9 numbers for which $N^2(N+1)$ will be a perfect square.

48. Go through the options

49. Any number of the format *abcabc* or *aaaaaa* will be divisible by 7, 11, and 13.

50. For minimum tiles, the sides of tiles should be the HCF of 3.78 m and 4.8 m.

HCF of 3.78 and 4.8 = 0.06 m

Hence, number of tiles $= \dfrac{\text{Area of floor}}{\text{Area of tile}}$

$$= \frac{3.78 \times 4.8}{0.06 \times 0.06} = 5040$$

51. Number should be like (multiple of LCM of 7, 11, and 21) + 3

Then, find the smallest five-digit multiple of LCM of 7, 11, and 21 and add 3 to that number.

52. Answer should be HCF of $(57 - x)$, $(129 - x)$, and $(177 - x)$.

In other words, the largest number that gives the same remainder when dividing 57, 129, and 177 is the answer.

Now, go through the options. Answer is 24 L.

53. Let us first find the number that is divided by 3, 4, and 5, which gives remainder 1, 1, and 0, respectively. It is equal to 25.

It is given that only 10 eggs are left now. It means 25 – 10 = 15 eggs has been broken.

54. There is only one set of prime number that satisfies the given condition, and the set of prime number is (2, 3).

55. Clearly, the two quantities are both integers, and therefore, we check the prime factorization of $2005 = 5 \times 401$. It can be seen that $(A, B, C) = (4, 0, 1)$ satisfies the relation. Hence, option (a) is the answer.

56. There are 17 factors of 3^{16} that are 3^0, 3^1, 3^2, 3^3, 3^4,…, 3^{16}. Product of factors $= 3^0 \times 3^1 \times 3^2 \times 3^3 \times 3^4 \ldots \times \ldots \times 3^{16} = 3^{(0+1+2+3+\ldots+16)} = 3^{136}$

57. Remainder obtained when $= \dfrac{90^{91}}{13} = \dfrac{(-1)^{91}}{13} \dfrac{-1}{13}$

Hence, remainder is –1 or 12.

58. $\dfrac{8641}{8640} \times \dfrac{8642}{8640} \times \ldots \times \dfrac{8650}{8640} = \dfrac{1 \times 2 \times 3 \times \ldots \times 10}{8640}$

$$= \frac{10!}{8640} = \frac{3{,}628{,}800}{8640}$$

This is divisible by 8640 as can be seen through the actual calculation. Hence, remainder = 0

59. **Method 1**

We find the number of numbers with a digit 4 and subtract from 2005. Fast counting tells us that there are 200 numbers with a digit 4 in the hundreds place, 200 numbers with a digit 4 in the tens place, and 201 numbers with a digit 4 in the units place (counting 2004). There are 20 numbers with a digit 4 in the hundreds and in the tens, and 20 for both the other two intersections. The intersection of all three sets is just 2. Therefore, we get:

2005 – (200 + 200 + 201 – 20 – 20 – 20 + 2) = 1462

Hence, option (b) is the answer.

Method 2

Alternatively, consider that counting without the number 4 is equivalent to counting in base 9; only, in base 9, number 9 is not counted. Since 4 is skipped, the symbol 5 represents 4 miles of travel, and we have travelled 2004_9 miles. By basic conversion, $2005_9 = 9^3(2) + 9^0(5) = 729(2) + 1(5) = 1458 + 5 = 1463 \Rightarrow 1463 - 1 = 1462$

Hence, option (b) is the answer.

60. These are exactly 18 numbers between 400 and 500, and 18 numbers between 500 and 600 where 8 occurs only once. Therefore, total number = 18 + 18 = 36

61. $n^2 = 123.45654321 = 12{,}345{,}654{,}321 \times 10^{-8}$ Therefore, $n^2 = (111{,}111 \times 10^{-4})^2$. Therefore, $n = 11.1111$

62. Let the load of mule is x and load of horse is y. Now, from the question

$2(x-1) = (y+1)$

$2(x-2) = (y+1) \rightarrow (2x - y) = 3$ (i)

and $(x+1) = (y-1)$

$(x-y) = -2$ (ii)

Now, from equation (i) and (ii) $x = 5$ and $y = 7$, then $(x+y) = 12$

64. Remainder obtained when $3x^2 - x^6 + 31x^4 + 21x + 5$ is divided by $(x+2)$ can be obtained by substituting $(x+2) = 0$ in the original expression.

Substituting $x = -2$ in the given expression:

$3x^2 - x^6 + 31x^4 + 21x + 5 = 3(-2)^2 - (-2)^6 + 31(-2)^4 + 21(-2) + 5 = 406$

Therefore, option (d) is the answer.

65. Let the numbers be a, b, c, and d

It is given that $a \times b \times c = 385$ (i)

and $b \times c \times d = 1001$ (ii)

Now, divide equation (ii) by equation (i)

$$\frac{b \times c \times d}{a \times b \times c} = \frac{1001}{385} \rightarrow \frac{d}{a} = \frac{13}{5}$$

Hence, the largest number =13

66. Remainder when $\dfrac{4^{44}}{15} = \dfrac{16^{22}}{15} = \dfrac{(1)^{22}}{15} = \dfrac{1}{15}$

Hence, remainder =1

67. Since 211 is a prime number, P and Q = (1 and 211) or (211 and 1).

Hence, HCF = 1

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68. We know that in every consecutive 100 numbers, every digit comes 10 times at units place and 10 times at tens place. Then, from 11 to 100, 6 will appear for 19 times, and from 100 to 400, 6 will appear for $3 \times 20 = 60$ times.

Hence, answer = 19 + 60 = 79 times

69. In every 100 consecutive natural numbers, every digit will appear in 19 numbers (a total of 20 times). Now, solve the question.

70. A number like *aaaaaa* is divisible by 7. It means 8 written 84 (6 × 14) times is divisible by 7. Now, divide the last four digits of the number by 7 and find the remainder.

Hence, remainder obtained $= \frac{8888}{7} = 5$

71. Do it from actual counting.

72. $\frac{7+7+777....(\text{till 100 terms})}{8} = \frac{7}{8} + \frac{77}{8} + \frac{777}{8} \ldots$ (till 100 terms)

$\frac{7+5+1+1+1...(\text{till 100 terms})}{8} = \frac{7+5+98}{8} = \frac{6}{8}$

Hence, remainder = 6

73. Total number of factors of any number = 1 + prime factors of that number + composite factors of that numbers. Now, we will verify the number of prime factors one by one.

If number of prime factor = 1, then total number of factors = 1 + 1 + 15 = 17.

If the number is like 2^{16}, it will have 17 factors. Hence, using one prime factor, it is possible to make total 17 factors (or total number of composite factors = 15).

If the prime factors are two, then number of total factor = 15 + 2 + 1 = 18

This is possible for $a^2 \times b^5$.

When prime factors are three, then the number of total factors = 19, which is not possible because 19 cannot be divided into three parts. When prime factors are four, total number of factors = 20, which is also not possible because 20 cannot be divided into four parts. Similarly, when prime factors are five, minimum number of factors of $a \times b \times c \times d \times e = 32$. Hence, maximum value of prime factors is 2.

74. $N = 204 \times 221 \times 238 \times 255 \times \ldots \times 850 = (17 \times 12) \times (17 \times 13) \times (17 \times 14) \times (17 \times 15) \times \ldots \times (17 \times 50)$

We are required to count the number of 5s in N = number of zeroes in N.

To count the number of 5s, we can count it from

$[(17 \times 1) \times (17 \times 2)\ldots(17 \times 12) \times (17 \times 13) \times (17 \times 14) \times (17 \times 15) \times \ldots (17 \times 50)]$ and then subtract the number of 5s in $[(17 \times 1) \times (17 \times 2)\ldots(17 \times 10) \times (17 \times 11)]$

Number of 5s in $[(17 \times 1) \times (17 \times 2) \ldots (17 \times 12) \times (17 \times 13) \times \ldots (17 \times 50)] = 12$

Number of 5s in $[(17 \times 1) \times (17 \times 2) \ldots (17 \times 11)] = 2$

Hence, number of 5s in $(17 \times 12) \times (17 \times 13) \times (17 \times 14) \times (17 \times 15) \times \ldots (17 \times 50) = 12 - 2 = 10$

75. $\frac{123,456,\ldots,125,126}{5625} = \frac{123,456,\ldots,125,000+126}{5^4 \times 3^2}$

$= \frac{123,456,\ldots,124,125 \times 10^3}{5^4 \times 3^2} + \frac{126}{5^4 \times 3^2}$

Now, number 12,345…24,125 is divisible by 9 because the sum of digits is divisible by 9 and it is also divisible by 5^4 because 10^3 is divisible by 5^3 and number 123,456…125 is divisible by 5.

Hence, the remainder = remainder obtained when $\frac{126}{5625} = 126$

76. Number will be like $3(4(7x + 4) + 1) + 2 = 84x + 53$. When this number is divided by 84, remainder obtained is 53.

Alternatively, go through the options.

77. Remainder, when $\frac{1714 \times 1715 \times 1717}{12} = \frac{10 \times 11?13}{12} = \frac{2}{12}$

Hence, the remainder is 2.

78. Following pattern can be observed:

$(11)^2 = 121$

$(111)^2 = 12,321$

$(1111)^2 = 12,34,321$

$(111111111)^2 = 12,345,678,987,654,321$

79. For the largest value of the product, difference between $(a + b)$ and $(c + d)$ should be as less as possible.

Then, for this condition, let $a = 12$, $b = 15$, $c = 13$, $d = 14$

Therefore, $(a + b)(c + d) = (12 + 15)(13 + 14) = 27 \times 27 = 729$

80. There is only one number of form *abbb*, which is $38^2 = 1444$

Moderate

1. In the case of decimal system, we obtain 10 by multiplying 5 and 2, and then, to find the number of zeroes, we search the exponents of 5. In the case of base 6, 10 will be obtained by multiplying 3 and 2. Therefore, here, we will check for the exponents of 3 to know about the number of zeroes. Obviously, it is 5[12/3 + 12/9].

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2. The remainder obtained when $2222^{5555} + 5555^{2222}$ is divided by 7 will be the same as the remainder when $3^{5555} + 4^{2222}$ is divided by 7. Now, find the individual remainder and solve it.

3. If we look at the numbers $100 < N < 105$, we see only 101 and 103 do not have their factors in N (because these are primes). Therefore, obviously, the new LCM will be $101 \times 103 \times N$.

4. $10^5 = 2^5 \times 5^5$

Now, all the factors of 10^5 that will end in one zero will be zero power of 2 and $(1 - 5)$ powers of 5 and vice versa. This will be equal to 9.

5. $1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10 = 55$

Therefore, by replacing the signs, we need to make 45 extra.

This is possible only if we write in this way:

$1 \times 2 + 3 \times 4 + 5 + 6 + 7 \times 8 + 9 + 10 = 55 + 45 = 100$

6. The key is the fact that in this country, only three symbols are used to write numbers.

Therefore, $6 = (20)_3 = (bc)_3$. $b = 2$, $c = 0$, and $a = 1$; $17 = (122)_3 = abb$

7. Let the number is AB

For perfect square $= AB + BA = (10A + B) + (10B + A) = 11(A + B)$

For being a perfect square, $(A + B)$ should be equal to 11. Then, $(A + B) = 11$. Now, find the sets of values of A and B.

8. Remainder of $({}_{32}32^{32}$ divided by 7) = Remainder of $({}_{4}32^{32}$ divided by 7)

Now, find cyclicity of remainder of $({}_{32}32^{n}$ divided by 7).

Remainder when ${}_{4}32^{1}$ divided by 7 = 2

Remainder when ${}_{4}32^{2}$ divided by 7 = 4

Remainder when ${}_{4}32^{3}$ divided by 7 = 2

Therefore, the cyclicity is 2, 4, 2, 4, and so on.

For every even value of n, remainder = 4

Therefore, answer is option (d).

9. $1001 = 7 \times 11 \times 13$.

We know that any digit written 6 times consecutively (such as 111,111 or 222,222) will be divisible by 3, 7, 11, 13, and 37. Therefore, this question is—what is the remainder when 11,111 is divided by 1001. Find using actual division method.

10. It is given that

$a + b + c = d + 8$ (i)

$a^2 + b^2 = c^2 + d^2 + 36$ (ii)

$ab + cd = 42$ (iii)

$a^3 = b^3 + c^3 + d$ (iv)

Now, go through the options.

11. Digital sum of $(19)^{100}$ = Digital sum of $(1 + 9)^{100}$ = Digital sum of $(10)^{100}$ = Digital sum of $(1 + 0)^{100}$ = Digital sum of $(1)^{100} = 1$

12. Use the formula given in the concepts.

13. **Method 1**

Note that $\frac{1}{11} = 0.\overline{09}$

Dividing by 3 gives $\frac{1}{33} = 0.\overline{03}$, and dividing by 9 gives $\frac{1}{99} = 0.\overline{01}$

$S = \{11, 33, 99\}$

$11 + 33 + 99 = 143$

The answer must be at least 143, but cannot be 155. Hence, option (d) is the answer.

Method 2

Let us begin by working with the condition $0.\overline{ab} = 0.ababab....$,

Let $x = 0.ababab....$, and therefore, $100x - x = ab$ or $x = \frac{ab}{99}$

For this fraction, x should be in the form $\frac{1}{n}$, and 99 must be a multiple of ab. Hence, the possibilities of ab are 1, 3, 9, 11, 33, and 99. Checking each of these, $\frac{1}{99} = 0.\overline{01}$, $\frac{3}{99} = \frac{1}{33} = 0.\overline{03}$, $\frac{9}{99} = \frac{1}{11} = 0.\overline{09}$, $\frac{11}{99} = \frac{1}{9} = 0.\overline{1}$, $\frac{33}{99} = \frac{1}{3} = 0.\overline{3}$, and $\frac{99}{99} = 1$. Hence, the only values of n that have distinct a and b are 11, 33, and 99. Therefore, $11 + 33 + 99 = 143$

14. Digits which can create confusion = 1, 6, 8, 9 (0 cannot create confusion because passwords has to be two-digit numbers).

Total two-digit numbers with distinct digit = 81

Two-digit numbers created by 1, 6, 8, 9 = 12

Therefore, total numbers left = 69

However, 69 and 96 will not create confusion (it looks same upside down), and hence, total numbers = 71

15. There is only one set possible, where $p = 3$, $(p + 2) = 5$, and $(p + 4) = 7$

In every other set, one number will be divisible by 3, and hence, that number will not be a prime number.

16. See the solution of CAT 2004 given at the end of this book.

17. For minimum number of students, who has/have solved at least five questions, the case is as follows:
Exactly one student has solved one question;
exactly one student has solved two questions;
exactly one student has solved three questions;
exactly six students have solved four questions;
and exactly one student has solved five questions.

18. $N!$ is having 37 zeroes at its end, and therefore, $N = 150$ (can be arrived at by a guess).

Obviously, $150 \le N < 155$ is the answer.

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19. From the previous question, we have found that the range of $N = 150 \le N < 155$

Then, odd values of $N = 151$, 153, and 155

20. There is no number having 30 zeroes at its end. Because 124! has 28 zeroes at its end, and 125! has 31 zeroes at its end.

21. There are 50 odd numbers and 50 even numbers. Every even number will be divisible by 4. Further, in odd numbers, half of them have 1 as the remainder and half of them have −1 as the remainder. Then, overall remainder is zero.

22. We cannot take 9 and 7 together because $9^3 + 7^3 = 1072$ (four-digit number)

We cannot 9 and 5 together, because $9^3 + 5^3 = 854$ (8 is an even number)

We cannot take 9, 3 and 1 together because $9^3 + 3^3 + 1^3 = 757$

We cannot take 7 and 5 together because $7^3 + 5^3 = 468$ (4 is an even number)

We can take 7, 3, 1 as the digits because $3^3 + 7^3 + 1^3 = 371$

23. The question is in which system of writing, (24 + 32) = 100. Go through the options.

24. $108 = 3^3 \times 4 = 27 \times 4$

Remainder obtained when 3^{450} is divided by 108 is same as the number obtained when 3^{450} is divided by 27 and 4. Remainder obtained when 3^{450} is divided by $3^3 = 0$; remainder obtained when 3^{450} is divided by 4 = 1.

Now, we are required to find a number that when divided by 27 gives 0 as the remainder and when divided by 4 gives 1 as the remainder = 81. Hence, option (d) is the answer.

25. $2P$ is having 28 (7×4) divisors but $3P$ is not having a total divisor, which is divisible by 7. Therefore, the first part of the number P will be 2^5.

Similarly, $3P$ is having 30 (3×10) divisors but $2P$ does not have a total divisor, which is divisible by 3. Therefore, second part of the number P will be 3^3. Therefore, $P = 2^5 \times 3^3$

26. pqr can be 370 or 371. Therefore, it is not possible to arrive at a unique answer.

27. Let the smaller number be x. Then, the larger number = $6x$.

Therefore, from the question, $6x + x = 504\ K$

Here the only value of K should be 2.

Then, $6x + x = 504 \times 2$. Hence, $x = 144$

Then, $6x - x = 864 - 144 = 720$

28. It is given that LCM of 12^{24}, 16^{18} and $N = 24^{24}$

Or, $3^{24} \times 2^{48}$, 2^{72} and $N = 2^{72} \times 3^{24}$

From here, the value of N can be $2^{0-72} \times 3^{0-24}$

Then, total number of value = (72 + 1) (24 + 1) = 1825

29. It is given that if $Q = 0$, then $R = 1$

However, $R = 0$, so $Q = 1$ (i)

It is also given that if $S = 0$, then $P = 1$.

However, for $R = 0$, $P = S$.

Therefore, $P = S = 1$ (ii)

Then, $(P + Q + R + S) = 1 + 1 + 0 + 1 = 3$

30. Mayank DOB = 1948 and Siddharth DOB = 1898

31. Let us discuss the fate of any particular cell number as per the algorithm given:

Cell Number 45

Initially – Closed

After Step 1	After Step 2	After Step 3	After Step 4	After Step 5	After Step 6
Open	Open	Close	Close	Open	Open

After Step 7	After Step 8	After Step 9	After Step 15	After Step 16	After Step 45
Open	Open	Close	Open	Open	Close

It can be seen that for cell number 45, only those step numbers that are factors of 45 will have any impact of the position of cell number 45. These are going to be—Step 1, Step 3, Step 5, Step 9, Step 15, and Step 45. Beyond Step 45, none of the steps will have any impact on cell number 45.

It can be concluded that the moment 1st factor is obtained (in the form of Step 1), the cell will be opened and so on

1st Factor	2nd Factor	3rd Factor	4th Factor
Open	Close	Open	Close
5th Factor	6th Factor	And so on	
Open	Close		

We can see that when 1st or 3rd or 5th factor or any odd number of factor is obtained, the cell gets opened.

However, when 2nd or 4th or 6th or any even number of factor is obtained, the cell will get closed.

Is it the case that a number having only odd number of factors, the cell would have remain opened? This is possible only if the cell number is perfect square.

Hence, cell numbers that will remain open = 1, 4, 9, 16, 25, 36, 49, 64, 81, 100. These 10 cells will remain open and 10 prisoners will be released.

32. In the given options, only 64 is a perfect square number; then, option (b) is the answer.

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33. Option (c) is the answer, because only perfect squares have odd number of factors.

34. We know that there are 10 perfect squares in the range $1 \le x \le 100$.

Then, number of prisoners = $1^2 + 2^2 + 3^2 + 4^2 + \ldots + 10^2 = 385$

35. Shopkeeper should return ₹1150 − ₹1143 = ₹7, but he returned ₹5.

Hence, it means that the base used to write the number should be less than 2 than the base 10.

Therefore, base should be 8.

36. $(N)_5 = 345_6 + 632_7 + 487_9 = 137_{10} + 317_{10} + 403_{10} = 857_{10} = 11{,}412_5$

Therefore, $N = 11{,}412$

37. $(124)_\Phi$ shows that value of Φ should be 5 or more than 5 (since 4 is a digit used in this system).

$(12\Phi)_4$ shows that value of Φ can be at max 3 (since the system = 4).

Now, this is a contradiction. Therefore, no such value exists. $N = A^2 \times B^3 \times C^4$, where A, B, and C are prime numbers.

38. Digital sum of $(4444)^{4444} = 7^{4444}$

Now, find the cycle of digital sum of 7

$7^1 = 7$, $7^2 = 13$, $7^3 = 10$, and $7^4 = 7$,

Which is 3. Now, divide 4444 by 3 and the remainder is 1.

Hence, digital sum = $7^1 = 7$

39. For the factors to be perfect squares, assume $A^2 = X$, $B^2 = Y$, and $C^2 = Z$.

Therefore, for the factors to be perfect square, we can write $N = XYZ^2$ and the total number of factors = (1 + 1) (1 + 1) (2 + 1) = 12

40. For cube $B^3 = x$ and $C^3 = y$.

Then, for a perfect cube $N = xy$.

Then, the number of factors = (1 + 1) (1 + 1) = 4

41. All the squares of prime numbers give 3 factors. We use this to solve the question.

42. It is given that LCM of M and $N = 2628 = 2^2 \times 3^2 \times 73$ and $(M + N) = 949$

From here, $M = 2^2 \times 73$ and $N = 3^2 \times 73$

Then, HCF of M and $N = 73$

43. From the previous question, HCF of M and N is 73.

Now, factors of 73 = 2 (namely 1 and 73)

Hence, the answer is 2.

44. Largest number = FFFF and smallest number = 1000

46. Numbers will be 642 and 246. It can be seen that the sum will be divisible by 111.

Then, sum = 642 + 246 = 888

47. Statement I: It is right and N has infinite value greater than 8.

Statement II: It is wrong because it is possible only on base 8.

Statement III: It is also wrong and A has only three values 6, 3, and 0 on base 8, 9, and 10, respectively.

48. For $x = 2$, there are five number present from 2^3 to 2^7; for $x = 3$, there are two numbers present from 3^4 and 3^5.

49. There is no number present for $x = 45$ and greater than 4 because $4^5 > 250$. Therefore, there are numbers for tens place digit, divide the expression by 100; however, we know that 10! and greater than 10! is divisible by 100 because there are at least two 5s in these numbers.

Now, divide 1! + 2! + 3! ... + 9! by 100 and find the remainder.

The remainder is 13, and therefore, tens place digit = 1

50. Let the digits be a and b.

Then, $K(a + b) = 10a + b$ (i)

$N(a + b) = 10b + a$ (ii)

By adding equation (i) and equation (ii), then $(a + b)$

$(K + N) = 11(a + b)$

$(N + K) = (11 \rightarrow N) = 11 - K$

51. If $\frac{32^{33^{34}}}{7} = \frac{(4)^{33^{34}}}{7} = \frac{(64)^{11^{34}}}{7} = \frac{(11)^{11^{34}}}{7} = \frac{1}{7}$,

then remainder = 1

52. The digital sum of any perfect square can be l, 4, 9, 7, 0 only. 100 times ls are given, 100 times 2s are given, and 100 times 3s are given. Digital sum of all these digits taken all at a time will be 6. Therefore, this will not give any perfect square.

53. Substitute the value of N = (l, 2, 3..). Find values of $N^5 - N$, which is always divided by 30.

54. Find the number, which when divided by 5 gives remainder 1 and when divided by 6 gives remainder 5.

55. The difference = 34,041 − 32,506 = 1535

56. Factorize $1535 = 307^1 \times 5^1$

57. Divide 7777…(56 times) by 19 and find a cycle of remainder.

58. For three consecutive odd number, −3, −1, and 1,

the greatest value of prime number = $-3 \times -1 \times 1 = 3$

59. The dimensions of floor = $72 \times 72 = (2^3 \times 3^2) \times (2^3 \times 3^2)$

Then, minimum dimension of tile = 2×3

Maximum dimension of tile = $(2^3 \times 3) \times (2^2 \times 3^2)$

Now, find the difference.

60. Let the number = $xxxx$ 9

Since, the last digit of $9^{even} = xxx\ldots$ 1

and the last digit of $9^{odd} = xx\ldots$ 9,

then the last digit of $(xxxx\ 9)^{even(>50)} = xx\ldots1$

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$17 \times (xx\ldots 1) = yy\ldots 7$

and $(yy\ldots 7)^{4n} = zz\ldots 1$

Therefore, the last digit of resulting number = 1

61. We know that all prime numbers are added except 2. Therefore, for odd number of prime number, there should be one even number.

62. Let number of guavas be x.

According to the question, $\frac{3}{4}x + \frac{3}{4} = x$. Hence, $x - \frac{3}{4}x = \frac{3}{4}$.

Therefore, $x = 3$.

63. Number of pen > number of pencils > number of erasers. Minimum number of each of the items = 11. Therefore, number of erasers = 11. [If we take it 12, then minimum value of pencils = 13, and minimum value of pens = 14; sum of these three exceeds 38.]

64. Number of pens = 15, number of pencils = 12, and number of erasers = 11

65. Different possibilities for the number of pencils = 12 or 13. Since it cannot be divided into his 4 brothers and sisters, it has to be 13. In that case, number of pens = 14.

66. Use Wilson's Theorem.

67. In subset B, if one number is 1, then, others would be 2, 3, 4…10, which are 9 numbers.

When one number is 2, then other numbers would be 3, 5, 7, 9 (which is 4 numbers).

When one number is 3, then other numbers would be 4, 5, 7, 8, 10 (which is 5 numbers).

When one number is 4, then other numbers are 5, 7, 9 (which are three numbers).

When one number is 5, then other numbers would be 6, 7, 8, 9 (which is 4 numbers).

When one number is 6, then other number would be 7 (which is one number).

When one number is 7, then other numbers would be 8, 9, 10 (which are three numbers).

When one number is 8, then other number would be 9 (which is one number).

When one number is 9, then other number would be 10 (which is one number).

Therefore, total number of subsets = 9 + 4 + 5 + 3 + 4 + 1 + 3 + 1 + 1 = 31

69. $AD4 \times E = A206$,

It is possible only when A = 1, D = 3, and E = 9. Then, $D = 3$

70. Then, $A + E = 1 + 9 = 10$

71. $P + Q + R + S + T = 482$

Sum of five prime numbers is even possible only if four of these are odd and one is even (all these cannot be odd).

Therefore, $P = 2$

72. $1075^{n-1} - 1075+^{n-1} = 1075^{n-1}(1075 - 1) = 1075^{n-1} \times 6 \times 179$

Now, substitute $n = 2$

Then, $1075^{n-1} \times 6 \times 179 = 1075 \times 6 \times 179 = 43 \times 25 \times 6 \times 179$

However, this number is not divisible by 23.

Hence, option (b) is the answer.

73. Let b be $xy = (10x + y)$

Then, a is $yx = (10y + x)$

Therefore, $(a + b) = xy + yx = 11x + 11y = 11(x + y)$

For $(a + b)$ being a perfect square, $(x + y)$ should be equal to 11.

Then, value of (x, y) would be (2, 9), (3, 8), (4, 7), (5, 6), (6, 5), (7, 4), (8, 3), (9, 2). Then, ordered pair = 8

74. $(a - b) = (10x + y) - (10y + x) = 9(x - y)$

For $(a - b)$ being a perfect square, $(x - y)$ should also be a perfect square.

Then, the value of $(x - y)$ can be 4 and 1.

Values of (x, y) are (9, 5), (8, 4), (7, 3), (6, 2), (5, 1); when $(x - y) = 4$ and (9, 8), (8, 7), …, (2, 1)

Then, ordered pair = (5 + 8) = 13

75. Assume some values of N and check it out.

Advanced

1. Factorize 96! into prime factors. Find the units digit of all the factors individually and multiply to get the units digit of 96!

$96! = 2^{92} \times 3^{46} \times 5^{22} \times \ldots$

Now, 5^{22} and 2^{22} can be eliminated, since these will result in zeroes. Find the units digit of all the remaining.

2. This number 1,212,121,212... 300 times is divisible by 9. Therefore, we can write 1,212,121,212...300 times = 9 N, where N is the quotient obtained when divided by 9.

Now, this question is like $\frac{1212121212....300 \text{ times}}{99}$

$= \text{Remainder } \frac{9}{9} \times \frac{134,680...\text{written 50 times}}{11}$

[121,212 = 13,468 × 9]

Now, we will have to find the reminder obtained when 134,680,134,680…50 times is divided by 11. For this, we are supposed to use the divisibility rule of 11 from right-hand side.

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[Using the divisibility rule from the left-hand side might give the incorrect remainder; similarly, if we find the remainder obtained when 12 is divided by 11, remainder $1 = (2-1) \neq (1-2)$]

Remainder $\frac{134680...\text{written 50 times}}{11} = 2$

Therefore, the total remainder = 18

Alternatively, divisibility rule of $10^n - 1$, $n = 2$ can be used to find the remainder in this case.

3. Whatever is the sign between two consecutive integers starting from 1 to 100, it will be odd. Therefore, we are getting 50 sets of odd numbers. Now, whatever calculation we do among 50 odd numbers, the result will always be even. Therefore, A will win always.

4. Using Fermat's Theorem, the remainder will be 1.

5. $S = (x_1, x_2, x_3, ..., x_n)$ where $n > 0$, $x_1 = 1$ and $x_n = x_{n-1} + 2_{n-1}$

Then, $x_2 = x_{2-1} + 2 \times (2-1) = x_1 + 3 = (1+3) = 4$ and $x_3 = x_{3-1} + 2 \times (3-1) = x_2 + 5 = 9$. Now, if $x_1 = 1$, $x_2 = 4$, and $x_3 = 9$, then it would be a cube series.

6. More than one value is possible, and therefore, we cannot determine any unique number.

7. Total number of factors of $10^5 = 36$

Total number of factors of 10^5 ending with no zero = 11

Hence, the total number of factors having at least one zero at its end = 25

8. It can be seen that $V_1 = V_5 = V_9 = ...$ and so on $V_2 = V_6 = V_{10}$ and so on

So, $V_3 = V_7 = ... = V_{99} = 9$

9. $V_5 = V_9 = ... = V_{89}$; therefore, we cannot find the value of V_{90}.

10. Since for all consecutive odd integers, one of every five is a multiple of 5 and one of every three is a multiple of 3, the answer is $3 \times 5 = 15$. Therefore, option (d) is the answer.

11. There are six values of K for which $(K+2)!$ has 2 more zeroes than $K!$

The values are 148, 149, 173, 174, 198, 199.

12. When we are adding the sum of page numbers of 25 pages, it will always be an odd number. Therefore, she could not have found the sum of pages as 1990.

13. Assume $S1 = \{1, 2, 3, 24\}$ and $S2 = \{25, 26, 27, 28, ..., 50\}$, and now, check the options. None of the first three options are correct.

14. Assume $S1 = \{1, 2, 3, ..., 24\}$ and $S2 = \{50, 49, 48, 47, ..., 25\}$. Now, even if we have interchanged 24 and 25, $S1$ continues to be in ascending order, and $S2$ continues to be in descending order. However, by choosing negative values of $a24$ and $a25$, we can show that $S1$ continues to be in ascending order, but $S2$ is no longer in descending order.

15. Do it by the method of elimination.

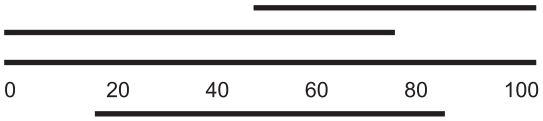
16. This is one classic example of pigeonhole principle. Since Mr Roy has ordered for 25 boxes and three different types of sweets, minimum 9 boxes of sweets will have the same type of sweets.

17. The useful number of shoes will be 100 when there are 150 left shoes and 50 right shoes of size 8, 200 right shoes of size 9, 150 left shoes, and 50 right shoes of size 10.

18.
$$\begin{array}{r} 2\ 3\ _\ 5\ _ \\ +\ \mathbf{1}\ _\ \mathbf{6}\ \mathbf{4}\ \mathbf{2} \\ \hline 4\ 2\ 4\ 2\ 3 \end{array}$$

Here, the key is the numbers written in bold letters. When 5 and 4 are added in decimal system, the value is equal to 9. However, its units digit here is represented by 2. From the given options, only option (c) is satisfying this equation.

19.



Easy problems = Q.40 to Q.60 [not both including] = 20

Difficult problem = Q.1 to Q.20 + Q.80 to Q.100 [not both including] = 40

Therefore, the answer = 40 − 20 = 20

21. 9699 − 7179 = 2520 and all the factors of this number will be the value of N.

24. $10{,}101 = 3 \times 7 \times 13 \times 37$

$ababab = 10{,}101 \times (10a + b)$

27. Prime numbers less than 10 = 2, 3, 5, 7.

Therefore, option (d) cannot be the answer.

Sum of digits is more than 13, and therefore, set of [2, 3, 5] is not possible.

Therefore, the option (a) cannot be the answer.

Now, check for options (b) and (c) by taking the values.

30. Go through the options.

Checking option (a) 1313... 1313 (100 digits). Therefore, for 1 digit, it should be equal to 1 only.

Now, taking 1st two digits of [111.. .111 (200 digits)] and 1 digit from [222.. .22 (100 digits)] $\rightarrow [11-2]^{1/2}$ = 3, and therefore, option (a) is not correct.

Similarly, we will check all the options.

31. Check it through the options.

Alternatively, answer will be LCM (2, 3, 4,..., 9) − 1 = 2520 − 1 = 2519

32. There is no rocket science involved in it. Simply write $(10^3 + 9^3)^{752} = (1729)^{752}$, and now, find the remainder from the given method of calculations.

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35. Going through the options, we get 8181/1818 = 4.5

36. His roll number is divisible by 1001, and the first three digits are 267. Hence, the last three digit will also be 267.

37. Go through the options.

Answers to Q.38 to 39:

The number is 125,367.

40. $1080 = 2^3 \times 3^2 \times 5^1$

Now, $N \times Q = 1080$ and $M \times (Q - 25) = 1080$. By trial-and-error method, $N = 27$

42. The given numbers are two consecutive even numbers, and therefore, their HCF = 2

Now, using LCM × HCF = product of two numbers

LCM × 2 = (…6) × (…8)

It can be seen now that the units digit of LCM = 4

43. $a + b + c = 2003$

It is possible only if all the three a, b, and c are odd, or two of the three are even, and one is odd.

Hence, E can have only two values.

45. In this question, we have to find the possible units digits of $(n!)^4$. This can be either 1 or 5 or 6.

47. Using options, the only possible value is 4112. The key here is that the sum of the digits equals the product of the digits.

49. The number of mistakes made by all the students will be between 0 and 14, that is, students are having a total of 15 options to make mistakes. Since the number of students = 16, at least two students will have the same number of mistakes (that can be zero also, i.e., two students are making no mistakes). Hence, option (a) is the answer.

50. There is only one number, 145, which exhibits this property.

52. Actually, the value of 9 here would become 10 and x represents 8.

Therefore, $(9x)_{11} = 10 \times 11^1 + 8 \times 11^0 = 118$

54. In this question, several restrictions are operating:

If S and S^2 are ending with the same units digit, then it can be 0, 1, 5, 6; however, it is given that none of the digits is equal to zero, so the units digit can be only 1, 5, 6. Next, units digit of the square of the number written in reverse order is 6, so tens place digit of the actual number should be either 4 or 6. Therefore, the actual numbers could be 41, 45, 46, 61, 65, 66.

Now, this square is less than 3000, so the only possibilities are 41, 45, 46.

55. The total number of numbers of four digits in octal system = $7 \times 8 \times 8 \times 8 = 3584$.

When we convert this number into octal system, this is equal to 7000.

60. Eliminate the options.

For example, option (a) can be eliminated by assuming $N = 2$

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BENCHMARKING TEST 1

This test paper contains 20 questions of 3 marks each. One-third of the mark allotted to a particular question will be deducted in the case of wrong answer

Time Given: 40 Minutes **Total Marks: 60** **Qualifying Marks: 20**

Break-up of the Expected Percentile

(assuming this to be a CAT paper)

Marks	Expected percentile
35+	99+
30	98
25	96
19.5	92
14	85

Q.1 Find the remainder when 25! is divided by 10^7.
(a) 2 (b) 4×10^6
(c) 6×10^6 (d) 2×10^6

Q.2 A box contains 90 balls of different colours: 13 yellow, 19 green, 27 red, 10 black, 7 brown, and 14 white. The smallest number N such that any N ball drawn from the box will contain at least 14 balls of the same colour is:
(a) 79 (b) 69
(c) 70 (d) 72

Q.3 At a bookstore, the words 'Modern Book Store' are flashed using neon lights. The words are individually flashed at the intervals of $2\frac{1}{2}$ s, $4\frac{1}{4}$ s and $5\frac{1}{8}$ s, respectively, and each word is off after a second. The least time after which the full name of the bookstore can be read again is:
(a) 49.5 s (b) 73.5 s
(c) 1744.5 s (d) 85.5 s

Q.4 A number has exactly 32 factors; from these factors, 4 are not composite. Product of these 4 factors is 30. How many such numbers are possible?
(a) 2 (b) 6
(c) 3 (d) Not possible

Q.5 The first 126 natural numbers are given side by side in the ascending order to form a large number: 123,456... 125,126. What will be the remainder when this large number is divided by 5625?
(a) 5126 (b) 126
(c) 26 (c) None of these

Q.6 P and Q are two distinct whole numbers and $(P + 1)$, $(P + 2)$, $(P + 3)$,... , $(P + 7)$ are integral multiples of $(Q + 1)$, $(Q + 2)$, $(Q + 3)$,..., $(Q + 7)$, respectively. What is the minimum value of P?
(a) 0 (b) 240
(c) 1080 (d) None of these

Q.7 A child was asked to add first few natural numbers (that is 1 + 2 + 3 +...) as long as his patience permitted. As he stopped, he gave the sum as 575. When the teacher declared the result wrong, the child discovered that he had missed one number in the sequence during addition. The number he missed was:
(a) less than 10 (b) 10
(c) 15 (d) more than 15

Q.8 $1000^7 - 17$ is written in decimal notation. What is the sum of its digits?
(a) 181 (b) 182
(c) 183 (d) None of these

Q.9 N is an integer. For how many distinct values of N, $N^4 - 20N^2 + 4$ is a prime number?
(a) 1 (b) 3
(c) 3 (d) None of these

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Q.10 Of 128 boxes of oranges, each box contains at least 120 and at most 144 oranges. The number of boxes containing the same number of oranges is at least:
(a) 5 (b) 6
(c) 103 (d) None of these

Q.11 What is the value of $\sum_{N=1}^{99} \frac{1}{\sqrt{N}+\sqrt{N+1}}$?
(a) 9 (b) $\sqrt{99}$
(c) $\sqrt{100}-\sqrt{99}$ (d) None of these

Q.12 a, b, c, d, and e are prime numbers, not necessarily consecutive. If $a+b+c+d+e=350$ and $a<b<c<d<e$, how many values of a^5 is/are possible?
(a) 1 (b) More than 1, but finite
(c) Infinite (d) Nothing can be said

Q.13 The 10-digit number 9,793,x61,60y is divisible by 11. If $0<x<y$, find the sum of remainders when the number is divided by $(x+y)$ and x, successively.
(a) 4 (b) 6 (c) 7 (d) 9

Q.14 What is the remainder when 128^{500} is divided by 153?
(a) 63 (b) 67
(c) 123 (d) None of these

Q.15 What is the minimum number of weights that enable us to weigh any integer number of grams of gold from 1 g to 100 g on a standard balance with two pans, weights can be put on either pan of balance?
(a) 4 (b) 5
(c) 6 (d) None of these

Q.16 $N=2^8\times3^{10}\times5^8\times7^2$. What is the number of factors of this number N which are factors of 360 but not the factors of 540?
(a) 120 (b) 210
(c) 114 (d) 144

Q.17 What is the remainder when 2^{104} is divided by 101?
(a) 1 (b) 2 (c) 16 (d) 64

Q.18 Given that:

$X=5$, $X_2=25$ and $X_{n+2}=\text{GCD}\,(X_{n+1},X_n)+X$. What is the LCM of ($X_{19}$ and X_{20})?
(a) 1840 (b) 2560
(c) 2160 (d) 2440

Q.19 $N_1+N_2+N_{3-}+\ldots+N_N=100$, where N_1, N_2, N_3 ... and N_N are $n(n>1)$ consecutive natural numbers such that $N_1<N_2<N_3<N_4\ldots<N_N$. If $N_1>8$, then how many values of n is/are possible?
(a) 4 (b) 8
(c) 2 (d) Infinity

Q.20 A teacher writes 101 numbers on the blackboard; from these numbers, 50 are zeros and 51 are ones. A student is asked to perform the following operation 100 times on the board: strike out any two numbers. If they are equal, write another zero. If they are unequal, write 1. What are the numbers left on the board?
(a) Single one (b) Single zero
(c) Two ones (d) None of these

Answers

Benchmarking Test 1

1. (b)	**2.** (c)	**3.** (b)	**4.** (b)	**5.** (b)	**6.** (d)	**7.** (d)	**8.** (b)	**9.** (d)	**10.** (b)
11. (a)	**12.** (a)	**13.** (d)	**14.** (b)	**15.** (b)	**16.** (d)	**17.** (c)	**18.** (b)	**19.** (c)	**20.** (a)

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MODULE

2 ARITHMETIC

- Percentage
- Simple Interest and Compound Interest
- Profit, Loss and Discount
- Average and its Applications
- Ratio, Proportion and Variation
- Time and Work
- Time, Speed and Distance

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CHAPTER

3

Percentage

LEARNING OBJECTIVES

After completion of this chapter, the reader should be able to understand:

- What is percentage?
- Definitions and properties
- Percentage–ratio equivalence
- Usage of percentage as a calculation device
- Methods of solving questions

INTRODUCTION

Percentage is of immense importance when it comes to problem solving in quantitative aptitude (QA) and data interpretation (DI). Despite the fact that the number of questions asked from this concept is not very high in the CAT, a sound understanding and good practice of percentage calculations comes handy in almost all the topics of QA and DI. However, percentage is an important concept for all the other B-school and aptitude examinations.

WHAT IS PERCENTAGE?

Before we talk of the utility of the concept 'percentage', we should be clear regarding when should we read '%' as per cent and when as percentage. Whenever % symbol is attached to any value, known or unknown, the word which we use for symbol % is per cent, e.g., x% is read as x per cent or 20% is read as 20 per cent, not 20 percentage. The word 'percentage' is used whenever % is not attached to any value.

This whole nomenclature can be understood with the help of the following example:

The salary of Abhishek is 20% more than the salary of Abhinav. By how much percentage is the salary of Abhinav less than the salary of Abhishek?

Let us first have some standard meaning of percentage:

- Percentage as a common platform
- Percentage as a rate measurer

Percentage as a Common Platform

This concept can be further understood with the help of the table given below, which gives us the marks obtained by different students in their Class 10 exam.

Student in country	Marks obtained
America	100
India	25
China	45
France	50

By just having this piece of information, we cannot decide that students of which country have performed the best and the data required here is the total marks obtained in each country.

Now, suppose if we add the following data to this question, then the whole situation looks like:

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Student in country	Marks obtained	Total marks	Marks obtained/100 marks
America	100	1000	10%
India	25	25	100%
China	45	300	15%
France	50	100	50%

Actually, here we are using percentage as a common platform to compare all the given values.

Percentage as a Rate Measurer

The concept of percentage is best-suited to find out the rate of change when two or more than two quantities are attached to each other by any common relationship. In this case, we get the respective change which brings in the value of quantities, when the value of other quantities are changing. We will see the application of percentage as a rate measurer while doing product stability ratio.

Before we move ahead, it is important to understand the basic statements used in percentage.

Basic statement 1

What is x% of y?

$$\Rightarrow \frac{x+y}{100}$$

It can also be seen that x% of $y = y$% of x.

For example, 4.5% of 20,000 = 20,000% of 4.5

This one simple fact can be used to divide or multiply any number by 50 or 25 or so. Let us see this with the help of an example. We are trying to find out the value of 25×32, which is nothing but $32 \times 100/4 = 800$. Similarly, if we have to divide any number by 50, then we should multiply the number by 100 and divide it by 2 finally.

Using this, we can see that if we have to calculate 24% of 25 (or any other calculation of similar nature), it is better to find out 25% of 24 $(= 24 \times 100/4) = 600$.

Example 1 What is 20% of 50% of 40% of 20?

Solution Percentage means 'per hundred'.

So, 20% of 50% of 40% of 20 =

$(20/100) \times (50/100) \times (40/100) \times 20 = 0.8$

What we can observe here is that even if we change the order of values, the final result will be the same.

Example 2 Statistics show that 20% of smokers get lung cancer and 80% of lung cancer patients are smokers. If 30% of the population smokes, then the percentage of population having lung cancer is

(a) 4 (b) 3 (c) 8 (d) 7.5

Solution Assume that the total population is 100.

30 people smoke, and out of them, 6 people will be having lung cancer. This 6 represents 80% of lung cancer patients (because they smoke). Hence, total percentage of population having lung cancer $= \frac{6}{80} \times 100 = 7.5\%$.

Basic statement 2

What percentage of x is y?

$$\Rightarrow \frac{y \times 100}{x}$$

(This can be easily deduced by assuming Z% of x is y and then proceeding as given in Statement 1.)

Some more examples of the same kind are given below:

1. What is the percentage growth in the sales value of the year 2002–03 over the sales value of 2003–04?
2. What is the percentage hike in the salary of Manoj this year compared to his salary hike last year?

Basic statement 3

Change in value

(a) Percentage change $= \frac{\text{Change}}{\text{Initial value}} \times 100$

(b) Percentage point change: It is the numerical difference between the values for which we have to calculate the change.

Let us assume some values to understand the above-mentioned concept.

Market share	2002–03	2003–04
Maruti	40%	48%
Honda	30%	26%

Percentage change in the market share of Maruti over the years $= \frac{48-40}{40} \times 100 = 20\%$

Percentage point change in the market share of Maruti over the years = 48% – 40% = 8%

Similarly, if we have to increase any quantity N by S%, then it is equal to $N\left(1+\frac{S}{100}\right)$ and when the same quantity N is to be decreased by S%, then final quantity $= N\left(1-\frac{S}{100}\right)$.

It is worth mentioning here that

$100 \rightarrow 20\%\uparrow \rightarrow 100 \times 1.2 = 120$ $\quad 100 \rightarrow 20\%\downarrow \rightarrow 100 \times 0.8 = 80$

$150 \rightarrow 30\%\uparrow \rightarrow 150 \times 1.3 = 195$ $\quad 150 \rightarrow 30\%\downarrow \rightarrow 150 \times 0.7 = 105$

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$210 \rightarrow 27\% \uparrow \rightarrow 210 \times 1.27 = 266.7$ $\quad$ $210 \rightarrow 27\% \downarrow \rightarrow 210 \times 0.73 = 153.3$

Hence, if the final value and percentage increase or percentage decrease are given and we have to find out the initial value, then it can be done in similar way.

Using $S \rightarrow 30\% \uparrow \rightarrow S \times 1.3 = 195$

So, if the final value 195 and 30%↑ is given, then the initial value $S = \frac{195}{1.3} = 150$.

Example 3 My Reliance India phone bill for the month of May is ₹B. Moreover, there is a service tax of S% which is to be levied upon this value. However, since they are overcharging their customers, they offer a discount of D% on it. So, now I have the following two options to make the payment:

₹B → S%↑→ D% ↓→ Final bill

₹B → D% ↓→ S%↑→ Final bill

Which option is beneficial for me if S > D?

Solution *Prima facie*, it might appear that the 1st one is better than the 2nd one or the 2nd one is better than the 1st one, but a close and deep inspection will reveal that the final bill is same in both the cases.

It can be checked with the help of assuming values also.

B = ₹100, S% = 20% and D% = 10%

₹100 – (20%↑) – ₹120→ (10%↓) – ₹108 (final bill)

₹100 – (10%↓) – ₹90→ (20%↑) – ₹108 (final bill)

So, both the values are same at the end.

Importance of base Whenever we are talking about percentage, it is important to specify what it is relative to, i.e., what is that total which corresponds to 100%. The following situation illustrates this point:

In a certain school, 60% of all students are male, and 10% of all students are computer science majors. If 5% of male students are computer science majors, then what percentage of computer science majors are male?

Here, we are asked to calculate the ratio of male computer science majors to all computer science majors. We know that 60% of all students are male, and among these, 5% are computer science majors, so we conclude that $0.6 \times 0.05 = 0.03$ or 3% of all students are male computer science majors. Dividing this by the 10% of all students who are computer science majors, we arrive at the answer: 3%/10% = 0.3 or 30% of all computer science majors are male.

In QA, choosing the right denominator is often not a problem, whereas sometimes it becomes very problematic in DI, because we are unsure as to which value should be taken as denominator or base. However, if we go by some keywords viz., of/than/over/to, the quantity, which is attached to these keywords, will be the denominator.

It can be seen through the following example:

Example 4 A student multiplies a number x by 5 instead of dividing it by 5. What is the percentage change in the result due to this mistake?

Solution Let us assume $x = 5$

So, the actual result should have been $100 \div 5 = 20$

However, the result obtained = $100 \times 5 = 500$

So, percentage change = $(500 - 20) \times 100/20 = 2400\%$

(It can be very well observed that if we take base as 500, then percentage change = 96%. What we should primarily take care of, with comparison to which number we are calculating percentage change?)

SUCCESSIVE PERCENTAGE CHANGE

Suppose we have to increase a quantity successively by 20% and 30%. Ideally, this should be done by taking 100 as the initial value and then changing this initial value first by 20% and then by 30%. It can be seen below:

$100 \rightarrow 20\% \uparrow \rightarrow 120 \rightarrow 30\% \uparrow \rightarrow 156$

So, net percentage increase = 56%

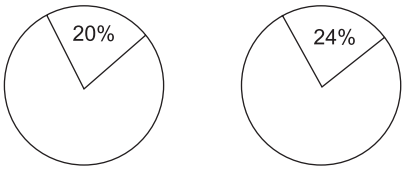
This is known as **straight line method** of solving the problems.

Alternatively, we can use successive change formula also $\left(a + b + \frac{ab}{100}\right)\%$.

Using this formula, we get $20 + 30 + \frac{600}{100} = 56\%$

This kind of calculation is very much important in DI questions, especially in pie chart which can be seen with the help of following example.

The following figure shows the sales figure of Due North Inc. for the year 2002–03 and 2003–04. It is also given that the share of North India in total sales figure is 20% and 24% in respective years.



Total sales for the year 2002–03 is ₹1202 crores and for the year 2003–04, it is ₹1381 crores.

Now the question is: What is the percentage growth in the sales value of North India in 2003–04 over the sales value of 2002–03?

Normal way of doing this question is:

Sales value of north India in 2002–03 = ₹240.4 crores

Sales value of north India in 2003–04 = ₹331.4 crores

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So, growth = ₹91 crore

Hence, the percentage growth

$$= \frac{\text{Growth}}{\text{Initial value}} \times 100 = 37.93\%$$

Alternatively, percentage increases in two factors in the given pie-charts viz., share is increasing from 20% to 24% and then total sales value is increasing from ₹1202 crores to ₹1381 crores.

Rather than calculating the sales value of individual years, we can directly calculate the net percentage change by calculating percentage changes independently in the above-mentioned two factors.

Percentage growth in share = 20%

$$\left(\frac{\text{Growth} \times 100}{\text{Initial value}}; \text{and remember that is not } 4\%\right)$$

Percentage growth in total value ≅ Approx. 15%

So, 100 → 20%↑ → 120 → 15%↑→ 138 or

100 → 15%↑→ 115 → 20%↑→ 138

Hence, net percentage growth = 38%

Example 5 The price of petrol is increased by 20%. However, expenses increase only by 10%. What is the percentage increase or decrease in consumption?

Solution If consumption remains constant, then the expenses should have also increased by 20%. However, since expenses increase by only 10%, consumption has been reduced.

These kinds of problems can be done in three ways:

(i) Straight line method

100 (Initial expenditure)→20%↑ in price→120→x%↓ in consumption→110 (final expenditure),

$$x\% = \frac{10}{120} \times 100 = 8.33\%$$

(ii) 10 × 10 method

We know price × consumption = expenditure

Assume initial price = initial consumption = ₹10

So, initial expenditure = ₹100

	Price		Consumption		Expenditure
Initially… →	10	×	10	=	100
Finally… →	12	×	C	=	110

So, C = 110/12 = ₹9.16

Hence, percentage reduction = 8.33%

(iii) Formula method

Here, a = 20%, b is not known and final percentage change = 10%

$20 + b + 20b/100 = 10$

$\Rightarrow 20 + b + b/5 = 10$

$\Rightarrow b = -8.33\%$

Hence, there is a decrease of 8.33% in the consumption.

PERCENTAGE–RATIO EQUIVALENCE

The essence of percentage–ratio equivalence lies in the fact that most of the percentage calculations like 25%, 37.5%, 33.33%, etc., hover around some particular ratios only. Having a good command over these ratios is definitely going to give a good percentage calculation speed which is very essential for both DI and QA calculations. (See more such calculation in ratio, proportion, and variation chapter.)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	100%	200	300	400	500	600	700	800	900	1000	1100	1200	1300	1400	1500	1600
2	50	100	150	200	250											800
3	33.33	66.66	100													533.33
4	25	50	75	100												400
5	20	40	60	80	100											
6	16.66	33.33	50	66.66	83.33	100										
7	14.28	28.56	42.85	57.14	71.42	85.71	100									
8	12.5	25	37.5	50	62.5	75	87.5	100								
9	11.11	22.22	33.33	44.44	55.55	66.66	77.77	88.88	100							
10	10	20	30	40	50	60	70	80	90	100						
11	9.09	18.18	27.27	36.36	45.45	54.54	63.63	72.72	81.81	90.90	100					
12	8.33	16.66	25	33.33	41.66	50	58.33	66.66	75	83.33	91.66	100				
13	7.69												100			
14	7.14	14.28	21.42	28.57	35.71	42.85								100		
15	6.66	13.33	20	26.66	33.33	40	46.66	53.33	60	66.66	73.33	80	86.66	93.33	100	
16	6.25	12.5	18.75	25	31.25	37.5	43.75	50	56.25	62.5	68.75	75	81.25	87.5	93.75	100

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PRODUCT STABILITY RATIO

In a number of topics and concepts, we encounter the relationship where the product of two quantities equals the third quantity.

For example, Speed × Time = Distance

Price × Consumption = Expenditure

Number of persons × Days = Work done

Length × Breadth = Area of rectangle

Apart form these examples, many times we see instances where one quantity is increased to get another quantity, e.g., if we increase cost price to obtain a certain profit, we obtain selling price or if we increase principal, we obtain amount.

If we generalize product stability ratio, then it can be as

$$A \times B = P$$

Now, if A is increased by a certain percentage, then B is required to be decreased by certain percentage so that the product (P) remains stable. For example, if we increase A by 25% and P has to be constant, then B is required to be decreased by 20%.

This procedure can be summed up in the following way:

Change in A	Change in B	Change in P
100%↑	50%↓	0%
50%↑	33.33%↓	0%
33.33%↑	25%↓	0%
25%↑	20%↓	0%
20%↑	16.66%↓	0%
16.66%↑	14.28%↓	0%
14.28%↑	12.5%↓	0%
12.5%↑	11.11%↓	0%
11.11%↑	10%↓	0%
10%↑	9.09%↓	0%
9.09%↑	8.33%↓	0%
8.33%↑	7.69%↓	0%
and so on.....		

So, if A is increased by 25%, then we need to decrease B by 20% to maintain the product stable.

This mathematical information can be used in so many forms:

1. **Percentage** If A is 25% more than B, then by how much percentage B is less than A?
2. **Profit and loss** An article is sold for ₹125 at a profit of 25%. What is the cost price of the article?
3. **Time, Speed, and Distance (TSD)** When speed of a car is increased by 25%, time taken reduces by 20 minutes in covering a certain distance. What is the actual time taken to cover the same distance by actual speed?
4. **TSD** Mayank goes to his office from his home at a speed of 20 kmph and gets late by 10 min. However, when he increases his speed to 25 kmph, he is 20 min early. What is the distance from his office to his home?
5. **Time and Work** Efficiency of Amit is 25% more than Vinit. Vinit takes 20 days to complete a work. How many days will Amit take to do the same work?
6. **Time and Work** 20 men can do some job in 50 days. In how many days will 25 men do the same job?
7. **Simple Interest (SI)** Rate of interest is 12.5% per annum SI. What is the principal if amount obtained after two years is ₹1250?
8. **Percentage** Due to a price hike of 25%, 5 kg less sugar is available for ₹100. What is the original price per kg?
9. **Mensuration** Length of a rectangle is increased by 25%. By what percentage should the breadth be decreased so that area remains constant?

In all the above -given written situations, just one mathematical information has been used, i.e., if A is increased by 25%, then B decreases by 20%. Let us see the solution of all the questions given above.

Solution 1 Normal Method

Let us assume B = 100, then A = 125

Now, B is 25 less than A.

Percentage of B is less than A = 25/125 × 100 = 20%

Product Stability Ratio Method

Using product stability rule, since A is 25% more than B, so B is 20% less than A.

Solution 2 Normal Method

CP × 1.25 = SP

So, CP = SP/1.25 = 125/1.25 = ₹100

Product Stability Ratio Method

If we increase CP by 25%, we will get SP.

So, if we decrease SP by 20%, we will get CP.

Hence, CP = ₹100

Solution 3 Normal Method

Since we know S = V × T (Distance = Speed × Time)

New speed = 1.25 V, so new time = T/1.25

So, reduction in time = T − T/1.25 = 0.25 T/1.25 = T/5

T/5 = 20 min ⇒ T = 100 min

Product Stability Ratio Method

Since speed has been increased by 25%, so time will reduce by 20%.

Now, 20% T (Time) = 20 min

So, Total time = 100 min

Solution 4 Normal Method

Let us assume that distance = D

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So, D/20 – D/25 = 30/60 h = 1/2
So, D = 50 km

Product Stability Ratio Method

S = V × T
25%↑20%↓
So, 20% T = 30 min
$\Rightarrow$ T = 150 min = $2\frac{1}{2}$
So, total distance = $20 \times 2\frac{1}{2} = 50$ km

Solution 5 Normal Method

Vinit is taking 20 days to complete the work, i.e., Vinit is doing 100% work in 20 days. So, Vinit is doing 5% work in a day. Since efficiency of Amit is 25% more than Vinit, so Amit is doing 6.25% work per day.

So, number of days taken by Amit = 100/6.25 days = 16 days

Product Stability Ratio Method

Efficiency of Amit is 25% more than Vinit. So, Amit will take 20% less days than Vinit.

So, number of days taken by Amit = 16 days

Solution 6 Normal Method

Using Work = Number of persons × Number of days
Work = 20 × 50 = 1000
Now, 1000 = 25 × D
So, D = 40

Product Stability Ratio Method

Number of persons increases by 25%, so number of days will decrease by 20%.

So, number of days = 40 days

Solution 7 Normal Method

Using the formula for SI = PRT/100
P = (SI × 100)/RT
Putting the values gives us P = ₹1000

Product Stability Ratio Method

Interest for two years = 25%

So, if we decrease the amount by 20%, then we will get the principal.

Hence, Principal = ₹1000

Solution 8 Normal Method

Let us assume that original price per kg = ₹P per kg
So, final price per kg = ₹1.25 P
Hence, (120/P) – (120/1.25P) = 5
Solving this equation gives P = ₹4 per kg

Product Stability Ratio Method

Since, the price hike is 25%, 20% less quantity of sugar will be available for ₹100.

20% = 5 kg $\Rightarrow$ 100% = 25 kg
So, 25 kgs were available for ₹100 initially.
So, Price = ₹4/kg

Solution 9 Normal Method

	Length		*Breadth*		*Area*
Initially…→	10	×	10	=	100
Finally… →	12.5	×	B	=	100

So, B = 8
Percentage decrease = 20%

Product Stability Ratio Method

Till now, it must have become very obvious that the breadth will decrease by 20% to keep the area constant.

Extension of Product Stability Ratio

This table is a two-way table, i.e., if we decrease A by 50%, then B is needed to be decreased by 100%.

If we express the percentage figures given in the product stability table in ratios, then it looks like the following:

Change in A	Change in B	Change in P
$\frac{1}{1}\uparrow$	$\frac{1}{2}\downarrow$	0%
$\frac{1}{2}\uparrow$	$\frac{1}{3}\downarrow$	0%
$\frac{1}{3}\uparrow$	$\frac{1}{4}\downarrow$	0%

So, $\frac{1}{50}\uparrow$ corresponds to $\frac{1}{51}\downarrow$. It means that if we increase A by 2%, then B is needed to be decreased by 1.96% (approx.) so that P remains constant.

Similarly, it can be done with all the reciprocals.

However, the problem that lies with this table is that it has values which are reciprocals only.

So, what are we required to do if we increase A by 15%?

Take this as:

Change in A	Change in B	Change in P
$15\% = \frac{3}{20}\uparrow \Leftrightarrow$	$\frac{3}{23}\downarrow$	0%
$35\% = \frac{7}{20}\uparrow \Leftrightarrow$	$\frac{7}{27}\downarrow$	0%

APPLICATION OF PERCENTAGE IN CALCULATION

Percentage is mostly helpful in multiplication and division. Let us learn this with the help of examples.

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Multiplication

To use this percentage method effectively for multiplying two values, we should be thorough with the equivalent ratio of the percentage figures. Besides, having a good addition speed will be an added advantage.

Example 8 63×72

Solution The moment we see any number, we should start mental scanning of the percentage–ratio equivalence. Here, $63 \times 72 = (62.5 + 0.5) \times 72$

$= 5/8 \times 100 \times 72 + 0.5 \times 72$ (5% of 72)

$= 4500 + 36$

$= 4536$

Example 9 76×24

Solution $= (75 + 1) \times 24 = 3/4 \times 100 \times 24 + 1 \times 24 = 1800 + 24 = 1824$

Division

Example 10 Dividing 243 by 50.

Solution $\frac{243}{50} = \frac{243}{100} \times 2 = 2.43 \times 2 = 4.86$

To divide any number by 50, we will divide it by 100 and multiply by 2.

Example 11 Dividing 312 by 25.

Solution $\frac{312}{15} = \frac{312}{100} \times 4 = 3.12 \times 4 = 12.48$

To divide any number by 25, we will divide it by 100 and multiply by 4.

Similarly, while dividing any number by any such number for which we can find out a comparable value in terms of 100 $\left(\text{as } 25 = \frac{100}{4}\right)$ should be used.

Practice Exercises

Warm Up

Q.1 What fraction is equal to 57.12% (approximately)?
(a) 4/8 (b) 5/8 (c) 5/7 (d) 4/7

Q.2 What is 50% of 40% of ₹1200?
(a) ₹450 (b) ₹100 (c) ₹300 (d) ₹240

Q.3 If a% of a + b% of b = 2% of ab, then what percentage of a is b?
(a) 50% (b) 75%
(c) 100% (d) Cannot be determined

Q.4 Subtracting 34% of x from x is equivalent to multiplying x by N. What is the value of N?
(a) 0.66 (b) 66 (c) 0.34 (d) 6.6

Q.5 Subtracting 48% of N from N is equivalent to multiplying N by
(a) 0.48 (b) 1.48 (c) 0.52 (d) 0.42

Q.6 Tipu's salary has increased from ₹7200 to ₹8100. What is the percentage increase in Tipu's salary?
(a) $33\frac{1}{3}\%$ (b) 25%
(c) $16\frac{2}{3}\%$ (d) $12\frac{1}{2}\%$

Q.7 42% of a number is subtracted from 61% of itself and the resulting value is 38. What is the number?
(a) 100 (b) 250 (c) 200 (d) 300

Q.8 The length of a rectangle is increased by 30% and the breadth is decreased by 25%. What is the percentage change in the area of the rectangle due to this?
(a) 2.5% increase (b) 25% decrease
(c) 25% increase (d) 2.5% decrease

Q.9 If a value is increased twice successively by 20%, then what is the net percentage increase in that value?
(a) 20% (b) 40% (c) 21% (d) 44%

Q.10 The numbers are, respectively, 30% and 40% of a third number. What percentage is the first number of the second number?
(a) 25% (b) 50% (c) 40% (d) 75%

Q.11 Two numbers are 15% and 20% less than a third number. What percentage is the first number as a percentage of the second number?
(a) $116\frac{2}{3}\%$ (b) $83\frac{1}{3}\%$
(c) $106\frac{1}{4}\%$ (d) $123\frac{2}{3}\%$

Q.12 Ten litres of water is added to 50 litres of a solution containing 20% of alcohol in water. What is the concentration of alcohol in the solution now?
(a) 20% (b) $16\frac{2}{3}\%$
(c) $12\frac{1}{2}\%$ (d) $33\frac{1}{3}\%$

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Q.13 A man spends 25% of his money on food. After spending 50% of the remaining, he is left with ₹375. How much money was with that man initially?
(a) ₹5000 (b) ₹4000
(c) ₹5800 (d) ₹1000

Q.14 A man earns ₹4000 in a month and saves 30% of his income. What is his expenditure in a year?
(a) ₹28,600 (b) ₹18,400
(c) ₹32,000 (d) None of these

Q.15 After spending 20% on clothes, 10% on books, 9% on purchasing gift for girl friend, and 7% on others, Chandra has a balance of ₹2700. How much money was there with him initially?
(a) ₹5000 (b) ₹6000
(c) ₹7000 (d) None of these

Q.16 A man spends 60% of his income. His income increases by 40%, and due to this, he increases his expenditure also by 40%. Find the percentage increase in his savings.
(a) 30% (b) 4%
(c) 4.44% (d) None of these

Q.17 If the numerator of a fraction is increased by 20% and its denominator decreased by 10%, then the fraction becomes 3/2. What is the value of the original fraction?
(a) $\frac{2}{3}$ (b) $\frac{7}{8}$ (c) $\frac{9}{8}$ (d) $\frac{6}{7}$

Q.18 In an exam of two papers Maths and Chemistry, 60% of the students pass in Maths and 70% pass in Chemistry. What is the minimum percentage of students who could have failed in both the subjects?
(a) 0% (b) 30%
(c) 40% (d) None of these

Q.19 Everyday a fruit seller sells 50% of his stock. 10% of the stock gets spoiled during the night. If a total of 1983 fruits got spoiled in three days period, how many fruits did he start with?
(a) 20,000 (b) 25,000
(c) 24,000 (d) None of these

Q.20 A certain sum becomes 7/3 times itself in 10 years under simple interest. Find the rate of interest.
(a) $7\frac{1}{2}$% p.a (b) $13\frac{1}{3}$% p.a
(c) 20% p.a (d) 10% p.a

Q.21 An amount is lent at x% p.a. simple interest for two years. Instead, had it been lent at $2x$% p.a. simple interest for y more years, then the interest would have been five times the earlier interest. Find the value of y.
(a) 2 (b) 3 (c) 4 (d) 5

Q.22 According to a new plan rolled out by HISP Bank, the rate of simple interest on a sum of money is 8% p.a. for the first two years, 10% p.a. for the next three years, and 6% p.a. for the period beyond the first five years. Simple interest accrued on a sum for a period of 8 years is ₹12,800. Find the sum.
(a) ₹24,000 (b) ₹16,000
(c) ₹15,000 (d) None of these

Q.23 Find the least number of integral years in which a sum of money invested at 20% compound interest per annum will become more than double itself.
(a) 1 (b) 4 (c) 2 (d) 3

FOUNDATION

Q.1 In the country of Tattoland, a new term *tercentage* has been introduced in the place of percentage. While percentage is per 100, tercentage is per 125. What will be the tercentage marks obtained by a student in Tattoland if his percentage of marks is 60?
(a) 48 tercentage (b) 75 tercentage
(c) 60 tercentage (d) Cannot be determined

Q.2 In the country of Tappoland, a new term *hercentage* has been introduced in the place of percentage. While percentage is per 100, hercentage is per 1000. What will be the hercentage marks obtained by a student in Tappoland if his percentage is 60?
(a) 6 hercentage
(b) 600 hercentage
(c) 166.66 hercentage
(d) Cannot be determined

Q.3 In a ΔABC, measurement of ∠ABC = 15°. What will be the value of ∠ABC when seen through a magnifying glass which can magnify any object by 20%?
(a) Value reduced by 20%
(b) Same value
(c) Value increased by 20%
(d) Value decreased by 16.66%

Q.4 In a maternity centre, 5% of all the childbirth cases result in twins. What is the approximate percentage of twins out of total children born?
(a) 5% (b) 7.6% (c) 9.5% (d) 10.4%

Q.5 Due to a price hike of 20%, 4 kg less sugar are available for ₹120. What is the initial price per kg of sugar?
(a) ₹5/kg (b) ₹4/kg
(c) ₹6/kg (d) ₹5.5/kg

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Q.6 A person has to pay 20% tax on first ₹4500 earned; for the amount earned between ₹4500 and ₹7500, the tax is 25%, and for the amount earned more than ₹7500, the tax is 30%. If he paid 23.33% of his income as tax, then what is his income?
(a) ₹9000 (b) ₹8500
(c) ₹10,000 (d) ₹10,500

Q.7 A Clothes shop offers a discount to all its customers on the basis of the value of the purchase. If the value of purchase is less than ₹3200, then the customer gets a discount of 14%. The customer gets 16% discount on purchase between ₹3200 and ₹6300. The customer gets a discount of certain % on the purchase exceeding ₹6300. A customer gets a total discount of ₹1520 which is equal to 16% of the purchase value. How much discount does the customer get for amount exceeding ₹6300?
(a) 16% (b) 18% (b) 20% (b) 22%

Q.8 Three friends P, Q, and R donate 8%, 7%, and 9% of their salary to a charitable hospital in the given order. Salary of P and Q is same and the difference of their donations is ₹74. The total donation by the P and Q is ₹525 more than R's donation. What is the percentage of the total salary of the three donations?
(a) 7.95% (b) 8.3%
(c) 6.34% (d) None of these

Q.9 Fresh watermelons contain 90% water by weight, whereas dry watermelons contain 20% water by weight. What is the weight of dry watermelon obtained from 20 kg of fresh watermelon?
(a) 2.4 kg (b) 2 kg
(c) 2.5 kg (d) Cannot be determined

Q.10 In solution of sugar and water, the ratio of sugar and water by weight is 1:4. This solution is heated, and in the process, it loses 50% weight. What is the ratio of weight of sugar and water in the final mixture?
(a) 4:1 (b) 2:1 (c) 2:3 (d) 3.5:2

Q.11 Rate of inflation is 1000% per annum. What is the value of an article two years from now if it costs ₹6 today?
(a) ₹66 (b) ₹126 (c) ₹660 (d) ₹726

Q.12 Dataman Infosys Systems is a Lucknow-based software company which is growing tremendously. It doubled its turnover in 1998 from ₹50 crores in 1997. Then it tripled its turnover in 1999 and grew by 50% in 2000. What is the turnover at the end of 2000?
(a) ₹250 cr (b) ₹450 cr
(c) ₹300 cr (d) ₹600 cr

Q.13 In the recently concluded CBSE board examination, a total of 600,000 students appeared. 40% of them were females, whereas the rest were males. Pass percentage among males is 75% and the overall pass percentage is 70%. What is the pass percentage among females?
(a) 37.5% (b) 50% (c) 62.5% (d) 70%

Q.14 Because of the budget presented by Mr Yashwant Sinha, the price of sugar increased by 40%. Verma family reduced its consumption so that expenditure on sugar is up by 12%. If the total consumption of sugar before the rise in price was 50 kg. What is the present consumption of sugar?
(a) 48 kg (b) 40 kg (c) 36 kg (d) 32 kg

Q.15 Tatto's working hours per day were increased by 25% and his wages per hour were increased by 20%. By how much per cent was his daily earnings increased?
(a) 20% (b) 25% (c) 50% (d) 45%

Q.16 A's income is reduced from ₹75,000 to ₹60,000, whereas B's income for the same period is increased from ₹60,000 to ₹75,000. What percentage of decrease in A's income is the increase in B's income?
(a) 125% (b) 75% (c) 133% (d) 100%

Q.17 18% of A plus 15% of B plus 19% of C is equal to 17% of the sum of A, B, and C. If A − B = 500 and A − C = 3400, then what is the value of A + B + C?
(a) 12,400 (b) 11,600
(c) 13,500 (d) None of these

Q.18 Out of 51,600 candidates who appeared in an examination, 35% failed, whereas 15% passed with honours. What is the number of candidates who passed the examination, but failed to obtain honours, assuming that no student who has failed can obtain honours?
(a) 25,000 (b) 30,000
(c) 27,309 (d) 25,800

Q.19 A and B have some guavas divided among themselves. A says to B, 'If I give you 25% of the guavas, then I will still have 2 more guavas than you have'. To this, B says, 'If you give me guavas equal to 70% of what I have now, then I will have 4 more guavas than you have'. What is the total number of guavas they have?
(a) 80 (b) 64 (c) 36 (d) 88

Q.20 Raghupati Raghav was able to score a total of 600 in 12 tests. He scored less than or equal to 80% of his average score per test in four of these tests. If he did not score more than 60 in any of the tests, then what is the minimum number of tests in which he should have scored more than 50?
(a) 8 (b) 4 (c) 3 (d) 2

Q.21 A locomotive engine runs at a speed of 50 km/h when no compartment is attached to it. For every new compartment that is attached to it, the speed of the train is reduced by 10% of the earlier speed. At most, how many compartments can be attached so that the train can cover a distance of 180 km in a maximum of 8 hours?
(a) 7 (b) 5 (c) 6 (d) 8

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Q.22 Babloo organized a party by taking contributions from the participants. 60% of the participants contributed 80% of the funds. The average contribution of all the people who attended is ₹50. What is the average contribution of the remaining 40% of the people?
(a) ₹100 (b) ₹150
(c) ₹250 (d) ₹25

Q.23 If the price of an article rose by 25% every odd year and fell by 20% every even year, then what would be the percentage change after 180 years?
(a) 10% increase (b) 25% increase
(c) No change (d) 20% decrease

Q.24 Pranav saves 10% of his total salary. Next year, he increases his expenses by 20%, but his percentage savings remain the same. What is the percentage increase in his salary next year?
(a) 10% (b) 20% (c) 40% (d) 16.66%

Q.25 In an examination, 80% students passed in philosophy and 70% students passed in Maths. At the same time, 15% failed in both the subjects. If 390 students passed in both the subjects, then how many students appeared in the examination?
(a) 500 (b) 400 (c) 800 (d) 600

Q.26 The salary of Anil and Vinit is 20% and 30% less than the salary of Dheeraj, respectively. By what percentage is the salary of Anil more than the salary of Vinit?
(a) 33.33% (b) 50%
(c) 10% (d) 14.28%

Q.27 In the recent census report of Patna, it was found that 40% of total male adult population and 55% of total female adult population is married. What percentage of total adult population of Patna is married? (Polygamy is not in practice in Patna.)
(a) 46.32% (b) 43.6%
(c) 49.81% (d) 40%

Q.28 Sona and Ellaraputtu are very close friends. One day, they were figuring out about their respective incomes and expenses. It was found that income of Sona and Ellaraputtu is 'S' and 'E', respectively. Sona spends 12% of her income and Ellaraputtu also spends the same amount. What percentage of her income is Ellaraputtu spending?
(a) E/(12S) (b) 12S/E
(c) 12 E/S (d) S/(12E)

Q.29 Sixty students appeared in Physics and Maths papers of an examination. Of these, 60% students passed in Physics and 70% passed in Maths. What is the maximum percentage of students who failed in both the subjects?
(a) 40% (b) 30% (c) 0% (d) 100%

Q.30 In an election, 10% of the voters on the voter's list did not cast their votes, and 60 voters cast their ballot papers blank. There were only two candidates. The winner was supported by 47% of all voters in the list and he got 308 votes more than his rival. The number of voters on the list were
(a) 3600 (b) 6200
(c) 4575 (d) 6028

Q.31 A rainy day occurs once in every 10 days. Half of the rainy days produce rainbows. What in the percentage of all the days when we have no rainbows?
(a) 95% (b) 10% (c) 50% (d) 5%

Q.32 One-third of Mayank's marks in history equals his marks in geography. If he obtained 180 marks in two subjects together, then how many marks did he get in geography?
(a) 40 (b) 45 (c) 60 (d) 36

Q.33 If the price of the sugar is increased by 20%, then one can buy 2 kg less sugar in ₹50. What is the quantity of sugar that could be bought before the price hike?
(a) 10 (b) 8 (c) 16 (d) 12

Q.34 The entry fee in an exhibition was ₹10. Later this was reduced by 25%, which increased the sale of tickets by 20%. Find the percentage increase in the number of visitors.
(a) 54 (b) 57 (c) 60 (d) 66

Q.35 A car bought for ₹13,350 depreciates each year by $12\frac{1}{2}$%. In how many years will its value be reduced to ₹6000?
(a) 4 years (b) 6 years
(c) 8 years (d) 10 years

Q.36 An electrical contractor purchased a certain amount of wire, of which 10% of which was stolen. After using 85% of the remainder, he had 47.25 m of wire left. How much wire did he purchase?
(a) 600 m (b) 103 m
(c) 325 m (d) 350 m

Q.37 Anoop got 273 marks in his graduation exam and got 5% more than the pass percentage. If Siddharth got 312 marks, then by what percentage above the pass marks did he pass the exam?
(a) 9% (b) 12.5% (c) 20% (d) 25%

Q.38 There were only two contestants A and B in the recent assembly election in Bihar at Raghopur constituency. B got 50% of the votes that A got. Had A got 200 votes less, then there would have been a tie. What is the total number of votes polled?
(a) 1,000 (b) 800
(c) 1,200 (d) Cannot be determined

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Q.39 A city had a population of 30,00,000 in the beginning of 1999. Its average growth rate is 4% per year, but due to a massive earthquake in 2001, its population is reduced by 8% in that year. However, it again maintained the same growth rate of 4% in following years. What will be the approximate population of the city at the end of 2003?
(a) 32,06,460 (b) 34,68,420
(c) 31,52,360 (d) 32,28,810

Q.40 A is an alloy of tin and copper, and B is an alloy of copper and zinc. A has 30% copper, whereas B has 50% copper. x gm of A is mixed with 30 gm of B to form another alloy that has 45% copper. What is the value of x?
(a) 5 (b) 10 (c) 15 (d) 20

Q.41 Rajesh Raj has ₹90,000 with him. He purchases a Bike, a home theatre, and a flat TV for ₹15,000, ₹13,000, and ₹35,000, respectively, and puts the remaining money in his bank account which pays 15 per cent per annum Compound Interest. After 2 years, he sells off the three items at 80% of their original price and also withdraws his entire money from the bank by closing the account. What is the total change in his asset?
(a) −4.5% (b) +3.5%
(c) −4.32% (d) +5.5%

Q.42 After three successive raises, Aftab's salary became equal to $\frac{378}{125}$ of his initial salary. By what percentage was the salary raised the first time if the third rise was twice as high (in percentage) as the second rise and the second raise was twice as high (in percentage) as the first rise?
(a) 10% (b) 15% (c) 20% (d) 25%

Q.43 One litre of water is evaporated from 6 L of solution containing 4% of sugar. The percentage of sugar in the remaining solution is
(a) 1.4% (b) 5.2% (c) 4.8% (d) 3.4%

Q.44 A's income is 60% of B's income, and A's expenditure is 70% of B's expenditure. If A's income is 75% of B's expenditure, then find the ratio of A's savings to B's savings.
(a) 5:1 (b) 1:5 (c) 3.5:1 (d) 2:7

Q.45 There are 110,000 books at Khuda Bakhsh Oriental Public Library, 40% of which are Science books. It was decided to add 20,000 new books to the library. How many of the new books should be Science books in order to bring the percentage of Science books in the library equal to 45%?
(a) 15,000 (b) 1500
(c) 1450 (d) 14,500

Q.46 If 33 untrained labourers can do a piece of work in 15 days of 12 hours each, then how many trained labourers can do 50% more work in 11 days of 9 hours each? (Assuming that it takes 2 trained labourers to do the work of 5 untrained labourers.)
(a) 42 (b) 36 (c) 90 (d) 100

Q.47 X's income is 75% of Y's income, and X's expenditure is 80% of Y's expenditure. If X's income is 90% of Y's expenditure, then find the ratio of X's savings to Y's savings.
(a) 1:2 (b) 2:1 (c) 1:4 (d) 2:3

Q.48 Kajol went to a fruit market with a certain amount of money. With the money she has, she can buy either 40 apples or 70 mangoes. She retains 15% of her money for auto fare. If she buys 35 mangoes, how many more apples can she buy?
(a) 15 (b) 20
(c) 14 (d) None of these

Q.49 In an election for the post of class representative at Pitampura Centre where two candidates were contesting, 10% of the students did not vote (because they were late, as usual) and 60 votes recorded were declared invalid (because stray marks were made on the ballot paper). Winning candidate won the election by 308 votes and it was found that he had been supported by 47% of the whole students of the class. What is the the number of valid votes recorded for each candidate?
(a) 2308, 2000 (b) 2904, 2596
(c) 2914, 2606 (d) 2464, 2772

Q.50 If the length and breadth of a rectangular plot are increased by 20% and 50%, respectively, Then the new area is how many times the original area?
(a) 5/8 (b) 9/5 (c) 3/7 (d) 7/3

Moderate

Q.1 There are fifty successive percentage discounts given in a series of 2%, 4%, 6%, 8%…and so on. What is the net discount?
(a) 98% (b) 2550%
(c) 100% (d) Infinite

Q.2 In Question 1, after how many such percentage discounts in succession will the effective discount be higher than 50%?
(a) 7th (b) 8th (c) 9th (d) 10th

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Q.3 My Alliance India phone bill for the month of May is ₹B. On top of this, there is a service tax of S%, which has to be levied upon this value. However, since they are having a guilt feeling of overcharging their customers fraudulently, they offer a discount of D% also on top of this. So, my billing goes like this

₹B→ S%↑→ D% ↓→ Final Bill

But surprisingly, my final bill is equal to ₹B itself. Now, which of the following is true regarding S and D (assume all the values to be positive)?

(a) S > D (b) S = D
(c) D > S (d) Cannot be determined

Q.4 Assuming percentage to be true for all the real values, which of the following can be true?

40% of A = 60% of B

(i) A > B (ii) B > A (iii) B = A
(a) (i) and (ii) only (b) (ii) and (iii) only
(c) (i) and (iii) only (d) None of these

Direction for Questions 5 and 6: Read the passage below and solve the questions based on it.

Alok has certain number of oranges with him. He sells each orange for ₹2. Shazmi bought 25% of the oranges and Sandeep bought one-third of what was left. Mohit bought 50% of what was left and Siddharth bought 4 oranges. Total amount paid by Shazmi and Mohit was ₹24.

Q.5 How many oranges did Sandeep purchase?

(a) 3 (b) 4 (c) 6 (d) 8

Q.6 How many oranges were left with Alok?

(a) 3 (b) 4
(c) 6 (d) None of these

Direction for Questions 7 and 8: Read the passage below and solve the questions based on it.

All students of class X took the pre-board test. One-third of the class received B, one-fourth received B+, one-sixth received B–, and one-eighth of the class failed (grade C). Everyone else got A.

Q.7 What can be the minimum number of students in class X?

(a) 8 (b) 48
(c) 24 (d) Cannot be determined

Q.8 How many students in class X got A?

(a) 3 (b) 6
(c) 4 (d) Cannot be determined

Q.9 A manufacturer of detergent cakes increased the price of his product, and as a result of which, there was a decrease of 20% in its sales. If the sales turnover increases by 20% due to this change, then what was the percentage increase in the price?

(a) 20% (b) 36% (c) 40% (d) 50%

Q.10 Ravi Shankar wishes to buy an air conditioner (AC) with the money in the bank, which currently is earning interest at the rate of 15 p.c.p.a compounded annually. However, Tanzar, his friend, forecasts that the inflation rate applicable to AC is going to be 14%, 15%, and 16%, respectively, for the next 3 consecutive years and advises Ravi Shankar to postpone the purchase by 3 years. Does Ravi Shankar gain monetarily, if he takes Tanzar's advice?

(a) Yes
(b) No
(c) He neither gains nor losses
(d) He gains only if the purchase is made in the second year.

Q.11 In the recently concluded referendum in New York, regarding calling back US troops from Iraq, 600 people voted on a referendum, but after some discussion, the opponents increased by 150%. The motion was then rejected by a majority two times as great as that by which it was formerly passed. How many people voted against the initial referendum?

(a) 400 (b) 300 (c) 200 (d) 500

Q.12 The number of students present in the class of Manoj Yadav is M, and out of those M students, 22.1875% are stupids. If the number of stupids in the class is N, then what can be the minimum value of M + N?

(a) 1,955 (b) 391
(c) 16,200 (d) 12,21,785

Q.13 If 75% of a class of 40 students can sing and only 20% cannot dance, the maximum number of students who can neither sing nor dance is

(a) 0 (b) 6
(c) 8 (d) Cannot be determined

Q.14 There are five contestants A, B, C, and D in the assembly election from the Bihar Sharif constituency. It is given that none of the contestants got less than 1% of the valid vote. Consider the following statements in isolation to each other:

(i) A got 49% of the total valid votes.
(ii) B got 55% of the total votes.
(iii) C got 46% of the total valid votes.
(iv) D got 48% of the total valid votes.

With how many of the following statements when used independently, it is possible to find out the winner of the election?

(a) 0 (b) 1 (c) 2 (d) 3

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Q.15 For admission in a graduation program of Delhi University, 90% of the candidates who appeared for the written test were males and the rest were females; 60% of the males and 80% of the females passed in the written test. Find the total number of students appeared for the written test, if the total number of passed candidates was 1240.
(a) 2500 (b) 2000 (c) 2750 (d) 3500

Q.16 In a laboratory experiment, a sample of air, which is a mixture of oxygen and water vapour is taken. Water vapour contains hydrogen and oxygen gases. If air contains a total of 70% oxygen (including that contained in the water vapour) by weight, whereas water vapour contains 16·66% of oxygen by weight, then how many kilograms of water vapour is present in 1 kg of air?
(a) 0.2 (b) 0.36 (c) 0.32 (d) 0.25

Q.17 In Patna University, out of every 7 students learning Science subject, 3 take literature as well; for every student taking at least one of these two courses, there are 3 students who take up neither. If 10% of the students take only Science, then the percentage of students who have taken Science is:
(a) $17\frac{1}{2}\%$ (b) $23\frac{1}{3}\%$
(c) $7\frac{1}{2}\%$ (d) Cannot be determined

Q.18 Sixty per cent of the students of a class are boys, and in an examination, 80% of the girls scored more than 40 marks (out of a maximum possible 150 marks). If 60% of the total students scored more than 40 marks in the same examination, then find the fraction of the boys who scored 40 marks or less.
(a) 8/15 (b) 7/15 (c) 4/5 (d) 1/5

Q.19 A thousand people voted on a resolution with 10% of the votes being invalid. After some discussion 1000 people voted again. This time there were 20% invalid votes. The opponents were increased by 50% while the motion was now rejected by a majority, which is 300% more than it was formerly passed by. How many people voted against the resolution after the discussion?
(a) 400 (b) 200 (c) 300 (d) 500

Q.20 Recently I had gone to a locality called Shadigarh for conducting a survey about the number of married persons in the locality. The population of the locality is 7200 and 11/18th of those are males and the rest were females. If 40% of the males are married, then find percentage of married females in the locality.
(a) $48\frac{1}{7}\%$ (b) $52\frac{4}{7}\%$
(c) $62\frac{6}{7}\%$ (d) $71\frac{1}{7}\%$

Q.21 Nitish was to get a 50% hike in his pay but the computer operator wrongly typed the figure as 80% and printed the new pay slip. He received this revised salary for three months before the organization realized the mistake. What percentage of his correct new salary will he get in the fourth month, if the excess paid to him in the previous three months is to be deducted from his fourth month?
(a) 20% (b) 25% (c) 40% (d) $33\frac{1}{3}\%$

Q.22 In the US opinion polls held during August, 60% of the respondents favoured Walter Mondale, whereas the rest favoured George W. Bush. It was found in September polls that 10% of Mondale supporters switched their preference to Bush, whereas the same percentage of Bush's supporters also switched their preference to Mondale. What percentage of the electorate should now switch their preference from Mondale to Bush so that they are at par?
(a) 6% (b) 8% (c) 10% (d) 14%

Q.23 An index of 12 shares contains, among others, the shares of Reliance, HLL, and Infosys with weightage of 7%, 13%, and 15%, respectively. What is the increase in the prices of other shares, if these three rise by 9%, 10%, and 4%, respectively, while the index rises by 6%?
(a) 5.34% (b) 4.91%
(c) 4.58% (d) Cannot be determined

Q.24 A community service organization has raised 80% of the donation that it needed for a new building by receiving an average donation of ₹400 per person from a group of people. To raise the remaining amount, the organization now intends approaching another group of people which has two-thirds the number of people as the first group. then What should be the average donation per person from the new group?
(a) ₹100 (b) ₹150
(c) ₹200 (d) ₹250

Q.25 Sixty per cent of the employees of Due North Inc. are female, and 75% of the men earn more than ₹25,000 per year. If 45% of the company's employees earn more than ₹25,000 per year, then what fraction of women employees earn ₹25,000 per year or less?
(a) 2/11 (b) 1/4 (c) 1/3 (d) 3/4

Q.26 The pollution in normal atmosphere is less than 0.01%. Due to leakage of methyl isocyanide (MIC) from a factory, the pollution is increased to 20%. If every day 80% of the pollution in the atmosphere is neutralized, in how many days will the atmosphere become normal?
(a) 5 days (b) 3 days
(c) 4 days (d) None of these

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Q.27 The marks obtained by Arpit in English, Maths, Science, and Hindi in Class 10 are as follows (Maximum marks per subject = 100):

(a) The marks obtained in Maths is 1.5 times the marks obtained in English.

(b) Arpit got 64% in these four subjects.

(c) Arpit got the maximum and minimum marks in Science and Hindi, respectively, with a difference of 48 marks between them.

(d) An addition of 50% of the marks obtained in English to the final score gives an overall percentage of 70%.

What would be his percentage of marks if only Maths and Science marks are counted?

(a) 14 (b) 82 (c) 84 (d) 86

Q.28 Amit puts some amount in bank and obtained an interest of ₹15 at the end of one year. He added ₹85 to this amount and put the whole amount again in the bank for another year. At the end of the 2nd year, Amit got ₹420 as total (interest + principal). What amount did he put in the beginning and what is the rate of interest offered by the bank if the minimum amount that can be deposited in the bank is ₹75?

(a) 250, 5% (b) 300, 5%

(c) 200, 8% (d) 350, 3.5%

Q.29 Amul, Binit, and Chand start a business by investing ₹70,000 that earns them a profit of ₹42,000 at the end of the year. Amul invests his share in the profit in a scheme that gives him 10% interest compounded annually and Binit invests his share in a scheme that gives him 20% interest compounded annually. Amul gets ₹2520 as interest at the end of 2 years and Binit gets an interest of ₹4200 at the end of one year. Find Chand's investment in the business.

(a) ₹160,000 (b) ₹15,000

(c) ₹17,520 (d) Cannot be determined

Q.30 A food processing unit received fresh consignment of berries from Himachal Berry Corporation containing 'a' kilogram of berries. After analysis these berries were found to contain 99% water. After 15 days, sample of berries from the same lot was found to contain 98% water. What must be the weight of same lot at that time?

(a) $\frac{a}{2}$kg (b) $\frac{2a}{3}$kg

(c) $\frac{4a}{5}$kg (d) $\frac{a}{3}$kg

Q.31 Kishore is an employee of a MNC firm. He needs ₹10 lakhs for buying a new flat in Patel Nagar. His company is offering housing loan at an interest rate of 15% for 5 years. The two banks ICICI and HFDC are offering loan for 5 year period at an interest rate of 18% and 13%, respectively. How much should Kishore borrow from the two banks so that he can pay the interest rate same as that of his own MNC firm?

(a) 5 lakhs from HDFC and 5 lakhs from ICICI

(b) 3 lakhs from HDFC and 7 lakhs from ICICI

(c) 6 lakhs from HDFC and 4 lakhs from ICICI

(d) 8 lakhs from HDFC and 2 lakhs from ICICI

Q.32 The population of towns A and B is in the ratio of 1 : 4. For the next 2 years, the population of A would increase and that of B would decrease by the same percentage every year. After 2 years, their population became equal. What is the percentage change in the population?

(a) 33.33% (b) 66.66%

(c) 25% (d) Not possible

Direction for Questions 33 to 35: Read the passage below and solve the questions based on it.

Petroleum product sales (in million tones)		
Petroleum product	**2002–03**	**2003–04**
Diesel	36.0	36.6
Petrol	7.6	7.9
LNG	8.3	9.3
Crude oil	11.9	11.7
ATF	2.3	2.5
Total	104.2	107.4

Choose Option (a) if only A is true.

Choose Option (b) if only B is true.

Choose Option (c) if both A and B are true.

Choose Option (d) if neither A nor B is true.

Q.33 A. The percentage share of ATF in petroleum product sales has increased in 2003–2004 over the previous year.

B. Percentage share of diesel in petroleum product sales has decreased in 2003–2004 over the previous year.

Q.34 A. LNG has experienced the maximum percentage increase in petroleum product sales in 2003–2004 over the previous year.

B. Change in percentage shares of crude oil in petroleum product sales in 2003–2004 over the previous years is approximately 0.2%.

Q.35 A. Percentage decrease in crude oil sales in 2003–2004 over the previous year is more than percentage increase in diesel sales during the same period.

B. LNG sales in 2003–2004 is less than 25% of diesel sales in 2003–2004.

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Direction for Questions 36 to 38: Read the passage below and solve the questions based on it.

In each question, there are two statements, A and B, either of which can be right or wrong on the basis of the information given below.

Direct taxes collected in different years (₹in crores)

Taxes	1998–99	1999–00	2000–01	2001–02	2002–03
Corporation tax	22,750	23,460	25,190	27,080	28,880
Income tax	18,210	20,460	23,790	26,110	29,080
Service tax	220	240	200	230	260
Expenditure tax	300	320	340	350	345
Interest tax	1,280	1,190	1,350	1,410	1,490
Water tax	8	15	20	22	25

Mark your answer as

(a) if only A is right.
(b) if only B is right.
(c) if both A and B are right.
(d) If both A and B are wrong.

Q.36 A. The growth rate for Service Tax from 1998–99 to 1999–2000 was higher than that for income tax for the same period.
B. The growth rate for Service Tax from 1998–99 to 1999–2000 was lower than that for the period 2001–02 to 2002–03.
(a) 4 (b) 2 (c) 3 (d) 1

Q.37 A. The ratio of income tax to Corporation tax was higher in 1998–99 than in 1999–2000.
B. Service Tax, as a percentage of expenditure tax, was greater in 2000–01 than in 2001–02.
(a) 1 (b) 2 (c) 3 (d) 4

Q.38 A. The average annual growth rate for Income tax during the period 1998–99 to 2002–03 was more than 1.5 times the same for Corporation tax during the same period.
B. The average annual growth rate for Interest tax was more than 5% during the given period.
(a) 1 (b) 3 (c) 2 (d) 4

Direction for Questions 39 and 40: Read the passage below and solve the questions based on it.

Abhay, Bhaskar, Chitra, and Danny work in the same office. Abhay's salary is 20% less than Bhaskar's salary. Chitra's salary is 25% more than Abhay's salary. Danny's salary is 20% more than Abhay's salary.

Q.39 If Abhay has a salary of ₹10,000 per month, who has a salary of ₹12,500 per month?
(a) Bhaskar (b) Chitra
(c) Danny (d) Both (a) and (b)

Q.40 If Chitra received a raise and her new salary is 10% higher than Danny's, what is her new monthly salary? (use the data from the previous question).
(a) ₹12,750 (b) ₹13,200
(c) ₹13,333 (d) ₹14,225

Direction for Questions 41 to 44: Read the passage below and solve the questions based on it.

There are two statements given in each question: A and B, either of which can be true or false on the basis of the information given below.

The following table gives us the detailed information regarding the admission process of a reputed B-School IIM-Jehanabad for the year 2003 and 2004.

Year	Gender	Number of applicants	Number of candidates appeared for written test	Number of candidates called for an interview	Number of candidates selected for the course
2003	Male	6100	5900	60	12
	Female	1900	1500	15	5
2004	Male	6300	6000	64	13
	Female	4000	3500	30	6

Choose Option (a) if only A is true.
Choose Option (b) if only B is true.
Choose Option (c) if both A and B are true.
Choose Option (d) if neither A nor B is true.

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Q.41 A. The success rate of moving from the written test to the interview stage for females was worse than for males in 2003.
B. The success rate of moving for the written test to the interview stage for males was better in 2003 than in 2004.

Q.42 A. The percentage of absentees in the written test among males decreased from 2003 to 2004.
B. The percentage of absentees in the written test in 2003 among females was larger than that among females in 2004.

Q.43 A. The percentage of the female candidates selected in 2004 from female applicants is more than the percentage of male candidates selected for year 2004 out of total male applicants of that year.
B. In 2003, from among those called for an interview, the males had a greater success rate than females.

Q.44 A. Difference in the total number of male and female applicants is more than 50% of the total candidates who appeared for the written test in 2004.
B. The total number selected for the job for both the years is more than 70% of the total female candidates called for an interview in both the years.

Q.45 In a laboratory, equal number of rats were injected with solution A. In a week, 55% of rats injected with solution A got infected with a disease and of that, equal number of rats were treated with vaccine C and D. Now, percentage of rats treated with vaccine C that recovered was 70% and that with vaccine D was 50%, whereas others died. Calculate the number of rats on whom the experiment was carried out in the laboratory, if the number of rats that died even after being treated with vaccines due to the disease is 77.
(a) 600 (b) 700 (c) 750 (d) 800

ADVANCED

Direction for Questions 1 to 3: Read the passage given below and solve the questions based on it.

The table gives the production of five agricultural crops between 1999–2000 and 2000–2001.

Crops	Target for 2001–2002 (in MT)	Output for 2000–2001 (in MT)	% increase in output over 1999–2000
Rice	162.7	160.4	25
Wheat	18.9	16.2	18
Tea	20.5	22.5	35
Cotton	14.7	18.9	32
Coffee	12.3	14.8	16

Q.1 The production of rice in 1999-2000 (in MT) is approximately:
(a) 124 (b) 128
(c) 132 (d) 136

Q.2 Output of which crop in 2000–01 is showing maximum percentage difference from the target set in 2001–02?
(a) Wheat (b) Coffee
(c) Cotton (d) Tea

Q.3 A new parameter is invented to judge the target: output ratio. That parameter is known as efficacy ratio and is defined as the ratio of target for 2001–02 and output for 2000–01. For how many given crops the efficacy ratio is more than 1?
(a) 1 (b) 3 (c) 2 (d) 4

Q.4 Some people form a joint account for one year with a condition that "each person will deposit an amount equal to the number of people in the account that month. Moreover, the person who withdraws from the account in between will get their amount at the end of the year. After 6 months, 25% of the people withdrew, and 33.33% of the remaining withdrew after 3 months. At the end of the year, there was an amount of ₹4860 in their account before paying off the withdrawn amounts. Find the number of people in the beginning of the year.
(a) 12 (b) 24
(c) 36 (d) None of these

Direction for Questions 5 to 9: Read the passage below and solve the questions based on it.

The number of taxi drivers and auto drivers working in three shifts are given below:

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Vehicle	Shift	Number of drivers	Working hours	Total collection (in 1,000 ₹)
Taxi	M1	425	10	132.1
	M2	672	8	240.2
	M3	135	6	42.3
Auto	M1	350	9	57.3
	M2	425	8	100.8
	M3	95	7	18.9

A taxi driver charges ₹13 for every 1.6 km, whereas an auto driver charges ₹8 for every 1.3 km. For the night shift, charges are 1.5 times the normal rate.

Q.5 By what percentage is the average collection per taxi driver more during the M2 shift than during the M1 shift?
(a) 4% (b) 15% (c) 34% (d) 82%

Q.6 What is the average distance travelled by the auto drivers during the M3 shift?
(a) 1356 km (b) 2045 km
(c) 3054 km (d) 2672 km

Q.7 Who get the highest returns for their efforts with respect to work hours?
(a) Auto drivers — M3 Shift
(b) Taxi drivers — M2 shift
(c) Taxi drivers — M3 shift
(d) Auto drivers — M1 shift

Q.8 What is the maximum difference between the average distances covered in an hour by drivers of any two shifts?
(a) 3100 km (b) 3300 km
(c) 3500 km (d) 3400 km

Q.9 Which of the following is false?
(a) Lowest contribution per driver is by auto drivers in the M1 shift.
(b) Lowest returns for the efforts are for auto drivers in the M1 shift.
(c) The percentage of drivers during the M3 shift is 21% of the total drivers.
(d) None of these

Q.10 Consider a big cube formed of 64 smaller cubes of which 40 are coloured. The smaller cubes are arranged such that the exposure of the coloured cubes to the outside is minimized. What is the percentage of the exposed area that is coloured?
(a) 62.5% (b) 50%
(c) 41.6% (d) None of these

Q.11 On May 1, 2002, a person sold all his investments for ₹108,000, incurring a loss of 10% with respect to the previous month. On April 1, 2002, he had invested all his money in shares, which represented a 20% increase over the sum he had on March 1, 2002, which represented a 20% decrease over the money he had on February 1, 2002, which represented a 25% increase over the money he had on January 1, 2002. The person had maximum money in
(a) February, 2002 (b) January, 2002
(c) April, 2002 (d) None of these

Direction for Questions 12 and 13: Read the passage below and solve the questions based on it.

Patna and Lucknow are the only cities of India that are having only adults. Female population in Patna grows by 50% to 60,000, but maintaining the same male to female ratio as previous year. Population in Lucknow increased by 12.5% to 90,000, but the ratio of male to female changed by 1:1 to 4:5. Male population in Patna was 10% more than the male population in Lucknow in the previous year.

Q.12 What is the female population in Lucknow in the second year?
(a) 55,000 (b) 45,000
(c) 400,000 (d) 50,000

Q.13 What is the total population in Patna after the increase (i.e., in the second year)?
(a) 80,000 (b) 100,000
(c) 20,000 (d) 26,000

Q.14 The owner of a flower shop follows a particular pattern for his business. During a period of inflation, he raises his price by P%, and during a slowdown, he decreases his existing prices by P%. After a year in which there was inflation first, followed by a slowdown, the cost of a red-rose bouquet decreases by ₹162. After another year, in which there was inflation once more followed by a slowdown, the cost of this bouquet reduced by a further ₹147.42. What was the original price of the red-rose bouquet?
(a) ₹1500 (b) ₹1000
(c) ₹1800 (d) ₹1111

Direction for Questions 15 to 17: Read the passage below and solve the questions based on it.

Three leading drug manufacturers Arpit Pharma, Bittu Pharma and Chintu Pharma each launch an antiseptic drug. Each company

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uses all or some of the 5 constituents A, B, C, D, and E. The table below gives the percentage composition of these constituents in the respective drugs manufactured by different companies.

	A	B	C	D	E
Arpit Pharma	12	15	20	28	25
Bittu Pharma	37	15	13	15	20
Chintu Pharma	26	15	10	34	15

The above values are in percentage of composition. All manufacturers produce their drug as a capsule of 300 mg.

Q.15 The effectiveness of constituents A, B, C, and D are in the ratio 5:6:4:3, whereas E is just a sweetener and does not have any curing effect. Which of the following company's capsule is most effective?
(a) Arpit Pharma (b) Bittu Pharma
(c) Chintu Pharma (d) Data Insufficient

Q.16 If the side effects caused by drugs are proportional to the ratio of C's composition to the sum of B's and E's composition, then which drug has least side effect?
(a) Arpit's drug
(b) Bittu's drug
(c) Chintu's drug
(d) All of them have equal side effect

Q.17 Arpit Pharma finds that its sales are dipping because its drug tastes too sweet. So, it decides to decrease the composition of E and increase that of C, keeping other composition unchanged. It also wants to make sure that the side effect is not more than double its previous value. What could be the maximum percentage of C's composition in Arpit's drug? (Use data from the previous questions)
(a) 25% (b) 30% (c) 34% (d) 37%

Q.18 If 3000 capsules are produced by these 3 companies per annum, then the consumption of component C annually would be:
(a) 60 kg (b) 90 kg
(c) 105 kg (d) Data Insufficient

Q.19 In Ranchi, only two newspapers *Dainik Jagran* and *Prabhat Khabar* are published. It is known that 25% of the city population reads *Dainik Jagran* and 20% reads *Prabhat Khabar*, whereas 8% reads both *Dainik Jagran* and *Prabhat Khabar*. It is also known that 30% of those who read *Dainik Jagran* but not *Prabhat Khabar* look into advertisement and 40% of those who read *Prabhat Khabar* but not *Dainik Jagran* look into advertisements, whereas 50% of those who read both *Dainik Jagran* and *Prabhat Khabar* look into advertisements. What is the percentage of the population who read an advertisement?
(a) 10.8% (b) 11.3% (c) 13.9% (d) 12.7%

Direction for Questions 20 to 22: Read the passage below and solve the questions based on it.

The following table gives the distribution of number of B-schools out of a sample of 100 B-Schools based on the scores obtained by them on five different parameters—infrastructure, placements, research, industry interface, and intellectual capitals. The maximum possible score in any of these parameters is 100 points.

Score/ Parameter	80 points and above	60 points and above	40 points and above	20 points and above
Infrastructure	16	30	50	65
Placements	12	28	37	50
Research	30	50	70	92
Industry interface	24	42	57	73
Intellectual capitals	27	43	47	70

For example, 16 out of 100 schools surveyed have scored 80 points or above in the parameter infrastructure.

Q.20 What is the maximum number of business schools that could have scored 40 points or more but less than 60 points in at least three of the five parameters?
(a) 20 (b) 18
(c) 12 (d) None of these

Q.21 What is the minimum number of business schools that could have scored 40 points or more but less than 80 points in at least two of the five parameters?
(a) 10 (b) 5
(c) 12 (d) None of these

Q.22 Which of the following is a possible figure for number of schools that scored 20 points or more but less than 80 points in each of the five parameters?
(a) 38 (b) 36
(c) 42 (d) None of these

Q.23 In D-segment car market, only three competitors (Honda, Toyota, and Hyundai) exist. Last year, the sales of Toyota were 10% more than Honda. In a year, both the firms Honda and Toyota increased their respective sales by 20%. This year, the sales of Toyota are five times that of Hyundai. How much were the sales of the firm Hyundai last year, if the total sales remained constant over the two years period?
(a) 46% (b) 25% (c) 40.33% (d) 38.67%

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Q.24 Chintu is given a quadratic equation $ax^2 + bx + c = 0$ and is asked to make another quadratic equation from this with $a = 1$. In addition, one root of the second quadratic equation is same as one of the roots of the first equation but opposite in sign and the other root of the second equation is two times the second root of the first equation. Find the percentage change in the constant term of the second equation as compared to the first equation?

(a) 200% increase (b) 300% decrease
(c) 400% increase (d) 100% decrease

Q.25 There are two investment schemes offered by two different financial institutions X and Y as follows:
Scheme X: Rate of interest = 10% per annum and period of maturity = 2 years
Amount is returned with compound interest compounded annually.
Scheme Y: Rate of interest = 11% per annum amount is returned with simple interest.
For what duration, should I invest ₹25,000 in scheme Y, so as to get same amount in return if I had invested in scheme X?

(a) $\frac{21}{11}$ Years (b) $\frac{31}{11}$ Years
(c) $\frac{19}{7}$ Years (d) $\frac{19}{7}$ Years

Direction for Questions 26 to 28: Read the passage below and solve the questions based on it.

Given below is the conversion rate of some of the currencies with respect to rupees.

	USD	Pound	Euro	Canadian dollar	Australian dollar
Rate of 1 unit	₹44	₹66	₹54	₹32	₹36

Q.26 Mr Kumar Kalyan Prasad Sinha is having two Pounds with him. How many Euros can he purchase for two Pounds?

(a) 2.33 (b) 2.44 (c) 2.55 (d) 2.66

Q.27 Value of USD is expected to rise by 10% against Rupee next year, whereas value of Rupee is expected to decline by 10% against Canadian Dollar next year. How many USD can be purchased for 5 Canadian Dollars?

(a) 3.67 (b) 2.97
(c) 4.27 (d) None of these

Q.28 Mrs Sanjila Sinha has ₹1000 with her and her husband Mr Kumar Kalyan Prasad Sinha has ₹2000 with him. They want to purchase maximum number of currency notes given above, but they are not ready to combine their money. What will be the maximum number of currency notes, as given above, other than rupees which they can have?

(a) 62 (b) 94 (c) 93 (d) 32

Direction for Questions 29 and 30: Read the passage below and solve the questions based on it.

The table given below shows the break-up of the percentage of people of different age groups frequenting bars in 4 different metro cities viz., Delhi, Hyderabad, Bangalore, and Patna in the year 2002.

	Percentage break-up for age groups (Years) in 2002						
Cities	Up to 15	15–20	20–25	25–30	30–35	35–40	Above 40
Delhi	8	13	24	21	11	17	6
Hyderabad	3	8	35	23	10	16	5
Bangalore	4	21	27	11	8	14	15
Patna	1	7	43	32	9	5	3

Q.29 Which city has the maximum number of people upto 25 years of age frequenting bars?

(a) Bangalore (b) Patna
(c) Both (a) and (b) (d) Cannot be determined

Q.30 Which age group contributes to the maximum number of people frequenting bars in the given cities?

(a) 20–25 (b) 25–30
(c) 30–35 (d) Cannot be determined

True/False

Q.1 Length of a rectangle is increased by 30% and its breadth is increased by 10%. Perimeter of this rectangle can increase by 25%.

State whether the above statement is true or false.

Q.2 Length of a rectangle is increased by 30% and its breadth is increased by 10%. Perimeter of this rectangle can increase by 32%.

State whether the above statement is true or false.

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Q.3 Length of a rectangle is increased by 40% and its breadth is increased by 20%. Percentage increase in the perimeter of the rectangle over the previous perimeter depends upon the dimension of the rectangle.

State whether the above statement is true or false.

Q.4 Length of a rectangle is increased by 20% and its breadth is increased by 10%. Percentage increase in the area of the rectangle over the previous area depends upon the dimension of the rectangle.

State whether the above statement is true or false.

Answers

Warm Up

1. (d)	**2.** (d)	**3.** (c)	**4.** (a)	**5.** (c)	**6.** (d)	**7.** (c)	**8.** (d)	**9.** (d)	**10.** (d)
11. (c)	**12.** (b)	**13.** (d)	**14.** (d)	**15.** (a)	**16.** (d)	**17.** (c)	**18.** (a)	**19.** (c)	**20.** (b)
21. (b)	**22.** (d)	**23.** (b)							

Foundation

1. (b)	**2.** (b)	**3.** (b)	**4.** (c)	**5.** (a)	**6.** (a)	**7.** (b)	**8.** (a)	**9.** (c)	**10.** (c)
11. (d)	**12.** (b)	**13.** (c)	**14.** (b)	**15.** (c)	**16.** (d)	**17.** (c)	**18.** (d)	**19.** (b)	**20.** (b)
21. (a)	**22.** (d)	**23.** (c)	**24.** (b)	**25.** (d)	**26.** (d)	**27.** (a)	**28.** (b)	**29.** (b)	**30.** (b)
31. (a)	**32.** (b)	**33.** (d)	**34.** (c)	**35.** (b)	**36.** (d)	**37.** (c)	**38.** (d)	**39.** (d)	**40.** (b)
41. (d)	**42.** (c)	**43.** (c)	**44.** (b)	**45.** (d)	**46.** (b)	**47.** (a)	**48.** (c)	**49.** (c)	**50.** (b)

Moderate

1. (c)	**2.** (b)	**3.** (a)	**4.** (d)	**5.** (c)	**6.** (d)	**7.** (c)	**8.** (d)	**9.** (d)	**10.** (a)
11. (c)	**12.** (b)	**13.** (c)	**14.** (b)	**15.** (b)	**16.** (d)	**17.** (a)	**18.** (b)	**19.** (a)	**20.** (c)
21. (c)	**22.** (d)	**23.** (a)	**24.** (b)	**25.** (b)	**26.** (a)	**27.** (b)	**28.** (b)	**29.** (b)	**30.** (a)
31. (c)	**32.** (a)	**33.** (c)	**34.** (a)	**35.** (a)	**36.** (c)	**37.** (b)	**38.** (b)	**39.** (d)	**40.** (b)
41. (a)	**42.** (b)	**43.** (d)	**44.** (b)	**45.** (b)					

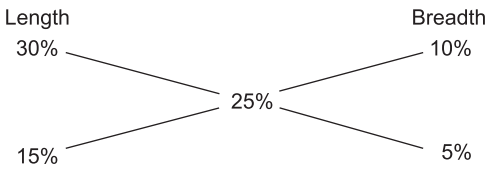
Advanced

1. (b)	**2.** (c)	**3.** (c)	**4.** (b)	**5.** (b)	**6.** (b)	**7.** (b)	**8.** (d)	**9.** (c)	**10.** (c)
11. (a)	**12.** (d)	**13.** (d)	**14.** (c)	**15.** (b)	**16.** (c)	**17.** (b)	**18.** (d)	**19.** (c)	**20.** (d)
21. (d)	**22.** (d)	**23.** (b)	**24.** (b)	**25.** (a)	**26.** (b)	**27.** (a)	**28.** (c)	**29.** (d)	**30.** (a)

True/False

1. True

Increase in perimeter has to be in between 10% and 30%. Hence, 25% increase in perimeter is possible. Actually it is a simple question of alligation. Ratio between length and breadth of rectangle should be 3:1.



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Hence, the ratio between length and breadth
= 15:5 = 3:1

2. False

Taking a cue from the above problem, 32% increase in perimeter is not possible.

3. True

Percentage increase in the perimeter of rectangle can be any value between 20% and 40%, and it depends upon the dimension of the rectangle (i.e., the length and breadth).

4. False

Percentage increase in the area of rectangle does not depend upon the dimension of the rectangle and it remains constant for all the values of length and breadth. In this case, percentage increase in the area = 32%.

Hints and Solutions

Warm Up

1. $\frac{4}{7} = 4 \times \frac{1}{7} = 4 \times 14.28 = 57.12\%$

2. $\frac{50}{100} \times \frac{40}{100} \times 1200 = 240$

3. $\frac{a}{100} \times a + \frac{b}{100} \times b = \frac{2}{100} ab$

$a^2 + b^2 = 2ab$ and $(a-b)^2 = 0$

$a = b$

4. $x - 0.34x = Nx$

or, $0.66x = Nx$

$N = 0.66$

5. $N - 0.48\,N = 0.52\,N$

6. $\frac{900}{7200} = \frac{1}{8} = 12.5\%$

7. $0.61x - 0.42x = 38$

or, $0.19x = 38$

$x = 200$

8. Let length = 100, breadth = 100

Hence, area = 10,000

Now, length = 130, breadth = 75

Hence, area = 130 × 75 = 9750

Required percentage $\frac{250}{10{,}000} \times 100$

2.5 % decrease

9. Let number = 100

1st 20% increment, number becomes = 120

2nd 20% increment, number becomes = 144

10. Let 3rd number = 100

1st number = 30 and 2nd number = 40

Hence, 1st is 75% of 2nd number

11. Let 3rd number = 100

1st number = 85

2nd number = 80

Required percentage = $\frac{85}{80} \times 100 = \frac{425}{5} = 106\frac{1}{4}\%$

12. Alcohol in solution = 20% of 50 L = 10 L

Concentration of alcohol in new solution

$= \frac{16}{40} \times 100 = 16\frac{2}{3}\%$

13. Assume that he had initially ₹100

After spending on food, money left = ₹75

After spending 50% of ₹75, he had ₹37.5

Money he had initially $= \frac{100}{37.5} \times 37.5 \times 10 = ₹1000$

14. Monthly expenditure = 70% of 4000 = ₹2800

Expenditure in a year = ₹2800 × 12 = ₹33600

15. Let initial amount = ₹100

Money left = 100 − (20 + 10 + 9 + 7)

= ₹(100 − 46)

= ₹54

Money he had $\frac{100}{54} \times 2700 = ₹5000$

16. Let Income = ₹100

Expenditure = ₹60

Savings = ₹40

Net Income = ₹140

New Expenditure = ₹84

New Savings = ₹56

Require increment = $\frac{16}{40} \times 100 = 40\%$

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17. Require increment

$$\frac{3-\frac{1}{2}}{\frac{18+2}{9}}=\frac{\frac{5}{2}}{\frac{20}{9}}=\frac{5}{2}\cdot\frac{9}{20}=\frac{9}{8}$$

18. Since, we have to find the minimum percentage, it can be possible that no one has failed in both the subjects.

19. It is better to approach this question using options.

Let us start with option (a) 20,000

Days	Start with	Sold on that day	Remaining left after selling	Spoiled during the night	Remaining on that day
On day 1	20,000	10,000	10,000	1000	9000
On day 2	9000	4500	4500	450	4050
On day 3	4050	2025	2025	202.5	1822.5
Total Spoilt in three days				1652.5	

It is given in the question that total number of spoilt apples in three days = 1983. Obviously option (a) is NOT the answer.

To get the answer now, we can use unitary method now:

When total spoilt in three days = 1652.5 apples, then number of apples on day 1 = 20,000.

Hence when total spoilt in three days = 1983 apples, then number of apples on day 1 $= \frac{1983}{1652.5}\times 20{,}000 = 24{,}000.$

Hence option (c) is the answer.

20. $\text{SI}=\left(\frac{7}{3}-1\right)=\frac{4}{3}$

$\text{SI}=\left(\frac{7}{3}-1\right)=\frac{4}{3}$

$R=\frac{40}{3}=13\frac{1}{3}\%$

21. $S\left(\frac{P\times x\times 2}{100}\right)=\frac{P\times 2x\times(y+2)}{100}$

$y=3$

22. $\frac{x\times 8\times 2}{100}+\frac{x\times 10\times 3}{100}+\frac{x\times 6\times 3}{100}=12{,}800$

$0.64\,x=12{,}800$

$x=20{,}000$

23. $\text{CI}=P\left(1+\frac{20}{100}\right)^n=p\left(\frac{6}{5}\right)^n$

When $n=4$, only then, the condition given is satisfied.

FOUNDATION

1. The numerator has been increased by 25%. Hence, the denominator will also increase by 25%.

Value in percentage = 60 × 1.25 = 75 percentage

2. Numerator has become 10 times; hence, denominator will also become 10 times.

3. There will be no change in the measurement of angles. It is the size that will increase and not the slope or the angle.

4. Number of twins cases out of 100 = 5

Number of twins out of 100 children = 10

$$\text{Percentage of twins}=\frac{10}{100}\times 100=\frac{100}{11}=9.5\%$$

5. Price Quantity

$\uparrow\frac{1}{5}$ $\frac{1}{6}\downarrow$

Or, $\frac{1}{6}x=4$

$X=24$

Initially, price per kg $=\frac{120}{4}=₹5/\text{kg}$

6. Going through choices is the best way. Option (a) is the answer.

7. Let the purchase value be ₹x.

$$\therefore\frac{16}{100}\,x=₹1520.\ \therefore x=\frac{₹1520\times 100}{16}=₹9500$$

8. Salary of P and Q = x and salary of R = y, then

$\frac{8x}{100}-\frac{7x}{100}=74$ and $x=7400$

then donation by R = ₹585

then $\frac{9}{100}y=585$

$y=\frac{585\times 100}{9}=₹6500$

$\text{Required percentage}=\frac{1695}{21{,}800}\times 100=7.95\%$

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9. In 1 kg of fresh watermelon,

Solid Part	Water Part
10%	90%
2 kg	18 kg

When watermelon dries, theoretically, solid part remains constant, and only water part gets dried.

In dried watermelon,

Solid Part	Water Part
80%	20%
2 kg (This will remain fixed vis a vis Fresh watermelon)	?

Since 80% of dried watermelon = 2 kg, hence, 100% of dried watermelon $= \frac{100}{80} \times 2 = 2.5\text{kg}$

10. Weight of solution = (1 + 4) = 5 units

weight of sugar = 1 unit

weight of water after heating = 1.5 units

Required ration = 1:1.5 = 2:3

11. It can be done by either of the methods of compound interest or straight line method of solving questions.

6 gets increased by 1000% = final value = 6 + 60 = 66

60 gets increased by 1000% = final value = 66 + 660 = 726

12. Turnover in 1998 = ₹100 cr

Turnover in 1999 = ₹300 cr

Turnover in 2000 = ₹450 cr

13. $\frac{40}{60} = \frac{75-70}{70-x}$ and $\frac{2}{3} = \frac{5}{70-x}$

and $140 - 2x = 15$

$2x = 125$

$x = 62.5\%$

14. Original price = ₹100

Expenditure = ₹50 × 100 = ₹5000

Expenditure after rise = ₹5600

Then, $140 \times x = 5600$

$X = 40$

15. Hours × wage = earnings

Previously, $x \times y = xy$

$\frac{5}{4}x \times \frac{6}{5}y = \frac{3}{2}xy$

Increment $= \frac{\frac{3}{2}xy}{xy} \times 100 = \frac{\frac{1}{2}xy}{xy} \times 100 = 50\%$

16. Decrease in A's income = ₹15,000

Increase in B's income = ₹15,000

Required ratio $\frac{1500}{15,000} \times 100 = 100\%$

17. $0.18A + 0.15B + 0.19C = 0.17A + 0.17B + 0.17C$

$0.01A - 0.02B + 0.02C = 0$ (i)

$A - B = 500$ (ii)

$A - C = 3400$ (iii)

Solving equations (i), (ii), and (iii), we get

A = 5800, B = 5300, and C = 2400

A + B + C = 13,500

18. Number of candidates failed $= \frac{35}{100} \times 51,600 = 18,060$

Number of candidates passed with honours

$= \frac{15}{100} \times 33,540 = 5031$

Note: 15% of the remaining

Number of students failed to obtain honours but got passed = 28,509.

19. Let A has x guavas and B has y guavas, the

$x - \frac{1}{4}x = \left(y + \frac{1}{4}x\right) + 2$ (i)

and $y - \frac{7}{10}y = x + \frac{1}{10}y - 4$ (ii)

Solving equations (i) and (ii), we get

$x = 44$ and $y = 20$

20. Total score by Raghav in 12 tests = 600

It is given that he scored less than or equal to 40 in 4 tests and he did not score more than 60 in any test.

For minimum number of tests with more than 50, we maximize the score of rest of the tests, i.e., he scored 40 in 4 tests, which is equal to 160.

In remaining 8 tests, he scored 600 − 160 = 440 marks.

Now, out of remaining 8 tests, he must have scored more than 50 in at least 4 tests [(4 × 60 = 240 marks; he can score equal to 60, more than 60 is not possible.) + (4 × 50 = 200 marks)] = 440 marks. If he scores more than 50 in three tests only, then it will not be possible to get the sum of 440 marks in 8 tests.

21. Let n = number of components attached, V_n = speed of train when 'n' compartments are attached

$V_0 = 50$

$V_1 = 45$

$V_2 = 40.5$ and so on

At $n = 7$, we get the answer.

22. x = number of participants

total fund = $50x$

then, $\frac{3}{5}x$ participates contributed $40x$

Required average $\frac{10x}{\frac{2}{5}x} = \frac{50}{2} = ₹25$

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23. Let the original price = ₹100
Price after 1st year = ₹125
Price after 2nd year = ₹100
Hence, after 180 year, there will be no change in the price

24. Salary = ₹100
Savings = ₹10, expense = ₹90
Next year, expense = ₹108,
90% of salary = ₹108
Salary $= \frac{10}{90} \times 108 = ₹120$
Hence, 20% increase in salary.

25. 15% failed in both the subjects ⇒ 85% passed in at least one of the subjects.
So, percentage of students passing both the subjects
$= 80 + 70 - 85 = 65\%$
So, 65% = 390 ⇒ 100% = 600

26. Salary of Dheeraj = ₹100
Salary of Anil = ₹80
Salary of Vinit = ₹70
Required per cent $= \frac{10}{70} \times 100 = 14.28\%3$

27. Let adult males = x and adult females = y,
then $0.55y = 0.4x$.
$x = \frac{0.55y}{0.40} = \frac{11}{8}y$
Now, required percentage
$= \frac{0.4x + 0.55y}{x + y} = 46.32\%$

28. Sona spends = $\frac{3}{25}$S and Ellaraputtu spends = $\frac{3}{25}$S.
Let Ellaraputtu spends x% of her income.
Then x % of $E = \frac{3}{25}x$
or $x = 12\frac{S}{E}$

29. Number of students passed in Physics = 60% of 60 = 36
Number of students passed in Maths = 70% of 60 = 42
∴ Number of students failed in Physics = 60 – 36 = 24
∴ Number of students failed in Maths = 60 – 42 = 18

30. Number of voters on the voter's list = x
Number of voters participated $= x - \left(\frac{x}{10} + 60\right)$
$= \frac{10x - x - 600}{10} = \frac{9x - 600}{10}$
Then $\frac{47}{100}x + \left(\frac{47x}{100} + 380\right) + \left(\frac{9x - 600}{10}\right) = x$
We get $x = 6200$

31. Percentage of rainy days = 10%
Percentage of days having rainbows = 5%
Hence, the required percentage = 95%

32. $x + y = 180$ (i)
$x - 3y = 0$ (ii)
Solving the equations, we get $x = 135$ and $y = 45$.

33. Price Quantity
$\uparrow \frac{1}{5}$ $\downarrow \frac{1}{6}$
Hence, $\frac{x}{6} = 2$ or $x = 12$

34. $10 \times 100 = 1000$, 100 = number of visitors
Now, 7.5 × number of visitors = 1200
Number of visitors = 160

35. $6000 = 13{,}350\left(1 - \frac{125}{1000}\right)^n$
$\frac{600}{1335} = \left(\frac{7}{8}\right)^n$

36. $\frac{15}{100}.\frac{9x}{10} = 47.25$
$1x = 350$

38. $x - 200 = \frac{x}{2} \rightarrow x = 400$
A got 400 votes and B got 200 votes.
However, the information is given regarding invalid votes.
So, option (d) is the answer.

39. Population after 2000 = 32,44,800
Population after 2001 = 29,85,216
Population at the end of 2003 = 32,28,810

40. $\frac{\frac{30}{100}x + 15}{x + 30} \times 100 = 45$
$x = 10$

41. Money spend in buying things = ₹63,000
Money kept in bank = ₹27,000
Money he got after selling the things = ₹50,400
Money he received from bank = ₹35,707.5
Loss = ₹3892.5
Loss % = 4.32%

42. $\frac{378}{125} = 3 + \frac{3}{125} = 302.4\%$
Let original salary be ₹100.
And now going through the options, we get Option (c) as answer.

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43. Sugar in original solution $= \frac{4}{100} \times 6 = 0.24$

Required percentage $= \frac{0.24}{5} \times 100 = 4.8\%$

44. Let B's income = ₹x

A's income = ₹$\frac{3}{5}x$

And B's expenditure = ₹y

A's expenditure = ₹$\frac{7}{10}y$

Also, $\frac{3}{5}x = \frac{3}{4}.\frac{7}{10}y$

$$\frac{\text{A's savings}}{\text{B's savings}} = \frac{x-y}{\frac{3}{5}x - \frac{7}{10}y} = \frac{\frac{7}{8}y - y}{\frac{3}{5}.\frac{7}{8}y - \frac{7}{10}y} = \frac{\frac{-y}{8}}{\frac{21y}{40} - \frac{7}{10}y}$$

$= \frac{5}{25} \approx 1:5$

45. Number of Science book initially $= \frac{40}{100} \times 110{,}000$

$= 44000$

Number of books later on = 130,000

Number of Science books $= \frac{45}{100} \times 130{,}000 = 58{,}500$

Required number of books = (58,500 – 44,000)

= 14,500

46. 2 trained labourers = 5 untrained labourers

33 untrained labourers $= \frac{2}{5} \times 33$ trained labourers

= 13.2 trained labourers

Number of labour	Days	Hours	Works
13.2	15	12	1
x	11	9	1.5

$\frac{13.2 \times 15 \times 12}{1} = \frac{x \times 11 \times 9}{1.5}$

$x = 36$

47. y's income = ₹A

x's income = ₹$\frac{3}{4}$A

y's expenditure = ₹B

x's expenditure = ₹$\frac{4}{5}$B

Also, $\frac{3}{4} = \frac{9}{10}$.B

A = $\frac{6}{5}$B.

Required ratio $= \frac{\frac{3}{4}A - \frac{4}{5}B}{A - B} = 1:2$

48. Let she had ₹100

70 mangoes – ₹100 and 40 apples – ₹100

35 mangoes = ₹50

₹15 retained for auto fare

Remaining money = ₹35

Number of apples $= \frac{40}{100} \times 35 = 14$

50. Let L = 100, B = 100

A = 10,000

New length = 120, New breadth = 150

New area = 18,000

MODERATE

1. Let MP = ₹100

CP after 1st discount = ₹98

CP after 2nd discount = ₹94.08

But discount cannot be more than 100%

2. Proceeding the solution of 1, we get (b) as the answer.

3. Let B = ₹100, service tax = ₹10

Final Bill = ₹100, discount = ₹10

S% = 10% D% = 9.09

4. $\frac{2}{5}A = \frac{3}{5}B$

2A = 3 B

5. and **6.** Let number of oranges with Alok = x

Shazmi bought = $\frac{1}{4}x$

Sandeep = $\frac{1}{3}.\frac{3}{4}x = \frac{1}{4}x$

Siddharth bought = $\frac{1}{4}x$

$\frac{1}{4}x = 4$ and $x = 16$

7. Let the class has x students.

Number of students receiving grade B = $\frac{1}{3}x$

Number of students receiving B + = $\frac{1}{4}x$

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Number of students receiving B $- = \frac{1}{6}x$

Number of students failed $= \frac{1}{8}x$

Number of students receiving

$A = x - \left(\frac{1}{3}x + \frac{1}{4}x + \frac{1}{6}x + \frac{1}{8}x\right)$

$= x - \left(\frac{21x}{24}\right) = \frac{3x}{24}$

For $\left(\frac{3x}{24}\right)$ to be an integer, minimum value of $x = 24$.

8. Since we do not know the number of students in the class.

$\therefore$ Option (d) is the answer.

9. Let original price = ₹100

Number of units sold = 100

Turnover = ₹10000

Now, number of units sold = 80

Turnover = ₹12000

Price per unit $= \frac{1200}{80} =$ ₹150

10. Let ₹x is kept in bank.

After 3 years, money will become

$x\left(1 + \frac{15}{100}\right)^3 =$ ₹$1.521x$

Price of AC after 3 years = ₹$1.52\,x$

Hence, Ravi will gain.

11. The best way to solve the question is through options.

12. N = 22.1875% of M

$N = \frac{221{,}875}{10{,}00{,}000}$

Values of M and N should be integer. So, minimum value of M + N = 391

13. Number of students who can sing = 30

Number of students who cannot dance = 8

Maximum number of students who can neither sing nor dance = 8

14. Using Statement (ii), we can find the winner. In case of Statement (i), there may be two contestants with 49% valid votes each and remaining two getting 1% valid votes each.

15. Let total number of students appeared = x

Number of males students appeared $= \frac{9}{10}x$

Number of female students appeared $\frac{x}{10}$

Now, $\frac{3}{5} \cdot \frac{9}{10}x + \frac{4}{5} \cdot \frac{x}{10} = 1240$

$\frac{31x}{50} = 1240$

$x = 2000$

16. Hydrogen = 300 g

or $\frac{5}{6}$ of water vapor = 300 g

Water vapor = 360 g = 0.36 kg

17. Number of students taking only Science = 4

or 10% of total students = 4

or total students = 40

Required percentage $= \frac{7}{40} \times 100 = 17.5\%$

18. Total number of students = x

Number of boys $= \frac{3}{5}x$

Number of girls $= \frac{2}{5}x$

Number of girls scoring more than 40 marks

$= \frac{4}{5} \cdot \frac{2}{5}x \cdot \frac{8}{25}x$

Total number of students scoring more than 40 marks $\frac{3}{5}x$

Required fraction $= \frac{7}{25} \cdot \frac{5}{3} = \frac{7}{15}$

19. The best way of solving this question would be the method of elimination, i.e., going through the options.

Obviously, option (d) cannot be the answer, because if 500 people are voting against the resolution, then motion cannot be passed.

Checking option (a); if 400 people are voting against the resolution for the first time, then 500 are voting in favour. Now, in the second voting, persons voting against the resolution would be= 600, so persons voting in favour = 200.

In the first voting, resolution was passed by 100 votes, and in the second voting, resolution was defeated by 400 votes. And it is 300% more than 100.

20. Number of males $= \frac{11}{18} \times 7200 = 4400$

Number of males married $= \frac{40}{100} \times 4400 = 1760$

Number of females married = 1760

Required percentage $= \frac{1760}{2800} \times = 62\frac{6}{7}\%$

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21. Original salary = ₹100

Actual hiked salary = ₹150

Salary for three months = ₹450

Salary to be deducted = ₹90

Required percentage $= \frac{60}{150} \times 100 = 40\%$

22. Let total number of respondents = 100

People favoured Mondale = 60

People favoured Bush = 40

New number of people favouring Mondale = 58

New number of people favouring Bush = 42

Required per cent = 14%

23. Let the value of Index = 100

Value of Reliance share = 7

Value of HLL = 13

Value of Infosys = 15

Value of remaining = 65

New value of Reliance = 7.63

New value of HLL = 14.3

New value of Infosys = 15.60

New value of remaining = 68.47

Hence, required percentage $= \frac{3.47}{65} \times 100 = 5.34\%$

24. Let total donation raised = ₹x

Number of people in group = y

then $\frac{4}{5}x = 400y$

$x = 500\,y$

Now, let z = average donation per person from the new group

$\left(\frac{2y}{3}\right)z = \frac{x}{5}$

$z = 150$

25. Total employees = x

Female employees $= \frac{3}{5}x$; male employees $= \frac{2}{5}x$

Number of male employees who earn more than ₹25,000 $= \frac{3}{4} \times \frac{2}{5}x = \frac{3}{10}x$

Number of employees who earn more than ₹25,000 $= \frac{9}{20}x$

Number of female employees who earn more than ₹25,000 $= \frac{3}{20}x$

Required fraction $= \frac{\frac{3x}{5} - \frac{3x}{20}}{\frac{3}{5}x} = \frac{3}{4}$

26. The question is based upon the concept of successive discount.

Percentage pollution at the end of one day = 20% (1 − 0.8) = 4%

Percentage pollution at the end of two days = 20% (1 − 0.8) (1 − 0.8) = 4%

Similarly, you can go ahead now.

27. $M = 1.5E$ (i)

$E + M + S + H = 64$ (ii)

$S = H + 48$ (iii)

$\frac{3}{2}E + M + S + H = 70$ (iv)

Using all the equations, we get

$M + \frac{2M}{3} + S + S - 48 = 64$

$5M + 6S = 336$ (A)

Similarly, using equations (i), (ii), and (iv), we get another equation (B).

Solving both (A) and (B), we get option (b) as the answer.

28. Amount = x

Rate = r

$15 = \frac{x \times r \times 1}{100}$ or $rx = 1500$

$\frac{(x+100) \times r \times 1}{100} = [420 - (x+100)]$

Solving, we get $x = 300$ and $r = 5$

29. $(x + 2520) = x\left(1 + \frac{10}{100}\right)^2$, x = Amul's profit

$x = 12{,}000$

y = Binit's profit

$4200 = \frac{y \times 20 \times 1}{100}$

$y = 21{,}000$

Chand's profit = ₹9000

Ratio of their profit

= 12,000:210,00:9000

= 12:21:9 = 4:7:3

Chand's share $= \frac{3}{14} \times 70{,}000 =$ ₹15,000

31. Go through the options.

32. $x\left(1 + \frac{r}{100}\right)^2 = 4x\left(1 - \frac{r}{100}\right)^2$

$\left(1 + \frac{r}{100}\right) = 2\left(1 - \frac{r}{100}\right)$

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$$\frac{3r}{100}=1$$

$$r=\frac{100}{3}=33\frac{1}{3}\%$$

33. For Statement A:

Share of ATF in 2002–03 $=\frac{2.3}{104}\times100=2.32\%$

Hence, A is true

For Statement B:

Share of diesel in 2002–03 $=\frac{36}{104.2}\times100=34.55\%$

Share of diesel in 2003–04 $=\frac{36.6}{107.4}\times100=34.0\%$

34. Diesel % change $=\frac{36.6-36}{36}\times100=1.67\%$

Petrol % change $=\frac{7.9-7.6}{7.6}\times100=3.95\%$

LNG % change $=\frac{9.3-8.3}{8.3}\times100=12.05\%$

Crude oil % change $=\frac{11.9-11.7}{11.9}\times100=1.68\%$

ATF % change $=\frac{2.5-2.3}{2.3}\times100=8.7\%$

Hence, Statement A is true and Statement B is false.

∴ Option (a) is the answer.

35. It can be seen from the given passage that Statement A is true.

25% of diesel sales in 2003–04 $=\frac{1}{4}\times36.6=9.15$

Whereas LNG sales in 2003–04 is 9.3.

Hence, Statement B is false.

∴ Option (a) is the answer.

36. % increase in income tax from 1998–99 to 1999–2000 $=\frac{20460-18210}{18210}\times100=12.36\%$

% increase in service tax from 1998–99 to 1999–2000 $=\frac{240-220}{220}\times100=9.09\%$

Hence, Statement A is false.

% increase in service tax from 2001–02 to 2002–03 $=\frac{260-230}{230}\times100=13.04\%$

Hence, Statement B is true.

∴ Option (b) is the answer.

37. Ratio of income tax to corporate tax in:

1998–1999 $=\frac{18210}{22750}=0.80$

1999–2000 $=\frac{20460}{23460}=0.87$

Hence, Statement A is false.

Service tax as a percentage of expenditure tax in

2000–01 $=\frac{200}{340}\times100=58.82\%$

2001–02 $=\frac{230}{350}\times100=65.71\%$

Hence, Statement B is false.

∴ Option (d) is the answer.

39. Salary of Abhay = ₹10,000

Salary of Bhaskar = ₹12,500

Salary of Chitra = ₹12,500

Salary of Danny = ₹12,000

40. Chitra's new salary = ₹13,200

ADVANCED

1. $\frac{1}{5}\times160.4=32.8$

Production in 1999–2000 = (160.4 − 32.8) MT = 128

2. Wheat $=\frac{2.7}{18.9}\times100=14.28\%$

Cotton $=\frac{4.2}{14.7}\times100=28\%$

Coffee $=\frac{2.5}{12.3}\times100=20\%$

Tea $=\frac{2}{20.5}\times100=10\%$

3. Rice $=\frac{162.7}{160.4}>1$, Coffee $=\frac{12.3}{14.3}<1$

Wheat $=\frac{18.9}{16.2}>1$

Tea $=\frac{20.2}{22.5}<1$

Cotton $=\frac{14.7}{18.9}<1$

4. $6x^2+\frac{27x^2}{16}+\frac{3}{4}x^2=4860$

x = number of people at the beginning

$x=24$

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5. Required percentage $= \frac{357.5 - 310.8}{310} \times 100$

$= \frac{46.7}{310.8} \times 100 = 15\%$

6. Required distance $= \frac{18.9 \times 1000}{1.3 \times 95}$

7. Auto drivers M3 shift $= \frac{18.9 \times 1000}{7} = 2700$

Taxi drivers M2 shift $= \frac{240,200}{8} = 30025$

Taxi drivers M3 shift $= \frac{42,300}{6} = 7050$

Auto drivers M1 shift $= \frac{57,300}{9} = 6366.66$

11. January 2002 = ₹x

February 2002 = ₹$\frac{5}{4}x$

March 2002 = ₹x

April 2002 = ₹1.2 x

May 2002 = ₹$\frac{54x}{50}$

12. Female population in Patna in the previous year = 40,000

For Lucknow, $x + \frac{1}{8}x = 90,000$

$\frac{9x}{8} = 90,000$

$x = 80,000$

Male = 40,000 and Female = 40,000

Next year

Male = 40,000 and Female = 50,000

13. Male population of Patna in the previous year = 44,000

Male:Female = 11:10

In the next year, Male:Female = 11:10

$\frac{\text{Male}}{60,000} = \frac{11}{10}$, hence, Male = 66,000

Total population = 66,000 + 60,000 = 126,000

15. For Arpit Pharma:

Constituent A $= \frac{12}{100} \times 300 = 36$g

Constituent B $= \frac{15}{100} \times 300 = 45$g

Constitutent C $= \frac{20}{100} \times 300 = 60$g

Constituent D $= \frac{28}{100} \times 300 = 84$g

For Bittu Pharma:

Constituent A $= \frac{37}{100} \times 300 = 111$g

Constituent B = 45g

Constituent C = 39g

Constituent D = 45g

For Chintu Pharma:

Constituent A = 78g

Constituent B = 45g

Constituent C = 30g

Constituent D = 102g

The most effective constituent is B, and B is present equally in all three capsules.

Now, it is turn of A.

16. For Arpit's drug, side effect $= \frac{60}{100} = \frac{1}{2} = 0.5$

For Bittu's drug, side effect $= \frac{39}{105} = \frac{13}{35} = 0.37$

For Chintu's drug, side effect $= \frac{30}{90} \times \frac{1}{3} = 0.33$

17. Go through the options:

Option (d): 2

In 6 tests, we assume that he scored 50, which is equal to 300.

Now, the remaining score of 140 he scored in 2 tests, which is not possible because he did not score more than 60 in any of the tests.

Option (c): 3

In 5 tests, we assume that he scored $50 \times 5 = 250$.

$\therefore$ Remaining 190 he scored in 3 tests, which is again not possible.

Option (b): 4

In 4 tests, we assume that he scored $50 \times 4 = 200$

$\therefore$ Remaining 240 he scored in 4 tests, i.e., 60 in each, which is possible.

Hence, option (b) is the answer.

18. $\frac{3000}{1000} \times [60 + 39 + 30] = 3 \times 129 = 387$kg

19. Let population = 100
People read *Dainik Jagran* = 25
People read *Prabhat Khabar* = 20
People reading both = 8
People reading only *Dainik Jagran* = 17
People reading only *Prabhat Khabar* = 12
Required percentage = (5.1 + 4.8 + 4) = 13.9%

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23. For last year

Honda = 100 cars

Toyota = 110 cars

This year

Honda = 120 cars

Toyota = 132 cars

Hyundai = 26.4 cars

Last year, Hyundai = 68.4 cars

Required percentage $= \frac{68.4}{278.4} \times 100 = 25\%$

24. Let the equations be

$x^2 - 2x + 1 = 0$ (i)

and $x^2 - x - 2 = 0$ (ii)

Required percentage $= \frac{1-(-2)}{1} \times 100 = 300\%$

25. $25{,}000\left(1+\frac{10}{100}\right)^2 - 25{,}000 = \frac{25{,}000 \times 11 \times x}{100}$

$25{,}000\left[\left(\frac{11}{10}\right)^2 - 1\right] - \frac{25{,}000 \times 11 \times x}{100}$

$\left(\frac{11}{10}\right)^2 - 1 = \frac{11x}{100} \Rightarrow \left(\frac{121}{100} - 1\right) = \frac{11x}{100}$

$\frac{21}{100} = \frac{11x}{100}$ and $x = \frac{21}{11}$

26. 2 Pounds = ₹132

₹54 = 1 Euro

₹132 $= \frac{1}{54} \times 132$ Euro $= \left(\frac{66}{27}\right)$ Euro $= \frac{22}{9}$ Euro

= 2.44 Euro

27. 1 Canadian Dollar = ₹35.2

1 USD = ₹48.4

Required number $= \frac{35.2 \times 5}{48.4} = 3.67$

28. Required number of notes

$= \frac{1000}{32} + \frac{2000}{32} = 93.47 = 93$ notes

29. Data is in percentage and question is about numbers, so it cannot be determined.

30. Through observation

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CHAPTER 4

Simple Interest and Compound Interest

LEARNING OBJECTIVES

After completion of this chapter, the reader should be able to understand:

- What is simple interest and what is compound interest
- How interest is calculated in each of the cases
- Types of problems and how to solve the same

INTRODUCTION

Interest is defined as 'Time Value of Money'. It works under the basic principle that value of money is not fixed, and it will keep changing over the period of time. In this chapter, we are going to study two types of interest, namely simple interest (SI) and compound interest (CI).

SIMPLE INTEREST

In case of simple interest, the interest as well as the principal remains fixed for every compounding period.

COMPOUND INTEREST

In case of compound interest, the interest will be added to the initial principal after every compounding period. Hence, compound interest keeps on increasing after every compounding period.

Differences between Simple Interest and Compound Interest

	Simple interest	Compound interest
1.	Principal remains fixed for the whole period.	Principal keeps on increasing.
2.	Interest is not added to the principal.	Interest is added to the principal after every compounding period.
3.	Interest remains fixed.	Interest keeps on increasing.
4.	Amount follows simple interest.	Amount follows geometric progression.
5.	Interest for next year is calculated over the original principal.	Interest for next year is calculated over the last year's amount.

In Case of SI

For example, if the rate of interest = 10% and the principal = ₹1000, then:
Interest for 1st year = 10% of ₹1000 = ₹100
Interest for 2nd year = 10% of ₹1000 = ₹100
Interest for 3rd year = 10% of ₹1000 = ₹100
It can be seen that interest generated every year = ₹100

	Principal	Rate	Interest
1st Year	1000	10%	100
2nd Year	1000	10%	100
3rd Year	1000	10%	100

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In Case of CI

Principal of 1st year (initially) = P
Principal of 2nd year = P + interest of 1st year
Principal of 3rd year = P + interest of 1st year + interest of 2nd year
For example, if the rate of interest = 10% and the principal = ₹1000, then
Interest for 1st year = 10% of ₹1000 = ₹100

	Principal	Rate	Interest
1st Year	1000	10%	100
2nd Year	1000 + 100 = 1100	10%	110
3rd Year	1000 + 100 + 110 = 1210	10%	121

Expression for Simple Interest and Compound Interest

$$SI = \frac{\text{Principal} \times \text{Rate of Interest} \times \text{Time}}{100}$$

$$CI = \text{Principal} \times \left[1 + \frac{R}{100}\right]^N - \text{Principal}$$

Principal = Sum invested or lent
R = Rate of interest per annum
N = Number of years

It should be noted that the unit of rate of interest and time should be same. So, if rate of interest is 'per year', then time should also be in 'year'.

In case of CI, if the compounding is not done annually, then formula changes like the following:

1. Half yearly compounding: It means that interest is given after every 6 months. In this case, after every 6 months, interest will be added to the principal.

Rate of interest	Compounding period	Interest in 6 months (Half year)	Number of compounding period in a year
R% per year	Half yearly	R%/2	2 (12 months/6 months)

$$CI = \text{Principal} \times \left[1 + \frac{R/2}{100}\right]^{2N} - \text{Principal}$$

2. Quarterly compounding: It means that interest is given after every three months. In this case, after every three months, interest will be added to the principal.

Rate of interest	Compounding period	Interest in 3 months (Quarter year)	Number of compounding period in a year
R% per year	Quarterly	R%/4	4 (12 months/3 months)

$$CI = \text{Principal} \times \left[1 + \frac{R/4}{100}\right]^{4N} - \text{Principal}$$

Remember

1. If the rate of interest = R% per annum for both CI and SI, then the difference between CI and SI for 2 yr will be equal to (R% of R)% of principal = $\frac{R^2}{100}$% of principal.
   In the above case, R = 10%, so the difference between CI and SI for 2 yr is 1%.
2. If a sum doubles itself in n years at SI, then rate of interest = $\frac{100}{n}$.
3. At SI, if a sum of money amount to n times in t years, then rate of interest = $\frac{(n-1)}{T}100$ %.

Comparison between CI and SI

Assume two different sums are getting double at their respective rates of SI and CI in 5 yr. Following table gives us the mechanism of getting money n times in the above situation:

	After 5 yr	After 10 yr	After 15 yr	After 20 yr
At SI	2 times	3 times	4 times	5 times
At CI	2 times	4 times	8 times	16 times

It happens because in case of SI, the amount follows arithmetic progression, and in case of CI, the amount follows geometric progression.

Example 1 At SI, a sum of money amounts to twice its original value in 8 yr. What is the rate of interest?

Solution Rate of interest in this case = $\frac{100}{8} = 12.5\%$

Example 2 At SI, a sum of money amounts to three times its original value in 5 yr. What is the rate of interest?

Solution

Method 1 Equation Method
Assume principal = ₹100
According to the question, amount = three times the original value = ₹300
Hence, interest = ₹300 – ₹100 = ₹200
Since interest remains same in case of SI, interest obtained/year = $\frac{200}{5} = 40\%$

Method 2 Formula Method

Rate of interest = $\frac{(n-1)\times 100}{T}$ % = $\frac{(3-1)}{5}\times 100 = 40\%$

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Example 3 At CI, amount at the end of 1st year is ₹2400 and amount at the end of 2nd year is ₹2880. Find (a) the rate of interest, (b) principal, (c) interest obtained in the 3rd year, and (d) amount at the end of 3rd year.

Solution In case of CI, interest for next year is calculated over the last year's amount.
In this case, interest obtained in 2nd year = ₹2880 – ₹2400 = ₹480

(a) Hence, rate of interest = $\frac{480}{2400} \times 100 = 20\%$

(b) To calculate principal, we'll simply use the CI formula.
Amount = Principal $\times \left[1+\frac{R}{100}\right]^N$, or, $2400 = P\left(1 \times \frac{20}{100}\right)^1$.
Solving it we get, P = ₹2000
Hence, principal = ₹2000

Note: A common mistake that students do in this question is to subtract 20% of 2400 from 2400 to get principal and form the following equation: 2400 – 20% of 2400 = P. This is wrong.

(c) Interest obtained in 3rd year = 20% of ₹2880 = ₹576
(d) Amount at the end of 3rd year = ₹2880 × ₹576 = ₹3456

Example 4 The difference between SI and CI on a sum of money at the rate of 5% per annum for 2 yr is ₹25. What is the principal?

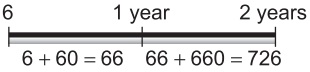
Solution Difference between CI and SI for 2 yr = $\frac{R^2}{100}$% of principal = $\frac{5^2}{100}$% of principal

$\Rightarrow \frac{1}{4}$% of principal = ₹25 $\Rightarrow$ 1% of principal = ₹100

$\therefore$ Principal = ₹10,000

Example 5 The rate of inflation is 1000%. What will be the cost of an article, which costs ₹6 today, 2 yr from now?

Solution 1000% inflation means an increase of 10 times.



Hence, price after 2 yr = ₹726

Example 6 SI on a certain sum at 5% per annum for 2 yr is ₹60. What is the CI on the same sum for the same period of time?

Solution

Method 1
Principal = $100 \times \frac{SI}{RT} = 100 \times \frac{60}{5 \times 2} =$ ₹600

$CI = P\left(1+\frac{R}{100}\right)^T - P$

$= 600\left(1+\frac{5}{100}\right)^2 - 600 = 600 \times \left(\frac{21}{20}\right)^2 - 600 = 661.5 - 600$
= ₹61.5

Method 2
The difference between CI and SI on ₹P for 2 yr at R% per annum = $R \times \frac{SI}{2 \times 100}$
Difference between CI and SI = $R \times \frac{SI}{2 \times 100} = 5 \times \frac{60}{2 \times 100}$ = 1.5
Hence, CI = SI + 1.5 = 60 + 1.5 = ₹61.5

Practice Exercises

Foundation

Q.1 A scheme invests ₹1000 at simple interest rate of 5% on the condition that interest will be added to the principal after 10 yr. In how many years will it amount to ₹2000?
(a) $16\frac{2}{3}$ yr (b) $16\frac{1}{4}$ yr (c) 16 yr (d) 11 yr

Q.2 A woman borrows ₹4000 from a bank at 7.50% CI. At the end of every year, she pays ₹1500 as a part repayment of loan and interest. How much does she still owe to the bank after three such installments?
(a) ₹123.25 (b) ₹125
(c) ₹400 (d) ₹469.18

Q.3 Anand borrowed a certain sum of money for 2 yr at 8% per annum on SI and immediately lent it to Ravish but at CI and gained ₹16. What amount did Anand borrow?
(a) ₹1600 (b) ₹2500
(c) ₹2400 (d) ₹1800

Q.4 The effective annual rate of interest corresponding to a CI rate of 8% per annum payable half yearly is:
(a) 8% (b) 8.01%
(c) 8.13% (d) 8.16%

Q.5 A money lender lent out ₹25,000 in two parts, one at 8% simple interest and the other at 8.5% simple interest.

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If the total annual income on the amount is ₹2031.25, then find the money lent at 8%.
(a) ₹12,500 (b) ₹6250
(c) ₹10,000 (d) ₹18,750

Q.6 Asmita invests an amount of ₹9535 at the rate of SI 4% per annum. For how many years did she invest the amount to double her sum?
a) 10 yr b) 25 yr c) 5 yr d) 4 yr

Q.7 The difference between the CI and SI on a certain sum of money for 2 yr at 15% per annum is ₹45. Find the sum.
(a) ₹7000 (b) ₹2750
(c) ₹2000 (d) ₹2250

Q.8 A man takes a loan of ₹10,000 and pays back ₹13,310 after 3 yr. What is the rate of CI?
(a) 8% (b) 9% (c) 10% (d) 11%

Q.9 The ratio of the amount for 2 yr under CI annually and for 1 yr under SI is 6:5. When the ratio of interest is same, find the value of the rate of interest.
(a) 12.5% (b) 18%
(c) 20% (d) 16.66%

Q.10 At the rate of 12%, the difference between SI and CI compounded annually on ₹5000 for 2 yr will be:
(a) ₹17.50 (b) ₹36 (c) ₹45 (d) ₹72

Q.11 In 3 yr, the difference between the simple and the compound interest on same principal amount at the rate of 20% is ₹48. Find the principal.
(a) ₹650 (b) ₹300 (c) ₹375 (d) ₹400

Q.12 I get an annual income of ₹688.25 from ₹10,000 invested partly at the rate 8% per annum and partly by 5% per annum at the SI. How much of my money is invested at the rate of 5%?
(a) ₹3725 (b) ₹4225
(c) ₹4800 (d) ₹5000

Q.13 A sum of money becomes eight times in 3 yr, if the rate is compounded annually. In how years the same amount at the same compound interest rate will become sixteen times?
(a) 6 yr (b) 4 yr (c) 8 yr (d) 5 yr

Q.14 The interests in the first two successive years were ₹400 and ₹420, respectively, when a sum of money is invested at CI. Find the sum.
(a) ₹8000 (b) ₹7500 (c) ₹8500 (d) ₹8200

Q.15 Anand borrows a certain sum from Manisha at a certain rate of SI for 2 yr. He lends this sum to Sunita at the same rate of interest but compounded annually for the same period, that is, 2 yr. At the end of 2 yr, he receives ₹2400 as compound interest, but paid ₹2000 as simple interest. What is the rate of interest?
(a) 40% (b) 30% (c) 20% (d) 10%

Q.16 If a sum on CI becomes three times in 4 yr, then with the same interest rate, the sum will become 81 times in:
(a) 12 yr (b) 18 yr (c) 16 yr (d) 14 yr

Q.17 A sum is invested for 3 yr compounded at 5%, 10%, and 20% per annum, respectively. In 3 yr, if the sum amounts to ₹1386, then find the sum.
(a) ₹1500 (b) ₹1400 (c) ₹1200 (d) ₹1100

Q.18 A sum of ₹6600 was taken as a loan. This is to be repaid in two annual installments. The rate of interest is 20%, which compounded annually. Find the value of each installment.
(a) ₹4320 (b) ₹2220 (c) ₹4400 (d) ₹4420

Q.19 If in a certain time period, ₹10,000 amount to ₹160,000 at CI. What is the amount in half at that time if principal is ₹10,000?
(a) ₹50,000 (b) ₹40,000
(c) ₹80,000 (d) ₹60,000

Q.20 What will be the amount if a sum of ₹10,000 is placed at CI for 3 yr while rate of interest for the first, second, and third years is 10%, 5%, and 2%, respectively?
(a) ₹11,781 (b) ₹11,244
(c) ₹11,231 (d) ₹11,658

Answers

Foundation

1. (a)	**2.** (a)	**3.** (b)	**4.** (d)	**5.** (d)	**6.** (b)	**7.** (c)	**8.** (c)	**9.** (c)	**10.** (d)
11. (c)	**12.** (a)	**13.** (b)	**14.** (a)	**15.** (a)	**16.** (c)	**17.** (d)	**18.** (a)	**19.** (b)	**20.** (a)

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Hints and Solutions

FOUNDATION

1. SI for 10 yr $= 1000 \times 5 \times \frac{10}{100} = 500$

Now, principal after 10 yr = 1500 and final amount = 2000

So, SI = 500

$500 = 1500 \times 5 \times \frac{T}{100}$

$T = \frac{500 \times 100}{1500 \times 5} = 6\frac{2}{3}$ yr

Hence, option (b) is the answer.

2. Amount remaining after:

1 yr $= 4000\left(1+\frac{7.5}{100}\right) - 1500 = 2800$

2 yr $= 2800\left(1+\frac{7.5}{100}\right) - 1500 = 1510$

3 yr $= 1510\left(1+\frac{7.5}{100}\right) - 1500 = 123.25$

Hence, option (b) is the answer.

3. Assume that Anand borrows ₹x

$x\left[\left(1+\frac{8}{100}\right)^2 - 1\right] - \frac{x \times 8 \times 2}{100} = 16$ or

$0.1664x - 0.16x = 16$

Therefore, $x = \frac{16}{0.0064} = 2500$

Hence, option (b) is the answer.

4. Rate of 8% per annum payable half yearly. So, effective rate = 4%

Effective annual rate $= 4 + 4 + \frac{4 \times 4}{100} = 8.16\%$

Hence, option (d) is the answer.

5. Let the amount lend at 8% be x. Then, amount lend at 8.5% be $(25{,}000 - x)$.

$x \times 8 \times \frac{1}{100} + (250{,}000 - x) \times 8.5 \times \frac{1}{100} = 2031.25$ or

$8x + 212{,}500 - 8.5x = 203{,}125$ or $-0.5x = -9375$

Therefore, $x = 18{,}750$

Hence, option (d) is the answer.

6. Let she invests for x years. A sum will be double when the interest is equal to the principal.

$9535 = 9535 \times 4 \times \frac{x}{100}$

$4x = 100$

Therefore, $x = 25$ yr

Hence, option (b) is the answer.

7. Difference between CI and SI for 2 yr $= \frac{R^2}{100}\%$ of principal $= \frac{15^2}{100}\%$ of principal

Or, 2.25% of principal = ₹45

Therefore, principal = ₹2000

Hence, option (c) is the answer.

8. Let the rate of CI be r.

Then, $10{,}000 \times \left(1+\frac{r}{100}\right)^3 = 13{,}310$

$\Rightarrow \left(1+\frac{r}{100}\right)^3 = \frac{1331}{1000} \Rightarrow 1+\frac{r}{100} = \frac{11}{10}$

$\Rightarrow \frac{r}{100} = \frac{1}{10} \Rightarrow r = 10\%$

Hence, option (c) is the answer.

9. $\frac{P\left(1+\frac{r}{100}\right)^2}{\left(1+\frac{Pr}{100}\right)} = \frac{6}{5} \Rightarrow \left(1+\frac{r}{100}\right) = \frac{6}{5} \Rightarrow r = 20\%$

Hence, option (c) is the answer.

10. $\left[5000\left(1+\frac{12}{100}\right)^2 - 5000\right] - \frac{5000 \times 12 \times 2}{100}$

$= 5000\left(\frac{28}{25} \times \frac{28}{25} - 1\right) - 1200$

$= 5000\left(\frac{784-625}{625}\right) - 1200 = ₹72$

Hence, option (d) is the answer.

11. Let the principal amount be 100.

Then, SI $= \frac{100 \times 20 \times 3}{100} = ₹60$ and CI $=$

$100\left(1+\frac{20}{100}\right)^3 - 100$

$= 100 \times \left(\frac{6}{3}\right)^3 - 100 = \frac{364}{5}$

$\therefore$ CI − SI $= \frac{364}{5} - 60 = \frac{64}{5}$

If difference is $\frac{64}{5}$, principal = 100

If difference is 48, principal $= \frac{100 \times 5}{64} \times 48 = ₹375$

Hence, option (c) is the answer.

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12. Let money invested at 5% be ₹x.

$$\therefore \frac{x\times1\times5}{100}+\frac{(1000-x)\times1\times8}{100}=688.25$$

$$\Rightarrow 5x-8x+8000=68{,}825$$

$$\Rightarrow 3x=11{,}175 \Rightarrow x=₹3725$$

Hence, option (a) is the answer.

13. Let the sum of money be ₹x.

Then, $8x=x\left(1+\frac{r}{100}\right)^3$

$$\Rightarrow\left(1+\frac{r}{100}\right)^3=(2)^3 \quad\Rightarrow\quad 1+\frac{r}{100}=2 \qquad \text{(i)}$$

Again, let the sum will become 16 times in n years.

Then, $16x=x\left(1+\frac{r}{100}\right)^n \Rightarrow 16=2^n \Rightarrow 2^4=2^n$ [From Eq. (i)]

$\Rightarrow n=4$ yr

Hence, option (b) is the answer.

14. Difference in interest between 2 yr = ₹420 − ₹400 = ₹20

So, rate % $=\frac{20}{400}\times100=5\%$

Given that ₹400 is the interest on the sum for 1st year.

$\therefore$ Sum $=100\times\frac{400}{5}\times1=₹8000$

Hence, option (a) is the answer.

15. Let the sum be x.

Simple interest on x for 2 yr = ₹2000

$$\text{SI}=\frac{PRT}{100} \Rightarrow 2000=\frac{x\times R\times2}{100} \Rightarrow xR=100{,}000 \qquad \text{(i)}$$

CI on x for 2 yr = 2400

$$P\left(1+\frac{R}{100}\right)T-P=2400 \quad x\left(1+\frac{R}{100}\right)2-x=2400$$

$$x\left(1+\frac{2R}{100}+\frac{2R}{10{,}000}\right)-x=2400$$

$$x\left(\frac{2R}{100}+\frac{2R}{10{,}000}\right)=2400$$

$$\frac{2xR}{100}+\frac{2xR}{10000}=2400 \qquad \text{(ii)}$$

Substituting the value of xR from (i) in (ii), we get

$2000+10R=2400 \quad 10R=400 \quad R=40\%$

Hence, option (a) is the answer.

16. Let the sum be P. The sum P becomes $3P$ in 4 yr on CI.

Method 1

$3P=P\left(1+\frac{R}{100}\right)^4 \Rightarrow 3=\left(1+\frac{R}{100}\right)^4$ Let the sum P becomes $81P$ in n years or

$$81P=P\left(1+\frac{R}{100}\right)^n$$

$$\Rightarrow 81=\left(1+\frac{R}{100}\right)^n \Rightarrow (3)^4=\left(1+\frac{R}{100}\right)^n$$

$$\Rightarrow\left(\left(1+\frac{R}{100}\right)^4\right)^4=\left(1+\frac{R}{100}\right)^n \Rightarrow\left(1+\frac{R}{100}\right)^{16}=\left(1+\frac{R}{100}\right)^n$$

$\therefore n=16$

i.e., the sum will become 81 times in 16 yr.

Method 2

We know that amount follows geometric progression in case of CI.

Thus, $P\xrightarrow{4\text{yr}}3P\xrightarrow{4\text{yr}}9P\xrightarrow{4\text{yr}}27P\xrightarrow{4\text{yr}}81P$

Hence, option (c) is the answer.

17. Amount $=1386=\left(1+\frac{5}{100}\right)\times\left(1+\frac{10}{100}\right)\times\left(1+\frac{20}{100}\right)$

or $1386=P\left(\frac{21}{20}\right)\times\left(\frac{11}{10}\right)\times\left(\frac{6}{5}\right)$

Solving it, we get $P=₹1000$

Hence, option (d) is the answer.

18. Present worth of ₹x due T years $=x/\left(1+\frac{R}{100}\right)^T$

Let x be the annual payment. Then, present worth of x due 1 yr hence + present worth of x due 2 yr hence = 6600

$$\left(\frac{x}{1+20{,}100}\right)+\left(\frac{x}{1+20{,}100}\right)^2=6600 \text{ or } \frac{x}{6/5}+\left(\frac{x}{6/5}\right)^2=6600$$

$$\frac{5x}{6}+\frac{25x}{36}=6600 \text{ or } \frac{55x}{36}=6600$$

$$\therefore x=6600\times\frac{36}{55}=4320$$

Hence, option (a) is the answer.

19. Let the rate of interest be $R\%$ per annum.

Assume that ₹10,000 amount to ₹160,000 in T years.

$$10{,}000\left(1+\frac{R}{100}\right)\text{T}=160{,}000$$

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$$\Rightarrow \left(1+\frac{R}{100}\right)^{T} = \frac{160{,}000}{10{,}000} = 16$$

$$\Rightarrow \left(1+\frac{R}{100}\right)^{T/2} = \sqrt{16} = 4$$

In $T/2$ yr, ₹10,000 amounts to $10{,}000\left(1+\frac{R}{100}\right)^{T/2}$

$= 10{,}000 \times 4 = 40{,}000$

Hence, option (b) is the answer.

20. When rates are different for different years, $A = P\left(1+\frac{R1}{100}\right)\times\left(1+\frac{R2}{100}\right)\times\left(1+\frac{R3}{100}\right)$

Amount after 3 yr $= 10{,}000 \times \left(1+\frac{2}{100}\right)\times\left(1+\frac{5}{100}\right)\times\left(1+\frac{10}{100}\right)$

$= 10{,}000 \times \left(\frac{102}{100}\right)\times\left(\frac{105}{100}\right)\times\left(\frac{110}{100}\right)$

$= 102 \times 105 \times \frac{11}{10} =$ ₹11,781

Hence, option (a) is the answer.

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CHAPTER

5

Profit, Loss, and Discount

LEARNING OBJECTIVES

After completion of this chapter, the reader should be able to understand:

- Basic terms involved
- Mechanism of profit generation
- Honest and dishonest ways
- Kinds of questions that are asked in the CAT
- Methods for solving questions

INTRODUCTION

As far as the CAT is concerned, this chapter is important because the application of the concept of profit, loss, and discount ranges from simple mathematical calculations in data interpretation (DI) to complex logical evaluations in quantitative ability (QA). This chapter can be considered as an extension of percentage with only difference in the terminology usage. The terms used in percentage are initial value, final value, percentage increase, etc., but the similar terms are used as cost price, selling price, profit percentage, etc. in profit, loss, and discount. Let us understand the different concepts and learn problem-solving methods for these concepts.

TERMINOLOGY

Cost Price

The price (amount) paid to purchase a product or the cost incurred in manufacturing a product is known as the cost price (CP) of that product.

Types of Cost

1. **Fixed cost:** It is that kind of cost that is fixed in all the cases.
2. **Variable cost:** Variable costs are costs that vary according to the number of units produced.
3. **Semivariable cost:** Semivariable costs are costs that are fixed in one particular strata, but the costs vary among the different stratas.

One good example of fixed cost, variable cost, and semi variable cost is the bill that we receive for the telephone connections at our home. A part of that bill, the rental, is fixed cost; and the rest of the part of the bill is calculated on the basis of the numbers of calls made.

Selling Price

The price at which a product sold is called selling price (SP) of the product.

Marked Price

The marked price or the mark-up price (MP) is the price that the shopkeeper/retailer fixes in the anticipation of some discount that they may be asked by a customer.

List Price

List price or the tag price, as the name suggests, is the price that is printed on the tag of the article.

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For our calculations related to the concept of PLD, till the moment nothing is stated in the questions, we would not see much difference between marked price and list price. This can be understood in the following example.

Suppose, the CP of an article is ₹100 and the shopkeeper wants to earn a profit of 20%, that is he wants to have a profit of ₹20. Now, he anticipates that a customer will ask for some discount and if he fixes the SP at ₹120, he would not be able to give any discount to the customer. So, he now marks up his price over ₹120 so that he can give some discount to his customer.

Remember that if nothing else is stated, then mark-up percentage is always calculated over CP.

$$MP \xrightarrow{\%\text{ decrease}} SP \xrightarrow{\%\text{Profit}\%\text{loss}} CP$$

Example 1 After selling an article at a discount of 50%, profit percentage obtained is 20%. What is the mark-up over CP?

Solution Let us assume that CP = ₹100.

So, SP = ₹120.

Now, after giving a discount of 50% over MP, ₹120 is the SP.

Using product stability ratio, 50%↓ = 100%↑

So, MP = 100%↑ over SP = ₹240

Hence, percentage mark-up = 140%

Alternatively, 0.5 MP = 1.2 CP

MP/CP = 1.2/0.5 = 2.4

So, percentage mark-up $= \left(\frac{MP}{CP} - 1\right) \times 100$

$= (2.4 - 1) \times 100 = 140\%$

Further, as we know

If SP > CP, then there will be a profit and if SP < CP, then there will be a loss.

SP − CP = Profit

CP − SP = Loss

Profit % = Profit/CP × 100

Loss % = Loss/CP × 100

Profit% or Loss % is always calculated upon CP, unless something else is mentioned.

It can be seen that whenever, there is a profit, then MP ≥ SP > CP.

However, when there is a loss, then we can not have some definite relationship between the above written three quantities.

Margin

When we calculate profit percentage as a percentage of SP, then it is known as margin.

Calculating CP/SP Profit % or Loss %

If we say that there is a profit of 20%, then

CP × 1.2 = SP

So, if CP = ₹120, then SP = ₹120 × 1.2 = ₹144

Or, if SP = ₹144, then CP = 144/1.2 = ₹120

(Needless to say that if there is a profit of 30%, then we will take 1.3 in the place of 1.2, and if there is a loss of 27%, then we will take 0.73 at the place of 1.2)

So, if there is a profit of R%, and CP = C, then,

$$CP = \frac{SP}{100 + R} \times 100$$

And in case of loss of R%, $CP = \frac{SP}{100 - R} \times 100$

Alternatively, we can use product stability ratio also to find out CP if SP is given.

Example 2 Two shopkeepers sell some article for ₹4000 each. A shopkeeper calculates his profit per cent on his CP and another calculates his profit per cent wrongly on SP. What is the difference in their actual profit if both claim to have a profit of 20%?

Solution

In first case,	In second case
SP = ₹4000	SP = ₹4000
Profit = 20% of CP	Profit = 20% of SP
CP = ₹3333.33	CP = ₹3200
Profit = ₹666.66	Profit = ₹800

So, the difference in profit = ₹133.33

PROFIT GENERATION

The purpose of studying profit, loss, and discount is to understand the mechanism of profit generation under different situations. Broadly, it can be seen in two ways:

Honest Ways

Questions that you will find here will be based either in terms of money or in terms of goods.

Type 1: Questions in Terms of Goods

In these types of questions, CP of a fixed number of goods is compared with the SP of another fixed number of goods. Let us see through an example.

Example 3 The CP of 30 articles is equal to the SP of 40 articles. What is the profit or loss percentage?

Solution To obtain the same amount of money, which was needed to purchase 30 articles, we need to sell 40 articles, which is more than what we have got for the same sum. It means that we need to arrange 10 more articles

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apart from the articles which we have purchased. So, there will be a loss.

Now, CP of 30 articles = SP of 40 articles

Or, CP/SP = 30/40 = 3/4

Or, 1 − CP/SP = 1 − 3/4 = 1/4

So, loss percentage

= (1 − CP/SP) × 100 = 1/4 × 100 = 25%

Alternatively,

CP of 30 articles = SP of 40 articles

= ₹120 (Assume)

So, CP of one article = ₹4

SP of one article = ₹3

Obviously, there is a loss of ₹1

Loss percentage = 1/4 × 100 = 25%

Or, we can use profit/loss percentage

$$= \frac{(\text{Good left/added})}{(\text{Good sold})} \times 100$$

Type 2: Questions in Terms of Money

Basically, questions in terms of money relate to the CP or SP. However, sometimes the results given below also come handy in solving problems.

Some Important Results

When SPs of two articles are same

i. First one is sold at a profit of x% and second one is sold at a profit of y%.
   Ratio of $CP_1:CP_2 = (100 + y):(100 + x)$
ii. First one is sold at a profit of x% and second one is sold at a loss of y%.
   Ratio of $CP_1:CP_2 = (100 - y):(100 + x)$
iii. First one is sold at a loss of x% and second one is sold at a loss of y%.
   Ratio of $CP_1:CP_2 = (100 - y):(100 - x)$
iv. First one is sold at a loss of x% and second one is sold at a profit of y%.
   Ratio of $CP_1:CP_2 = (100 + y):(100 - x)$

Example 4 SPs of two articles are same. One is sold at a loss of 20% and another one at a profit of 20%. What is the net loss/profit in the whole transaction?

Solution Assume that SPs of each of the article = ₹100

So, CP_1 (For the article which is sold at a loss of 20%) = 100/0.8 = ₹125

CP_2 (For the article which is sold at a profit of 20%) = 100/1.2 = ₹83.33

So, net CP = ₹125 + ₹83.33 = ₹208.33

As we can observe now that SP < CP, so, there is a loss.

Loss % = (8.33/208.33) × 100 = (1/25) × 100 = 4%

Alternatively, if SPs of two articles are same, and one is sold at a profit of x% and another is sold at a loss of x%, then in that case there will be a loss always. And loss percentage $= \frac{x^2}{100}\%$

Example 5 Two articles are sold at ₹12,000 each. One is sold at a profit of 20% and another one at a loss of 20%. What is the net loss?

Solution SP_1 = ₹12,000

CP_1 = ₹12,000/1.2 = ₹10,000

SP_2 = ₹12,000

CP_2 = ₹12,000/0.8 = ₹15,000

So, total CP = ₹25,000 and total SP = ₹24,000

So, loss = ₹1,000.

Alternatively, since there is a loss of 4%. So, for every ₹100 invested, ₹96 is coming back and ₹4 is lost. In our case, 96% = ₹24,000; so, 4% = ₹1,000.

Example 6 Due North Inc. is the number 1 idea developer company worldwide. One day, they sold Idea 1 to CL and Idea 2 to AMS at a profit percentage of 20% and 30%, respectively. If the sum of developing Idea 1 and Idea 2 is ₹25,000, what is the developing cost of Idea 1?

Solution The ratio of developing cost of Idea 1 and Idea 2 = 130:120

So, developing the cost of Idea 1 $= \frac{130}{250} \times 25{,}000 =$ ₹13,000

Dishonest Ways

Dishonest ways of earning profit includes adulterating or using faulty balance or both of these simultaneously.

Example 7 A shopkeeper sells his articles at his CP but uses a faulty balance which reads 1000 g for 800 g. What is his actual profit percentage?

Solution This question can be solved in several methods.

Method 1 Shopkeeper's net profit = 200 g

CP of 1000 g = SP of 800 g

So, profit % = 200/800 × 100 = 25%

Method 2 Since while selling, 800 g = 1 kg

So, 200 g = 1/4 kg

So, profit = 1/4 = 25%

Method 3 While selling, since the shopkeeper is branding 800 g as 1 kg, so 1000 g will be branded as 1000/800 = 1.25 kg and while purchasing, he has paid for just 1000 g.

So, net profit = (1.25 − 1) = 0.25 kg

So, profit percentage = 0.25/1 × 100 = 25%

Example 8 A shopkeeper marks up his goods by 40% and gives a discount of 10%. Apart from this, he uses a faulty balance also, which reads 1000 g for 800 g. What is his net profit percentage?

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Solution Let us assume his CP/1000 g = ₹100
So, his SP/kg (800 g) = ₹126
So, his CP/800 g = ₹80
So, profit = ₹46
So, profit percentage = 46/80 × 100 = 57.5%

Some More Examples

Example 9 A cloth store is offering 'Buy 3, get 1 free'. What is the net percentage discount being offered by the store?

Solution Suppose price of one article is ₹1. So, price of 4 articles = ₹4. Now, the whole scene can be understood as: we are paying only ₹3 at the place of ₹4. So, discount = ₹1
So, discount percentage = 1/4 (discount/total price) = 25%

Example 10 A shopkeeper sells his goods at its CP only. But still he manages to gain a profit of 40% because he has manipulated his weights. Find how many grams he is actually selling at the place of 1000 g.

Solution To earn a profit of 40%, shopkeeper needs to make 1.4 kg out of 1 kg. So, he will be selling $\frac{1000}{1.4}$ g = 728 g at the place of 1 kg.

Example 11 When an article is sold at 20% discount, SP is ₹24. What will be SP when the discount is 30%?

Solution 20% discount = 0.8 of MP = ₹24
So, MP = ₹30
So, SP when discount is 30% = ₹21

Practice Exercises

Warm Up

Q.1 An article is bought for ₹600 and sold for ₹750. What is the profit percentage?
(a) 20% (b) 25%
(c) 30% (d) None of these

Q.2 By selling a VCD player for ₹1950, I got a profit of 30%. At what price should I have sold it in order to get a profit of 40%?
(a) ₹2000 (b) ₹2100
(c) ₹2500 (d) None of these

Q.3 The profit obtained by selling a book for ₹56 is the same as the loss obtained by selling this book for ₹42. What is the cost price of the book?
(a) ₹40 (b) ₹50
(c) ₹49 (d) None of these

Q.4 A man bought a shirt at 3/4th of its list price and sold it at 50% above its list price. What is his percentage gain in the transaction?
(a) 50% (b) 70% (c) 80% (d) 100%

Q.5 By selling a CD for ₹150, a shop owner lost 1/16 of what it costs. What is the cost price of CD?
(a) ₹120 (b) ₹140
(c) ₹150 (d) ₹160

Q.6 Saurabh sold 250 cycles and had a gain equal to the selling price of 50 cycles. What is his profit percentage?
(a) 25% (b) 20%
(c) 10% (d) None of these

Q.7 If profit as a percentage of the selling price is 25%, then what is profit as a percentage of the cost price?
(a) 25% (b) $33\frac{1}{3}\%$
(c) 20% (d) 15%

Q.8 A man sold an article at a profit of 10%. If he had charged ₹45 more his profit percentage would have been 25%. What is the cost price?
(a) ₹200 (b) ₹300
(c) ₹250 (d) None of these

Q.9 Richa purchased a pen drive at 4/5th of its list price and sold it at 20% above the original list price. What is Richa's profit percentage?
(a) 25% (b) 40%
(c) 50% (d) None of these

Q.10 A man buys 200 oranges for ₹10. How many oranges a rupee can he sell so that his profit percentage is 25%?
(a) 10 (b) 14 (c) 16 (d) 20

Q.11 The CP of 24 apples is the same as the SP of 18 apples. Find the percentage gain.
(a) $12\frac{1}{2}\%$ (b) $14\frac{2}{7}\%$
(c) $16\frac{2}{3}\%$ (d) $33\frac{1}{3}\%$

Q.12 A manufacturer earns ₹20 on selling his product at 10% above his cost. What is the cost price?

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(a) ₹125 (b) ₹150
(c) ₹200 (d) None of these

Q.13 By selling an article for ₹360, the loss incurred is 10%. At what minimum price should he sell that article to avoid loss?
(a) ₹320 (b) ₹324 (c) ₹396 (d) ₹400

Q.14 Nitika buys a kinetic for ₹16,000 and sells it for ₹18,500. If she wants to gain 40%, how much more should she charge for the kinetic?
(a) ₹3800 (b) ₹4000
(c) ₹4200 (d) None of these

Q.15 The owner of Book Point does not get either profit or loss by selling 15 books for ₹225. How many books should he sell for the same amount to gain 25% profit?
(a) 10 (b) 11 (c) 12 (d) 13

Q.16 A reduction 20% in the price of the rice enables to get 6 kg more of rice for ₹120. What is the initial price of rice (₹/kg)?
(a) 4 (b) 5
(c) 6 (d) None of these

Q.17 CP of 12 apples is equal to the SP of 9 apples and the discount on 10 apples is equal to the profit on 5 apples. What is the percentage difference between the CP and SP of apples?
(a) 20% (b) 25%
(c) 16.66% (d) None of these

Q.18 A shopkeeper professes to sell his articles at CP but gives only 800 g in the place of 1000 g. What is his profit percentage?
(a) 20% (b) 25%
(c) 16.66% (d) 33.33%

Q.19 Cost price of 30 apples is same as selling price of 25 apples. What is the profit percentage or loss percentage?
(a) 20% loss (b) 20% profit
(c) 25% loss (d) 25% profit

Q.20 A shopkeeper professes to sell his articles at CP but gives only N g in the place of 1000 g. If his profit percentage is 100%, what is the value of N?
(a) 750 (b) 500
(c) 0 (d) Not possible

FOUNDATION

Q.1 A car is sold for ₹2400 at a profit of 20%. What is the CP?
(a) ₹2100 (b) ₹2000
(c) ₹1800 (d) ₹2200

Q.2 A car is sold for ₹2400 at a profit of 20% over SP. What is the CP?
(a) ₹2000 (b) ₹1920
(c) ₹1980 (d) ₹1800

Q.3 A car is sold for ₹2400 at a profit of 20% over SP. What is the actual profit percentage?
(a) 16.66% (b) 25%
(c) 21.21% (d) 14.28%

Q.4 Apples are purchased at 10 apples/₹How many apples should be sold for ₹1 to obtain a profit of 25%?
(a) 6 (b) 8 (c) 12 (d) 4

Q.5 Sharat sells some articles to Chandra at a profit of 20%. Chandra now sells this article to Mayank at a loss of 30% and Mayank sells this article at a profit of 20%. If CP of Chandra is ₹150, then what is the SP of Mayank?
(a) ₹105 (b) ₹87.5 (c) ₹125 (d) ₹126

Q.6 Petrol is purchased at ₹5/L and sold at 5 L/₹ What is the profit/loss %?
(a) Loss of 96% (b) No profit, no loss
(c) Profit of 2400% (d) None of these

Q.7 Despite giving a discount of 11.11% on the mark-up price, profit obtained is equal to 14.28%. What is the per cent mark-up over CP?
(a) 14.28% (b) 28.56%
(c) 25% (d) 50%

Q.8 A shopkeeper wants to earn a profit of 20% and at the same time, the minimum discount which he wants to offer is 25%. What should be the minimum percentage mark-up over CP?
(a) 60% (b) 42.5% (c) 62.5% (d) 35%

Q.9 In Q. No. 8, what should be the maximum percentage mark-up over CP?
(a) 50% (b) 100%
(c) 71.4% (d) Cannot be determined

Q.10 An article when sold at 10% discount on the marked price gave a profit of ₹70. What is the CP?
(a) ₹700 (b) ₹350
(c) ₹125 (d) Cannot be determined

Q.11 100 kg of gold is purchased for ₹1100. It is sold in such a way that after selling the whole quantity, the quantum of loss is equal to the amount obtained by selling 20 kg of gold. What is the selling price?
(a) ₹9.16 (b) ₹18.32 (c) ₹11.11 (d) ₹25

Q.12 A milkman buys two cows for ₹750. He sells first cow at a profit of 22% and the second cow at a loss of 8%.

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What is the SP of second cow if in the whole transaction there is no profit no loss?
(a) ₹312 (b) ₹506 (c) ₹484 (d) ₹532

Q.13 Two cars are sold for ₹24,000 each. One is sold at a profit of 20% and another at a loss of 20%. What is the net profit percentage/loss percentage, and amount of profit/loss??
(a) 4% profit, ₹2000 profit
(b) 4% loss, ₹2000 loss
(c) 1% profit, ₹500 profit
(d) 1% loss, ₹500 loss

Q.14 A supplier sells 20 pencils at the marked price of 16 pens to a retailer. The retailer, in turn, sells them at the marked price. What is the percentage profit or percentage loss of the retailer?
(a) Loss 25% (b) Profit 25%
(c) Loss 20% (d) Profit 20%

Q.15 Sum of the CP's of two cars is ₹1,00,000. First car is sold at a profit of 20% and second car is sold at a loss of 20%. However, their SPs are same. What is the CP of 1st car?
(a) ₹40,000 (b) ₹60,000
(c) ₹52,400 (d) ₹47,600

Q.16 Sum of CP's of two cows is ₹13,000. Both the cows are sold at a profit of 20% and 40%, respectively, with their SP's being the same. What is the difference of CPs of both the cows?
(a) ₹1000 (b) ₹2000 (c) ₹1500 (d) ₹2500

Q.17 Buy three, get one free. What is the percentage discount being offered here?
(a) 33.33% (b) 25% (c) 20% (d) 28.56%

Q.18 CP of 40 articles is equal to the SP of 30 articles. What is the profit/loss percentage?
(a) 25% profit (b) 33.33% profit
(c) 25% loss (d) 33.33% loss

Q.19 Due to a price hike of 20%, 4 kg less tea is available for ₹120. What is the original price of tea?
(a) ₹4/kg (b) ₹5/kg (c) ₹6/kg (d) ₹4.5/kg

Q.20 A shopkeeper sells his goods at its CP only. But he uses 750 g weight at the place of 1000 g weight for a kg. What is his net profit percentage?
(a) 25% (b) 20% (c) 16.66% (d) 33.33%

Q.21 A shopkeeper sells his goods at its CP only. But still he manages to gain a profit of 30% because he has manipulated his weights. Find out how many grams he is actually selling at the place of 1000 g.
(a) 700 (b) 769 (c) 800 (d) 820

Q.22 Which of the following discount options is better for a customer?
(i) Successive discounts of 20%, 30% and then pay a service tax of 10%.
(ii) Successive discounts of 30%, 20% and then pay a service tax of 10%.
(iii) Pay service tax of 10% first, then successive discounts of 20% and 30%.
(a) (i) or (ii)
(b) (iii)
(c) Either of these three
(d) (i) and (iii) or (ii) and (iii)

Q.23 A wholeseller supplies few chips to a retailer every year. Each chip costs ₹10,000 to the wholeseller. 5% of the chips are defective and they are to be replaced without charging anything extra. If the wholeseller still makes a profit of 20%, at what price is he selling it to retailer?
(a) ₹12,300 (b) ₹12,600
(c) ₹13,200 (d) None of these

Q.24 A seller calculated his intended selling price at 6% profit on the cost of a product. However, owing to some mistake while selling, the units and tens digits of the selling price got interchanged. This reduced the profit by ₹9 and profit percentage to 2.4%. What is the cost price of the product?
(a) ₹240 (b) ₹250 (c) ₹400 (d) ₹480

Q.25 A shopkeeper wrongly calculates his profit on SP and finds it to be 25%. What is the actual profit percentage?
(a) 20% (b) 33.33% (c) 40% (d) 25%

Q.26 There are two shopkeepers selling the same article at the same price for same quantity. One day, first shopkeeper offers a price discount of 25% for the same quantity, whereas second shopkeeper offers 25% more quantity for the same price. From a customer's point of view, which deal is better?
(a) First shopkeeper's deal
(b) Second shopkeeper's deal
(c) Both are equal
(d) Cannot be determined

Q.27 A shopkeeper sells two cows in such a way that CP of first cow is equal to the SP of second cow and SP of first cow is equal to the CP of second cow. Which of the following is final result after the whole transaction?
(a) There is a loss (b) There is a profit
(c) No profit, no loss (d) Cannot be determined

Q.28 When an article is sold for ₹180, a profit of 20% is made. At what price should the article be sold so that profit percentage is double?
(a) ₹200 (b) ₹210 (c) ₹192 (d) ₹240

Q.29 Vinit sells a car to Amit at a profit of 20%. Now, Amit sells this car to Vicky at a profit of 12%. And finally Vicky sells this car to Nishu at a loss of 21%. What is the sum of CPs of Amit and Nishu?
(a) ₹250 (b) ₹475
(c) ₹540 (d) Cannot be determined

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Q.30 Anoop sells a book to Mayank at a profit of 20% and Mayank sells this book to Siddharth at a profit of 25%. Now Siddharth sells this book at a loss of 10% to Shishir. At what percentage, loss should Shishir sell this book now so that his SP becomes equal to Anoop's CP?
(a) 36.68% (b) 25.92%
(c) 48.66 (d) Cannot be determined

Q.31 When an article is sold for ₹X, loss percentage is equal to L%. However, when the same article is sold for ₹Y, profit percentage is equal to P%. What is the CP of that article?
(a) 100 (P + L)/(Y − X) (b) (P + L)/(Y − X)
(c) 100 (Y − X)/(P + L) (d) (Y − X)/(P + L)

Q.32 A milkman professes to sell milk at its CP only. But still he is making a profit of 20% since he has mixed some amount of water in the milk. What is the percentage of milk in the mixture?
(a) 80% (b) 83.33% (c) 75% (d) 66.66%

Q.33 There is some profit when an article is sold for ₹720. However, when the same article is sold for ₹420, there is some loss. If the quantum of loss is two times the quantum of profit, find the cost price of the article.
(a) ₹620 (b) ₹700
(c) ₹520 (d) None of these

Q.34 While I was shopping in Kamla Nagar market for my baby girl Tatto, I came across two shopkeepers who were giving me a discount of same amount for a top. While first shopkeeper was offering a discount of 14.28%, second shopkeeper was offering a discount of 12.5% over their respective mark-ups. Whose cost price is higher?
(a) First shopkeeper
(b) Second shopkeeper
(c) Either of them depending upon their respective market price (MP)
(d) Cannot be determined

Q.35 A seller offers discounts on the basis of the number of articles purchased. He gives a discount of 10% when 5 articles are bought and a discount of 12.5% when 8 articles are bought. If the profit he makes in each case is the same, find the ratio of the marked price to the cost price of the article.
(a) 4:3 (a) 3:4
(c) 6:5 (d) None of these

Q.36 A shopkeeper purchases his goods at a discount of 10% on the list price. He normally gives a discount of 6% to his customers. If during a special scheme, he gets an additional discount of 2% on his purchase price, find his approximate profit percentage.
(a) 8.8% (b) 7.8%
(c) 6.6% (d) None of these

Q.37 A shopkeeper purchases a packet of 50 pencils at ₹10 per pencil. He sells a part of the packet at a profit of 30%. On the remaining part, he incurs a loss of 10%. If his overall profit on the whole packet is 10%, find the number of pencils he sold at profit.
(a) 25 (b) 30 (c) 20 (d) 15

Q.38 The costs of a pen and a book are ₹3 and ₹4 respectively. A student has to spend a sum of exactly ₹100 to buy pens and books, buying at least one pen and one book. How many combinations of books and pens can he buy?
(a) 6 (b) 8 (c) 12 (d) 10

Q.39 Anam bought a jute bag at 30% discount on the list price. He then sold it at a price which is 160% of the list price thereby making a profit of ₹81. What is the list price of the bag?
(a) ₹90 (b) ₹100 (c) ₹180 (d) ₹200

Q.40 What is the profit percentage of a dishonest cloth merchant who professes to sell his articles at CP but uses a scale which weighs less by 16.67%?
(a) 20 (b) 15
(c) 16 (d) None of these

Q.41 A shopkeeper calculated his profit on the marked price and finds it to be 30%. He forgets the fact that he gave a discount of 20%. What is his actual profit percentage?
(a) $14\frac{2}{7}\%$ (b) 20% (c) 25% (d) $28\frac{4}{7}\%$

Q.42 Santa Singh, the local fruit vendor, buys a certain number of oranges at ₹7 a dozen and equal number at ₹6 a dozen. He sells the oranges at ₹7.50 a dozen and makes a profit of ₹80. How many oranges does he buy?
(a) 20 dozens (b) 30 dozens
(c) 40 dozens (d) 80 dozens

Q.43 A milkman who wants to make a fast buck by using a false measure, measures only 900 ml instead of a L. If his cost price and selling price per litre of milk are ₹10 and ₹12, respectively, what is his daily profit and profit percentage if he sells 20 L in one day?
(a) ₹60, 20% (b) ₹60, $33\frac{1}{3}\%$
(c) ₹40, 25% (d) ₹40, $33\frac{1}{3}\%$

Q.44 Two lots of oranges with equal quantity, one costing ₹20 per dozen and the other costing ₹30 per dozen, are mixed together and the whole lot is sold at ₹24 per dozen. Then what is the profit or loss?
(a) 4% profit (b) 5.6% profit
(c) 4% loss (d) 5.6% loss

Q.45 A shopkeeper marks up the price of his product by 40%. If he increases the discount from 5% to 10%, the profit

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would decrease by ₹14. How much profit would he earn if he gives a discount of 20% on the marked price?
(a) ₹56 (b) ₹28 (c) ₹32 (d) ₹24

Q.46 Navneet is a bananawallah who buys a gross of bananas at ₹10 a dozen. During transport and storage, two dozen bananas get spoilt. If Navneet spends 10% of his outlay on transport costs and wants an overall profit of 20%, at what price must he sell a dozen bananas?
(a) ₹14.64 (b) ₹15.84 (c) ₹16.24 (d) ₹17.54

Q.47 Krishna sells his watch at a loss of 5%. If he had sold it for ₹56.25 more he would have gained 10%. What will be his gain or loss percentage if he sells it for ₹450?
(a) 10% (b) 15% (c) 20% (d) 30%

Q.48 A person sold his watch for ₹24 and got a percentage of profit equal to the cost price; then the cost price is:
(a) ₹145 (b) ₹18 (c) ₹22 (d) ₹20

Q.49 A merchant buying goods abroad at a discount of 33.33% on the catalogue price has to pay import duty of 20% on the net cost of goods, realize a profit of 25% on his outlay, what must he charge for an article priced ₹228 in the catalogue?
(a) 228
(b) 322
(c) Both (a) and (b)
(d) Cannot be determined

Q.50 A group of students prepared stuffed toys as part of their group activity. They spent ₹100 on velvet, ₹5 on thread and needle and ₹27 on miscellaneous items. They made 30 toys, 50% of which were purchased by force by some senior students causing them a loss of 50%. At what % profit should they sell the remaining toys so as to gain 50% on their total cost?
(a) 100% (b) 150%
(c) 75% (d) None of these

Moderate

Direction for Questions 1 to 5: Read the passage below and solve the questions based on it.

Patna-mart, sister concern of Wal-mart, is planning to establish a mechanism to give discount. It is actually a Khul Ja Simsim kind of game where if you win any round, you will get some discount specific to that round, and if you lose, you will be losing all the discounts so far won.

Now, Round 1 is having discount coupons of 2%.
Round 2 is having discount coupons of 4%.
Round 3 is having discount coupons of 6%.
And so on

There are infinite rounds and you can participate in all the rounds, but you can participate in any round only once, and before participating in round *N* (any particular round), you should have participated in Round 1: Round *N*-1 (all the rounds before that particular round.)

All the discounts won will be successive, for example if you win Round 1 and then Round 2 also, you will get a successive discount of 2% and then 4%. But again as the rule goes on, if you lose Round 3, you will be losing all the discount coupons won so far. But despite losing Round 3, you can participate in round 4 and hence.

Q.1 What is the minimum number of rounds to obtain maximum possible discount?
(a) 50 (b) 100
(c) Infinite (d) Cannot be determined

Q.2 What is the maximum possible discount?
(a) 1000% (b) 50%
(c) 100% (d) Infinite

Q.3 What is the minimum possible discount after 20 rounds provided that customer has won a minimum of 18 rounds?
(a) 100% (b) 50%
(c) 0% (d) None of these

Q.4 In the above question, what is the maximum possible discount?
(a) 100% (b) 50%
(c) 0% (d) None of these

Q.5 In a bid to attract more customers, Patna-mart is on a mass-compaigning move. With a new punch line, 'You can always check-out, but you can never leave', they are organizing several contact programmes across the city. As a result, there is an increase in sale of 10% and 20%, respectively, on weekdays and weekends (Saturday and Sunday). If each of the articles sold by Patna-mart is sold at a profit of 40%, what is the maximum possible percentage growth in the profit of Patna-mart?
(a) 32% (b) 72%
(c) Minimum 50% (d) Cannot be determined

Direction for Questions 6 to 8: Read the passage below and solve the questions based on it.

Had the CP been 10% less and SP been 10% more, profit % would have been double than that of earlier case.

Q.6 What is the actual CP?
(a) ₹70 (b) ₹100
(c) ₹140 (d) Cannot be determined

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Q.7 What is the profit % in original case?
(a) 20% (b) 28.56%
(c) 14.28% (d) Cannot be determined

Q.8 How many values for actual SP is/are possible?
(a) 1 (b) 0 (c) 3 (d) Infinite

Direction for Questions 9 and 10: Read the passage below and solve the questions based on it.

King Nandan is the king of Divyagarh. One day he decides to sell all the animals living in his big kingdom. Now, there are five states in his kingdom and the only animals which his kingdom is having are horses, cows, and sheeps. King Nandan sells all the animals to eight neighbouring states. Each of the neighbouring states purchases same number of animals at the following rate.

Horse – ₹1700 each
Cow – ₹200 each
Sheep – ₹200 each
King Nandan receives ₹28,500 in all?

Q.9 What can be the maximum number of animals?
(a) 80 (b) 130
(c) 160 (d) None of these

Q.10 What are the respective numbers of horses, cows, and sheeps in any state?
(a) (3, 109, 8) (b) (1, 37, 3)
(c) (2, 74, 6) (d) Cannot be determined

Q.11 Sunny marks up his goods by 40% and gives a discount of 10%. Apart from this, he uses a faulty balance which reads 800 g for 1000 g. What is his net profit/loss percentage?
(a) 8% (b) 57.2%
(c) 37.6% (d) None of these

Q.12 A shopkeeper marks up his goods by 20% and then gives a discount of 20%. Besides he cheats both his supplier and customer by 100 g, that is, he takes 1100 g from his supplier and sells only 900 g to his customer. What is his net profit percentage?
(a) 24.5% (b) 17.33% (c) 25% (d) 32.5%

Q.13 In Question number 12, shopkeeper takes a discount of 20% from his supplier which he disregards while preparing his mark-up. What is his net profit percentage?
(a) 37.5% (b) 46.66% (c) 52.33% (d) 58.33%

Q.14 In addition to all the data of Question number 13, now he adulterates his goods in the ratio of 1:5 (adulterant: original). Assuming that adulterant comes free of cost, what is the net profit percentage?
(a) 62.5% (b) 66.66%
(c) 72.08% (d) None of these

Q.15 In Question number 14, if the cost of adulterant is 20% the cost of original substance, what is the net profit percentage?
(a) 77.77% (b) 81.08%
(c) 86.75% (d) None of these

Q.16 Some mangoes are purchased at the rate of 8 mangoes/₹ and some more mangoes at the rate of 6 mangoes/₹, investment being equal in both the cases. Now, the whole quantity is sold at the rate of 7 mangoes/₹ What is the net percentage profit/loss?
(a) 0.6% profit (b) 0.6% loss
(c) 1.2% loss (d) No profit/no loss

Q.17 Some olives are purchased at the rate of 8 olives/₹ and same number of olives at the rate of 6 olives/₹. Now, the whole quantity is sold at the rate of 7 olives/₹. What is the net percentage profit/loss?
(a) 0.6% profit (b) 0.6% loss
(c) 1.2% loss (d) No profit/no loss

Direction for Questions 18 to 22: Read the passage below and solve the questions based on it.

A shop has the following discount layers

First layer – Buy 1, get 2 free
Second layer – Buy 2, get 3 free
Third layer – Buy 3, get 4 free
And so on there are infinite layers on the same pattern.

Q.18 For how many of these discount layers, discount percentage offered is more than 50%?
(a) 4 (b) 27
(c) 99 (d) None of these

Q.19 The shopkeeper wishes to lessen the percentage of discount by clubbing any two layers. Which two layers should be clubbed so that discount is minimized by maximum percentage?
(a) 1st + 2nd (b) 2nd + 3rd
(c) 3rd + 4th (d) None of these

Q.20 What is the net percentage discount if we club first three layers?
(a) 43.33% (b) 46.66%
(c) 53.33% (d) None of these

Q.21 A new condition has been introduced by the shop that anybody can avail one layer discount only once (i.e., if you are planning to purchase 6 articles, then you can not purchase 2 articles in first layer and get rest four free. Since you have availed first layer discount, you can now ask for only other layer discounts.) Sona is purchasing 8 articles from the store. What is the maximum possible discount that she can avail?
(a) 72% (b) 26%
(c) 38% (d) None of these

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Q.22 In Q.21, what is the minimum percentage discount that she can avail?
(a) 0% (b) 22%
(c) 44% (d) None of these

Direction for Questions 23 and 24: Read the passage below and solve the questions based on it.

To produce a certain type of engine, two components are imported— Component P is imported from USA and component Q is imported from Japan. P and Q account for 30% and 40% of the production cost of the engine. A profit of 25% is to be made but discount of 10% is to be offered to customers for promotional reasons.

Q.23 If the price of US Dollar appreciates by 20% and the price of Japanese Yen appreciates by 30%, what percentage above the original CP should the product be marked?
(a) 33.3% (b) 48.5%
(c) 63.88% (d) 71.4%

Q.24 US Dollar appreciates by 4% and Japanese Yen by 32%. If marked price remains same, then what is the change in profit percentage?
(a) 4.38% (b) 3.33%
(c) 6.25% (d) 7.69%

Direction for Questions 25 and 26: Read the passage below and solve the questions based on it.

A shopkeeper sells two types of custom confiscated goods, A and B, both at same price of ₹120, whereas they cost him ₹90 and ₹160, respectively. Sale for A started from 1 for the first day and then went on increasing every day by 4 units, whereas sale of article B was 450 on the first day and went on decreasing every day by 6 units.

Q.25 Shopkeeper has planned the things in such a way that he starts earning profit exactly on diwali which is after N days. What is the value of N?
(a) 103 (b) 105 (c) 107 (d) 109

Q.26 On which day will he earn profit for the first time on day-to-day basis?
(a) 49 (b) 51 (c) 53 (d) 55

Q.27 In the year 2002, the cost price of an item was 90% of the selling price while in 2003, the cost price was 95% of the selling price. If the gross profit remains the same for both the years, what is the percentage increase in the cost price of the item from 2002 to 2003?
(a) 60% (b) 80%
(c) 100% (d) None of these

Direction for Questions 28 to 33: Read the passage below and solve the questions based on it.

Mittal has recently acquired four companies viz., Bank of Bozoland (BOB), My Own Bank (MOB), Zany Obliterated Bank (ZOB), and Dogmatically Obscure Bank (DOB). He noticed that the sales of DOB are half that of BOB, whereas the profits of DOB are double that of BOB. The expenses of ZOB are ₹3 crores less than that of DOB, whereas the profits of MOB is ₹1 crore less than that of ZOB. The expenses of BOB are three times that of DOB. It is also known that the sales of ZOB are ₹15 crore or one-fourth that of MOB. All figures are for 1992-1993. An insider further informs Mittal that the sales of DOB are ₹10 crores more than that of ZOB and the expenses of BOB are 90% of its own sales.

Sales – Expenses = Profit

Q.28 The total sales of all the four companies is (₹crores):
(a) 200 (b) 150 (c) 125 (d) 160

Q.29 The expenses of BOB exceed that of ZOB by (₹crores):
(a) 31 (b) 43 (c) 33 (d) 62

Q.30 Which bank had the maximum profit?
(a) BOB (b) MOB (c) ZOB (d) DOB

Q.31 Total profits of the four companies (₹crores) is:
(a) 51 (b) 52
(c) 53 (d) None of these

Q.32 What is the total expense (₹crores) of all the four banks together?
(a) 110 (b) 130 (c) 150 (d) 160

Q.33 The profits of MOB form what percentage of the expenses of ZOB?
(a) 16% (b) 20% (c) 15% (d) 23%

Q.34 By selling a watch at a profit of 10 per cent, a man got ₹15 more than half its price. What is the price of the watch?
(a) 10 (b) 15 (c) 25 (d) 5

Q.35 A man would gain 25% by selling a chair for ₹47.5 and would gain 15% by selling a table for ₹57.5. He sells the chair for ₹36; what is the least price for which he must sell the table to avoid any loss on the two together?
(a) ₹51.2 (b) ₹58.5
(c) ₹62.5 (d) None of these

Q.36 A bookseller marks his books at an advance of 69% on the actual cost of production. He allows a discount of 15% and also given a copy free for every dozen sold at a time. What rate per cent profit does the bookseller make, if books are sold in lots of 12?
(a) 32.6 (b) 47.5
(c) 24.9 (d) None of these

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Q.37 While returning from Dubai festival, Mr Sanjay Singh purchased a number of articles, all at a discount of $33\frac{1}{3}$% of the list price. However, he is required to pay a duty of 20% on the his cost of goods. If he realizes a profit of 25% on his outlay, what must he charge for an article priced ₹228 as the list price?

(a) 228 (b) 322
(c) Neither (a) nor (b) (d) Data insufficient

Q.38 A certain manufacturer sells a product to the distributor at 10% profit. Then, the distributor sells it to the dealer and the dealer sells to the retailer at a mark-up of 10% and 20%, respectively. The retailer marks up his cost by 20% and then offers a 10% discount to the customer. If the customer had bought it from the distributor directly at the distributor's selling price, then how much reduction in price would he have got with respect to buying it from the retailer?

(a) 29.6% (b) 18% (c) 32% (d) 22.8%

Direction for Questions 39 and 40: Read the passage below and solve the questions based on it.

Machine *X* produces articles at the rate of 50 units/h. SP of each article is ₹100 and the cost of production is ₹40 each. However, 20% of the articles produced are defective and hence, cannot be sold. The rate of production of machine can be increased, but with every increase of *N* units/h production cost would increase by $2x$% and the number of defectives would become $(20 + 1.5x)$%.

Q.39 Find the maximum increase in production that can be undertaken without incurring losses.

(a) 14 units (b) 15 units (c) 16 units (d) 17units

Q.40 What will be the profit in 1 hour if the machine produces 58 units per hour?

(a) ₹1839 (b) ₹1178 (c) ₹1253 (d) ₹1624

Direction for Questions 41 and 42: Read the passage below and solve the questions based on it.

There is a growing demand of platinum in the Indian market. A firm dealing in platinum wants to earn a profit to the tune of ₹5 lacs per annum. Platinum can be produced from a mine. It is required to employ 25 people whose average salary is ₹2000 per month. Cost of the raw material, that is, platinum ore is ₹200 per ton of pure platinum produced. However, the fixed cost associated with the plant is ₹1,00,000 annually.

Q.41 If the production rate is 20 ton/month, then what should be the price per ton of platinum to earn the required profit?

(a) ₹4300 (b) ₹5200
(c) ₹6300 (d) None of these

Q.42 If the price per ton is ₹3000, then how much ton of platinum is required annually to fulfill the profit target?

(a) 300 (b) 350
(c) 400 (d) None of these

Q.43 Two pieces of the same cloth together cost ₹910. On selling from the first piece as many metres as the second piece originally contained and from the second piece half as many metres as the first piece contained originally. The remaining of the first piece turned out to exceed the remaining of the second piece by 10 m. Each metre of the cloth costs ₹14. How many metre of cloth did each piece originally contain?

(a) 45 m; 20 m (b) 45 m; 25 m
(c) 36 m; 29 m (d) 40 m; 25 m

Q.44 Rupesh marks up an article by p%, gives a discount of $\frac{p}{4}$% and gets a profit of $\frac{p}{4}$%. Had he marked up by $\frac{p}{2}$% and given a discount $\frac{p}{6}$%, what would be his profit percentage?

(a) 25% (b) $33\frac{1}{3}$%
(c) 50% (d) $66\frac{2}{3}$%

Q.45 Amit brought two cars. He then sold the first car at 10% profit and the second one at 25% profit. The selling price of the second car is 25% more than the selling price of the first car. What is the approximate profit per cent in both the cars together?

(a) 17.85% (b) 18.36%
(c) 16.19% (d) Cannot be determined

ADVANCED

Q.1 A publisher printed 3000 copies of *'Future Shock'* at a cost of ₹2400. He gave 500 copies free to different philanthropic institutions. He allowed a discount of 25% on the published price and gave one copy free for every 25 copies bought at a time. He was able to sell all the copies in this manner. If the published price is ₹3.25, then what is his overall gain or loss percentage in the whole transaction?

(a) 113% (b) 130%
(c) 162% (d) 144%

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Q.2 A car manufacturing company XYZ can make 300 cars per month at a cost of ₹2 lakhs per car and pay an excise duty of 30% per car. It has 500 cars in its storage room in a month and is able to sell 300 cars per month at a rate of ₹3 lakhs per car. In the budget in the last week of February, the excise duty was reduced by 10% and it will take effect from 1st April. But, the customers cannot wait till 1st April. So the company decides to give this rebate to every customer who buys a car in March as well, but for this particular month the company has to pay 30% duty. The company can reduce or increase manufacturing of car per month by a maximum of 25%. What is the profit or loss in the revenue to the company in the month of March?

(a) 15 lakh (b) 60 lakh (c) 45 lakh (d) 30lakh

Q.3 A shopkeeper uses a 'point of sales' software which prepares the bill for customers based on the number of units per unit. Both of the above are 2-digit numbers. While entering the data, he erroneously swaps the digits of both quantity of units and price. As a result, the closing stock balance is lowered by 72 units, and the sales value shown is ₹1,368. How many units have actually been sold?

(a) 27 (b) 91
(c) 83 (d) None of these

Direction for Questions 4 and 5: Read the passage below and solve the questions based on it.

Given below is the production of fertilizers bags by ABC fertilizers during different days of the week.

Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
650	750	700	800	750	600

There are two modes of transporting of these bags to warehouse. A truck has a capacity of 800 bags and cost ₹1100 per trip. A Matador has a capacity of 500 bags and cost ₹700 per trip. The cost of storing the material in case it is not transported is ₹1 per bag per day. The cost of transportation includes the storing cost.

Q.4 In order to minimize transportation cost, how many trips should the Matador and the Truck, respectively, make to the warehouse during the week?

(a) 3, 4 (b) 2, 4 (c) 1, 5 (d) 4, 3

Q.5 What should be the minimum cost of storing 1 bag/day, at which it is best to hire a truck everyday?

(a) ₹0.4/day (b) ₹1.4/day
(c) ₹1.6/day (d) ₹1.2/day

Q.6 Mohit goes to furniture shop to buy a sofa set and a centre table. He bargains for a 10% discount on the centre table and 25% discount on sofa set. However, the shopkeeper, by mistake, interchanged the discount percentage figures while making the bill and mohit paid accordingly. When compared to what he should pay for his purchases, what percentage did mohit pay extra given that the center table costs 40% as much as the sofa set.

(a) 12.3% (b) 7.2% (c) 8.1% (d) 6.3%

Q.7 In a laboratory experiment, a sample of air which is a mix of only oxygen and vapour is taken. Water contains hydrogen and oxygen gases. If air contains 70% oxygen (including that contained in the water vapour) by weight while water vapour contain 16.66% oxygen by weight, how many kilograms of water vapour is present in 1kg of water?

(a) 0.25 (b) 0.34 (c) 0.36 (d) 0.3

Q.8 A shopkeeper purchases a packet of 50 pens at ₹10 per pen. He sells a part of the packet at a profit of 30%. On the remaining part, he incurs a loss of 10%. If his overall profit on the whole packet is 10%, find the number of pens he sold at profit.

(a) 25 (b) 30 (c) 20 (d) 15

Q.9 Amar sold his moped to Bharat at 20% profit and Bharat sold it to Sridhar at 10% profit. Sridhar sold the same to a mechanic and received ₹2316. If Amar had sold the same moped to the mechanic and receive the same amount the mechanic paid to Sridhar, what profit percentage would Amar have made?

(a) 52% (b) 48%
(c) 33.3% (d) Cannot be determined

Direction for Questions 10 to 13: Read the passage below and solve the questions based on it.

Newspaper major *The Asian Times* is trying to enter the city of Paperabad, where already two newspapers *The Deccan City* and *Good Morning Paperabad* are in circulation. The population of the city is 35 lakhs of which, according to the recent survey, 60% read newspaper. The two evening newspapers sell at ₹1.75 per copy on weekdays and ₹3.00 per copy on Sundays. *The Asian Times* promised its readers newspapers at ₹1.50 per copy per day. Since people were aware of the quality of the paper, the management thought that it can capture a major chunk of the newspaper reading population. On weekdays, the operational cost of the *The Asian Times* were ₹62,500 per day and the cost for the material and printing charges was ₹2.75 per newspaper. For the Sunday edition the cost of the newspaper per copy was ₹3.00 and the operational cost were ₹75,000. The number of the copies of the newspaper *The Asian Times* sold on Sunday was 16.66% less than the number of copies sold on any of the week days. The number of copies sold on any weekday being the same. The amount collected through

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the advertisement on any weekday was the same and equaled ₹2,50,000 per day while for Sunday it was 4,35,000.

Q.10 If *The Asian Times* wanted to break even, how many copies of the newspaper are required to be sold every week?
(a) 21,25,800 (b) 16,97,143
(c) 47,50,500 (d) 50,50,500

Q.11 What percentage of the newspaper readers should read *The Asian Times* so that the newspaper makes a profit of ₹12,500 every weekday?
(a) 7.61% (b) 12.4 %
(c) 12.8 % (d) None of these

Q.12 If 50,000 of the readers of *The Deccan City* now read *The Asian Times* on all weekdays, how much profit would *The Asian Times* expect to make every week?
(a) 2,55,000 (b) 3,85,000
(c) 4,57,000 (d) Cannot be determined

Q.13. How many copies of the newspaper should *The Asian Times* sell on Sunday so that there is a profit of ₹60,000 every Sunday?
(a) 1,88,000 (b) 2,40,000
(c) 3,20,000 (d) 3,76,000

Direction for Questions 14 to 16: Read the passage below and solve the questions based on it.

In the 'Get-One' supermarket, a new software was installed in the computerized billing section which made the calculations of each of the item sold very easy on the first day of the operations. To verify the workability of the software only one of the item was billed on the computer.

Due to some technical problem in the software, it reversed both the numbers of the item sold and price per piece as a result of which the closing of the stock items showed 63 items less and the value on the items still left was shown ₹936. It was known that both the stocks left and the price per piece are two digit numbers.

Q.14 How many items were actually left at the end of the day?
(a) 29 (b) 81
(c) 92 (d) None of these

Q.15 What is the correct collection for the day if 44 pieces were sold?
(a) ₹2288 (b) ₹1744
(c) ₹1444 (d) None of these

Q.16 By how much would the incorrectly calculated value of the items sold be more/less than the correctly calculated value?
(a) ₹887 more (b) ₹887 less
(c) None of these (d) Cannot be determined

Q.17 Mahesh and Umesh purchased a radio each for the same price and both marked-up their respective radios by the same amount. Mahesh gave a discount of ₹20 followed by another discount of 20% on the reduced price, while Umesh gave a discount of 20% followed by a discount of ₹20. If Mahesh's profit percentage is equal to thrice of Umesh's loss percentage, what is the profit (in ₹) of Mahesh on his radio?
(a) ₹2 (b) ₹3 (c) ₹4 (d) ₹5

Q.18 Vinod is a very shrewd shopowner. He adjusted his electronic weighing scale in a typical way. Error percentage of his sale is directly proportional to the displayed weight of the commodity. But he gives a discount of 10% in every transaction. His weighing scale shows 5 kg for 4 kg. What is the actual profit % when the displayed weight is 10 kg?
(a) 35% (b) 30% (c) −5% (d) 50%

Direction for Questions 19 and 20: Read the passage below and solve the questions based on it.

A shopkeeper believes in running his business on the sentiments of market, that is, quick response to the demand-supply relationship. He starts every morning by selling product X at ₹500 for the first hour. If the number of customers for any hour is more than 10, he will increase the price by ₹10 over the price of the previous hour. If less than 10, he will decrease the price by ₹10 over the previous hour. If it is equal to 10, then he will maintain the same price level of the previous hour. CP of product X is ₹440.

Q.19 If the customer for the first four hours are 15, 10, 6, and 12, then find the profit made by the shopkeeper.
(a) ₹2050 (b) ₹2270
(c) ₹2740 (d) None of these

Q.20 The shopkeeper never sells for a loss. If he shuts his shop after 10 hours due to this principle and there is exactly one hour when the number of customers was 10, then the minimum number of hour when the number of customers is more than 10 is:
(a) 2 (b) 1 (c) 4 (d) 3

Q.21 In a joint venture, Ramesh invested ₹6 lakhs as fixed cost. While Suresh had to pay the variable cost for every three months in advance (i.e., for the first three months at the start of first month and for the next three months at the start of the fourth month and so on).
Table for costs is given below:

Quantity produced/month	Variable cost per unit (₹)	Fixed cost (₹) for 6 months
0–1,00,000	1.5	6,00,000
1,00,001–2,00,000	1.25	6,00,000
2,00,001–3,00,000	1	9,00,000

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For the first three months, they produced 1,00,000 units per month, and for the next three months, they produced 1,50,000 units per month.
All units produced were sold at ₹4 each. After six months, they sold their company for ₹2,00,000 and divided total profit according to their investments. Find out the total profit.
(a) 15.875 lakhs (b) 15.125 lakhs
(c) 10.125 lakhs (d) None of these

Q.22 In the above question, what is the share of Suresh?
(a) 8.7 lakhs (b) 6.2 lakhs
(c) 4.3 lakhs (d) 5.6 lakhs

Q.23 In an orchard, there are only mango and apple trees. Fruits available on each tree are equal to the total number of trees of the same kind. Also, mango trees are twice in number as compared to the apple trees. Only 40% of total mangoes and 70% of total apples are in good condition. These have to be stored before selling. This year the farmer could sell only 75% of the stored mangoes and $\frac{6}{7}$ of the stored apples. The rest was wasted. Selling price per mango is ₹10 and per apple is ₹5. Investment to store (consider only this as the cost price) is ₹20 per mango and ₹10 per apple. In this transaction, the farmer lost ₹15,000. What is the total number of trees in the garden?
(a) 75 (b) 80 (c) 25 (d) 60

Q.24 A spring balance reads 1 kg when actually 800 g is weighed on it. The shrewd grocer has already marked up the price of his goods by 20%. One day, a police inspector detects this and orders him to repair his spring balance and revert to the original list price of the goods. Moreover, he asked him to give a discount of 10% to all the customers for a month from then onwards as a punishment. A day after, Ramu paid ₹16 for 1 kg rice. Before the raid, how much he would have paid for the same quantity?
(a) ₹26.66 (b) ₹27.5 (c) ₹24 (d) ₹28

Q.25 The evergreen shrubs at Ravi's nursery are planted in rows on a square plot of land measuring 2401 ft^2. The shrubs are planted in such a manner that the centers of the adjacent shrub are 7 ft apart and the outer shrubs are planted along the edges of the plot, with a shrub at each corner. Ravi spent $ 896 to cover all the costs necessary for raising this crop of the evergreen shrubs. If Ravi succeeds in selling each shrub for $ 35, his profit will be what percentage of his total cost?
(a) 50% (b) 90% (c) 150% (d) 200%

Q.26 A retailer keeps Reebok shoes, which are listed at a mark-up of 10% above their factory price. However, on Monday, he realized the shortage of demand and hence, decreased the list price by 10% on the very next day, that is Tuesday, he realized he is making a loss so, he increased the list price by 10% again. He continues this trend indefinitely. On which day, he will suffer a loss of more than 10% per shoes for the first time?
(a) Monday (b) Tuesday
(c) Thursday (d) Sunday

Direction for Questions 27 and 28: Read the passage below and solve the questions based on it.

A person purchased 50,000 cups of glasses at a rate of ₹10 per cup. For transporting these cups to his shop, he has four options as follows:

Mode of transportation	Rent for one trip	Capacity of number of glass in one trip	Breakages during transportation
Truck	₹2000	20,000	8%
Small truck	₹1000	12,000	10%
Jeep	₹750	10,000	15%
Rickshaw	₹500	5000	10%

Even if a transport vehicle is not full to its capacity, rent for the trip will remain the same.

Q.27 Which mode of transportation is most profitable?
(a) Truck (b) Small truck
(c) Jeep (d) Rickshaw

Q.28 If the most profitable mode of transportation is chosen, then what should be his selling price per cup so as to get the profit of 10%?
(a) ₹12.45 (b) ₹14.67
(c) ₹13.80 (d) ₹12.10

Q.29 The CP of a dozen mangoes is equal to the selling price of 10 mangoes. The SP of apples per piece is the cost price of a mango (per piece). If the profit % made on selling an apple is half that of selling a mango, then what is the cost of an apple if the selling price of one mango is ₹X?
(a) ₹$\frac{X}{6}$ (b) ₹$\frac{5X}{6}$
(c) ₹$\frac{25X}{33}$ (d) ₹$\frac{33X}{25}$

Direction for Questions 30 to 32: Read the passage below and solve the questions based on it.

The following table shows the relationship between production volumes, unit price and unit cost of a particular product.

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Refer to the following table to answer the questions that follow:

Contribution = volume x (price − cost)

Production volumes (Number of Units)	Unit price (₹)	Unit cost (₹)
1000	140	100
2000	125	95
3000	110	90
4000	100	80
5000	90	75
6000	75	70
7000	65	60

Q.30 At what volume will the contribution be maximum?
(a) 2000 units (b) 3000 units
(c) 4000 units (d) 5000 units

Q.31 If the unit cost goes down by ₹10 at all levels of production, what will be the maximum possible contribution (₹)?
(a) ₹75,000 (b) ₹1,25,000
(c) ₹1,20,000 (d) ₹80,000

Q.32 A breakeven point is the volume of production at which for the first time the contribution equals fixed costs. If the fixed costs are ₹60,000, then the production level at which the break even reached is:
(a) 2000 units (b) 4000 units
(c) 5000 units (d) 6000 units

Answers

Warm Up

1. (b)	**2.** (b)	**3.** (c)	**4.** (d)	**5.** (d)	**6.** (a)	**7.** (b)	**8.** (b)	**9.** (c)	**10.** (c)
11. (d)	**12.** (c)	**13.** (d)	**14.** (d)	**15.** (c)	**16.** (b)	**17.** (d)	**18.** (b)	**19.** (b)	**20.** (b)

Foundation

1. (b)	**2.** (b)	**3.** (b)	**4.** (b)	**5.** (d)	**6.** (a)	**7.** (b)	**8.** (a)	**9.** (d)	**10.** (d)
11. (a)	**12.** (b)	**13.** (b)	**14.** (b)	**15.** (a)	**16.** (a)	**17.** (b)	**18.** (b)	**19.** (b)	**20.** (d)
21. (b)	**22.** (c)	**23.** (b)	**24.** (b)	**25.** (b)	**26.** (a)	**27.** (c)	**28.** (b)	**29.** (d)	**30.** (b)
31. (c)	**32.** (b)	**33.** (c)	**34.** (d)	**35.** (c)	**36.** (d)	**37.** (a)	**38.** (a)	**39.** (a)	**40.** (a)
41. (a)	**42.** (d)	**43.** (b)	**44.** (c)	**45.** (d)	**46.** (b)	**47.** (c)	**48.** (d)	**49.** (a)	**50.** (b)

Moderate

1. (a)	**2.** (c)	**3.** (c)	**4.** (d)	**5.** (d)	**6.** (b)	**7.** (b)	**8.** (d)	**9.** (d)	**10.** (d)
11. (a)	**12.** (b)	**13.** (b)	**14.** (c)	**15.** (d)	**16.** (d)	**17.** (b)	**18.** (d)	**19.** (a)	**20.** (d)
21. (d)	**22.** (a)	**23.** (c)	**24.** (a)	**25.** (c)	**26.** (b)	**27.** (d)	**28.** (b)	**29.** (c)	**30.** (d)
31. (d)	**32.** (b)	**33.** (a)	**34.** (c)	**35.** (d)	**36.** (a)	**37.** (a)	**38.** (d)	**39.** (d)	**40.** (c)
41. (b)	**42.** (d)	**43.** (d)	**44.** (c)	**45.** (a)					

Advanced

1. (d)	**2.** (c)	**3.** (b)	**4.** (c)	**5.** (c)	**6.** (c)	**7.** (a)	**8.** (a)	**9.** (d)	**10.** (b)
11. (d)	**12.** (d)	**13.** (b)	**14.** (b)	**15.** (d)	**16.** (b)	**17.** (b)	**18.** (a)	**19.** (c)	**20.** (b)
21. (d)	**22.** (a)	**23.** (a)	**24.** (a)	**25.** (c)	**26.** (d)	**27.** (a)	**28.** (d)	**29.** (c)	**30.** (c)
31. (b)	**32.** (a)								

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Hints and Solutions

Warm Up

1. Profit percentage $= \frac{150}{600} \times 100 = 25\%$

2. CP $= \frac{₹1950}{1.3} = ₹1500$

Hence, required SP $= ₹1500 \times 1.4 = ₹2100$

3. Let us assume that the cost price $= ₹X$

$56 - x = x - 42$

$2x = 98$

Hence, $x = 49$

4. MP $= x$

CP $= 3\frac{x}{4}$

SP $= \frac{3}{2}x$

Profit percentage $= \frac{\frac{3}{2}x - \frac{3}{4}x}{\frac{3}{4}x} \times 100 = 100\%$

5. Assume CP $= ₹X$

$x - 150 = \frac{x}{16} \qquad \Rightarrow x = 160$

6. Profit percentage $= \frac{50}{200} \times 100 = 25\%$

7. SP $= x$

Profit percentage $= \frac{\text{profit}}{x} \times 100$

$25 = \frac{\text{Profit}}{x} \times 100$

Profit $= \frac{1}{4}x$

CP $= x - \frac{1}{4}x = \frac{3}{4}x$

Profit percentage $= \frac{\frac{1}{4}x}{\frac{3}{4}x} \times 100 = 33\frac{1}{3}\%$

8. CP $= ₹x$

Profit $= \frac{x}{10}$

Now, $\frac{\frac{x}{10} + 45}{x} \times 100 = 25$

$X = 300$

9. List price $= x$

CP $= \frac{4}{5}x$, SP $= \frac{6}{5}x$

Profit percentage $= \frac{\frac{6x}{5} - \frac{4x}{5}}{\frac{4}{5}x} \times 100 = 50\%$

10. CP of 1 orange $= ₹\frac{10}{200} = ₹\frac{1}{20}$

SP of 1 orange $= ₹\left(\frac{1}{20} + \frac{1}{4} \cdot \frac{1}{20}\right) = ₹\frac{1}{16}$

So, he should sell 16 oranges in a rupee to make a profit of 25%.

11. Profit percentage $= \frac{6}{18} \times 100 = 33\frac{1}{3}\%$

12. $\frac{1}{10}x = 200$

$x = 200$

13. CP $= 360 + 360 \cdot \frac{1}{9} = ₹400$

14. Profit in order to gain 40% $= ₹6400$

More money charged $= (₹6400 - 2500) = ₹3900$

15. CP of 1 book $= ₹15$

SP of 1 book in order to gain 25% profit $= ₹18.75$

Required number of books $= \frac{225}{18.75} = 12$

16. Suppose initially x kg of rice was bought.

Now, $\frac{1}{4}x = 6$

$x = 24$

Initial price $= ₹\frac{120}{24} = ₹5$

17. CP of 1 apple $= ₹1$

CP or 12 apples $= ₹12$, SP of 9 apples $= ₹12$

Required difference $= \frac{3}{9} \times 100 = 33\frac{1}{3}\%$

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18. Let CP of 1000 g = ₹1000

CP of 800 g = ₹800

SP of 800 g = ₹1000

$$\text{Profit percentage} = \frac{200}{800} \times 100 = 25\%$$

19. $$\text{Profit percentage} = \frac{(30-25)}{25} \times 100 = \frac{5}{25} \times 100 = 20\%$$ profit

20. **Method 1**

Let CP = ₹1000/1000 g, so, CP of 1 g = ₹1 and CP of N g = ₹N

But, SP of N g = ₹1000

$\therefore$ Profit = $1000 - N$

$$\text{Profit \%} = \frac{1000-N}{N} \times 100$$

According to the question, profit % = 100%

$$\therefore\ 100 = \frac{1000-N}{N} \times 100$$

Hence, $N = 1000 - N$, or, $2N = 1000$, or, $N = 500$ g

Method 2

To obtain 100% profit, he should have made 2 kg out of 1 kg [You may think of this situation as – doubling the quantity by adding adulterant = Original. In this way, total quantity will be double the original quantity.]

Hence, $\frac{1000}{x} = 2$ kg $\Rightarrow x = 500$ g

FOUNDATION

1. $$\text{CP} = 2400 - 2400 \times \frac{1}{6} - 2400 - 400 - 2000$$

2. $$\text{Profit percentage} = \frac{2400 \times 20}{100} = 480$$

CP = ₹(2400 – 480) = ₹1920

3. $$\text{Actual profit percentage} = \frac{480}{1920} \times 100 = 25\%$$

4. CP of 10 apples = ₹1

$$\text{CP of 1 apple} = ₹\frac{1}{10}$$

$$\text{SP of 1 apple} = ₹\left(\frac{1}{10} + \frac{1}{4} \cdot \frac{1}{10}\right) = \frac{1}{8}$$

5. CP of Mayank = CP of Chandra × 0.7 = ₹150 × 0.7 = ₹105

SP of Mayank = ₹126

6. CP of 1 L petrol = ₹5

$$\text{SP of 1 L petrol} = ₹\frac{1}{5}$$

$$\text{Loss \%} = \frac{5 - \frac{1}{5}}{5} \times 100 = \frac{24}{25} \times 100 = 96\%$$

7. MP = x

$$\text{SP} = x - \frac{1}{9}x = \frac{8}{9}x$$

$$\text{CP} = \frac{8}{9}x - \frac{8}{9}x \cdot \frac{1}{8} = \frac{8x}{9} - \frac{x}{9} = \frac{7x}{9}$$

$$\text{Required percentage} = \frac{x - \frac{7x}{9}}{\frac{7}{9}x} \times 100 = \frac{\frac{2x}{9}}{\frac{7x}{9}} \times 100$$

= 28.56%

8. Let CP = ₹100

SP = ₹120

MP = x

$\frac{3}{4}x = 120$

x = ₹160

Hence, 60%

9. Maximum can be anything beyond 60%.

10. Without knowing the profit percentage, we cannot determine the CP.

11. Assume x is the SP per kg.

$1100 - 100x = 20x$

$120x = 1100$

$$x = \frac{110}{12} = 9.16$$

12. $$\frac{22}{100}x - (750 - x) \times \frac{8}{100}$$

$x = 200$

SP of 2nd cow = (550 – 44) = ₹506

13. $$\text{Net loss percentage} = \left(\frac{20}{10}\right)^2 = 4\%$$

For every ₹100 invested, ₹4 is the loss, or, for every ₹96 realized through selling, ₹4 is the loss.

$$\text{Loss amount} = ₹\left(48{,}000 \times \frac{1}{24}\right) = ₹2000$$

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14. MP of 1 pencil = ₹1

For supplier, SP of 20 pencils = ₹16

For retailer, SP of 20 pencils = ₹20

Profit percentage $= \frac{4}{16} \times 100 = 25\%$

15. $x + \frac{1}{5}x = (100{,}000 - x) - \frac{1}{5}(100{,}000 - x)$

$x + \frac{1}{5}x = 100{,}000 - x - 20{,}000 + \frac{x}{5}$

$x = 40{,}000$

16. $x + \frac{1}{5}x = (13{,}000 - x) + \frac{2}{5}(13{,}000 - x)$

$x + \frac{1}{5}x = 13{,}000 - x + 5200 - \frac{2}{5}x$

$2x + \frac{3}{5}x = 18{,}200$

Or, $\frac{13}{5}x = 18{,}200$

$x = 7000$

17. Required difference = 7000 − 6000 = 1000

18. CP of 1 article = ₹1

CP of 40 articles = ₹40

SP of 30 articles = ₹40

Profit percentage $= \frac{10}{30} \times 100 = 33.33\%$

19. Due to price hike of 20%, quantity of less tea available = $\frac{1}{6}x = 4$ $x = 24$ (where x = initial quantity)

Original price = $₹\frac{120}{24} = ₹5$

20. Profit percentage $= \frac{250}{750} \times 100 = 33.33\%$

21. Weight he is selling instead of 1000 g = 1000/1.3 = 769 g.

22. Let MP = ₹100

(i) Customer has to pay = ₹61.6

(ii) Customer has to pay = ₹61.6

(iii) Customer has to pay = ₹61.6

23. Let number of chips supplied = 100

For whole seller, Net CP = (10,000 × 100 + 5 × 10,000) = ₹10,50,000

Profit = $₹\left(1050000 \times \frac{1}{5}\right) = ₹210{,}000$

SP for whole seller = ₹12,60,000

SP of 1 chip = ₹12,600

24. Going through the options.

Option (a) ₹240 is the CP.

$\therefore$ SP $= 240 + \frac{6}{100} \times 240 = 240 + 14.40 = ₹254.40$

Profit = ₹14.40

$\therefore$ New SP = 245.40.

$\therefore$ New profit = ₹5.40

Profit % $= \frac{5.40 \times 100}{240 \times 100} = 2.25\%$

Option (b) ₹250 is the CP.

$\therefore$ SP $= 250 + \frac{6}{100} \times 250 = ₹265$

Profit = ₹15

Now, new SP = ₹256

$\therefore$ New profit = ₹6.

Profit % $= \frac{6}{250} \times 100 = \frac{12}{5} = 2.4\%$

Hence, option (b) is the answer.

25. Let SP = ₹100

Profit = ₹25

CP = ₹75

Actual profit percentage $= \frac{25}{75} \times 100 = 33.33\%$

26. Let SP of 1 kg article = ₹100

For 1st shopkeeper

SP of 1 kg article = ₹75

For 2nd shopkeeper

SP of 1.25 kg article = ₹100

SP of 1 kg article = ₹80

27. Let CP of 1st cow = ₹x

CP of 2nd cow = ₹Y

Hence, net CP = ₹$(x + y)$

SP of 1st cow = ₹y

SP of 2nd cow = ₹X

Net SP = ₹$(x + y)$

Hence, no profit no loss

28. CP $= ₹\left(180 - 180 \cdot \frac{1}{6}\right) = ₹150$

29. In the whole question, there is nowhere the mention of any rupees figures. Since amount is not given, we cannot calculate any value (in ₹).

30. Let CP of book for Anoop = ₹100

CP of book for Mayank = ₹120

CP of book of Siddharth = ₹150

CP of book of Shishir = ₹135

Required loss percentage $= \frac{35}{135} \times 100 = 25.92\%$

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31. Let CP = ₹700

x = ₹90

L = 10%

Y = ₹110

P = 10%

Putting these values, we get (c) as the answer.

32. CP of 100% Pure Milk = ₹100

$$\text{CP of the Mixture} = ₹\left(100 - 100 \times \frac{1}{6}\right) = ₹83.33$$

Hence, 83.33% milk is present 1 L of mixture.

Alternatively, if he has initially 10 L of milk, he must have made it 12 L of mixture to get a profit of 20% (since SP per litre = CP per litre). Hence, out of 12 L of mixture, 10 L is milk and 2 L is water.

33. Let us assume that CP = ₹x

$720 - x = 2(x - 420)$

$720 - x = 2x - 840$

$3x = 1560$

$x = 520$

34. According to the question, 14.28% of mark-up of 1st shopkeeper = 12.5% of mark-up of 2nd shopkeeper. With this we can compare the mark-up prices.

However, nothing can be said about the cost prices of the shopkeepers.

35. Let MP = ₹100 and CP = ₹X

then, $450 - 5x = 700 - 8x$

$$x = \frac{250}{3}$$

$$\text{Required ratio} = \frac{100}{\frac{250}{3}} = 6:3$$

36. Since, cost price and selling price are not known therefore, we cannot find profit percentage.

37. $13x + 9(50 - x) = 550$

$4x = 100$

$x = 25$

38. According to the question, we can make out that $3x + 4y = 100$

Also, x has to be even since total of the two values is even.

So, values of x can be 4, 8, 12, 16, 20, 24, 28, 32.

39. Assume that MP = ₹100

then CP of Anam = ₹70

and his SP = ₹160

So, his profit = ₹90

Now, using unitary method, when profit of Anam = ₹90, then his MP = 100. So, when his MP = ₹81, his MP = 90

40. CP of 100 m = ₹100

$$\text{CP of } \frac{500}{6}\text{ m} = ₹\frac{500}{6}$$

$$\text{Profit Percentage} = \frac{100 - \frac{500}{6}}{\frac{500}{6}} \times 100 = 20\%$$

Alternatively, profit is 16.66% by selling 83.33% quantity.

$$\text{Hence, profit percentage} = \frac{16.66 \times 100}{83.33}\% = 20\%$$

41. Let MP = ₹100

CP = ₹70 and SP = ₹80

Hence, profit percentage = 14.28%

42. $(7.50 \times 2x) - (7x + 6x) = 80$

$2x = 80$

$x = 40$

43. CP of 1000 ml milk = ₹100

CP of 900 ml milk = ₹9

Profit = ₹(3 × 20) = ₹60

$$\text{Profit percentage} = \frac{60}{9 \times 20} \times 100 = 33.33\%$$

44. Net CP = ₹50 x

Net SP = ₹48x

$$\text{Loss percentage} = \frac{2x}{50x} \times 100 = 4\%$$

45. Let CP = ₹100

MP = ₹140

Profit, when discount is 5% = ₹33

Profit, when discount is 10% = ₹26

$$\text{Hence, CP} = \frac{100}{7} \times 14 = ₹200$$

MP = ₹280

SP = ₹224

46. 1 Gross = 12 dozen

Net CP = ₹(120 + 12) = ₹132

Net SP = ₹158.4

$$\text{Required price} = ₹\frac{158.4}{10} = ₹15.84$$

47. 15% of CP = ₹56.25, Hence, CP = ₹375

New SP = ₹450, Hence, profit = ₹75

Profit percentage = 20%

48. $x = \frac{24 - x}{x} \times 100$, So, $X = 20$

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49. Net CP $= \left(228 - \frac{1}{3} \times 228 + \frac{1}{5} \times 152\right)$

$= ₹(152 + 30.4) = ₹182.4$

$SP = ₹\left(182.4 + \frac{1}{4} \times 182.4\right) = ₹228$

50. Net CP = ₹132

Money paid by seniors = ₹33

Required percentage $= \frac{99}{66} \times 100 = 150\%$

MODERATE

1. Maximum possible discount in any case can be only 100% and any discount more than 100% is actually 100% only. Now, whatever is the discount, till the moment we get a discount of 100%, value of the article cannot be zero (even infinite discounts of 99.99% will not make it equal to a discount of 100%) So, number of rounds needed to obtain 100% discount = 50

2. As discussed above, maximum possible discount can be only 100%.

3. Despite the fact that a customer has won 18 rounds out of the first 20 rounds, if he looses in the 20th round, net discount = 0%

6.

Actual CP	C
New CP	0.9C
Actual SP	S
New SP	1.1 S
Actual profit %	100 (S – C)/C
New profit %	100 (1.1 S – 0.9 C)/0.9 C

So, [S/C] – 1 = 2/7

Profit percentage = 28.56%

Since in the question, nowhere the term ₹ has been used, the value of CP cannot be calculated.

7. Profit percentage as calculated above = 28.56%

8. Since there can be any value of CP, the number of values of SP can also be anything, provided profit percentage = 28.56%.

11. Assume CP/1000 g = ₹1000

So, MP = ₹1400

SP = ₹1260

Now, when he sells 1000 g, he actually obtains the money for only 800 g (Go through the statement carefully).

So, when he sells articles worth ₹1000, money obtained after selling will be equal to = ₹1260 × 0.8 = ₹1008

So, profit percentage = 8%

12. Assume his CP = ₹1000/1100 g

MP = ₹1200 and SP = ₹960/900 g

So, SP/1100 g = ₹1173.33

So, profit = ₹173.33

Profit percentage = 17.33%

18. Since in all these cases, more articles are given free than the purchased article, so the net discount is more than 50%. Hence, in all these layers, the discount offered is more than 50%.

(If the articles given free is equal to the articles purchased, discount percentage will be equal to 50%).

20. If we club the first three layers, it is like: buy 6, get 9 free. So, net discount = 60%

Answers to Q.28 to 33:

	Sales	Profit	Expenses
BOB	$2x = 50$	$y = 5$	$3z = 45$
MOB	60	$t - 1$	$60 - 2 = 58$
ZOB	15	$t = 3$	$15 - 3 = 12$
DOB	$x = 25$	$2y = 10$	$25 - 10 = 25$

34. Price = ₹X

SP = ₹1.1 x = 0.5 x + 15, So, 0.6 x = 15

So, x = 25

Alternatively, we can do this question very easily by using options.

36. CP = ₹X/dozen = 0.833/copy

So, MP = 1.69 x/dozen

SP = ₹1.4365x /13 copies = 0.1105 x/copy

So, profit = 32.6%

37. List price = 228

CP (before duty) = ₹152

CP (after duty) = ₹152 × 1.2 = ₹182.4

SP = ₹152 × 1.2 × 1.25 = ₹228

38. This whole sequence goes like this:

100 →10% UP → 110 →10% UP → 121→20% UP→145.2→ UP by 20% and down by 10%→ 156.816

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Distributors' SP = 121

So, percentage reduction = 22.8%

43. Go through the options.

44. This question can be done by forming equation by assuming the CP to be ₹X.

Alternatively, doing a bit of hit and trial gives the value of p = 200.

Using this, we get the answer as option (d) = 66.66%

ADVANCED

1. Cost = ₹2400

Published Price ₹3.25

SP = 75/100 × 3.25 = ₹2.4375

Number of free copy = (3000/25) = 120 + 500 = 620

So, total SP = 2380 × ₹2.4375 = ₹5801.25

Hence, percentage gain = 5801.25 – 2400/2400 – 100 = 144%

3. $9(x - y) = 72$, $x - y = 8$

Therefore, only possibility is 19 or 91.

4. Matador's capacity is just 500, so it will be used minimum, i.e., 1 day.

So, truck will be used for the rest 5 days.

6. Assume the cost of a sofa set = ₹100. So, price of centre table = ₹40

For Mohit, following is the bargained price:

Cost of centre table = 90% of ₹40 = ₹36

Cost of sofa set = 75% of ₹100 = ₹75

So, the bargained value = ₹111

But, he actually pays:

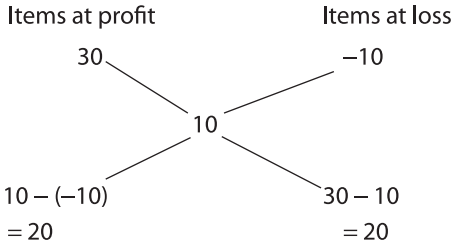
Cost of centre table = 75% of ₹40 = ₹30

Cost of sofa set = 90% of ₹100 = ₹90

So, the amount paid = ₹120

So, percentage extra that he paid = $\frac{120-111}{111} \times 100 = \frac{9}{111} \times 100 = 8.18\%$

8. Using the method of alligation,



So, the ratio will be 1:1.

Hence, number of pens sold at profit = number of pens sold at loss = 25 pens

9. Amar – 100, Bharat – 120, Sridhar – 132

No profit or loss is mentioned about the deal between Sridhar and the mechanic. So, the answer cannot be determined.

Answers to Q.10 to 13:

10.

	Weekdays	Weekend
Fixed cost	₹3,75,000	₹75,000
Variable cost	2.75 × 6x	(5x/6) × 3
Total cost	₹3,75,000 + 2.75x	₹75,000 + (5x/6) × 3
Reveneue through advertisement	₹15 lakhs	₹4,35,000

Where x is the number of copies sold any week day.

For breaking even,

$[₹375,000 + 16.5x] + [₹75,000 + (5x/6) \times 3] = ₹19,35,000 + 10.25x$

$₹450000 + 19x = ₹19,35,000 + 10.25x$

So, x = 16,97,142.28

Hence, to obtain break-even point, minimum 16,97,143 copies are to be sold.

12. Since we are not having any information regarding the total number of copies sold (the information is talking about the reader of 'The Deccan City'), we cannot determine the profit.

13. ₹75, 000 + 3x = ₹4,35,000 + 1.5x, where x is the number of copies sold on Sunday.

Answers to Q.14 to 16:

14. Number of items – Shown number of items = 63

$9(x - y) = 63$, $(x - y) = 7$ where x and y are its Ten's and Unit's digits

So, the number may be 81 and 18 or 92 and 29.

However value = ₹936, which is divisible by 18 only.

So, actually left items = 81

15. Rate = 936/18 = 52

It is given that price/piece was also reversed.

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So, price/piece = ₹25

Collection = ₹25 × 44 = ₹1100

27.

Mode of transportation	Rent for one trip	Breakages during transportation	Total cost
Truck	₹6000	₹40,000	₹46,000
Small truck	₹5000	₹50,000	₹55,000
Jeep	₹3750	₹75,000	₹78,750
Rickshaw	₹5000	₹50,000	₹55,000

Obviously, truck is the most profitable mode.

28. For 10% profit, he should obtain 1.1 × (₹5,00,000 + ₹6000) = ₹5,56,600

Number of cups left = 46,000

So, SP per cup = ₹12.1

30. Using the options,

(a) 2000 units, C = 2000 × 30 = ₹60,000

(b) 3000 units, C = 3000 × 20 = ₹60,000

(c) 4000 units, C = 4000 × 20 = ₹80,000

(d) 5000 units, C = 5000 × 15 = ₹75,000

31. At 5000 units, C = 5000 × 25 = ₹1,25,000

32. From Question number 38, at 2000 units, C = ₹60,000

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CHAPTER

6

Average and its Applications

LEARNING OBJECTIVES

After completion of this chapter, the reader should be able to understand:

- Average and its various interpretations
- Definitions and properties
- Applications of average in different cases such as alligation
- Different cases of mixtures and alligation
- Methods of solving questions

INTRODUCTION

Average and its applications were once an important topic for CAT, but the number of questions asked from this topic has been reduced in the last couple of years. However, this concept is used vigorously in other exams. As a concept, average as well as its applications should be seen more as a logical concept of group formation than a pure mathematical concept.

AVERAGE

Traditionally, average is calculated by dividing the sum of all the numbers by the number of numbers.

$$\text{Average} = \frac{\text{Sum of numbers}}{\text{Number of numbers}}$$

For example, the average of the four numbers 214, 215, 219, and 224 will be:

$$\text{Average} = \frac{214+215+219+224}{4} = 218$$

Central Value Meaning of Average

Average can also be seen as the central value of all the given values.

Applying this definition for the above example, let us assume the central value of all the given numbers = 214

Now, find the deviations of all the numbers from 214

214 215 219 224

When assumed central value is (214), the sum of the deviations

= 0 + 1 + 5 + 10

Now, finding the average of deviations gives us

$$\frac{0+1+5+10}{4} = \frac{16}{4} = 4$$

Hence, average = assumed central value + average of deviations = 214 + 4 = 218

Therefore, we can assume any value to be the assumed average and then find the average of all the deviations; and when we add all the numbers and divide it by number of numbers, 0 is assumed to be the central value.

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Example 1 Average age of A, B and C is 84 years. When D joins them the average age of A, B, C, and D becomes 80 years. A new person E, whose age is 4 years more than D, replaces A and the average age of B, C, D, and E becomes 78 years. What is the age of A?

Solution Since the average age of A, B, and C is 84 years so, we can assume that age of A, B, and C is 84 years.

A = 84 years

B = 84 years

C = 84 years

After D has joined them,

	Initially	*Finally*
A	84 years	80 years
B	84 years	80 years
C	84 years	80 years
D	…..	80 years

Decrease in the age of A, B, and C can be attributed to the increase in the age of D. So, after getting 12 years in total (4 years each from A, B, and C) D is at 80 years. The original age of D = 80 – 12 = 68 years.

Hence, age of E = 72 years

Now, the average age of A, B, C, and D = 80 years; A + B + C + D = 320

And average of B, C, D, and E = 78 years; B + C + D + E = 312

(Since the average difference between the age of A and E is 2 years.)

Difference (A – E) = 2 × 4 = 8 years

Since E = 72 years, so A = 80 years

By using central value method of averages, every question of average can be done by mental calculation only.

Example 2 Average of 10 two-digit numbers is S. However, when we reverse one of the numbers AB as BA from the given 10 numbers, then the average becomes S + 1.8. What is the value of B – A?

Solution Average of 10 numbers is increasing by 1.8, so it can be assumed that 1.8 has been added to all the numbers.

Hence, BA is 1.8 × 10 = 18 more than AB.

There are so many two-digit numbers that satisfy the above condition. Using hit and trial method, the numbers can be 13, 24, 35, 46, 57, 68, and 79. In every case, difference between the digits = 2

Otherwise, we can use the formula

(BA – AB) = 9 × (B – A)

Where BA and AB are two-digit numbers.

Example 3 The average score of Rahul Dravid after 25 innings is 46 runs per innings. If after the 26th innings, his average runs increased by 2 runs, then what is his score in the 26th inning?

Solution Runs in 26th inning = Total runs after 26th innings – Total runs after 25th innings

$$= 26 \times 48 - 25 \times 46 = 98$$

Alternatively, this question can be done by the above given central value meaning of average. Since the average increases by 2 runs per innings, we can assume that 2 runs have been added to his score in each of the first 25 innings. Now, the total runs added in these innings have been contributed by the runs scored in the 26th inning, which must be equal to 25 × 2 = 50 runs.

And after contributing 50 runs, his score in the 26th inning is 48 runs.

Hence, runs scored in the 26th inning = new average + old innings × change in average

$$= 48 + 25 \times 2 = 98$$

To have a mental mapping, we can see the whole situation as:

Number of innings	Average in the 1st 25 innings	Average in the 1st 26 innings	Addition
1	46	48	2
2	46	48	2
3	46	48	2
…	…	…	…
…	…	…	…
…	…	…	…
25	46	48	2
26		48	

Properties of Average

1. Average always lies in between the maximum and the minimum value. It can be equal to the maximum or minimum value if all the numbers are equal.
   For example, $A1$, $A2$, $A3$, and $A4$ are four numbers given where $A1 > A2 > A3 > A4$.
   Average of these four numbers will always lie in between $A1$ and $A4$.
   However, if all the four numbers are equal ($A1 = A2 = A3 = A4$), then the average will be equal to each of these numbers.
   Average = $A1 = A2 = A3 = A4$
2. Average is the resultant of net surplus and net deficit, as used in the central tendency method.
3. When weights of different quantities are same, then simple method is used to find the average. However, when different weights of different quantities are taken, then, it is known as weighted average. Here, the method of weighted average is used to find the average.

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For example, assume per capita income of India is USD 500 and per capita income of US is USD 200. Now, if we merge India and United States into one country, then it is observed that per capita income of this new country will not be equal to $\frac{500+200}{2}$ = USD 350.

4. If the value of each quantity is increased or decreased by the same value S, then the average will also increase or decrease, respectively, by S.
5. If the value of each quantity is multiplied by the same value S, then the average will also be multiplied by S.
6. If the value of each quantity is divided by the same value S ($S \neq 0$), then the average will also be divided by S.

Example 4 The average of 4 positive numbers is A and the average of all the possible triples formed out of these four positive numbers is B. Which of the following is true regarding A and B?

(a) $A = B$ (b) $A > B$
(c) $A < B$ (d) Cannot be determined

Solution Let us assume that the numbers are 1, 2, 3, and 4

Average of 1, 2, 3, and 4: $1 + 2 + 3 + 4 = 10/4 = 2.5$

The triplets are 1, 2, and 3; and the average = $\frac{6}{3}$

1, 2, and 4; and the average = $\frac{7}{3}$

1, 3, and 4; and the average = $\frac{8}{3}$

2, 3, and 4; and the average = $\frac{9}{3}$

Average of these four averages = $\frac{\frac{6}{3}+\frac{7}{3}+\frac{8}{3}+\frac{9}{3}}{4} = 2.5$

Hence, option (a) is the answer.

Central Value Method

It should be observed here that when we find the average of all the possible triplets, all the numbers (1, 2, 3, 4) are added thrice. So effectively we are adding 12 numbers. Hence, the average should be equal to:

$$\frac{3(1+2+3+4)}{12} = 2.5$$

Therefore, the average will be equal in all the cases.

Extension to this problem The average of four positive numbers is A and the average of all the possible pairs formed out of these four positive numbers is B. Which of the following is true?

(a) $A = B$ (b) $A > B$
(c) $A < B$ (d) Cannot be determined

Some Special Cases

1. Average Involving Time, Speed, and Distance

$$\text{Average speed} = \frac{\text{Total distance}}{\text{Total time}}$$

However, while solving the questions involving time, speed, and distance, we should assume some distance, preferably the LCM of all the given speeds.

Example 5 Lovely goes to Patna from New Delhi at a speed of 40 km/h and returns with a speed of 60 km/h. What is her average speed during the whole journey?

Solution Assuming that the total distance between Patna and New Delhi is 120 km (LCM of 40 and 60) the total time taken (Patna – New Delhi and New Delhi – Patna) = 3 + 2 = 5 h

Hence, average speed = $\frac{240}{5}$ = 48 km/h

2. Average Involving Age

Average of a group of n persons given at any point of time can be calculated in the following way

5 years ago	10 years ago	Now	10 years later	5 years later
$N-5$	$N-10$	N	$N+10$	$N+5$

Example 6 The average age of the five members of a family is 20 years. The youngest member of the family is 4 years old. At the time of his birth, the average age of the rest of the family was N years. What is the average age of the family (in terms of N) excluding the youngest member?

Solution Sum of ages of all the members of the family = 100

Sum of ages of all the members of the family excluding the youngest number = 100 – 4 = 96

Hence, average age of all the members of the family excluding the youngest number = 96/4 = 24 = N

The average age of the family(in terms of N) excluding the youngest member = $N + 4$

3. Average Involving Number System

1. Average of 1st n consecutive natural numbers = $\frac{n+1}{2}$

   For example, the average of 1st five natural numbers = 3

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2. The average of 1st n consecutive even natural numbers $= n + 1$
   Sum of 1st n consecutive even natural numbers $= n(n + 1)$
   For example, the average of 1st five even natural numbers = 6
3. The average of 1st n consecutive odd natural numbers $= n$
   Sum of 1st n consecutive odd natural numbers $= n^2$
   For example, the average of 1st five odd natural numbers = 5

Weighted Average

It is observed that the average can be calculated only if the weights of all the factors are same. Hence, the weighted average is a more generalized form of average. This can be further understood with the following illustration.

	Class A	Class B
No.of students	10	10
Average age	12 years	16 years

Now, if we combine both these classes, then the average age of all the students $= \frac{12+16}{2} = \frac{28}{2} = 14$ years. This is one standard example of average.

Let us see another example:

	Class A	Class B
No.of students	12	16
Average age	10 years	14 years

Now, if we combine these two classes, then the average can not be calculated by the above mentioned method, since the weights attached to different averages are different.

Some More Cases of Weighted Average

1. As we have observed above in the case of average, if per capita income of India is USD 500 and per capita income of US is USD 200, and if we merge India and US into one country, then the per capita income of this new country (India + US) cannot be found by just adding the per capita income of both the countries and dividing it by 2.
   The weights, that is, the population attached to the different averages, that is, the per capita income would also have to the considered.
2. Average speed cannot be calculated by just adding the different speeds and then dividing it by 2. This can be understood by the following example:

   A person goes to A from B at a speed of 40 km/h and returns with a speed of 60 km/h, then the average speed for the whole journey can not be equal to 50 km/h.

   We know that average speed $= \frac{\text{Total distance}}{\text{Total time}}$

Finding Expression for Weighted Average

Let us assume that there are N groups with the following structure:

Group no.	No. of members	Avg. age of the group
G_1	N_1	A_1
G_2	N_2	A_2
G_3	N_3	A_3
...	...	...
...	...	...
G_N	N_N	A_N

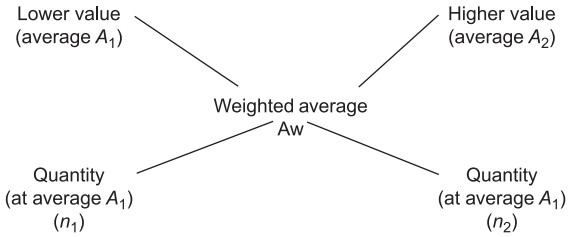
If we combine all these groups, then the average age of all the members $= (N_1 \times A_1 + N_2 \times A_2 \ldots + N_N \times A_N)/(N_1 + N_2 + N_3 + \ldots + N_N)$

$$= \frac{\sum AN}{N}$$

Group no.	No. of members	Avg. age of the group
G_1	N_1	A_1
G_2	N_2	A_2

Considering that there are only two groups and both the groups are combined, then the average age of all the members $= (N_1 \times A_1 + N_2 \times A_2)/(N_1 + N_2) = A_w$

Simplifying the above written expression, we get the conventional criss-cross method as given below.



And we write this as: $\frac{n_1}{n_2} = \frac{A_2 - A_w}{A_w - A_1}$

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i.e.,

$$\frac{\text{Quantity (Lower priced)}}{\text{Quantity (Higher priced)}} = \frac{\text{Higher price} - \text{Average price}}{\text{Average price} - \text{Lower price}}$$

It is quite obvious that the ratio of the number of persons /items in different groups is proportionate to the deviations of their average from the average of all the people combined.

This average of all the members combined is known as weighted average and is denoted by A_w. This process of mixing the two groups is also referred as alligation.

Elements of Weighted Average

As we can see from the above derivation, there are five quantities:

i. Number of members in 1st group (n_1)
ii. Number of members in 2nd group (n_1)
iii. Average of 1st group (A_1)
iv. Average of 2nd group (A_2)
v. Weighted average (A_w)

Normally, in the case of weighted average, we get questions in which one of these five elements is missing, and with the help of the remaining four quantities, the value of that missing quantity is found. Different possibility (situations) are given below (Y represents – data given, N represents – data not given):

Situations	n_1	n_2	A_1	A_2	A_w
First	Y	Y	Y	Y	N
Second	Y	Y	Y	N	Y
Third	Y	Y	N	Y	Y
Fourth	Y	N	Y	Y	Y
Fifth	N	Y	Y	Y	Y
Sixth	N	N	Y	Y	Y

First Situation

Example 7 10 kg of rice priced at ₹12 per kg is mixed with 6 kg of rice priced at ₹16 per kg. What is the average price of the whole mixture?

Solution Lower priced value = ₹12 per kg and its quantity = 10 kg

Higher priced value = ₹16 per kg and its quantity = 6 kg

Using alligation,

$$\frac{10}{6} = \frac{16 - A_w}{A_w - 12}, \text{ or, } A_w = ₹13.5/\text{kg}$$

However, in my opinion, in this situation, it is better to use the normal method rather than using the weighted average method of finding A_w.

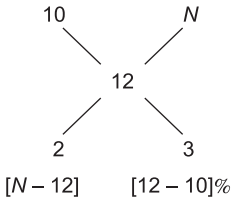
Normal method – Total value = $12 \times 10 + 16 \times 6 = 216$

Hence, average price = $\frac{216}{16} = ₹13.5/\text{kg}$

Second/Third Situation

Example 8 Two varieties of rice are mixed in the ratio 2:3. The price of the mixture is ₹12 per kg and the price of the variety having lower weight is ₹10 per kg. Find the price of the other variety.

Solution



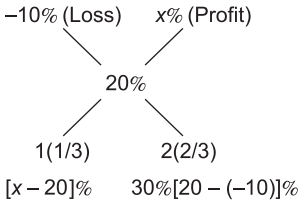
Now, $\frac{2}{3} = \frac{(N-12)}{(12-10)} = \frac{N-12}{2}$

Hence, N = ₹13.33 per kg

Fourth/Fifth Situation

Example 9 Some articles are purchased for ₹450. 1/3rd of the articles are sold at a loss of 10%. At what percentage profit should the remaining articles be sold to obtain a net profit of 20% on the whole transaction?

Solution



Now, = $[20 - (-10)]/[x - 20] = \frac{2}{1}$

Hence, x = 35%

It is seen that the quantities are in the ratio of 1 : 2, so the deviation from mean percentage profit in the loss percentage and profit percentage will also be same.

Two corresponds to 30%, and 1 will correspond to 15%. Hence, x = 35%

Sixth Situation

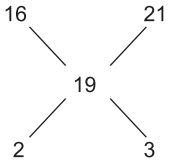
Example 10 Two different qualities of sugar are mixed in some ratio. The price of one quality of sugar is ₹16/kg and that of another quality is ₹21/kg. In what ratio have

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the sugar of two qualities been mixed if the price of the mixture is ₹19/kg?

Solution



Hence, the ratio of quantity of sugar of different qualities = 2:3

MIXTURES

When two or more than two pure substances/mixtures are mixed in a certain ratio, they create a mixture. Here, we shall confine ourselves to mostly homogenous mixtures in view of the questions commonly asked in CAT.

Mixing without Replacement

In this particular type of mixing, two or more than two substances are mixed without any part of any mixture being replaced.

Example 11 In a mixture of 420 L, the ratio of milk and water is 6:1. Now, 120 L of the water is added to the mixture. What is the ratio of milk and water in the final mixture?

Solution Volume of milk = 360 L and volume of water = 60 L.

When 120 L of water is added, volume of water = 180 L

Hence, the ratio of milk water = 2:1

Example 12 A milkman mixes 20 L of water with 80 L of milk. After selling one-fourth of this mixture, he adds water to replenish the quantity that he had sold. What is the current proportion of water to milk?

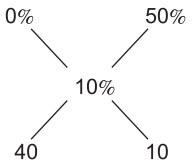
Solution Ratio of milk and water = 20:80

When one-fourth of this mixture is sold, total volume of mixture will be reduced by 25%, so 25% of milk and water both will reduce. So, volume of milk and water after selling out one-fourth of mixture = 60 L and 15 L, respectively. Addition of 25 L, of water will finally give us the following: volume of milk = 60 L and volume of water = 40 L. Hence, the ratio of water and milk is 40:60 = 2:3.

Example 13 How many litres of fresh water should be mixed with 30 L of 50% milk solution so that resultant solution is a 10% milk solution?

Solution

Method 1 *Using Alligation*



So, the ratio of fresh water added: milk solution = 4:1

Hence, 120 L of fresh water should be added.

Method 2 *Principle of constant volume of one component*

Since we add fresh water, the volume of milk will be constant.

Now, volume of milk = 15 L = 10% of the new mixture.

So, 100% of the new mixture = 150 L

So, volume of fresh water added = 150 − 30 = 120 L.

Method 3 *Principle of inverse proportion*

We know that concentration is inversely proportional to the volume of solute added.

So, in this case $30 \times 50\% = 10\% \times (30 + x)$, where x is the volume of water added.

So, $x = 120$ L

Method 4 *Using equation*

In the final mixture, $\frac{\text{Milk}}{\text{Total}} = 10\% = \frac{15}{30 + x}$

So, $x = 120$ L

Mixing with replacement In this particular type of mixing, two or more than two substances are mixed by replacing some part of a mixture. In these types of questions, total volume may or may not be the same and information regarding the same can be obtained from the question.

Case I When the quantity withdrawn and the quantity replaced are of the same volume.

Initially, there are 40 L of milk, and 4 L of milk is replaced with 4 L of water.

Obviously, there will be 36 L of milk and 4 L of water.

Now, 4 L of mixture is replaced with 4 L of water.

The quantity of milk and water being withdrawn here will be in the ratio of 9:1 (36:4). So, quantity of milk withdrawn = 9/10 × 4 = 3.6 l.

So, the volume of milk = 32.4

And the volume of water = 7.6

Now, again 4 L of mixture is replaced with 4 L of water

The quantity of milk and water being withdrawn here will be in the ratio of 81:19 (32.4:7.6). So, the quantity of milk withdrawn = (81/100) × 4 = 3.24

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So, the volume of milk = 29.16
And the volume of water = 10.84
If we summarize the above values, then it looks like:

	1st operation		2nd operation		3rd operation	
	Taken out	Left	Taken out	Left	Taken out	Left
Milk	4	36	3.6	32.4	3.24	29.16
Water	0	4	0.4	7.6	0.76	10.84

It can be seen that the quantity of water or milk withdrawn is 10% of the existing volume of milk or water because only 10% of the total volume of 40 L taken out.

With this, we can deduce a standard formula for these kinds of calculations.

If V is the initial volume of milk (or any liquid), and x litres of milk is always replaced by water, then quantity of milk left after n such operations $= V\left(1-\frac{x}{v}\right)^n$

This formula is very similar to the standard formula we have seen in the case of compound interest $\left[p\left(1+\frac{R}{100}\right)n\right]$. The only difference between the two formulae is that while the interest is being added every year (or for the given time-period), volume of milk gets reduced after every operation.

Using the values of the above example, quantity of milk left after three operations $= 40\times\frac{36}{40}\times\frac{36}{40}\times\frac{36}{40} = 19.16$ L

The same problem can be solved with straight-line approach of percentage also Since 10% of existing volume is taken out every time, the percentage of milk in the final mixture after the third operation = 72.9%

$$\left(100\% \xrightarrow{10\%\downarrow} 90\% \xrightarrow{10\%\downarrow} 81\% \xrightarrow{10\%\downarrow} 72.9\%\right)$$

Since 100% = 40, so 72.9% = 29.16 L

Case II When the quantity withdrawn and the quantity replaced are of the same volume, but the total volume before replacement does not remain the same.

Initially, there are 40 L of milk, and 4 L of milk is taken out and 4 L of water is poured in

So, there will be 36 L of milk and 4 L of water.

Now, 5 L of mixture is taken out and 5 L of water is poured in.

The quantity of milk and water being withdrawn here will be in the ratio of 36:4. So, the quantity of milk withdrawn $= \frac{36}{40}\times 5$

Milk left $= 40\times\frac{36}{40}\times\frac{35}{40}$

Again, if now 6 L of mixture is taken out and 6 L of water is poured in,

Milk left $= 40\times\frac{36}{40}\times\frac{35}{40}\times\frac{34}{40}$

Case III When the quantity withdrawn and the quantity replaced are not of the same volume.

Initially, there are 40 L of milk, and 4 L of milk is taken out and 5 L of water is poured in.

Obviously, there will be 36 L of milk and 5 L of water.

Now, 5 L of mixture is taken out and 6 L of water is poured in then the quantity of milk and water being withdrawn will be in the ratio of 36:5. So, the quantity of milk withdrawn $= \frac{36}{41}\times 5$

Milk left $= 40\times\frac{36}{40}\times\frac{36}{41}$

Again 6 L of mixture is taken out and 7 L of water is poured in.

Therefore, the volume of milk in the final mixture $= 40\times\frac{36}{40}\times\frac{36}{41}\times\frac{36}{42}$

Example 14 Two vessels A and B of equal capacities contain mixtures of milk and water in the ratio 4:1 and 3:1, respectively. 25% of the mixture from A is taken out and added to B. After mixing it thoroughly, an equal amount is taken out from B and added back to A. The ratio of milk to water in vessel A after the second operation is:

(a) 79:21 (b) 83:17
(c) 77:23 (d) 81:19

Solution Assume there is 20 L of the mixture in both the vessels.

In vessel A, milk = 16 L and water = 4 L
25% from A to B = milk in B = 15 + 4 = 19 L
= water in B = 5 + 1 = 6 L
ratio = 19:6
Equal amount from vessel B to vessel A

$= \text{milk in } A = 12+\frac{19}{5} = \frac{79}{5}$

$= \text{water in } A = 3+\frac{6}{5} = \frac{21}{5}$

Hence, the ratio is 79:21

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Practice Exercises

Warm Up

Q.1 Find the average age of a family of five members, whose ages are 42, 49, 56, 63, and 35 years, respectively.
(a) 50 years (b) 49 years
(c) 45 years (d) 48 years

Q.2 Rahul Ghosh walks from A to B at 8 km/h and comes back from B to A at 12 km/h. What is his average speed for the entire journey?
(a) 8.8 km/h (b) 9.6 km/h
(c) 10.2 km/h (d) 11.4 km/h

Q.3 The average of five numbers is 39.20 and the average of three of these numbers is 41. Find the average of the remaining two numbers.
(a) 35.5 (b) 36.5 (c) 37.5 (d) 38.5

Q.4 In a class of 60 students, the average height of 30 students is x cm and that of the remaining students is y cm. Find the average height of the whole class.
(a) $(x + y)$ cm (b) 30 cm
(c) $\frac{x+y}{2}$ cm (d) None of these

Q.5 What is the average of the first 15 multiples of 3?
(a) 21 (b) 27 (c) 24 (d) 15

Q.6 The average age of 7 members of a family is 25. If one of the members whose age is 28 years, is excluded, then others are in the ratio 1:2:3:4:5:6. Find the age of the eldest member of the family.
(a) 28 (b) 35 (c) 14 (d) 42

Q.7 The average of seven consecutive even numbers is 10. If the next three even numbers are included, then what is the new average?
(a) 10 (b) 12 (c) 13 (d) 14

Q.8 Average weight of three students is 40 kg. If the first student is twice the second and the second student is thrice the weight of the third student, then what is the weight of the student with lightest weight?
(a) 18 (b) 20
(c) 14 (d) None of these

Q.9 In a test, the average marks of five students was found to be 50. However, during cross checking of the papers, it was discovered that the mark of one student was misread as 84 instead of 48. Find the correct average.
(a) 57.2 (b) 42.8 (c) 46.5 (d) 43.2

Q.10 The average weight of seven members of a family is 18 kg. If the head of the family is not considered, the average weight of the others would be 5 kg less. The weight of the head of the family is:
(a) 42 kg (b) 48 kg
(c) 52 kg (d) 36 kg

Q.11 In a prep school, the average weight of 30 girls in a class of 50 students is 16 kg and that of the remaining students is 15.5 kg. What is the average weight of the all the students in the class?
(a) 15.2 kg (b) 15.8 kg
(c) 15.4 kg (d) None of these

Q.12 The average of the amount of money that 10 girls have is ₹45. When two more girls join, the average increases by ₹2. Find the average amount of the two new girls.
(a) ₹57 (b) ₹54
(c) ₹48 (d) None of these

Q.13 In what ratio must 35% spirit solution be mixed with pure spirit to get a resultant solution of 56% spirit?
(a) 35:44 (b) 44:21 (c) 8:3 (d) 56:9

Q.14 A beggar has 50 coins with an amount of ₹15 in the denominations of 25 paise and 50 paise. How many 25 paise coins are there with the beggar?
(a) 36 (b) 40 (c) 30 (d) 25

Q.15 Two litres of pure spirit is added to 6 L of a spirit solution containing $16\frac{2}{3}\%$ spirit. What is the concentration of the spirit in the resultant solution?
(a) 12.5% (b) 25%
(c) 37.5% (d) None of these

Q.16 Three litres of pure milk is added to 10 L of a milk solution containing 9% milk. Find the concentration of the resultant solution.
(a) $33\frac{1}{3}\%$ (b) 40%
(c) 50% (d) 30%

Q.17 In what ratio must a quality of sugar @ ₹15.50/kg be mixed with another quality of sugar @ ₹21.80/kg so that selling the mixture at ₹20.02/kg results in 10% profit?
(a) 5:4 (b) 3:2 (c) 2:1 (d) 4:3

Q.18 Parul Jaiswal has ₹11,200 in ₹100 and ₹500 denomination. What is the ratio of the number of notes, of the two types, if she has a total of 40 notes with her?
(a) 3:2 (b) 7:3 (c) 8:7 (d) 11:9

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Q.19 What is the average of all the odd numbers in between 50 and 80?
(a) 66 (b) 64
(c) 65 (d) 67

Q.20 A shrewd milkman mixes water and milk in the ratio of 2:3. What part of this mixture should be removed and replaced with water so that the solution contains water and milk in the ratio 1:1?
(a) 1/6 (b) 1/4 (c) 1/3 (d) 1/2

Q.21 A milkman sells his mixture of milk and water at the cost price of milk itself and thus he gains a profit of 20%. What is the ratio of milk and water in the mixture?
(a) 4:1 (b) 5:1
(c) 1:4 (d) None of these

FOUNDATION

Q.1 Neo is going to NDLS from PNBE by his car at a speed of 40 km/h. While coming back, he returns with a speed of x km/h. What should be the value of x so that his average speed during the intrez journey is 80 km/h?
(a) 160 km/h (b) 40 km/h
(c) 120 km/h (d) It is not possible

Q.2 There are 125 middle level employees in Due North Inc. The average monthly salary of the middle level employees is ₹5500 and that of the senior level employees is ₹14,000. If the average monthly salary of all these employees is ₹8687.5, find the total number of employees in the company, if middle and senior level employees of Due North Inc. form 80% of their total employees.
(a) 175 (b) 200 (c) 220 (d) 250

Q.3 I went to a hotel along with 12 friends. I paid ₹145 and all the others paid an equal amount. In the end, we did some calculations and found that the average sum paid by all of us was ₹5 more than what was originally paid by each of my friends. How much money did each of my friends pay?
(a) ₹120 (b) ₹100
(c) ₹80 (d) ₹70

Q.4 Saketprakash married ten years ago at the age of 27 years. His wife was 23 years old then. Six years after their marriage, the average age of Saketprakash, his wife and their son was 22 years. After how many years of Saketprakash's marriage was his son born?
(a) 6 years (b) 3 years
(c) 2 years (d) 4 years

Q.5 Mr Anant Roy, the renowned author, recently got his new novel released. To his utter dismay, he found that for the 1007 pages on an average there were two mistakes every page. While, in the first 612 pages there were only 434 mistakes, they seemed to increase for the latter pages. Find the average number of mistakes per page for the remaining pages.
(a) 6 (b) 4
(c) 2 (d) None of these

Q.6 The class X of Nav Yug Vidhyalaya has four sections—A, B, C and D. The average weight of the students of A, B, C together and A, C, D together are 45 kg and 55 kg, respectively, while the average weight of the students of A, B, D together and B, C, D together are 50 kg and 60 kg, respectively. Which of the following could be the average weight of the students of all the four sections together?
(a) 47.6 kg (b) 52.5 kg
(c) 53.7 kg (d) 56.5 kg

Q.7 The average score of a cricketer in a certain number of innings is 44. He then played another eight innings in which he got out at 97, 3, 23, 0, 68, 40, 50, and 71 runs, respectively. Then, which of the following statements is true regarding his new average score in all the innings together?
(a) The new average is more than the old average
(b) The new average is less than the old average
(c) The new average is the same as the old average
(d) Cannot conclude any of the above unless we know the number of innings played

Q.8 Average age of A, B, and C is 84 years, however, when D joins them, then the average comes down to 80. Now, a new person E whose age is 3 years more than D replaces A and the new average is 79 years. What is the age of A?
(a) 75 (b) 79 (c) 81 (d) 77

Direction for Questions 9 and 10: Read the passage below and solve the questions based on it.

The average age of the students in a class is 35 years.

Q.9 If a student whose age was 25 is absent from the class, the average of those present goes up by 1. How many students were there in the class originally?
(a) 9 (b) 10 (c) 11 (d) 12

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Q.10 If two students whose ages were 30 years and 20 years left, by what value would the average of the remaining increase? (Use the result from Question 9.)

(a) $\frac{20}{9}$ (b) 2

(c) $\frac{20}{11}$ (d) None of these

Q.11 If the average of m numbers is a, and on adding x to the m numbers, the average of the $1 + m$ numbers is b. Then find the value of x.

(a) $m(b-a)+b$ (b) $m(b+a)+a$

(c) $m(a-b)+a$ (d) None of these

Q.12 Brian Lara, the famous batsman, scored 6000 runs in certain number of innings. In the next five innings, he was out of form and hence could make only a total of 90 runs, as a result of which his average fell by 2 runs. How many innings did he play in all, if he gets out in all the innings?

(a) 105 (b) 95 (c) 115 (d) 104

Q.13 The average weight of A, B, and C is x kg. A and C lose y kg each after dieting and B gains $\frac{y}{2}$ kg. After this their average weight decreases by 1 kg. Find y.

(a) 1 kg (b) 2 kg

(c) 3 kg (d) Cannot be determined

Q.14 One out of the five weights A, B, C, D, and E is of a different weight. A person makes a measurement and finds the following: $A + B = C + D + E$

Which of the following should be known to find out which one is of a different weight?

(a) $A + C > B + D + E$

(b) $A + D + E < B + C$

(c) $A + B + C > D + E$

(d) $A + C < B + D + E$

Q.15 In a class, with a certain number of students, if another student weighing 50 kg joins, then the average weight of the class increases by 1. If one more student weighing 50 kg joins then the average weight of the class increases by 1.5 over the original average. What is the original average weight of the class?

(a) 46 (b) 4 (c) 2 (d) 47

Q.16 The average weight of the books carried by Kuku to school is 1.5 kg and 2.5 kg is the average weight of the exercise books. If Kuku is carrying only text books and exercise books in the ratio of 3:4, what is the total weight of his school bag? (Ignore any other weights such as that of school bag, stationary, etc.)

(a) 10.5 kg (b) 1 kg

(c) 6 kg (d) Data insufficient

Q.17 The average of n numbers is 32. If 3/4 of the numbers are increased by 4 each and the remaining are decreased by 6 each, then what is the new average?

(a) 32 (b) 32.5 (c) 33.5 (d) 34.5

Q.18 The average of eight numbers is 25, that of the first two is 20 and of the next three is 26. The sixth number is less than the seventh by 4, and less than the eighth by 6. The last number is:

(a) 30 (b) 32 (c) 40 (d) 36

Q.19 The average amount of sales per day in a shop for 5 days starting from Monday is ₹1800. The average amount of sales per day for 5 days from Tuesday is ₹1890. By how much does the sale on Saturday exceed that on Monday?

(a) ₹450 (b) ₹500

(c) ₹600 (d) ₹650

Q.20 Three years ago, the average age of a family of five members was 17 years. Inspite of the birth of a child in the family, the present average age of the family remains the same. The present age of the child is:

(a) 3 years (b) 1 year

(c) 2 years (d) 1.5 years

Q.21 A batsman has a certain average of runs for 16 innings. In the 17th inning, he makes a score of 85 runs, thereby increasing his average by 3 runs. What is the average after the 17th inning?

(a) 58 (b) 37

(c) 35 (d) None of these

Q.22 If the average marks of 'a' students in a class is 'c' and that of the remaining 'b' students is 'd', what is the average marks of the class?

(a) $\frac{ac+bd}{b+d}$ (b) $\frac{ab+cd}{a+d}$

(c) $\frac{ac+bd}{a+b}$ (d) $\frac{ad+cd}{b+d}$

Q.23 The average marks of a class of 48 students is 35. Of them, two score zero, of the rest, the first 30 scored an average of 40, the next fourteen scored an average of 20. If the remaining two scored equal marks, what are their individual marks?

(a) 80 (b) 75 (c) 90 (d) 100

Q.24 The total salary of A, B, and C is ₹444. If they spend 80%, 85%, 75% of their salaries, respectively, their savings are as 7:6:9. Then, salary of B is:

(a) ₹140 (b) ₹160

(c) ₹144 (d) None of these

Q.25 The average age of two-third of the class is 17. What should be the average the age of the remaining

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one-third students so that the average age of the entire class is 20?
(a) 21 (b) 22 (c) 24 (d) 26

Q.26 Sidharth Tripathi travels from Delhi to Lucknow at a speed of 75 km/h and returns to Delhi at a speed of 50 km/h. He again leaves for Lucknow immediately at a speed of 30 km/h and returns back to Delhi at a speed of 60 km/h. What is his average speed for the entire journey?
(a) 54 km/h (b) 48 km/h
(c) 56 km/h (d) Cannot be determined

Q.27 Average marks of a class of n students is 64. When eight new students with an average marks of 73 join the class, the new average of the entire class is a whole number. Find the number of students in the class given that n lies between 25 to 60.
(a) 44 (b) 32 (c) 36 (d) 72

Q.28 Average of five consecutive integers is N. What will be the new average when next two integers are also included?
(a) $N + 2$
(b) $N + 2/7$
(c) $N + 1$
(d) $N + K/7$, where K depends upon the values taken.

Q.29 The average of 10 two-digit positive integers is Z. However, one number AB is taken as BA, then the average increases to $Z + 2.7$. What is the value of $|B - A|$?
(a) 1 (b) 2 (c) 3 (d) 4

Q.30 In the above question, how many set of values for A and B are possible?
(a) 5 (b) 6 (c) 7 (d) 8

Q.31 In a country, the average salary per month is calculated on the basis of 14 months and average expenses is calculated on the basis of 10 months. What will happen to the average savings per month with respect to actual savings (calculated on the basis of 12 months)?
(a) Decrease (b) Increase
(c) No change (d) Cannot be determined

Q.32 In the above question, what will be the net percentage change in the savings with respect to the actual savings?
(a) 11.11% (b) 22.22%
(c) 33.33% (d) Cannot be determined

Q.33 The average rainfall for the first three days out of five days was recorded to be 0.45 inches. The rainfall on the last two days was in the ratio 2:3. The average of five days was 0.40 inches. What was the rainfall on the last day?
(a) 0.45 inches (b) 0.39 inches
(c) 0.35 inches (d) 0.30 inches

Q.34 The average height of the girls of a class is 155 cm and the average height of the boys of the class is 4 cm more than the average of the class. If the number of girls is 20% less than the number of boys, what is the average height of the boys?
(a) 140 cm (b) 162 cm
(c) 152 cm (d) None of these

Q.35 The average price of a share is the average of five readings taken at regular intervals in a day. The index price is taken by a weighted arithmetic average price of a class A and class B stock. The respective weights are 1.1 and 0.9 for the two kinds of stocks. If the five readings of a class A stock were 19, 26, 31, 35, 39 and for a class B stock the readings were 7, 8, 17, 20, 23 then what was the index price that day?
(a) 46.5 (b) 25 (c) 23.25 (d) 45

Q.36 Ankit travels from New Delhi to Patna. He stops at equidistant places, that is Lucknow and Mirzapur during his journey. He covers the first one-third distance at a speed of 50 km/h, the second one-third distance at a speed of 40 km/h and the last one-third distance at a speed of 60 km/h. However, while returning, his uniform speed is 75 km/h and he stops only at Jaunpur, which is exactly at the mid-point between New Delhi and Patna. What is the average speed (in km/h) of Ankit during the whole journey, not considering his halt time?
(a) 59.8 (b) 59.01
(c) 60.8 (d) None of these

Q.37 The average age of 24 students and the class teacher is 16 years. If the age of the class teacher is excluded the average reduces by 1 year. What is the age of the class teacher?
(a) 38 (b) 39
(c) 40 (d) None of these

Q.38 First ten multiples of 1, 2,… 10 are taken. What is the average of all these 100 numbers?
(a) 25.5 (b) 50.5
(c) 75.5 (d) None of these

Q.39 There are 20 students in Mr Rahul Ghosh's class. He conducts an examination out of 100 and then arranged the marks in an ascending order. He found Chandan, the topper of the class, had slipped to the tenth position. When he was adding the scores of the last 11 students the average was 64 and that of the top 10 was 67. If the average marks obtained by all the students of his class was 65, how many marks did Chandan score?
(a) 64 (b) 67
(c) 74 (d) Cannot be determined

Q.40 The average of 38, 84, 63, 45, 32 and n lies between 50 and 60. If n is an integer greater than the average of the above numbers, then n lies between:

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(a) $53 < n < 97$ (b) $50 < n < 100$
(c) $52 < n < 98$ (d) None of these

Q.41 Milawati Sharma, the local grocer mixed three qualities of tea T_1, T_2, and T_3 priced at ₹74 per kg, ₹68 per kg and ₹63 per kg in the ratio of 1:2:4. He added some more quantity of T_1 to 4 kg of this mixture. He sold this new mixture for ₹84 per kg, thereby making a profit of 20%. How much of T_1 did he mix with the mixture?
(a) 4 kg (b) 6 kg
(c) 8 kg (d) None of these

Q.42 Sixty litres of a 75% solution of wine is taken into a laboratory. Four L of the solution is flushed and 4 L of water is added to compensate the loss. Then, 5 L of the solution is flushed again and 5 L of water is added. Again 6 L of the solution is flushed and 6 L of water is added. The final concentration of wine in the solution is:
(a) 255% (b) 57.75%
(c) 60.25% (d) 59%

Q.43 *A* and *B* are two alloys of gold and copper prepared by mixing the respective metals in the proportions of 7:2 and 7:11, respectively. If the alloys are melted and mixed to form a third alloy *C*, which has an equal proportion of gold and copper. What is the ratio of alloys *A* and *B* in the mixture?
(a) 2:5 (b) 5:2 (c) 11:2 (d) 1:2

Q.44 A vessel contains a mixture of two liquids *A* and *B* in the ratio of 4:1. When 10 L of the mixture is replaced with liquid *B*, the ratio becomes 2:3. How many litres of liquid *A* was present in the jar initially?
(a) 12 L (b) 14 L
(c) 16 L (d) 10 L

Q.45 From a vessel containing pure milk, 9 L is taken out. It is replaced with an equal quantity of water. This operation is repeated twice. The ratio of milk and water now in the vessel is 16:9. What was the volume of milk in the vessel initially?
(a) 54 L (b) 36 L
(c) 42 L (d) 45 L

Q.46 There are only deers and ducks in a zoo. When heads are counted, it is 204 and when the legs are counted, it is 512. How many deers are there in the zoo?
(a) 52 (b) 152 (c) 176 (d) 26

Q.47 Sharat invests a total of ₹10,000 in two parts, some money @ 6% per annum and rest of the money @ 8% per annum of SI. After two years, he got a total of ₹1440 as interest. What is the amount put at 6% per annum interest?
(a) ₹4000 (b) ₹6000
(c) ₹3000 (d) ₹7000

Q.48 *A* vessel contains a mixture of two liquids *A* and *B* in the ratio of 7:5. When 9 L of the mixture is taken out and the vessel is filled with *B*, the ratio of *A* and *B* is now 7:9. How many litres of liquid *A* was there in the vessel initially?
(a) 24 L (b) 21 L
(c) 18 L (d) 27 L

Q.49 A milkman dilutes 36 L of pure milk with water. The percentage of milk in the solution is now 75%. How many litres of water did he add?
(a) 6 L (b) 8 L
(c) 12 L (d) 16 L

Q.50 Doodhimal, the local milkman was notorious for the quality of milk that he used to sell. Initially he had 60 L of milk in his can. He removed one-third of it and replaced that with water. He then removed one-third of the mixture and again added water to it. How many times should this process be repeated if Doodhimal wants the concentration of milk to be just below 20%?
(a) 3 (b) 4 (c) 5 (d) 7

Moderate

Q.1 The average marks of the girls in a class is equal to the number of boys and the average marks of boys is equal to the number of girls. If the class average is 4 less than the average of both the boys' and the girls' average marks, what will be the number of students in the class?
(a) 24 (b) 48
(c) 50 (d) 64

Q.2 In the above question, what can be the minimum number of students in the class?
(a) 0 (b) 2
(c) 8 (d) None of these

Q.3 *A*, *B*, *C*, *D*, and *E* are five consecutive integers and the average of these five numbers is less than $\frac{1}{4}$ th of *A*. Then, *A* is:
(a) Positive (b) Negative
(c) Even (d) Data Sufficient

Q.4 Find the difference between the average of the first *n* even numbers and the average of all the even numbers up to *n* (*n* is an even number).
(a) $\frac{n-1}{2}$ (b) $\frac{n+1}{2}$
(c) $\frac{n}{2}$ (d) None of these

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Q.5 I was born 30 years after my father was born. My sister was born 25 years after my mother was born. The average age of my family is 26.25 years right now. My sister will get married 4 years from now and will leave the family. Then, the average age of the family will be $\frac{107}{3}$ years. What is the age of my father?

(a) 30 years (b) 35 years
(c) 40 years (d) 45 years

Direction for Questions 6 to 9: Read the passage below and solve the questions based on it.

India is supposed to send its boxing team for Olympics in each of the following 10 weight group divisions.

A (48 kg – 52 kg) B (52 kg – 56 kg)
C (56 kg – 60 kg) D (60 kg – 64 kg)
E (64 kg – 68 kg) F (68 kg – 72 kg)
G (72 kg – 76 kg) H (76 kg – 80 kg)
I (80 kg – 84 kg) J (84 kg – 88 kg)

After selecting one player from each group, their average weight comes out to be 68 kg. If one of the players named *X* leaves the team, their average weight comes down to 66.5 kg.

Q.6 Player *X* is from group:
(a) *A* (b) *E*
(c) I (d) None of these

Q.7 If *X* leaves the team, and two new players join the group, then their average weight increases to 68 kg. These players can be from group:
(a) *A*, *C* (b) *D*, *J*
(c) Both from *G* (d) Either 2 or 3

Q.8 What is the average weight (in kg) of all the players taken together?
(a) 56.7 (b) 58.8
(c) 61.4 (d) Cannot be determined

Q.9 In the average of all the groups together, which group contributes most in overall average?
(a) F (b) G
(c) H (d) Cannot be determined

Q.10 A shipping clerk has to weigh six distinct packets. He weighs them four at a time, weighing all the possible combinations of the packets from the six. The average weight of all the weighing combinations is found to be 500 g. What is the combined weight of all the six packets?
(a) 375 g (b) 1250 g
(c) 750 g (d) 1000/3 g

Direction for Questions 11 to 21: Read the passage below and solve the questions based on it.

There are three sections in class *X* at A. R. Academy. Below is the table indicating the number of students, and the range of marks obtained by the students of that particular section.

	Section *A*	Section *B*	Section *C*
No. of students	25	30	20
Minimum marks	22	24	20
Maximum marks	30	28	26

The minimum and the maximum marks provides us with only the range of marks which can be obtained by the students of that particular section.

Q.11 What is the average marks obtained by all the students of class *X* together?
(a) 24.25 (b) 26.33
(c) 26.75 (d) Cannot be determined

Q.12 If we shift five students from section *A* to section *B* and then '*x*' students from section *B* to section *C*, then what should be the value of '*x*' so that the minimum possible average of section *B* and the maximum possible average of section *C* are the same?
(a) 6 (b) 7
(c) 9 (d) Not possible

Q.13 If we combine section *A* and section *C*, then what will be the ratio of the maximum possible average of this combined section and the average of section *B*?
(a) 2:1 (b) 3:2
(c) 1:2 (d) Cannot be determined

Q.14 What is the maximum number of students that can be transferred from section *A* to section *C*, so that the maximum possible average of section *A* is still more than the maximum possible average of section *C*?
(a) 20 (b) 22
(c) 24 (d) None of these

Q.15 What can be the maximum possible average of class *X* when all the three sections are combined?
(a) 26.33 (b) 28.13 (c) 27.28 (d) 28.5

Q.16 We take a sample size of the following numbers from different sections:

	Section *A*	Section *B*	Section *C*
Total no. of students selected	10	10	10

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	Section A	Section B	Section C
No. of students of minimum marks	5	6	4
No. of students of maximum marks	5	4	6

What is the average of this sample size?
(a) 25.06 (b) 26.16 (c) 25.83 (d) 26.71

Q.17 In the above question, which section has contributed the maximum in the average of the whole sample size?
(a) A
(b) B
(c) C
(d) Two sections contributed equally

Q.18 If we shift five students from section A to section B and then 10 students from section B to section C, what will be the maximum average of section C?
(a) 27 (b) 26.66 (c) 26 (d) 27.33

Q.19 In the above question, what can be the minimum average of section C?
(a) 20 (b) 21 (c) 21.33 (d) 22

Q.20 10 students are shifted from each of the sections A and C to section B and then 10 students each are shifted from section B to each of the section A and section C. Now what is the ratio of maximum possible average of section A and section C?
(a) 1:1 (b) 15:14
(c) 14:15 (d) Cannot be determined

Q.21 In the above question, what is the ratio of the minimum possible average of section A and section C?
(a) 11:12 (b) 12:11
(c) 1:1 (d) Cannot be determined

Q.22 The average of five positive numbers is A. The average of the average of all the possible distinct triplets of these five numbers is B. Which of the following is true?
(a) $A > B$ (b) $A < B$
(c) $A = B$ (d) Cannot be determined

Q.23 The average of five positive numbers is A. The average of the average of all the possible distinct pairs of these five numbers is B. Which of the following is true?
(a) $A > B$ (b) $A < B$
(c) $A = B$ (d) Cannot be determined

Direction for Questions 24 to 26: Read the passage below and solve the questions based on it.

The following operators have been defined for two numbers A and B.

@(A, B) = Average of A and B
*(A, B) = Product of A and B
/(A, B) = A divided by B

Q.24 If $A = 2$ and $B = 4$, then the value of @ (/(*(A, B), B), A) would be:
(a) 2 (b) 4 (c) 6 (d) 16

Q.25 The sum of A and B is given by:
(a) *(@ (A, B), 2) (b) /(@ (A, B), 2)
(c) @(*(A, B), 2) (d) @ (/(A, B) 2)

Q.26 The sum of A, B, and C is given by:
(a) * (@ (* (@ (B, A), 2), C), 3)
(b) * (@ (* (@ (B, A), 2), C), 2)
(c) /(* (@ (* (B, A), 2), C), 3)
(d) None of these

Q.27 The average age of Mr and Mrs Sinha at the time of their marriage in 1972 was 23 years. On the occasion of their anniversary in 1976, they observed that the average age of their family had come down by 4 years compared to their average age at the time of their marriage. This was due to the fact that their son Vicky was born during that period. What was the age of Vicky in 1980?
(a) 6 (b) 7 (c) 8 (d) 5

Q.28 A man makes 60 articles in the first hour. His efficiency decreases by 25% in the second hour, increases by 40% in the third hour, decreases by 33% in the fourth hour and increases by 50% in the fifth hour. If he has to work for more than one hour, then in which hour the average number of articles produced per hour then would be minimum?
(a) Second hour (b) After fifth hour
(c) Third hour (d) None of these

Q.29 Sachin Tendulkar has a certain batting average N (a whole number) in his career of 86 innings. In the 87th inning, he gets out after scoring 270 runs which increases his batting average by a whole number. The batting average is calculated by dividing the total number of runs scored by the total number of innings played by the player. How many values of his new average is/are possible?
(a) 0 (b) 1
(c) 2 (d) None of these

Q.30 In a school, there are two sections for class X – section A and section B. The number of students in section A and section B are 60 and 70, respectively, and their respective average weight is 35 kg and 38 kg. If one of the students from section A is shifted to section B then the average weight of both the sections decreases. What can be said about the weight of that student?

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(a) His weight is less than 35 kg.
(b) His weight is more than 38 kg.
(c) His weight is more than 35 kg and less than 38 kg.
(d) Cannot be determined

Direction for Questions 31 and 32: Read the passage below and solve the questions based on it.

The average score of a batsman for a certain number of innings was 21.75 per inning. He played 3 innings more and scored 28, 34 and 37 runs, respectively, thus increasing his average by 1.25.

Q.31 How many innings did he play in all?
(a) 18 (b) 21
(c) 27 (d) None of these

Q.32 What was his total score?
(a) 520 (b) 601
(c) 621 (d) None of these

Q.33 A curious student of statistics calculated the average height of all the students of his class as A. He also calculated the average of the average heights of all the possible pairs of students (two students taken at a time) as B. Further, he calculated the average of the average heights of all the possible triplets of students (three students taken at a time) as C. Which of the following is true of the relationship among A, B, and C?
(a) $A + 2B = C$ (b) $A + B = 2C$
(c) $A = B = 3C$ (d) None of these

Q.34 A succession of numbers is said to be in the Arithmetic Progression (AP) if the difference between any given term and the previous term is constant throughout. In a AP $n_1, n_2, n_3, \ldots, n_{46}$, the sum of odd numbered terms $n_1, n_3, n_5, \ldots, n_{45}$, is 1272. What is the sum of all the 46 terms of this AP?
(a) 2491 (b) 2500
(c) 2400 (d) Cannot be determined

Q.35 There are two vessels of milk of different prices with the volume of 220 L and 180 L, respectively. Equal amounts of milk were poured off simultaneously from the two vessels and the milk poured off from the first vessel was poured into the second vessel and the milk poured off from the second vessel was poured into the first vessel. Then, the price of milk in both the vessels becomes the same. How much milk was poured from one vessel into the other?
(a) 90 (b) 80 (c) 160 (d) 99

Q.36 There are two solutions of sulphuric acid (acid + water) with concentration of 50% and 80%, respectively. They are mixed in a certain ratio to get a 62% sulphuric acid solution. This solution is mixed with 6 L of water to get back 50% solution. How much of the 80% solution has been used in the entire process?
(a) 15 L (b) 12 L
(c) 10 L (d) None of these

Q.37 Two-fifths of the volume of the mixture of the milk and water is of the ratio of 4:3 is converted into a mixture of the ratio of 5:3 by the method of replacement with milk. The 4:3 mixture was prepared from a 4:1 mixture by the method of 'addition of the substances of the mixture'. If the replacement volume is 14 L, what is the volume of the substance added?
(a) 12 L (b) 60 L (c) 80 L (d) 24 L

Q.38 Rakesh has a stock of 1000 L of pure milk. He decides to decrease the concentration to half, using the method of successive dilution, replacing 200 L of contents by water, in each step of dilution. After the fourth dilution, he realizes that the concentration is less than the required value. Each of the contents removed during the dilution process is stored separately. He also has ample supply of water. Will he be able to make the mixture of desired concentration?
(a) Not possible
(b) Possible
(c) Cannot be determined
(d) None of these

Q.39 An industrial solvent of 90% strength is prepared and stored in a 150 L capacity container. The container is filled to 80% of its capacity. It is required to decrease the concentration of the solvent in the container to less than 60%. For doing this, either pure water can be added to it and/or solution can be removed. But removal can only be done if there is no place for adding water. At a given time, only 10 L of the solution can be added or removed. How many times should water be added to the solution?
(a) 4 (b) 5 (c) 9 (d) 6

Q.40 Four gallons are drawn from a case full of wine. It is then filled with water. Four gallons of mixture are again drawn and the cask is re-filled with water. The ratio of the quantity of wine now left in the cask to that of the mixture in it is 36:49. How much does the cask hold?
(a) 30 gallons (b) 25 gallons
(c) 35 gallons (d) 28 gallons

Q.41 There is a vessel holding 40 L of milk. Four litres of milk is initially taken out from the vessel and 4 L of water is then poured in. After this, 5 L of mixture is replaced with 5 L of water. And finally 6 L of the mixture is replaced with 6 L of water. How much milk (in litres) is there in the vessel?
(a) 26.775 (b) 29.16
(c) 24.72 (d) 27.42

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Q.42. There is a vessel holding 40 L of milk. Four litres of milk is initially taken out from the vessel and 5 L of water is poured in. After this, 5 L of the mixture from this vessel is replaced with 6 L of water. And finally 6 L of mixture from this vessel is replaced with 7 L of water. How much of the milk (in litres) is there in the vessel now?

(a) 22.42 (b) 27.09
(c) 24.72 (d) 29.42

Q.43 There is a vessel holding 40 L of milk. Four litres of milk initially is taken out from the vessel and 5 L of water is poured in. After this, 6 L of mixture from this vessel is replaced with 7 L of water. And finally 8 L of mixture from this vessel is replaced with 9 L of water. How much of the milk (in litres) is there in the vessel now?

(a) 22.82 (b) 20.92
(c) 26.78 (d) 24.87

Q.44 There are three vessels A, B and C having a mixture of milk and water. Concentration of milk in the vessels is C_1, C_2, and C_3, respectively. When a part of vessel A is replaced with the mixture in vessel B, then the concentration of milk in the final mixture is less than C_1. However, when vessels B and C are mixed in the ratio of 1 : 1, then the quantity of milk in the final mixture is double than the quantity of milk in the vessel B. Which of the vessels is having the mixture of maximum concentration of milk?

(a) A (b) B
(c) C (d) Cannot be determined

Q.45 In the above question, which of the vessels is having the mixture of minimum concentration of milk?

(a) A (b) B
(c) C (d) Cannot be determined

Q.46 In question number 44, if we arrange the concentration of milk in different vessels in ascending order, then which of the following is true?

(a) ABC (b) BAC
(c) CAB (d) Cannot be determined

Q.47 In two alloys, the ratios of copper to zinc is 5 : 2 and 3 : 4. How many kilograms of the first alloy and the second alloy, respectively, should be melted together to obtain 28 kg of a new alloy with equal copper and zinc?

(a) 8 kg and 20 kg (b) 4 kg and 24 kg
(c) 3 kg and 25 kg (d) 7 kg and 21 kg

Q.48 From a 3 : 5 solution of milk and water, 20% is taken out and replaced by milk. How many times should this process be done to make the ratio of milk to water as 17 : 8?

(a) Once (b) Twice
(c) Thrice (d) Four times

Advanced

Direction for Questions 1 and 2: Read the passage below and solve the questions based on it.

In the national baseball league (NBL), the batting average (AB) of a hitter is computed from the home runs scored, innings played, and completed innings and incomplete innings.

R_1 = Number of home runs scored in complete innings

N_1 = Number of complete innings

R_2 = Number of home runs scored in incomplete innings

N_2 = Number of incomplete innings

$$AB = \frac{R_1 + R_2}{N_1}$$

To accurately assess a hitter's accomplishment, the NBL has devised two other statistical computations called the quick average (AQ) and the comprehensive average (AC), which are defined as follows:

$$AQ = \frac{R_1}{N_1} + \frac{N_2}{N_1} \max\left[0, \left(\frac{R_2}{N_2} - \frac{R_1}{N_1}\right)\right]$$

$$AC = \frac{R_1 + R_2}{N_1 + N_2}$$

Q.1 Based on the information, which of the following is true?

(a) $AQ \le AB \le AC$ (b) $AB \le AC \le AQ$
(c) $AC \le AB \le AQ$ (d) None of these

Q.2 An experienced hitter with no incomplete innings has AB of 50. The next time he bats, the inning is incomplete and he scores 45 runs. It can be inferred that

(a) AB and AQ will increase, while AC will decrease.
(b) AB will increase, while AC will decrease.
(c) AB will increase and we cannot comment on the changes in AQ and AC.
(d) None of these

Q.3 If the average marks of 17 students in a class is A. The marks of the students when arranged in either an ascending or a descending order was found to be in arithmetic progression. The class teacher found that the students who were ranked 3rd, 7th, 9th, 11th, and

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15th had copied in the exam and hence, got all of them rusticated. The average of the remainder of the class was *B*. Then

(a) $A = B$ (b) $A > B$
(c) $A < B$ (d) Data insufficient

Q.4 The average weight of the students in four sections *A*, *B*, *C*, and *D* is 60 kg. The average weights of the students of *A*, *B*, *C*, and *D* individually are 45 kg, 50 kg, 72 kg, and 80 kg, respectively. If the average weight of the students of section *A* and *B* together is 48 kg and that of the students of *B* and *C* together is 60 kg, what is the ratio of the number of students in sections *A* and *D*?

(a) 4:3 (b) 3:2 (c) 8:5 (d) 12:7

Q.5 Illara, Puttu, and Munchun had three bottles containing equal amounts of orange juice, mango juice, and water, respectively. Since Illara and Puttu wanted to dilute their drinks, Munchun gave one-fourth of the water, that she had, to each of them. After mixing their bottles well, each of them gave to Munchun the same quantity of the solution that they had borrowed from her. Which of the following statements is true?

i. The concentration of water in Munchun's drink is more than the concentration of orange juice in Illara's drink.
ii. The concentration of orange juice in Illara's drink is more than the concentration of mango juice in Puttu's drink.
iii. The concentration of orange is more than the concentration of mango in Munchun's drink.

(a) i and ii (b) i, ii, and iii
(c) i, ii, and iii (d) None of these

Q.6 We write down all the digits from 1–9 side by side. Now, we put '+' between as many digits as we wish to, so that the sum of numbers become 666. It is explained below:

1 2 3 4 5 6 7 8 9 = 666

Now suppose we put plus signs at following places:

12 + 345 + 67 + 89 = 513

Since there are four numbers, so the average can be calculated by dividing the sum by 4. What is the average if the sum is 666?

(a) 166.5 (b) 111
(c) 133.2 (d) Cannot be determined

Q.7 In the above question, what is the average if the sum is 261?

(a) 54 (b) 52.2
(c) 55 (d) None of these

Direction for Questions 8 to 11: Read the passage below and solve the questions based on it.

Go through the following definitions:

$A + B$ = All *A* are *B*
$A * B$ = Some *A* are *B*
$A @ B$ = No *A* are *B*
$A \# B$ = Only *A* are *B*

Following table gives the average age break-up of all the students at IIM-A (2005 batch):

Group of students	Average age (years)	No. of persons in the group
Males M	23	140
Females F	22	100
Engineers E	25	160
Fresher R	20	80
Working Professionals W	27	160

Assume that all the students mentioned above are either males or females. The minimum entry age in the IIM is 20 years (at the time of taking the entrance exam to qualify for IIM-A). No fresher can be a working professional and males/females/engineers can be either a fresher or a working professional. Everybody is either a fresher or a working professional.

Q.8 Following statement is definitely true: $M + E$. What is the average age (in years) of all the engineers who are not male?

(a) 39 (b) 40
(c) 38 (d) Cannot be determined

Q.9 Total number of working professionals is divided in the ratio of 3:2 among males and females. If $A * B$, then what is the range of the average age (x) of all the males who are engineers?

(a) $32 \le x \le 38$ (b) $39 \le x \le 43$
(c) $44 < x \le 49$ (d) Cannot be determined

Q.10 $R + M$ and $F + W$ are definitely true. What is the average age (in years) of all the males who are working professionals?

(a) 35 (b) 35.33
(c) 36 (d) None of these

Q.11 Consider the following statements – $F @ E$. What is the average age (in years) of all the working females?

(a) 23 (b) 24
(c) 24.5 (d) Data inconsistent

Direction for Questions 12 to 14: Read the passage below and solve the questions based on it.

Ten different samples of milk solution— *A, B, C, D, E, F, G, H, I,* and *J* are taken and the concentration of each sample (total

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quantity of milk as a percentage of total quantity of solution) was found out and the results were tabulated as follows:

Sample	A	B	C	D	E	F	G	H	I	J
Concentration (%)	78	57	82	84	98	66	34	87	79	71

Q.12. If two samples of milk are taken to form a sample whose concentration is more than 80%, then a maximum of how many distinct pairs of samples are there, which will never give the desired result for any ratio of their volumes?
(a) 30 (b) 48 (c) 12 (d) 15

Q.13 A triplet is a set of three distinct samples taken in some ratio of their volumes. How many such distinct triplets are there, which when mixed can give a sample with more than 80% concentration?
(a) 50 (b) 100 (c) 150 (d) 200

Q.14 There are how many distinct samples such that if two or mote than two such samples are mixed in any ratio, the resulting solution is always a sample of more than 80% concentration?
(a) 4 (b) 6 (c) 8 (d) 10

Q.15 There are three containers C_1, C_2, and C_3 having milk and water solutions in the ratio of $a:b$, where $a \in (1, 2, 3)$ and $b \in (4, 5, 6)$. Containers C_1, C_2 and C_3 can take any value of a and b. For how many different set of values of concentration of containers C_1, C_2, and C_3, the resultant mixture will have the ratio of milk and water as 1:1?
(a) 12 (b) 0
(c) Infinite (d) None of these

Q.16 In the above question, $a \in (1, 2, 6)$ and $b \in (4, 5, 3)$. Containers C_1, C_2, and C_3 can take any value of a and b. For how many different set of values of concentration of containers C_1, C_2, and C_3, the resultant mixture will have the ratio of milk and water as 1:1?
(a) 12 (b) 0
(c) Infinite (d) None of these

Q.17 The average weight of a class of 25 students is 30 kg. The average weight of the girls is 5 kg more than that of the boys. If the class teacher's weight, which is between 64 kg and 106 kg, is included, the average weight of the male members of the class equals that of the female members. If the average weight of the boys (in kg) is an integer, then what is the number of girls in the class?
(a) 20 (b) 15
(c) 10 (d) Cannot be determined

Q.18 A succession of numbers is said to be in arithmetic progression (AP) if the difference between any given term and the previous term is constant throughout. In an AP $n_1, n_2, n_3, \ldots, n_{47}$, the sum of odd numbered terms $n_1, n_3, n_5, \ldots, n_{47}$ is 1272. What is the sum of all the 47 terms of this AP?
(a) 2491 (b) 2500 (c) 2400 (d) 2591

Direction for Questions 19 to 21: Read the passage below and solve the questions based on it.

In the entrance examination of IIMs, there were 200 questions, each of which carried the same marks. *A* correct answer gets 2 marks and there is 100% negative marking. *A* total of 70 candidates took the exam and it was later found that the average marks obtained by these 70 candidates was 240. The candidates were not required to attempt all the questions. None of the candidates got more number of incorrect answers than correct answers.

Q.19 Which of the following is always true?
(a) None of the applicants had a net score less than 240
(b) The net scores of the individuals would always be an even integer
(c) The net scores of any candidate would always be a positive integer
(d) The net scores of the topper could be exactly 24% more than the average of the entire group

Q.20 If the topper had a net score of 380 and the sixth ranker had a net score of 370 and the net scores of the top six rankers were distinct, then what would be the average net score of the 64 students who got ranks from the seventh to the last?
(a) 225 (b) 245
(c) 275 (d) None of these

Q.21 When the scores of the top four students are deleted, the average score of the remaining 66 students falls by 6 marks. Assume that it is possible for two or more students to have the same net score. What is the minimum score possible for the fourth ranking student if no student got a net score of more then 352?
(a) 280 (b) 300 (c) 308 (d) 320

Q.22 Two alloys composed of gold and silver together weigh 20 kg. One lump contains 75% gold and 31.25 g kg silver. Another alloy contains 85% gold and 30 g per kg silver. The total quantity of silver in two lumps is 617.5 g. If the two lumps are melted and formed into one, what percentage of gold will it contain?
(a) 50% (b) 89% (c) 78% (d) 67%

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Q.23 Let N be a set of real numbers such that p is any real number in the set. There exists two numbers in N whose average is p, then
(a) N is a finite set.
(b) N is a set containing all real numbers.
(c) N is a set of all numbers in the interval (2, 3).
(d) None of these

Direction for Questions 24 and 25: Read the passage below and solve the questions based on it.

There are two brands of TV, Onida and Videocon, in the market. Onida is an old brand launched in 1990 and Videocon is a new brand launched in 1997. For both these brands, 30% of the TV's purchased every year are disposed off as out-dated exactly two years later. It is known that 10 Onida TV's were disposed off in 1997. The following table gives us the number of TV's in operation at the end of the year from 1995 to 2000.

	1995	1996	1997	1998	1999	2000
Onida	120	162	182	222	236	236
Videocon	0	0	30	80	124	134

Q.24 What is the average number of Videocon TVs disposed every year by 2000?
(a) 4 (b) 12
(c) 8 (d) None of these

Q.25 What is the average number of Onida TVs purchased every year till 1995?
(a) 20 (b) 24
(c) 28 (d) Cannot be determined

True/False

1. For any given series of 100 terms, if we know the last term and the first term, we can always calculate the average of all the 100 terms.
State whether True or False

2. A, B, C, D, E, and F are six numbers, not necessarily natural numbers. They are divided into two groups— each group having three numbers each and we do not know which number belongs to which group. Final average of all the numbers can be calculated by calculating the average of individual average of the two groups.
State whether True or False

3. In the above question, final average will depend upon the set of numbers forming the group.
State whether True or False

4. 10 natural numbers are taken. Average of all the possible pairs of numbers formed out of these ten natural numbers are calculated. It is possible to determine the average of all the ten natural numbers using this information.
State whether True or False

Answers

Warm Up

1. (b)	**2.** (b)	**3.** (b)	**4.** (c)	**5.** (c)	**6.** (d)	**7.** (c)	**8.** (d)	**9.** (b)	**10.** (b)
11. (b)	**12.** (a)	**13.** (b)	**14.** (b)	**15.** (c)	**16.** (d)	**17.** (d)	**18.** (d)	**19.** (c)	**20.** (a)
21. (b)									

Foundation

1. (d)	**2.** (d)	**3.** (c)	**4.** (c)	**5.** (b)	**6.** (b)	**7.** (c)	**8.** (a)	**9.** (c)	**10.** (a)
11. (a)	**12.** (a)	**13.** (b)	**14.** (d)	**15.** (d)	**16.** (d)	**17.** (c)	**18.** (a)	**19.** (a)	**20.** (c)
21. (b)	**22.** (c)	**23.** (d)	**24.** (b)	**25.** (d)	**26.** (b)	**27.** (c)	**28.** (c)	**29.** (c)	**30.** (b)
31. (a)	**32.** (d)	**33.** (b)	**34.** (d)	**35.** (c)	**36.** (b)	**37.** (c)	**38.** (d)	**39.** (c)	**40.** (a)
41. (d)	**42.** (b)	**43.** (a)	**44.** (c)	**45.** (d)	**46.** (a)	**47.** (a)	**48.** (b)	**49.** (c)	**50.** (b)

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Moderate

1. (c)	**2.** (d)	**3.** (a)	**4.** (d)	**5.** (d)	**6.** (c)	**7.** (d)	**8.** (d)	**9.** (d)	**10.** (c)
11. (d)	**12.** (d)	**13.** (d)	**14.** (c)	**15.** (b)	**16.** (a)	**17.** (a)	**18.** (a)	**19.** (b)	**20.** (d)
21. (d)	**22.** (c)	**23.** (c)	**24.** (a)	**25.** (a)	**26.** (b)	**27.** (b)	**28.** (d)	**29.** (d)	**30.** (c)
31. (c)	**32.** (c)	**33.** (d)	**34.** (d)	**35.** (d)	**36.** (c)	**37.** (c)	**38.** (b)	**39.** (d)	**40.** (d)
41. (a)	**42.** (b)	**43.** (d)	**44.** (a)	**45.** (d)	**46.** (d)	**47.** (d)	**48.** (c)		

Advanced

1. (d)	**2.** (b)	**3.** (a)	**4.** (a)	**5.** (d)	**6.** (d)	**7.** (b)	**8.** (a)	**9.** (d)	**10.** (b)
11. (d)	**12.** (d)	**13.** (b)	**14.** (b)	**15.** (b)	**16.** (c)	**17.** (b)	**18.** (a)	**19.** (b)	**20.** (d)
21. (b)	**22.** (c)	**23.** (b)	**24.** (a)	**25.** (d)					

True/False

1. False

Formula to find out the average = (first term + last term)/2 can be used only if the series is in AP (i.e., the difference between any two consecutive terms is same).

2. True

Average is only the central value of the numbers, and it does not matter what ways the numbers has been taken.

3. False

4. True

Average is only the central value of the numbers, and it does not matter what ways the numbers has been taken—numbers are taken independently or they are first paired up and then taken together to find out the average.

Hints and Solutions

Warm up

1. $\frac{42+49+56+63+35}{5}=\frac{245}{5}=49$

2. Average speed $=\frac{28.12}{80+12}=\frac{192}{20}=9.6$ km/h

3. Required average $=\frac{39.2\times5-41\times3}{2}=\frac{196-123}{2}$

$=\frac{73}{2}=36.5$

4. $\frac{30x+30y}{60}=\frac{x+y}{2}$

5. $\frac{15}{2}[2.3+14.3]=\frac{1}{2}[6+42]=24$

6. $25\times7-28=21x$

$21x=147$

$x=7$

Age of eldest member $=6\times7=42$ years

7. $7x+42=70$

$7x=28$

$x=4$

Required average $=\frac{10x+90}{10}=x+9=13$

8. $6x+3x+x=120$

$10x=120$

$x=12$

9. Correct average $=\frac{250-84+48}{5}=\frac{214}{5}$

10. Total weight of the whole family $=7\times18=126$

Weight of family excluding head of the family $=13\times6=78$ kg

Weight of head of the family = 48 kg

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11. $\frac{30.16+20\times15.5}{50}=\frac{480+310}{50}=\frac{790}{50}=15.8$ kg

12. $\frac{47.12-45.10}{2}=\frac{564-450}{2}=\frac{114}{2}=57$

13. $\frac{x}{y}=\frac{100-56}{56-35}=\frac{44}{21}$

14. $x+y=50$

$\frac{1}{4}x+\frac{1}{2}y=15$

Solving both the equations, we get $x=40$ and $y=10$

15. $\frac{1+2}{6+2}=\frac{3}{8}=37.5\%$

16. $\frac{3+\frac{9}{100}\times10}{10+3}\times=\frac{3.9}{13}\times100=30\%$

17. $CP=₹\left(20.02-20.02\times\frac{1}{11}\right)=₹(20.02-1.82)$

$=₹18.20$

18. $\frac{x}{y}=\frac{20,000-11,200}{11,200-4000}=\frac{88,000}{7200}=\frac{22}{18}=\frac{11}{9}$

19. Average of any arithmetic progression

$=\frac{\text{First number}+\text{last number}}{2}=\frac{51+79}{2}=65$

20. Let x part of mixture is removed,

then $\frac{2-\frac{2}{5}x+x}{3-\frac{3}{5}x}=1$

$x=\frac{5}{6}$

21. **Method 1** Let CP of milk be ₹100,
then CP of mixture = ₹

$\left(100-\frac{1}{6}\times100\right)=₹\frac{250}{3}$

Using alligation, we get

$\frac{\text{Milk}}{\text{Water}}=\frac{\frac{250}{3}-0}{100-\frac{40}{3}}=\frac{5}{1}$

Method 2 In these questions, answer can be calculated as follows:
Ratio of milk and water = 100%:profit% = 100%:20% = 5:1

FOUNDATION

1. $80=\frac{2.x.40}{40+x}$

$40+x=x$

Hence, not possible.

Alternatively, the average speed for the whole journey has to be less than double the speed of one side. In this case, speed in one direction = 40 kmph, hence, average speed for the whole journey has to be less than 80 kmph.

2. Let x be number of senior level employees.

Then, $\frac{125\times5500+x\times14,000}{125+x}=8687.5$

$x=75$

$\frac{4}{5}$ of total employees $=(125+75)=200$

Hence, number of total employees $=\frac{5}{4}.200=250$

3. $\frac{140+12x}{13}=x+5$

$145+12x=13x+65$

$x=80$

4. After 6 years from marriage

Age of Saketprakash = 33 years

Age of his wife = 29 years

Let the age of their son be x years.

Then, $\frac{33+29+x}{3}=22$

$62+x=66$

$x=4$

Hence, option (c) is the answer.

5. Total mistakes $=1007\times2=2014$

Let x be average mistake per page for the remaining pages.

$434+395x=2014$

$395x=1580$

$x=4$

6. Let a, b, c, and d be the number of students in section A, B, C, and D, respectively.

Then,

$=\frac{45(a+b+c)+55(a+c+d)+50(a+b+d)+60(b+c+d)}{3(a+b+c+d)}$

$=50+\frac{5b+10c+15d}{3(a+b+c+d)}$

Clearly, a, b, c, and d are natural number; put $a=b=c=d=1$

Then, required average $=50+\frac{30}{12}=50+2.5$

$=52.5$

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7. Old average = 44

Average of 8 new innings

$$= \frac{97+3+23+0+68+40+50+71}{8}$$

$$= \frac{352}{8} = 44$$

8. Age of $D = 80 \times 4 - 84 \times 3 = 320 - 252 = 68$

Age of $E = 71$ years

Age of $A = 80 \times 4 - 79 \times 4 + 71$

$= 75$

9. Let x students were in the class.

$$\frac{35x-25}{x-1} = 36$$

Or, $35x - 25 = 36x - 36$

$x = 11$

10. $$\frac{11\times35-30-20}{9} = \frac{385-50}{9} = \frac{335}{9} = 37\frac{2}{9}$$

Required number $37\frac{2}{9} - 35 = \frac{335-315}{9} = \frac{20}{9}$

11. $$\frac{ma+x}{1+m} = b \Rightarrow ma + x = b + bm$$

$x = b + m\,(b - a)$

12. Go through the options.

For option (a):

He played 100 matches previously.

His old average = $\frac{6000}{100} = 60$

His new average = 6090/105 = 58

Option (a) is satisfied. Hence, option (a) is the answer.

13. $$\frac{3x - y - y + \frac{y}{2}}{3} = x - 1$$

$$3x - \frac{3y}{2} = 3x - 3$$

$$\frac{3y}{2} = 3, y = 2$$

14. Assign some numerical values, then proceed; we will get (d) as answer.

15. Let the number of students in the class be x and the average weight of the class be N.

Therefore, weight of these x students = Nx kg

Now, $\frac{Nx+50}{x+1} = N + 1$

Or, $Nx + 50 = (N + 1)(x + 1)$

Or, $Nx + 50 = Nx + N + x + 1$

$N + x = 49$ (i)

Also, $\frac{Nx+50+50}{x+2} = N + 1.5$

Or, $Nx + 100 = (N + 1.5)\,(x + 2)$

Or, $Nx + 100 = Nx + 2N + 1.5x + 3$

Or, $97 = 2N + 1.5x$

Or, $97 = N + 49 + 0.5x$, or, $48 = N + 0.5x$

Or, $48 = N + 0.5\,(49 - N)$, or, $48 = N + 24.5 - 0.5N$

Or, $23.5 = 0.5N$. Hence, $N = \frac{23.5}{0.5} = 47$

Hence, option (d) is the answer.

16. No data regarding the numbers of books is given, hence, we cannot determine the weight of school bag.

Hence, option (d) is the answer.

17. $$\frac{32\times n + \frac{3}{4}n.4 - \frac{1}{4}n.6}{n} = \frac{70n-3n}{2n} = 33.5$$

18. $40 + 78 + x - 6 + x - 2 + x = 200$

$3x = 90$

$x = 30$

19. Total sales from Monday to Friday = ₹9000

Sales from Tuesday to Saturday = ₹9450

Thus, a sale of ₹450 more was on Saturday than on Monday.

20. Total age of family 3 years ago = $17 \times 5 = 85$ years

Total age of family now = $17 \times 6 = 102$ years

Total age of family excluding the child now = (85 + 15) = 100 years

Age of child = 2 years

21. Let a be the average till 16th inning.

$$\frac{16a+85}{17} = a + 3$$

$16a + 85 = 17a + 51$

$a = 34$

Required average = 37

22. $$\frac{\text{Total marks}}{\text{Total number of students}} = \frac{ac+bd}{a+b}$$

23. $$\frac{2\times0+30\times40+14.20+2x}{48} = 35$$

$1200 + 280 + 2x = 1680$

$2x = 2007$

$x = 100$

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24. Let salary of A, B, and C be ₹a, b, and c, respectively.

Given that, $a + b + c = 444$

Expense of $A = \frac{80}{100}\ a = 0.8a$, so, savings of $A = a - 0.8a = 0.2a$

Expense of $B = 0.85b$, so, savings of $B = b - 0.85b = 0.15b$

Expense of $C = 0.75c$, so, savings of $C = c - 0.75c = 0.25c$

Ratio of savings of $A:B:C = 7:6:9$

Hence, $0.2a:0.15b:0.25c = 7:6:9 = 7k:6k:9k$

Therefore, $0.2a = 7k$, $0.15b = 6k$, $0.25c = 9k$

Therefore, $a = 35k$, $b = 40k$, $c = 36k$

Hence, $35k + 40k + 36k = 444$, or, $111k = 444$, or, $k = 4$

Therefore, salary of $B = 40k =$ ₹160

Hence, option (b) is the answer.

25. Let n = total number of students

$$\frac{\frac{2}{3}n.14 + \frac{1}{3}n.x}{n} = 20$$

$$\frac{34n + nx}{3n} = 20$$

$$34n + nx = 60n$$

$$nx = 26x$$

$$x = 26$$

26. Average speed for first journey

$$= \frac{2.75.50}{125} = 60 \text{ km/h}$$

Average speed for second journey

$$= \frac{2.30.60}{90} = 40 \text{ km/h}$$

Average speed for culture journey

$$= \frac{2.60.40}{100} = 48 \text{ km/h}$$

27. $$\frac{64n + 8.73}{n+8} = \frac{64n + 584}{n+8}$$

Put the values of n from the options to get the answer.

28. Let the integers be 1, 2, 3, 4, 5.

$$N = \frac{1+2+3+4+5}{5} = \frac{15}{5} = 3$$

When next two integers are also included, then new average $= \frac{28}{7} = 4 = N + 1$

29. $10Z - (10.A + B) + (10B + A) = 10\ (Z + 2.7)$

$10Z - 10\ A - B + 10\ B + A = 10Z + 27$

$9\ B - 9\ A = 27$

$B - A = 3$

$|B - A| = 3$

30. Clearly, $B > A$, and none of them is zero.

A	1	2	3	4	5	6
B	4	5	6	7	8	9

31. Think like this:

To calculate the average salary, total salary is divided by 14, or, total salary is distributed among 14 persons. Hence, it would be less than what it would have been had it been distributed among 12 persons.

To calculate the average expenses, total expense is divided by 10, or, total expense is distributed among 10 persons. Hence, it would be more than what it would have been had it been distributed among 12 persons.

Average savings = Average salary − Average expenses

So, average savings would be less than what it would have been if number of months were 12.

Hence, option (a) is the answer.

33. $$\frac{3 \times 0.45 + 2x + 3x}{5} = 0.40$$

$1.35 + 5x = 2$

$5x = 0.65$

$x = 0.13$

Rainfall on last day $= 3x = 3 \times 0.13 = 0.39$

34. Let the number of boys in a class be x. Therefore, number of girls $= 0.8x$

Average height of girls = 155 cm

Therefore, total of height of girls $= 155 \times 0.8x$

Total number of students $= x + 0.8x = 1.8x$

Let the average height of boys be B cm

Therefore, total of height of boys $= Bx$ cm

$\frac{155 \times 0.8x + Bx}{1.8x}$ = Average height of all students or average height of class

Therefore, $\frac{155 \times 0.8x + Bx}{1.8x} + 4x = B$

$$155 \times 0.8x + Bx + 7.2x = 1.8Bx$$

$$155 \times 0.8x + 7.2x = 0.8Bx$$

$$0.8x(155 + 9) = 0.8x \times B$$

$$B = 164 \text{ cm}$$

Therefore, (d) is the answer.

35. Average price of $A = \frac{19+26+31+35+39}{5}$

$= \frac{150}{5} = 30$

Average price of $B = \frac{7+8+17+20+23}{5} = \frac{150}{5} = 30$

Index price $= \frac{30 \times 1.1 + 15 \times 0.9}{2} = 23.25$

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36. Average speed when Ankit travels from New Delhi to Patna $= \dfrac{d}{\frac{-d/3}{50}+\frac{d/3}{40}+\frac{d/3}{60}}$

$= \dfrac{d}{\frac{d}{150}+\frac{d}{120}+\frac{d}{180}} = \dfrac{1800d}{12d+15d+10d}$

$= \dfrac{1800d}{37d} = \dfrac{1800}{37}$ km/h

Required average speed $= \dfrac{2.\frac{1800}{37}.75}{\frac{1800}{37}+75}$

$= \dfrac{270,000}{4575} = 59.01$

37. $\dfrac{25.16-x}{24} = 15, x$ = age of a teacher

$x = 25.16 - 24.15 = 400 - 360 = 40$

38. Average of first 10 multiples of

$1 = \dfrac{1+2+3+\cdots+10}{10} = 5.5$

Average of first 10 multiples of

$2 = \dfrac{2+4+\cdots+20}{10} = 11$

Average of first 10 multiples of

$3 = \dfrac{2+6+\cdots+30}{10} = 16.5$

Hence, required average

$= \dfrac{5.5+11+16.5+\ldots \text{ to 10 terms}}{10}$

$= \dfrac{\frac{10}{2}[2\times 5.5+9\times 5.5]}{10}$

$= \dfrac{1}{2}\times 60.5 = 30.25$

39. Score of Chandan $= 64 \times 11 + 10 \times 67 - 20 \times 65$

$= 704 + 670 - 1300$

$= 1374 - 1300 = 74$

40. $\dfrac{38+84+63+45+32+n}{6} = \dfrac{262+n}{6}$

$= \dfrac{262+n}{6} = 43.66 + \dfrac{n}{6}$

Only (c) satisfies the condition

41. Type of T_1 tea in 4 kg mixture $= \dfrac{1}{7}\times 4\text{kg} = \dfrac{4}{7}$ kg

Type of T_2 tea in 4 kg mixture $= \dfrac{2}{7}.4\text{kg} = \dfrac{8}{7}$ kg

Type of T_3 tea in 4 kg mixture $= \dfrac{4}{7}\times 4 = \dfrac{16}{7}$ kg

CP of 4 kg tea mixture $= ₹\left(\dfrac{296\times 504\times 1008}{7}\right)$

$= ₹\dfrac{1808}{7}$

SP of new mixture = ₹84

CP of new mixture $= ₹\left(84 - 84.\dfrac{1}{4}\right)$

= ₹63 per kg

Suppose x kg of T_1 tea is added to 4 kg mixture.

CP of $(4 + x)$ kg tea mixture $= (4 + x)\ 63 = ₹(252 + 63x)$

Also, CP of $(4 + x)$ kg tea mixture

$= ₹\left(\dfrac{1808}{7}+\dfrac{x}{7}.74\right)$

$= ₹\left[\dfrac{1808+74x}{7}\right]$

Therefore, $252 + 63x = \dfrac{1808+74x}{7}$

$1764 + 441x = 1808 + 74x$

$367x = 47$

$x = \dfrac{47}{367}$

42. Initial concentration of wine = 75%

Concentration of wine after first flushing

$= \dfrac{45 - 75\% \text{ of } 4\times 100}{60} = 70\%$

Concentration of wine after second flushing

$= \dfrac{42 - 70\% \text{ of } 5\times 100}{60} = 63.66\%$

Concentration of wine after third flushing

$= \dfrac{38.5 - 63.66\% \text{ of } 38.5\times 100}{60} = 57.75\%$

43. x = amount of alloy B

y = amount of alloy A

In the alloy

Take the quantity of gold in both the alloy under consideration and then take alligation.

$\dfrac{x}{y} = \dfrac{\frac{7}{9}-\frac{1}{2}}{\frac{1}{2}-\frac{7}{18}} = \dfrac{\frac{14-9}{18}}{\frac{9-7}{18}} = \dfrac{\frac{5}{18}}{\frac{2}{18}} = 5:2$

Hence, required ratio = 2:5

44. Suppose x litre was the initial volume of mixture. Then, volume of liquid $A = \dfrac{4}{5}x$, and volume of liquid $B = \dfrac{x}{5}$

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Now, $\dfrac{\frac{4}{5}x-\frac{4}{5}.10}{\frac{1}{5}x-\frac{1}{5}.10+10}=\dfrac{2}{3}$

$$\frac{\frac{4}{5}x-8}{\frac{1}{5}x-2+10}=\frac{2}{3} \Rightarrow \frac{\frac{4}{5}n-8}{\frac{1}{5}n+8}=\frac{2}{3}$$

$12x - 120 = 2x + 80$

$10x = 200$

$x = 20$

Volume of liquid $= \dfrac{4}{5}.20 = 161$

45. Let $25x$ be the volume of pure milk present in vessel.

Applying the formula,

Amount left = Initial amount $\left(1-\dfrac{\text{Amount take out}}{\text{Initial amount}}\right)^n$

n = number of times operation carried out

$$16x = 25x\left(1-\frac{9}{25x}\right)^2$$

$$x=\frac{9.5}{25}=\frac{9}{5}$$

Hence, volume of milk present initially $= 25.\dfrac{9}{5} = 45$ L

46. Let x = number of deer

y = number of ducks

Total number of heads are 204,

i.e., $x + y = 204$ (i)

Total number of legs = 512

i.e., $4x + 2y = 512$ (ii)

Solving 1 and 2

$x = 52$

$y = 152$

47. Net interest earned by him $= \dfrac{1}{2}\times\dfrac{1440}{1000}\times 100 = 7.2\%$

Let ₹x is invested at 6% p.a. and Rest invested at 8% p.a.

$$\frac{x}{y}=\frac{8-7.2}{7.2-6}=\frac{.8}{1.2}=\frac{2}{3}$$

Required amount $= \dfrac{2}{5}\times 10{,}000 = ₹4000$

48. Let $7x$ and $5x$ volume of liquids A and B are present in vessel then,

$$\frac{7x-\frac{7}{12}.9}{5-\frac{5}{12}.+9}=\frac{7}{9}$$

$x = 3$

Hence, $7 \cdot 3 = 21$ L of liquid A was present

49. Let x = volume of water he added

$$\frac{36}{36+x}=\frac{3}{4}$$

$144 = 108 + 3x$

$3x = 36$

$x = 12$

50. After first replacement concentration of milk

$= \dfrac{40}{60}\times 100 = 66.66\%$

After second replacement concentration of milk

$= \dfrac{80/3}{60}\times 100 = 44.44\%$

A sequence is found: 66.66%, 44.44%, and 22.22%; thus, this process is done for 4 times.

MODERATE

1.

Number of Boys	B	Average of Girls
Number of Girls	G	Average of Boys

And class average $= [BG + BG]/2$

Average of boys' average and girls' average $= [B + G]/2$

Now, $[BG + BG]/2 = [B + G]/2 - 4$

Or, $[B - G]^2 = 8\,[B + G]$

Since LHS is a perfect square, RHS should also be a perfect square. Now, see the options that satisfy this.

2. Now different values of $[B + G]$ possible are – $0(2 \times 0^2)$, $2(2 \times 1^2)$, $8(2 \times 2^2)$, $18(2 \times 3^2)$, etc.

We cannot take 0, since number of boys + number of girls cannot be equal to zero.

We cannot take 2, since in this case, $B - G$ will exceed $B + G$.

We cannot take 8, since in this case, $B - G$ will become equal to $B + G$, making $G = 0$. And this will result into making all the equations wrong.

So, the minimum possible value = 18

3. Take five consecutive integers such that it satisfies the condition.

4. Sum of first n even number $= n(n + 1)$

Average of first n even number $= \dfrac{n(n+1)}{n} = n + 1$

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Let $n = 2m$

Average of all the even number upto n, i.e., $2m$

$$\frac{2m(2m+1)}{2m} = (2m+1)$$

$= (n + 1)$

Hence, required difference = 0

5. Let present age of father = x year

and present age of mother = y year

Therefore, present age of son = $(x - 30)$

and present age of daughter = $(y - 25)$

Sum of their ages = $4 \times 26.25 = 105$ years

i.e., $x + y + x - 30 + y - 25 = 105$

$2x + 2y = 160$

$x + y = 80$ (i)

After 4 years, their total ages will be (excluding the daughter)

$x + 4 + y + 4 + x - 30 + 4 = 107$

$2x + y - 18 = 107$

$2x + y = 125$ (ii)

Solving (i) and (ii)

$x = 45$

6. Total weight of all players initially = $68 \times 10 = 680$ kg

Total weight of players when 1 player left the team = 598.5

Different in weight = $(680 - 598.5)$ kg = 81.5

7. Total weight of 11 players (68×11) kg = 748 kg

Increase in weight = $(748 - 598.5)$ kg = 149.5

8. Since the exact weight is not known, we cannot find out the average weight of all the players taken together.

9. Exact weight of players are not known; hence, option (d) is the answer.

10. If he is taking 4 packages at a time, then he is taking each of packet 15 times ($^6C_2$). So, total weight taken this way $= 15 \times 500$. So, actual weight = $[7500 \times 3/2]/15$

11. What is given here is the range, and not the exact marks.

13. Since we do not have the exact average, we cannot calculate this.

14. Try with options. Also, maximum students that can be shifted cannot be more than 24.

When (c) then average of section A = 30

and average of section C

$$= \frac{24 \times 22 + 1 \times 20 + 19 \times 36}{44} = 23.6$$

15. Maximum possible average of all three sections

$$= \frac{(25 \times 30 + 30 \times 28 + 20 \times 26)}{(25 + 30 + 20)} = 28.13$$

16. Required average

$$= \frac{5 \times 22 + 5 \times 30 + 6 \times 24 + 4 \times 28 + 20 \times 4 + 6 \times 24}{(30)}$$

$= 25.06$

17. Marks contributed by Section A

$= 5 \times 22 + 5 \times 30 = 260$

Marks contributed by Section B

$= 6 \times 24 + 4 \times 28 = 256$

Marks contributed by Section C

$= 20 \times 4 + 6 \times 26 = 236$

18. Now, shift 5 students from A to B having the maximum marks then, shift 5 students from B having maximum marks and 5 those came from A.

Required average

$$= \frac{5 \times 30 + 5 \times 28 + 1 \times 20 + 19 \times 26}{30} = 27$$

19. Now, shift 5 students from A having minimum marks and shift 5 from B having minimum marks and 5 those came from A.

Required average

$$= \frac{5 \times 22 + 5 \times 24 + 19 \times 20 + 1 \times 26}{30} = \frac{636}{30}$$

$= 21.2 \approx 21$

22. Let the five numbers be 1, 2, 3, 4, and 5

Average of these five numbers $= \frac{1+2+3+4+5}{5} = 3$

Distinct triplets are (1, 2, 3), (2, 3, 4), (3, 4, 5), (1, 2, 4), (1, 2, 5), (1, 3, 4), (1, 3, 5), (1, 4, 5), (2, 4, 5), (2, 3, 5).

[Total number of distinct triplets = 5C3 = 10]

In the above triplets, each of the numbers occur 6 times. For example, 1 will occur six times in total. Instead of calculating the average of average of all the triplets, what we can do is – calculate the average of all the 10 triplets at one go.

There is a total of 30 numbers [3 numbers in each triplets $\times$ 10 triplets]

Average of all the ten triplets $= \frac{6(1+2+3+4+5)}{30}$

$$= \frac{6 \times 15}{30} = 3$$

Hence, $A = B$

23. Let the numbers be 1, 2, 3, 4, and 5, and then solve.

24. Let $A = 1$, $B = 2$

Therefore, sum of A and $B = 3$

Now, going through the options

(a) $*(@(A, B), 2) = * (1.5, 2) = 3$

Hence, option (a) is the answer.

25. Check the options to find the answer as option (a).

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26. Let $A = 1, B = 2, C = 3$

Sum $(A, B, C) = 6$

Now, going through the options

(a) *(@ (*(@ (B, A), 2), C), 3)

(@ ((1.5, 2), C), 3)

* (@ (3, C), 3)

* (3, 3) = 9

(b) *(@(*(@ (B, A), 2) C), 2)

(@ ((1.5, 2), C), 2)

*(@ (3, C), 2)

*(3, 2) = 6

Hence, option (b) is the answer.

27. Sum of ages of Mr and Mrs Sinha in 1972 = 46 years

Sum of age of their family in 1976 = 19 × 3 = 57 years

Sum of ages of Mr and Mrs Sinha in 1976 = (46 + 8) years = 54 years

Their son's age in 1976 = (57 − 54) years = 3 years

28. Number of articles made in first hour = 60

Number of articles made in second hour = 45

Number of articles made in third hour = 63

Number of articles made in fourth hour = 42

Number of articles made in fifth hour = 63

So, obviously articles made in fourth hour is minimum.

29. Total number of runs scored till 86th inning = 86 N

Now, 86 N + 270 = 87 (N + S), where S is the increase in batting average.

Different values of S possible now is 0, 1, 2, and 3.

30. Since a boy is increased in Section B and the average is decreased ⇒ Hence, his weight is less than 38 kg. Again, the average weight is also decreased in both the sections. Hence, his weight is more than 35 kg. Hence, option (c) is the answer.

31. Let the number of innings played be N

then, $\frac{21.5N + 28 + 34 + 37}{N + 3} = \text{N} + 1.25$

Solving, we get $N = 27$

32. Total score = 21.75 × 24 + 28 + 34 + 37 = 621

33. Let the height of four students be 150, 160, 170, and 180 cm, then

$$A = \frac{150 + 160 + 170 + 180}{4} = \frac{660}{4} = 165 \text{ cm}$$

$$B = \frac{\frac{150+160}{2} + \frac{160+170}{2} + \frac{170+180}{2} + \frac{150+180}{2} + \frac{150+170}{2} + \frac{160+180}{2}}{6}$$

$$= \frac{155 + 165 + 175 + 165 + 160 + 170}{6} = \frac{990}{6} = 175 \text{ cm}$$

$$= \frac{670}{4} = 167.5$$

Similarly, $C = 167.5$

Now, go through the options.

34. Given $n_1 + n_3 .... + n_{45} = 1273$

From above, we cannot find the value of d.

35. $V_1 = 220$ L $\qquad V_2 = 180$ L

Price = P_1 $\qquad$ Price = P_2

Solving through the options,

Option (a): 92 L

Volume in V_1 after pouring 90 L from V_2

= 130 L (milk of P_1 value) + 90 L (milk of P_2 value)

Volume in V_2 after pouring 90 L from V_1

= 90 L (P_1) + 90 L (P_2)

Option (b) Sol

V_1 price = 140 P_1 + 80 P_2

V_2 price = 100 P_2 + 80 P_1

Option (c)

160 L V_1 price = 60 P_1 + 160 P_2

V_2 price = 20 P_2 + 160 P_1

Option (d): 99 L V_1 price = 121 P_1 + 99 P_2

V_2 price = 101 P_2 + 99 P_1

36. Let x L of 50% solution and y L of 80% solutions are used.

$$\frac{x}{y} = \frac{80 - 62}{62 - 50} = \frac{18}{12} = \frac{x}{y} = \frac{3}{2}$$

Solution get mixed in the ratio 3:2

Now, suppose the value of acid is Z litres ⇒

$$\frac{0.62z}{z + 6} = \frac{1}{2}$$

$$\Rightarrow 1.24Z = Z + 6 \quad \Rightarrow 0.24Z = 6$$

$$\Rightarrow Z = 25$$

Hence, required rate = $\frac{2}{5} \times 25 = 10$ litres

37. Solve with the help of options.

38. It is possible because each content removed during the dilution process is stored separately.

39. Value of solvent in container = $\left(\frac{4}{5} \times 150\right)\text{L} = 120$ L

Concentration of solvent = $\left(\frac{9}{10} \times 120\right)\text{L} = 108$ L

When first time water is added, the concentration of solvent

$$\frac{108}{130} \times 100 = 83\%$$

When second time water is added, the concentration of solvent

$$\frac{108}{140} \times 100 = 77\%$$

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When third time the water is added, the concentration of solvent $= \frac{108}{140} \times 100 = 72\%$

Now, 10 L solution should be removed

Solvent present in container = (108 – 7.2)L = 100.8 L

Again 10 L water is added and 10 L solvent is removed. To get the required solution, water is added 6 times.

40. Let the initial volume be $85x$.

$$36x = 49x\left(1-\frac{4}{49x}\right)^2$$

$$\left(\frac{6}{7}\right)^2 = \left(1-\frac{4}{49x}\right)^2$$

$$1-\frac{4}{49x} = \frac{6}{7}$$

$\Rightarrow x = 4/7$

Required volume $= 49 \times 4/7 = 28$ g

41. After first operation, milk left =

$$40\left(1-\frac{4}{40}\right) = 40\left(1-\frac{1}{10}\right) = 40.\frac{9}{10} = 36 \text{ L}$$

After second operation, milk left $= 36\,[1 - (5/40)] = 31.5$

After third operation, milk left $\left(36.5 - 6 \times \frac{31.5}{40}\right)$

$= (26.775$ L)

42. When 4 L of milk is taken out, volume of milk left in the vessel = 36 L[40{1 – (4/40)}]

When 5 L of mixture is taken out, volume of milk left in the vessel = [40{1 – (4/40)}{1 – (5/41)}

When 6 L of mixture is taken out, volume of milk left in the vessel = [40{1 – (4/40)}{1 – (5/41) {1 – (6/42)}]

43. Volume of milk left $= 40 \times \left(1-\frac{4}{40}\right) \times \left(1-\frac{6}{4}\right) \times \left(1-\frac{8}{42}\right)$

= 24.87 L.

44. When a part of vessel A is replaced with the mixture in vessel B, then the concentration of milk in the final mixture is less than C_1. It implies that $C_1 > C_2$.

When vessels B and C are mixed in the ratio 1:1, then the quantity of milk in the final mixture is double than the quantity of milk in the vessel B. It implies that $C_2 = C_3$

Hence, C_1 is greatest.

45. Obviously, $C_2 = C_3$ is the right answer.

46. When a part of vessel A is replaced with the mixture in vessel B, then the concentration of milk in the final mixture is less than C_1. It implies that $C_1 > C_2$.

When vessels B and C are mixed in the ratio 1:1, then the concentration of milk in the final mixture is double than the concentration of milk in the vessel B. It implies that $C_3 > C_2$. Now, we do not know which of C_1 or C_3 is greater. So answer is cannot be determined.

47. Let x and y kg of first alloy and second alloy are taken.

$$\frac{x}{y} = \frac{\frac{5}{7}-\frac{1}{2}}{\frac{1}{2}-\frac{3}{7}} = \frac{\frac{10-7}{14}}{\frac{7-6}{14}} = \frac{3}{1}, \frac{x}{y} = \frac{3}{1}$$

Hence, option (d) is the answer.

48. Let the value of milk = $3x$ L and volume of water = $5x$ L

When 20% of mixture is taken out

Volume of milk left $= \left(3x - \frac{1}{5} \times 3x\right) = \frac{125}{5}$ L $= 2.4\,x$ L

Volume of water left $= \left(5x - \frac{1}{5} \times 5x\right) = 4x$ L

And $1.6\,x$ L of milk is added, New ratio = 1:1

By proceeding similarly, we get option (c) as answer.

ADVANCED

1. Obviously, in this case, $AC \le AB$. Now we are required to compare AB and AQ.

$$AB = \frac{R_1}{N_1} + \frac{R_2}{N_1} \text{ and } AQ = \frac{R_1}{N_1} + \frac{R_2}{N_1} \max\left[0, \left(\frac{R_2}{N_2} - \frac{R_2}{N_1}\right)\right]$$

So, basically we have to compare $\frac{R_2}{N_1}$ and $\frac{R_2}{N_1} \max\left[0, \left(\frac{R_2}{N_2} - \frac{R_2}{N_1}\right)\right]$

Since we do not have any information regarding values of R_2, R_1, N_2, and N_1, we cannot determine anything. So, answer is option (d).

2. Since total number of runs is increasing without increase in the number of completed innings, AB will increase. However, since runs scored is less than 50 in the last inning, so AC will decrease.

3. This question is one standard example of the definition of average in terms of surplus and deficit of the values. (Refer to the definition given.)

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Now, what we were expected to mark here was the fact that whatever is the average (in this case, it is A), surplus generated by the marks of 3rd student will be same as deficit incurred due to 15th student. So, rusticating both of them is not going to create any difference on average marks of the class (remember marks are in AP). And similar will be the impact of rusticating 7th student and 11th student and then finally 9th student.
So, $A = B$

4. Let the number of students in section A, B, C and D be a, b, c, and d, respectively.
Sum of weight of students of section $A = 45a$
Sum of weight of students of section $B = 50b$
Sum of weight of students of section $C = 72c$
Sum of weight of students of section $D = 80d$
Average of section A and B together:

$$48 = \frac{45a + 50b}{a + b}$$

$$48a + 48b = 45a + 50b$$

$$3a = 2b \quad \text{(i)}$$

Average of section B and C together

$$\frac{50b + 72c}{b + c} = 60$$

$$50b + 72c = 60b + 60c$$

$$12c = 10b$$

$$6c = 5b \quad \text{(ii)}$$

Also, $\frac{45a + 50b + 72c + 80d}{a + b + c + d} = 60$

$$45a + 50b + 72c + 80d = 60a + 60b + 60c + 60d$$

$$12c + 20d = 15a + 10b \quad \text{(iii)}$$

$12c = 10b$ from equation (ii)

$$10b + 20d = 15a + 10b$$

$$20d = 15a$$

$$4d = 3a$$

Therefore, $\frac{a}{d} = \frac{4}{3}$

Hence, option (a) is the answer.

5. Basically, in these types of situations, we should start with an attempt of refuting the statements given. The statement that cannot be refuted will be true, otherwise false.

Checking Statement (i), since Munchun is donating 1/2 of water from her bottle and Illara is donating only 1/4 of mixture from her bottle, concentration of water in Munchun's drink will be less than the concentration of orange juice in Illara's drink. So, Statement (i) is not true.

Checking Statement (ii), since Illara and Puttu are receiving and then donating the same amount of liquid, so concentration of orange juice in Illara's drink will be equal to the concentration of mango juice in Puttu's drink. Hence, Statement (ii) is also not true.

Checking Statement (iii) concentration of orange will be equal to the concentration of mango in Munchun's drink. Therefore, Statement (iii) is also wrong.
Hence, option (d) is the answer.

6. We can get the sum 666 in two ways:
$1 + 2 + 3 + 4 + 567 + 89 = 666$, or,
$123 + 456 + 78 + 9 = 666$
Therefore, average cannot be uniquely determined.

7. 261 is possible only if we take $123 + 45 + 6 + 78 + 9$
Hence, the average will be $261/5 = 52.2$

8. It is equal to (sum of ages of all the engineers – sum of ages of all the males)/(number of engineers – number of males) = 780/20 = 39

9. Since the relationship here is: Some A are B, so it cannot be determined.

10. Required average = (sum of ages of all working professional – sum of ages of all females)/(number of working professional – number of females)

$$= \frac{27.160 - 22.100}{160 - 100} = \frac{4320 - 2200}{40} = 35.33$$

11. If no females (F) are engineers (E), then it is not possible to have 160 engineers from 140 males. So, F @ E is not possible.

12. The required result will be produce when we choose sample of A, B, F, G, I, J

$$\text{Number of pairs} = {}^{6}C_{2} = \frac{6.5}{1.2} = 15$$

13. There are ten different solution of milk. So total different triplet sets possible $= 10C_{3} = 120$ (i)

To obtain the concentration of mixture as less than 80%, we are required to mix those solutions which concentration is less than 80%.

So, if we mix the solutions whose concentration is less than 80% in any ratio of their volume, we will get the concentration of mixture less than 80%.

Following solutions have their concentration as less than 80%: A, B, F, G, I and J

So, different triplet sets for which concentration is less than 80% = Any three solutions taken from the above list $= {}^{6}C_{3} = 20$ (ii)

So total sets possible = 120 − 20 = 100

14. The required result well be produced by taking the samples C, D, E and H.

Hence, required $= {}^{4}C_{2} = 6$

15. All the ratios formed from a and b will be resulting into a ratio less than 1 (like 1:4, 2:5, etc). To obtain a resultant ratio of 1:1, we should have at least one of the ratios greater than 1. Hence, we cannot get the desired ratio of 1:1.

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16. When three solutions are taken together, we can have infinite values.

17. Total of weights of all students = 25 × 30 = 750 kg

Let the number of girls in a class be x.

Therefore, number of boys = $25 - x$

Let the average weight of girls be M kg.

Therefore, total weight of all girls = Mx kg

Therefore, total weight of boys = $750 - Mx$

Average of boys $= \dfrac{750 - Mx}{25 - x}$

It is known that $M - \dfrac{750 - Mx}{25 - x} = 5$

$25M - Mx - 750 + Mx = 125 - 5x$

$25M - 750 = 125 - 5x$

$25M - 5x = 875$

$5M - x = 175$

It is given that $= \dfrac{750 - Mx + \text{class teacher weight}}{26 - x}$

$26M - Mx = 750 - Mx +$ class teacher's weight.

$26M = 750 +$ class teacher's weight.

Therefore, class teacher's weight = 82 kg

Therefore, $M = 32$. Now, it can be calculated.

18. Average of the sum of $n_1, n_3, n_5, \ldots n_{47}$ terms $= \dfrac{1272}{24} = 53$

Required sum = average of odd terms × number of terms = 53 × 47 = 2491

Answers to Q.19 to 21:

19. All the additions in the net score would be in the multiple of 2, and similarly, all the subtractions will also be in the multiple of 2. Hence, net score will also be a multiple of 2.

20. We know that for every right answer, there is +2 marks and for every wrong answer, there is –2 marks.

It means every student can get an even number of marks only.

It is given that topper got 380 marks and sixth ranker got 370 marks.

So, second ranker will get 378, third ranker will get 376, fourth ranker will get 274, and fifth ranker will get 272 marks.

Now, let the average of last 64 student is x.

Then, 380 + 378 + 376 + 374 + 372 + 370 + 64 × x = 70 × 240

Or, 2250 + 64x = 16,800. Hence, x = 227.34

21. The sum of scores of top four students = 1356

Maximum possible score possible for top three students = 352 × 3 = 1056

Hence, the minimum possible score for the 4th student = 300

22. Eliminating the option, we get (c) as answer because average always lies between the greatest and lowest values.

23. A set of all real number can give the above conditions. Hence, option (b) is the answer.

24. Total number of TVs disposed off in 1999 = 20% of 30 = 6

Total number of TVs disposed off in 2000 = 20% of 50 = 10

Therefore, total number of TVs disposed off by 2000 = 16

Hence, average number of TVs disposed off = $\dfrac{16}{4} = 4$

25. Since we do not have the figure of disposed off TVs every year, we cannot determine the average number of TVs purchased every year till 1995.

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CHAPTER

7

Ratio, Proportion, and Variation

LEARNING OBJECTIVES

After completion of this chapter, the reader should be able to understand:

- Ratio, proportion, and variation
- Definitions and properties
- Different applications of ratio, proportion, and variation
- Kinds of questions asked in the CAT
- Methods of solving questions

INTRODUCTION

Ratio and proportion is important not only for the QA paper but also for DI, especially for calculations. Questions from this topic are based on conceptual clarity and their different applications are often tested in the CAT. However, this is a favourite topic for all the non-CAT and non-XAT level examinations. While going through this chapter, students should try to internalize both the mathematical and the logical methods for solving the problems.

RATIO, PROPORTION, AND VARIATION

Ratio is a quantity that expresses the relationship between two similar quantities. It expresses a magnitude by which one quantity is a part of or a multiple of another quantity.

If the value of A and B are 8 and 6, respectively, then they are in the ratio 8:6 (read as 8 is to 6). Ratio can be understood also as the relationship which one quantity bears with the other of the same kind. Due to this reason, we cannot compare salary of one person with the percentage expenditure of another person.

The ratio of two quantities A and B is written as A:B. Here, A is known as an antecedent and B is known as a consequent. It can also be said that A:B = kA:kB, where k is any constant known as constant of proportionality, $k \neq 0$.

If the antecedent is more than the consequent (or, the numerator is more than the denominator), then the ratio is known as an improper ratio.

For example, 5/3, 55/29, etc.

If the antecedent is less than the consequent (or, the numerator is less than the denominator), then the ratio is known as a proper ratio.

For example, 3/7, 7/18, etc.

Since ratio compares two similar quantities, it cannot have any units.

Example 1 Consider any ratio $\frac{a}{b}$. Now, x is added to the numerator and the denominator of this fraction. Which of the following is greater: $\frac{a+x}{b+x}$ or $\frac{a}{b}$?

Solution It depends upon the following two factors:

i. If the ratio is proper or improper.
ii. x is positive or negative.

If $\frac{a}{b} > 1$ and $x > 0$, or, $\frac{a}{b} < 1$ and $x < 0$

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$$\frac{a}{b} > \frac{a+x}{b+x}$$

and if $\frac{a}{b} > 1$ and $x < 0$, or, $\frac{a}{b} < 1$ and $x > 0$

$$\frac{a}{b} < \frac{a+x}{b+x}$$

RATIO

Ratio can be understood in the following two ways:

1. Ratio as a bridging element
2. Ratio as a multiplier

Ratio as a Bridging Element

Ratio as a bridging element helps us in establishing the relationship between more than two quantities. This can be further understood with the following example:

Suppose conversion rate of our currency Rupee is given with respect to US dollar and also with respect to Pound sterling. If we have to find the conversion ratio of US dollar with respect to pound sterling, we can do it by making rupee as the bridge between US dollar and pound sterling.

Example 2 The ratio of the age of A and B is 2:5 and ratio of the age of B and C is 3:4. What is the ratio of the age of A, B, and C?

Solution Since B is the common platform that associates A and C, so we will try to make B equal in both the cases.

Age of A:Age of B = [2:5] × 3
Age of B:Age of C = [3:4] × 5
Or, Age of A:Age of B = 6:15 (i)
Age of B:Age of C = 15:20 (ii)

Since ratio of B is same in both the cases, hence, age of A:Age of B:Age of C = 6:15:20.

Example 3 Given that

Salary of A:Salary of B = 1:2
Salary of B:Salary of C = 3:4
Salary of C:Salary of D = 5:6
Salary of D:Salary of E = 7:8
Salary of E:Salary of F = 9:10
What is the ratio of the salaries of A, B, C, D, E, and F?

Solution Salary of A:Salary of B:Salary of C:Salary of D:Salary of E:Salary of F = (1 × 3 × 5 × 7 × 9):(2 × 3 × 5 × 7 × 9):(2 × 4 × 5 × 7 × 9):(2 × 4 × 6 × 7 × 9):(2 × 4 × 6 × 8 × 9):(2 × 4 × 6 × 8 × 10)

(Understand the above mechanism with the help of the method given in Example 2. In these cases, this method can be used as a shortcut to find the ratios in the following way: For A, take all the leftmost digits, and now keep shifting towards the right digits by quitting one by one all the leftmost digits. So, B = Right digit of 1st ratio and so on for C, D, E, and F.)

Example 4 If A:B = 3:4,
B:C = 5:7
C:D = 10:11
What is the ratio of A:D?

Solution A = 3 × 5 × 10 and D = 4 × 7 × 11
So, the ratio = 150:308
Alternatively, (A/B) × (B/C) × (C/D) = (3/4) × (5/7) × (10/11) = (3 × 5 × 10)/(4 × 7 × 11) = 150:308

Example 5 A, B, C, and D purchase a gift worth ₹60. A pays 1/2 of what others are paying, B pays 1/3rd of what others are paying and C pays 1/4th of what others are paying. What is the amount paid by D?

Solution Since A is paying 1/2 of what others are paying, so A is paying 1/3rd of the total amount.

(To understand this, let us assume that B, C, and D are paying ₹2x. So, A is paying ₹x. The total amount being paid by A, B, C, and D = 3x = ₹60, hence, the amount paid by A = x/3x = 1/3rd of the total.)

So, the amount paid by A = 60/3 = ₹20

Similarly, B is paying 1/4th of the total and C is paying 1/5th of the total.

Hence, the amount paid by B and C are ₹15 and ₹12, respectively.

So, the amount paid by D = ₹13

Ratio as a Multiplier

The moment we say that the ratio of two numbers A and B is 5:1, what we mean to say that A is 5 times of B.

It can also be seen that A:B:C in A/2:B/3:C/4 = K is not same as A:B:C = 1/2:1/3:1/4 since multiplier of A, B, and C are not the same in both the cases.

Ratio of A:B:C in A/2:B/3:C/4 = K can be calculated in the following way:

Since A/2 = B/3 = C/4 = K, so A = 2K, B = 3K, and C = 4 K

Hence, the ratio of A:B:C = 2:3:4

While calculating the ratio of A, B, and C in A:B:C = 1/2:1/3:1/4, we will multiply each of A, B, and C by the LCM of the denominator of all the ratios, that is, 12.

So, A:B:C = 6:4:3

Example 6 Ten persons can cut 8 trees in 12 days. How many days will 8 persons take to cut 6 trees?

Solution Let us see this question in a changed perspective.

Suppose if the question is—10 persons can cut 8 trees in 12 days. How many days will 10 persons take to cut 4 trees?

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Answer to this question is:Since the amount of work is getting halved, so the number of days will also get halved.

There are three factors, namely the number of men, the number of days and the number of trees, which are responsible for the final answer.

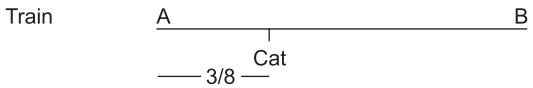
Since the number of men are less in the final situation, so more number of days will be required. Hence, multiplier = 10/8 (had there been 12 persons, multiplier would have been 10/12).

The number of trees are less in the final situation, so less number days will be required. So, multiplier = 6/8

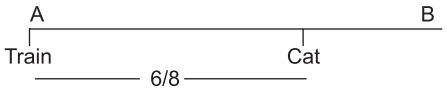
Hence, the total number of days = 12 × 10/8 × 6/8 = 90/8 = 11.25 days

Example 7 A train approaches a tunnel AB. Inside the tunnel, a cat is sitting at a point that is 3/8th of the distance of AB measured from the entrance A. When the train whistles, the cat runs. If the cat moves to the entrance of the tunnel A, the train catches the cat exactly at the entrance. If the cat moves to the exit B, the train catches the cat exactly at the exit. What is the ratio of the speed of the train and the speed of the cat?

Solution



Initially, this was the position of the train and the cat. Now, let us assume that the cat is moving towards exit B. The moment the cat covers 3/8th of AB distance in the direction of exit B, the train will be at the entrance A.



Now, if the cat moves in the direction of exit B, the train is catching up with the cat at the exit B. So, in the time cat covers 2/8th distance, the train is covering the whole distance from A to B.

So, the ratio of the distance covered by train and the distance covered by the cat = 4:1

So, the ratio of speed = 4:1

Example 8 Pranesh can do a work in 15 days. In how many days, will the work be completed by his brother Saket if efficiency of Saket is 60% more than that of Pranesh?

Solution Since the ratio of efficiency of Pranesh and Saket = 100:160 = 5:8, the number of days taken by Pranesh and Saket will be in the ratio of 8:5.

Since Pranesh takes 15 days to do this work, Saket will take 15 × 5/8 = 9.37 days

Comparison of Ratio

This is one of the most important calculations and is extensively used in DI. On an average, if somebody does 100 calculations in DI at least 8 to 10 calculations will be from comparing the ratios. Normally, there are two methods to compare two or more than two ratios:

Cross Multiplication Method

Example Let us compare $\frac{11}{15}$ and $\frac{13}{18}$.

$$\frac{11}{15} \times \frac{13}{18}$$

Cross multiplying numerator of first fraction with the denominator of second fraction and denominator of first fraction with the numerator of second fraction,

11×18 $\quad$ 13×15

198 $\quad$ 195

Since, 198 is greater than 195 the first fraction $\left(\frac{11}{15}\right)$ is greater than the second fraction $\left(\frac{13}{18}\right)$.

Decimal Calculation

$$\frac{11}{15} = 0.733 \qquad \frac{13}{18} = 0.722$$

Obviously, here the first fraction $\left(\frac{11}{15}\right)$ is greater than the second fraction $\left(\frac{13}{18}\right)$.

However, if we have to compare $\frac{3156}{5438}$ and $\frac{3423}{5822}$, then using any of the above two methods becomes cumbersome and time-consuming.

Here, we will compare ratios with the help of percentage.

Percentage Comparison

Let us first understand this with the help of the following ratios:

First case $\quad \frac{10}{15} \xrightarrow[100\%]{} \xrightarrow{100\%} \frac{20}{30}$

Second case $\quad \frac{10}{15} \xrightarrow[100\%]{} \xrightarrow{200\%} \frac{30}{30}$

Third case $\quad \frac{10}{15} \xrightarrow[300\%]{} \xrightarrow{200\%} \frac{30}{60}$

In the first case, percentage change in numerator (100%↑) = percentage change in denominator (100%↑), so ratios are equal.

In the second case, percentage change in the numerator (200%↑) > percentage change in the denominator (100%↑), so the second ratio is greater than the 1st ratio.

In the third case, percentage change in the numerator (200%↑) < percentage change in the denominator (300%↑), so the 1st ratio is greater than the 2nd ratio.

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This particular example can also be seen as a general rule for determining the order of ratios.

PROPORTION

In case of an analogy, two quantities share same kind of relationship. For example, what Macbeth is to William Shakespeare, Dr Zivago is to Boris Pasternak.

In QA, the same is true for proportion. It is basically the equality of the two ratios.

$$\frac{A}{B} = \frac{C}{D}$$

When A, B, C, and D are in proportion, then A and D are known as 'extremes', and B and C are known as 'means'.

Therefore, we can say,

Product of extremes = Product of means

Example 9 What is the value of x in the following expression?

$$\frac{5}{8} = \frac{x}{12}$$

Solution $\frac{5}{8} = \frac{x}{12}$

$\Rightarrow \quad x = \frac{60}{8} = 7.5$

It can be calculated with the help of percentages also. In this question, the percentage increase in the denominator is 50%, so the numerator will also increase by 50%.

Standard Results/Definitions on Ratio/Proportion

1. Continued proportion

   a, b, and c are said to be in continued proportion

   if $\frac{a}{b} = \frac{b}{c}$

   So, $b^2 = ac$. Here, b is known as the mean proportion. Similarly, if a, b, c, and d are in continued proportion, then we get:

   $$\frac{a}{b} = \frac{b}{c} = \frac{c}{d}$$

2. Componendo

   if $\frac{a}{b} = \frac{c}{d}$, then $\frac{a+b}{b} = \frac{c+d}{d}$

3. Dividendo

   If $\frac{a}{b} = \frac{c}{d}$, then $\frac{a-b}{b} = \frac{c-d}{d}$

4. Componendo and Dividendo

   If $\frac{a}{b} = \frac{c}{d}$, then $\frac{a+b}{a-b} = \frac{c+d}{c-d}$

5. Invertendo

   If $\frac{a}{b} = \frac{c}{d}$, then $\frac{b}{a} = \frac{d}{c}$

6. Alterando

   If $\frac{a}{b} = \frac{c}{d}$, then $\frac{a}{c} = \frac{b}{d}$

7. $\frac{a}{b} = \frac{c}{d} = \frac{a+c}{b+d}$

   In general, if $\frac{a}{b} = \frac{c}{d} = \frac{e}{f} = ... = K$

   Then, $\frac{a}{b} = \frac{c}{d} = \frac{e}{f} = ... = K = \frac{a+c+e+....}{b+d+f+....}$

   = (any combination of the numerator/any combination of the corresponding denominator)

   For example, 1/2 = 3/6 = 4/8 =…= (1 + 3 + 4)/ (2 + 6 + 8) = (3 + 4)/(6 + 8)

8. If we multiply the numerator and the denominator of a ratio by any number N (N ≠ 0), then the ratio remains same.

   A/B = NA/NB

9. If we divide the numerator and the denominator of a ratio by any number N (N ≠ 0), then the ratio remains same.

   A/B = (A/N)/(B/N)

10. If a/b, c/d, e/f …etc., are all unequal ratios, then the value of $(a+c+e+\ldots)/(b+d+f+\ldots)$ lies in between the minimum and the maximum of all these ratios.

VARIATION

Two quantities A and B are said to be varying with each other if there exists some relationship between A and B such that the change in A and B is uniform and guided by some rule.

Some typical examples of variation:

- Area (A) of a circle = πR^2, where R is the radius of the circle.

  Area of a circle depends upon the value of the radius of a circle, or, in other words, we can say that the area of a circle varies as the square of the radius of a circle.
- At a constant temperature, pressure is inversely proportional to the volume.
- If the speed of any vehicle is constant, then the distance traversed is proportional to the time taken to cover the distance.

Direct Proportion

If A varies directly to B, then A is said to be in direct proportion to B.

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It is written as A ∝ B.

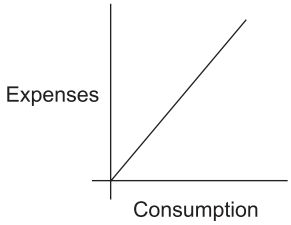
It can be understood with the typical example of percentage relating to expenses, consumption, and price of the article.

If the price of a article is constant, then Consumption ∝ Expenses.

⇒ Consumption = *K*. Expenses, where *K* is proportionality constant

If we increase consumption by 20%, then the expenses will also increase by 20%.

At a constant price, if a graph is drawn between consumption and expenses by taking them at X-axis and Y-axis, respectively, then this graph will be a straight line.



This is what we mean to say with direct proportion.

Inverse Proportion

If A varies inversely to *B*, then *A* is said to be in inverse proportion to *B*.

It is written as A ∝ 1/B.

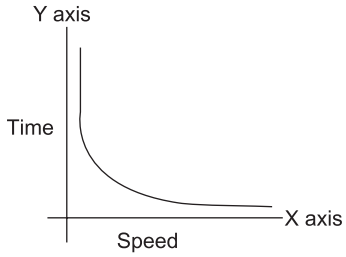
It can be understood with a time-speed-distance example, where if the distance is constant, then speed ∝ 1/time.

Assuming the distance between New Delhi to Patna is 1000 km then consider the following table:

Speed (km/h)	Time (in h)
500	2
2 × 500	1/2 × 2
3 × 500	1/3 × 2
4/3 × 500	3/4 × 2

It can be seen here that the multiplier of time is reciprocal of the multiplier of speed.

For any fixed distance, if we draw a graph between speed and time by taking them at X-axis and Y-axis, respectively, then this graph will be a curve.



Therefore, when the speed is minimum, the time is maximum and when the speed is maximum, the time is minimum.

Example 10 The height of a tree varies as the square root of its age (between 5 and 17 years). When the age of a tree is 9 years, its height is 4 feet. What will be the height of the tree at the age of 16?

Solution Let us assume the height of the tree is *H* and its age is *A* years.

So, $H \propto \sqrt{A}$, or, $H = K \times \sqrt{A}$

Now, $4 = K \times \sqrt{9}$

$\Rightarrow K = 4/3$

So, height at the age of 16 years = $H = K \times \sqrt{A} = 4/3 \times 4 = 16/3 = 5$ feet 4 inches

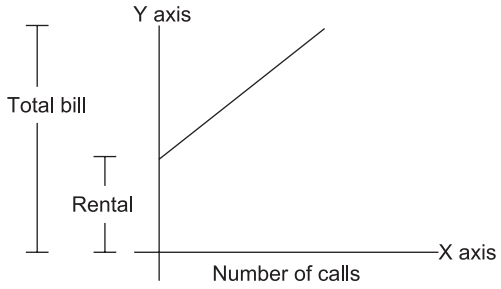
Direct Relation

A is directly related to *B* and as *B* changes, *A* also changes but not proportionally.

It is written as $A = C + K.B$, where *C* and *K* are constants.

One classical example of direct relation can be seen as the telephone connection in a house. In a telephone connection, we pay some money as the rent along with the phone charges according to the rate and number of calls made. So, the total bill = rental + *K* (number of calls).

If we draw a graph between the number of calls and the total bill by taking them at X-axis and Y-axis, respectively, this graph will be a straight line in the following way:



Example 11 Total expenses at a hostel is partly fixed and partly variable. When the number of students is 20, total expense is ₹15,000 and when the number of students is 30, total expense is ₹20,000. What will be the expense when the number of students is 40?

Solution Expenses = $F + K.V$; where *F* is the fixed cost and *V* is the number of students.

₹15,000 = $F + K.20$ (i)

₹20,000 = $F + K.30$ (ii)

Solving equations (i) and (ii), we get

₹5000 = $10.K \Rightarrow K$ = ₹500

So, F = ₹5000

So, $F + 40K$ = ₹5000 + 40 × 500 = ₹ 25,000

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APPLICATION OF RATIO, PROPORTION, AND VARIATION

Income–Expense Ratio

When the ratio of incomes and expenses of two persons are given and their savings is being asked.

Example 12 The ratio of the incomes of Mr Vinay Singh and Mr Arun Sharma is 3:5 and the ratio of their expenses is 1:3. Who is saving more?

Solution Let us assume the values of income and expenses of A and B.

	Income	Expense	Savings
Vinay Singh	₹3	₹1.5	₹1.5
Arun Sharma	₹5	₹4.5	₹0.5

So, Vinay saves more than Arun.
In the other case,

	Income	Expense	Savings
Vinay Singh	₹3	₹1	₹2
Arun Sharma	₹5	₹3	₹2

So, savings of both of them is equal.

	Income	Expense	Savings
Vinay Singh	₹3000	₹1600	₹1400
Arun Sharma	₹5000	₹4800	₹200

So, in this case, Mr Singh is saving less than Mr Sharma. Therefore, it is difficult to determine who is saving more.

The concept tells us: If the value of the ratio of income is more than the value of the ratio of expenses, then we cannot determine who is saving more. If the value of the ratio of expenses is more than the value of the ratio of income, then we can determine who is saving more. (Ratio should be taken in such a way that the value of ratio is less than 1, i.e., the numerator should be less than the denominator.)

In the above case, the value of the ratio of income = 3/5 = 0.6 and value of ratio of expenses = 1/3 = 0.33

Since the value of ratio of expenses < value of the ratio of income, we cannot determine who is saving more.

However, in the above question, if we take the ratio of income of Vinay and Arun as 3:5 and the ratio of their expenses as 3:1, then Arun is saving more.

Partnership

If two or more than two persons are investing their money in a joint venture, then the profit or loss incurred is distributed among everybody in the ratio of the time period for which the money has been invested by each of them.

It can be observed here that the partnership is a typical case of direct variation.

Example 13 A, B, C,…, K, L invested money in the ratio of 12:11:10:…: 2:1. The duration for which they invested the money is in the ratio of 1:2:3 :….: 11:12. Who will obtain the maximum profit at the end of year?

Solution Profit will be distributed in the ratio of product of the time and money of every individual.

So, the ratio of profit = 12 × 1:11 × 2:10 × 3:9 × 4:8 × 5:7 × 6:6 × 7:5 × 8:4 × 9:3 × 10:2 × 11:1 × 12

So, the maximum profit will be obtained by F and G both.

Practice Exercises

Warm Up

Q.1 The speed of three buses are in the ratio 2:3:4. The ratio between the time taken by these buses to travel the same distance is:
(a) 2:3:4 (b) 4:3:2 (c) 4:3:6 (d) 6:4:3

Q.2 The difference between the two positive numbers is 10 and the ratio between them is 5:3. Find the product of the two numbers.
(a) 375 (b) 325 (c) 275 (d) 125

Q.3 If 30 oxen can plough 1/7th of a field in 4 h, in how many hours will 18 oxen take to do the remaining work?
(a) 30 h (b) 20 h
(c) 15 h (d) 40 h

Q.4 Four numbers in the ratio of 1:3:4:7 add up to give a sum of 75. Find the value of the biggest number.
(a) 42 (b) 35
(c) 49 (d) 63

Q.5 In the above question, what is the difference between the biggest and the smallest number?
(a) 42 (b) 30 (c) 49 (d) 63

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Q.6 When we increase both the numerator and the denominator by 7, a fraction changes to 3/4. Find the original fraction.
(a) 5/12 (b) 7/9 (c) 2/5 (d) 3/8

Q.7 According to the Boyle's law, at a constant temperature, pressure of a definite mass of gas is inversely proportional to the volume. If the pressure is reduced by 20%, find the respective change in volume.
(a) –33.33% (b) +25%
(c) –25% (d) +33.33%

Q.8 What number must be subtracted from both the numerator and the denominator 27:35, so that it becomes equal to 2:3?
(a) 8 (b) 12
(c) 6 (d) None of these

Q.9 P is directly proportional to Q, and $Q = 7$ when $P = 15$, find P when $Q = 14$.
(a) 36 (b) 54 (c) 30 (d) 60

Q.10 The concentration of three milk solutions A, B, and C are 10%, 20%, and 30%, respectively. They are mixed in the ratio 2:3:N resulting in a 23% concentration solution. Find N.
(a) 7 (b) 6 (c) 5 (d) 4

Q.11 Which of the following will have the maximum change in their values if 10 is added to both the numerator and the denominator of all the fractions?
(a) 3:4 (b) 2:3 (c) 4:7 (d) 5:7

Q.12 If $p/q = 3/4$, then find the value of the expression $(5p - 3q)/(7p + 2q)$.
(a) 3/29 (b) 5/27 (c) 7/31 (d) 5/58

Q.13 In a mixture of milk and water of volume 30 L, the ratio of milk and water is 7:3. Find the quantity of water to be added to the mixture to make the ratio of milk and water 1:2.
(a) 30 (b) 32 (c) 33 (d) 35

Q.14 A and B invested ₹12,000 and ₹18,000, respectively, in a business for the whole year. At the year end, there was a total profit of ₹2000. What is the share of A in the profit?
(a) ₹800 (b) ₹1200
(c) ₹1600 (d) None of these

Q.15 A and B invested the same capital in a business. At the year end, they share the profit in the ratio of 3:2. If A has invested his capital for the whole year, for how many months B has invested his capital?
(a) 6 months (b) 8 months
(c) 9 months (d) None of these

Q.16 A and B invest ₹12,000 and ₹16,000, respectively, in a business. At the year end, they share the profit in the ratio of 3:1. If A has invested his capital for the whole year, for how many months B has invested his capital?
(a) 4 months (b) 3 months
(c) 6 months (d) 8 months

FOUNDATION

Q.1 King Dashrath decided to distribute gold coins to his three queens in the following way: second queen would get 5/7th of what the first queen would get and the third queen would get 3/5th of what the second queen would get. First queen got 60 gold coins more than the 3rd queen. How many gold coins were distributed to the three queens?
(a) 120 (b) 175 (c) 225 (d) 250

Q.2 The sum of the ages of the 4 members of Sinha family is 140 years. 5 years ago the ages of the 4 members Nishu, Vicky, Mrs Sinha and Mr Sinha were in the ratio of 2:3:7:8. After how many years would Nishu be as old as the present age of his mother?
(a) 10 yrs (b) 17 yrs (c) 30 yrs (d) 32 yrs

Q.3 In the prime minister office, one-third of the workers are women, half of the women are married and one-third of the married women have children. If three-fourth of the men are married and two-third of the married men have children, what part of the persons working at the PMO are without children?
(a) 5/18 (b) 4/9 (c) 11/18 (d) 17/36

Q.4 Two-fifth of Anil's salary is equal of Bhuvan's salary and seven-ninth of Bhuvan's salary is equal to Chandra's salary. The sum of the salary of all of them is ₹770. Which of the following is the salary of each?
(a) 300, 225, 250 (b) 500, 425, 375
(c) 450, 180, 140 (d) 520, 610, 475

Q.5 Inspector Vijay Shankar spends one-fourth of his salary on the house rent, one-third on food and one-sixth on travel. After spending one-tenth of the remaining amount on clothes, he is left with ₹13,500. What is the difference between the amounts spent on the house rent and food?
(a) ₹7500 (b) ₹10,000 (c) ₹4000 (d) ₹5000

Q.6 Bidhan is planning to buy a bike worth ₹35,000, provided his brother agrees to lend him 3/2 times the money. Bidhan contributes and the financer provides 3 times

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the brother's contribution. How much does Bidhan's brother contribute?
(a) 5000 (b) 7500 (c) 10,000 (d) 7000

Q.7 A group of students decided to buy a book jointly which costs between ₹170 to 195. However, at the last moment, two students decided not to contribute and so, each of the remaining students had to pay one rupee each more. What was the price of the book if the students paid equal shares?
(a) ₹182 (b) ₹188.12 (c) ₹192.4 (d) ₹180

Q.8 An alloy of manganese, tin and bronze contains 90% bronze, 7% manganese, and 3% tin. A second alloy of bronze and tin is melted with the first and the mixture contains 85% of bronze, 5% of manganese, and 10% of tin. What is the percentage of bronze in the second alloy?
(a) 67.5% (b) 72.5% (c) 77.5% (d) 82.5%

Q.9 Vinit is carrying some money with him. He gives 25% of the money to his friend, Sachin. He spends $\frac{1}{3}$rd of the remaining, on a movie ticket. With 40% of the remaining money he does shopping. After all this, he meets his friend Gita who gives him 66.66% of the money which he was having at that time. Finally, Vinit gives $\frac{4}{5}$th of the total money to his mother. What is left is the square root of the initial amount he was carrying. How much money does Gita give him?
(a) 40 (b) 25 (c) 10 (d) 20

Q.10 Two cogged wheels of which one has 32 cogs and other 54 cogs, work into each other. If the latter turns 80 times in three quarter of a minute, how often does the other turn in 8 seconds? (Assume equal size cogs and equispaced)
(a) 24 (b) 16 (c) 32 (d) 8

Q.11 Kunal, Saurav, and Gopi assemble for a dutch party. Kunal brings 3 guavas, whereas Saurav brings 5. Since Gopi did not have any guavas, he contributed ₹8. How many rupees should Kunal and Saurav, respectively, get, assuming each of the three consumes an equal portion of the guavas?
(a) 1, 6 (b) 2, 5
(c) 5, 7 (d) None of these

Q.12 A bag contains ₹600 in the form of one rupee, 50 paise and 25 paise coins in the ratio of 3:4:12. Find the total number of 25 paise coins in the bag.
(a) 75 (b) 200 (c) 300 (d) 900

Q.13 A certain sum of money was divided among Anand, Bidhan, and Chandra in the following way: Chandra got half as much as Anand and Bidhan together got. Anand got one-third of what Bidhan and Chandra together got. What is the ratio of Anand's share to that of Chandra's share?
(a) 1:2 (b) 3:5
(c) 4:7 (d) None of these

Q.14 Anil started a manufacturing unit with a certain amount of money. After a few months, Dheeraj became his partner, contributing three times of what Anil had contributed. At the end of the year, each was entitled to half the total profit. If Anil started the unit in January, then when did Dheeraj join as a partner?
(a) August (b) September
(c) July (d) October

Q.15 Pawan, Qureshi, and Ravi entered into partnership, and provided capital of ₹22,000, ₹26,000 and ₹34,000, respectively. Some months later ₹10,000 extra capital was supplied by Qureshi. At the end of 12 months, the total profit was ₹50,274, and Pawan's share was ₹12,474. When did Qureshi supply the extra capital?
(a) After 4 months (b) After 5 months
(c) After 6 months (d) None of these

Q.16 A's salary is $\frac{5}{4}$ of B's salary, and B's salary is $\frac{4}{3}$ of C's salary, whereas M's salary is equal to the sum of salaries of A, B, and C. If salary of C is ₹1500, What is the salary of M?
(a) 5500 (b) 6500 (c) 7000 (d) 6000

Q.17 Four years ago, the ratio of the ages of A and B was 13:9 and eight years hence, it would be 4:3. The difference of their present ages is:
(a) 56 years (b) 40 years
(c) 16 years (d) 24 years

Q.18 The work done by a man, a woman and a child is in the ratio of 3:2:1. There are 20 men, 30 women and 36 children in a factory. Their weekly wages amount to ₹780, which is divided in the ratio of work done by the men, women and children. What will be the wages of 15 men, 21 women, and 30 children for 2 weeks?
(a) ₹585 (b) ₹292.5 (c) ₹1170 (d) ₹900

Q.19 On the eve of the foundation day party of Due North Inc., the ratio of males to females was 5:3. However, when 10 males left the party, this ratio got changed to 1:1. How many people were there originally at the party?
(a) 48 (b) 32 (c) 64 (d) 40

Q.20 What is the angle in degrees made by a sector, the ratio of whose area with the area of the semicircle is equal to 1:10?
(a) 36 (b) 18 (c) 24 (d) 9

Q.21 Two candles of the same length are lighted at 12 noon. The first is consumed in 6 h and the second in 4 h. Assuming that each candle burns at a constant rate, in

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how many hours after being lighted, was the first candle twice the length of the second?
(a) 3 pm (b) 2 pm
(c) 1:30 pm (d) 2:30 pm

Q.22 The value of a diamond varies directly as the square of its weight. If a diamond worth ₹10,000 is divided into 2 pieces in the ratio of 4:6, what is the loss in value?
(a) 52% (b) 48%
(c) 36% (d) None of these

Q.23 The monthly income of A and B are in the ratio of 3:2, and their expenditures are in the ratio of 5:3. If each of them saves ₹1000 a month, then what is the income of A?
(a) ₹4000 (b) ₹6000
(c) ₹9000 (d) None of these

Q.24 If ab, bc, x, and c^2 are in proportion, then find x, if none of these are equal to zero.
(a) a^2c (b) c^2a
(c) ac (d) None of these

Q.25 The first, second, and third class fares between two railway stations, Patna and Lucknow were 10:8:3 and the number of first, second, and third class passengers between the two stations was is 3:4:10. If total sales of the ticket is ₹16,100, find the money obtained by the sales of second class tickets.
(a) ₹5250 (b) ₹5600 (c) ₹6400 (d) ₹6650

Q.26 If $A:B = 3:4$, then what will be the value of $(3A^2 + 4B)/(3A - 4B^2)$?
(a) 43:55
(b) −43:55
(c) Either of (a) or (b)
(d) Cannot be determined

Q.27 The sum total of the salary of Atul, Binod, and Charu is ₹200. If they spend 60%, 80%, and 75% of their respective salaries, then their savings are in the ratio of 8:8:5. Find the salary of Atul.
(a) ₹50 (b) ₹100 (c) ₹200 (d) ₹250

Q.28 Two horses cost as much as 5 dogs, 6 dogs as much as 8 oxen, 10 oxen as much as 50 sheep, 14 sheep as much as 9 goats. If the price of one goat is ₹70, how much will one horse cost?
(a) ₹670 (b) ₹730 (c) ₹830 (d) ₹750

Q.29 A pond has a rare breed of lotus that triples in number every minute. If $\frac{1}{3}$rd of the pond is full of lotus in 30 min, what is the total time taken for the whole pond to be full?
(a) 1 min (b) 31 min
(c) 90 min (d) 60 min

Q.30 Satish used 6 L of oil paint to paint a map of India, 6 m high. How many litres of paint would Santosh need for painting a proportionally scaled map of the height of 18 m?
(a) 54 (b) 18
(c) 30 (d) Cannot be determined

Q.31 A is proportional to B. B is inversely proportional to C. C is proportional to the square of D. D is directly proportional to the cube root of E. Assuming positive integers, if A increases then E:
(a) Increases
(b) Decreases
(c) Cannot say
(d) Could increase or decrease

Q.32 Measurement of the temperature is carried out using thermometers in which the mercury expands linearly with the change in temperature. If the temperature range from the melting point of water to the boiling point of water is divided into 0 to 100 in Celsius scale and 32 to 212 in Fahrenheit scale, what is the temperature at which both the scales indicate the same value?
(a) 0 (b) 40 (c) 32 (d) −40

Q.33 The sum of four numbers is 255. The ratio of the first number to the second number is 2:3. The ratio of the second number to the third number is 5:6. The ratio of the third number to the fourth number is 8:9. What is the average of the second number and the third number?
(a) 72 (b) 132
(c) 60 (d) None of these

Q.34 The population of the Balearic Island situated in Mediterranean Sea is 18,000. Balearic Island has three ports A, B, and C. Every year, the entire population of each port moves to the other two ports, half going to one and the remaining half going to the other. The current populations of A, B, and C are 2000, 6000, and 10,000, respectively. Then four years from now, the population of Port A will be:
(a) 5000 (b) 6500 (c) 6000 (d) 5750

Q.35 Arun has a certain amount of money in the denomination of 1 rupee and 10 rupee notes. The number of 1 rupee notes multiplied by the number of 10 rupee notes is equal to the total money (in rupees) that he has. What is the total number of ten rupee notes that he can have?
(a) 11 (b) 13
(c) 15 (d) None of these

Q.36 Divya and Nandan go to a bakery and buy some black currant pastries. Divya buys 9 black currants more than Nandan. Later in the day, they go to the bakery again and buy some butter scotch pastries. In the entire day, Divya and Nandan buy a total of 12 and 21 pastries, respectively, but spend the same amount of money. Find the ratio of the price of a black current to that of a butter scotch.
(a) 3:2 (b) 2:1 (c) 5:2 (d) 3:1

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Q.37 Rohan purchases some pastries of two varieties— one variety costing ₹3 per piece and the other ₹7 per piece. He wanted to spend exactly 50 rupees for the same. What is the maximum number of pastries he can purchase of the costlier variety?
(a) 6 (b) 5 (c) 4 (d) 2

Q.38 Three friends A, B, and C started a venture with capitals in the ratio of 4:1:15. At the end of every quarter A halves his capital, whereas B doubles his capital, and C leaves his capital untouched. This process is repeated till the end of the year. If at the end of the year B's share of the profit is ₹22,000, what is the total profit?
(a) ₹88,000 (b) ₹1,10,000
(c) ₹1,21,000 (d) None of these

Q.39 Rakesh, Shyam, and Prashant have some money with them. If Rakesh gives ₹15 to Shyam then he would have the same amount. If Prashant takes half of Shyam's money, he would have 5 times as much money left with Shyam. After this, the ratio of money with Rakesh, Shyam and Prashant is 5:2:4. What is the total amount with them?
(a) ₹55 (b) ₹110 (c) ₹165 (d) ₹231

Q.40 King Fahd of Saudi Arabia distributed his property equally among his 6 sons as follows— to his eldest son he gave 1 lakh gold coins and 1/7th of the remaining coins. To the second son, he gave 2 lakh gold coins and 1/7th of the remaining gold coins and so on and to his youngest son he gave 6 lakh gold coins. The entire number of gold coins was distributed without any remainder. What was the total number of gold coins with the king?
(a) 22 lakh (b) 50 lakh (c) 71 lakh (d) 36 lakh

Q.41 Abhishek bought a total of 'N' dozen fruits consisting of apples and bananas in the ratio of 3:7. The number of dozen of apples he bought is equal to the cost per dozen of bananas and vice versa. Total amount spent by him is equal to ₹1050. Find the number of dozen of apples and the number of dozen of bananas he bought, respectively.
(a) 12 dozen and 38 dozen
(b) 14 dozen and 36 dozen
(c) 15 dozen and 35 dozen
(d) 18 dozen and 32 dozen

Q.42 If $xy = r$, $yz = r^3$, $xz = r^2$, also $x + y + z = 13$ and $x^2 + y^2 + z^2 = 91$, then what is the value of $z:y$?
(a) 3 (b) $\frac{7}{3}$ (c) 13 (d) $\frac{13}{3}$

Q.43 Gandhiji owns cows, some black and some white. He finds that 4 black cows and 3 white cows gave the same amount of milk in 5 days as 3 black cows and 5 white cows gave in 4 days. What is the ratio of milk given by a black cow in a day to that given by a white cow in a day?
(a) 8:5 (b) 5:8 (c) 3:5 (d) 5:3

Q.44 Two software developers Vinit and Harpal share a mainframe and the final cost is ₹74,750. Vinit uses 5 type I terminals for 7 weeks and 22 type II terminals for 4 weeks. Harpal uses 10 type I terminals for 6 week and 8 type II terminals for 6 weeks. If cost of operating each of type I and type II terminal is in a ratio of 2:3, find the share of Vinit and Harpal in the final cost.
(a) ₹41,750, ₹33,000 (b) ₹40,000, ₹34,000
(c) ₹41,000, ₹3,80,000 (d) ₹50,000, ₹24,750

Q.45 A metal trader buys 2 kinds of silver foils, the ratio of their prices being 1:4. He sells the alloy at ₹90 per kg so that he can make a profit of 20%. If the ratio of their quantities present in a alloy is 6:1, respectively, find the purchase price of the foil present in lesser quantity.
(a) ₹52.5 (b) ₹55 (c) ₹47.5 (d) ₹45

Q.46 A heap of topaz, rubies, and emeralds is required to be formed. A topaz is worth ₹5 crores, a ruby ₹2 crores and an emerald ₹3 crores. The weight of a topaz is 0.7 kg, weight of a ruby is 0.2 kg and that of an emerald is 0.3 kg. If the total weight allowed is 8 kgs, then which combination maximizes the worth of the entire heap?
(a) 10 topaz, 2 rubies, and 2 emeralds
(b) 6 topaz, 1 ruby, and 12 emeralds
(c) 11 topaz and 1 emerald
(d) 26 emeralds and 1 ruby

Direction for Questions 47 and 48: Read the passage below and solve the questions based on it.

A book having pages between 4000 and 5000 is divided into four parts, each part being divided into chapters. The total number of pages in each of the four parts is the same. The ratio of the chapters across all the parts is 6:5:10:14. The number of chapters in the fourth part is 70.

Q.47 What is the total number of pages in the book?
(a) 4000 (b) 4800 (c) 4200 (d) 4600

Q.48 What is the total number of chapters in the book?
(a) 113 (b) 226
(c) 127 (d) None of these

Q.49 A rat takes 5 leaps for every 4 leaps of a cat, but 3 leaps of the cat are equal to 4 leaps of the rat. What is the ratio of the speed of the rat to that of the cat?
(a) 11:15 (b) 15:11 (c) 16:15 (d) 15:16

Q.50 The ratio of the age of Sita and her mother is 2:3. N years from now, ratio of their ages will become 3:5. What is the value of N?
(a) Infinite (b) 1
(c) More than 1 but finite (d) Not possible

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MODERATE

Direction for Questions 1 to 3: Read the passage below and solve the questions based on it.

Two of my friends Nishant and Rajesh, have the same number of children. Each son of Nishant has 5 times as many sisters as the number of brothers, whereas each son of Rajesh has as many sisters as the number of brothers. Each daughter of Nishant has twice the number of the brothers but each daughter of Rajesh has twice the number of brothers as the number of sisters.

Q.1 How many children did Rajesh have?
(a) 3 (b) 4 (c) 6 (d) 7

Q.2 How many sons does Nishant have?
(a) 1 (b) 2
(c) 4 (d) Cannot be determined

Q.3 What is the ratio of the number of sons to the number of daughters in the 2 families?
(a) 1:3 (b) 3:1
(c) 7:2 (d) None of these

Direction for Questions 4 to 6: Read the passage below and solve the questions based on it.

Three persons, Anny, Berry, and Cherry, started playing a game. They had money in the ratio of 3:2:1. It was decided that if Anny lost the game he would distribute 1/4th of the money that he had prior to the round, equally between Berry and Cherry. If Berry lost the game he would distribute half of his money equally between Anny and Cherry and if Cherry lost the game he would distribute 2/3rd of his money prior to that round equally between Anny and Berry. Berry lost the game first, and then Cherry lost the next game followed by Anny and then followed by Berry again. In the process Anny gained ₹10.

Q.4 What was the total sum of money they had initially?
(a) ₹60 (b) ₹120 (c) ₹150 (d) ₹240

Q.5 After which round the game would have been stopped for Anny to gain the maximum profit?
(a) Round 1 (b) Round 2
(c) Round 3 (d) Round 4

Q.6 Which of these did not have any chance of making profit in any of the rounds?
(a) Both Berry and Cherry
(b) Both Anny and Cherry
(c) Only Cherry
(d) Only Berry

Direction for Questions 7 to 9: Read the passage below and solve the questions based on it.

Anu, Bunty, Christy, and Dorthy are four salesmen. In the first year, they received a commission of ₹4200 from their company and divided it in the ratio of 2:3:4:5. In the second year, the commission doubled, the amount was divided in the ratio of 3:4:5:2. In the third year, the commission was tripled when compared to the first year and they shared in the ratio of 4:5:3:2, and in the fourth year, the commission was half compared to the previous year and they shared in the ratio of 4:3:5:2.

Q.7 What was the average yearly income of Christie over the period?
(a) ₹2287.5 (b) ₹2252.5
(c) ₹1680.25 (d) None of these

Q.8 How much should Bunty have earned so that his average yearly income would have been ₹3500?
(a) 2450 (b) 3250 (c) 4850 (d) 5750

Q.9 The ratio of the rate of tomatoes to the rate of brinjals is:
(a) 1:2 (b) 1:3
(c) 1:5 (d) None of these

Direction for Questions 10 and 11: Read the passage below and solve the questions based on it.

Narendrajee is the office assistant at LC's North Campus centre. One day he realized that (i) among all the visitors the ratio of the number of men and women was the same as that between women and children, (ii) maximum number of the visitors to the centre were children. At the end of the day, he finds that 7 women visited the centre.

Q.10 How many children visited the centre that day?
(a) 44 (b) 49 (c) 52 (d) 57

Q.11 How many men visited the centre that day?
(a) 25 (b) 68
(c) 49 (d) None of these

Q.12 *P* works twice as fast as *Q*, whereas *P* and *Q* together can work three times as fast as *R*. If *P*, *Q*, and *R* together work on a job, in what ratio should they share the earnings?
(a) 2:1:1 (b) 4:2:1 (c) 4:3:2 (d) 4:2:3

Direction for Questions 13 to 15: Read the passage below and solve the questions based on it.

Sachin has three children Ten, Dul, and Kar. One day, he brought 117 sweets for his children and planned to divide the

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sweets in such a manner that two times the number of sweets that Ten would have got should be equal to 2/3 times that Dul got and 4 times that Kar got. Since Sachin was poor at counting, he divided the sweets in the ratio of 2:3:4.

Q.13 How many sweets did Kar get as a result of wrong sharing?
(a) 16 (b) 20
(c) 22 (d) None of these

Q.14 If Kar were to get the same number of sweets that Dul got now after the wrong distribution had the distribution being done correctly, how much should the total number of sweets be?
(a) 150 (b) 169
(c) 216 (d) None of these

Q.15 If X bulbs consume X^2 units of electricity in X^3 days, find the time in which Y bulbs will consume Y units of electricity.
(a) X^1 days (b) X^2 days
(c) X^3 days (d) X^0 days

Direction for Questions 16 and 17: Read the passage below and solve the questions based on it.

There are certain number of apples, guavas and oranges in a basket. The number of each variety is more than one. The ratio of the number of apples to the number of guavas is equal to the ratio of the number of guavas to the number of oranges.

Q.16 If the total number of fruits is 61, then find the number of guavas.
(a) 16 (b) 20
(c) 25 (d) Cannot be determined

Q.17 If the number of guavas is 21, then which of the following can be the total number of fruits in the basket?
(a) 63 (b) 89 (c) 101 (d) 117

Q.18 The time taken by a jet, to reach a certain place while flying at a speed, slower by 300 m/h than the normal, is double the time taken by it while flying at a speed faster by 300 m/h than normal. What is the ratio of the 'delay time' while flying slower to the 'early time' while flying faster?
(a) 3:2 (b) 1:2
(c) 2:1 (d) Cannot be determined

Q.19 Anand and Bhavana have a few one rupee, 50 paise and 25 paise coins each. The total amount with each of them is ₹5. The number of coins of two of the three types of coins that Anand has are equal and same is the case for the coins that Bhavana has. If Anand has the maximum possible total number of coins, then find the maximum possible total number of coins that Bhavana can have given that she has fewer coins than Anand.
(a) 16 (b) 12 (c) 10 (d) 9

Q.20 In the above question, what is the number of 50 paise coins with Anand?
(a) 1 (b) 2 (c) 3 (d) 4

Q.21 In a flood-affected village called 'Waterland', there are 500 adult residents. They planned to construct a dam to overcome the situation. Each of the men contributed a sum equal to the number of women in the village and each woman a sum equal to the number of men. The village-chief came to know about this and contributed an amount equal to the contribution of the villagers. The total contribution then became ₹2.4 lakhs. Find the number of men in 'Waterland'.
(a) 250 (b) 300
(c) 350 (d) Cannot be determined

Q.22 If the man initially had ₹82, instead of ₹100, when he went to the market, then the ratio of the rate of tomatoes to the rate of cucumber was:
(a) 19:4 (b) 2:3
(c) 4:19 (d) None of these

Q.23 The sides of a right angled triangle are a, $a + x$ and $a + 2x$ with a and x both positive. The ratio of a to x is:
(a) 3:1 (b) 5:2 (c) 2:1 (d) 3:4

Q.24 If $\frac{a}{b+c} = \frac{b}{c+a} = \frac{c}{a+b}$, then each fraction is equal to
i. -1 ii. 1/2 iii. 1
(a) i and ii only (b) ii and iii only
(c) i, ii, and iii only (d) None of these

Direction for Questions 25 and 26: Read the passage below and solve the questions based on it.

Ujala Jewellers is a well-known diamond jewellery shop. Owner of the shop has priced his diamond necklaces such that the cost varied as the square of the number of diamonds, for example the necklace placed at the first right top corner has 8 diamonds and costs ₹1,60,000. There are lots of necklaces and the price of all necklaces have been found by the same method.

Q.25 What would be the price of the bracelet with 10 diamonds?
(a) 1,75,000 (b) 2,25,000
(c) 2,50,000 (d) 4,00,000

Q.26 Vijay Shanker asked for diamonds in a necklace to be divided in the ratio of 5:3:2. Owner of Ujala Jewellers knew that he would suffer some loss. What is the %

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decrease in the price that he would incur, if he has to oblige Vijay Shanker?
(a) 22% (b) 38% (c) 62% (d) 78%

Q.27 Between two stations Kanpur and Mumbai the first, second and third class fares were fixed in the ratio of 8:6:3. However, when Mr L Prasad, the railway minister, presented the railway budget, he reduced the first class fares by 1/6 and the second class by 1/12 in order to rationalize the fares vis-à-vis the second-class fares. In a year, ratio of the number of first, second and third class passengers were, respectively, 9:12:26 and the money at the booking offices was ₹1,08,800. How much was paid by the first class passengers?
(a) ₹32,080 (b) ₹16,800
(c) ₹24,400 (d) None of these

Direction for Questions 28 and 29: Read the passage below and solve the questions based on it.

In the famous Hawthorne studies by Elton Mayo, employees of the organization were given the options of going to several places for entertainment. One-seventh of the employees preferred to go to the polo ground, a number equal to the square root of the remaining employees went to the museum. One-fifth of the remaining employees went to the fair. One-third of the remaining to the Science exhibition and the rest to watch movie ASHOKA, which was running house full in the nearby theatre. The total number of employees who went to the polo ground and the museum is twice that of the employees who went to the fair.

Q.28 What is the strength of the organization?
(a) 14 (b) 35 (c) 42 (d) 49

Q.29 The entry tickets for museum, fair, exhibition and the movie were ₹5, ₹4, ₹2, and ₹25, respectively. How much money did the employees pay for all these on that Sunday?
(a) ₹275 (b) ₹320 (c) ₹388 (d) ₹470

Direction for Questions 30 and 31: Read the passage given and solve the questions based on it.

In the last year's 'Tycoon Contest', nine teams from different colleges participated. Navneet Anand, the quiz master, asked the nine teams— 'I have 9 coins, the ratio of the weight of the coins are same, except one. I give the different numbers to these 9 coins starting from 11 to 19. The team who can find the coin with different weights in the least number of weighings can take that coin and will be the winner'.

Q.30 Team number 5 said that they knew the answer and if they had won the coin, what was their answer?
(a) One weighing (b) Two weighings
(c) Three weighings (d) Four weighings

Q.31 Navneet Anand now said that the coin whose weight was different from the rest was the heaviest of the all. What is the minimum number of weighings required to find that coin? Contestant 8 gives the correct answer and his answer is:
(a) One weighing (b) Two weighings
(c) Three weighings (d) Four weighings

Q.32 The cost of building a wall is ₹1347. Wages of workmen is increased by 1/8 of the former wages and working hours per day have been increased by 1/20 of the former duration. What is the new cost (approximately) of building a wall, the length of which is two times the length of this wall and the other dimensions of this wall are same?
(a) ₹2692 (b) ₹2724 (c) ₹2886 (d) ₹2484

Q.33 If $a^2 = by + cz$, $b^2 = cz + ax$, $c^2 = ax + by$, then $\frac{x}{a+x}+\frac{y}{b+y}+\frac{z}{c+z}$?

(a) abc (b) 1 (c) $\frac{abc}{xyz}$ (d) $\frac{xyz}{abc}$

Q.34 If 29 goats can feed on a field of uniformly grown grass in 7 days or 25 goats can feed on the same field in 9 days, how many goats will feed in 6 days?
(a) 36 (b) 44 (c) 42 (d) 32

Q.35 Given, $x:y = a:b = 1:5$. Then, the value of $\frac{x^2+a^2}{y^2+b^2}$?

(a) $\frac{1}{5}$ (b) $\frac{1}{50}$
(c) $\frac{1}{250}$ (d) None of these

Q.36 a, b, and c are three positive numbers and $s=\frac{(a+b+c)}{2}$. If $(s-a):(s-b):(s-c) = 1:7:4$, the ratio of $a:b:c$?
(a) 8:10:6 (b) 9:4:12 (c) 7:8:9 (d) 11:5:8

Q.37 In an election, each voter may vote for two candidates. Half of the voters vote for A, but divide their votes between B, C, and D in the ratio of 3:2:1. Of the remaining, half vote for B and divides their votes between C and D in the ratio of 2:1. Of the remaining, half vote for C and D and the remaining 840 do not vote for anyone. How many votes were received by A, B, C and D, respectively?
(a) 3360, 3360, 3080, 1960
(b) 3360, 3360, 1960, 3080
(c) 3450, 3210, 4120, 1210
(d) 5000, 5000, 2500, 1250

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Direction for Questions 38 and 39: Read the passage below and solve the questions based on it.

Two friends Pankaj and Pranav decide to play a game where the person losing that round gives half of his money to the one who wins. They start playing with ₹100 each.

Q.38 At the end of 3 rounds, which of the following cannot be the amount that Pankaj has?
(a) ₹137.5 (b) ₹162.5 (c) ₹112.5 (d) ₹125

Q.39 What is the maximum amount that changes hands in the game in 3 rounds?
(a) ₹100 (b) ₹75 (c) ₹11.5 (d) ₹87.5

Q.40 The square of the time taken by a planet to complete one revolution around the sun varies as the cube of its mean distance from the sun. If the ratio of the mean distances of Mars and Saturn from the sun is 16:3. Find the number of days on Saturn that are equal to a year on Mars.
(a) 4496 days (b) 4294 days
(c) 5598 days (d) 4696 days

Q.41 In a forest reserve, the ratio of the number of deers, bear and fox is 3:7:5. If the difference between the number of deers and bears is a multiple of 3 as well as 7, what is the minimum number of animals in the park?
(a) 315 (b) 310
(c) 45 (d) Cannot be determined

Q.42 A watermelon is cut into two pieces in the ratio of 3:5 by weight. The bigger of the two is further cut in the ratio of 5:7 by weight. Find the ratio of each of the three pieces.
(a) 3:5:7 (b) 36:25:35
(c) 15:25:56 (d) None of these

Q.43 Shalaj and Associates, a shoe-manufacturing company, has three machines producing shoes. If machine X produces $\frac{1}{4}$th as many as machine Y produces in the same time, and machine Y produces twice as many as machine Z produces in the same time, then during a fixed time, what fraction of the total number of shoes is produced by machine Z?
(a) $\frac{1}{14}$ (b) $\frac{1}{3}$ (c) $\frac{2}{7}$ (d) $\frac{1}{2}$

Direction for Questions 44 and 45: Read the passage below and solve the questions based on it.

A man went to the market with ₹100 in his pocket. He bought 3 kgs of brinjals, y kgs of tomatoes and 4 kgs of cucumber. The amount spent on tomatoes was double than that spent on cucumbers and also is same as one-third of that spent on brinjals. After shopping, the man was left with ₹10 in his pocket.

Q.44 Which of the following statements is true?
(a) The ratio of the rate of tomatoes to the rate of brinjals is 1:4.
(b) The ratio of the rate of tomatoes to the rate of cucumber is 1:8.
(c) The ratio of the rate of cucumber to the rate of brinjals is 8:1.
(d) None of these

Q.45 If the man was left with ₹19 instead of ₹10 and the total weight of vegetables bought is 10 kgs then, which of the following statement is true?
(a) The value of y is 4 kg.
(b) The value of y is one more than the rate of tomatoes.
(c) The value of y is one-sixth of the amount spent on tomatoes.
(d) None of these

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Q.1 The Road Transport Corporation has hired a consultancy firm to determine the standard time that should be taken for a journey between two local destinations. The consultant after a detailed study and discussion submitted in his report that the time of journey is proportional to the product of the square root of number of passengers and the distance to be covered. For two distinct trips, the times taken are in the ratio of 2:1 and the distances covered are 80 km and 30 km, respectively. How many passengers are there in the first bus, if the second bus has 64 passengers?
(a) 9 (b) 16 (c) 36 (d) 24

Q.2. The inhabitants of Hawaii Island are taxed in a peculiar manner. They have to pay a fixed sum irrespective of their income strata. In addition to this, they have to pay an amount which is proportional to the excess of their salary over ₹50,000. Mr Kalyan pays a total tax of ₹6200 when his annual salary is ₹60,000 and his wife Mrs Kalyan pays a total tax of ₹7700 when her annual salary is ₹75,000. What is the yearly salary of Mr Ajay who pays an annual tax of ₹8200?
(a) ₹80,000 (b) ₹82,000
(c) ₹90,000 (d) None of these

Q.3 Rohit finds that the distance he covers is equal to the sum of two quantities, one of which varies with time 'T' and the second varies with the square of the time 't' that has elapsed from the start. After 5 seconds Rohit has covered a distance of 100 metres and after 6 seconds,

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he has covered a distance of 138 metres. How far is he from the starting point after a time of 20 seconds?
(a) 450 m (b) 880 m
(c) 1220 m (d) None of these

Q.4 While developing Tejas, the Light Combat Aircraft developed by DRDO to replace MIG series aircraft, it was found that the distance a man falls under gravity is $5t^2$ m in t s and the distance a man falls with an open parachute under gravity is $2t^2 - 3t$ m in t s. Squadron leader Ajay Rathod jumps out of a Tejas at a height of 2300 m with a closed parachute. Find the height at which he can open the parachute so as to touch the ground in exactly 40 s.
(a) 1050 m (b) 1775 m
(c) 1175 m (d) 1250 m

Q.5 A manufacturing unit is to be set up at Geneva. The cost of setting up a unit is proportional to the square of the raw material it handles. If for all the processes four separate units are deployed, then the raw material that the individual unit is going to handle is in the ratio of 6:3:3:1. But if a single unit is deployed, then the total raw material that is to be handled is summation of raw material that is to be handled by an individual unit. By installing separate units, the company saved ₹57 lakhs. What is the cost of installing a single unit?
(a) ₹84.5 lakhs (b) ₹285 lakhs
(c) ₹338 lakhs (d) None of these

Direction for Questions 6 and 7: Read the passage given and solve the questions based on it.

At 70° F, 100 litres of milk can dissolve a maximum of 1.5 kg of sugar. For every increase in temperature by 1° F, the solubility of milk increases by 0.1 L. For example, at 71° F, 100 litre of milk can dissolve 1.6 kg of sugar. The boiling point of milk is 100° F, i.e., when the temperature of milk reaches 100° F, it stops increasing further and the milk starts evaporating @ 200 g/min.

Q.6 If 20 L of milk at 80° F with 180 g sugar is heated till it starts to evaporate, then what is the maximum amount of sugar that can be added so that it does not sediment?
(a) 900 g (b) 520 g (c) 720 g (d) 770 g

Q.7 At 75° F, 75 L of milk with 1 kg of sugar is heated till it starts to evaporate, and appropriate quantity of sugar is added so that it becomes fully saturated. Thereafter, it is heated for 10 min. How much sugar would have deposited at the bottom after 10 min?
(a) 120 g (b) 90 g (c) 100 g (d) 60 g

Q.8 Pranav, Raj, and Santosh entered into a partnership with a sum of ₹6200, ₹9300, and ₹12,400, respectively. It was decided that Pranav will manage the business for the whole year. But after some months due to some personal problems, Pranav handed over his responsibility of managing the business to Raj for the rest of the year. The person who manages the business gets 10% of the profit as commission. If the total income of Pranav at the end of the year is ₹290 and the total profit is ₹1200, for how many months did Pranav manage the business?
(a) 2 (b) 4
(c) 6 (d) None of these

Q.9 The total profit made by running a train between New Delhi to Lucknow consists of two parts: (i) a constant amount of ₹24 lakh per trip and (ii) it varies as the square of the number of coaches attached to the engine that amounts to ₹$7P^2$, where P is the number of coaches attached to the engine in the trip. If the average profit per coach per trip should not fall below ₹169 lakh, then what is the minimum number of coaches that have to be attached to the engine?
(a) 23 (b) 24
(c) 25 (d) None of these

Direction for Questions 10 and 11: Answer the questions based on the following information.

A simple weighing balance having two pans is in perfect balance when no weights are kept in the two pans even though both the pans are not of the same weight. Further even when 10 kg is kept in the first pan and 5 kg is kept in the second pan, it remains in balance.

Q.10 What is the ratio of the weight of the first pan to the second pan?
(a) 1:2 (b) 2:1
(c) 3:1 (d) Cannot be determined

Q.11 What weight should be kept in the second pan if a weight of 5 kg is kept in the first pan so that the weighing scale is in balance?
(a) 2.5 kg (b) 1.25 kg
(c) 7.5 kg (d) None of these

Q.12 There are two alloys of gold and copper. In the first alloy there is twice as much gold as copper and in the second alloy there is 5 times less gold than copper. How many times of the second alloy should be taken than the first, in order to obtain a new alloy in which there would be twice as much copper as gold?
(a) Two times (b) Three times
(c) Four times (d) None of these

Q.13 The average age of the students in a class of 50 is 13. The weight of each student is directly proportional to the height. A 165 cm tall student has a weight of 33 kg. The average weight of the class is:
(a) 33 kg (b) 39 kg
(c) 36 kg (d) Data insufficient

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Q.14 Nishit Transport Corporation (NTC) operates buses between Kolkata and Patna. To increase the profit, NTC has sought the services of Dipanjan Das and Sons consultants. After a thorough study of the mechanism of the whole business, Dipanjan Das and Sons suggested that—(i) profit in running a bus is directly proportional to the distance when the number of passengers is constant, and (ii) directly proportional to the number of passengers in excess of a certain fixed number when the distance is constant. The profit is ₹80 when 30 passengers are carried over a distance of 40 km, and ₹180 when 35 passengers are carried over 60 km. What should be the advice, furnished by Dipanjan Das and Sons consultants, regarding the minimum number of passengers to be carried, so that there is no loss?
(a) 19 (b) 18 (c) 20 (d) 25

Q.15 During the acceleration and breaking test by two men for a new bike launched by Kajaj, the 1st man starts the scooter and accelerates upto 40 km/h for a distance of 30.2 m and drives the remaining distance at 40 km/h. Second man starts and accelerates upto 60 km/h for a distance of 80.9 m and travels with a speed of 60 km/h and then applies brakes to 0 km/h at the distance of 24.2 m. The total distance travelled is 200 m. If both of them started at the same time, then who will reach the 200 m mark first?
(a) First man (b) Second man
(c) Same time (d) Data insufficient

Q.16 Two alloys of chromium have different percentage of chromium in them. The first one weighs 6 kg and second one weighs 12 kg. One piece each of equal weight was cut off from both the alloys and the first piece was alloyed with the second alloy and the second piece was alloyed with the first one. As a result, the percentage of chromium became the same in the resulting two new alloys. What was the weight of each cut-off piece?
(a) 4 kg (b) 2 kg (c) 3 kg (d) 1 kg

Q.17 The cost C of manufacturing a brick can be estimated by the formula $C = 0.03\ ABt^2$, where A and B are the amounts in ₹of the two major ingredients: sand and clay, and t is the production time in hours. If A is increased by 50%, B is increased by 20%, and t is decreased by 30% by what percentage will the estimated cost of manufacturing the brick change?
(a) 12.4% (b) 11.8% (c) 14.6% (d) 18.2%

Q.18 During the bank wiring experiment to understand the impact of pattern of informal relationship among employees in any organization, it is found that the quantity of work done by a man in an hour is directly proportional to his pay per hour and inversely as the square root of the number of hours he works per day. He can finish a piece of work in six days when he works 9 h a day at ₹1 per hour. How many days will he take to finish the same work when working 16 h a day at the rate of ₹1.50 per hour?
(a) 10 days (b) 3 days (c) 8 days (d) 4 days

Q.19 According to the personnel administration prevalent in India, people are awarded pensions in proportion to the square root of the number of years they have served. One has served 9 years longer than the other and receives a pension greater by ₹250. If the length of service of the first had exceeded that of the second by $4\frac{1}{4}$ years their pensions would have been in the ratio of 9:8. How long have they served, respectively?
(a) 16 years and 25 years (b) 9 years and 16 years
(c) 9 years and 25 years (d) None of these

Q.20 The speed of man while walking, running, and driving are in the ratio of 1:3:9. If he travels equal distance while walking, running, and driving, what is the ratio of $\frac{\text{time taken}}{\text{velocity}}$ in the above three cases?
(a) 9:3:1 (b) 1:1:1 (c) 81:9:1 (d) 27:9:1

Q.21 Given that $2^x = 4^y = 8^z$ and $xyz = 288$. What is the value of $\frac{1}{2x} + \frac{1}{4y} + \frac{1}{8z}$?
(a) $\frac{18}{74}$ (b) $\frac{18}{96}$ (c) $\frac{11}{74}$ (d) $\frac{11}{96}$

Q.22 According to census 2001 of India, population of 5 states: State 1, State 2, State 3, State 4, State 5, showed that their population figures are in the ratio of 1:3:2:3:2. Further, the ratio of the percentage of women in the population in these states is 1:3:2:5:4. If the total population of women in the five states is 3.7 million, then what is the excess number of women in State 2 over that of State 5?
(a) 10 lakh (b) 1 lakh
(c) 2.2 lakh (d) Cannot be determined

Q.23 The time period of oscillation T for a pendulum is given by $T = k\sqrt{\frac{l}{g}}$, where l is the length of the pendulum and g is the acceleration due to gravity, 'k' is any const. If 'k' and 'g' remain unchanged under any condition, what should be the percentage change in l such that the time period T increases by 10%?
(a) 10% increase (b) 10% decrease
(c) 20% increase (d) 21% increase

Q.24 A tap A can fill an acid into a vessel in 30 min. Another tap B can fill alkali into a vessel in 20 min and a leak C at the bottom of the vessel can empty the vessel in 12 min. Initially, both the taps A and B are opened for 6 min. Then, tap B is closed and leak C is opened.

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For how many more minutes should both these taps A and C be kept opened so that the proportion of acid and alkali is same? (Assume tap C removes acid and alkali in exactly the same ratio as they were present in the solution when tap C was just opened and no reaction takes place.)

(a) 1 min (b) 2 min
(c) 3 min (d) 4 min

Q.25 If after the operation of Question 24 is performed (till the ratio is 1:1), tap C is closed and B is opened again and then taps A and B are kept open till the vessel is full, what will be the ratio of acid and alkali in the full vessel?

(a) 19:11 (b) 11:19 (c) 14:11 (d) 11:14

Direction for Questions 26 to 29: Read the passage below and solve the questions based on it.

On 13th of November, I kept some money in a magical purse in which on every alternate day starting from the 13th, money increases with respect to the closing amount on the evening of the previous day, that the ratio of money becomes 2:1 but on every alternate day starting from the 14th, the money decreases by ₹10, when compared to the closing amount on the evening of the previous day. I had ₹500 on the 19th and the magical purse kept on multiplying money for me till the 25th of that November.

Q.26 What is the initial amount of money that I kept into the purse?

(a) 50 (b) 80
(c) 140 (d) Cannot be determined

Q.27 How much money was there on the 22nd of November?

(a) 440 (b) 360
(c) 240 (d) None of these

Q.28 Had I stopped on the 21st, how much less money would I have received with respect to the money which I would have obtained on the 25th?

(a) 2880 (b) 3260
(c) 4270 (d) None of these

Q.29 Had I started with ₹100, how much more/less money would I have received than I am receiving now?

(a) 400 (b) 760 (c) 1280 (d) 1640

Q.30 The square of the ratio of the number of sweets with Aldrin's sons and daughters is equal to the cube of the ratio of the number of sweets with Bladimir's sons and daughters. The total number of chocolates with all the children of Aldrin and Bladimir is a prime number The number of sweets with Bladimir's sons equals the number of sweets with Aldrin's daughters. The number of sweets with Bladimir's daughters is 1/32 of the number of sweets with Aldrin's sons. What is the total number of sweets with all the children?

(a) 31 (b) 41 (c) 61 (d) 101

Direction for Questions 31 and 33: Read the passage below and solve the questions based on it.

Surya Pratap Singh has four children (two daughters who are older to the two sons). He decides to divide his property among his children such that both the daughters together get the same share as the two sons together. Elder son gets more than the younger son and the elder daughter gets more than the younger daughter. His entire property consists of cash, gold coin and copper plate. Each gold coin is worth ₹5000 and copper plate in worth ₹1000. The eldest daughter got 1/3rd of the gold and 1/5th of the copper, whereas elder son got ₹1,50,000 in cash, 50% of his wealth as gold and 250 copper plates. The younger daughter got 18% of the copper and 15% of the cash. The younger son got cash equal to 90% of the cash the elder daughter got and 1/6th of the gold and 12% of the copper. Surya Pratap Singh has gold worth three times that of copper.

Q.31 Who got the maximum wealth?

(a) Elder son (b) Elder daughter
(c) Younger son (d) Both (a) and (b)

Q.32 How much cash did the younger daughter get?

(a) ₹50,000 (b) ₹60,000
(c) ₹75,000 (d) Cannot be determined

Q.33 If the younger son wants to sell all the copper he has, who is capable of buying it by paying cash?

(a) Elder son (b) Elder daughter
(c) Younger son (d) All the three

Q.34 Two items having the same cost price and marked price are given two different discounts, which differ by 20 percentage points. If their profits differ by 32 percentage points, what is the ratio of their cost price to the marked price?

(a) 5:8 (b) 4:7
(c) 2:3 (d) Cannot be determined

Q.35 A, B, and C run around a circular track of the length 300 m. A and B run with speeds of 10 m/s and 12 m/s in the same direction, respectively, and C runs in the opposite direction with a speed of 15 m/s. If all the three start at the same time, which of the following is true?

(a) For a given period of time, the ratio of the number of meeting points of A and C to the number of meeting points of B and C is more than 1.
(b) For a given period of time, the ratio of the number of meeting points of B and C to the number of meeting points of A and C is more than 1.

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(c) For a given period of time, the ratio of the number of meeting points of A and C to the number of meeting points of B and C is equal to 1.
(d) None of the above can be concluded.

Direction for Questions 36 to 38: Read the passage below and solve the questions based on it.

In a class, every student plays exactly one of the following games—squash, table tennis, hockey, handball, and cricket. Exactly 1/12th of the girls and 1/8th of the boys play squash; 1/15th of the girls and 1/6th of the boys play hockey; 1/4th of the girls and 1/12th of the boys play hand-ball; 1/5th of the girls and 3/8th of the boys plays table tennis, and the remaining boys and girls play cricket. The total strength of the class is less than 120.

Q.36 How many girls are there in the class?
(a) 60 (b) 48
(c) 90 (d) Cannot be determined

Q.37 Which of the following cannot be the ratio of boys and girls in the class?
(a) 2:5 (b) 4:5
(c) 3:4 (d) Cannot be determined

Q.38 What is the total strength of the class?
(a) 84 (b) 108
(c) 132 (d) Either (a) or (b)

Direction for Questions 39 and 40: Read the passage below and solve the questions based on it.

There are four friends Sharat, Chandra, Mayank, and Sid living in the same village of Bhagalpur. Once Sharat had nothing to eat, so, Chandra, Mayank, and Sid collected their rotis and gave a part of their rotis to Sharat. It was later found that the number of rotis given by Chandra is one more than the rotis given by Mayank, and the number of rotis given by Mayank is one more than the rotis given by Sid. Now, Sharat was required to pay the cost of rotis in the ratio of the rotis contributed by all three of them. It was found that Chandra should get twice as much money as Mayank. Price of one roti = ₹5.
(All the values, number of rotis and money are in integers.)

Q.39 How much money did Chandra get?
(a) ₹8 (b) ₹12
(c) ₹16 (d) None of these

Q.40 How many rotis were consumed by all of them?
(a) 6 (b) 9 (c) 12 (d) 15

Answers

Warm Up

1. (d)	**2.** (a)	**3.** (d)	**4.** (b)	**5.** (b)	**6.** (c)	**7.** (b)	**8.** (d)	**9.** (c)	**10.** (c)
11. (b)	**12.** (a)	**13.** (c)	**14.** (a)	**15.** (b)	**16.** (b)				

Foundation

1. (c)	**2.** (c)	**3.** (c)	**4.** (c)	**5.** (d)	**6.** (b)	**7.** (d)	**8.** (b)	**9.** (d)	**10.** (a)
11. (d)	**12.** (d)	**13.** (d)	**14.** (b)	**15.** (d)	**16.** (d)	**17.** (c)	**18.** (c)	**19.** (d)	**20.** (b)
21. (a)	**22.** (b)	**23.** (b)	**24.** (c)	**25.** (b)	**26.** (d)	**27.** (a)	**28.** (d)	**29.** (b)	**30.** (a)
31. (b)	**32.** (d)	**33.** (d)	**34.** (d)	**35.** (d)	**36.** (b)	**37.** (b)	**38.** (d)	**39.** (b)	**40.** (d)
41. (c)	**42.** (a)	**43.** (b)	**44.** (a)	**45.** (a)	**46.** (d)	**47.** (c)	**48.** (d)	**49.** (d)	**50.** (d)

Moderate

1. (d)	**2.** (d)	**3.** (d)	**4.** (b)	**5.** (b)	**6.** (d)	**7.** (d)	**8.** (c)	**9.** (d)	**10.** (b)
11. (d)	**12.** (a)	**13.** (d)	**14.** (d)	**15.** (c)	**16.** (b)	**17.** (a)	**18.** (c)	**19.** (b)	**20.** (b)
21. (b)	**22.** (a)	**23.** (a)	**24.** (a)	**25.** (c)	**26.** (b)	**27.** (d)	**28.** (c)	**29.** (d)	**30.** (c)
31. (b)	**32.** (c)	**33.** (b)	**34.** (d)	**35.** (d)	**36.** (d)	**37.** (a)	**38.** (d)	**39.** (d)	**40.** (a)
41. (a)	**42.** (b)	**43.** (c)	**44.** (d)	**45.** (c)					

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Advanced

1. (c)	**2.** (a)	**3.** (d)	**4.** (c)	**5.** (d)	**6.** (c)	**7.** (b)	**8.** (d)	**9.** (b)	**10.** (b)
11. (a)	**12.** (a)	**13.** (b)	**14.** (c)	**15.** (b)	**16.** (a)	**17.** (b)	**18.** (b)	**19.** (a)	**20.** (c)
21. (d)	**22.** (b)	**23.** (d)	**24.** (b)	**25.** (d)	**26.** (b)	**27.** (d)	**28.** (a)	**29.** (c)	**30.** (b)
31. (a)	**32.** (b)	**33.** (d)	**34.** (a)	**35.** (b)	**36.** (a)	**37.** (d)	**38.** (d)	**39.** (d)	**40.** (c)

Hints and Solutions

Warm Up

1. Required ratio = 1/2:1/3:1/4

To find the simple ratio, we will multiply each of these by the LCM of 2, 3, and 4.

= 12 (1/2:1/3:1/4) = 6:4:3

2. Let the numbers be $5x$ and $3x$.

then, $5x - 3x = 10$

$2x = 10, x = 5$

Required product = $5x \times 3x = 5 \times 5 \times 3 \times 5 = 375$

3.

Number of Oxen	Work	Time
30	$\frac{1}{7}$	4
18	$\frac{6}{7}$	x

Applying the variation, we get

$30 \times 4 \times 7 = \frac{18 \times x \times 7}{6}$ $x = 40$ h

4. Let the number be x, $3x$, $4x$, and $7x$.

then, $x + 3x + 4x + 7x = 75$

$15x = 75 \quad \Rightarrow x = 5$

Required value = $7 \times 5 = 35$

5. Required difference = 35 − 5 = 30

6. Go through the options, and option (c) is the answer.

7. p = pressure, v = volume

$p = \frac{k}{v}$ k = constant term

$k = pv$

p is reduced by 20% i.e., by $\frac{1}{5}$

then v is increased $\frac{1}{4}$ i.e., 25%

8. $\frac{27 - x}{35 - x} = \frac{2}{3}$

$81 - 3x = 70 - 2x, x = 11$

9. $p \propto q \Rightarrow p = k\,q$

$15 = k \times 7, k = \frac{15}{7}$

$P = kq = 14 \times 15/7 = 30$

10. Go through the options.

11. Go through the options.

12. $\frac{5p - 3q}{7p + 2q} = \frac{5.\frac{3}{4}q - 3q}{7.\frac{3}{4}q + 2q} = \frac{15q - 12q}{21q + 8q} = 3/29$

13. Volume of milk in mixture = $\frac{7}{10} \times 30 = 21$ L

Volume of water = 9 L

Let x L of water is added.

$\frac{21}{9 + x} = \frac{1}{2}$

$42 = 9 + x$

$x = 33$

14. Ratio of their investment = 12,000:18,000 = 2:3

A's share in profit = $(2/5) \times 2000$ = ₹800

15. Suppose B invested for n months

Hence, $\frac{12 \times x}{n \times x} = \frac{3}{2}$ x = their investment

$\frac{12}{n} = \frac{3}{2}, n = 8$

16. Let B invested for n months

Then, $\frac{12{,}000 \times 12}{16{,}000 \times n} = \frac{3}{1}$

$n = 3$

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FOUNDATION

1. Let x gold coins were given to first queen. Then, number of gold coins given to second queen $= \frac{5}{7}x$

And number of gold coins given to third queen $= \frac{3}{7}x$

Then, $x - \frac{3}{7}x = 60 \Rightarrow \frac{4x}{7} = 60$

$x = 105$

Required number $= 105 + \frac{5}{7} \times 105 + \frac{3}{7} \times 105 = 225$

2. Let their ages 5 years ago be $2x$, $3x$, $7x$, and $8x$.

Their ages now $2x + 5$, $3x + 5$, $7x + 5$, $8x + 5$.

or $20x + 20 = 140$

$\Rightarrow 20x = 120 \Rightarrow x = 6$

Present age of Nishu $= 2 \times 6 + 5 = 17$ years

Present age of mother $= 7 \times 6 + 5 = 47$ years

Hence, required years $(47 - 17)$ years $= 30$ years

3. Let x = total number of workers

Number of women workers $= \frac{1}{3}x$

Number of men workers $= \frac{2}{3}x$

Number of married women $= \frac{1}{6}x$

Number of women having children $= \frac{1}{3}.\frac{1}{6}x = \frac{1}{18}x$

Number of women without children $= \frac{5x}{18}$

Number of married men $= \frac{3}{4}.\frac{2}{3}x = \frac{1}{2}x$

Number of men having children $= \frac{x}{3}$

Number of men without children $= \left(\frac{2}{3}x - \frac{1}{3}x\right) = \frac{x}{3}$

Required fraction $= \frac{\frac{5x}{18} + \frac{x}{3}}{x} = \frac{\frac{5x+6x}{18}}{4} = \frac{11}{18}$

4. Let salary of Anil = ₹x

Bhuvan's salary $= \frac{2}{5}x$

Chandra's Salary $= \frac{7}{9}.\frac{2}{5}x = \frac{14x}{45}$

$x + \frac{2}{5}x + \frac{14x}{45} = 770$

$\Rightarrow \frac{77x}{45} = 770$

$x = 450$

5. Let his salary be ₹x

Money spent on house rent $= \frac{1}{4}x$

Money spent on food $= \frac{1}{3}x$

Money spent on travel $= \frac{1}{6}x$

Money spent $= \frac{1}{4}x + \frac{1}{3}x + \frac{1}{6x} = \frac{6x - 8x + 4x}{24} = \frac{18x}{24} = \frac{3}{4}x$

Money left $= \frac{x}{4}$

Money spent on clothes $= \frac{1}{10}.\frac{x}{4} = \frac{x}{40}$

Money left $= \frac{x}{4} - \frac{x}{40} = \frac{10x - x}{40} = \frac{9x}{40}$

i.e., $\frac{9x}{40} = 13{,}500$

$x = 1500 \times 40 = 60{,}000$

Required amount = 5000

6. Let ₹x be Bidhan' s contribution. Then, his brother's contribution $= \frac{3}{2}x$

Financers contribution $= 3.\frac{3}{2}x = \frac{9x}{2}$

then, $x + \frac{3}{2}x + \frac{9x}{2} = 35{,}000$

then, $\frac{14x}{2} = 35{,}000 \Rightarrow x = \frac{35{,}000 \times 2}{14} = 5000$

His brother's contribution = ₹7500

7. Go through the options.

8. Let x and y be the mass of first alloy and second alloy, respectively.

Bronze in first alloy $= \frac{9}{10}x$

Manganese in first alloy $= \frac{7}{100}x$

Tin in first alloy $= \frac{3}{100}x$

For manganese, we have

$\frac{\frac{7}{100}x}{x+y} \times 100 = 5 \Rightarrow 7x = 5x + 5y \Rightarrow 2x = 5y$

Hence, $\frac{x}{y} = \frac{5}{2}$

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Applying alligation,

t = % of bronze in second Alloy

$\frac{y}{x} = \frac{5}{85-t}$

$\frac{2}{5} = \frac{S}{85-t}$, $t = 72.5$

9. Let ₹x was initially with Vineet.

Money given to Sachin = $\frac{1}{4}x$

Money spent on movie ticket = $\frac{1}{3}.\frac{3}{4}x = \frac{1}{4}x$

Money left = $\frac{1}{2}x$

Money spent on shopping = $\frac{2}{5}.\frac{1}{2}x = \frac{1}{5}x$

Money given by Gita = $\frac{2}{3}.\frac{3x}{10} = \frac{x}{5}$

Total money he had = $\frac{3x}{10} + \frac{x}{5} = \frac{5x}{10} = \frac{1}{2}x$

Money given to mother = $\frac{4}{5}.\frac{1}{2}x = \frac{2x}{5}$

Money left = $\frac{1}{2}x - \frac{2}{5}x = \frac{x}{10}$

Given that,

$\sqrt{x} = \frac{x}{10} \Rightarrow x = \frac{x^2}{100}$

$100x - x^2 = 0 \Rightarrow x(100 - x) = 0 \Rightarrow x = 100$

Money given by Gita = $\frac{100}{5}$ = ₹20

10. Number of turns that wheels make is inversely proportional to the number of cogs in them.

Number of turns that 1st cog will make in 45 s = $80 \times \frac{54}{32}$ turns

Number of turns that 1st cog will make in 8 s = $80 \times \frac{54}{32} \times \frac{8}{45}$ turns = 24 turns

11. Total number of guavas with all of them = 8 (Kunal has three guavas and Saurav has five guavas).

Each of them consumes equal quantity of guavas. Hence, the quantity of guavas consumed by each of them = $\frac{8}{3}$ guavas

Since Gopi did not bring any guavas on his own, so it means he has got the guavas from Kunal and Saurav.

Kunal has three guavas, out of which he consumed $\frac{8}{3}$ guavas. Hence, the quantity of guavas contributed by him = $3 - \frac{8}{3} = \frac{1}{3}$ guavas

Saurav has five guavas, out of which he consumed $\frac{8}{3}$ guavas. Hence, the quantity of guavas contributed by him = $5 - \frac{8}{3} = \frac{7}{3}$ guavas

So, the ratio in which they contributed the guavas = 1:7. Payments made to them will be in the same ratio. Hence, payment made to Kunal = ₹1 and payment made to Saurav = ₹7.

Hence, option (d) is the answer.

12. Let the number of coins denomination ₹1, 50 paise and 25 paise be $3x$, $4x$, and $12x$, respectively.

Then, $3x.1 + 4x.\frac{1}{2} + 12x.\frac{1}{4} = 600$

$3x + 2x + 3x = 600 \Rightarrow 8x = 600 \Rightarrow x = 75$

Number of 25 paise coins = $12 \times 75 = 900$

13. Let Anand, Bidhan, and Chandra got ₹x, y, and z, respectively, then Chandra's Share = $\frac{1}{2}(x + y)$ and Anand's Share = $\frac{1}{3}(y + z)$

We do not know the values of x, y, and z; therefore, option (d) is the answer.

14. Let Anil invested ₹X, then Dheeraj's investment = ₹$3x$

Let Dheeraj join him for n month.

Then, $\frac{x \times 12}{3x \times n} = 1 \Rightarrow x \times 12 = 3x \times n \Rightarrow n = 4$

Hence, Dheeraj joined him is September.

15. Since, we do not know the share of either of them, we cannot find the required time.

16. Let salary of C = ₹x = ₹1500

Salary of B = $\frac{4}{3}x = \frac{4}{3} \times 1500$ = ₹2000

Salary of A = $\frac{5}{4}.\frac{4}{3}x = \frac{5}{3}x =$ ₹$\frac{5}{3} \times 1500$ = ₹2500

Salary of M = (₹(1500 + 2000 + 2500) = ₹6000

17. Four years ago, let ages of A and B be $13x$ and $9x$ years, respectively.

Then, $\frac{13x + 12}{9x + 12} = \frac{4}{3} \Rightarrow 39x + 36 = 36x + 48$

or $3x = 12 \Rightarrow x = 4$

Hence, present age of A = $13x + 4 = 13 \times 4 + 4 = 56$

And present age of B = $9 \times 4 + 4 = 40$

18. Let the wages of a man, a woman and a child he $3x$, $2x$, and x, respectively,

then, $20 \times 3x + 30 \times 2x + x \times 36 = 780$

$60x + 60x + 36x = 780$

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$\Rightarrow 156x = 780 \Rightarrow x = 5$

Required Amount $= 2(15 \times 3 \times 5 + 21 \times 2 \times 5 + 30 \times 5)$

$= 2\,(225 + 210 + 150) = ₹1170$

19. Let $5x$ and $3x$ be the number of males and females originally at the party.

Now, $\frac{5x-10}{3x} = 1$, $5x - 10 = 3x$

$\Rightarrow 2x = 10 \Rightarrow x = 5$

Number of people originally at the party $= 8x = 40$

20. Area of semicircle $= \frac{1}{2}\pi r^2$

Area of sector $= \frac{\theta}{360} \times \pi r^2$

$$\frac{\frac{\theta}{360}}{\frac{1}{2}\pi r^2} \times \pi r^2 = \frac{1}{10}, \frac{\theta}{180} = \frac{1}{10} = \theta = 18°$$

21. Assume the length of each candles = 12 cm (LCM of 4 and 6)

Hence, rate of burning of 1st candle = 2 cm/h and rate of burning of 2nd candle = 3 cm/h

Assume that after t hours of burning, 1st candle is twice the length of 2nd candle.

According to the question, $(12 - 2t):(12 - 3t) = 2:1$ or $12 - 2t = 2(12 - 3t)$ or $12 - 2t = 24 - 6t$

Solving it we get, $t = 3$ h

Hence, option (a) is the answer.

22. $V \propto W^2$

$V = KW^2$, V = Value of diamond and W = Weight of diamond

K = constant

$10000 = K \,.\, (10)^2 \quad \Rightarrow K = 100$

Hence, $V = 100 \times 4^2 = 1600$

$V = 100.6^2 = 3600$

Net loss = (₹(10,000 − 5200) = ₹4800, i.e., 48%.

23. Monthly income of A and B be $3x$ and $2x$, respectively.

Monthly expenditure of A and B be $5y$ and $3y$, respectively.

$3x - 5y = 1000$, $2x - 3y - 1000$

Solving, we get, $y = 1000$, $x = 2000$

Income of A = ₹6000

24. $\frac{ab}{bc} = \frac{x}{c^2}$ and $x = \frac{ab.c^2}{bc} = ac$

25. Let the fare of 1st, 2nd, and 3rd class be $10x$, $8x$, and $3x$, respectively, and let the number of passengers of 1st, 2nd, and 3rd class be $3y$, $4y$, and $10y$, respectively.

then $30xy + 32xy + 30xy = 16{,}100 \Rightarrow 92xy = 16{,}100$

Hence, $xy = 175$

Required amount $= 32xy = 32 \times 175 = 5600$

26. $$\frac{3A^2 + 4B}{3A - 4B^2} = \frac{3.\left(\frac{3}{4}B\right)^2 + 4B}{4.\frac{3}{4}B - 4B^2} = \frac{\frac{27B^2}{16} + 4B}{3B - 4B^2}$$

$$= \frac{27B^2 + 64B}{16(3B - 4B^2)} = \frac{27B + 64}{16(3 - 4B)}$$

Further, it cannot be solved. Hence, option (d) is the answer.

Note: Had there been no square terms, we could have calculated the final value by just putting A = 3 and B = 4.

27. Assume that the salary of Atul = ₹A, salary of Binod = ₹B, and salary of Charu = ₹C

Given that A + B + C = ₹200 (i)

After deducting their respective expenses from their respective salaries, we get savings of them as follows:

Savings of Atul = 40% of A = 0.4A

Savings of Binod = 20% of B = 0.2B

Savings of Charu = 25% of C = 0.25C

Given that 0.4A:0.2B:0.25C = 8:8:5

Or, we get A:B = 1:2 and B:C = 2:1

Hence, A:B:C = 2:4:2 = 1:2:1 (ii)

Solving (i) and (ii), we get A = ₹50, B = ₹100, and C = ₹50

28. Price of one goat = ₹70 ⇒ Price of 9 goats = ₹9 × 70 = ₹630

Price of 14 sheeps = ₹630 ⇒ Price of 1 sheep = $₹\frac{630}{14} = ₹45$

Price of 50 sheeps = ₹45 × 50 = ₹2250

Price of 10 oxen = ₹2250

Price of 1 oxen = ₹225 ⇒ Price of 8 oxen = ₹1800

Price of 6 dogs = ₹1800 ⇒ Price of 1 dog = 300

Price of 5 dogs = ₹1500 ⇒ Price of two horses = ₹1500

Price of 1 horse = ₹750

29. Since in 30 min pond is $\frac{1}{3}$ full, in next 1 min it will triple, i.e., the pond well be full. Hence, option (b) is the answer.

30. Map is a two-dimensional figure with length being proportionate to height.

Since it takes 6 l of oil paint to paint a 6-m high map, quantity of oil paint it will take to paint a 18-m high map $= \frac{18}{6} \times \frac{18}{6} \times 6 = 54$ l of paint. Hence, option (a) is the answer.

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31. $A \propto B$

$B \propto (1/C)$

$C \propto D^2$

$D \propto E^{1/3}$

When A increase $\Rightarrow$ B also increase $\Rightarrow$ C decreases

Decrease in C results in decrease in D and decrease in D results decrease in E.

32. Using the formula $\frac{C}{5} = \frac{F-32}{9}$

We well get [−40] as answer

33. Let the 1st number be $2x$ and 2nd number = $3x$

3rd number = $\frac{18}{5}x$

4th number = $\frac{81x}{20}$

So, $x + 3x + \frac{18}{5}x + \frac{81x}{20} = 233$

$\frac{20x + 60x + 72x + 81x}{20} = 233$

$x = \frac{233 \times 20}{233} \Rightarrow x = 20$

2nd number = $3x = 3 \times 20 = 60$

3rd number = $\frac{1820}{2} = 72$

Average = $\frac{60+72}{2} = \frac{132}{2} = 66$

34. After 1st year

Population of Port A = 8000

Population of Port B = 6000

Population of Port C = 4000

After 2nd year

Population of Port A = 5000

Population of Port B = 6000

Population of Port C = 7000

After 3rd year

Population of Port A = 6500

Population of Port B = 6000

Population of Port C = 5500

After 4th year

Population of A = 5750

35. Let x = number of ₹1 notes and y = number of ₹10 notes

So, $xy = x + 10y \quad \Rightarrow xy = x + 10y$

Now, go through the options.

36. At first let Nandan buys x black currents, then number of pastries bought by Divya = $x + 9$

Now, let Divya bought y butter scotches, then number of butter scotches bought by Nandan = $y + 18$

$\frac{x+y+9}{x+y+18} = \frac{12}{21} \Rightarrow x + y = 3$

x and y cannot be zeroes.

When $x = 1$, $y = 2$ (i)

When $x = 2$, $y = 1$ (ii)

Let the price of black current be ₹A/unit and price of butter scotch be ₹B/unit.

For Case (i),

$10a + 2b = a + 20b \Rightarrow 9a = 18b \Rightarrow a:b = 2:1$

37. Go through the options.

38. Ratio of their investments for 1 year

$= 4x \times 3 + 2x \times 3 + x \times 3 + \frac{x}{2} \times 3 : x \times 3 + 2x \times 3 + 4x \times 3 + 8x \times 3 : 15x \times 12$

$\frac{45x}{2} : 45x : 180x = 45x : 90x : 360x = 1:2:4$

B's share in profit = $\frac{2}{7}$ of profit

Let y be profit, then $\frac{2}{7}y = 22000 \; y = 77000$

42. $y = \frac{r}{x}, \; z = \frac{r^2}{x^2}$

Then, $x + y + z = 13$

$x + \frac{r}{x} + \frac{r^2}{x^2} = 13$

$x^2 + xr + r^2 = 13$ (i)

And, $x^2 + \frac{r^2}{x^2} + \frac{r^4}{x^4}$ (ii)

With the help of equations (i) and (ii), we get option (a) as answer.

43. Let 1 black cow gives x L milk in 1 day and 1 white cow gives y L milk in 1 day. Then,

$5(4x + 3y) = 4(3x + 5y) \Rightarrow 20x + 15y = 12x + 20y$

$\Rightarrow 8x = 5y$; hence, $\frac{x}{y} = \frac{5}{8}$

44. Let the cost of type I terminal be ₹x per week and cost of type II terminal be ₹y per week.

$5 \times x \times 7 + 22 \times 4 \times y + 10 \times 6x + 8 \times 6x = 74750$ (i)

Also, given $\frac{x}{y} = \frac{2}{3}$ (ii)

Using equations (ii) and (i), we get option (a) as answer.

45. SP of alloy = ₹90

Profit = 20% = $\frac{1}{5}$

CP = ₹$\left(90 - \frac{1}{60}.90\right) =$ ₹75

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Prices of foils are in ratio 1:4.

Hence, price of cheaper quality = $\frac{1}{5} \times 75 = ₹15$

and price of dearer quality = $\frac{4}{5} \times 75 = ₹60$

46. Go through the options.

47. Let the number of chapters be $6x$, $5x$, $10x$, and $14x$.

Also, $14x = 70$, $x = 5$

Therefore, chapters are 30, 25, 50, and 70.

We can easily eliminate (a) as number of pages has to be in between 4000 and 5000.

For (b) Each parts well have 1200 pages which is not possible for part having 70 chapters. Only option (c) satisfies the conditions.

48. Total number of chapters = $(6x + 5x + 10x + 14x) = 35x = 35 \times 5 = 175$

49. Required ratio = 15/16

50. Let their present ages be $2x$ and $3x$.

Given that

$$\frac{2x+N}{3x+N} = \frac{3}{5}$$

$10x + 5N = 9x + 3N$

$X = -2N$ which 18 not possible.

Hence, option (d) is the answer.

MODERATE

Answers to Q.1 to 3:

	Brother	Sister
Mr Nishant's son	X	5X
Mr Rajesh' son	Y	Y
Mr Nishant's daughter	2Z	Z
Mr Rajesh' daughter	2A	A

Each son of Mr Rajesh has as many sisters as the number of brothers and each daughter of Mr Rajesh has twice the number of brothers as the number of sisters. It is possible only if the number of sons = 4 and number of daughters = 3. However, nothing concrete can be said about the children of Mr Nishant.

Answers to Q.10 and 11:

Number of men:number of women = Number of women: number of children

10. Only the second option is satisfying the values.

11. Since this ratio is less than 1, the number of men should be less than the number of females visited. So, number of males should be less than 7. So, the answer is option (d).

12. If P is taking 3 days to do some work, then Q takes 6 days to do the same work. Now, both of them will take 2 days to complete the work. So, R takes 6 days to complete the same work.

Hence, earning should be distributed in the ratio of their efficiency, i.e., 2:1:1.

Answers to Q.13 to 15:

Assume that Ten, Dul, and Kar are getting T, D, and K sweets, respectively.

$2T = 2/3\ D = 4\ K = 4P$ (assume)

So, $T = 2P$, $D = 6P$ and $K = P$

So, $T:D:K = 2:6:1$

13. Kar would get – 117(4/9). So, the answer is option (d).

15. x bulbs consume $1/x$ units everyday.

So, 1 bulb will consume $1/x^2$ units of electricity everyday.

Y bulbs consume y/x^2 units everyday.

Now, y/x^2 units is consume in 1 day.

Hence, y units will be consumed in x^2 days.

16. Number of apples/Number of guavas = Number of guavas/Number of oranges

(Number of guavas)^2 = Number of apples × Number of oranges

This gives us a hint that Number of apples × Number of oranges should be a prefect square.

Now, start using the options.

Option (b) satisfies the relationship (16/20 = 20/25).

17. We have to check all the possibilities in this question. If we take the ratio to be equal number of apples/number of guavas = number of guavas/number of oranges = 1, then number of apples = number of guavas = number of oranges. So, one possible number of total fruits = 63.

In 16 and 17, if we start using the concept of GP, then it will be difficult to find the number of fruits by GP as this questions contains a lot many variables.

19. Village-chief contribution = ₹1.2 lakhs, so villagers' contribution = ₹1.2 lakhs

Now, go through the options.

24. $\frac{a}{a+b} = \frac{b}{c+a} = \frac{c}{a+b} = \frac{a+b+c}{2(a+b+c)}$

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Now, if $a+b+c=0$, then $a+b=-c$

So, $\frac{a+b+c}{2(a+b+c)}=-1$

And if $a+b+c\neq 0$, then $\frac{a+b+c}{2(a+b+c)}=\frac{1}{2}$

28. Go through the options.

29.

No. of people	6	6	8	16
Event	Museum	Fair	Exhibition	Movie
Rate	5	4	2	25

36. Using options is the best way of solving this question.

37. Using options is the best way of solving this question.

38. It is all about taking different values and checking which option cannot satisfy the money with Pankaj.

41. Number of deers, bears, and foxes = $3x:7x:5x$

Difference between the number of deers and bears = $4x$

Now, $4x$ has to be a multiple of 21. So, x has to be a multiple of 21.

So, total number of animals = $15x = 15 \times 21 = 315$

42. Weight of two pieces of watermelon

$= (3 \times 12y):(5 \times 12y)$

(We are assuming the weight to be the multiple of 12 because the bigger of two pieces has to be cut in the ratio 5:7.)

Now, $5 \times 12y$ has to be cut in the ratio 5:7. So, the two parts will be $25y$ and $35y$.

So, finally the ratio is 36:25:35.

43. Assume that Y produces $4P$ units in unit time. Then, X and Z produce P and $2P$ units in the same time. So, total shoes produced = $7P$, Fraction of the total number of shoes produced by machine $Z = 2/7$

ADVANCED

1. Time $\propto$ (√Number of passengers) × Distance

Expressing this in symbols, $T \propto (\sqrt{N}) \times D$

Or, $T1/T2 = (\sqrt{N}1/N2) \times D1/D2$

Or, $2/1 = (\sqrt{N}1/64) \times 80/30$

2. Tax = Fixed + Variable

Since Mr Kalyan pays ₹6200 and Mrs Kalyan pays ₹7700, so it can be inferred that for ₹10,000 (Difference between their salaries), one has to pay ₹1500 as the variable tax. So, Fixed component of tax = ₹5200. The point worth to be noted here is the fact that tax is to be paid over an above ₹50,000. So, the variable tax percentage above ₹50,000 = 10%. Now, the answer can be calculated using options.

Answers to Q.6 and 7:

Generate the equations first and then use options for faster calculations.

9. Total profit = ₹24 lakh per trip + ₹$7P^2$, where P is the number of coaches

Now, go through the options.

10. When 10 kg is kept in first pan and 5 kg is kept in second pan, it remains in balance. It means that 1st pan is heavier than the second pan (or, more difficult to pull down), so the ratio of weight of pans = 2:1

11. Do as given in the solution of 10.

12.

	Gold	Copper
First Alloy	2	1
Second Alloy	2	7
Finally	1	2

It is possible if we mix first alloy and second alloy in the ratio 1:3. So, the second alloy will be mixed two times more than the first alloy.

13. Weight $\propto$ Height and Height $\propto$ Age

19. Go through the options.

20. Assume the distance = 9 units. So, the ratio of the time taken = 9:3:1

So, the answer = 9/1:3/3:1/9

22. Population State 1, State 2, State 3, State 4, State 5 = $x:3x:2x:3x:2x$. The ratio of the percentage of women in the population = $y:3y:2y:5y:4y$.

Now, use the above values to establish the relationship between xy and the total population of women = 3.7 million.

23. It is possible if new length of the pendulum become 1.21 times of the old length of the pendulum.

Answers to Q.26 to 29:

Using the options given in Question 26, we can see that it is possible to have ₹500 on 19th only if he starts with ₹80 on 13th of November.

Date	Money
13th	₹80
14th	₹70
15th	₹140
16th	₹130
17th	₹260

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Date	Money
18th	₹250
19th	₹500
20th	₹490
21st	₹980
22nd	₹970
23rd	₹1940
24th	₹1930
25th	₹3860

27. It is ₹970 on 22nd.

28. It is obvious from the table given above.

29. The new table would have been like this:

Date	Money
13th	₹100
14th	₹90
15th	₹180
16th	₹170
17th	₹340
18th	₹330
19th	₹660
20th	₹650
21st	₹1300
22nd	₹1290
23rd	₹2580
24th	₹2570
25th	₹5140

Now, it can be calculated.

35. Total number of distinct meeting points for B and C = 9 and Total number of distinct meeting points for A and C = 5. Now, it can be checked using the options.

Answers to Q.36 to 38:

Whatever is the total number of boys, it should be divisible by 8, 6, and 12, and total number of girls should be divisible by 12, 15, and 4. Now, use the options.

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CHAPTER

8

Time and Work

LEARNING OBJECTIVES

After completion of this chapter, the reader should be able to understand:

- Time and work equivalence
- Relationship between time and efficiency
- Different types of work
- Kinds of questions asked at the CAT
- Methods of solving questions

INTRODUCTION

A brief glance of the last 15 years' CAT paper shows that 'time and work' is an important topic for this examination. Almost every year, there have been questions from this topic. The number of questions, however, has varied between 1 and 3. A basic understanding of the concept of time and work will enable you to solve the problems with considerable ease.

TIME AND WORK

There is a definite relationship between the amount of time taken, the number of persons doing the work, and the unit of work done. In this topic, we will study the phenomenon of accomplishment of a work with relation to time. There are certain formulae that demonstrate the abovementioned relationship and exhibit how each of these factors influence the other two.

TIME–WORK EQUIVALENCE

The essence of time–work equivalence lies in the fact that it exhibits the most fundamental relationship between the three factors as mentioned previously viz., work, time, and the agent, which completes the work. That is,

$$\text{Work done} = \text{number of days} \times \text{number of men}$$
$$W = M \times D$$

This gives us an important concept of man-days.

Suppose, there are 20 persons working for 10 days to complete a job, then the total work done is equal to 200 man-days. Now, if we change the number of days in which the work has to be completed, then the other factor, that is, the number of persons will change accordingly, and hence, the product of the factors becomes equal to 200 man-days. Product–stability ratio (Chapter 3) is a very effective tool to calculate this.

Example 1 Seven persons can clean 7 floors by 7 mops in 7 days. In how many days, can 5 persons clean 5 floors by 5 mops?

Solution This problem can be solved through several methods.

Method 1 To clean 7 floors, we need to have $7 \times 7 = 49$ man-days.

Therefore, to clean 5 floors, we need to have 35 man-days. So, $35 = D \times 5$. Hence, $D = 7$ days

Method 2 Using ratio proportion, less work and less men are involved here.

Hence, the number of days $= 7 \times \frac{5}{7} \times \frac{7}{5} = 7$ days

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Method 3 Let us try to have a mental imagination of this situation: There is a building with seven floors, namely F_1, F_2,..., F_7, and seven persons P_1, P_2,..., P_7 are cleaning this building in such a way that one floor is being cleaned by each one of them. Since it takes 7 days to complete the whole work, it can be inferred that everybody is taking 7 days to clean his respective floor. Hence, if there are just five floors and five persons are cleaning these five floors, then it will take them 7 days to clean (assuming that the top two floors have been demolished).

Now, depending upon different situations, three conditions are possible in the relationship: $W = MD$ (where W = quantity of work, M = number of persons, and D = number of days).

Condition 1: *W* is constant

$$M \times \mathrm{D} = \text{Constant}$$

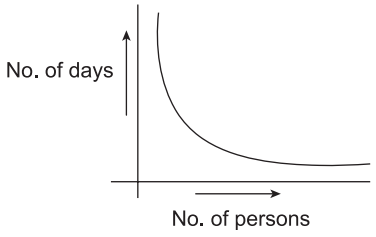
$$M \propto 1/D$$

It can be observed that if the work done is constant, then the number of persons doing the work is inversely proportional to the number of days, which means that the multipliers of M and D will be reciprocal. Extending this situation, if 10 persons can do a work in 20 days, then five persons can do the same work in 40 days or 20 persons will do the same work in 10 days. Furthermore, it can be summarized as follows:

W	=	M	×	D
200	=	10	×	20
200	=	$5(10 \times 1/2)$	×	$40(20 \times 2)$
200	=	$20(10 \times 2)$	×	$10(20 \times 1/2)$

Therefore, it can be said that the multiplier of M and multiplier of D are reciprocal to each other.

It can be seen with the help of the graph given below.



Example 2 Yadavjee contractor undertakes a work to get it done in 50 days by 50 labourers. After 40 days, he realizes that only 50% of the work is done. How many more men should be employed so that the work is complete on time?

Solution

W	=	M	×	D	
50%		50		40	(i)
Rest 50%		$(50 + M)$		10	(ii)

Since work is constant in both the cases, the number of men and the number of days will be reciprocal to each other. As the number of days left in (ii) is one-fourth of initial period (i), then the number of persons will become four times the initial number of persons.

Hence, the number of persons $= 50 \times 4 = 200$. Therefore, $M = 150$ men

Example 3 In the above example, if the schedule can go behind by 10 days, then how many extra men are required to complete the work?

Solution So, now we will have to complete the work in 60 days.

W	=	M	×	D	
50%		50		40	(i)
Rest 50%		$(50 + M)$		20	(ii)

Since work is constant in both the cases, the number of men and the number of days will be reciprocal to each other. As the number of days left in (ii) is half of initial period (i), the number of persons will become two times the initial number of persons.

Hence, the number of persons $= 50 \times 2 = 100$. Therefore, $M = 50$ men

Example 4 In Example 2, the contractor realizes after 40 days that work is only 20% complete. How many extra men should be employed now so that the work is completed on time?

Solution

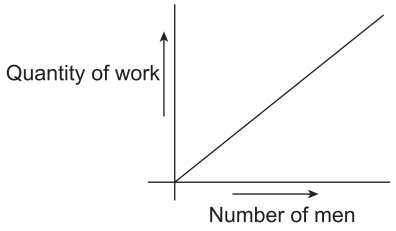
	W	=	M	×	D	
	20%		50		40	(i)
	20%		200		10	(ii)
So,	80%		800		10	(iii)

Hence, 750 more persons are needed to complete the job on time.

Condition 2: *D* is constant

$$W \propto M$$

More work will be done if we employ more men and vice versa. It means that multiplier of W and M will be same. It can be seen with the help of the graph given below.



Condition 3: *M* is constant

$$W \propto D$$

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In general, we can summarize that

$$\frac{M_1 D_1}{W_1} = \frac{M_2 D_2}{W_2}$$

Example 5 Twelve persons can cut 10 trees in 16 days. In how many days can 8 persons cut 12 trees?

Solution Here, $W_1 = 10$ $W_2 = 12$
$M_1 = 12$ $M_2 = 8$
$D_1 = 16$ $D_2 = ?$

Putting the values in the equation $W_1/W_2 = (M_1/M_2) \times (D_1/D_2)$

We get $\frac{10}{12} = \frac{12}{8} \times \frac{16}{D_2} \Rightarrow D_2 = 28.8$ days

INDIVIDUAL WORK AND INDIVIDUAL EFFICIENCY

Individual Work

If Amit can do a certain work in 10 days, then he will finish 1/10th of the work in one day.

Example 6 Amit can do some work in 12 days and Vinit can do the same work in 15 days. In how many days will both of them do the work when working together?

Solution Assume total work = 1 unit

Work done by Amit in one day = 1/12 unit

Work done by Vinit in one day = 1/15 unit

Work done by both of them in one day when working together = (1/12) + (1/15) = 9/60 unit

Hence, they will be doing the whole work in 60/9 days $= 6\frac{2}{3}$ days

LCM Method of Solving Time and Work Questions

This can be understood in terms of the above example in the following way:

Let us assume that total work to be equal to the LCM of the days taken by Amit and Vinit (i.e., of 10 and 15).

Assume work = 60 units

Work done by Amit in one day = 5 units

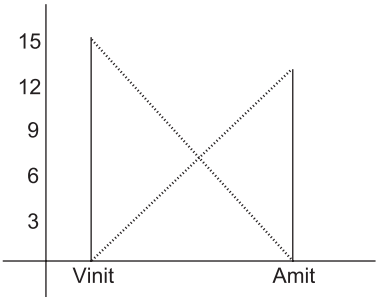
Work done by Vinit in one day = 4 units

Work done by both of them in one day when working together = 9 units

So, the number of days taken by both of them when working together = $60/9 = 6\frac{2}{3}$ days

Alternatively, we can do this problem with the help of a graph also.

We have drawn the pillars of the number of days of Amit and Vinit. Now, the point of intersection of the top of the



first pillar to the bottom of the second pillar and the top of second pillar to the bottom of the first pillar is the number of days taken by both of them when working together. In the above-drawn graph, it can be clearly seen that the point of intersection of both the straight lines is a bit more than 6.

Despite graphical method appearing easier than the earlier two methods, the usage of this method should be avoided due to the complexity of denoting the points on the graph paper.

Example 7 *A*, *B*, and *C* can do a piece of work individually in 8, 12, and 15 days, respectively. *A* and *B* start working, but *A* quits after working for two days. After this, *C* joins *B* till the completion of work. In how many days will the work be complete?

Solution Let us assume that the work = LCM (8, 12, 15) = 120 units

So, the work done by *A* in one day = 15 units

Work done by *B* in one day = 10 units

Work done by *C* in one day = 8 units

Work done by *A* and *B* in two days = 2 × 25 = 50 units

Remaining work = 70 units

Work done by *C* and *B* in one day = 18 units

Time taken to complete the remaining work by *C* and $B = \frac{70}{18} = 3\frac{16}{18}$ days

Hence, the total number of days = $5\frac{16}{18}$ days

Example 8 *A* and *B* together can do a work in 12 days. *B* and *C* together can do the same work in 10 days, and *A* and *C* together can do the same work in 8 days. In how many days will the work be complete if *A*, *B*, and *C* are working together?

Solution Let us assume work = LCM of (12, 10, 8) = 120 units

So, *A* and *B* are doing 10 units in one day, *B* and *C* are doing 12 units in a day, and *A* and *C* are doing 15 units in a day.

Adding all these, 2 (*A* + *B* + *C*) are doing 37 units in a day.

$\Rightarrow$ (*A* + *B* + *C*) are doing $\frac{37}{2} = 18.5$ units in a day

Hence, time taken to complete the work $\frac{120}{18.5}$ days = 6.48 days

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Individual Efficiency

Efficiency is also known as work-rate.

If A is taking less number of days with respect to B to complete the same work, then we can say that the efficiency of A is more than the efficiency of B.

So, more the efficiency, less will be the number of days, and less the efficiency, more will be the number of days to do a certain work. We have observed individual efficiency in case of percentage also (product–stability ratio).

Now, assume A takes 20 days to complete a work and B takes 25 days to complete the same work. It means A is doing 5% (100%/20) work in one day and B is doing 4% (100%/25) work in a day. So, efficiency of A is 25% more than efficiency of B.

General Expression Correlating Time Taken and Efficiency

If efficiency of A is x% more than the efficiency of B and B takes 'B' days to complete the work, then A will take $\left(\frac{B}{100+x}\times 100\right)$ days to complete the same work.

If efficiency of A is x% less than the efficiency of B and B takes 'B' days to complete the work, then A will take $\left(\frac{B}{100-x}\times 100\right)$ days to complete the same work.

So, if A is 20% more efficient than B and B takes 'B' days to complete the work, then A will take $\frac{B}{1.2}$ days to do the same work.

With this, it can also be observed that if work is constant, then time taken is inversely proportional to efficiency.

Example 9 John is thrice as efficient as Abraham and hence completes a work in 60 days less than the number of days taken by Abraham. What will be the number of days taken by both of them when working together?

Solution Since John is thrice as efficient as Abraham, the number of days taken by him will be one-third the number of days taken by Abraham. If John is taking x days, then Abraham will take $3x$ days to complete the same work.

Now, $3x - 2x = 60$ days

So, $x = 30$ days and $3x = 90$ days

Let us assume that the total work = 90 units (LCM of 30 and 90)

So, the total work done by both of them in one day = 3 + 1 = 4 units of work

Hence, the total number of days = $\frac{90}{4}$ days = 22.5 days

Example 10 A can do a work in 4 days. Efficiency of B is half the efficiency of A, efficiency of C is half the efficiency of B, and efficiency of D is half the efficiency of C. After they have been grouped into two pairs, it is found that the total number of days taken by one group is two-third the time taken by the other group. Which of the following is a possible group? (*CAT 2001*)

(a) AB (b) BC (c) CD (d) AC

Solution Number of days taken by A = 4 days

Number of days taken by B = 8 days

Number of days taken by C = 16 days

Number of days taken by D = 32 days

Assume that the total work = 32 units

So, the work done by A = 8 units

Work done by B = 4 units

Work done by C = 2 units

Work done by D = 1 units

It can be observed that the work done by B and C together in one day is two-third the work done by A and D. So, the groups are AD and BC.

Example 11 In a nuts and bolts factory, one machine produces only nuts at the rate of 100 nuts per min and needs to be cleaned for 5 min after production of every 1000 nuts. Another machine produces only bolts at the rate of 75 bolts per min and needs to be cleaned for 10 min after production of every 1500 bolts. If both the machines start their production at the same time, then what is the minimum duration required for producing 9000 pairs of nuts and bolts?

(a) 130 min (b) 135 min (c) 170 min (d) 180 min

Solution

Machine I

Number of nuts produced in one min = 100

Time required to produce 1000 nuts = 10 min

Cleaning time for nuts = 5 min

Overall time to produce 1000 nuts = 15 min

Over all time to produce 9000 nuts = 138 min – 5 min = 133 min (i)

Machine II

Time required to produce 75 bolts = 1 min

Time required to produce 1500 bolts = 20 min

Cleaning time for bolts = 10 min

Effective time to produce 1500 bolts = 30 min

Effective time to produce 9000 bolts = 30 × 6 – 10 = 170 min (ii)

From equations (i) and (ii),

Minimum time = 170 min

Example 12 A, B, and C are assigned a piece of work which they can complete by working together in 15 days. Their efficiencies (measured in terms of rate of doing work) are in the ratio of 1:2:3. After one-third of the work is completed, one of them has to be withdrawn due to budget constraint. Their wages per day are in the ratio of 3:5:6. The number of days in which the remaining two persons can complete the work (at optimal cost) is:

(a) 18 (b) 20 (c) 15 (d) 12

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Solution A, B, and C together in 15 days
= A alone in 90 days, B alone in 45 days, and C alone in 30 days.
Wages per day per unit work for A, B, and C are $\frac{3}{1}:\frac{5}{2}:\frac{6}{3}$

Hence, A is the least efficient and hence, must be done away with.

For B and C, the whole work can be finished in 18 days, and hence, remaining two-third of the work can be finished in 12 days only.

COLLECTIVE WORK AND COLLECTIVE EFFICIENCY

When people of different efficiencies start working together, the method of time–work equivalence to find the time or the amount of work done cannot be used. In those cases, we will be required to relate time with the efficiency of a group.

For example, if 5 men and 8 women can do a piece of work in 10 days, from this information, we cannot find out that in how many days a man or a woman can do the work individually, since we are not aware of their individual efficiencies. However, if we get a similar equation like x men and y women can do the same work in 'p' days, then we can correlate these two equations to find the number of days taken by one man and one woman to do the work.

Example 13 10 men and 9 women can do a piece of work in 20 days. The same work can be done by 6 men and 12 women in 30 days. In how many days, can the same work be done by 1 man and 1 woman?

Solution 10 men and 9 women can do a work in 20 days
$\Rightarrow 20 \times$ (10 men and 9 women) can do the same work in 1 day.

Similarly, 6 men and 9 women can do the same work in 30 days $\Rightarrow 30 \times$ (6 men and 12 women) can do the same work in 1 day.

So, efficiency of $20 \times$ (10 men and 9 women) = efficiency of $30 \times$ (6 men and 12 women)

Or, 200 men + 180 women = 180 men + 360 women
Or, 200 men = 180 women $\Rightarrow$ 1 man = 9 women

So, the total work = $20 \times$ (10 men and 9 women) = $20 \times (90 + 9)$ women = $99 \times 20 = 1980$ units

Total number of persons employed = 1 man + 1 women = 10 women

So, the time taken = $\frac{1980}{10}$ = 198 days

EXTENSION OF THE CONCEPT OF TIME AND WORK

Pipes and Cisterns

Pipes and cisterns is just another application of the concept of time and work. While we see only positive work being done in normal cases of time and work, in case of pipes and cisterns, negative work is also possible.

Given that pipes A and B can fill a tank in 20 min and 25 min working individually $\Rightarrow$ this statement is similar to 'A can do a work in 20 min and B can do the same work in 25 min'.

Again, given that pipe C can empty a tank in 40 min we can say this statement is similar to 'C can demolish a wall in 40 min (assuming that the work is building or demolishing the wall).

Let us understand this with the help of an example.

Example 14 A and B are two taps which can fill a tank individually in 10 min and 20 min, respectively. However, there is a leakage at the bottom, which can empty a filled tank in 40 min. If the tank is empty initially, how much time will both the taps take to fill the tank (leakage is still there)?

Solution Let us assume the units of work = LCM of (10, 20, 40) = 40 units

Work done by Tap A/min = 4 units/min (Positive work)
Work done by Tap B/min = 2 units/min (Positive work)
Work done by leakage/min = 1 unit/min (Negative work)
Net work done/min = 5 units/min
Hence, time taken = 8 min

Example 15 Pipe A can fill a tank in 3 h. But there is a leakage also, due to which it takes 3.5 h for the tank to be filled. How much time will the leakage take in emptying the tank if the tank is filled initially?

Solution Assume the total units of work = 10.5 units
Work done by Tap A/h = 3.5 units/h (Positive work)
Work done by leakage/h = 3 units/h (Negative work)
Net work done/h = 0.5 units/h

So, the time taken = $\frac{10.5}{0.5}$ = 21 h

Alternatively, due to the leakage, the pipe is required to work for an extra half an hour. So, the quantity filled by pipe in half an hour is being emptied by the leakage in 3.5 h. Hence, the quantity filled by pipe in 3 h will be emptied by the leakage in 21 h.

Variable Work

The concept of variable work comes from the possibility
$\Rightarrow$ that the rate of working can be different or
$\Rightarrow$ can be dependent upon some external agent.
In these cases, the rate of work will be proportional to some external factor.

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Understand this with the help of a simple statement: The rate of the flow of water from a pipe is directly proportional to the area of the cross section of the pipe.

Example 16 There are three inlet taps whose diameters are 1 cm, 2 cm, and 3 cm, respectively. The rate of flow of the water is directly proportional to the square of the diameter. It takes 9 min for the smallest pipe to fill an empty tank. Find the time taken to fill an empty tank when all the three taps are opened.

Solution The rate of flow of a diameter^2, or, rate of flow = $K \times$ diameter^2 (where K is a constant)

For first tap, rate of flow = $K \times 1$

For second tap, rate of flow = $K \times 4$

For third tap, rate of flow = $K \times 9$

We know that the quantity filled will be equal to the product of the rate of flow and time.

So, the quantity filled by the smallest pipe = $K \times 1 \times 9$ = 9 K = Capacity of tank

Quantity of water filled by all the taps together in 1 min = 9 K + 4 K + 1 K = 14 K

Assume that all the taps working together take 't' min.

So, 14 $K \times t$ = 9 K

So, the time taken $t = \dfrac{9K}{14K} = \dfrac{9}{14}$ min

Alternate Work

The concept of alternate work is analogous to the concept of man-days. As we have seen in the concept of man-h that if 20 men can do a work in 10 days, then this work is equivalent to 200 man-days. However, in the case of alternate work, two or more than two people of different efficiencies work alternately or in some particular pattern.

Example 17 Navneet can build a wall in 30 days and Rakesh can demolish the same wall in 40 days. If they work on alternate days with Navneet starting the job on the 1st days, then in how many days will the wall be built for the first time?

Solution Let us assume the total units of work = 120 units

so, the wall built by Navneet in one day = 4 units

And wall demolished by Rakesh in one day = 3 units

So, effectively in two days, total wall built = 1 unit

Now, they work on alternate days, so days taken to build 116 units = 116 days

On 117th day, Navneet will add another 4 units and so completing the construction of wall in 117 days.

(This problem can be understood well with another very traditional problem—A frog climbs up a pole 4 inches in 1 h and slips 3 inches next hour. If height of the pole is 120 inches, then what is the time taken by the frog to reach the top of the pole?)

Practice Exercises

Warm Up

Q.1 Anil can do a piece of work in 18 days and Nishit can do the same work in 12 days (Nishit is more efficient than Anil). In how many days, both of them will complete the work when working together?

(a) 7.2 days (b) 6 days

(c) 5.2 days (d) None of these

Q.2 A can do a piece of work in 'a' days and B can do the same piece of work in 'b' days, then the number of days in which both of them will complete the work working together is N. Which of the following can never be true?

(a) 2N is the Arithmetic Mean of a and b

(b) 2N is the Geometric Mean of a and b

(c) 2N is the Harmonic Mean of a and b

(d) None of these

Q.3 In the context of the above question, if $a \neq b$, then which of the following is true?

(a) 2N is the Arithmetic Mean of a and b

(b) 2N is the Geometric Mean of a and b

(c) 2N is the Harmonic Mean of a and b

(d) None of these

Q.4 Aftab can do a piece of work in 12 days and Ravi can do the same work in 18 days. If Aftab works for three days and then quits, in how many days will the remaining work be completed by B alone?

(a) $13\frac{1}{2}$ (b) 9 (c) $16\frac{1}{2}$ (d) 12

Q.5 Efficiency of Ravi and Mayank in doing the same work is 3:2. If together they can complete a work in 18 days, then in how many days Ravi can complete the work alone?

(a) 45 (b) 30 (c) 24 (d) 40

Q.6 A tap can fill an empty tank in 12 h and a leakage can empty half the tank in 10 h. If the tap and the leakage

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are working simultaneously, how long would it take for the empty tank to be filled to half its capacity?
(a) 60 h (b) 30 h (c) 15 h (d) 12 h

Q.7 Pipe A can fill an empty tank in eight hours and pipe B can empty the full tank in four more hours. If both the pipes are opened simultaneously, then how long would it take to fill an empty tank?
(a) 12 h (b) 24 h (c) 18 h (d) 30 h

Q.8 Sanjay can wash a truck in 45 min and he works for six hours a day. How many trucks can he wash in three days?
(a) 24 (b) 12 (c) 21 (d) 15

Q.9 Sumit Niyogi can build a wall measuring 20 ft × 3 ft × 15 ft in 6 days. Vartika Dikshit can build a wall measuring 30 ft × 4 ft × 10 ft in 5 days. What is the ratio of the work done by Sumit Niyogi and Vartika Dikshit in one day?
(a) 5:8 (b) 6:5 (c) 6:7 (d) 7:5

Q.10 Nandan is twice as good a workman as Divya and together they complete the work in 15 days. In how many days, can Divya alone complete the work?
(a) 45 days (b) 40 days
(c) 35 days (d) None of these

Q.11 N men can do a piece of work in 8 days. The same work can be done by $(N + 4)$ men in 6 days. Find the value of N.
(a) 10 (b) 12
(c) 14 (d) None of these

Q.12 Harveen can do a piece of work in 18 days. He worked for 12 days and left. Deepak finished the remaining work in 8 days. In how many days, can Deepak alone complete the work?
(a) 21 (b) 22 (c) 23 (d) 24

Q.13 Munchun can do a work in 20 days and Chunmun can do it in 10 days. Both worked together and were paid ₹1800. The share of Chunmun is:
(a) ₹1200 (b) ₹1500
(c) ₹900 (d) None of these

Q.14 Shahid Bhai can do two-third of the work in 18 days. In how many days, can he finish one-fourth of the work?
(a) $6\frac{1}{2}$ (b) 6.75 (c) 6 (d) 13.5

Q.15 A can do half as much work as B in one day. B alone can do a certain work in 12 days. In how many days, can A and B together finish that work?
(a) 6 days (b) 7 days
(c) 8 days (d) None of these

Q.16 If 35 persons can do a piece of work in 6 days, in how many days, can 15 persons do it?
(a) 12 (b) 14 (c) 16 (d) 18

Q.17 A and B can complete a work in 15 days. A is 50% more efficient than B. How long would A take to complete the work alone?
(a) 20 days (b) 22 days
(c) 24 days (d) None of these

Q.18 Forty persons can finish a work in 60 days. If 40 persons start the work and at the end of every 10 days 40 more persons of the same efficiency join them, then how long will it take to complete the work?
(a) 30 days (b) 20 days
(c) 40 days (d) None of these

Q.19 A group of 18 men and 12 women can do a work in 18 days. A woman takes twice as much time as a man to do the work. How many days will 8 men take to finish the same work?
(a) 45 (b) 48
(c) 54 (d) None of these

Q.20 A team has a food stock for N number of days. After 20 days, one-fourth of the team quits and the food lasted for another N number of days for the remaining persons in the team. What is the value of N?
(a) 50 (b) 60 (c) 70 (d) 80

FOUNDATION

Q.1 To complete a work, A and B take 30 days, B and C take 24 days, and C and A take 20 days. They all start together and work for 5 days after which B and C leave. How many more days will A take to finish the remaining work?
(a) 11 (b) 28 (c) 33 (d) 18

Q.2 A water tank normally takes 7 h to be filled by a tap but because of the leak, it takes another 2 h. In how many hours will the leak empty a full water tank?
(a) 20.5 h (b) 24.5 h (c) 30 h (d) 31.5 h

Q.3 The wages of 10 men and 8 boys amount to ₹37. If 4 men together receive ₹1 more than 6 boys, what are the wages of each man and boy, respectively?
(a) 2.5 and 1.5 (b) 11 and 9
(c) 3 and 4 (d) 6 and 5

Q.4 Fourteen machines can produce 500 units in 12 h. After 4 h of production, two more machines were started. How many more hours will it take to complete the production of 500 units?
(a) 4 (b) 5 (c) 6 (d) 7

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Q.5 Pramod can do two-fifth of the work in 8 days and Binod can finish the remainder in 18 days. How long would they take to finish it together?
(a) 8 (b) 10 (c) 12 (d) 14

Q.6 A cistern can be filled separately by two pipes *P* and *Q* in 45 min and 35 min, respectively. A tap *R* at the bottom can empty the full cistern in 30 min. If the tap *R* is opened 7 min after the two pipes *P* and *Q* are opened then after what time from the opening of pipe *P* and *Q* the cistern becomes full?
(a) 40th min (b) 61st min
(c) 37th min (d) None of these

Q.7 *A* can do a job in 20 days, *B* in 30 days, and *C* in 60 days. If *A* is helped by *B* and *C* every third day. How long will it take for them to complete the job?
(a) 12 days (b) 4 days (c) 15 days (d) 18days

Q.8 Saket and Prakash take 7.5 days working together. However, if Saket and Prakash work individually to complete the equal amount of work, it takes them 20 days to complete the job. How long will Prakash alone take to do the job if Saket is more efficient than Prakash?
(a) 18 days (b) 24 days (c) 30 days (d) 27days

Q.9 Four machines *A*, *B*, *C*, and *D* are to be used for producing 10,000 units. Machine *A* alone can produce 50 units in a day, *B* is half as efficient as *A*, *C* is as efficient as *A* and *D* is twice as efficient as *A*. In how many days, would the work be complete if all the four machines start working together?
(a) 19 days (b) 29 days
(c) 46 days (d) None of these

Q.10 In an organization, regular working day is of 8 h and regular week is of 5 working days. A man is paid ₹3.20 per hours overtime. If he has earned ₹432 in 4 weeks, what is the total number of hours he worked in those 4 weeks?
(a) 180 (b) 175
(c) 160 (d) Data insufficient

Q.11 Tap A fills a tank in 20 min while *C* empties it at one-third the rate at which A fills it. At 12:00 noon, *A* and *C* are simultaneously started and when the tank is 50% full, tap A is turned off. At what time will the tank be empty?
(a) 12:35 pm (b) 12:45 pm
(c) 12:30 pm (d) 12:55 pm

Q.12 Two pipes *X* and *Y* can fill a tank in 6 h and 9 h, respectively. If they are turned on alternately for one hour each starting with *X*, then what is the time taken to fill the tank?
(a) 6.25 h (b) 5.66 h (c) 7 h (d) 8 h

Q.13 A 10 hectare field is reaped by 2 men, 3 women, and 4 boys in 10 days. If a man, a woman and a boy work in the ratio of 5:4:2, then in how much time 6 men, 4 women and 7 boys reap a 16 hectare field?
(a) 5 days (b) 6 days (c) 8 days (d) 7 days

Q.14 Four plumbers had to repair 364 taps. At the end of 3 days working 7 h per day they had repaired 168 taps. In order to speed up the work, three more plumbers were added who also worked 7 h daily. If efficiency of each of the plumbers is same, how many additional days would it take to complete the task?
(a) 2 days (b) 3 days
(c) 4 days (d) 1 1

Q.15 Three factories are capable of manufacturing a particular part of the bike. Each of the two factories can produce 1,00,000 parts in 15 days. The third factory can produce them 30% faster. How many days (approx.) would it take to produce a million parts with all the three working simultaneously?
(a) 43 (b) 45 (c) 48 (d) 52

Q.16 Five persons working 8 h daily can complete a wall in 10 days. When they have worked for 5 and half days, five more persons are brought to work. In how many more days will the wall be complete?
(a) One more day (b) More than 2 days
(c) More than 4 days (d) More than 3 days

Q.17 Two filling pipes can fill a tank in 15 h and 12 h, respectively, while a third pipe C can empty it in 20 h when filled. If the tank is empty and all the three pipes are opened, in how much time will the tank be full?
(a) 14 h (b) 10 h (c) 6 h (d) 8 h

Q.18 A work can be finished by Arnold, Bill, and Kennedy in 20 days, 24 days, and 30 days, respectively, provided the working hours are 10 h per day. If all of them are engaged to complete this work and they finished the work in 10 days, how many hours per day did they work?
(a) 7 h (b) 10 h (c) 8 h (d) 9 h

Q.19 If A completes a work in *t* days, B in 1/*r* days, and C in *s* days then in how many days will they complete the work together?
(a) $st/(s + t + rst)$ (b) $rt/(r + t + rst)$
(c) $rst/(rs + st + tr)$ (d) $rs/(r + s + rst)$

Q.20 A certain job was assigned to a group of people to complete it in 20 days. But 12 of them did not turn up for the job and the remaining persons did the job in 32 days. What is the original number of men in the group?
(a) 32 (b) 34
(c) 36 (d) None of these

Q.21 A machine *P* can print one lakh books in 8 h machine *Q* can print the same number of books in 10 h while

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machine *R* can print them in 12 h. If all of them work together for 6 h, how many books will be printed?
(a) 60,000 (b) 1,80,000
(c) 1,85,000 (d) 2,85,000

Q.22 A pipe can fill a cistern in 6 h. Due to a leak in its bottom, it is filled in 7 h. When cistern is full, in how much time will it be emptied by the leakage?
(a) 42 h (b) 40 h (c) 43 h (d) 45 h

Q.23 Tap A can fill a tank in 16 min and tap B can empty it in 8 min. If the tank is already 1/2 full and both the taps are opened together, the tank will be either filled or emptied? How long will it take before the tank is either filled or emptied?
(a) Emptied; 16 min (b) Filled; 8 min
(c) Emptied; 8 min (d) Filled; 12 min

Q.24 Three-fourth of a tank is full of water. If 5 L is added to it, then four-fifth of the tank becomes full. What is capacity of the tank?
(a) 75 L (b) 80 L (c) 100 L (c) 120 L

Q.25 A 6-cm long cigarette will burn itself up in 15 min if no puff is taken. For every puff, it burns three times as fast as during the duration of the puff. If the cigarette burns itself in 13 min, then how many puffs has the smoker taken if his average puff lasted 3 sec?
(a) 30 (b) 20
(c) 15 (d) None of these

Q.26 Two taps A and B can fill a vessel in 12 min and 15 min, respectively, but a third tap C can empty the full tank in 60 min. A and B is kept open for 5 min in the beginning and then C is also opened. In what time will the vessel be emptied?
(a) 30 min (b) 33 min
(c) 45 min (d) None of these

Q.27 Two pipes A and B can separately fill a cistern in 15 and 10 min, respectively, and a waste pipe C can carry off 7 L per min. If all the pipes are opened when the cistern is full, it is emptied in 2 h. How many litres does the cistern hold?
(a) 30 L (b) 40 L (c) 35 L (d) 45 L

Q.28 One pipe can fill a tank three times as fast as another pipe. If together the two pipes can fill the tank in 36 min, then how much time will the slower pipe take to fill the tank working alone?
(a) 81 min (b) 108 min
(c) 144 min (d) 192 min

Q.29 Two pipes A and B can separately fill a cistern in 10 min and 5 min, respectively, and a waste pipe C can carry off 5 L per min. If all the pipes are opened when the cistern is full, it is emptied in 1 h. What is the capacity of the cistern?
(a) 40 L (b) 35 L
(c) 95 L (d) None of these

Q.30 Sanjay can do as much work in 16 h as Balram can do in 24 h. Pranesh can do the same work in 32 h. If all of them work together, they can complete the same work in 24 h. In how many hours can Pranesh alone complete the work?
(a) 80 (b) 88 (c) 104 (d) 112

Q.31 One fill-pipe A is 10 times faster than the second fill-pipe B. If B can fill a cistern in 55 min, then find the time when the cistern will be full if both the fill-pipes are opened together?
(a) 5 min (b) 6 min (c) 7 min (d) 3 min

Q.32 A tank can be filled by one tap in 20 min and by another in 25 min. Both the taps are kept open for 5 min and then the second tap is turned off. In how much more time will the tank be filled?
(a) 11 min (b) 13 min (c) 15.5 min (d) 18 min

Q.33 Fifteen men take 21 days working 8 h each to do a piece of work. How many days of 6 h each would 21 women take, if 3 women can do as much work as two men can do?
(a) 20 (b) 30 (c) 21 (d) 28

Q.34 Three hundred workers are set to build a dam. In 21 weeks, they have done 60% of the work, but subsequent rains lasting 4 weeks washed away 25% of what they had done. The job is resumed by 200 workers. What is the additional time it will take to complete the work?
(a) 12 (b) 24
(c) 36 (d) None of these

Q.35 Tap A can fill a tank completely in 4 h, while tap B can empty it in 8 h. However, the operator forgot to close the tap B, and as a result, both the taps remained open. After 4 h, the person realized his mistake and immediately closed the tap B. How much more time would the tank take to get full now?
(a) 1 h (b) 2 h (c) 3 h (d) 4 h

Q.36 If work done by 2 men are equivalent to work done by 5 women and work done by 2 women are equivalent to work done by 5 children, then in how many days can the work be completed by one man alone if the same work can be completed by 100 children in 10 days?
(a) 120 days (b) 80 days
(c) 100 days (d) 160 days

Q.37 A certain number of people completed a job in 24 days. If there were 2 persons less, the work would have taken 6 more days. How many persons were there initially?
(a) 12 (b) 16 (c) 18 (d) 10

Q.38 Twelve men complete a work in 9 days. After they have worked for 6 days, 6 more men join them.

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How many days will they take to complete the remaining work?
(a) 2 days (b) 3 days (c) 4 days (d) 5 days

Q.39 A certain number of men can finish a piece of work in 100 days. If however there were 10 men less, it would take another 10 days to finish the work. How many men were there originally?
(a) 50 (b) 75 (c) 82 (d) 110

Q.40 A Zamindar hired a certain number of men to plough his 480 acres of fields. Some of the men did not turn up at the job, and consequently each of those who reported had to plough 20 acres more of the field each than originally planned by the zamindar, assuming equal distribution of labour. How many men were initially appointed and how many abstained from the job?
(a) 15, 5 (b) 12, 8 (c) 10, 3 (d) 16,10

Q.41 Anoop takes 5 h more than Mayank to build a wall. However, Anoop takes 9 h more than Manoj. Anoop and Mayank together can do the same work in the same time as Manoj. How many hours will Anoop take to do it?
(a) 12 h (b) 15 h (c) 18 h (d) 24 h

Q.42 A tank can be filled in by two taps A and B in 3 min. However, due to a leakage at the bottom of the tank, it takes 3.5 min to fill the tank. If tank is full, then how much time will the leakage take to empty the tank?
(a) 21 min (b) 12 min (c) 24 min (d) 18 min

Q.43 A road construction agency takes up the work of making a road 189 km long in 15 months. He engages 256 men, but after one-fifth of the time, agency finds that only one-sixth of the work has been completed. How many more men should be now employed to complete the work in time?
(a) 57 (b) 59 (c) 64 (d) 67

Q.44 Twenty-five men are employed to do a piece of work in 24 days. After 15 days, 10 more men are engaged and the work was finished a day earlier than actually planned. In what time could they finish the work if extra men were not employed?
(a) 29.5 days (b) 24.6 days
(c) 26.2 days (d) 21.7 days

Q.45 Twelve men or 18 women can do some work in 7 days. In what time, can 4 men and 8 women do the same work?
(a) 4 days (b) 8 days (c) 7 days (d) 9 days

Q.46 A man works for 10 h per day to finish a work in 3 days. A boy finishes the same in 5 days working 18 h per day. How many boys are required to do the job in the time taken by one man?
(a) 1 (b) 2 (c) 3 (d) 4

Q.47 A tap can fill a tank in 60 min. Tap is turned on to fill the tank. When the tank is half full, it springs a leak at the bottom and the water level falls to one-fourth of the tank in 10 min. The tap is now closed. What is the time taken by the leakage to empty the tank now?
(a) 7 min (b) 4 min
(c) 8 min (d) None of these

Q.48 A piece of work can be done by *A* and *B* in 10 days by *B* and *C* in 15 days and by *C* and *A* in 20 days. All three worked for 6 days and then *A* withdrew. *B* and *C* worked for another 4 days and then *B* stopped working. How many more days will *C* take to complete the Remaining work?
(a) 12 (b) 12.5 (c) 10.33 (d) 10

Direction for Questions 49 and 50: Read the passage given below and solve the questions based on it.

A group of 30 people can complete a job by working for 10 h a day in 15 days. The group starts the work. But at the end of every day, starting from the first day, one person leaves the group and the remaining people work for 20 min less on the next day.

Q.49 On which day will the work be completed?
(a) 28th (b) 29th
(c) 30th (d) None of these

Q.50 What part of the work, approximately remains to be done at the end of 15 days?
(a) 0.51 (b) 0.61 (c) 0.49 (d) 0.39

Moderate

Q.1 Mayank can do 50% more work than Shishu in the same time. Shishu alone can do a piece of work in 30 h. Shishu starts working and he had already worked for 12 h when Mayank joins him. How many hours should Shishu and Mayank work together to complete the remaining work?
(a) 6 (b) 12 (c) 4.8 (d) 7.2

Q.2 A man works twice as fast as a woman. A woman works twice as fast as a child. If 16 men can complete a job in 12 days, how many days would be required for 32 women, and 64 boys together to complete the same job?
(a) 2 day (b) 3 days
(c) 4 days (d) 6 days

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Q.3 A tank with capacity T L is empty. If water flows into the tank from pipe X at the rate of X L per min and water is pumped out by pipe Y at the rate of Y L per min and $X > Y$, then in how many min will the tank be filled?
(a) $T/(Y-X)$ (b) $T/(X-Y)$
(c) $(T-X)\,Y$ (d) $(X-Y)/60T$

Q.4 A can do a piece of work in 60 days, B in 40 days, and C in 12 days. They work for a day each in turn, that is, first day A does it alone, second day B does it alone, and third day C does it alone. After that the, cycle is repeated till the work is finished. They get ₹270 for this job. If the wages are divided in proportion to the work each had done, find the amount A will get.
(a) 14 (b) 36 (c) 24 (d) 27

Q.5 There are 12 pipes connected to a tank. Some of them are fill pipes and the others are drain pipes. Each of the fill pipes can fill the tank in 8 h and each of the drain pipes can drain completely in 6 h. If all the pipes are kept open, an empty tank gets filled in 24 h. How many of the 12 pipes are fills pipes?
(a) 5 (b) 6 (c) 7 (d) 8

Direction for Questions 6 and 7: Read the passage below and solve the questions based on it.

The tank at a water supply station is filled with water by several pumps. At first, three pumps of the same capacity are turned on; 2.5 h later, two more pumps (both the same) of a different capacity are set into operation. After 1 h, the additional pumps were set into operation; the tank was almost filled to its capacity (15 m^3 were still lacking); in another h, the tank was full. One of the two additional pumps could have filled the tank in 40 h.

Q.6 What is the volume of the tank?
(a) 60 m^3 (b) 80 m^3 (c) 75 m^3 (d) 90 m^3

Q.7 How much water does one of the first three pumps fill in an hour?
(a) 5 m^3 (b) 4 m^3 (c) 3 m^3 (d) 2 m^3

Direction for Questions 8 and 9: Read the passage below and solve the questions based on it.

A person can dig a trench 40 metres in depth in 4 days working 8 h a day. However, after every day he finds that one-fifth of the depth got filled up with mud again.

Q.8 In how many days can he actually dig a trench if it is of 30 metres in depth?
(a) Exactly 4 days (b) On the 5th day
(c) Exactly 5 days (d) On the 6th day

Q.9 What was the depth of the trench at the beginning of the fourth day?
(a) 14.4 m (b) 19.52 m
(c) 23.616 m (d) None of these

Direction for Questions 10 to 12: Read the passage below and solve the questions based on it.

There are three taps A, B, C, and an outlet pipe D. A, B, and C can fill the tank in the Panikam locality in 10, 20, and 25 h, respectively. The outlet pipe can empty the same tank in 100 h. There are 2000 houses in the locality. The tank has a capacity of 50,000 L.

Q.10 If all the taps and the outlet pipe are opened simultaneously, how much water is thrown into the tank every hour?
(a) 8000 L (b) 9000 L
(c) 1000 L (d) Cannot be determined

Q.11 All the three taps were open and the emptying pipe is closed. At the time, when the tank was supposed to be full, it was found that only two-fifth of the tank was full. It was discovered that all the residents had kept their water taps open during this period. At what rate, were the residents of each house getting water? Consider that each house has only one tap.)
(a) 1.1 L/h (b) 2.22 L/h
(c) 2.85 L/h (d) 4.46 L/h

Q.12 If the average requirement of every house in the area per day is 40 litres, for how many hour, for the second time should the taps be opened? (Consider that all the three tapes were opened on both occasions.)
(a) $3\frac{3}{19}$h (b) $4\frac{1}{17}$h
(c) 5 h (d) None of these

Direction for Questions 13 and 14: Refer to the data below and answer the questions that follow.

Fifty cows can graze a field for 40 days and 40 cows can graze the same field for 80 days. Some grass is already there and the grass grows at a constant rate.

Q.13 For how many days, can 60 cows graze the same field? (growth of grass is also there)
(a) 20.66 (b) 50.66 (c) 60.66 (d) 26.66

Q.14 What is the maximum number of cows that can graze the same field for an infinite time?
(a) 30 (b) 25 (c) 8 (d) 35

Q.15 Anand got an order from a boutique for 480 shirts. He brought 12 sewing machines and appointed some expert tailors to do the job. However, many did not report to duty.

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As a result, each of those who reported had to stitch 32 more shirts than was originally planned by Anand, with equal distribution of work. How many tailors had been appointed earlier and how many had not reported to work?

(a) 12, 4 (b) 10, 3
(c) 10, 4 (d) None of these

Direction for Questions 16 to 18: Refer to the data below and answer the questions that follow.

Anoop and Sandeep can dig a well each in 10 and 5 days, respectively. But on a particularly difficult terrain the work is such that due to fatigue every subsequent day the efficiency of a worker falls by 10%.

Q.16 If Anoop is given a task of digging one such well in the difficult terrain, then in how many days will he finish the work?

(a) 12th day (b) 15th day
(c) 11th day (d) Never

Q.17 If Sandeep is given the task of digging one such well in the difficult terrain, then in how many days will he finish the work?

(a) 9th day (b) 7th day
(c) 8th day (d) Never

Q.18 If both Anoop and Sandeep work together to finish two such wells then, in how many days will the work finish?

(a) 11th day (b) 12th day
(c) 8th day (d) Never

Direction for Questions 19 and 20: Refer to the data below and answer the questions that follow.

Anoop was writing the reading comprehension sections in the DOG entrance examinations. There were four passages of exactly equal length in terms of number of words and the four passages had 5, 8, 8 and 6 questions following each of them, respectively. It is known that Anoop can answer exactly 12 questions in the time he takes to read any one of the four passages. Assume that his rate of reading and answering questions remains the same throughout the section.

Q.19 Anoop took 13 min more to finish the first three passages than the time he took to finish the last passage. Assuming that Anoop answered all the questions in each passage, what percentage of the total time did he spend on the first passage?

(a) 24.5% (b) 25.4% (c) 22.6% (d) 26.2%

Q.20 By what per cent should Anoop increase his reading speed if he has to cut down on his total time spent on the section by 20%? Assume that the time spent on answering the questions is constant and as given in the directions.

(a) 36.36% (b) 54.54%
(c) 50.50% (d) 45.45%

Q.21 The kingdom of Charles had a circular wall around it and four-city gates pointing north, south, east, and west, respectively. Now, the Shire of Nottingham and the Shire of Birmingham decided to race on the shortest route touching all the four gates. They start at the west gate and one of the shires rides five times as fast as the second. They will meet for the 19th time after they start, at the

(a) the north gate.
(b) the south gate.
(c) exactly midway between the north and the west gate.
(d) exactly midway between the south and the east gate.

Q.22 The packing process in a garment factory involves processing by three machines. The first machine, *A*, inserts the garment in a packet, after which the second machine, *B*, seals the packet and finally the third machine, *C*, attaches a price tag to the packet. Machine *A* takes at least 10 per unit. Machine *B* takes at least 15 per unit. Machine *C* takes at least 12 per unit. The factory starts at 8 am and runs non-stop till 6 pm, What is the maximum number of packets that can be processed in a one-hour period?

(a) 360 (b) 240 (c) 300 (d) 900

Q.23 *P*, *Q*, and *R* can each complete a certain work in 16, 20, and 30 days, respectively. The three of them start the work together. *P* leaves after 4 days; *Q* leaves 4 days before the work is finished. How long did the work last?

(a) 12.6 days (b) 9.2 days
(c) 16.4 days (d) 11.4 days

Q.24 A certain work was started by 4 men and 10 women who completed 50% of the work in 6 days. Then, another 2 men and 2 women joined them and they could complete two-third of the remaining work in three days. How many men along with 6 women are required to complete the remaining work in two more days?

(a) 3 (b) 5 (c) 6 (d) 9

Q.25 A can work half as fast as *B* and *C* together. *A* and *B* together can work three times as fast as *C*. If all three of them complete a job in 15 days, how long would *C* take to complete the same work last?

(a) 45 days (b) 54 days
(c) 60 days (d) 65 days

Q.26 A cistern can be filled by two taps A and B coming from the same main pipeline in 20 and 30 min, respectively. Both the taps are opened at the same time but due to

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partial closing of a valve in the main pipeline, tap A was supplying only four-fifth of its capacity and tap B, 5/6th of its capacity. After some time, the valve in the main pipeline was opened fully, thereby enabling the two taps to supply at full capacity. It took another 5 min to fill the cistern completely, how long (approx.) was it before the valve on the main pipeline opened?
(a) 7 min (b) 9 min (c) 11 min (d) 13 min

Q.27 In what time would a cistern be filled by three pipes of diameter of 1 cm, 2 cm, and 3 cm if the largest pipe alone can fill the cistern in 49 min, the amount of water flowing through each pipe being proportional to the square of its diameter?
(a) 31.5 min (b) 63 min
(c) 126 min (d) None of these

Q.28 Three workers—*A*, *B*, and *C*—working simultaneously, produced 144 dyes in four hours. In one hour, *A* can produce as many dyes more than *B* as *B* can produce more than *C*. During a period of five hours, *A* can produce as many dyes as *C* can in seven hours. What is the difference between the number of dyes produced by *B* and *C* per hour?
(a) 1 (b) 2 (c) 3 (d) 4

Q.29 *A* can do a piece of work in 36 days, *B* in 54 days, and *C* in 72 days. All three began the work together, but *A* left after 8 days and *B* 12 days before the completion of the work. In all for how many days did *C* work?
(a) 24 days (b) 28 days
(c) 32 days (d) 36 days

Q.30 Three men with 5 boys can do a piece of work in 2 days and 4 men and 16 boys can complete the job in one day. How much time will it take for 1 boy together with a woman who can work twice as fast as the boy to complete a job that is three times as time consuming?
(a) 24 days (b) 28 days
(c) 32 days (d) 36 days

Q.31 Tatto, Bubby, and Tappo are doing a work in 10 days. However, Tappo only worked for the first three days till 37/100th of total work is done. The remaining work was done by Tatto and Bubby alone. Efficiency of Tatto is 25% more than the efficiency of Bubby. What is the time taken by the fastest worker to finish the work alone?
(a) 16.66 days (b) 30 days
(c) 25 days (d) 20 days

Q.32 *P* works twice as fast as *Q*, whereas *P* and *Q* together can work three times as fast as *R*. If *P*, *Q*, and *R* work together on a job, in what ratio should they share the earnings?
(a) 2:1:1 (b) 4:2:1 (c) 4:3:2 (d) 4:2:3

Q.33 Rakesh and Sid can complete a piece of work in 12 and 24 days, respectively. Rakesh started the work and after a few days Sid joined him and the work was completed in 10 days. After how many days, did Sid join Rakesh?
(a) 6 (b) 7 (c) 8 (d) 10

Q.34 Sagar is 20% more efficient than Pranesh. If Pranesh can complete a piece of work in 30 days, then in how many days can both Sagar and Pranesh complete the work?
(a) (11 + 5/11) days (b) (13 + 7/11) days
(c) (15 + 5/11) days (d) (17 + 7/11) days

Q.35 Manoj and Harvinder can complete a piece of work in 16 days, which Harvinder can complete in 32 days. If Manoj and Harvinder work on alternate days, starting with Harvinder, in how much time would the work be completed?
(a) 17.25 days (b) 17.66 days
(c) 20 days (d) 32 days

Q.36 A factory manufactures dyes. It has 12 men and two machines which can be operated by all of its men. It takes 4 h to manufacture one dye on the machine with the operator. The machines can work continuously without a break. Without the machine each of the men can manufacture a dye in 8 h. The policy is such that the production is maximized and the men are ready to work in three shifts of 8 h per day. What will be the average cost incurred per dye if 1 man hour costs ₹20 and 1 machine hour costs ₹15?
(a) ₹140 (b) ₹160
(c) ₹147 (d) Cannot be determined

Q.37 Ramesh and Suresh could finish the garden cleaning in '*p*' days. They worked together for '*q*' days; then, Ramesh fell sick and the remaining cleaning was completed by Suresh in '*r*' days. In how many days could each of them clean the garden?

(a) Ramesh = $\frac{pr}{r-p+q}$ days; Suresh = $\frac{pr}{p-q}$ days

(b) Ramesh = $\frac{p}{r-p+q}$ days; Suresh $\frac{p}{p-q}$ days

(c) Ramesh = $\frac{pr}{r+p-q}$ days; Suresh $\frac{pr}{q-p}$ days

(d) Ramesh = $\frac{pr^2}{r-p+q}$ days; Suresh $\frac{pr^2}{q-p}$ days

Q.38 *A*, *B*, *C*, *D*, and *E* are five taps. The capacity of *B* is two times that of *A*, the capacity of *C* is three times that of *A*. Capacities of *D* and *E* are 4 and 5 times that of *A*, respectively. In the first case *A*, *C*, and *E* act as input pipes and *B* and *D* act as output pipes.

In the second case, *C*, *D*, *E* act as input pipes and *A* and *B* act as output pipes.

If *A* and *B* working together as input pipes can fill the tank in 4 h, then what is the difference in the time

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required to fill the tank in the first and second case stated above?

(a) 4.44 h (b) 3.33 h
(c) 2.22 h (d) 1.11 h

Q.39 A tank has 5 input pipes, *A*, *B*, *C*, *D*, and *E*. Among them, *A* takes the maximum time to fill the tank. Their capacities to fill the tank are in 'Geometric Progression (GP)' in the given order with common ratio 2. The tank also has 5 output pipes *F*, *G*, *H*, *I*, and *J*. Among them, *F* takes the maximum time to empty the tank. *G*'s capacity to empty the tank is two times that of *F* and so on. Working together *A* and *F* can fill the tank in 100 h and all the taps working together can fill the tank in $\left(\frac{100}{63}\right)$ h.

How much time will *E* and *J* take to fill the tank when working together?

(a) $\frac{100}{62}$ (b) $\frac{100}{38}$

(c) $\frac{100}{48}$ (d) Cannot be determined

Q.40 A man makes 60 articles in the first hour. His efficiency decreases by 25% in the second hour, increases by 40% in the third hour, decreases by 33% in the fourth hour and increases by 50% in the fifth hour. If he has to work for more than 1 h, then in which h the average number of articles produced per hour till then would be minimum?

(a) The second hour
(b) After the fifth hour
(c) The second and the fourth hour
(d) None of these

Q.41 A worm climbs a vertical bamboo stick 32 cm tall. The worm climbs 3 cm in the day and slips 2 cm during the night each time. However, there are cavities in the stick at 2 cm, 4 cm, 16 cm from the bottom that would not let the worm slip further. If the worm climbs only during the daytime and slips only during the night, on which day from the start will it reach the top?

(a) 29 days (b) 28 days
(c) 27 days (d) 26 days

Q.42 Anil, Bhuvan and Chandan take 10, 20, and 25 days to complete a job. What is the minimum time required to finish the job if not more than two of them work together on a single day and no two consecutive days have the same pair of people working?

(a) 6.5 days (b) $6\frac{13}{15}$ days

(c) 7.5 days (d) $7\frac{21}{28}$ days

Q.43 A swimming pool is fitted with three pipes. The first two pipes working simultaneously fill the pool in the same time as the third pipe alone. The second pipe alone fills the pool 5 h faster than the first pipe and 4 h slower than the third pipe. In what time will the second and the third pipe together fill the pool?

(a) 3.25 h (b) 6.5 h
(c) 5 h (d) None of these

Q.44 A contractor needs to complete a given work in 24 days. For every day that he finishes ahead of time, he gets an incentive '*I*' and for every day that he finishes behind time, he needs to pay a penalty '*P*'. He has a choice of two workers who work as follows: A completes the work in 24 days for 75% of the time and in 30 days for 25% of the time. B completes the work in 20 days for 60% of the time and in 30 days for 40% of the time. At what ratio of *P* to *I* will the contractor be indifferent of using A or B?

(a) $\frac{8}{3}$ (b) 2 (c) $\frac{7}{3}$ (d) $\frac{7}{4}$

Q.45 Anuj can do a piece of work in a certain number of days. To do the same piece of work, Bhanu takes thrice the number of days as Anuj takes whereas Chandu takes thrice as many days as Bhanu does and Dodo takes thrice as many days as Chandu does. Now, they are paired and two groups are formed. The first pair takes one-third the time taken by the second pair to complete the work. which is the first pair?

(a) Anuj and Dodo
(b) Anuj and Chandu
(c) Chandu and Bhanu
(d) Bhanu and Dodo

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Direction for Questions 1 to 5: Read the passage below and solve the questions based on it.

There are infinite pipes attached to a very big tank. Pipes are numbered like P_1, P_2, P_3... and so on. Its also known that efficiency of every subsequent pipe is half the efficiency of earlier pipe, that is, the efficiency of P_2 pipe is half the efficiency of P_1 pipe and so on. However, in case of P_1 pipe, this rule is not true, since there is no earlier pipe. When all the pipes are working together, tank gets filled in 2 h.

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Q.1 What time will P_4 take to fill the tank working alone?
(a) 8 (b) 16
(c) 32 (d) Cannot be determined

Q.2 What is the difference in the time taken by P_5 and P_6?
(a) 32 (b) 64
(c) 128 (d) Cannot be determined

Q.3 How many pipes can fill the tank within 100 h working alone?
(a) 4 (b) 5 (c) 6 (d) Infinite

Q.4 Due to some technical problem, only four pipes P_1, P_2, P_3, and P_4 are in working condition. These pipes are now grouped to obtain two pairs of pipes. It is found that one pair of the pipe is taking two-third of the time taken by the other pair to fill the tank independently. Which of the following is one of the two pairs?
(a) P_1, P_2 (b) P_1, P_3
(c) P_1, P_4 (d) Cannot be determined

Q.5 What time will P_2 take to fill the tank working alone?
(a) 4 (b) 8 (c) 16 (d) 32

Direction for Questions 6 and 7: Read the passage below and solve the questions based on it.

There are two filling taps A and B and an emptying tap C which are used in a laboratory to fill or empty bottles. The three taps are attached to a computer and they function according to the commands given by the computer. The following are the commands that the computer gives:

Command	Effect
$F(A, x, t_1)$	Fill the bottle with tap *A* at x cc/min for t_1 min
$F(B, y, t_2)$	Fill the bottle with tap *B* at y cc/min for t_2 min
$E(C, z, t_3)$	Empty the bottle with tap *C* at z cc/min for t_3 min

Q.6 The following commands, in the given order, are used to fill an empty bottle and each command can be executed only after the previous command is completed. What is the final volume after these commands?
$F(A, 10, 600)$, $E(C, 50, 20)$
$F(A, 10, 400)$, $F(B, 20, 200)$
$F(A, 50, 100)$, $F(B, 20, 50)$
$E(C, 10, 50)$, $F(B, 20, 75)$
(a) 15 (b) 20
(c) 22 (d) None of these

Q.7 The commands $F(A, 10, 500)$ and $(B, 40, 500)$ can fill an empty bottle. Which of the following commands can now empty the bottle?
(a) $E(C, 20, 500)$ (b) $E(C, 50, 500)$
(c) $E(C, 100, 200)$ (d) None of these

Q.8 A group of workers can complete a certain job in 9 days. But it so happens that every alternate day starting from the second day, two workers are withdrawn from the job and every alternate day starting from the third day, one worker is added to the group. In such a way, the job is finished by the time, there is no worker left. If it takes the double time to finish the job now, find the number of workers who started the job?
(a) 5 (b) 10 (c) 15 (d) 20

Q.9 A student studying the weather for *d* days observed that (i) it rained on 7 days, morning or afternoon, (ii) when it rained in the afternoon, it was clear in the morning, (iii) there were five clear afternoons, and (iv) there were six clear mornings. Then, *d* equals:
(a) 3 (b) 7 (c) 11 (d) 9

Q.10 Tanveer constructs a wall working in a special way and takes 12 days to complete it. If S_n is the length of the wall (in m) that he constructs on the nth day, then
$S_n = 2n$, $0 \le n \le 4$
$S_n = 8$, for $n = 5$
$S_n = 3n - 7$, $6 \le n \le 12$
Find the total length of the wall he constructs in the first 10 days.
(a) 31 m (b) 35 m
(c) 113 m (d) None of these

Direction for Questions 11 and 12: Read the passage below and solve the questions based on it.

There are infinite number of taps that are filling a certain tank. The taps are numbered 1, 2, 3, and so on. Any tap numbered p ($p \ge 2$) can fill the tank in $(p^2 - 1)$ min while the first tap does not work at all.

Q.11 If all the taps from the second tap onwards are opened simultaneously, then what is the shortest time in which the tank can be filled?
(a) 11/3 min (b) 22/3 min
(c) 2/3 min (d) 11/5 min

Q.12 The minimum number of taps that must be opened so that the tank is filled within 106 sec is:
(a) 6 (b) 3 (c) 5 (d) 4

Q.13 Alok, Mithilesh, and Bimlesh started a work and after completing 1/5th of the work Bimlesh left. Alok and Mithilesh then worked for 20 days. Bimlesh then took over from Alok and Mithilesh and completed the remaining portion of the work in 12 days. If Bimlesh takes 40 days to complete the work, in how many days would Alok alone or Mithilesh alone complete the work if the efficiencies with which they work is the same?

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(a) 40 (b) 60
(c) 100 (d) None of these

Q.14 Sixty-four men working 8 h a day plan to complete a piece of work in 9 days. However, 5 days later they found that they had completed only 40% of the work. They now wanted to finish the remaining portion of the work in 4 more days. How many hours per day should they need to work in order to achieve the target?
(a) 11 (b) 12 (c) 13 (d) 15

Direction for Questions 15 and 16: Read the passage below and solve the questions based on it.

The employees of ABC Corporation put forward demands for a shorter working week and a better pay. They were paid by the hour and worked 5 days a week. Their demands were:

(i) the working days to be shorter by 1 h and
(ii) in the course of the shorter working day that they are demanding, payment for a working day be the same as the number of hour needed to earn ₹49.

At present, a full day's work brings in ₹1 less than that and as many rupees as it is necessary to work and earn ₹40.

Q.15 For how many hours, were the employees proposing to work?
(a) $8\frac{1}{2}$ (b) 10 (c) $10\frac{1}{2}$ (d) 9

Q.16 What was the revised rate demanded per hour?
(a) $₹1\frac{1}{4}$ (b) $₹2\frac{1}{3}$ (c) $₹2\frac{3}{4}$ (d) $₹2\frac{1}{2}$

Direction for Questions 17 to 19: Read the passage given below and solve the questions based on it.

Mr. Bose can do a job in two days if his four secretaries *A*, *B*, *C*, and *D* are all present. His efficiency got reduced to $\frac{1}{6},\frac{1}{4},\frac{1}{3},\frac{1}{2}$, respectively, as *A*, *B*, *C*, and *D* remain absent. In any day, only one secretary remains absent. Also, Mr. Bose finishes his work exactly in integral number of days

Q.17 Which of the following secretaries cannot be absent for an odd number of days?
(a) *A* (b) *B* (c) *C* (d) *D*

Q.18 If each secretary remained absent for at least one day, then in how many days will Mr. Bose finish the work?
(a) 5, 6, or 7 days (b) 6, 7, or 8 days
(c) 6, 8, or 9 days (d) 7, 8, or 9 days

Q.19 If Mr. Bose finished his work in 6 days with one secretary being present every day, then the secretary was:
(a) *A* (b) *B*
(c) *C* (d) Cannot be determined

Direction for Questions 20 to 22: Read the passage given below and solve the questions based on it.

A team of *N* workers are selected to do a certain job. Each worker can individually complete the job in 10 days. The team follows the following procedure for any job:

The work is started by one person on the first day. Then, from the next day onwards, till the work is completed, everyday one new person joins the work.

The work is such that, on any day, it starts at 9:00 am and stops at 5:30 pm with a 30-min break for lunch from 1:00 pm to 1:30 pm. At whatever stage, a worker joins the work, he does not leave till the job is over.

Answer each question in this set independently of the other questions.

Q.20 If the entire team works at only 50% efficiency, and the worker who is supposed to join on the last day does not report for work, then the work is completed
(a) exactly at the end of the 6th day.
(b) before lunch, on the 6th day.
(c) after lunch, before the end of the 6th day.
(d) before lunch, on the 7th day.

Q.21 If the team takes up a task which is twice as time consuming as the original work, then the work will be completed on the:
(a) 5th day (b) 6th day
(c) 7th day (d) 8th day

Q.22 If the team takes up a task which is twice as time consuming as the original work, then at what time on the concluding day will the work be completed?
(a) 3:40 pm (b) 4:20 pm
(c) 5:30 pm (d) 4:10 pm

Q.23 The rate of flow of water (in litre per min) of three pipes are 2, *N* and 3, where $2 < N < 3$. The lowest and the highest flow rates are both decreased by a certain quantity *x*, while the intermediate rate is left unchanged. If the reciprocals of the three flow rates, in the order given above, are in arithmetic progression both before and after the change, then what is the quantity *x* (in litre per min)? (Negative flow rates indicate that the pipes act as emptying pipes instead of filling pipes.)
(a) 2.6 (b) –3.2 (c) 3.8 (d) –2.6

Direction for Questions 24 to 26: Read the passage below and solve the questions based on it.

Two products A and B can be made on either of the two extrusion machines M_1 or M_2. The time in min, taken by these products according to the machine used, are as given below. Assume a day has 8 working hours.

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	M_1	M_2
A	8	10
B	6	6

Q.24 If A and B are produced in the ratio of 3:1, then what is the maximum total of the products A and B that can be produced in a day?
(a) 120 (b) 160
(c) 116 (d) None of these

Q.25 If A is produced only on Machine M_1 and B produced only on machine M_2, then what is the unutilized time on M_2 in a day, if A and B are produced in 1:1 ratio?
(a) 20 min (b) 60 min
(c) 150 min (d) None of these

Q.26 If each product requires working on both the machines and the time required on each machine is as indicated in the table, then what is the maximum production possible in a day, assuming only A is produced?
(a) 80 (b) 180
(c) 48 (d) None of these

Direction for Questions 27 to 30: Read the passage below and solve the questions based on it.

There are three bottles of water – A, B, and C whose capacities are 5 L, 3 L, and 2 L, respectively. For transferring water from one bottle to another and to drain out the bottles, there exists a programme which is computer-controlled. The computer that controls the flow through these pipes can be fed with three types of instructions, as explained below:

Instruction type	Explanation of the instruction
Fm(x,y)	Fill bottle labelled *x* from the water in bottle labelled *y*, where the remaining capacity of *x* is less than or equal to the amount of water in *y*.
Empty (x, y)	Empty out the water in bottle labelled *x* into bottle labelled *y*, where the amount of water in *x* is less than or equal to remaining capacity of *y*.
Drain (x)	Drain out all the water contained in bottle labelled *x*.

Initially, A full of water and B and C are empty.

Q.27 First instruction: Fill (B, A), Second instruction: Fill (C, B).
What is the quantity of water in B after executing all the instructions?
(a) 1 L (b) 2 L
(c) 3 L (d) None of these

Q.28 After executing a sequence of three instructions, bottle A contains one litre of water. The first and the third of these instructions are shown below.
First instruction: Fill (C, A).
Third instruction: Fill (C, A).
Then, which of the following statements about the instructions is true?
(a) The second instruction is Fill (B, A)
(b) The second instruction is Empty (C, B)
(c) The second instruction transfers water from B to C
(d) The second instruction involves using the water in bottle A

Q.29 Consider the same sequence of three instructions and the same initial state mentioned in the last question. Three more instructions are added at the end of the above sequence to have A contain 4 L of water. In this total sequence of six instructions, the fourth one is Drain (A). This is the only Drain instruction in the entire sequence. At the end of the execution of the above sequence, how much water is contained in C?
(a) 1 L (b) 2 L
(c) 0 (d) None of these

Q.30 Initially, A had 5 L, B had 2 L, and C had 1 L. What is the minimum number of times Fill (x, y) is required to be used so that A is empty, provided we cannot use Empty (x, y) instruction?
(a) 1 (b) 2
(c) 3 (d) None of these

Direction for Questions 31 to 35: Read the passage below and solve the questions based on it.

Modern Boutique received a large order for stitching uniforms from A. R. Academy and T. M. High School. He has two cutters who will cut the fabric, five tailors who will do the stitching, and two assistants to stitch the buttons and button holes. Each of these nine persons will work for exactly 10 h a day. Each of the A. R. Academy uniform requires 20 min for cutting the fabric, 1 h for doing the stitching and 15 min for stitching the buttons and the button holes. The T. M. High School uniform requires 30 min, 1 h and 30 min, respectively, for the same activities.

Q.31 Find the maximum number of T. M. High School uniforms that Modern Boutique can complete in a day.
(a) 30 (b) 35 (c) 40 (d) 36

Q.32 On a particular day, Modern Boutique decided to complete 20 T. M. High School uniforms. How many A. R. Academy uniforms can it complete on that day?
(a) 30 (b) 35 (c) 40 (d) 36

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Q.33 If Modern Boutique decided to complete 30 T. M. High School uniforms only and no other uniform on that particular day, how many total man-hours will go idle?
(a) 30 (b) 60 (c) 45 (d) 36

Q.34 If Modern Boutique hires one more assistant, what is the maximum number of A. R. Academy uniforms that can be completed in a day?
(a) 60 (b) 48 (c) 120 (d) 50

Q.35 Modern Boutique has the option to hire one more employee of any category. Whom should it hire to get the maximum increase in the production capacity, assuming that it needs to stitch only A. R. Academy uniforms on that day?
(a) One cutter
(b) One tailor
(c) One assistant
(d) Either of the tailor or the cutter

Q.36 Two poles, one 78 m in height above the ground and the other 91 m in height above the ground, are at some distance from each other. Two strings are tied, one from the top of one pole to the bottom of the other and the other from the top of the second pole to the bottom of the first. What is the height above the ground at which the string meets?
(a) 52 m (b) 86 m (c) 42 m (d) 48 m

Q.37 The daily work of 2 men is equal to that of 3 women or that of 4 youngsters. By employing 14 men, 12 women, and 12 youngsters a certain work can be finished in 24 days. If it is required to finish it in 14 days and as an additional labour, only men are available, how many of them will be required?
(a) 20 men (b) 18 men
(c) 48 men (d) 28 men

Q.38 Two Pipes A and B can fill a cistern in $12\frac{1}{2}$ and 15 h, respectively. If the pipes can be opened or closed only after every 30 min (i.e., at 6.00, 6.30, etc.) and the tank is to be filled by using both the pipes without any overflow then,
(a) pipe B should be turned on later.
(b) A should be turned on later.
(c) A should be turned off after some time.
(d) both the pipes should remain on till the tank is full.

Q.39 A, B, and C are assigned a piece of job that they can complete by working together in 15 days. Their efficiencies (measured in terms of rate of doing job) are in the ratio of 1:2:3. After one-third of the job is completed, one of them has to be withdrawn due to budget constraint. Their wages per day are in the ratio of 3 : 5 : 6. The number of days in which the remaining two persons can finish the job (at optimal cost) is:
(a) 9 (b) 12 (c) 15 (d) 18

Direction for Questions 40 and 41: Read the passage below and solve the questions based on it.

A group of 10 people, each with a different capacity to work, is assigned to complete a certain work. They work in such a way that only two people work on any given day, with no group of the same two people working on more than one day. By working in this manner, they could finish only 3/4th of the work by the time all the possible pairs of people had worked once. The remaining part of the work was then finished by all of them working together.

Q.40 What is the number of days for which all of them worked together?
(a) 135 days (b) 120 days
(c) 45 days (d) 3 days

Q.41 If on the last day before the end of the work, the work assigned to the group is doubled, then find the total number of days from the start that the group took to complete the work, assuming that all the people continue to work together?
(a) 240 days (b) 160 days
(c) 90 days (d) 60 days

Direction for Questions 42 to 46: Read the passage below and solve the questions based on it.

ABC is a firm that deals with furniture. Manufacturing of a table requires three levels of assembly. The finished table is at the first level. The leg assembly and the tabletop are at the second level. The pieces that go into the leg assembly are at the third level which consist of short rails, long rails, and legs. One unit of the table requires one unit of the tabletop and one unit of the leg assembly. One unit of the leg assembly requires two units of short rails, two units of long rails, and four units of legs. The orders are placed just in time to minimize storage.

The lead time for activities are (lead time is the waiting time required to complete one activity):

Parts	Weeks
Assemble table	1
Finished leg assembly	1
Purchase legs	1
Purchase short rails	1
Purchase long rails	1
Purchase table top	2

The availability of the parts at the present time:

Parts	Units
Table	50
Leg assembly	100
Legs	150
Short rails	50
Long rails	0
Table top	50

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Demand of the finished tables:

Details	Week 4	Week 5	Week 6
Demand (units)	200	150	100

Q.42 For meeting the demand of 200 units of the finished table of week 4, when should the first order of the tabletops be placed?
(a) Week 1 (b) Week 3
(c) Week 5 (d) Week 5

Q.43 What is the net requirement of legs for meeting the demand of the finished table of week 4?
(a) 200 (b) 50 (c) 400 (d) 800

Q.44 When and how many units of short rail would be placed for meeting the demand of the finished table of week 6?
(a) 100 units in week 1 (b) 200 units in week 3
(c) 300 units in week 6 (d) Data insufficient

Q.45 Due North Inc. needs a new kind of table in which 8 legs and 4 short rails would be needed for manufacturing a single table. What would be the net requirement of legs and short rails for meeting the total demand of the finished table (including the additional demand of 200 tables for due North inc.) in week 5?
(a) 1800, 900 (b) 2200, 1100
(c) 1600, 800 (d) 800, 400

Q.46 The supplier of longrails has shifted its manufacturing units to a new location and owing to this, the delivery time of the long rail has been increased by 1 week. When should the order of longrails be placed to meet the week 5 demand of the finished table?
(a) Week 0 (b) Week 1
(c) Week 2 (d) Week 3

Q.47 Labour allocation is a very important process. A particular weaving section has 20 looms and with five labourers, loom efficiency is 75%. The production of a loom at 100% efficiency is 10 m/h. Salary of a labourer is ₹11,000 per month. I removed one labourer due to which the efficiency came down to 70%. How much do I gain or loose due to this action? (Assume that the profit on 1 m cloth is ₹4 and the looms are working for 30 days in a month and 10 hours per day.)
(a) ₹1000 profit (b) ₹1500 loss
(c) ₹1500 profit (d) ₹1000 loss

Q.48 To make an article, it takes 40 h for a workman who is paid ₹1.80 per hour. 20% the material of is wasted in the course of working, which costs ₹22.5 per kilogram. At what price, the article must be sold so as to yield a profit of 33.33%, if its weight was measured to be 8 kg?
(a) ₹360 (b) ₹404
(c) ₹396 (d) None of these

Answers

Warm Up

1. (a)	**2.** (d)	**3.** (c)	**4.** (a)	**5.** (b)	**6.** (c)	**7.** (b)	**8.** (a)	**9.** (a)	**10.** (a)
11. (b)	**12.** (d)	**13.** (a)	**14.** (b)	**15.** (c)	**16.** (b)	**17.** (d)	**18.** (a)	**19.** (c)	**20.** (d)

Foundation

1. (c)	**2.** (d)	**3.** (a)	**4.** (d)	**5.** (c)	**6.** (c)	**7.** (c)	**8.** (c)	**9.** (d)	**10.** (d)
11. (b)	**12.** (c)	**13.** (c)	**14.** (a)	**15.** (b)	**16.** (b)	**17.** (b)	**18.** (c)	**19.** (a)	**20.** (a)
21. (c)	**22.** (a)	**23.** (c)	**24.** (c)	**25.** (d)	**26.** (c)	**27.** (d)	**28.** (c)	**29.** (c)	**30.** (c)
31. (a)	**32.** (a)	**33.** (b)	**34.** (b)	**35.** (d)	**36.** (d)	**37.** (d)	**38.** (a)	**39.** (d)	**40.** (b)
41. (b)	**42.** (a)	**43.** (c)	**44.** (c)	**45.** (d)	**46.** (c)	**47.** (d)	**48.** (d)	**49.** (b)	**50.** (d)

Moderate

1. (d)	**2.** (d)	**3.** (b)	**4.** (b)	**5.** (c)	**6.** (a)	**7.** (b)	**8.** (b)	**9.** (b)	**10.** (b)
11. (c)	**12.** (a)	**13.** (d)	**14.** (a)	**15.** (c)	**16.** (d)	**17.** (b)	**18.** (c)	**19.** (c)	**20.** (d)
21. (c)	**22.** (b)	**23.** (d)	**24.** (c)	**25.** (c)	**26.** (b)	**27.** (a)	**28.** (b)	**29.** (a)	**30.** (b)
31. (d)	**32.** (a)	**33.** (a)	**34.** (b)	**35.** (d)	**36.** (c)	**37.** (a)	**38.** (a)	**39.** (b)	**40.** (c)
41. (d)	**42.** (b)	**43.** (d)	**44.** (a)	**45.** (b)					

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Advanced

1. (c)	**2.** (b)	**3.** (b)	**4.** (c)	**5.** (b)	**6.** (d)	**7.** (b)	**8.** (a)	**9.** (d)	**10.** (c)
11. (a)	**12.** (c)	**13.** (c)	**14.** (d)	**15.** (d)	**16.** (b)	**17.** (b)	**18.** (b)	**19.** (d)	**20.** (a)
21. (b)	**22.** (d)	**23.** (a)	**24.** (a)	**25.** (c)	**26.** (d)	**27.** (a)	**28.** (b)	**29.** (c)	**30.** (b)
31. (c)	**32.** (a)	**33.** (a)	**34.** (d)	**35.** (b)	**36.** (c)	**37.** (a)	**38.** (a)	**39.** (b)	**40.** (d)
41. (d)	**42.** (a)	**43.** (b)	**44.** (b)	**45.** (b)	**46.** (b)	**47.** (d)	**48.** (c)		

Hints and Solutions

Warm Up

1. Let total work be LCM (18, 12) = 36 units

Therefore, Anil completes $= \frac{36}{18} = 2$ units per day

Nishit completes $= \frac{36}{12} = 3$ units per day

Therefore, Anit and Nishit can together complete (2 + 3) which is 5 units per day.

Hence, they will together complete the work in $\frac{36}{5} = 7.2$ days

2. If a and b are equal, then all the statements can be true.
Hence, option (d) is the answer.

3. Work done by A and B in one day $= \frac{1}{A} + \frac{1}{B} = \frac{A+B}{AB}$

Hence, time taken by A and B in finishing the work $= N = \frac{AB}{A+B}$

So, $2N = \frac{2AB}{A+B}$

Hence, option (c) is the answer.

4. Let the total work be LCM (12, 18) = 36 units
Therefore, Aftab completes 36/12 = 3 units
And Ravi completes 36/18 = 2 units per day
Work completed by Aftab in 3 days = 3 × 3 = 9 units
Therefore, remaining work = 36 − 9 = 27 units
Therefore, Ravi will complete the remaining work in $\frac{27}{2} = 13\frac{1}{2}$ days
Hence, option (a) is the answer.

5. Since in the question, relative efficiency of Ravi and Mayank is given, let us assume that Ravi does 3 units in a day and Mayank does 2 units of work per day. Therefore, work done by them in one day = 2 + 3 = 5 units per day working together. It is given that they completed the work in 18 days.

Therefore, total work = 5 units per day × 18 days = 90 units

Hence, Ravi alone can complete 90 units in $\frac{90}{3} = 30$ days

Hence, option (b) is the answer.

6. Since the leakage can empty half the tank in 10 h, it can empty full tank in 20 h.

Let the capacity of tank be LCM (12, 20) = 60 L

Therefore, tap fills = 60/12 = 5 L per hour

Leakage empties = 60/20 = 3 L per hour

When both tap and leakage work simultaneously, tank gets filled by 5 − 3 = 2 L per hour

Therefore, to fill 30 L, tap 2 leakage will together take $\frac{32}{2} = 15$ h

Hence, option (c) is the answer.

7. Let the capacity of tank be LCM (8, 12) = 24 L

Therefore, Pipe A fills 24/8 = 3 L per hour

And pipe B empties 24/12 = 2 L per hour

If both the pipes work simultaneously, then the tank is 3 − 2 = 1 L filled per hour

Hence, it will take 24/1 = 24 h to fill an empty tank if both work together.

Hence, option (b) is the answer.

8. Sanjay washes $= \frac{6}{3} \times 4 = 8$ trucks per days

Therefore, in 3 days, he will wash = 3 × 8 = 24 trucks

Hence, option (a) is the answer.

9. In one day, Sumit Niyogi builds $\frac{20 \times 3 \times 15}{6}$ cubic feet

and Vartika Dikshit builds $= \frac{30 \times 4 \times 10}{5}$ cubic feet

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Therefore, Ratio of $\frac{\text{Sumit Niyogi work}}{\text{Vartika Dikshit work}}$

$= \frac{20\times3\times15}{6}\times\frac{5}{30\times4\times10} = 5:8$

Hence, option (a) is the answer.

10. Let work done by Divya in one day = 1 unit
Therefore, Nandan completes = 2 units per day
Working together they complete → 2 + 1 = 3 units per day
Therefore, total work = 3 × 15 = 45 units
Therefore, Divya alone can complete the work in 45/1 = 45 days
Hence, option (a) is the answer

11. Total work = Number of men × Number of days = $N \times 8 = 8N$ units.
Now, $8N = (N + 4) \times 6$, or, $8N = 6N + 24$, or, $2N = 24$
So, $N = 12$
Hence, option (b) is the answer.

12. Let the total work be 36 units.
Therefore, Harveen completes $\frac{36}{18} = 2$ units per day
Therefore, in 12 days, he completed = 2 × 12 = 24 units
Remaining work = 36 – 24 = 12 units which is completed by Deepak in 8 days.
Therefore, Deepak completes $\frac{12}{8} = 1.5$ units per day
Hence, to complete 36 units, Deepak will take $\frac{36}{1.5} = 24$ days
Hence, option (d) is the answer.

13. Let the total work be LCM (20, 10) = 20 units
Therefore, Munchun completes $\frac{20}{20} = 1$ unit per day
And Chunmun completes $\frac{20}{10} = 2$ unit per day
Efficiency of Munchun to Chunmun are in the ratio = 1:2.
We understand that payment is to be done in ratio of efficiency. More efficient person will be paid more, and less efficient person will be paid less.
Hence, share of Chunmun = $\frac{2}{3}\times1800 = ₹1200$
Therefore, option (a) is the answer.

14. Let the total work be W units.
$\frac{2}{3}W = 1 \text{ man} \times 18 \text{ days}$
$W = \frac{18\times3}{2} = 27$ units
Since Shahid bhai completes 18 units in 18 days, he does 1 unit of work per day.
Therefore, to complete $\frac{27}{4}$ units he will take $\frac{27}{4}$ days.
Hence, option (b) is the answer.

15. Let B does 2 units of work in 1 day.
Therefore, A does 1 unit of work in 1 day.
Since B completes the work in 12 days,
Total work = 2 × 12 = 24 units
Both of them together completes 2 + 1 = 3 units per day
Therefore, they will take $\frac{24}{3} = 8$ days to finish the work.
Hence, option (c) is the answer.

16. Work done = Number of men × Number of days
Work done = 35 × 6 = 210 units
Hence, 210 = 15 × Number of days
So, number of days = 14 days
Hence, option (b) is the answer.

17. Let B does 2 units of work per day.
Therefore, A does 3 units of work per day.
Together they do 2 + 3 = 5 units per days.
Since both of them complete the work in 15 days, total work = 5 × 15 = 75 units
Working alone, A will take $\frac{75}{3} = 25$ days
Hence, option (d) is the answer.

18. Total work = Number of men × Number of days
Total work = 40 × 60 = 2400 units
Work done by 40 men in 10 days = 400 units
Therefore, remaining work = 2400 – 400 = 2000 units
Work done by 80 men in 10 more days = 80 × 10 = 800 units
Therefore, remaining work = 2000 – 800 = 1200 units
Now, work done by 120 men in 10 more days = 120 × 10 = 1200 units
Therefore, remaining work = 1200 – 1200 = 0
So, the work will be completed in 30 days.
Hence, option (a) is the answer.

19. Let men does two units of work per day.
Hence, women does 1 unit of work per day.
Work done by 18 men in 18 days = 18 × 18 × 2 units
Work done by 12 women in 18 days = 12 × 18 × 1 units
Therefore, total work = (18 × 18 × 2) + (12 × 18 × 1) units = 18(36 + 12) = 18 × 48 units
Since 8 men completes 16 units per days,
Therefore, they will take = $\frac{18\times48}{16} = 54$ days
Hence, option (c) is the answer.

20. Go through the options.

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FOUNDATION

1. Method 1

$(A + B)'$s 1 day work $= \frac{1}{30}$ (i)

$(B + C)'$s 1 day work $= \frac{1}{24}$ (ii)

$(C + A)'$s 1 day work $= \frac{1}{20}$ (iii)

Adding (i), (ii), and (iii)

$2(A + B + C)'$s 1 day work $= \frac{1}{30} + \frac{1}{24} + \frac{1}{20} = \frac{4+5+6}{120}$

$= \frac{15}{120} = \frac{1}{8}$

$(A + B + C)'$s 1 day work $= \frac{1}{16}$ of total work

$(A + B + C)'$s 5 day work $= \frac{5}{16}$ of total work

Remaining work $= 1 - \frac{5}{16} = \frac{11}{16}$

Now, work done by A in one day = work done by $(A + B + C)$ in one day − work done by $(B + C)$ in one day

$\left[\frac{1}{16} - \frac{1}{24}\right]$ of total work $= \frac{1}{48}$ of total work

So, time taken by A in completing the remaining work $= \frac{11}{16} / \frac{1}{48} = 33$ days

Method 2

Assume total work = LCM (30, 24, 20) = 120 units

Hence, work done by $(A + B)$ in one day $= \frac{120}{30} = 4$ units/day (i)

Work done by $(B + C)$ in one day $= \frac{120}{24} = 5$ units/day(ii)

Work done by $(A + C)$ in one day $= \frac{120}{20} = 6$ units/day (iii)

Adding all these three, we get:

$2(A + B + C)$ will do $(4 + 5 + 6)$ units/days = 15 units/ day

Hence, $A + B + C$ will do $\frac{15}{2} = 7.5$ units/day (iv)

From (ii) and (iv), we get work done by A in one day= 7.5 units/day − 5 units/day = 2.5 units/day.

It is given that all three of them have worked for 5 days.
Hence, workdone by them in 5 days = 7.5 units/days × 5 = 37.5 units.
Remaining units of work = 120 units − 37.5 units = 82.5 units
Time taken by A in doing remaining work $= \frac{82.5}{2.5} = 33$ days.

2. Tap alone can fill a water tank in 7 h.
Tap + Leakage can fill a tank in 9 h.
Let the capacity of tank be LCM (7, 9) = 63 L
Therefore, tap fills 63/7 = 9 L in 1 h. Tap + Leakage fills 63/9 = 7 L in 1 h
Hence, Leakage empties 2 L in 1 h.
Therefore, leak can empty full tank in $\frac{63}{2} = 31.5$ h
Hence, option (d) is the answer.

3. Wage of 10 men and 8 boys = ₹37
$10M + 8B = 37$ ..... × 6
$4M - 6B = 1$ ..... × 8
$60M + 48B = 222$
$32M - 48B = 8$
$92M = 230$

$M = \frac{230}{92} = \frac{115}{46}$

4. Since 14 machine produces 500 units in 12 h.
Therefore, 14 machines produced $\frac{500}{12} \times 4$ units in 4 h.

5. Work done = Number of men × Number of days.
$\frac{2}{5}$ Work = 1 × 8
Therefore, total work = 20 units
Therefore, Pramod completed 8 units in 8 days
Hence, he completes 1 unit per day.
Remaining work = 20 − 8 = 12 units
Binod completes $\frac{10}{12} = 1.5$ units per day
Together Pramod and Binod completes $1 + 1.5 = 2.5$ units per day.
Together, they will finish the work in $\frac{18}{2.5} = 8$ days
Hence, option (a) is the answer.

6. Let the capacity of cistern be LCM (45, 35, 30) = 630 L
Therefore, Pipe A fills $\frac{630}{45} = 14$ L per min
Pipe B fills $\frac{630}{35} = 18$ L per min
Therefore, Pipe A + Pipe B fills 14 + 18 = 32 L per min
In 7 min, they fill = 32 × 7 = 224 L
Remaining volume of cistern → 630 − 224 = 406 L
Tap R empties $= \frac{630}{30} = 21$ L per min
When all three are opened simultaneously
Cistern is 14 + 18 − 21 = 11 L per min filled.
To fill 406 L, it will take $\frac{406}{11} = 36\frac{10}{11}$ min
Therefore, the cistern will get filled in 37th min
Hence, option (c) is the answer.

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7. Let the total work be LCM (20, 30, 60) = 60 units

Therefore, A completes $\frac{60}{20} = 3$ units every day

B completes $\frac{60}{30} = 2$ units every day

C completes $\frac{60}{60} = 1$ unit every day

Total work done in 3 days = $(3 \times 2) + (6 \times 1) = 12$ units

Hence, 60 units of work will be completed in $\frac{60}{12} \times 3 = 15$ days

Hence, option (c) is the answer.

8. Let the total work be 30 units.

Together Saket and Prakash complete $\frac{30}{7.5} = 4$ units per day

Let Saket does x units per day.

Therefore, Prakash completes $(4 - x)$ units per day.

$\frac{15}{x} + \frac{15}{4-x} = 20$

$\frac{60 - 15x + 15x}{x(4-x)}$

$60 = (20x)(4 - x)$

$60 = 80x - 20x^2$

$20x^2 - 80x + 60 = 0$, or, $x^2 - 4x + 3 = 0$,

or $x^2 - 3x - x + 3 = 0$

Hence, $x(x - 3) - 1(x - 3) = 0$, and $x = 3, 1$

Since Saket is more efficient than Prakash, Saket completes 1 unit per day and Prakash completes 3 units per day.

Working alone, Prakash will take $\frac{30}{1} = 30$ days to complete the work.

9. Machine *A* produces 50 units per day.
Machine *B* produces 25 units per day.
Machine *C* produces 50 units per day.
Machine *D* produces 100 units per day. All four working together produces 225 units per day.

To produce 10,000 units, it will take $\frac{10,000}{225} = 44.44$days

Hence, option (d) is the answer.

10. Since we do not know the rate of normal working hours, we cannot find out the total hours worked.

Hence, option (d) is the answer.

11. Let the capacity of tank be 60 L.

Therefore, tap A fills $\frac{60}{20} = 3$ L per min

Tap C empties $6\frac{1}{2} = 1$ L per min

Working together 3 − 1 = 2 L of water is filled per min into the tank.

When the tank is 30 L filled time is 12:15 noon.
To empty 30 L, tap C will take 30 min.
Therefore, the time will be 12:45 pm.

Hence, option (b) is the answer.

12. Let the capacity of tank be LCM (6, 9) = 18 L
Tap x fills 18/6 = 3 L per hour
Tap y fills 18/9 = 21 per hour
After every 2 h tank is 3 + 2 = 5 L filled.
Therefore, time taken to fill the tank is 7 h.
Hence, option (c) is the answer.

14. Remaining taps to be repaired = 364 − 168 = 196 taps
Four plumbers repaired 168 taps in 21 h.

Therefore, in 1 h, they repaired $\frac{168}{21} = 8$ taps.

Since, efficiency of all plumbers is the same; each plumber repaired 2 taps per hour.

Therefore, 7 plumbers will repair 14 taps per hour.

Therefore, in 7 working hours, they will repair = $14 \times 7 =$ 98 taps

Hence, they will complete the task in $\frac{196}{98} = 2$ days

Hence, option (a) is the answer.

16. Total work = Number of men × Number of days × Number of hours they worked per day.

Therefore, total work = $5 \times 10 \times 8 = 400$ units

Work completed in $5\frac{1}{2}$ days = $5 \times \frac{11}{2} \times 8 = 220$ units

Remaining work = 400 − 220 = 180 units

Therefore, Remaining work will be completed in 180 = $10 \times$ days $\times 8$

Hence, required number of days = $\frac{180}{8} = 22.5$ days

17. Let the capacity of tank be LCM (12, 15, 20) = 60 L

Therefore, first pipe fills $\frac{60}{12} = 5$ L per hour

Second pipe fills $\frac{60}{15} = 4$ L per hour

Third pipe empties $\frac{60}{20} = 3$ L per hour

If all three are opened simultaneously, then the tank is 5 + 4 − 3 = 6 L filled per hour.

Hence, the tank will be filled in $\frac{60}{6} = 10$ h

18. Let the total work be LCM (20, 24, 30) × 10 = 1200 units

Therefore, Arnold completes $\frac{1200}{20} = 60$ units per day

Bill completes $\frac{1200}{24} = 50$ units per day

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Kennedy completes $\frac{1200}{30} = 40$ units per day.

Working together all of them complete = 60 + 50 + 40 = 150 units per day. This is equivalent to 15 units per hour.

Therefore, number of hours required to complete 1200 units = $\frac{1200}{15} = 80$ h

Also, they had worked these 80 h in 10 days @ 8 h per day.

Hence, option (c) is the answer.

19. A's 1 day work = $\frac{1}{t}$

B's 1 day work = r

C's 1 day work = $\frac{1}{8}$

Therefore, $(A+B+C)$'s 1 day work = $\frac{1}{t} + r + \frac{1}{8} = \frac{s + rst + t}{st}$

Hence, option (a) is the answer

20. Let the number of men be M.

Total work = $M \times 20$.

Also, $20M = (M - 12) \times 32$.

$20M = 32M - 12 \times 32$.

$12 \times 32 = 12M$, or, $M = 32$

Hence, option (a) is the answer.

21. Machine P can print $\frac{100{,}000}{8} = 12{,}500$ copies per hour.

Machine Q can print $\frac{100{,}000}{10} = 10{,}000$ copies per hour.

Machine R can print $\frac{100{,}000}{12} = 8333\frac{1}{3}$ copies per hour.

All of them together can print = $30{,}833\frac{1}{3}$ copies per hour.

Therefore, in 6 h they can print $30{,}833.33 \times 6 = 185{,}000$ copies.

Hence, option (c) is the answer.

22. Let the capacity of tank be LCM (6, 7) = 42 L

Therefore, Pipe fills $\frac{42}{6} = 7$ L per hour

Pipe + leak fills $\frac{42}{7} = 6$ L per hour.

Hence, leak empties 1 L per hour.

Therefore, the cistern will be emptied in $\frac{42}{1} = 42$ h

Hence, option (a) is the answer.

23. Let the capacity of tank be LCM (16, 8) = 16 L

Therefore, tap A fills 16/16 = 1 L per min

Tap B empties 16/8 = 2 L per min. If both are opened simultaneously tank is emptied at the rate of 2 – 1 = 1 L per min

Since the tank was 1/2 full, that is, 8 L full it will be emptied in 8 min.

Hence, option (c) is the answer.

24. Let the capacity of tank be x L.

$\frac{3}{4}x + 5 = \frac{4}{5}x$

$\frac{4}{5}x - \frac{3}{4}x = 5$ or, $16x - 15x = 100$, or $x = 100$ L

Hence, option (c) is the answer.

26. Let the capacity of vessel be LCM (12, 15, 6) = 60 L

Therefore, Tap A fills $\frac{60}{12} = 5$ L per min

Tap B fills $\frac{60}{15} = 4$ L per min

Tap C empties $\frac{60}{6} = 10$ L per min

Tap A + tap B fills = 5 + 4 = 9 L per min

Tap $A + B + C = 9 - 10 = 11$ per min empty

Therefore, in 5 min they filled = 45 L

Therefore, the vessel will be emptied in 45 min.

27. Solving through options.

(a) 30 L → Tap A fills = 2 L per min

Tap B fills = 3 L per min

Working together all three will empty 7 – (2 + 3) = 2 L per min

Therefore, tank is emptied in $\frac{30}{2} = 15$ min

(b) 40 L → Tap A fills = $\frac{40}{15} = 2\frac{2}{3}$ L per min

Tap B fills $\frac{40}{10} = 4$ L per min

All three empties $6\frac{1}{3}$ L per min

Hence, tank is emptied in 120 min.

28. Let the first pipe fills 1 L per min.

Therefore, second pipe fills 3 L per min.

Together they fill = 1 + 3 = 4 L per min

Since, the tank is filled in 36 min.

Capacity of tank = $4 \times 36 = 144$ L

Therefore, slower pipe will take 144 min to fill the tank working alone.

Hence, option (c) is the answer.

29. Assume capacity of tank = x L

Tank is emptied in one hour ⇒ it means that both the taps and leakage are active for one hour.

Pipe A can file one tank in 10 min. Hence, water filled by pipe A in one hour = Equivalent to $\frac{60}{10} = 6$ tanks = $6x$

and water filled by pipe B in one hour = Equivalent to

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$\frac{60}{5} = 12$ tanks = $12x$. So, both the pipes together have filled in water = equivalent to 18 tanks = $18x$
Initial cistern is full.
So, total water drained out by leakage in one h = Water filled by pipes A and B + initially the tank was full = $18x + x = 19x = 12$ L/min
Hence, $x = \frac{720}{19}$ L
Hence, option (d) is the answer.

30. Let the total work be LCM (16, 24, 32) = 96 units
Therefore, Sanjay completes $\frac{96}{24} = 6$ units per hour
Balram completes $\frac{96}{24} = 4$ units per hour
Pranesh completes $\frac{96}{32} = 3$ units per hour

31. Let Pipe B fills 1 L per min.
Therefore, Pipe A fills 10 L per min.
Therefore, capacity of cistern = $1 \times 55 = 55$ L
Both working together fill 11 L per min.
Hence, cistern will be fill in $\frac{55}{11} = 5$ min working together.
Hence, option (a) is the answer.

32. Let the capacity of tank be LMC (20, 28) = 100 L
Therefore, first pipe fills = $\frac{100}{20} = 5$ L per min
Second pipe fills = $\frac{100}{25} = 4$ L per min
Therefore, together they fill 9 L per min.
In 5 min, tank is 45 L filled.
Remaining liquid = 100 − 45 = 55 L in tank
Therefore, first pipe can fill 55 L in $\frac{55}{5} = 11$ min
Hence, option (a) is the answer.

33. Since, 3 women = 2 men
21 women = 14 men
Total work = $15 \times 21 \times 8$ units
$15 \times 21 \times 8 = 14 \times D \times 6$, or, $D = \frac{15 \times 21 \times 8}{14 \times 6} = 30$
Hence, option (b) is the answer.

34. $\frac{3}{5}$ of work = 300×21
Total work = $\frac{300 \times 21 \times 5}{3} = 10{,}500$
Work washed away → $\frac{1}{4} \times \frac{3}{5} \times 10{,}500 = 1575$ units
Work completed till now = 4725 units
Remaining work = 5775 units
Therefore, 5775 = 200 × Number of weeks
Hence, option (d) is the answer.

35. Let the capacity of tank be LCM (4, 8) = 8 L
Therefore, Tap A fills = $\frac{8}{4} = 2$ L per min
Tap B empties = $\frac{8}{8}$ 1 L per min
Therefore, in four hours tank is $(2 - 1) \times 4 = 4$ L filled
Therefore, remaining tank to be filled = 8 − 4 = 4 L
Hence, the tank will get filled in $\frac{4}{2} = 2$ h
Therefore, option (b) is the answer.

36. 2 men = 5 women, 2 women = 5 children.
16 men = 40 women, 40 women = 100 children
16 men = 100 children
Total work = $16 \times 10 = 160$ units. Or, $160 = 1 \times D$,
or, $D = 160$
Hence, option (d) is the answer.

37. Total work = $M \times 24$ (i)
$24M = (M - 2) \times 30$ (ii)
$24M = 30M - 60$ or $6M = 60$ or $M = 10$
Hence, option (d) is the answer.

38. Total work = $12 \times 9 = 108$ units
Work done in 6 days = $12 \times 6 = 72$ units
Remaining work = 36 units.
$36 = 12 \times D$ or $D = 3$
Hence, option (b) is the answer.

39. Total work = $M \times 100$
$100M = (M - 10) \times 110$
$100M = 110M - 1100$
$10M = 1100$, or, $M = 110$
Hence, option (d) is the answer.

40. Solving through options.

(a) 15, 5
Work to be completed by each men = $\frac{480}{15} = 32$ acres
Did not turn up = 5
Remaining 10 ploughed = $52 \times 5 = 260$ acres
This is not possible.

(b) 12, 8
$\frac{480}{12} = 40$ acres
Therefore, remaining 4 ploughed = $8 \times 60 = 480$ acres
Hence, option (b) is the answer.

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MODERATE

1. Ratio of efficiency of Mayank and Shishu = 3/2

So, ratio of time taken by Mayank and Shishu = 2/3

So, if Shishu takens 30 h, then Mayank will take 20 h

Shishu in 6 h = 1/5th work

Remaining work = 1 − 1/5 = 4/5th work,

Shishu and Mayank together = $\frac{1}{20}+\frac{1}{30}+\frac{1}{12}$

So, required time = $\frac{4/5}{1/12}=9.6$ h

2. 1 men = 2 women and 1 women = 2 boys

So, 1 men = 2 women = 4 boys

Alternatively, assume work done by a child = x units/ day, then work done by a woman and a man will be equal to $2x$ units per day and $4x$ units per day. Now, 16 men are completing the work in 12 days.

Therefore, total work done = $4x \times 16 \times 12$. Equate this with the supplied work force to obtain the answer.

4. Let the total amount of work = 360 units (LCM of 60, 40, 12)

Hence, work done by A in one day = $\frac{360}{60}$ = 6 units per day

Work done by B in one day = $\frac{360}{40}$ = 9 units per day

Work done by C in one day = $\frac{360}{12}$ = 30 units per day

Hence, work done by all of them in one day = (6 + 9 + 30) = 45 units per day

So, total number of days that each one of them work to finish the work = $\frac{360}{45}$ = 8 days

So, the wages paid to A = $\frac{6}{45}\times$ ₹270 = ₹36

6. First three pumps are on for 4.5 h and two additional pumps for 2 h. Let any one of the first three pipes takes x h to fill the tank and it is given that any one of the two additional pumps take 40 h to fill the tank.

So, according to question, $\frac{4.5}{x}+\frac{4.5}{x}+\frac{4.5}{x}+\frac{2}{40}+\frac{2}{40}=1$

10. Water filled into the tank by all pipes/hour

$\frac{1}{10}+\frac{1}{20}+\frac{1}{25}-\frac{1}{100}$

11. If emptying pipe is closed, then it takes 100/19 h to fill the tank.

But, only 2/5th of 50,000 = 20,000 L is filled. So, in 100/19 h, 30,000 L water is drained.

Hence, in 1 h, 5700 L water is drained.

So, rate at which each house gets water is

= $\frac{5700}{2000}$ = 2.85 L/h

12. For the second time, the total amount of water needed.

= $40 \times 2000 - 50{,}000 = 30{,}000$ L, this amount in 3/4th of total

Let pipes A, B, and C are opened for x h.

So, x (1/10 + 1/20 + 1/25) = 3/5 $\Rightarrow x = 60/19$

15. Solve using options.

16. Let us assume that digging one well = 40 units work

So, A does = 4 units/day and B does = 8 units/day

Now, efficiency is falling by 10%, so, it is a case of Geometric progression.

$4, \frac{18}{5}, \frac{81}{25}, \ldots$ Where common ratio = 9/10

Now, sum of an infinite GP = $a/(1-r)$

= $\frac{4}{1-9/10}=40$

Now, it is clear that 'A' takes infinite time, and so, he will never finish the work.

17. B does = 8 units/day 4

Using $\frac{4[1-(9/10)^2]}{1-9/10}=40 \Rightarrow n \cong 7$

18. A and B together do = 4 + 18 = 12 units/day

Now, proceed the same way as in the previous question.

19. Let us assume that time to read one passage = x

So, according to the question, $3x + 21x/12 - 13 = 6x + x/12 = x = 4$

Therefore, total time taken for answering all questions

= $4x4 + 4x\ 27/12 = 25$ min

Time spent on first passage = $4 + 4x\ 5/12 = 17/3$ min

Required percentage = 22.6%

20. As we have found in the previous question that total time spent = 25 min

If total time spent is cut down by 20%, then time = 20 min

Now, let us assume increased reading speed = x

So, $20 = 4x + 4x\ 27/12 \Rightarrow x = 11/4$

Hence, percentage increase = 45.45%

25. Let us assume that

A finishes the work alone in a days

B finishes the work alone in b days

C finishes the work alone in c days

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So, according to the question,

$$\frac{1}{a}=\frac{1}{2}\left(\frac{1}{b}+\frac{1}{c}\right) \quad \text{(i)}$$

$$\frac{1}{c}=\frac{1}{3}\left(\frac{1}{c}+\frac{1}{b}\right) \quad \text{(ii)}$$

$$\text{and } \frac{1}{a}+\frac{1}{b}+\frac{1}{c}=\frac{1}{15} \quad \text{(iii)}$$

Using these options, $C = 60$ days

26. Let us assume the cistern's capacity = 60 L
So, A supplies = 3 L/min and B supplies = 2 L/min
Now, due to closing of valve,
A supplies = $3 \times 4/5 = 12/5$ L/min
B supplies = $2 \times 5/6 = 5/3$ L/min
In last 5 min, cistern is filled at full capacity of pipes, and it is filled by = $5 \times 5 = 25$ L
Remaining capacity = $60 - 25 = 35$ L
So, required time ≈ 9 min

27. Since the amount of water flowing through each pipe is proportional to square of its diameter, if efficiency of longest pipe (3 cm) = 1/49
Then, efficiency of pipe (2 cm) = $4/(49 \times 9)$
and efficiency of pipe (1 cm) = $1/(49 \times 9)$
Now, let cistern is filled by all three pipes in x min.

$$\text{So, } \frac{1}{49}+\frac{4}{49\times 9}+\frac{1}{49\times 9}=\frac{1}{x}$$

$\Rightarrow x = 31.5$ min

28. In four hours, $A + B + C = 144$ (i)
And given, $A - B = B - C = A + C - 2B = 0$ (ii)
From (i) and (ii), $B = 48$, Hence, $A = 56$ and $C = 40$
So, $B - C = 48 - 40 = 8$ h
It is for 4 h so per hour = $8/4 = 2$ h

32. Ratio of efficiency, $P = 2Q$ (i)
And $P + Q = 3R$ (ii)
From (i) and (ii), $Q = R$
$P = 2Q = 2R$
Hence, $P:Q:R = 2:1:1$

34. Pranesh takes 30 days to finish the work.
Since Sagar is 20% more efficient than Pranesh, he will take 16.66% less time.
So, Pranesh takes 25 days
So, work done by both of them in one day = (1/30) + (1/25)

35. Let us assume the total amount of work = 32 units
So, Manoj and Harvinder does = 2 unit/day
Harvinder does = 1 unit/day
So, Manoj does = 1 unit/day
So, 32 units of work will be completed in 32 days.

36. To maximize the production, four men will work in each shift.
Two men will work with machines and two men work alone.
Total cost incurred in one hour

$$=\frac{2\times 20+(2\times 20+2\times 15)}{6}=110/6$$

So, total cost incurred in 8 h (per die) = $(110/6) \times 8 = 146.66$

37. Let Suresh take x days to complete the work. So, in one day, Suresh does = $\frac{1}{x}$
Given total efficiency of Ramesh and Suresh in oneday = $\frac{1}{P}$
So, Ramesh = $\frac{1}{P}-\frac{1}{x}$
According to question = $\frac{q}{P}+r\left(\frac{1}{x}\right)=1$
So, $x=\frac{Pr}{P-q}$
So, Suresh takes = $\frac{Pr}{P-q}$
So, Ramesh takes = $\frac{Pr}{r-P-q}$

38. Let B take x h to fill the tank,
So according to question, time required by each
$A = 2x$
$B = x$
$C = 2x/3$
$D = 2x/4$
$E = 2x/5$
A and B together fill the tank in 4 h

$$\frac{1}{2x}+\frac{1}{x}+.....=\frac{1}{4}\Rightarrow x=6$$

In first case, when A, C, and E act as input pipes and B and D as output pipes,
Total time = 1/12 + 1/4 + 5/12 − 1/6 − 1/3 = 4 h
In second case, when C, D, and E act as input pipes and A and B as output pipes,
Total time = 1/4 + 1/3 + 5/12 − 1/12 − 1/6 = 4/3 h

40. Solve using options.

42. Assume total amount of work = 100 units
A does = 10 units/day, B does = 5 units/day, and C does = 4 units/day
Possible Pairs:
$A + B = 15$ units/day, $A + C = 14$ units/day, $B + C = 9$ units/day

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To minimize time, we will use the first two pairs.

So, 15 + 14 + 15 + 14 + 15 + 14 + 15 = 102 units

So, 7 days are required.

45. Let Anuj do the work in x days,

So, Bhanu = $3x$ days, Chandu = $9x$ days, and Dodo = $27x$ days

Now, using options,

Anuj + Chandu = $1/x + 1/9x = 10/9x$, so, they take $9x/10$ days

Bhanu + Dodo = $1/3x + 1/27x = 10/27x$, so, they take $27x/10$ days

Now, one-third of $27x/10 = 9x/10$; so, Anuj and Chandu is first pair.

ADVANCED

Answers to Q.1 to 4:

Let efficiency of pipe $P_1 - e$

Let efficiency of pipe $P_2 - e/2$

Let efficiency of pipe $P_3 - e/4$

Let efficiency of pipe $P_4 - e/8$ and so on

Now, e, $e/2$, $e/4$, $e/8$... are in geometric progression

So, total efficiency $= \frac{e}{2} = \frac{e}{4} + ... \frac{e}{8} + ... + ... + ... \infty$

All pipes together means can fill the tank in 2 h.

So, pipe P_1 with efficiency e will take = $2 \times 2 = 4$ h

Similarly, $P_2 = 8$ h

$P_3 = 16$ h

$P_4 = 32$ h

$P_5 = 64$ h

$P_6 = 128$ h

6. $F(A, 10, 600), E(C, 50, 20) = +6000 - 1000$

$F(A, 10, 400), E(B, 20, 200) = +4000 + 4000$

$F(A, 50, 100), F(B, 20, 50) = +5000 + 1000$

$E(C, 10, 50), F(B, 20, 75) = \underline{500 + 1500}$

Final volume $= 20{,}000$

7. $F(A, 10, 500), F(B, 40, 500) = 5000 + 20{,}000$

$= 25{,}000$

Now, option (b) $E(C, 50, 500)$ gives same amount 25,000 to be emptied.

9. Let x = Number of days it rained in the morning and had clear afternoons.

y = Number of days it rained in the afternoon and had clear mornings.

z = Number of days it rained in the morning or afternoon.

So, according to question, $x + y = 7$

$x + z = 5$

$y + z = 6$

Adding all three equations, $x + y + z = 9$

So, $d = 9$ days

10. Let us make a table of the units of work everyday:

Day	1	2	3	4	5	6	7	8	9	10
Length	2	4	6	8	8	11	14	17	20	23

Total length = 113 m

13. Ratio of efficiency of Sanjay:Balram:Pranesh = 1/16:1/24:1/32 = 6:4:3

Work finished in 24 h, so in 1 h = 1/24 part of work

Pranesh's contribution = $3/13 \times 1/24 = 1/104$ part of work

So, Pranesh will take 104 days.

Hence, option (c) is the answer.

14.

Men	Working hours	Day	Work done
64	8	5	40%
64	n (say)	4	60%

40% work in 40 h → 60% work in 60 h.

Hence, working hours = 60/4 = 15 h

17. Combined efficiency of four secretaries = ½

$$\frac{1}{A} + \frac{1}{B} + \frac{1}{C} + \frac{1}{D} = \frac{1}{2}$$

As it is given in the question that boss's efficiency reduces, respectively, according as A, B, C, and D remain absent, it means their efficiencies are in descending order by trial and error.

20. A worker in a day completes 10% work.

Since entire team works at their 50%

Day 1	Day 2	Day 3	Day 4	Day 5	Day 6
5%	10%	15%	20%	25%	25%

(Day 6 = 25% because that day worker is absent)

Total = 5 + 10 + 15 + 20 + 25 + 25 = 100%

21. Amount of work = 200%

Day 1	Day 2	Day 3	Day 4	Day 5	Day 6
10%	20%	30%	40%	50%	60%

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Total = 10 + 20 + 30 + 40 + 50 + 60 = 210%

Hence, work is finished on 6th day.

22. We can use the data of previous question. On the last day, 50% work is required but efficiency to finish 60% work is there, therefore, 60% work in 8 h.

50% work in (8/60) × 50 = 6 h 40 min

They start working at 9:00 am and work till 1:00 pm

So, 4 h passed till 1:00 pm.

Now, 2 h 40 min is left, they again started working at 1:30 pm.

So, 1.30 + 2 h 40 min

23. According to the question, 1/2, 1/a, 1/3 are in AP

So, 'a' can be calculated.

Now, $\frac{1}{2-x}, \frac{P}{a}, \frac{1}{3-x}$

So, x can be calculated.

$x = 2.6$

36. Let the height above the ground = $EF = h$

$\Delta ECF \sim \Delta ACB$

So, $\frac{h}{78} = \frac{FC}{BF}$ (i)

$\Delta BEF \sim \Delta BCD$

So, $\frac{h}{91} = \frac{FC}{BF + FC}$ (ii)

$\Delta ECF \sim \Delta ACB$

(i) ÷ (ii) $= \frac{FC}{BF} \Rightarrow \frac{91}{78}\ 7BF = 6FC$ (iii)

By putting values of equation (iii) in equation (9), h = 42 m

40. Total numbers of possible pairs = ${}^{10}C_2 = 45$

So, each is working 9 times to finish three-fourth of the work.

In one day, all of them working together will do = 1/9 of 3/4 = 1/12th work

Remaining work = 1/4

So, the required days = 3 days

41. As we have found in a previous problem that working together, they do 1/12th work in a day, that is, they take 12 days to complete the work working together.

Now, till the last day, they have worked for 45 + 3 = 48 days

Now, for the same work again, they will take 12 more days working together.

Hence, total number of days required from start = 48 + 12 = 60 days

42. Fifty table and 50 tabletops are available, and so, we need to purchase 100 more tabletops. Lead time for purchasing table top in 2 week and lead time for assemble table is 1 week so order should be placed 3 weeks in advance, that is, Week 1.

47. At 75% efficiency, loom works 7.5 m/h.

So, 20 looms @ 10 h/day in 30 days will make = 7.5 × 20 × 10 × 30 = 45,000 m cloths

At 70% efficiency, loom works = 7 m/h

So, 20 looms @ 10 h/day in 30 days will make =

7 × 10 × 20 × 30 = 42000 m cloths

So, production of cloths decreases by 45,000 – 42,000 = 3000 m

So, loss = 3000 × 4 = ₹1,200,000

Saving on a labourer by removing him = 11,000

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CHAPTER

9

Time, Speed, and Distance

LEARNING OBJECTIVES

After completion of this chapter, the reader should be able to understand:

- Basic relationship between time, speed, and distance
- Different situations such as motion in a straight line, trains, races, boats and streams, circular track, and clocks
- Concepts attached to these motions
- Kinds of questions asked in the CAT
- Methods of solving questions

INTRODUCTION

Time, speed, and distance (TSD) is one of the most popular topics in the CAT. Almost 5–10% of questions in the paper are asked from this chapter every year. The concept of time, speed, and distance is also used extensively for questions relating to different areas in various other examinations. Some of these areas are the following: motion in a straight line, boats and streams/escalators, races, circular motion, trains, and clocks. Students should try to understand the inter-relationship between the factors time, speed, and distance.

RELATIONSHIP BETWEEN TIME, SPEED, AND DISTANCE

As we know, distance = speed × time

It means that if a person is running at a speed of 20 km/h and he runs for 2 h, he will be covering a total distance of 40 km.

That is, distance = 20 × 2 = 40 km

Distance

When an object is moving with a certain speed in a particular time, the displacement made by an object is called the distance.

Unit of Distance

Kilometre (km) and metre (m) is usually taken as the unit of distance. Sometimes, mile or feet, etc., can be found as the unit.

Time

Time is defined as a quantity, which governs the order or sequence of an occurrence. In the absence of time, the actual sequence of any occurrence or incident would be lost. If we did not have the concept of time, we would not be able know in what period or in what order something took place.

Therefore, time could be seen as a big building with a number of floors where all the floors are designated according to the occurrence of incidents/events on the respective floors. In our case, time shall be seen as the duration of happening of any event.

Unit of Time

Mostly, hour and second are considered as the unit of time; however, day or minute can also be used as units.

Speed

Speed is defined as the distance covered per unit time. In other words, it is the rate at which the distance is covered.

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Unit of Speed

Commonly, km/h and m/s are the units of speed. Yet, any unit of distance upon any unit of time can be treated as the unit of speed. For example, mile/h, feet/s, mile/s, feet/h, etc.

Conversion from m/s to km/h and Vice Versa

If speed is given in m/s and it is required to convert it into km/h, then we multiply it by $\frac{18}{5}$, and when speed is given in km/h and we have to convert it into m/s, then we multiply it by $\frac{5}{18}$

$$36 \text{ km/h} = 36 \times \frac{5}{18} = 10 \text{ m/s}$$

$$20 \text{ m/s} = 20 \times \frac{18}{5} = 72 \text{ km/h}$$

MOTION IN A STRAIGHT LINE

This is one of the primary areas of application of time, speed, and distance. By using the basic relationship between time, speed, and distance, the following three different cases are possible:

Case I When S (distance) is constant

$$v \propto \frac{1}{T}$$

So, $V_1/V_2 = T_2/T_1$

It is read as V is inversely proportional to T.

It can be understood in terms of the reciprocal–ratio–multiplication relationship between V and T.

Suppose distance = 1000 km and speed = 100 km/h

Speed = 100 km/h	Time = 10 h
Speed = 100 × 2	Time = 1/2 × 10
Speed = 100 × 3	Time = 1/3 × 10
Speed = 100 × 1/2	Time = 2 × 10

To simplify it, the product stability ratio can be further used. Hence, the more the speed, the lesser is the time taken and the lesser the speed, more will be the time taken.

Example 1 A man cycles with a speed of 10 km/h and reaches his office at 1 pm. However, when he cycles with a speed of 15 km/h, he reaches his office at 11 am at what speed should he cycle so that he reaches his office at 12 noon? **(*CAT 2004*)**

Solution Using the product stability ratio,

The speed is being increased by 50%, and therefore, the time taken will reduce by 33.33%.

So, 33.33 % of time = 2 h

Hence, total time = 6 h

So, distance = 10 × 6 = 60 km

This distance is to be covered in 5 h

So, speed = 60/5 = 12 km/h

Alternatively, it can be seen that time taken in three (given) situations are in arithmetic progression (AP). Hence, speeds will be in harmonic progression (HP).

Required speed = Harmonic mean of two speeds

So, required speed $= \frac{2 \times 15 \times 10}{15 + 10} = 12$ km/h

Example 2 Siddharth goes by a bike to pick up his girlfriend every day from college and then drops her at her house. College timings are till 5 pm daily, but today, the college at 4 pm. His girlfriend, not finding Siddharth at the college gate, starts walking towards her house. Siddharth, unaware of this fact, leaves his house as usual meets his girlfriend on way, picks her up and drops her at her house. At night, Siddharth realizes that he had saved 40 min that day. What is the ratio of the speed of Siddharth to that of his girlfriend (both of them live in the same building)?

Solution Let us see the following schematic representation:



The usual route of Siddharth is home–college–home. His route today is home–meeting point–home. In this way, 40 min are saved. So, he takes 20 min to cover the distance between the meeting point and the college. It can be further concluded that he usually reaches college at 5 pm, but he reached at 4:40 today (20 min are saved) and his girlfriend took 40 min (she starts at 4 pm) to cover the distance between her college to the meeting point.

The ratio of time of Siddharth and his girlfriend = 20:40 = 1:2

The ratio of the speed of Siddharth and his girlfriend = 2:1

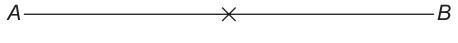
Case II When T (time) is constant S at V

So, $S_1/S_2 = V_1/V_2$

The higher is the speed, the more will be the distance covered and the lower the speed, the lesser will be the distance covered.

We will see that T constant is a situation specific to meeting point cases.

Example 3 Distance between two points AB = 110 km. Manoj starts running from A at a speed of 60 km/h and Ravi starts running from B at a speed of 40 km/h at the same time. They meet at a point X, somewhere on the line AB. What is ratio of AX to BX?



Solution Since both Manoj and Ravi are running for the same time, T is constant. Hence, the ratio of the

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distance covered by them will be the same as the ratio of their speed.

So, AX/BX = Speed of Manoj/speed of Ravi = 60/40 = 3:2

Some Typical Meeting Point Cases

When two persons are running between the ends of a linear track for infinite time:

Example 4 Two robots Mango and Bango start from the opposite ends A and B of a linear track, respectively, and keep running between the ends for infinite time. They meet for the first time at a point 60 m from A. If AB = 100 m, which point is their point of 4th meeting?

Solution First meeting point



The ratio of the speed of Mango and Bango is 60:40 = 3:2. Now, Mango is moving towards B and Bango is moving towards A.

For the second meeting, Bango reaches A after covering 60 m, Mango must have covered 90 m in the same time. So, he is at the mid-point of A and B. Now, the distance between Mango and Bango is 50 m. They will cover this distance of 50 m in the ratio of 3:2.

So, the point of their second meeting is



It can be seen here that the sum of the distance covered by both Mango and Bango from the starting till now is 300 m, and the difference between the distance covered between the first meeting and the second meeting is 200 m.

So, again, they will meet for the third time, when they have covered a total distance of 200 m together. Mango and Bango will cover this distance of 200 m in the ratio of 3:2.

Distance covered by Mango = 120 m and distance covered by Bango = 80 m.

So, the third meeting point is point B.

To have a fourth meeting, they will again have to cover a total distance of 200 m. So, the point of their fourth meeting is 20 m from A.

So, we can now generalize the above situation as follows:

Distance covered by both of them for the first meeting = 100 m

Distance covered by both of them for the second meeting = 100 m + 200 m = 300 m

Distance covered by both of them for the third meeting = 300 m + 200 m = 500 m

Distance covered by both of them for the fourth meeting = 500 m + 200 m = 700 m

The ratio of the speed of Mango and Bango is 3:2, so the distance covered by Mango = 420 m

Hence, they will meet at 20 m from A.

Example 5 Two persons, Ram and Mohan, start from the same end A of a linear track AB and keep running to and fro for infinite time. They meet for the first time at a point 20 m from B. If AB = 100 m, which point is their point of fourth meeting?

Solution Using the above generalization, distance covered by both of them for the first meeting = 200 m

Distance covered by both of them for the second meeting = 200 m + 200 m = 400 m

Distance covered by both of them for the third meeting = 400 m +200 m = 600 m

Distance covered by both of them for the fourth meeting = 600 m + 200 m = 800 m

The ratio of the speed of Ram and Mohan is 3:2, so the distance covered by Ram = 480 m

Hence, they will meet at 80 m from A.

Limitation of Above Generalization

For the meeting to occur after every 200 m, the ratio of the speed of the two runners should be less than 2. If it is more than or equal to 2, then the problems can only be evaluated on the basis of actual calculation.

Case III When V (Speed) is constant $S \propto T$

So, $S/S_2 = T/T_2$

In layman terms, if a person is running with a speed of 20 km/h, then the ratio of the distance covered in 1 h to the distance covered in 2 h will be 1:2.

BOATS AND STREAMS/ ESCALATOR



Boats and streams should be ideally seen as just a logical extension of the motion in a straight line with distance being constant.

As we know, if the distance is constant, then $V \propto 1/T$.

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Basic Terminology

Downstream Movement

When the direction of the movement of a river and a boat is the same, their collective movement is known as the downstream movement. And the distance covered by boat is known as downstream distance.

If the speed of the river = R and the speed of the boat = B, hen downstream speed = $B + R$

Upstream Movement

When the direction of the movement of the river and a boat is opposite, they are said to be in upstream movement. The distance covered in this case is known as upstream distance.

If the speed of the river = R and the speed of the boat = B, then upstream speed = $B - R$ (Conventionally, the speed of one boat is taken more than the speed of river; otherwise, the boat would not be able to go back.)

Now, speed of boat = 1/2 (downstream speed + upstream speed) = 1/2 $(B + R + B - R) = B$

And speed of river = 1/2 (downstream speed – upstream speed) = 1/2 $(B + R - B + R) = R$

Hence, if downstream speed and upstream speed are given as 20 km/h and 10 km/h, respectively, then the speed of the boat = 15 km/h and speed of the river = 5 km/h.

In most cases of boats and streams, the distances covered downstream and upstream are the same. In those cases, the ratio of the time taken becomes inverse of the ratio of the speeds.

Time taken downstream:time taken upstream = upstream speed:downstream speed.

Example 6 The speed of the boat in still water is 6 km/h and the speed of the river is 1.2 km/h. Boat takes a total of 10 h to go to a place and come back. What is the total distance covered in the whole process?

Solution Let us assume D is the distance. Upstream Speed = 4.8 km/h Downstream speed = 7.2 km/h. According to the question, $D/4.8 + D/7.2 = 10$. So, $D = 28.8$ km, and hence, the total distance = 57.6 km, and alternatively, the ratio of downstream speed:upstream speed = 3:2

Ratio of the downstream time:upstream time = 2:3

The time taken in the downstream movement is 4 h and the time taken in the upstream movement is 6 h.

So, the distance covered = $4 \times 7.2 = 6 \times 4.8 = 28.8$ km

Hence, the total distance = 57.6 km

In case of escalators, moving staircase works like an external agent as the river works for boats and streams. The speed of an escalator and the person will be added when the staircase is going up and the person walking up with it have the same direction of the movement.

Now, if the direction of the movement of an escalator and the person are opposite, then the resultant speed (or, the relative speed) will be equal to the speed of the person to the speed of an escalator.

Example 7 A man can walk up in a moving escalator (upwards) in 30 seconds (s). The same man can walk down this moving 'up' escalator in 90 s. Assume that this walking speed is the same both upwards and downwards. How much time will he take to walk up the escalator when it is not moving?

Solution Let us assume that the speed of the man = m steps/s and the speed of the escalator = e steps/s

Distance covered while going up = $30m + 30e$

Distance covered while going down = $90m - 90e$

Now, these two are equal.

So, $30m + 30e = 90m - 90e$

Or, $60m = 120e$

Hence, $1m = 2e$

So, the total length of escalator = 45m

So, the time taken by the man to cover the whole escalator = distance/speed = $45m/m = 45$s

Alternatively, answer would be harmonic mean of the given time = $\frac{2 \times 30 \times 90}{30 + 90} = 45$s

[See the solution of Example 1 on page 1.203]

RACES

Basic Statements

1. A gives a start of 10 m to B—When B has already run 10 m, then A starts running.



Example 8 In a race of 100 m, A gives a start of 10 m to B. Despite this, A wins the race by 20 m. What is the ratio of the speed of A and B?

Solution Time taken by A to cover 70 m = Time taken by B to cover 100 m

Since the distance is constant, the ratio of speed of A and B = 10:7

2. A gives a start of 10 s to $B \rightarrow B$ has already run for 10 s, now A starts running.



Where v m/s is the speed of B.

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Example 9 In a 100 m race, Tom runs at a speed of 1.66 m/s. If Tom gives a start of 4 m to Jerry and still beats him by 12 s, what is the speed of Jerry?

Solution Time taken by Tom to cover 100 m = 60 s

Now, since Tom beats Jerry by 12 s, time taken by Jerry = 72 s

And the distance covered by Jerry = 96 m

So, speed = 96/72 = 1.33 m/s

Example 10 Karan and Arjun run a 100 m race where Karan beats Arjun by 10 m. To do a favour to Arjun, Karan starts 10 m behind the starting line in a second 100 m race. They both run at their earlier speeds. Which of the following is true in connection with the second race?

1. Karan and Arjun reach the finishing line simultaneously.
2. Arjun beats Karan by 1 m.
3. Arjun beats Karan by 11 m.
4. Karan beats Arjun by 1 m.

Solution

Situation (I)

In whatever time Karan covers a distance of 100 m, Arjun covers 90 m in the same time.

Situation (II)

Karan is 10 m behind the starting point. Once again to cover 100 m from this new point Karan will take the same time as before. In the same time, Arjun will cover only 90 m. This means that both of them now will be at the same point, which is 10 m away from the end point. Since both of them are required to cover the same distance of 10 m and Karan has a higher speed, he will beat Arjun. There is no need for calculations as option (4) is the only such option.

CIRCULAR MOTION

In the case of races and motions in straight line, we have observed that if the two bodies or persons are moving with different speeds in a straight line in one direction, then they will never meet. This is due to the fact that with the passage of time, the distance between them is increasing constantly.

Circular motion should be seen as a logical extension of races where runners are running on a circular track. Since it is an enclosed track (by virtue of it being circular), runners are bound to meet at some point or the other.

Case I When two or more than two persons are running around a circular track in the same direction.

Example 11 When will they meet for the first time anywhere on the track?

Solution To understand the situation completely, let us assume that there are two persons A and B. Speed of A = 20 m/s, speed of B = 10 m/s, length of the track is 1000 m and they are running in the same direction. It can be seen in Figure 1 that initially both of them are at the same point, that is, the starting point.

They will be meeting for the first time only if the faster runner A has taken one more round of the track than the slower runner B. This can be interpreted as – A will have to cover 1000 m more than B.

It is understood with the figures given previously that the distance will keep on increasing between them with the passage of time. And the moment distance between them becomes equal to 1000 m, and they will be at the same point.

$$\text{So, the time taken} = \text{distance/relative speed} = \frac{1000}{10} = 100 \text{ s}$$

Or, this can be done by using unitary method also: distance of 10 m is created in 1 s and therefore, the distance of 1000 m will be created in 100 s Now, let us assume that there are three persons A, B, and C running with following speeds in the same direction:

Speed of A = 30 m/s

Speed of B = 20 m/s

Speed of C =10 m/s

To calculate when will they meet for the first time, we are required to find the time taken by the fastest runner to take one round over the other runners.

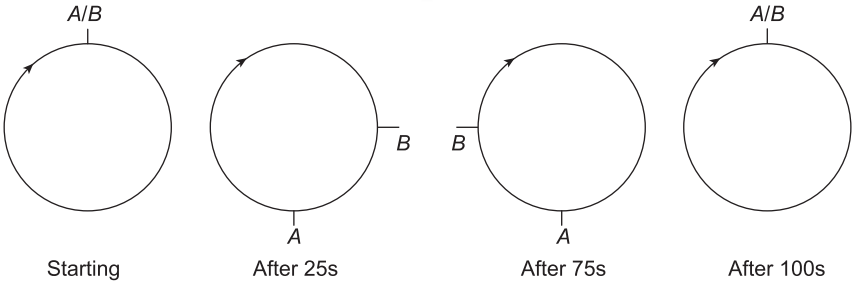
Time taken by A to take one round over $B = t_{A\text{-}B}$ = 1000/10 = 100 s

Time taken by A to take one round over $C = t_{A\text{-}c}$ = 1000/20 = 50 s

Now, the LCM of these two values $t_{A\text{-}B}$ and $t_{A\text{-}c}$ will give us the time after which all of them will be meeting at the same place.

LCM = (100, 50) = 100 s

It can also be seen that they will be meeting after every 100 s.



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Example 12 When will they meet for the first time at the starting point?

Solution To calculate this, we will use the concept of LCM (usage of LCM and HCF, Chapter 2, case II).

Find the time taken by each individual to take one round and then calculate LCM of these values.

Assume that there are three persons A, B, and C with a respective speed of 30 m/s, 20 m/s, and 10 m/s running in the same direction. Length of the circular track is 1000 m.

Time taken by A to take one round = t_1 = 1000/30 = 33.33 s

Solution Time taken by B to take one round = t_2 = 1000/20 = 50 s

Time taken by C to take one round = t_3 = 1000/10 = 100 s

LCM of t_1, t_2, t_3 = 100 s

Example 13 At how many different points of the track will they be meeting?

Solution Let us assume that the speed of A = 25 m/s and the speed of B = 10 m/s and the length of the track = 1000 m

They will be meeting for the first time after a time–gap of 1000/15 = 66.66 s

Till this time, A has covered 1666.66 m and B has covered 666.66 m. This point is 666.66 m from the starting point. Now, this point can be assumed to be the starting point.

So, they will meet at a distance of 666.66 m from here. This is the second meeting point, at a distance of 333.33 m from the starting point. Next meeting point will be 666.66 m from here. This point will be nothing but the starting point again (Figures 1 and 4 are same).

This can be seen through the figures given below:

So, there are a total of three distinct meeting points on the track.

In general, number of meeting points = difference of ratio of the speed of A and B in its simplest form. Ratio of speed of A and B = 5:2

So, the number of different meeting points = 5 − 2 = 3 points

Case II When two or more than two persons are running around a circular track in the opposite direction.

Here again, there are two persons A and B with a speed of 20 m/s and 10 m/s, respectively, and the length of track is 1000 m.

Example 14 When will they meet for the first time anywhere on the track?

Solution Since they are running in the opposite direction, relative speed = 10 + 20 = 30 m/s

So, time taken = distance/relative speed = 1000/30 = 33.33 s

Example 15 When will they meet for the first time at the starting point?

First, we will calculate the time taken by each individual to take one round and then calculate the LCM of those values.

Time taken by A to take one round = t_1 = 1000/20 = 50 s

Time taken by B to take one round = t_2 = 1000/10 = 100 s

LCM of (t_1, t_2) = 100 s

Example 16 At how many different points of the track will they be meeting?

Solution They are meeting after 33.33 s for the first time. Till this time, A has covered 666.66 m and B has covered 333.33 m. So, obviously they are meeting at a distance of 666.66 m from starting point in the direction of A. Next point will be again 666.66 m ahead of this point. And, the next point will be another 666.66 m ahead of this point, which will be the starting point.

So, a total of three points will be there.

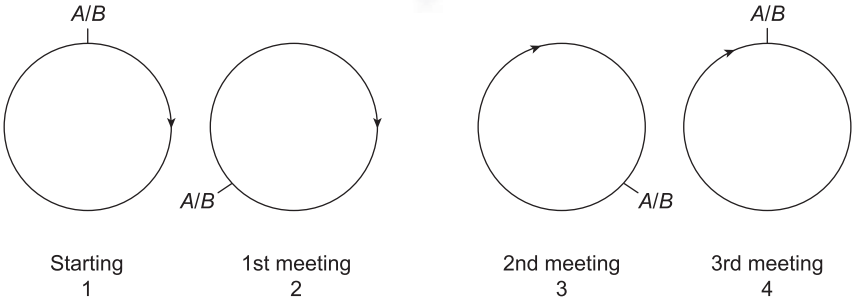
In general, number of distinct meeting points = addition of the ratio of the speed of A and B in its simplest form.

The ratio of speed of A and B = 2:1

So, the number of different meeting points = 2 + 1 = 3 points

Example 17 Anup and Shishir start running from the same point of a circular track at the same time. After how much time will Anup and Shishir, who are running with a speed of 35 m/s and 40 m/s, respectively, meet at diametrically opposite point?

Solution The simplest ratio of speed of Anup and Shishir = 7:8



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So, if they are running in the same direction, they will meet at one point and if they are running in the opposite direction, they will meet at 15 different points.

Now, for them to meet at a diametrically opposite point, there should be at least two meeting points or the number of meeting points should be a multiple of 2.

Since, they would meet either at one point or at 15 different points, depending on the direction of their movement, they will, therefore, not meet at a diametrically opposite point.

TRAINS

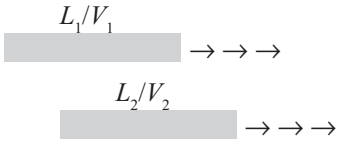


We know that when the direction of the movement of a boat and a river is the same, the relative speed is obtained by adding the speeds of both, the boat and the river. But if two trains are moving in the same direction, then what is the relative speed?

Let us see some cases:

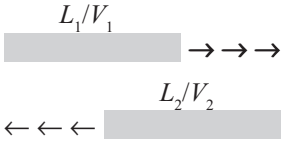
1. When two trains of length L_1 and L_2 and speed V_1 m/s and V_2 m/s, respectively, are crossing each other:

   i. The direction of the movement of both the trains are the same:

   Relative speed = $|V_1 - V_2|$
   Total distance covered = $L_1 + L_2$

   ii. The direction of the movement of both the trains are opposite:

   Relative speed = $|V_1 + V_2|$
   Total distance covered = $L_1 + L_2$

2. When a train is crossing a stationary object:

   i. When the train is crossing a pole or a stationary human being:

   Figure 1

   Figure 2

   Let us assume that A is a pole. In Figure 1, the front of the train is about to cross the pole and in Figure 2, the tail of the train has just crossed the pole. It is understood here that the train has crossed its whole length with respect to the pole. So, when the train is crossing any stationary object of negligible width, total distance covered is its own length.

   Relative speed = $V_1 + V_2$, since $V_2 = 0$, then, the relative speed = V_2

   Total distance covered = $L_1 + L_2$, since L_2 (width of the pole) is negligible with respect to L_2 (Length of the train), so we do not consider it while calculating the quantities. Therefore, distance = L_1

   However, it should be remembered that this is mathematically not correct and all the solutions are on the assumption that the width of the pole is zero, which is obviously not true.

   ii. When the train is crossing a platform or a standing train:

   Relative speed = $V_1 + V_2$

   Where V_1 is the speed of the moving train and V_2 is the speed of the standing train or the platform.

   Since $V_2 = 0$, so the relative speed = V_1

   Total distance covered = $L_1 + L_2$

   Where L_1 is the length of the moving train and L_2 is the length of the standing train or the platform.

Example 18 A train takes 10 s to cross a pole and 20 s to cross a platform of length 200 m. What is the length of train?

Solution The train takes 10 s to cross its own length and 20 s to cross its own length and length of the platform. So, it is inferred that the train takes 10 s to cross the platform and 10 s to cross its own length.

Since the time taken to cross the platform = time taken to cross its own length

So, length of the platform = length of the train = 200 m

Example 19 Speed of a train is 36 km/h. It takes 25 s to cross a pole. What is the length of this train?

Solution Speed of train = 10 m/s ($36 \times 5/18$)

Distance covered = $10 \times 25 = 250$ m

So, the length of train = 250 m

Some Special Cases

Case I Two trains are moving in an opposite direction with a speed of V_1 and V_2. Their lengths are L_1 and L_2. Now, see the whole situation from the point of view of a person sitting on the window seat of the first train.

Relative speed = $V_1 + V_2$ (This person can be assumed to be running with a speed of V_2.)

Relative distance = L_2

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Case II A train is running with a speed V_1 and a person X is running inside the train with a speed of V_2 in the direction of the movement of train. Now, if a person Y is watching this from the outside of the train, then the relative speed of Y with respect to $X = V_1 + V_2$

Speed of person X with respect of another person Z who is sitting in the train = V_2

There is also a person P who is outside the train and is moving with a speed of V_3 in the opposite direction of train.

Relative speed of P with respect to person $X = V + V_2 + V_3$

Had this person P been running in the same direction as that of the train, then the relative speed of P with respect to person $X = |V_1 + V_2 - V_3|$

CLOCKS



A clock is a typical example of a circular motion where length of the track is equal to 60 km (assume 1 min = 1 km). Now, on this track, two runners, that is, hour hand and minute hand are running with a speed of 5 km/h and 60 km/h, respectively. Since the direction of their movement is the same, the relative speed = 55 km/h.

Example 20 When will the hour hand and the minute hand of a clock be together between 1 and 2?

Solution Hands have to be together in between 1 O'clock to 2 O'clock.

At 1 O'clock, the distance between hour and minute = 5 km

And the relative speed = 55 km/h

Time = 5/55 h = 1/11 h = 60/11 min = 5 5/11 min = 5 300/11 s = 5:27.27 s

So, the hour hand and the min hand will be together at 1:05:27.27 s

Students can learn this value five min 27.27 s as a standard result.

Therefore, both the hands will meet at:

1: 05:27.27 – Between 1 O'clock and 2 O'clock

2:10:54.54 – Between 2 O'clock and 3 O'clock:

3:16:21.81 – Between 3 O'clock and 4 O'clock and so on

Example 21 How many times in a day will the hands of a clock be together?

Solution Using the data from the above question, hands of a clock meet at a regular interval of five min 27.27 s. So, the number of times they will the meet = $\frac{60}{5:27:27} = 11.3$ times

Therefore, the hands will meet for a total of 11 times.

However, it can also be observed that the hands of a clock meet once every hour except in between 11 and 1. They meet just once in between 11 and 1. Therefore, they are meeting 11 times.

Degree Concept of Clocks

Total angle subtended at the centre of a clock = 360°

Angle made by an hour hand at the centre per hour = 30° per hour or 0.5° per min

Angle made by the minute hand at the centre per hour = 360° per hour or 6° per min

Solving 20 by this method,

Angle between an hour hand and the minute hand at 1 O'clock = 30°

Relative speed (in terms of angle) = 5.5°/h

Hence, time taken = 30°/5.5° = 60/11 min

Example 22 Mr Binod Kumar Roy goes to a market between 4 pm and 5 pm. When he comes back, he finds that the hours hand and the minutes hand have interchanged their positions. For how much time, was he out of his house?

Solution Since hands are interchanging their position, minute hand is taking the place of an hour hand and an hour hand is taking the place of minute hand. So, sum of the angles formed by hour hand and minute hand = 360°

Let us assume that he was out of house for 't' min.

So, the angle formed by minutes hand = $6 \times t$ and by hour hand = $0.5 \times t$

Therefore, $0.5 \times t + 6t = 360$

Or, $6.5 \times t = 360$

Hence, $t = 55.38$ min

Important Derivations

→The number of times hands of a clock are in a straight line (either at 0° or at 180°) in 24 h = 44

→The number of times hands of a clock are at a right angle (at 90°) in 24 h = 44

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→Both the hands of clock are together after every $65\frac{5}{11}$ min.

(Therefore, if both the hands of the clock are meeting after every 65 min or anything less than $65\frac{5}{11}$ min, then the clock is running fast and if both the hands of the clock are meeting after every 66 min or anything more than $65\frac{5}{11}$ min, then clock is running slow.)

Practice Exercises

Warm up

Q.1 A man is running at the speed of 36 km/h, then how many *m* does he cover in 40 s?
(a) 400 m (b) 300 m (c) 2400 m (d) 100 m

Q.2 The speed of a train is 20 m/s. The time that it takes to cross a telephone pole is 10 s. What is the length of the train?
(a) 100 m (b) 1.5 km (c) 150 m (d) 200 m

Q.3 A train having a length of 400 m, travelling at a speed of 15 m/s crosses a platform in one min. What is the length of the platform?
(a) 2000 m (b) 500 m
(c) 400 m (d) Cannot be determined

Q.4 Speed of a train is 88 km/h. What time will it take to cross a platform 180 m long if the length of the train is 150 m?
(a) $11\frac{1}{2}$ s (b) $10\frac{1}{2}$ s (c) $13\frac{1}{2}$ s (d) $14\frac{1}{2}$ s

Q.5 A certain distance is covered at a speed *V* km/h. If half of the same the time, then the ratio for the former speed to that of the latter is:
(a) 4:1 (b) 1:4 (c) 2:1 (d) 1:2

Q.6 A person can row a distance of one kilometre upstream in 10 min and downstream in four min. What is the speed of the stream?
(a) 4 km/h (b) 9 km/h
(c) 5.6 km/h (d) 4.5 km/h

Q.7 Anil calculated that it will take 45 min to cover a distance of 60 km by his car. How long will it take to cover the same distance if the speed of his car is reduced by 15 km/h (approximate)?
(a) 36 min (b) 55 min (c) 48 min (d) 40 min

Q.8 A car travels from Patna to Jehanabad at a speed of 65 km/h in 1 h. If the speed is reduced by 15 km/h then, how much more time will the car take to cover the same distance?
(a) 12 min (b) 16 min (c) 18 min (d) 44 min

Q.9 In a kilometre race, *A* beats *B* by 10 m. In a two kilometres race, *A* beats *B* by:
(a) 10 m (b) 20 m (c) 40 m (d) 25 m

Q.10 A train, 225 m long, is running at 45 km/h. In what time does it cross a man running at 18 km/h in the same direction?
(a) 20 s (b) 30 s (c) 25 s (d) 15 s

Q.11 Two trains started from a station, at the same time in the same direction, at speeds of 70 km/h and 60 km/h. What will be the distance between the two trains after 6 h?
(a) 42 km (b) 54 km (c) 60 km (d) 84 km

Q.12 A man can row half a kilometre against the current in 12 min and returns in six min. Find the speed of the current.
(a) 1.25 km/h (b) 1.5 km/h
(c) 2.5 km/h (d) 3 km/h

Q.13 In a kilometre race, Amit beats Bahadur by 100 m and Bahadur beats Chandra by 200 m. By how many metres does Amit beat Chandra in the same race?
(a) 100 m (b) $83\frac{1}{3}$ m (c) 68 m (d) 280 m

Q.14 On a river, *Q* is the mid-point between two points *P* and *R* on the same bank of the river. A boat can go from *P* to *Q* and back in 12 h and from *P* to *R* in 16 h 40 min. How long would it take to go from *R* to *P*?
(a) $3\frac{1}{3}$ h (b) 5 h
(c) $6\frac{2}{3}$ h (d) None of these

Q.15 Two trains are running on a parallel line in the same direction at a speed of 50 km/h and 30 km/h, respectively. Given that the faster train crosses a man in the slower train in 18 s, what is the length of the faster train?
(a) 170 m (b) 100 m (c) 98 m (d) 85 m

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Q.16 In a kilometre race, Ajay beats Bijay by 100 m and Bijay beats Chand by 100 m. By how many m does Ajay beat Chand in the same race?
(a) 100 m (b) 200 m (c) 190 m (d) 119 m

Q.17 Two persons are walking with the speed A and B, respectively. If the first person takes 10 min less to cover a distance, what is the time taken by the second person to cover the same distance provided $A:B = 3:2$?
(a) 20 min (b) 30 min (c) 10 min (d) 50 min

Q.18 A train of length 200 m crosses a pole in 15 s and crosses a platform in 60 s. What is the length of the platform?
(a) 450 m (b) 600 m
(c) 800 m (d) None of these

Q.19 Hands of a clock are meeting after every 65 min. We can conclude that the clock is running:
(a) Fast
(b) Slow
(c) Cannot be determined

Q.20 In 12 h, how many times the hour and the minutes hands of a clock will be at 0° to each other?
(a) 11 (b) 12
(c) 13 (d) None of these

FOUNDATION

Q.1 When the speed of a train is increased by 20%, it takes 20 min less to cover the same distance. What is the time taken to cover the same distance with the actual speed?
(a) 2 h (b) 1 h (c) 1.5 h (d) 2.5 h

Q.2 Difference between the time taken in covering a certain journey decreases by 45 min when the speed is increased from 10 km/h to 20 km/h. What is the difference between the time taken when the same distance is covered at a speed of 40 km/h and 25 km/h?
(a) 12.5 min (b) 13.5 min
(c) 18 min (d) None of these

Q.3 Shramjeevi express goes to Patna from New Delhi at a speed of 60 km/h. At what speed (in km/h) should Shramjeevi express return from Patna to New Delhi so that the average speed during the whole journey is 120 km/h?
(a) 180 (b) 240
(c) 120 (d) This is not possible

Q.4 Sona travels 600 km to his house partly by train and partly by car. He takes 8 h if he travels 120 km by train and the rest by car. He takes 20 min more if he travels 200 km by train and the rest by car. Find the speed of the train in km/h.
(a) 45 (b) 60
(c) 75 (d) None of these

Q.5 Sama travelled the first half of the distance from her house to school at a speed of 10 km/h and for exactly half of the remaining time she travelled at 15 km/h and the rest of the time at 20 km/h. Find her average speed.
(a) 13 km/h (b) $12\frac{8}{11}$ km/h
(c) 15 km/h (d) None of these

Q.6 Shilu can go 30 km upstream and 44 km downstream in 10 h. Also, she can go 40 km upstream and 55 km downstream in 13 h. What is her speed in still water?
(a) 6 km/h (b) 12 km/h
(c) 8 km/h (d) 16 km/h

Q.7 A train travels a distance of 192 km at an average speed of 80 km/h from station P to another station Q. It then travels back to P taking 5 h 36 min. Find the average speed of the train on the return journey to P.
(a) 72 km/h (b) 36 km/h
(c) 20 km/h (d) 60 km/h

Q.8 Two persons, 27 km apart, start at the same time and are together in 9 h if they walk in the same direction. But, if they walk in the opposite directions they are together in 3 h. What are their speeds (in km/h)?
(a) 5, 4 (b) 4, 3 (c) 6, 3 (d) 8, 7

Q.9 In a race of 300 m, Abhishek beats Bijay by 30 m, while Bijay beats Chandan by 50 m. By what distance should Abhishek beat Chandan in the same 300 m race?
(a) 80 m (b) 40 m (c) 160 m (d) 75 m

Q.10 What is the speed of a train, which overtakes a cow walking at a speed of 5 km/h in 30 s, if the train is 274 m long?
(a) 51.88 km/h (b) 27.88 km/h
(c) 37.88 km/h (d) 21.67 km/h

Q.11 In 11 h, C walks 12.5 km less than what D does in 12 h; and in 5 h, D walks 3.25 km less than what C does in 7 h. How many kilometre does each walk per hour?
(a) 3 and 4 (b) 3.5 and 4.25
(c) 2.33 and 3.4 (d) None of these

Q.12 The ship Vikrant starts from a point P towards a point Q at noon and at 1:00 pm ship, Viraat starts from Q towards P. If ship Vikrant is expected to complete the voyage in 6 h and ship Viraat is moving at a speed of 2/3rd of that of ship Vikrant, at what time are the two ships expected to meet?
(a) 4 pm (b) 4:30 pm (c) 3 pm (d) 2:30 pm

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Q.13 Amit Kumar starts driving from Patna to Quillon at 9 am and Bhartendu starts driving from Quillon to Patna at 10 am Bhartendu is 50% faster than Amit Kumar. At what time, will they meet if Patna and Quillon are 300 km apart and Amit Kumar's speed is 50 km/h?
(a) 12:30 pm (b) 12 noon
(c) 11 am (d) 11:30 am

Q.14 A ship, 156 km away from the seacoast springs a leak, which admits $2\frac{1}{3}$ metric tons of water in $6\frac{1}{2}$ min, but the pumps throw out 15 metric tons of water in 1 h. Seventy metric tons would be sufficient to sink the ship. What should be the average speed (in km/h) of the ship so that it may just reach the shore as she begins to sink?
(a) 14.5 (b) 15 (c) 18 (d) 20

Q.15 A man rows a boat a certain distance upstream and then returns to the same place. If the time taken by him in going upstream is twice the time taken in rowing downstream, find the ratio of the speed of the boat in still water and the speed of the stream.
(a) 2:1 (b) 3:2 (c) 5:3 (d) 3:1

Q.16 Excluding stoppages, the speed of a bus is 54 km/h and including stoppages it is 45 km/h. What is the stoppage time of the bus (in min) per h?
(a) 10 (b) 12 (c) 9 (d) 20

Q.17 For covering a distance of P km, a man takes t h. He takes 40 min less if he moves 3 km/h faster, But if he moves slower by 2 km/h, he takes 40 min more. Then, P is equal to:
(a) 22.5 km (b) 32.5 km
(c) 42.5 km (d) None of these

Q.18 Two trains start at the same time from A and B and proceed towards B and A at 36 km/h and 42 km/h, respectively. When they meet, it is found that one train has moved 48 km more than the other. Then, the distance between A and B (in km) is:
(a) 624 (b) 636 (c) 544 (d) 460

Q.19 Anand can beat Bidhan by 5 m in a 100 m race and Bidhan can beat Chandan by 10 m in a 200 m race. Then, in a race of 400 m, Anand can beat Chandan by:
(a) 40 m (b) 39 m (c) 15 m (d) 10 m

Q.20 A man can swim with the stream at the rate of 3 km/h and against the stream at the rate of 2 km/h. How long will he take to swim 7 km in still water?
(a) 3 h (b) 2.8 h (c) 2.6 h (d) 3.2 h

Q.21 Points P and Q are 120 km apart. A motorcyclist starts from P to Q along a straight road P-Q with a speed of 30 km/h. At the same time, a cyclist starts along a road perpendicular to road PQ, with a speed of 10 km/h. After how much time will the distance between them be the least?
(a) 3 h (b) 3.4 h
(c) 4 h (d) Cannot be determined

Q.22 A and B start at the same time from P and Q (55 km apart) to Q and P at 3 and $2^{\frac{1}{2}}$ km/h respectively. They meet at R and then reach Q and P to return immediately and meet again at S. The distance from R to S is:
(a) 10 km (b) 12 km (c) 15 km (d) 16 km

Q.23 A runs 1.75 times as fast as B. If A gives B a start of 60 m, how far should the winning post be so that A and B reach there at the same time?
(a) 140 m (b) 70 m (c) 210 m (d) 175 m

Q.24 X, Y, and Z walk at 6, 12, and 18 km/h, respectively. They start from Jhumritalaiya towards Delhi at 2, 5, and 7 pm, respectively, when Y catches up with X, Y sends X back with a message to Z, when will Z get the message?
(a) 9:45 pm (b) 9 pm (c) 8:45 pm (d) 9:15 pm

Q.25 A train 75 m long overtook a person who was walking at the speed of 6 km/h in the opposite direction and passed him in $7\frac{1}{2}$ s. Subsequently, it overtook a second person, walking in the same direction as the first person and passed him in $6\frac{3}{4}$ s. At what rate was the second person travelling?
(a) $11\frac{3}{4}$ km/h (b) 10 km/h
(c) 8 km/h (d) $4\frac{1}{2}$ km/h

Q.26 Alok rows a boat against a stream flowing at 2 km/h for a distance of 9 km and then turns around and rows back with the current. If the whole trip lasts 6 h, find Alok's rowing speed in still water.
(a) 4 km/h (b) 3 km/h (c) 2 km/h (d) 5 km/h

Q.27 A man walked 12 km at a certain rate, and then, 6 km farther at a rate of $\frac{1}{2}$ km/h faster. If he had walked the whole distance at a faster rate, his time would have been 20 min less. How long did he really take to walk the distance of 18 km?
(a) 4.5 h (b) 5.33 h
(c) 5.66 h (d) None of these

Q.28 Shramjeevi Express leaves Patna for New Delhi at 10:10 am and reaches New Delhi at 3:30 pm. Magadh Express

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leaves New Delhi at 9:48 am and reaches Patna at 2:36 pm. If routes of both the trains are the same, then at what time will both the trains meet each other?
(a) 12:52 pm (b) 1:12 pm
(c) 1:32 pm (d) None of these

Q.29 *A* can give a start of 20 m to *B* and *B* can give a start of 10 m to *C* in a race of 200 m. By how many metres can *A* beat *C* in the same race?
(a) 21 m (b) 28 m (c) 29 m (d) 31 m

Q.30 Find the distance between two towns when by increasing the speed to 7 km/h, a train finishes the journey in 1 h less, and when the speed is reduced to 5 km/h, the train finishes the journey in 1 h more.
(a) 210 km (b) 180 km
(c) 240 km (d) 164.4 km

Q.31 Preeti and Uma stay next to each other and study in the same school. Preeti and Uma start walking towards their school at the speed of 2 km/h and 3 km/h, respectively. The faster of the two reaches the school first, turns around and starts walking back. If Preeti and Uma meet 200 m away from the home, what is the distance between their residences and the school?
(a) 260 m (b) 240 m (c) 200 m (d) 250 m

Q.32 Vinit has to cover a certain distance of 7 km in 1 h 12 min. If he walks some distance at 4 km/h and rides a horse for some distance at 10 km/h, he reaches his destination in time. How far should he walk on foot?
(a) 14/3 km (b) 11/3 km
(c) 13/3 km (d) 10/3 km

Q.33 *A* starts driving from *P* to *Q* at 9 am and *B* starts driving from *Q* to *P* at 10 am *B* is 50% faster than *A*. What is the time when they meet if *P* and *Q* are 300 km apart and *A*'s speed is 50 km/h.
(a) 12:30 pm (b) 12 noon
(c) 11:00 am (d) 11:30 am

Q.34 Binod and Neeraj start from one end of a 1000 m track while Deleep starts from the other end. Binod is 50% faster than Neeraj and Deleep is 150% faster than Neeraj. If Binod meets Deleep in 25 s, how long does Neeraj take to meet Deleep?
(a) 30 s (b) 35 s (c) $28\frac{4}{7}$ s (d) $26\frac{3}{7}$ s

Q.35 Vinod and Kaurvaki start travelling in the same direction at 8 km/h and 13 km/h, respectively. After 4 h, Vinod doubled his speed and Kaurvaki reduced his speed by 1 km/h and reached the destination together. How long did the entire journey last?
(a) 3 h (b) 6 h (c) 9 h (d) 12 h

Q.36 A bus B_1, left a town T_1 for another town T_2 at 6:00 am at 7:30 am. Another bus B_2 also left T_1 for T_2 and the speed of this bus was 5 km/h more than that of B_1. At 10:30 pm, on the same day, the buses were 21 km apart. Find the speed of bus B_1.
i. 36 ii. 64 iii. 72
(a) i or ii only (b) i or iii only
(c) Either i or ii or iii (d) None of these

Q.37 Two men leave places *A* and *B* simultaneously. One of them left *A* for *B* and the other *B* for *A*. Both travel at their own uniform velocity. Having arrived at their destination, they turn back without stopping and travel to their starting points. For the first time, they met at 12 km from *B*, the second time at 6 km from *A*. Find the distance *AB*.
(a) 72 km (b) 30 km (c) 45 km (d) 60 km

Q.38 *P* and *Q* start running simultaneously—one from point *A* to *B* and the second from point *B* to *A*. *P*'s speed is 6/5th of *Q*'s speed. If after crossing *Q*, *P* takes 2 1/2 h to reach *B*, how much time does *Q* take to reach *A* after crossing *P*?
(a) 3:36 min (b) 3:48 min
(c) 4:12 min (d) None of these

Q.39 *A* beats *B* by 20 m in a 100 m race and *B* beats *C* by 20 m in a 100 m race. How much start should *A* give to *C* in a 100 m race so that both of them reach the winning post at the same time?
(a) 30 m (b) 44 m (c) 32 m (d) 36 m

Q.40 Mahima beats Kareena in 1000 m race by 50 s and Bhumika by 450 m. If Kareena and Bhumika run 1000 m race, Kareena wins by 40 s. How much time does Bhumika take to run 1000 m?
(a) 210 s (b) 225 s (c) 180 s (d) 200 s

Q.41 Two trains have respective lengths as 230 m and 190 m. They cross each other completely in 21 s, if they are travelling in the opposite direction and in 42 s, if they are travelling in the same direction. Find the ratio of the speeds of the two trains.
(a) 3 :1 (b) 4:1
(c) 3:2 (d) None of these

Q.42 A car after travelling a distance of 110 km develops a problem in the engine and proceeds at 3/4th of its former speed and arrives at the destination 60 min late. Had the problem developed 30 km further on, the car would have arrived 12 min sooner. Find the original speed of the car.
(a) 45 km/h (b) 60 km/h
(c) 50 km/h (d) None of these

Q.43 A stone allowed to fall under gravity falls $5t^2$ m in '*t*' s. When a stone is dropped into a well, the sound of the stone striking the water was heard 65/9 s after it was dropped. Find the depth of the well if sound travels 324 m in a second?
(a) 360 m (b) 180 m (c) 210 m (d) 240 m

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Q.44 A watch which gains uniformly is five min slow at 8:00 am on Sunday and is five min 48 s fast at 8:00 pm on the following Sunday. When did it show the correct time?
(a) 7:20 pm Thursday (b) 7:20 am Wednesday
(c) 7:20 pm Wednesday (d) 7:20 am Thursday

Q.45 Anil and Dheeraj run a 4 km race on a circular course 400 m long in the same direction. If their speeds are in the ratio of 3:2, how many times does Anil pass Dheeraj?
(a) 2 times (b) 3 times
(c) 4 times (d) 5 times

Q.46 At what time between 8 and 9 O'clock, will the hours and min hand make an angle of 72° with each other?
(a) 8 h 306/11 (b) 8 h 568/11
(c) 1 or 2 (d) None of these

Q.47 At what time between 2 and 3 O'clock, will the hands of a clock be in a straight line facing away from each other?
(a) $2:42:\frac{9}{11}$ (b) $2:42:\frac{49}{77}$
(c) $2:43:\frac{9}{11}$ (d) $2:43:\frac{49}{77}$

Q.48 A man can row a distance of 2 km upstream in 15 min and returns the same distance in 10 min. How much time will he take to row the same distance upstream if due to a tide the speed of the current gets doubled?
(a) 25 min (b) 45 min
(c) 20 min (d) None of these

Q.49 Two points A and B are diametrically opposite points on a 6 km long circular road. A cyclist started from A and made two rounds. He made the first round with a certain uniform speed and then decreased his speed by 3 km/h. The interval between his two passages through point B is 50 min. Find the speed with which the cyclist made the first round.
(a) 6 km/h (b) 9 km/h
(c) 12 km/h (c) 16 km/h

Q.50 Minute hand of a clock overtakes the hour hand at the intervals of 63 min of the correct time. How much does the clock lose or gain in a day?
(a) $56\frac{24}{77}$ min (b) $55\frac{8}{77}$
(c) $56\frac{8}{77}$ min (d) None of these

Moderate

Q.1 A train of 300 m is travelling with the speed of 45 km/h, when it passes point A completely. At the same time, a motorbike starts from point A with the speed of 70 km/h. When it exactly reaches the middle point of the train, the train increases its speed to 60 km/h and motorbike reduces its speed to 65 km/h. How much distance will the motorbike travel while passing the train completely?
(a) 2.52 km (b) 2.37 km
(c) 2 km (d) None of these

Q.2 Two persons A and B are standing diametrically opposite on a circular track of with a radius of 3.5 units. They start running in an opposite direction on the circular track and their speeds are four units/min and seven units/min, respectively. On meeting each other, they take rest for half a min and then start running with twice the original speed. In five min, how many times will they take rest together?
(a) 6 times (b) 7 times
(c) 10 times (d) None of these

Q.3 Two riders on the horseback with a gun and a bullet proof shield were moving towards each other at a constant speed of 20 km/h and 5 km/h, respectively. When they were 100 km apart, they started firing bullets at each other at the speed of 10 km/h. When a bullet of rider 1 hits the shield of rider 2, rider 2 fires a bullet and the process continues vice versa. Neglecting the time lag at the instant when the bullet hits the shield and the rider fires the shot, find the total distance covered by all the bullets shot by both the riders.
(a) 50 km (b) 40 km
(c) 25 km (d) None of these

Q.4 A group of soldiers are marching with a speed of 5 m/s. The distance between the first and the last row of soldiers is 100 m. A dog starts running from the last row and moves towards the first row, turns and comes back to the last row. If the dog has travelled 400 m, the speed of the dog is:
(a) $5\sqrt{2}$ m/s (b) $3\sqrt{5}$ m/s
(c) $6\sqrt{5}$ m/s (d) $6\sqrt{2}$ m/s

Q.5 A train approaches a tunnel PQ which is 16 m long. Two rabbits A and B are standing at points which are 12 m and 8 m inside the tunnel with respect to the entrance P. When the train is x m away from P, A starts running towards P and B towards Q. Difference between the ratios of the speed of A to that of the train and the ratio of the speed of B to that of train is 1/8. How much can

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the distance x be, if both of them get caught at the ends of the tunnel?

(a) 30 m (b) 16 m
(c) 48 m (d) None of these

Q.6 Pranesh wants to travel to Pune. He has three options.
Route-1 Travel by a train from Chiplun to Mumbai and then from Mumbai to Pune. The trains take 4.5 h and 3 h, respectively.
Route-2 Travel by a bus via Bhor. The distance covered by this route is 300 km at 40 km/h and there is a half an hour break in between.
Route-3 Travel by a bus via Koyana. The distance covered by this route is 500 km at 60 km/h. If Pranesh goes by train, there is a time span of 30 min between the respective trains. Which route should he take to save his time?

(a) Route 1 (b) Route 2
(c) Route 3 (d) [1] or [2]

Q.7 The numbers expressing (in metres) the length of a fast and slow trains are three times the numbers expressing (in km/h) the speed of the slow and the fast train respectively. The speed of the slow train and that of the fast train consists of the same two digits, but in a reverse order. The time taken by the two trains to pass each other, when travelling in an opposite direction on parallel lines will be:

(a) 10.8 s (b) 5.4 s (c) 9.77 s (d) 5/18 s

Q.8 Michael Johnson is a great sprinter. He can climb 12 stairs of a staircase in 3 s. But after climbing 480 stairs his speed reduces to eight stairs in 3 s. The Empire State Building in NY has 90 floors and the lift takes 4 s per floor. To get to the next floor, one has to climb 12 stairs. If Michael is on the ground floor how should he reach the 90th floor to be the first?

(a) By lift
(b) Running
(c) Run till 50th floor and then take the lift
(d) None of these

Q.9 A car goes 20 miles on a gallon of gas driving at 60 miles/h. If the car is driven at 75 miles/h, it goes as far as 70%. How many gallons of gas will be required to travel 210 miles/h.

(a) 6.17 (b) 22.5 (c) 12 (d) 15

Q.10 Rakesh and Siddharth row on a river simultaneously from the same point. Rakesh rows downstream and Siddharth rows upstream. In 15 min, they are 2.25 km apart.Rakesh then turns to follow Siddharth and after 30 min from the beginning, the boats have rowed together 3.5 km. If the speeds of Rakesh and Siddharth and the stream are constant at how many km/hour does the stream flow?

(a) 3 km (b) 2.5 km (c) 2 km (d) 5 km

Direction for Questions 11 and 12: Refer to the data below and answer the questions that follow.

There is a race between Sagar and Sapna. Both of them were going to Delhi. Sagar started on a bike with a speed of 40 km/h and Sapna has started in a car with a speed of 60 km/h from Mumbai to Delhi. After $3\frac{1}{2}$ h of the journey there was a snag in the car. She tried to repair the car but in vain. After half an hour, she got a lift for another 500 km in a truck, which was travelling with a speed of 45 km/h. From there Delhi was at a distance of 200 km on road, instead, Sapna took a shorter route which was only 100 km away from Delhi. She started running at the speed of 30 km/h on the shorter route to reach Delhi.

Q.11 How much time did Sapna take to reach Delhi?

(a) 15.5 h (b) 16.5 h
(c) 17.5 h (d) None of these

Q.12 If there was no snag in the car, by how much distance Sapna would have defeated Sagar?

(a) 252 km (b) 264 km
(c) 303 km (d) 321 km

Q.13 A 200-m-long train passes a 350-m-long platform in 5 s. If a man is walking at a speed of 4 m/s along the track and the train is 100 m away from him, how much time will it take to reach the man?

(a) Less than 1 s (b) 1.04 s
(c) More than 2 s (d) Data insufficient

Q.14 A man waiting at a point P has to reach point Q at some distance from P. He is considering three plans of action.

A. Wait at P itself and catch the first taxi going towards Q.
B. Start walking towards Q and catch the first taxi going towards Q.
C. Start walking away from Q and catch the first taxi going towards Q.

He walks at a constant rate of 5 km/h and taxi travel at a constant rate of 50 km/h. The arrival time of the taxi is totally random. Which of the following statement is then true? (Assume that he is the only person who is looking for a taxi on that road.)

(a) He will always reach Q first, if he chooses plan A.
(b) He will always reach Q at the same time regardless of which plan he chooses.
(c) He will sometimes reach Q at the earliest if he chooses plan B.
(d) He will always reach Q at the earliest if he chooses plan C.

Q.15 Three cockroaches John, Mac, and Roe decide to take a race from one corner of the room to the diagonally opposite corner of the room. John can fly, Mac can walk and Roe can walk only along the edges. If all of

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them reach the destination at the same time by taking the smaller paths, what is the ratio of the speeds of John, Mac, and Roe assuming that the room is cubical in shape?

(a) $\sqrt{3}+1:3:\sqrt{3}$ (b) $\sqrt{3}+5:3$

(c) $\sqrt{3}+\sqrt{2}:3$ (d) None of these

Q.16 Four men start to move from a particular point. Two of them go in the direction opposite to each other at an equal speed. The other two also move in the opposite directions at an equal speed, but in a direction that is perpendicular to that of the original two. All of them stop at the same time, if the final positions of the men are joined, then it forms.

(a) A parallelogram (b) A rectangle

(c) A rhombus (d) A square

Q.17 Saket runs twice on a circular track at a uniform speed. Bidhan starts at the same time as Saket and after Bidhan has completed one lap, he stops and Sunny starts and completes the second lap at a uniform speed. It is given that they start the race exactly at 2:30 pm and the track is 500 m in circumference. Sunny runs at a speed of 5 km/h greater than that of Bidhan and Saket's speed is 2 km/h greater than that of Bidhan. When does Saket finish the second lap if Sunny also finishes at the same time?

(a) 2:32 pm (b) 2:35 pm

(c) 2:33 pm (d) 2:36 pm

Q.18 Two persons A and B are running between two points P and Q to and from infinitely. A is starting from P and B is starting from Q. They meet for the first time at $0.6D$ from point P, where D is the distance between the points AB. At how many distinct points, will they meet till their 10th meeting?

(a) 10 (b) 3 (c) 4 (d) 6

Direction for Questions 19 and 20: Refer to the data below and answer the questions that follow.

When BMW introduced its most advanced engine, it was found that the engine of the car followed certain mileage characteristics in a speed limit of 25 km/h to 80 km/h. The car can travel a distance of 15 km in one litre of petrol while speeding at 50 km/h. From this point onwards for every increase in 5 km/h in the speed of the car the mileage is increased by 1 km per litre. Similarly, for every decrease of 5 km/h in the speed of the car, the mileage is decreased by 1 km per litre. Assume that the cars operate in this speed limit only.

Q.19 Rahul Ghosh has filled exactly 10 l of petrol in his car. He drives at a speed of 45 km/h for first 2 h. What is the maximum distance that he can cover before his tank goes dry?

(a) 80 km (b) 67.37 km

(c) 75 km (d) None of these

Q.20 In the previous question, what is the maximum span of time for which Rahul Ghosh can enjoy the drive?

(a) 24/7 h (b) 15/4 h

(c) 37/15 h (d) None of these

Q.21 Two rabbits A and B are running a race in which they have to go up to 50 m mark and then come back to the starting point. A runs in a sequence of three jumps in which the distance travelled in the first one is twice that of the other two and B runs in a sequence of three jumps in which the distance travelled in the first jump is 1.5 that of the distance travelled in the second jump and distance travelled in the second jump is twice the distance travelled in the third jump. Distance travelled by A in two jumps and by B in three jumps equal to 6 m. In the second race, they interchange their individual sequence of jumps. Who will win the first and the second match if on completing the 50 m mark they turn back and start from the first jump?

(a) A, A (b) B, B

(c) A, B (d) None of these

Q.22 There are two clocks. One of them gains two min in 12 h and another one loses two min in 36 h. Both are set right at 12 noon on Tuesday. What will be the correct time when both of them show the same time for the next time?

(a) 12 night (b) 1:30 am

(c) 10:30 pm (d) 12 noon

Q.23 A train's journey is disrupted due to an accident on its track after it has travelled 30 km. Its speed then come down to four-fifth of its original, and consequently, it runs 45 min late. Had the accident taken place 18 km farther away, it would have been 36 min late. Find the original speed of the train.

(a) 25 km/h (b) 36 km/h

(c) 30 km/h (d) 20 km/h

Q.24 A passenger train departs from Ahmadabad at 6 pm for Bombay. At 9 pm, an express train, whose average speed exceeds that of the passenger train by 15 km/h, leaves Bombay for Ahmedabad. Two trains meet each other mid-route. At what time, do they meet, given that the distance between the cities is 1080 km?

(a) 4 pm (b) 2 pm

(c) 12 mid-night (d) 6 am

Q.25 In a race on a circular track, there were 12 laps of 150 m each. Akhil and Bharat were very close till the end of the third lap, but thereafter Bharat went on to win the race by 81 m. If Bharat gained uniformly over Akhil in all the remaining laps, the ratio of the speed of Akhil and Bharat for each of other laps was:

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(a) 129:150 (b) 150:141
(c) 150:129 (d) 141:150

Q.26 Magadh Express and Vikramshila Express run between Patna and New Delhi. Every night each of these trains leaves Patna for Delhi. Similarly, every night each of the trains leaves Delhi for Patna. The train journey lasts 36 h and the trains are identical in all aspects. If the train that leaves in the night is like the one that arrived in the morning, what is the minimum number of trains required?
(a) 4 (b) 8
(c) 16 (d) None of these

Q.27 An athlete runs to and fro between points *A* and *B* at a speed of 10 km/h. A second athlete simultaneously runs from point *B* to *A* and back at a speed of 15 km/h. If they cross each other 12 min after the start, after how much time will they cross each other?
(a) 18 min (b) 24 min (c) 36 min (d) 48 min

Q.28 There are two swimmers *A* and *B* who start swimming towards each other from opposite banks of the lake. They meet at a point 900 ft from one shore for the first time. They cross each other, touch the opposite bank and return. They meet each other again at 300 ft from the other shore. What is the width of the lake?
(a) 2400 ft (b) 1800 ft
(c) 2700 ft (d) 3600 ft

Q.29 DMRC is plying trains between Vishwa Vidyalaya and Central Secretariat, which are at a distance of 120 km. There are three more stations in between Vishwa Vidyalaya and Central Secretariat and each train stops at these two stations for two min. What is the minimum numbers of trains DMRC should put in this route, if the trains are travelling at a speed of 40 km/h and it is desirable to have a regular interval of six min between the departures of two trains from both the stations?
(a) 32 (b) 63
(c) 64 (d) None of these

Q.30 The front wheels of a tractor make 36 revolutions per min, while the rear wheels make 18 revolutions per min. After how many seconds, will the two wheels be in the same position relative to each other as they were when they started moving?
(a) 2.22 (b) 3.33 (c) 1.66 (d) 4.44

Q.31 A train and a car start at the same time, parallel to each other in the same direction. The car starts its journey from the rear end of the train. The car reaches the front end of the train and comes back to the back end of the train. In the meantime, the mail train travels a distance of 1 km. If the speed and the length of the mail train is 1 km/h and 1 km, respectively, then what is the distance covered by the car?
(a) 2 km (b) $1+\sqrt{2}$ km
(c) $2+\sqrt{2}$ km (d) cannot be determined

Direction for Questions 32 to 34: Refer to the data below and answer the questions that follow.

A, *B*, and *C* participate in a triathlon. The triathlon consists of three parts. The first part consists of running, the second part cycling, and the third part swimming. *A* can cycle thrice as fast as he can run and swim one-third as fast as he can run. *B* can cycle four times as fast as he can run and swim one-fourth as fast as he can run. *C* can swim half as fast as he can run and cycle four times as fast he can run. *A* can run 10% faster than *B* and 20% faster than *C*. The ratio of the length of the running track to that of the cycling is half and to that of swimming tracks is two.

Q.32 Who is leading after the second part is over?
(a) *A* (b) *B*
(c) (d) Cannot be determined

Q.33 Who will win the triathlon?
(a) *A* (b) *B*
(c) *C* (d) Cannot be determined

Q.34 *C* beats *B* by 100 m finally. What is the length of the cycling tracks given that the length of the swimming track is more than 100 m?
(a) 580 m (b) 622 m (c) 518 m (d) 576 m

Q.35 Pramod can row at 3 km/h in still water. He wants to cross a river that is flowing at the speed of 4 km/h in the shortest possible time. Find the ratio of the distance between the point he leaves the bank on one side and the point that he reaches on the other side to the distance between the shortest possible distance to cross the river?
(a) 1 :1 (b) 1:2
(c) 2:3 (d) None of these

Q.36 Vinit's rowing at a speed of 4 km/h in still water. He has to reach a destination that is 130 km downstream of a river. The river flows at a speed of 2 km/h but due to tides every 2 h, the velocity doubles and then after another 2 h, it halves. If Vinit starts at the change of the tide when the river was flowing at 2 km/h, then after how much time will he reach his destination?
(a) $8\frac{1}{4}$h (b) $18\frac{1}{2}$h
(c) $9\frac{1}{2}$h (d) None of these

Q.37 Ramesh and Somesh are competing in a 100 m race. Initially, Ramesh runs at twice the speed of Somesh for the first 50 m. After the 50 m mark, Ramesh runs at one-fourth his initial speed, while Somesh continues to run at his original speed. If Somesh catches up with

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Ramesh at a distance of 'N' m from the finish line, then N is equal to:
(a) 35 (b) 10
(c) 45 (d) None of these

Direction for Questions 38 and 39: Read the passage given below and solve the questions based on it.

Two cars A and B start simultaneously from two different Cities P and Q and head for the Cities Q and P, respectively. As soon as car A reaches the City Q, it turns and starts for City P and as soon as it reaches P again, it leaves for City Q and so on. Similarly, car B travels from Q to P, from P to Q and so on. The speeds of the car A and B are in the ratio of 3:4.

Q.38 If car B travelled 1040 m when the two cars meet for the seventh time, what is the distance between the two Cities P and Q?
(a) 140 m (b) 280 m
(c) 182 m (d) Cannot be determined

Q.39 If car A travelled 1860 m, how many times did the cars A and B meet? (Take the distance PQ as calculated in the previous question.)
(a) 14 (b) 15
(c) 16 (d) Cannot be determined

Q.40 The Howrah Mail leaves Trivandrum every day at 12:00 noon and reaches Howrah exactly three days later. Also, the Trivandrum Mail leaves Howrah at 12:00 noon every day and reaches Trivandrum exactly three days later. Not counting the trains just starting or just leaving the platform when a train leaves or arrives, how many Howrah Mails will one cross when travelling by Trivandrum Mail?
(a) 2 (b) 3 (c) 5 (d) 9

Q.41 A person starts writing natural numbers, starting from 1, on a blackboard, at the rate of 60 digits per min. At the same time, another person begins to erase the digits from 1 onwards, at the rate of 40 digits per min. Find the difference between the leftmost digit and the rightmost digit on the blackboard immediately after 1 h.
(a) 3 (b) 5
(c) 2 (d) None of these

Q.42 Two persons Prabhat and Vinay are walking around a circular park of the length 960 m. Prabhat walks at the rate of 80 m/min, while Vinay walks at the rate of 60 m/min. If both of them start from the same starting point at the same time in the same direction, when will they be together?
(a) 24 min (b) 48 min
(c) 96 min (d) 120 min

Q.43 A 100-m-long train is approaching an unmanned railway crossing. The train is running at a uniform speed of 90 km/h and is 1 km away from the crossing. At the same time, a packed jeep, also approaching the crossing, is 700 m away from it. Assuming that the jeep is also moving at a uniform speed of N, what is the range of N, for which the jeep will not collide with the train?
(a) 59 km/h < N < 61 km/h
(b) 57 km/h < N < 63 km/h
(c) 55 km/h < N < 65 km/h
(d) 53 km/h < N < 67 km/h

Q.44 A road from A to B is $11\frac{1}{2}$ km long, first goes uphill, then crosses a plain, and then goes downhill. A person walking from A to B covered this road in 2 h 54 min, and the return journey took him 3 h 6 min. His speed uphill is 3 km/h, on the plain is 4 km/h and downhill is 5 km/h. What is the length of the plain part of the journey?
(a) 4 km (b) 6 km
(c) $4\frac{1}{2}$ km (d) None of these

Q.45 Two cyclists start from the same place to ride in the same direction. Aflatoon starts at noon with a speed of 8 km/h and Bablajoon starts at 2 pm with a speed of 10 km/h. At what times, Aflatoon and Bablajoon will be 5 km apart?
(a) 7:30 pm same day and 1:30 am on the next day
(b) 7:30 pm same day and 12:30 am on the next day
(c) 8:30 pm same day and 1:30 am on the next day
(d) 8:30 pm same day and 12:30 am on the next day

Q.46 Mrs and Mr Sinha travel from Patna to Lucknow and break their journey at Buxer. Somewhere between Patna and Lucknow, Mrs Sinha asks, 'How far have we travelled?' Mr Sinha replies, 'Half as far as the distance from here to Lucknow'. Somewhere between Buxer and Lucknow, exactly 300 km from the point where she asked the first question, Mrs Sinha asks, 'How far have we to go?' Mr Sinha replies, 'Half as far as the distance from Buxer to here'. The distance between Patna to Lucknow is:
(a) 750 km (b) 600 km
(c) 450 km (d) None of these

Q.47 At his normal speed, Rishu can go 18 km downstream in a flowing river in 9 h less than what he takes to go the same distance upstream. The downstream journey would take 1 h less than what the upstream journey would take provided he doubles his rate of rowing. Find the speed of the river (in km/h)?
(a) $6\frac{2}{3}$ km (b) 8 km
(c) $9\frac{1}{2}$ km (d) 12 km

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Q.48 Umesh and Harish are running on a circular track of length 1200 m in opposite direction. Every time they meet, they exchange their speed as well as their directions. What is the shorter distance along the track between their 13th and 23rd meeting points, if the ratio of their original speeds is 1:5?
(a) 200 m (b) 400 m
(c) 800 m (d) 1000 m

Q.49 Two friends Manoj and Sagar start running simultaneously in an opposite direction on a circular track of the length 2100 m with speed of 7 m/s and 3 m/s, respectively. With every meeting, they exchange their speed and they meet for a total of 10 times. How much distance did Sagar cover in total?
(a) 1400 m (b) 700 m
(c) 1050 m (d) None of these

Q.50 *A* and *B* start simultaneously at one end of a swimming pool whose length is 50 m. The swimming race is a race of 1000 m. If *A* beats *B* and meets him, 17 times during the course and *A*'s speed is 5 m/s, then the speed of *B* could be:
(a) 1 m/s (b) 3 m/s
(c) 4 m/s (d) 6 m/s

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Direction for Questions 1 to 5: Read the passage given below and solve the questions based on it.

There are eight days in a week from Sunday to Saturday and another day called Funday on planet North. There are 36 h in a day and each hour has 90 min, while each min has 60 s.

Q.1 Find the approximate angle between the hands of a clock on North when the time is 16:50 am?
(a) 189 (b) 131 (c) 320 (d) 165

Q.2 At what time between 14 O'clock and 15 O'clock will the two hands of the clock be at an angle of 60°?
(a) $14:89\frac{16}{17}$ (b) $14:37\frac{13}{17}$
(c) $14:58\frac{14}{17}$ (d) $14:55\frac{5}{17}$

Q.3 Bubby, an inhabitant of the planet North leaves home between 7 am and 8 am and returns between 1 pm and 2 pm to find that the min and hours hands have interchanged their positions. How long was bubby out of the house?
(a) 6 h 28 min (b) 12 h 39 min
(c) 12 h 28 min (d) 12 h 22 min

Q.4 A watch which is correctly set at 16 O'clock in the morning shows 10 min after 13 O'clock in the evening when the correct time is 14 O'clock. What is the time when the watch indicates 15 O'clock in the evening?
(a) 16:00 (b) $16:32\frac{5}{17}$
(c) $16:16\frac{16}{17}$ (d) $16:33\frac{6}{17}$

Q.5 A watch on planet North which loses time uniformly is 15 min fast at 14 am on Sunday and is 45 min slow at 8 pm on the next Sunday. When did the clock show the correct time?
(a) 6 am on Saturday (b) 8 am on Sunday
(c) 17 am on Tuesday (d) 11 am on Sunday

Q.6 In the shooting of the movie 'AKS', Manu, the actor, is running towards a vertical mirror with a speed of 10 m/s on a line which is perpendicular to the plane of the mirror. The mirror is moving towards Manu with the speed of 6 m/s along the same line. Find the speed of Manu's image with respect to Manu (the image of Manu forms on the other side of the mirror such that the distance between the image of Manu and the mirror is equal to the distance between Manu and the mirror).
(a) 15 m/s (b) 25 m/s
(c) 30 m/s (d) None of these

Q.7 A rabbit, a tortoise, and a mongoose decide to run a closed circuit race. If the rabbit overtakes the mongoose two times per round, with the second overtake being at the starting point and the mongoose overtakes the tortoise four times per round, the fourth overtake being at the starting point, then find the ratio of the speed of the rabbit to that of the tortoise.
(a) 15:1 (b) 8:1 (c) 6:1 (d) 10:1

Q.8 Anil left his house for school *S* min late but he travelled at 4/3th of his usual speed and reached school *S* min early. Next day, he left home *S* min earlier than the previous day and travelled at 8/11th of his usual speed. He reached his office:
(a) *S* min late
(b) 2*S* min late
(c) Exactly on time
(d) Cannot be determined

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Direction for Questions 9 and 10: Read the passage given below and solve the questions based on it.

There are two cities Nagpur and Pune along a straight line 120 km apart. *A* and *B* start at the same time from Nagpur and Pune, respectively, with the speed of 40 km/h and 60 km/h, respectively. They travel towards each other, and after they meet for the first time, they reverse directions and also interchange their speeds. After reaching their respective starting points, they reverse their directions and start proceeding towards each other again.

Q.9 How many hours from the beginning do they meet for the second time?

(a) 2 h (b) $2\frac{1}{5}$ h (c) $2\frac{1}{3}$ h (d) $3\frac{3}{5}$ h

Q.10 How much distance had *B* covered till the time he met *A* for the second time?

(a) 216 km (b) 180 km
(c) 174 km (d) 168 km

Direction for Questions 11 and 12: Read the passage below and solve the questions based on it.

A number of runners, numbered 1, 2, 3 ,..., *N* and so on, start simultaneously at the same point on a circular track and keep on running continuously in the same direction, around the track. They run in such a way that the speed of the runner numbered *N* ($N > 1$) is *N* times that of the runner numbered 1.

Q.11 If there are exactly six runners, then at how many distinct points on the track is the runner numbered 1 over taken by any of the other five runners?

(a) 15 (b) 11 (c) 9 (d) 10

Q.12 If there are exactly four runners, then at how many distinct points on the track do two or more runners meet?

(a) 3 (b) 4 (c) 5 (d) 6

Direction for Questions 13 and 14: Read the passage given below and solve the questions based on it.

Amit intended to travel a certain distance at a certain uniform speed. But after 1 h, he increased his speed by 25%. As a result, in the remaining part of the time that he originally planned for the journey, he could now cover as much distance as he initially thought he would be able to cover.

Q.13 What is the total time taken for the journey?

(a) 4 h (b) 5 h
(c) 6 h (d) None of these

Q.14 After Amit increased his speed, if he decided to terminate his journey after covering the distance he initially intended to cover and not cover the extra distance as given in the data, what is the total time taken for the journey?

(a) 4 h 12 min (b) 5 h 24 min
(c) 3 h 36 min (d) None of these

Direction for Questions 15 and 16: Read the passage given below and solve the questions based on it.

A train started at 9:00 am from station *X* with a speed of 72 km/h. After 2 h, another train started from station *Y* towards *X* with a speed 90 km/h. The two trains are expected to cross each other at 1:30 pm. Owing to a signal problem arising at 12 noon, the speed of each of them was reduced by the same quantity and they crossed each other at 4:30 pm.

Q.15 What is the new speed of the train that started from station *X*?

(a) 18 km/h (b) 36 km/h
(c) 45 km/h (d) 54 km/h

Q.16 If the signal problem had occurred at 1:00 pm instead of 12 noon, at what time would the two trains cross each other?

(a) 2:30 pm (b) 3:00 pm
(c) 2:00 pm (d) 2:30 pm

Direction for Questions 17 to 19: Read the passage given below and solve the questions based on it.

Bihar Couriers (BC) sends a person on motorcycle every day to the airport to collect the courier. This person reaches the airport exactly when the plane lands at its scheduled time. One day, *BC* came to know that the flight would be late by 50 min. So, the *BC* person also started from his office 50 min later than the usual time. However, the plane landed earlier than was anticipated, and therefore, the airport authorities dispatched the courier through a cyclist to *BC*. The cyclist meets the *BC* personnel after 20 min and hands over the courier to the motorcyclist. As a result, the *BC* person reached office 10 min earlier than expected. (Assume that everybody travels at their own uniform speeds)

Q.17 What was the delay in the landing of the plane when compared to the normal schedule?

(a) 30 min (b) 35 min
(c) 20 min (d) 25 min

Q.18 If the cyclist covered a distance of 2.5 km before he met the motorcyclist, what is the speed of the motorcyclist?

(a) 60 km/h (b) 45 km/h
(c) 30 km/h (d) Cannot be determined

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Q.19 If on that particular day when the cyclist covered 2.5 km before he met the motorcyclist, the motor cyclist spent a total of 30 min for his complete journey, what is the distance from *BC*'s office to the airport?

(a) 20 km (b) 15 km
(c) 10 km (c) Cannot be determined

Q.20 My digital clock is peculiar. It counts 10 s of a normal clock as 1 min and 60 such minutes of itself as 1 h. It has also a display which shows the day. It was at par with the normal clock at 12 noon on Monday. At 3 pm of the same day (Actual time), I just read the time display of my weird clock. I read:

(a) 6:00 pm Monday
(b) 6:00 pm Tuesday
(c) 6:00 am Tuesday
(d) 4:00 am Tuesday

Direction for Questions 21 and 22: Read the passage below and solve the questions based on it.

Every morning Ravi and Beena start walking from *A* and *B*, respectively, towards each other. They meet at *C*. Ravi always starts at 5:30 am and Beena starts at 4:45 am. Ravi's speed is 6 km/h, whereas Beena's speed is 5 km/h.

Q.21 One fine morning, Beena started late by a few minutes and hence met Ravi who started on time at *D* in between *CB*. *CD* = 2.5 km. By how many minutes was Beena late in starting on that day?

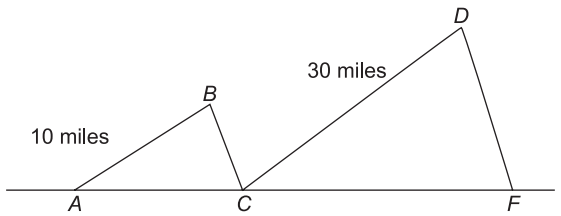
(a) 25 min (b) 30 min
(c) 55 min (d) Data insufficient

Q.22 One morning Ravi got delayed but Beena was on time and that day the two met at 7:00 am at *E* which was 2.5 km away from *C*. What is the distance between *A* and *B*?

(a) 20.25 km (b) 17.25 km
(c) 14.75 km (d) Data insufficient

Direction for Questions 23 and 24: Read the following passage and solve the questions based on it.

Given below is the structure of roads from *A* to *F*. *ABC* and *CDF* are two similar mountains with ideal straight lines *AB* and *CD* as their slant heights. $\angle BAC = \angle DCF = 30°$ and $\angle BCA = DFC = 60°$. Speed of Manoj Kumar uphill is 20 miles/h and downhill is 40 miles/h.



Q.23 What is the total time taken by him to travel from *A* to *F*?

(a) 3 h 12 min (b) 2 h 21 min
(c) 2 h 35 min (d) Cannot be determined

Q.24 Jonathan Livingston Seagull is flying between the points *B* and *D*. What is the distance travelled by Jonathan?

(a) $10\sqrt{\frac{28}{3}}$ miles (b) $10\sqrt{\frac{7}{3}}$ miles

(c) $\frac{10}{3}$ miles (d) None of these

Q.25 Ram and Rahim left Bombay simultaneously towards Pune. Speed of Ram is 15 km/h and the speed of Rahim is 12 km/h. Half an hour later, Mohan started from Bombay towards Pune on the same road in the same direction. After some time, he overtook Rahim and 90 mins further, he overtook Ram. What is Mohan's speed?

(a) 18 km/h (b) 24 km/h
(c) 32 km/h (d) 36 km/h

Direction for Questions 26 and 27: Read the passage below and solve the questions based on it.

Ankit and Sanket are travelling by bus from Ahmedabad to Kolkatta via Bangalore in a bus which leaves Ahmedabad at 8:00 am. The bus reaches Bangalore at 10:00 am, just in time for them to catch the last bus to Kolkata. However, Ankit decides to get off earlier (at point P_1) and travels to Bangalore by car. Because of this he reaches 15 min late. If he had got off at a distance 16 km ahead (at point P_2), he would have reached only 7 min late. Sanket also gets off at a point 16 km from Bangalore, whereas the bus continues at its normal speed of 40 km/h and the ratio of speeds of Ankit and Sanket is 15:16.

Q.26 If the bus takes 45 min to reach Bangalore from point P_1, Sanket will reach:

(a) Early by 6 min (b) Late by 6 min
(c) On time (d) Cannot be determined

Q.27 Had Sanket got off at P_2, at what time would he reach Bangalore?

(a) 10.05.15 (b) 10.05.05
(c) 9.54.45 (d) Cannot be determined

Q.28 A candidate misinterpreted the timings of an interview and reached the venue 35 min before the scheduled time. After 25 min, he presumed that the interview might have been cancelled and so he started walking back home at the speed of 1 km/h. But then he realized

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that he had left his bag there. So, he comes back with the same speed and enters the office along with the interviewer, who is 5 min late. How much distance had he moved away from the office?
(a) 7.5 km (b) 250 m (c) 125 m (d) 750 m

Q.29 Two cyclists are travelling at a speed of 3 km/h and 10 km/h around a circular track. What is the ratio of number of meeting points that would meet on that track if they travel in the same direction first and in the opposite direction next?
(a) 3:13 (b) 7:13
(c) 3:7 (c) Cannot be determined

Direction for Questions 30 and 31: Read the passage below and solve the questions based on it.

A jetfighter moving 3 km above the ground with a speed of 360 km/h tries to cross the border. A radar station notices it at distance of 6 km from the Jet. But it can fire anticraft missile only when the plane enters in its border. The border is at a distance of $\sqrt{3}$ km from radar station. (Assume that the fighter moves in a straight line in the same plane with the same speed)

Q.30 If it fires the missile at the fighter as soon as it notices the jetfighter such that it will hit the fighter as soon as it enters in the borderline, then at what speed should be missile fired?
(a) 360 km/h (b) 240 km/h
(c) $200\sqrt{3}$ km/h (d) None of these

Q.31 The radar station can fire 10 missiles simultaneously in such a way that the first missile will meet the fighter exactly at the border if the fighter moves with its original speed. The other missiles will reach up to the line of movement of the jet at same time, but at a distance of 100 m from each other such that one of them will hit the fighter even if the fighter increases its speed. If jetfighter has to escape the attack, at what increased speed it should move?
(a) 400 km/h (b) 545 km/h
(c) 464 km/h (d) 380 km/h

Q.32 Two trains going on a parallel line in opposite directions take 10 s to cross each other. But if they are going in the same direction, the longer train crosses the shorter train in 30 s. If the length of the longer train is decreased by 50%, the time taken to cross the shorter train while going in the same direction decreases by 8 s. Find the time taken by the longer train to cross a tunnel twice its length, if the difference between the length of the trains is 25 m?
(a) 30 s (b) 24 s (c) 40 s (d) 36 s

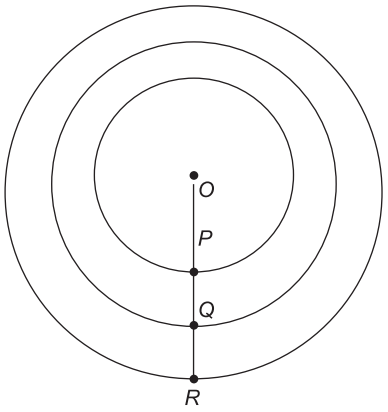
Q.33 A runner, Ajay runs twice on a circular track at a uniform speed. Vijay start at the same time as Ajay and after Vijay has completed one lap, he stops and Sanjay starts and completes the second lap at a uniform speed. It is given that they start the race exactly at 2:30 pm and the track is 500 m is circumference. Sanjay runs at a speed of 5 km/h greater than that of Vijay and Ajay's speed is 2 km/h greater than that of Vijay. When does Ajay finish the second lap, if Sanjay also finishes at the same time?
(a) 2:32 pm (b) 2:33 pm (c) 2:35 pm (d) 2:36 pm

Q.34 A man travels 1200 km from *A* to *B* by air, train and ship. The distance travelled by ship is twice the distance travelled by train. The average speed of the aircraft is twice the overall speed and the time he was on the ship was thrice the time he was on the train. If the man travels 10 h by train out of the total travel time of 50 h, then what is the total distance covered by the ship?
(a) 720 km (b) 630 km
(c) 480 km (d) 570 km

Q.35 Three friends Rupesh, Nishit, and Vijay planned to race with each other on their bikes. Rupesh has a lot of riding experience, while Nishit had just learned how to ride. The race seemed unfair. Therefore, they decided that the excellent biker Rupesh will form a team with Nishit. They would bike against the other rider Vijay, each riding half the total distance of 50 km. Also, Nishit can ride only half as fast as Vijay and Rupesh travels four times as faster as Nishit. What is the ratio of the time taken by the winner to that of the looser?
(a) 4:5 (b) 5:2 (c) 2:5 (d) 4:3

Direction for Questions 36 and 37: Read the passage below and solve the questions based on it.

Three persons *A*, *B*, and *C* start running simultaneously on three different concentric circular tracks from three collinear points *P*, *Q*, and R, respectively, which are collinear with the centre *O* and are on the same side of the centre as shown. The speeds of *A*, *B*, and *C* are 5 m/s, 9 m/s, and 8 m/s, respectively. The lengths of the tracks on which *A*, *B*, and *C* are running are 400 m, 600 m, and 800 m, respectively.



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Q.36 If A and B run in a clockwise direction and C in an anticlockwise direction, after how much time will the positions of A, B, and C be collinear with the centre (and on the same side of the centre), for the first time after they start?
(a) 200 s (b) 400 s (c) 600 s (d) 800 s

Q.37 If A runs in the clockwise direction and C in the anticlockwise direction, after how much time will the positions of A and C be collinear with the centre, (and on the same side), for the fourth time after they start?
(a) $177\frac{7}{9}$ s (b) 160 s (c) $145\frac{5}{11}$ s (d) 1

Direction for Questions 38 and 39: Read the passage below and solve the questions based on it.

A train started at 9:00 am from station X with a speed of 72 km/h. After 2 h, another train started from station Y towards X with a speed of 90 km/h. The two trains are expected to cross each other at 1:30 pm. Owing to a signal problem arising at 12 noon, the speed of each of them was reduced by the same quantity and they crossed each other at 4:30 pm.

Q.38 What is the new speed of the train that started from station X?
(a) 18 km/h (b) 36 km/h
(c) 45 km/h (d) 54 km/h

Q.39 If the signal problem had occurred at 1:00 pm instead of 12 noon, at what time would the two trains have crossed each other?
(a) 3:30 pm (b) 3:00 pm
(c) 2:00 pm (d) 2:30 pm

Direction for Questions 40 to 42: Read the passage below and solve the questions based on it.

P, Q, and R are three villages, located in the same order, along a highway. An absent minded person, Andrews, intending to reach R, starts from Q and travels at a speed of 60 km/h in the direction of QP instead of QR. At the same time as Andrews starts, Bill leaves R to reach Q and travels at the speed of 90 km/h in the direction of RQ. After travelling for 1 h in the direction of QP, Andrews realizes his mistake and reverses his direction and increases his speed to 90 km/h. After reaching Q, Bill reverses his direction and proceeds back to R at a reduced speed of 60 km/h. After reaching R, Andrews starts back towards Q with no change in speed. The distance PQ and QR are 162 km and 108 km, respectively.

Q.40 At what distance from Q, do Andrews and Bill meet for the first time?
(a) 72 km (b) 90 km (c) 84 km (d) 96 km

Q.41 At what distance from R, do the two meet for the second time?
(a) 9.6 km (b) 2.4 km (c) 4.8 km (d) 7.2 km

Q.42 Which of the two persons travelled more distance up to the point of their second meeting, and by how much?
(a) Bill by 33.6 km (b) Bill by 31.2 km
(c) Andrews by 16.8 km (d) Andrews by 21.6 km

Q.43 In a certain race, the winner beats the other six contestants by 10, 20, 30, 40, 50, 60 m, respectively. Find by what distance the third runner-up beats the fourth runner-up, if he was 11.11 m ahead of the fourth runner-up, when the first runner-up finished the race.
(a) 14.14 m (b) 14.28 m
(c) 16.66 m (d) None of these

Q.44 Distance between Lucknow and Patna is 300 km. Mayank leaves at a speed of x km/h from Lucknow towards Patna. After 3 h, Sharat leaves at the speed of $(x + 10)$ km/h from Lucknow towards Patna. If x and the number of hours taken to meet after Sharat starts are integers, how much distance can Mayank cover before they meet?
(a) 174 km (b) 60 km (c) 150 km (d) 180km

Q.45 Two champion swimmers start a two-length swimming race at the same time, but from the opposite ends of the pool. They swim at constant but different speeds. They first pass at a point 18.5 m from the deep end. Having completed one length, each swimmer takes a rest at the edge of the pool for 45 s. After setting off on the return length, the swimmers pass for the second time just 10.5 m from the shallow end. Therefore, the length of the pool is:
(a) 90 m (b) 45 m
(c) 26.5 m (d) Data insufficient

Q.46 A tortoise lays eggs on the beach and goes back to the sea where its mate is waiting. One way to reach the mate is going down 12 km on a straight line perpendicular to the sea, turning 90° and swimming for 5 km on a straight line. On its way down to the sea, the tortoise can cut the water at any point. The speed of the tortoise on land is 1 km/h and in water is 2 km/h. If the tortoise takes the described route, at what distance from the point where it had laid the eggs will the tortoise cut the water, if the total time taken is 12 h?
(a) 7 km (b) 8 km (c) 5 km (d) 4 km

Direction for Questions 47 and 48: Read the passage below and solve the questions based on it.

Trains X and Y start towards each other at the same time from points A and B for destinations B and A, respectively, that are 300 km apart. They meet each other 120 km from point A.

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Q.47 Another express train Z starts from point A to point B, 1 h after X starts. This express train meets train Y, 1.5 h after Z starts. If this express train is at least 20 km/h faster than the faster of X and Y, which of the following is true? (Ignore train lengths in calculations.)
(a) The minimum possible speed of Y is 60 km/h.
(b) The minimum possible speed of X is 45 km/h.
(c) The maximum possible speed of Y is 60 km/h.
(d) The maximum possible speed of X is 45 km/h.

Q.48 What is the minimum speed for train Z to overtake train X, before it meets train Y? (Use the data from previous question, if necessary.)
(a) 45 km/h
(b) 50 km/h
(c) 60 km/h
(d) Train Z will always overtake train X, before it meets train Y.

Q.49 A, B, and C start running a race from the same starting point at the same time in the same direction. A's speed around a path which is an equilateral triangle. B's path is a square and C's path is a regular hexagon. One edge of the triangular path, square path and hexagonal path completely overlaps with each other. If all of them complete one round at the same time then which of the following is true?
(a) Speed of C is twice that of the speed of B.
(b) Speed of A is half that of C.
(c) Speed of B is 50% more than that of A.
(d) None of these

Q.50 On a river, there is no current from A to B, but a current from B to C. A man rows down a stream from A to C in 3 h, and upstream from C to A in $3\frac{1}{2}$ h; had there been the same current in all the ways as from B to C his journey down stream would have occupied $2\frac{3}{4}$ h; find the length of the time, his return journey would have taken under the same circumstances.
(a) 4 h (b) 4.2 h
(c) $3\frac{17}{20}$ h (d) 2 h 24 min

True/False

1. If the ratio of speed of two trains and ratio of their lengths are given, we can find out the time taken by them to cross each other while passing in the same direction.
   State whether true or false.
2. Sum of lengths of two trains are given. Speed of the longer train is less than the speed of the smaller train and both the speeds are given. We can find out the time taken by these trains to cross each other while passing in the opposite direction.
   State whether true or false.
3. Two runners are running in opposite direction on a circular track starting from the same point in the same direction same time. Length of track is given. We can find out the distinct meeting points of the runners if the ratio of their speeds are given.
   State whether true or false.
4. Two runners A and B are running on a linear track. Length of linear track is given. B gives a start of 10 s to A and both of them finish the race in dead heat. With the information given we can find out the ratio of speed of A and B.
   State whether true or false.
5. There are five runners running on a circular track starting from the same point in the same direction same time. If we know the speeds of each of the runners and the distances between each of the possible distinct meeting points between any two of them is given, we can find out the length of track.
   State whether true or false.

Answers

Warm Up

1. (a)	**2.** (d)	**3.** (b)	**4.** (c)	**5.** (a)	**6.** (d)	**7.** (b)	**8.** (c)	**9.** (b)	**10.** (b)
11. (c)	**12.** (a)	**13.** (d)	**14.** (d)	**15.** (b)	**16.** (c)	**17.** (b)	**18.** (b)	**19.** (a)	**20.** (a)

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Foundation

1. (a)	**2.** (b)	**3.** (d)	**4.** (b)	**5.** (b)	**6.** (c)	**7.** (d)	**8.** (c)	**9.** (d)	**10.** (c)
11. (b)	**12.** (a)	**13.** (b)	**14.** (b)	**15.** (d)	**16.** (a)	**17.** (d)	**18.** (a)	**19.** (b)	**20.** (b)
21. (d)	**22.** (a)	**23.** (a)	**24.** (c)	**25.** (b)	**26.** (a)	**27.** (d)	**28.** (d)	**29.** (c)	**30.** (a)
31. (d)	**32.** (d)	**33.** (b)	**34.** (c)	**35.** (c)	**36.** (a)	**37.** (b)	**38.** (a)	**39.** (d)	**40.** (d)
41. (a)	**42.** (c)	**43.** (b)	**44.** (c)	**45.** (b)	**46.** (c)	**47.** (d)	**48.** (c)	**49.** (b)	**50.** (c)

Moderate

1. (b)	**2.** (d)	**3.** (b)	**4.** (a)	**5.** (c)	**6.** (d)	**7.** (a)	**8.** (d)	**9.** (d)	**10.** (c)
11. (d)	**12.** (c)	**13.** (a)	**14.** (c)	**15.** (d)	**16.** (a)	**17.** (b)	**18.** (b)	**19.** (c)	**20.** (a)
21. (d)	**22.** (d)	**23.** (c)	**24.** (c)	**25.** (d)	**26.** (b)	**27.** (c)	**28.** (a)	**29.** (c)	**30.** (b)
31. (b)	**32.** (b)	**33.** (c)	**34.** (c)	**35.** (d)	**36.** (d)	**37.** (d)	**38.** (a)	**39.** (c)	**40.** (c)
41. (d)	**42.** (b)	**43.** (b)	**44.** (a)	**45.** (b)	**46.** (d)	**47.** (a)	**48.** (b)	**49.** (d)	**50.** (b)

Advanced

1. (b)	**2.** (c)	**3.** (c)	**4.** (a)	**5.** (b)	**6.** (d)	**7.** (a)	**8.** (b)	**9.** (a)	**10.** (a)
11. (d)	**12.** (b)	**13.** (b)	**14.** (a)	**15.** (a)	**16.** (a)	**17.** (d)	**18.** (c)	**19.** (d)	**20.** (b)
21. (c)	**22.** (c)	**23.** (c)	**24.** (a)	**25.** (a)	**26.** (b)	**27.** (a)	**28.** (c)	**29.** (b)	**30.** (a)
31. (c)	**32.** (b)	**33.** (c)	**34.** (c)	**35.** (a)	**36.** (b)	**37.** (a)	**38.** (a)	**39.** (d)	**40.** (c)
41. (c)	**42.** (d)	**43.** (d)	**44.** (b)	**45.** (b)	**46.** (a)	**47.** (d)	**48.** (d)	**49.** (b)	**50.** (c)

True/False

1. False	**2.** True	**3.** True	**4.** False	**5.** True

Hints and Solutions

Warm up

1. Speed = 36 km/h = $36 \times \frac{5}{8}$ m/s = 10 m/s

In 40 s, he will cover 400 m.

2. Length of the train = Speed of train × time taken to cross a telephone pole = 20 × 10 = 200 m

3. Let length of platform = x

Distance = Length of train + Length of the platform = $400 + x$

$D = S \times T$

$400 + x = 15 \times 60$

$x = 900 \text{ m} - 400 \text{ m} = 500 \text{ m}$

4. Speed of train = 88 km/h = $88 \times \frac{5}{18}$ m/s

Distance = (150 + 180) m = 330 m.

Time = $\frac{\text{Distance}}{\text{Speed}} = \frac{330 \times 18}{88 \times 5} = 13.5$ s

5. Former's speed = v km/h

Latter's speed = $\frac{v}{4}$

Therefore, former speed:latter speed = 4:1

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6. Let speed of the person = x km/h.

Speed of the stream = y km/h

Upstream speed = $(x - y)$ km/h = 6 km/h

Downstream speed = $(x + y)$ km/h = 15 km/h. On solving, we get $x = 10.5$, $y = 4.5$

Hence, option (d) is the answer.

7. Speed of Anil's car = $\frac{60}{45} \times 60$ km/h = 80 km/h

New speed = 65 km/h

$$\text{Time} = \frac{\text{Distance}}{\text{Speed}} = \frac{60}{65} \times 60 \text{ min} = 55 \text{ min}$$

8. Distance covered by the car in 1 h = 65 km

New speed = 50 km/h

$$\text{Time} = \frac{\text{Distance}}{\text{Speed}} = \frac{60}{65} = 1 \text{ h } 18 \text{ min}$$

Hence, option (c) is the answer.

9. In a km, A beats B by 10 m. Therefore, in two kilometres A beats B by 20 m.

Hence, option (b) is the answer.

10. Speed of train = $45 \times \frac{5}{18}$ m/s = 12.5 m/s

Speed of man = $18 \times \frac{5}{18}$ 5 m/s

Relative speed = (12.5 − 5) m/s = 7.5 m/s

$$\text{Time} = \frac{\text{Distance}}{\text{Speed}} = \frac{225}{7.5} = 30 \text{ s.}$$

11. In same direction, their relative speed = (70 − 60) km/h = 10 km/h

Therefore, distance between the two trains = 10 × 6 = 60 km

12. Let speed of the man = x km/h

And speed of the current = y km/h

Therefore, against current relative speed = $(x - y)$ km/h = 2.5 km/h

And with current = $(x + y)$ km/h = 5 km/h.

On solving, we get $y = 1.25$ km/h

13. Distance covered by Amit = 1000 m and that covered by Bahadur = 900 m

Therefore, speed of Amit:speed of Bahadur = 1000:900 = 10:9

Similarly, speed of Bahadur:speed of Chandra = 1000:800 = 10:8

Therefore, speed of Amit:Bahadur:Chandra = 100:90:72

Therefore, in a kilometre race, Amit will travel 1000 m and Chandra will travel 720 m.

Hence, option (d) is the answer.

15. Relative speed = (50 − 30) km/h = 20 km/h = $20 \times \frac{5}{18}$ m/s = $\frac{100}{18}$ m/s

Length of the faster train = Speed × Time = $\frac{100}{18} \times 18$ m =100 m

16. Ratio of speed of Ajay:Bijay = 1000:900 = 10:9

Ratio of speed of Bijay:Chand = 1000:900 = 10:9

Therefore, ratio of speed of Ajay:Bijay:Chand = 100:90:80

Therefore, in a kilometre race Ajay beats Chand by 190 m.

Hence, option (c) is the answer.

17. Let us consider distance be = m.

And speed of A and B = 30 m/min and 20 m/min

$$\frac{d}{20} - \frac{d}{30} = 10$$

So, $D = 600$ m

Therefore, second person will take = 600/20 = 30 min.

18. Speed of train = $\frac{200}{15}$ m/s

Let length of platform = x m

Distance = Length of platform + Length of train = $x + 200$

Distance = Speed × Time = $\frac{200}{15} \times 60$

Or, $x + 200 = 800$

Hence, $x = 600$ m

Hence, option (b) is the answer.

20. Consider speed of hour and minute hand be 5 km/h and 60 km/h, respectively.

So, the relative speed = (60 − 5) km/h = 55 km/h

FOUNDATION

1. Using product stability ratio method: $D = S \times T$

Since speed has been increased by 20% or $\frac{1}{5}$, so time will reduce by $\frac{1}{6}$.

Now, $\frac{1}{6}$ of T (Time) = 20 min

$T = 120$ min = 2 h

2. Using product stability ratio method:

Case I Since speed has been increased from 10 km/h to 20 km/h, that is, by 100%.

So, time will reduce by 50%.

Now, 50% of T = 45 min.

Hence, $T = 90$ min

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Distance = $10 \times \frac{90}{60} = 15$ km

Case II Time taken to travel 15 km @ speed of 4 km/h

$= \frac{15}{40} \times 60$ min $= 22.5$ min

Time taken to travel 15 km @ speed of 25 km/h $= \frac{25}{15} \times 60$ min $= 36$ min

Therefore, difference between the time taken to travel will be = 13.5 min

3. Average speed $= \frac{\text{Total distance}}{\text{Total time taken}} = \frac{2ab}{a+b}$

[Where a, b are two speeds]

Hence, average speed of 120 km/h is not possible.

4. Here, total distance is 600 km.

Let speed of train be x km/h and speed of car be y km/h.

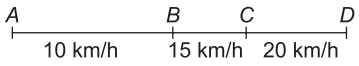
Now, $\frac{120}{x} + \frac{480}{y} = 8$

$\frac{200}{x} + \frac{400}{y} + \frac{25}{3}$

On solving, we get $x = 60$ km/h, $y = 80$ km/h

5. Let us assume that total distance is 60 km

Now,



Such that $BC = CD = \frac{1}{2}AB$

Now, AB will be 30 km, $t_1 = \frac{30}{10}$ 3 h

$BC = 15$ km, $t_2 = \frac{15}{15} = 1$ h

$CD = 20$ km $t_1 = \frac{15}{20} = 0.75$ h

Since, average speed $= \frac{\text{Total distance}}{\text{Total time}} = \frac{60}{3+1+0.75} = \frac{60}{4.75}$

$= 12.63$ km/h

6. Let the speed of Shilu in still water be x km/h, and speed of stream be y km/h.

Upstream speed = $(x - y)$ km/h

Downstream speed = $(x + y)$ km/h

According to the question,

$\frac{30}{x-y} + \frac{44}{x+y} = 10$ h

$\frac{40}{x-y} + \frac{55}{x+y} = 13$ h

On solving, we get $x = 8$ km/h and $y = 3$ km/h

7. Average speed $= \frac{\text{Total distance}}{\text{Total time taken}}$

Total distance = 192 + 192 = 384 km

$t_1 = \frac{192}{80} = 2.4$ h, $t_2 = 5$ h 36 min $= 5.6$ h

Total time taken = 5.6 + 2.4 = 8 h

Therefore, average speed $= \frac{384}{8} = 48$ km/h

8. Distance between them is 27 km.

Let the speed of first person be x km/h and the speed of second person be y km/h.

Case I In same direction

Relative speed = $x - y$ km/h

Therefore, $\frac{27}{x-y} = 9$, $x - y = \frac{27}{9} = 3$

Case II In opposite direction,

Relative speed = $(x + y)$ km/h

Therefore, $\frac{27}{x+y} = 9$, $x + y = \frac{27}{3} = 9$

Now, $x + y = 9$

$x - y = 3$

On solving, we get $x = 6$ km/h and $y = 3$ km/h

9. Abhishek beats Bijay by 30 m in a race of 300 m,

That is, ratio of speed of Abhishek:Bijay = 300:270 = 10:9.

Similarly, ratio of speed of Bijay:Chandan = 300:250 = 6:5

Therefore, ratio of speed of Abhishek:Bijay:Chandan = 20:18:15

Therefore, in a race of 300 m, Abhishek will travel 300 m and Chandan will travel 225 m.

Hence, Abhishek beats Chandan by 75 m.

11. Let speed of C be x km/h and speed of D be y km/h.

Now, $12y - 11x = 12.5$.

So, $7x - 5y = 3.25$

On solving, we get $x = 3.5$ km/h and $y = 4.25$ km/h

12. Let the distance between P and Q be 90 km.

Now, speed of Vikrant = $\frac{90}{6}$ = 15 km/h and speed of

Virat $= \frac{2}{3} \times 15 = 10$ km/h

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Since Vikrant starts at noon and Virat starts at 1:00 pm, distance travelled by Vikrant in 1 h = 15 km.

Now, distance between them is 90 − 15 = 75 km.

Relative speed = 15 + 10 = 25 km/h

Time = $\frac{75}{25}$ = 3 h

Therefore, the two ships expected to meet at 4 pm.

13. Distance between Patna and Quillon is 300 km.

Speed of Amit is 50 km/h and speed of Bhartendu will be = 75 km/h

Since Amit starts at 9 am and Bhartendu starts at 10 am, distance travelled by Amit in 1 h = 50 × 1 = 50 km

Now, distance between them = 300 − 50 = 250 km. Relative speed = 125 km

So, time taken = $\frac{250}{125}$ = 2 h

Hence, they will meet at 12 noon.

15. Let speed of boat = x km/h

Speed of stream = y km/h

Upstream speed = $(x - y)$ km/h

Downstream speed = $(x + y)$ km/h

Since $\frac{D}{x-y} = \frac{2D}{x+y}$

$\frac{x+y}{x-y} = \frac{2}{1}$

$x:y = 3:1$.

17. Let speed be x km/h

Case I $\frac{p}{x} - \frac{p}{x+3} = \frac{40}{60}$

Case II $\frac{p}{x-2} - \frac{p}{x} = \frac{40}{60}$

$\frac{p}{x} - \frac{p}{x+3} = \frac{p}{x-2} - \frac{p}{x}$

$\frac{2p}{x} = P\left(\frac{1}{x-2} + \frac{1}{x+3}\right)$

$\frac{2}{x} = \frac{2x+1}{x^2+x-6}$

$2x^2 + 2x - 12 = 2x^2 + x$

x =12 km/h

Therefore, $\frac{P}{12} - \frac{P}{15} = \frac{2}{3}$, or, $\frac{3}{60} = \frac{2}{3}$

Hence, P = 40 km.

18. Let distance between A and B be x km.

Speed of two trains are 36 km/h and 42 km/h.

Relative speed = 78 km/h

Time taken = $\frac{x}{78}$ km/h

Now, first train has travelled = $\frac{x}{78} \times 36 = \frac{36x}{78}$ km

Second train has travelled $\frac{42x}{78}$ km.

Hence, $\frac{42x}{78} - \frac{36x}{78} = 48$

or $\frac{6x}{78} = 48$

Hence, $x = 78 \times 8 = 624$ km

19. Ratio of speed of Anand:Bidhan = 100:95 = 20:19
Ratio of speed of Bidhan:Chandan = 200:190 = 20:19
Therefore, ratio of speed of Anand:Bidhan:Chandan = 400:380:361

Hence, Anand can beat Chandan by 39 m.

20. Let speed of man be x km/h and speed of stream be y km/h.

Given that $x + y = 3$ and $x - y = 2$

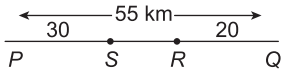
Hence, $x = \frac{5}{2}$ km/h.

Therefore, to travel 7 km in still water, he will take

$\frac{7}{5/2} = 2.8$ h

21. As we do not know, the exact positions where the road is perpendicular to PQ, it cannot be determined.

22.



Here, relative speed will be = 5.5 km/h

First meeting point is 'R'.

That is, = $\frac{55}{5.5\text{km/h}} = 10\text{h}$

That means PR = 30 km, and QP = 25 km

Second meeting point is 'S' = $\frac{110}{5.5} = 20$ h

That means A will travel 60 km from 'R',

That is, RS = 10 km

23. As $A:B = 1.75:1 = 7:4$

A gives B a start of 60 m

Let the winning post be x km

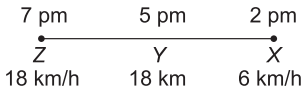
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Speed of $A:B = x:(x-60)$

That is, $\frac{x}{x-60} = \frac{7}{4}$, So, $x = 140$ km

24.



Distance travelled by X in 3 h $= 6 \times 3 = 18$ km

Relative speed between X and $Y = 18 - 6 = 12$ km/h

Therefore, they will meet at $= \frac{18}{12} = 1.5$ h

Hence, 6:30 pm is the answer.

25. **Case I** Relative speed $= \frac{\text{Length of train}}{\text{Time taken}} = \frac{75}{15/2} = 10\text{m/s}$

$10\,\text{m/s} \times \frac{18}{5} = 36$ km/h

Speed of train $= 36 - 6 = 30$ km/h

Now, distance travelled by X in 1.5 h is $= 6 \times \frac{3}{2} = 9$ km

Therefore, at 6:30 pm, X has travelled = (9 + 18) km = 27 km

Case II With second person:

Relative speed $= \frac{75}{27/4} = \frac{300}{27}$ m/s

Or, $\frac{300}{27}\frac{\text{m}}{\text{s}} = \frac{300}{27} \times \frac{18}{5}\text{km/h} = 40\text{km/h}$

Thus, speed of second person $= 40 - 30 = 10$ km/h

26. Speed of stream = 2 km/h

Let speed of boat = x km/h

According to condition $\frac{9}{x-2} + \frac{9}{x+2} = 6$

On solving, we get $x = 4$ km/h

27. Let speed of man be x km/h

Time taken to travel 12 km $= \frac{12}{x}$h

Time taken to travel further 6 km $= \frac{6}{x+0.5}$ h

Now, according to question, Time taken to travel at faster rate $= \frac{18}{x+0.5}$

$\frac{12}{x} + \frac{6}{x+0.5} - \frac{18}{x+0.5} = \frac{20}{60}$

$\frac{12}{x} - \frac{12}{x+0.5} = \frac{1}{3}$

$12\left(\frac{1}{x} - \frac{1}{x+0.5}\right) = \frac{1}{3}$

$\frac{x+0.5-2}{x(x+0.5)} = \frac{1}{36}$

Or, $0.5 \times 36 = x(x+0.5)$

$18 = x^2 + \frac{x}{2}$

$2x^2 + x - 36 = 0$

On solving we get $x = 4$

Thus, he will take $\frac{18}{4} = 4.5$ h.

29. Ratio of speed of $A:B = 200:180 = 10:9$

Ratio of speed of $B:C = 200:190 = 20:19$

Thus, ratio of speed of $A:B:C = 200:180:171$

Therefore, A beats C by 29 m.

31. Let the distance between their residence and the school be x m.

Speed of Preeti is 2 km/h and speed of Uma is 3 km/h

As given, Preeti has travelled 200 m.

Time taken to travel 200 m will be $\frac{200 \times 2}{1000 \times 2} = 0.1$ h

Then, distance covered by Uma in 0.1 h $= 3 \times 0.1 = 0.3$ km $= 300$ m

Total distance between school and residence will be $\frac{300+200}{2} = \frac{500}{2} = 250\,\text{m}$

32. Here total distance = 7 km, and total time = 1 h 12 min

Let us assume he walks x km.

Time taken to cover x km $= \frac{x}{4}$

Time taken to cover the remaining distance $= \frac{(7-x)}{10}$

$\frac{x}{4} + \frac{7-x}{10} = 1\,\text{h } 12\,\text{min}$

On solving, we get $x = \frac{10}{3}$km

33. A's speed is 50 km/h and B's speed will be 75 km/h.

Since A starts at 9 am, the distance travelled by $A = 50$ km

Relative speed = 50 + 75 = 125 km/h

Now, distance between them at 10 am $= 300 - 50 = 250$ km

Time $= \frac{250}{125}$ 2 h

Thus, they will meet at 12 noon.

34. Ratio of speed of Binod:Neeraj = 3:2

Ratio of speed of Deleep:Neeraj = 5:2

Ratio of speed of Binod:Neeraj:Deleep = 3:2:5

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Let their speed be $3x$, $2x$, and $5x$.

Relative speed of Binod and Deleep will be = $3x + 5x = 8x$

And $\frac{1000}{8x} = 25\text{s}$

On solving ($x = 5$)

Thus, their speed will be 15, 10, and 25.

Now, Neeraj and Deleep will meet at = $\frac{1000}{35} = 28\frac{4}{7}$ s

35. Distance travelled by Vinod in 4 h = $8 \times 4 = 32$ km

And distance travelled by Kaurvaki in 4 h = $13 \times 4 = 52$ km

New speed of Vinod = 16 km/h

And also new speed of Kaurvaki = 12 km/h

Now, distance between Vinod and Kaurvaki = 52 km = 32 km = 20 km

Relative speed = 4 km/h

Time = $\frac{20}{4} = 5$ h

Total time = 4 + 5 = 9 h.

39. Ratio of speed of A:B = 10:8

Ratio of speed of B:C = 10:8

Ratio of speed of A:B:C = 100:80:64

Thus, A should give 36 m head start to C.

41. Let the speed of two trains be x m/s and y m/s

Case I In opposite direction:

Relative speed will be = $(x + y)$ m/s

That is, $\frac{230+190}{x+y} = 21\text{s}$.

$\frac{420}{x+y} = 21$

$x + y = 20$ (i)

Case II In same direction:

Relative speed = $(x - y)$ m/s

That is, $\frac{420}{42} = x - y$

Hence, $x - y = 10$ (ii)

On solving (i) and (ii), we get $x = 15$ m/s, $y = 5$ m/s

Hence, the ratio = 3:1

48. Let speed of man be x km/h

And speed of current be y km/h

Upstream speed = $(x - y)$ km/h

And downstream = $(x + y)$ km/h

Now, $\frac{2}{x-y} = 15$ min

or $x - y = \frac{2}{15} \times 60$. Hence, $x - y = 8$ km/h

Similarly, $\frac{2}{x+y} = 10$ min

Hence, $x + y = \frac{2}{10} \times 60$. So, $x + y = 12$ km/h

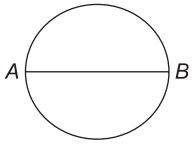
On solving, we get $x = 10$ km/h, $y = 2$ km/h

Now, if the speed of current is doubled,

Upstream speed = 10 − 4 = 6 km/h

Therefore, time $\frac{2}{6} \times 60$ min = 20 min

49.



Let the speed with which the cyclist made the first round be x km/h

Then, the speed for second round = $(x - 3)$ km/h

Time taken to reach 'B' $\frac{3}{x}$

Again, the time taken to reach 'B' in the second round

$= \frac{3}{x} + \frac{3}{x-3}$

As given x $\frac{3}{x} + \frac{3}{x-3} = \frac{50}{60}$

On solving we get, $x = 9$ km/h

Moderate

1. Speed of train while passing point $A = 70 \times (5/18)$ m/s = V_1
Speed of bike initially = $70 \times (5/18)$ m/s = V_2

Time taken by the bike to reach at the mid-point of the train = $150/(V_2 - V_1)$

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Again find out the new speeds of train and bike and calculate the time taken by the bike to cover the rest 150 m distance relative to the train.

2. As $r = \frac{7}{2}$ units

Circumference $2\pi r = 2 \times \frac{22}{7} \times \frac{7}{2} = 22$ units

First, they will cover 11 units with relative speed = 11 units/min

Time $= \frac{11}{11} = 1$ min

Now, they will cover 22 units with twice their relative Speed. Hence,

Time

First meeting rest 1 min $\frac{1}{2}$ min	Second meeting rest 1 min $\frac{1}{2}$ min	Third meeting rest $\frac{1}{2}$ min $\frac{1}{2}$min
Fourth meeting rest $\frac{1}{4}$ min $\frac{1}{2}$ min	Fifth meeting rest $\frac{1}{8}$ min $\frac{1}{2}$min	

Therefore, in 5 min, they will take rest five times.

Hence, option (d) is the answer.

3. This question has actually got nothing to do with bullets initially. We can see that it takes them 4 h to reach each other. And this is the same time for which bullets will cover some distance.

So, the total distance covered by the bullet = $4 \times 10 = 40$ km

4. Form the equations first and then use the options.

5. Go through the actual calculation method.

6. Route 1: Total time = 4.5 h + 3 h + 0.5 h = 8 h

Route 2: Total time = $\frac{300}{40}$ + 0.5 h = 7.5 h + 0.5 h = 8 h.

Route 2 will take more than 8 h.

Hence, option (d) is the answer.

7. Let the speed of fast train be = x km/h

And the speed of slow train = y km/h

Then, length of fast train = $3y$ m

Length of slow train = $3x$ m

Relative speed = $x + y$ km/h

$(x + y) \times \frac{18}{5}$ m/s.

Time $= \frac{3x + 3y}{(x + y)18/5}$

Time = 10.8 s

8. He takes 3 s to get to the next floor.

Up to 480 stairs or 40 floors, his speed = 3 sec/floor. For next 50 floors, his speed = 4.5 second/floor. Therefore, he should run till 40th floor and then take the lift.

9. With 60 miles/h, it goes 20 miles on a gallon @ 70 miles/h.

It will go 14 miles on a gallon.

Therefore, for 210 miles, 15 gallon of gas will be required.

10. Let the speed of Siddharth and Rakesh be x and y km/h, respectively.

And the speed of stream = a km/h

Case I As Rakesh rows downstream and Siddharth rows upstream:

That is, $(x - a) \times \frac{15}{60} + (y + a)\frac{15}{60} = 2.25$

$x - a + y + a = 9$ (i)

$x + y = 9$

Case II $(x - a) \times \frac{15}{60} + (y - a)\frac{15}{60} = 1.25$

$x - a + y - a = 5$

$(x + y - 2a = 5)$ (ii)

On solving (i) and (ii)

We get $a = 2$ km/h

11. Total time taken by Sapna

Car	Repair	Truck	Running
$3\frac{1}{2}$ h	$\frac{1}{2}$ h	11.11 h	3.33 h

Therefore, total time taken = 3.5 + 0.5 + 11.11 + 3.33 = 18.44 h

12. Total distance from Mumbai to Delhi = 910 km. Time taken by Sapna to cover this distance had there been no snag = 910/60

Since their speeds are in the ratio 3:2, Manoj would have covered [910 − (910 × 40)/60] km less than Sapna.

13. The train can cover (200 + 350) m distance in 5 s which means the speed of the train is 110 m/s. Relative speed of man and train is 114 m/s. To cover the distance of 100 m, it will take less than 1 s.

14. Go through the options.

15. Let us consider the room has dimension $1 \times 1 \times 1$ m^3

Now, diagonal of the cube = $\sqrt{3}$ m

Distance covered by John = $\sqrt{3}$ m

Distance covered by Mac = $\sqrt{1+4} = \sqrt{5}$ m

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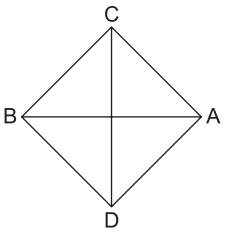
Distance covered by Roe = 1 + 1 + 1 = 3 m

Since time taken by all the three cockroaches is same.

Hence, ratio of their speed = $\sqrt{3} : \sqrt{5} : 3$.

Hence, option (d) is the answer.

16.



Since diagonals bisect at 90° and diagonals are equal.

Hence, it will be a square.

17. Let the speed of Bidhan be x km/h

Then, speed of Saket = $(x + 2)$ km/h

And speed of Sunny = $(x + 5)$ km/h

Time taken by Saket = $\frac{2 \times 0.5}{x+2}$ h

Time taken by Bidhan = $\frac{0.5}{x}$ h

Time taken by Sunny = $\frac{0.5}{x+5}$ h

Now, $\frac{2 \times 0.5}{x+2} = \frac{0.5}{x} + \frac{0.5}{(x+5)}$

$\frac{1}{x+2} = \frac{1}{2x} + \frac{1}{2(x+5)}$

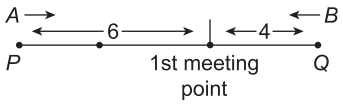
On solving, we get

$x = 10$ km/h

Hence, Saket's speed = 12 km/h

Therefore, Saket will finish the lap at 2:35 pm.

18.



Speed of $A:B$ = 3:2

Let us consider D = 10 m

First meeting point = 6 m from P

Distance covered by both of them for second meeting

10 m + 10 m = 20 m

Second meeting point = $\frac{20}{5} = 4$

That is, A will travel 12 m and B will travel 8 m.

Second meeting point will be 2 m from P.

Similarly, distance covered by both of them for third meeting = 20 + 10 = 30 m

Third meeting point = $\frac{30}{5} = 6$ m

That is, A will travel 18 m.

Therefore, we will get three distinct points.

Hence, option (b) is the answer.

19. With the speed of 45 km/h, mileage is 14 km in 1l.

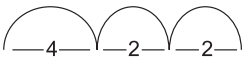
Therefore, in 2 h, he has covered 90 km.

In 90 km, he has used $6\frac{6}{14}$l of petrol.

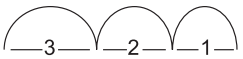
Now, maximum mileage is 21 km with 80 km/h. Therefore, he can cover 75 km further.

Hence, option (c) is the answer.

21. A's jump



B's jump



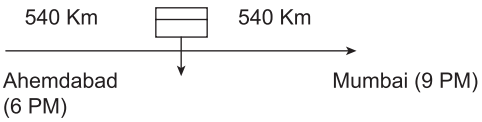
Therefore, distance covered by A in three jumps is 8 m. And distance covered by B = 6 m.
Therefore, A will win the first race and B will win the second race.

Hence, option (c) is the answer.

23. Let the original speed be X km/h

According to the question, $18/(4/5x) - 18/x = 9/60$ h $x = 30$ km/h

24.



Now, the result is obtained using the options.
Take the option 6 am which means the train from Ahmedabad takes 12 h to cover 540 km. In this way, the speed will be 45 km/h and train from Mumbai takes 9 h to cover 540 km which means the speed is 60 km/h. It is written in the question that the difference between the speed of the train from Ahmedabad and that from Mumbai is 15 km/h. Hence, this is the answer.

25. Total distance covered by Bharat and Akhil = 1800 m and 1719 m, respectively.

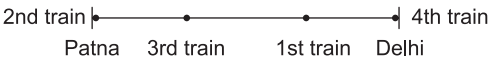
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Therefore, speed of Akhil:Bharat = 1719:1800 = 0.95
Also, 141:150 = 0.95

Hence, option (d) is the answer.

26. If the duration is more than 24 h between any two stations, then at least four trains will be required.



27.



Both the trains are crossing each other after 12 min, which means that the distance between them is 5 km. It will be easy to go through the ratio of the speed which is 2:3. Hence, the answer is 36 min.

28. Let us assume that the width of the lake = x. So, when one of the runners A covers 900 m, the other one B is covering $(x - 900)$ m. To meet next time, A will be covering $(x - 900 + 300)$ m, whereas B will be covering $(900 + x - 300)$ m.

Now, $900/(x - 900) = (x - 900 + 300)/(x + 900 - 300)$
Now, use options to find the answer.

30. Front wheels make 0.6 revolution per second. And rear wheels make 0.3 revolution per second. Therefore, in 3.33 s, rear wheel will complete one revolution while front wheel will complete two revolutions.

Hence, option (b) is the answer.

31. Speed of train = 1 km/h

And distance covered by train = 1 km.
That means the total duration was 1 h.
Let us consider speed of car be x km/h.

Total time taken by car $= \frac{1}{x+1} + \frac{1}{x-1} = 1$

On solving, we get $(x = \sqrt{2} + 1)$ km/h

Therefore, distance covered in 1 h by car $= \sqrt{2} + 1$ km

Hence, option (b) is the answer.

35. The ratio should be greater than 1.

Hence, option (d) is the answer.

36. Go through the options.

37. This question gives us the freedom to assume any value of speeds of Ramesh and Somesh. Let us assume the initial speed of Somesh = 20 m/s, then the initial speed of Ramesh = 40 m/s. Till 50 m, they are running with this speed only. Time taken by Ramesh in covering 50 m = 1.25 s. In the same time, Somesh is covering 25 m.

After this stage, speed of Somesh is 20 m/s, whereas speed of Ramesh = 10 m/s. Now, relative speed = 10 m/s and distance = 25 m. At 75 m from the starting, both of them will be meeting.

38. For seventh meeting, both of them together must have covered a total of 13 D distance, where D is the distance between P and Q. Now, ratio of their speeds is 3:4, and so, distance covered will also be in the same ratio.

Hence, distance covered by $B = 13D \times \frac{4}{7} = 1040$

Hence, D = 140 m

39. If A covers 1860 m, then in the same time, B covers 2480 m (ratio of speed 3:4). So, total distance covered = 4340 m = 140 m + 30 × 140 m = 31 × 140 m, this is the distance covered for 16th meeting.

40. Assume that the Howrah mail is leaving on 13th November at 12 noon. It will be reaching its destination on 16th November at 12 noon. Now, any train that leaves Trivandrum between the 10th and 16th November (excluding extremes) will be meeting Howrah mail on the track. These are five trains.

42. Relative speed of Vinay and Prabhat will be 20 m/min to cover the track of 960 m. It will take 48 min.

43. The train has to cover 1100 m @ 90 km/h and to cover that it will take 44 s to cover the distance 700 m by jeep and not to collide with each other the jeep has to take either less than 44 s or more than that.

45. At 2 O'clock Aflatoon has already covered 16 km @ 8 km/h, Bablajoon starts running in the same direction @ 10 km/h. The relative speed is 2 km/h. They will be 5 km apart at 7:30 pm the same day and 12:30 am on the next day.

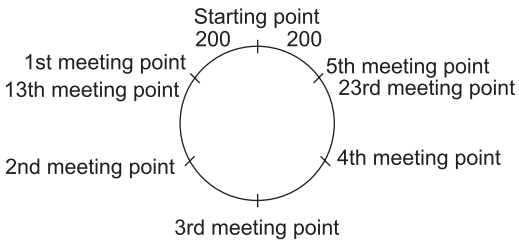
47. Let speed of Rishu be x km/h and speed of the river be y km/h.

Case I $= \frac{18}{x+y} - \frac{18}{x+y} = 9$

Case II $= \frac{18}{2x-y} - \frac{18}{2x+y} = 1$

On solving we get $y = \frac{20}{3}$ km/h

48.



Thus, the shorter distance = 400 m.

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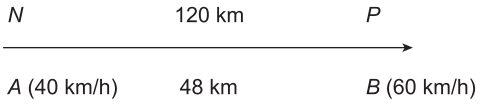
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1. Since there are 36 h in a day, 18 h dial will be there in the clock.

At 16:50, the hour hand will be in between 10 and 11 of the actual clock and the minute hand will be in between 6 and 7 of the actual clock. Now, we can eliminate the options.

8. Assume any value of S and solve.

Answers to Q.9 to 12:



Ratio of relative speed	2:3
Distance travelled by A	$[2/5] \times 120 = 48$ km
Total distance travelled by B	$[3/5] \times 120 = 72$ km

9. According to the question:

First time meet at 48 km and reverse the direction with interchanging speed

$T = [48 \text{ km}/40]\ 60 = 72$ min (reverse T) and second time $= [7260]/60 = 72$

10. First time B covers 72 km. In reverse, B covers $(72 + 72)$ km. Hence, the total distance is 216 km.

11. Assume that the track length is 1000 m.

Now, runner 1 and runner 2 will meet at one point, that is, the starting point.

Runner 1 and runner 3 will meet at two points, at 500 m and at the starting point.

Runner 1 and runner 4 will meet at three points, at 333.33 m, at 666.66 m and at the starting point.

Runner 1 and runner 5 will meet at four points, at 250 m, 500 m, 750 m and at starting point.

Runner 1 and runner 6 will meet at five points, at 200 m, at 400 m, at 600 m, at 800 m and at the starting point.

These are 10 distinct points.

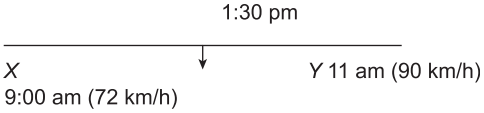
12. Do as done in Q.11.

13. If Amit would have increased his speed by 25% in the beginning, he would have saved 1 h in covering the actual planned distance. So, $1/5\ T = 1$ h. (where T is the actual planned time). Hence, $T = 5$ h

14. Saving in time would be in the later 4 h part. Since speed is increase by 25%, time taken to cover the same distance would reduce by 20%. Hence, he would save 48 min.

Answers to Q.15 and 16:

15.



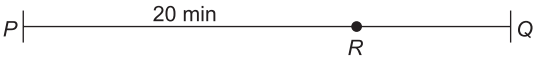
They cross each other at 1:30 pm which means the distance is 549 km.

At 12 noon, a signal was down, distance travelled by x and y at 12 noon is 306 km. Rest of the distance will be covered in 4.5 h which mean the relative speed is 54 km. After reducing an equal speed from the speed of both the trains, the equation is like this: $72 - p + 90 - p = 54$, so, $p = 54$.

Hence, the answer is 18 km/h.

16. Suppose that the signal problem occurred at 1 pm, the distance travelled by them is $(72 \times 4 + 19 \times 2)$ km. Rest of the distance will be covered by reduced speed.

17.



Let us consider BC courier and airport at P and Q, respectively.

And the cyclist meets BC person at 'R'.

Time taken by BC person to cover $PR = 20$ min.

Since he saves 10 m, therefore, he can cover the total distance in 50 min.

Therefore, the delay in the landing of the plane $= (20 + 5) = 25$ min

Hence, option (d) is the answer.

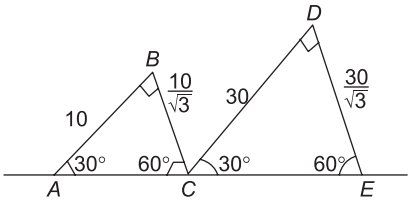
18. Ratio of speed of motorcyclist:cyclist = 4:1

$$\text{Speed of cyclist} = \frac{2.5}{20} \times 60 \text{ km/h} = 7.5 \text{ km/h}$$

Speed of motorcyclist $= 7.5 \times 4 = 30$ km/h

Hence, option (c) is the answer.

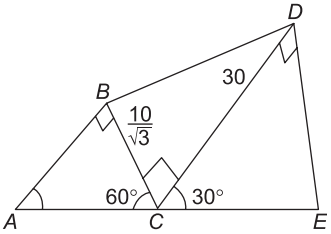
23.



$$\text{Total time taken} = \frac{10}{20} + \frac{10/\sqrt{3}}{40} + \frac{30}{20} + \frac{30/\sqrt{3}}{40} + \frac{40}{20} + \frac{40/\sqrt{3}}{40}$$

$= 2 + 0.58 = 2.58$ h $= 2$ h 35 min

24.



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Using Cos $C = \dfrac{BC^2 + CD^2 - BD^2}{2BC\,CD}$

$$\text{Cos}\,90^\circ = \frac{\left(\frac{10}{\sqrt{3}}\right)^2 + 30^2 - BD^2}{2 \times \frac{10}{\sqrt{3}} \times 30}$$

or $0 = \dfrac{100}{3} + 900 - BD^2$

or $BD^2 = \dfrac{2800}{3}$

or $BD = 10\sqrt{\dfrac{28}{3}}$ miles

Hence, option (a) is the answer.

25. Distance covered by Ram and Rahim in half an hour is 7.5 km and 6 km, respectively. Let speed of Mohan be x km/h as Mohan overtook Rahim and then Ram.

So, $\dfrac{7.5}{x-15} - \dfrac{6}{x-12} = \dfrac{90}{60}$ h

On solving, we get $x = 18$ km/h

Hence, option (a) is the answer.

28. The total duration he walked = 10 + 5 = 15 min. Let 'd' be the distance he moved away from the office.

Therefore, $\dfrac{d}{1} + \dfrac{d}{1} = 15$ min, or, $d = \dfrac{1}{8}$ km = 125 m

Hence, option (c) is the answer.

29. In same direction:

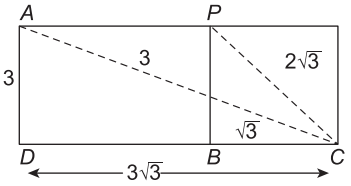
Number of meeting points = difference between their speed = 10 − 3 = 7. In opposite direction:

Number of meeting points = Sum of their speed = 10 + 3 = 13

Therefore, ratio = 7:13

Hence, option (b) is the answer.

30.



Let 'A' be the point the radar station notices the jet plane.

C and B be the points at which the radar station and its border, respectively.

Since $AC = 6$ km, $AQ = 3$ km

Therefore, $QC = 3\sqrt{3}$ km. And also $PC = 2\sqrt{3}$ km

$AP = QC - BC = 3\sqrt{3} - \sqrt{3}$

$AP = 2\sqrt{3}$ km

Therefore, distance travelled by jet fighter = $AP = 2\sqrt{3}$ km.

And also distance travelled by missile = $PC = 2\sqrt{3}$ km

Therefore, speed of missile = speed of jet fighter [as both cover same distance in equal internal of time]

Hence, option (a) is the answer.

32. Let the length of shorter and longer trains be x, $x + 25$ m

And their speed be y and x m/s, respectively

Case I In opposite direction

$$\frac{d + d + 25}{10} = x + y$$
$$\frac{2d + 25}{10} = x + y \qquad \text{(i)}$$

Case II In same direction $2d$

$$\frac{2d + 25}{30} = x - y \qquad \text{(ii)}$$

Case III In same direction with decreased length

$$\frac{d + \frac{d + 25}{2}}{22} = x - y \qquad \text{(iii)}$$

On solving (i), (ii), and (iii)

We get $d = 175$ m, $x = 25$ m/s, $y = 12.5$ m/s

Therefore, the length of two trains be 175,200 m.

Now, the length of platform = 2 × 200 = 400 m

Therefore, distance = 400 + 200 m

Time taken to cross tunnel = $\dfrac{600}{25} = 24$ s

Hence, option (b) is the answer.

36. It does not create any difference whether they are running in the clockwise direction or anticlockwise direction.

44. One of the ways of solving this question is going through equations. But after a certain stages, we will be required to start assuming the values because all the data are not given.

Another way of doing this problem is: start working by assuming some values. Let us assume the speed of Mayank = 10 km/h. In 3 h, he has covered 30 km. Now, Sharat starts with a speed of 20 km/h. He will take 3 h to meet Mayank. Till that time, the total distance covered by Mayank will be 60 km.

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BENCHMARKING TEST 2

This test paper contains 30 questions of 4 marks each. One-third of the marks allotted to a particular question will be deducted in case of wrong answer.

Time Given: 70 Minutes **Total Marks: 120** **Qualifying Marks: 55 Marks**

Break-up of the Expected Percentile

(assuming this to be a CAT paper)

Marks	Expected percentile
88+	99+
83	98.5+
78	98
55	92
40	85
30	80

Direction for Questions 1 and 2: Read the passage given below and solve the questions based on it.

A pedestrian having travelled from *A* to *B* found that if his rate had been 1 km/h quicker, he would have completed his journey in 11/2 h less and if it had been 1 km/h slower, he would have taken 2 h more to complete it.

Q.1 The distance from *A* to *B* is:
(a) 12 km (b) 20 km (c) 26 km (d) 30 km

Q.2 The rate at which, the pedestrian covered the distance is:
(a) 2 km/h (b) 3 km/h (c) 4 km/h (d) 6 km/h

Q.3 There were 500 seats in PVR placed in similar rows. After the reconstruction of the hall, the total number of seats became 450. The number of rows was reduced by 5, but each row contained five seats more than the earlier setting. What is the total number of rows and total numbers of seats in a row initially in the hall?
(a) 30 and 15 (b) 25 and 20
(c) 20 and 25 (d) 50 and 10

Q.4 Krishna owns several cows, some black and some white. He finds that 4 black cows and 3 white cows provide the same amount of milk in 5 days as 3 black cows and 5 white cows provide in 4 days. What is the ratio of milk provided by a black cow in a day to that by a white cow in a day?
(a) 8 :5 (b) 5:8 (c) 3:5 (d) 5:3

Q.5 Two Cities *P* and *Q* are 60 km apart. A college is to be built to serve 150 students in City *P* and 50 students in City *Q*. If the total distance to be travelled by all 200 students is to be as small as possible, and then the college should be built at:
(a) City *P* (b) City *Q*
(c) 45 km from City *P* (d) 45 km from City *Q*

Q.6 When ACP Rathod was going to Wahed from New Delhi to arrest Mirchi Seth, he observed few surprising things through his rear-view mirror:
i. In *n*th hour, he saw inscribed on a milestone—New Delhi—*xy* km, where *x* and *y* are the 10s and units place digit of the distance from that place to New Delhi (remember, he was observing it in rear-view mirror).
ii. In $(n+1)$th hour, he saw inscribed on a milestone—New Delhi—*yx* km, where *x* and *y* are representing the same digits.
What is the speed of ACP Rathod?
(a) 9 km/h (b) 12 km/h
(c) 18 km/h (d) Cannot be determined

Q.7 *A* and *B* start from the same point and in the same direction at 7 am to walk around a rectangular field 400 m × 300 m. *A* and *B* walk at the rate of 3 km/h and 2.5 km/h, respectively. How many times shall they cross each other, if they continue to walk till 12:30 pm?
(a) Not even once (b) Once
(c) Twice (d) Thrice

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Q.8 Left pan of a faulty balance weighs 100 g more than its right pan. A shopkeeper keeps the weight measure in the left pan while buying goods but keeps it in the right pan while selling his goods. He uses only 1 kg weight measure. If he sells his goods at cost price, what is his gain percentage?

(a) $1\frac{1}{2}\%$ (b) $\frac{100}{11}\%$

(c) $\frac{1000}{9}\%$ (d) $\frac{200}{9}\%$

Q.9 Four different candles, which can last for 5 h, 4 h, 3 h, and 2 h of burning, respectively, are lit in a room at the same instance and allowed to burn till the time the three candles in the room get extinguished. The cost of burning each one of these candles is 75 paise per h. What is the total cost involved?

(a) ₹2.75 (b) ₹3.75

(c) ₹9.75 (d) None of these

Q.10 Two ladies simultaneously leave Cities *A* and *B* connected by a straight road and travel towards each other. The first lady travels 2 km/h faster than the second lady and reaches *B* 1 h before the second lady reaches *A*. The two Cities *A* and *B* are 24 km. apart. What is the difference in their speeds/h?

(a) 4 km/h (b) 3 km/h (c) 2 km/h (d) 1 km/h

Q.11 In a company, 605 of the employees are men. Of these 40% are drawing more than ₹50,000 per year. If 36% of the total employees of the company draw more than ₹50,000 per year, what is the percentage of women who are drawing less than ₹50,000 per year?

(a) 70 (b) 60

(c) 40 (d) None of these

Q.12 The following table shows the percentage change in the consumption of electricity by five towns *P*, *Q*, *R*, *S*, and *T* from 1986 to 1988:

Town	Percentage change	
	From 1986 to 1987	From 1987 to 1988
P	+8	−18
Q	−15	+11
R	+6	+9
S	−7	−5
T	+13	−6

In town *T*, where 50,000 units consumed in 1986, how much did it consume in 1988?

(a) 37,100 units (b) 53,110 units

(c) 55,110 units (d) 57,100 units

Q.13 Water is filled in a container in such a manner that its volume doubles after every 5 min. If it takes 30 min for the container to be full, in how much time will it be one-fourth full?

(a) 7:30 min (b) 10 min (c) 20 min (d) 25 min

Q.14 Amar, Akbar, and Anthony are friends, being looked after by a matron Farah, Amar weighs 50% more than Akbar and Anthony weighs 25% less than Amar. Farah weighs a third of the combined weight of the three boys. All four together weigh 232 kg. What is the correct sequence of the persons in the ascending order of their weights?

(a) Anthony, Akbar, Farah, Amar

(b) Anthony, Akbar, Amar, Farah

(c) Akbar, Anthony, Amar, Farah

(d) Akbar, Anthony, Farah, Amar

Q.15 In a family, a couple has a son and a daughter. The age of the father is three times that of his daughter and the age of the son is half of his mother. The wife is nine years younger to her husband and the brother is seven years older than his sister. What is the average age of the mother and son?

(a) 40 years (b) 45 Years

(c) 50 years (d) 60 years

Direction for Questions 16 and 17: Read the passage below and solve the questions based on it.

Several runners, numbered 1, 2, 3,... and so on, start running simultaneously from the same point on a circular track and run continuously, in the same direction. They run such that the speed of the runner numbered *n* ($n > 1$) is n times that of the runner numbered 1 and so on.

Q.16 If there are exactly six runners, then at *N* distinct points on the track is the runner numbered 1 overtaken by any of the other five runners. What is the value of *N*?

(a) 15 (b) 11 (c) 9 (d) 10

Q.17 If there are exactly four runners, then at how many distinct points on the track do two or more runners meet?

(a) 3 (b) 4 (c) 5 (d) 6

Q.18 Sharat is walking down a descending escalator and takes 40 steps to reach the bottom. Chandra start simultaneously from the bottom, taking two steps for every one step taken by Sharat. Time taken by Sharat to reach the bottom from the top is the same as the time taken by Chandra to reach the top from the bottom. How many more steps did Chandra take than Sharat before they crossed each other on the escalator?

(a) 20 steps (b) 25 steps

(c) 40 steps (d) 60 steps

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Q.19 A hare and a tortoise decide to run around a pond, in opposite directions from the same point at the same time. The hare meets the tortoise at a certain pine tree for the first time. The second time, they meet at a banyan tree and the third time at a mango tree. The fourth time, the hare meets the tortoise exactly at the first pine tree. How many of the following statements is/are true?

i. One of them runs three times as fast as the other.
ii. One of them is twice as fast as the other.
iii. One of them is six times as fast as the other.
iv. The starting points and starting times are required to comment about the ratio of their speeds.

(a) 0 (b) 1
(c) 2 (d) Cannot be determined

Q.20 Chunmun is running at a uniform speed from Rajender Nagar to Patna Junction along the railway tracks. After every 12 min, a train moving from Rajender Nagar to Patna Junction overtakes him. Also, after every 4 min, he meets a train travelling from Patna Junction to Rajender Nagar. Trains leave from Rajender Nagar for Patna Junction and Patna Junction for Rajender Nagar at a regular interval of '*S*' min. What is the value of '*S*'?

(a) 4 (b) 6 (c) 8 (d) 12

Q.21 Two brothers Kumar Kalyan and Amar Kumar are walking around a circular track of length 1800 m, with a speed of 40 m/min and 50 m/min, respectively. Both of them start in the same direction simultaneously, but Amar Kumar reverses his direction every time he completes one round. After how much time (in minute) from the start, will they meet for the first time?

(a) 120 (b) 40 (c) 80 (d) 160

Q.22 Trains are coming from Patna to New Delhi and New Delhi to Patna at a regular interval of 1 h. They take 5 h to complete the journey. How many trains coming from New Delhi will cross a train coming from Patna that started at 10 pm?
(Given that the trains start from both the stations at the same time.)

(a) 5 (b) 6 (c) 10 (d) 11

Q.23 Two runners are running in a clockwise direction on a circular race track of n $(n > 1)$ km long. They meet for the second time at a point C which is 200 m from their starting point in the anticlockwise direction on the circular track. If both of the runners started at the same time, then the location of their first meeting point is:

i. 100 m from the starting point in the anticlockwise direction.
ii. 100 m from the point diagonally opposite to the starting point in anticlockwise direction.

(a) Definitely 1 (b) Definitely 1
(c) Either 1 or 2 (d) Neither 1 nor 2

Q.24 In a 4000 m race, around a circular stadium having a circumference of 1000 m, the fastest runner and the slowest runner reach the same point at the end of the 5th min, for the first time after the start of the race. All the runners have the same starting point and each runner maintains a uniform speed throughout the race. If the fastest runner runs at twice the speed of the slowest runner, what is the time taken by the fastest runner to finish the race?

(a) 5 min (b) 10 min
(c) 15 min (d) None of these

Q.25 In a race of 1000 m, A beats B by 100 m or 10 s. If they start a race of 1000 m simultaneously from the same point and if B gets injured, after running 50 m less than half the race length and due to which his speed gets halved, then by how much time will A beat B?

(a) 45 s (b) 50 s (c) 60 s (d) 65 s

Q.26 A train, 100 m long is approaching an unmanned railway crossing. The train is travelling at a uniform speed of 90 km/h and is 1 km away from the crossing. At the same time, a bus, also approaching the crossing, is 700 m away from it. Assuming the bus is also travelling at a uniform speed of B, for what range of values of N will the bus collide with the train?

(a) 59 km/h < N < 61 km/h
(b) 57 km/h < N < 63 km/h
(c) 55 km/h < N < 65 km/h
(d) 53 km/h < N < 67 km/h

Q.27 Two friends Karan and Arjun start running around a circular track in the same direction. Karan and Arjun take one full round in 7 min and 4 min, respectively. After how much time, will they meet at diametrically opposite point of their starting point?

(a) 120 s (b) 240 s
(c) They will never meet (d) None of these

Q.28 The analysis of the recently held CBSE exams shows that Chennai zone, Delhi zone and Ajmer zone have performed the best. The average marks of the Ajmer zone and the Delhi zone put together is 71%. The average marks of the Chennai zone and the Delhi zone put together is 76% and the average marks of the Ajmer zone and the Chennai zone put together is 79%. What is the range of the average marks of all the three zones (Assume N) put together?

(a) 71% < N < 78% (b) 70.5% < N < 77.5%
(c) 73.5% < N < 77.5% (d) 73% < N < 79%

Q.29 At the end of the year 1998, a shepherd bought nine dozen goats. Henceforth, every year he added p% of the goats at the beginning of the year and sold q% of the goats at the end of the year, where $p > 0$ and $q > 0$. If shepherd had nine dozen goats at the end of the year

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2002, after making the sales for that year, which of the following is true?
(a) $p = q$ (b) $p < q$ (c) $p > q$ (d) $p = q/2$

Q.30 Two fuel tanks, one shaped like a cylinder and other like a cone, contain fuel. The cylindrical tank holds 500 litres more fuel than the conical fuel. After 200 litres of fuel have been taken out from each tank, the cylindrical tank contains twice the quantity of fuel in the conical tank. How many litres of fuel did the cylindrical tank have when it was full?
(a) 700 (b) 1000 (c) 1100 (d) 1200

Answers

Benchmarking Test 2

1. (d)	**2.** (c)	**3.** (b)	**4.** (b)	**5.** (a)	**6.** (a)	**7.** (b)	**8.** (a)	**9.** (c)	**10.** (c)
11. (a)	**12.** (b)	**13.** (c)	**14.** (b)	**15.** (a)	**16.** (d)	**17.** (d)	**18.** (a)	**19.** (b)	**20.** (b)
21. (b)	**22.** (d)	**23.** (c)	**24.** (b)	**25.** (d)	**26.** (b)	**27.** (c)	**28.** (c)	**29.** (c)	**30.** (d)

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MODULE 3

X+2 MATHS

- Basics of Algebra
- Equations
- Sequence and Series
- Function
- Graphs and Maxima Minima
- Logarithm
- Permutation and Combination
- Probability

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CHAPTER

10

Basics of Algebra

LEARNING OBJECTIVES

After completion of this chapter, the reader should be able to understand:

- Basic terms involved in algebra
- Different types of solving the equations
- System of equations
- Kinds of questions in the CAT
- Methods of solving questions

INTRODUCTION

Algebra is important for CAT in terms of weightage of the questions asked from this area. In this chapter, basics of algebra have been covered for those students who have 'passed' through it at some point of time in their life but have no much idea about it currently. Initially, students are advised to see this chapter without any connection with other chapters/concepts of algebra. Once you complete this chapter thoroughly, you will start seeing the connection between the basics given in this chapter with the concepts given ahead.

WHAT IS ALGEBRA?

Algebra is a branch of Mathematics that substitutes letters for numbers. It simply means that in arithmetic, we have statements like 'Ram is having 5 apples', whereas in algebra, we will be having statements like 'Ram is having x apples', where the value of x can be deduced from the given conditions.

Moving from Arithmetic to Algebra

In arithmetic, it will be:

$$3 + 4 = 3 + 4$$

In algebra, it will be:

$$x + y = y + x$$

While in arithmetic, we mostly deal with numbers (which have a constant fixed value always), in algebra, we mostly deal with the variables (which do not have a fixed value and can take different values according to the conditions governing it). For example, if we say that 'Ram is having ₹100', we get a clear picture that how much money Ram is having. Even if next day Ram is having 100 apples instead of ₹100, the number '100' has the same meaning in this different situation.

Now, replace '100' by 'x' ⇒ 'Ram is having ₹x'; this does not give us a clear picture that how much money Ram is having. If next day Ram is having x apples, the letter 'x' may not have the same value in a different situation.

ALGEBRAIC NOTATIONS AND OPERATIONS

(i) Notations

Algebraic notations	Meaning	Example
$x = y$	x is equal to y.	$1 = 1$

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$x \neq y$	x is not equal to y.	$1 + 1 \neq 1$
$x < y$	x is less than y or y is greater than x.	$2 < 5$
$x > y$	x is greater than y or y is less than x.	$4 > 3$
$x \leq y$	y is greater than or equal to x or x is less than or equal to y.	
$x \geq y$	x is greater than or equal to y or y is less than or equal to x.	
$x \approx y$	x is approximately equal to y.	$2.99999 \approx 3$

(ii) Additions/Subtractions

Statement 1: Ram is having 5 more apples than Shyam.

$\Rightarrow$ Number of apples with Ram = Number of apples with Shyam + 5

Assume that number of apples with Shyam = x, then number of apples with Ram = $x + 5$

Alternatively, if we assume that the number of apples with Ram = y, then number of apples with Shyam = $y - 5$

(iii) Multiplication

When there are two or more equal numbers to be added together, the expression of their sum may be abridged. For example,

$$x + x = 2 \times x = 2x$$

$$x + x + x = 3 \times x = 3x$$

$$x + x + x + x = 4 \times x = 4x$$

In this manner, we may form an idea of multiplication, and it is to be observed that,

$2 \times x$ signifies 2 times x or twice x

$3 \times x$ signifies 3 times x or thrice x

Furthermore, we can multiply such products again by other numbers. For example:

$$2y \times 5 = 10y$$

$$2y \times 5z = 10yz$$

$$2y \times 6y = 12y^2$$

(iv) Linear Equation

An algebraic equation such as $y = 2x + 7$ or $3x + 2y - z = 4$ in which the highest degree term in the variable or variables is of the first degree. The graph of such an equation is a straight line if there are two variables.

The general form of linear equation in two variables x and y is $ax + by + c = 0$, $a \neq 0$, $b \neq 0$, and a, b, and c are real numbers. Here, a and b are known as co-efficients of x and y, respectively, and c is a constant. A solution of such an equation is a pair of values: one for x and the other for y, which makes LHS and RHS of the equation equal.

For a linear equation in two variables x and y:

$$ax + by + c = 0 \Rightarrow by = -ax - c \Rightarrow y = \frac{-ax - c}{b}$$

In the above equation, for every real x, there exists a real number y corresponding to x.

Therefore, every linear equation in two variables has infinitely many solutions, i.e., infinitely many pairs (x, y). For example, the equation $2x + 3y = 10$ will have infinite solution.

All these solutions are represented by points on a certain line. Due to this fact, only this equation is called LINEAR because the graph of the equation on the x–y Cartesian plane is a straight line.

For example, consider the equation

$$3x + 2y = 8 \Rightarrow 2y = 8 - 3x \Rightarrow y = \frac{8 - 3x}{2} \qquad \text{(i)}$$

To find a solution of a linear equation with two variables, we assign any value of one of the two variables and determine the value of the other variable, from the given equation (i).

Therefore, taking $x = 1$, we get corresponding value of $y = \frac{5}{2}$

Similarly, taking $x = 0$, we get $y = 4$, and so on.

The following table lists six possible values for x and the corresponding values for y, that is, six solutions of the equation:

x	–2	–1	0	1	2	8/3
y	7	11/2	4	5/2	1	0

Taking any two pairs of given equation, we plot corresponding points, say, P and Q. The line PQ passing through these points is related with the given equation in the following manner:

1. Every solution $x = p$ and $y = q$ of the given equation determines a point (p, q) that lies on this line.
2. Every point (x_i, y_i) lying on the line PQ determines a solution $x = x_p$ $y = y_i$ of the given equation.

The line PQ is said to be the graph of the given equation. It is worth noting the following points:

1. We can add or subtract any number on both sides of the equation without affecting the equation and its solution.
2. We can multiply or divide both sides of an equation by a non-zero number without affecting the equation and its solution.

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If we plot the solutions of the equation $3x + 2y = 8$, which is represented in the table above, then we notice that they all lie on the same line. We call this line the graph of the equation, as it corresponds precisely to the solution set of the equation.

SYSTEM OF TWO LINEAR EQUATIONS IN TWO UNKNOWNS

A system of two linear equations in two unknowns is a system of two equations of the form:

$$a_1 x + b_1 y = c_1$$
$$a_2 x + b_2 y = c_2$$

where a_1, a_2, b_1, b_2, c_1, and c_2 are arbitrary real numbers.

The solution set of the system of linear equation in two unknowns is a pair of real numbers (x_0, y_0) which satisfies each of the equations of the system. In general,

$$\Rightarrow x = \frac{c_1 b_2 - c_2 b_1}{a_1 b_2 - a_2 b_1}$$

x is defined if $a_1 b_2 - a_2 b_1 \neq 0$ (i.e., for the existence of a unique solution of the system of linear equations).

In other words, the system of linear equations has a unique solution if $(a_1 b_2 - a_2 b_1) \neq 0$

$$\Rightarrow \frac{a_1}{a_2} \neq \frac{b_1}{b_2} \text{ or } \Rightarrow \frac{a_1}{b_1} \neq \frac{a_2}{b_2}$$

Now, let us consider the geometric interpretation of the equation given above. We know that the set of points in a plane whose coordinates satisfy an equation of the form $ax + by = c$, where either a or b are non-zero, constitutes a straight line. Therefore, to solve a system, every equation has at least one unknown means to find the common point of two straight lines. Hence, a system of linear equations has a unique solution if the lines intersect.

Example 1 Consider the system of linear equations

$$4x + 6y = 10 \text{ and } 4x - 2y = 2$$

Solution Point (1, 1) is the intersection of two straight lines. Hence, $x = 1$, $y = 1$ is the unique solution of system of linear equations.

We can say that the system of linear equations has a unique solution if $a_1 b_2 - a_2 b_1 \neq 0$

What if $a_1 b_2 - a_2 b_1 = 0$?

If $a_1 b_2 - a_2 b_1 = 0$

or $$a_1 b_2 = a_2 b_1$$

or $\frac{a_1}{a_2} = \frac{b_1}{b_2}$, then there are two situations:

Situation (i)

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

If Situation (i) exists, then the system of linear equations has an infinite number of solutions.

Situation (ii)

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

If Situation (ii) exists, then the system of linear equations has no solutions.

For example, consider following equations:

$$x + y = 10 \quad \text{(i)}$$
$$2x + 2y = 25 \quad \text{(ii)}$$

In the given equations, $a_1/a_2 = b_1/b_2$; hence, unique solution is not possible. To have a better understanding, it can be seen that LHS of equation (i) $\times 2$ will give us the LHS of equation (ii). Hence, there will not be any point of intersection of these two graphs drawn on X–Y axis. In other words, lines will be parallel.

Summarizing the whole discussion:

In geometric terms	In algebraic terms
(i) The lines intersect	System of linear equations has a unique solution, i.e., $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$ (known as system is determinate)
(ii) The lines are parallel	The system is inconsistent, i.e., the system of linear equations has no solution, i.e., $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$
(iii) The lines are coincident.	System of linear equations has infinitely many solutions, i.e., $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$.

SOLVING A SYSTEM OF LINEAR EQUATIONS

Graphical Method

We plot the straight lines correspondence to each of the given linear equations in two variables. The point of intersection

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is the solution for x and y (i.e., two variables) in the system of linear equation.

Example 2 Solve the two equations $2x - y = 1$ and $2x + 3y = 5$

Solution We will draw the graph of both the equations. We can see that the straight lines intersect at the point (1, 1). Hence, the set of solutions of the given equations is (1, 1).

Elimination Method

Step 1 Multiply the co-efficients of the equations by suitable numbers so that the co-efficients of one of the variables becomes same in both the equations.

Step 2 Add or subtract to get one of the variables (which has become same in Step 1) cancelled. This will lead to obtaining the value of other variable.

Step 3 Substitute this value in one of the equations to obtain the value of the other variable.

Example 3 Solve $3x - 5y = 1$
$x + 4y = 6$

Solution

Given equations are $3x - 5y = 1$ (i)
$x + 4y = 6$ (ii)

Step 1 If we multiply equation (ii) by 3, then co-efficient of x will become same in both the equations. Alternatively, if we multiply equation (i) by 4 and equation (ii) by 5, co-efficient of y will become same in both the equations. We are making co-efficient of x same in both the equations.

$$3x - 5y = 1 \quad \text{(iii)}$$

$$3x + 12y = 18 \quad \text{(iv)}$$

Step 2 Subtracting equation (iii) from equation (iv), we get,

$$(3x + 12y) - (3x - 5y) = 18 - 1$$

$$\Rightarrow \quad 17y = 17 \quad \Rightarrow \quad y = 1$$

Step 3 Substitute the value of y in equation 1.

$$3x - 5 \times 1 = 1 \text{ or } x = 2$$

Alternative Method

Suppose we have to solve the system of equations:

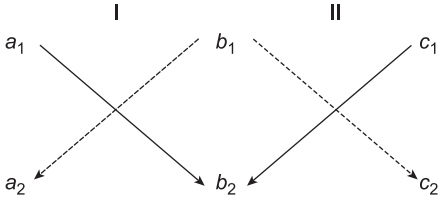
$$a_1x + b_1y = c_1$$

$$a_2x + b_2y = c_2$$

Write the co-efficients of x, y, and the constants in the following manner (taking notice of the signs, i.e., positive and negative values, of the co-efficients).

I		II
a_1	b_1	c_1
a_2	b_2	c_2

Now, do the cross-multiplication in Part I and Part II as given below:



Note that there are two types of cross-multiplication in each of the Part I and Part II, which are shown in bold lines and by dotted lines.

Now, subtract the value obtained by the cross-multiplication of dotted lines from that the value obtained by the cross-multiplication of bold lines in each of the parts. Suppose the values obtained from Part II is m and the value obtained from Part I is n, then the value of the variables corresponding to co-efficient a_1 and a_2 is $\frac{m}{n}$;

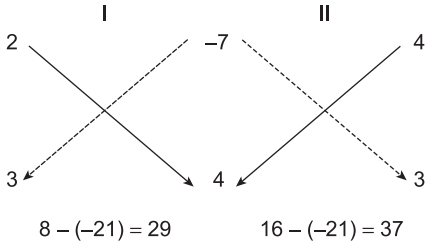
$$\text{i.e., } x = \frac{m}{n}.$$

Let us see an example:

Given equations are

$$2x - 7y = 4$$

$$3x + 4y = 3$$



$x = \frac{37}{29}$ and y can be obtained by putting this value of x in any of the equations $\left[y = \frac{2x - 4}{7} \right]$

Substitution Method

Assume that equations are in terms of x and y.

Step 1 Express x in terms of y in one of the equations.

Step 2 Put the value of x obtained from Step 1 in the 2nd equation. Calculate the value of 'y' now.

Step 3 Substitute the value of 'y' in any of the equations to find the value of x.

Example 4 Solve $x + 7y = 15$

$$7x - 3y = 1$$

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Solution We have $x + 7y = 15$ (i)

$$7x - 3y = 7 \quad \text{(ii)}$$

From equation (i), $x = (15 - 7y)$

Substituting this value of x in equation (ii), we get

$$7 \times (15 - 7y) - 3y = 1$$

$\Rightarrow y = 2$ and corresponding value of $x = 1$

REMAINDER AND FACTOR THEOREM

Remainder Theorem

Remainder theorem gives us a method for finding the remainder without actual division.

Let us understand this with the following examples:

Example 5 Let $p(x) = x^4 - 3x^2 + 2x + 5$. Find the remainder when $p(x)$ is divided by $(x - 1)$.

Solution Rather going for the actual division, Let us compute $p(1)$, i.e., value of $p(x)$ when x is replaced by 1, we have $p(1) = 1 - 3 + 2 + 5 = 5$

We find that the remainder when $p(x)$ is divided by $(x - 1)$ is equal to $p(1)$, i.e., the value of $p(x)$ at $x = 1$

Example 6 Find the remainder when $p(y) = y^3 + y^2 + 2y + 3$ is divided by $y + 2$

Solution Compute $p(-2)$ = value of $p(y)$ when y is replaced by -2, we have

$$P(-2) = (-2)^3 + (2)^2 + 2(-2) + 3 = -8 + 4 - 4 + 3 = -5$$

From these two examples, it is noted that the remainder obtained when $p(x)$ is divided by $(x - a)$ is equal to $p(a)$, i.e., the value of $p(x)$ at $x = a$

The above result is stated as remainder theorem (see Chapter 2—Number System for more reference).

Note:

1. If a polynomial $p(x)$ is divided by $(x + a)$, then the remainder is the value of $p(x)$ at $x = -a$, i.e., $p(-a)$
2. If a polynomial $p(x)$ is divided by $(ax - b)$, then the remainder is the value of $p(x)$ at $x = b/a$, i.e., $P(b/a)$
3. If a polynomial $p(x)$ is divided by $(ax + b)$, then the remainder is the value of $p(x)$ at $x = -b/a$, i.e., $p(-b/a)$
4. If a polynomial $p(x)$ is divided by $(b - ax)$, then the remainder is equal to the value of $p(x)$ at $x = b/a$, i.e., $p(b/a)$

Example 7 Determine the remainder when the polynomial $p(x) = x^4 - 3x^2 + 1$ is divided by $x - 1$

Solution By remainder theorem, the required remainder is equal to $p(1)$.

Now, $p(x) = x^4 - 3x^2 + 2x + 1$

$\Rightarrow p(1) = (1)^4 - 3 \times 1^2 + 2 \times 1 + 1 = 1 - 3 + 2 + 1 = 1$

Hence, the required remainder $= p(1) = 1$

Factor Theorem

If $g(x)$ divides $f(x)$, then we say that $f(x)$ is divisible by $g(x)$ or $g(x)$ is a factor of $f(x)$.

Factor theorem gives us a method to determine whether a polynomial $g(x)$ is a factor of a polynomial $f(x)$ or not without actual division.

Let $f(x)$ be a polynomial of degree greater than or equal to 1 and 'a' be a real number such that $f(a) = 0$. Then, $(x - a)$ is a factor of $f(x)$. Conversely, if $(x - a)$ is a factor of $f(x)$, then $f(a) = 0$

Note:

1. $(x + a)$ is a factor of a polynomial $f(x)$ if $f(-a) = 0$
2. $(ax - b)$ is a factor of a polynomial $f(x)$ if $f(b/a) = 0$
3. $ax + b$ is a factor of polynomial $f(x)$ if $f(-b/a) = 0$
4. $(x - a)(x - b)$ is a factor of a polynomial $f(x)$ if $f(a) = 0$ and $f(b) = 0$

Example 8 Show that $(x - 3)$ is a factor of the polynomial

$$x^3 - 3x^2 + 4x - 12$$

Solution Let $f(x) = x^3 - 3x^2 + 4x - 12$ be the given polynomial. By factor theorem, $(x - a)$ is a factor of a polynomial $f(x)$ if $f(a) = 0$. Therefore, in order to prove that $(x - 3)$ is a factor of $f(x)$, it is sufficient to show that $f(3) = 0$

Now, $f(x) = x^3 - 3x^2 + 4x - 12$

$\Rightarrow f(3) = 3^3 - 3 \times 3^2 + 4 \times 3 - 12 = 27 - 27 + 12 - 12 = 0$

Hence, $(x - 3)$ is a factor of $f(x)$.

Example 9 Find out if $(x - 1)$ is a factor of (i) x^{10-1} and (ii) x^{11-1}.

Solution Let $f(x) = x^{10} - 1$ and $g(x) = x^{11} - 1$.

To find out that $(x - 1)$ is a factor of both $f(x)$ and $g(x)$, it is sufficient to show that $f(1) = 0$ and $g(1) = 0$

Now, $f(x) = x^{10} - 1$ and $g(x) = x^{11} - 1$

$\Rightarrow f(1) = 1^{10} - 1 = 0$ and $g(1) = 1^{11} - 1 = 0$

$\Rightarrow (x - 1)$ is a factor of both $f(x)$ and $g(x)$

More Worked Out Problems

Example 10 Solve the equation $3x - 2y = 7$

Solution Let us first find one particular solution and then we will extend that. Its not too difficult to find one solution using hit and trial. One such solution is (3, 1). Alternatively, we can find it using $[x = (7 + 2y)/3]$

To find the next solution, let us re-write the equation as $3x = 2y + 7$

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Using the concepts of remainder, the above equation can be explained as:

LHS = A multiple of 3

RHS = A number that gives remainder 7 when divided by 2, which means a number that gives remainder 1 when divided by 2.

Obviously, the next number will be obtained after the LCM (2, 3) = 6. To move ahead by 6, we will cover 2 multiples of 3 (2×3) and 3 multiples of 2 (3×2). Hence, the next solutions are (5, 4), (7, 7), (9, 10), (11, 13), etc.

Note: Since it is a linear equation, there will be obviously infinite solution.

Example 11 Find all integral roots of the equation $21x + 48y = 5$

Solution LHS is divisible by 3, but RHS is not divisible by 3. Hence, no integral solution is possible.

Statement I

(i) $1990x - 173y = 11$ has no solution in integers for x and y.
(ii) $3x - 12y = 7$ has no solution in integers for x and y.

Which of the following is true?

(a) Statement (i) is wrong and Statement (ii) is right.
(b) Statement (i) is right and Statement (ii) is wrong.
(c) Statements (i) and (ii) both are right.
(d) Statements (i) and (ii) both are wrong.

Statement II

The co-efficients in the equation are large enough to find a particular solution. However, we can see that the numbers 1990 and 173 are relatively prime and this helps us reach the conclusion.

Hence, we can find the solution of $1990x - 173y = 11$ in integers for x and y.

Hence, Statement (i) is wrong.

Statement III

In the linear equation $3x - 12y = 7$; co-efficients of x and y, i.e., 3 and 12, are not relatively prime. HCF of (3, 12) = 3 Also, constant 7 is not exactly divisible by 3. In other words, LHS is divisible by 3, but RHS is not divisible by 3.

Hence, $3x - 12y = 7$ has no solution for x and y in integers.

Hence, Statement (ii) is wrong.

Therefore, option (d) is true.

Practice Exercises

Q.1 If $x + y + z = 0$, then $x^3 + y^3 + z^3$ is equal to:
(a) 0 (b) $3xyz$
(c) $\frac{xy + yz + zx}{xyz}$ (d) $xyz\,(xy + yz + zx)$

Q.2 If $x - \frac{1}{x} = 2$, then the value of $x^4 + \frac{1}{x^4}$ is:
(a) 4 (b) 8 (c) 12 (d) 34

Q.3 If $\left(x + \frac{1}{x}\right) = 3$, then the value of $\left(x^6 + \frac{1}{x^6}\right)$ is:
(a) 927 (b) 414 (c) 364 (d) 322

Q.4 Let $f(x) = x^2 - 27x + 196$. If $(x) = x$, then what is the value of x?
(a) 28 (b) 14 (c) 7 (d) 4

Q.5 Consider the following statement: The equations $ax + by = 1$ and $ax - by = 1$ have a unique solution for:
(a) $a = 1, b = 0$ (b) $a = 0, b = 1$
(c) $a = 0, b = 0$ (d) $a = 1, b = 1$

Q.6 Find the numerical value of 'c', if the expression $xy - 3x + 5y + c$ can be factorized.
(a) 9 (b) −15 (c) 15 (d) −9

Q.7 If x is less than 2, then which of the following statements is always true?
(a) x is negative.
(b) $2x$ is greater than or equal to x.
(c) x^2 is greater than or equal to x.
(d) None of these

Q.8 If $x - y = 8$, then which of the following must be true?
I Both x and y are positive.
II If x is positive, y must be negative.
III If x is negative, y must be negative.
(a) I only (b) II only (c) I and II (d) IIIonly

Q.9 Five tables and eight chairs cost ₹7350. Three tables and five chairs cost ₹4475. What is the price of one table?
(a) ₹950 (b) ₹325 (c) ₹925 (d) ₹350

Q.10 The cost of 3 apples and 2 guavas is ₹23 and that of one apple and four guavas of the same type is ₹21. What is the cost of 5 guavas?
(a) ₹4 (b) ₹20 (c) ₹12 (d) ₹25

Q.11 If the two digits of the age of Mr Manoj Sagar are reversed, then the new age so obtained is the age of his wife. If Mr Manoj Sagar is elder to his wife, then find the difference between their ages.

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(a) 9 years (b) 10 years
(c) 8 years (d) Cannot be determined

Q.12 Find the remainder when the polynomial $f(x) = 2x^4 - 6x^3 + 2x^2 - x + 2$ is divided by $x + 2$.
(a) 102 (b) −102 (c) 92 (d) −92

Q.13 Find the remainder when $f(x) = 4x^3 - 12x^2 + 14x - 3$ is divided by $g(x) = x - 1/2$.
(a) 2/3 (b) −2/3 (c) 3/2 (d) −3/2

Q.14 Find the remainder when $f(x) = x^3 - 6x^2 + 2x - 4$ is divided by $g(x) = 3x - 1$.
(a) 107/27 (b) −107/27 (c) −4 (d) 4

Q.15 Find the remainder when $f(x) = x^3 - 6x^2 + 2x - 4$ is divided by $g(x) = 1 - 3x$.
(a) 107/27 (b) −107/27 (c) −4 (d) 4

Q.16 If the polynomials $ax^3 + 4x^2 + 3x - 4$ and $x^3 - 4x + a$ leave the same remainder when divided by $(x - 3)$, then find the value of a.
(a) 1 (b) 0
(c) −1 (d) None of these

Q.17 The polynomials $ax^3 + 3x^2 - 13$ and $2x^3 - 5x + a$ are divided by $x + 2$. If the remainder in each case is the same, the find the value of a.
(a) 5/9 (b) −5/9 (c) 2/7 (d) −2/7

Q.18 What is the value of a, if $(x - a)$ is a factor of $(x^3 - a^2x + x + 2)$?
(a) 2 (b) −2 (c) 0 (d) 1

Q.19 Find the values of $(a + b)$ if the polynomial $x^3 - ax^2 - 13x + b$ has $(x - 1)$ and $(x + 3)$ as factors.
(a) 2 (b) 15
(c) 18 (d) None of these

Q.20 What is the value of p, if $x + 3$ is a factor of $3x^2 + px + 6$?
(a) 9 (b) −9 (c) 11 (d) −11

Answers

1. (b)	**2.** (d)	**3.** (d)	**4.** (b)	**5.** (d)	**6.** (b)	**7.** (d)	**8.** (d)	**9.** (a)	**10.** (b)
11. (d)	**12.** (c)	**13.** (c)	**14.** (b)	**15.** (b)	**16.** (c)	**17.** (a)	**18.** (b)	**19.** (c)	**20.** (c)

Hints and Solutions

1. $x + y = -z \Rightarrow (x + y)^3 = (-z)^3$
$\Rightarrow x^3 + y^3 + 3xy(x + y) + z^3 = 0$
Hence, $x^3 + y^3 + z^3 = 3xyz$

2. $x - \frac{1}{x} = 2 \Rightarrow \left[x - \frac{1}{x}\right]^2 \Rightarrow x^2 + 1/x^2 = 6$
or $[x^2 + 1/x^2]^2 = 36$
Hence, $x^4 + \frac{1}{x^4} = 34 = 34$

3. $x + \frac{1}{x} = 3$
Do square of both sides
$\left(x + \frac{1}{x}\right)^2 = (3)^2 \rightarrow x^2 + \frac{1}{x^2} = 9 - 2 = 7$
Do cube of both sides
$\left(x^2 + \frac{1}{x^3}\right)^3 = 7^3 \rightarrow x^2 + \frac{1}{x^2} + 3\left[x^2 + \frac{1}{x^2}\right] = 343$
$x^6 + \frac{1}{x^6} + 3 \times 7 = 343$ [since $x^2 + \frac{1}{x^2} = 7$]
$x^6 + \frac{1}{x^6} = 343 - 21 = 322$

4. $f(x) = x^2 - 27x + 196 = x$ or $x^2 - 28x + 196 = 0$
$\Rightarrow (x - 14)(x - 14) = 0$
Hence, $x = 14$

5. $a_1x + b_1y = 1$
$a_2x - b_2y = 1$
For unique solution, we know $\frac{a_1}{a_2} \neq \frac{b_1}{-b_2}$
Now, go through the options: a and b cannot be zero. Hence, options (a), (b), and (c) are rejected. Only option (d) is the answer.

6. $xy - 3x + 5y + c$ can be factorized (given) as
$x(y - 3) + 5(y - 3) \Rightarrow xy - 3x + 5y - 15$
After comparing, we get $(c = -15)$

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7. Statement (a) is obviously not always true.
Statement (b) is not true for negative values of x.
Statement (c) is not true for $0 < x < 1$.

8. If $x = 10$ and $y = 2$, $x - y = 8$. Hence, Statement II is rejected.
If $x = 6$ and $y = -2$, $x - (-2) = 6 + 2 = 8$. Hence, Statement I is rejected.
If $x = -$ve, for $x - y = +$ve, y must be $-$ve as
$x = -2$ and $y = -10 \Rightarrow x - y = -2 - (-10) = -2 + 10 = 8$
So, Statement III is true.

9. 5T (Table) + 8C (chair) = 7350 (given) (i)
3T + 5C = 4475 (given) (ii)
$5 \times$ (i) $- 8 \times$ (ii)
$\Rightarrow$ 25T + 40C = 36,750
24T + 40C = 35, 800
T = ₹950

10. Assume that price an apple = ₹x and price of a guava = ₹y
Given that $3x + 2y = 23$ and $x + 4y = 21$
By solving we get $y = 4$, Hence, cost of 5 guavas = ₹20

11. Assume age of Manoj Sagar = $10x + y$, then age of his wife = $10y + x$
Difference between their ages = $9x - 9y = 9(x - y)$
It can be seen that $(x - y)$ can take multiple values. Hence, the answer cannot be determined.

12. We have, $x + 2 = x - (-2)$. So, by remainder theorem, when $f(x)$ is divided by $\{x - (-2)\}$, the remainder is equal to $f(-2)$.
Now, $f(x) = 2x^4 - 6x^3 + 2x^2 - x + 2$
$\Rightarrow f(-2) = 2(-2)^4 - 6(-2)^3 + 2(-2)^2 - (-2) + 2$
$\Rightarrow f(-2) = 2 \times 16 - 6 \times -8 + 2 \times 4 + 2 + 2$
$\Rightarrow f(-2) = 32 + 48 + 8 + 2 + 2 = 92$
Hence, the required remainder = 92

13. Using remainder theorem, we know that $f(x)$ when divided by $g(x) =$ gives a remainder equal to $f\left(\frac{1}{2}\right)$.
Now, $f(x) = 4x^3 - 12x^2 + 14x - 3$
$\Rightarrow f\left(\frac{1}{2}\right) = \frac{3}{2}$
Hence, the required remainder $= f\left(\frac{1}{2}\right) = \frac{3}{2}$

14. We have $g(x) = 3x - 1 = 3\left(x - \frac{1}{3}\right)$. Therefore, by remainder theorem, when $f(x)$ is divided by $g(x) = 3\left(x - \frac{1}{3}\right)$, the remainder is equal to $f\left(\frac{1}{3}\right)$.
Now, $f(x) = x^3 - 6x^2 - 2x - 4$
$\Rightarrow f\left(\frac{1}{3}\right) = \left(\frac{1}{3}\right)^3 - 6\left(\frac{1}{3}\right)^2 + 2\left(\frac{1}{3}\right) - 4$
$= \frac{1}{27} - \frac{6}{9} + \frac{2}{3} - 4 = -\frac{107}{27}$
Hence, the required remainder $= -\frac{107}{27}$

15. Using remainder theorem, when $f(x)$ is divided by $g(x) = 1 - 3x$, the remainder is equal to $f(1/3)$.
Now, $f(x) = x^3 - 6x^2 + 2x - 4$
$f\left(\frac{1}{3}\right) = \frac{1}{27} - \frac{2}{3} + \frac{2}{3} - 4 = \frac{-107}{27}$
Hence, the required remainder $= \frac{-107}{27}$

16. Let $f(x) = ax^3 + 4x^2 + 3x - 4$ and $g(x) = x^3 - 4x + a$ be the given polynomials. The remainders when $f(x)$ and $g(x)$ are divided by $(x - 3)$ are $f(3)$ and $g(3)$, respectively.
According to the question, $f(3) = g(3)$
$\Rightarrow 27a + 36 + 9 - 4 = 27 - 12 + a$
$\Rightarrow 26a + 26 = 0 \Rightarrow a = -1$

17. Let $f(x) = ax^3 + 3x^2 - 13$ and $g(x) = 2x^3 - 5x + a$ be the given polynomials. The remainder when $p(x)$ and $q(x)$ are divided by $(x + 2)$ are $f(-2)$ and $g(-2)$, respectively.
Given that $f(-2) = g(-2)$
$\Rightarrow a(-2)^3 + 3(-2)^2 - 13 = 2(-2)^3 - 5(-2) + a$
$\Rightarrow -8a + 12 - 13 = -16 + 10 + a$
$\Rightarrow -8a - 1 = a - 6$
$\Rightarrow a = \frac{5}{9}$

18. Let $f(x) = x^3 - a^2x + x + 2$ be the given polynomial. By factor theorem, $(x - a)$ is a factor of $f(x)$ if $f(a) = 0$
Now, $f(a) = 0 \Rightarrow a^3 - a^2 \times a + a + 2 = 0$
$\Rightarrow a + 2 = 0 \Rightarrow a = -2$
Hence, $(x - a)$ is a factor of the given polynomial, if $a = -2$.

19. Let $f(x) = x^3 - ax^2 - 13x + b$ be the given polynomial. If $(x - 1)$ and $(x - 3)$ are factors of $f(x)$, then
$f(1) = 0$ and $f(-3) = 0$
$\Rightarrow 1^2 - a \times 1^2 - 13 \times 1 + b = 0$
and $(-3)^2 - 13 \times (-3) + b = 0$
$\Rightarrow 1 - a - 13 + b = 0$
and $-27 - 9a + 39 + b = 0$
$\Rightarrow -12 - a + b = 0$
and $-9a - b = 12$
Solving the equations, we get $\Rightarrow a = 3$
Putting $a = 3$ in $a - b = -12$, we get $b = 15$
Hence, $a = 3$ and $b = 15$

20. $f(-3) = 0$
$\Rightarrow 3(-3)^2 + p(-3) + 6 = 0$
$\Rightarrow 27 - 3p + 6 = 0 \Rightarrow p = 11$

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CHAPTER

11

Equations

LEARNING OBJECTIVES

After completion of this chapter, the reader should be able to understand:

- Polynomials and different equations
- Definitions and properties of those equations
- Concepts attached to the these equations and their maxima–minima
- Inequalities
- Kind of questions that are asked in the CAT
- Method/s of solving questions

INTRODUCTION

Quadratic equation has been one of the favourite topics for the CAT. Almost every year, 1 to 3 questions are asked form this topic. Normally, the questions are asked using the simple concepts of quadratic equation only, but it is desirable to have a good knowledge of the cubic equation and other polynomials also. Most of the questions asked in the CAT from this topic are logical in nature, that is, can be done with the help of other alternative methods viz., using options, or assuming some values, etc.

DEFINITIONS

Polynomials

Assume that a_1, a_2, a_3, a_4, … are real numbers and x is a real variable.

Then, $f(x) = a_1x^n + a_2x^{n-1} + a_3x^{n-2} + \ldots + a_{n-1}x + a_nx$ is called a polynomial.

Example $5x^5 + 3x^4 + \ldots + x$ is a polynomial in x, where x is a real variable.

Degree of a Polynomial

Polynomial $f(x) = a_1x^n + a_2x^{n-1} + a_3x^{n-2} + \ldots + a_{n-1}x + a_nx$ is a polynomial of degree n, where $a_1 \neq 0$.

Example $5x^5 + 3x^4 + \ldots + x$ is a polynomial of degree 5.
$2x^4 + x^3 + 4x^2 + 2x + 10$ is a polynomial of degree 4.
$4x^3 + 4x^2 + 2x + 10$ is a polynomial of degree 3.
$4x^2 + 2x + 10$ is a polynomial of degree 2.

Remember

1. Degree of polynomial is defined for both real and complex polynomials.
2. Degree of polynomial cannot be fractional.

Polynomial Equation

If $f(x)$ is a polynomial, real or complex, then $f(x) = 0$ is the corresponding equation. Simply put, when we equate any polynomial with zero, then it becomes equation.

Example $5x^5 + 3x^4 + \ldots + x = 0$ is an equation.
$2x^4 + x^3 + 4x^2 + 2x + 10 = 0$ is an equation.

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Polynomial Inequation

If $f(x)$ is a polynomial, real or complex, then $f(x) \neq 0$ is the corresponding inequation. Simply put, if any polynomial is not equal to zero, that is, either greater than zero or less than zero, then it known to be as inequation.

Example $2x^4 + x^3 + 4x^2 + 2x + 10 > 0$ is an inequation.
$4x^2 + 2x + 10 < 0$ is an inequation.

Roots of an Equation

All the values of variable satisfying the equation are known as roots of the equation.

It is known to us that an equation of degree n will have n roots, real or imaginary. Roots can be same or distinct. In case of same roots, roots are known as repeated roots.

QUADRATIC EQUATION

Any equation of degree 2 is known as a quadratic equation.

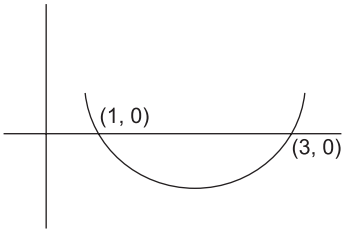
$ax^2 + bx + c = 0$ is known to be the standard equation of quadratic equation. It can be seen that this equation will have two roots, real or imaginary.

Geometrical Meaning of Roots

For any given equation $y = f(x) = 0$, number of times graph of this equation cuts X-axis is equal to the distinct real roots of this equation.

Example $y = (x-1)(x+2)(x-2) = 0$ will intersect X-axis at three distinct points, namely $x = 1, -2$, and 2.

For any quadratic equation $f(x) = (x-1)(x-3)$, graph of this equation (can be seen below) will intersect X-axis at two distinct points, namely $x = 1$ and $x = 3$.



Therefore, the roots of the equation $f(x) = x^2 - 4x + 3 = 0$ will have two roots, that is, two values of $x = 1$ and $x = 3$, which will satisfy this equation.

Therefore, if we say that the two roots 'p' and 'q' of the quadratic equation $f(x) = ax^2 + bx + c = 0$ are such that $f(2) \times f(-2) > 0$, then either both the roots of this equation 'p' and 'q' will lie inside -2 and 2 or both the roots 'p' and 'q' will lie outside this range of -2 and 2.

Similarly, if we say that the two roots 'p' and 'q' of the quadratic equation $f(x) = ax^2 + bx + c = 0$ are such that $f(2) \times f(-2) < 0$, then one of the two roots of this equation 'p' and 'q' will lie inside this range and another will lie outside this range -2 and 2.

Properties of Roots

1. A polynomial equation of degree n will have n roots, real or imaginary.
2. Complex roots or surds always occur in pair.
   Example If $2 + \sqrt{5}$ is a root of any equation, then $2 - \sqrt{5}$ will also be a root of that equation.
3. Every equation of an even degree with coefficient of highest degree term positive and constant term negative will have at least two real roots. Out of these two real roots, one will be positive and other one will be odd.
4. Any equation with sum of all of its coefficients equal to zero, will have 1 as its one of the roots.
   Example $x^2 - 5x + 4 = 0$ will have 1 as one of its roots.
5. If all the terms of an equation are positive and it does not involve any odd powers of x, then this equation will not have any real root.
   Example $f(x) = x^4 + x^2 + 1 = 0$ will not have any real root. Obviously, $x^4 \geq 0$, $x^2 \geq 0$, and 1 is positive. And we know that sum of three positive numbers can never be equal to zero for any real value of x.

Example 1 How many real roots will be there of the quadratic equation $f(x) = x^2 + 5|x| + 6 = 0$?

Solution

Method 1 If $x > 0$, then $f(x) = x^2 + 5x + 6 = 0$

Or, $x^2 + 5x + 6 = (x + 2)(x + 3) = 0$, Therefore, $x = -2$ and $x = -3$

But, we have taken $x > 0$, so negative values of x are not admissible.

Now, if $x < 0$, then $f(x) = x^2 - 5x + 6 = 0$

Or, $x^2 - 5x + 6 = (x - 2)(x - 3) = 0$, therefore, $x = 2$ and $x = 3$

But, we have taken $x < 0$, so positive values of x are not admissible.

And $x = 0$ is not possible.

Therefore, no real value of x can be obtained.

Method 2 It is worth observing that all the terms of $f(x)$, viz., x^2 and $5|x|$ and 6 are positive. Therefore, sum of these three terms cannot be equal to zero for any real value. Hence, no real value of x can be obtained.

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Descartes' Rule of Signs of Roots

Descartes' rule tells us the maximum number of positive roots in any equation. Maximum number of positive roots in $f(x)$ is equal to the change of signs from positive to negative and negative to positive.

With the help of this, we can find out the maximum number of real roots also.

To find out negative roots, we apply the same procedure with $f(-x)$.

Example Let us see the roots of the quadratic equation $x^2 + x - 2 = 0$.

Signs of different terms: x^2 is having positive sign, x is having + sign, constant term (2) is having negative sign.

$f(x) = x^2 + x - 2$, and their signs are

$+ \quad + \quad -$

As there is one sign change, we can conclude that this equation will have one positive real root.

Now, $f(-x) = (-x)^2 + (-x) - 2 = x^2 - x - 2$

Again, there is just one sign change, and hence, there will be one negative root.

Remember That there is nothing like positive or negative imaginary roots. Only real numbers can be negative or positive.

Example 2 How many real roots will be there of the equation $x^4 + x^2 + 1 = 0$?

Solution To find out positive roots

$f(x) = x^4 + x^2 + 1$, and signs are

$+ \quad + \quad +$

There is no sign change between any two terms. Hence, no roots will be positive.

To find out negative roots

$f(-x) = (-x)^4 + (-x)^2 + 1 = x^4 + x^2 + 1$

Again, there is no sign change. Therefore, no roots of this equation will be negative.

And $x = 0$ is also not a root of this equation.

Hence, no real roots are there, which can satisfy this equation.

It can also be inferred here that all the four roots of this equation will be imaginary.

Alternatively, we have done this question while seeing the properties of roots (v) also.

Using graph to find the real/imaginary number of roots: Some typical questions

As we have discussed earlier, number of distinct real roots of an equation is equal to the number of times the graph of equation intersects with the X-axis.

Example 3 How many distinct real roots are possible for the equation $x^{11} + 7x^3 - 5 = 0$?

Solution It can be seen that $f(0) < 0$ and $f(1) > 0$. It means graph of this equation will cut X-axis at some point between 0 and 1.

Hence, there will be a root of this equation between 0 and 1.

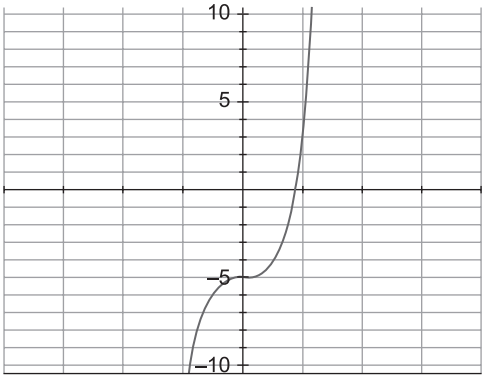
Next, for all the values of $x < 0$ (i.e., negative values), equation will give negative value because x^{11} will become a huge negative number.

And for all the values of $x > 1$ (i.e., positive values), equation will give positive value because x^{11} will become a huge positive number.

Hence, graph of the equation will never intersect the X-axis again. Therefore, it will not have any real root for $x > 1$ or $x < 0$.

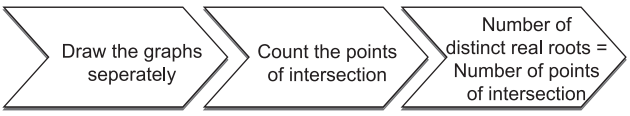
Hence, total number of real roots = 1

Note: You are not required to plot the graph of this equation to understand the method to derive the number of real roots. It is more related to observation than the actual plotting of graph. However, just to make the points given above clear, I have given the graph of the equations below.



Example 4 Find the number of imaginary roots of the equation $x^3 + 2x - 4 = 0$.

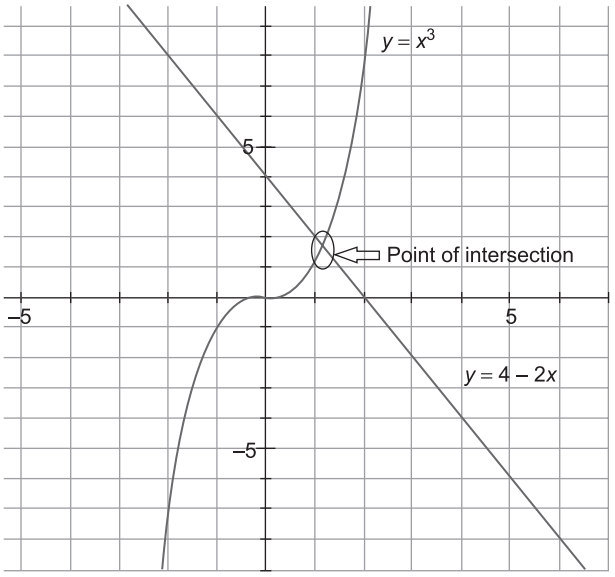
Solution Instead of drawing $x^3 + 2x - 4 = 0$, this equation can be written as $x^3 = 4 - 2x$. We will further draw the graph of $y = x^3$ and $y = 4 - 2x$ as drawing the graph of x^3 and $(4 - 2x)$ are easier. Further, we will observe the points of intersection of these two graphs. Number of points of intersection will be equal to the number of distinct real roots.



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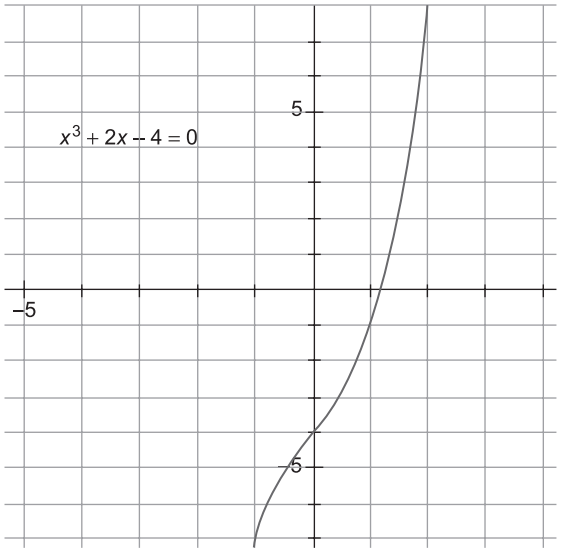
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Following are the graphs:



There is only one point of intersection (given in the circle in the above figure), and hence, only one distinct real root will be there.

It can also be seen with the help of the graph of original equation $x^3 + 2x - 4 = 0$ that it intersects X-axis at only one point. Hence, there will be only one distinct real root.



Since degree of the equation = 3, it will have total three roots. Of which, one root is real ⇒ hence, two roots are imaginary.

ROOTS OF A QUADRATIC EQUATION AND THEIR NATURE

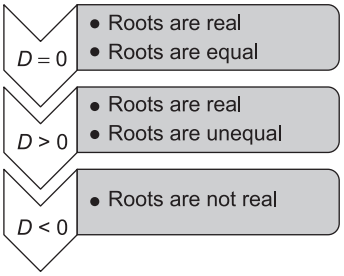
Given is the quadratic equation $ax^2 + bx + c = 0$, where $a \neq 0$.

Assuming that α and β are the roots of this equation.

Then, $\alpha = \dfrac{-b - \sqrt{b^2 - 4ac}}{2a}$ and $\beta = \dfrac{-b - \sqrt{b^2 - 4ac}}{2a}$

It can be seen that nature of both of these two roots depends upon the value of $b^2 - 4ac$.

$(b^2 - 4ac)$ is also known as **discriminant (*D*)** of the quadratic equation $ax^2 + bx + c = 0$.



If D is a perfect square (including $D = 0$) and a, b, and c are rational, then roots will be rational.

Example 5 Given is the quadratic equation $ax^2 + bx + 1 = 0$, where a, $b \in (1, 2, 3, 4)$. For how many set of values of (a, b), quadratic equation $ax^2 + bx + 1 = 0$ will have real roots? ***(CAT 2003)***

Solution For roots to be real, $D \geq 0$

$D = b^2 - 4a \geq 0$

Forming the table for the above-written condition:

B	A
1	Not possible
2	1
3	1
4	2
4	1
4	2
4	3
4	4

Hence, there are seven set of values for (a, b).

Graph of a Quadratic Equation

$f(x) = ax^2 + bx + c = 0$, where $a \neq 0$, and b and c are real coefficients.

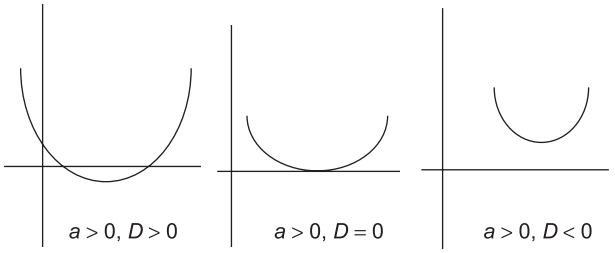
Structure and placement of graph of $y = f(x) = ax^2 + bx + c$ is decided by two factors:

I. Value of 'a' decides that graph will be upward opening or downward opening.

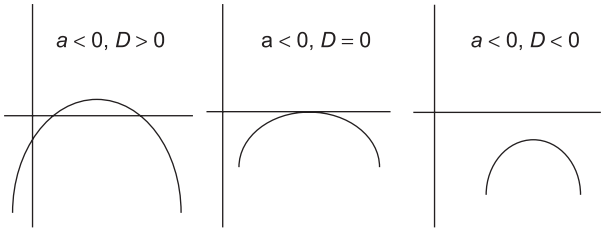
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(a) If $a > 0$, then graph of $y = ax^2 + bx + c$ will be upward opening.



(b) If $a < 0$, then graph of $y = ax^2 + bx + c$ will be downward opening.



II. Value of discriminant ($D = b^2 - 4ac$) decides that at how many points the graph will be intersecting the X-axis.

If $D > 0 \Rightarrow$ There will be two distinct real roots $\Rightarrow$ Graph will intersect X-axis at two distinct points.

If $D = 0 \Rightarrow$ There will be two equal real roots, $\Rightarrow$ Graph will intersect (rather touch) X-axis at one distinct point.

If $D < 0 \Rightarrow$ There will be no real root $\Rightarrow$ Graph will intersect X-axis at no point.

Value of 'c' is the Y-intercept.

Hence, for an equation $x^2 - 6x + 5 = 0$, following observations can be made:

$a > 0 \Rightarrow$ graph will be upward opening.

$D = b^2 - 4ac = 6^2 - 4 \times 1 \times 5 = 36 = 20 = 16 > 0 \Rightarrow$ Graph will intersect X-axis at two distinct points = Equation will have two distinct real roots.

$c = 5 \Rightarrow$ It means graph will intersect Y-axis at point (5, 0).

Obvious from the above graphs, when $D > 0$, there are two points at which this curve is intersecting X-axis. And we know that there are two distinct values of x, which satisfy $f(x)$. In other words, there are two distinct real roots of $f(x) = 0$.

When $D = 0$, there is one point at which this curve is intersecting X-axis. And we know that there is one distinct value of x, which satisfies $f(x)$. In other words, there are two real roots of $f(x) = 0$ and both roots are equal.

When $D < 0$, there is no point at which this curve is intersecting X-axis. Hence, there are no real values of x, which can satisfy $f(x)$. In other words, there is no real root of $f(x) = 0$.

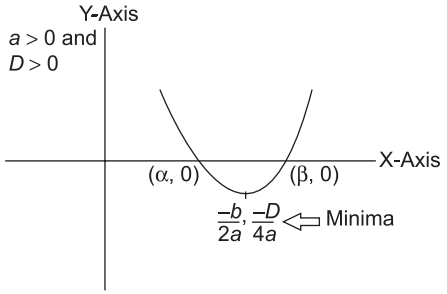
Maxima and Minima of Quadratic Function $y = f(x) = ax^2 + bx + c$

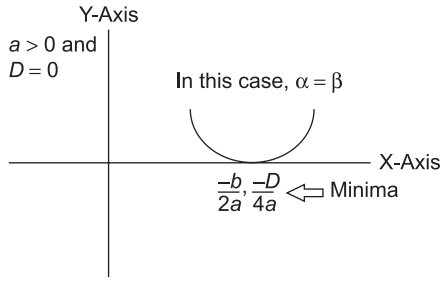
Method 1 *Through coordinates/graphs*

Graph of $y = f(x) = ax^2 + bx + c$ will intersect X-axis at following points with mentioned maxima and minima:

	$D > 0$	$D = 0$	$D < 0$	Maxima	Minima
$a > 0$	Will intersect X-axis at two points	Will intersect X-axis at one point	Will not intersect X-axis.	Infinity (∞)	$-D/4a$
$a < 0$	Will intersect X-axis at two points	Will intersect X-axis at one point	Will not intersect X-axis.	$-D/4a$	Negative infinity ($-\infty$)

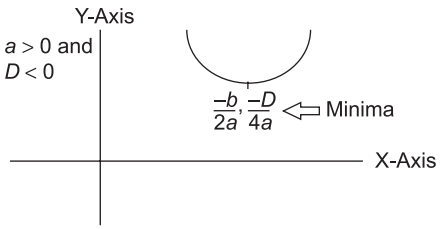
Note: Technically, if the maxima is infinity (∞), we say that maxima does not exist. Similarly, when minima is negative infinity ($-\infty$), we say that minima does not exist. Minima or maxima is said to exist only if a certain number is attained as minima or maxima.

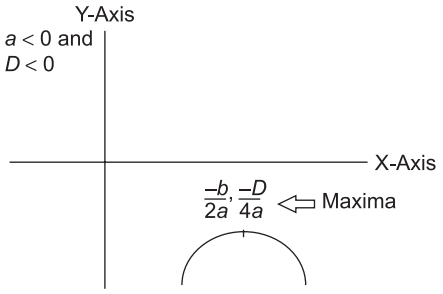


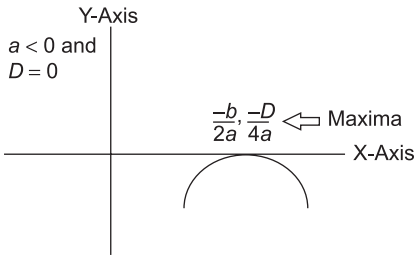


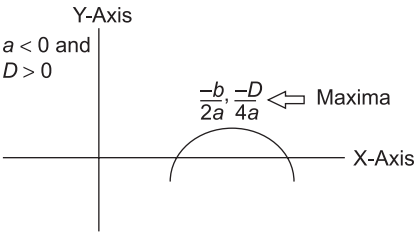
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It can be seen that:

Coordinates of maxima (when $a < 0$) and coordinates of minima (when $a > 0$) is same $= \frac{-b}{2a}, \frac{-D}{4a}$.

It is read as:

At $x = \frac{-b}{2a}$, we obtain minimum value of y (for $a > 0$) which is equal to $\frac{-D}{4a}$, or

At $x = \frac{-b}{2a}$, we obtain maximum value of y (for $a < 0$) which is equal to $\frac{-D}{4a}$.

Where D = Discriminant $= b^2 - 4ac$

Method 2 Through breaking the given equation in a perfect square.

Objective is to break the equation into a format of $(P)^2 \pm Q$. Now, as we know that any perfect square can have a minimum value of 0, so minimum value of $f(x) = \pm Q$.

Let us assume $f(x) = x^2 - 5x + 6$

$$f(x) = \left[x^2 - 2 \times \frac{5}{2} \times x + \left(\frac{5}{2}\right)^2\right] - 0.25$$

$$= \left(x - \frac{5}{2}\right)^2 - 0.25$$

Since $\left(x - \frac{5}{2}\right)^2$ can have a minimum value of zero, the minimum value of $f(x)$ here is (–0.25).

Using graphical method, at $x = -b/2a = 5/2$, minimum value of $f(x) = -D/4a = -1/4 = (-0.25)$

In my perception, any of these methods is better than using differential calculus method.

Reiterating the Points Mentioned above:

(i) If $a > 0$

As we keep on increasing the value of x, value of y also keep on increasing. Hence, maximum value of y is infinity.

It is quite obvious from the graph also that it is going upwards till infinity.

(ii) If $a < 0$

Finding minima:

As we keep on increasing the value of x, value of y keeps on decreasing (since $a < 0$). So, minimum value of y is minus infinity.

It is quite obvious from the graph also that it is going downward till negative of infinity.

Example 6 Consider the quadratic function: $y = 2x^2 + 5x - 6$.

(a) For what value of x, minima of y will be obtained?
(b) What is the minimum value of y?
(c) What is the maximum value of y?

Solution

(a) Minima of y will be obtained for $x = \frac{-b}{2a} = \frac{-5}{2 \times 2} - \frac{-5}{4}$
(b) Minimum value of y:

Method 1 As seen in (a) above, if we put $x = \frac{-5}{4}$, we will obtain the minima.

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At $x = \frac{-5}{4}$, $y = (2x^2 + 5x - 6)$ will be equal to $= 2 \times \left(\frac{-5}{4}\right)^2 + 5 \times \frac{-5}{4} - 6 = 2 \times \frac{25}{16} + \frac{-25}{4} - 6 = \frac{25}{8} - \frac{25}{4} - 6$

$= \frac{25 - 50 - 48}{8} = \frac{-73}{8}$

Method 2 $Y_{\min} = \frac{-D}{4a} = \frac{-(b^2 - 4ac)}{4a}$

$= \frac{[(5)^2 - 4 \times 2 \times 2(-6)]}{4 \times 2} = \frac{-73}{8}$

(c) It can be seen that if we increase the value of x, value of y will keep on increasing.
Hence, maximum value of y = infinite $(+\infty)$

Example 7 For the polynomial $f(x) = x^2 - 13x + 36$, find the following:
(a) The range of values of x for which $f(x) < 0$
(b) Minimum value of $f(x)$
(c) Value of x for which $f(x)$ is minimum
(d) x intercept and y intercept of $f(x)$

Solution
(a) $x^2 - 13x + 36 < 0 \Rightarrow (x - 9)(x - 4) < 0 \Rightarrow 4 < x < 9$

(b) Minimum value of $f(x) = \frac{-D}{4a} = \frac{-(b^2 - 4ac)}{4a}$

$= \frac{-[(13)^2 - 4 \times 1 \times 36]}{4 \times 1} = \frac{-25}{4}$

(c) Value of x for which $f(x)$ is minimum $= \frac{-b}{2a} = \frac{-(-13)}{2 \times 1}$

$= \frac{13}{2}$

(d) X intercept will be obtained when we put $y = f(x) = 0$
$x^2 - 13x + 36 = 0 \quad \Rightarrow (x - 9)(x - 4) = 0$
$\Rightarrow x = 9$ or $x = 4$
Y intercept will be obtained when we put $x = 0$
Given that $f(x) = x^2 - 13x + 36 \Rightarrow f(0) = 0^2 - 13 \times 0 + 36 = 36$. This is the point at which curve intersects Y-axis.

Example 8 $f(x) = ax^2 + bx + c = 0$ has two real roots. Given that $c > 0$ and minima of quadratic function $f(x) = 9a + 3b + c$. How many roots of $f(x)$ are positive?
(a) 0 (b) 1
(c) 2 (d) Cannot be determined

Solution Question mentions that $f(x)$ obtains a minima $\Rightarrow$ Hence, $a > 0$.

Further, it can be seen that $f(3) = 9a + 3b + c \Rightarrow f\left(\frac{-b}{2a}\right) = f(3) \Rightarrow \left(\frac{-b}{2a}\right) = 3 \Rightarrow b < 0$ (as $a > 0$)

We can conclude that sum of roots $= \frac{-b}{a} > 0$ and product of roots $= \frac{c}{a} > 0$.

Sum of roots as well as product of roots are positive $\Rightarrow$ both the roots are positive.

Hence, answer is option (c).

ALGEBRAIC CALCULATIONS ON ROOTS

$f(x) = ax^n + bx^{n-1} + cx^{n-2} + \ldots + K$

Sum of roots taking one at a time = (coefficient of x^{n-1}/coefficient of x^n) $\times (-1)^n$, where n is the number of roots taken at a time.

Sum of roots taking two at a time = (coefficient of x^{n-2}/coefficient of x^n) $\times (-1)^n$, where n is the number of roots taken at a time.

Sum of roots taking three at a time = (coefficient of x^{n-3}/coefficient of x^n) $\times (-1)^n$, where n is the number of roots taken at a time.

…

Product of roots, taken all at a time = (constant term/coefficient of x^n) $\times (-1)^n$, where n is the total number of roots.

Quadratic Equation

$f(x) = ax^2 + bx + c = 0$

Assume that the roots are α and β.

$\alpha + \beta$ = (coefficient of x^{n-1}/coefficient of x^n) $\times (-1)^n = \frac{b}{a}(-1)^1 = -\frac{b}{a}$

$\alpha\beta$ = (constant term/coefficient of x^n) $\times (-1)^n = \frac{c}{a}(-1)^2 = \frac{c}{a}$

Cubic Equation

$f(x) = ax^3 + bx^2 + cx + d = 0$

Assume that the roots are α, β, and γ.

$\alpha + \beta + \gamma$ = (coefficient of x^{n-1}/coefficient of x^n) $\times (-1)^n$

$\frac{b}{a} = (-1)^1 = -b/a$

$\alpha\beta + \gamma\alpha + \beta\gamma$ = (coefficient of x^{n-2}/coefficient of x^n) $\times (-1)^n = \frac{c}{a}(-1)^2 = \frac{c}{a}$

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$\alpha\ \beta\ \gamma$ = (constant term/coefficient of x^n) × $(-1)^n$ $= \frac{d}{a}(-1)^3 = -\frac{d}{a}$

Bi-quadratic Equation

$f(x) = ax^4 + bx^3 + cx^2 + dx + e = 0$

Assume that the roots are α, β, γ, and δ.

$\alpha+\beta+\gamma+\delta$ = (coefficient of x^{n-1}/coefficient of x^n) × $(-1)^n$ $= \frac{b}{a}(-1)^1 = -\frac{b}{a}$

$\alpha\beta+\gamma\delta+\alpha\delta+\beta\gamma+\alpha\gamma+\delta\beta$ = (coefficient of x^{n-2}/coefficient of x^n) × $(-1)^n = \frac{c}{a}(-1)^2 = \frac{c}{a}$

$\alpha\beta\gamma+\delta\alpha\beta+\gamma\delta\alpha+\beta\gamma\delta$ = (coefficient of x^{n-3}/coefficient of x^n) × $(-1)^n = \frac{d}{a}(-1)^3 = -d/a$

$\alpha\ \beta\ \gamma\ \delta$ = (constant term/coefficient of x^n) × $(-1)^n$ $= \frac{e}{a}(-1)^4 = \frac{e}{a}$

Example 9 If the polynomial $ax^4 + bx^3 + cx^2 + dx + e$ has the property that the product of all the roots, taken all at a time, is 1/3rd of the sum of the product of roots, taking two at a time. What is the relationship between e and c?

Solution Product of all the roots, taking at a time $= \frac{e}{a}$

Sum of the product of roots, taking two at a time $= \frac{c}{a}$

Now, $\frac{e}{a} = 1/3\ \frac{c}{a}$

So, $c = 3 \times e$

Example 10 If a, b, and c are the roots of the equation $x^3 - 3x^2 + 2x + 1 = 0$, then what is the value of $\frac{1}{a}+\frac{1}{b}+\frac{1}{c}$?

Solution $\frac{1}{a}+\frac{1}{b}+\frac{1}{c} = \frac{ab+bc+ca}{abc}$

$ab + bc + ca = 2$ and $abc = -1$

$\frac{1}{a}+\frac{1}{b}+\frac{1}{c} = \frac{ab+bc+ca}{abc} = -2$

Example 11 $f(x) = ax^2 + bx + c = 0$ has two real roots. If $ac < 0$, then how many of the roots of $f(x) = 0$ will be negative?

Solution If $ac < 0$, then signs of a and c will be opposite (one will be positive and other will be negative).

Hence, product of the roots $= \alpha\beta = \frac{c}{a} < 0$

Hence, α and β will have opposite signs. Therefore, one of these two will be positive and other will be negative.

Hence, one root of $f(x) = 0$ will be negative.

Example 12 $f(x) = ax^2 + bx + c = 0$ has two real roots. If $ac > 0$, then how many of the roots of $f(x) = 0$ will be negative?

Solution If $ac > 0$, then, signs of a and c will be same (both will be either positive or both will be negative).

Hence, product of the roots $= \alpha\beta = \frac{c}{a} > 0$

Hence, α and β will have same signs. Therefore, both the roots will be either positive or both will be negative.

Formation of Quadratic Equation from its Roots

Case I If α and β are the roots of the equation $ax^2 + bx + c = 0$, then we can write

$ax^2 + bx + c = x^2 - (\alpha+\beta)x + \alpha\beta = x^2 -$ (sum of roots) $x +$ product of roots $= 0$

Or, $ax^2 + bx + c = a(x - \alpha)(x^2 - \beta) = 0$

In general, if $x_1, x_2, x_3, \ldots, x_n$ are the roots of the equation $f(x) = ax^n + bx^{n-1} + cx^{n-2} + \ldots + K = 0$

Then, $f(x) = ax^n + bx^{n-1} + cx^{n-2} + \ldots + K = a(x - x_1)(x - x_2)(x - x_3)\ldots(x - x_n)$

Case II p and q are the roots of the equation $ax^2 + bx + c = 0$, and we will have to find out the quadratic equation with $(1/p)$ and $(1/q)$ as the roots:

Put $x = 1/p$

$a(1/p)^2 + b(1/p) + c = 0$

Or, $cp^2 + bp + a = 0$

Hence, quadratic equation is $cx^2 + bx + a = 0$

Case III p and q are the roots of the equation $ax^2 + bx + c = 0$, and we will have to find out the quadratic equation with $(-p)$ and $(-q)$ as the roots:

Put $x = -p$

$a(-p)^2 + b(-p) + c = 0$

Or, $ap^2 - bp + c = 0$

Hence, quadratic equation is $ax^2 - bx + c = 0$

Example 13 $f(x) = ax^2 + bx + c = 0$ has 2 and 3 as its roots. What is the value of $a + b + c$?

Solution Since 2 and 3 are the roots of the equation, hence, equation will be

$a(x - 2)(x - 3) = 0 \Rightarrow$ This does not give us the unique value of a, b, and c. Hence, unique value of $(a + b + c)$ cannot be determined.

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Flaw Detector – Some students might solve this question like the following:

Since 2 and 3 are the roots, hence, equation is: $(x-2)(x-3)=0 \Rightarrow x^2-5x+6=0$. Comparing it with $ax^2+bx+c=0$ gives $a=1$, $b=-5$, and $c=3$. Hence, $a+b+c=1-5+3=-1$

In fact, $(x-2)(x-3)=0$ is **only one** of the equations out of infinite such equations that will have 2 and 3 as the roots. Another such equation will be $2(x-2)(x-3)=0$. In this case, $a=2$, $b=-10$, and $c=6$ giving $a+b+c=-2$

Hence, it is wrong to say that $a+b+c=-1$

In fact, general quadratic equation with roots as 2 and 3 will be:

$a(x-2)(x-3)=0$, where a is the coefficient of x^2 in $ax^2+bx+c=0$

Example 14 What will be the quadratic equation having the roots opposite in sign as that of the quadratic equation $x^2+5x+6=0$?

Solution Putting $(-x)$ at the place of x will give us the solution.

Desired equation is $(-x)^2+5(-x)+6=0$

Therefore, equation is $x^2-5x+6=0$

Alternatively, roots of the quadratic equation $x^2+5x+6=0$ are -2 and -3.

Therefore, roots of the desired equation should be 2 and 3.

Hence, the quadratic equation is $(x-2)(x-3)=0$, or, $x^2-5x+6=0$

Case IV p and q are the roots of the equation $ax^2+bx+c=0$, and we will have to find out the quadratic equation with $(p)^2$ and $(q)^2$ as the roots:

Let x be a root of original equation and y be a root of transformed equation.

$y=x^2$, therefore, we will put $x=\sqrt{y}$

$a(\sqrt{y})^2+b(\sqrt{y})+c=0$

Solving this will give us the desired equation.

SOLUTION OF INEQUATIONS

Case I $f(x)=x^2-5x+6\geq 0$

$f(x)=x^2-5x+6=(x-2)(x-3)\geq 0$



We have got three intervals of values here: $-\infty$ to 2, 2 to 3 and 3 to $+\infty$.

To check the values of x satisfying the above written inequation, we will take any one value of x from any of the intervals. If this value satisfies the inequation, then inequality will be satisfied in the alternate intervals.

Taking $x=10$, $f(x)=(x-2)(x-3)=(10-2)(10-3)=+\text{ve}$

Therefore, inequality is satisfied in this interval. Hence, it will not be satisfied in the interval $2\leq x\leq 3$ and again will be satisfied in the interval $-\infty<x\leq 2$.

Hence, values of x satisfying the above written inequality lies outside 2 and 3, that is, $x\leq 2$ or $x\geq 3$.

Case II $f(x)=x^2-5x+6\leq 0$



It can similarly be observed that values of x are lying in between 2 and 3.

Hence, $2\leq x\leq 3$

In general, if $(x-a)(x-b)\leq 0$, then $a\leq x\leq b$

And if $(x-a)(x-b)\geq 0$ and $a<b$, then $x\leq a$ or $x\geq b$

Example 15 What values of x will satisfy the quadratic inequation $f(x)=-x^2+3x+4>0$?

Solution $f(x)=-x^2+3x+4>0$, or, $x^2-3x-4<0$

or, $(x+1)(x-4)<0$

So, values of x satisfying $f(x)=-1<x<4$

Newton's Sums of Roots

If $\alpha, \beta, \gamma, \delta \ldots$ are the roots of a polynomial $f(x)=0$, then Newton's Sums of Roots formula gives the sum of $(\alpha^n+\beta^n+\gamma^n+\delta^n+\ldots)$.

General Expression

Consider a polynomial $P(x)$ of degree n:

$$P(x)=a_nx^n+a_{n-1}x^{n-1}+\ldots+a_1x+a_0$$

Let $p(x)=0$ have roots $x_1, x_2, \ldots, x_n$.

Define the following sums:

$$P_1=x_1+x_2+\ldots+x_n$$

$$P_2=x_1^2+x_2^2+\ldots+x_n^2$$

$\ldots$

$$P_k=x_1^k+x_2^k+\ldots+x_n^k$$

Newton's sum of roots formula tell us that,

$$a_nP_1+a_{n-1}=0$$

$$a_nP_2+a_{n-1}P_1+2a_{n-2}=0$$

$$a_nP_3+a_{n-1}P_2+a_{n-2}P_1+3a_{n-3}=0$$

and so on

Example 16 Consider the equation $P(x)=x^3+2x^2+5x-8=0$. If a, b and c are the roots of this equation, find the value of $a^2+b^2+c^2$.

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Solution Assume $a + b + c = P_1$, $a^2 + b^2 + c^2 = P_2$ and $a^2 + b^2 + c^2 = P_3$

Given that $a_n = 1$ = Coefficient of x^3

$a_{n-1} = 2$ = Coefficient of x^2

$a_{n-2} = 5$ = Coefficient of x^1

a_{n-4} = Constant term = -8

Using Newton's sum of roots formula:

$a_n P_1 + a_{n-1} = 0 \Rightarrow 1 \times P_1 + 2 = 0$. Hence, (i) $P_1 = a + b + c = -2$

Note: To calculate $P_1 = a + b + c$, we could have simply used sum of roots $= \frac{-b}{a} = \frac{-2}{1} = -2$. [I used Newton's sum of roots formula to calculate this, so as to put you in 'groove' of using this formula.]

Next, $1 \times P_2 + 2 \times P_1 + 2 \times 5 = 0$

$\Rightarrow P_2 + 2 \times (-2) + 2 \times 5 = 0$

[Using the value of P_1 from (i) as derived above]

$\Rightarrow$ Hence, (ii) $P_2 = -6$

Note: Since $P_2 = a^2 + b^2 + c^2$ = Sum of squares of three numbers = negative number, hence, we can conclude that two roots of this equation are imaginary. All three cannot be imaginary as imaginary roots always occur in pairs.

Some More Worked-Out Problems

Example 17 If the roots of the quadratic equation $x^2 - 4x - \log_3 N = 0$ are real, then find the minimum value of N?

Solution Since the roots of the given equation are real, so $D \geq 0$.

Or, $16 + 4 \log_3 N \geq 0$, or, $\log_3 N \geq -4$

Or, $N \geq 3^{-4}$

So, least value of $N = 1/81$

Example 18 Find the real values of N for which the quadratic equation $2x^2 - (N^3 + 8N - 1)x + N^2 - 4N = 0$ will have roots of opposite sign.

Solution The roots of given equation will be of opposite signs only if they are real and the product of the roots is negative.

Or, $D \geq 0$, and product of roots < 0

Or, $(N^3 + 8N - 1)^2 - 8(N^2 - 4N) \geq 0$ and $\frac{N^2 - 4N}{2} < 0$

Or, $N^2 - 4N < 0$. Hence, $0 < N < 4$

Example 19 If p and q are the roots of the quadratic equation $2x^2 + 6x + N = 0 (N < 0)$, then what is the maximum value of $\frac{p}{q} + \frac{q}{p}$?

Solution $p + q = -3$, and $pq = N/2$

Now, $N < 0$, hence, $D = 36 - 4N > 0$. So, p and q are real.

Again, $\frac{p}{q} + \frac{q}{p} = \frac{(p+q)^2}{pq} - 2 = \frac{18}{N} - 2$

Therefore, maximum value of $\frac{p}{q} + \frac{q}{p} = -2$

Example 20 For which value of k does the following pair of equations yield a unique solution of x such that the solution is positive? ***(CAT 2005)***

$X^2 - y^2 = 0$

$(x - k)^2 + y^2 = 1$

(a) 2 (b) 0 (c) $\sqrt{2}$ (d) $\sqrt{2}$

Solution (c) $y^2 = x^2$

$2x^2 - 2kx + k^2 - 1 = 0$

As there is unique solution, $D = 0$

$\Rightarrow 4k^2 = 8k^2 - 8$

$\Rightarrow 4k^2 = 8 \Rightarrow k = \sqrt{2}$

Example 21 Let

$$x = \sqrt{4 + \sqrt{4 - \sqrt{4 + \sqrt{4 - \ldots \text{to infinity.}}}}}$$ Then x equals

(a) 3 (b) $\left(\frac{\sqrt{13}+1}{2}\right)$

(c) $\left(\frac{\sqrt{13}-1}{2}\right)$ (d) $\sqrt{3}$ ***(CAT 2005)***

Solution (b)

$x = \sqrt{4 + \sqrt{4 - x}} \Rightarrow x^2 - 4 + \sqrt{4 - x}$

$(x^2 - 4) = \sqrt{4 - x}$

Now, put the values from options.

Only option (b) satisfies the condition.

Alternatively, value of x will be more than 2, which is given there only in option (a) and option (b). Since it is slightly more than 2, option (b) will be the answer.

Example 22 The polynomial $p(x) = x^3 + ax^2 + bx + c$ has the property that the average of its roots, the product of its roots, and the sum of its coefficients are all equal. The y-intercept of the graph of $y = p(x)$ is 2. Which of the following is equal to b?

(a) -11 (b) -10

(c) -9 (d) 1

Solution Since, y-intercept of the graph of $y = p(x)$ is 2 it can be concluded that $c = 2$. Therefore, the product of the roots is $\frac{-c}{1} = -2$.

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This also tell us that $\frac{-a}{3}$ is the average of the root, so $\frac{-a}{3} = -2$, or, $a = 6$. We are also given that the sum of the coefficients is also equal to -2. Therefore, $1 + 6 + b + 2 = -2$

Hence, $b = -11$. Hence, option (a) is the answer.

Practice Exercises

Warm Up

Q.1 What is the value of k when the equation $x^2 - (k + 6)x + 2(2k - 1) = 0$ has sum of the roots equal to half of their product?
(a) -2 (b) 7 (c) 9 (d) 12

Q.2 What is the value of P when the sum of the squares of roots of the equation $x^2 - (p - 2)x - p - 1 = 0$? Assume the least value.
(a) -2 (b) 1 (c) 9 (d) 12

Q.3 If the equation $x^2 + 2(p + 1)x + 9p - 5 = 0$ has only positive roots, then which of the following is true?
(a) $p \geq 6$ (b) $p \leq 0$
(c) $p \leq -6$ (d) None of these

Q.4 The quadratic equations $x^2 - ax - 21 = 0$ and $x^2 - 3ax + 35 = 0$, where $a > 0$ have a common root. What is the value of a?
(a) 0 (b) 1 (c) 4 (d) 6

Q.5 A, B, and C are real values such that $A + B + C = 2$, $A^2 + B^2 + C^2 = 6$ and $A^3 + B^3 + C^3 = 8$, then find the value of $A^4 + B^4 + C^4$?
(a) 6 (b) 12 (c) 18 (d) 24

Q.6 What is the sum of the roots of quadratic equation $3x^2 + 2x - 1 = 0$?
(a) 1/3 (b) 2/3 (c) 3 (d) $-2/3$

Q.7 What is the product of the roots of quadratic equation $3x^2 + 2x - 1 = 0$?
(a) $-1/3$ (b) 2/3 (c) 3 (d) $-2/3$

Q.8 What is the sum of the roots of cubic equation $x^3 + 2x^2 + x - 1 = 0$?
(a) -2 (b) 2/3 (c) -1 (d) 2

Q.9 Which of the following quadratic equations will not have real roots?
(a) $3x^2 + 2x - 1 = 0$
(b) $x^2 + 2x + 1 = 0$
(c) $3x^2 + 2x + 5 = 0$
(d) $2x^2 + 3x + 1 = 0$

Q.10 Roots of the quadratic equation $ax^2 + bx + c = 0$ are 1 and 2, for some certain value of a, b, and c. What will be the sum of the roots of equation having the roots opposite in sign of the equation $ax^2 + bx + c = 0$?
(a) 3 (b) 6
(c) -3 (d) None of these

Q.11 The roots of equation $a^2x^2 + abx = b^2$, where $a \neq 0$, are:
(a) Real but unequal (b) Real but equal
(c) Not real (d) None of these

Q.12 Which of the following satisfies the equation: $-x^2 + 6x - 8 > 0$?
(a) $3 < x < 5$ (b) $2 < x < 4$
(c) $x < 2$ and $x > 4$ (d) $x < 3$ and $x > 5$

Q.13 For what value of K, equations $2x + 3y = 5$ and $4x + Ky = A$ will not have any solution?
(a) 2
(b) -2
(c) 6
(d) More than one value is possible

Q.14 Five burgers, six pizzas, and seven cold drinks cost ₹178, and six burgers, four pizzas, and two cold drinks cost ₹124. What is the cost of 3 (pizzas + burgers + cold drinks)?
(a) ₹60
(b) ₹62.5
(c) ₹90
(d) Cannot be determined

Q.15 A and B are the roots of equation $x^2 - 13x + N = 0$. If $A - B = 3$, what is the value of N?
(a) 20 (b) 27
(c) 36 (d) None of these

Q.16 If $x < 0$, then what is the maximum value of $\frac{9}{x} + \frac{x}{9}$?
(a) 2 (b) -2
(b) $+\infty$ (d) None of these

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Q.17 If the coefficient of x^2 and the constant term of a quadratic equation are interchanged, then which of the following will not get changed?
(a) Sum of the roots
(b) Product of the roots
(c) Roots of the equation
(d) None of these

Q.18 If $(x+2)^2 = 9$ and $(y+3)^2 = 25$, then what is the maximum value of $\frac{x}{y}$?
(a) $-5/8$ (b) $5/8$ (c) $3/7$ (d) $-3/7$

Q.19 If one of the roots of quadratic equation $2x^2 - 7x + q = 0$ is 3, then what is the other root of this equation?
(a) 1/3 (b) 1/2 (c) 1 (d) 2

Q.20 What is the remainder when the polynomial $x^4 - 3x^2 + 7x - 10$ is divided by $x - 2$?
(a) 12 (b) 1 (c) -1 (d) 8

FOUNDATION

Q.1 How many real roots will be there of the equation $x^4 + x^2 = 0$?
(a) 0 (b) 2
(c) 4 (d) None of these

Q.2 If $(x) + (1/x) = 2$, then what is the value of $(x)^3 + (1/x)^3$?
(a) 1 (b) -2 (c) -1 (d) 2

Q.3 Given that $as^4 + bs^3 + cs^2 + ds + e = 0$ is a bi-quadratic equation in s and $a \neq 0$, what is the value of $(1 - s_1)(1 - s_2)(1 - s_3)(1 - s_4)$?
(a) 1
(b) $(a + b + c + d + e)/a$
(c) 0
(d) $a + b + c + d + e$

Q.4 If $(x+2)$ is a factor of $x^4 - 4x^2 + 2ax + 3 = 0$, then what is the value of a?
(a) -2 (b) 3 (c) 3/4 (d) -3

Q.5 $x^2 - 5x + 6 = 0$ and $x^2 + mx + 3 = 0$ have a root in common. What is the value of m?
(a) -4
(b) $-7/2$
(c) Both 1 and 2
(d) None of these

Q.6 Find the number of integral roots of equation $x^3 - x^2 + 2x - 17 = 0$.
(a) 1 (b) 2
(c) 3 (d) None of these

Q.7 The number of real solutions of $x - \frac{1}{x^2 - 4} = 2 - \frac{1}{x^2 - 4}$ is:
(a) 0 (b) 1 (c) 2 (d) Infinite

Q.8 The number of real solutions of the equation $e^x = x$ is:
(a) 1 (b) 2
(c) 0 (d) None of these

Q.9 The sum of the real roots of equation $x^2 + |x| - 6 = 0$ is:
(a) 4 (b) 0
(c) -1 (d) None of these

Q.10 If one root of the equation $(k^2 + 1)x^2 + 13x + 4k = 0$ is reciprocal of the other, then k has the value:
(a) $-2 + \sqrt{3}$ (b) $2 - \sqrt{3}$
(c) 1 (d) None of these

Q.11 If α, β are roots of the equation $(x - a)(x - b) = c$, $c \neq 0$, then find the roots of the equation $(x - \alpha)(x - \beta) + c = 0$.
(a) a, c (b) b, c
(c) a, b (d) $a + c, b + c$

Q.12 If the sum of the roots of quadratic equation $ax^2 + bx + c = 0$ is equal to the sum of the squares of their reciprocals, then $\frac{b^2}{ac} + \frac{bc}{a^2}$ is equal to:
(a) 2 (b) -2 (c) 1 (d) -1

Q.13 If a, b are roots of $x^2 + px + q = 0$ and g, d are the roots of $x^2 + px - r = 0$, then $(a - g)(a - d)$ is equal to:
(a) $q + r$
(b) $q - r$
(c) $-(q + r)$
(d) $-(p + q + r)$

Q.14 If α, β, γ be the roots of the equation $x(1 + x^2) + x^2(6 + x) + 2 = 0$, then find the value of $\alpha^{-1} + \beta^{-1} + \gamma^{-1}$.
(a) -3 (b) $\frac{1}{2}$
(c) $-\frac{1}{2}$ (d) None of these

Q.15 If the equations $2x^2 - 7x + 1 = 0$ and $ax^2 + bx + 2 = 0$ have a common root, then:
(a) $a = 2, b - 7$
(b) $a = \frac{7}{2}, b = 1$
(c) $a = 4, b = -14$
(d) None of these

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Q.16 If $(x-1)^3$ is a factor of $x^4+ax^3+bx^2+cx-1$, then find the other factor.
(a) $x-3$ (b) $x+1$
(c) $x+2$ (d) None of these

Q.17 If $x^2-bx+c=0$ has equal integral roots, then
(a) b and c are integers.
(b) b and c are even integers.
(c) b is an even integer and c is a perfect square of a positive integer.
(d) (a) and (c) both

Q.18 Find the number which exceeds its reciprocal by 35/6.
(a) 4 (b) 5 (c) 6 (d) 7

Q.19 If a and b are the roots of equation $2x^2-3x-2=0$, find the value of (a^3+b^3) without finding the roots of the equation.
(a) $\frac{8}{3}$ (b) $\frac{64}{3}$ (c) $\frac{63}{8}$ (d) 16

Q.20 If the roots of the equation $3ax^2+2bx+c=0$ are in the ratio of 2:3, then:
(a) $8ac=25b$ (b) $8ac=9b^2$
(c) $8b^2=9ac$ (d) $8b^2=25ac$

Q.21 In writing a quadratic equation of the form $x^2+bx+c=0$, a student writes the coefficient of x incorrectly and finds the roots as 7 and 8. Another student makes a mistake in writing the constant term and gets the roots as 8 and -3. Find the correct quadratic equation.
(a) $x^2-5x+56=0$ (b) $x^2-5x+21=0$
(c) $x^2-21x+56=0$ (d) $x^2-7x+21=0$

Q.22 The quadratic equation, whose roots are reciprocal of the roots of $1-3x-2x^2=0$, is:
(a) $2-3x-2x^2=0$ (b) $2x^2+3x-1=0$
(c) $x^2-3x-2=0$ (d) $x^2-2x-3=0$

Q.23 Find the value of K in the quadratic equation $Kx^2-3x+2=0$, such that one root is double the other.
(a) -1 (b) 1 (c) -2 (d) 2

Q.24 The denominator of a positive fraction is 1 more than the square of the numerator. If 4 is added to both the numerator and the denominator, the fraction becomes 1/2. Find the original fraction.
(a) 1/3 (b) 2/5 (c) 4/17 (c) 3/10

Q.25 Ramesh bought certain number of chairs for ₹2400. If the price of each chair is reduced by ₹20, then 10 more chairs can be purchased for the same amount. Find the number of chairs he purchased initially.
(a) 20 (b) 30 (c) 40 (d) 50

Q.26 If $x^2-kx+5=0$ has 3 as a root, then find the value of K.
(a) 17/5 (b) 13/2 (c) 14/3 (d) 16/5

Q.27 If a, b, and c are real numbers such that $a+b+c=0$, then find the value of $a^3+b^3+c^3$.
(a) 0 (b) $ab^2+bc^2+ca^2$
(c) $3abc$ (d) None of these

Q.28 One root of the equation $x^3-ax^2+11x-6=0$ is 2. Find other two roots.
(a) 0, 1 (b) 1, 3 (c) 2, 3 (d) 1, 2

Q.29 If $x^4+\frac{1}{x^4}=47$, then find the value of $x^3+\frac{1}{x^3}$.
(a) 9 (b) 18 (c) 12 (d) 27

Q.30 If the roots of the equation $px^2+rx+r=0$ are in the ratio of $a:b$, then find the value of $\sqrt{b/a}+\sqrt{a/b}$.
(a) $\sqrt{r/p}$ (b) $-\sqrt{r/p}$
(c) $\sqrt{1/p}$ (d) $\sqrt{1/r}$

Q.31 How many numbers in the set $\{-4,-3,0,2\}$ satisfy the conditions $|y-4|<6$ and $|y+4|<5$?
(a) 3 (b) 1
(c) 2 (d) None of these

Q.32 If $x^3-ax^2+bx+10$ is perfectly divisible by $(x+5)$ and $x^4+x^3+bx^2-ax+42$ is perfectly divisible by $(x-3)$, find the value of $(a-3b)$.
(a) $\frac{450}{7}$ (b) 50
(c) 100 (d) None of these

Q.33 If α and β are the roots of $x^2+x+2=0$, then $\frac{\alpha^{10}+\beta^{10}}{\alpha^{-10}+\beta^{-10}}$ is equal to:
(a) 4096 (b) 2048 (c) 1024 (d) 512

Q.34 A man born in 1900s realized that in 1980 his age was the square root of the year of his birth. What is his birth year?
(a) 1929 (b) 1949 (c) 1936 (d) 1946

Q.35 In solving a problem on quadratic equation, one student makes a mistake in the constant term of the equation and gets -3 and -2 for the roots. Another student makes a mistake in the coefficient of the first degree term and finds -1 and -2 for the roots. The correct equation was:
(a) $x^2-5x+2=0$ (b) $x^2+5x+2=0$
(c) $x^2+5x-2=0$ (d) $x^2-5x-2=0$

Q.36 What is the solution set of the inequation $x^3-x^2+x-1>0$?

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(a) $x < -1$ (b) $-1 < x < 1$
(c) $x > 1$ (d) $x > 0$

Q.37 The number of distinct real roots of the equation $|V^2 + V - 6| - 3V + 7 = 0$ is:
(a) 0 (b) 4 (c) 2 (d) 3

Q.38 In solving an equation of the form $ax - b = 0$ (a and b having only 1 as the common factor), A made a mistake of copying 'b' and got $\frac{7}{3}$ as the root, whereas B made a mistake of copying 'a' and got $\frac{8}{5}$ as the root. The correct root is:
(a) $\frac{7}{5}$ (b) $\frac{3}{8}$
(c) $\frac{8}{3}$ (d) Cannot be determined

Q.39 If one of the roots of the cubic expression $x^3 - ax^2 + 11x - 6$ is 3, what are the other roots?
(a) 6 and 2 (b) −1 and 2
(c) 1 and 2 (d) −2 and −1

Q.40 If m and n are natural numbers, then:
(a) $m^{100} + n^{100} > (m+n)^{100}$
(b) $m^{100} + n^{100} \geq (m+n)^{100}$
(c) $m^{100} + n^{100} < (m+n)^{100}$
(d) $m^{100} + n^{100} \leq (m+n)^{100}$

Moderate

Q.1 How many real roots will be there of the equation $= x^2 + 5|x| + 6 = 0$?
(a) 0 (b) 2
(c) 4 (d) None of these

Q.2 If $s = 2 + 2^{1/3} + 2^{2/3}$, then what is the value of $s^3 - 6s^2 + 6s$?
(a) 4 (b) 2 (c) 1 (d) 8

Direction for Questions 3 and 4: Read the passage below and solve the questions based on it.

x_1, x_2, and x_3 are the roots of two distinct quadratic equations (with coefficient of x^2 being 1 in both the equations) in which x_1 is a common root.

Q.3 If x_1, x_2, and x_3 are in AP and the coefficient of x in the equation with x_1 and x_2 as its root is −4, then find one of the root.
(a) 1 (b) 3
(c) 5 (d) Cannot be determined

Q.4 If x_1, x_2, and x_3 are in AP and the coefficient of x in the equation with x_1 and x_3 as its root is −6, then find one of the root.
(a) 1 (b) 3
(c) 5 (d) Cannot be determined

Direction for Questions 5 and 6: Read the passage below and solve the questions based on it.

One of the +ve roots of a quadratic equation $ax^2 + mx + c = 0$, where $a = 1$, is twice that of the other. The sum of the coefficient of x and constant term is 2.

Q.5 What is the value of m in the equation?
(a) −4 (b) 4 (c) −6 (d) 6

Q.6 If x_1 and x_2 are the roots of this equation, then find the value of $|x_1 - x_2|$.
(a) 2 (b) 4
(c) 6 (d) None of these

Q.7 The two roots of the quadratic equation $ax^2 + bx + c = 0$ are both two-digit integers with the same units digit, but D/a^2 (where D is the discriminant given as $b^2 - 4ac$) is not a multiple of 100. Which of the following is a possible value of the unit digit of $b/2a$?
(a) 4 (b) 5 (c) 6 (d) 7

Q.8 The polynomial $(ax^2 + bx + c)(ax^2 - bx - c)$, $ac \neq 0$, has:
(a) Four real zeros
(b) At least two real zeros
(c) At most two real zeros
(d) No real zeros

Q.9 The number of real solutions of $1 + |e^x - 1| = e^x(e^x - 2)$ is:
(a) 0 (b) 1 (c) 2 (d) 4

Q.10 If $y \neq 0$, then the number of values of the pair (x, y) such that $x + y + \frac{x}{y} = \frac{1}{2}(x+y)\frac{x}{y} = -\frac{1}{2}$, is:
(a) 1 (b) 2
(c) 0 (d) None of these

Q.11 The equation $\sqrt{x+1} - \sqrt{x-1} = \sqrt{4x-1}$ has.
(a) No solution
(b) One solution
(c) Two solutions
(d) More than two solutions

Q.12 Find the number of real solutions of the equation $\frac{6-x}{x^2-4} = 2 + \frac{x}{x+2}$.

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(a) Two (b) One
(c) Zero (d) None of these

Q.13 If $x \in Z$ is (the set of integers) such that $x^2 - 3x < 4$, then find the number of possible values of x.
(a) 3 (b) 4
(c) 6 (d) None of these

Q.14 If x is an integral satisfying $x^2 - 6x + 5 \leq 0$ and $x^2 - 2x > 0$, then find the number of possible value of x.
(a) 3 (b) 4 (c) 2 (d) Infinite

Q.15 The polynomial $p(x) = x^3 + ax^2 + bx + c$ has the property that the average of its zeros, the product of its zeros, and the sum of its coefficients are all equal. The y-intercept of the graph of $y = p(x)$ is 2, what is b?
(a) −11 (b) −10 (c) −9 (d) 1

Q.16 Find the number of solutions of $\log_4(x-1) = \log_2(x-3)$.
(a) 2 (b) 1 (c) 3 (d) 0

Q.17 If $a > 1$, the roots of the equation $(1 - a)\, x^2 + 3ax - 1 = 0$ are:
(a) One positive and one negative
(b) Both negative
(c) Both positive
(d) Both non-real complex

Q.18 If the roots of the equation $x^2 - 2ax + a^2 + a - 3 = 0$ are less than 3, then:
(a) $a < 2$ (b) $2 \leq a \leq 3$
(c) $3 < a \leq 4$ (d) $a > 4$

Q.19 The number of integral values for which $x^2 - (a - 1)\, x + 3 = 0$ has both the roots positive and $x^2 + 3x + 6 - a = 0$ has both the roots negative is:
(a) 0 (b) 1 (c) 2 (d) Infinite

Q.20 If the equations $ax^2 + bx + c = 0$, $cx^2 + bx + a = 0$, $a \neq c$ have a negative common root, then find the value of $a - b + c$.
(a) 0 (b) 2
(c) 1 (d) None of these

Q.21 Find the number of values of k, for which $(x^2 - (k-2)\, x + k^2)\, (x^2 + kx + (2k-1))$ is a perfect square.
(a) 1 (b) 2
(c) 0 (d) None of these

Q.22 If $x^2 + ax + b = 0$ and $x^2 + bx + a = 0$, $a \neq b$, have a common root, then which of the following is true?
(a) $a + b = 1$ (b) $a + 1 = 0$
(c) $a = 0$ (d) $a + b + 1 = 0$

Q.23 If 2 and 4 are two roots of the expression $x^4 + 2x^3 + mx^2 + nx + 3 = 0$ (m and n are constants), then what is the value of $2m + 3n$?
(a) 106 (b) −106 (c) 53 (d) −53

Q.24 If $p + q + r = 1$, $p^2 + q^2 + r^2 = 9$ and $p^3 + q^3 + r^3 = 1$, then what is the value of $\frac{1}{p} + \frac{1}{q} + \frac{1}{r}$?
(a) 1 (b) −4 (c) −1 (d) −8

Q.25 If the roots of the equation $x^2 + (3k - 36)\, x + k^2 - 24k + 144 = 0$ are reciprocal to each other, then find the value of k.
(a) $k = 11$ or $k = 13$ (b) $k = -11$ or $k = -13$
(c) $k = 12$ (d) $k = -12$

Q.26 Find the number of common roots of $x^2 + 3x - 4 = 0$ and $x^3 - 4x = 0$.
(a) 0 (b) 1 (c) 2 (d) 3

Q.27 If $Z + \frac{1}{Z} = 1$, then $Z^{64} + \frac{1}{Z^{64}}$ is equal to:
(a) 0 (b) 1 (c) −1 (d) −2

Q.28 Find the maximum and the minimum values of the function $\frac{x^2 - x + 1}{x^2 + x + 1}$ for real values of x.
(a) 3 and −3 (b) $\frac{1}{3}$ and $-\frac{1}{3}$
(c) 3 and $\frac{1}{3}$ (d) None of these

Q.29 Consider the equation of the form $x^2 + bx + c = 0$. The number of such equations that have real roots and have coefficients of b and c in the set {1, 2, 3, 4, 5, 6}, (b may be equal to c), is:
(a) 20 (b) 18 (c) 17 (d) 19

Q.30 If a, b, c, d are four consecutive terms of an increasing AP, then the roots of the equation $(x - a)\, (x - c) + 2\, (x - b)\, (x - d) = 0$ are:
(a) Real and distinct
(b) Non-real complex
(c) Real and equal
(d) Integers

Q.31 If $2a^{-1/3} + 2a^{1/3} = 5$, then find the value of a.
(a) 8 or $\frac{1}{8}$ (b) 2 or $\frac{1}{2}$
(c) 3 or $\frac{1}{3}$ (d) None of these

Q.32 If α, β, and γ are the roots of the equation $9x^3 - 7x + 6 = 0$, then the equation whose roots are $3a + 2$, $3b + 2$ and $3\gamma + 2$ is:
(a) $x^3 - 6x^2 + 5x + 24 = 0$ (b) $9x^3 - x + 16 = 0$
(c) $2x^3 - 27x - 8 = 0$ (d) None of these

Q.33 If the roots of equation $(q - r)\, x^2 + (r - p)\, x + p - q = 0$ are equal, then p, q, r are in:
(a) AP (b) GP
(c) HP (d) None of these

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Q.34 Both the roots of the equation $(x-b)(x-c) + (x-a)(x-c) + (x-a)(x-b) = 0$ are always:
(a) +ve (b) −ve
(c) Real (d) None of these

Q.35 If $a(b-c)x^2 + b(c-a)xy + c(a-b)y^2$ is a perfect square, then, a, b, c are in:
(a) AP (b) GP
(c) HP (d) None of these

Q.36 $x^{3^n} + y^{3^n}$ is divisible by $x + y$, if
(a) n is any integer ≥ 0.
(b) n is an odd positive integer.
(c) n is an even positive integer.
(d) n is a rational number.

Q.37 The condition that both the roots of quadratic equation $ax^2 + bx + c = 0$ are positive is
(a) a and c have an opposite sign that of b.
(b) b and c have an opposite sign that of a.
(c) a and b have an opposite sign that of c.
(d) None of these

Q.38 The condition that both the roots of the equation $ax^2 + bx + c = 0$ are zero is:
(a) $b \neq 0$
(b) $c \neq 0$
(c) Either b or c (not both) $= 0$
(d) $b = c = 0$

Q.39 There is a quadratic equation of the form $ax^2 + bx + c = 0$ along with 3 conditions.
I. If only the value of a is correct in the equation, the roots are $\frac{1}{3}$ and 1.
II. If only the value of b is correct, the roots are 6 and 5.
III. If only the value of c is correct, the roots are -3 and -2.
What is the equation of a quadratic equation opposite in sign to the roots of the given quadratic equation?
(a) $3x^2 + 11x + 6 = 0$ (b) $3x^2 - 7x - 6 = 0$
(c) $x^2 - 11x + 30 = 0$ (d) $x^2 - 6x - 1 = 0$

Q.40 Quadratic equations $(2p-1)z^2 + (2p+1)z + c = 0$ and $(q+1)y^2 + (4q+1)y + 3c = 0$ have the same pair of roots. Given that $c \neq 0$, what is the value of $(p+q)$?
(a) 3 (b) 4
(c) 2 (d) Cannot be determined

ADVANCED

Q.1 Quadratic equation $x^2 + bx + c =$ has roots a and b, such that LCM of $(a, b) = 24$ and HCF of $(a, b) = 2$. What is the total number of such quadratic equations if a and b are natural numbers?
(a) 1 (b) 2 (c) 3 (d) 4

Q.2 If $(p^2 + q^2)x^2 - \sqrt{2}(p+q)x + 1 = 0$, what must be the relation between q and p if the equation has equal and real roots?
(a) $p = q$ (b) $p > q$
(c) $p < q$ (d) None of these

Q.3 How many positive integers p, q, and r exist such that $p/q + q/r + r/p = 2$?
(a) 1 (b) 3
(c) 15 (d) None of these

Q.4 If $x^2 + y^2 + z^2 = 1$, then the value of $xy + yz + zx$ lies in the interval of:
(a) $\left[\frac{1}{2}, 2\right]$ (b) $[-1, 2]$ (c) $\left[-\frac{1}{2}, 1\right]$ (d) $\left[-1, \frac{1}{2}\right]$

Q.5 If $a \in R$, $b \in R$, then the factors of the expression $a(x^2 - y^2) - bxy$ are:
(a) Real and different
(b) Real and identical
(c) Complex
(d) None of these

Q.6 If the sum of the roots of the quadratic equations $ax^2 + bx + c = 0$ is equal to the sum of the squares of their reciprocals, then $\frac{b^2}{ac} + \frac{bc}{a^2} =$
(a) 0 (b) −1 (c) 1 (d) 2

Q.7 If equations $x^2 + 3x + 10 = 0$ and $ax^2 + bx + c = 0$ have a common root, then $a:b:c =$
(a) 2:3:10 (b) 1:3:5
(c) 1:3:10 (d) None of these

Q.8 Sum of all the real roots of the equation $|x-2|^2 + |x-2| - 2 = 0$ is:
(a) 0 (b) −4 (c) 4 (d) 2

Q.9 If $f(x) = \sum_{r=1}^{x} \frac{1}{r}$, $f(10) = \frac{a}{b}$, $f(20) = \frac{c}{d}$ and $f(30) = \frac{m}{n}$, where HCF of a and b = HCF of c and d = HCF of m and $n = 1$, then the roots of the equation $ax^2 + cx + m = 0$ are:

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(a) Imaginary (b) Rational
(c) Cannot be rational (d) Integers

Q.10 If pth, qth, and rth term of a GP be 27, 8, and 12, respectively, then the equation $px^2 + 2qx - 2r = 0$ has
(a) only one root in (0, 1).
(b) no root in (0, 1).
(c) both roots in (0, 1).
(d) imaginary roots

Q.11 If a, b be the roots of equation $x^2 + ax + b = 0$ and m, n, p, q are distinct integers in AP such that $a < m < n < p < q < b$ and $q = m^2 + n^2 + p^2$, then which of the following can be true?
(a) $b + 1 + |a| > 0$ (b) $b + 1 + |a| = 0$
(c) $b + |a| = 0$ (d) $1 + b + |a| < 0$

Q.12 For $a, b, c \in R$ if $ax^2 + bx + c$ is real for real values of x and imaginary for imaginary values of x, then:
(a) $a > 0$ (b) $a < 0$
(c) $a = 0$ (d) None of these

Q.13 If $x + \frac{1}{x} = 1$ and $p = x^{4000} + \frac{1}{x^{4000}}$ and q be the digit at units place in the number $2^{2^n} + 1$, n being a natural number greater than 1, then $p + q =$
(a) 2 (b) 4
(c) 6 (d) None of these

Q.14 If the roots of $ax^2 + bx + b = 0$ be in the ratio $p:q$, then $\sqrt{\frac{p}{q}} + \sqrt{\frac{q}{p}} + \sqrt{\frac{b}{a}} =$
(a) 1 (b) −1 (c) 2 (d) 0

Q.15 There are two quadratic expressions $a_1x^2 + b_1x + c_1$ and $a_2x^2 + b_2x + c_2$. Both of them have the same roots. If the ratio of a_1 to a_2 is 1:2, what is the ratio of the maximum values of the two quadratic expressions? $(a_1, a_2 < 0)$
(a) 1:3 (b) 3:1
(c) 1:5 (d) None of these

Q.16 If the two equations $ax^2 + bx + c = 0$ and $cx^2 + bx + a = 0$ have a common root, then how many of the following statements can be true?
I. $a = c$
II. The common root is 1
III. $a - b + c = 0$
(a) 0 (b) 1 (c) 2 (d) 3

Q.17 Given that P and Q are the roots of $Ax^2 - 4x + 1 = 0$ and R and S are the roots of the equation $Bx^2 - 6x + 1 = 0$. What is the value of A so that P, Q, R, and S are in HP?
(a) 1 (b) 2 (c) 3 (d) 4

Q.18 If $a < b < c < d$, then which of the following is true about the roots of the quadratic equation $(x - a)(x - c) + K(x - b)(x - d) = 0$, for the real value of K?
(a) Both the roots of this equation will have a positive sign.
(b) Both the roots of this equation will have a negative sign.
(c) Both the roots of this equation will be real.
(d) Cannot be determined

Q.19 If $a + b + c = 0$ and a, b, c are rational, then which of the following is true about the roots of the quadratic equation $(b + c - a)x^2 + (c + a - b)x + (a + b - c) = 0$?
(a) Both the roots of the equation are rational.
(b) Both the roots of the equation are irrational.
(c) One of the roots of the equation is rational and the other is irrational.
(d) None of these

Q.20 Find all the values of p for which one root of the equation $x^2 - (p + 1)x + p^2 + p - 8 = 0$, is greater than 2 and the other root is smaller than 2.
(a) $-2 < p < 3$ (b) $-3 < p < 2$
(c) $-3 < p$ (d) $p < 2$

Q.21 For how many values of a, $a > 0$, both the roots of $ax^2 - (a + 1)x + (a - 2) = 0$, are greater than 3?
(a) 0 (b) 2 (c) 5 (d) Infinite

Q.22 The number of real values of the parameter k for which $(\log_{16}x)^2 - \log_{16}x + \log_{16}k = 0$ with real coefficients will have exactly one solution is:
(a) 2 (b) 1
(c) 4 (d) None of these

Q.23 If the roots of the equation $(x - a)(x - b) + (x - b)(x - c) + (x - c)(x - a) = 0$, (where a, b, c are real numbers) are equal, then:
(a) $b^2 - 4ac = 0$ (b) $a = b = c$
(c) $a + b + c = 0$ (d) None of these

Q.24 If the equation $ax^2 + bx + c = 0$ has a root less than −2 and a root greater than 2, and $a > 0$, then which of the following is true?
(a) $4a + 2|b| + c < 0$
(b) $4a + 2|b| + c > 0$
(c) $4a + 2|b| + c = 0$
(d) None of these

Q.25 The equations $x^2 + x + a = 0$ and $x^2 + ax + 1 = 0$
(a) cannot have a common real root for any value of a.
(b) have a common real root for exactly one value of a.
(c) have a common real root for exactly two values of a.
(d) have a common real root for exactly three values of a.

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Answers

Warm Up

1. (b)	**2.** (b)	**3.** (d)	**4.** (c)	**5.** (c)	**6.** (d)	**7.** (a)	**8.** (a)	**9.** (c)	**10.** (c)
11. (d)	**12.** (b)	**13.** (c)	**14.** (c)	**15.** (d)	**16.** (b)	**17.** (d)	**18.** (b)	**19.** (b)	**20.** (d)

Foundation

1. (b)	**2.** (d)	**3.** (b)	**4.** (c)	**5.** (c)	**6.** (d)	**7.** (a)	**8.** (c)	**9.** (b)	**10.** (b)
11. (c)	**12.** (a)	**13.** (c)	**14.** (c)	**15.** (c)	**16.** (b)	**17.** (d)	**18.** (c)	**19.** (c)	**20.** (d)
21. (a)	**22.** (c)	**23.** (b)	**24.** (d)	**25.** (b)	**26.** (c)	**27.** (a)	**28.** (d)	**29.** (b)	**30.** (b)
31. (b)	**32.** (b)	**33.** (c)	**34.** (c)	**35.** (b)	**36.** (c)	**37.** (a)	**38.** (d)	**39.** (c)	**40.** (c)

Moderate

1. (a)	**2.** (b)	**3.** (d)	**4.** (d)	**5.** (c)	**6.** (d)	**7.** (b)	**8.** (b)	**9.** (b)	**10.** (b)
11. (a)	**12.** (b)	**13.** (b)	**14.** (a)	**15.** (a)	**16.** (b)	**17.** (c)	**18.** (a)	**19.** (c)	**20.** (a)
21. (a)	**22.** (d)	**23.** (a)	**24.** (a)	**25.** (a)	**26.** (a)	**27.** (c)	**28.** (c)	**29.** (d)	**30.** (a)
31. (a)	**32.** (a)	**33.** (a)	**34.** (c)	**35.** (c)	**36.** (a)	**37.** (a)	**38.** (d)	**39.** (a)	**40.** (a)

Advanced

1. (b)	**2.** (a)	**3.** (b)	**4.** (c)	**5.** (a)	**6.** (d)	**7.** (c)	**8.** (c)	**9.** (d)	**10.** (a)
11. (a)	**12.** (c)	**13.** (c)	**14.** (d)	**15.** (d)	**16.** (d)	**17.** (c)	**18.** (c)	**19.** (a)	**20.** (a)
21. (a)	**22.** (b)	**23.** (b)	**24.** (a)	**25.** (b)					

Hints and Solutions

Warm Up

1. $K + 6 = 2K - 1$

$K = 7$

2. Let roots = α, β

Therefore, $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = (P - 2)^2 + 2(p + 1)$

$= p^2 - 4p + 4 + 2p + 2 = (P - 1)^2 + 5$

Hence, value of p for the least value = 1

3. $\alpha - \beta = -2(p + 1)$; $\alpha\beta = 9p - 5$

$\therefore \alpha > 0$ and $\beta > 0$

$\therefore \alpha + \beta > 0 \Rightarrow -2(p - 1) > 0 \Rightarrow p < -1$

$\therefore \alpha\beta > 0 \Rightarrow 9p - 5 > 0 \Rightarrow p > 5/9$

Hence, option (d) is the answer.

4. Let common root = α

$\therefore \alpha^2 - a\alpha - 21 = 0$

$\alpha^2 - 3a\alpha + 35 = 0$

Solving the two equations, we get $a = 4$

5. Taking the values of A, B, and C as 1, 2, and –1.

We get $A^4 + B^4 + C^4 = 18$

6. Sum of roots = $-2/3(-b/a)$

7. Product of roots = $-1/3(c/a)$

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8. In a cubic equation $ax^3 + bx^2 + cx + d = 0$, sum of Roots $= -b/a$

$\therefore$ Here, sum of roots $= -2$

9. Calculating $D = b^2 - 4ac$ for each of the given option and for a non-real root, D should be less than 0.

10. For roots with opposite sign replace x with $-x$ in the given equation.

We get, $ax^2 - bx + c = 0$

Now, the sum of roots $= b/a$

For the new equation, sum of roots $= -3$

11. $a^2x^2 + abx = b^2$

$$\left(ax + \frac{b}{2}\right)^2 = \frac{5b^2}{4}$$

$$ax + b/2 = \pm\frac{\sqrt{5}}{4}b$$

If $b = 0$, then real and equal roots, else real and unequal.

Hence, option (d) is the answer.

12. $-x^2 + 6x - 8 > 0$

Or, $x^2 - 6x + 8 < 0 \Rightarrow (x - 2)(x - 4) < 0$

Hence, x will lie in between 2 and 4.

13. $2x + 3y = 5$

$4x + ky = A$

For no solution,

$$\frac{2}{4} = \frac{3}{k} \neq \frac{5}{A}$$

$\therefore k = 6$

14. $5b + 6p + 7c = 178$ (i)

$6b + 4p + 2c = 124$ (ii)

Multiplying equation (i) by 2 and adding to equation (ii):

$16b + 16p + 16c = 480$

$b + p + c = 30$

$\therefore 3b + 3p + 3c = 90$

15. $x^2 - 13x + N = 0$ with roots A and B

$\therefore A + B = 13$

and $A - B = 3$

$\therefore A = 8$ and $B = 5$

$\therefore N = AB = 40$

16. let $x = -a$ where $a > 0$

For any two positive numbers $\frac{9}{a}, \frac{a}{9}$

AM $\geq$ GM

$$\Rightarrow \frac{\frac{9}{a} + \frac{a}{9}}{2} \geq \sqrt{\frac{9}{a} \times \frac{a}{9}}$$

$$\Rightarrow \frac{9}{a} + \frac{a}{9} \geq 2$$

$$\Rightarrow -\left(\frac{a}{9} + \frac{9}{a}\right) \leq -2$$

17. Take a quadratic equation: $x^2 + 3x + 2 = 0$

$\therefore$ Sum of roots $= -3$

Product of roots $= 2$

Roots $= -1, -2$

Now, new quadratic equation: $2x^2 + 3x + 1 = 0$

$\therefore$ Sum of roots $= -\frac{3}{2}$

Product of roots: $\frac{1}{2}$

Roots $= -1, -\frac{1}{2}$

Therefore, sum, product, and roots will change. Hence, option (d) is the answer.

18. $(x + 2)^2 = 9, (y + 3)^2 = 25$

$\therefore x = 1, -5; y = 2, -8$

$\therefore$ Maximum value of $\frac{x}{y} = \frac{5}{8}$

19. Sum of roots $= \frac{7}{2}$ and one of the roots $= 3$

Therefore, other root $= \frac{7}{2} - 3 = \frac{1}{2}$

20. Using remainder theorem; put $x = 2$ in the main polynomial:

$\therefore p(2) = 16 - 12 + 14 - 10 = 8$

$\therefore$ Remainder $= 8$

FOUNDATION

1. $x^4 + x^2 = 0$ can be written as $\Rightarrow x^2(x^2 + 1) = 0$

Therefore, either $x^2 = 0$ or $(x^2 + 1) = 0$

Case I If $x^2 = 0$, then $x = 0, 0$ (two solutions)

Case II $(x^2 + 1) = 0 \Rightarrow x^2 = -1$

Minimum value of $x^2 = 0$, and $x^2 = -1$ is not possible.

Hence, a total of two real solutions are possible.

2. Method 1

Using the formula $a^3 + b^3 = (a + b)^3 - 3ab(a + b)$

$(x^3 + 1/x^3) = (x + 1/x)^3 - 3(x).(1/x)(x + 1/x) = 2^3 - 3 \times 1 \times (2) = 2$

Method 2

$(x + 1/x) = 2$

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Or, $(x + 1/x) - 2 = 0$, or, $x^2 - 2x + 1 = 0$

$\Rightarrow (x-1)^2 = 0 \Rightarrow x = 1$

Hence, $(x^3 + 1/x^3) = 2$

Method 3

$x + \frac{1}{x} = 2$. Going by hit and trial, we can see that $x = 1$

$\therefore\ x^3 + \frac{1}{x^3} = 1 + 1 = 2$

3. $(1 - S_1)(1 - S_2)(1 - S_3)(1 - S_4)$

$= 1 - (S_1 + S_2 + S_3 + S_4) + (S_1S_2 + S_2S_3 + S_3S_4 + S_4S_1)$

$- (S_1S_2S_3 + S_2S_3S_4 + S_3S_4S_1 + S_4S_1S_2) + S_1S_2S_3S_4$

$= 1 + \frac{b}{a} + \frac{c}{a} + \frac{d}{a} + \frac{e}{a}$

$= \frac{a+b+c+d+e}{a}$

Alternate solution:

$as^4 + bs^3 + cs^2 + ds + e = 0$

$\Rightarrow a(s - s_1)(s - s_2)(s - s_3)(s - s_4) = 0$ (because S_1, S_2, S_3 and S_4 are roots)

Now, putting $s = 1$ in LHS

$\Rightarrow P(1) = a(1 - s_1)(1 - s_2)(1 - s_3)(1 - s_4)$

$\Rightarrow (1 - s_1)(1 - s_2)(1 - s_3)(1 - s_4) = (a + b + c + d + e)/a$

4. Since $(x + 2)$ is a factor of $P(X)$, so $p(-2) = 0$

$\Rightarrow 16 - 16 - 4a + 3 = 0 \Rightarrow a = \frac{3}{4}$

5. $x^2 - 5x + 6 = 0 \Rightarrow$ Roots = 2, 3

When 2 is the common root, then $p(2) = 0 \Rightarrow 2^2 + 2m + 3 = 0$

$\Rightarrow m = -7/2$

When 3 is the common root, then $P(3) = 0 \Rightarrow 3^2 + 3m + 3 = 0$

$\Rightarrow m = -4$

6. $x^3 - x^2 + 2x - 17 = 0$

Let roots be α, β, and γ

$\therefore\ \alpha\beta\gamma = 17$

Now, 17 is having only two factors 1 and 17, so the only possible integral roots = 1 and 17. Checking for these two values we find no integral root possible.

7. $x - \frac{1}{x^2 - 4} = 2 - \frac{1}{x^2 - 4}$

$\Rightarrow x = 2$. Although we can see that for $x = 2, \frac{1}{x^2 - 4}$ will not hold.

Hence, no value of x is possible.

8. Draw the graphs of both LHS and RHS to find and there are no intersection points.

9. $x^2 + |x| - 6 = 0 \Rightarrow x^2 + x - 6 = 0$ where ≥ 0, $\therefore$ root = 2

Else $x^2 - x - 6 = 0$ if $x < 0$, $\therefore$ root = -2

Hence, sum of roots = 0

10. Product of roots = 1

$\Rightarrow \frac{4k}{k^2 + 1} = 1 \Rightarrow k = 2 \pm \sqrt{3}$

11. $(x - a)(x - b) = c$

$\Rightarrow x^2 - (a + b)x + ab - c = 0$; roots = α, β

$(x - \alpha)(x - \beta) + c = 0$

$\Rightarrow x^2 - (a + \beta)x + \alpha\beta + c = 0$

$\Rightarrow x^2 - (a + b) + ab - c + c = 0$

(putting values of $(\alpha + \beta)$ and $\alpha\ \beta$ from previous equation)

Hence, new equation = $x^2 - (a + b) + ab = 0$. Therefore, the roots are a and b.

12. $\alpha + \beta = \frac{1}{\alpha^2} + \frac{1}{\beta^2}$

$\Rightarrow \frac{-b}{a} = \frac{\alpha^2 + \beta^2}{\alpha^2\beta^2} \Rightarrow \frac{-b}{a} = \frac{(\alpha+\beta)^2 - 2\propto\beta}{\alpha^2\beta^2}$

$\Rightarrow \frac{-b}{a} = \frac{(b^2/a^2) - \frac{2c}{a}}{c^2/a^2}$

$\Rightarrow \frac{b^2}{ac} + \frac{bc}{a^2} = 2$

13. $x^2 + px + q = 0$

Given roots are (a, b)

Then, $a + b = -p$ and $ab = q$ (i)

And $x^2 + px - r = 0$ and the roots are (g, d)

Then, $g + d = -p$ and $gd = -r$

But, in first equation, $-p = a + b$

Then, $a + b = g + d$ (ii)

Then, $(a - g)(a - d) = a^2 - a[g + d] + gd$

$= a^2 - a[a + b] + gd$ [$\because\ a + b = g + d$]

$= a^2 - a^2 - ab + gd = -q - r = (q + r)$

Hence, option (c) is the answer.

14. $x(1 + x^2) + x^2(6 + x) + 2 = 0$

$\Rightarrow 2x^3 + 6x^2 + x + 2 = 0$

For roots: α, β, γ

$\alpha + \beta + \gamma = -\left(\frac{6}{2}\right) = -3$

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$$\alpha\beta+\beta\gamma+\gamma\alpha=\frac{1}{2}$$

$$\alpha\,\beta\,\gamma=-\left(\frac{2}{2}\right)=-1$$

$$\therefore\ \frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma}=\frac{\alpha\beta+\beta\gamma+\gamma\alpha}{\alpha\beta\gamma}=\frac{1/2}{-1}=-1/2$$

15. $2x^2-7x+1=0$ and $ax^2+bx+2=0$

In the first equation, the roots are

$$\frac{7+\sqrt{41}}{4}\text{ and }\frac{7-\sqrt{41}}{4}\left[\frac{-b\pm\sqrt{b^2-4ac}}{2a}\right]$$

Because they are irrational roots, they always occur in pairs. If one root is common, then second root is also common.

Because constant term is double in second equation, then value of a and b should also be doubled.

Then, $a=4$ and $b=-14$

16. As $(x-1)^3$ is a factor of the polynomial, 1 is a repeated root (3 times) of the given equation.

Let the fourth root be x, Therefore, 1.1.1. $x=-1$

$\Rightarrow x=-1\Rightarrow -1$ is another root

Hence, $(x+1)$ is a factor.

17. Let 'α' be the equal integral root,

$\therefore x^2-bx+c=0$

$(x-\alpha)^2=0$

$\Rightarrow x^2-2\alpha x+\alpha^2=0$

Now comparing the two equations,

$\therefore 2\alpha=b$ and $c=\alpha^2$

$\therefore b$ is even integer and c is an integer as well as a perfect square.

18. $x-\frac{1}{x}=\frac{35}{6}$

$\therefore x=6$

19. $\alpha^3+\beta^3=(\alpha+\beta)^3-3\alpha\beta\,(\alpha+\beta)$

$$\left(\frac{3}{2}\right)^2+3\times1\times\frac{3}{2}=\frac{63}{8}$$

20. Assume that roots of the equation $3ax^2+2bx+c=0$ are α,β.

$\alpha+\beta=-\frac{2b}{3a}$, $\alpha\beta=\frac{c}{3a}$ and $\frac{\alpha}{\beta}=\frac{2}{3}$ (given)

$$\therefore\ \frac{\alpha}{\beta}+\frac{\beta}{\alpha}=\frac{2}{3}+\frac{3}{2}$$

$$\frac{\alpha^2+\beta^2}{\alpha\beta}=\frac{13}{6}\Rightarrow\frac{(\alpha+\beta)^2-2\alpha\beta}{\alpha\beta}=\frac{13}{6}$$

Now putting the values of $(\alpha+\beta)$ and $\alpha\beta$ and then solving,

We get, $8b^2=25ac$.

21. Coefficient of $x^2=2$

In first case, the product of the roots $=8\times7=56$, which is the constant term. Hence, $c=56$.

In the second case, the sum of the roots $=-b=8-3=5$

$\Rightarrow b=-5$

$\therefore$ The correct quadratic equation is $x^2-5x+56=0$

22. For reciprocal roots, replace x with $1/x$ in LHS of

$1-3x-2x^2=0$

$\Rightarrow x^2-3x-2=0$

23. Let the roots be α and 2α

Therefore, sum of roots $=\alpha+2\alpha=3\alpha=\frac{3}{k}\Rightarrow\alpha=\frac{1}{k}$

and product of roots $\Rightarrow 2\alpha^2=\frac{2}{k}\Rightarrow\frac{2}{k^2}=\frac{2}{k}$

Hence, $k=1$

24. **Method 1**

Let fraction is $\frac{x}{y}$

$x^2+1=y$ (i)

and $\frac{x+4}{y+4}=\frac{1}{2}$

Further, $2x+8=y+4$ and $-2x+y-4=0$ (ii)

Put $y=x^2+1$ [From 1st equation]

$x^2-2x+1-4=0$

$x^2-3x+x-3=0$

Therefore, $x=3,-1$

But, x is positive, so $y=x^2+1=10$

Hence, the fraction is $\frac{3}{10}$.

Method 2

Go through options.

25. Let the number of chairs bought initially $=n$.

$$\therefore\ \frac{2400}{n}-\frac{2400}{n+10}=20$$

$$\Rightarrow\ 120\left[\frac{10}{n(n+10)}\right]=1$$

$\therefore\ n(n+10)=1200\Rightarrow n=30$

26. Here, sum of roots $=k$ and product of roots $=5$

One root is 3 and the other root is $\frac{5}{3}$.

$$\therefore\ k=3+\frac{5}{3}=\frac{14}{3}$$

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27. We know

$a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$

As, $a + b + c = 0$

$\therefore \quad a^3 + b^3 + c^3 = 3abc$

28. Let the roots be 3, α, and β.

$\therefore 3 + \alpha + \beta = a;\ 3\alpha + 3\beta + \alpha\beta = 11;\ 3\alpha\beta = 6$

Therefore, $\alpha\beta = 2$ and $3(\alpha + \beta) + 2 = 11$

$\Rightarrow 3(\alpha + \beta) + 2 = 11 \Rightarrow \alpha + \beta = 3$

$\therefore \alpha, \beta = (1, 2)$

29. $x^4 \dfrac{1}{x^4} = 47$

$$\Rightarrow \left(x^2 + \frac{1}{x^2}\right)^2 = 47 \Rightarrow x^2 + \frac{1}{x^2} = 7$$

$$\Rightarrow \left(x^2 + \frac{1}{x^2}\right)^2 - 2 = 7$$

$$\Rightarrow \left(x^2 + \frac{1}{x^2}\right)^2 = 9$$

$$\Rightarrow x - \frac{1}{x} - 3 \Rightarrow \left(x + \frac{1}{x}\right)^3 = 27$$

$$\Rightarrow x^3 - \frac{1}{x^3} - 3\left(x + \frac{1}{x}\right) = 27$$

$$\Rightarrow \quad x^3 + \frac{1}{x^3} + 3 \times 3 = 27 \Rightarrow x^3 + \frac{1}{x^3} = 18$$

30. Let roots be ak and bk.

$\therefore (a + b)\, k = -r/p$

$\therefore abk^2 = r/p$

$$\text{Now, } \sqrt{\frac{b}{a}} + \sqrt{\frac{a}{b}} = \frac{a+b}{\sqrt{ab}} = \frac{(a+b)k}{\sqrt{abk^2}} = -\frac{r/p}{\sqrt{r/p}}$$

$$= -\sqrt{r/p}$$

31. $|y - 4| \le 6 \Rightarrow -2 \le y \le 10$

$|y + 4| < 5 \Rightarrow -9 < y < 1$

$\therefore$ Combining the two region, we get $-2 \le y < 1$

$\therefore$ Only one element satisfies the condition, i.e., = 0

Alternative Solution:

Go through the options.

32. $p(x) = x^3 - ax^2 + bx + 10$; since it is divisible by $(x + 5)$

$\therefore \quad p(-5) = 0$

$\Rightarrow (-5)^3 - 25a - 5b + 10 = 0$

$\Rightarrow \quad 5a + b = -23 \qquad$ (i)

$\Rightarrow \quad Q(x) = x^4 + x^3 + bx^2 - ax + 42 = 0$

$\therefore \quad Q(3) = 0$

$\Rightarrow \quad 81 + 27 + 9b - 3a + 42 = 0$

$\Rightarrow \quad a - 3b = 50$

33. $x^2 + x + 2 = 0$

$\therefore \alpha + \beta = -1$

$\alpha\beta = 2$

$$\text{Now, } \frac{\alpha^{10} + \beta^{10}}{\alpha^{-10} + \beta^{-10}} = \frac{\alpha^{10} + \beta^{10}}{\frac{\alpha^{10} + \beta^{10}}{(\alpha\beta)^{10}}} = (\alpha\beta)^{10} = 2^{10} = 1024$$

34. Go through the options.

For option (c), $44^2 = 1936$ and $1980 - 44 = 1936$.

35. In the correct equation,

Sum of roots $= -3 - 2 = -5$ and the product of roots $= (-1) \times (-2) = 2$

Hence, the correct equation is $= x^2 + 5x + 2 = 0$

36. $x^3 - x^2 + x - 1 > 0$

$x^2(x - 1) + (x - 1) > 0$

$(x^2 + 1)(x - 1) > 0$

As, $x^2 + 1 > 0$ for all x, so $x > 1$

37. $|V^2 + V - 6| - 3V + 7 = 0$

If $V^2 + V - 6$ is positive, then $V^2 + V - 6 - 3V + 7 = 0$

$V^2 - 2V + 1 = 0\ [V - 1] = 0$. Hence, $V = 1$

$\therefore$ $V^2 + V - 6$ is positive, so $V = 1$ does not satisfies the condition.

If $V^2 + V - 6$ is negative, then $-V^2 - V + 6 - 3V + 7 = 0$

Or, $V^2 + 4V - 13 = 0$

It can be seen that $b^2 - 4ac < 0$. Therefore, there is no real value of V.

38. $ax - b = 0$

$$\therefore \quad x = \frac{b}{a}$$

Since 'b' and 'a' have only '1' as a common factor,

$\therefore$ We can directly equate with a rational number in its empirical form.

$\therefore$ For A, $X = \dfrac{7}{3}$ with incorrect 'b'

$\therefore$ Correct the value of 'a' = 3 or any multiple of 3.

Similarly, from second statement, $x = \dfrac{8}{5} \Rightarrow$ we get correct value of 'b' = 8 or any multiple of 8.

Since we do not get any unique value of a and b, we cannot find unique value of $\dfrac{b}{a}$.

39. Let the roots be 3, α, and β.

$\therefore 3\,\alpha\,\beta = 6 \Rightarrow \alpha\,\beta = 2$

and $3x + 3\,\beta + \alpha\,\beta = 11$

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$\Rightarrow 3(\alpha+\beta)+2=11$

$\therefore \alpha+\beta=3$

$\therefore \alpha=1, \beta=2$

40. $(m+n)^{100}=m^{100}+{}^{100}c_1 m^{99}.n+{}^{100}c_2 m^{98}.n+.....+n^{100}$

$=m^{100}+n^{100}+k$

[where $k={}^{100}C_1 m^{99}n+{}^{100}C_2 m^{98}.n+.....+{}^{100}C_{99} mn^{99}$]

$\therefore$ $k>0$ for m and n belonging to natural number

$\therefore$ $(m+n)^{100}>m^{100}+n^{100}$

MODERATE

1. $x^2+5|x|+6=0$

$\because$ All the terms in LHS are positive.

Hence, no real root is possible.

2. $S-2=2\frac{1}{3}+2^{2/3}\Rightarrow (S-2)^2=2^{2/3}+2.2^{1/3}+2.2$

Now, required $=2^{2/3}-2.2^{1/3}+2.2-(2.2^{1/3}+2.2^{2/3})-2=2$

3. Given that x_1, x_2, and x_3 are in AP.

Then, $2x_2=x_1+x_3$ (i)

It is also given that sum of the roots

$x_1+x_2=4$ (ii)

Here, with both equations, we can find neither x_1 nor x_2.

Then, answer is (d)

4. x_1, x_2, and x_3 are in AP.

Therefore, $2x_2=x_1+x_3$ (i)

It is given that sum of the roots

$$x_1+x_3=-\frac{b}{a}=\frac{6}{1}=6 \quad \text{(ii)}$$

With both equations, we can find the value of $x_2=3$ only.

But, we cannot find neither x_1 nor x_3 as asked in the question.

Hence, answer is option (d).

5. Let the roots be α, 2α. Where $\alpha>0$

$\therefore \alpha+2\alpha=-m\Rightarrow m=-3\alpha$

and, $2\alpha^2=C$

Now, since $m+c=2\Rightarrow 2\alpha^2-3\alpha=2$

$\alpha=-1/2, 2$

$\because \alpha>0$

$\therefore \alpha=2$ $\therefore m=-3\alpha=-6$

6. In the above question, the roots are α and 2α, that is, (2, 4)

$\therefore |x_1-x_2|=|\alpha-2\alpha|=2$

7. **Method 1**

Let p, q be the roots with same unit digit.

$$\therefore \frac{D}{a^2}=\frac{b^2-4ac}{a^2}=\left(\frac{b}{a}\right)^2-\frac{4c}{a}=(p+q)^2-4pq=(p+q)^2$$

Now, given that p and q have same unit digit still $(p-q)^2$ is not divisible by 100, which means p and q have to be of opposite signs.

Now, $\frac{b}{2a}$ is nothing but $=\frac{-1}{2}(p+q)$

Now, since p and q are of opposite signs with same unit digit.

Therefore, unit digits of $p+q=0$

Therefore, unit digits of $\frac{1}{2}(p+q)$ will be either 0 or 5.

Out of the given options, second option is the answer.

Method 2

Let the roots be –21 and 11.

Equation is $X^2+10X-231=0$

$(X+21)(X-11)=0$

$$\frac{b^2-4ac}{a^2}=\frac{100+924}{1}=1024 \text{ (not multiple of 100)}$$

$b/2a=10/2=5$

8. $(ax^2+bx+c)(ax^2-dx-c)=0$

$\therefore$ Either $ax^2+bx+c=0$ or, $ax^2-dx+c=0$ or both

$\therefore$ Roots of $ax^2+bx+c=0$ will be real, if

$b^2-4ac>0$

Similarly, for ax^2-dx-c, roots will be real, if

$d^2+4ac>0$

Now, at least one of the two conditions will hold true since either $4ac$ will be greater than zero or less than zero or equal to zero.

$\therefore$ At least 2 real zeroes will be there.

9. $1+|e^x-1|=e^x(e^x-2)$

When $x\geq 0$

$1+e^x-1=e^x(e^x-2)$

$\Rightarrow e^x-e^x(e^x-2)=0$

$\Rightarrow e^x(1-e^x+2)=0$

$\Rightarrow e^x=0$ $\quad e^x=3$

for $x\geq 0$ $\quad e^x=0$ is not possible; therefore, $e^x=3$ is one root

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When $x < 0$

$1 - e^x + 1 = e^x(e^x - 2)$

$\Rightarrow 2 - e^x + (2 - e^x)e^x = 0$

$(2 - e^x)(1 + e^x) = 0$

$\Rightarrow e^x = 2\ e^x = -1$

For $x < 0$, $e^x = 2$ is not possible and e^x can never be equal to -1.

$\therefore$ Number of roots = 1

10. $(x+y)\left(\frac{x}{y}\right) = \frac{1}{2}$ and $(x+y)\frac{x}{y} = \frac{-1}{2}$

Solving these two equations, the values of

$(x+y)$ and $\left(\frac{x}{y}\right)$ will be $(1, -1/2)$

When $x + y = 1$ and $\frac{x}{y} = -1/2$

$(x, y) = (2, -1)$

When $x + y = -1/2$ and $\frac{x}{y} = 1$

$(x, y) = \left(\frac{-1}{4}, -\frac{1}{4}\right)$

$\therefore$ Number of possible pairs = 2

11. $\sqrt{x+1} - \sqrt{x-1} = \sqrt{4x-1}$

$\Rightarrow (x + 1) + (x - 1) - 2\sqrt{x^2 - 1} = 4x - 1$

$\Rightarrow -2\sqrt{x^2 - 1} = 2x - 1$

$\Rightarrow 4(x^2 - 1) = 4x^2 + 1 - 4x$

$\Rightarrow x = \frac{5}{4}$ which when put in the main equation does not satisfy it.

Hence, no solution is possible.

12. $\frac{6-x}{x^2-4} = \frac{2(x+2)+x}{x+2}$ with $x + 2 \neq 0$ and $x - 2 \neq 0$

$\Rightarrow \frac{6-x}{x-2} = 3x + 4$

$\Rightarrow 6 - x = (3x + 4)(x - 2)$

$\Rightarrow 3x^2 - x - 14 = 0$

$\Rightarrow x = -2$ or $x = 7/3$

But, $(x + 2) \neq 0$, hence, only one root is possible.

13. $x^2 - 3x - 4 < 0$

$(x - 4)(x + 1) < 0$

$\therefore$ Possible values = (0, 1, 2, 3)

Hence, there are four values.

14. $x^2 - 6x + 5 \leq 0 \Rightarrow (x - 5)(x - 1) \leq 0 \Rightarrow x \in (1, 5)$

$x(x - 2) > 0 \Rightarrow (-\infty, 0) \cup (2, \infty)$

Combining these two regions will give following integers -3, 4, and 5.

Hence, there are three values.

15. We are given $c = 2$. Therefore, the product of the roots is $-c = -2$ by Vieta's formulas. These also tell us that $\frac{-a}{3}$ is the average of the zeros, so $\frac{-a}{3} = -2 \rightarrow a = 6$. We are also given that the sum of the coefficients is -2; so, $1 + 6 + b + 2 = -2 \rightarrow b = -11$. Hence, the answer is (a).

16. $\text{Log}_4(x - 1) = \log_2(x - 3)$

$\Rightarrow \frac{1}{2}\log_2(x-1) = \log_2(x-3)$

$\Rightarrow (x - 1) = (x - 3)^2$

$\therefore x = 5, 2$

Now, $x = 2$ is not possible as $\log(x - 3) = \log(-1)$ is not possible.

17. $\alpha + \beta = \frac{3a}{a-1}, \alpha\beta = \frac{1}{a-1}$ where α and β are the roots.

$\because a > 1$

$\therefore \alpha + \beta > 0$ and $\alpha\beta > 0$

$\therefore \alpha > 0$ and $\beta > 0$

Hence, both the roots are positive.

18. $x^2 - 2ax + a^2 + a - 3 = 0$

$\because$ Roots are less than 3.

$\therefore$ Their sum will be less than 6.

$\therefore 2a < 6 \Rightarrow a < 3$

Out of the given options, only option (a) satisfies it.

19. The equation $x^2 - (a - 1)x + 3 = 0$ has both the roots positive.

Then, sum of the root $\left(-\frac{b}{a}\right)$ should also be greater than 0.

$\therefore \quad a - 1 > 0 \rightarrow a > 1$ (i)

In second equation, $x^2 + 3x + 6 - a = 0$ has both the roots negative.

$\therefore$ Product of the roots $\left(\frac{c}{a}\right)$ should greater than 0.

$\therefore 6 - a > 0 \rightarrow 6 > a$ (ii)

From first and second equations, a should be 2, 3, 4, or 5.

But, 2 and 3 do not hold for first equation because product of the roots is 3.

If we take $a = 2$, then sum of the roots is $a - 1 = 2 - 1 = 1$

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For positive numbers, if sum of them is 1, then product cannot be 3.

For the same reason, $a = 3$

Hence, the integral values for a are 4 and 5.

20. $ax^2 + bx + c = 0$ and $cx^2 + bx + a = 0$ have roots which are reciprocal to each other.

That is, if the roots of $ax^2 + bx + c = 0$ are α, β, then the roots of $cx^2 + bx + c = 0$ are $\frac{1}{\alpha}, \frac{1}{\beta}$

Now, it is given that $\alpha < 0$ and $\alpha - \frac{1}{\alpha}$

$\therefore$ $a = -1$

$\therefore$ The common root is -1.

$\therefore$ Putting $x = -1$ in any of the equations, we get, $a - b + c = 0$

21. $(x^2 - (K-2) + K^2)\,(x^2 + Kx + (2K-1))$

For the expression to be a perfect square, there can be two possible ways:

(i) When both the quadratic expression are perfect square for a particular value.

For that to happen, $K - 2 = 2K \Rightarrow K = -2$

Now, for $K = -2$ in the second expression, we get, $x^2 - 2x - 5$, which is not a perfect square.

(ii) The other is when both the quadratic equations are same.

$\therefore -(K-2) = K \Rightarrow K = 1 + K^2 = 2K - 1 \Rightarrow K = 1$

$\therefore$ At $K = 1$, the expression is a perfect square.

22. Let the common root be x.

$\therefore \alpha^2 + \alpha x + b = 0$

and $\alpha^2 + b\alpha + a = 0$

$\Rightarrow (a-b)\alpha - (a-b) = 0$

$\Rightarrow (a-b)(\alpha - 1) = 0 \Rightarrow \alpha = 1$ $\{\because \alpha \neq b\}$

$\therefore$ Putting $\alpha = 1$ is any of the equations,

$a + b + 1 = 0$

23. $P(x) = x^4 + 2x^3 + mx^2 + nx + 3$

Now, $P(2) = 0$

$\Rightarrow 16 + 16 + 4m + 2n + 3 = 0$ (i)

$\Rightarrow 4m + 2n + 35 = 0$

and, $P(4) = 0$

$\Rightarrow 256 + 128 + 16m + 4n + 3 = 0$

$\Rightarrow 16m + 4n + 387 = 0$ (ii)

Multiplying 5 in equation (i) and then subtracting from equation (ii)

$4m + 6n - 212 = 0$

$\therefore 2m + 3n = 106$

24. Putting $p = 2$, $q = -2$, $r = 1$; we can get the value of

$\frac{1}{p} + \frac{1}{q} + \frac{1}{r} = 1$

25. $x^2 + (3K - 36)x + K^2 - 24K + 144 = 0$

Since roots are reciprocal, product of roots = 1

$\Rightarrow K^2 - 24K + 144 = 1$

$\therefore K = 11, 13$

26. $x^2 + 3x - 4 = 0$

$x^2 + 4x - x - 4 = 0$

$x[x+4] - 1[x+4] = 0 \rightarrow (x+4)(x-1) = 0$

Then, the roots are $(1, -4)$. (i)

In second equation, $x^3 - 4x = 0$

$x[x^2 - 4] = 0$

$\therefore$ $x = 0$ and $x^2 - 4 = 0 \rightarrow x^2 = 4$

$x = \pm 2$

Hence, roots are $0, 2, -2$ (ii)

But, there is no common root; hence, answer is zero.

27. $Z + \frac{1}{Z} = 1$

$\Rightarrow Z^2 - Z + 1 = 0$

$\Rightarrow (Z+1)(Z^2 - Z + 1) = 0$ {if $Z \neq -1$}

$\Rightarrow (Z^3 + 1) = 0$

$\therefore Z^3 = -1$ and $Z \neq -1$

Now, $Z^{64} + \frac{1}{Z^{64}} = Z^{63}.Z + \frac{1}{Z^{63}.Z}$

$\Rightarrow (Z^3)^{21}.Z + \frac{1}{(Z^3)^{21}.Z} = -Z - \frac{1}{Z} = -1$

28. Let $\frac{x^2 - x + 1}{x^2 + x + 1} = y$

$x^2 - x + 1 = y[x^2 + x + 1]$

$x^2 - x + 1 = yx^2 + yx + y$

$yx^2 - x^2 + yx + x + y - 1 = 0$

$x^2[y-1] + x[y+1] + y - 1 = 0$

For real values of $D^2 \geq 0$

Then, $b^2 - 4ac \geq 0 \rightarrow$

$(y+1)^2 - 4(y-1)^2 \geq 0$ $(y^2 + 2y + 1) - 4(y^2 - 2y + 1) \geq 0$

Or, $y^2 + 2y + 1 - 4y^2 + 8y - 4 \geq 0 - 3y^2 + 10y - 3 \geq 0$

Or, $3y^2 - 10y + 3 \leq 0$

Or, $3y^2 - 9y - y + 3 \leq 0$

Or, $3y[y-3] - 1[y-3] \leq 0$ $(3y-1)(y-3) \leq 0$

Hence, $3y - 1 \leq 0$ and $y - 3 \leq 0$

$y \leq \frac{1}{3}$ and $y \leq 3$

Hence, maximum value of y is 3 and minimum value of y is 1/3.

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29. $X^2 + bx + c = 0$

For real roots $b^2 - 4c \geq 0$

$\Rightarrow b^2 \geq 4c$

Now, for each values of 'b' from the given set checking the number of values of 'c' possible in the same set.

$b = 1$ number of values of $c = 0$

$b = 2$ number of values of $c = 1$ (1)

$b = 3$ number of values of $c = 2$ (1, 2)

$b = 4$ number of values of $c = 4$ (1, 2, 3, 4)

$b = 5$ number of values of $c = 6$ (1, 2, 3, 4, 5, 6)

$b = 6$ number of values of $c = 6$ (1, 2, 3, 4, 5, 6)

Therefore, total possible sets = 1 + 2 + 4 + 6 + 6 = 19

30. Let $a, b, c, d, = -1, 0, 1, 2$ (respectively)

$\therefore$ Given expression comes out to be $(x - 1)(x + 1) + 2(x - 2)x = 0$

$\Rightarrow 3x^2 - 4x - 1 = 0$; $\therefore D = 4^2 + 12 = 28$

Hence, it will have real and distinct roots.

31. $2\left[a^{1/3} + \frac{1}{a^{1/3}}\right] = 5$

$\Rightarrow \quad 2a^{2/3} - 5a^{1/3} + 2 = 0$

$\Rightarrow \quad (a^{2/3} - 2)(2a^{1/3} - 1) = 0$

$\therefore \quad a^{1/3} = 2, a^{1/3} = 1/2$

$\Rightarrow \quad a = 8, a = 1/8$

32. Since each of the roots are changing symmetrically from $\alpha \to 3\alpha + 2$

Therefore, to get the required equation, just replace x with $\frac{x-2}{3}$ in the given equation.

Therefore, we get $9\left(\frac{x-2}{3}\right)^3 - 7\left(\frac{x-2}{3}\right) + 6 = 0$

$\Rightarrow x^3 - 6x^2 + 5x + 24 = 0$

33. Since the roots are equal, $D = 0$

$\Rightarrow (r - p)^2 - 4(q - r)(p - q) = 0$

$\Rightarrow [(r - q) + (q - p)]^2 - 4(r - q)(q - p) = 0$

$\Rightarrow [(r - q) - (q - p)]^2 = 0$

$\Rightarrow r - q = q - p$

$\Rightarrow p + r = 2q$, i.e., p, q, r are in AP

34. $(x - b)(x - c) + (x - c) + (x - a)(x - b) = 0$

$\Rightarrow 3x^2 - 2(a + b + c)x + ab + bc + ca = 0$

$\therefore X = 4(a + b + c)^2 - 4(ab + bc + ca) \times 3$

$\Rightarrow D = 4[a^2 + b^2 + c^2 - (ab + bc + ca)]$

Now, we know $(a - b)^2 > 0$

$\Rightarrow a^2 + b^2 > 2ab$

Similarly, $b^2 + c^2 > 2bc$

and, $c^2 + a^2 > 2ca$

$\therefore a^2 + b^2 + c^2 > ab + bc + ca$

$\Rightarrow a^2 + b^2 + c^2 - (ab + bc + ca) > 0$

$\Rightarrow 4(a^2 + b^2 + c^2 - (ab + bc + ca)) > 0$

$\Rightarrow D > 0$

$\therefore$ The roots are real.

35. Applying $D = 0$ in the given equation

$b^2(c - a)^2 = 4ac(a - b)(b - c)$

$\Rightarrow \frac{b^2(c-a)^2}{ac} = 4(a - b)(b - c)$

$\Rightarrow b^2ac\left(\frac{c-a}{ac}\right)^2 = 4\left(\frac{a-b}{ab}\right)\left(\frac{b-c}{bc}\right).ab^2c$

$\Rightarrow \left(\frac{1}{a} - \frac{1}{c}\right)^2 = 4\left(\frac{1}{b} - \frac{1}{a}\right)\left(\frac{1}{c} - \frac{1}{b}\right)$

$\Rightarrow \left[\left(\frac{1}{a} - \frac{1}{b}\right) + \left(\frac{1}{b} - \frac{1}{c}\right)\right]^2 - 4\left(\frac{1}{a} - \frac{1}{b}\right)\left(\frac{1}{b} - \frac{1}{c}\right) = 0$

$\Rightarrow \left[\left(\frac{1}{a} - \frac{1}{b}\right) - \left(\frac{1}{b} - \frac{1}{c}\right)\right]^2 = 0$

$\Rightarrow \frac{1}{a} - \frac{1}{b} = \frac{1}{b} - \frac{1}{c}$

$\Rightarrow \frac{2}{b} = \frac{1}{c} + \frac{1}{a}$. So, a, b, and c are in HP.

36. We know $(x + y)$ is a factor of $x^p + y^p$ if p is odd.

Therefore, here, $p = 3^n$, which is odd integer when 'n' is an integer ≥ 0.

37. For both roots: (α, β) to be positive

$\alpha + \beta > 0$ and $\alpha\beta > 0$

$\Rightarrow \frac{-b}{a} > 0$ and $\frac{c}{a} > 0$

that is, b and a are of opposite sign and c and a are of same sign.

38. $ax^2 + bx + c = 0$

for both roots to be zero, sum of roots $= \frac{-b}{a} = 0$

$\therefore b = 0$

and product of roots $= \frac{c}{a} = 0$ $\therefore c = 0$

39. $ax^2 + bx + c = 0$

Now, in first case, when root is 1/3 and 1

The equation is $(3x - 1)(x - 1) = 0$

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$\therefore$ Correct value of $a = x\ 3 \times 1 = 3$

Similarly, in second case, the equation is $(x-6)(x-5)=0$

$\therefore$ Correct value of $b = -(5+6) = -11$

and in third case, the equation is $(x+3)(x+2)=0$

$\therefore$ Correct equation is $3x^2 + 11x + 6 = 0$

40. For the equations to have same pair of roots

$$\frac{2p-1}{q+1} = \frac{2p+1}{4q+1} = \frac{c}{3c}$$

$\therefore 3(2p-1) = q+1 \Rightarrow 6p - q = 4$

and, $3(2p+1) = 4q+1 \Rightarrow 6p - 4q = -1$

Solving two equations $q = 2$ and $p = 1$

$\therefore (p+q) = 3$

ADVANCED

1. HCF of $\alpha, \beta = 2$; therefore, we can assume $\alpha = 2x$ and $\beta = 2y$

Hence, $2x \times 2y = 24 \times 2$, or, $xy = 12$

Different values of x and y possible are (12, 1), (4, 3). and hence, different values of roots of the equation will be (24, 2) and (8, 6).

Now, since we are talking about LCM and HCF of α, β, roots of the equation has to be non-negative and real.

Therefore, two such quadratic equations are possible.

Note: This is a question on Number System primarily, and not on Quadratic Equation.

2. If roots are real and equal, then $D = 0$

$D = [\sqrt{2}(p+q)]^2 - 4(p^2+q^2) \times 1$

$= 2(p^2+q^2+2pq) - 4(p^2+q^2) = -2(p^2+q^2-2pq)$

$= -2[(p-q)^2] = 0$

Hence, $p = q$

4. Due to symmetry, we can say that the maximum value of $xy + yz + zx$ will be at $x = y = x$

Now, $x^2 + y^2 + z^2 = 1$

$\Rightarrow x = y = z = 1/\sqrt{3}$

$\therefore xy + yz + zx \le 1$ which is present only in one option.

5. $ax^2 - bxy - ay^2 = 0$

$D = b^2y^2 + 4a^2y^2 = y^2\{b^2 + (2a)^2\} > 0$ but not a perfect square. Hence, factors are real and different.

6. Let the roots be α and β.

$$\therefore \alpha + \beta = \frac{1}{\alpha^2} + \frac{1}{\beta^2}$$

$$\Rightarrow \frac{-b}{a} = \frac{\alpha^2+\beta^2}{\alpha^2\beta^2} \Rightarrow \frac{-b}{a} = \frac{(\alpha+\beta)^2 - 2\alpha\beta}{\alpha^2\beta^2}$$

$$\Rightarrow \frac{-b}{a} = \frac{(b/a)^2 - \frac{2c}{a}}{c^2/a^2}$$

$$\Rightarrow \frac{-b}{a} = \frac{b^2-2ca}{c^2} \Rightarrow b^2a + bc^2 = 2ca^2$$

$$\Rightarrow \frac{b^2}{ac} + \frac{bc}{a^2} = 2$$

7. $ax^2 + bx + c = 0$

$x^3 + 3x + 10 = 0$

For common roots, $a = 1$, $b = 3$, $c = 10$ (ratio)

$a:b:c = 1:3:10$

8. Taking $(x-2) > 0$, we get $x = 0, 3$. Therefore, $x = 3$

Taking $(x-2) < 0$, we get $x = 1, 4$. Therefore, $x = 1$

Hence, sum of roots = 4

Alternatively,

Assuming $|x-2| = Z$, this equation is now $z^2 + z - 2 = 0$

Or, $(z+2)(z-1) = 0$; hence, $z = -2$ or 1

Taking z = −2 is not possible since $z = |x-2|$

Taking $z = 1$ or $|x-2| = 1$, so $x = 3$ and $x = 1$

And these are the only real roots of this equation. Hence, sum of real roots = 1 + 3 = 4

12. It is possible only if the given quadratic equation becomes linear equation. Hence, $a = 0$

13. $x + \frac{1}{x} = 1 \Rightarrow x^2 - x + 1 = 0$

Now, as $x \neq -1 \Rightarrow (x+1)(x^2-x+1) = 0$

$\Rightarrow x^3 + 1 = 0$

$\therefore x^3 = -1 \Rightarrow x^{4000} = (x^3)^{1333}.\ x = -x$

$$\therefore P = x^{4000} + \frac{1}{x^{4000}}$$

$$= -x - \frac{1}{x} = -1 \Rightarrow P = -1$$

Now, let $n = 2$

$\therefore$ p = unit digit of 17, that is, 7. So, $p + q = 7 - 1 = 6$

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14. Let the roots of the equation $ax^2 + bx + b = 0$ be pk and qk.

$\therefore (p + q)\, K = \dfrac{-b}{a}$ and $pqk^2 = \dfrac{b}{a}$

$\therefore \sqrt{\dfrac{p}{q}} + \sqrt{\dfrac{q}{p}} + \sqrt{\dfrac{b}{a}} = \dfrac{p+q}{\sqrt{pq}} + \sqrt{\dfrac{b}{a}}$

$\dfrac{(p+q)^k}{\sqrt{pak^2}} + \sqrt{\dfrac{b}{a}} = \dfrac{-b/a}{\sqrt{b/a}} + \sqrt{b/a} = 0$

15. $a_1x^2 + b_1x + c_1 = 0$ and $a_2x^2 + b_2x + c_2 = 0$

For same pair of roots $\dfrac{a_1}{a_2} = \dfrac{b_1}{b_2} = \dfrac{c_1}{c_2} = \dfrac{1}{2}$ (given)

Expression (2) $= a_2x^2 + b_2x + c_2 = 2(a_1x^2 + b_1x + c_1)$

$= 2$ [Expression (1)]

Then, the ratio of the maximum value is 1:2

16. We have to check that if the statements can be true.

Checking Statement I, if $a = c$, then obviously both the equation will have common root.

Checking Statement II, if the common root is 1, then both the equations are giving $a + b + c = 0$.

Checking Statement III, if the common root is -1, then the given condition is true.

Hence, all the three statements are true.

18. Solving equation, we get $(1 + k)\, X^2 - \{a + c + (b + d)\, k\}\, x + (ac + bdk) = 0$

Here, $D \geq 0$, hence roots of equation will be real.

19. Any operation of rational numbers will always result in rational numbers.

20. Let the roots of the given equation be α and β.

Now, for roots $(\alpha - \beta)$, $(\beta - 2)$, the equation can be deduced by replacing x with $(x + 2)$.

$\therefore$ The deduced equation would be

$\Rightarrow (x + 2)^2 - (p + 1)\,(x + 2) + p^2 + p - 8 = 0$

$\Rightarrow x^2 + (3 - p)\, x + p^2 - p - 6 = 0$

$\Rightarrow x^2 + (p - 3)\, x + (p + 2)\,(p - 3) = 0$

Now, $\alpha > 2$ and $\beta < 2$

$\therefore (\alpha - 2) > 0$ and $(\beta - 2) < 0$

$\therefore (\alpha - 2)\,(\beta - 2) < 0 \Rightarrow (p + 2)\,(p - 3) < 0$

$\therefore (\alpha - 2)\,(\beta - 2) < 0 \Rightarrow (p + 2)\,(p - 3) < 0$

$\therefore -2 < p < 3$

21. Solving equation, we get $D = -5a^2 + 10a + 1$

Hence, we can conclude that no value of a fulfils the equation.

22. If there is only one solution of any quadratic equation, then it means both the roots of the equation are equal or $D = 0$.

Alternatively, put $x = 4$ and $k = 2$.

23. Solving equation, we get $3x^2 - 2(a + b + c)\, x + ab + bc + ca = 0$

For roots to be equal, $a = b = c$

24. Let the equation, be $x^2 - x - 12 = 0$ $(x = -3, 4)$

Here, $a = 1$, $b = -1$, $c = -12$

25. Putting $a = 1$, we get one common real roots for both equations.

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CHAPTER

12

Sequence and Series

LEARNING OBJECTIVES

After completion of this chapter, the reader should be able to understand:

- The concept of sequences and series
- Progression and their types
- Definitions and properties of progressions
- Kinds of questions asked in the CAT
- Methods of solving questions

INTRODUCTION

Sequence and series is a mathematical concept that draws majorly from the basic number system and the simple concepts of arithmetic. This is the reason that makes it an important topic for this examination. On an average, one to three questions have been asked from this topic in the CAT, almost every time in the last 12 years. Besides the CAT, this topic is also important for other examinations such as IIFT, SNAP, XAT, and MAT. One of the important features of these problems, from this chapter, is that they can be solved simply by the application of logic or by some very simple concepts of calculation.

SEQUENCE AND SERIES

Let us consider the following series:

- 1, 4, 9, 16, …
- 2, 6, 12, 20, …

Here, it can be observed that each of these two series shares some or the other common property:

Series (i) is → 1^2, 2^2, 3^2, 4^2 …

Series (ii) is → $1^2 + 1$, $2^2 + 2$, $3^2 + 3$, $4^2 + 4$ …

With this, any term or in general t_n, for either of the two series can be very easily found out.

For series (i), $t_{10} = 10^2$

For series (i), $t_{10} = 10^2 + 10$

If the terms of a sequence are written under some specific conditions, then the sequence is called a *progression*.

With respect to preparation for the CAT, we will confine ourselves only to the following standard series of progression:

1. Arithmetic progression
2. Geometric progression
3. Harmonic progression

1. Arithmetic Progression

A succession of numbers is said to be in an arithmetic progression (AP), if the difference between any term and the previous term is constant throughout. In other words, the difference between any of the two consecutive terms should be the same. This difference that is common between any two consecutive terms is known as common difference of this AP and is denoted by 'd'.

For example → Series (i) 1, 2, 3, …

Series (ii) → 2, 5, 7, 10, …

Series(iii) → a, $a + d$, $a + 2d$, …

Common difference (d) of series (i) = 1

Common difference (d) of series (ii) = 3

Common difference (d) of series (iii) = d

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nth term of an arithmetic progression

First term $t_1 = a = a + (1 - 1)d$
Second term $t_2 = a + d = a + (2 - 1)d$
Third term $t_3 = (a + d) + d = a + 2d = a + (3 - 1)d$
Fourth term $t_4 = a + 3d = a + (4 - 1)d$
nth term $t_n = a + (n - 1)d$, where a is the first term, d is the common difference, and n is the number of terms.

Important Points

- t_n is also known as the general term of AP.
- If in any question, some particular term is given (like t_4 or t_{10}), then we should assume those terms in the form of t_n. However, if the total number of terms are given, then we should assume the terms in the following way:

If three terms or any odd number of terms are involved, then we should assume these terms as $a - d$, a, $a + d$, and so on.

If four terms or any even number of terms are involved, then we should assume these terms as $a - 3d$, $a - d$, $a + d$, $a + 3d$, and so on.

Example 1 The sum of three numbers in an AP is 27 and the sum of their squares is 293. Find the numbers.

Solution Let the numbers be $a - d$, a, $a + d$.
Given is $(a - d + a + a + d) = 27$
So, $a = 9$
Also, $(a - d)^2 + a^2 + (a + d)^2 = 293$
$\Rightarrow d^2 = 25$
$\Rightarrow d = \pm 5$
When $d = +5$, then the terms are 4, 9, 14.
When $d = -5$, then the terms are 14, 9, 4.
Alternatively, this question can be worked out very easily with the help of options.

Example 2 If the sum of the first 11 terms of an arithmetic progression equals that of the first 19 terms, then what is the sum of the first 30 terms?

(a) 0 (b) −1
(c) 1 (d) Not unique

Solution Given

$$t_1 + t_2 + \ldots + t_{11} = t_1 + t_2 + \ldots + t_{19} \text{ (for an AP)}$$

$$\Rightarrow \frac{11}{2}[2a + (11-1)d] = \frac{19}{2}[2a + (19-1)d]$$

$$\Rightarrow 22a + 110\,d = 38a + 342d$$

$$\Rightarrow 16a + 232\,d = 0$$

$$\Rightarrow 2a + 29\,d = 0$$

$$\Rightarrow \frac{30}{2}[2a + (30-1)d] = 0$$

$$\Rightarrow S_{30 \text{ terms}} = 0$$

Properties of AP

If $a, b, c, d, \ldots$ are in AP, then

1. $a + k, b + k, c + k, d + k \ldots$ will be in AP, where k is any constant.
2. $a - k, b - k, c - k, d - k \ldots$ will be in AP, where k is any constant.
   In the above two cases, the common difference will be the same as earlier.
3. $ak, bk, ck, dk \ldots$ will be in AP, where k is any constant.
   In this case, new common difference will be k times the earlier common difference.
4. $\frac{a}{k}, \frac{b}{k}, \frac{c}{k}, \frac{d}{k}$ will be in AP, where $k \neq 0$.
   In this case, new common difference will be $\frac{1}{k}$ times the earlier common difference.
5. For any AP, if $S_n = S_m$, then $S_{m+n} = 0$.
   (For any AP, if sum of 1st n terms is equal to the sum of 1st m terms, then sum of 1st $(m + n)$ terms = 0.)

Example 3 If a, b, c are in AP, then $b + c$, $c + a$, $a + b$ will be in

(a) AP
(b) GP
(c) HP
(d) Cannot be determined uniquely

Solution a, b, c are in AP, then $a - (a + b + c)$, $b - (a + b + c)$, $c - (a + b + c)$ will be in AP.
$\Rightarrow -(b + c)$, $-(a + c)$, and $-(a + b)$ will be in AP.
$\Rightarrow (b + c)$, $(a + c)$, and $(a + b)$ will also be in AP.
Alternatively, let us assume a, b, c to be 1, 2, 3. Then, $(b + c) = 5$, $(a + c) = 4$ and $(a + b) = 3$, which are obviously in AP.

Example 4 If x, y, z are in GP, then $1/(1 + \log_{10}x)$, $1/(1 + \log_{10}y)$ and $1/(1 + \log_{10}z)$ will be in:

(a) AP
(b) GP
(c) HP
(d) Cannot be determined uniquely

Solution Let us go through the options.
Checking option (a), the three will be in AP if the second expression is the average of the first and the third expressions.
This can be mathematically written as:

$$2/(1 + \log_{10}y) = 1/(1 + \log_{10}x) + 1/(1 + \log_{10}z)$$

$$= \frac{[1 + (1 + \log_{10}x) + 1 + (1 + \log_{10}z)]}{[(1 + \log_{10}x)(1 + \log_{10}z)]}$$

$$= [2 + \log_{10}xz]/(1 + \log_{10}x)\,(1 + \log_{10}z)$$

Obviously, this will not give us the answer.
Checking option (b),

$$[1/(1 + \log_{10}y)]^2 = [1/(1 + \log_{10}x)]\,[1/(1 + \log_{10}z)]$$

$$= [1/(1 + \log_{10}(x + z) + \log_{10}xz)]$$

Again, no solution is found.

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Checking option (c),

$1/(1 + \log_{10}x)$, $1/(1 + \log_{10}y)$ and $1/(1 + \log_{10}z)$ are in HP, then $1 + \log_{10}x$, $1 + \log_{10}y$ and $1 + \log_{10}z$ will be in AP.

So, $\log_{10}x$, $\log_{10}y$ and $\log_{10}z$ will also be in AP.

Hence, $2\log_{10}y = \log_{10}x + \log_{10}z$

$\Rightarrow y^2 = xz$ which is given.

Hence, (c) is the answer.

Alternatively, we can also apply the following process:

Assume $x = 1$, $y = 10$, and $z = 100$ as x, y, z are in GP.

So, $1 + \log_{10}x = 1$, $1 + \log_{10}y = 2$, and $1 + \log_{10}z = 3$.

$\Rightarrow$ Thus, we find that since 1, 2, and 3 are in AP, we can assume that

$$1 + \log_{10}x,\ 1 + \log_{10}y,\text{ and } 1 + \log_{10}z \text{ are in AP.}$$

Hence, by definition of an AP, we have that $1/(1 + \log_{10}x)$, $1/(1 + \log_{10}y)$, and $1/(1 + \log_{10}z)$ are in AP. Hence, option (c) is the answer.

Sum of *n* terms of an arithmetic progression

$S_n = \frac{n}{2}[2a + (n-1)d]$, where n = number of terms, a = first term and d = common difference.

Example 5 Find the sum of AP 3, 5, 7, …, 50 terms and find its sum.

Solution Here, $n = 50$, $d = 2$, and $a = 3$

Using formula, $S_n = \frac{n}{2}[2a + (n-1)d]$

$= 25 \times [2 \times 3 + (50 - 1)2)]$

$= 25 \times 104 = 2600$

However, we can find out the sum of any AP in a better way through average also.

The last term of this series = 101, so, the average

$$= \frac{3+101}{2} = 52$$

So, sum = average × number of numbers

$= 52 \times 50 = 2600$

Hence, $S_n = \frac{F+L}{2} \times n$, where F is the first term, L is the last term and n is the number of terms of that AP.

Example 6 What is the sum of all the two-digit numbers which when divided by 7 gives a remainder of 3?

(*CAT 2003*)

Solution This series is like → 10, 17, 24, …, 94

Here, $n = 13$, $d = 7$, and $a = 10$

Using the formula for the sum

$S_n = \frac{n}{2}[2a + (n-1)d]$, sum = 676

Alternatively, using the average method, average = (first number + last number)/2

Average $= \frac{10+94}{2} = 52$

So, the sum = average × number of numbers

$= 52 \times 13 = 676$

Example 7 Find the value of the expression $1 - 4 + 2 - 5 + 3 - 6 + \ldots$ to 100 terms.

(a) −250 (b) −500

(c) −450 (d) −300

Solution We can write the given expression $(1 - 4 + 2 - 5 + 3 - 6 + \ldots$ to 100 terms) as follows:

(1 + 2 + 3 + … to 50 terms) − (4 + 5 + 6 + … to 50 terms)

Both of these are AP's with different values of 'a' and 'd'. $a = 1$, $n = 50$, and $d = 1$ for first series and $a = 4$, $n = 50$, and $d = 1$, respectively.

Using the formula for the sum of AP

$= \frac{n}{2}[2a + (n-1)d]$, we get

$= \frac{50}{2}[2.1 + (50-1).1] - \frac{50}{2}[2.4 + (50-1).1]$

$= 25(2 + 49) - 25(8 + 49)$

$= 25(51 - 57) = -150$

Alternatively, logically, this question can be done a bit faster by assuming (1 − 4), (2 − 5), etc., as one unit.

$$1 - 4 = 2 - 5 = \ldots = -3$$

Therefore, the above series is equivalent to a series of 50 (−3)s added to each other.

So, (1 − 4) + (2 − 5) + (3 − 6) + … 50 terms = −3 × 50 = −150

2. Geometric Progression

A succession of numbers is said to be in a geometric progression if the ratio of any term and the previous term is constant. This constant ratio that is common to any of the two terms is known as the common ratio and is denoted by 'r'.

Example i. 1, 2, 4, 8, …

ii. 20, 10, 5, …

iii. a, ar, ar^2, …

Common ratio of series (i) is 2.

Common ratio of series (ii) is 0.5.

Common ratio of series (iii) is r.

***n*th term of a geometric progression**

First term	$t_1 = a = ar^{1-1}$
Second term	$t_2 = ar = ar^{2-1}$
Third term	$t_3 = ar^2 = ar^{3-1}$
Fourth term	$t_4 = ar^3 = ar^{4-1}$
nth term	$t_n = ar^{n-1}$

where a is the first term, r is the common ratio, and n is the number of terms.

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Important Points

- t_n is also known as the general term of GP.
- In any question, if some particular term is given like t_4 or t_{10}, then we should assume those terms in the form of t_n. However, if the total number of terms are given, then we should assume the terms in the following way:
  If three terms or any odd number of terms are involved, then we should assume these terms as $\frac{a}{r}$, a, ar and so on.

Example 8 The seventh term of a GP is 8 times the fourth term and the fifth term of the same GP is 48. Find the sixth term of this GP.

Solution Given $t_7 = 8 \times t_4$
Or, $ar^6 = 8 \times ar^3$
$\Rightarrow r^3 = 8$, or, $r = 2$
Now, $ar^4 = 48$, So, $a = 3$
So, sixth term $= ar^5 = 3 \times 2^5 = 96$

Properties of GP

If $a, b, c, d, \ldots$ are in GP, then

1. $ak, bk, ck, dk \ldots$ will be in GP, where k is any non-zero constant.
2. $\frac{a}{k}, \frac{b}{k}, \frac{c}{k}, \frac{d}{k}$, will be in GP, where k is any non-zero constant.
   In the above two cases, the common ratio will be the same as the earlier.
3. If a GP of any even number of terms is given, then its common ratio will be the same as the ratio of the sum of all even terms and the sum of all odd terms.

Sum of *n* terms of a geometric progression

$S_n = \frac{a(1-r^n)}{1-r}$ when $r \neq 1$
$S_n = na$ when $r = 1$
where n = number of terms, a = first term, and d = common difference.

Sum of Infinite Geometric Progression

So far, we have done the summation of n terms of a GP. Now, there is also a need of a separate expression for the sum of infinite GP. In case of AP, since either of the terms are always decreasing and going till $-\infty$ or are always increasing and going till $+\infty$. So, the summation of the infinite terms in AP will be either $-\infty$ or $+\infty$. However, the case is not the same in GP.

The need of a formula for infinite GP can be seen with the following example:

Find the sum of the series $\rightarrow 2 + 1 + \frac{1}{2} + \frac{1}{4} + \ldots$ till infinite terms.

In the above-written expression, the number of terms is not given, and therefore, we cannot find out the sum using the formula for finding the sum of n terms of a GP.

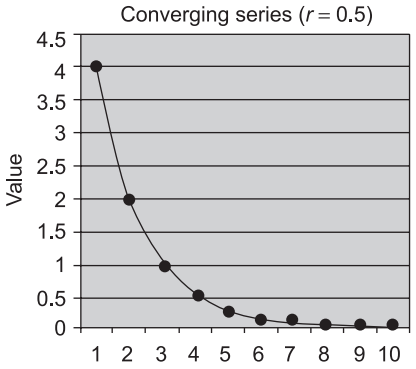
This can be further seen with the help of the following graphs:

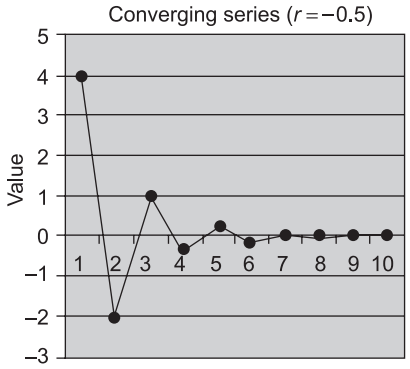
If ***r*** is outside the range of -1 to 1, the terms of the series get bigger and bigger (even if they change the sign), and the series ***diverges***. If ***r*** is within the range of -1 to 1, the terms get smaller and smaller (closer to 0) and the series ***converges***.

Sum of Infinite GP $= \frac{a}{1-r}$, where $-1 < r < 1$

Example 9 What is the sum of the following series:
$1 + 2 + 4 + \ldots$ till infinity.

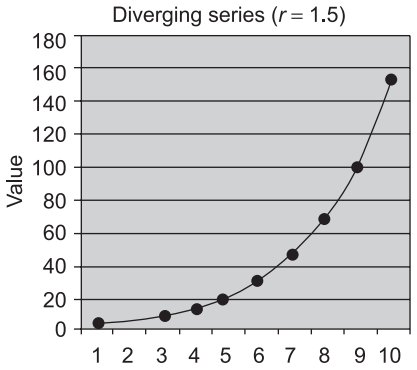
Solution It is very obvious that the sum is going to be $+\infty$. This formula is applicable only for $-1 < r < 1$.

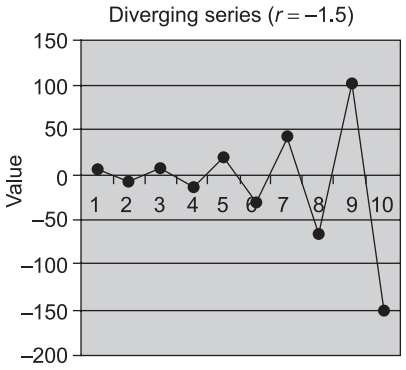




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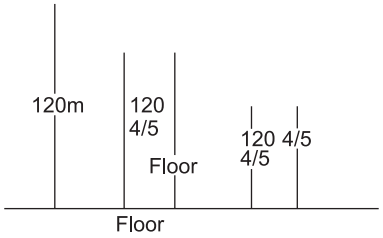
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Example 10 After striking the floor, a ball rebounds to 4/5th of the height from which it has fallen. What is the total distance that it travels before coming to rest if it is gently dropped from a height of 120 m?

Solution The distance covered before the first rebound = 120 m



Then, the ball bounces back to a height of $120 \times \frac{4}{5}$ and then falls from the same height. Next time, the ball will go up by $120 \times \frac{4}{5} \times \frac{4}{5}$ m and then it will fall from the same height.

So, the total distance covered

$$= 120 + 2 \times 120 \times \frac{4}{5} + 2 \times 120 \times \frac{4}{5} \times \frac{4}{5} + \ldots \infty.$$

$$= 120 + 2 \times 120 \left(\frac{4}{5} + \frac{4}{5} \times \frac{4}{5} + \ldots \infty \right)$$

$$= 120 + 240 \left[\frac{\frac{4}{5}}{1 - \frac{4}{5}} \right] = 1080 \text{ m}$$

Alternatively, if the ball rebounds to $\frac{a}{b}$ th of the original height H, then the total distance covered $= H \times \frac{a+b}{a-b}$

In the above example, the total distance covered

$$= 120 \times \frac{4+5}{4-5} = 1080 \text{ m}$$

Example 11 On 1 January 2004, two new societies S_1 and S_2 are formed, each of n numbers. On the first day of each subsequent month, S_1 adds b members, while S_2 multiples its current numbers by a constant factor r. Both the societies have the same number of members on 2 July 2004. If $b = 10.5n$, what is the value of r? ***(CAT 2004)***

(a) 2.0 (b) 1.9 (c) 1.8 (d) 1.7

Solution There will be an increase of six times. The number of members in S_1 will be in an AP.

On 2 July 2004, S_1 will have $n + 6b$ members

$= n + 6 \times 10.5\, n, = 64n$

The number of members in S_2 will be in a GP.

On 2 July 2004, the number of members in $S_2 = nr^6$

They are equal, hence, $64\, n = nr^6$

$\Rightarrow 64 = r^6 \Rightarrow r = 2$

3. Harmonic Progression

Unequal numbers $a, b, c, \ldots$ are said to be in a harmonic progression (HP) if reciprocals of these terms, i.e., $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}, \ldots$ are in an AP. It is noteworthy, that no term of a HP can be equal to zero.

nth term of a harmonic progression

nth term of HP = 1/(nth term of the corresponding AP)

Example 12 If $a, b,$ and c are in HP, then $\frac{a}{b+c}, \frac{b}{c+a}, \frac{c}{a+b}$ are in:

(a) AP
(b) GP
(c) HP
(d) Cannot be determined uniquely

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Solution a, b, c are in HP, so $\frac{1}{a}, \frac{1}{b}$, and $\frac{1}{c}$ will be in AP.

Or, $\frac{a+b+c}{a}, \frac{a+b+c}{b}$, and $\frac{a+b+c}{c}$ will be in AP.

Or, $1+\frac{b+c}{a}, 1\frac{a+c}{b}$, and $1+\frac{a+b}{c}$ will be in AP.

Hence, $\frac{b+c}{a}, \frac{a+c}{b}$, and $\frac{a+b}{c}$ are in AP.

So, $\frac{a}{b+c}, \frac{b}{c+a}$, and $\frac{c}{a+b}$ will be in HP.

Alternatively, these kinds of problems can also be done by assuming values.

Let us take 1, 1/2, 1/3 (which are in HP)

$$\frac{a}{b+c}=\frac{6}{5}, \frac{b}{a+c}=\frac{3}{8}, \frac{c}{a+b}=\frac{2}{9}$$

Now, when we check these values for AP, GP, and HP, we find that $\frac{8}{3}$ is the AM of $\frac{5}{6}$ and $\frac{9}{2}$.

So, obviously, the given terms are in HP.

Properties of HP

If a, b, c, and d are in HP, then

1. $a + d > b + c$
2. $ad > bc$

Sum of *n* terms of a harmonic progression

There is no standard formula for finding the sum of n terms of a HP.

ARITHMETICO GEOMETRIC SERIES

A series is said to be in arithmetico geometric series if each of its term is the product of the corresponding terms of an AP and a GP.

For example, $1, 2x, 3x^2, 4x^3, \ldots$

In the above series, the first part of this series is in an AP $(1, 2, 3, 4, \ldots)$ and the second part is $(x^0, x^1, x^2, x^3, \ldots)$ in a GP.

Sum of *n* Terms of Any Arithmetico Geometric Series (AGS)

The sum of n terms of any AGS $a, (a + d)r, (a + 2d)r^2, \ldots$ is given by

$$S_n = \frac{a}{1-r} + dr\frac{(1-r^{n-1})}{1-r} - \frac{[a+(n-1)d]}{1-r}.\ r^n, \text{ if } r \neq 1$$

$$S_n = \frac{n}{2}[2a + (n-1)d], \text{ if } r = 1$$

Sum of Infinite Terms of Any Arithmetico Geometric Series (AGS)

$$S_\infty = \frac{a}{1-r} + \frac{d \times r}{(1-r)^2}, |r| < 1$$

However, I would suggest students to desist from using these formulae. They should use the standard process to find out the sum of any AGS which is given below:

Let N be the sum of the arithmetico geometric series. Then, each term of the series is multiplied by r (the common ratio of GP) and is written by shifting each term one step rightward, and then by subtracting rN from N to get $(1 - r)N$. Therefore, N is finally obtained.

Example 13 What is the sum of the following series till infinity: $1 + 2x + 3x^2 + 4x^3 + \ldots, |x| < 1$

Solution Assume $S = 1 + 2x + 3x^2 + 4x^3 + \ldots$ (i)

Multiplying S by x, $x.\ S = x + 2x^2 + 3x^3 + 4x^4 + \ldots$ (ii)

Subtracting (ii) from (i)

$$S - x\ S = 1 + (x + x^2 + x^3 + \ldots \infty)$$

$$S(1-x) = 1 + (x + x^2 + x^3 + \ldots \infty) = 1 + \frac{x}{1-x}$$

$$S = \frac{1}{1-x} + \frac{x}{(1-x)^2} = \frac{1}{(1-x)^2}$$

MEANS

Arithmetic Mean

It two numbers a and b are in AP, then their arithmetic mean

$$(\text{AM}) = \frac{a+b}{2}$$

In general, if $a, b, c, \ldots n$ terms are in AP, then their

$$\text{AM} = \frac{a+b+c+\ldots n \text{ terms}}{n}$$

Its imperative to mention here is that, in simple terms, AM of n terms is nothing but the average of n terms.

Hence, the sum of n AMs between P and Q

$$= \left(\frac{P+Q}{2}\right)n$$

Example 14 If a, b, c, d, e, and f are the AMs between 2 and 12, then find the value of $a + b + c + d + e + f$?

Solution The sum of n AMs between P and Q

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$$= \left(\frac{P+Q}{2}\right)n$$

Hence, the sum $= \left(\frac{2+12}{2}\right)6 = 42$

Inserting a Given Number of Arithmetic Means Between Two Given Quantities

Let p and q be the given quantities and n be the number of means which is to be inserted. After inserting n means, the total number of terms including the extremes will be equal to $n + 2$. Now, we have to find a series of $n + 2$ terms in AP, of which p is the first, and q is the last term.

Let d be the common difference.

Then, q = the $(n + 2)$th term $= p + (n + 1)\,d$

Hence, $d = \frac{(q-p)}{(n+1)}$

And the inserted means are as follows:

$$p+\frac{(q-p)}{(n+1)}, p+2\frac{(q-p)}{(n+1)}, ..., p+n\frac{(q-p)}{(n+1)}$$

Geometric Mean

If two numbers a and b are in GP, then their geometric mean

$(GM) = \sqrt{ab}$

In general, if a, b, c, … n terms are in GP, then their

$GM = \sqrt[n]{a \times b \times c \times ...n \text{ terms}}$

So, if three terms a, b, and c are in GP, then their

$GM = b = \sqrt[3]{a \times b \times c}$

Inserting a Given Number of Geometric Means Between Two Given Quantities

Let p and q be the given quantities and n be the required number of geometric means to be inserted between p and q. In all, there will be $n + 2$ terms, and so, we have to find a series of $n + 2$ terms in a GP of which p is the first term and q is the last term.

Let us assume that r is the common ratio.

Then, q is the $(n + 2)$th term.

$q = pr^{n+1}$

So, $r^{n+1} = \frac{q}{p}$; $r = \frac{q^{\frac{1}{n+1}}}{p}$

Hence, the required means are $p\,\frac{q^{\frac{1}{n+1}}}{p}$, $p\,\frac{q^{2\frac{1}{n+1}}}{p}$, $...p\frac{q^{n\frac{1}{n+1}}}{p}$

The above written means are nothing but $pr, pr^2, \ldots, pr^n$,

where $r = \frac{q^{\frac{1}{n+1}}}{p}$

Harmonic Mean

If two numbers a and b are in HP, then their harmonic mean

$(HM) = \frac{2ab}{a+b}$

The process of inserting n HMs between two given numbers is quite similar to the process of inserting AM's between two given numbers.

Relationship among AM, GM, and HM

Now, we know that for any two given numbers a and b,

$AM = \frac{a+b}{2}$

$GM = \sqrt{ab}$

$HM = \frac{2ab}{a+b}$

1. AM, GM, and HM will be in a GP.
   So, GM is the geometric mean of this series.
   And, $GM = \sqrt{AM \times HM}$ [true only for two terms]
2. AM ≥ GM ≥ HM [always true for any number of terms]
3. The equation having a and b as its roots is $x^2 - 2Ax + G^2 = 0$.

Sum of N Terms of Some Special Series

In this part of progression, the sum of some other special sequences will be discussed.

1. The sum of the first n natural numbers:

   $S_n = \frac{n(n+1)}{2}$

   Hence, $\sum_{S=1}^{n} S = 1 + 2 + 3 + \ldots + n = \frac{n(n+1)}{2}$

2. The sum of the squares of the first n natural numbers:

   $S_n = \frac{n(n+1)(2n+1)}{6}$

   Hence, $\sum_{S=1}^{n} S^2 = 1^2 + 2^2 + 3^2 + \ldots + n^2$

   $= \frac{n(n+1)(2n+1)}{6}$

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3. The sum of the cubes of the first n natural numbers:
S_n = (Sum of the first n natural numbers)^2
$= \left(\frac{n(n+1)}{2}\right)^2$
It can be seen that $(1 + 2 + 3)^2 = 1^3 + 2^3 + 3^3$
Hence, $\sum_{S=1}^{n} S^3 = 1^3 + 2^3 + 3^3 + \ldots + n^3$
$= \left(\frac{n(n+1)}{2}\right)^2$

4. $2^2 + 4^2 + 6^2 + \ldots + n$ terms $= 1/4 \frac{n(n+1)(2n+1)}{6}$

Solved Problems

Example 15 An investor starts with $ 500 in an investment account, and each month, it earns a constant interest of $ 32. After how many months will the sum exceed $ 700 in the account?
(a) 2 (b) 3 (c) 6 (d) 8

Solution Here, we know that $a + (n - 1)d = 700$ where $a = 500$ and $d = 32$. The reason for not using the sum formula is that the amount in the account, after each successive month, represents each term of the series and not the added up terms.

The equation becomes $500 + (n - 1) \times 32 = 700$. This solution gives $n = 7.25$, which is rounded up to 8. After eight months, the account will hold in excess of $ 700.

Hence, option (d) is the answer.

Example 16 The sum of an AP, consisting of 12 terms, is 354. The ratio of the sum of the odd terms to the sum of the even terms is 27:32. What is the common difference of this AP?
(a) 2 (b) 3 (c) 4 (d) 5

Solution Let the common difference be 'd' and the terms are $T_1, T_2, T_3 \ldots$
So, $T_2 = T_1 + d$
And $T_4 = T_3 + d$ and so on.
So, $T_2 + T_4 + T_6 + T_8 + T_{10} + T_{12} = T_1 + T_3 + T_5 + T_7 + T_9 + T_{11} + 6d$ (i)
Sum of the even terms:Sum of the odd terms = 32 : 27 and their sum = 354
So, the sum of the even terms = 192 and of the odd terms = 162, Using (i), $d = 5$

Example 17 Find the sum of n terms of the series $11 + 103 + 1005 + \ldots$
(a) $10/9(10^n - 1) - 1$ (b) $100/99(10^n - 1) + n^2$
(c) $10/9(10^n - 1) + n^2$ (d) None of these

Solution Ideally, in these kinds of problems, instead of going by the mathematical process of solving, we should use options.
Checking option (a), Put $n = 1$
$10/9(10^n - 1) - 1 = 9$, and therefore, it is not correct.
Checking option (b), Put $n = 1$
$100/99(10^n - 1) + n^2$ is not equal to 11, and so, this is also not correct.
Checking option (c), Put $n = 1$
$10/9(10^n - 1) + n^2 = 11$. But just because this option satisfies $n = 1$, it should not be assumed to be correct. Let us check it for $n = 2$.
Option (c) gives us 104. So, this is the answer.
Normally, in these cases, checking the options till $n = 2$ guarantees the answer, but sometimes we need to check it till $n = 3$.

Practice Exercises

Warm Up

Q.1 Find the sum of the first 15 terms of the series whose nth term is $(4n + 1)$.
(a) 485 (b) 495
(c) 505 (d) 630

Q.2 What is the sum of the first 7 terms of the series 1/3, 1/2, 3/4, …?
(a) 2059/164 (b) 2050/192
(c) 2059/164 (d) None of these

Q.3 In a GP, the first term is 7 and nth term is 448, and the sum of the n terms is 889. What is the common ratio of this GP?
(a) 2 (b) 4 (c) 1 (d) 3

Q.4 The sum of 15 terms of an AP is 600, and the common difference is 5. Find the first term.
(a) 4 (b) 5
(c) 3 (d) None of these

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Q.5 A man arranges to pay off a debt of ₹3600 by 40 annual installments which form an AP. When 30 of the installments were repaid, he died leaving third of the debt unpaid. What is the value of the first installment?
(a) 50 (b) 51 (c) 52 (d) 53

Q.6 An AP has 23 terms, the sum of the middle three terms is 144, the sum of the last three terms is 264. What is the 16th term?
(a) 102 (b) 64
(c) 82 (d) None of these

Q.7 Find A, B, and C between 2 and 18 such that (i) their sum is 25, (ii) 2, A and B are consecutive terms of an AP and (iii) B, C and 18 are the consecutive terms of a GP.
(a) 5, 8, 14 (b) 4, 8, 12
(c) 5, 8, 12 (d) None of these

Q.8 If the sum of an AP is the same for p terms as for the q terms, find the sum for $(p + q)$ terms.
(a) 2 (b) 0
(c) 4 (d) None of these

Q.9 The sum of the three numbers in GP is 70. If the two extreme terms are multiplied by 4, and the middle term by 5, the resultants are in an AP. Find the numbers.
(a) 10, 25, 35 (b) 10, 20, 40
(c) 15, 30, 35 (d) 12, 24, 34

Q.10 Find the first term of the infinite GP whose first two terms add up to 5, and each term is three times the sum of the all terms that follow it.
(a) 1 (b) 2 (c) 3 (d) 4

Q.11 If $x = 1 + a + a^2 + a^3 + \ldots$ to ∞ $(|a| < 1)$,
$y = 1 + b + b^2 + b^3 + \ldots$ to ∞ $(|b| < 1)$,
Then, find $1 + ab + a^2b^2 + a^3b^3 + \ldots$ to ∞
(a) $x + y/x + y - 1$ (b) $xy/x + y$
(c) $xy/x + y - 1$ (d) None of these

Q.12 If the $(n + 1)$th term of a harmonic progression is twice the $(3n + 1)$th term, find the ratio of the first term to the $(n + 1)$th term.
(a) 1 (b) 2 (c) 3 (d) 4

Q.13 If the sum of the reciprocals of the first seven terms of a harmonic progression is 70, find the fourth term of the HP.
(a) 2/15 (b) 1/10 (c) 3/7 (d) 5/12

Q.14 If the mth term of a HP is n and the nth term is m, what is the value of the $(m + n)$th term?
(a) $m/(m + n)$ (b) $mn/m + n$
(c) $n/m + n$ (d) None of these

Q.15 If $(x^{n+1} + y^{n+1})/(x^n + y^n)$ is the harmonic mean of x and y, find the value of n.
(a) 1 (b) −1 (c) 2 (d) 4

Q.16 A series is simultaneously in an AP and a HP. Which of the following is true?
(a) The set of common ratios of all the possible GPs can have only one element.
(b) The set of common ratios of all the possible GPs can have more than one element, but there will be a finite value.
(c) The set of common ratios of all the possible GPs can have infinite elements.
(d) None of these

Q.17 A series is simultaneously in an AP and a HP. Which of the following is true?
(a) The set of common differences of all the possible APs can have only one element.
(b) The set of common differences of all the possible APs can have more than one element, but there will be a finite value.
(c) The set of common differences of all the possible APs can have infinite elements.
(d) None of these

Q.18 Find the sum of all the numbers divisible by 6 in between 100 to 400.
(a) 12,550 (b) 12,450
(c) 11,450 (d) 11,550

FOUNDATION

Q.1 Let t_r denote the rth term of an AP. If $t_m = \frac{1}{n}$ and $t_n = \frac{1}{m}$, then t_{mn} equals:
(a) $\frac{1}{mn}$ (b) $\frac{1}{m} + \frac{1}{n}$ (c) 1 (d) 0

Q.2 Let x, y, z be three positive prime numbers. The progression in which $\sqrt{x}, \sqrt{y}, \sqrt{z}$ can be three terms (not necessarily consecutive) is:
(a) AP (b) GP
(c) HP (d) None of these

Q.3 Let $f(x) = 2x + 1$. Then, the number of real values of x for which the three unequal numbers $f(x)$, $f(2x)$, and $f(4x)$ are in a GP is:
(a) 1 (b) 2
(c) 0 (d) None of these

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Q.4 If a, b, and c are in a GP, then $a+b$, $2b$, $b+c$ are in:
(a) AP (b) GP
(c) HP (d) None of these

Q.5 If a, b, and c are in an AP, then $\frac{a}{bc}, \frac{1}{c}, \frac{2}{b}$ are in:
(a) AP (b) GP
(c) HP (d) None of these

Q.6 In the sequence 1, 2, 2, 4, 4, 4, 4, 8, 8, 8, 8, 8, 8, 8, …, where n consecutive terms have the value n, the 1025th term is:
(a) 2^9 (b) 2^{10} (c) 2^{11} (d) 2^8

Q.7 If the pth, qth, and rth terms of an AP are in a GP, then find the common ratio of the GP.
(a) $\frac{p+q}{r+q}$ (b) $\frac{r-1}{q-p}$
(c) $\frac{p-r}{p-1}$ (d) None of these

Q.8 The sum of an infinite geometric series is 4 and the sum of the cubes of the terms of the same GP is 192. The common ratio of the original geometric series is:
(a) $\frac{1}{2}$ (b) $-\frac{1}{2}$ (c) $\frac{1}{4}$ (d) $\frac{1}{6}$

Q.9 The 10th common term between the series 3 + 7 + 11 + … and 1 + 6 + 11 + … is:
(a) 191 (b) 193
(c) 211 (d) None of these

Q.10 If a, b, c, and d are four numbers such that the first three are in an AP, while the last three are in a HP, then:
(a) $bc = ad$ (b) $ac = bd$
(c) $ab = cd$ (d) None of these

Q.11 If a, b, c, and d are positive real numbers such that a, b, c, and d form an increasing arithmetic sequence and a, b, and d form a geometric sequence, then $\frac{a}{d}$ is:
(a) $\frac{1}{4}$ (b) $\frac{1}{6}$ (c) $\frac{1}{3}$ (d) $\frac{1}{2}$

Q.12 In a GP of even number of terms, the sum of all the terms is 5 times the sum of the odd terms. The common ratio of the GP is:
(a) $\frac{-4}{5}$ (b) $\frac{1}{5}$
(c) 4 (d) None of these

Q.13 If a, b, and c are in an AP, then $a+\frac{1}{bc}, b+\frac{1}{ca}, c+\frac{1}{ab}$ are in:
(a) AP (b) GP
(c) HP (d) None of these

Q.14 The ratio of HM and GM for two positive numbers is 4:5. What is the ratio of the numbers?
(a) 4:1 (b) 3:2 (c) 3:4 (d) 2:3

Q.15 The arithmetic mean of the nine numbers in the set {9, 99, 999, 9999, ..., 999999999} is a 9-digit number M, all of whose digits are distinct. The number M does not contain the digit:
(a) 2 (b) 2 (c) 0 (d) 6

Q.16 The 288th term of the series a, b, b, c, c, c, d, d, d, d, e, e, e, e, e, … is:
(a) u (b) v (c) w (d) x

Q.17 The product of n positive numbers is 1. Their sum is:
(a) A positive integer
(b) Divisible by n
(c) Equal to $n+\frac{1}{n}$
(d) Greater than or equal to n

Q.18 If t_n denotes the nth term of the series 2 + 3 + 6 + 11 + 18 + …, then t_{50} is:
(a) $49^2 - 1$ (b) 49^2
(c) $50^2 + 1$ (d) $49^2 + 2$

Q.19 If $a^x = b^y = c^z$ and x, y, z are in GP, then $\log_c b$ is equal to:
(a) $\log_b a$ (b) $\log_a b$
(c) $\frac{z}{y}$ (d) None of these

Q.20 The numbers 1, 4, and 16 can be three terms (not necessarily consecutive) of:
(a) no AP
(b) only one GP
(c) infinite number of APs
(d) more than one but finite number of GPs

Q.21 If a, b, c, and d are four positive numbers, then which of the following is not true?
(a) $\left(\frac{a}{b}+\frac{b}{c}\right)\left(\frac{c}{d}+\frac{d}{e}\right) \geq 4.\sqrt{\frac{a}{e}}$
(b) $\left(\frac{a}{b}+\frac{b}{c}\right)\left(\frac{b}{c}+\frac{d}{e}\right) \geq 4.\sqrt{\frac{a}{e}}$
(c) $\frac{a}{b}+\frac{b}{c}+\frac{c}{d}+\frac{d}{e}+\frac{e}{a}+ \geq 5$
(d) $\frac{b}{a}+\frac{c}{b}+\frac{d}{c}+\frac{e}{d}+\frac{a}{e}+ \geq \frac{1}{5}$

Q.22 Find the 45th term of the series 12, $13\frac{1}{2}$, 15.
(a) 57 (b) 77.5 (c) 74 (d) 78

Q.23 The 5th term of an AP is 15 and the 9th term is 23. Find the 14th term.
(a) 31 (b) 33 (c) 35 (d) 37

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Q.24 If 9 times the 9th term in an AP is equal to 15 times, the 15th term in the AP what is the 24th term?
(a) −1 (b) −3 (c) 0 (d) 1

Q.25 Find the sum of an AP of 14 terms whose first and the last terms are 8 and 99, respectively.
(a) 707 (b) 749 (c) 789 (d) 747

Q.26 A gets as many marks less than B as B gets less than C. If the total marks secured together by them is 270 and C got 90 marks more than A, find the marks secured by them individually.
(a) 30, 60, 180 (b) 45, 90, 135
(c) 40, 80, 150 (d) 35, 70, 165

Q.27 Find the sum of all the two-digit numbers that leave remainder 1 when divided by 5.
(a) 963 (b) 968 (c) 967 (d) 960

Q.28 Find the common ratio of GP whose first term is 3, the last is 3072 and the sum of the series is 4095.
(a) 2 (b) 3 (c) 4 (d) 6

Q.29 The least value of n, for which $1 + 4^2 + 4^4 + \ldots$ to n terms is greater than 1000, is:
(a) 5 (b) 4 (c) 3 (d) 2

Q.30 The number of terms in an AP is odd. If the sum of all the terms is 45 and the number of terms is 4 more than the middle term, find the middle term.
(a) 9 (b) 4 (c) 6 (d) 5

Q.31 If $\frac{a^{n+1}+b^{n+1}}{a^n+b^n}$ is the arithmetic mean between a and b, then n is:
(a) 0 (b) 1
(c) 1/2 (d) None of these

Q.32 In an infinite GP, every term is equal to the sum of all the terms that follow. Find the common ratio.
(a) 1/2 (b) 1/3 (c) 1/4 (d) 1/6

Q.33 The difference between the two numbers is four and the AM between them is six. The product of the numbers is:
(a) 24 (b) 12 (c) 32 (d) 48

Q.34 Which of the following statements is always true? (Statements are related to AP and the terms are having their general notations.)
(a) $t_5 + t_8 = t_1 + t_{12}$ (b) $t_8 = a + 8d$
(c) $t_1 + t_2 = t_3$ (d) Only a and b

Q.35 The number of terms common between the series $1 + 2 + 4 + 8 + \ldots$ to 100 terms and $1 + 4 + 7 + 10 + \ldots$ to 100 terms is:
(a) 6 (b) 4
(c) 5 (d) None of these

Q.36 If the geometric mean of two non-negative numbers is equal to their harmonic mean, then which of the following is necessarily true?
I. One of the numbers is zero.
II. Both the numbers are equal.
III. One of the numbers is one.
(a) I only (b) Either I or II
(c) II only (d) I or III

Q.37 Consider three numbers in an AP. Which of the following will definitely divide the sum of the cubes of all the three numbers?
(a) Middle number
(b) Difference of the first and last numbers
(c) 3
(d) (a) and (c)

Q.38 My neighbour, Mr Makrant Chatpatia, is a nice person. Last night he hosted a party for me. He told me that he would retire next year. He also added that when he joined his firm he was 19 years old and his basic salary was ₹100 and after every two years there was an increase of ₹60. What must be the basic salary of Mr Chatpatia now, if the age of retirement is 60 years?
(a) ₹1990 (b) ₹1240
(c) ₹1200 (d) ₹1300

Q.39 Given is $54 + 51 + 48 + \ldots$ till n terms = 513. How many values of 'n' are possible?
(a) 1 (b) 2
(c) 0 (d) None of these

Q.40 In a decreasing AP the sum of all its terms, except the first term, is equal to −36, the sum of all its terms, except the last term, is zero, and the difference of the tenth and the sixth term is equal to −16. What will be first term of this series?
(a) 16 (b) 20 (c) −16 (d) −20

Moderate

Q.1 A group of friends have some money that was in an increasing GP. The total money with the first and the last friend was ₹66 and the product of the money that the second friend had and that the last but one friend had was ₹128. If the total money with all of them together was ₹126, then how many friends were there?
(a) 6 (b) 5
(c) 3 (d) Cannot be determined

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Q.2 If the sum of the 10th, 20th, and 30th terms of an AP is equal to the 58th term, what is the ratio of the sum of the 10th, 20th, and 30th terms to the sum of the 5th, 10th and 15th terms; when the common difference $\neq 0$?
(a) 2:1 (b) 20:9
(c) 19:9 (d) Cannot be determined

Q.3 Corresponding terms of two arithmetic progressions (AP) are multiplied. As a result, we obtain the following sequence 1440, 1716, 1848, … What is the 8th term of this sequence?
(a) 408 (b) 4168
(c) 348 (d) 2276

Q.4 If a, b, c, and d are non-zero real numbers such that $(a^2 + b^2 + c^2)(b^2 + c^2 + d^2) \leq (ab + bc + cd)^2$, then a, b, c, and d are in:
(a) AP (b) GP
(c) HP (d) None of these

Q.5 If $\log\left(\frac{5c}{a}\right), \log\left(\frac{3b}{5c}\right)$ and $\log\left(\frac{a}{3b}\right)$ are in an AP, where a, b, and c are in a GP, then a, b, and c, are the lengths of sides of:
(a) An isosceles triangle
(b) An equilateral triangle
(c) A scalene triangle
(d) None of these

Q.6 The coefficient of x^{15} in the product $(1 - x)(1 - 2x)(1 - 2^2.x) \ldots (1 - 2^{15}.x)$ is equal to:
(a) $2^{105} - 2^{121}$ (b) $2^{121} - 2^{105}$
(c) $2^{120} - 2^{104}$ (d) None of these

Q.7 The coefficient of x^{49} in the product $(x - 1)(x - 3) \ldots (x - 99)$ is:
(a) -99^2 (b) 1
(c) -2500 (d) None of these

Q.8 The AM of two given positive numbers is 2. If the larger number is increased by 1, the GM of the numbers becomes equal to the AM of the given numbers. Then, the HM of the given numbers is:
(a) $\frac{3}{2}$ (b) $\frac{2}{3}$
(c) $\frac{1}{2}$ (d) None of these

Q.9 Let a and b be two positive numbers, where $a > b$ and $4 \times GM = 5 \times HM$ for the numbers. Then, a is:
(a) $4b$ (b) $\frac{1}{4}b$ (c) $2b$ (d) b

Q.10 The sequence P_n is in a GP with $P_2/P_4 = 1/4$ and $P_1 + P_4 = 108$. What will be the value of P_3?
(a) 42 (b) 48 (c) 44 (d) 56

Q.11 Let $t_n = n.(n!)$. Then, $\sum_{n=1}^{15} t_n$ is equal to:
(a) $15! - 1$ (b) $15! + 1$
(c) $16! - 1$ (d) None of these

Q.12 In a GP, the product of the first four terms is 4 and the second term is the reciprocal of the fourth term. The sum of the GP upto infinite terms is:
(i) 8 (ii) -8 (iii) $\frac{8}{3}$ (iv) $-\frac{8}{3}$
(a) Only i and ii (b) Only ii and iii
(c) Only i, ii, and iii (d) i, ii, iii, and iv

Q.13 Let a_n = product of the first n natural numbers. Then, for all $n \in N$,
(i) $n^n \geq a_n$ (ii) $\left(\frac{n+1}{2}\right)^n \geq n!$
(iii) $n^n \geq a_{n+1}$ (iv) None of these
(a) i and ii only (b) ii and iii only
(c) i, ii and iii only (d) None of these

Q.14 A GP consists of 1000 terms. The sum of the terms occupying the odd places is P_1 and the sum of the terms occupying the even places is P_2. Find the common ratio of this GP.
(a) P_2/P_1 (b) P_1/P_2
(c) $(P_2 - P_1)/P_1$ (d) $(P_2 + P_1)/P_2$

Q.15 The harmonic mean of the roots of the equation $(5 + \sqrt{2})x^2 - (4 + \sqrt{5})x + 8 + 2\sqrt{5} = 0$ is:
(a) 2 (b) 4 (c) 6 (d) 8

Q.16 Let A, G, and H be the AM, GM, and HM of two positive numbers a and b. The quadratic equation whose roots are A and H is:
(a) $Ax^2 - (A^2 + G^2)x + AG^2 = 0$
(b) $Ax^2 - (A^2 + H^2)x + AH^2 = 0$
(c) $Hx^2 - (H^2 + G^2)x + HG^2 = 0$
(d) Both (a) and (c)

Q.17 If the roots of $x^3 - 12x^2 + 39x - 28 = 0$ are in an AP, then find their common difference.
(a) ± 1 (b) ± 2 (c) ± 3 (d) ± 4

Q.18 In an infinite GP, each term is equal to four times the sum of all the terms that follow. Find the common ratio.
(a) 1/4 (b) 1/5 (c) 1/8 (d) 1/7

Q.19 If the terms of one geometric progression are multiplied by the corresponding terms of another geometric progression, the sequence obtained will be in:
(a) AP (b) HP
(c) GP (d) Cannot be determined

Q.20 In a network system, each person has to include four more persons under him and such a chain should continue.

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The person at any level would get ₹1 commission per person below him in his group. If a person earns ₹84, find the number of persons under him, in his group earning zero amount.
(a) 32 (b) 64 (c) 128 (d) 256

Q.21 If the nth term of AP is p and the mth term of the same AP is q, then find $(m+n)$th term of AP.
(a) $p+q$ (b) $p+q-a+d$
(c) $p+q-d-a$ (d) $p+q-d+a$

Q.22 Two sequences of numbers {1, 4, 16, 64 …} and {3, 12, 48, 192, …} are mixed as follows:
{1, 3, 4, 12, 16, 48, 64, 192, …}. One of the numbers in the mixed series is 1048576. Then, the number immediately preceding it is:
(a) 7, 86, 432 (b) 2, 62, 144
(c) 8, 14, 572 (d) 7, 86, 516

Q.23 For each positive integer n, consider the set S_n defined as follows:
$S_1 = \{1\}, S_2 = \{2, 3\}, S_3 = \{4, 5, 6\}$ …, and in general, S_{n+1} consists of $n+1$ consecutive integers the smallest of which is one more than the largest integer in S_n. Then, the sum of all the integers in S_{21} equals.
(a) 1113 (b) 5336 (c) 5082 (d) 4641

Q.24 If $a_1, a_2, a_3, a_4, \ldots, a_{24}$ are in an arithmetic progression and $a_1 + a_5 + a_{10} + a_{20} + a_{24} = 225$, then find the sum of the series $a_1 + a_2 + a_3 + \ldots + a_{22} + a_{23}$.
(a) 909 (b) 75 (c) 750 (d) 1035

Q.25 The harmonic mean of two positive real numbers is 4. Their arithmetic mean A and their geometric mean G satisfy the relation $2A + G^2 = 27$. Find the two numbers.
(a) 4 and 4 (b) 2 and 6
(c) 3 and 6 (d) 5 and 10/3

Q.26 Find the sum of the 37th bracket of the following series.
$(1) + (7 + 7^2 + 7^3) + (7^4 + 7^5 + 7^6 + 7^7 + 7^8) + (7^9 + 7^{10} + \ldots + 7^{15}) \ldots$
(a) $\frac{7^{37}}{6}(7^{37}-1)$ (b) $\frac{(7^{73}-1)}{6}$
(c) $\frac{7^{71}}{6}(7^{73}-1)$ (d) None of these

Q.27 If a, b, and c are positive integers, then find the product of $(a+b)(b+c)(c+a)$.
(a) $>8\,abc$ (b) $<8\,abc$
(c) $=8\,abc$ (d) None of these

Q.28 The sum the of 'nth' terms of two different arithmetic progression is zero. The sum of a certain number of terms (beginning with the first) of one of the progression is 386. What is the sum of the same number of terms (from the beginning) of the second progression?
(a) 386 (b) +193 (c) −193 (d) −386

Q.29 An infinite GP has the first term 'X' and sum '5', then X belongs to:
(a) $X < -10$ (b) $-10 < X < 0$
(c) $0 < X < 10$ (d) $X > 10$

Q.30 $f(x) = 2x + 1$, where $x = 1, 2, \ldots, 100$. $g(x) = 3x - 2$, where $x = 1, 2, \ldots, 100$. For how many values of x, $f(x) = g(x)$?
(a) 31 (b) 32 (c) 33 (d) 34

Q.31 Find the sum of the series:
$1.2 + 2.2 + 3.2^2 + 4.2^3 + \ldots + 100.2^{99}$
(a) $99 \times 2^{100} - 1$ (b) $99 \times 2^{101} + 1$
(c) $99 \times 2^{101} - 1$ (d) $99 \times 2^{101} + 1$

Q.32 The mid-points of the adjacent sides of a square are joined. Again the mid-points of the adjacent sides of the newly formed figure are connected and this process is repeated again and again. Calculate the sum of the areas of all such figures given that the diagonal of outermost square is $6\sqrt{2}$ cm.
(a) 35 cm^2 (b) 44 cm^2 (c) 72 cm^2 (d) 58 cm^2

Q.33 The sum of the first n terms of an AP is $n(n-1)$. Then, find the sum of the squares of these terms.
(a) $n^2(n-1)^2$ (b) $\frac{n}{6}(n-1)(2n-1)$
(c) $\frac{2}{3}n(n-1)(2n-1)$ (d) $\frac{1}{3}n(n-1)(2n+1)$

Q.34 What is the sum of the following series?
$7 + 26 + 63 + 124 + \ldots + 999$
(a) 3014 (b) 3013
(c) 3015 (d) None of these

Q.35 In ΔABC, points P_1, Q_1, and R_1 divide the lines AB, BC and AC respectively in the ratio of 2:1. In $\Delta P_1Q_1R_1$, the points P_2, Q_2 and R_2 divide the sides P_1Q_1, Q_1R_1 and P_1R_1 in the ratio of 2:1. In every such new triangle, a new triangle is generated by joining the points on the sides that divide these sides in the ratio of 2:1. Find the sum of the areas of all such triangles formed till infinity. (Area of ΔABC = 'A' sq. units)
(a) $\frac{A}{2}$ sq. units (b) $2A$ sq. units
(c) $\frac{A}{3}$ sq. units (d) $3A$ sq. units

Q.36 Find the sum of the series $-1 + 1^2 - 2 + 2^2 - 3 + 3^2 + \ldots n + n^2$.

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(a) $\frac{-n(n+1)}{3}$ (b) $\frac{n(n+1)(n-1)}{3}$

(c) $\frac{n(n-1)}{3}$ (d) None of these

Q.37 There are three numbers in an arithmetic progression. If the two larger numbers are increased by one, then the resulting numbers are prime. The product of these two primes and the smallest of the original numbers is 598. Find the sum of the three numbers.
(a) 45 (b) 29 (c) 42 (d) 36

Q.38 If three successive terms of a GP with the common ratio $r > 1$ form the sides of a triangle and $[r]$ denotes the integral part of x, then find $[r] + [-r]$.
(a) 0 (b) 1
(c) −1 (d) None of these

Q.39 In how many ways, can we select three natural numbers out of the first 10 natural numbers so that they are in a geometric progression with the common ratio greater than 1?
(a) 2 ways (b) 3 ways
(c) 4 ways (d) 5 ways

Q.40 If m times the mth term of an AP is equal to p times the pth term, find the $(m+p)$th term.
(a) 0 (b) $a^2 - b^2$
(c) $a - b$ (d) 1

ADVANCED

Q.1 If $a_1, a_2, a_3, \ldots, a_n$ $(n \geq 3)$ are in an AP, then $\left(\frac{1}{a_1a_2} + \frac{1}{a_2a_3} + \ldots + \frac{1}{a_{n-1}a_{n-2}}\right)$ will be equal to:

(a) $\frac{n^2}{a_1a_n}$ (b) $\frac{(n-1)}{a_1a_n}$

(c) $\frac{2n}{a_1a_n}$ (d) None of these

Q.2 Three distinct numbers x, y, z, form a GP in that order and the numbers $x + y, y + z, z + x$ form an AP in that order. Find the common ratio of the GP.
(a) 1 (b) −2
(c) 2 (d) Either (a) or (b)

Direction for Questions 3 and 4: Answer the questions based on the following information.

A series of positive integers have the following terms.
First term: 1; Second term: (2 + 3);
Third term: (4 + 5 + 6), … so on.

Q.3 The last number in the nth term of the series is:

(a) $\frac{(n^2 - n + 2)}{2}$ (b) $\frac{(n^2 + n)}{2}$

(c) $\frac{(2n^2 + 3)}{4n}$ (d) None of these

Q.4 The sum of the numbers of the nth term is:

(a) $n^2 + n$ (b) $\frac{n(n^2 + 1)}{2}$

(c) $\frac{(n+n)^2}{3n}$ (d) None of these

Q.5 An arithmetic series consists of $2n$ terms, and the first term equals the value of the common difference. If a new series is formed taking the 1st, 3rd, 5th, … $(2n - 1)$th term of the old series, find the ratio of the sum of the new series to that of the sum of the terms of the old series.

(a) $\frac{(n+1)}{2(2n+1)}$ (b) $\frac{n}{2n+1}$

(c) $\frac{1}{2}$ (d) Cannot be determined

Q.6 If the sum of the first $2n$ terms of the AP 2, 5, 8 … is equal to the sum of first n terms of the AP 57, 59, 61 …, then what is the value of n?
(a) 7 (b) 9 (c) 11 (d) 13

Q.7 Let the positive numbers a, b, c, and d be in AP. Then, abc, abd, acd, bcd are:
(a) Not in AP or GP or HP
(b) In AP
(c) In GP
(d) In HP

Q.8 Given that $(m + 1)$th, $(n + 1)$th, and $(r + 1)$th term of an AP are in GP and m, n, r are in HP, then find the ratio of the first term of AP to its common difference in terms of n.
(a) 2:n (b) n:2 (c) 2n:3 (d) 3:4n

Q.9 Let x be the arithmetic mean and y and z be the two geometric means between any two positive numbers, then the value of $\frac{y^3 + z^3}{xyz} = ?$

(a) xy (b) 2 (c) 0 (d) xyz

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Q.10 If rational numbers a, b, c, and d are in GP, then the roots of the equation $(a-c)^2x^2+(b-c)^2x+(b-d)^2=(a-d)^2$ are necessarily:
(a) Imaginary (b) Irrational
(c) Rational (d) Real and unequal

Q.11 Let $a = 111\ \ldots\ 1$ (55 digits),
$b = 1 + 10 + 10^2 + 10^3 + 10^4$
$c = 1 + 10^5 + 10^{10} + 10^{15} + \ \ldots\ + 10^{50}$, then:
(a) $a = b + c$ (b) $a = bc$
(c) $b = ac$ (d) $c = ab$

Q.12 The first four terms of an arithmetic sequence are p, 9, $3p - q$, and $3p + q$. What is the 2010th term of this sequence?
(a) 8047 (b) 8043 (c) 8045 (d) 8041

Q.13 If a geometric mean of two non-negative numbers is equal to their harmonic mean, then which of the following is necessarily true?
I. One of the numbers is zero.
II. Both the numbers are equal.
III. One of the numbers is one.
(a) I and III only (b) Either I or III
(c) III only (d) None of these

Q.14 The middle points of the sides of a triangle are joined forming a second triangle. Again a third triangle is formed by joining the middle points of this second triangle and this process is repeated infinitely. If the perimeter and the area of the outer triangle are P and A, respectively, find the sum of the areas of all the triangles.
(a) $4/5A$ (b) $4/3A$ (c) $3/4A$ (d) $5/4A$

Q.15 A series in which any term is the sum of the preceding two terms is called a Fibonacci series. The first two terms are given initially and together they determine the entire series. If the difference of the squares of the ninth and the eighth terms of a Fibonacci series is 715 then, what is the 12th term of that series?
(a) 157 (b) 142
(c) 144 (d) Cannot be determined

Q.16 In a sequence of terms, a_1, a_2, a_3 …, the nth term, a_n (for $n > 1$), is given by $a_n = a_{n-1} + a_{n+1}$. If $a_7 = 6$ and $a_{14} = 7$, find S_{26}/S_{13}, where S_i is the sum of all the terms from a_1 to a_i in the sequence.
(a) $\frac{1}{6}$ (b) $-\frac{1}{6}$
(c) $\frac{13}{6}$ (d) Cannot be determined

Q.17 The sum of an infinite GP is 162 and the sum of its first n terms is 160. If the inverse of its common ratio is an integer, then how many values of common ratio is/are possible, common ratio is greater than 0?
(a) 0 (b) 1 (c) 2 (d) 3

Q.18 In the above question, which of the following is the value of n?
(a) 4 (b) −2
(c) −1 (d) None of these

Q.19 The first term of an AP = the common ratio of a GP and the first term of the GP = common difference of the AP. If the sum of the first two terms of the GP is equal to the sum of the first 2 terms of the AP, then the ratio of the first term of the GP to the first term of an AP is
(a) > 1
(b) independent of first term of AP.
(c) independent of first term of GP.
(d) < 1

Direction for Questions 20 to 23: Read the passage below and solve the questions based on it.

Let there be a series 'S' with it is nth term be equal to $n(x)^n$. Also, S_n denotes the sum of the first n terms of the series S.

Q.20 What is $S_5 - S_4$ equal to?
(a) $3x^3$ (b) $4x^4$
(c) $5x^5$ (d) None of these

Additional information for Questions 21 to 23:

If the $(n + 1)$th term of S is equal to 64 and the nth term is equal to 24 then:

Q.21 What is the value of n?
(a) 2 (b) 3
(c) 4 (d) Cannot say

Q.22 What is the value of x?
(a) 2 (b) 3
(c) 4 (d) None of these

Q.23 What is the value of S_3?
(a) 20 (b) 14
(c) 34 (d) None of these

Q.24 Let S_n be defined as the sum of n terms of the following series: $S_n = 2^{66} - 2^{65} - 2^{64}$. What is the sum of S_{20}?
(a) 2^{47} (b) 2^{12}
(c) 2^{19} (d) None of these

Q.25 The odd natural numbers are arranged in a pyramidical shape given below:

1
3 5
7 9 11
13 15 17 19
… … … … …
and so on.

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What is the sum of the numbers in the nth row of this structure?

(a) $2n^2 - 1$ (b) n^3
(c) $(n-1)^2 + 1$ (d) $2n^2 - 1$

Q.26 Set X is having 2000 elements in AP with the first term = 1, and the common difference = 1. What is the maximum number of elements Set Y can have such that sum of no two elements of Y is equal to 9 or a multiple of 9?

(a) 890 (b) 891
(c) 892 (d) None of these

Q.27 In an AP of even number of terms, the sum of the odd terms is 24 while that of the even terms is 30. Find the number of terms if the last term exceeds the first by 10.5.

(a) 7 (b) 9 (c) 6 (d) 8

Q.28 Find the sum of the products of the integers, 1, 2, 3, ... n taking two at a time.

(a) $1/12\ n(n^2+1)\ (n+1)$
(b) $1/6\ n(n^2+1)\ (n+1)$
(c) $\frac{1}{12} n\ (n^2-1)\ (3n+2)$
(d) None of these

Q.29 The sum of the first 10 terms of an AP is equal to 155 and the sum of the first two terms of a GP is 9. How many values for the first term of an AP is possible if the first term of an AP is equal to the common ratio of a GP and the first term of a GP is equal to the common difference of a AP.

(a) 0 (b) 1 (c) 2 (d) Infinite

Q.30 The sum upto the n terms of the following series:

$$\frac{1}{1.3.5} + \frac{1}{3.5.7} + \frac{1}{5.7.9} \cdots$$

(a) $\frac{(n+2)(n+4)}{n(n+1)(n+3)}$ (b) $\frac{(n+2)(n+4)}{(2n+1)(n-2)}$

(c) $\frac{n(n+2)(n+4)}{3(2n+1)(n+3)}$ (d) $\frac{n(n+2)}{3(2n+1)(2n+3)}$

Direction for Questions 31 and 32: Read the passage below and solve the questions based on it.

30 Alphonso mangoes are to be distributed among Tom, Dick, and Harry. Harry's share is less than that of Dick. Further, they have agreed to divide the mangoes such that the number of mangoes with the three of them are in an arithmetic progression.

Q.31 If none of them is to receive more than 15 mangoes, in who many ways can the mangoes be distributed?

(a) 10 (b) 20 (c) 5 (d) 15

Q.32 Which of the following statements must be true?

(a) The number of mangoes Dick receives is at least 11.
(b) The number of mangoes Tom receives is at most 9.
(c) One person gets exactly 10 mangoes.
(d) More than one of the above.

Q.33 The sum of the first n terms ($n > 1$) of an AP is 153 and the common difference is 2. If the first term is an integer, then find the number of possible values of n.

(a) 2 (b) 3 (c) 4 (d) 5

Q.34 The positive numbers a, b, c, and d are in a HP and $a \neq b$, then

(a) $a + d > b + c$ is always true.
(b) $a + b > d + c$ is always true.
(c) $a + c > b + d$ is always true.
(d) None of these

Q.35 The number of ways in which three distinct numbers in an AP can be selected from 1, 2, …, 24 is:

(a) 112 (b) 124
(c) 132 (d) 146

Q.36 Numbers are placed on the vertices of a polygon with n (>4) sides in such a way that each of them equals the arithmetic mean of its neighbours. Then, it can be concluded that all the numbers are equal

(a) for every n (>4).
(b) only for even n (>4).
(c) only for odd n (>4).
(d) only for odd prime n (>4).

Q.37 In an infinite geometric progression, the nth term is equal to three times the sum of all the terms that follow it and the sum of the first two terms is 15. What is the sum of the whole series?

(a) 32 (b) 16
(c) $+\infty$ (d) None of these

Direction for Questions 38 and 39: Read the passage below and solve the questions based on it.

$N_1 + N_2 + N_3 + \ldots + N_N = P$, where N_1, N_2, N_3 … and N_N are $n(n > 1)$ consecutive natural numbers such that $N_1 < N_2 < N_3 < N_4 \ldots < N_N$.

Q.38 If $P = 100$, then how many values of N is/are possible?

(a) 0 (b) 1 (c) 2 (d) Infinite

Q.39 If $P = 100$, then what is the highest value of N?

(a) 5 (b) 10 (c) 20 (d) 8

Q.40 An employee joined a company on 01.04.2004 in the salary grade of ₹8000–500–9500–750–12,500 with a basic salary of 9000. He is due to retire on 31.03.2007. He

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contributes 10% of his basic salary to an EPF scheme. His employer contributes an equal amount. If, on retirement, he gets the full amount of his share of the EPF and 50% of the employer's share, what is the amount of EPF received by him, ignoring any interest on the deposits?

(a) 34,500 (b) 17,250
(c) 51,750 (d) 62,500

True/False

1. If we know any two terms of an AP, we can definitely find out the remaining terms of the same AP.
State whether true or false.
2. If we know any two terms of a GP, we can definitely find out the remaining terms of the same GP.
State whether true or false.
3. If we know any two terms of a GP, we can definitely find out the common ratio of the same GP.
State whether true or false.
4. If we know any two terms of an AP, we can definitely find out the common difference of the same AP.
State whether true or false.
5. Sum of all the odd terms (viz., first term, third term, etc.) and sum of all the even terms (viz., 2nd term, 4th term, etc.) of a GP with 'n' terms are given. We can find out the common ratio of this GP with this information.
State whether true or false.
6. If a series is in both AP and GP, then sum of first 10 terms will be equal to sum of any other set of consecutive 10 terms.
State whether true or false.

Answers

Warm Up

1. (b)	**2.** (d)	**3.** (a)	**4.** (b)	**5.** (b)	**6.** (b)	**7.** (c)	**8.** (b)	**9.** (b)	**10.** (d)
11. (c)	**12.** (b)	**13.** (b)	**14.** (b)	**15.** (b)	**16.** (a)	**17.** (a)	**18.** (b)		

Foundation

1 (c)	**2.** (d)	**3.** (a)	**4.** (c)	**5.** (d)	**6.** (b)	**7.** (b)	**8.** (b)	**9.** (a)	**10.** (a)
11. (a)	**12.** (c)	**13.** (a)	**14.** (a)	**15.** (c)	**16.** (c)	**17.** (d)	**18.** (d)	**19.** (a)	**20.** (c)
21. (d)	**22.** (d)	**23.** (b)	**24.** (c)	**25.** (b)	**26.** (b)	**27.** (a)	**28.** (c)	**29.** (b)	**30.** (d)
31. (a)	**32.** (a)	**33.** (c)	**34.** (a)	**35.** (c)	**36.** (b)	**37.** (d)	**38.** (d)	**39.** (b)	**40.** (a)

Moderate

1. (a)	**2.** (c)	**3.** (c)	**4.** (b)	**5.** (d)	**6.** (a)	**7.** (c)	**8.** (a)	**9.** (a)	**10.** (b)
11. (c)	**12.** (d)	**13.** (a)	**14.** (a)	**15.** (b)	**16.** (d)	**17.** (c)	**18.** (b)	**19.** (c)	**20.** (b)
21. (b)	**22.** (a)	**23.** (d)	**24.** (d)	**25.** (c)	**26.** (d)	**27.** (a)	**28.** (d)	**29.** (c)	**30.** (c)
31. (d)	**32.** (c)	**33.** (d)	**34.** (c)	**35.** (b)	**36.** (d)	**37.** (c)	**38.** (c)	**39.** (b)	**40.** (a)

Advanced

1. (b)	**2.** (d)	**3.** (b)	**4.** (b)	**5.** (b)	**6.** (c)	**7.** (d)	**8.** (b)	**9.** (b)	**10.** (c)
11. (b)	**12.** (d)	**13.** (d)	**14.** (b)	**15.** (c)	**16.** (c)	**17.** (c)	**18.** (a)	**19.** (c)	**20.** (c)

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21. (b)	**22.** (a)	**23.** (c)	**24.** (a)	**25.** (b)	**26.** (b)	**27.** (d)	**28.** (c)	**29.** (c)	**30.** (d)
31. (d)	**32.** (c)	**33.** (d)	**34.** (d)	**35.** (c)	**36.** (a)	**37.** (b)	**38.** (c)	**39.** (d)	**40.** (c)

True/False

1. True **2.** True **3.** True **4.** True **5.** False **6.** True

Hints and Solutions

Warm Up

1. Method 1 Using Summation Formula

Sum of 15 terms $= \sum_{n-1}^{15} 4n + \sum_{n-1}^{15} 1 = 4[1 + 2 + 3 + 4 + 5 \ldots + 15] + 15 = 4 \times 120 + 15 = 495$

Method 2 Using Summation of AP Formula

Terms are: 5, 9, 13, …, 61.

Hence, sum of AP

$$= \frac{\text{First term} + \text{Last term}}{2} \times \text{Number of terms}$$

$$= \frac{5+61}{2} \times 15 = 33 \times 15 = 495$$

2. It is GP whose common ratio $= \frac{3}{2}$

Then, sum of first 7 terms $= a\left[\frac{r^n - 1}{r-1}\right] = \frac{1}{3}\left[\frac{(3/2)^7 - 1}{3/2 - 1}\right]$

$$= \frac{2059}{192}$$

3. nth term of a GP $= ar^{n-1} = 448$

$7r^{n-1} = 448$, or, $r^{n-1} = 64$

Sum of a GP $= a\left[\frac{r^n - 1}{r-1}\right] = 889$

$$7\left[\frac{64r - 1}{r-1}\right] = 889$$

$64r - 1 = 127r - 127 \rightarrow r = 2$

4. Sum of an AP $= \frac{n}{2}[2a + (n-1)d]$

$600 = \frac{15}{2}[2a + 14 \times 5]$, or, $2a + 70 = 80$. So, $a = 5$

5. Let the first installment be a and common difference be d.

Then, $3600 = \frac{40}{2}[2a + 39d]$

$2a + 39d = 180$ (i)

And $\frac{3600}{3} = \frac{10}{2}[2(a + 30d) + 9d]$

$1200 = 5[2a + 69d]$

$2a + 69d = 120$ (ii)

From equations (i) and (ii)

$a = 51$ and $d = 2$

6. $T_n = a + (n-1)d$

Since $T_{11} + T_{12} + T_{13} = 144$ and $T_{21} + T_{22} + T_{23} = 264$

$\therefore a + 10d + a + 11d + a + 12d = 144$ $\quad a + 20d + a + 21d + a + 22d = 264$

$3a + 33d = 144$ $\quad 3a + 63d = 264$

$a + 11d = 48$ (i) $\quad a + 21d = 88$ (ii)

On solving (i) and (ii)

We get $a = 4$ and $d = 4$

$T_{16} = a + 15d$

$4 + 15 \times 4 = 64$.

7. Since $A + B + C = 25$ (i)

Since 2, A, B are in AP and B, C, 18 are in GP.

$\therefore 2A = B + 2$ (ii)

$\therefore C^2 = 18B$ (iii)

On solving above equations, we get

$A = 5, B = 8, C = 12$

8. If $S_p = S_q$, then $[S_{p+q} = 0]$.

You may use the result of this question as a standard result.

9. Since, three numbers are in GP, option (b) satisfies it.

10. $a + ar = 5$

$S_1 = 3(S_2 + S_3 + \ldots S_\infty)$

Since $S_\infty = \frac{a}{1-r}$

$\therefore S_1 = \frac{3 \times ar}{1-r}$, i.e., $a = \frac{3ar}{1-r}$

$r = \frac{1}{4}$. On solving, we get $a = 4$

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11. Since $S_\infty = \frac{a}{1-r}$

$x = \frac{1}{1-a}$; $y = \frac{1}{1-b}$

$\therefore a = \frac{x-1}{x}, b = \frac{y-1}{y}$

$S = 1 + \frac{1}{ab} + \frac{1}{a^1b^2} + \ldots \infty$

$S = \frac{1}{1-ab}$

Substitute the values of a and b

We get $S = \frac{xy}{x+y-1}$

12. $T_n = \frac{1}{n\text{th term of AP}}$

$T_n = \frac{1}{a+(n-1)d}$

Since $T_{n+1} = 2\ T_{3n+1}$

$\therefore \frac{1}{a+nd} = \frac{2}{a+3nd}$

$a - nd = 0 \rightarrow a = nd$

$\frac{T_1}{T_{n+1}} = \frac{a+nd}{a} = \frac{2a}{a} \rightarrow$ '2'

13. Since $S_7 = 70$ $\{S_n = \frac{n}{2}[2a + (n-1)\,d]\}$

Given that $\frac{7}{2}(2a + 6d) = 70$, or, $2a + 6d = 20$, or, $a + 3d = 10$. Hence, $T_4 = 10$

14. $T_m = n$, $T_n = m$

$\frac{1}{a+(m-1)d} = n$, $\frac{1}{a+(n-1)d} = m$

$a + (m-1)d = \frac{1}{n}$ … (i), $a + (n-1)d = \frac{1}{m}$ (ii)

On solving (i) and (ii)

We get $a = 1/mn$, $d = \frac{1}{mn}$, $\left(T_{m+n} = \frac{mn}{m+n}\right)$

15. HM of x and $y = \frac{2xy}{x+y}$

$\frac{x^{n+1} + y^{n+1}}{x^n + y^n} = \frac{2xy}{x+y}$

Substituting $n = -1$ gives us the result.

16. If a series is simultaneously in an AP and a HP, then it means that every term of this series will be equal. Hence, option (a) is the answer.

17. Since every term of this series will be equal.
Then, the set of common differences of all the possible APs can have only one element.

18. First number = 102 and last number = 396. There will be 50 terms in this series.

Then, sum $= \frac{n}{2}$ (First term + Last term) $= \frac{50}{2}[102 + 396]$

$= 12450$

FOUNDATION

1. Given that $t_m = \frac{1}{n}$, $t_n = \frac{1}{m}$

$t_m = a + (m-1)\,d = \frac{1}{n}$ (i)

$t_n = a + (n-1)\,d = \frac{1}{m}$ (ii)

Solving (i) and (ii), we get $(m-n)\,d = \frac{1}{n} - \frac{1}{m}$

$d = \frac{1}{mn}$, $a = \frac{1}{mn}$

$t_{mn} = a + (mn-1)\,d = \frac{1}{mn} + \frac{(mn-1)}{mn} = 1$

2. Assume the prime numbers to be 2, 3, 5 or 11, 23, 29.
Hence, option (d) is the answer.

3. $f(x) = 2x + 1, f(2x) = 4x + 1, f(4x) = 8x + 1$
Since, $f(x)$, $f(2x)$, and $f(4x)$ are in GP.
Therefore, $(4x + 1)^2 = (8x + 1)\,(2x + 1)$
On solving, we get $x = 1$.

4. Since a, b, c are in GP, assume that a, b, c are 2, 4, 8, respectively.

Then, $a + b = 6$, $2b = 8$, $b + c = 12$

Thus, 6, 8, and 12 are in HP.

5. Since a, b, c are in AP

Consider a, b, c as 2, 4, 6.

Then, $\frac{a}{bc}, \frac{1}{c}, \frac{2}{b}$ will be $\frac{1}{12}, \frac{1}{6}, \frac{1}{4}$.

6. 1, 2, 2, 4, 4, 4, 4, 8, 8, 8, 8, 8, 8, 8, 8

First term = 1, second term = 2, fourth term = 4, eighth term = 8

Therefore, terms are in GP.

$\therefore S_n = \frac{a(r^n - 1)}{r-1} = \frac{1(2^n - 1)}{2-1}$

For $n = 10$ $S_{10} = 1023$

Therefore, 1025th term will be 2^{10}.

7. Let us consider a AP.

2, 4, 6, 8, 10 …

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Let pth term be 3 = 2, that is, $t_p = 2$

qth term be = 4, $t_q = 4$

rth term be = 8, $t_r = 8$

Therefore, $p = 1$, $q = 2$, $r = 4$

Now, check the options.

Since none of the options satisfy, hence, option (d) is the answer.

8. $S_\infty = \frac{a}{1-r}$

$\frac{a}{1-r} = 4$, $\frac{a^3}{1-r^3} = 192$

$\frac{a^3}{(1-r)^3} = 64$, $\frac{a^3}{1-r^3} = 192$

$\frac{(1-r)^3}{1-r^3} = 3$

On solving, we get $= r = -\frac{1}{2}$

9. $3 + 7 + 11 + \ldots$

$1 + 6 + 11 + \ldots$

Common series will be

11, 31, 51

$a = 11$, $d = 20$

$T_{10} = 11 + 19 \times 20 = 191$

10. Since abc are in AP and bcd are in HP.

Consider a b c and d as 4, 6, 8, 12

Therefore, $bc = ad$

11. **Method 1**

We can let $a = 1$, $b = 2$, $c = 3$, and $d = 4$. $\frac{a}{d} = \left(\frac{1}{4}\right)$

Method 2

As a, b, d is a geometric sequence, let $b = ka$ and $d = k^2a$ for some $k > 0$.

Now, a, b, c, d is an arithmetic sequence. Its difference is $b - a = (k - 1)a$. Therefore, $d = a + 3(k - 1)a = (3k - 2)a$.

Comparing the two expressions for d we get $k^2 = 3k - 2$. The positive solution is $k = 2$, and $\frac{a}{b} = \frac{a}{k^2a} = \frac{1}{k^2} = \left(\frac{1}{4}\right)$.

Hence, option (a) is the answer.

Method 3

Letting n be the common difference of the arithmetic progression, we have $b = a + n$, $c = a + 2n$, $d = a + 3n$.

We are given that $\frac{b}{a} = \frac{d}{b}$, or $\frac{a+n}{a} = \frac{a+3n}{a+n}$

Crossmultiplying, we get

$a^2 + 2an + n^2 = a^2 + 3an$

Or, $n^2 = an$

Or, $n = a$

So, $\frac{a}{d} = \frac{a}{a+3n} = \frac{a}{4a} = \left(\frac{1}{4}\right)$. Hence, option (a) is the answer.

12. Consider a GP.

a, ar, ar^2, ar^3 …

$S_n = \frac{a(r^n - 1)}{r - 1}$

Sum of odd term will be $= \frac{a(r^2)^{m/7} - 1}{r^2 - 1} = \frac{a(r^n - 1)}{r^2 - 1}$

Since $S_n = 5 \times$ Sum of odd terms

$\frac{a(r^n - 1)}{r - 1} = 5 \times \frac{a(r^n - 1)}{(r^2 - 1)}$

On solving, we get $r = 4$.

13. Let a, b, and c be 1, 2, 3

$a + \frac{1}{bc} = 1 + \frac{1}{2 \times 3} = \frac{7}{6}$

$b + \frac{1}{ca} = 2 + \frac{1}{3} = \frac{7}{3}$

$c + \frac{1}{ab} = 3 + \frac{1}{2} = \frac{7}{2}$

14. $\text{HM} = \frac{2ab}{a+b}$, $\text{GM} = \sqrt{ab}$

Let two numbers be $4x$ and x

$\frac{HM}{GM} = \frac{4}{5}$

$\frac{\text{HM}}{\text{GM}} = \frac{4}{5}$

$\frac{2ab}{(a+b)} \times \frac{1}{\sqrt{ab}} = \frac{4}{5}$, $\frac{2\sqrt{ab}}{a+b} = \frac{4}{5}$

Therefore, $a{:}b = 4{:}1$

15. We wish to find $\frac{9 + 99 + \ldots + 999999999}{9}$, or $\frac{9(1 + 11 + 111 + \ldots + 111111111)}{9} = 123456789$

This does not have the digit 0.

Hence, option (c) is the answer.

16. a, b, b, c, c, c, d, d, d, d

$1a$, $2b$, $3c$, $4d$, $5e$ …

$T_n = \frac{n(n+1)}{2}$

$T_{23} = 276$

Therefore, 288th term of the series will be 23rd alphabet.

17. **Method 1**

AM ≥ GM

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In this case, given that GM = 1. Hence, AM ≥ 1.

Now, AM = $\text{AM} = \frac{\text{Sum of numbers}}{n}$

Hence, $\text{AM} = \frac{\text{Sum of numbers}}{n} \geq 1$

Hence, sum of numbers ≥ n

Method 2

Since the product of n positive number is 1.

i.e., $\frac{1}{2}$, 1, 2 = 1, where $n = 3$

Sum will be $\frac{7}{2}$, i.e., greater than 'n'

If we consider 1, 1, 1,

Product is = 1

$n = 3$, Sum = 3

i.e., equal to 'n'

Therefore, their sum is greater than or equal to 'n'.

18. t_n – 2, 3, 6, 11, 18

$T_n = (n-1)^2 + 2$

For $n = 1$, $T_1 = 2$

$n = 2$, $T_2 = 3$

Therefore, $T_{50} = 49^2 + 2$

19. $a^x = b^y = c^z$

$x \log a = y \log b = z \log c$ and $\frac{\log b}{\log c} = \frac{z}{y}$

Since, x, y, z are in GP, $y^2 = xz$ or $\frac{y}{x} = \frac{z}{y}$

$\frac{\log a}{\log b} = \frac{y}{x} = \frac{z}{y} = \frac{\log b}{\log c}$

Therefore, $\log_b a = \log_c b$

20. The numbers 1, 4, and 16 can be a set of 'n' natural numbers or can be a set of integers which are in AP.

Thus, infinite number of APs.

21. Take any four positive numbers and check the options.

22. 12, 13.5, 15 are in AP.

a = 12, $d = 1.5$. Thus, $T_{45} = 12 + 44 \times 1.5 = 78$

23. $T_5 = 15$, $T_9 = 23$

$a + 4d = 15$, $a + 8d = 23$

$d = 2$, $a = 7$

$T_{14} = 7 + 13 \times 2 = T_{14} = 33$

24. 9th term = $a + 8d$ and 15th term = $a + 14d$

Given that, $9(a + 8d) = 15(a + 14d)$

$\rightarrow 90 + 72d = 15a + 210d$

$\rightarrow 6a + 138d = 0 \rightarrow a + 23d = 0$

This is nothing but t_{24} (24th term) of the same AP.

Hence, option (c) is the answer.

25. $a = 8$, $l = 99$

$S_n = \frac{n}{2}(a + l) = \frac{14}{2}(8 + 99) = 7 \times 107 = 749$

26. $A + B + C = 270$

$B - A = C - B$

$2B = A + C$, and $C = A + 90$

On solving, we get,

$A = 45$, $B = 90$, $C = 135$

27. Series will be

11, 16, 21, 26, …, 96

$n = 18$

$S = \frac{18}{2}(11 + 96) = 963$

28. $a = 3$, last term = 3072,

$S_n = 4095$

$T_n = ar^{n-1}$

$ar^{n-1} = 3072$

$S_n = \frac{a(r^n - 1)}{r - 1} = 4095$

On solving, we get $r = 4$

29. $1 + 4^2 + 4^4 + \ldots$ n term

Here, $a = 1$, $r = 4^2$

1 + 16 + 256 + 4096 > 1000

Therefore, $n = 4$.

30. $S_n = n \times$ middle term

Middle term = $(n + 4)$

Since $S_n = 45$

Therefore, $n \times$ middle term = 45

Hence, $n(n + 4) = 45$

Therefore, $[n = 5]$

31. $\text{AM} = \frac{a+b}{2}$

Substitute $n = 0$

In $\frac{a^{n+1} + b^{n+1}}{a^n + b^n} = \frac{a+b}{2}$

32. Since $a = ar + ar^2 + \ldots\infty$, $[r < 1]$

$S_\infty = \frac{a}{1-r}$

$a = \frac{ar}{1-r}$. Hence, $r = \frac{1}{2}$

33. Since $a - b = 4$ and $\frac{a+b}{2} = 6$. Thus, $a = 8$, $b = 4$

Hence, $ab = 32$

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34. $T_5 = a + 4d$, $T_8 = a + 7d$

$T_1 = a$, $T_{12} = a + 12d$

Therefore, $T_5 + T_8 = T_1 + T_{12}$

35. $1 + 2 + 4 + 8 + 16 + 32 + 64 + 128 + 256 + \ldots$

$1 + 4 + 7 + 10 + \ldots$

The second series is in AP. $T_{100} = 1 + 99 \times 3 = 298$

Therefore, common terms will be 1, 4, 16, 64, 256, i.e., 5 terms.

36. For GM = HM, both the numbers should be equal.

37. Take any three numbers that are in AP and then check the options.

38. He will get his last increment at the age of 59 years; there will not be any increment at the 60th year of his age.

Difference in age = 59 years − 19 years = 40 years, and therefore, he will get 20 increments of ₹60 (for every 2 years, there is an increment of ₹60).

So, total increment = ₹60 × 20 = ₹1200

Given that initial salary = ₹100

Hence, salary at the time of retirement = ₹1200 + ₹100 = ₹1300

Hence, option (d) is the answer.

39. $S_n = \frac{n}{2}[2a + (n-1)d]$

Here, $a = 54$, $d = -3$

Therefore, $\frac{n}{2}[2 \times 54 + (n-1) \times -3] = 513$

$[108 - 3n + 3] = 513$

It is a quadratic equation, and so, two values of n will satisfy it.

40. $S_n = \frac{(a+l)}{2} \times n$

Therefore, $(a + l - d) = -36$

$(a + d + l)\frac{n}{2} = 0$

And $T_{10} - T_6 = -16$, or, $4d = -16$, or, $d = -14$

Therefore, on solving above equations, we get $a = 16$

Moderate

1. The sum of money with the first and the last friend = 66. This can be used as a hint. Let us assume the first friend was having ₹2 and the last friend was having ₹64. So, the money can be in the sequence 2, 4, 8, 16, 32, 64. It satisfies the given conditions.

Alternatively, this can be done by using the formula for t_n of GP also.

2. Sum of the 10th, 20th, 30th terms of an AP is equal to the 58th term. Hence, $3a + 57d = a + 57d$

So, $a = 0$ and d can be anything not equal to zero.

Now, we have to find out the ratio $3a + 57d$ and $3a + 27d$, or, we have to find out the ratio $57d$:$27d = 19$:9.

3. T_n for any AP $= a + (n-1)d$ which is a linear equation.

When corresponding terms of the two APs are multiplied, the series we obtain will be a quadratic equation.

$t_n = an^2 + bn + c$ [this expression will give the terms of the sequence obtained by multiplying the corresponding terms of two APs]. Putting $n = 0$, 1, and 2 will give first, second, and third term of this sequence.

$T_1 = 1440 = a + b + c$ [Put $n = 1$]

$T_2 = 1716 = 4a + 2b + c$ [Put $n = 2$]

$T_3 = 1848 = 9a + 3b + c$ [Put $n = 3$]

Solving the above equations, we obtain $a = -72$, $b = 348$ and $c = 1440$

So, $t_n = -72n^2 + 348n + 1440$

So, $t_8 = -72 \times 7^2 + 348 \times 7 + 1440 = 348$

Hence, option (c) is the answer.

4. Let us consider a, b, c, and d as 1, 2, 3, 4.

Now, $(a^2 + b^2 + c^2)(b^2 + c^2 + d^2) \leq (ab + bc + cd)^2$

$14 \times 2a > 20^2$, Hence, not AP.

Again consider a, b, c, and d as 1, 2, 4, and 8

$(a^2 + b^2 + c^2)(b^2 + c^2 + d^2) \leq (ab + bc + cd)^2$

$1764 = 1764$

Hence, GP

Also, consider a, b, c, and d as HP.

Therefore, option (b) is the answer.

5. For a, b, c to be the length of the sides of the triangle, it is AM should not be 0.

6. $(1 - x)(1 - 2x)(1 - 4x) \ldots (1 - 2^{15}x)$

Let us consider $(1 - x)(1 - 2x) = 1 - 3x + 2x^2$

Coefficient of x is − 3.

i.e., $= 2^0 - 2^2$

For $(1 - x)(1 - 2x)(1 - 4x)$

Coefficient of x^2 is 14.

i.e., $= 2^4 - 2^1 \rightarrow -(2^1 - 2^4)$

Therefore, coefficient of x is $2^0 - 2^2$

Coefficient of x^2 is $-(2^1 - 2^4)$.

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Coefficient of x^3 is $(2^3 - 2^7)$

Coefficient of $x^4 = -(2^5 - 2^{11})$

.

.

.

Coefficient of $x^{15} = 2^{105} - 2^{121}$

Hence, option (a) is the answer.

7. $(x - 1)(x - 3) \ldots (x - 99)$

Coefficient of x^{49} will be sum of

$-1, -3, -5 \ldots -99 = -2500$

Hence, option (c) is the answer.

8. Take two numbers such that their AM is 2 like 1 and 3. And go according to the conditions given in the question.

9. Using option (a), let $a = 2$, $b = 8$ and solve. If it does not satisfy, use other options.

10. Let us consider a GP = $a, ar, ar^2, ar^3 \ldots$

$P_1 = a, P_2 = ar, P_3 = ar^2, P_4 = ar^3$

$\frac{P_2}{P_4} = \frac{1}{4} \rightarrow \frac{1}{r^2} = \frac{1}{4}$, so, $r = 2$

$P_1 + P_4 = 108, \rightarrow a(1 + r^3) = 108$

$a \times 9 = 108$, so, $a = 12$

$P_3 = ar^2 = 48$

Hence, option (b) is the answer

11. Solve through options.

12. Let us consider a GP

a, ar, ar^2, ar^3

$a \times ar \times ar^2 \times ar^3 = 4$

or, $a^4 r^6 = 4$, Hence, $a^2 r^3 = \pm 2$

Also, $ar = \frac{1}{ar^3}$

$a^2 r^4 = 1$, hence, $ar^2 = \pm 1$

On solving, we get, $a = 4, -4, r = \frac{1}{2}, -\frac{1}{2}$

Now, when $a = 4$, $r =$ when $a = 4$, $r = -\frac{1}{2}$

$S_\infty = 8$ $\qquad$ $S_\infty = \frac{8}{3}$

When $a = -4$, $r =$ when $a = -4$, $r = -\frac{1}{2}$

$S_\infty = -8$ $\qquad$ $S_\infty = -\frac{8}{3}$

Hence, option (d) is the answer.

13. $a_1 = 1, a_2 = 2, a_3 = 6, a_n = n!$

Now, (i) $n^n \geq a_n$

$2^2 \geq a_2$, $4 \geq 2$. Hence, true

(ii) $\left(\frac{n+1}{2}\right)^n \geq n!$

$\left(\frac{3+1}{2}\right)^3 \geq 3!$

$2^3 \geq 3! \rightarrow 8 \geq 6$. Hence, true

(iii) $n^n \geq a_{n+1}$

$2^2 \geq a_3$

$4 < 6$ not true

Hence, option (a) is the answer.

14. Take any small sample of 4 terms of any GP viz. 1, 2, 4, and 8. Its common ratio = 2 = (8 + 2)/(1 + 4).

And this will be true for any number of terms as long as we are taking total number of terms as even (in this question, it is given as 1000). However, if the total number of terms is given as odd, we cannot find the common ratio in this way, because the number of odd terms ≠ number of even terms.

15. We can easily find out the values: $2ab$ and $a + b$, where a and b are the roots of the equation given.

16. Solve through options.

17. Factorize the equation and we get $(x - 1)(x - 4)(x - 7)$

18. $S_\infty = \frac{a}{1-r}$

Since $a = 4\ (ar + ar^2 + ar^3 \ldots \infty)$

i.e., $a = a = 4\frac{ar}{1-r}$

So, $r = \frac{1}{5}$

Hence, option (b) is the answer.

19. Take two series and solve.

20. Make the three and solve though options.

21. $T_n = p = a + (n - 1)d$

$T_m = q = a + (m - 1)d$

The two and solve through the given options.

22. 1048576 is divisible by 4.

So, previous number is divisible by 3.

23. Sum = $n(n^2 + 1)/2$, where n is the number of terms in S.

24. $a_1 + a_5 + a_{10} + a_{20} + a_{24} = 225$

$5a + 55d = 225$

$a + 11d = 45$, or, $a_1 + a_2 + \ldots a_{23} = \frac{23}{2}[2a + 22d]$

$= \frac{23}{2} \times 2(a + 11d) = 23 \times 45 = 1035$

Hence, option (d) is the answer.

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25. Solve through options.

26. Solve through options.

27. Put $a = 1$, $b = 2$, $c = 3$ and check through options.

28. Assume the first AP as 1, 1, 1, … or 2, 2, 2, …
and the second AP as –1, –1, –1, or –2, –2, –2

29. $S_\infty = \frac{a}{1-r}$ as $(r < 1)$

Here, $\frac{x}{1-r} = 5$

$x = 5\ (1 - r)$

$\frac{x}{5} = 1 - r$

Or, $r = 1 - \frac{x}{5}$

x cannot be negative as r should be less than 1 and $x = 1$ satisfy the condition.

Hence, option (c) is the answer.

30. Get the terms of sets $f(x)$ and $g(x)$ and check how many terms are common.

31. Sum $= (n - 1)2^n + 1$

32. The side of the outermost square is 6 cm. Area of the square one in GP with first term 36 and common ratio = 1/2

33. Let us consider a series

1, 2, 3 $\qquad S_n = n\ (n - 1)$

$1 \times 2 \times 3 = 6 \quad = 3 \times 2 = 6$

Now, $1^2 + 2^2 + 3^2 = 1 + 4 + 9 = 14$

Check the options.

Hence, option (d) is the answer.

34. $7 + 26 + 63 + 124 + \ldots + 999$

$1^3 - 1, 2^3 - 1, 3^3 - 1, 4^3 - 1, \ldots 10^3 - 1$

$T_n = n^3 - 1$

Sum $= \sum_{n=1}^{10} T_n = \sum_{n=1}^{10} n^3 - 1,$

We know that $\left[1^3 + 2^3 + 3^3 \ldots + n^3 = \left(\frac{n(n+1)^2}{2}\right)\right]$

So, sum $= \sum_{n=1}^{10} n^3 - 1 = \left(\frac{n(n+1)^2}{2}\right) - n = \left(\frac{10 \times 11}{2}\right)^2 - 10$

$= 55^2 - 10 = 3025 - 10 = 3015$

36. Break the series in two sequence 1, 2, 3, 4,…n and 1^2, 2^2, 3^2, …, n^2 and solve.

37. Factorize 598 as the product of two prime numbers and an even number.

$598 = 2 \times 13 \times 23$

38. Take any value of $r > 1$ and then solve.

39. Make pair of three numbers. Only sequence with 2 and 3 as the common ratio is possible.

40. Take any series for which the given condition is true.

One such series can be: 4, 3, 2, 1, 0, …

ADVANCED

1. Since a_1, a_2, a_3 are in AP, so $a_2 - a_1 = a_3 - a_2 = d$

$$\left[\frac{1}{a_1 a_2} + \frac{1}{a_2 a_3} + \ldots \frac{1}{a_{n-1} a_{n-2}}\right]$$

$$= \frac{1}{d}\left[\frac{1}{a_1} - \frac{1}{a_2} + \frac{1}{a_2} - \frac{1}{a_3} + \ldots\right]$$

2. Since x, y, z are in GP, so, $y/x = z/y$ and also $x + y$, $y + z$, $z + x$ are in AP. So, $2(y + z) = 2x + y + z$, i.e., $z = 2x - y$. So, $y/x\ (2x - y)/y \Rightarrow y/x = (2x/y) - 1$

Since y/x is the ratio, assume $y/x = r$ and solve the equation $r^2 + r - 2 = 0$

3. Go through the options.

4. Go through the options.

5. The series consist of $2n$ terms,

First term $= a$, Common difference $= a$, No of terms $= 2n$

Sum of all terms $= \frac{2n}{2}[2a + (2n - 1)a]$ (i)

for the new series taking first, third, fifth, $(2n - 1)$th term of old series.

First term $= a$, common difference $= 2a$, number of terms $= n$

Sum of all terms $= \frac{n}{2}[2a + (2n - 1)2a]$ (ii)

Dividing (i) by (ii) gives the required ratio.

6. The equation can be written as follows:

$$\frac{2n}{2}[4 + (2n - 1)3] = \frac{n}{2}[114 + (n - 1)2]$$

Solving this equation gives us the value of n.

7. Divide each term by $abcd$ will give $1/d$, $1/c$, $1/b$, and $1/a$, respectively, which are in HP (since a, b, c, d are in AP).

Alternatively, assume values of a, b, c, and d such that they are in AP.

8. Since the $(m + 1)$th, $(n + 1)$th, and $(r + 1)$th term of an AP are in GP. So,

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$(a + nd)^2 = (a + md)(a + rd)$ (i)

[assume a and d as the first term and common difference of an AP]. Also, m, n, r are in HP so,

$$\frac{2n}{2} = \frac{m+r}{mr} \quad \text{(ii)}$$

By solving the equation, you will get,

$\frac{a}{d} = \frac{n^2 - mr}{m + r - 2r}$, put the $m + r = 2\ mr/n$ from equation (iii)

You will get $a/d = -n/2 \approx n{:}2$ [–ve sign indicates that either common difference or first term is –ve].

9. Assume any two positive numbers a, b. So, according to the question a, x, b are in AP.

So, $x = \frac{a+b}{2}$, also, a, y, z, b are in GP

So, $y^2 = az$, i.e., $y^3 = ayz$ and $z^3 = byz$

Now, solve $\frac{y^3 + z^3}{xyz}$.

10. In a quadratic equation, $ax^2 + bx + c = 0$. If $a + b + c = 0$, then roots will be rational. Use this property. Also a, b, c, d are in GP, i.e., $b/a = c/b = d/c$.

11. Since $a = 1111 \ldots 1$ (55 digits)

$b = 1 + 10 + 10^2 + 10^3 + 10^4 = \frac{1(10^5 - 1)}{10 - 1} = \frac{10^5 - 1}{9}$ and

$c = 1 + 10^5 + 10^{10} + 10^{15} + \ldots 10^{50} = 1\ \frac{\left[(10^5)^{11} - 1\right]}{10^5 - 1}$

$bc = (10^5 - 1)/9 \times (10^{55} - 1)/(10^5 - 1) = (9999\ldots$ 55 digits$)/9 = a$

12. $3p - q$ and $3p + q$ are consecutive terms, and so, the common difference is $(3p + q) - (3p - q) = 2q$.

Or, $p + 2q = 9$, or, $9 + 2q = 3p - q$

Hence, $q = 2$ and $p = 5$

The common difference is 4. The first term is 5 and the 2010th term is $5 + 4(2009) = 8041$.

Hence, option (d) is the answer.

13. Assuming two non-negative a and b, then according to the question $\sqrt{ab} = \frac{2ab}{a+b}$

It is true only if $a = b$.

14. Area of the triangle formed by joining the middle point of the triangle will be 1/4th of the original triangle, and so, the series will be:

A + 1/4 A + 1/16 A + 1/64 A + … ∞.

15. The series is like 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144 …
The difference is 715 and the 12th term is 144.

17. $a/(1 - r) = 162$ and $a(1 - r^n)/1 - r = 160$
$\Rightarrow 1 - r^n = 160/162 \Rightarrow r^n = 1/81$
Hence, there will be only two values of r, i.e., 2 and 4.

18. See the hints of the above question.

19. $(\alpha - \beta)\ (\alpha)\ (\alpha - \beta)$ AP
(a, ar, ar^2) GP
$\alpha - \beta = r + a = \beta$
$a + ar = 2\alpha - \beta$
$a + ar = 2\alpha - 2\beta + \beta$
$a(1 + r) = 2(r + a) + r + a$
$a(1 + r) = 3(r + a)$
$a + r = a(1 + r)/3$
$\frac{\alpha}{\alpha - \beta} = ?$

$a/(a + r) = (a.3)/a(1 + r)$
$= 3/1 + r$
This is independent of the first term of GP.

20. $S_5 = x + 2x^2 + 3x^3 + 4x^4 + 5x^5$ and $S_4 = x + 2x^2 + 3x^3 + 4x^4$

21. Series is $1x, 2x^2, 3x^3, 4x^4$
Since options are 2, 3, 4, check the options.
Let us consider $3x^3 = 24$, or, $x^3 = 8 \Rightarrow x = 2$
Now, check $4x^4 = 4 \times 2^4 = 64$
Therefore, $(x = 2)$, $(n = 3)$

23. $S_3 = 1x + 2x^2 + 3x^3 = 1 \times 2 + 2 \times 4 + 3 \times 8 = 34$
Hence, option (c) is the answer.

25.

1
3 5
7 9 11
13 15 17 19

Check the options.
For $n = 2$, second row is obtained. Sum is 8.
For $n = 3$, third row is obtained. Sum is 27.
Therefore, nth row sum n^3
Hence, option (b) is the answer.

27. $a = 0.75$, $d = 1.75$

28. Go through the options.

29. Put values like (3, 6)
Alternatively, this question can be done by going through options also.

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30. Put $n = 1$ and check the options.

31.

D	H	T	T	H	D	T	H	D
10	5	15	5	10	15	10	5	15

.....

.....

.....

Total = 15

32. Assume that they are getting $a - d$, a and $a + d$ mangoes irrespective of the fact who gets what numbers.

Now, $a - d + a + a + d = 30$, So, $a = 10$

Regarding who gets how many mangoes, we have only one information, i.e., D > H. So, we cannot decide about the first two options.

33. $S_n = 153$, $d = 2$

$\frac{n}{2}(2a + (n-1)\,d) = 153$

Or, $\frac{n}{2}\,2\,[a + n - 1] = 153$, or, $n \times (a + n - 1) = 153$

Now, factorize 153, i.e., 17×9, 15×3, 153×1

Now, check $n\,(a + n - 1) = 153 \Rightarrow$ For $n = 9$, $a = 9$

For $n = 3$, $a = 49$

Therefore, five values satisfy.

34. Consider an HP a, b, c, $d = 6, 8, 12, 24$

Now, check the options.

Hence, option (d) is the answer.

35. … $= 1 \times 12 \times 13$ = Number of ways = 132.

36. Go through the options.

37. The series is 12, 3, 3/4, 3/16 …

38. $N_1 + N_2 + N_3 + \ldots + N_N = 100$ is possible only in two ways:

$23 + 24 + 25 + 26 + 27 = 100$, or, $9 + 10 + 11 + 12 + 13 + 14 + 15 + 16 = 100$

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CHAPTER

13

Function

LEARNING OBJECTIVES

After completion of this chapter, the reader should be able to understand:

- What is function?
- Definitions and properties
- Nature of function
- Ways of representing a function and questions associated
- Method/s of solving questions

INTRODUCTION

If we go by the standards of the CAT paper of last two to three years, then we can conclude that function has become an important topic for CAT. Number of questions being asked from this topic are almost constant and we do not have any reason to see any change in this situation for the next few years. Most of the questions asked in the CAT from this area can be done through just knowing the basics of functions. What is more important is the fact that if a child at the age of 14 to 15 years can learn function, then a man/woman at the age of 20-odd years can also do this, if he/she is not succumbing to the mental block that function is all about core of Maths.

FUNCTION

Function can be understood as a machine that takes an input and produces an output. For example, consider an automated vending machine as shown below.

Through a pre-defined program, if you press certain keys, then you would get your desired object. In this case, the output (your desired object) is related to the input (the buttons pressed) through an algorithm. It can also be seen that for a particular input, they always produce the same output. Now, we will define function:



Function is a rule that associates two or more than two variables. Function can be understood with the help of some very basic examples:

(i) Area of circle $(A) = \pi r^2$, where r is the radius.
So, area of circle is dependent upon value of radius of the circle. We can write this mathematically as $A = f(r)$.

(ii) If the distance (d) is constant, then time taken (t) to cover that distance (d) will be dependent on the value of speed (v). This can be written mathematically as $t = f(v)$.

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Domain and Range of $y = f(x)$

Let us go back to the same automated vending machine.

Example: Assume that the vending machine accepts currency notes only in the denominations of ₹10, ₹20, and ₹50. In other words, one can insert the notes of only ₹10, ₹20, and ₹50 in the machine. So, these are the inputs. Mathematically, we know these as domain of the function.

This vending machine is capable of selling potato wafers of brand X, chocolates of brand Y, and biscuits of brand Z. This is the output. Mathematically, we know these as the range of the function. One cannot buy a burger or soft drinks from this particular machine. So, these are not the range of this function.

In case of $y = f(x)$, values of y are dependent upon the corresponding values of x.

Here, y is known as a dependent variable and x is known as an independent variable.

All the real values of x for which at least one real value of y exists are known as domain of this function $y = f(x)$.

All those possible real values of y are known as range of this function $y = f(x)$.

In short, for any function, all that can go inside is domain and all that can come outside is range.

While finding out the domain, the variable generally take a range of values unlike the case of equations where the variable will, mostly, take one value or a discrete set of values 'interval notation'. Hence, it is important to understand the standard notations that are used to represent the solution set of the variable as well as the interval notation.

- (a, b) read as 'open interval a, b' means all real numbers between a and b, excluding a and b; $a < b$.
- $[a, b]$ read as 'closed interval a, b' means all real numbers between a and b, including a and b; $a < b$.
- $[a, b)$ means all numbers between a and b, with a being included and b excluded; $a < b$.
- $(a, b]$ means all numbers between a and b, with a being excluded and b included; $a < b$.

Example 1 Represent all real numbers between 1 and 10 in the interval from where

(i) 1 and 10 are included,
(ii) 1 and 10 are excluded,
(iii) 1 is included and 10 is excluded, and
(iv) 1 is excluded and 10 is included.

Solution

(i) $[1, 10]$, that is, $1 \le x \le 10$ and x is a real number.
(ii) $(1, 10)$, that is, $1 \le x \le 10$ and x is a real number.
(iii) $[1, 10)$, that is, $1 \le x \le 10$ and x is a real number.
(iv) $(1, 10]$, that is, $1 \le x \le 10$ and x is a real number.

Note: In case of $y = f(x)$ to be a function,

(a) One or more than one value of x can have a single corresponding mapping in y.
(b) However, converse cannot be true. It means that one value of x cannot have more than one mapping in Y. For example: $x^2 + y^2 = 16$ cannot be a function $y = f(x)$ because for one value of x, there are two corresponding values in y.
(c) It is possible that there are elements in Y that does not have any corresponding value in X.

Theorems Related to Number of Functions

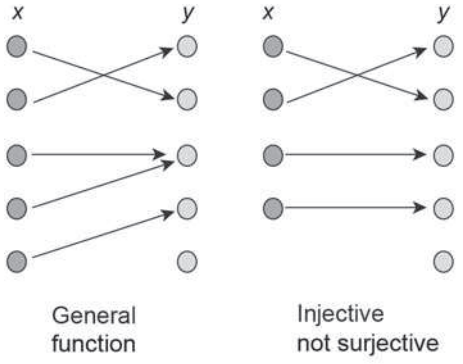
(a) For two sets A and B, the number of functions from A to B is $|B|^{|A|}$.
For example, Let $A = \{1, 2, 3, 4\}$ and $B = \{v, w, x, y, z\}$
Number of functions = There are $|B|^{|A|} = |5|^{|4|} = 625$ functions from A to B.

(b) For two sets A and B, the number of relations from A to B is $|2|^{|A||B|}$
For example, let $A = \{1, 2, 3, 4\}$ and $B = \{v, w, x, y, z\}$
Number of functions = There are $|B|^{|A|} = |5|^4 = 625$ functions from A to B.
Number of relations = There are $|2|^{|A||B|} = |2|^{|4||5|} = 2^{20}$ relations from A to B.

(c) For two sets A and B, the number of one-to-one functions from A to B is:
$$\frac{|B|!}{(|B|-|A|)!}$$
For example, Let $A = \{1, 2, 3, 4\}$ and $B = \{v, w, x, y, z\}$
Number of one-to-one functions from A to B is:
$$\frac{|5|!}{(|5|-|4|)!} = 120$$

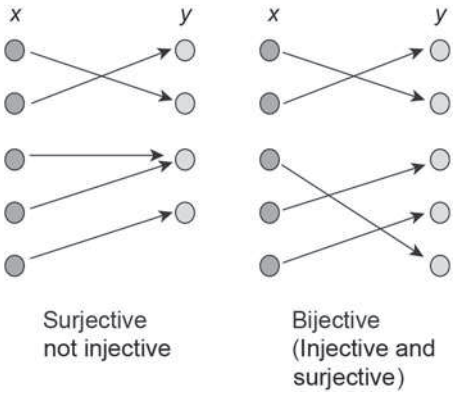
INJECTIVE, SURJECTIVE, AND BIJECTIVE FUNCTIONS

Let us consider a function from X to Y [$Y = f(X)$] and understand the types of function:



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(a) General function—We have already discussed this. Two or more elements of X can have same mapping in Y. For example, $y = f(x) = x^2$. In this function, two values of x will point to same value of y. ($x = 2$ or $x = -2$ both will give $y = 4$)

(b) Injective—In this type of function, each element in X has its unique corresponding mapping in Y. Y may have some elements without any correspondence in X. For example, $y = f(x) = 3x + 2$

In this case, you would not have two values of X having same mapping in Y. So, 'many to one' is not possible. **(One to many is obviously not possible because it is a function. In a function, one *y* cannot have two mapping in one *x*.)**

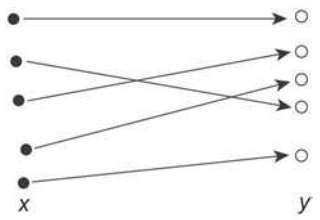
(c) Surjective—In this type of function, each element in Y should be mapped by one or more than one element in X.

There would not be an element in 'Y' left out. Surjective functions are also known as onto function.

(d) Bijective—It means both injective and surjective. So, each element of Y will be mapped by a unique element in X.

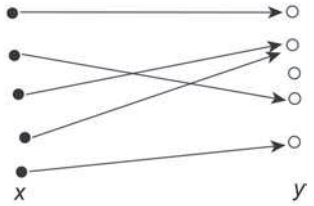
Example 2 Identify what type of function is this (x to y)?

(a)



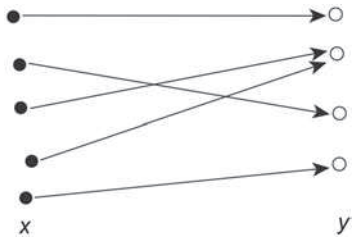
(i) A general function
(ii) Injective
(iii) Surjective
(iv) Bijective

(b)



(i) A general function
(ii) Injective
(iii) Surjective
(iv) Bijective

(c)



(i) A general function
(ii) Injective
(iii) Surjective
(iv) Bijective

Solution

(a) Bijective is the answer.

(b) A general function is the answer because many to one mapping is there.

(c) It cannot be injective because two X are pointing to single Y. Since no Y is left out, hence, surjective is the answer.

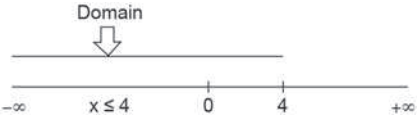
Process to find out the domain of $y = f(x)$

1. Denominator should not be equal to zero.
2. Expression under square root should be non-negative.

Example 3 Find the domain of the definition of function $y = f(x) = \sqrt{4x - x} + \sqrt{x - 6}$.

Solution Expression under square root should be non-negative.

Hence, $4 - x \geq 0$, or, $x \leq 4$ is the domain.



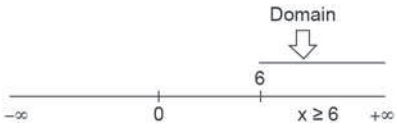
Example 4 Find the domain of the definition of function $Y = f(x) = \sqrt{x - 6}$.

Solution Expression under square root should be non-negative.

Hence, $x - 6 \geq 0$, or, $x \geq 6$ is the domain.

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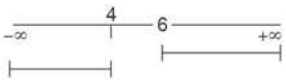
Example 5 Find the domain of the definition of function $y = f(x) = \sqrt{4x - x} + \sqrt{x - 6}$.

Solution Expression under square root should be non-negative.

So, $4 - x \geq 0$ or $x \leq 4$

$x - 6 \geq 0$ or $x \geq 6$

Arranging these values on number line gives the following:



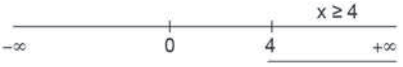
So, no value of x is common. Hence, domain of $y = f(x)$ is not defined.

Example 6 Find the domain of the definition of function $y = f(x) = \sqrt{x - 4} + \sqrt{6 - x}$.

Solution Expression under square root should be non-negative.

$$x - 4 \geq 0 \Rightarrow x \geq 4$$

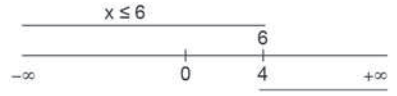
Arranging this value on number line gives the following:



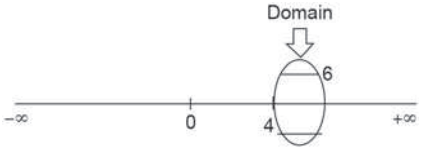
Next,

$6 - x \geq 0 \Rightarrow x \leq 6$

Arranging this value on number line gives the following:



Following is common between both the ranges (given in the circle):



Hence, domain of the function is $4 \leq x \leq 6$.

Example 7 Find the domain of $y = f(x) = \dfrac{1}{\sqrt{|x| - x}}$.

Solution For $y = f(x)$ to be defined, $|x| - x > 0$ or $|x| > x$

It is true only if $x < 0$. So, the domain is $-\infty < x < 0$.

Example 8 Find the domain of $y = f(x) = \dfrac{1}{\sqrt{x - |x|}}$.

Solution For $y = f(x)$ to be defined, $x - |x| > 0$ or $x > |x|$

This is not possible.

Hence, the domain of $f(x)$ is not defined.

Example 9 Find the domain of $y = f(x) \dfrac{1}{\sqrt{x - [x]}}$, where $[x]$ = greatest integer value of x.

Solution There are both conditions operating here: Expression under square root is in denominator.

Hence, $x - [x] > 0 \Rightarrow x > [x]$. This is possible for all the real values except the integers. Hence, domain of the function is = all real number minus integers.

Example 10 Are the following functions identical?

$y = f(x) = x$

$y = h(x) = \dfrac{x^2}{x}$

Solution Domain of $f(x)$ = all the real values

Domain of $h(x)$ = all the real values except at $x = 0$

because domains are not the same for $f(x)$ and $h(x)$; so, $f(x)$ and $h(x)$ are not identical.

Some Special Functions

1. Modulus Function

$$y = |x|$$

It is defined as follows:

$y = x$; if $x > 0$

$y = -x$; if $x < 0$

$y = 0$; if $x = 0$

Despite that in the above equations, we are finding out a negative value of x if $x < 0$; its absolute value can never be negative. This can be seen through the following example:

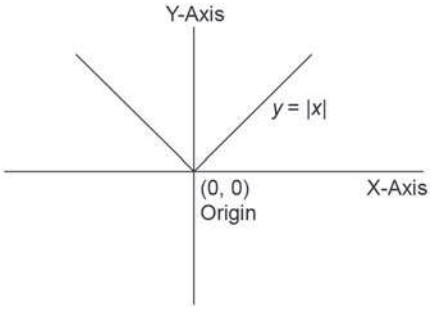
We are finding out the value of $y = |-5|$

Assuming that $-5 = x$; so, $y = |x|$

Now, since $x < 0$; so, $y = -x = -(-5) = 5$

Graphical Representation of Modulus Function

This is the graph of $y = |x|$



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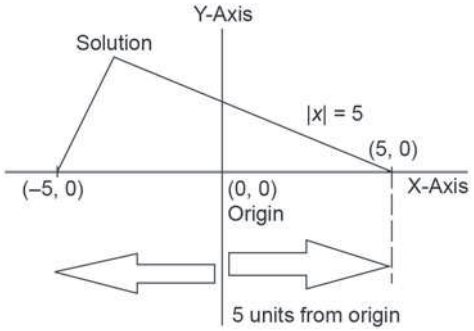
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Alternative Way of Understanding Modulus Function

Modulus function can also be understood as the absolute distance from the origin on the X-axis. $|x - a|$ refers to the distance of the point x from a, on a number line.

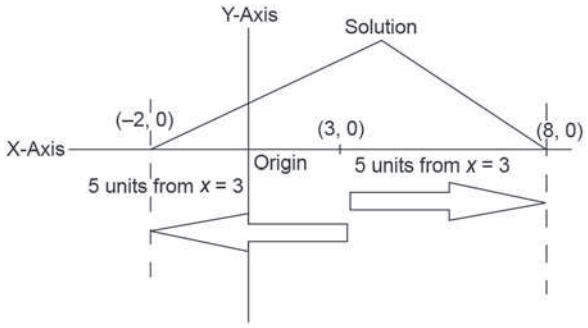
For example,

(a) $|x| = 5$ means 'A person is standing at a distance of five units from origin'. There are two different ways a person can stand at a distance of five units from origin—if she/he is standing either at $x = -5$ or $x = 5$.



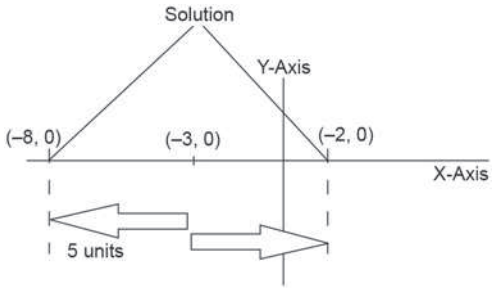
So, solution of $|x| = 5$ is $x = \pm 5$

(b) $|x - 3| = 5$ means 'A person is standing at a distance of five units from $x = 3$ (on X-axis)'. There are two different ways that a person can stand at a distance of five units from $x = 3$ (on X-axis)—if she is standing either left-hand side at $x = -2$ on X-axis or right-hand side at $x = 8$ on X-axis.



So, solution of $|x - 3| = 5$ is $x = -2$ or 8

(c) $|x + 3| = 5$ means 'A person is standing at a distance of five units from $x = -3$ (on X-axis)'. There are two different ways that a person can stand at a distance of five units from $x = -3$ (on X-axis)—if she/he is standing either left-hand side at $x = -8$ on X-axis or right-hand side at $x = 2$ on X-axis.



So, solution of $|x + 3| = 5$ is $x = -8$ or 2

It can be observed that for every value of x, be it positive or negative, value of y cannot be negative.

Example 11 What is the value of x if $|2x + 3| = 9$?

Solution

Method 1 Equation Method

Working method of questions involving modulus involves first taking the value of expression under modulus as positive and then as negative.

Case I When $(2x + 3) > 0$, or, $x > \frac{-3}{2}$, then $= |2x+3| = 2x + 3$; So, $2x + 3 = 9$, or, $2x = 6$

So, $x = 3$

Case II When $(2x + 3) < 0$, or, $x < \frac{-3}{2}$, then $= |2x+3| = -(2x + 3)$

So, $-(2x + 3) = 9$, or, $-2x = 12$

So, $x = -6$

Method 2 (a) Graphical Alternative Method

First convert $|2x + 3|$ in such a way that co-efficient of x becomes 1. This can be done by dividing LHS and RHS by 2.

$$\frac{|2x+3|}{2} = \frac{9}{2}$$

Hence, $\left|x+\frac{3}{2}\right| = \frac{9}{2} = 4.5$

$\left|x+\frac{3}{2}\right|$ means a person is standing at $x = \frac{-3}{2} = -1.5$ (on X-axis)

$\left|x+\frac{3}{2}\right| = \frac{9}{2} = 4.5$ means finding a point that is at a distance of 4.5 units on X-axis from $x = -1.5$. It can be on the left-hand side (towards negative X-axis) or on the right-hand side (towards positive X-axis).

It will be $(-1.5 + 4.5)$ or $(-1.5 - 4.5) = 3$ or -6

Hence, $x = 3$ or -6 are the solution.

Method 2 (b) Graphical Alternative Method

Instead of trying to get the co-efficient of x as 1, $|2x + 3| = 9$ can also be solved like:

$2x = 9 - 3 = x = 3$

or $2x = -9 - 3 = x = -6$

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Note from Author: In my opinion, best way of solving the above question is through **Method 2 (b): Graphical Alternative Method**.

Example 12 What is the value of 'x' if $x^2 + 5|x| + 6 = 0$?

Solution

Method 1 Equation Method

Taking $x > 0$, $x^2 + 5|x| + 6 = x^2 + 5x + 6 = (x + 2)(x + 3) = 0$
or $x = -2$ and $x = -3$

However, as we have assumed that $x > 0$; so, $x = -2$ and -3 are not admissible.

Taking $x < 0$, $x^2 + 5|x| + 6 = x^2 - 5x + 6 = (x - 2)(x - 3) = 0$
or $x = 2$ and $x = 3$

However, as we have assumed that $x < 0$; so, $x = 2$ and 3 are not admissible.

So, no real value of x is there that can satisfy this equation.

Method 2 Logical Observation Method

It can be seen that x^2 and $5|x|$ and 6, all three are positive values. So, sum of these three can never be equal to 0. So, no real value of x is possible.

Modulus-based Inequality Questions

$|x - a| < N$ means a person is standing *within* a distance of N units from $x = a$ on X-axis.

$|x - a| > N$ means a person is standing *outside* a distance of N units from $x = a$ on X-axis.

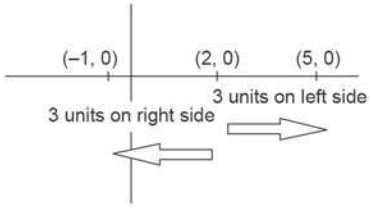
Example 13 If $|x - 2| < 3$, then solve it for x.

Solution

Method 1 Graphical Solution

$|x - 2| < 3$ means a person is standing within a distance of three units from $x = 2$ on X-axis.

If we go left-hand side, then three units from $x = 2$ will be $x = -1$, and on the right-hand side, three units from $x = 2$ will be $x = 5$.



Hence, $(-1 < x < 5)$ is the solution.

Method 2 Inequality Method

$|x - a| < N$ simply means that $\Rightarrow -N < (x - a) < N$. So, $|x - 2| < 3$ means $\Rightarrow -3 < (x - 2) < 3$

So, solve this for x, we will add two (adding 2 to $x - 2$ will make it equal to x) to the whole inequality.

$(-3 + 2) < (x - 2) + 2 < (3 + 2) \Rightarrow -1 < x < 5$

Method 3 Equation Method

$|x| < 3 \Rightarrow -3 < x < 3$

Inequality relation between $|x|$ and x

$|x| > x \Rightarrow x < 0$
$|x| < x \Rightarrow$ It is not possible.
$|x| \geq x \Rightarrow$ This is always true. In this case, x can be any real number.
$|x| = x \Rightarrow x \geq 0$

Example 14 If $|x^2 - 5x + 6| > (x^2 - 5x + 6)$, then find the values of x.

Solution It can be seen above that if $|N > N$, then $N < 0$. Assume $N = x^2 - 5x + 6$.

So, $x^2 - 5x + 6 < 0$
or $(x - 2)(x - 3) < 0$
So, $2 < x < 3$

2. Greatest Integer Value Function

$$y = [x]$$

It is defined as the largest integral value of x, which is less than or equal to x.

Given that $y = [3.23]$ and we have to find out the greatest integer value of y.

Taking second part of the definition, that is, integer less than or equal to 3.23, what we get is a set of integers less than or equal to 3.23 $\Rightarrow$ 3, 2, 1, 0, −1, and so on.

Largest integer among all these integers = 3. So, greatest integer value of [3.23] = 3.

Similarly, if we find out the greatest integer value of $y = [-2.76]$, then all the integers less than this value (−2.76) = {−3, −4, −5, −6,}.

Now, greatest integer among all these integers given in the above set is −3.

It can be seen through tabular presentation also:

x	y
0–1 (excluding at $x = 1$)	0
1–2 (excluding at $x = 2$)	1
2–3 (excluding at $x = 3$)	2

and so on.

Example 15 What is the value of x in the following expression:

$$[x]^2 \leq 16?$$

Solution $-4 \leq [x] \leq 4$

$-4 \leq [x]$ or $-4 \leq x$

and $[x] \leq 4$ or $x < 5$

So, value of x is: $-4 \leq x < 5$

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3. Logarithmic Function

$$y = \log_a x$$

It is known to us that value of x has to be positive here. However, y can have negative values.

We will discuss more about it in Logarithm Chapter.

4. Exponential Function

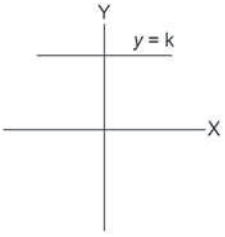
$$y = e^x$$

Exponential function is the inverse of logarithmic function.

Again, it can be observed that value of y cannot be negative, be what the value of x in $y = e^x$.

5. Constant Function

$f(x) = k$, (where k is any constant) is known as constant function.



WAYS OF REPRESENTING FUNCTIONS

There are three standard methods of representing functions:

1. Equation method
2. Tabular method
3. Graphical method

1. Equation Method

It is the most commonly used form of writing any function. This form of representation involves a simple equation written in terms of the variable.

Example: $y = f(x) = 5x^2 + 8$

2. Tabular Method

In this form of representing a function, we form a table of values of x and y, with or without mentioning the equation of function.

Example: A function $y = f(x)$ is represented by the table given below:

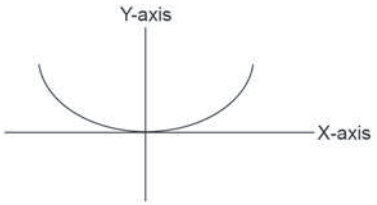
x	1	2	–1	–2
y	1	4	1	4

Obviously, the function which will satisfy the above written table is $y = f(x) = x^2$

3. Graphical Method

A graph is considered to be the most revealing way of representing a function. Apart from giving an inkling about the function $y = f(x)$, its nature and shape, it also gives us information about the maximum and minimum value of the function.

Example:



It is quite evident here that for both the negative and positive values of x, y is having the same value, and the shape of graph is parabolic. So, this graph is being represented by the function $y = f(x) = x^2$

NATURE OF FUNCTION (In Terms of Equation)

On the basis of symmetric nature of functions, we define the graphs to be of three natures:

1. Even Function

For $y = f(x)$,

Mathematically, even functions are defined as $y = f(x) = f(-x)$. In layman's term, if we replace 'x' from the given equation by '$-x$', then equation should be same.

Example: $y = f(x) = x^2 + 8$

Here, $f(-x) = (-x)^2 + 8 = x^2 + 8 = f(x)$

So, $y = f(x) = x^2 + 8$ is an even function.

Properties of Even Function

(i) Sum or difference or product or division of two even functions is an even function.

(ii) For $y = f(x)$, graph of even functions are symmetrical to Y-axis.

2. Odd Function

For $y = f(x)$,

Mathematically, odd functions are defined as $y = f(x) = -f(-x)$. In layman's term, if we replace 'x' from the given equation by '$-x$', then equation should be same with opposite sign.

Example:

$y = f(x) = x^3$ is an odd function.

Here, $f(-x) = (-x)^3 = -(x^3) = -f(x)$.

So, $y = f(x) = x^3$ is an odd function.

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Properties of Odd Function

(i) Sum or difference of two odd functions is an odd function.
(ii) For $y = f(x)$, graph of odd functions are symmetrical to origin.

3. Neither Odd nor Even Function

For $y = f(x)$,

If any equation is not satisfying the conditions of either odd or even function, then it is said to be neither odd nor even function.

In layman's term, if we replace 'x' from the given equation by '$-x$', then equation should not be same, either with same sign or with opposite sign.

Example: $y = f(x) = x^3 + 1$

Here, $f(-x) = (-x)^3 + 1 = -x^3 + 1$

It is now quite clear that $f(-x)$ is neither equal to $f(x)$ or $-f(x)$.

So, $y = f(x) = x^3 + 1$ is neither an even function nor an odd function.

Properties of Neither Odd nor Even Function

i. Sum of difference of odd and even functions are neither odd nor even function.

To understand the mechanism of sum/difference/product of two functions, we can take even function as a positive number and odd function as a negative number.

Example: x^2(even function) × x^3(odd function) = odd function

x^2(even function) + x^3 (odd function) = neither odd nor even function

Remember That no function can be both even and odd function simultaneously.

Example 16 Find out the nature of following functions:

(a) $y = f(x) = e^{-x}$
(b) $y = f(x) = x^2 + x^4$
(c) $y = f(x) = \frac{1}{x} + x^3$
(d) $y = f(x) = x^2 + x^3$
(e) $y = f(x) = \log x^2$
(f) $y = f(x) = \log_a x$

Solution

(a) Neither Odd Nor Even Function
(b) Even Function
(c) Odd Function
(d) Neither Odd Nor Even Function
(e) Even Function
(f) Neither Odd Nor Even Function

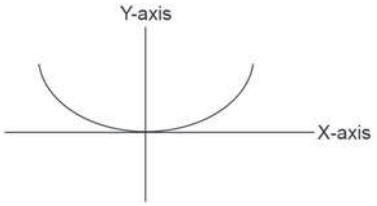
NATURE OF FUNCTION (In Terms of Graph)

For $y = f(x)$

Even Function

Graph of even function will be symmetrical to Y-axis. It simply means that graph on the left side of Y-axis will be same as graph on the right side of Y-axis.

As we have seen in the case of $y = f(x) = x^2$, graph on the left-hand side of Y-axis is same in shape and size with respect to the graph on the right-hand side of Y-axis.



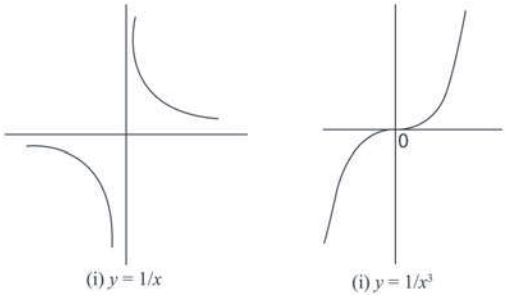
Properties of Graph of Even Function

(i) Even function graphs will always be in adjacent quadrants (i.e., either first quadrant and second quadrant or second quadrant and third quadrant or third quadrant and fourth quadrant or fourth quadrant and first quadrant).
If graph is in first quadrant and second quadrant or third quadrant and fourth quadrant, then function will be $y = f(x)$. If graph is in second quadrant and third quadrant or fourth quadrant and first quadrant, then function will be $x = f(y)$.
(ii) Even function graphs are one-fold graphs. That is, we can make both the parts of graphs overlap on each other by just folding the graph paper once.
Whenever we fold the paper, we fold it along the axis, either X-axis or Y-axis.

Odd Function

Graph of odd function will be symmetrical to the origin. It simply means that graph on one side of origin will be same as graph on the other side of origin.

As we have seen in the case of $y = f(x) = 1/x$, graph on the one side of origin is same in shape and size with respect to the graph on the other side of origin.



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Properties of Graph of Odd Function

(i) Odd function graphs will always be in opposite quadrants (i.e., either first quadrant and third quadrant or second quadrant and fourth quadrant).

(ii) Odd function graphs are two-fold graphs. That is, we can make both the parts of graphs overlap on each other, by folding the graph paper twice, once along X-axis and then along Y-axis or vice versa.

Whenever we fold the paper to overlap the graph, we fold it along the axis, either X-axis or Y-axis.

Composite Function

As the name suggests us, composite function is the kind of function which is composed, into one, of two or more than two functions.

Example: If $y = f(x) = 5x + 3$ and $y = g(x) = x^2$ is given, then $f(g(x))$ or $g(f(x))$ is the composite function.

In the above example, $f(g(x)) = 5g(x) + 3 = 5x^2 + 3$

Similarly, $g(f(x)) = f(x)]^2 = (5x + 3)^2$

Example 17 If $g(x) = \dfrac{1}{f(x)}$ and $f(x) = x$, $x \neq 0$ is given, then which of the following is true?

(a) $f(f(f(g(g(f(x)))))) = g(g(f(g(f(x)))))$

(b) $f(f(g(g(g(f(x)))))) = g(g(f(g(f(x)))))$

(c) $f(g(f(g(g(f(g(x))))))) = g(g(f(g(f(x)))))$

(d) $f(f(f(g(g(f(x)))))) = f(f(f(g(f(x)))))$

Solution $f(x) = x$ and $g(x) = 1/x$

Since applying 'f' any number of times is not changing the final result, but applying 'g' is making the value reciprocal. So, LHS can be equal to RHS only if the number of 'g' on both the sides is either even or odd.

Hence, option (b) is the answer.

Inverse Function

If two functions f and g satisfy $g(f(x)) = x$ for every x in the domain of f, and similarly, $f(g((x)) = x$ for every x in the domain of g, then f is said to be the inverse of g and vice versa. We write this as f^{-1} and g^{-1}. For example, $y = \log_e x$ and $y = e^x$ are inverse to each other.

Example 18 $y = f(x) = \dfrac{x+1}{x+2}$. Find $f^{-1}(x)$.

Solution $y = \dfrac{x+1}{x+2}$

or $y - 1 = \dfrac{x+1}{x+2} - 1 = \dfrac{-1}{x+2}$

or $x + 2 = \dfrac{1}{1-y}$

or $x = \dfrac{1}{1-y} - 2 = \dfrac{2y-1}{1-y}$

$f^{-1}(x) = \dfrac{2x-1}{1-x}$

Two Important Rules

Rule 1:

If $f(x-1) + f(x+1) = f(x)$, then

(a) $f(x+6) = f(x)$ and so on. In general, cycle of recurrence will be of length 6. Hence, every term after six terms will be equal.
For example, $f(7) = f(1) = (13) = f(19)\ldots; f(8) = (2) = f(14)\ldots$

(b) $f(x) = -f(x+3)$
For example, $f(1) = -f(4) = f(7) = -f(10) = f(13)\ldots$ and so on

(c) $f(x) + (x+1) + f(x+2) + f(x+3) + f(x+4) + f(x+5) = 0$. In general, sum of any six consecutive terms will be equal to zero.
For example, $f(1) + f(2) + f(3) + f(4) + f(5) + f(6) = 0$

Rule 2:

If $f(x-1) - f(x+1) = f(x)$, then

(a) $f(x+3) = f(x)$ and so on. In general, cycle of recurrence will be of length 3.
Hence, every term after three terms will be equal.
For example, $f(4) = f(1) = (7) = f(10)$; $f(2) = (5) = f(8)\ldots$

Example 19 Let $g(x)$ be a function such that $g(x + 1) + g(x - 1) = g(x)$ for every real x. Then, for what value of p is the relation $g(x + p) = g(x)$ necessarily true for every real x? **(*CAT* 2005)**

(a) 3 (b) 3 (c) 2 (d) 6

Solution

Method 1

It can be seen that the condition given in this question satisfies Rule 1 (as given above). Hence, $g(x) = g(x+p) = g(x+6)$.

So, $p = 6$. Hence, option (d) is the answer.

Method 2

Given that $g(x + 1) + g(x - 1) = g(x)$.

Putting $(x + 1)$ at the place of x gives us the following:

$g(x + 2) + g(x) = g(x + 1)$

Adding these two equations, we get

$g(x + 2) + g(x - 1) = 0 \Rightarrow g(x + 3) + g(x) = 0$

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$\Rightarrow g(x+4)+g(x+1)=0$
$\Rightarrow g(x+5)+g(x+2)=0$
$\Rightarrow g(x+6)+g(x+3)=0$
$\Rightarrow g(x+6)-g(x)=0$. Hence, $g(x+6)-g(x)$
So, $p=6$
Hence, option (d) is the answer.

Example 20 Consider the sequence of numbers a_1, a_2, a_3, … to infinity where $a_1 = 81.33$ and $a_2 = -19$ and $a_j = a_{j-1} - a_{j-2}$. for $j \geq 3$.

What is the sum of the first 6002 terms of this sequence?

(a) −100.33 (b) −30.00
(c) 62.33 (d) 119.33

Solution The sequence given above $a_j = a_{j-1} - a_{j-2}$ can be written as: $a_{j-1} = a_{j+1} + a_{j-1}$

Replacing j by $j+1$ (or writing the next term in the sequence) gives the following: $a. = a_{j+1} + a_{j-1}$

It can be observed that the sequence given follows Rule 1 as given.

Hence, $a_1 + a_2 + a_3 + a_4 + a_5 + a_6 = 0$

So, sum of first 6002 terms = Sum of first two terms = $81.33 + (-19) = 62.33$

Hence, option (c) is the answer.

Recursive Function

In such functions, iteration of functions occurs. See the following example to understand it more clearly:

Example 21 Consider sequences of positive real numbers of the form x, 2000, y, ... in which every term after the first is 1 less than the product of its two immediate neighbours. For how many different values of x does the term 2001 appear somewhere in the sequence?

(a) 1 (b) 2 (c) 3 (d) 4

Solution Let us compute few terms of the sequence in order to get a feel how it looks like.

In our case, the definition is that for all $n > 1$, $a_n = a_{n-1}a_{n+1} - 1$

This can be rewritten as:

$$a_{n+1} = \frac{a_n + 1}{a_{n-1}}$$

We have $a_1 = x$ and $a_2 = 2000$, and we compute:

$$a_3 = \frac{a_2 + 1}{a_2} = \frac{2001}{x}$$

$$a_4 = \frac{a_3 + 1}{a_2} = \frac{\frac{2001}{x} + 1}{2000} = \frac{2001 + x}{2000x}$$

$$a_5 = \frac{a_4 + 1}{a_3} = \frac{\frac{2001 + x}{2000x} + 1}{\frac{2001}{x}}$$

$$= \frac{2001 + 2001x}{2000 \times 2001} = \frac{1 + x}{2000}$$

$$a_6 = \frac{a_5 + 1}{a_4} = \frac{\frac{1+x}{2000} + 1}{\frac{2001+x}{2000x}} = \frac{\frac{2001+x}{2000}}{\frac{2001+x}{2000x}} = x$$

$$a_7 = \frac{a_6 + 1}{a_5} = \frac{x+1}{\frac{1+x}{2000}} = 2000$$

It can be clearly seen that the sequence will become periodic where every sixth term will be equal like the one given ahead: $a_6 = a_1$, $a_7 = a_2$, and each subsequent term is uniquely determined by the previous two.

So, if 2001 appears, then it has to be one of a_1 to a_5. As $a_2 = 2000$, we only have four possibilities left for 2001.

Clearly, $a_1 = 2001$ for $x = 2001$, and $a_3 = 2001$ for $x = 1$

Equation $a_4 = 2001$ reduces to $x = \frac{2001}{2000 \times 2001 - 1}$ and the equation $a_5 = 2001$ to $x = (2000 \times 2001) - 1$

Further, it can be seen that no two values of x calculated are equal. Hence, there are four different values of x for which the sequence contains the value 2001.

Example 22 A polynomial $f(x)$ with real coefficient satisfies the functional equation $f(x)$. If $f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right)$ if $f(2) = 9$, then $f(4)$ is:

(a) 82 (b) 17
(c) 65 (d) None of these

Solution Assume $f(x) = x^n + 1$ and $f\left(\frac{1}{x}\right) = \frac{1}{x^n + 1}$

So, $f(x) \cdot f\left(\frac{1}{x}\right) = (x^n + 1)\left(\frac{1}{x^n}\right) + 1$

$$= (1 + x^n) + \left(\frac{1}{x^n + 1}\right)$$

Now, $f(x) = x^n + 1 = 9$

or $x^n = 8$ or $n = 3$ Hence, $f(x) = x^3 + 1 = 4^3 + 1 = 65$

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Practice Exercises

Warm Up

Q.1 The domain of the function $f(x) = \sqrt{|x|}$ is:
(a) $x \in (-\infty, \infty)$
(b) $x \in (0, \infty)$
(c) $x \in (-\infty, 0) \cup (0, \infty)$
(d) $x \in (0, \infty)$

Q.2 The domain of the function $f(x) = \sqrt{|x|}$ is:
(a) $x \in (-\infty, \infty)$
(b) $x \in (0, \infty)$
(c) $x \in (0, \infty)$
(d) None of these

Q.3 The domain of the function $f(x) = \frac{1}{\sqrt{x}}$ is:
(a) $-\infty < x < \infty$
(b) $0 < x < \infty$
(c) $-\infty < x < \infty$ excluding $x = 0$
(d) $0 \le x < \infty$

Q.4 The domain of the function $f(x) = \sqrt{x^2}$ is:
(a) $-\infty < x < \infty$
(b) $0 < x < \infty$
(c) $-\infty < x < \infty$ excluding $x = 0$
(d) None of these

Q.5 The domain of the function $f(x) = \sqrt{4x-3} + \sqrt{2x-6}$ is:
(a) $[0, \infty]$ (b) $\left[\frac{3}{4}, \infty\right]$ (c) $\left[\frac{4}{3}, \infty\right]$ (d) $[3, \infty]$

Q.6 The domain of the function $f(x) = \log(5x - 6 - x^2)$ is:
(a) (5, 6) (b) (2, 3)
(c) (2, ∞) (d) None of these

Q.7 The domain of $y = \log_x 5$ is:
(a) $x > 5$ (b) $(0, \infty)$
(c) $(0, \infty)$ excluding $x = 1$ (d) $(5, \infty)$

Q.8 If $f(x) = \frac{x}{\sqrt{1+x^2}}$, then $f(f(f(x)))$ is:
(a) $\frac{3}{\sqrt{1+x^2}}$ (b) $\frac{x}{\sqrt{1+3x^2}}$
(c) $\frac{x^3}{\sqrt{1+x^2}}$ (d) $\frac{x}{\sqrt{3+3x^2}}$

Q.9 Let $f(x) = \frac{x}{x+3}$, then $f(x+1)$ is equal to:
(a) $\frac{3x+2}{x+2}$ (b) $\frac{x+1}{x+4}$
(c) $\frac{x+1}{x+3}$ (d) $\frac{2x+3}{x+3}$

Q.10 If $f(x) = \frac{x+1}{x-1}$, $x \neq 1$, then find $f(f(f(f(f(2)))))$.
(a) 1 (b) 2 (c) 3 (d) 4

Q.11 Let $f(x) = \sqrt{x^5}$, then $f(5x)$ is equal to:
(a) $5x\sqrt{5x^5}$ (b) $25x\sqrt{5x^5}$
(c) $5x$ (d) None of these

Q.12 If $f(x) = 4x - 5$, $g(x) = x^2$ and $h(x) = \frac{1}{x}$, then $f(g(h(x)))$ is:
(a) $\frac{4}{x-5}$ (b) $\frac{1}{(4x-5)^2}$
(c) $\left(\frac{x}{4} - 5\right)$ (d) None of these

Q.13 Let f be a function satisfying $f(xy) = \frac{f(x)}{y}$ for all positive real numbers x and y. If $f(500) = 3$, then what is the value of $f(600)$?
(a) $\frac{5}{2}$ (b) 2 (c) 1 (d) 3

Q.14 Which of the following is an even function?
(a) $\text{In}\left(\frac{1+x^3}{1-x^3}\right)$ (b) $x\frac{e^{ax}-1}{e^{ax}+1}$
(c) $\frac{(1+2^x)^2}{2^x}$ (d) None of these

Q.15 Which of the following function is an even function?
(a) $f(x) = \log\frac{1-x}{1+x}$
(b) $f(x) = x\frac{a^x+1}{a^{x-1}}$
(c) $f(x) = \log\left[x + \sqrt{(1+x^2)}\right]$
(d) $f(x) = \sqrt{(1+x+x^2)} - \sqrt{(1-x+x^2)}$

Q.16 The function $f(x) = \frac{x}{e^x - 1} + \frac{x}{2} + 1$ is:
(a) Even (b) Odd
(c) Neither even nor odd (d) Cannot be determined

Q.17 Given $f(x) = \frac{1}{(1-x)}$, $g(x) = f(f(x))$ and $h(x) = f(f(f(x)))$, then the value of $f(x).\ g(x).\ h(x)$ is:
(a) 0 (b) −1 (c) −2 (d) 2

Q.18 If $f(x) = x^2 + 2$, then $f^{-1}(x)$ is:
(a) x^5 (b) $\sqrt{x+2}$
(c) $\sqrt{x} + 2$ (d) None of these

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Q.19 If $f(x) = \frac{x}{x+1}$, then $f^{-1}(x)$ is:

(a) $\frac{x+1}{x}$ (b) $\frac{x}{1-x}$

(c) $\frac{x-1}{x}$ (d) None of these

Q.20 If $f(x) = \frac{x-1}{x+1}$, then $f^{-1}(x)$ is:

(a) $\frac{x+1}{1-x}$ (b) $\frac{x+1}{1-x}$

(c) $\frac{1}{x+1}$ (d) None of these

FOUNDATION

Q.1 Given that $f(x) = x^2 + 4x + 4$ and $g(x) = x^2 + 4x + 3$, then find the values of x such that $f(g(x)) = g(f(x))$.

(a) $x = -1$ (b) $x = -2$

(c) $x = -3$ (d) None of these

Q.2 For $y = f(x)$, even functions are defined as $f(-x) = f(x)$ and odd functions are defined as $f(-x) = -f(x)$ and

$f(x) = \log_{10}\left(x + \sqrt{x^2+1}\right)$ is:

(a) An odd function

(b) An even function

(c) Both even and odd function

(d) Neither odd nor even function

Q.3 A function $f(x)$ is such that $f(x) + f(y) = f(xy)$. Which of the following could be $f(x)$?

(a) a^x (b) $\sqrt{x}$ (c) x^2 (d) $\log ax$

Q.4 Let $f(x) = 3x^2 - 2x + 4$ and $g(x) = x + 1$. Find $f(g(f(1)))$.

(a) 69 (b) 100

(c) 96 (d) None of these

Q.5 If $f(x) = x^2$ and $f[g(x)] = g[f(x)]$, then which of the following can never be equal to $g(x)$?

(a) $\sqrt{x}$ (b) x^2 (c) x (d) $2x$

Q.6 Let $f(x) = |x - 1|$, then

(a) $f(x^2) = f(x)]^2$ (b) $f(x + y) = f(x) + f(y)$

(c) $f(|x|) = f|(x)|$ (d) None of these

Direction for Questions 7 to 9: Read the passage below and solve the questions based on it.

For any $y = f(x)$,

If $f(-x) = f(x)$, then this function is known as an even function.

If $f(-x) = -f(x)$, then this function is known as an odd function.

Q.7 $f(x) = |x|^3$ is:

(a) An odd function

(b) An even function

(c) Neither odd nor even function

(d) Cannot be determined

Q.8 The sum of the two odd functions is always

(a) an odd function.

(b) an even function.

(c) neither odd nor even function.

(d) nothing can be said

Q.9 The sum of an odd function and an even function is always

(a) an odd function.

(b) an even function.

(c) neither odd nor even function.

(d) nothing can be said

Q.10 $f(x) = c.x + 1$ and $g(x) = 3x + 2$. If $f(g(x)) = g(f(x))$, then what is the value of c?

(a) 1 (b) 2 (c) 3 (d) 4

Q.11 If $f(x) = 2^x - 2^{-x}$, then the value of $2f(x) - 5f(x-1) + 2f(x-2)$ is:

(a) 1 (b) −3

(c) 15 (d) None of these

Direction for Questions 12 to 14: Read the passage below and solve the questions based on it.

If $(x, y) = x^{y^2}$

$g(x\ y) = y^{x^2}$

$h(x, y)$ = Remainder when x is divided by y

$i(x, y)$ = Quotient when x is divided by y

Q.12 The value of $g[h\{f\{i(16, 3), 2\}, 7\}, 3]$ is:

(a) 16 (b) 25 (c) 49 (d) 81

Q.13 The value of $\frac{h[g\{i(76,28),2\},3]}{h[f\{i(16,3),2\},7]}$ is:

(a) 1 (b) $\frac{1}{4}$ (c) $\frac{1}{3}$ (d) $\frac{1}{2}$

Q.14 The value of $f[h\{g(2,2), 3\}, 2] - g[h[f(5,2), 7], 3]$ is:

(a) −100 (b) −90 (c) −85 (d) −80

Q.15 If $f(x) = 2x + 3$, then the value off $[f\{f(x)\}]$ is:

(a) $-6x + 7$ (b) $7x + 8$

(c) $8x + 21$ (d) $9x + 10$

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Q.16 If $f(x) = e^x + e^{-x}$, then $f(x)$is:
(a) An odd function (b) An even function
(c) Neither odd nor even (d) None of these

Q.17 If $f(x) = \frac{x^2 - 2x}{1+x}$, then the value of $f\left(\frac{1}{-x}\right)$ is:
(a) $\frac{x^2 - 2x}{1+2x}$ (b) $\frac{1-2x^2}{1+x}$
(c) $\frac{1-2x}{x(x-1)}$ (d) None of these

Q.18 If one root of the equation $x^2 - 3ax + f(a) = 0$ is double of the other, then which of the following is equal to $f(x)$?
(a) x^2 (b) $2x$ (c) $2x^2$ (d) $\frac{x^2}{2}$

Q.19 If $f(x, y) = 3x^2 - 2xy - y^2 + 4$, then $f(f(2, 3), f(-1, 1))$ is equal to:
(a) -68 (b) 95 (c) 251 (d) 232

Q.20 If $y = f(x) = \frac{x+2}{x-1}$, then it is incorrect to say:
(a) $x = \frac{y+2}{y-1}$ (b) $f(0) = -2$
(c) $f(1) = 0$ (d) $f(-2) = 0$

Direction for Questions 21 and 22: Read the passage below and solve the questions based on it.

A function $f(x)$ is defined as $f(x) = \log(g(x))$ where $g(x)$ is any function of x.

Q.21 For which of the values of $g(x)$, can $f(x)$ be expressed as $f1(x) + f2(x)$, where $f1(x)$ and $f2(x)$ are any two functions of x?
(a) $g(x) = e^x$ (b) $g(x) = x^2$
(c) $g(x) = \log(x)$ (d) None of these

Q.22 For what value of $g(x)$, $g(x) = g(f(x))$?
(a) e (b) $\log x$
(c) e^x (d) None of these

Q.23 For what value of $g(x)$ is the function $f(x) = g(x)$?
(a) e^{-1} (b) $\log x^2$
(c) e^{2x} (d) None of these

Q.24 If $f(t) = \frac{t^3 + 2t}{t-1}$, then $\frac{f(t) + f(-t)}{f(t) - f(-t)}$ is equal to:
(a) t^3 (b) $\frac{1}{t}$ (c) t (d) t^2

Q.25 If $f(x) = 2x^2 + 6x - 1$ and $g(x) = |x + 5|$, then the value off $[g\{f(g(-6))\}]$ is:
(a) 225 (b) 249 (c) 341 (d) 359

Q.26 If $f(x) = x^3 + 1$ and the value of the expression $f(2x) - 2f(x) = 383$, the value of x is:
(a) 1 (b) 0 (c) 4 (d) 6

Direction for Questions 27 and 28: Read the passage below and solve the questions based on it.

$f(x) = 0$ when $x = 1$
$= 1$ when $x = 0$
$= 0.5$ when $x = 0.5$
$g(x) = 1$ when $x = 1$
$= 0.5$ when $x = 0$
$= 0$ when $x = 0.5$

Q.27 If $f[g[f[g(t)]]] = 0$, the value of t is:
(a) 1 (b) 0
(c) 0.5 (d) None of these

Q.28 If $f[g(t) + f(t)] + g[g(t)] = 1$, the value of t is:
(a) 1 (b) 0
(c) 0.5 (d) None of these

Q.29 If $f(x) = x^2$ and $h(x) = \sqrt{x}$, then
(i) h of $(-2) = 4$
(ii) h of $(4) = 4$
(iii) h of $(3) = 6$
(a) (i) is correct but other two are incorrect.
(b) (ii) is correct but other two are incorrect.
(c) (i) and (iii) are correct but (ii) is incorrect.
(d) None of the above is true.

Q.30 Let $h(x) = \frac{1-x}{1+x}$, find the domain of $h^{-1}(x)$.
(i) $(-\infty, +\infty)$ (iii) $(-\infty, -1)$
(ii) $-1, +\infty)$ (iv) $[-1, +1]$
(a) (i) alone is correct
(b) (ii) $\cup$ iii are correct, where $\cup$ represents union of the sets.
(c) (ii) $\cup$ (iii) $\cup$ (iv) are correct, where $\cup$ represents union of the sets.
(d) None of these

Q.31 If $f(n) = 2n^2 - n$ for $n > 5$, $= n^2 + n$ for $n \le 5$, then $f(1) + f(2) + ... + f(10)$ equals:
(a) 705 (b) 440
(c) 690 (d) 715

Q.32 If $y = f(x) = (ax + b)/(cx - a)$, then which of the following is equal to $f(y)$?
(a) x (b) $2x$ (c) $x/2$ (d) x^2

Q.33 If $y = f(x) = (x + 2)/(x - 1)$, then which of the following is equal to $f(y)$?
(a) x (b) $2x$ (c) $x/2$ (d) x^2

Q.34 If $f(x) = \frac{x}{1-x}$, then find the value of $\frac{f\left(\frac{a}{b}\right)}{f\left(\frac{b}{a}\right)}$.
(a) a/b (b) $-b/a$ (c) $-a/b$ (d) b/a

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Q.35 Find the domain of the function $y=f(x)$, which is defined as $f(x)\frac{1}{\sqrt{|x|+x}}$.
(a) $(-\infty, +\infty)$ (b) $(0, +\infty)$
(c) $(0, +\infty)$ (d) $(1, +\infty)$

Q.36 Find the domain of the function $y=f(x)$, which is defined as $f(x)=\frac{1}{\sqrt{\{x+[x]\}}}$=; $[x]$ is the greatest integer function.
(a) x is any real number other than integers.
(b) Any real value of x
(c) All the natural numbers
(d) None of these

Q.37 Are these two functions identical?
$f(x) = \log (x -2) + \log (x - 3)$ and $g(x) = \log (x - 2)(x - 3)$
(a) Yes (b) No
(c) Cannot be determined (d) None of these

Q.38 Are these two functions identical?
$f(x) = 2 \log x$ and $g(x) = \log x^2$
(a) Yes (b) No
(c) Cannot be determined (d) None of these

Q.39 The domain of the function $f(x) = \log_{10} \log_{10}(1 + x^3)$ is:
(a) $(-1, +\infty)$ (b) $(0, +\infty)$
(c) $(0, +\infty)$ (d) $(-1, 0)$

Q.40 The set of real values of x satisfying $|x - 1| \le 3$ and $|x - 1| \ge 1$ is:
(a) $[2, 4]$ (b) $(-\infty, 2) \cup [4, +\infty]$
(c) $[-2, 0] \cup [2, 4]$ (d) None of these

Moderate

Q.1 If $b=f(a)$ and $f(a)=\frac{a-1}{a+1}$, then which of the following is true?
(a) $f(2a)=f(a)+1$ (b) $f\left(\frac{1}{a}\right)=-f(a)$
(c) $a=f(b)+f\left(\frac{1}{a}\right)$ (d) $a=f(b)$

Q.2 Define $x \blacklozenge y$ to be $|x - y|$ for all real numbers x and y. Which of the following statements is not true?
(a) $x \blacklozenge y = y \blacklozenge x$ for all x and y
(b) $2(x \blacklozenge y) = (2x) \blacklozenge (2y)$ for all x and y
(c) $x \blacklozenge 0 = x$ for all x
(d) $x \blacklozenge x = 0$ for all x

Direction for Questions 3 and 4: Read the passage below and solve the questions based on it.

$[x]$ = Greatest integer less than or equal to x.
$\{x\}$ = Smallest integer more than or equal to x.

Q.3 Given that 'x' is not an integer, then what is the value of $[x] - \{x\}$?
(a) -1
(b) 1
(c) 0
(d) Depends upon the value of x

Q.4 Given that 'x' is not an integer, then $[x] + \{x\}$ is:
(a) An even integer
(b) An odd integer
(c) Positive number
(d) Non-positive integer

Q.5 let $f(n)=\left[\frac{1}{2}+\frac{n}{100}\right]$, where $[x]$ denotes the integral part of x. Then, the value of $\sum_{n=1}^{100} f(n)$ is:
(a) 50 (b) 51
(c) 1 (d) None of these

Q.6 If $0 < x < 1000$ and $\left[\frac{x}{2}\right]+\left[\frac{x}{3}\right]+\left[\frac{x}{5}\right]=\frac{31}{30}x$, where $[x]$ is the greatest integer less than or equal to x, then the number of possible values of x is:
(a) 34 (b) 32
(c) 33 (d) None of these

Q.7 If $f(x) = 1/x$, $x \neq 0$; $f^{n+1}(x) = f^n(f(x))$, find the value of the product $f^{13}(1) \times f^{15}(1) \times f^{19}(1) f^{21}(1)$.
(a) 34 (b) 16 (c) 1 (d) 17

Q.8 The set of real values of x satisfying $||x - 1| - 1| \le 1$ is:
(a) $[-1, 3]$ (b) $[0, 2]$
(c) $[-1, 1]$ (d) None of these

Q.9 A function is defined as $f^n(x) = f\{f^{n-k}(x)\}$
If n is odd, $k = 1$.
If n is even, $k = 2$ and $f^0(x) = 1$.
Find $f^7(x)$ at $x = 1$, if $f(x) = x^2 + 1$.
(a) 676 (b) 677 (c) 577 (d) 1076

Q.10 $(x^2 - 1)$ is a factor of $f(x) = (x^5 + ax^4 + bx^3 + cx^2 + x + d)$. The graph of $f(x)$ intersects Y axis at $(0, -3)$. Find the value of $(a + c)$.
(a) 0 (b) 3 (c) -3 (d) -1

Q.11 Let $f(x)$ denote the sum of the digits of the positive integer x. For example, $f(8) = 8$ and $f(123) = 1 + 2 + 3 = 6$. For how many two-digit values of x is $f(f(x)) = 3$?
(a) 3 (b) 4 (c) 10 (d) 9

Q.12 If $f(x+1)+f(x-1)=2f(x)$ and $f(0)=0$, then $f(n)$, $n \in N$ is:
(a) $n f(1)$ (b) $\{f(1)\}^n$
(c) 0 (d) None of these

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Q.13 Let $f(x) = 1 + |x|, x < -1$
$[x], x \geq -1$, where [.] denotes the greatest integer function. Then, $f\{f(-2.3)\}$ is equal to:
(a) 4 (b) 2 (c) −3 (d) 3

Q.14 The largest set of real values of x for which $f(x) = \sqrt{(x+2)(5-x)} - \frac{1}{\sqrt{x^2-4}}$ is a real function is:
(a) $[1, 2] \cup [2, 5]$ (b) $[2, 5]$
(c) $[3, 4]$ (d) None of these

Q.15 Let a, b, and c be fixed positive real numbers. Let $f(x) = \frac{ax}{b+cx}$ for $x \geq 1$. Then, as x increases,
(a) $f(x)$ increases.
(b) $f(x)$ decreases.
(c) $f(x)$ increases first and then decreases.
(d) None of these

Q.16 For any real number x, let (x) denote the largest integer less than or equal to x and $< x > = x - [x]$, that is, the fractional part of x. For arbitrary real numbers x, y, and z, only one of the following statements is correct. Which one is it?
(a) $[x + y + z] = [x] + [y] + [z]$
(b) $[x + y + z] = [x + y] + [z] = [x] + [y + z] + [y]$
(c) $< x + y + z > = y + z - [z + z] + < x >$
(d) $[x + y + z] = [x + y] + [z + < y + x >]$

Q.17 The domain of the function $f(x) = {}^{16x}C_{2x-1} + {}^{20-3x}P_{4x-5}$ where the symbols have their usual meanings, is the set:
(a) [1, 2, 3, 4, 5] (b) {2, 3, 4}
(c) {2, 3} (d) None of these

Q.18 The domain of the real valued function $f(x) = \log_e |\log_e x|$ is:
(a) $(1, +\infty)$ (b) $(0, +\infty)$
(c) $(e, +\infty)$ (d) None of these

Q.19 The domain of the function $f(x) = \log_e(x - [x])$, where [.] denotes the greatest integer function, is:
(a) R (b) $R - Z$
(c) $(0, +\infty)$ (d) None of these

Q.20 Let $f(x) = \log_{x2} 25$ and $g(x) = \log_x 5$, then $f(x) = g(x)$ holds for x belonging to:
(a) All the real values
(b) $(0, 1) \cup (2, +\infty)$
(c) ϕ
(d) None of these

Q.21 Let $f(x) = |x - 2| + |x - 3| + |x - 4|$ and $g(x) = f(x + 1)$. Then,
(a) $g(x)$ is an even function.
(b) $g(x)$ is an odd function.
(c) $g(x)$ is neither even nor odd.
(d) None of these

Q.22 If $f(x) = x^n$, $n \in N$ and g of $(x) = n\, g(x)$, then $g(x)$ can be:
(a) $n|x|$ (b) $3.\sqrt[3]{x}$ (c) e^x (d) $\log |x|$

Q.23 The inverse function of the function $f(x) = f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$ is:
(a) $\frac{1}{2}\log\frac{1+x}{1-x}$ (b) $\frac{1}{2}\log\frac{2+x}{2-x}$
(c) $\frac{1}{2}\log\frac{1-x}{1+x}$ (d) None of these

Q.24 The graph of the function $y = f(x)$ is symmetrical about the line $x = 2$. Then:
(a) $f(x + 2) = f(x - 2)$
(b) $f(2 + x) = f(2 - x)$
(c) $f(x) = f(-x)$
(d) None of these

Q.25 If $f(x) = \log \frac{(1+x)}{1-x}$ and $g(x) = \frac{3x + x^3}{1 + 3x^2}$, then $f[g(x)]$ is equal to:
(a) $[f(x)]3$ (b) $-f(x)$
(c) $[f^3(x)]$ (d) $3[f(x)]$

Q.26 If $f(x) = 2x^2 + 3x + 4$ and $g(x) = 5 - x^2$, then the minimum $[f(x)]$ − maximum $[g(x)]$ is:
(a) 0 (b) 5 (c) $\frac{17}{8}$ (d) $-\frac{17}{8}$

Q.27 If $f(x) = x^3 - x^2 + x + f(x)$, where x is a whole number less than or equal to 15. For how many values of x, $f(x)$ will not be a whole number?
(a) 7 (b) 8
(c) 5 (d) None of these

Q.28 For any real number x, define Mag(x) = Maximum $(x, -x)$ and Nag(x) = Minimum $(x, -x)$. What is the value of Mag (8, (Nag (4, (Nag (7, −7))))) − (Mag (6,(Nag (7, −7)))?
(a) 7 (b) 0 (c) 2 (d) −14

Q.29 If $f(x) = \frac{4^x}{4^x + 2}$, then find the value of $f\left(\frac{1}{1999}\right) + f\left(\frac{2}{1999}\right) + \ldots + f\left(\frac{1998}{1999}\right)$.
(a) 1998 (b) 1999 (c) 998 (d) 999

Q.30 If $f(x) = 2x^2 + 6x - 1$, then the value of $\frac{f\left(\frac{3}{4}\right)+1}{f\left(\frac{3}{4}\right)-1}$ is:
(a) 11/13 (b) 35/3
(c) 45/29 (d) None of these

Q.31 Let $\{x\}$ and $[x]$ denote the fractional and integral part of a real number x, respectively. Solve $4\{x\} = x + |x|$.
(a) 2/3 (b) 4/7 (c) 8/3 (d) 5/3

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Direction for Questions 32 and 33: Read the passage below and solve the questions based on it.

Consider a function $f(n)$ defined for non-negative integer values of n.

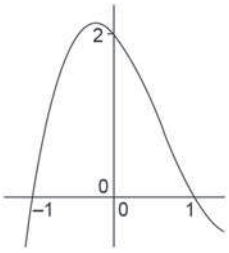
$f(n) = (-1)^{sn+1} + (-1)^{S_n} + f(n-1)$, where $f(0) = 1$ and si = 1 + 2 + 3 + 4 + … + i, for all $i \in N$

Q.32 Find $f(111)$.
(a) 1 (b) −1 (c) 2 (d) 0

Q.33 Evaluate $f(1) + f(2) + f(3) + \ldots f(25)$.
(a) −26 (b) −24
(c) −22 (d) None of these

Q.34 Let $f(x)$ be the greatest integer function and $g(x)$ be the modulus function, then what is the value of $(gof)\left(-\frac{5}{3}\right) + (fog)\left(-\frac{5}{3}\right)$?
(a) 3/5 (b) 11/3 (c) 2/7 (d) 3

Q.35 Part of the graph of $f(x) = ax^3 + bx^2 + cx + d$ is shown. What is b?



(a) −4 (b) −2 (c) 0 (d) 2

Q.36 In the sequence 2001, 2002, 2003,..., each term after the third is found by subtracting the previous term from the sum of the two terms that precede that term. For example, the fourth term is 2001 + 2002 − 2003 = 2000. What is the 2004th term in this sequence?
(a) −2004 (b) −2 (c) 0 (d) 4003

Q.37 If $f(x) = \log \frac{1+x}{1-x}$, then $f(x) + f(y)$ is:
(a) $f(x+y)$ (b) $f\left(\frac{x+y}{1+xy}\right)$
(c) $(x+y) f\left(\frac{1}{1+xy}\right)$ (d) None of these

Q.38 If $f(x, y) = |x + y|$ and $g(x, y) = |x - y|$, how many ordered pairs of the form (x, y) would satisfy $f(x, y) = g(x, y)$?
(a) 1 (b) 2
(c) 4 (d) Infinitely many

Q.39 $f(x) = y + \frac{1}{y}$, where $y > 0$. If y increases in value, then $f(x)$
(a) fluctuates up and down in value.
(b) decreases in values.
(c) increases in value but at a much higher rate than y.
(d) increases in value and tends towards y.

Q.40 A function H is defined for all the positive integers that satisfy the following condition:
$H(1) + H(2) + H(3) + \ldots + H(x) = x^2 H(x)$.
If $H(1) = 2006$, then find the value of $H(2005)$.
(a) $\frac{2}{2005}$ (b) $\frac{2}{2006}$ (c) $\frac{2}{2006!}$ (d) $\frac{2}{2005!}$

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Q.1 If $a < b < c < d < e$ and $f(x) = (x-a)^2 (x-b)(x-c)(x-d)(x-e)$, then which of the following is true?
(a) $f(x) > 0$, for $(x < a)$ as well as for $(d < x < e)$
(b) $f(x) < 0$, for $(a < x < b)$ as well as for $(d < x < e)$
(c) $f(x) < 0$, for $(b < x < c)$ as well as for $(d < x < e)$
(d) None of these

Q.2 The graph of the polynomial $f(x) = x^5 + ax^4 + bx^3 + cx^2 + dx + e$ has five distinct x-intercepts, one of which is at (0, 0). Which of the following coefficients cannot be zero?
(a) a (b) b (c) c (d) d

Q.3 $[x]$ is defined as the greatest integer less than or equal to x, and $\{x\}$ is defined as the least integer greater than or equal to x, for all real values of x. Consider the four statements.
1. $[x] + [-x] = -1$, if x is not an integer.
2. $\{x\} + [-x] = 0$
3. $\{[x]\} - [\{x\}] = -1$
4. $[2x] + \{3x\} \le 5x$

How many of the above statements are always true for all the real values of x?
(a) 1 (b) 2 (c) 3 (d) 4

Q.4 If $f(x) = \frac{1-x}{1+x}$, then $f(ax)$ in terms of $f(x)$ is equal to:
(a) $\frac{f(x)+a}{1+af(x)}$ (b) $\frac{(a-1)f(x)+a+1}{(a+1)f(x)+a-1}$
(c) $\frac{(a+1)f(x)+a-1}{(a-1)f(x)+a+1}$ (d) None of these

Q.5 Let $f(1) = 1$ and $f(n) = \sum_{r=1}^{n-1} f(r)$. Then, $\sum_{r=1}^{m} f(n)$ is equal to:
(a) 3^{m-1} (b) $3m$
(c) 3^{m-1} (d) None of these

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Direction for Questions 6 and 7: Read the passage below and solve the question based on it.

A formula is given by $a(n) = a(a(n-1)) + a(n - a(n-1))$, where n is any integer greater than 2.

Q.6 If it is known that $a(1) = 1$ and $a(2) = 1$, find the value of $a(4)$.
(a) 1 (b) 2 (c) 3 (d) 4

Q.7 If it is known that $a(2) = 1$ and $a(3) = 2$, find the value of $a(1)$.
(a) -1 (b) 0
(c) 1 (d) Cannot be determined

Q.8 Which of the following functions satisfies the condition? $f\left(\frac{x+y}{x-y}\right) = \frac{f(x)+f(y)}{f(x)-f(y)}$?
(a) $f(x) = x$ (b) $f(x) = ax + b$
(c) $f(x) = 2^x$ (d) $f(x) = x^2$

Q.9 $[N]$ denotes the largest integer less than or equal to N. x is a positive integer and $\left[\frac{x}{5}\right] - \left[\frac{x}{7}\right] = 1$. If the minimum value of x is r and the maximum value of x is s, then the value of $r + s$ is equal to:
(a) 40 (b) 33 (c) 35 (d) 34

Q.10 $f(x) = |x| + |y|$
$g(x) =$ maximum $(x + y, x - y)$
$h(x) =$ minimum $(x + y, x - y)$
(i) $g(x) \geq f(x)$
(ii) $g(x) + h(x) \geq f(x)$
(iii) $g(x) > f(x)$
Which of the following are not necessarily true?
(a) (i) and (ii) (b) (i) and (ii)
(c) (ii) and (iii) (d) (i), (ii) and (iii)

Direction for Questions 11 and 12: Read the instructions given below and solve the questions based on it.

Following are the certain functions:
$H(a, b, c)$ = Greatest common divisor of a, b, and c
$L(a, b, c)$ = Least common multiple of a, b, and c
$A(a, b, c)$ = Average of a, b, and c
Minimum (a, b, c) = Smallest value among a, b, and c
Maximum (a, b, c) = Largest among a, b, and c

Q.11 Which of the following is true? (a, b, and c are distinctly different positive numbers.)
(a) $H(a, b, c) \times L(a, b, c) = abc$
(b) $H(a, b, c) > A(a, b, c)$
(c) $H(a, b, c) >$ Minimum (a, b, c)
(d) $H(a, b, c) < A(a, b, c) < L(a, b, c)$

Q.12 If max(a, b, c) = minimum(a, b, c), then:
(a) $A(a, b, c) = H(a, b, c)$
(b) $A(a, b, c) = L(a, b, c)$
(c) $A(a, b, c) =$ minimum(a, b, c)
(d) All of these

Q.13 If the area of a circle increases at a uniform rate, then the rate of increase in the perimeter is
(a) directly proportional to the radius.
(b) inversely proportional to the radius.
(c) directly proportional to the square of the radius.
(d) inversely proportional to the square of the radius.

Q.14 If $f(x) = 3^x + 4^x + 5^x - 6^x$, then $f(x) < f(3)$ for
(a) only one value of x.
(b) no value of x.
(c) only two values of x.
(d) infinitely many values of x.

Direction for Questions 15 and 16: Read the passage below and solve the questions based on it.

For a natural number x, if x has two or more than two digits, then $f(x)$ = the sum of all digits of x and $f(x) = x$, when x has only one digit. Also, given that $f^n(x) = f\{f^{n-1}(x)\}$, where $n \geq 2$.

Q.15 If $x = 123456^{654321}$, then $f^{\infty}(x)$ is equal to:
(a) 9 (b) 3 (c) 1 (d) 7

Q.16 If $x = 7^{183}$, then $f(x)$ is equal to:
(a) 7 (b) 4 (c) 1 (d) 9

Q.17 If $f(x) = ax + b$ and $f^{-1}(x) = bx + a$ with a and b real, what is the value of $a + b$?
(a) 0 (b) -1 (c) -2 (d) 1

Q.18 Let f be a linear function for which $f(6) - f(2) = 12$ What is $f(12) - f(2)$?
(a) 12 (b) 18 (c) 24 (d) 30

Direction for Questions 19 and 20: Read the passage below and solve the questions based on it.

A function f is defined for $x, y \in$ such that
$f(0, y) = y^2 - 1$
$f(x + 1, y) = f(x, y + 4)$

Q.19 Find the value off (7, 3).
(a) 728 (b) 840
(c) 960 (d) Cannot be determined

Q.20 Which of the following is true?
(a) $f(8, 4) = f(0, 32)$
(b) $f(0, 0) = f(-5, 20)$
(c) $f(-2, 9) = f(3, 29)$
(d) All of these

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Q.21 $f(x)$ = Sum of the base-10 logarithms of the divisors of a number x.
Given that the $f(10^n) = 792$. What is n?
(a) 10 (b) 11 (c) 13 (d) 14

Q.22 The set of all real numbers x for which $\log_{2004}(\log_{2003}(\log_{2002}(\log_{2001}$ is defilled is $\{x|x > c\}$. What is the value of c?
(a) 0 (b) 2003^{2004}
(c) 2002^{2003} (d) 2001^{2002}

Q.23 The following computer programme is a very standard example of an iterative function:
A computer performs a certain algorithm to compute an output S for an input N, with both S and N being real values. The algorithm is as follows:
Step 1 $S = 0$
Step 2 If ($N \le -1$ or $1 \le N$), then $S = S + 1$ else go to Step 5
Step 3 $N = N \div 10$
Step 4 Go to step 2
Step 5 Print S
Which of the following best describes the algorithm?
(a) It calculates the highest power of 10 that can perfectly divide N.
(b) It calculates the number of digits in N.
(c) It calculates the number of zeros on the extreme right of N.
(d) It calculates the number of digits to the left of the decimal point in N.

Q.24 Let f be a function with the following properties:
(i) $f(1) = 1$ and
(ii) $f(2n) = n \times f(n)$, for any positive integer n.
What is the value of $f(2^{100})$?
(a) 2^{4950} (b) 2^{99}
(c) 2^{100} (d) 2^{9999}

Q.25 If $f(x, y) = 2x^2 + 4xy - y^2$, then what is the value of $f(x + 1, y - 1)$?
(a) $2y^2 + 4xy - x^2 - 6y + 3$
(b) $2x^2 + 4xy - y^2 + 6y - 3$
(c) $2y^2 + 4xy - x^2 + 6y - 3$
(d) $2x^2 - 2xy + 2y^2 - 6y + 3$

Direction for Questions 26 to 29: Read the passage below and solve the questions based on it.

If $P(x, y, z)$ = maximum (x, y) + minimum (y, z)
$Q(x, y, z)$ = maximum (y, z) + minimum (x, y)
$R(x, y, z)$ = maximum (x, y, z)
$S(x, y, z)$ = minimum (x, y, z)
$T(x, y, z)$ = maximum (x, z) − minimum (y, z)
$U(x, y, z)$ = maximum (y, z) − minimum (x, y)

Q.26 For $x = 1$, $y = 2$, and $z = 3$, which of the following is equal to 1?
(a) $\dfrac{p(x,y,z) + S(x,y,z)}{2 \times U(x,y,z)}$ (b) $\dfrac{Q(x,y,z) + S(x,y,z)}{2 \times T(x,y,z)}$
(c) $\dfrac{Q(x,y,z) + U(x,y,z)}{2 \times R(x,y,z)}$ (d) $\dfrac{R(x,y,z) + S(x,y,z)}{2 \times P(x,y,z)}$

Q.27 For $x = 1$, $y = 2$, and $z = 3$, which of the following is greater than 1?
(a) $\dfrac{P(x,y,z)}{2 \times S(x,y,z)}$ (b) $\dfrac{U(x,y,z)}{Q(x,y,z)}$
(c) $\dfrac{S(x,y,z)}{R(x,y,z)}$ (d) $\dfrac{T(x,y,z)}{R(x,y,z)}$

Q.28 For $x = 1$, $y = 2$, and $z = 3$, which of the following is equal to zero?
(a) $P(x, y, z) + Q(x, y, z)$
(b) $Q(x, y, z) - R(x, y, z) - S(x, y, z)$
(c) $R(x, y, z) - S(x, y, z) - T(x, y, z)$
(d) $T(x, y, z) - U(x, y, z) - S(x, y, z)$

Q.29 For $x = 1$, $y = 2$, and $z = 3$, which of the following is less than 1?
(a) $\dfrac{Q(x,y,z)}{R(x,y,z) + S(x,y,z)}$ (b) $\dfrac{Q(x,y,z) - U(x,y,z)}{2S(x,y,z)}$
(c) $\dfrac{Q(x,y,z) + U(x,y,z)}{R(x,y,z) + T(x,y,z)}$ (d) $\dfrac{U(x,y,z)}{R(x,y,z) + S(x,y,z)}$

Q.30 $f(x) = x^3 + ax^2 + bx + c$ and $g(x) = x^2 + 1$, where a, b, and c belongs to $\{1, 2, 3, ...,10\}$. What are the number of polynomials which are divisible by $g(x)$?
(a) 0 (b) 1 (c) 10 (d) 100

Direction for Questions 31 to 34: Read the passage below and solve the questions based on it.

The pages of a book are numbered 0, 1, 2, ... P, $P > 0$. There are four categories of instructions that direct a person in positioning the pages in the book. The instruction types with their meanings are given below:
(i) Open position of the book at page number 1.
(ii) Close position of the book at page number 0.
(iii) Forward, n from the current page, move forward by n pages. If, in this process, page number P is reached, stop at P.
(iv) Backward, n from the current page, move backward by n pages. If, in this process, page number 0 is reached, stop at page number 0.

In each of the following questions, you will find a sequence of instructions. Let $p1$ be the page number before the instructions are given and $p2$ be the page number after executing all the given instructions.

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Q.31 Forward, 30; Backward, 12. Which of the following statements is true?
(a) $p1 = p2$, if $P = 12$ and $p1 = 0$
(b) $P = 25$, provided $p1 > 0$
(c) $p1 > 35$, provided $P = 100$
(d) $p1 = 40$, provided $P = 30$

Q.32 Backward, 7; Forward, 7. Which of the following statements is true?
(a) $p1 = p2$, if $p1 > 7$
(b) $p1 = p2$, if $p1 > 0$
(c) $p2 = 7$, provided $P > 0$
(d) $p1 > p2$, provided $P > 0$

Q.33 Forward, 15; Forward, 15. Which of the following statements is true?
(a) $p2 - p1 = 30$, only if $p1 = 0$
(b) $p2 - p1 = 30$, only if $p1 = 1$ and $P > 30$
(c) $p2 - p1 = 15$, only if $p1 = 0$ and $P = 31$
(d) $p2 > p1$, if $P > 0$

Q.34 Forward, 5; Backward. Which of the following statements is true?
(a) $p2 = p1 + 4$ provided $1 < p1 < 7$
(b) $p2 = p1$, if $P < 6$
(c) $p2 = p1 + 1$, if $P - p1 \geq 5$
(d) $p2 - p1 < 0$, if $P > 0$

Q.35 If $f(x) = a^{1+x}$, then $f(x).\ f(y).\ f(z)$ is equal to?
(a) $f(x + y + z)$
(b) $f(x + y + z + 1)$
(c) $f(x + y + z + 2)$
(d) $f(x + y + z + 3)$

Q.36 A polynomial function $f(x)$ satisfies $f(x)\,f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right)$. If $f(10) = 1001$, then what is the value of $f(20)$?
(a) 2002 (b) 8004
(c) 8001 (d) None of these

Answers

Warm Up

1. (d)	**2.** (c)	**3.** (b)	**4.** (a)	**5.** (d)	**6.** (b)	**7.** (c)	**8.** (b)	**9.** (b)	**10.** (c)
11. (d)	**12.** (d)	**13.** (a)	**14.** (a)	**15.** (b)	**16.** (c)	**17.** (b)	**18.** (d)	**19.** (b)	**20.** (a)

Foundation

1. (d)	**2.** (d)	**3.** (d)	**4.** (b)	**5.** (d)	**6.** (d)	**7.** (b)	**8.** (a)	**9.** (c)	**10.** (b)
11. (d)	**12.** (d)	**13.** (d)	**14.** (d)	**15.** (c)	**16.** (b)	**17.** (c)	**18.** (c)	**19.** (b)	**20.** (c)
21. (d)	**22.** (d)	**23.** (d)	**24.** (c)	**25.** (d)	**26.** (c)	**27.** (c)	**28.** (a)	**29.** (d)	**30.** (b)
31. (c)	**32.** (c)	**33.** (a)	**34.** (c)	**35.** (b)	**36.** (a)	**37.** (b)	**38.** (b)	**39.** (b)	**40.** (c)

Moderate

1. (b)	**2.** (c)	**3.** (a)	**4.** (b)	**5.** (a)	**6.** (c)	**7.** (c)	**8.** (a)	**9.** (b)	**10.** (b)
11. (c)	**12.** (a)	**13.** (d)	**14.** (b)	**15.** (a)	**16.** (d)	**17.** (c)	**18.** (d)	**19.** (b)	**20.** (d)
21. (c)	**22.** (d)	**23.** (a)	**24.** (b)	**25.** (d)	**26.** (d)	**27.** (b)	**28.** (c)	**29.** (d)	**30.** (d)
31. (d)	**32.** (a)	**33.** (d)	**34.** (d)	**35.** (b)	**36.** (c)	**37.** (b)	**38.** (d)	**39.** (d)	**40.** (a)

Advanced

1. (c)	**2.** (d)	**3.** (b)	**4.** (c)	**5.** (b)	**6.** (b)	**7.** (c)	**8.** (a)	**9.** (d)	**10.** (d)
11. (d)	**12.** (d)	**13.** (a)	**14.** (d)	**15.** (a)	**16.** (c)	**17.** (c)	**18.** (d)	**19.** (c)	**20.** (c)
21. (b)	**22.** (d)	**23.** (a)	**24.** (a)	**25.** (b)	**26.** (c)	**27.** (a)	**28.** (b)	**29.** (d)	**30.** (c)
31. (a)	**32.** (a)	**33.** (b)	**34.** (c)	**35.** (c)	**36.** (c)				

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Hints and Solutions

Warm Up

1. As it is a modulus function, for every values of x, it will become positive.

So, for every value of x, we can find out the value of $f(x)$.

Hence, the domain is $-\infty < x < \infty$. In other words, for all the real value of x.

2. Expression under square root should be non-negative. So, we cannot take negative values.

Hence, the domain is $0 \le x < \infty$.

3. Expression under square root should be non-negative. So, we cannot take negative values.

We cannot take zero also because for zero, the value of $f(x)$ will become infinity. So, the domain is $0 < x < \infty$.

4. We can take every value of x because x^2 will become a positive number every time.

$\therefore$ The domain is $-\infty < x < \infty$.

5. For $\sqrt{4x-3}$, $4x - 3 \ge 0$

[since we cannot find under root of negative values.]

or $4x \ge 3$. So, $x \ge \frac{3}{4}$

For $\sqrt{2x-6}$, $2x - 6 \ge 0 \rightarrow x \ge 3$

So, for both the equations, x should be greater than or equal to 3. The domain is $[3, \infty)$. Hence, the domain is $3 \le x < \infty$.

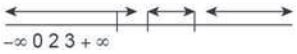
6. $f(x) = \log(5x - 6 - x^2)$

Logarithmic function is defined only for positive values.

So, $5x - 6 - x^2 > 0$ or $x^2 - 5x + 6 < 0$

Roots of this equation are 2 and 3.

Put them on a number line.



Take a value from the internal 2 to 3 (say 2.5), it satisfies the inequality.

It can be seen that value of x lies in between 2 and 3. So, the domain is $(2, 3) = 2 < x < 3$.

7. $y = \log_x 5$

To find log, base should be positive $\neq 1$.

Hence, the domain is $0 < x < \infty$ except $x = 1$.

8. $f(x) = \dfrac{x}{\sqrt{1+x^2}}$

So, $f(f(f(x))) = f\left(f\left(\dfrac{x}{\sqrt{1+x^2}}\right)\right)$

Put $x = \dfrac{x}{\sqrt{1+x^2}}$

So, $f(f(f(x))) = f\left(f\left(\dfrac{x}{\sqrt{1+x^2}}\right)\right) \rightarrow f\left(\dfrac{x/\sqrt{1+x^2}}{\sqrt{1+\left(\dfrac{x}{\sqrt{1+x^2}}\right)^2}}\right)$

$f\left(\dfrac{x/\sqrt{1+x^2}}{\sqrt{1+x^2/1+x^2}}\right) \rightarrow f\left(\dfrac{x/\sqrt{1+x^2}}{\sqrt{1+2x^2/1+x^2}}\right)$

$\rightarrow f\left(\dfrac{x/\sqrt{1+x^2}}{\sqrt{1+2x^2/\sqrt{1+x^2}}}\right)$ ($\sqrt{1+x^2}$ is called from both sides).

$= f\left(\dfrac{x}{\sqrt{1+2x^2}}\right)$

Again put $x = \dfrac{x}{\sqrt{1+2x^2}}$

So, $f(f(f(x))) = \dfrac{x/\sqrt{1+2x^2}}{\sqrt{1+2\left(\dfrac{x}{\sqrt{1+2x^2}}\right)^2}} = \dfrac{x}{\sqrt{1+3x^2}}$

9. $f(x) = \dfrac{x}{x+3}$

Then, $f(x+1) = \dfrac{x+1}{(x+1)+3} = \text{put } x = (x+1) = \dfrac{x+1}{x+4}$

10. $f(x) = \dfrac{x+1}{x-1}$

Then, $f(f(f(f(f(2))))) = f\left(f\left(f\left(f\left(\dfrac{2+1}{2-1}\right)\right)\right)\right)$ (put $x = 2$)

$\rightarrow f(f(f(f(3)))) = f\left(f\left(f\left(\dfrac{3+1}{3-1}\right)\right)\right)$ (put $x = 2$)

$= f\left(f\left(f\left(\dfrac{4}{2}\right)\right)\right) = f(f(f(2)))$

$= f(f(3))$ [since $f(f(2)) = f(3)$, we have found it]

$= f(f(3)) = f(2)$ [since $f(f(3)) = f(2)$, we have found it]

$= f(2) = 3$

11. $f(x) = \sqrt{x^5}$

Then, $f(5x) = \sqrt{(5x)^5}$ (put $x = 5x$)

$= \sqrt{(5x)^4 . 5x} = (5x)^2\sqrt{5x}$

$= 25x^2 = \sqrt{5x}$

$\therefore$ Answer is none of these.

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12. $f(x) = 4x - 5$ and $g(x) = x^2$ and $h(x) = \frac{1}{x}$

Let us now compute $f(g(h(x)))$.

Since $h(x) = \frac{1}{x}$, $g(h(x)) = g\left(\frac{1}{x^2}\right)\left(\frac{1}{x}\right)^2$ since $g(x) = x^2$]

Hence, $f(g(h(x))) = f\left(\frac{1}{x^2}\right)$ $\left[\text{since } g(h(x)) = \frac{1}{x^2}\right]$

$$= 4\left[\frac{1}{x^2}\right] - 5 \left[\begin{matrix}\text{since} = 4x - 5 \\ \text{put } x = \frac{1}{x^2}\end{matrix}\right]$$

$$= \frac{4}{x^2} - 5$$

Hence, option (d) is the answer.

13. $f\left(500 \times \frac{6}{5}\right) = \frac{3}{\frac{6}{5}} = \frac{5}{2}$; so, the answer is (a).

14. If $f(-x) = f(x)$, then it is a even function. After verifying the options, we can see that option (b) is an even function.

Because $f(x) = x = \frac{e^{ax} - 1}{e^{ax} + 1}$

Then, $f(-x) = (-x) = \frac{e^{a(-x)} - 1}{e^{a(-x)} + 1} = -x\frac{e^{-ax} - 1}{e^{-ax} + 1}$

$$= -x\frac{\frac{1}{e^{ax}} - 1}{\frac{1}{e^{ax}} + 1} = -x\frac{(1 - e^{ax})/e^{ax}}{(1 + e^{ax})/e^{ax}}$$

$$= -x\frac{(1 - e^{ax})}{1 + e^{ax}}$$

(Take '–' sign out of bracket)

$$= -x\frac{-(e^{ax} - 1)}{(e^{ax} + 1)} = x\frac{e^{ax} - 1}{e^{ax} + 1} = f(x)$$

15. If $f(-x) = f(x)$, then it is a even function. After verifying the options, we can see that option (b) is an even function.

Option (b) will be even function because

$f(x) = x\frac{a^x + 1}{a^x - 1}$

Then, $f(-x) = -x\frac{a^{-x} + 1}{a^{-x} - 1}$

$$= -x\frac{\frac{1}{a^x} + 1}{\frac{1}{a^x} - 1} = -x\frac{(1 + a^x)/a^x}{(1 - ax)/a^x}$$

$$= -x\frac{(1 + a^x)}{(1 - a^x)} = -x\frac{a^3 + 1}{-(a^3 - 1)})$$

$$= x\frac{a^3 + 1}{a^3 - 1} = f(x)$$

16. $f(x) = \frac{x}{e^x - 1} + \frac{x}{2} + 1$

find $f(-x) = \frac{-x}{e^{-x} - 1} + \frac{-x}{2} + 1$

$$= -\frac{x}{\frac{1}{e^{-x}} - 1} - \frac{x}{2} + 1$$

$$= -\frac{x}{(1 - e^x)/e^x} - \frac{x}{2} + 1$$

$$= -\frac{xe^x}{1 - e^x} - \frac{x}{2} + 1$$

which is neither $f(x)$ nor $-f(x)$.

So, it is neither even nor odd function.

17. $f(x) = \frac{1}{1 - x}$ (i)

$g(x) = f(f(x)) \rightarrow f\left(\frac{1}{1 - x}\right)$ $\left[\text{since } f(x) = \frac{1}{1 - x}\right]$

$$= \frac{1}{1 - \left(\frac{1}{1 - x}\right)} \left[\text{put } x = \frac{1}{1 - x}\right]$$

$$= \frac{1}{(1 - x - 1)/(1 - x)} = \frac{1 - x}{-x} = \frac{x - 1}{x} \quad \text{(ii)}$$

$h(x) = f(f(f(x))) = f(g(x)$ [since $g(x) = f(f(x))$] $= f(g(x)) =$ $f\left(\frac{x - 1}{x}\right)$ [from equation (ii)]

$$= \frac{\left(\frac{1}{1 - x}\right) - 1}{\left(\frac{1}{1 - x}\right)} \left[\text{put } x = \frac{1}{1 - x}, \text{ from equaiton (i)}\right]$$

$$= \frac{1 - (1 - x)}{\frac{1 - x}{1}} = \frac{1 - 1 + x}{1} = x$$

Then, $f(x).\ g(x).\ h(x) = \frac{1}{1 - x} \times \frac{x - 1}{x} \times x = -1$

18. $f(x) = x^2 + 2$

Let $y = x^2 + 2 - y - 2 = x^2 \rightarrow x = \sqrt{y - 2}$

Then, put $y = x$

$f^{-1}(x) = \sqrt{x - 2}$

This is not matching with the given conditions. So, the answer is none of these.

19. $f(x) = y = \frac{x}{x + 1}$

$= y = \frac{x}{x + 1}$ or, $y - 1 = \frac{x}{x + 1} - 1$

[subtract 1 from both sides]

$y - 1 = \frac{x - x - 1}{x + 1} = \frac{1}{x + 1}$

$x + 1 = -\frac{1}{y - 1} = \frac{1}{1 - y}$

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$$x = \frac{1}{1-y} - 1 = \frac{1-1+y}{1-y} = \frac{y}{1-y}$$

So, $f^1(x) = \frac{x}{1-x}$

20. $f(x) = y = \frac{x-1}{x+1}$

$$y - 1 = \frac{x-1}{x+1} = -1 \qquad \text{[subtract 1 from both sides]}$$

$$y - 1 = \frac{x-1-x-1}{x+1} = -\frac{2}{x+1}$$

$$x + 1 = -\frac{2}{y-1} = \frac{2}{1-y}$$

$$x = -\frac{2}{1-y} - 1 = \frac{2-1+y}{1-y} = \frac{1+y}{1-y}$$

So, $f^1(x) = \frac{1+y}{1-x}$

FOUNDATION

1. $f(x) = x^2 + 4x + 4$ (i)

$g(x) = x^2 + 4x + 3$ (ii)

Then, $f(g(x)) = f(x^2 + 4x + 3)$

$= (x^2 + 4x + 3)^2 + 4(x^2 + 4x + 3) + 4$ [put $x = x^2 + 4x + 3$ in equation (i)]

$= (x^4 + 8x^3 + 22x^2 + 24x + 9) + (4x^2 + 16x + 12) + 4$

$= x^4 + 8x^3 + 26x^2 + 40x + 25$ (iii)

$g(f(x)) = g(x^2 + 4x + 4)$

$= (x^2 + 4x + 4)^2 + 4\ (x^2 + 4x + 4) + 3$

$= (x^4 + 8x^3 + 24x^2 + 32x + 16) + (4x^2 + 16x + 16) + 3$

$= x^4 + 8x^3 + 28x^2 + 48x + 35$ (iv)

Then, $f(g(x)) = g(f(x))$

$x^4 + 8x^3 + 26x^2 + 40x + 25 = x^4 + 8x^3 + 28x^2 + 48x + 35$

$2x^2 + 8x + 10 = 0 \rightarrow x^2 + 4x + 5 = 0$

It can be seen that no real value of x is possible because discriminant $= b^2 - 4ac < 0$

2. $f(x) = \log_{10}\left(x + \sqrt{x^2+1}\right)$

Then, $f(-x) = \log_{10}\left(-x + \sqrt{(x)^2+1}\right)$

$$= \log_{10}\left(\sqrt{x^2+1} - x\right)$$

$$= \log_{10}\left[\frac{\sqrt{x^2+1}-x}{(\sqrt{x^2+1}+x)} \times (\sqrt{x^2+1}+x\right]$$

$$= \log_{10}\left[\frac{(\sqrt{x^2+1})^2 - (x)^2}{(\sqrt{x^2+1}+x)}\right]$$

$$= \log_{10}\left[\frac{x^2+1-x^2}{\sqrt{x^2+1}+x}\right]$$

$$= \log_{10}\left[\frac{1}{\sqrt{x^2+1}+x}\right] = \log_{10}(\sqrt{1+x^2}+x)^{-1}$$

$$= -\log_{10}(\sqrt{1+x^2}+x) = -f(x)$$

∴ It is an odd function.

3. Going through the options, we can see that none of (a), (b), and (c) are matching.

Because $\log ax + \log ay = \log a^2xy$

∴ $\log ax$ could be $f(x)$. Hence, option (d) is the answer.

4. $f(x) = 3x^2 - 2x + 4$ (i)

$g(x) = x + 1$

Then, $f(g(f(1))) = f(g(3(1)^2 - 2(1) + 4)) = f(g(5))$

$= f(5 + 1)$ (since $g(x) = x + 1$)

$= 3(6)^2 - 2(6) + 4 = 108 - 12 + 4 = 100$

5. Except option (d), every option could be $g(x)$.

Reason—if we put $g(x) = 2x$, then $f(g(x)) = f(2x)$

$= (2x)^2 = 4x^2$ (i)

$g(f(x)) = g(x^2) = 2x^2$ (ii)

Here, $f(g(x)) = g(f(x))$

So, the answer is $2x$.

6. Let us verify the options:

Option (a): For $f(x^2) = f(x)]^2$

$f(x^2) = |x^2 - i|$ and $f(x)]^2 = (|x - i|)^2$, which is incorrect.

Option (b): For $f(x + y) = f(x) + f(y)$

$f(x + y) = |x + y - 1|$ and $f(x) - f(y) = |x - 1| + |y - 1|$

It is also incorrect.

Option (c): For $f(|x|) = |f(x)|$

$f(|x|) = ||x| - 1|$

$|f(x)| = ||x - 1||$

So, the answer is option (d).

7. $f(x) = |x|^3 = x^3$

Then, $f(-x) = |-x|^3 = x^3 = f(x)$

It is an even function.

8. Sum of the two odd function is always an odd function. See the properties of odd function.

9. The answer is option (c)—neither odd nor even.

See the properties of odd and even functions.

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10. $f(x) = c.x + 1$

$g(x) = 3x + 2$

Then, $f(g(x)) = f(3x + 2) = c(3x + 2) + 1$ (i)

$g(f(x)) = g(c.x + 1) = 3(c.x + 1) + 2$ (ii)

Then, $f(g(x)) = g(f(x))$

$c(3x + 2) + 1 = 3\ (c.x + 1) + 2$

$3cx + 2c + 1 = 3cx + 3 + 2$

$2c + 1 = 5$, or, $2c = 5 - 1 = 4$, or, $c = 2$.

11. $f(x) = 2^x - 2^{-x}$

Then, $2f(x) - 5f(x - 1) + 2f(x - 2)$

$= 2[2^x - 2^{-x}] - 5\ [2^{x-1} - 2^{1-x}] + 2\ [2^{x-2} - 2^{2-x}]$.

Hence, the answer is option (d).

12. $g[h\{f\{i(16, 3), 2\}, 7\}, 3]$

$= g[h\{f\{5, 2\}, 7\}, 3]$ [since $i(x, y)$ = Quotient, when x is divided by y.]

$= g[h\{625, 7\}, 3]$ [since $f(x, y) = xy^2$]

$= g[2, 3]$ [since $h(x, y)$ = Remainder, when x is divided by y.]

$= 3^{2^2} = 3^4 = 81$ [since $g(x, y) = y^{x^2}$]

13. The value of $h[g\{i(76, 28), 2\}, 3] = h[g\{20, 2\}, 3]$

$= h[2^{20^2}, 3] = [2^{400}, 3] = h[1, 3] = 1$ (i)

The value of $h[f\{i(16, 3), 2\}, 7] = hf\{5, 2\}, 7] = h[5^{2^2}, 7]$

$= h[5^4, 7] = h[625, 7] = 2$ (ii)

Then, the value of

$$\frac{h[g\{i(76, 28), 2\}, 3]}{h[f\{i(16, 3), 2\}, 7]} = \frac{1}{2}$$ [from equations (i) and (iii)]

14. The value of $f[h\{g(2, 2), 3\}, 2]$

$= f[h\{16, 3\}, 2] = f[1, 2] = 1^{2^2} = 1$ (i)

The value of $g[h\{f(5, 2), 7\}, 3] = g[h\{5^4, 7\}, 3]$

$= g[2, 3] = 3^{2^2} = 3^4 = 80$ (ii)

Then, the value of $f[h\{g(2, 2), 3\}, 2] - g[h\{f(5, 2), 7\}, 3]$

$= 1 - 81 = -80$

15. $f(x) = 2x + 3$

Then, $f\{f(f(x))\} = f[f\{2x + 3\} = f[2[2x + 3] + 3]$

$= f[4x + 6 + 3] = f[4x + 9] = 2[4x + 9] + 3$

$= 8x + 18 + 3 = 8x + 21$

16. $f(x) = e^x + e^{-x}$

Then, $f(-x) = e^- + e^{-(-x)} = e^{-x} + e^x = f(x)$

If $f(-x) = f(x)$, then it is an even function.

17. $f(x) = \dfrac{x^2 - 2x}{1 + x}$

Then, $f\left(\dfrac{1}{-x}\right) = \dfrac{(1/-x)^2 - 2(1/-x)}{1 + (1/-x)}$

$$= \frac{(1/x^2) + (2/x)}{(-x + 1)/x}$$

$$= \frac{\dfrac{1 + 2x}{x^2}}{\dfrac{1 - x}{x}} = \frac{(1 + 2x)}{x^2} \times \frac{x}{(1 - x)} = \frac{1 + 2x}{x(1 - x)}$$

18. $x^2 - 3ax + f(a) = 0$

Let α and 2α are the roots, then sum of the roots

$= \alpha + 2\alpha = \dfrac{-b}{a} = 3a$

$= 3\alpha = 3\alpha \rightarrow \alpha = a$ (i)

Product of the roots $= \alpha \times 2\alpha = c/a = f(a)$

$f(a) = 2\alpha^2$ (ii)

Put the value of α from (i) equation

$f(a) = 2a^2$

Then, $f(x) = 2x^2$

19. $f(x) = 3x^2 - 2xy - y^2 + 4$

Then, $f(f(2, 3), f(-1, 1)]$

The value of $f(2, 3) = 3(2^2) - 2 \times 2 \times 3 - (3)^2 + 4$

$= 12 - 12 - 9 + 4 = -5$ (ii)

$f(-1, 1) = 3(-1)^2 - 2 \times -1 \times 1 = (1)^2 + 4$

$= 3 + 2 - 1 + 4 = 8$ (iii)

Then, $f(f(2, 3), f(-1, 1)] = f[-5, 8]$

$= 3(-5)^2 - 2 \times -5 \times 8 - (8)^2 + 4$

$= 75 + 80 - 64 + 4 = 95$

20. $y = f(x) = \dfrac{x + 2}{x - 1}$

$\Rightarrow$ It is incorrect to say that $f(1) = 0$

Because $f(1) = \dfrac{(1) + 2}{(1) - 1} = \dfrac{3}{0} = \infty$

Hence, the answer is option (c).

24. $f(t) = \dfrac{t^3 + 2t}{t - 1}$

Then, $\dfrac{f(t) + f(-t)}{f(t) - f(-t)} = \dfrac{\dfrac{(t)^3 + 2t}{t - 1} + \dfrac{(-t)^3 + 2(-t)}{(-t) - 1}}{\dfrac{t^3 + 2t}{t - 1} - \dfrac{(-t)^3 + 2(-t)}{(-t) - 1}}$

$$= \frac{\dfrac{t^3 + 2t}{t - 1} + \dfrac{-t^3 - 2t}{-t - 1}}{\dfrac{t^3 + 2t}{t - 1} - \dfrac{-t^3 - 2t}{-t - 1}}$$

$$= \frac{\dfrac{t^3 + 2t}{t - 1} + \dfrac{-(t^3 - 2t)}{-t + 1}}{\dfrac{t^3 + 2t}{t - 1} - \dfrac{-(t^3 - 2t)}{-(t + 1)}}$$

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$$= \frac{\frac{(t^3+2t)(t+1)+(t^3+2t)(t-1)}{(t-1)(t+1)}}{\frac{(t^3+2t)(t+1)-(t^3+2t)(t-1)}{(t+1)(t-1)}}$$

$$= \frac{t^3+2t[t+1]+(t-1)]}{t^3+2t[t+1)-(t-1)]} = \frac{2t}{2} = t$$

25. $f(x) = 2x^2 + 6x - 1$ and $g(x) = |x + 5|$

Then, the value of $f[g(f(g(-6))\}]$

$= f[g(f(|(-6) + 5|)\}]$ [since $g(x) = |x + 5|$]

$= f[g\{f(1)\}] = f[g\{2(1)^2 + 6(1) - 1\}] = f[g\{2 + 6 - 1\}] = f[g\{7\}]$

$= f[1|7 + 5|] = f[12]$

$= 2(12)^2 + 6(12) - 1 = 2 \times 144 + 72 - 1 = 288 + 71 = 359$

26. $f(x) = x^3 + 1$ (i)

$f(2x) - 2f(x) = 383$

Put the value of $f(x)$ from (i) equation

or $(2x)^3 + 1 - 2(x^3 + 1) = 383$

or $8x^3 + 1 - 2x^3 - 2 = 383$

or $6x^3 = 383 + 1 = 384$

$x^3 = \frac{384}{6} = 64$ Hence, $x = 4$

27. It is given that $f(x) = 0$ when $x = 1$

$= 1$ when $x = 0$

$= 0.5$ when $x = 0.5$

$g(x) = 1$ when $x = 1$

$= 0.5$ when $x = 0$

$= 0$ when $x = 0.5$

$f[g[f[g(t)]]] = 0$

It is given that when $f(x) = 0$, then $x = 1$

$\therefore g[f[g(t)]] = 1$

When $g(x) = 1$, then $x = 1$

So, $f[g(t)] = 1$

But, when $f(x) = 1$, then $x = 0$

So, $g(t) = 0$

Then, $t = 0.5$ [since $g(x) = 0$ when $x = 0.5$]

28. Let us verify the options to solve this question:

When $t = 1$, then $f(g(1) + f(1)] + g[g(1)] = 1$

$f[1 + 0] + g[1] = 1 - f(1) + g(1) = 1$

or $0 + 1 = 1$

or $1 = 1$

Since LHS = RHS, the value of t is 1.

29. $f(x) = x^2$ and $h(x) = \sqrt{x}$, then

(i) h of $(-2) = 4$

Take LHS h of $(-2) = h[(-2)^2] = h[4]$

$= \sqrt{4} = \pm 2$

Since LHS ≠ RHS, this is incorrect.

(ii) h of $(4) = 4$

Take LHS h of $(4) = h[(4)^2] = h[16] = \pm 4$

Since LHS = RHS, this is correct.

(iii) h of $(3) = 6$

Take LHS h of $(3) = h[(3)^2] = h[9] = \pm 3$

Since LHS ≠ RHS, it is incorrect.

So, only (ii) is correct.

30. $h(x) = y = \frac{1-x}{1+x}$

$y + 1 = \frac{1-x}{1+x} + 1$ (add 1 both side)

$y + 1 = \frac{1-x+1+x}{1+x} = \frac{2}{1+x}$

$1 + x = \frac{2}{y+1} \rightarrow x = \frac{2}{y+1} - 1$

$x = \frac{2-y-1}{y+1} = \frac{1-y}{1+y}$

So, $h^{-1}(x) = \frac{1-x}{1+x}$

In this function, $1 + x$ should not be equal to zero, which means x should not be equal to -1.

So, the domain of x should be all real number except -1.

$\therefore$ Hence, the answer is (b).

31. $f(n) = 2n^2 - n$ for $n > 5$

$= n^2 + n$ for $n \leq 5$

Then, $f(1) + f(2) + f(3) \ldots f(10)$

$= [(1)^2 + 1] + [(2)^2 + 2][(3)^2 + 3] + [(4)^2 + 4] + [(5)^2 + 5] + [2(6)^2 - 6] + [2(7)^2 - 7] + [2(8)^2 - 8] + [2(9)^2 - 9] + [2(10)^2 - 10]$

$= (2) + (6) + (12) + (20) + (30) + (66) + (91) + (120) + (153) + (190)$

$= 690$

32. $y = f(x) = \frac{ax+b}{cx-a}$

Then, $f(y) = f[f(x)] = f\left[\frac{ax+b}{cx-a}\right]$

$$= \frac{a\left[\frac{ax+b}{cx-a}\right]+b}{c\left[\frac{ax+b}{cx-a}\right]-a} = \frac{a(ax+b)+(cx-a)}{c(ax+b)-a(cx-a)}$$

$= a^2x + ab + bcx - ab$

33. $y = f(x) = \frac{x+2}{x-1}$

Then, $f(y) = f[f(x)] = f\left[\frac{x+2}{x-1}\right]$

$= \left(\frac{x+2}{x-1} + 2\right) \Big/ \left(\frac{x+2}{x-1} - 1\right)$

$= (x + 2 + 2x - 2)/(x + 2 - x + 1) = 3x/3 = x.$

34. $f(x) = \frac{x}{x-1}$

Then, $f\left(\frac{a}{b}\right)/f(b/a) = \left(\frac{a/b}{\frac{a}{b}-1}\right) / \left(\frac{b/a}{\frac{b}{a}-1}\right)$

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$$= \left(\frac{\frac{a/b}{a-b}}{b}\right) / \left(\frac{\frac{b/a}{b-a}}{a}\right) = \left(\frac{a}{a-b}\right) / \left(\frac{b}{b-a}\right)$$

$$= \frac{a}{a-b} \times \frac{b-a}{b} = \frac{a}{a-b} \times \frac{-(a-b)}{b} = -a/b$$

35. $f(x) = \dfrac{1}{\sqrt{|x|+x}}$

In this function, $(|x| + x)$ should not be a negative number because under root does not hold negative values and also $(|x| + x)$ should not be equal to zero because denominator should not be zero.

So, $(|x| + x) > 0$

In this case, we can only take positive values.

So, the domain is $(0, +\infty) = 0 < x < +\infty$

36. $y = f(x) = \dfrac{1}{\sqrt{\{x-[x]\}}}$

Again, $\{x - [x]\}$ should be neither negative nor zero.

So, the domain is all the real numbers other than integers.

37. $f(x) = \log(x-2) + \log(x-3)$

Here, the domain should be greater than 3 because log does not hold negative values.

So, the domain is $(3, +\infty)$.

$g(x) = \log(x-2)(x-3) = \log(x^2 - 5x + 6)$

Here, x should not be $\{2, 3\}$.

So, the domain is $(-\infty, 2) \cup (3, +\infty)$.

It can be observed that their domain is not same, and hence, they are not identical functions.

39. $(x) = \log_{10} \log_{10} (1 + x^3)$

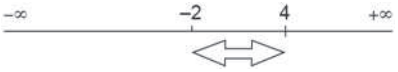
Here, x cannot be negative number and zero.

Because $f(x) = \log_{10}\log_{10}(1+(0)^3) = \log_{10}\log_{10}(1) = \log_{10}(0)$ [since $\log_{10} 1 = 0$]

But, we cannot find log of zero.

So, domain is $(0, +\infty)$.

40. $|x-1| \le 3$ or $-3 \le (x-1) \le 3$ or $(-3+1) \le (x-1+1) \le (3+1)$ or $-2 \le x \le 4$



For $|x-1| \ge 1$

Solving this, we get

$(x-1) \le -1$ or $(x-1) \ge 1$ or

$x \le 0$ or $x \ge 2$



Combining both, we get



So, common set is [–2, 0] U [2, 4].

MODERATE

1. Go through the options.

The answer is option (b).

Since $f\left(\frac{1}{a}\right) = -f(a)$

Take LHS

$$f\left(\frac{1}{a}\right) = \frac{(1/a)-1}{(1/a)+1} = \frac{(1-a)/a}{(1+a)/a} = \frac{1-a}{1+a}$$

$$= \frac{a-1}{a+1} = -f(a) = \text{RHS}$$

2. Examining Statement (c):

$x \blacklozenge 0 = |x - 0| = |x|$

$|x| \ne x$ when $x < 0$, but Statement (c) says that it does for all x.

Therefore, the statement that is not true is (c): $x \blacklozenge 0 = x$ for all x.

Answers to Q.3 and 4:

$[x]$ = Greatest integer less than or equal to x.

$\{x\}$ = Smallest integer more than or equal to x.

Take $x = P$ such that $P = I + F$, where I is the integral and $F (F > 0)$ is the fractional part of the number.

$P (P$ cannot be an integer).

$[x] = [I + F] = I$, and similarly, $\{x\} = \{I + F\} = I + 1$

3. Hence, $[x] - \{x\} = -1$

4. $[x] + \{x\} = I + I + 1 = 2I + 1 =$ odd integer

Alternatively, we can do both the problems by assuming any non-integral value of x (like 3.5 or 4.2 or –2.6, etc.).

5. For all the values of $n < 50$, $f(n) = 0$ and

For all the $n \ge 50$, $f(n) = 1$ Hence, 51 such values are there.

6. $x/2 + x/3 + x/5 = 31x/30$

It means $[x/2] = x/2$, $[x/3] = x/3$ and $[x/5] = x/5$

Now, $[N] = N$ is possible only if N is an integer.

Hence, $x/2$, $x/3$, and $x/5$ are integers. So, x is divisible by 2, 3, and 5, or x is divisible by 30.

Total 33 values are possible.

7. We have to find out some powers of $f(1)$, which is in all the cases equal to 1.

8. $||x-1|-1| < 1$

It is possible when $|x-1| \le 2$

So, x could be ≤ 3 and ≥ -1

So, set of real values of x is $[-1, 3]$.

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9. A function is defined as $f^n(x) = f\{f^{n-k}(x)\}$.
If n is odd, $k = 1$
If n is even, $k = 2$ and $f(x) = 1$ and
$f(x) = x^2 + 1$
Then, $f^7(x)$ at $x = 1$ is:
$f^7(1) = f(f^{7-1}(1)] = f(f^6(1)]$
$= f[f[f^{6-4}(1)]]$ [since if n is even $\therefore k = 2$]
$= f[f[f^4(1)]] = f(f[f[f^{4-2}(1)]]]$
$= f[f[f[f[f^{2-2}(1)]]]] = f[f[f[f[[f[f^{2-2}(1)]]]]]]$
$= f[f[f[f[f[f^0(1)]]]]]$
$= [f[f[f[f(1)]]]]$ [since $f^0(x) = 1$]
$= f[f[f(2)]]]$ [since $(1)^2 + 1 = 2$]
$= f[f(5)]]$ [$(2)^2 + 1 = 5$]
$= f[26]$ [$(5)^2 + 1 = 26$]
$= (26)^2 + 1 = 676 + 1 = 677$

10. $x = 1$ and $x = -1$ are two roots of $f(x)$. Similarly, if we put $x = 0$, then $f(x) = -3$. Now, solve the equation to get the value of $a + c$.

11. Let a and b be the digits of x.
$f(f(x)) = a + b = 3$
Clearly, $f(x)$ can only be 3, 12, 21, or 30 and only 3 and 12 are possible to have two digits sum.
If $f(x)$ sums to 3, then there are three different solutions: 12, 21, or 30.
If $f(x)$ sums to 12, then there are seven different solutions: 39, 48, 57, 66, 75, 84, or 93.
The total number of solutions is $3 + 7 = 10$.

12. $f(x + 1) + f(x - 1) = 2f(x)$ and
$f(0) = 0$
Then, $f(n)$, when n is a integer.
Let $x = 1$
Then, $f(1 + 1) + f(1 - 1) = 2f(1)$
$f(2) + f(0) = 2f(1)$ [since $f(0) = 0$]
$f(2) = 2f(1)$ (i)
When $x = 2$, then
$f(2 + 1) + f(2 - 1) = 2f(2)$
$f(3) + f(1) = 2f(2)$
$f(3) + f(1) = 2[2f(1)]$ [since $f(2) = 2f(1)$]
$f(3) = 4f(1) - f(1)$
$\therefore f(3) = 3f(1)$ (ii)
When $x = 3$
$f(3 + 1) + f(3 - 1) = 2f(3)$
$f(4) + f2) = 2f(3)$
Since $f(2) = 2f(1)$ and $f(3) = 3f(1)$
$\therefore f(4) + 2f(1) = 2[3f(1)]$
$f(4) = 6f(1) - 2f(1)$
$f(4) = 4f(1)$ (iii)
From equations (i), (ii), and (iii), we can conclude that $f(n) = n(f(1)$
$\therefore$ Hence, option (a) is the answer.

13. $f(x) = 1 + |x|$, $x < -1$ and $[x]$, $x \geq -1$,
So, $f(-2.3) = 1 + |-2.3| = 1 + 2.3 = 3.3$
$f\{f(-2.3)\} = f(3.3) = [3.3] = 3$

14. Expression under square root should be non-negative and the denominator cannot be equal to zero. Now, find the common values.
Alternatively, we can do this problem by using options.

15. For any increase in the value of x, increase in the numerator will be more than the increase in the denominator. This can be verified by taking a few values of x. Alternatively, this can be verified by plotting the graphs. The slope of the graph of the numerator will be more than the slope of the graph in the denominator.
Hence, $f(x)$ will increase.

16. Assume some values of x, y, and z and then go through the options.

17. For ${}^{16-x}C_{2x-1}$ to be defined, $(16 - x) \geq (2x - 1)$, and similarly, for ${}^{20-3x}P_{4x-5}$ to be defined, $(20 - 3x) \geq (4x - 5)$

18. $f(x) = \log_e |\log_e x|$
For any logarithmic expression, value of x should be a positive number.
However, for this expression, we cannot take 1 as a value of x. [Because if we take $x = 1$, then $f(x) = \log_e |\log_e 1| = \log e |0| = \log_e 0$]
But, we can find log of positive values.
So, the domain is all the positive numbers except 1.

20. $f(x) = \log_{x^2} 25$

For this function, we can take positive values as well as negative values. Because square of any number equals to a positive number. However, we cannot take x equals to zero as well as 1.
So, the domain is $(-\infty, +\infty)$ except 0 and 1 and for $g(x) = \log_x 5$
The domain should be all positive numbers except 1.
So, common domain is $(0, 1) \cup (1, +\infty)$

21. $g(x) = f(x + 1) = |x - 2 + 1| + |x - 3 + 1| + |x - 4 + 1| = |x - 1| + |x - 2| + |x - 3|$
Obviously, this is neither odd nor even.
Alternatively, we know the graph of this function will neither be symmetrical to axis or origin. [See the topic Graphs and Maxima Minima.]

22. $f(x) = x^n$, n e N
$g(x) = ng(x)$
Then, $g(x)$ can be $\log|x|$

23. $$f(x) = y = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{e^x - (1/e^x)}{e^x + (1/e^x)}$$

$$= y = \frac{(e^{2x} - 1)/e^x}{(e^{2x} + 1)/e^x} = \frac{e^{2x} - 1}{e^{2x} + 1}$$

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Subtract 1 from both sides.

$$y-1=\frac{(e^{2x}-1)}{(e^{2x}+1)}-1=\frac{-2}{e^{2x}+1}$$

$$e^{2x}+1=\frac{-2}{y-1}=\frac{2}{1-y}$$

$$e^{2x}=\frac{2}{1-y}-1=\frac{1+y}{1-y}$$

$$2x=\log\frac{1+y}{1-y}$$

$$x=\frac{1}{2}\log\frac{1+y}{1-y}$$

Then, $f^{-1}(x)=\frac{1}{2}\log\frac{1+x}{1-x}$

25. $f(x)=\log\frac{1+x}{1-x}$ and $g(x)=\frac{3x+x^3}{1+3x^2}$

Then, $f[g(x)]=f\left[\frac{3x+x^3}{1+3x^2}\right]$

$$=\log\frac{1+(3x+x^3)/(1+3x^2)}{1-(3x+x^3)/(1+3x^2)}$$

$$=\log\frac{1+(3x^2+3x+x^3)/(1+3x^2)}{1+(3x^2-3x-x^3)/(1+3x^2)}$$

$$=\log\frac{(1+x)^3}{(1-x)^3}=\log\left(\frac{1+x}{1-x}\right)^3$$

$$=3\log\frac{1+x}{1-x}=3f(x)$$

26. $f(x)=2x^2+3x+4$

Then, minimum of $f(x)=-\frac{D}{4a}=\frac{-(b^2-4ac)}{4a}$

$=\frac{-(9-32)}{4\times2}=\frac{23}{8}$ and

$g(x)=5-x^2$

Then, maximum of $g(x)=5-(0)^2=5$ since x^2 is a positive number or zero.

Then, smallest value of x^2 should be zero.

Then, minimum $f(x)$ – maximum $g(x)=\frac{23}{8}-5=\frac{23-40}{8}$

$=-\frac{17}{8}$

27. $f(x)=x^3-x^2+x-f(x)$

$2f(x)=x^3-x^2+x$

$f(x)=\frac{x^3-x^2+x}{2}$

For $f(x)$ not to be whole number, x should not be an even number because

$\frac{(e)^3-(e)^2+e}{2}=\frac{e}{2}$ = whole number

So, for all the odd value of x till 15, $f(x)$ cannot be a whole number.

So, $x=8$

28. Let us find out (Mag (6, (Nag (7, –7)))

Nag (7, –7) = –7; so, (Mag (6, (Nag (7, –7))) = 6

Mag (8, (Nag (4, (Nag (7, –7))))) = 8

Hence, the answer is option (b).

30. Applying componendo and dividendo in $\frac{f\left(\frac{3}{4}\right)+1}{f\left(\frac{3}{4}\right)-1}$

$= [2.f(3/4)]/2 = f(3/4)$

$f(3/4) = 2(3/4)^2 + 6(3/4) - 1$

$f(x) = 2x^2 + 6x - 1$, then the value of $\frac{f\left(\frac{3}{4}\right)+1}{f\left(\frac{3}{4}\right)-1}$ is $\frac{74}{16}$

Hence, the answer is option (d).

31. $4\{x\} = x + [x]$, where x is a number such that $[x]$ and $\{x\}$ denote the integral and fractional part of this number. Then, do this question from options and the answer is 5/3.

Because $4\{x\} = x + [x]$ [here, $\{x\} = \frac{2}{3}$ and $[x] = 1$]

Take LHS $= 4\{4\} = 4\times\frac{2}{3}=\frac{8}{3}$ (i)

RHS $= x + [x] = \frac{5}{3}+1=\frac{8}{3}$ (ii)

Since LHS = RHS

Hence, the answer is 5/3.

32. $f(x) = (-1)^{S_{n+1}} + (-1)^{S_n} + f(n-1)$ where $f(0) = 1$

Here, S_{n+1}, and S_n will be consecutive numbers.

So, if S_{n+1} is a even number,

Then, $(-1)^{S_{n+1}} = 1$

S_n should be a odd number [$\therefore$ both are consecutive].

Then, $(-1)^{S_n} = -1$

$\therefore\ (-1)^{S_{n+1}} + (-1)^{S_n} = 0$

If S_{n+1} is a odd number, then vice versa is also true.

$\therefore f(n) = f(n-1)$

Then, $f(111) = f(111-1) = f(110)$

$= f(109) = f(108)$ [since $f(n) = f(n-1)$]

$= f(0) = 1$ [since $f(0) = 1$]

33. $f(1) + f(2) + f(3) + ... + f(25) = ?$

In the previous question, we have proved that

$f(n) = f(0) = 1$

Then, $f(1) = 1, f(2) = 1, f(3) = 1$ and $f(25) = 1$

So, $f(1) + f(2) + \ldots + f(25) = 1 + 1 + 1 + \ldots + 1$ (till 25) = 25

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34. It is given that $f(x) = [x]$ and $g(x) = |x|$

Then, $gof\left(-\frac{5}{3}\right) = gof\,[-1.66]$

$= g(-2) = 2$ (i)

$fog\left(-\frac{5}{3}\right) = fog\,[-1.66]$

$= f(1.66) = 1$ (ii)

Then, $gof\left(-\frac{5}{3}\right) + fog\left(-\frac{5}{3}\right) = 2 + 1 = 3$

35. Solution 1

Since

$-f(-1) = a - b + c - d = 0 = f(1) = a + b + c + d$

It follows that $b + d = 0$

Also, $d = f(0) = 2$;

So, $b = -2$

Solution 2

Two of the roots of $f(x) = 0$ are ± 1, and let the third one be n. Then,

$a(x - 1)(x + 1)(x - n) = ax^3 - anx^2 - ax + an = ax^3 + bx^2 + cx + d = 0$

Notice that $f(0) = d = an = 2$; so, $b = -an = -2$

36. Method 1

We already know that $a_1 = 2001$, $a_2 = 2002$, $a_3 = 2003$, and $a_4 = 2000$. Let us compute the next few terms to get the idea how the sequence behaves. We get $a_5 = 2002 + 2003 - 2000 = 2005$, $a_6 = 2003 + 2000 - 2005 = 1998$, $a_7 = 2000 + 2005 - 1998 = 2007$, and so on.

We can now discover the following pattern: $a_{2k+1} = 1999 + 2k$ and $a_{2k} = 2004 - 2k$. This is easily proved by induction. It follows that $a \times 1002 = (0)$

Method 2

Note that the recurrence $a_n + a_{n+1} - a_{n+2} = a_{n+3}$ can be rewritten as $a_n + a_{n+1} = a_{n+2} + a_{n+3}$

Hence, we get that $a_1 + a_2 = a_3 + a_4 = a_5 + a_6 = \ldots$ and also $a_2 + a_3 = a_4 + a_5 = a_6 + a_7 = \ldots$

From the values given in the problem statement, we see that $a_3 = a_1 + 2$

From $a_1 + a_2 = a_3 + a_4$, we get that $a_4 = a_2 - 2$

From $a_2 + a_3 = a_4 + a_5$, we get that $a_5 = a_3 + 2$

Following this pattern, we get $a_{2004} = a_{2002} - 2 = a_{2000} - 4 = \ldots = a_2 - 2002 = (0)$

37. If $f(x) = \log\frac{1+x}{1-x}$

Then, $f(x) + f(y) = \log\frac{1+x}{1-x} + \log\frac{1+y}{1-y}$

$= \log\left[\frac{(1+x)(1+y)}{(1-x)(1-y)}\right]$

$= \log\left[\frac{1+x+y+xy}{1-x-y+xy}\right]$ (i)

Then, check the options.

Option (b) is the answer because

$f\left(\frac{x+y}{1+xy}\right) = \log\frac{1+(x+y)/(1+xy)}{1-(x+y)/(1+xy)}$

$= \log\frac{1+(xy+x+y)/(1+xy)}{1+(xy-x-y)/(1+xy)}$

$= \log\left[\frac{1+x+y+xy}{1-x-y+xy}\right]$

$= f(x) + f(y)$ [from equation (i)]

38. $f(x, y) = g(x, y)$ is possible for all the sets of $(x, y) = (x, 0)$.

39. Draw the graph of y or see it by assuming different values of y.

40. Using $H(1) + H(2) + H(3) + - + H(x) = n^2 H(x)$

For $n = 1$: $H(1) = 1.\ H(1)$

For $n = 2$: $H(1) + H(2) = 4 \times H(2)$ or $H(2) = [H(1)]/3$.

For $n = 3$: $H(1) + H(2) + H(3) = 9 \times H(3)$ or $H(3) = [H(1)]/2.3$

Now, go through the same method to find $H(2005)$.

ADVANCED

1. Use the following:

If $a < x$, then $(x - a) > 0$, and if $b < x$, then $(x - b) > 0$.

If $c < x$, then $(x - c) > 0$, and if $d < x$, then $(x - d) > 0$, and if $e < x$, then $(x - e) > 0$.

If $a > x$, then $(x - a) < 0$; similarly, other also can be calculated. Now, use the options. Hence, the answer is option (d).

2. Method 1

Let the roots be $r_1 = 0, r_2, r_3, r, r_5$.

We have $d = r_1r_2r_3r_4 + r_1r_2r_3r_5 + r_1r_2r_4r_5 + r_1r_3r_4r_5 + r_2r_3r_4r_5$. The first four terms contain $r_1 = 0$ and are, therefore, zero; thus, $d = r_2r_3r_4r_5$. This is a product of four non-zero numbers; therefore, d must be non-zero.

Method 2

Clearly, since (0,0) is an intercept, e must be 0. However, if d was 0, then x^2 would divide the polynomial, which means it would have a double root at 0, which is impossible, since all five roots are distinct.

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3. Assume different values of x in an attempt to refute the statements given. After a bit of hit-and-trial, we can observe that none of the statements can be refuted. Hence, the answer is option (d).

4. $f(x)=\dfrac{x-1}{x+1}$

Then, $f(ax)=\dfrac{ax-1}{ax+1}$

Solve this question from options, and the answer is option (c).

Because $\dfrac{(a+1)f(x)+a-1}{(a-1)f(x)+a+1}=\dfrac{(a+1)\left(\dfrac{x-1}{x+1}\right)+(a-1)}{(a-1)\left(\dfrac{x-1}{x+1}\right)+(a+1)}$

$$=\frac{\dfrac{(a+1)(x-1)+(a-1)(x+1)}{x+1}}{\dfrac{(a-1)(x-1)+(a+1)(x+1)}{x+1}}$$

$$=\frac{ax-a+x-1+ax-x+a-1}{ax-a-x+1+ax+x+a+1}$$

$$=\frac{2ax-2}{2ax+2}=\frac{2(ax-1)}{2(ax+1)}$$

$$=\frac{ax-1}{ax+1}=f(ax) \qquad \text{[from equation (i)]}$$

6. $a(n)\ a(a(n-1))+a(n-a(n-1))$, where n is any integer greater than 2.

It is known that $a(1)=1$ and $a(2)=1$

Put $n=3$

$a(3)=a(a(3-1))+a(3-a(3-1))$

$=a(a(2))+a(3-a(2))$

$=a(1)+a(3-1)$ [since $a(2)=1$]

$=a(1)+a(2)$

$=1+1=2$ (i)

and

$a(4)=a(a(4-1))+a(4-a(4-1))$

$=a(a(3))+a(4-a(3))$

$=a(2)+a(4-2)$ [since $a(3)=2$]

$=1+a(2)=1+1=2$

7. It is given that $[a(x)=a(a(n-1))+a(n-a(n-1))]$. Where $a(2)=1$, $a(3)=2$ and $a(1)=?$

Put $n=3$

$a(3)=a(a(3-1))+a\ (3-a(3-1))$

$a(3)=a(a(2))+a\ (3-a(2))$

$a(3)=a(1)+a(3-1)$ [since $a(2)=1$]

$a(3)=a(1)+a(2)$

$2=a(1)+1$ [since $a(2)=1$ and $a(3)=2$]

$a(1)=1$

8. Using the options, if $f(x)=x$, then $f[(x+y)/(x-y)]=(x+y)/(x-y)$, and similarly, it can be seen that RHS will be same as LHS.

9. Minimum value of $x=5$ and maximum value of $x=29$. So, sum = 34

10. (i) $g(x)\geq f(x)$

It is not true for negative values of x and y.

(ii) $g(x)+h(x)\geq f(x)$

It is also not true for negative values of x and y, Since

$g(x)=$ maximum $[(x+y),(x-y)]$

Put $x=-x$ and $g=-y$

Then, $g(x)=$ maximum $[-x-y,-x+y]$

$=-x+y$ (i)

$h(x)=$ minimum $[-x-y,-x+y]$

$=-x-y$ (ii)

and $f(x)=|-x|+|-y|=x+y$ (iii)

Since $g(x)+h(x)=-x+y-x-y=-2x$

Which is less than $f(x)$

Then, $g(x)+h(x)\geq f(x)$ is not true.

(iii) $g(x)>f(x)$

It is also not true for negative values of x and y.

11. When a, b, and c are distinctly different positive numbers. Then,

(a) $H(a,b,c)\times L(a,b,c)=abc$

is not true for the numbers, which are not a prime to each other.

(b) $H(a,b,c)>(a,b,c)$

It is not true for every distinctly different positive number. Because HCF should be less than or equal to the minimum number from a, b, c and average should be greater than minimum number from a, b, and c.

(c) $H(a,b,c)>$ minimum (a,b,c)

This is not true for numbers like (8, 10, 12).

(d) $H(a,b,c)<$, $A(a,b,c)<L(a,b,c)$

It is true.

12. If maximum $(a,b,c)=$ minimum (a,b,c)

It means that a, b, and c are equal to each other.

Then, $H(a,b,c)=A(a,b,c)=L(a,b,c)=$ minimum (a,b,c)

So, all the options are true.

13. If the area of a circle increases at a uniform rate, then the rate of increase of the perimeter is directly proportional to the radius.

14. $f(x)=3^x+4^x+5^x-6^x$

$f(3)=3^3+4^3+5^3-6^3=0$

For $x>3$, $f(x)$ would be a negative number.

Then, for infinite values of x, $f(x)$ would be less than $f(3)$.

15. This question requires finding out the digital sum of x given.

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Digital sum of $x = 123456^{654321}$ is same as digital sum of $x = (1+2+3+4+5+6)^{654321} = f^{\infty}(x) = 21^{654321} = 3^{654321}$

Now, digital sum of any power of 3 greater than 1 is 9. So, the digital sum of $3^{654321} = 9$

16. Digital sum of $7^1 = 7$

Digital sum of $7^2 = 49 = 4 + 9 = 1 + 3 = 4$

Digital sum of $7^3 = 343 = 1$

Digital sum of $7^4 = 2401 = 7$

Digital sum of $7^5 = 16{,}807 = 4$

So, digital sum of $7^{183} = 1$

17. By the definition of an inverse function, $x = f(f^{-1}(x)) = a(bx + a) + b = abx + a^2 + b$. By comparing coefficients, we have $ab = 1 \rightarrow b = \frac{1}{a}$ and

$a^2 + b = a^2 + \frac{1}{a} = 0$

Simplifying, $a^3 + 1 = 0$ and $a = b = -1$

Therefore, $a + b = -2$

Hence, option (c) is the answer.

18. Since f is a linear function with slope m,

$m = \frac{f(6) - f(2)}{\Delta x} = \frac{12}{6-2} = 3$

$f(12) - f(2) = m\Delta x = 4(12 - 2) = 30$

19. $f(0, y) = y^2 - 1$ (i)

$f(x + 1, y) = f(x, y + 4)$

$\rightarrow f(x, y) = f(x - 1, y + 4)$ (ii)

Then, $f(7, 3) = f[7 - 1, 3 + 4] = f(6, 7)$

Then, $f(6, 7) = f[6 - 1, 7 + 4] = f(5, 11)$

Repeat this process, and after some process, we will find $f(7, 3) = f(0, 31)$

Then, $f(0, 31) = 31^2 - 1$ [from (ii) equation]

$f(7, 3) = 961 - 1 = 960$

21. Every factor of 10^n will be of the form $2^a \times 5^b, a < n, b \le n$. Using the logarithmic property $\log(a \times b) = \log(a) + \log(b)$, it suffices to count the total number of 2's and 5's running through all possible (a, b). For every factor $2^a \times 5^b$, there will be another $2^b \times 5^a$, so it suffices to count the total number of 2's occurring in all factors (because of this symmetry, the number of 5's will be equal). Since $\log(2) + \log(5) = \log(10) = 1$, the final sum will be the total number of 2's occurring in all factors of 10^n.

There are $n + 1$ choices for the exponent of 5 in each factor, and for each of those choices, there are $n + 1$ factors (each corresponding to a different exponent of 2), yielding $0 + 1 + 2 + 3 \ldots + n \frac{n(n+1)}{2}$ total 2's.

So, the total number of 2's $= \frac{n \times (n+1)^2}{2} = \frac{n^3 + 2n^2 + n}{2}$

Now, going through the options, we get $n = 11$.

Hence, option (b) is the answer.

22. We know that the domain of $\log_k n$, where k is a constant is $n > 0$. So, $\log_{2003}(\log_{2002}(\log_{2001} x)) > 0$.

By the definition of logarithms, we then have $\log_{2002}(\log_{2001} x)) > 2003 = 1$

Then, $\log_{2001} x > 2002^1 = 2002$ and $x > 2001^{2002}$

24. $f(2^{100}) = f(2 \times 2^{99}) = 2^{99} \times f(2^{99}) = 2^{99} \times 2^{98} \times f(2^{98}) = \ldots$

$= 2^{99}2^{98} \ldots 2^1.1.f(1)$

$= 2^{99+98+\ldots+2+1} = 2^{\frac{99(100)}{2}} = 2^{4950}$

Hence, option (a) is the answer.

26. Option (c) will be the answer because $\frac{Q[1,2,3) + U(1,2,3)}{2 \times R(1,2,3)}$

$= \frac{(3+1)+(3-1)}{2 \times 3} = \frac{4+2}{6} = \frac{6}{6} = 1$

27. Option (d) is the answer because $\frac{p(1, 2, 3)}{2 \times S(1, 2, 3)} = \frac{2+2}{2 \times 1} = \frac{4}{2} = 2$

28. Option (b) will be the answer because $Q(1, 2, 3) - R(1, 2, 3) - S(1, 2, 3) = 4 - 3 - 1 = 0$

29. Option (d) is the answer because $\frac{U(1,2,3)}{R(1,2,3) + S(1,2,3)}$

$\frac{2}{2+1} = 2/3 = 0.66 (< 1)$

31. Checking the options:

Option (a): If $p1 = 0$, then after forwarding 30, page number 12 will be obtained. After this, if we apply backward 12, then we will again get back to $0 = p1 = p2$.

32. If $p1$ is greater than or equal to 7 pages, then $p1 = p2$.

33. Use the options.

35. $f(x).f(y).f(z) = a^{1+x} \times a^{1+y} \times a^{1+z} = a^{3+x+y+z}$

This is equal to $f(x + y + z + 2)$

36. Let $f(x) = x^n + 1$, so $f(1/x) = (1/x^n) + 1$

Checking it for $f(x), f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right)$ satisfies the value of $f(x)$ assumed.

$f(x) = x^n + 1 = 1001$, so $x^n = 1000$

Hence, $n = 3$

$f(20) = 20^3 + 1 = 8001$

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CHAPTER

14

Graphs and Maxima Minima

LEARNING OBJECTIVES

After completion of this chapter, the reader should be able to understand:

- Maxima and minima, and how they are calculated
- Some standard graphs and their shifting
- Definitions and properties
- Kinds of questions asked in the CAT
- Methods of solving questions

INTRODUCTION

In case of maxima, minima, and graphs, the biggest mistake is to view this chapter on purely mathematical concept. However, there is some mathematics involved, as far as the questions asked in the CAT are concerned, with good logic and common sense, they can be handled without much effort. This chapter will help the students understand the basic concepts on which the problem-solving techniques are built.

MAXIMUM AND MINIMUM VALUES

Maximum Value

Let $f(x)$ be a function with the domain D $\subset$ R. Then, $f(x)$ is said to have attained the maximum value at a point 'a' if $f(a) \geq f(x)$ for all the values of x. In this case, 'a' is called the point of maximum and $f(a)$ is known as the maximum value or the greatest value or the absolute maximum value of $f(x)$.

For example, in case of $y = f(x) = |x|$, we will have the maximum value of y as $+\infty$.

Minimum Value

Let $f(x)$ be a function with domain D $\subset$ R. Then, $f(x)$ is said to attain the minimum value at a point 'a' if $f(x) \geq f(a)$. In such a case, the point 'a' is called the point of minimum and $f(a)$ is known as the minimum value or the least value or the absolute minimum value of $f(x)$.

Let us first see the maximum and minimum values of some very basic functions:

	$y = f(x) = x$	$y = f(x) = x^2$	$y = f(x) = \frac{1}{x}$	$y = f(x) = \log_e x$	$y = f(x) = \lvert x - 2 \rvert$
Maximum	$+\infty$	$+\infty$	$+\infty$	$+\infty$	$+\infty$
Minimum	$-\infty$	0	$-\infty$	$-\infty$	0

Example 1 Find the maximum and the minimum values of $y = f(x) = -|x - 1| + 4$.

Solution We have $f(x) = -|x - 1| + 4$ for all the real values of x.

$|x - 1| \geq 0$ for all real values of x

Hence, $-|x - 1| \leq 0$

$|x - 1| + 4 \leq 4$

So, the maximum value of $y = 4$

Now, $f(x) = 4 \Rightarrow -|x - 1| + 4 = 4 \Rightarrow |x - 1| = 0 \Rightarrow x = 1$

Hence, $f(x)$ attains the maximum value 4 at $x = 1$.

To find out the minimum value, we can make $-|x - 1|$ as small as possible by taking different values of x. So, the minimum value of y will be $-\infty$.

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Example 2 What is the maximum and minimum value of $y = f(x) = |x + 3|$?

Solution We have $y = f(x) = |x + 3|$.

$|x + 3| \geq 0$

$\Rightarrow \quad f(x) \geq 0$ for all $x \in R$

So, the minimum value of $f(x)$ is 0, which is attained at $x = -3$.

To find out the maximum value, it can be seen that $y = f(x) = |x + 3|$ can be made as big as possible. So, the maximum value of y is $+\infty$.

Finding Maxima and Minima in Some Typical Cases

1. Maxima and minima of quadratic equation

The standard methods of finding the maxima and the minima of all the quadratic equations are as follows:

Let us discuss these methods with respect to a particular equation $y = x^2 - 5x + 6$.

(a) *Graphical method*

As we have seen in the concepts of quadratic equation, graph of $y = ax^2 + bx + c = 0$ will have its minimum value at $x = \frac{-b}{2a}$ and $y_{min} = \frac{-D}{4a}$.

So, the minimum value of $y = x^2 - 5x + 6$ will be $y_{min} = \frac{-D}{4a} = \frac{-1}{4}$.

(b) *Quadratic equation method*

Suppose we have found the minimum value of $y = x^2 - 5x + 6$, then we can find the minimum value by breaking this equation into the sum of a whole square and whatever is left, i.e., $P^2 + Q$.

$$x^2 - 5x + 6 = \left[x^2 - 2.x.\frac{5}{2} + \left(\frac{5}{2}\right)^2\right] - \frac{1}{4} = \left(x - \frac{5}{2}\right)^2 - \frac{1}{4}$$

We know that the minimum value of $\left(x - \frac{5}{2}\right)^2$ is equal to zero, so the minimum value of $y = -\frac{1}{4}$.

Example 3 Find the minimum value of $y = (x - a)(x - b)$.

(a) ab (b) $\frac{(a-b)^2}{4}$

(c) 0 (d) $-\frac{(a-b)^2}{4}$

Solution $x^2 - x(a + b) + ab$

$$= \left[x^2 - 2\frac{(a+b)}{2}x + \left(\frac{a+b}{2}\right)^2\right] - \frac{(a-b)^2}{4}$$

$$= \left[x - \left(\frac{a+b}{2}\right)\right]^2 - \frac{(a-b)^2}{4}$$

Now, we know that the minimum value of any perfect square is 0.

So, the minimum value of $y = -\frac{(a-b)^2}{4}$

2. Maxima and minima of any modulus

We have seen above how to find the maxima and the minima of any modulus. Here, we will discuss some more situations involving modulus. Let us see this with the help of an example:

Finding the maximum and the minimum value of $y = f(x) = |x + 3| + |x - 5| + |x - 7|$.

Maximum value: Obviously, the maximum value can be extended upto $+\infty$.

Minimum value: In this case, all the three parts of $f(x)$, i.e., $|x + 3|$, $|x - 5|$, and $|x - 7|$ cannot be simultaneously equal to zero. So, we are required to find the critical points here at first, and then these points in $f(x)$ to see that which one gives us the minimum value.

To obtain critical points, put all the three components $|x + 3|$, $|x - 5|$, and $x - 7|$ one by one equal to zero.

$|x + 3| = 0$ so $x = -3$

$|x - 5| = 0$ so $x = 5$

and $|x - 7| = 0$, so $x = 7$

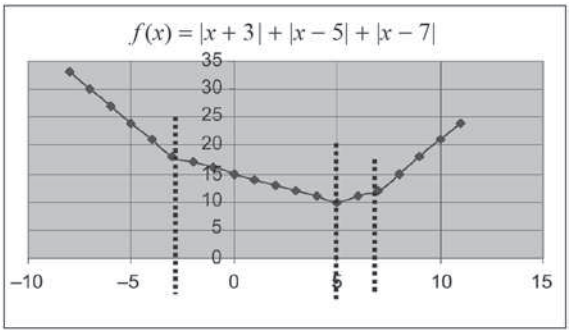
Now, putting these values in $f(x)$ gives us the following result:

At $x = -3$, $f(x) = 18$

At $x = 5$, $f(x) = 10$

At $x = 7$, $f(x) = 12$

So, the minimum value of $f(x) = 10$



Alternatively, drawing the graph gives us the clear picture regarding the movement of the graph which gives us the minimum value of $f(x)$.

The dotted lines in the above graph show the movement of the graph. Obviously, at $x = 5$, graph is at its lowest point, which is the minimum value of the graph.

The inspection of the graph gives us the maximum value of $f(x)$ which is $+\infty$.

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3. Finding the maximum or the minimum value of the product/sum of two or more than two variables when the sum/product of these variables is given.

1. If sum of two or more than two variables is given, then the product will be maximum when the value of all the variables are equal.

Example 4 Find the maximum and the minimum value of xy subject to $x + y = 8$.

(a) 8 (b) 16 (c) 20 (d) 24

Solution The maximum value of xy will occur when $x = y = 4$.

So, the maximum value of $xy = 4 \times 4 = 16$

To obtain the minimum value, we can take either x or y to be negative, and then the product of xy will be negative. This process continues till $-\infty$. So, the minimum value of $xy = -\infty$.

2. If the product of two or more than two positive variables is given, then there sum will be the minimum when the value of all the variables are equal.

Example 5 What is the minimum value of $f(x) \frac{x^3 + x + 2}{x}$, $= x > 0$?

Solution Since $\frac{x^3 + x + 2}{x} = x^2 + 1 + \frac{1}{x} + \frac{1}{x}$

The product of all the terms x^2, 1, $\frac{1}{x}$, and $\frac{1}{x}$ is 1, the sum of these terms will have the minimum value for $x^2 = 1 = \frac{1}{x}$, that is, for $x = 1$. Hence, the minimum value of $f(x)$ is 4.

Example 6 If $ab^2c^3 = 27 \times 2^8$, then find the minimum value of $a + b + c$ (given that $a, b, c > 0$).

Solution To find the minimum value of $a + b + c$, we need to know the product of a, b, and c.

Assuming $a = x$, $b = 2y$, and $c = 3z$, we have $x \times 4y^2 \times 27z^3 = 27 \times 2^8$. So, now we have to calculate the minimum value of $x + 2y + 3z$.

Thus, $ab^2c^3 = 27 \times 2^8 \Rightarrow x \times y^2 \times z^3 = 2^6$

Now, $x + 2y + 3z = x + y + y + z + z + z$ and we know that $x \times y \times y \times z \times z \times z = 2^6$. So, the least value will occur when all are equal to 2. Thus, the least sum will be 12.

Corollary: For any positive value of x, the minimum value of the sum of x and its reciprocal will be 2.

$$\left(x + \frac{1}{x}\right)_{\min} = 2$$

Example 7 If $a > 1$ and $b > 1$, then the minimum value of $\log_b a + \log_a b$ is:

(a) 0 (b) 1

(c) 2 (d) None of these

Solution Assume $\log_a b = N$, so $\log_b a = 1/N$

So, the minimum value of $(\log_b a + \log_a b)$ = minimum value of $(N + 1/N) = 2$

4. Using arithmetic mean, geometric mean, and harmonic mean to find the maxima and the minima

(i) AM ≥ GM ≥ HM

(ii) $(GM)^2 = AM \times HM$ (for two numbers only)

Example 8 If a, b, c, and d are positive real numbers such that $a + b + c + d = 2$, then which of the following is true regarding the values of $N = (a + b)(c + d)$?

(a) $0 \le N \le 1$ (b) $1 \le N \le 2$

(c) $2 \le N \le 3$ (d) $0 \le N \le 1$

Solution Using AM ≥ GM

$$\frac{1}{2}[(a + b) + (c + d)] \ge [(a + b)]^{1/2} = \frac{1}{2}(2) \ge (N)^{1/2}$$

Also, $(a + b)(c + d) \ge 0$

So, $0 \le N \le 1$

GRAPHS

Sketching a Graph

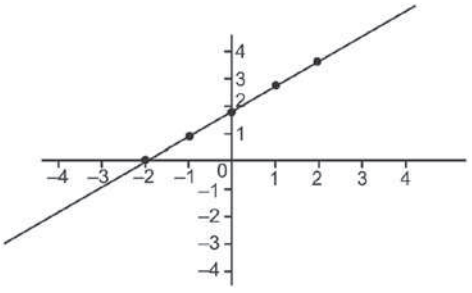
To sketch any graph, we make a table of points on the X-axis and the corresponding points on the Y-axis.

1. $y = x + 2$

Making a table of points for x and y, we can have the graph as given below.

y	x
2	0
3	1
4	2
1	–1
0	–2

$y = x + 2$



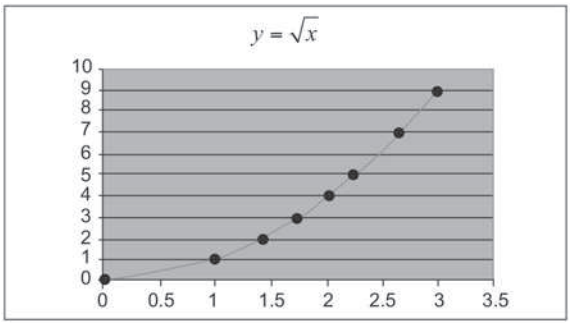
2. $y = \sqrt{x}$

Making a table of points for x and y, we can have the graph as given below.

y	x
0	0
1	1
1.414214	2
1.732051	3
2	4

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3. $y = |x - 1| + |x - 2|$

To draw the graph of any modulus function, we need to work with the help of the critical points, which help us in determining the intervals in which the modulus will be +ve or –ve.

Arranging the critical points on the number line:

$-\infty$ 1 2 $+\infty$

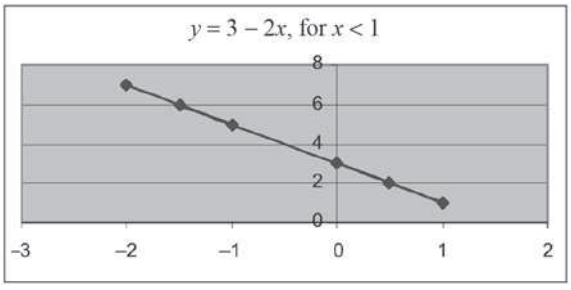
For all the values $-\infty < x < 1$,

$|x - 1| + |x - 2| = (1 - x) + (2 - x) = 3 - 2x$

So, the table for values in between this interval is given below.

$y = 3 - 2x$	x
3	0
2	0.5
6	–1.5
5	–1
7	–2

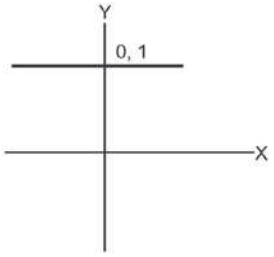
So, the graph for this interval will look like:



For all the values $1 < x < 2$,

$|x - 1| + |x - 2| = (x - 1) + (2 - x) = 1$

So, the graph for this interval will look like:



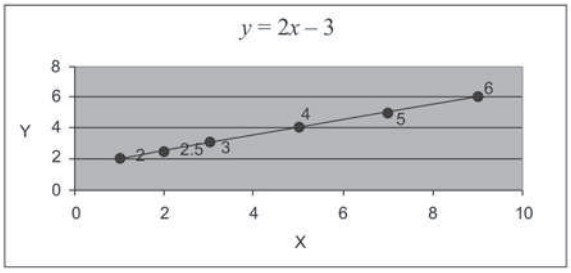
For all the values $x > 2$,

$|x - 1| + |x - 2| = (x - 1) + (x - 2) = 2x - 3$

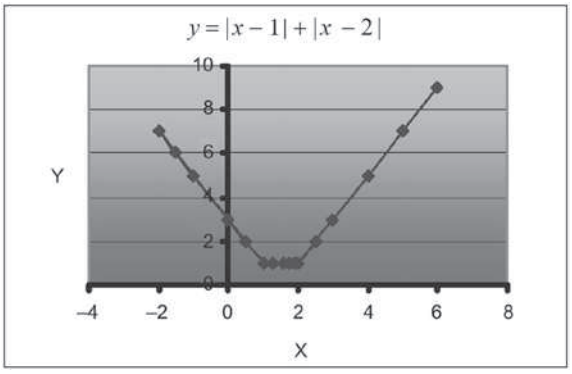
So, the table for values in between this interval is given below.

$y = 2x - 3$	x
2	2.5
1	2
3	3
5	4
7	5
9	6

So, the graph will look like:

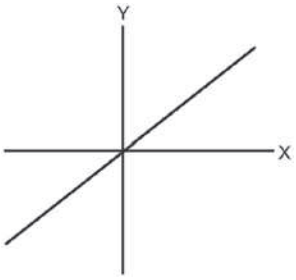


Combining all these graphs, we finally get the below graph.



Some Standard Graphs

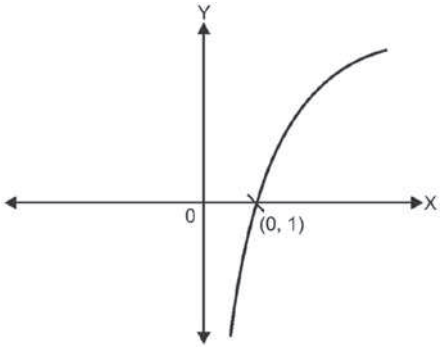
1. $y = x$



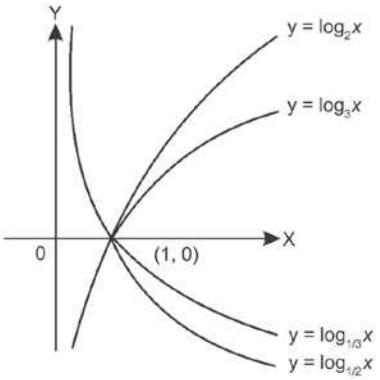
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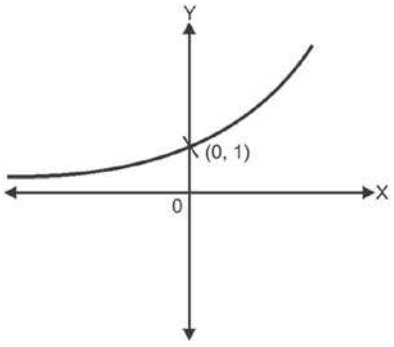
2. $y = \log_e x$



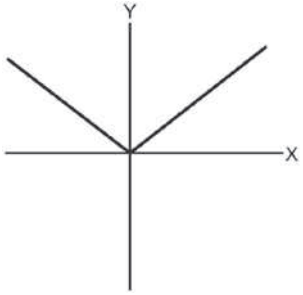
3. $y = \log_a x$



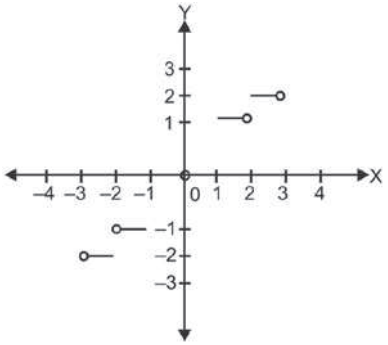
4. $y = e^x$



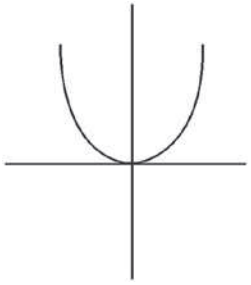
5. $y = |x|$



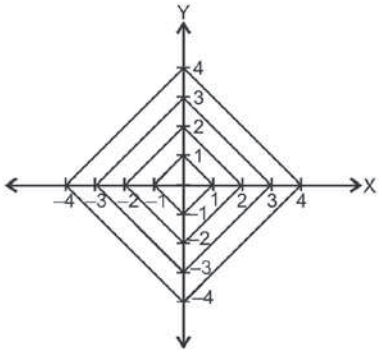
6. $y = [x]$, where $[x]$ represents the greatest integer less than or equal to x.



7. $y = x^2$



8. $|x| + |y| = K$



Shifting of a Graph

To know how a graph would shift, we should know the following two things:

(i) The shifting is either on the X-axis or on the Y-axis or on both of them.

(ii) The shifting is in the +ve direction or in the –ve direction.

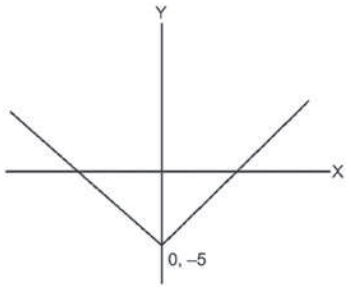
Let us understand this phenomena with the help of an example $y = f(x) = |x|$.

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Case I

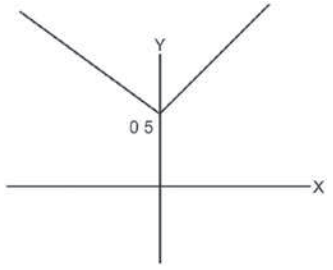
If $y = f(x)$ is changed to	Impact on the graph of $y = f(x)$
$Y = f(x) - 5$	It will shift 5 units downwards, i.e., 5 units on the –ve Y-axis.



$y = |x| - 5$

Case II

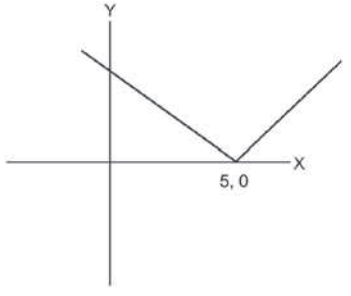
If $y = f(x)$ is changed to	Impact on the graph of $y = f(x)$
$Y = f(x) + 5$	It will shift 5 units upwards, i.e., 5 units on the +ve Y-axis.



$y = |x| + 5$

Case III

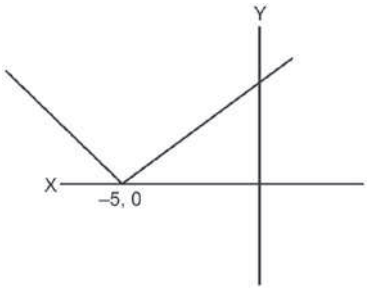
If $y = f(x)$ is changed to	Impact on the graph of $y = f(x)$
$Y = f(x) - 5$	It will shift 5 units rightwards, i.e., 5 units on the +ve X-axis.



$y = |x - 5|$

Case IV

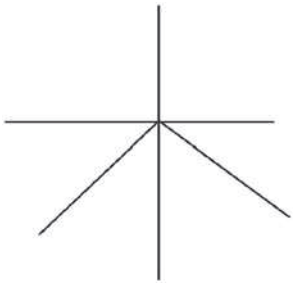
If $y = f(x)$ is changed to	Impact on the graph of $y = f(x)$
$Y = f(x) - 5$	It will shift 5 units leftwards, i.e., 5 units on the +ve X-axis.



$y = |x + 5|$

Case V

If $y = f(x)$ is changed to	Impact on the graph of $y = f(x)$
$Y = - f(x)$	It will be reflected along the X-axis as if X-axis is a mirror.



$y = -|x|$

Case VI

If $y = f(x)$ is changed to	Impact on the graph of $y = f(x)$
$Y = \|f(x)\|$	Graph below the X-axis will be reflected along the X-axis on its above as if X-axis is a mirror.

We can summarize the whole phenomena of shifting of a graph as follows:

1. If there is a change in the value of X-axis, then it will move on the X-axis, and it is similar for the Y-axis also. The graph of $y = f(x - 5)$ has a change in the values of x where as graph of $y = f(x) - 5$, which can be written as $y + 5 = f(x)$ has a change in the values of y.

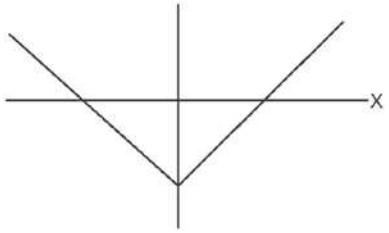
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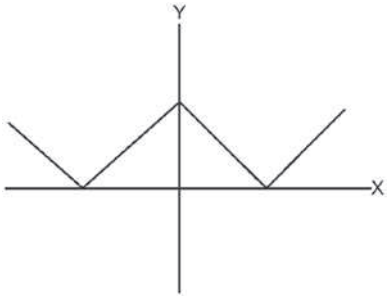
2. If there is an addition, then the graph will move on the −ve side, either on the X-axis or on the Y-axis, and if there is a subtraction, then graph will move on the +ve side.

Example 9 The graph of $y = |x|$ is given. Draw the graph of $y = ||x| - 2|$.

Solution Let us first sketch $|x| - 2$.

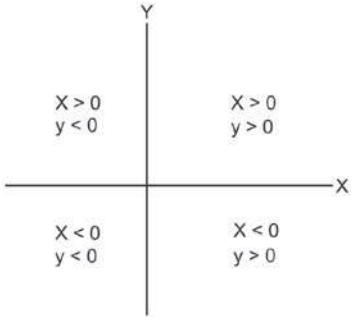


Now, $||x| - 2|$ will be having all the negative part of $|x| - 2$ on the positive side of Y-axis or above X-axis.



Area of a Graphs

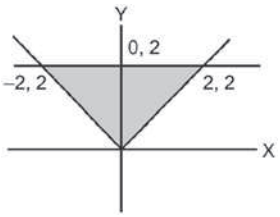
Before we proceed ahead with calculating the area of the combination of graphs, we should be clear with the quadrants and the signs of X and Y in the same.



To find the area of graphs, we first need to sketch the graphs of the equations and then by using geometry/co-ordinate geometry, we can find the area of the enclosed figure.

Example 10 Find out the area of the region enclosed by $y = |x|$ and $y = 2$.

Solution Following is the area enclosed by the equations given above:



The area of the enclosed figures will be,
Area = $1/2 \times 4 \times 2 = 4$ sq. units

Example 11 In the X–Y plane, the area of the region bounded by the graph $|x + y| + |x - y| = 4$ is:
(a) 8 (b) 12 (c) 16 (d) 20

Solution Let $x \geq 0$, $y \geq 0$, and $x \geq y$
Then, $|x + y| + |x - y| = 4$
$\Rightarrow x + y + x - y = 4 \Rightarrow x = 2$
Similarly, $x \geq 0$, $y \geq 0$, $x \leq y$
$x + y + y - x = 4 \Rightarrow y = 2$
The area in the first quadrant is 4.
By using symmetry, the total area in all the four quadrants = $4 \times 4 = 16$ sq. units

Example 12 If p, q, and r are any real numbers, then:
(a) maximum $(p, q) <$ maximum (p, q, r)
(b) minimum $(p, q) = \frac{1}{2}(p + q - |p - q|)$
(c) minimum $(p, q) <$ minimum (p, q, r)
(d) None of these

Solution If we take $r < p, q$, then options (a) and (c) cannot hold. For (b), if $p \geq q$, then $|p - q| = p - q$

$$\frac{1}{2}(p+q-|p-q|)=\frac{1}{2}(p+q-p+q|)=q=\text{minimum}\,(p,q)$$

Similarly, if $p < q$, then $|p - q| = q - p$

$$(p+q-|p-q|)=\frac{1}{2}(p+q-q+p)=p=\text{minimum}\,(p,q)$$

So, option (b) is the answer.

Example 13 If $a\ b\ c\ d = 1$, $a > 0$, $b > 0$, $c > 0$, $d > 0$, then what is the minimum value of $(a + 1)\ (b + 1)\ (c + 1)\ (d + 1)$?
(a) 1 (b) 8
(c) 16 (d) None of these

Solution The minimum value will occur when $a = b = c = d = 1$
So, the minimum value of $(a + 1)\ (b + 1)\ (c + 1)\ (d + 1) = 16$

Example 14 Consider a triangle drawn on the X-Y plane with its three vertices (41, 0), (0, 41), and (0, 0), and each vertex being represented by its (X,Y) coordinates. What is

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the number of points with integer coordinates inside the triangle (excluding all the points on the boundary)?
(a) 780 (b) 800 (c) 820 (d) 741

Solution The equation formed from the data is $x + y < 41$.
The values which will satisfy this equation are:
(1, 39), (1, 38), …(1, 1)
(2, 38), (2, 37), …(2, 1)
(39, 1)
So, the total number of cases are $39 + 38 + 37 \ldots + 1$
$= \frac{39 \times 40}{2} = 780$

Example 15 A telecom service provider engages male and female operators for answering 1000 calls per day. A male operator can handle 40 calls per day, whereas a female operator can handle 50 calls per day. The male and the female operators get a fixed wage of ₹250 and ₹300 per day, respectively. In addition, a male operator gets ₹15 per call he answers and female operator gets ₹10 per call she answers. To minimize the total cost, how many male operators should the service provider employ assuming he has to employ more than 7 of the 12 female operators available for the job?
(a) 15 (b) 14 (c) 12 (d) 10

Solution First let us form both the equations:
$40\,m + 50\,f = 1000$
$250\,m + 300\,f + 40 \times 15\,m + 50 \times 10 \times f = A$
$850\,m + 8000\,f = A$
Where m and f are the number of males and females, and A is the amount paid by the service provider.
Then, the possible values for f are 8, 9, 10, 11, and 12
If $f = 8$, then $m = 15$
If $f = 9$, 10, and 11, then m will not be an integer, whereas $f = 12$, then m will be 10.
By putting $f = 8$, $m = 15$, and $A = 18{,}800$. When $f = 12$ and $m = 10$, then $A = 18{,}100$
Hence, the number of males will be 10.

Example 16 If a, b, and c are the sides of a triangle, then what is the maximum value of the expression $\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b}$?
(a) 1 (b) 3/2 (c) 2 (d) 5/2

Solution Assume $2s = a + b + c$. We know that $b + c > a$, so we get $2(b + c) > a + b + c = 2s$
Hence, $b + c > s$
Similarly, $c + a > s$, $a + b > s$
Hence, $\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} < \frac{a+b+c}{s} = \frac{2s}{s} = 2$

Example 17 At how many distinct points the graphs of $y = x^{-1}$ and $y = \log_e x$ intersect? ***(CAT 2003)***

Solution We can see that the graphs of $y = x^{-1}$ and $y = \log_e x$ intersect just once.

Example 18 How many integral solution is/are possible for the equation $|y - 18| + |y - 9| + |y + 9| + |y + 18| = 54$?

Solution Here, $|y - N|$ should be seen as nothing but the distance of the point y from the point N on the number line, a person standing at a point N. Hence, $|y - 18| + |y - 9| + |y + 9| + |y + 18|$ is the sum of the distances of the point y from 18, 9, –9, and –18. Now, for any point y, where $p \le y \le q$, the sum of the distances from p and q is $q - p$. So, for point y, where $-9 \le y \le 9$, the sum of the distances of y from –18, –9, 9, and 18 is $[18 - (-18)] + [9 - (-9)] = 54$
For points outside this limitation, this expression will have different values. Hence, the required numbers are –9, –8, …, 8, and 9. So, there will be a total of 19 values.

Some Special Cases

(a) Given that $a + b + c + d \ldots = K$ (constant)
Maximum value of $(a^x.b^y.c^z.d^t \ldots..)$ will be obtained if $\frac{a}{x} = \frac{b}{y} = \frac{c}{z} = \frac{d}{t} = \ldots$

Example 19 Given that $a + b + c = 30$, and a, b, and c are all positive. What is the maximum value of $a^2 \cdot b^3 \cdot c^1$?

Solution Maximum value of $a^2 \cdot b^3 \cdot c^1$ will be obtained if $\frac{a}{2} = \frac{b}{3} = \frac{c}{1} \Rightarrow a:b:c = 2:3:1$
Assume a, b, and c are $2x$, $3x$, and x.
Given that $a + b + c = 30 \Rightarrow 2x + 3x + x = 6x = 30$
$\Rightarrow x = 5$
Hence, $a = 2x = 10$, $b = 3x = 15$, and $c = x = 5$
Maximum value of $a^2 \cdot b^3 \cdot c^1 = 10^2 \cdot 15^3 \cdot 5^1 = 16{,}87{,}500$

(b) Given that $ax + by = K$ (constant)
Maximum value of $x^p y^q$ is obtained when $\frac{ax}{p} = \frac{by}{q}$.
Converse of the above expression is also true.
When the expression $x^p y^q$ is constant, then minimum value of $ax + by$ is obtained when $\frac{ax}{p} = \frac{by}{q}$.

Example 20 Given that $5x + 2y = 50$. What is the maximum value of (x^2y^3)?

Solution Maximum value of (x^2y^3) will be obtained when $\frac{5x}{2} = \frac{2y}{3} \Rightarrow x = \frac{4y}{15}$

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$$5x+2y=5\times\frac{4y}{15}+2y=50\Rightarrow\frac{4y}{3}+2y=50\Rightarrow 10y=50$$

$$\Rightarrow y=5.\ \text{Hence, } x=\frac{4\times5}{15}=\frac{4}{3}$$

$$\text{Hence, the maximum value of } (x^2y^3)=\left(\frac{4}{3}\right)^2\times(5)^2=\frac{2000}{9}$$

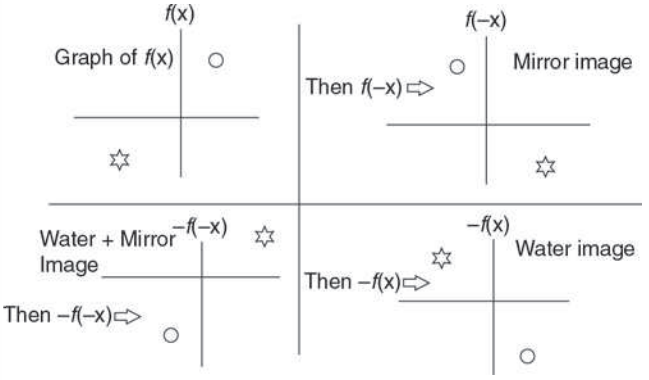
MIRROR IMAGE AND WATER IMAGE

If graph of $f(x)$ is given, then graph of $f(-x)$ will be the mirror image of the graph of $f(x)$ ⇒ Mirror image means it will rotate around Y-axis.

If graph of $f(x)$ is given, then graph of $-f(x)$ will be the water image of the graph of $f(x)$ ⇒ Water image means it will rotate around X-axis.

If graph of $f(x)$ is given, then graph of $-f(-x)$ will be both water and mirror images of the graph of $f(x)$ ⇒ Mirror image and water image means it will rotate around both the Y-axis and X-axis.

Let us understand it through the pictures given below.



Note: These pictures are given just to give you a better understanding of the graphs mentioned above.

Practice Exercises

Warm Up

Q.1 What is the minimum value of the expression $x^2+8x+10$?
(a) 0 (b) $+\infty$ (c) -6 (d) $+6$

Q.2 What is the maximum value of the expression $1-6x-x^2$?
(a) 0 (b) 10
(c) $-\infty$ (d) None of these

Q.3 What is the maximum value of the function $y=\dfrac{2x^2+3x+4}{x^2+x+3}$?
(a) 11/23 (b) 23/11 (c) 4/7 (d) 7/4

Q.4 In the above question, what is the minimum value of y?
(a) 0 (b) 1
(c) -1 (d) Cannot be determined

Q.5 If $f(x)$ = maximum $(4x+3, 3x+6)$ for $x\in[-6, 10]$, then find the maximum value of $f(x)$?
(a) $+\infty$ (b) 36
(c) 43 (d) None of these

Q.6 What is the maximum value of $f(x)$ = minimum $(4-5x, x-3)$ for every $x\in(0, 4)$?
(a) -1 (b) 2
(c) 4 (d) None of these

Direction for Questions 7 to 11: Go through the following definitions and solve the questions based on it.

A (x, y, z) = minimum $(x+y, y+z, z+x)$
B (x, y, z) = minimum $(x-y, y-z, z-x)$
C (x, y, z) = maximum (A (x, y, z), B(x, y, z))
D (x, y, z) = minimum (A(x, y, z), B (x, y, z))

Q.7 What is the value of C (1, 2, 3)?
(a) 1 (b) 2 (c) 3 (d) 4

Q.8 Find the value of D (0, 1, 2).
(a) -1 (b) 1
(c) 0 (d) None of these

Q.9 Find the value of A (2, 3, 4) + B (2, 3, 4).
(a) 13 (b) 4
(c) 7 (d) None of these

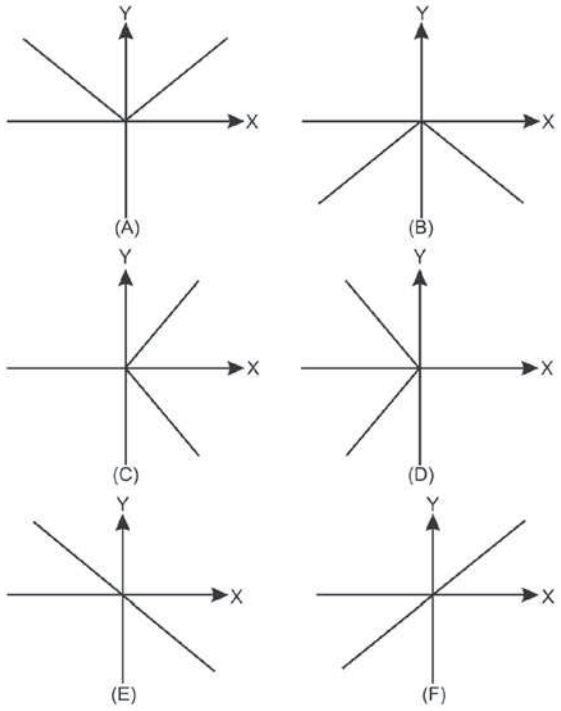
Q.10 Find the value of C (0, 1, D (1, 2, 3)).
(a) 1 (b) 0
(c) -2 (d) None of these

Q.11 What is the value of D (1, 2, C (0, 1, 2))?
(a) -1 (b) 2 (c) 3 (d) 0

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Direction for Questions 12 to 16: Following graphs represent various functions. Match the figure with the appropriate function.



Q.12 $f(x) = -x$
(a) B (b) C (c) E (d) D

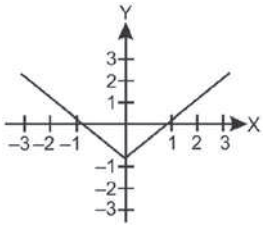
Q.13 $f(x) = |x|$
(a) A (b) B (c) C (d) D

Q.14 $f(x) = -|x|$
(a) E (b) B (c) A (d) F

Q.15 $f(x) = x$
(a) E (b) F (c) A (d) B

Q.16 $f(x) = |x| - x$
(a) A (b) B
(c) C (d) None of these

Q.17 Which of the following represents the following graph?



(a) $|x - 1|$ (b) $|x + 1|$ (c) $|x| - 1$ (d) $|x| + 1$

Q.18 The graph of $y = (x + 2)^2 - 3$ is the graph of $y = x^2$.
(a) Shift 2 units up and 3 units rightwards towards +ve X-axis.
(b) Shift 2 units rightwards towards +ve X-axis and 3 units downwards towards –ve Y-axis.
(c) Shift 2 units leftwards towards –ve X-axis and 3 units downwards towards –ve Y-axis.
(d) Shift 2 units up and 3 units rightwards towards +X-axis.

Q.19 The graph of the function $y = f(x)$ is symmetrical about the line $x = 2$, then:
(a) $f(x + 2) = f(x - 2)$ (b) $f(2 + x) = f(2 - x)$
(c) $f(x) = f(-x)$ (d) None of these

Q.20 If $x + y + z = 24$, then the maximum value of xyz is:
(a) 215 (b) 512
(c) 125 (d) 576

Foundation

Q.1 The minimum value of $4^x + 4^{1-x}$, $x \in R$, is:
(a) 2 (b) 4
(c) 1 (d) None of these

Q.2 a, b, and c are three positive numbers and abc^2 has the greatest value $\frac{1}{64}$, then:
(a) $1 = b = \frac{1}{2}, c = \frac{1}{4}$ (b) $a = b = \frac{1}{4}, c = \frac{1}{2}$
(c) $a = b = c = \frac{1}{3}$ (d) None of these

Q.3 The sum of the products of the ten numbers $\pm 1, \pm 2, \pm 3, \pm 4, \pm 5$ taking two at a time is:
(a) 165 (b) –55
(c) 55 (d) None of these

Q.4 If $ab = 2a + 3b$, $a > 0$, $b > 0$, then the minimum value of ab is:
(a) 12 (b) 24
(c) $\frac{1}{4}$ (d) None of these

Q.5 If $a > 1$ and $b > 1$, then find the minimum value of $\log_b a + \log_a b$.
(a) 0 (b) 1
(c) 2 (d) None of these

Q.6 What is the maximum value of $\frac{x}{y}$ for $(x - 2)^2 = 4$, and $(y - 3)^2 = 25$?
(a) $\frac{1}{2}$ (b) $\frac{5}{8}$ (c) $-\frac{1}{8}$ (d) $-\frac{5}{2}$

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Q.7 If a and b are both positive real values, then find the minimum value of $(a+b)\left(\frac{1}{a}+\frac{1}{b}\right)$.

(a) 0 (b) 2 (c) −2 (d) 4

Q.8 If $|X+2|>3$, then find the solution set for X.

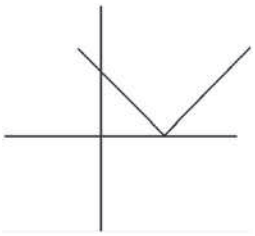
(a) $X>-1$ (b) $X>1$
(c) $X<-5$ (d) $X>1$ or $X<-5$

Q.9 If a and b are both positive real values, then find the maximum value of $(a+b)\left(\frac{1}{a}+\frac{1}{b}\right)$.

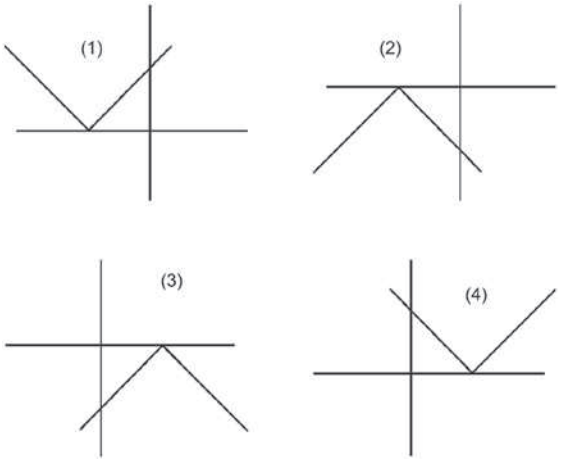
(a) 0 (b) 2 (c) 4 (d) ∝

Direction for Questions 10 to 13: Read the passage below and solve the questions based on it.

The graph of $y=f(x)$ is given as:



Following are the four graphs:



Q.10 Which of the following defines Graph (1)?

(a) $f(x)-1$ (b) $f(x-1)$
(c) $f(x)+1$ (d) $f(-x)$

Q.11 Which of the following defines Graph (2)?

(a) $f(x)-1$ (b) $f(x-1)$
(c) $f(x)+1$ (d) None of these

Q.12 Which of the following defines Graph (3)?

(a) $-f(x)$ (b) $f(x-1)$
(c) $f(x)+1$ (d) $f(x+1)$

Q.13 Which of the following defines Graph (4)?

(a) $f(x)-1$ (b) $f(x)$
(c) $f(x)+1$ (d) $f(x+1)$

Q.14 What is the maximum value of $y=|x-5|+|x-7|$?

(a) −2 (b) 2 (c) 0 (d) ∝

Q.15 If a lies between 2, and 3, both included, and b lies between 4, and 6, both included, then what is the ratio of the minimum and the maximum limits of a^2-b^2?

(a) −4 (b) 4 (c) $\frac{32}{7}$ (d) $-\frac{28}{6}$

Q.16 If $y=|x|-5$, then the minimum possible value of y is:

(a) 5 (b) −5
(c) 0 (d) Both a and b

Q.17 Given that $(x-2)^2=9$, and $(y-3)^2=25$. What is the minimum value of $\frac{x}{y}$?

(a) $\frac{1}{2}$ (b) $\frac{5}{8}$ (c) $-\frac{1}{8}$ (d) $-\frac{5}{2}$

Q.18 When $x^2+4xy+4y^2$ takes a minimum value, then:

(a) $x=-2y$ (b) $x=2y$
(c) $2x=y$ (d) $-2x=y$

Q.19 A man finds that it costs ₹$(300+4N)$ per day to make N articles. If the selling price of each article is ₹7, then what is the minimum number of articles to be produced per day if he has to make a profit?

(a) 300 (b) 150
(c) 100 (d) None of these

Q.20 If a, b, and c are all the real numbers, then the minimum value of $ab+bc+ca$ is:

(a) 1 (b) $-\frac{1}{2}$
(c) $-\frac{3}{2}$ (d) None of these

Q.21 $|x+y|=10$, where x and y are integers, then what is the minimum value of x^2+y^2?

(a) 75 (b) 100
(c) 50 (d) None of these

Q.22 If $|x|+|y|=7$, then what is the sum of the minimum and the maximum values of $x+y$?

(a) $\frac{3}{2}$ (b) −7 (c) 7 (d) 0

Q.23 If $|x|-|y|=13$, then which of the following cannot be the value of $x-y$?

(a) −19 (b) −9
(c) −17 (d) None of these

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Q.24 What is the minimum value of $\frac{(a+1)^2}{a}+\frac{(b+1)^2}{b}+\frac{(c+1)^2}{c}$?

(a) 1 (b) 4 (c) 8 (d) 12

Q.25 Let y = maximum $(x+3), (7-2x)$ what is the minimum value of y for $2 \geq x \geq 1$?

(a) 4.50 (b) 4.00 (c) 4.67 (d) 4.33

Q.26 The minimum value of $3x + 3y + z$ subjected to the condition $xyz = 24$, where x, y, and z are all positive real numbers, is:

(a) $14 \times 3^{1/3}$ (b) 18 (c) 216 (d) 12

Q.27 Find the maximum and the minimum values of the function $\frac{x^2-x+1}{x^2+x+1}$ for real values of x.

(a) 3 and –3 (b) $\frac{1}{3}$ and $-\frac{1}{3}$

(c) 3 and $\frac{1}{3}$ (d) None of these

Q.28 A group of children walked to the Ramnarayan's shop and bought pastries for a total of ₹168 and ice candies for a total of ₹126. Each child has one pastry and one ice candy. How many children could be there in the group?

(a) 28 (b) 24 (c) 42 (d) 36

Q.29 Let x, y, and z be three positive numbers such that $x + y + z = 1$. The minimum value of $\left(x+\frac{1}{x}\right)^2+\left(y+\frac{1}{y}\right)^2+\left(z+\frac{1}{z}\right)^2$ will be:

(a) 16 (b) 24

(c) 33 (d) None of these

Q.30 If a, b, and c are the positive real numbers, then find the greatest value of $a^2\ b^3\ c^4$ subject to the condition $a + b + c = 18$.

(a) $+\infty$ (b) $4^2.6^3.8^4$

(c) 16686 (d) None of these

Q.31 What is the least value of $(x - 2)\ (x - 4)^2\ (x - 6)\ (x + 6)$, for real values of x?

(a) 6 (b) $\frac{4}{3}$

(c) 4 (d) None of these

Q.32 Consider the following functions:

$f(x) = x^2 + 3x$

$g(x) = 3x + 4$

For what value of x will $f[g(x)]$ have its minimum value?

(a) 0 (b) $\frac{-11}{2}$

(c) $\frac{-11}{6}$ (d) None of these

Q.33 Suppose July of year N has five Mondays. Which of the following will definitely occur five times in the August of the same year?

(a) Monday (b) Tuesday

(c) Thursday (d) Friday

Q.34 Four distinct circles are drawn in a plane. What is the maximum number of points where at least two of the circles intersect?

(a) 8 (b) 9 (c) 10 (d) 12

Q.35 The least value of $\frac{1}{x}+\frac{1}{y}+\frac{1}{z}$ for positive x, y, and z, satisfying the condition $x + y + z = 9$ is:

(a) $\frac{15}{7}$ (b) $\frac{1}{9}$ (c) 3 (d) 1

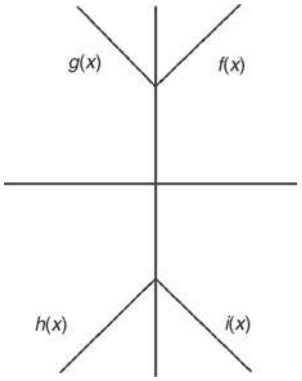
Q.36 The minimum value of $3x + 4y$, subject to the conditions $x^2y^3 = 6$ and x and y are positive, is:

(a) 10 (b) 14 (c) 13 (d) 13

Q.37 The sum of 18 consecutive positive integers is a perfect square. The smallest possible value of this sum is:

(a) 225 (b) 289 (c) 361 (d) 441

Direction for Questions 38 to 40: Go through the graph given below and solve the questions based on it (neglect fine errors).



Q.38 Which of the following is true?

(a) $f(x) = g(-x)$ (b) $f(x) = -g(x)$

(c) $f(x) = -g(-x)$ (d) None of these

Q.39 Which of the following is true?

(a) $i(x) = g(-x)$ (b) $i(x) = -g(x)$

(c) $i(x) = -g(-x)$ (d) None of these

Q.40 How many of the following statements are true?

A. $g(x) = h(-x)$ B. $f(x) = -h(-x)$

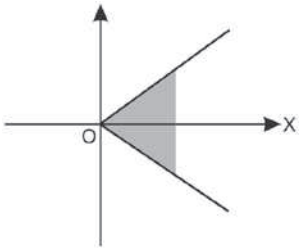
C. $h(x) = -i(x)$ D. $f(x) = -i(x)$

(a) 1 (b) 2 (c) 3 (d) 4

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Q.41 The shaded region in the diagram represents the relation:

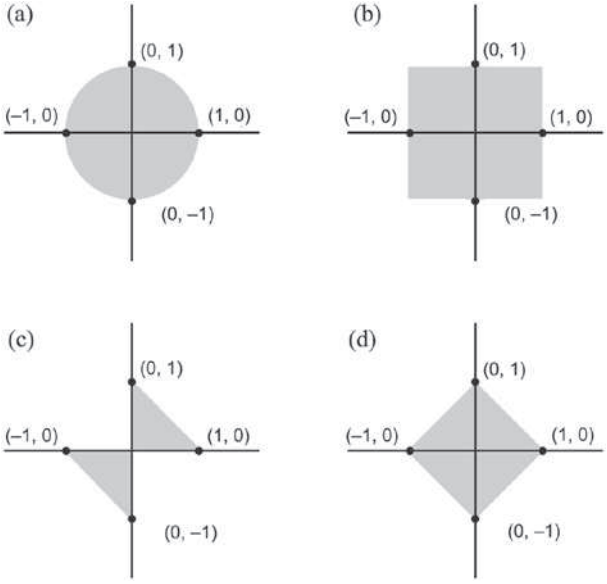


(a) $y \le x$ (b) $|y| < |x|$
(c) $y \le |x|$ (d) $|y| \le x$

Q.42 If P, Q, and R are three vertices with coordinates (1, 4), (4, 2), and $(m, 2m - 1)$, respectively, then the value of m for which PR + RQ is minimum is:

(a) $\frac{17}{8}$ (b) $\frac{5}{2}$ (c) $\frac{7}{4}$ (d) $\frac{11}{8}$

Q.43 The set $\{(x, y): |x| + |y| \le 1\}$ is represented by the shaded region in one of the four figures. Which one is it?



Q.44 If x, y, and z are three integers satisfying $x + y + z = 13$ and $xy + xz = 42$, then what is the minimum possible value of $y^2 + z^2$?

(a) 18 (b) 22.5 (c) 16 (d) 24.5

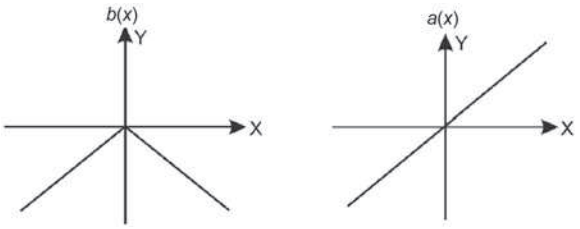
Q.45 What will be the maximum value of $\left(\frac{1}{x}\right)^x$?

(a) e (b) e^{-1} (c) $e^{1/e}$ (d) $e^{-1/e}$

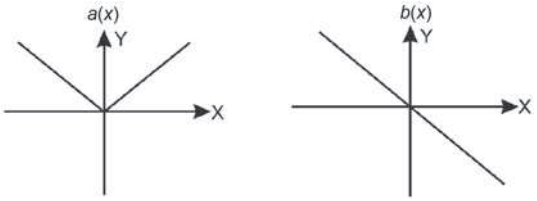
Direction for Questions 46 to 48: Read the passage below and solve the questions based on it.

Mark option (a) as the answer if $a(x) = b(-x)$ for $x < 0$, and $a(x) = -b(-x)$ for $x > 0$.
Mark option (b) as the answer if $a(x) = -b(x)$ for $x < 0$, and $a(x) = b(-x)$ for $x > 0$.
Mark option (c) as the answer if $a(x) = -b(x)$ for all the values of x.
Mark option (d) as the answer if none of the above is true.

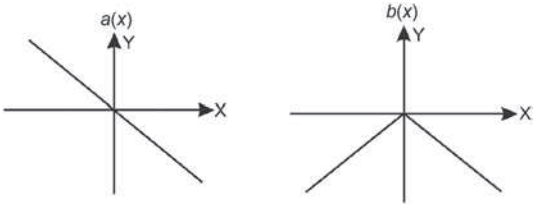
Q.46



Q.47



Q.48



Q.49 Let a, b, c, and d be the real numbers. If G = maximum {minimum (a, b), minimum(c, d)} and H = minimum {maximum (a, c), maximum(b, d)}, then

(a) G ≤ H, for all values of a, b, c, and d
(b) G = H, for all values of a, b, c, and d
(c) G ≥ H, for all values of a, b, c, and d
(d) Any one of the above three options can be true for some particular values of a, b, c, and d.

Q.50 If $x = \frac{5}{a+b}$ and $y = 5(a + b)$, and if both x and y are greater than zero, then $(x + y)$ is:

(a) <5 (b) ≥10
(c) Between 5 and 10 (d) None of these

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MODERATE

Q.1 We have $|a + b| = |a| + |b|$, $|c + d| < |c| + |d|$, $|e+f| < |e| + |f|$. If $a, b, c, d, e, f \in$ R, then the product *abcdef* is:
(a) Equal to zero
(b) Necessarily non-negative
(c) Necessarily non-positive
(d) Either positive or negative

Q.2 For how many values of x, is the equality $|||x - 2|-3|-1| = 0$, true?
(a) 0 (b) 2
(c) 3 (d) None of these

Q.3 Let $f(x) = |x - a| + |x - 2a| + |x - 3a|$, where $a > 0$. Let the minimum value of $f(x)$ be denoted by m and the corresponding value of x be denoted by b. Which of the following is true?
(a) $m = a; b = a$ (b) $m = a; b = 2a$
(c) $m = 2a; b = a$ (d) $m = 2a; b = 2a$

Q.4 The minimum of $f(x) = |x - 1| + |x - 4| + |x - 5|$ is attained at $x =$
(a) 1 (b) 3 (c) 4 (d) 5

Q.5 Let $y = k^2(x^2 + 1) + 2k(3x + 4) + 27$. If k and x are real numbers, what is the minimum value of y?
(a) 1 (b) 3
(c) 4 (d) None of these

Q.6 If $x < -3$ and $y > 2$, then which of the following is true for all real values of x and y?
(a) $x + 2y < -1$ (b) $3y + 2x > 0$
(c) $2x - xy > 0$ (d) None of these

Q.7 If $a_1, a_2, \ldots a_n$ are n integers such that $a_1 a_2 \ldots a_n = 1$, then which of the following must be true?
(a) The minimum value of $(a_1 + a_2 + \ldots + a_n)$ is $-(n - 1)$, if n is odd.
(b) The minimum value of $(a_1 + a_2 + \ldots + a_n)$ is $(n - 2)$, if n is even.
(c) The minimum value of $(a_1 + a_2 + \ldots + a_n)$ is $(2 - n)$, if n is even.
(d) None of these

Q.8 If a and b are positive real values, and $ab + (a + b) = 21$, what is the minimum value of $(a + b)$?
(a) 6 (b) 8
(c) 12 (d) Cannot be determined

Q.9 The least integral value of k for which $(k - 2)x^2 + 8x + k + 4 > 0$ for all $x \in R$, is:
(a) 5 (b) 4
(c) 3 (d) None of these

Q.10 If $f(x) = 2x^2 + 3x + 4$ and $g(x) = 5 - x^2$, then the minimum $[f(x)]$ − maximum $[g(x)]$ is:
(a) 0 (b) 5 (c) $\frac{17}{8}$ (d) $-\frac{17}{8}$

Q.11 If $a + b + c = 20$ and $\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) = 30$, then the value of $\frac{a}{b} + \frac{b}{c} + \frac{a}{c} + \frac{c}{a} + \frac{b}{c} + \frac{c}{b}$ is:
(a) 597 (b) 8 (c) 350 (d) 441

Q.12 The minimum value of the expression $\frac{1+V^2}{1+V}$, for positive values of V, is:
(a) $\sqrt{2}$ (b) $\frac{2}{\sqrt{2}+1}$
(c) 1 (d) None of these

Q.13 If p, q, r, and s are real numbers such that $p^2 + q^2 = 1$, $r^2 + s^2 = 1$, then the maximum value of $pr + qs$ would be:
(a) −1 (b) 2 (c) −2 (d) 1

Q.14 If p, q, and r denote the sides of a triangle, then $\frac{p}{q+r} + \frac{q}{r+p} + \frac{r}{p+q}$ always lie between (both end values inclusive):
(a) 3 and 4 (b) 2 and 3
(c) 3.5 and 5 (d) 1.5 and 2

Q.15 p, q, r, and s are any four positive real numbers, the minimum value of $\frac{p}{q} + \frac{q}{r} + \frac{r}{s} + \frac{s}{p}$ is:
(a) 1 (b) 2 (c) $2\sqrt{2}$ (d) 4

Q.16 p, q, and r are all positive numbers. If $p + q + r = 1$, then the least value of $\left(\frac{1}{p} - 1\right)\left(\frac{1}{q} - 1\right)\left(\frac{1}{r} - 1\right)$ is:
(a) 10 (b) 25
(c) 24 (d) None of these

Q.17 Find the maximum value of $y = 5 - |x + 1| - |x - 3|$.
(a) 1 (b) 5 (c) 4 (d) ∞

Q.18 If three positive real numbers x, y, and z are in an AP such that $xyz = 4$, then what will be the minimum value of y?
(a) $2^{1/3}$ (b) $2^{2/3}$ (c) $2^{1/4}$ (d) $2^{3/4}$

Q.19 Find the least value of the expression $2\log_{10} x - \log_x (.01)$ for $x > 1$.
(a) 4 (b) 2
(c) 1 (d) None of these

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Q.20 When the curve $y = x^2 + 4x - 5$ and $y = \frac{1}{(1+x^2)}$ are drawn in the $x - y$ plane, how many times do they intersect for values of $x > 0$?
(a) Never (b) Once
(c) Twice (d) More than twice

Q.21 If p, q, and r are three real numbers such that $p + q + r = 4$ and $p^2 + q^2 + r^2 = 6$, then what can be the maximum value of q?
(a) 1 (b) 2/3
(c) 4 (d) None of these

Q.22 In the above question, what will be the value of p when q is minimum?
(a) 2 (b) 1/3
(c) 2/3 (d) None of these

Q.23 Given y = minimum $(x - 5, 7 - x)$. What is the minimum value of y?
(a) 0 (b) $-\infty$ (c) $+\infty$ (d) 1

Q.24 Given y = minimum $(x - 5, 7 - x)$. What is the maximum value of y?
(a) 0 (b) $-\infty$ (c) $+\infty$ (d) 1

Q.25 Given y = maximum $(x - 5, 7 - x)$. What is the minimum value of y?
(a) 0 (b) $-\infty$ (c) $+\infty$ (d) 1

Q.26 Given y = maximum $(x - 5, 7 - x)$. What is the maximum value of y?
(a) 0 (b) $-\infty$ (c) $+\infty$ (d) 1

Q.27 Let y be the maximum of $\{(x + 5), (3 - x)$. What is the minimum value of y, if $-2 \le x \le 0$?
(a) 3 (b) 2
(c) 1 (d) None of these

Q.28 The maximum possible value of $x^2 + 3y^2 + 9z^2$, subject to $x + 2y + 3z = 12$, where x, y, and z are real numbers, is:
(a) 48 (b) 216
(c) 240 (d) None of these

Q.29 At how many different points $y = x^{-1}$ and $y = \log_e x$ intersect?
(a) 0 (b) 1 (c) 2 (d) Infinite

Q.30 A rectangle of the largest area is inscribed in a semicircle of radius r. What is the area of the rectangle?
(a) $1.5 \times r^2$ (b) $r^2 \times \sqrt{2}$
(c) r^2 (d) $r^2\sqrt{3}$

Q.31 The circumference of a circle and the perimeters of a pentagon and an octagon are the same. Their areas are denoted by C, P, and O, respectively. Which of the following is true for their areas?
(a) C > P > O (b) O > P > C
(c) O > C > P (d) C > O > P

Q.32 At what value of x, $\left(\frac{1}{x}\right)^x$ will attain the maximum value?
(a) e (b) e^{-1} (c) e^{-2} (d) e^2

Q.33 The altitude of a right circular cone of the minimum volume circumscribed about a sphere of the radius r is:
(a) $2r$ (b) $3r$
(c) $4r$ (d) None of these

Q.34 The minimum value of $(x - a)(x - b)$ is:
(a) ab (b) 0
(c) $\frac{(a-b)^2}{4}$ (d) $-\frac{(a-b)^2}{4}$

Q.35 If x, y, and z are arbitrary positive real numbers satisfying the equation $4xy + 6xz + 8zx = 9$, then find the maximum possible value of the product xyz.
(a) $\frac{1}{2\sqrt{2}}$ (b) $\frac{\sqrt{3}}{4}$
(c) $\frac{3}{8}$ (d) None of these

Q.36 The minimum value of the quantity $\frac{(a^2+3a+1)(b^2+3b+1)(c^2+3c+1)}{abc}$, where a, b, and c are positive real numbers, is:
(a) $\frac{11^3}{2^3}$ (b) 125 (c) 25 (d) 27

Q.37 A right circular cylindrical container closed on both the sides contains a fixed volume of motor oil. Suppose its base has the diameter d and its height is h. The overall surface area of the container is minimum when:
(a) $h = \frac{4}{3}\pi d$ (b) $h = 2d$
(c) $h = d$ (d) None of these

Q.38 A rectangular box of volume 48 cu. ft is to be constructed, so that its length is twice its width. The material to be used for the top and the four sides is three times costlier per ft^2 than that used for the bottom. Then, the height (in ft) of the box that minimizes the cost is equal to:
(a) $\frac{8}{27}$ (b) $\frac{8\sqrt[3]{4}}{3}$ (c) $\frac{4}{27}$ (d) $\frac{8}{3}$

Q.39 A truck is to be driven for 300 km on a highway at a constant speed of x km/h. The speed rules of the highway requires $30 \le x \le 60$. The fuel costs ₹10 per litre and is consumed at the rate of $2 + \frac{x^2}{600}$ litres per hour. The wages of the driver are ₹200 per hour. The most economical speed to drive the truck, in km/h, is:
(a) 30 (b) 60
(c) $30\sqrt{3.3}$ (d) $20\sqrt{33}$

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Q.40 A square tin sheet of side 12 inches is converted into a box with an open top in the following steps: The sheet is placed horizontally. Then, equal sized squares, each of side x inches, are cut from the four corners of the sheet. Finally, the four resulting sides are bent vertically upwards in the shape of a box. If x is an integer, then what is the value of x that maximizes the volume of the box?

(a) 3 (b) 4 (c) 1 (d) 2

Q.41 How many integral points are there within the graph of $|x| + |y| \le 4$?

(a) 32 (b) 41
(c) 9 (d) None of these

Q.42 The sides of a rectangle of the greatest area that can be inscribed into an ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$ is:

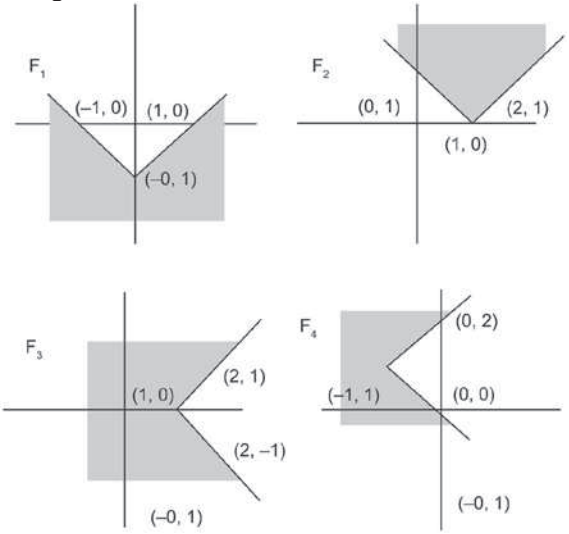
(a) 5, 3 (b) $5\sqrt{2}$, 3
(c) 5, $3\sqrt{2}$ (d) $5\sqrt{2}$, $3\sqrt{2}$

(See geometry chapter for ellipse)

Q.43 The sets

$\{[x, y]: |y - 1| - x \ge 1\}$
$\{[x, y]: |x| - y \ge 1\}$
$\{[x, y]: |x - 1| - y \le 1\}$
$\{[x, y]: |y - 1| x - 1 | \ge 0\}$

are represented by the shaded regions in the figures given below in some order.



The correct order of the figures is:

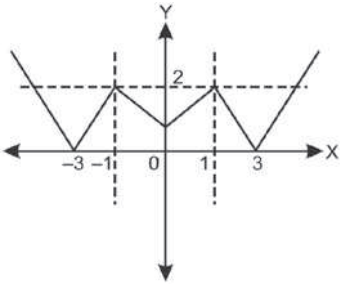
(a) F_4, F_1, F_2, F_3 (b) F_4, F_2, F_3, F_1
(c) F_1, F_4, F_3, F_2 (d) F_4, F_1, F_3, F_2

Direction for Questions 44 to 46: Read the passage below and solve the questions based on it.

In each question, choose the function that best describes the graph.

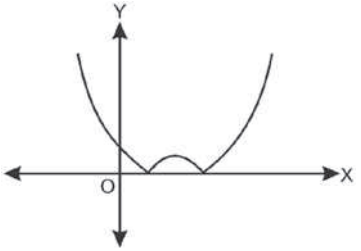
$f(x) = (|x| - 1)(|x| - 2)$
$g(x) = ||x - 1| - 2| - 2$
$h(x) = ||\,|x| - 1| - 2|$
$j(x) = (|x| + 1)(|x| - 2)$
$k(x) = |x - 1|\,|x - 3|$
$l(x) = ||\,|x| - 2| - 1|$

Q.44



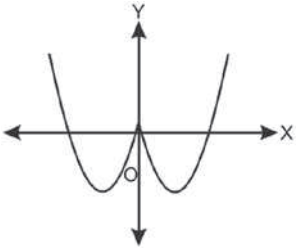
(a) $g(x)$ (b) $l(x)$ (c) $h(x)$ (d) $k(x)$

Q.45



(a) $f(x)$ (b) $l(x)$ (c) $j(x)$ (d) $k(x)$

Q.46



(a) $f(x)$ (b) $k(x)$ (c) $j(x)$ (d) $h(x)$

Q.47. From a square tin sheet of side 12 feet, a box with its top open is made by cutting equal squares at the four corners and then bending the tin sheet so as to form the sides of the box. The side of the removed square for which the box has the maximum possible volume in feet is:

(a) 3 (b) 1
(c) 2 (d) None of these

Q.48. From a circular sheet of paper of radius a, a sector with a central angle is cut out and folded into the shape of a conical funnel. The volume of this funnel is maximum when θ equals:

(a) $\frac{2\pi}{\sqrt{2}}$ (b) $2\pi\sqrt{\frac{2}{3}}$
(c) $\frac{\pi}{2}$ (d) π

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Answers

Warm Up

1. (c)	**2.** (b)	**3.** (b)	**4.** (b)	**5.** (c)	**6.** (d)	**7.** (c)	**8.** (a)	**9.** (c)	**10.** (d)
11. (a)	**12.** (c)	**13.** (a)	**14.** (b)	**15.** (b)	**16.** (d)	**17.** (c)	**18.** (c)	**19.** (b)	**20** (b)

Foundation

1. (b)	**2.** (b)	**3.** (b)	**4.** (b)	**5.** (c)	**6.** (a)	**7.** (d)	**8.** (d)	**9.** (d)	**10.** (d)
11. (d)	**12.** (a)	**13.** (b)	**14.** (d)	**15.** (c)	**16.** (c)	**17.** (b)	**18.** (a)	**19.** (d)	**20.** (d)
21. (c)	**22.** (d)	**23.** (b)	**24.** (d)	**25.** (d)	**26.** (b)	**27.** (c)	**28.** (c)	**29.** (d)	**30.** (b)
31. (d)	**32.** (c)	**33.** (c)	**34.** (d)	**35.** (d)	**36.** (a)	**37.** (a)	**38.** (a)	**39.** (c)	**40.** (b)
41. (d)	**42.** (a)	**43.** (d)	**44.** (a)	**45.** (c)	**46.** (d)	**47.** (d)	**48.** (b)	**49.** (a)	**50.** (b)

Moderate

1. (b)	**2.** (d)	**3.** (d)	**4.** (c)	**5.** (d)	**6.** (b)	**7.** (d)	**8.** (a)	**9.** (a)	**10.** (d)
11. (a)	**12.** (b)	**13.** (d)	**14.** (d)	**15.** (d)	**16.** (d)	**17.** (a)	**18.** (b)	**19.** (a)	**20.** (d)
21. (d)	**22.** (d)	**23.** (b)	**24.** (d)	**25.** (d)	**26.** (c)	**27.** (d)	**28.** (d)	**29.** (b)	**30.** (c)
31. (d)	**32.** (b)	**33.** (d)	**34.** (d)	**35.** (c)	**36.** (b)	**37.** (c)	**38.** (d)	**39.** (b)	**40.** (d)
41. (b)	**42.** (d)	**43.** (d)	**44.** (b)	**45.** (d)	**46.** (c)	**47.** (c)	**48.** (b)		

Hints and Solutions

Warm up

1. Minimum value of the expression $x^2 + 8x + 10$ is $\frac{-D}{4a}$

$= \frac{-(b^2 - 4ac)}{4a} = \frac{-[8^2 - 4\times1\times10]}{4\times1} = \frac{-24}{4} = -6$

2. For expression like $-x^2 - 6x + 1$,

Maximum value $= \frac{-D}{4a} = -\frac{(-6)^2(4\times-1\times1)}{4\times-1} = -\frac{36+4}{-4}$

$= 10$

3. $\frac{2x^2+3x+4}{x^2+x+3} = y$

$2x^2 + 3x + 4 = yx^2 + yx + 3y$

$x^2(y - 2) + x(y - 3) + 3y - 4 = 0$

For this equation, $b^2 - 4ac$ should be greater than or equals to zero.

Then, $b^2 - 4ac \geq 0 \rightarrow (y - 3)^2 - 4(y - 2)(3y - 4) \geq 0$

$-11y^2 + 34y - 23 \geq 0$

$-11y^2 + 11y + 23y - 23 \geq 0$

$(11y - 23)(1 - y) \geq 0$, or, $11y - 23 \geq 0$, or, $y \geq \frac{23}{11}$

From here, the maximum value of y should be $\frac{23}{11}$.

4. From the previous question,

$(11y - 23)(1 - y) \geq 0$

$(1 - y) \geq 0 \rightarrow y - 1 \leq 0$

Then, $y \leq 1$

From here, the minimum value of y should be 1.

5. $f(x)$ = maximum $(4x + 3, 3x + 6)$ for $x \in [-6, 10]$

It is clear from the function that for the maximum value of $[4x + 3, 3x + 6]$, the value of x should be the biggest positive number, which is +10.

So, for $x = 10$, $f(x)$ = maximum $(4 \times 10 + 3, 3 \times 10 + 6)$ = maximum $(43, 36) = 43$

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6. For maximum value of $f(x)$ = minimum $(4 - 5x, x - 3)$, where $x \in (0, 4)$

x should be 1.

So, maximum value of $f(x)$ = minimum $(4 - 5 \times 1, 1 - 3)$ = minimum $(-1, -2) = -2$, which is the maximum value of $f(x)$.

7. It is given that

$C(x, y, z)$ = maximum $(A(x, y, z), B(x, y, z))$

Then, $A(1, 2, 3)$ = minimum $(1 + 2, 2 + 3, 3 + 1) = 3$

$B(1, 2, 3)$ = minimum $(1 - 2, 2 - 3, 3 - 1) = -1$

$C(1, 2, 3)$ = maximum $(3, -1) = 3$

8. $A(0, 1, 2)$ = minimum $(0 + 1, 1 + 2, 2 + 0) = 1$

$B(0, 1, 2)$ = minimum $(0 - 1, 1 - 2, 2 - 0) = -1$

Then, $D(0, 1, 2)$ = minimum $(1, -1) = -1$

9. $A(2, 3, 4) + B(2, 3, 4)$ = minimum $(2 + 3, 3 + 4, 4 + 2)$ + minimum $(2 - 3, 3 - 4, 4 - 2) = 5 - 1 = 4$

10. $C(0, 1, D(1, 2, 3)) = ?$

Here, $D(1, 2, 3)$ = minimum $(A, (x, y, z), B(x, y, z))$ = minimum $[3, -1] = -1$

Then, $C(0, 1, -1)$ = maximum $(A(0, 1, -1), B(0, 1, -1))$

$C(0, 1, -1)$ = maximum $[-1, -1] = -1$

11. $D(1, 2, C(0, 1, 2)) = ?$

Here, $C(0, 1, 2)$ = maximum $(A(0, 1, 2), B(0, 1, 2)$ = maximum $(1, -1) = 1$

Then, $D(1, 2, 1)$ = minimum $(A(1, 2, 1), B(1, 2, 1))$ = minimum $(2, -1) = -1$

12. $y = f(x) = -x$

Here, if $x = 1$, then $y = -1$

If $x = 0$, then $y = 0$

If $x = -1$, then $y = 1$

Put these coordinates on graph, which is graph (E).

So, the answer is option (c).

13. $y = f(x) = (x)$

Here, if $x = -1$, then $y = 1$

If $x = 0$, then $y = 0$

If $x = 1$, then $y = 1$

These are the coordinates of graph (A).

14. $y = f(x) = -|x|$

Here, if $x = 1$, then $y = -1$

If $x = 0$, then $y = 0$

If $x = -1$, then $y = -1$

Therefore, these are the coordinates of graph (B).

15. $y = f(x) = x$

If $x = 1$, then $y = 1$

If $x = 0$, then $y = 0$

If $x = -1$, then $y = -1$

Therefore, these are the coordinates of graph (F).

16. $y = f(x) = |x| - x$

If $x = 1$, then $y = 0$

If $x = 2$, then $y = 0$

If $x = -1$, then $y = 2$

Here, these coordinates are not of any graphs.

17. Do it from options, and the answer is option (c) because

$y = f(x) = |x| - 1$

If $x = 1$, then $y = 0$

If $x = 0$, then $y = -1$

If $x = -1$, then $y = 0$

Therefore, these are the coordinates of the given graph.

20. $x + y + z = 24$

If x, y, and z all are equal, then the value of xyz will be maximum.

So, $x = y = z = 8$

Then, $x\,y\,z = 8 \times 8 \times 8 = 512$

FOUNDATION

1. We know that AM ≥ GM

$$\frac{4^x + 4^{1-x}}{2} \geq [4^x \times 4^{1-x}]^{1/2} = [4^{(x+1-x)}]^{1/2}$$

$$\text{or } \frac{4^x + 4^{1-x}}{2} \geq [4^{(x+1-x)}]^{1/2}$$

$$\text{or } \frac{4^x + 4^{1-x}}{2} \geq 2, \text{ or, } 4^x + 4^{1-x} \geq 4$$

Hence, minimum value = 4

2. Assume $a = x$, $b = x$, and $c = 2x$

Then, $abc^2 = \frac{1}{64}$

$$x \times x \times (2x)^2 = \frac{1}{64} \rightarrow x \times x \times 4x^2 = \frac{1}{64}$$

$$x^4 = \frac{1}{512} \quad \text{or } x = \frac{1}{4}$$

So, $a = \frac{1}{4}$, $b = \frac{1}{4}$, and $c = 2x = \frac{1}{2}$

4. For minimum value of ab, $2a$ should equal to $3b$.

Then, $2a = 3b$ or $\frac{a}{b} = \frac{3}{2}$

Then, $a = 3k$ and $b = 2k$

Put $k = 1$, then $a = 3$ and $b = 2$

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However, for these values, $ab \neq 2a + 3b$

Then, put $k = 2$, then $a = 6$ and $b = 4$

For $a = 6$ and $b = 4$, $ab = 2a + 3b$

So, the minimum value of $ab = 4 \times 6 = 24$

5. Method 1

$\log_a b = 1/\log_b a$

Minimum value of $x + \frac{1}{x} = 2$ if $x > 0$

Method 2

$\log_b a + \log_a b = ?$ When $a > 1$, $b > 1$

Put $a = b$ (>1)

Then, $\log_b b + \log_b b = 1 + 1 = 2$

If we take $a > b$ or $b > a$, then the value of the function will be greater than 2.

So, the minimum value of $\log_b a + \log_a b = 2$

6. $(x - 2)^2 = 4 = (\pm 2)^2$

$x - 2 = \pm 2$

Then, $x = +4$ or 0 (i)

$(y - 3)^2 = 25 = (\pm 5)^2$

$y - 3 = \pm 5$

$y = 8$ or -2 (ii)

From these values of x and y, the maximum value of $\frac{x}{y} = \frac{4}{8} = \frac{1}{2}$

7. $(a+b)\left(\frac{1}{a}+\frac{1}{b}\right) = \frac{a}{a}+\frac{a}{b}+\frac{b}{a}+\frac{b}{b} = \frac{a}{b}+\frac{b}{a}+2$

$= \left(\frac{a}{b}+\frac{b}{a}\right)+2$

The minimum value of $\frac{a}{b}+\frac{b}{a}$ will be, when $(a = b)$

$= \frac{a}{b}+\frac{b}{a}+2 = \frac{b}{b}+\frac{b}{b}+2 = 4$

If we put $a > b$ or $b > a$, then it will be greater than 2.

8. $|x + 2| > 3$, when $(x + 2)$ is a positive number.

Then, $x + 2 > 3 \rightarrow x > 3 - 2$

$x > 1$ (i)

When $(x + 2)$ is a negative number,

Then, $-(x + 2) > 3 \rightarrow x + 2 < -3$

$x < -5$ (ii)

Then, the solution set of x is ($x > 1$ or $x < -5$).

9. For positive values of a and b,

Maximum value of $(a + b)\left(\frac{1}{a}+\frac{1}{b}\right) = \frac{a}{b}+\frac{b}{a}+2$

Put $a > b$ or $b > a$, if we increase the gap between a and b, then the value of $\left(\frac{a}{b}+\frac{b}{a}\right)$ will also increase. So, the maximum value of $\left(\frac{a}{b}+\frac{b}{a}+2\right)$ would be (∞).

Answers to Q.10 to 13:

The given graph is a graph of function

$f(x) = |x - a|$

10. Graph (1) has shifted some points towards $-x$ axis. So, graph could be $f(x + a)$, which is like option (d). Hence, option (d) is the answer.

14. It is clearly seen that if we increase the value of x in the function, then the value of y will also increase. Hence, the maximum value of the function should be (∞).

15. The value of $a = [2, 3]$ and the value of $b = [4, 6]$

When the value of a is minimum and value of b is maximum, the minimum value of $a^2 - b^2 = (2)^2 - (6)^2 = -32$.

When the value of a is maximum and value of b is minimum, the value of $a^2 - b^2$ would be maximum.

So, the maximum value of $a^2 - b^2 = (3)^2 - (4)^2 = -7$

Then, ratio of minimum $(a^2 - b^2)$ and maximum $(a^2 - b^2) = \frac{-32}{-7} = \frac{32}{7}$

16. $y = |x| - 5$. Then, the minimum value of y will be -5 because the minimum value of $|x| = 0$.

17. $(x - 2)^2 = 9 = (\pm 3)^2$

$x - 2 = \pm 3$

$x = 5$ or -1 and

$(y - 3)^2 = 25 = (\pm 5)^2$

$y - 3 = \pm 5$

$y = 8$ or -2

Then, minimum value of $\frac{x}{y} = \frac{5}{-2} = -\frac{5}{2}$ (which is minimum).

18. $x^2 + 4xy + 2y^2 = (x + 2y)^2$

Since the minimum value of a perfect square is zero, then $x + 2y = 0 \rightarrow x = -2y$

So, at $x = -2y$, $x^2 + 4xy + 2y^2$ will be minimum.

19. Cost of N article $= 300 + 4N$ and selling price of a article $= 7$

If he makes no profit and no loss, then the cost of N article = selling price of N article

$300 + 4N = 7 \times N = 7N$

$300 = 3N$

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Hence, $N = 100$

At $N = 100$, he makes no profit and no loss.

So, to make a minimum profit, he would have to make 101 articles.

20. For real numbers a, b, and c, the minimum value of $ab + bc + ca$ will be attained when one number is a negative number and others are positive, and it could be $-\infty$.

21. $|x + y| = 10$

Minimum value of $x^2 + y^2$ will be the option, when $x = y = 5$

So, $x^2 + y^2 = (5)^2 + (5)^2 = 50$

The maximum value of $x^2 + y^2$ could be (∞).

22. $|x| + |y| = 7$

The minimum value of $x + y$ will be attained, when both are negative.

So, minimum of $x + y = -7$

The maximum value of $x + y$ will be attained when both are positive.

So, maximum of $x + y = 7$

Then, maximum of $(x + y)$ and minimum of $(x + y) = 7 + (-7) = 0$

25. $y = \text{maximum } (x + 3), (7 - 2x)$

The minimum value of y should be, when

$x + 3 = 7 - 2x$

$x + 2x = 7 - 3$

$3x = 4 \rightarrow x = \frac{4}{3}$

Then, at $x = \frac{3}{4}$, The value of

$y = \text{maximum } \left(\frac{4}{3} + 3\right), \left(7 - 2 \times \frac{4}{3}\right) = \text{maximum } (4.33), (4.33) = 4.33$

26. Minimum value of $3x + 3y + z$ will be attained, when $3x = 3y = z = k$

Then, $xyz = 24 \rightarrow \frac{k}{3} \times \frac{k}{3} \times k = 24$

$k^3 = 6^3 \rightarrow k = 6$

Then, $3x + 3y + 2 = 6 + 6 + 6 = 18$

28. Question is asking about common factors.

Then, factor of $168 = 2 \times 2 \times 2 \times 3 \times 7$ and factors of $126 = 2 \times 3 \times 3 \times 7$

Then, the common factors = 1, 2, 3, 6, 7, 14, 21, 42

So, the number of children could be one of the numbers, which is 42.

31. Least value of $(x - 2)(x - 4)^2(x - 6)(x + 6) = ?$

Take x as a negative number.

If we decrease the value of x, then the value of this function will also decrease.

Hence, the least value of this function is $(-\infty)$.

32. $f(x) = x^2 + 3x$

$g(x) = 3x + 4$

Then, $f[g(x)] = (3x + 4)^2 + 3(3x + 4) = 9x^2 + 24x + 16 + 9x + 12 = 9x^2 + 33x + 28$

Then, the value of x for which $f[g(x)]$ has its minimum value $= -\frac{b}{2a} = \frac{-33}{2 \times 9} = \frac{-11}{6}$

33. If there are five Mondays, then there are only three possibilities for their dates: (1, 8, 15, 22, 29), (2, 9, 16, 23, 30), and (3, 10, 17, 24, 31).

In the first case, August starts on a Thursday, and there are five Thursdays, Fridays, and Saturdays in August.

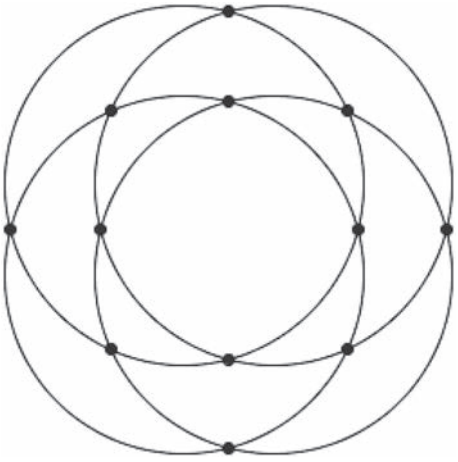
In the second case, August starts on a Wednesday, and there are five Wednesdays, Thursdays, and Fridays in August.

In the third case, August starts on a Tuesday, and there are five Tuesdays, Wednesdays, and Thursdays in August.

The only day of the week that is guaranteed to appear five times is, therefore, Thursday. Hence, the answer is option (c).

34. **Method 1**

For any given pair of circles, they can intersect at most 2 times. Since there are 6 ($^4C_2$) pairs of circles, the maximum number of possible intersections is $6 \times 2 = 12$. We can construct such a situation as below. Hence, the answer is option (d).



Method 2

Maximum total number of points of intersection among N circles $= Np_2$

In this case, number of circles = 4

Hence, maximum total number of points of intersection = $4p_2 = \frac{4!}{2!} = 12$. Hence, option (d) is the answer.

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35. $x + y + z = 9$

For the minimum value of $\frac{1}{x}+\frac{1}{y}+\frac{1}{z}$, the value of x, y, and z should be equal.

Then, $x = y = z = a$

$\therefore\ a + a + a = 9$, so $a = 3$

So, $\frac{1}{x}+\frac{1}{y}+\frac{1}{z}=\frac{1}{3}+\frac{1}{3}+\frac{1}{3}=1$

37. Let a, $a + 1$, …, $a + 17$ be the consecutive positive integers. Their sum $18a + \frac{17(18)}{2} = 9(2a + 17)$ is a perfect square. Since 9 is a perfect square, it follows that $2a + 17$ is a perfect square. The smallest possible perfect square is 25 when $a = 4$, and the sum is 225. Hence, option (a) is the answer.

MODERATE

1. $|a + b| = |a| + |b|$ is possible only if both a and b are having the same signs or $a = b = 0$.

$|c + d| < |c| + |d|$ is possible only if both c and d are having opposite signs, and the same is true about $|e + f| < |e| + |f|$.

Sign of ab = +ve or $ab = 0$

Sign of cd = –ve and sign of ef = –ve

Hence, the sign of $abcdef$ = non-negative

2. $||x - 2| - 3| - 1 = 0$ or $||x - 2| - 3| = 1$ or

$|x - 2| - 3 = 1$ or $|x - 2| = 4$

If $x - 2 > 0$, then $x = 6$, and if $x - 2 < 0$, then $-(x - 2) = 4$, so $x = -2$

Similarly,

If $x - 2 > 0$, then $|x - 2| = (x - 2)$

or $|x - 2| - 3 = -1$ or $(x - 2) = 2$ or $x = 4$

If $x - 2 < 0$, then $|x - 2| = -(x - 2)$

Therefore, $|x - 2| - 3 = -1$ or $-(x - 2) = 2$ or $2 - x = 2$, so $x = 0$

Hence, four values of x are possible.

3. Obviously, the minimum value will be attained at the median of the critical points, which is $x = 2a$, and the minimum value will be $f(x) = 2a$.

4. Minimum value of $f(x)$ will be attained at the median of the critical points of $f(x)$, i.e., $x = 4$.

6. Checking option (b): $3y > 6$ and $2x < -6$, so $3y + 2x > 0$

8. $ab + 2(a + b) = 21 \rightarrow ab + 2(a + b) + 4 = 21 + 4$

So, $(a + 2)(b + 2) = 25$

For minimum value of $(a + b)$, a and b should be equal.

Hence, $(a + 2) = (b + 2) = 5$ (We will not take –5 because it will give negative values of a and b.)

So, $a = b = 3$

Alternatively, the quadratic equation can also be used to find out the minimum value of $(a + b)$.

Minimum value will occur if $a = b$.

So, $ab + 2(a + b) = 21$ will be equal to $a^2 + 2(2a) - 21 = 0$

or $(a + 7)(a - 3) = 0$; hence, $a = 3 = b$

9. $D = 8^2 - 4(k - 2)(k + 4) > 0$

or $(k + 6)(k - 4) < 0$

or $4 < k < 6$. So, $k = 5$

10. Minimum $[f(x)]$ = minimum $(2x^2 + 3x + 4) = 46/16$

maximum $[g(x)] = 5$

11. Use AM ≥ GM ≥ HM for all the four fractions.

15. Use AM ≥ GM ≥ HM for all the four fractions.

17. Find the maximum value of $y = 5 - [|x + 1| + |x - 3|]$

Assume $[|x + 1| + |x - 3|] = z$

So, $y = 5 - z$; y will attain the maximum value at minimum value of z.

Now, minimum of z will occur at $x = -1$ [Using critical points]

So, minimum value of $z = 4$

Maximum value of $y = 1$

18. $2y = x + z$

Now, AM of x, y, and z will be more than their GM.

$Y \geq (xyz)^{1/3}$

Hence, $y_{min} = 4^{1/3} = 2^{2/3}$

19. $2\log_{10} x - \log_x (.01) = \log_{10} x^2 - \log_x 10^{-2} = 2[\log_{10} x + 1/\log_{10} x]$

Minimum value of $[\log_{10} x + 1/\log_{10} x] = 2$

23. Drawing the graphs of $x - 5$ and $7 - x$ tells us that both the graphs are heading towards $+\infty$ and $-\infty$, depending upon the values. So, the minimum of y will be $-\infty$.

24. Now, the direction of movement of both the graphs are opposite (one is going towards $+\infty$ and other one is going towards $-\infty$ for a particular set of values and vice versa).

For $x < 6$, $x - 5$ will provide us the minimum value, and for $x > 6$, $7 - x$ will provide us with the minimum value, and at $x = 1$, we will get the values equal. This $x = 6$ is the point for which y_{max} = minimum $(x - 5, 7 - x)$ is obtained.

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28. Any of x, y, or z can be taken in such a way that $x^2 + 3y^2 + 9z^2$ can go to $+\infty$. Actually, it was a problem of common sense, rather than maxima and minima (refer to introduction of this topic).

29. Drawing the curves tells us that it will intersect at one and only one point.

31. We know that for a fixed perimeter of polygons, polygon with the more number of sides will have more area with respect to the polygon with the less number of sides. Similarly, if circumference of a circle and perimeter of polygon are same, then the area of circle will be more than the area of any polygon (circle can be treated to be a polygon of n sides, n tending to infinity).

32. Go through the options.

34. Make a quadratic equation in the format of $z^2 + y$. Minimum value of $z^2 = 0$; so, minimum value $= y$

36. $$\frac{(a^2+3a+a)(b^2+3b+1)(c^2+3c+1)}{abc}$$

$= [a + (1/a) + 3]\ [b + (1/b) + 3]\ [c + (1/c) + 3]$

Minimum value of $[a + (1/a)] = [b + (1/b)] = [c + (1/c)] = 2$

39. Go through the options.

44. At $x = 0$, $y = 1$ is the key. Otherwise, you do not have to draw the graph. Just check the options using the points given on the graph.

45. Since no part of the graph is negative, function has to involve an overall modulus. So, it cannot be $f(x)$ or $j(x)$. Now, this graph is having two points on the +ve side of X-axis.

Checking option (b), we get

$l(x) = 0 = ||x| - 2| - 1|$ or

$|x| - 2| = 1$ or

$|x| - 2 = \pm 1$ or $|x| - 2 = +1$ or $|x| = 3$ or $x = \pm 3$

$|x| - 2 = -1$ or $|x| = 1$ or $x = \pm 1$

We get four points of x for $l(x) = 0$, so it is not possible.

Checking option (d), we get

$k(x) = |x - 1|\ |x - 3| = 0$, Hence, $x = 1$ or $x = 3$

So, we get two points on +ve X-axis for $k(x) = 0$.

Hence, option (d) is the answer.

Otherwise, since we have eliminated the first three options, the fourth option has to be the answer.

46. Here, we get two values of x, one +ve and other one −ve, for $y = 0$.

Checking the options, we get

Option (a): $f(x) = (|x| - 1)\ (|x| - 2)$

We are getting four points on X-axis for $f(x) = 0$, so it is not possible.

Option (b): $k(x)$ is given in 45. Please go through option (c).

Option (c): $j(x) = (|x| + 1)\ (|x| - 2) = 0$

So, either $(|x| + 1) = 0$ or $(|x| - 2) = 0$

If $(|x| + 1) = 0$, then $|x| = -1$, which is not possible.

If $(|x| - 2) = 0$, then $x = \pm 2$.

Hence, option (c) is the answer.

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CHAPTER

15

Logarithm

LEARNING OBJECTIVES

After completion of this chapter, the reader should be able to understand:

- What is the meaning of log and their bases?
- Definitions and properties
- Different types of questions
- Methods of solving the questions

INTRODUCTION

This chapter is quite important while considering the number of questions being asked in the CAT paper. Usually 1 to 2 questions are asked from this topic. The concept of logarithm is simple, and a simple formula is sufficient to solve the questions without getting into lengthy derivations.

LOGARITHM

If any number N is expressed in the form a^x, then the index 'X' is called the logarithm of the number N to the base 'a'.

Therefore, if $N = a^x$, then $x = \log_a N$. It is read as log of N to the base a.

A simple way (though not exactly mathematical) to recollect the formula:

$N = a^x$ $\quad$ $N = a^x$

$x = \log_a N$ $\quad$ $x = \log_a N$

(i) Base of index will become the base of log.

(ii) Left hand number will go to the right hand side and vice versa.

For example, $10^2 = 100$ will be written in logarithm format as follows:

$2 = \log_{10} 100$

1.(a) Write the following in the index format $x = \log_5 6$.

Solution Base of log will become the log of index and left-hand side number (x) will go to the right-hand side and vice versa.

$x = \log_5 6$

$\Rightarrow 6 = 5^x$

(b) Write the following in the logarithm format $3^4 = 81$.

Solution Base of index will become the base of logarithm and left-hand side number will go to the right-hand side number and vice versa.

$3^4 = 81$

$\Rightarrow \log_3 81 = 4$

(c) Write the following in the index format $x = \log_y z$.

Solution $y^x = z$

(d) Write the following in the logarithm format $a^b = c$.

Solution $b = \log_a c$

Generally, logarithm of any number is calculated to the base 10. When base is not mentioned, it should be taken as 10.

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Restrictions with Logarithm of Any Number

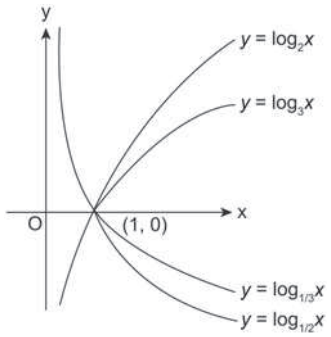
For logarithm of any number to be defined, the number should be greater than zero and base should be positive and not equal to 1.

⇒ $\log_a x$ to be defined as $x > 0$ and $a > 0$ and $a \neq 1$.

⇒ log of negative number is not defined. For example, log (−10) is not defined.

⇒ log to the base of any negative number or log to the base = 1 is not defined. For example, $\log_{(-5)} x$ is not defined. Similarly, $\log_1 x$ is not defined.

It can be seen with the help of a graph (log x is also given alongside).



Following observations can be made from this graph:

1. Value of y can be negative for some value of x.
2. Value of x cannot be negative in any case.
3. For constant x, if base is lying in between 0 and 1, then log x becomes decreasing function. Otherwise, it is an increasing function.

Some Important Properties

In case of all the properties given ahead, we will be using the following standard restrictions on logarithm:

1. $\log_a(XY) = \log_a X + \log_a Y$
   If we do not use the restrictions given regarding log of any number, then we can see a good number of contradictions about numbers. One of the examples of similar nature can be seen here:
   log (12) = log (−4 × −3) = log (−4) + log(−3)
   Now, on the LHS, we have a defined value, but on the RHS, we get a value which is not defined.
   It is all owing to the fact that $\log_a(XY) = \log_a X + \log_a Y$ is possible only if $X > 0$ and $Y > 0$.
2. $\log_a(X/Y) = \log_a X - \log_a Y$
3. (a) $\log_a(X^k) = k \log_a X$

   (b) $\log_a \sqrt[k]{X} = \frac{1}{k} \times \log_a x$
4. $\log_a 1 = 0$ (As $a^0 = 1$ provided $a \neq 1$)
5. $\log_x X = 1$
6. $\log_a X = \frac{1}{\log_x a}$
7. **Base change rule:**
   $\log_a x = \log_a b \times \log_b x = \log_b x \times \log_a b$
8. $a^{(\log_a x)} = x$
9. If $a > 1$ and $x > a$, then $\log_a x > 0$
10. **Increasing and decreasing nature of logarithm function:**
    (a) If 0 < base (assume to be a) < 1
    If $x > y$, then $\log_a x < \log_a y$
    For example, $\log_{0.5} 10 < \log_{0.5} 5$
    (b) If base (assume to be a) > 1
    If $x > y$, then $\log_a x > \log_a y$
    For example, $\log_5 10 > \log_5 5$
11. $\log_a 1 = 0$
12. $\log_a b = \frac{\log b}{\log a}$

Characteristics and Mantissa

The integral part of logarithm is called characteristic and its decimal part is called mantissa. Logarithms to the base 10 are called common logarithms. The characteristic of common logarithm can be found out by visual inspection. The characteristics of the logarithm (base 10) of a number greater than 1 is less by one than the number of digits in the integral part and is positive. However, if a decimal fraction number is less than 1 but positive, its characteristic will greater by unity than the number of consecutives zeros immediately after the decimal point and is negative.

Worked Examples

Example 1 What is the value of $\log_{125} 625$?

Solution $\log_{125} 625 = \log_{5^3} 5^4 = 4 \times \frac{1}{3}(\log_5 5) = \frac{4}{3} \times 1 = \frac{4}{3}$

Example 2 What is the value of $\log_3 2 \cdot \log_4 3 \cdot \log_5 4 \ldots \log_{16} 15$?

Solution $\log_3 2 = \frac{\log 2}{\log 3}, \log_4 3 = \frac{\log 3}{\log 4}$, and so on

$\log_3 2 \cdot \log_4 3 \cdot \log_5 4 \ldots \log_{16} 15$

$= \frac{\log 2}{\log 3} \cdot \frac{\log 3}{\log 4} \cdots \frac{\log 15}{\log 16} = \frac{\log 2}{\log 16}$

$= \log_{16} 2$

$\Rightarrow \log_{16} 2 = \log_{2^4} 2 = \frac{1}{4}(\log_2 2) = \frac{1}{4} \times 1 = \frac{1}{4}$

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Example 3 What is the value of x in the following expression?

$$x + \log_{10}(1 + 2^x) = x \log_{10} 5 + \log_{10} 6$$

(a) 1 (b) 0 (c) −1 (d) 3

Solution Ideally, these questions should be solved using the options.

Checking option (a): Putting $x = 1$

$$\text{LHS} = 1 + \log_{10}(1 + 2^1) = 1 + \log_{10}(3)$$
$$\text{RHS} = \log_{10} 5 + \log_{10} 6 = \log_{10}(6 \times 5)$$
$$= \log_{10} 30 = \log_{10} 10 + \log_{10} 3 = 1 + \log_{10}(3)$$
$$\text{LHS} = \text{RHS}$$

Hence, option (a) is the answer.

Note from the Author—Some Thumb Rules:

(a) If you get plus sign (+) in any logarithm question, you should try to use formula number (1) as mentioned previously and vice versa.
$[\log_a(XY) = \log_a X + \log_a Y]$

(b) If you get minus sign (−) in any logarithm question, you should try to use formula number (2) as mentioned previously and vice versa.
$[\log_a(X/Y) = \log_a X - \log_a Y]$

Example 4 What is the number of digits in 2^{50} (given that log 2 = 0.301)?

Solution Taking log gives us 50 log 2 = 50 × 0.301 = 15.05

Number of digits will be immediate next integer if the result obtained is not an integer. In this case, we obtain the product = 15.05; hence, the number of digits = 16

Example 5 What is the number of digits in 3^{60} (given that log 2 = 0.477)?

Solution Taking log gives us 60 log 3 = 60 × 0.477 = 28.62

Number of digits will be immediate next integer if the result obtained is not an integer. In this case, we obtain the product = 28.62; hence, the number of digits = 29

Example 6 If $5 \log_{27}(y) + 2 \log_9(81y^2) = 20$, then y is equal to:

(a) 1/7 (b) 81 (c) 729 (d) 243

Solution

$$5\log_{27}(y) + 2\log_9(81y) = \frac{5}{3}\log_3(y) + \frac{2}{2}\log_3(81y)$$
$$= \frac{5}{3}\log_3(y) + \log_3(81y)$$
$$= \frac{5}{3}\log(y) + \log_3(81) + \log_3(y) = \left(\frac{5}{3} + 1\right)\log_3(y)$$
$$+ \log_3(81)... \quad \text{[Taking } (\log_3 y) \text{ common]}$$
$$= \frac{8}{3}\log_3 y + 4\log_3 3 = \frac{8}{3}\log_3 y + 4$$

Given that this value = 20

So, $\frac{8}{3}\log_3 y + 4 = 20$

$\Rightarrow \frac{8}{3}\log_3 y = 20 - 4 = 16$

$\Rightarrow \log_3 y = 6 \Rightarrow y = 3^6 = 729$

Hence, option (c) is the answer.

Example 7 If $x \geq y$ and $y > 1$, then the value of the expression $\log_x\left(\frac{x}{y}\right) + \log_y\left(\frac{y}{x}\right)$ can never be equal to:

(a) −1 (b) −0.5 (c) 0 (d) 1

Solution Assume $N = \log_x\left(\frac{x}{y}\right) + \log_y\left(\frac{y}{x}\right)$

$$= \log_x x - \log_x y + \log_y y - \log_y x$$
$$= 1 - \log_x y + 1 - \log_y x = 2 - \log_x y - \log_y x$$

Assume that $\log_x y = t$

$$N = 2 - \frac{1}{t} - t = -\left(\frac{1}{t} + t - 2\right) = -\left(\sqrt{t} + \frac{1}{\sqrt{t}}\right)^2$$
$$= -(\text{Perfect square})$$

Minimum value of a perfect square = 0. Since negative of a perfect square cannot be equal to a positive value ⇒ can never be equal to 1. Hence, option (d) is the answer.

Logarithmic Inequality

Case I If base (assume to be N) > 1

(i) If $x > y$, then $\log_N x > \log_N y$

(ii) Vice versa of the above rule is also true, i.e., if $\log_N x > \log_N y \Rightarrow x > y$

Case II If base = N is $0 < N < 1$

(i) If $x > y$, then $\log_N x < \log_N y$

(ii) Vice versa of the above rule is also true, i.e., if $\log_N x > \log_N y \Rightarrow x < y$

Note: For base > 1, log is an increasing function. For base between 0 and 1, log is a decreasing function.

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Rules for Solving Questions Based upon Logarithmic Inequality

In addition to the rules given above, we need to take care of the following rules also:

For $\log_N x$ to be defined:

(a) $N > 0$ and $N \neq 0, N \neq 1$

(b) $x > 0$

Questions Based upon Logarithmic Inequality

Example 8 Solve the following equations for x: $\log_{10}(2x-1) < \log_{10} x$.

Solution Given that the base > 10.

Hence, $2x - 1 < x \Rightarrow x < 1$ (i)

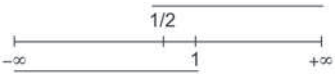
Let us mark this on number line.



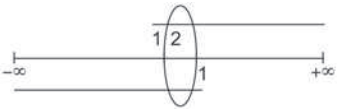
Then, $2x - 1 > 0 \Rightarrow x > 1$

$$\Rightarrow x > \frac{1}{2} \quad \text{(ii)}$$

Marking this on number line gives us the following:



It can be seen that the circled area (as shown in the next diagram) is common and it is the answer.



Hence, the answer is $\frac{1}{2} < x < 1$.

Example 9 Solve the equations for x: $\log_{10} x + \log_{10}(x-1) < \log_{10} 6$.

Solution Base > 1: Hence, the given logarithmic arguments will be in increasing order.

As the base is greater than 1,

$\Rightarrow \quad x(x-1) < 6$

or $x^2 - x - 6 < 0 \Rightarrow (x-3)(x+2) < 0$

Hence, range of value of $x = -2 < x < 3$ (i)

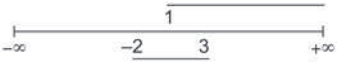
Let us mark this on the number line.



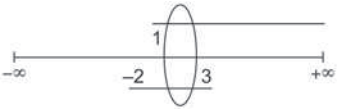
Next, for log to be defined:

$x > 0$ and $x - 1 > 0 \Rightarrow x > 1$ (ii)

Again mark this on the number line.



It can be seen that the circled area (as shown in the next diagram) is common and it is the answer.



Hence, the answer is $1 < x < 3$.

Practice Exercises

Warm Up

Q.1 What is the value of $\frac{1}{\log_2 n} + \frac{1}{\log_3 n} + \ldots + \frac{1}{\log_{40} n}$?

(a) $\frac{1}{\log_{40!} n}$ (b) $\log_n 40!$

(c) 1 (d) 0

Q.2 What is the number of digits in the expansion of 5^{20} given that $\log 5 = 0.698$?

(a) 12 (b) 13 (c) 14 (d) 15

Q.3 What is the value of $\log_{3\sqrt{3}} 27$?

(a) 2 (b) 3 (c) 4 (d) 5

Q.4 Evaluate $\log (36\sqrt{6})$ to the base 6.

(a) 1/2 (b) 5/2 (c) 3/2 (d) 7/2

Q.5 Find the value of $3\log\frac{81}{80} + 5\log\frac{25}{24} + 7\log\frac{16}{15}$.

(a) log 2 (b) log 3

(c) 1 (d) None of these

Q.6 If $[N]$ = the greatest integer less than or equal to N, then $[\log_{10} 6730.4]$ is equal to:

(a) 6 (b) 4 (c) 3 (d) 7

Q.7 What is the value of x in the expression

$\log_7 \log_5[\sqrt{(x+5)} + \sqrt{x}] = 0$?

(a) 1 (b) 2 (c) 3 (d) 4

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Q.8 Arrange the following in ascending order:
$A = \log_7 2401$, $B = \log_{7\sqrt{7}} 343$, $C = \log_{\sqrt{6}} 216$, $D = \log_2 32$
(a) ABCD (b) BDCA (c) BDAD (d) BADC

Q.9 Given that x, y, and z are three angles of a triangle. Which of the following will be one possible set of value for x, y, and z such that $\log(x \times y \times z) = 3 \log X + 4 \log 2$ (x, y, and z are integers)?
(a) 30°, 70°, and 80° (b) 30°, 60°, and 90°
(c) 20°, 80°, and 80° (d) 30°, 50°, and 100°

Q.10 If $2[\log(x+y) - \log 5] = \log x + \log y$, then what is the value of $x^2 + y^2$?
(a) $20xy$ (b) $23xy$ (c) $25xy$ (d) $28xy$

Q.11 What is the value of $\log_{32} 27 \times \log_{243} 8$?
(a) $\frac{\log 9}{\log 4}$ (b) $\frac{\log 3}{\log 2}$
(c) 1 (d) None of these

Q.12 If $x > 1$, $y > 1$, and $z > 1$ are three numbers in geometric progression, then $\frac{1}{1+\log x}, \frac{1}{1+\log y}$, and $\frac{1}{1+\log z}$ are in:
(a) Arithmetic progression
(b) Harmonic progression
(c) Geometric progression
(d) None of these

Q.13 If $\log_k N = 6$ and $\log_{25k}(8N) = 3$, then k is:
(a) 12.5 (b) $(12.5)^2$ (c) $\frac{5^{\frac{2}{3}}}{2}$ (d) $(12.5)^{\frac{2}{3}}$

Q.14 What is the value of x if $\log_3 x + \log_9 x + \log_{27} x + \log_{81} x = \frac{25}{4}$?
(a) 9 (b) 27
(c) 81 (d) None of these

Q.15 What is the value of x in the expression $\log_2(3-x) + \log_2(1-x) = 3$?
(a) 1 (b) 0
(c) −1 (d) Not possible

Q.16 The logarithm of a number to a certain base is 9. The logarithm of 64 times the number to a base which is 11 times the original base is 6. Find the actual base.
(a) $\frac{11}{2}$ (b) $\frac{121}{2}$
(c) $\frac{121}{4}$ (d) Cannot be determined

Q.17 If $\log N^2 - \log 2N = 3 \log 3 - \log 6$, then what is the value of N?
(a) 1 (b) 2
(c) 9 (d) More than one value

Q.18 If $x = \log_a(bc)$, $y = \log_b(ca)$, and $z = \log_c(ab)$, then what is the value of $\frac{1}{1+x} + \frac{1}{1+y} + \frac{1}{1+z}$?
(a) 0 (b) 1 (c) xyz (d) −1

Q.19 If $\log_4 5 = a$ and $\log_5 6 = b$, then what is the value of $\log_3 2$?
(a) $\frac{1}{2a+1}$ (b) $\frac{1}{2b+1}$
(c) $2ab + 1$ (d) $\frac{1}{2ab-1}$

Q.20 The number of solutions of $\log_2(x+5) = 6 - x$ is:
(a) 2 (b) 0 (c) 3 (d) 1

Q.21 Solve the following inequality for x: $2\log_{0.5}(x-4) < \log_{0.5} 2x$.
(a) $x > 4$ (b) $x > 8$
(c) $x > 2$ (d) None of these

Q.22 Solve the following equations for x: $\log_{10}(3-x) > \log_{10}(x-1)$.
(a) $x > 2$ (b) $1 < x < 2$
(c) $0 < x < 1$ (d) $1 < x < 3$

Q.23 Solve the inequality for x: $2\log_{10}(x-4) < \log_{10}(x+8)$.
(a) $x > 4$ (b) $-8 < x < 4$
(c) $4 < x < 8$ (d) $x < 8$

Q.24 Solve the inequality for x: $\log_{0.5}(x-2) - \log_{0.5}(x+1) > 1$.
(a) $x > 2$ (b) $2 < x < 4$
(c) $2 < x < 5$ (d) None of these

Q.25 Solve the inequality for x: $\log_{10}(6-x) > \log_{10}(x-2)$.
(a) $x > 2$ (b) $2 < x < 4$
(c) $x < 4$ (d) None of these

FOUNDATION

Q.1 If $3^{x+1} = 6^{\log_2 3}$, then x is:
(a) 2 (b) 3 (c) $\log_3 2$ (d) $\log_2 3$

Q.2 If $\log_4 5 = a$ and $\log_5 6 = b$, then what is the value of $\log_3 2$?
(a) $\frac{1}{2a+1}$ (b) $\frac{1}{2b+1}$ (c) $2ab + 1$ (d) $\frac{1}{2ab-1}$

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Q.3 $\log_5 P \times \log_P x = 2$. What is the value of x?
(a) 125 (b) P^2 (c) 25 (d) $5^{1/2}$

Q.4 Find the value of $3\log\frac{81}{80}+5\log\frac{25}{24}+7\log\frac{16}{15}$.
(a) log 2 (b) log 3
(c) 1 (d) None of these

Q.5 If $[N]$ = the greatest integer less than or equal to N, then $[\log_{10} 6730.4]$ is equal to:
(a) 6 (b) 4
(c) 5 (d) None of these

Q.6 What is the value of x in the following expression?

$$1-\log_{10}5=\frac{1}{3}\left(\log_{10}\frac{1}{2}+\log_{10}x+\frac{1}{3}\log_{10}5\right)$$

(a) 1 (b) $16\times5^{-1/3}$
(c) $16\times5^{1/3}$ (d) None of these

Q.7 What is the number of zeros coming immediately after the decimal point in the value of $(0.2)^{25}$? ($\log_{10}2 = 0.30103$)
(a) 15 (b) 16
(c) 17 (d) None of these

Q.8 What is the value of $\log_{32} 27\times\log_{243}8$?
(a) $\frac{\log 9}{\log 4}$ (b) $\frac{\log 3}{\log 2}$
(c) log 27 (d) None of these

Q.9 Find the value of $\log_{64}(1/16)$.
(a) −2/3 (b) 2/3 (c) 3/2 (d) −3/2

Q.10 Find the value of $\frac{1}{\log_2 X}+\frac{1}{\log_3 X}+\frac{1}{\log_4 X}+\frac{1}{\log_5 X}+\cdot\frac{1}{\log_7 X}+\frac{1}{\log_6 X}$.
(a) 1 (b) $\log_{27} X$
(c) $\log_{5040} X$ (d) $\log_x 5040$

Q.11 If $\log_y N = 5$ and $\log_{2y} 8N = 4$, then find the value of N.
(a) 16 (b) 32 (c) 5 (d) 25

Q.12 If $\log_y{}^3 N = \log_N{}^3 y$, then what is the value of $\log_N{}^3 y$?
(a) 1/3 (b) −1/3
(c) Either (a) or (b) (d) None of these

Q.13 If $\frac{\log X}{\log 4}=\frac{\log 343}{\log 49}=\frac{\log y}{\log 64}$, then what is the value of $x+y$?
(a) 520 (b) 740
(c) 880 (d) Cannot be determined

Q.14 Arrange the following in an ascending order:
$A=\log_7 2401$, $B=\log_7\sqrt{7}\times343$, $C=\log_{\sqrt{6}}216$, $D=\log_2 32$
(a) ABCD (b) BDCA (c) BDAD (d) BADC

Q.15 If $\log_{10}\left\{1-\left[1-\left(1-x^2\right)^{-1}\right]^{-1}\right\}^{-1/2}=1$, then which of the following is the value of x?
(a) 1 (b) 2
(c) 10 (d) None of these

Q.16 What is the value of N in the expression $4^{\log 2^8}+27^{\log 27^{81}}=144+\log_{10} N$?
(a) 1 (b) 10 (c) 12 (d) 100

Q.17 What is the value of x if $\log_3 x+\log_9 x+\log_{27} x+\log_{81} x=\frac{25}{4}$?
(a) 9 (b) 27
(c) 81 (d) None of these

Q.18 If $\log_y x = 8$ and $\log_{10y} 16x = 4$, then find the value of y.
(a) 1 (b) 2 (c) 3 (d) 5

Q.19 What is the value of x in the expression

$$\log_7\log_5\left[\sqrt{(x+5)}+\sqrt{x}\right]=0?$$

(a) 1 (b) 2 (c) 3 (d) 4

Q.20 What is the value of x in the expression $\log_2(3-x)+\log_2(1-x)=3$?
(a) 1 (b) 0
(c) −1 (d) Not possible

Q.21 If $\log N^2 - \log 2N = 3\log 3 - \log 6$, then what is the value of N?
(a) 1 (b) 2
(c) 9 (d) More than one value

Q.22 What is the value of $\log_{3\sqrt{3}} 27$?
(a) 2 (b) 3 (c) 4 (d) 5

Q.23 Evaluate $\log(36\sqrt{6})$ to the base 6.
(a) 1/2 (b) 5/2 (c) 3/2 (d) 7/2

Q.24 What is the value of P if $\log_e 2\cdot\log_p 625=\log_{10}16\cdot\log_e 10$?
(a) 2 (b) 4 (c) 5 (d) 7

Q.25 If $3\log_{(3x^2)}27-2\log_{(3x)}9=0$, then what is the value of x?
(a) 1/243 (b) 1/7
(c) 1/49 (d) None of these

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MODERATE

Q.1 $N = n!$, where $n > 2$. Find the value of $(\log_2 N)^{-1} + (\log_3 N)^{-1} + (\log_4 N)^{-1} + \ldots (\log_n N)^{-1}$.
(a) 0 (b) 1 (c) 10 (d) N

Q.2 What is the value of x in the expression $x + \log_{10}(1 + 2^x) = x \log_{10} 5 + \log_{10} 6$?
(a) 1 (b) 0 (c) −1 (c) 3

Q.3 If $a_1, a_2, a_3, \ldots$ are positive numbers in GP, then $\log a_n$, $\log a_{n+1}$, and $\log a_{n+2}$ are in:
(a) AP (b) GP
(c) HP (d) None of these

Q.4 If $x = \log_a (bc)$, $y = \log_b (ca)$, and $z = \log_c (ab)$, then which of the following is equal to 1?
(a) $x + y + z$
(b) $(1 + x)^{-1} + (1 + y)^{-1} + (1 + z)^{-1}$
(c) xyz
(d) None of these

Q.5 If $\log x : \log y : \log z = (y - z) : (z - x) : (x - y)$, then:
(a) $X^y . Y^z . Z^x = 1$ (b) $X^x Y^y Z^z = 1$
(c) $x\sqrt{x}, y\sqrt{y}, z\sqrt{z} = 1$ (d) None of these

Q.6 If $\log_{0.04}(x - 1) \geq \log_{0.2}(x - 1)$, then x belongs to the interval:
(a) (1, 2] (b) (−∞, 2]
(c) [2, +∞) (d) None of these

Q.7 If $\log_k x \cdot \log_5 k = 1 k \neq 1$, $k > 0$, then x is equal to:
(a) k (b) $\frac{1}{5}$
(c) 5 (d) None of these

Q.8 If $\frac{\log a}{b - c} = \frac{\log b}{c - a} = \frac{\log c}{a - b}$, then $a^a b^b c^c$ is:
(a) 0 (b) 1 (c) 0.5 (d) 2

Q.9 If $x > 1$, $y > 1$, and $z > 1$ are three numbers in GP, then $\frac{1}{1 + \log x}, \frac{1}{1 + \log y}, \frac{1}{1 + \log z}$ are in:
(a) AP (b) HP
(c) GP (d) None of these

Q.10 If $\log_k N = 6$ and $\log_{25k}(8N) = 3$, then k is:
(a) 12.5 (b) $(12.5)^2$
(c) $\frac{5^{\frac{2}{3}}}{2}$ (d) $(12.5)^{\frac{2}{3}}$

Q.11 If a, b, and c are distinct positive numbers $(\neq 1)$ such that $(\log_b a \log_c a - \log_a a) + (\log_a b \log_c b - \log_b b) + (\log_a c \log_b c - \log_c c) = 0$, then what is the value of abc?
(a) 1 (b) 0
(c) −1 (d) None of these

Q.12 If $4^{\log_9 3} + 9^{\log_2 4} = 10^{\log_x 83}$, $x \varepsilon R$, then how many values can x take?
(a) 0 (b) 3
(c) 2 (d) None of these

Q.13 How many values of x $(x > 1)$ satisfy the following equation:
$\log_2 \times \log_4 \times \log_6 x = \log_2 x \cdot \log_4 x + \log_2 x \cdot \log_6 x + \log_4 x \cdot \log_6 x$?
(a) 0 (b) 1
(c) 2 (d) More than 2

Q.14 If $a^{\log b} = k\, b^{\log a}$, then find the value of $\log_{ab} k$, where $a, b > 1$.
(a) 1 (b) 0
(c) 2 (d) Cannot be determined

Q.15 What is the value of x in the expression $\log_{3/4} \log_2 (x^2 + 7) \log_{1/4} (x^2 + 7)^{-1} = -2$?
(a) +3 (b) −3
(c) ±3 (d) None of these

Q.16 If $2[\log (x + y) - \log 5] = \log x + \log y$, then what is the value of $x^2 + y^2$?
(a) $20 - xy$ (b) $23xy$ (c) $25 - xy$ (d) $28xy$

Q.17 The logarithm of a number to a certain base is 9. The logarithm of 64 times the number to a base which is 11 times the original base is 6. Find the actual base.
(a) 11/6 (b) 2.33/7
(c) 121/4 (d) Cannot be determined

Q.18 If $\log_{30} 3 = x$ and $\log_{30} 5 = y$, then $\log_8 30$ is equal to:
(a) $3(1 - x - y)$ (b) $\frac{1}{3(1 - x - y)}$
(c) $\frac{3}{(1 - x - y)}$ (d) $\frac{1 - x - y}{3}$

Q.19 What is the value of $\frac{1}{\log_2 n} + \frac{1}{\log_3 n} + \ldots + \frac{1}{\log_{40} n}$?
(a) $\frac{1}{\log(40)! n}$ (b) $\log(40!) n$
(c) 1 (d) None of these

Q.20 The number of solutions of $\log_2 (x + 5) = 6 - x$ is:
(a) 2 (b) 0
(c) 3 (d) None of these

Q.21 x, y, and z are three angles of a triangle. Which of the following set of values of x, y, and z satisfies $\log (x \times y \times z) = 3 \log X + 4 \log 2$, given that x, y, and z are integers?
(a) 30°, 70°, and 80° (b) 30°, 60°, and 90°
(c) 20°, 80°, and 80° (d) 30°, 50°, and 100°

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Q.22 What is the value of $\log_3 2 \cdot \log_4 3 \cdot \log_5 4 \ldots \log_{16} 15$?
(a) 1/2 (b) 1/3 (c) 2/3 (d) 1/4

Q.23 If $\log_{12} 18 = a$ and $\log_{24} 54 = b$, find the value of $ab + 5(a - b)$.
(a) −1 (b) 1 (c) 0 (d) 2

Q.24 What is the value of x in the expression $x \log_{10}(10/3) + \log_{10} 3 = \log_{10}(2 + 3^x) + x$?
(a) 1 (b) −2
(c) 0 (d) 2

Answers

Warm Up

1. (a)	**2.** (c)	**3.** (a)	**4.** (b)	**5.** (a)	**6.** (c)	**7.** (d)	**8.** (d)	**9.** (a)	**10.** (c)
11. (b)	**12.** (b)	**13.** (a)	**14.** (d)	**15.** (c)	**16.** (c)	**17.** (c)	**18.** (b)	**19.** (d)	**20.** (d)
21. (b)	**22.** (b)	**23.** (c)	**24.** (c)	**25.** (b)					

Foundation

1. (d)	**2.** (d)	**3.** (c)	**4.** (a)	**5.** (d)	**6.** (b)	**7.** (c)	**8.** (d)	**9.** (a)	**10.** (d)
11. (b)	**12.** (c)	**13.** (a)	**14.** (d)	**15.** (c)	**16.** (b)	**17.** (d)	**18.** (d)	**19.** (d)	**20.** (c)
21. (c)	**22.** (a)	**23.** (b)	**24.** (c)	**25.** (a)					

Moderate

1. (b)	**2.** (a)	**3.** (a)	**4.** (b)	**5.** (b)	**6.** (c)	**7.** (c)	**8.** (b)	**9.** (b)	**10.** (a)
11. (a)	**12.** (d)	**13.** (b)	**14.** (b)	**15.** (c)	**16.** (c)	**17.** (c)	**18.** (b)	**19.** (a)	**20.** (d)
21. (c)	**22.** (d)	**23.** (b)	**24.** (c)						

Hints and Solutions

Warm Up

1. Given expression can be written as:
$\log_n 2 + \log_n 3 + \log_n 4 + \ldots + \log_n 40 = \log_n(2 \times 3 \times 4 \times \ldots 40) = \log_n 40!$
It is equal to $\frac{1}{\log_{40!} n}$.
Hence, option (a) is the answer.

2. $5 \log 20 = 5 \times 0.698 = 13.96$. Therefore, the number of digits = 14. Hence, option (c) is the answer.

3. $\log_{3\sqrt{3}} 27 = \log_{3\sqrt{3}}(3\sqrt{3})^2 = 2 \times \log_{3\sqrt{3}}(3\sqrt{3}) = 2.$
Hence, option (a) is the answer.

4. $\log_6 36\sqrt{6} = \log_6 6^{2.5} = 2.5(\log_6 6) = 2.5$
Hence, option (b) is the answer.

5. $3\log\frac{81}{80} = \log\left[\frac{81}{80}\right]^3$; $5\log\frac{25}{24} = \log\left[\frac{25}{24}\right]^5$;
$7\log\frac{16}{15} = \log\left[\frac{16}{15}\right]^7$
So, $3\log\frac{81}{80} + 5\log\frac{25}{24} + 7\log\frac{16}{15} = \log$

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$$\left(\left[\frac{81}{80}\right]^3 \times \left[\frac{25}{24}\right]^5 \times \left[\frac{16}{15}\right]^7\right) = \log 2$$

Hence, option (a) is the answer.

6. Assume that the value of $\log_{10} 6730.4 = z$

It can be seen that $1000 < 6730.4 < 10000$.

Hence, $\log_{10}(10^3) < \log_{10} 6730.4 < \log_{10}(10^4)$

Taking antilog, $3 < z < 4$. So, the value of z lies between 3 and 4.

Therefore, $[z]$ = greatest integer less than or equal to $z = 3$. Hence, option (c) is the answer.

7. In these questions, it is always advisable to go through the options.

$\log_7 \log_5[\sqrt{(x+5)} + \sqrt{x}] = 0$

Since one of the components in the question is $\sqrt{(x+5)}$ and $\sqrt{x}$, it is advisable to start with an option that will give us the square root.

Let us start with option (d). Putting $x = 4$, we get

$\log_7 \log_5[\sqrt{(4+5)} + \sqrt{4}] = \log_7 \log_5[\sqrt{9} + \sqrt{4}]$

$= \log_7 \log_5 5 = \log_7 1 = 0$

Hence, option (d) is the answer.

8. $A = \log_7 2401 = \log_7 7^4 = A = 4\log_7 7 = 4$

$B = \log_{7\sqrt{7}} 343 = 7\sqrt{7}^2 = 2$

$C = \log_{\sqrt{6}} 216 = 9$

$D = \log_2 32 = 5$

Hence, the ascending order is BADC. Hence, option (d) is the answer.

9. **Method 1**

RHS = $3 \log X + 4 \log 2 = \log (x^3 \times 2^4) = \log (16x^3)$

Let us go through the options.

Checking option (a):

LHS = $\log (x \times y \times z) = \log (30 \times 70 \times 80) = \log (168 \times 1000)$

RHS = $\log (16x^3) = \log (16 \times 30 \times 30 \times 30)$

It can be seen that LHS ≠ RHS

Similarly, option (b) will not satisfy.

Checking option (c):

LHS = $\log (x \times y \times z) = \log (20 \times 80 \times 80) = \log (128 \times 1000)$

RHS = $\log (16 \times 20^3) = \log (128 \times 1000)$

LHS = RHS

Hence, option (c) is the answer.

Method 2

It can be seen that RHS has a term in x but not in y and z. It means that y and z can be used interchangeably $\Rightarrow y$ and z are equal, which is given only in option (c). Hence, option (c) is the answer.

10. $2[\log (x+y) - \log 5] = \log x + \log y$ can be written as:

$[\log (x+y) - \log 5] = \frac{1}{2} (\log x + \log y)$

or $\log \frac{x+y}{5} = \log \sqrt{xy}$

Taking antilog on both the sides, we obtain the following:

$\frac{x+y}{5} = \sqrt{xy} \Rightarrow x + y = 5\sqrt{xy}$

Squaring both the sides, we get $(x+y)^2 = 25xy \Rightarrow x^2 + y^2 + 2xy = 25xy \Rightarrow x^2 + y^2 = 23xy$. Hence, option (b) is the answer.

11. $\log_{32} 27 = \log_{2^5} 3^3 = \frac{3}{5}\log_2 3$

$\log_{243} 8 = \log_{3^5} 2^3 = \frac{3}{5}\log_3 2$

Therefore, $\log_{32} 27 \times \log_{243} 8 = \frac{3}{5}\log_2 3 \times \frac{3}{5}\log_3 2 = \frac{9}{25}$

Hence, option (d) is the answer.

12. In this question, it is better to assume values and verify the options.

Assume $x = 10$, $y = 100$, and $z = 1000$

$$\frac{1}{1+\log x} = \frac{1}{1+\log 10} = \frac{1}{1+1} = \frac{1}{2}$$

$$\frac{1}{1+\log y} = \frac{1}{1+\log 100} = \frac{1}{1+2} = \frac{1}{3}$$

$$\frac{1}{1+\log z} = \frac{1}{1+\log 1000} = \frac{1}{1+3} = \frac{1}{4}$$

Now, it can be clearly seen that 1/2, 1/3, and 1/4 are in harmonic progression. Hence, option (b) is the answer.

13. $\log_k N = 6 \Rightarrow N = k^6$ (i)

Similarly, $\log_{25k}(8N) = 3$

$\Rightarrow (8N) = (25k)^3$ (ii)

Dividing equations (i) by (ii), we get

$$\frac{N}{8N} = \frac{k^6}{(25k)^3}$$

$$k = \frac{25}{2} = 12.5$$

Hence, option (a) is the answer.

14. Go through the options.

15. Go through the options.

It can be seen that $x = -1$ satisfies the equations. Hence, option (c) is the answer.

16. Assume $\log_n y = 9 \Rightarrow y = n^9$ (i)

Given that $\log_{11n} 64y = 6 \Rightarrow 64y = (11n)^6$ (ii)

Substituting the value of y from equation (i) in equation (ii):

$64 \times n^9 = 11^6 \times n^6 \Rightarrow n^3 = \frac{11^6}{2^6} \Rightarrow n = \frac{11^2}{2^2} = \frac{121}{4}$

Hence, option (c) is the answer.

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17. Again, in this question, we have to find out the value of N. It is better to go through the options.

$N = 9$ satisfies the equation. Hence, option (c) is the answer.

18. $1 + x = \log_a a + \log_a (bc) = \log_a (abc)$

Similarly, $1 + y = \log_b (abc)$ and $1 + z = \log_c (abc)$

$$\frac{1}{1+x} = \log_{abc} a$$

$$\frac{1}{1+y} = \log_{abc} b$$

$$\frac{1}{1+z} = \log_{abc} c$$

$$\frac{1}{1+x} + \frac{1}{1+y} + \frac{1}{1+z} = \log_{abc} a + \log_{abc} b + \log_{abc} c$$

$= \log_{abc} abc = 1$

Hence, option (b) is the answer.

19. $ab = \log_4 5 \times \log_5 6 = \log_4 6 = \frac{1}{2} \log_2 6 = \frac{1}{2} (\log_2 2 + \log_2 3)$

$$= \frac{1}{2} (1 + \log_2 3)$$

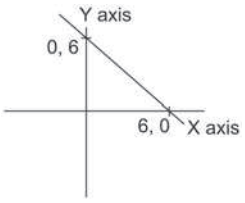
$\Rightarrow \quad 2ab = (1 + \log_2 3)$

$\Rightarrow \quad \log_2 3 = 2ab - 1$

$$\Rightarrow \quad \log_3 2 = \frac{1}{2ab-1}$$

Hence, option (d) is the answer.

20. Graph of $(6 - x)$ will be a straight line like this:



Whereas graph of $\log_2 (x + 5)$ will be a curve moving upwards.

Therefore, they will intersect at only one point. Hence, option (d) is the answer.

21. $2\log_{0.5} (x - 4)$ can be written as $\log_{0.5} (x - 4)^2$

So, $\log_{0.5} (x - 4)^2 < \log_{0.5} 2x$

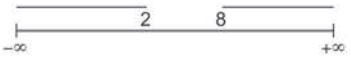
As the base is smaller than 1, it will be a decreasing function. Hence, $(x - 4)^2 > 2x$

or $x^2 - 8x + 16 > 2x$

$\Rightarrow \quad x^2 - 10x + 16 > 0 \Rightarrow (x - 8)(x - 2) > 0$

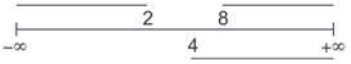
$x < 2$ or $8 < x$ (i)

Marking this on number line:

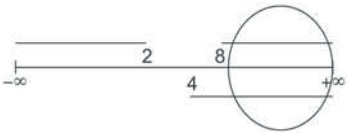


Next, for log to be defined:

$X - 4 > 0$ and $x > 0 \Rightarrow x > 4$ (ii)



It can be seen that the circled area (as shown in the next diagram) is common and it is the answer.



Therefore, the range of values of x: $x > 8$. Hence, option (a) is the answer.

22. For log to be defined:

$3 - x > 0$; so, $x < 3$, and $x - 1 > 0$; so, $1 < x$

Hence, $1 < x < 3$ (i)

As the base > 1, therefore, $3 - x > x - 1 \Rightarrow 2x < 4$

$\Rightarrow x < 2$ (ii)

Combining equations (i) and (ii), we get the following range of values of x: $1 < x < 2$. Hence, option (b) is the answer.

23. For log to be defined:

$x - 4 > 0$ and $x + 8 > 0 \Rightarrow x > 4$ (i)

Further, according to the question,

$\log_{10}(x - 4)^2 < \log_{10}(x + 8)$

As the base is greater than 1, log is an increasing function.

$(x - 4)^2 < x + 8$ or

$x^2 - 8x + 16 < x + 8$, or, $x^2 - 9x + 8 < 0 \Rightarrow (x - 1)(x - 8) < 0$

Hence, $1 < x < 8$ (ii)

Combining equations (i) and (ii), we obtain the following range of values of x: $4 < x < 8$.

24. For log to be defined:

$x - 2 > 0$; so, $x > 2$, and $x + 1 > 0$; so, $x > -1$

Hence, $x > 2$ (i)

Now, from the question, move the negative signed term to the other side.

$\log_{0.5} (x - 2) > 1 + \log_{0.5} (x + 1)$

or $\log_{0.5} (x - 2) > \log_{0.5} \frac{1}{2} + \log_{0.5} (x + 1)$ or $\log_{0.5} (x - 2) > \log_{0.5} \frac{1}{2} (x + 1)$

As the base $\frac{1}{2}$ is less than 1, it is a decreasing function.

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Hence, $(x-2) < \frac{1}{2}(x+1) \Rightarrow 2x-4 < x+1 \Rightarrow x < 5$ (ii)

Combining equations (i) and (ii), we obtain the following range of values of x: $2 < x < 5$. Hence, option (c) is the answer.

25. For log to be defined:

$6-x>0$ or $x<6 \Rightarrow x-2>0$

Hence, $x > 2$

Therefore, $2 < x < 6$ (i)

As the base is greater than 1, it is an increasing function and not required to switch the inequality sign.

$6-x > x-2 \Rightarrow 2x < 8$

Hence, $x < 4$ (ii)

Combining equations (i) and (ii), we obtain the following range of values of x: $2 < x < 4$. Hence, option (b) is the answer.

FOUNDATION

1. $3^{x+1} = (3\times 2)^{\log_2 3}$

$\Rightarrow 3^{x+1} = 3^{\log_2 3} \times 2^{\log_2 3}$

$\Rightarrow 3^{x+1} = 3^{\log_2 3} \times 3$

$\Rightarrow \quad 3^x = 3^{\log_2 3}$

$\Rightarrow x = \log_2 3$

2. $\log_4 5 = a$ and $\log_5 6 = b$

$\Rightarrow \log_4 5 \times \log_5 6 = ab$

$\Rightarrow \log_4 6 = ab \Rightarrow \frac{1}{2}\log_2 6 = ab$

$\Rightarrow (1+\log_2 3) = 2ab$

$\therefore \quad \log_2 3 = 2ab-1$

$\therefore \quad \log_3 2 = \frac{1}{2ab-1}$

3. $\log_5 P \times \log_P x = 2$

$\Rightarrow \log_5 x = 2$

$\therefore \quad x = 5^2 = 25$

4. $\log\left(\frac{81}{80}\right)^3 + \log\left(\frac{25}{24}\right)^5 + \log\left(\frac{16}{15}\right)^7$

$= \log\left(\frac{3^{12}\times 5^{10}\times 2^{28}}{2^{12}\times 5^3\times 2^{15}\times 3^5\times 3^7\times 5^7}\right)$

$= \log 2$

5. $\log_{10} 6730.4 = \log_{10} 6.7304 \times 10^3$

$= 3 + \log_{10} 6.7304$

$= 3$ [Value of $\log_{10} n$ will lie in between 0 and 1 when $1 < n < 10$.]

6. $1-\log_{10} 5 = \frac{1}{3}\left[\log_{10}\left(\frac{x.5^{1/3}}{2}\right)\right]$

$= 3[\log_{10} 10 - \log_{10} 5] = \log_{10}\left(\frac{x.5^{1/3}}{2}\right)$

$\Rightarrow \log_{10} 2^3 = \log_{10}\left(\frac{x.5^{1/3}}{2}\right)$

$\Rightarrow 2^3 = \frac{x.5^{1/3}}{2}$

$\therefore x = 16\times 5^{-1/3}$

7. $\log (0.2)^{25} = 25\log\left(\frac{2}{10}\right)$

$= 25(\log 2 - \log 10)$

$= -25(1-0.30103)$

$= -25\times 0.69897 = -17.47425$

$\therefore$ Number of zeros after the decimal point is 17.

8. $\log_{32} 27 \times \log_{243} 8 = \log_{2^3} 3^3 \times \log_{3^5} 2^3$

$= \frac{3}{5}\log_2 3 \times \frac{3}{5}\log_3 2$

$= \left(\frac{3}{5}\right)^2 \log_2 3 \times \log_3 2 = \frac{9}{25}$

9. $\log_{64}(1/16) = \log_{4^3} 4^{-2} = \frac{-2}{3}$

10. Expression is equal to $\log_x 2\times 3\times 4\times 5\times 6\times 7 = \log_x 5040$.

11. Go through the options.

14. $A = \log_7 2401 = 4$

$B = \log_7(\sqrt{7.343})$

$\Rightarrow B = \log_7(7^{1/2}\times 7^3) = 3\frac{1}{2}$

$C = \log_{\sqrt{6}} 216 = 2\cdot\log_6 6^3 = 6$

$D = \log_2 32 = 5$

$\therefore$ Arranging in ascending order: BADC

15. $\log_{10}\{1-[1-(1-x^2)^{-1}]^{-1}\}^{-\frac{1}{2}} = 1$

$\Rightarrow \log_{10}\left[1-\left(1-\frac{1}{1-x^2}\right)^1\right]^{-\frac{1}{2}} = 1$

$\Rightarrow \log_{10}\left[1-\left(\frac{-x^2}{1-x^2}\right)^1\right]^{-\frac{1}{2}} = 1$

$\Rightarrow \left[\frac{1}{x^2}\right]^{-\frac{1}{2}} = 10 \Rightarrow x = 10$

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17. $\log_3 x + \log_9 x + \log_{27} x + \log_{81} x = \frac{25}{7}$

$\Rightarrow \log_3 x + \frac{1}{2}\log_3 x + \frac{1}{3}\log_3 x + \frac{1}{4}\log_3 x = \frac{25}{4}$

$\Rightarrow \log_3 x\,[1 + 1/2 + 1/3 + 1/4] = \frac{25}{4}$

$\Rightarrow \frac{25}{12} \times \log_3 x = \frac{25}{4}$

$\Rightarrow \log_3 x = 3 \Rightarrow x = 27$

18. $\log_y x = 8 \Rightarrow y^8 = x$ (i)

$\log_{10y} 16x = 4 \Rightarrow 10^4 \cdot y^4 = 16x$ (ii)

Dividing equations (ii) by (i), we get $10^4 \cdot y^{-4} = 16$

$\Rightarrow y = 5$

19. Go through the options. (Use $\log 1 = 0$)

20. Going through the options.

Option (a) is not possible.

Checking option (c): For $x = -1$, expression $= \log_2 4 + \log_2 2 = \log_2 8 = 3\log_2 2 = 3$

21. $\log N^2 - \log 2N = 3\log 3 - \log 6$

or $\log\left(\frac{N^2}{2N}\right) = \log\frac{3^3}{6}$ or $\frac{N}{2} = \frac{9}{2}$

$\therefore N = 9$

22. $\log_{3\sqrt{3}} 27 = \log_{3\sqrt{3}} (3\sqrt{3})^2 = 2$

23. $\log_6 36\sqrt{6} = \log_6 36 + \log_6 \sqrt{6}$

$= 2 + 0.5 = 2.5$

24. $\log_e 2 \cdot \log_p 625 = \log_{10} 16 \cdot \log_e 10$

$\Rightarrow \log_p 625 = \log_{10} 2^4 \cdot \frac{\log_e 10}{\log_e 2}$

$\Rightarrow \log_p 5^4 = 4 \log_{10} 2 \cdot \log_2 10$

or $4 \log_p 5 = 4$

$\therefore \log_p 5 = 1 \Rightarrow P = 5$

25. $3\log_{3x^2} 27 - 2\log_{3x} 9 = 0 \Rightarrow 3\log_{3x^2} 3^3 - 2\log_{3x} 3^2 = 0$

$\Rightarrow 9\log_{3x^2} 3 = 4 \log_{3x} 3$

$\Rightarrow \frac{9}{\log_{3x} 3} = \frac{4}{\log_{3x^2} 3}$

$\Rightarrow 9 \log_3 3x = 4 \cdot \log_3 3x^2$

$\Rightarrow \log_3 (3x) 9 = \log_3 (3x^2)^4$

$\Rightarrow 3^9 \times x^9 = 3^4 \times x^8 \Rightarrow x = 3^{-5}$

$\therefore x = \frac{1}{243}$

MODERATE

1. $N = n!$ (given)

$$\frac{1}{\log_2 N} + \frac{1}{\log_3 N} + \frac{1}{\log_4 N} + \cdots + \frac{1}{\log_n N}$$

$= \log_N 2 + \log_N 3 + \log_N 4 + \ldots + \log_N n$

$= \log_N n! = \log_N N = 1$

2. $x + \log_{10}(1 + 2^x) = x \log_{10} 5 + \log_{10} 6$

$\Rightarrow \log_{10} 10^x + \log_{10}(1 + 2^x) = \log_{10} 5^x \cdot 6$

$\Rightarrow \quad 10^x (1 + 2^x) = 5^x \cdot 6 \Rightarrow 2^x(1 + 2^x) = 6$

$\therefore \quad 2^x = 2$, so $x = 1$

3. $a_1 = a_1$; $a_2 = a_1 r$; $a_3 = a_1 r^2$; and $a_4 = a_1 r^3$ are in GP with common difference 'r'.

$\therefore \quad \log a_1 = \log a_1$

$\log a_2 = \log a_1 + \log r$

$\log a_3 = \log a_1 + 2\log r$

$\log a_4 = \log a_1 + 3\log r$

$\log a_{n+1} = \log a_1 + n \log r$

$\log a_{n+2} = \log a_1 + (n + 1) \log r$

$\therefore \log a_1, \log a_2, \log a_3, \log a_4, \ldots, \log a_{n+2}$ are in AP with common difference '$\log r$'.

Alternatively,

Assume $a_1 = 10$, $a_2 = 100$, and $a_3 = 1000$

To find $\log a_n$, $\log a_{n+1}$, and $\log a_{n+2}$, take $n = 1$

Hence, we have to find that log 10, log 100, and log 1000 are in which series.

log 10 = 1, log 100 = 2, and log 1000 = 3

So, log 10, log 100, and log 1000 are in AP.

4. $x = \log_a bc \quad \therefore 1 + x = \log_a abc$

$y = \log_b ca \quad \therefore 1 + y = \log_b abc$

$z = \log_c ab \quad \therefore 1 + z = \log_c abc$

$\therefore \frac{1}{1+x} + \frac{1}{1+y} + \frac{1}{1+z} = \log_{abc} a + \log_{abc} b + \log_{abc} c$

$= \log_{abc} abc = 1$

5. $\log x : \log y : \log z = (y - z) : (z - x) : (x - y)$

$\therefore \frac{\log x}{y - z} + \frac{\log y}{z - x} + \frac{\log z}{x - y} = k$ (say)

$\therefore \log x = k\,(y - z)$

$\Rightarrow x \log x = kx\,(y - z)$

Similarly, $y \log y = ky\,(z - x)$

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$z \log z = kz(x - y)$

Adding all 3 equations, we get

$x \log x + y \log y + z \log z = 0 \Rightarrow \log X^x \cdot Y^y \cdot Z^z = 0$

$\therefore X^x \cdot Y^y \cdot Z^z = 1$

6. $\log_{0.04}(x-1) \geq \log_{0.2}(x-1)$

$$\Rightarrow \frac{\log_e(x-1)}{\log_e 0.04} \geq \frac{\log_e(x-1)}{\log_e 0.2}$$

$$\Rightarrow \frac{\log_e(x-1)}{\log_e (0.2)^2} \geq \frac{\log_e(x-1)}{\log_e 0.2}$$

$$\Rightarrow \frac{\log_e(x-1)}{2\log_e(0.2)} \geq \frac{\log_e(x-1)}{\log_e 0.2}$$

$\because \log_e 0.2 < 0$

$\therefore \log_e(x-1) \leq 2\log_e(x-1)$

$\therefore x - 1 \geq 1$

$\therefore x \geq 2 \Rightarrow x \in [2, \infty]$

7. $\log_k x \cdot \log_5 K = 1$

$\Rightarrow \log_5 x = 1 \Rightarrow x = 5$

9. Solution 1

x, y, and z are in GP.

$\therefore \log x$, $\log y$, and $\log z$ are in AP as shown in solution of question number 3 in moderate section.

$\therefore 1 + \log x$, $1 + \log y$, and $1 + \log z$ are in AP.

$$\therefore \frac{1}{1+\log x}, \frac{1}{1+\log y}, \text{ and } \frac{1}{1+\log z} \text{ are in HP.}$$

Solution 2

Since the numbers are in GP, let us assume the numbers x, y, and z to be equal to 10, 100, and 1000.

$$\text{Then, } \frac{1}{1+\log x} = \frac{1}{2}, \frac{1}{1+\log y} = \frac{1}{3}, \text{ and } \frac{1}{1+\log z} = \frac{1}{4}$$

$\therefore$ These are in HP.

10. $\log_K N = 6 \quad \Rightarrow K^6 = N$

$\log_{25K} 8N = 3 \quad \Rightarrow (25K)^3 = 8N$

$$\therefore \frac{25^3 K^3}{K^6} = 8 \Rightarrow 25^3 \cdot K^{-3} = 2^3$$

$$\therefore K = \frac{25}{2} = 12.5$$

11. $\log_b a \log_c a + \log_a b \log_c b + \log_a c \log_b c = 3$

$$\Rightarrow \frac{(\log a)^2}{\log b \cdot \log c} + \frac{(\log b)^2}{\log a \cdot \log c} + \frac{(\log c)^2}{\log a \cdot \log b} = 3$$

$$\Rightarrow \frac{(\log a)^3 + (\log b)^3 + (\log b)^3}{\log a \cdot \log b \cdot \log c} = 3$$

$\Rightarrow (\log a)^3 + (\log b)^3 + (\log c)^3 = 3 \log a \cdot \log b \cdot \log c$

We know, $x^3 + y^3 = z^3 = 3xyz$ when $x + y + z = 0$

$\log a + \log b + \log c = 0 \Rightarrow \log abc = 0 \Rightarrow abc = 1$

13. $\log_2 x \cdot \log_4 x \cdot \log_6 x = \log_2 x \cdot \log_4 x + \log_2 x \cdot \log_6 x + \log_4 x \cdot \log_6 x$

Dividing LHS and RHS by $\log_2 x \cdot \log_4 x \cdot \log_6 x$

$$\Rightarrow 1 = \frac{1}{\log_6 x} + \frac{1}{\log_4 x} + \frac{1}{\log_2 x}$$

$\Rightarrow 1 = \log_x 6 + \log_x 4 + \log_x 2$

$\Rightarrow \log_x 6 \times 4 \times 2 = 1$

$\Rightarrow x = 48 \ \therefore$ only 1 value of x

14. $a^{\log b} = kb^{\log a}$

$\Rightarrow \log b \cdot \log a = \log k + \log a \cdot \log b$ {taking log on both sides}

$\Rightarrow \log k = 0$

$\therefore K = 1$

$\therefore \log_{ab} k = 0$

15. Go through the options.

16. $2[\log(x+y) - \log 5] = \log x + \log y$

$$\Rightarrow \log \frac{(x+y)^2}{5^2} = \log xy$$

$$\Rightarrow \frac{(x+y)^2}{25} = xy \Rightarrow x^2 + y^2 + 2xy = 25xy$$

$\therefore x^2 + y^2 = 23xy$

17. Let the base be 'b'.

$\therefore \log_b N = 9$ and $\log_{11b} 64N = 6$

$\Rightarrow N = b^9$ and $64N = (11b)^6$

$$\therefore 64 = \frac{11^6 b^6}{b^9} \qquad \therefore b^3 = \frac{11^6}{64}$$

$$\therefore b = \frac{11^2}{2^2} = \frac{121}{4}$$

18. $\log_{30} 3 = x$, $\log_{30} 5 = y$

$\therefore x + y = \log_{30} 15$

$\Rightarrow x + y = \log_{30}(30/2)$

$\Rightarrow (x+y) = 1 - \log_{30} 2 \Rightarrow \log_{30} 2 = 1 - x - y$

$\Rightarrow 3\log_{30} 2 = 3(1 - x - y) \Rightarrow \log_{30} 8 = 3(1 - x - y)$

$$\therefore \log_8 30 = \frac{1}{3(1-x-y)}$$

19. $\frac{1}{\log_2 n} + \frac{1}{\log_3 n} + \frac{1}{\log_4 n} + \cdots \frac{1}{\log_{40} n}$

$= \log_n 2 + \log_n 3 + \log_n 4 + \ldots + \log_n 40$

$= \log_n (40!)$

$$= \frac{1}{\log_{(40!)} n}$$

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20. $\log_2(x+5) = 6 - x$

Draw the graph of $\log_2(x+5)$ and $(6-x)$.

$\therefore$ The two graphs will intersect at only one point.

21. It can be simply seen that in the expression $\log(x \times y \times z) = 3 \log X + 4 \log 2$, when we have replaced y and z on the LHS, RHS contains an equation in only one variable that is x. It means that y and z have to be replaced with each other. Only option 3 satisfies this relationship.

22. $\log_3 2 \cdot \log_4 3 \cdot \log_5 4 \ldots 3\log_{16} 15$

$$\frac{\log 2}{\log 3} \cdot \frac{\log 3}{\log 4} \cdot \frac{\log 4}{\log 5} \cdots \frac{\log 15}{\log 16} = \frac{\log 2}{\log 16} = \frac{\log 2}{\log 2^4}$$

$$= \frac{\log 2}{4 \log 2} = \frac{1}{4}$$

23. Put the values of a and b in $ab + 5(a-b)$ after converting them to same base.

24. Use options to solve this question.

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CHAPTER

16

Permutation and Combination

LEARNING OBJECTIVES

After completion of this chapter, the reader should be able to understand:

- Difference between permutation and combination
- Different cases of permutation such as word formation
- Different cases of combination such as selection of a team
- Application of P and C to derive more formulae
- Kinds of questions asked in the CAT
- Methods of solving questions

INTRODUCTION

Permutations and combinations has lately emerged as an important topic for management entrance examinations. This is primarily because questions from this topic require analytical skills and a logical bent of mind. Even students who do not have mathematics as a subject can handle them if they have a fairly good understanding of the concepts and their application. However, a point to note is that questions from permutations and combinations asked in the CAT are not based on mathematical concepts. Hence, anyone who is well-versed in different methods of counting and basic calculations will be able to solve these problems easily.

MEANING OF PERMUTATION AND COMBINATION

If we go by the dictionary meaning of the words permutation and combination, then permutation is the number of ways in which a set or a number of things can be put in an order or arranged, and combination refers to the number of ways in which a group of things can be chosen from a larger group without regard to their arrangement.

Let us go through an example. Suppose there are four different batsmen A, B, C, and D and we have to select a group of three batsmen out of these four. Now, we can select any combination of three batsmen so that no set of batsmen has all the same three batsmen. These set of batsmen will be — ABC, BCD, ABD, and ACD. This is a case of combination as for every set of selection of three batsmen, and order of selection does not play any role (i.e., we can select anybody—first or second or third—and it does not create any difference in the final selection as well as in the total number of selections).

Now, if we try to define their batting order also, that is who bats first and second, and so on, then corresponding to every selection of a set of three batsmen, we will have six different arrangements of their batting order. It can be seen below that corresponding to the selection of ABC as a team, following is the list of different batting orders:

ABC, ACB, BAC, BCA, CAB, and CBA

This is a case of permutation since the order of occurrence has become important. As there are four different ways of selecting a group of three batsmen and every selection can be arranged in 6 different ways, the total number of ways of arranging 3 batsmen (or distinct things) out of 4 batsmen (or distinct things) = $4 \times 6 = 24$ ways.

Permutation and combination can be better understood through the examples of hand-shakes and gift exchanges also. Assume that there are 20 persons in a party and

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everybody shakes hand with each other and also presents a gift. Now, if we take a case of two persons A and B, then the event of shaking hand between them is a case of combination because when A shakes hand with B or B shakes hand with A, the number of hand shake is just one. So, there is no order as such, and hence, it is a case of combination.

Similarly, the event of presenting the gift is a case of permutation because the gift given to B by A and the gift given to A by B are two different gifts. So, the order of case plays a role here, and hence, it is a case of permutation.

$n!$ = Product of all the natural numbers from n to $1 = n(n-1)(n-2)(n-3) \ldots \times 3 \times 2 \times 1$
$0! = 1$
Factorials are defined only for whole numbers and not for negative numbers or fractions (≠ whole numbers).

FUNDAMENTAL PRINCIPLES OF COUNTING: TWO BASIC THEOREMS

1. Multiplication Theorem

If there are two jobs in such a way that one of them can be done in m ways and when it is completed in any of the m ways, and the second job can be completed in n ways, then the whole job can be done in $m \times n$.

2. Addition Theorem

If there are two jobs in such a way that one of them can be done in m ways and the second one can be done in n ways independently, then either of the jobs can be done in $(m+n)$ ways.

Basically, there is one point where these two theorems differ—in multiplication, the job does not get completed, whereas in addition, it gets completed. In a layman's language, we multiply the number of ways when the job has not been completed and we add the number of ways when the job has been completed.

Example 1 There are 10 girls and 15 boys in a class. In how many ways can

(i) a class representative be selected?
(ii) a team of two students be chosen with one girl and one boy?

Solution

(i) A class representative can be a girl or a boy. Now, one girl can be selected from 10 girls in 10 ways (any of the girls can be selected) and one boy can be selected from 15 boys in 15 ways (any of the boys can be selected).
So, the ways of selecting a class representative includes either selecting a boy or a girl. Hence, the total number of ways of selecting a class representative = 10 + 15 = 25.
In this case, the moment a girl gets selected, the job is completed. There are some more ways of doing this by selecting a boy. So, it is a case of addition.

(ii) One girl can be chosen from 10 girls in 10 ways. Now corresponding to every selected girl, any one of the 15 boys can be selected in 15 ways.
It can be seen in the following presentation:
Girl selected (assume name of the girls are G_1, G_2, G_3, … G_9, G_{10}) – G_1
Boy selected (assume the names of the boys are B_1, B_2, B_3, … B_{14}, B_{15}) – B_1 or B_2 or B_3 or B_{15}
So, corresponding to G_1, the total number of selection of a boy = 15
Corresponding to G_2, the total number of selection of a boy = 15
Corresponding to G_3, the total number of selection of a boy = 15

… … … … …

Corresponding to G_{15}, the total number of selection of a boy = 15

So, the total number of ways of selecting a team of one boy and a girl = the total number of ways of selecting a girl × the total number of ways of selecting a boy = 10 × 15 = 150

In this case, just by selecting a girl or a boy, work has not been completed. So, it is a case of multiplication.

Another example of multiplication theorem: If there are three cities A, B, and C located in such a way that there are 3 roads joining A and B, and 4 roads joining B and C, then the number of ways one can travel from A to C is 3 × 4, i.e., 12.

PERMUTATIONS

As we have seen, the arrangements made by taking some or all elements out of a number of things is called a permutation. Permutation implies "arrangement" where "order of the things" is important.

The permutations of three things, a, b, and c, taken two at a time are ab, ba, ac, ca, cb, and bc. Since the order in which the things are taken is important, ab and ba are counted as two different arrangements.

The number of permutations of n things taking r at a time is denoted by ${}^nP_r$.

COMBINATIONS

As we have seen, the groups or selections made by taking some or all elements out of a number of things is called a

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combination. In combination, the order in which the things are taken is not important.

The combination of three things, *a*, *b*, and *c*, taken two at a time are *ab*, *bc*, and *ca*. Here, *ab* and *ba* are same because the order in which *a* and *b* are taken is not important. What is required is only a combination including *a* and *b*. The words "combination" and "selection" can be used without any differentiation.

The number of combinations of *n* things taking *r* at a time is denoted by ${}^nC_r$.

Approaching a Problem

Mostly the questions asked in the CAT are self-explanatory, that is, they clearly mention what process is to be used—permutation or combination.

In case, the question does not specify this, you should try to find out whether it is a case of permutation or a case of combination. Sometimes the problem very clearly states whether it is the number of permutations (or arrangements) or the number of combinations (or selections) that has to be found out. The questions can be as follows:

For permutations:

'What is the number of permutations that can be done…' or 'What is the number of arrangements that can be made…' or 'Find the different numbers of ways in which something can be arranged, etc'.

For combinations:

'What is the number of combinations that can be done…' or 'What is the number of selections that can be made…' or 'Find the different numbers of ways in which things can be selected, etc'.

Some other standard examples of permutation and combination are as follows:

Permutation Word formation, number formation, circular permutation, etc.

Combination Selection of a team, forming geometrical figures, and distribution of things (except some particular cases)

However, sometimes the questions may not explicitly state what you have to find—permutation or combination. In that case, the nature of what is to be found out will decide whether it is the number of permutations or the number of combinations. See the example given below:

I have to invite two of my eight friends to my anniversary party. In how many different ways can I do this?

Assume my eight friends are A, B, C, D, E, F, G, and H. Whether the two friends that I call for the party A and B or B and A, does not make any difference. As discussed earlier, what matters the most in case of permutation is the order of occurrence of things. As order does not play any role here, it is clearly the case of combination.

Meaning and Derivation of ${}^nP_r$ and ${}^nC_r$

Number of permutations of n different things taking r at a time = ${}^nP_r$

In this statement, we take the following two assumptions:

1. All the *n* things are distinct (or no two things are of the same type).
2. Each thing is used at most once (i.e., nothing is repeated in any arrangement).

Let us assume that there are *r* boxes and each of them can hold one thing. When all the *r* boxes are filled, what we have is an arrangement of *r* things taken from the given *n* things. So, each time we fill up the *r* boxes with things taken from the given *n* things, we have an arrangement of *r* things taken from the given *n* things without repetition. Hence, the number of ways in which we can fill up the *r* boxes by taking things from the given *n* things is equal to the number of permutations of *n* things taking *r* at a time.

The first box can be filled in *n* ways (because this box can be filled by any one of the *n* things given). After filling the first box, we now have only $(n-1)$ things to fill the second box; any one of these things can be used to fill the second box, and hence, the second box can be filled in $(n-1)$ ways. Similarly, the third box can be filled in $(n-2)$ ways and so on. The *r*th box can be filled in $(n-(r-1)$ ways, that is $[n-r+1]$ ways. Hence, from the fundamental rules of counting, all the *r* boxes together can be filled up in $\rightarrow n \cdot (n-1) \cdot (n-2)\ldots(n-r+1)$ ways.

Hence, ${}^nP_r = n \cdot (n-1) \cdot (n-2)\ldots(n-r+1)$

This can be simplified by multiplying and dividing the right-hand side by $(n-r)\,(n-\overline{r-1})$ … 3.2.1.

$$ {}^nP_r = n\,(n-1)(n-2)\ldots[n-\overline{r-1}] $$

$$ = \frac{n!}{(n-r)!} $$

The number of arrangements of *n* distinct things taken *r* things at a time is:

$$ {}^nP_r = \frac{n!}{(n-r)!} $$

If we take *n* things at a time, then we get ${}^nP_n$. From the discussion similar to that we had for filling the *r* boxes above, we can find that ${}^nP_n$ is equal to *n*! The first box can be filled in *n* ways, the second one in $(n-1)$ ways, the third one in $(n-2)$ ways, and so on, then the *nth* box in 1 way; hence, all the *n* boxes can be filled in:

$$ {}^nP_n = n! $$

However, if we substitute $r = n$ in the formula for ${}^nP_n$, then we get ${}^nP_n = \frac{n!}{0!}$; since we already found that ${}^nP_n = n!$

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We can conclude that 0! = 1.

The number of combinations of n distinct things taking r at a time $= {}^nC_r$

Let the number of combinations ${}^nC_r$ be S. Consider one of these S combinations. Since this is a combination, the order of the r things is not important. If we now impose the condition that order is required for these r things, then we can get $r!$ arrangement from this one combination. Hence, each combination can give rise to $r!$ permutations. S combinations will, thus, give rise to $S \times r!$ permutations. However, since these are all permutations of n things taking r at a time, this must be equal to ${}^nP_r$.

Therefore,

$$S \times r! = {}^nP_r = \frac{n!}{(n-r)!}$$

So,
$$S = {}^nC_r = \frac{n!}{(n-r)!} \times \frac{1}{r!}$$

It can also be deduced from here that the number of selections of n distinct things taken all at a time will be equal to 1 (since there is only one way in which all the articles can be selected).

$$\text{Alternatively, } {}^nC_n = \frac{n!}{0! \times n!} = 1$$

Out of n things kept in a bag, if we select r things and remove them from the bag, then we are left with $(n - 1)$ things inside the bag, that is whenever r things are selected out of n things, we automatically have another selection of $(n - 1)$ things. Hence, the number of ways of making combinations taking r out of n things is the same as selecting $(n - r)$ things out of n given things, that is,

$${}^nC_r = {}^nC_{n-r}$$

Before we move ahead, let us once again make it clear that whenever we are using ${}^nC_r$ and ${}^nP_r$, our assumption is that all the things are distinct, that is no two of them are same.

Example 2 Munchun has 10 children. She takes 3 of them to the zoo at a time, as often as she can, but she does not take the same three children to the zoo more than once. How many times Munchun will be required to go to the zoo?

(a) 120 (b) 45 (c) 90 (d) 180

Solution Number of times (read ways) 3 children (read distinct things) can be selected from 10 children (read distinct things) $= {}^{10}C_3$.

So, she will be required to go to the zoo ${}^{10}C_3$ times.

So, option (a) is the answer.

Example 3 In the above question, how many times a particular child will go?

(a) 72 (b) 45 (c) 90 (d) 36

Solution Consider the case for any particular child C_1. Since C_1 has already been selected, so out of the rest 9 children Munchun will be required to select 2 more children. This can be done on ${}^9C_2$ ways.

So, option (d) is the answer.

Example 4 In a chess competition involving some boys and girls of a school, every student has to play exactly one game with every other student. It was found that in 45 games, both the players were girls, and in 190 games, both were boys. The number of games in which one player was a boy and the other was a girl is:

(a) 200 (b) 216 (c) 235 (d) 256

Solution Let there be m boys and n girls.

$${}^nC_2 = 45 = \frac{n(n-1)}{2} \Rightarrow n(n-1) = 90 \Rightarrow n = 10$$

$${}^mC_2 = 190 = \frac{m(m-1)}{2} = 190 \Rightarrow m(m-1) = 380 \Rightarrow m = 20$$

Number of games between one boy and one girl
$= {}^{10}C_1 \times {}^{20}C_1 = 10 \times 20 = 200$

Hence, option (a) is the answer.

Example 5 In how many ways can three persons be seated on five chairs?

Solution This question is a very fundamental problem of arrangements without repetition. The first person can sit in 5 ways (into any of the five chairs), the second person can take place in 4 ways (into any of the remaining 4 chairs), and the third person can sit in 3 ways.

So, the total number of ways in which these 3 persons can arrange themselves on 5 chairs is $5 \times 4 \times 3 = 60$.

Some Important Derivations

While deriving an expression for ${}^nP_r$, we imposed two constraints, viz. distinct things and repetition being not allowed over it and learned how to find the number of permutations. Let us now see what will happen if we do not impose these two restrictions on ${}^nP_r$.

Number of arrangements of *n* things of which *p* are of one type, *q* are of a second type, and the rest are distinct When all the things are not distinct, then we cannot use the general formula for ${}^nP_r$ for any value of r. If we want to find out ${}^nP_r$ for a specific value of r in that given problem, then we will be required to use it on the basis of the given situation.

The number of ways in which n things may be arranged taking all of them at a time, when p of the things are exactly alike of one kind, q of them are exactly alike of another kind,

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r of them are exactly alike of a third kind, and the rest all are distinct, is:

$$\frac{n!}{p!q!r!}$$

Number of permutations of n distinct things where each one of them can be used for any number of times (i.e., repetition allowed) Derivation for this is based upon common sense. If I have 5 friends and 3 servants, and I have to send the invitation letters to all my friends through any of my servants, then I obviously have 3 options for the invitation card to be sent to friend 1, the same 3 options for the invitation card to be sent to friend 2, and similarly, 3 options for the invitation card to be sent to each of the friends. So, the total number of ways of sending the invitation letters = 3^5, and it will not be 5^3, as friends are not going to the servants to get the letter.

In general, the number of perambulations of n things, taking r at a time when each of the thing may be repeated once, twice, … up to r times in any arrangement is n^r.

Total Number of Combinations

Out of n things, the number of ways of selecting one or more things:

where we can select 1 or 2 or 3 … and so on, n things at time; hence, the number of ways is ${}^nC_1 + {}^nC_2 + {}^nC_3 + \ldots {}^nC_n = 2^n - 1$, where n is the number of things.

Above derivation can also be understood in the following manner:

Let there be n bags.

The first bag can be dealt in two ways—it is either included or not included. Similarly, the second bag can be dealt in two ways, the third one in two ways, and so on, the *nth* bag in two ways. Using multiplication theorem of counting, the number of ways of dealing with all the bags together is $2 \times 2 \times 2 \times \ldots$ n times $= 2^n$ ways. However, out of these, there is one combination where we do not include any of the bags. This is not allowed because we have to select at least one thing.

Hence, the number of ways of selecting one or more things from n given things is $2^n - 1$.

Distributing the given things $(m + n)$ into two groups where one group is having m things and other one n things If we select m things (which can be done in ${}^{m+n}C_m$ ways), then we will be left with n things, that is we have two groups of m and n things, respectively. So, the number of ways of dividing $(m + n)$ things into two groups of m and n things. respectively. is equal to ${}^{m+n}C_m$.

$$= {}^{m+n}C_m = \frac{(m+n)!}{n!m!}$$

If we take $m = n$, then the above expression will denote 'distributing $2m$ things' equally between two distinct groups

$$= {}^{2m}C_m = \frac{(2m)!}{m!m!}$$

However, when the groups are identical, then we will be required to divide the above result by 2!

Hence, in that case, it becomes $\frac{(2m)!}{2!m!m!}$.

(Refer to word formation examples)

All the above derivations with their different applications can be seen below in a summarized form.

1. Fundamental Principle of counting
   (a) *Multiplication rule* If a work is done only when all the number of works are done, then the number of ways of doing that work is equal to the product of the number of ways of doing separate works.
   (b) *Addition rule* If a work is done only when any one of the number of works is done, then the number of ways of doing that work is equal to the sum of the number of ways of doing separate works. Therefore, if a work is done when exactly one of the number of works is done, then the number of ways of doing this work = sum of the number of ways of doing all the works.
2. If ${}^nC_x = {}^nC_y$, then either $x = y$ or $x + y = n$.
3. $\lfloor n = 1 \cdot 2 \cdot 3 \ldots n$; $\lfloor 0 = 1$
4. (a) The number of permutations of n different articles taking r at a time is denoted by ${}^nP_r$ and ${}^nP_r = \frac{\lfloor n}{\lfloor n-r}$.
   (b) The number of permutations of n different articles taking all at a time is equal to $\lfloor n$.
   (c) The number of permutations of n articles, out of which p are alike and are of one type, q are alike and are of second type, and rest are all different = $\frac{\lfloor n}{\lfloor p \times \lfloor q}$.
5. The number of permutations (arrangements) of n different articles taking r at a time when articles can be repeated any number of times = $n \times n \times \ldots$ r times = n^r.
6. Circular permutations:
   (a) The number of circular permutations (arrangements) of n different articles = $\lfloor n-1$.
   (b) The number of circular arrangements of n different articles when clockwise and anticlockwise arrangements are not different, i.e., when the observation can be made from both the sides = $\frac{\lfloor n-1}{2}$.
7. The number of combinations of n different articles taking r at a time is denoted by ${}^nC_r$ and ${}^nC_r = \frac{\lfloor n}{\lfloor r \lfloor n-r}$.

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8. The number of selections of r articles ($r \le n$) out of n identical articles is 1.
9. Total number of selections of zero or more articles from n distinct articles = ${}^nC_0 + {}^nC_1 + {}^nC_2 + \ldots {}^nC_n = 2^n$.
10. Total number of selections of zero or more articles from n identical articles = 1 + 1 + 1 + … to $(n + 1)$ terms = $n + 1$.
11. The number of ways of distributing n identical articles among r persons when each person may get any number of articles = ${}^{n+r-1}C_{r-1}$.
12. The number of ways of dividing $m + n$ different articles in two groups containing m and n articles, respectively ($m \ne n$),

$$= {}^{m+n}C_n \times {}^mC_m = \frac{\lfloor m+n}{\lfloor m \lfloor n}$$

13. The number of ways of dividing $2m$ different articles each containing m articles = $\dfrac{\lfloor 2m}{\lfloor m \lfloor m \lfloor 2}$.
14. The number of ways of dividing $3m$ different articles among three persons and each is getting m articles = $\dfrac{\lfloor 3m}{\lfloor m \lfloor m \lfloor m \lfloor 3} \lfloor 3$.
15. The number of ways of selecting n distinct articles taken r at a time when p particular articles are always included = ${}^{n-p}C_{r-p}$.
16. ${}^nC_{r-1} + {}^nC_r = {}^{n+1}C_r$
17. ${}^np_r = r.{}^{n-1}p_{r-1} + {}^{n-1}P_r$

SOME STANDARD FORMATS OF QUESTIONS

Word Formation

As we know that order of occurrence of letters decide the formation of words, so word formation is one standard example of permutation.

Let us understand word formation with the help of some examples.

Example 6 How many words can be formed with the word 'LUCKNOW', which have:

(i) No restriction
(ii) L as the 1st letter of the word
(iii) L and W as the terminal letters
(iv) All the vowels together
(v) L always occuring before U
(vi) L always occuring before U and U always occuring before W

Solution

(i) The total number of distinct letters = 7 (L, U, C, K, N, O, W).
So, the total number of words that can be formed is 7!
(ii) Now, we can arrange only 6 letters (as place of L is restricted).
So, the total number of words that can be formed is 6!
(iii) Now, we can arrange only 5 letters (as place of L and W are restricted),
So, the number of arrangements = 5!
However, the place of L and W can be interchanged between themselves.
So, the total number of words that can be formed = 5! × 2!
(iv) U and O should be together, so we will assume that these two letters to be tied up with each other.
Now, we have 6 distinct things to be arranged—(L, U, O, C, K, N, W)
So, the number of arrangements = 6!
However, the place of U and O can be interchanged between themselves.
So, the total number of words that can be formed = 6! × 2!
(v) There is an equal likelihood occurrence of all the letters in the word, so in half of the cases, L will occur before U, and in the remaining half, U will occur before O.
So, the total number of words that can be formed = 7!/2!
(vi) There are six possible arrangements (3!) corresponding to L, U, and W. However, only one out of these six will be in the prescribed order: L always occurs before U, and U always occurs before W.
So, corresponding to 7! arrangements, the number of ways in which the condition will be satisfied = 7!/3! ways.

Example 7 How many new words can be formed with the word 'PATNA'?

Solution From total number of letters, P, T, and N occur once, whereas A occurs twice.

So, the total number of words that can be formed = 5!/2! = 60

Total number of new words = 60 – 1 = 59

Example 8 How many words can be formed with the word 'ALLAHABAD'?

Solution Letters are: A—Four times
L—Twice
H, B, and D occur once.
So, the total number of words = $\dfrac{9!}{4!2!}$

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Example 9 How many 4-lettered distinct words can be formed from the letters of the word 'EXAMINATION'?

Solution Letters are: A—Twice
I—Twice
N—Twice
E, X, M, T, O—Once
Words will be of three types:
(i) All distinct, (ii) two same and two distinct, and (iii) two same and of one kind; two same and of other kind.

(i) All distinct = ${}^{8}P_{4}$ (distinct letters are — A, I, N, E, X, M, T, and O)

(ii) Two same and two distinct
Selection of one pair out of the three pairs (A, I, N) can be chosen in ${}^{3}C_{1}$ ways. Now, rest of the two distinct letters can be chosen in ${}^{7}C_{2}$ ways.
Total number of words = ${}^{3}C_{1} \times {}^{7}C_{2} \times \frac{4!}{2!}$

(iii) Two same and are of one kind, and two same and are of other kind = out of the three pairs of letters (A, I, N), we can select two pairs in ${}^{3}C_{2}$ ways.
Total number of words = ${}^{3}C_{2} \times \frac{4!}{2! \times 2!}$

Number Formation

Number formation is another standard example of permutation. Here, we will discuss the box diagram method of solving the questions.

If a three-digit number is to be constructed, then we will use the following:

Hundred's place	Ten's place	Unit's place

If a four-digit number is to be constructed, then we will use the following:

Thousand's place	Hundred's place	Ten's place	Unit's place

While solving the questions related to number formation, we should know two things very clearly:

- While using the box diagram, we should start with the digit which has restriction, that is some condition is imposed on that digit.
- When nothing is mentioned about the repetition of digits in the question, then we will assume that the repetition is allowed.

Example 10 How many different three-digit numbers can be formed using the digits 1, 2, 3, 4, and 5?

(i) When repetition is not allowed.
(ii) When repetition is allowed.

Solution The box given below represents the respective positioning of digits in a three-digit number.

Hundred's place	Ten's place	Unit's place

(i) Since repetition of the digits is not allowed, we can fill the unit's place in 5 ways, ten's place in 4 ways, and hundred's place in 3 ways.

3	4	5

Using multiplication theorem, the total number of numbers that can be formed = $5 \times 4 \times 3 = 60$
Alternatively, 3 digits can be selected out of 5 digits in ${}^{5}P_{3} = 60$

(ii) Since repetition of the digits is allowed here, we can fill each of the hundred's, ten's, and unit's place in 5 ways.

5	5	5

Using multiplication theorem, the total number of numbers that can be formed = $5 \times 5 \times 5 = 125$

Example 11 How many 4-lettered numbers divisible by 4 can be formed from the digits 0, 1, 2, 3, 4, and 5?

Solution Any number divisible by 4 will have the number formed by its last two digits divisible by 4.

In this case, last two digits of the number can be 00, 04, 12, 20, 24, 32, 40, 44, and 52.

Corresponding to any one of 00, 04, 12, 20, 24, 32, 40, 44, and 52, we can have the following digits at its hundred's and thousand's place:

5	6	1	1

Thousand's place cannot be filled by 0, so it can be filled in 5 ways.

Hundred's place can be filled by any of the 0, 1, 2, 3, 4, and 5; hence, 6 ways.

So, corresponding to any one of 00, 04, 12, 20, 24, 32, 40, 44, and 52, the total number of ways = $5 \times 6 = 30$

So, the total number of numbers that can be formed = $30 \times 9 = 270$

Example 12 In the above example, how many numbers can be formed if repetition of the digits is not allowed?

Solution Last two digits of this number can be—04, 12, 20, 24, 32, 40, and 52.

At this point, we will have to divide the process of solving this question—one part will have those numbers that contain'0' as one of its last two digits viz., 04, 20, and 40, and other part will have the remaining numbers viz., 12, 24, 32, and 52.

1st part—last two digits are 04, 20, and 40.

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4	3	1	1

$= 4 \times 3 = 12$ ways

Hence, the total number of numbers that can be formed $= 12 \times 3 = 36$

2nd part — last two digits are 12, 24, 32, and 52.

'0' cannot occur at thousand's place.

3	3	1	1

$= 3 \times 3 = 9$ ways

Hence, the total number of numbers that can be formed $= 9 \times 4 = 36$

Total numbers = 36 + 36 = 72

Example 13 How many odd integers from 1000 to 8000 have none of its digits repeated?

Solution There are two restrictions operating in this questions:

(i) For a number to be odd, unit digit should be either 1 or 3 or 5 or 7 or 9.

(ii) Thousand's place cannot be filled with 8 or 9.

For unit's digit, when it is filled with 9, thousand's place can be filled in 7 ways, namely any digit from 1 to 7, and the remaining two places can be filled in $8 \times 7 = 56$ ways.

So, the total number of numbers formed in this way $= 56 \times 7 = 392$

Now, if the unit's place is filled with any of the four digits 1, 3, 5, or 7, then the thousand's place can be filled in 6 ways (0 will be excluded), and the remaining two places can be filled in $8 \times 7 = 56$ ways.

So, the total number of numbers formed in this way $= 56 \times 6 \times 4 = 1344$

So, the total number of numbers = 392 + 1344 = 1736

Example 14 How many integers from 6000 to 6999 have at least one of its digits repeated?

Solution Total number of numbers = none of its digits with repeated numbers + at least one of its digits with repeated number (i.e., either the digits will be repeated or not repeated).

Total numbers with none of its digits repeated $= 1 \times 9 \times 8 \times 7 = 504$

So, the numbers having at least one of its digits repeated $= 1000 - 504 = 496$

Example 15 How many natural numbers less than a million can be formed using the digits 0, 7, and 8?

Solution The numbers formed would be of a single digit, two digits, three digits, four digits, five digits, and six digits.

Single-digit numbers = 7 and 8

For two-digit numbers,

2	3

$= 2 \times 3 = 6$ numbers

For three-digit numbers,

2	3	4

$= 2 \times 3 \times 3 = 18$ numbers

For four-digit numbers,

2	3	3	3

$= 2 \times 3 \times 3 \times 3 = 54$ numbers

For five-digit numbers,

2	3	3	3	3

$= 2 \times 3 \times 3 \times 3 \times 3 = 162$ numbers

For six-digit numbers,

2	3	3	3	3	3

$= 2 \times 3 \times 3 \times 3 \times 3 \times 3 = 486$ numbers

So, the total number of numbers = 728

CIRCULAR PERMUTATION

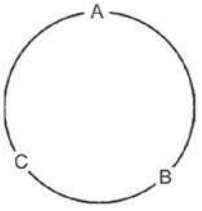
When n distinct things are to be arranged in a straight line, we can do this in $n!$ ways. However, if these n things are arranged in a circular manner, then the number of arrangements will not be $n!$

Let us understand this:

The number of ways A, B, and C can be arranged in a straight line $= 3! = 6$.

The possible arrangements are ABC, ACB, BAC, BCA, CAB, and CBA.

Now, arrange these three people A, B, and C in a circle.



What we observe here is that the arrangements ABC, BCA, and CAB are the same, and similarly, the arrangements ACB, CBA, and BAC are the same.

So, there are only two permutations in this case of circular permutation.

To derive the formula for circular permutation, we first fix the position of one thing, then the remaining $(n - 1)$ things can be arranged in $(n - 1)!$ ways.

Hence, the number of ways in which n distinct things can be arranged in a circular arrangement is $(n - 1)!$

It can be seen in the following way also:

If n things are arranged along a circle, then corresponding to each circular arrangement, the number of linear arrangement $= n$.

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So, the number of linear arrangements of n different things = $n \times$ (number of circular arrangements of n different things)

Hence, the number of circular arrangements of n different things = $(1/n) \times$ number of linear arrangements of n different things = $(1/n) \times n! = (n-1)!$

Clockwise and Anti-clockwise Circular Arrangements

If we take the case of four distinct things A, B, C, and D sitting around a circular table, then the two arrangements ABCD (in clockwise direction) and ADCB (the same order but in anti-clockwise direction) will be different and distinct. Hence, we can conclude that the clockwise and anti-clockwise arrangements are different. However, if we consider the circular arrangement of a necklace made of four precious stones A, B, C, and D, then the two arrangements discussed as above will be the same because we take one arrangement and turn the necklace around (front to back), and then we get the other arrangement. Hence, the two arrangements will be considered as one arrangement because the order of stones is not changing with the change in the side of observation. So, in this case, there is no difference between the clockwise and the anti-clockwise arrangements.

Summarizing the above discussion, the number of circular arrangements of n distinct things is $(n-1)!$ if there is a difference between the clockwise and anti-clockwise arrangements and $(n-1)!/2$ if there is no difference between the clockwise and anti-clockwise arrangements.

Example 16 In how many ways 5 Indians and 4 Americans can be seated at a round table if:

(i) There is no restriction.
(ii) All the 4 Americans sit together.
(iii) No 2 Americans sit together.
(iv) All the 4 Americans do not sit together.

Solution

(i) Total number of persons = 9. These 9 persons can be arranged around a circular table in 8! ways.

(ii) Assuming all the Americans to be one group, we have 6 things (5 Indians + 1 group) to be arranged around a circular table which can be arranged in 5! ways. However, these 4 Americans can be arranged in 4! ways among themselves.
So, the total number of arrangements = $5! \times 4!$

(iii) Since there is no restriction on Indians, the 5 Indians can be seated around a table in 4! ways. The Americans will now be seated between two Indians, that is 5 places. Four Indians can be seated on these 5 places in ${}^5P_4$ ways.

(iv) The total number of arrangements when there is no restriction = 8! and the number of arrangements when all the four Americans sit together = $5! \times 4!$
So, the total number of arrangements when all the four Americans do not sit together = $8! - 5! \times 4!$

PERMUTATION AND COMBINATION IN GEOMETRY

It is quite difficult to quantify the importance of P and C in geometry. A considerable number of P and C questions that use the concepts of geometry (and vice versa) have been asked in the CAT and other premier B-school exams.

Example 17 How many diagonals will be there in an n-sided regular polygon?

Solution An n-sided regular polygon will have n vertices. When we join any of these two vertices (${}^nC_2$), we get a straight line, which will be either a side or a diagonal.

So, ${}^nC_2$ = Number of sides + number of diagonals = n + number of diagonals

Hence, the number of diagonals = ${}^nC_2 - n = \dfrac{n(n-3)}{2}$

The above result can be used as a formula also.

Example 18 Ten points are marked on a straight line and 11 points are marked on another parallel straight line. How many triangles can be constructed with vertices among these points?

Solution Triangles will be constructed by taking one point from the 1st straight line and two more points from the 2nd straight line, and vice versa.

So, the total number of Δ formed = ${}^{10}C_2 \times {}^{11}C_1 + {}^{11}C_2 \times {}^{10}C_1 = 1045$

Example 19 There is an n-sided polygon ($n > 5$). Triangles are formed by joining the vertices of the polygon. How many triangles can be constructed that will have no side common with the polygon?

Solution An n-sided polygon will have n vertices. Triangles constructed out of these n vertices will be of the following three types:

(i) Having two sides common with the polygon
(ii) Having one side common with the polygon
(iii) Having no side common with the polygon, and the total number of triangles formed will be ${}^nC_3$.

(i) Having two sides common with the polygon—Out of total n vertices, any combination of three consecutive vertices will give us the triangle that has two sides common with polygon = n.

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(ii) Having one side common with the polygon—Number of selection of three vertices out of which two are consecutive (if we select A_5 and A_6 as the two vertices, then A_7 or A_4 should not be the third vertex because it will constitute the two sides of the common triangle) $= n \times {}^{(n-4)}C_1$

(iii) So, the total number of triangles having no side common with polygon $= {}^nC_3 - n \times {}^{(n-4)}C_1 - n$

Some Important Results

1. Maximum number of points of intersection among n straight lines $= {}^nC_2$
2. Maximum number of points of intersection among n circles $= {}^nP_2$

FINDING THE RANK OF A WORD

To find the rank of a word out of all the possibilities using all the letters given in the word is nothing but the extension of the concept of alphabetically arranging the words in a dictionary. However, unlike the case of the dictionary, we can have 'meaningless' words also in the case of finding the rank.

Example 20 All the letters of the word 'LUCKNOW' are arranged in all possible ways. What will be the rank of the word LUCKNOW?

Solution Alphabetical order of occurrence of letters—C, K, L, N, O, U, W.

Number of words starting with C = 6!
Number of words starting with K = 6!
All the words starting with LC = 5!
All the words starting with LK = 5!
All the words starting with LN = 5!
All the words starting with LO = 5!
Next, word will start with LU-C-K-N-O-W.
So, rank of LUCKNOW $= 2 \times 6! + 4 \times 5! + 1 = 1921$.

Example 21 In a chess competition involving some boys and girls of a school, every student has to play exactly one game with every other student. It was found that in 45 games, both the players were girls, and in 190 games, both were boys. Find the number of games in which one player was a boy and the other was a girl.

(a) 200 (b) 216 (c) 235 (d) 256

Solution Let there be m boys and n girls.

$${}^nC_2 = 45 = \frac{n(n-1)}{2} \Rightarrow n(n-1) = 90 \Rightarrow n = 10$$

$${}^mC_2 = 190 = \frac{m(m-1)}{2} = 190 \Rightarrow m(m-1) = 380 \Rightarrow m = 20$$

Number of games between one boy and one girl

$= {}^{10}C_1 \times {}^{20}C_1 = 10 \times 20 = 200$

Hence, option (a) is the answer.

Example 22 If each permutation of the digits 1, 2, 3, 4, 5, and 6 is listed in the increasing order of the magnitude, then the 289th term will be:

(a) 326,541 (b) 341,256
(c) 356,241 (d) 314,256

Solution $289 = (2 \times 5!) + (2 \times 4!) + 1$

So, the number will be 341,256.

Example 23 There are 12 intermediate stations between two places A and B. In how many ways can a train be made to stop at 4 of these 12 intermediate stations provided no two of them are consecutive?

Solution

Method 1 Let $S_1, S_2, \ldots, S_8$ denote the stations where the train does not stop. The four stations where the train stops should be at any four of the nine places indicated by cross.

$$\therefore \text{Required number} = {}^9C_4 = \frac{9 \cdot 8 \cdot 7 \cdot 6}{\lfloor 4} = 126$$

Method 2 Let S_1, S_2, S_3, S_4 be the four intermediate stations where the train stops.

$$\boxed{a}\ S_1\ \boxed{b}\ S_2\ \boxed{c}\ S_3\ \boxed{d}\ S_4\ \boxed{e}$$

Let $a, b, c, d,$ and e be the number of stations between A and S_1, S_1 and S_2, S_2 and S_3, S_3 and S_4, S_4 and B, respectively.

Then, $a + b + c + d + e = 8$ (i)

where $a \geq 0, b \geq 1, c \geq 1, d \geq 1$, and $e \geq 0$

Let $x = a, y = b - 1, z = c - 1, t = d - 1$, and $w = e$

Now, $x + y + z + t + w = a + b + c + d + e - 3 = 8 - 3 = 5$

or $x + y + z + t + w = 5$, where $x, y, z, t, w \geq 0$ (ii)

Required number = number of non-negative integral solutions

$= {}^{n+r-1}C_r = {}^{5+5-1}C_5 = {}^9C_5 = 126$

Example 24 Find the number of integral solutions of equation $x + y + z + t = 25, x > 0, y > 1, z > 2$, and $t \geq 0$.

Solution Given, $x + y + z + t = 25$, (i)

where $x \geq 1, y \geq 2, z \geq 3, t \geq 0$

Let $p = x - 1, q = y - 2, r = z - 3$, and $s = t$

Then, $p + q + r + s = x + y + z + t - 6 = 25 - 6 = 19$, where $p, q, r, s \geq 0$

$\therefore p + q + r + s = 19, p, q, r, s \geq 0$ (ii)

$\therefore$ Required number = number of ways in which 19 identical things can be distributed among 4 persons when each person can get any number of things

$= {}^{n+r-1}C_{r-1} = {}^{22}C_3$

Example 25 There are 4 oranges, 5 apricots, and 6 alphonso in a fruit basket. In how many ways can a person make a selection of fruits from among the fruits in the basket?

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Solution Whenever we are talking about fruits, we assume them to be identical. However, when we are talking about men, we treat them to be distinct.

Zero or more oranges can be selected out of 4 identical oranges in $4 + 1 = 5$ ways.

Zero or more apricot can be selected out of 5 identical apricots in $5 + 1 = 6$ ways.

Zero or more can be selected out of 6 identical alphanso in $6 + 1 = 7$ ways.

$\therefore$ The total number of selections when all the three types of fruits are selected (the number of any type of fruit may also be zero)

$= 5 \times 6 \times 7 = 210$

However, in one of these selections the number of each type of fruit is zero, and hence, this selection must be excluded.

$\therefore$ Required number $= 210 - 1 = 209$

Example 26 Twelve different letters of alphabet are given. Words with six letters are formed from these given letters. Find the number of words that have at least one letter repeated.

Solution The total number of letters is 12. Words of six letters are to be framed.

The total number of words of 6 letters when any letter may be repeated any number of times (This also includes the number of words formed when no letter is repeated).

$$= 12 \times 12 \times 12 \times 12 \times 12 \times 12 = 12^6$$

Number of words of 6 letters when no letter is repeated $= {}^{12}P_6$.

So, number of words of 6 letters that have at least one letter repeated $= 12^6 - {}^{12}P_6$

Practice Exercises

Warm Up

Q.1 How many numbers between 100 and 1000 can be formed using the digits 0, 2, 4, 6, 8, and 5, if repetition of digits in a number is not allowed?
(a) 99 (b) 82 (c) 100 (d) 120

Q.2 How many numbers between 100 and 1000 can be formed using the digits 0, 2, 4, 6, 8, and 5, if repetition of digits in a number is allowed?
(a) 164 (b) 180 (c) 192 (d) 100

Q.3 Two out of six paper sets for an examination are of mathematics. What is the number of ways in which the papers can be set so that the two mathematics papers are not together?
(a) 480 (b) 520 (c) 492 (d) 512

Q.4 In a letter lock, each of the three rings is marked with 15 letters. What is the maximum number of unsuccessful attempts that one has to make before the lock is opened?
(a) 3374 (b) 5284 (c) 8457 (d) 8101

Q.5 A certain code consists of 5 variables, with each variable having 4 different constant values possible. What is the total number of coded messages that can be sent with 5 constants one from each variable?
(a) $1024 \times 5!$ (b) $1024 \times 4!$
(c) $1024 \times 3!$ (d) None of these

Q.6 From 3 different soft drinks, 4 Chinese dishes, and 2 ice-creams, how many different meals are possible if at least one of each of the three items is to be included, depending upon the number of people likely to turn up?
(a) 315 (b) 282
(c) 864 (d) None of these

Q.7 The governing council of an institute has 15 members and wants to hold its annual meeting. In how many ways can the council be seated around a round table if the chairman and the vice-chariman of the council are always seated together?
(a) $10 \times 12!$ (b) $14 \times 10!$
(c) $13!$ (d) None of these

Q.8 Akshay is planning to give a birthday party at his place. In how many ways can he invite one or more of five friends and seat them at a circular table?
(a) 84 (b) 89 (c) 78 (d) 81

Q.9 In the above question, In how many ways can he invite one or more of his five friends and seat them at a circular table with him?
(a) 325 (b) 205 (c) 265 (d) 320

Q.10 A cricket team of 11 is to be chosen from 8 batsmen, 6 bowlers, and 2 wicket-keepers. In how many ways can the team be chosen if there must be at least four batsmen, at least four bowlers, and exactly one wicket-keeper?
(a) 1681 (b) 5304
(c) 1652 (d) None of these

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Q.11 How many words can be formed using the letters of the word 'CORRESPONDENCE'?
(a) $[(14!)/(2!)^4]$ (b) 14!
(c) $[(14!)/(2!)^4(3!)]$ (d) $[(14!)/(2!)^3(3!)]$

Q.12 How many words can be formed using the letters of the word 'CORRESPONDENCE' if the consonants are always written together?
(a) 182 (b) 184
(c) $216 \times 9!$ (d) None of these

Q.13 If six persons are selected out of ten, in how many ways will a particular person be found among those six?
(a) 124 (b) 126 (c) 144 (d) 84

Q.14 A committee of five has to be chosen among six men and four ladies. In how many ways can this be done in order to include at least one lady?
(a) 252 (b) 246 (c) 244 (d) 152

Q.15 How many ways are there to pick two different cards from a deck of 52 cards such that the first card is an ace and the second is not a queen?
(a) 188 (b) 198
(c) 164 (d) None of these

Q.16 In the above question, what is the number of ways such that the first card is a spade and the second is not a queen?
(a) 612 (b) 612
(c) 164 (d) None of these

Q.17 How many different license plates involving three letters and three digits are there if the three letters appear together, either at the beginning or at the end of the license?
(a) $2 \times 26^3 \times 10^3$ (b) 54102
(c) $4 \times 25^2 \times 10^4$ (d) None of these

Q.18 A joint committee of students and teachers of 5 members has to be formed with 4 teachers, 3 male students, and 5 female students. How many different committees can be formed if the committee must consist of at least 2 teachers, 1 male student, and 2 female students?
(a) 170 (b) 152 (c) 180 (d) 104

Q.19 There are 8 different locks, with exactly one key for each lock. All the keys have been mixed up. What is the maximum number of trials required in order to determine which key belongs to which lock?
(a) 44 (b) 28 (c) 24 (d) 32

Q.20 Hoppers' Stop stocks four styles of trousers. For each style, there are ten different possible waist sizes, six different trousers lengths and four colour choices. How many different types of trousers could the store have?
(a) 1024 (b) 960 (c) 921 (d) 924

Foundation

Q.1 In a group of boys, the number of arrangements of 4 boys is 12 times the number of arrangements of 2 boys. The number of boys in the group is:
(a) 10 (b) 8
(c) 6 (d) None of these

Q.2 From a group of persons, the number of ways of selecting 5 persons is equal to that of 8 persons. The number of persons in the group is:
(a) 13 (b) 40 (c) 18 (d) 21

Q.3 For Board of Cricket Control of India (BCCI), a selection committee is to be chosen consisting of 5 ex-cricketers. Now there are 10 representatives from four zones. It has further been decided that if Kapil Dev is selected, Sunil Gavaskar will not be selected and vice versa. In how many ways can this be done?
(a) 140 (b) 112 (c) 196 (d) 56

Q.4 At a get-together, it was found that a total of 66 handshakes took place at the conclusion of the party. If each guest shook hand only once with all the others, how many people were present.
(a) 33 (b) 22 (c) 12 (d) 13

Q.5 A committee is to be formed comprising of 7 members such that there is a majority of men and at least 1 woman in every committee. The shortlist consists of 9 men and 6 women. In how many ways can this be done?
(a) 3724 (b) 3630 (c) 3526 (d) 4914

Q.6 A number of six-digit numbers that can be made with the digits 0, 1, 2, 3, 4, and 5, so that even digits occupy odd places, is:
(a) 24 (b) 36
(c) 48 (d) None of these

Q.7 The Total number of ways in which six '+' and four '–' signs can be arranged in a line such that no two '–' signs occur together, is:
(a) $\frac{7!}{3!}$ (b) $6! \times \frac{7!}{3!}$
(c) 35 (d) None of these

Q.8 Let A be the set of four-digit numbers a_1, a_2, a_3, a_4, where $a_1 > a_2 > a_3 > a_4$, then how many values of A are possible?
(a) 126 (b) 84
(c) 210 (d) None of these

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Q.9 A number of five-digit numbers in which no two consecutive digits are identical is:
(a) $9^2 \times 8^3$ (b) 9×8^4
(c) 9^5 (d) None of these

Q.10 The total number of selections of at most n things from $(2n + 1)$ different things is 63. Find the value of n.
(a) 3 (b) 2
(c) 4 (d) None of these

Q.11 From 4 gentlemen and 6 ladies, a committee of five is to be formed. The number of ways in which the committee can be formed so that the gentlemen are in majority is:
(a) 66 (b) 156
(c) 60 (d) None of these

Q.12 The number of all four digit numbers, which are divisible by 4, that can be formed from the digits 1, 2, 3, 4, and 5 is:
(a) 125 (b) 30
(c) 95 (d) None of these

Q.13 The total number of integral solutions for (x, y, and z) such that $xyz = 24$ is:
(a) 36 (b) 90
(c) 120 (d) None of these

Q.14 The number of ways to give 16 different things to three persons A, B, and C, so that B gets 1 more than A, and C gets 2 more than B, is:
(a) $\frac{16!}{4!5!7!}$ (b) 4!5!7!
(c) $\frac{16!}{3!5!8!}$ (d) None of these

Q.15 The number of positive integral solutions of $x + y + z = n$, $n \in N$, $n > 3$ is:
(a) ${}^{n-1}C_2$ (b) ${}^{n-1}P_2$
(c) $n(n-1)$ (d) None of these

Q.16 If ${}^nC_4$, ${}^nC_5$, and ${}^nC_6$ are in an AP, then find n .s:
(a) 8 (b) 9 (c) 14 (d) 10

Q.17 There are 20 questions in a question paper. If no two students solve the same combination of questions but solve equal number of questions, then the maximum number of students who appeared in the examination is:
(a) ${}^{20}C_9$ (b) ${}^{20}C_{11}$
(c) ${}^{20}C_{10}$ (d) None of these

Q.18 The number of ways to distribute 32 different things equally among 4 persons is:
(a) $\frac{32!}{(8!)^3}$ (b) $\frac{32!}{(8!)^4}$
(c) $\frac{1}{4}(32!)$ (d) None of these

Q.19 In a dinner party, there are 10 Indians, 5 Americans, and 5 Englishmen. In how many ways can they be arranged in a row so that all persons of the same nationality sit together?
(a) $10! \times 5! \times 5!$ (b) 20!
(c) $3! \times 10! \times 5! \times 5!$ (d) 20! 3!

Q.20 In how many ways can 15 I.Sc. and 13 B.Sc. candidates be arranged in a line so that no two B.Sc. candidates may occupy consecutive positions?
(a) $15! \times 13!$ (b) $15! \times 16p_{12}$
(c) $13! \times {}^{16}P_{12}$ (d) $2! \times 15! \times 13!$

Q.21 How many new words can be formed from the letters of the word 'Circle' taken together?
(a) 720 (b) 719 (c) 360 (d) 359

Q.22 How many different words can be formed with the letters of the word 'Vice-Chancellor' so that the vowels are together?
(a) $10 \times 5!$ (b) $10! \times 5!$ (c) $5 \times 10!/2$ (d) $5 \times 10!$

Q.23 In how many ways can the letters of the word 'Lucknow' be arranged so that the two vowels do not come together?
(a) 720 (b) 1440
(c) 3600 (d) None of these

Q.24 How many words can be formed out of the letters of the word 'Article' so that the vowels occupy the even places?
(a) 72 (b) 144 (c) 288 (d) 36

Q.25 How many numbers greater than four millions (40,00,000) can be formed with the digits 2, 3, 0, 3, 4, 2, and 5?
(a) 280 (b) 380
(c) 360 (d) None of these

Q.26 How many signals can be made by hoisting 2 blue, 2 red, and 5 black flags on a pole at the same time?
(a) 378 (b) 1512
(c) 756 (d) None of these

Q.27 Find the number of arrangements of the letters of the word 'Maruti' if u always comes before a.
(a) 60 (b) 180 (c) 420 (d) 360

Q.28 In a class of students, there are 6 boys and 4 girls. In how many ways can they be seated, around a table so that all the 4 girls sit together?
(a) $6! \times 5!$ (b) $2 \times 6! \times 4!$
(c) $5! \times 4!$ (d) $6! \times 4!$

Q.29 A round table conference is to be held among 20 delegates of 20 countries. In how many ways can they and the host be seated if two particular delegates are always to sit on either side of the host?
(a) $19! \times 2!$ (b) 19!
(c) $18! \times 2!$ (d) $20! \times 2!$

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Q.30 Four gentlemen and four ladies are invited to a certain party. Find the number of ways of seating them around a table so that only ladies are seated on the two sides of each gentleman.
(a) 72 (b) 144 (c) 36 (d) 56

Q.31 Five men, 6 boys, and 7 women are to be seated in a row so that the men, women, and boys may not be separated in:
(a) $5! \times 6! \times 7!$ ways (b) 6!6!7! ways
(c) 2!5!6!7! ways (d) 3!5!6!7! ways

Q.32 In how many ways 12 different books can be distributed equally among 4 persons?
(a) 195 (b) 154
(c) 210 (d) None of these

Q.33 Eight chairs are numbered 1 to 8. Two women and three men wish to occupy one chair each. First two women choose the chairs from amongst the chairs marked 1 to 4, and then, the men selected the chairs from amongst the remaining. Find the number of possible arrangement.
(a) 720 (b) 1440
(c) 2110 (d) None of these

Q.34 How many numbers between 100 and 1000 can be formed with the digits, 1, 2, 3, 4, 5, 6, and 7, with no digit being repeated?
(a) 210 (b) 420 (c) 105 (d) 320

Q.35 How many different numbers greater that 5000 can be formed with the digits 0, 1, 5, and 9, with no digit being repeated?
(a) 18 (b) 12 (c) 24 (d) 6

Q.36 How many numbers less than 1000 and divisible by 5 can be formed in which no digit occurs more than once in the same number?
(a) 154 (b) 162 (c) 176 (d) 132

Q.37 Find the number of 4 digits formed with the digits 1, 2, 3, 4, and 5 in which 3 occurs in the thousands place and 5 occurs in the units place.
(a) 4 (b) 8 (c) 12 (d) 6

Q.38 Find the sum of all four-digit numbers that can be formed with the digits 3, 2, 3, and 4.
(a) 40,982 (b) 39,996
(c) 41,682 (d) None of these

Q.39 In how many ways can three prizes be given away to 5 students when each student is eligible for any of the prizes?
(a) 243 (b) ${}^5C_3$ (c) ${}^5P_3$ (d) 125

Q.40 A round table conference is to be held between 20 delegates. If two particular delegates are always to sit together, then they can be seated in:
(a) $2 \times 19!$ ways (b) 19! ways
(c) 20! ways (d) $2 \times 18!$ ways

Q.41 How many quadrilaterals can be formed joining the vertices of a polygon of n sides?
(a) ${}^nC_2 \times 2$ (b) ${}^nC_4$
(c) ${}^nC_4$ (d) None of these

Q.42 A delegation of 6 members is to be sent abroad out of the 12 members. In how many ways can the selection be made so that a particular member is included?
(a) 376 (b) 424 (c) 448 (d) 462

Q.43 The sides AB, BC, and CA of a triangle ABC have 3, 4, and 5 interior points, respectively, on them. Find the number of triangles that can be constructed using the given interior points as vertices.
(a) 210 (b) 195 (c) 205 (d) ${}^{12}C_3$

Q.44 Let S be the set of five-digit numbers formed by digits 1, 2, 3, 4, and 5, using each digit exactly once such that exactly two odd position are occupied by odd digits. What is the sum of the digits in the rightmost position of the numbers in S?
(a) 228 (b) 216 (c) 294 (d) 192

Q.45 Triplet x, y, and z are chosen from the set {1, 2, 3, ... 24, 25} such that $x \le y < z$. How many such triplets are possible?
(a) 25_{C_2} (b) 600
(c) $25_{C_2} + 25_{C_3}$ (d) 1200

Q.46 From 8 gentlemen and 4 ladies, a committee of 5 is to be formed. In how many ways can this be done so as to include at least one lady?
(a) 736 (b) 728 (c) 280 (d) 792

Q.47 Out of 7 men and 4 ladies, a committee of 5 is to be formed. In how many ways can this be done so as to include at least 3 ladies?
(a) 84 (b) 91 (c) 90 (d) 102

Q.48 A person has 12 friends of whom 8 are relatives. In how many ways can he invite 7 friends such that at least 5 of them may be relatives?
(a) 284 (b) 384 (c) 456 (d) 512

Q.49 A person wishes to throw as many different parties as he can out of his 20 friends such that each party consists of the same number of persons. The number of friends he should invite at a time is:
(a) 5 (b) 10
(c) 8 (d) None of these

Q.50 If all permutations of the letters of the word AGAIN are arranged as in dictionary, then the 15th word is:
(a) NAAGI (b) NAGAI
(c) NAAIG (d) NAIAG

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MODERATE

Q.1 The number of distinct rational numbers x such that $0 < x < 1$ and $x = \frac{p}{q}$, where $p, q \in (1, 2, 3, 4, 5, 6)$, is:
(a) 15 (b) 13 (c) 12 (d) 11

Q.2 The total number of nine-digit numbers of different digits is:
(a) 10 (9!) (b) 8 (9!)
(c) 9 (9!) (d) None of these

Q.3 How many three-digit numbers with distinct digits can be formed such that the product of the digits is the cube of a positive integer?
(a) 21 (b) 24 (c) 36 (d) 30

Q.4 Seven different lecturers are to deliver lectures in seven periods of a class on a particular day. A, B, and C are three of the lecturers. The number of ways in which a routine for the day can be made such that A delivers his lecture before B and B before C is:
(a) 420 (b) 120 (c) 210 (d) 840

Q.5 Let $A = \{x \mid x \text{ is a prime number and } x < 30\}$. The number of different rational numbers whose numerator and denominator belong to A is:
(a) 90 (b) 180
(c) 91 (d) None of these

Q.6 Find the number of non-negative integral solutions to the system of equations $x + y + z + u + t = 20$ and $x + y + z = 5$.
(a) 228 (b) 336 (c) 448 (d) 528

Q.7 Find the number of integral solution to $|x| + |y| + |z| = 15$.
(a) 720 (b) 728 (c) 734 (d) 904

Direction for Questions 8 and 9: Read the passage below and solve the questions based on it.

There are 6 boys and 4 girls sitting for a photo session. They were posing for the photograph standing in 2 rows one behind the other. There were 5 people sitting in the front row and 5 standing in the back row.

Q.8 If the boys were divided equally among the front and back rows, then in how many ways can the photo session be arranged?
(a) $5! \times 5!$
(b) ${}^6C_3 \times {}^4C_2 \times 5! \times 5!$
(c) ${}^6C_3 \times {}^4C_2 \times 10!$
(d) ${}^6C_3 \times {}^4C_2 \times 5! \times 5! \times 2!$

Q.9 In how many ways can the photos be taken such that no two boys and no two girls are standing or sitting together?
(a) ${}^6C_3 \times {}^4C_2 \times 10!$ (b) ${}^6C_3 \times {}^4C_2 \times (3!)2(2!)2$
(c) ${}^6C_3 \times {}^4C_2$ (d) None of these

Q.10 A teacher takes 3 children from her class to the zoo at a time as often as she can, but she does not take the same three children to the zoo more than once. She finds that she goes to the zoo 84 times more than a particular child goes to the zoo. The number of children in her class is:
(a) 12 (b) 10
(c) 60 (d) None of these

Q.11 ABCD is a convex quadrilateral. 3, 4, 5, and 6 points are marked on the sides AB, BC, CD, and DA, respectively. The number of triangles with vertices on different sides is:
(a) 270 (b) 220 (c) 282 (d) 342

Q.12 The number of triangles that can be formed with 10 points as vertices, n of them being collinear, is 110. Find the value of n.
(a) 3 (b) 4 (c) 5 (d) 6

Q.13 If x, y, and z are integers and $x \geq 0$, $y \geq 1$, $z \geq 2$, $x + y + z = 15$, then the number of values of the ordered triplet (x, y, z) is:
(a) 91 (b) 455
(c) ${}^{17}C_{15}$ (d) None of these

Q.14 If a, b, and c are positive integers such that $a + b + c \leq 8$, then the number of possible values of the ordered triplet (a, b, and c) is:
(a) 84 (b) 56
(c) 83 (d) None of these

Q.15 The product of r consecutive integers is necessarily divisible by:
(a) r (b) $\sum_{k=1}^{r-1} k$
(c) $(r + 1)!$ (d) None of these

Q.16 A candidate is required to answer 6 out of 10 questions which are divided into groups, each containing five questions and he is not permitted to attempt more than 4 from any group. In how many ways can he make up his choice?
(a) 184 (b) 192
(c) 200 (d) None of these

Q.17 In an examination, the question paper contains three different Sections A, B, and C containing 4, 5, and 6 questions, respectively. In how many ways, a candidate can make a selection of 7 questions, selecting at least two questions from each section?
(a) 890 (b) 1420 (c) 2700 (d) 3140

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Q.18 A tea party is arranged for 16 people along the two sides of a long table with 8 chairs on each side. Four men wish to sit on one particular side and two on the other side. In how many ways can they be seated?
(a) ${}^{10}C_4 \times 8!$ (b) ${}^{10}P_4 \times (8!)^2$
(c) ${}^{10}C_4 \times (8!)2$ (d) $4! \times 2! \times (8!)^2$

Q.19 A candidate is required to answer 7 questions out of 12 questions, which are divided into two groups, each containing 6 questions. He is not permitted to attempt more than 5 from either of the group. In how many different ways can he choose the seven questions?
(a) 640 (b) 780 (c) 920 (d) 1020

Q.20 The number of parallelograms that can be formed from a set of four parallel lines intersecting another set of three parallel lines is:
(a) 6 (b) 18 (c) 12 (d) 9

Q.21 We are required to form different words with the help of the word INTEGER. Let m_1 be the number of words in which *I* and *N* are never together and m_2 be the number of words which begin with *I* and end with *R*, then m_1/m_2 is equal to:
(a) 42 (b) 30 (c) 6 (d) 50

Q.22 In an election, there are five candidates and three members are to be elected, and an elector can vote for any number of candidates not greater than the number to be elected. Find the number of ways in which an elector may vote.
(a) 25 (b) 30
(c) 32 (d) None of these

Q.23 The greatest possible number of points of intersection of 8 straight lines and 4 circles is:
(a) 32 (b) 64 (c) 76 (d) 104

Q.24 A lady gives a dinner party to 5 guests to be selected from 9 friends. The number of ways of forming the party of 5, given that two particular friends A and B will not attend the party together is:
(a) 56 (b) 126
(c) 91 (d) None of these

Q.25 In a plane, there are 37 straight lines of which 13 pass through the point A and 11 pass through the point B. Besides, no three lines pass through one point, no line passes through both points A and B, no two are parallel. Find the number of intersection points the lines have.
(a) 535 (b) 601
(c) 728 (d) None of these

Q.26 If *a* denotes the number of permutations of $x+2$ things taking all at a time, *b* the number of permutations of *x* things taking 11 at a time, and *c* the number of permutations of $x-11$ things taking all at a time such that $a = 182bc$, then the value of *x* is:
(a) 15 (b) 12 (c) 10 (d) 18

Q.27 There are 4 letters and 4 directed envelopes. The number of ways in which all the letters can be put in the wrong envelope is:
(a) 8 (b) 9
(c) 16 (d) None of these

Q.28 The total number of natural numbers of six digits that can be made with digits 1, 2, 3, and 4 if all digits have to appear in the same number at least once is:
(a) 1560 (b) 840 (c) 1080 (d) 480

Q.29 In a club election, the number of contestants is one more than the number of maximum candidates for which a voter can vote for. If the total number of ways in which a voter can vote is 62, then what is the number of candidates?
(a) 7 (b) 5
(c) 6 (d) None of these

Q.30 Seven women and seven men are to sit round a circular table such that there is a man on either side of every women, then the number of seating arrangement is:
(a) $(7!)^2$ (b) $(6!)^2$ (c) 6! x 7! (d) 7!

Q.31 How many 10-digit numbers can be written by using the digits 1 and 2:
(a) ${}^{10}C_1 + {}^{9}C_2$ (b) 210
(c) ${}^{10}C_2$ (d) 10!

Q.32 All possible two factors products are formed from the numbers 1, 2, 3, 4, …, 200. The number of factors obtained out of the total which are multiples of 5 is:
(a) 5040 (b) 7180
(c) 8150 (d) None of these

Q.33 How many different flags can be made with 5 different stripes using them any number at a time?
(a) 3^5 (b) 5^3 (c) 385 (d) 625

Q.34 Three apples and two bananas have to be distributed among three boys—Anuj, Bairag, and Chandan, and two girls—Mugdha and Vaidehi, such that each person gets one fruit. In how many different ways can this be done if at least one girl gets an apple?
(a) 6 (b) 5
(c) 9 (d) None of these

Q.35 Find the distinct number of seven-digit numbers the sum of whose digits is even.
(a) 90×10^6 (b) 45×105
(c) 640000 (d) None of these

Q.36 Find the number of non-congruent rectangles that can be found on a chessboard normal 8×8 chessboard.
(a) 24 (b) 36
(c) 48 (d) None of these

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Q.37 A dinner menu is to be designed out of 5 different starters, 6 identical main courses, and 4 distinct desserts. In how many ways menu be designed such that there is at least one of each of the starters, main courses, and desserts?
(a) $31 \times 6 \times 15$ (b) $32 \times 6 \times 16$
(c) $31 \times 7 \times 15$ (d) $5 \times 6 \times 4$

Q.38 A man has 7 relatives. Four of them are ladies and 3 gentlemen; his wife also has 7 relatives. Three of them are ladies and 4 gentlemen. In how many ways can they invite 3 ladies and 3 gentlemen for a dinner party so that there are 3 of the man's relatives and 3 of the wife's relatives?
(a) 395 (b) 465 (c) 485 (d) 515

Q.39 Find the number of integral solutions of equation $x + y + z + t = 29$, $x > 0$, $y > 1$, $z > 2$ and $t \geq 0$.
(a) ${}^{27}C_3$ (b) ${}^{28}C_3$ (c) 2600 (d) ${}^{29}C_4$

Q.40 Find the number of numbers between 2×10^4 and 6×10^4 having sum of the digits even.
(a) 20,000 (b) 19,999 (c) 24,000 (d) 25,000

Q.41 Eight straight lines in a set are parallel to each other and the distance between any two adjacent lines is 1 cm. Another set of 6 straight lines are parallel to each other and the distance between any two adjacent lines is 1 cm. These 6 straight lines of second set intersect with the first set of 8 straight lines to form parallelograms. How many of such formed parallelograms will not be rhombuses?
(a) 285 (b) 365
(c) 335 (d) None of these

Q.42 Out of 10 guests, three are ladies, and they are to be seated in a row. The ladies insist on sitting together, whereas two of the gentlemen refuse to take consecutive seats. In how many ways can the guests be seated?
(a) 256 (b) 512
(c) 8!.3! – 7!.2! (d) 8!.3! – 7!.3!.2!

Q.43 Find the number of non-negative integral solutions of $2x + 2y + z = 10$.
(a) ${}^{12}C_2$ (b) ${}^{14}C_4$
(c) ${}^{11}C_2$ (d) None of these

Q.44 How many committees of 11 persons can be made out of 50 persons if 3 particular persons A, B, and C are not to be included together in the committee?
(a) ${}^{50}C_{11} - {}^{47}C_8$ (b) ${}^{50}C_{11} - {}^{47}C_9$
(c) ${}^{50}C_{11} \times {}^{47}C_8$ (d) None of these

Q.45 If each of the m points on the straight line be joined to each of the n points on the other straight line, then excluding the points on the given two lines, the number of points of intersection of these lines is:
(a) $\frac{1}{4} mn(m-1)(n-1)$
(b) ${}^{m+n}C_2$
(c) ${}^{mn}C_2$
(d) None of these

ADVANCED

Q.1 There is a regular decagon. Triangles are formed by joining the vertices of the polygon. What is the number of triangles that have no side common with any of the sides of the polygon?
(a) 50 (b) 300 (c) 44 (d) 294

Q.2 There are 12 intermediate stations between two places A and B. In how many ways can a train be made to stop at 4 of these 12 intermediate stations that no two stations are consecutive?
(a) ${}^{15}C_3$ (b) ${}^{11}C_3$ (c) ${}^{9}C_4$ (d) ${}^{9}C_3$

Q.3 A conference attended by 200 delegates is held in a hall. The hall has 7 doors, marked A, B, … G. At each door, an entry book is kept and the delegates entering through that door sign it in the order in which they enter. If each delegate is free to enter any time and through any door he likes, then how many different sets of 7 lists would arise in all? (Assume that every person signs only at his first entry.)
(a) ${}^{206}C_6$ (b) ${}^{199}P_5$ (c) ${}^{199}C_5$ (d) ${}^{206}P_6$

Q.4 In an election for the post of president of students union, every student can vote in 56 ways. Every student votes for at least one candidate and a student can vote for more than one candidate. If the total number of candidates exceeds the maximum number that can be elected by 2, then what is the number of candidates?
(a) 5 (b) 7
(c) 6 (d) None of these

Q.5 The number of positive integral solution of $abc = 30$ is:
(a) 27 (b) 81
(c) 243 (d) None of these

Q.6 A train going from Cambridge to London stops at 9 intermediate stations. Six persons enter the train during the journey with 6 different tickets of the same class. How many different sets of tickets would they have?
(a) ${}^{11}C_5$ (b) ${}^{35}C_6$
(c) ${}^{45}C_6$ (d) None of these

Q.7 Five balls of different colours are to be placed in 3 boxes of different sizes. Each box can hold all 5 balls. In how

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many different ways can we place the balls so that no box remains empty?

(a) 119 (b) 150 (c) 180 (d) 210

Q.8 An eight-oared boat is to be manned by a crew chosen from 11 men of whom 3 can steer but cannot row and the rest cannot steer. In how many ways can the crew be arranged if two of the men can only row on bow side?

(a) ${}^3C_1.{}^4C_2.6!$ (b) ${}^3P_1.{}^4P_2$

(c) ${}^3P_1.{}^4P_2.6!$ (d) ${}^3C_1.{}^4C_2$

Q.9 Find the number of n-digit number formed with the digits 1, 2, 3, …, 9 in which no two consecutive digits are the same.

(a) 9^{n-1} (b) 9.8^{n-1} (c) 9.8^n (d) 9^n

Q.10 If the number of ways in which n different things can be distributed among n persons so that at least one person does not get any thing is 232, then what is the value of n?

(a) 3 (b) 4

(c) 5 (d) None of these

Q.11 If m be the greatest value of k for which 100! is divisible by 3^k, then the number of ways of distributing k different things among 5 persons so that two of them get 10 things each, two get 12 things each, and the fifth one gets 4 things is:

(a) $\frac{48!}{(101)^2(121)^2 41}$ (b) $\frac{48!}{(10!)^2(12!)^2}$

(c) $\frac{48!}{(10!)^2(12!)^2 4}$ (d) $\frac{48!}{(10!)^2(12!)^2}\cdot\frac{5}{4}$

Q.12 A bag contains n white and n red balls. Pairs of balls are drawn without replacement until the bag is empty. If the number of ways in which each pair consists of one red and one white ball is 14,400, then what is the value of n?

(a) 5 (b) 10 (c) 15 (d) 20

Direction for Questions 13 to 15: Read passage below and solve the questions based on it.

In a horticultural show, a participant arranged a total of 1000 fruits of four types in a long straight line. He first placed 1 apple, then 2 mangoes, 3 oranges, 4 custard apples, 5 apples, 6 mangoes, 7 oranges, 8 custard apples, and so on (with apples followed by mangoes, oranges, custard apples in that order) upto the 1000th fruit.

Q.13 What was the 1000th fruit?

(a) Apple (b) Mango

(c) Orange (d) Custard apple

Q.14 What was the position of the 100th mango?

(a) 387 (b) 437 (c) 417 (d) 363

Q.15 Among the last 700 fruits, which type of fruit occurred the greatest number of times?

(a) Apple (b) Mango

(c) Orange (d) Custard apple

Q.16 There is a framework of a cuboid of length 6, breadth 5, and height 7 units. The cuboid is only composed of a skeleton of 210 cubes of side 1 units. An insect is on one corner of the cube and it wants to travel to the opposite end of the longest diagonal. It can only travel along the sides of the small cubes and it always takes the shortest possible route.

Find the number of choices the insect has.

(a) $\frac{18!}{7!6!5!}$ (b) $\frac{15!}{6!5!4!}$ (c) $\frac{17!}{7!6!5!}$ (d) $\frac{16!}{6!5!4!}$

Q.17 In the above question, if the insect suddenly realizes that one of the faces on the opposite side of the cuboid having maximum area has been sprayed with pesticides due to which it cannot reach the original destination and if the insect still wants to reach the opposite end of longest diagonal now possible, then in how many ways can it do so?

(a) $\frac{16!}{7!6!5!}$ (b) $\frac{18!}{6!5!4!}$ (c) $\frac{15!}{6!6!5!}$ (d) $\frac{16!}{6!5!5!}$

Direction for Questions 18 and 19: Read the passage below and solve the questions based on it.

James Bond has to crack a password of 3 symbols. The symbols can be +, –, or ×. Every time he makes an attempt to crack the password, a feedback is given in the form of 'R' and 'W'. 'R' indicates that one of the symbols is in the right place. 'W' indicates that one of the digits is in the wrong position. All the codes are used.

This is his first two attempts with their respective feedbacks:

Attempt 1			Feedback
+	+	×	R R W
Attempt 2			**Feedback**
–	+	×	R W W

Q.18 What is the minimum number of additional attempts required to crack the code?

(a) 0 (b) 1 (c) 2 (d) 3

Q.19 What is the right code?

(a) ×–+ (b) +–× (c) ++– (d) –+×

Q.20 At the beginning of a party, each person present shook hands with all other people present, and in total, there were 28 handshakes. In the midst of the party, 2 persons left due to an emergency. Now, the number of males and females present in the party was equal. At the end, each

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female shook hands only with every female present and each male shook hands only with every male present. What is the total number of handshakes that took place at the party?
(a) 52 (b) 40 (c) 34 (d) 36

Q.21 Consider $S = (1, 2, 3, \ldots, 10)$. In how many ways two numbers from S can be selected so that the sum of the numbers selected is a double-digit number?
(a) 36 (b) 16
(c) 29 (d) ${}^{9}C_{2} - {}^{5}C_{2}$

Q.22 In a chess tournament, every person played one game with every other person in the group. The total number of games that men played between themselves exceeded those played by men with women by 18. If there were 4 women in the tournament, then in total, how many games were played in the tournament?
(a) 114 (b) 66 (c) 78 (d) 120

Q.23 The number of ways of painting the faces of a cube with six different colour is:
(a) 1 (b) 6
(c) 6! (d) None of these

Q.24 Find the number of non-negative integer solutions to the system of equations $a + b + c + d + e = 20$ and $a + b + c = 5$.
(a) 240 (b) 336 (c) 672 (d) 1008

Q.25 If the number of ways of selecting k coupons out of an unlimited number of coupons bearing the letters A, T, and C so that they cannot be used to spell the word CAT is 93, then what is the value of k?
(a) 2 (b) 5
(c) 8 (d) None of these

Q.26 In a test of 10 multiple choice questions of one correct answer, each having 4 alternative answers, then the number of ways to put ticks at random for the answers to all the questions is:
(a) 4^{10} (b) 10^{4} (c) $4^{10} - 4$ (d) 10^{4}–10

Q.27 The streets of a city are arranged like the lines of a chessboard. There are m streets running north-south and n streets running in east-west direction. What is the number of ways in which a man can travel on these streets from the north-west to the south-east corner, going by the shortest possible distance?
(a) $(m + n - 2)!$
(b) $(m + n - 1)!/ [(m - 1)!]$
(c) $(m + n - 1)!/[(n - 1)!]$
(d) $(m + n - 2)!/[(m - 1!)(n - 1)!]$

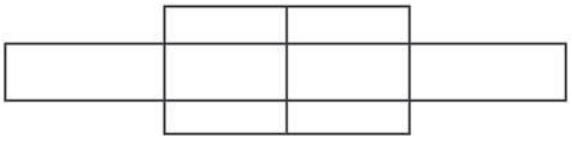
Q.28 Let $(1 + x + x^2)^9 = \alpha_0 + \alpha_1 x + \ldots + \alpha_{18}x^{18}$, then which of the following is true?
(a) $\alpha_0 + \alpha_2 + \ldots + \alpha_{18} = \alpha_1 + \alpha_3 + \ldots + \alpha_{17}$.
(b) $\alpha_0 + \alpha_2 + \ldots + \alpha_{18}$ is even.
(c) $\alpha_0 + \alpha_2 + \ldots + \alpha_{18}$ is divisible by 9.
(d) $\alpha_0 + \alpha_2 + \ldots + \alpha_{18}$ is divisible by 3 but not by 9.

Q.29 If N be an element of the set A = {1, 2, 3, 5, 6, 10,15, 30}, and P, Q, and R are integers such that PQR = N, then the number of positive integral solutions of PQR = N is:
(a) 32 (b) 64 (c) 96 (d) 128

Q.30 In how many ways can one arrange letters the in the word 'INSTITUTION' such that no two same letters comes together along with the following conditions:
(A) There is no 'T' which is immediately preceded as well as followed by 'N'.
(B) There is no 'T' which is immediately preceded as well as followed by 'T'.
(a) 32,256 (b) 40,320
(c) 60,480 (d) None of these

Q.31 How many positive integers less than 1000 are 6 times the sum of their digits?
(a) 0 (b) 2
(c) 1 (d) 4

Q.32 Six X's have to be placed in the squares of the figure such that each row contains at least one X.



In how many different ways can this be done?
(a) 28 (b) 27
(c) 26 (d) None of these

Q.33 Number of ways in which n distinct things can be distributed among n persons so that at least one person does not get anything is 232. Find n.
(a) 3 (b) 4 (c) 5 (d) 6

Q.34 Six persons A, B, C, D, E, and F are to be seated at a circular table. In how many ways can this be done if A must always have either B or C on his immediate right and B must always have either C or D on his immediate right?
(a) 16 (b) 24 (c) 30 (d) 18

Q.35 The number of employees in a nationalized bank in a small town is 10, out of which 4 are female and the rest are male. A committee of 5 is to be formed. If m be the number of ways to form such a committee in which there is at least one female employee and n be the number of ways to form such a committee which includes at least 2 male employees, then find the ratio m:n.
(a) 3:2 (b) 5:1 (c) 1:1 (d) 0.8:0.9

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Q.36 A question paper is split into two parts—Part A and Part B. Part A contains 5 questions and part B has 4 questions. Each question in Part A has an alternative. A student has to attempt at least one question from each part. Find the number of ways in which the student can attempt the question paper?
(a) 1830 (b) 2440 (c) 3630 (d) 4260

Q.37 How many integers between 1 and 100,000 have the sum of their digits equal to 18?
(a) 8993 (b) 25,927 (c) 36,592 (d) 24,987

Q.38 A person invites his 10 friends for a party and places them 5 at one round table, and 5 on the other round table. Find the number of ways in each case in which he can arrange the guests.
(a) ${}^{10}P_5/25$ (b) ${}^{10}P_5 \times {}^{5}P_5/25$
(c) 10!/25 (d) None of these

Q.39 In the above question, find the number of ways in which he can arrange 4 at one round table and 6 at the other round table.
(a) ${}^{9}P_5/24$ (b) ${}^{9}P_5 \times {}^{5}P_5/24$
(c) 10!/24 (d) None of these

Q.40 In how many ways 4 men and 4 women can be seated in a row so that men and women are alternate?
(a) 28 (b) 36 (c) 4!.4! (d) 2.4!.4!

Answers

Warm Up

1. (c)	**2.** (b)	**3.** (a)	**4.** (a)	**5.** (a)	**6.** (a)	**7.** (d)	**8.** (b)	**9.** (a)	**10.** (d)
11. (c)	**12.** (d)	**13.** (b)	**14.** (b)	**15.** (a)	**16.** (a)	**17.** (a)	**18.** (c)	**19.** (b)	**20.** (b)

Foundation

1. (c)	**2.** (a)	**3.** (c)	**4.** (c)	**5.** (d)	**6.** (a)	**7.** (b)	**8.** (c)	**9.** (c)	**10.** (a)
11. (a)	**12.** (a)	**13.** (c)	**14.** (a)	**15.** (a)	**16.** (c)	**17.** (c)	**18.** (b)	**19.** (c)	**20.** (d)
21. (d)	**22.** (d)	**23.** (c)	**24.** (b)	**25.** (c)	**26.** (c)	**27.** (d)	**28.** (d)	**29.** (c)	**30.** (b)
31. (d)	**32.** (d)	**33.** (b)	**34.** (a)	**35.** (b)	**36.** (a)	**37.** (d)	**38.** (b)	**39.** (d)	**40.** (d)
41. (b)	**42.** (d)	**43.** (c)	**44.** (b)	**45.** (b)	**46.** (a)	**47.** (b)	**48.** (c)	**49.** (b)	**50.** (c)

Moderate

1. (d)	**2.** (c)	**3.** (d)	**4.** (d)	**5.** (c)	**6.** (b)	**7.** (a)	**8.** (b)	**9.** (b)	**10.** (b)
11. (d)	**12.** (c)	**13.** (a)	**14.** (b)	**15.** (a)	**16.** (c)	**17.** (c)	**18.** (c)	**19.** (b)	**20.** (b)
21. (b)	**22.** (c)	**23.** (c)	**24.** (d)	**25.** (a)	**26.** (b)	**27.** (b)	**28.** (a)	**29.** (c)	**30.** (c)
31. (b)	**32.** (b)	**33.** (c)	**34.** (c)	**35.** (b)	**36.** (b)	**37.** (a)	**38.** (c)	**39.** (c)	**40.** (a)
41. (c)	**42.** (d)	**43.** (a)	**44.** (a)	**45.** (a)					

Advanced

1. (a)	**2.** (c)	**3.** (d)	**4.** (c)	**5.** (c)	**6.** (c)	**7.** (b)	**8.** (c)	**9.** (b)	**10.** (b)
11. (d)	**12.** (a)	**13.** (a)	**14.** (b)	**15.** (d)	**16.** (a)	**17.** (d)	**18.** (a)	**19.** (b)	**20.** (c)
21. (c)	**22.** (d)	**23.** (a)	**24.** (b)	**25.** (b)	**26.** (a)	**27.** (d)	**28.** (b)	**29.** (b)	**30.** (c)
31. (c)	**32.** (c)	**33.** (b)	**34.** (d)	**35.** (c)	**36.** (c)	**37.** (b)	**38.** (c)	**39.** (c)	**40.** (d)

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Hints and Solutions

Moderate

1. Method 1

In this case, $0 < x < 1$, and so, we must have $p < q$. Therefore, we can find the total such cases by first fixing a value for q from 2 to 6. For each value of q, p has values from 1 to $q - 1$. So, the possible pairs for (p, q) are totally 15, which are (1,2), (1,3), (2,3), (1,4), (2,4), (3,4), (1,5), (2,5), (3,5), (4,5), (1,6), (2,6), (3,6), (4,6), and (5,6).

But, $x = 1/2$ for $(p, q) = (1, 2)$, $(2, 4)$, and $(3, 6)$ and

$x = 1/3$ for $(p, q) = (1, 3)$, and $(2, 6)$ and

$x = 2/3$ for (p, q) = (2, 3)and (4, 6)

So, we have 11 distinct values of x as 4 values are repeating as seen above.

Hence, the correct option is (d).

Method 2

Total sets possible = ${}^6C_2 = 15$

But, $1/2 = 2/4 = 3/6$

$1/3 = 2/6$

$2/3 = 4/6$

So, distinct values of $x = 11$

2. It will be a number of 9 digits with the following box diagram:

$9 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 = 9 \times 9!$

3. We first need to identify the cube of positive integers that can be formed by the product of 3 distinct digits.

$1^3 = 1$—It cannot be expressed as product of 3 distinct numbers.

$2^3 = 8$—It can be expressed as product of 3 distinct numbers, which are 1, 2, and $4(2^2)$.

$3^3 = 27$—It can be expressed as product of 3 distinct numbers, which are 1, 3, and $9(3^2)$.

$4^3 = 64$—It can be expressed as product of 3 distinct numbers, which are 2, $4(2^2)$, and $8(2^3)$.

$5^3 = 125$—It cannot be expressed as product of 3 distinct numbers.

$6^3 = 216$—It can be expressed as product of 3 distinct numbers, which are $4(2^2)$, $6(2 \times 3)$, and $9(3^2)$.

It can also be expressed as product of 3 distinct numbers, which are 3, $8(2^3)$, and $9(3^2)$.

$7^3 = 343$—It cannot be expressed as product of 3 distinct numbers.

$8^3 = 512$—It can't be expressed as product of 3 distinct numbers.

$9^3 = 729$—It cannot be expressed as product of 3 distinct numbers.

So, we have 5 cases where we have to find possible permutations of 3 numbers taken 3 at a time.

Total possible cases = $5 \times {}^3P_3 = 5 \times 3!/0! = 5 \times 6 = 30$

Hence, the correct option is (d).

4. Method 1

Without any restriction, 7 can be arranged in 7! Ways.

Now, keeping 4 of them fixed, A, B, and C can be arranged in 3! ways, that is, out of every 3! arrangement, we have one choice for which A–B–C order is followed.

So, [7!]/[3!] is the answer.

Note: It can be seen that if only two are fixed, then the answer is [7!]/[2!].

- For A–B–C (or any three) = [7!]/[3!]
- For A–B–C–D (or any four) = [7!]/[4!]
- For A–B–C–D–E (or any five) = [7!]/[5!]
- For A–B–C–D–E–F (or any six) = [7!]/[6!]
- For A–B–C–D–E–E–F = [7!]/[7!]

Method 2

In this case, we have to arrange only 4 positions from the given 7 positions and the remaining 3 will be filled so that A is before B, and they are before C. So, our answer is ${}^7P_4 = 7!/(7 - 4)! = 7!/3! = 7 \times 6 \times 5 \times 4 = 840$

Hence, the correct option is (d).

Alternative Solution: We can look this situation in a different way too. We can arrange 7 lectures and then divide it by those ways in which the 3 lectures A, B, and C are arranged with respect to each other. As there is only 1 possible order between them, the total number of possible ways to made routine for the day = $7!/3! = 7 \times 6 \times 5 \times 4 = 840$

Hence, the correct option is (d).

5. The possible values of A are the prime numbers below 30, which are 2, 3, 5, 7, 11, 13, 17, 19, 23, and 29.

So, we have 10 distinct values for A. Each rational number can have any of these as numerator and denominator.

Different rational number whose numerator and denominator belong to A = rational number with distinct numerator and denominator + rational number with same numerator and denominator.

Rational number with distinct numerator and denominator is ${}^{10}C_2 = 90$.

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Now, rational number with same numerator and denominator is 1 2/2 = 3/3 = 5/5….upto 29/29, but the value of each of them is 1).

So, the total number of such rational numbers = 90 + 1 = 91

Hence, the correct option is (c).

6. Given $x + y + z + u + t = 20$

$x + y + z = 5$

Given, system of equations is equivalent to

$x + y + z = 5$ (i)

and $u + t = 15$ (ii)

Number of non-negative integral solutions of equation (i)

$= {}^{n+r-1}C_r = {}^{3+1-1}C_5 = {}^{7}C_5$

Number of non-negative integral solution of equation (ii)

$= {}^{n+r-1}C_r = {}^{2+15-1}C_{15} = {}^{16}C_{15}$

$\therefore$ Required number $= {}^{7}C_5 . {}^{16}C_{15} = 336$

7. The number of non-negative integral solutions is ${}^{n+r-1}C_{r-1}$ and the number of positive integral solutions is ${}^{n-1}C_{r-1}$.

Here, $n = 15$ and $r = 3$

So, the number of non-negative integral solutions is ${}^{15+3-1}C_{3-1} = {}^{17}C_2 = 136$. The number of non-negative solutions (which includes 0 too) is ${}^{15-1}C_{r-1} = {}^{14}C_2 = 91$. So, the number of solutions that includes 0 too (1 or 2 zeros) = 136 − 91 = 45. Now, there cannot be a solution with 3 zeros, and so, we have 3 cases of 2 zeros and 42 cases of 1 zero only.

Now, as we have modulus function |x| + |y| + |z|, and now, for all positive integral solutions, we have 91 × 6 = 546. All of x, y, and z could have both positive and negative values.

For, 42 cases of 1 zero, we have 42 × 4 = 168 possible solutions where both integers can have positive and negative values.

Now, for 3 cases of 2 zeros, we have 3 × 2 = 6 possible solutions. Here, each integer in solution has 2 possible values (1 positive and 1 negative).

So, total number of possible solutions = 546 + 168 + 6 = 720 possible solutions

8. In first row, we have 5 positions that can be filled in 5! ways. Now, we must select 3 boys (from total 6) and 2 girls (from the total 4) in first row. So, total number of ways the seating can be done for first row is ${}^{6}C_3 \times {}^{4}C_2 \times 5!$

Now, for row 2, we again have 5 positions to be filled in 5! ways. Now, in second row, we have 3 boys (from remaining 3, as there were total 6 and 3 are already seated in first row) and 2 girls (from remaining 2, as there were total 4 and 2 are already seated in second row). So, number of ways the seating for second row can be done is ${}^{3}C_3 \times {}^{2}C_2 \times 5!$

So, total number of ways seating in first and second rows is done in $({}^{6}C_3 \times {}^{4}C_2 \times 5!) \times ({}^{3}C_3 \times {}^{2}C_2 \times 5!)$ ways.

But, ${}^{3}C_3 = {}^{2}C_2 = 1$. So, total number of ways $= {}^{6}C_3 \times {}^{4}C_2 \times 5! \times 5!$

Hence, the correct option is (b).

9. No two boys and no two girls would be standing or sitting together only if in both the rows the boys are at 1st, 3rd, and 5th positions and girls are at 2nd and 4th positions. So, for each row, instead of arranging 5 positions together, we must arrange 3 and 2 positions separately. The number of ways for it is 3! × 2! Now, in 1st row, we have 3 boys from 6 who can be selected in ${}^{6}C_3$ ways and 2 boys from 4 boys can be selected in ${}^{4}C_2$ ways. So, total number of ways 5 students in 1st row can be arranged as ${}^{6}C_3 \times {}^{4}C_2 \times 3! \times 2!$ Now, in 2nd row, we can arrange 3 and 2 positions like row 1 in 3! × 2! ways. Now, we have to select 3 boys from 3 remaining boys, and similarly, we have to select 2 girls from remaining 2. So, the total number of ways in which seating of 2nd row can be done is ${}^{3}C_3 \times {}^{2}C_2 \times 3! \times 2!$ Now, the total number of ways in which seating of 1st and 2nd row can be done is

$({}^{6}C_3 \times {}^{4}C_2 \times 3! \times 2!) \times ({}^{3}C_3 \times {}^{2}C_2 \times 3! \times 2!)$.

But, ${}^{3}C_3 = {}^{2}C_2 = 1$. So, the total number of possible ways $= {}^{6}C_3 \times {}^{4}C_2 \times (3!)^2 \times (2!)^2$

Hence, the correct option is (b).

10. ${}^{n}C_3 - {}^{n-1}C_2 = 84$, where n is the number of students.

Now, use the options.

11. The triangles must be formed by the 1 marked point, each on any 3 consecutive sides, for examples, AB, BC, and CD (and so on). Points A, B, C, and D cannot be included in this case as we cannot form triangle using any of these points. So, total number of triangles that can be formed = 3 × 4 × 5 + 4 × 5 × 6 + 5 × 6 × 3 + 6 × 3 × 4 = 60 + 120 + 90 + 72 = 342

Hence, the correct option is (d).

12. According to the question, ${}^{10}C_3 - {}^{n}C_3 = 110$. Using options, we get $n = 5$.

13. This question can be rewritten as $x \geq 0, y \geq 0, z \geq 0, x + y + z = 12$. So, we have already given 1 to y and 2 to z. Now, number of non-negative solutions is ${}^{n+r-1}C_{r-1}$, where $n = 12$ and $r = 3$. So, number of possible triplets is ${}^{12+3-1}C_{3-1} = {}^{14}C_2 = 91$

Hence, the correct option is (a).

14. As the minimum value of a, b, and c is 1, the minimum value of $a + b + c = n$, which is 3. Here, $r = 3$, and n could have values 3, 4, 5, 6, 7, and 8.

As we know positive integral value of $a + b + c = n$, which is ${}^{n-1}C_{r-1}$. Now, we have $r = 2$ and n varies from 3 to 8. So, total number of possible solutions $= {}^{3-1}C_{3-1} + {}^{4-1}C_{3-1} + {}^{5-1}C_{3-1} + {}^{6-1}C_{3-1} + {}^{7-1}C_{3-1} + {}^{8-1}C_{3-1}$

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$= {}^2C_2 + {}^3C_2 + {}^4C_2 + {}^5C_2 + {}^6C_2 + {}^7C_2 = 1 + 3 + 6 + 10 + 15 + 21 = 56$

Hence, the correct option is (b).

16. We have 3 options that satisfy the required conditions, which are as follows:

1. Select 2 questions from 1st group and 4 questions from 2nd group.
2. Select 3 questions from 1st group and 3 questions from 2nd group.
3. Select 4 questions from 1st group and 2 questions from 2nd group.

The total number of ways it can be done $= {}^5C_2 \times {}^5C_4 + {}^5C_3 \times {}^5C_3 + {}^5C_4 \times {}^5C_2 = 2 \times {}^5C_2 \times {}^5C_4 + {}^5C_3 \times {}^5C_3$

$= 2 \times (5 \times 4)/(2 \times 1) \times 5 + (5 \times 4)/(2 \times 1) \times (5 \times 4)/(2 \times 1)$

$= 2 \times 10 \times 5 + 10 \times 10 = 100 + 100 = 200$

Hence, the correct option is (c).

17. There are 3 possible ways:

1. Selecting 3 question from 1st section and 2 questions each from 2nd and 3rd sections.
2. Selecting 3 question from 2nd section and 2 questions each from 1st and 3rd sections.
3. Selecting 3 question from 3rd section and 2 questions each from 1st and 2nd sections.

Total number of ways it can be done $= {}^4C_3 \times {}^5C_2 \times {}^6C_2 + {}^4C_2 \times {}^5C_3 \times {}^6C_2 + {}^4C_2 \times {}^5C_2 \times {}^6C_3 = 4 \times (5 \times 4)/(2 \times 1) \times (6 \times 5)/(2 \times 1) + (4 \times 3)/(2 \times 1) \times (5 \times 4)/(2 \times 1) \times (6 \times 5)/(2 \times 1) + (4 \times 3)/(2 \times 1) \times (5 \times 4)/(2 \times 1) \times (6 \times 5 \times 4)/(3 \times 2 \times 1)$

$4 \times 10 \times 15 + 6 \times 10 \times 15 + 6 \times 10 \times 20 = 600 + 900 + 1200 = 2700$

Hence, the correct option is (c).

18. Four persons have chosen to sit on one particular side (assume side A) and 2 persons on the other side (assume side B). So, we are supposed to select 4 persons for side A from the remaining 10 persons and remaining 6 persons will be sitting on side B.

Number of ways 4 persons can be selected from 10 persons $= {}^{10}C_4$

Number of ways 6 persons can be selected from the remaining 6 persons $= {}^6C_6$

Number of ways 8 persons can be arranged on side A = 8!

Number of ways 8 persons can be arranged on side B = 8!

Total number of ways $= {}^{10}C_4 \times {}^6C_6 \times 8! \times 8!$

19. These are the possible cases:

1. Two questions from 1st group and 5 questions from 2nd group
2. Three questions from 1st group and 4 questions from 2nd group
3. Four questions from 1st group and 3 questions from 2nd group
4. Five questions from 1st group and 2 questions from 2nd group

So, total number of possible cases $= {}^6C_2 \times {}^6C_5 + {}^6C_3 \times {}^6C_4 + {}^6C_4 \times {}^6C_3 + {}^6C_5 \times {}^6C_2 = 2({}^6C_2 \times {}^6C_5 + {}^6C_3 \times {}^6C_4)$

$= 2(90 + 300) = 2(390) = 780$

Hence, the correct option is (b).

20. This is nothing but ${}^4C_2 \times {}^3C_2 = 18$.

21. m_1 can be calculated by subtracting number of cases in which I and N are together from total number of possible combinations.

So, $m_1 = 7!/2! - (6!/2!) \times 2! = 7!/2! - 6!$

For m_2, we are left to arrange only 5 letters in which E is repeated twice as position of I and N are fixed.

So, $m_2 = 5!/2!$

Hence, $m_1/m_2 = (7!/2! - 6!)/(5!/2!) = (7! - 6!2!)/5! = 7 \times 6 - 6 \times 2 = 42 - 12 = 30$

Hence, the correct option is (b).

22. There are total 5 candidates and 3 members to be elected. Now, according to the given condition, an elector can vote for 1 candidate or 2 candidates or 3 candidates. Hence, the total number of ways in which an elector can vote $= {}^5C_1 + {}^5C_2 + {}^5C_3 = 5 + 10 + 10 = 25$

Hence, the correct option is (a).

23. Maximum total number of points of intersection among n straight lines $= {}^nC_2$

Maximum total number of points of intersection among n circles $= {}^nP_2$

Using this:

Maximum total number of points of intersection among 8 straight lines $= {}^8C_2 = 28$

Maximum total number of points of intersection among 4 circles $= {}^4P_2 = 12$

One straight line will intersect one circle at two points. Hence, maximum total number of points of intersection where 8 straight lines will cut 4 circles $= 2 \times 8 \times 4 = 64$

So, maximum total number of points of intersection $= 28 + 12 + 64 = 104$

24. **Method 1**

We can find the answer by subtracting the number of ways when both of the friends who cannot attend together are together (assuming) from total number of ways in which 5 can be selected from 9 friends.

The number of ways $= {}^9C_5 - {}^7C_3 = 9!/(5!4!) - 7!/(3!4!) = 126 - 35 = 91$

Hence, the correct option is (c).

Method 2

This can be done in three ways:

1. A is selected but B is not selected—A has already been selected, rest 4 is to be selected from 7 persons $= {}^7C_4 = 35$

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2. B is selected but A is not selected = ${}^{7}C_{4} = 35$

3. Neither A nor B is selected = ${}^{7}C_{5} = 21$

Total = 91

25. Number of intersection points between lines passing through point A and point B = 13 × 11 = 143

Out of 37 points, 13 pass through point A and 11 pass through point B. Hence, total number of points that does not pass through either A or B = 13 (i)

Points of intersection among these 13 points passing through point A = 13 × 13 = 169

Points of intersection among these 13 points and 11 points passing through point A = 13 × 11 = 143

Besides, point A and point B are two points.

Hence, total number of points of intersection = 143 + 169 + 143 + 2 = 457

Besides, these 13 points [as calculated in (i)] will intersect among themselves in ${}^{13}C_{2} = 78$ ways.

So, total number of points = 457 + 78 = 535

Hence, the correct option is (a).

26. $a = (x + 2)!$, $b = x!/(x - 11)!$, and $c = (x - 11)!$

As $a = 182bc = 182 \times x!/(x - 11)! \times (x - 11)! = 182x!$

So, $(x + 2)! = 182x!$

$(x + 2)(x + 1)x! = 182x!$

$(x + 2)(x + 1) = 182$

Solving this quadratic equation, we get $x = 12$

Hence, the correct option is (b).

27. It is the simple case of derangement with $n = 4$.

Use the following formula:

$$n!\left(1-\frac{1}{1!}+\frac{1}{2!}-\frac{1}{3!}+\ldots+(-1)^{n}\frac{1}{n!}\right)$$

Hence, answer = 4! (1 − 1 + ½ − 1/6 + 1/24) = 9

29. Here, number of ways an elector can vote = ${}^{n}C_{1} + {}^{n}C_{2} + \ldots + {}^{n}C_{n-1}$, where n is the number of candidates.

This question can be solved by using all the options.

Let $n = 7$, then number of possible ways = ${}^{7}C_{1} + {}^{7}C_{2} + {}^{7}C_{3} + {}^{7}C_{4} + {}^{7}C_{5} + {}^{7}C_{6} = 7 + 21 + 35 + 35 + 21 + 7 = 126$

Let $n = 7$, then number of possible ways = ${}^{5}C_{1} + {}^{5}C_{2} + {}^{5}C_{3} + {}^{5}C_{4} = 5 + 10 + 10 + 5 = 30$

Let $n = 6$, then number of possible ways = ${}^{6}C_{1} + {}^{6}C_{2} + {}^{6}C_{3} + {}^{6}C_{4} + {}^{6}C_{5} = 6 + 15 + 20 + 15 + 6 = 62$

Hence, the correct option is (c).

30. We have total of 14 positions in the circle and we have to 1st give 7 alternate positions to 7 women, which can be seen like we are giving 7 positions in a circle to 7 women and it can be done in (7 − 1)! = 6!

Now, we have 7 positions for 7 men, and this time it is no longer a circular arrangement. But, it is like a linear arrangement, and so, it can be done in 7! ways.

So, total number of ways = 6! × 7!

Hence, the correct option is (c).

31. Following is the answer:

$2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 2^{10}$

32. There are 40 numbers between 1 to 200 that are multiples of 5 (i.e., 5, 10, 15, …, 200).

Now, the 2 factor products that are multiples of 5 can be of 2 types, which are as follows:

1. Products of any 1 of these 40 (multiples of 5) and other 160 numbers. Such factors are 40 × 160 = 6400
2. Products of 2 numbers that are both multiples of 5. Such factors are ${}^{40}C_{2} = 780$

So, total such factors that are multiples of 5 = 6400 + 780 = 7180

Hence, the correct option is (b).

33. **Case I** When a boy sits at the first place:

Possible arrangement will be of the below form.

B	G	B	G	B	G	B	G

Now, there are four places, namely 1st, 3rd, 5th, and 7th for four boys; therefore, four boys can be seated in $\lfloor 4$ ways. Again, there are four places, namely 2nd, 4th, 6th, and 8th for four girls.

Four girls can be seated in $\lfloor 4$ ways.

Number of ways in this case = $\lfloor 4 \times \lfloor 4$

Similarly, G B G B G B G B

So, total = 2 × 4! × 4!

34. We can select a group of any 3 children to distribute apples as ${}^{5}C_{3} = 10$ ways. Now, if none of the girls get an apple, then all 3 boys get apples and both girls get bananas. So, it has 1 way only. So, number of ways that at least 1 girl gets apple = total number of ways − number of ways in which no girl gets apples = 10 − 1 = 9 ways

Hence, the correct option is (c).

35. As we know that the sum of digits for any 2 consecutive numbers will be odd and even, respectively (one of the properties of numbers), students are advised to try it for any numbers of consecutive numbers to check the sum of digits is odd and even alternatively for them. We know that total number of 7-digit numbers are 9×10^{6} (starting from 10,00,000 and ending on 99,99,999). Now, from these numbers, half numbers must be having even number as sum of digits. So, number of 7-digit numbers with the even numbers as sum of digits = $(9 \times 10^{6})/2 = 45 \times 10^{5}$

Hence, the correct option is (b).

36. Find 1 × 2 rectangles, 2 × 3 rectangles, 3 × 4 rectangles, 4 × 5 rectangles, 5 × 6 rectangles, 6 × 7 rectangles, and 7 × 8 rectangles.

37. At least one out of five different starters can be chosen in following ways:

${}^{5}C_{1}$ or ${}^{5}C_{2}$ or ${}^{5}C_{3}$ or ${}^{5}C_{4}$ or ${}^{5}C_{5}$ ways

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At least one out of six identical main courses can be chosen in 6 ways.

At least one out of four different desserts can be chosen in following ways:

${}^4C_1$ or ${}^4C_2$ or ${}^4C_3$ or ${}^4C_4$

Therefore, the total number of ways = $({}^5C_1 + {}^5C_2 + {}^5C_3 + {}^5C_4 + {}^5C_5) \times (6) \times ({}^4C_1 + {}^4C_2 + {}^4C_3 + {}^4C_4) = (2^5 - 1) \times (6) \times (2^4 - 1) = 31 \times 6 \times 15$

Hence, option (a) is the answer.

38. There are 4 ways in which the guests may be invited for dinner:

1. Three ladies from man's relatives and 3 gentlemen from his wife's relatives. It can be done in ${}^4C_3 \times {}^3C_0 \times {}^4C_3 \times {}^3C_0 = 4 \times 4 = 16$ ways.
2. Two ladies and 1 gentleman from man's relatives and 2 gentlemen and 1 lady from his wife's relatives. It can be done in ${}^4C_2 \times {}^3C_1 \times {}^4C_2 \times {}^3C_1 = 6 \times 3 \times 6 \times 3 = 324$ ways.
3. One lady and 2 gentlemen from man's relatives and 1 gentleman and 2 ladies from his wife's relatives. It can be done in ${}^4C_1 \times {}^3C_2 \times {}^3C_2 \times {}^4C_1 = 4 \times 3 \times 3 \times 4 = 144$ ways.
4. Three gentlemen from man's relatives and 3 ladies from his wife's relatives. It can be done in ${}^3C_3 \times {}^4C_0 \times {}^3C_3 \times {}^4C_0 = 1$ way.

So, total number of ways is 16 + 324 + 144 + 1 = 485 ways.

Hence, the correct option is (c).

39. It is similar to $x + y + z + t = 26$ where $x > 0$, $y > 0$, $z > 0$, and $t \geq 0$. Now, we can first find the number of cases when all of x, y, z, and t are more than 0. Then, we can add the cases when $t = 0$.

Case I $x + y + z + t = 26$ and it must be positive integral solution, that is, all of the variables are greater than 0.

Number of possible solutions = ${}^{n-1}C_{r-1}$ (where $r = 4$ and $n = 26$) = ${}^{26-1}C_{4-1} = {}^{25}C_3 = 2300$

Case II $t = 0$. Then, $x + y + z = 26$, where all are positive integers = ${}^{n-1}C_{r-1} = {}^{26-1}C_{3-1} = {}^{25}C_2 = 300$

So, total number of possible cases = 2300 + 300 = 2600

Hence, the correct option is (c).

40. There are 40,000 numbers between 2×10^4 and 6×10^4, that is, 20,001 to 60,000. As we know for any 2 consecutive numbers, the sum of digits will be odd and even alternatively. Hence, half of the numbers must be having even numbers as sum of digits. So, total number of required numbers = 40,000/2 = 20,000

Hence, the correct option is (a).

42. There are total 10 guests. If 3 ladies are always together, then we have to arrange 8 positions which have 8! ways. Now, 3 ladies can be seated in 3! ways. So, there are total 8! × 3! ways. Now, if 2 gentlemen are together, then it can be done as keeping both together, and so, we have 7 positions (3 ladies are together and 2 men are together), and they can be seated in 7! ways. Now, 3 ladies can be seated among themselves in 3! ways and 2 men who are together can be arranged in 2! ways. So, the total number of ways both men are consecutive = 7! × 3! × 2!

Now, required number of ways = total number of ways − number of ways in which both gentlemen are together

= 8! × 3! − 7! × 3! × 2!

Hence, the correct option is (d).

43. In this problem, we have to give values to z, and then, the possible values for x and y can be obtained.

Let $z = 0$, then we have $2x + 2y = 10$, which is same as $x + y = 5$

Number of non-integral solutions for it = ${}^{5+2-1}C_{2-1} = {}^6C_1 = 6$

If $z = 1$, then we have $2x + 2y = 9$, and it is not possible to have any integral solution for it.

Similarly, z cannot be 3, 5, 7, or 9.

If $z = 2$, then $x + y = 4$

Number of solutions = 5 (${}^5C_1$).

Similarly, for $z = 4$, number of solutions = 4

For $z = 6$, number of solutions = 3

For $z = 8$, number of solutions = 2

For $z = 10$, number of solution = 1 ($x = y = 0$)

So, total number of non-negative integral solutions = 6 + 5 + 4 + 3 + 2 + 1 = 21

44. Total number of ways of selecting 11 members from 50 is ${}^{50}C_{11}$. If 3 members A, B, and C are all selected together, then 8 members are to be selected from remaining 47 members. So, number of ways = ${}^{47}C_8$

Required number of ways = total number of ways – number of ways in which A, B, and C are together

= ${}^{50}C_{11} - {}^{47}C_8$

Hence, the correct option is (a).

45. Each of the m points of 1st straight line will form intersection points with the $n - 1$ points of 2nd line. Similarly, each of n points of 2nd line will make intersection points with the $m - 1$ points on other line (here, we have $m - 1$ and $n - 1$ as each point cannot make intersection point with the point it already joined with). However, due to the repetition of these points twice, we need to divide both by 2.

So, total number of points of intersection of these line is $[(1/2) \times m(n-1)] \times [(1/2) \times n(m-1)] = ¼\, mn(m-1) \times (n-1)$

Hence, the correct option is (a).

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1. Total number of triangles = triangles having no sides common + triangles having one side common + triangles having two sides common + triangles having three sides common. Now, take the help of Worked out Example 19.

2. Let $S_1, S_2, \ldots, S_8$ denote the stations where the train does not stop. The four stations where the train stops should be at any four of the 9 places indicated by cross.

$\therefore$ Required number $= {}^9C_4 = \dfrac{9.8.7.6}{\lfloor 4} = 126$

5. If $a = 1 = b$, then $c = 30$. Similarly, we can have $a = c = 1$ and $b = c = 1$. These are three sets.

Now, take $a = 1$, then $bc = 30 = 2^1 \times 3^1 \times 5^1$

Possible sets of b and c for the product $bc = 30$ is 13.

So, total number of sets for any one of these to be equal to zero = 39

Now, take $a = 2$, then $bc = 15 = 3^1 \times 5^1$. Now, again find the sets of (b, c), and similarly, check all the possibilities.

10. Go through the options using formula.

Answers to Q.13 to 15:

Fruits are ordered as:

1. Apple 2. Mangoes 3. Oranges 4. Custard apple

Fruits are being kept as one, then two, then three, then four, and so on.

13. $1 + 2 + 3 + \ldots + 44 = 990$

So, the 45th fruit will be the 100th fruit. Since we are having a set of four different types of fruits, 45th will be apple.

14. Mangoes will come like 2, 6, 10, 14, etc.

100th mango will come $2 + 6 + 10 + 14 + 18 + 22 + 26 + 2 = 100$

100th mango will come in its 8th turn.

Before that, apple must have got its 8th turn. Orange and custard apple must have got their 7th turn.

Total apples displayed till now $= 1 + 5 + 9 + \ldots + 29 = 120$

Total oranges displayed till now $= 3 + 7 + 11 + \ldots + 27 = 105$

Total custard apples displayed till now $= 4 + 8 + 12 + \ldots + 28 = 112$

So, total fruits displayed till now (other than mangoes) = 337

So, the position of the 100th mango = 437

Answers to Q.18 and 19:

Use the options of Q.19 to solve both the questions simultaneously.

21. If the sum is a two-digit number, then both the values simultaneously cannot be less than 5. Now, start making the sets by taking different values greater than or equal to at least one of the numbers.

22. ${}^MC_2 - {}^MC_1 {}^wC_1 = 18$, where M and W are the number of men and women, respectively.

24. Given $a + b + c + d + e = 20$ (i)

$a + b + c = 5$ (ii)

Given, system of equations is equivalent to $a + b + c = 5$ (iii)

and $d + e = 15$

Number non-negative integral solutions of equation (iii)

$= {}^{n+r-1}C_r = {}^{3+1-1}C_5 = {}^7C_5$

Number of non-negative integral solution of equation (iv)

$= {}^{n+r-1}C_r = {}^{2+15-1}C_{15} = {}^{16}C_{15}$

$\therefore$ Required number $= {}^7C_5 . {}^{16}C_{15} = 336$

26. Each of the questions can be answered in 4 ways. So, 10 questions can be answered in 4^{10} ways.

28. Eliminate the options by assuming several values of $x = 1$ or 2, etc.

29. This is nothing but the application of the concept of number of factors (see Number System).

31. Sum of the digits is at most $9 + 9 + 9 = 27$. Therefore, the number is at most $6 \times 27 = 162$. Out of the numbers 1 to 162, the one with the largest sum of digits is 99, and the sum is $9 + 9 = 18$. Hence, the sum of digits will be at most 18.

Also, each number with this property is divisible by 6; therefore, it is divisible by 3, and its sum of digits is also divisible by 3.

We only have six possibilities left for the sum of the digits: 3, 6, 9, 12, 15, and 18. These lead to the integers 18, 36, 54, 72, 90, and 108. But, for 18, the sum of digits is $1 + 8 = 9$, which is not 3; therefore, 18 is not a solution. Similarly, we can throw away 36, 72, 90, and 108, and we are left with just one solution: the number 54.

Hence, option (c) is the answer.

33. $n^n - n! = 232$. Use options now.

34. If B is sitting on the immediate right of A, then C or D can sit on the immediate right of B, and the rest 3 places can be filled in 3! ways. So, the total number of ways when B is sitting to the immediate right of A $= 2 \times 3! = 12$.

Similarly, when C is sitting to the immediate right of A, then D has to sit on the immediate right of B, so B-D becomes a pair. Now, the rest of 2 can be seated at 3 different positions in two ways each (this is dependent of

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the positions of B-C). Number of arrangements possible here = 6

So, total arrangements = 18

35.

Males	Females
6	4

Now, take different cases to fulfill the conditions given.

40. **Case I** When a boy sits at the first place:

Possible arrangement will be of the form:

B	G	B	G	B	G	B	G

Now, there are four places namely 1st, 3rd, 5th, and 7th for 4 boys; therefore, 4 boys can be seated in $\lfloor 4$ ways. Again there are four places, namely 2nd, 4th, 6th, and 8th for 4 girls.

$\therefore$ Four girls can be seated in $\lfloor 4$ ways.

$\therefore$ Number of ways in this case = $\lfloor 4 \lfloor 4$

Similarly, another possible arrangement will be like:

B	G	B	G	B	G	B	G

Number of ways of doing this = $\lfloor 4 \lfloor 4$

So, total ways = $2 \lfloor 4 \lfloor 4$

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CHAPTER 17

Probability

LEARNING OBJECTIVES

After completion of this chapter, the reader should be able to understand:

- What is probability?
- Meaning and definitions of different terms
- Different theorems
- Methods of solving questions

INTRODUCTION

Probability is an expectation of the happening of some event. In our day-to-day life, we often predict the outcome of several events, and often hear phrases such as 'probably it will be a hot day today' or 'it will probably rain tomorrow'. These statements involve an element of uncertainty, and hence, questions such as 'how to measure the uncertainty of the happening of something arises'. Probability is just an attempt to quantify this method of predicting the future events with the help of some concepts. The theory of probability deals with those situations in which a particular result or outcome is not certain, but it can be any one of the several possible outcomes given to us for consideration. As a decision maker, we will be required to use this tool of probability to obtain the so-called 'satisficing output', thereby keeping all the variables in mind.

PROBABILITY AND ITS APPLICATION

Probability is the measure of uncertainty. Let us see one of the problems that is widely held responsible for the development of the theory of probability, namely, that of throwing a dice. As all of us know, a dice is a well-balanced cube with six faces marked with numbers from 1 to 6, one number on one face.

Let us throw a dice once. What are all the possible outcomes? Clearly, a dice can fall with any of its faces uppermost. The number on each of the faces is, therefore, a possible outcome, and all the outcomes are equally probable. Hence, it is as likely to show up a number, say 2, as any other number such as 1, 3, 4, 5, or 6.

As there are six equally likely outcomes in a single throw of a dice and there is only one way of getting a particular outcome (say 2), the chance of the number 2 coming up on the dice is 1 by 6. In other words, the same phenomenon is known as probability of getting 2 in a single throw of dice is 1/6.

We write this as $P(2) = 1/6$

Similarly, when an ordinary coin is tossed, it may show up either head (H) or tail (T). Hence, the probability of getting a head in a single toss of a coin is given by

$$P(H) = 1/2$$

Before we define the process to finding out the probability, it is essential to understand the various terms associated with probability.

Trial and Elementary Events

When we repeat a random experiment under identical conditions, then the experiment is known as trial and the possible outcomes of the experiment are known as elementary events.

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For example,

- tossing of a coin is a trial and getting head or tail is an elementary event.
- throwing of a dice is a trial and getting 5 on its upper face is an elementary event.

Compound Events

Events obtained by combining two or more elementary events are known as compound events. A compound event is said to occur if one of the elementary events associated with it occurs.

Exhaustive Number of Cases

The total number of possible outcomes of a random experiment in a trial is known as exhaustive number of cases.

For example, in throwing of a dice, the exhaustive number of cases is 6, since any one of the six faces marked with 1, 2, 3, 4, 5, or 6 may appear on its upper face.

Mutually Exclusive Events

Events are said to be mutually exclusive if the occurrence of any one of them prevents the occurrence of all the others, that is no two or more than two events can occur simultaneously in the same trial.

Equally Likely Events

Events are equally likely if there is no reason for an event to occur in preference to any other event. For example, while throwing a dice, chances of occurring of Head or Tail are an equally likely event.

Favourable Number of Cases

The number of cases favourable to an event in a trial is the number of elementary events such that if any one of them occurs, we say that the event happens.

In other words, the number of cases favourable to an event in a trial is the total number of elementary events such that the occurrence of any one of them ensures the happening of the event.

For example, in throwing of a dice, the number of cases favourable to the appearance of a prime number is 3 viz., 2, 3, and 5.

Independent Events

Events are said to be independent if the happening (or non-happening) of one event is not affected by the happening (or non-happening) of others.

For example, if two cards are drawn from a well-shuffled pack of 52 cards one after the other with replacement, then getting an ace in the first draw is independent of getting a jack in the second draw. However, if the first card drawn in the first draw is not replaced, then the second draw is dependent on the first draw.

Sample Space

This is the most important factor in probability and is defined as the set of all possible outcomes of a random experiment associated with it.

These examples suggest the following definition of probability (assuming that outcomes are equally likely).

Now, Probability of an event E, written as $P(E)$, is defined as

$$P(E) = \frac{\text{Number of outcomes favourable to } E}{\text{Total number of possible outcomes (sample space)}}$$

Remember

1. $0 \le P(E) \le 1$
2. $P(E) + P'(E) = 1$

Example 1 In a single throw of two dice, what is the probability that the sum on the top face of both the dice will be more than 9?

Solution When two dice are thrown, sum of the numbers appearing on the faces can be anything from 2 to 12. To find the number of favourable cases, we will be required to find the cases in which the sum is more than 9.

Following are the cases—(5, 5), (6, 4), (4, 6), (6, 5), (5, 6), and (6, 6).

So, the total number of favourable cases = 6

The total number of possible outcomes = $6 \times 6 = 36$

Hence, probability = $6/36 = 1/6$

Example 2 Six dice are thrown simultaneously. Find the probability that all of them show the same face.

Solution Sample space of throwing six dice = $6 \times 6 \times 6 \times 6 \times 6 \times 6 = 6^6$

All dice are showing the same face implies that we are getting same number on the entire six dice. The number of ways for which is ${}^6C_1$.

Hence, the required probability = $\frac{{}^6C_1}{6^6} = \frac{1}{6^5}$

Example 3 In the above question, find the probability that all of them show a different face.

Solution The total number of ways in which all the dice show different numbers on their top-faces is the same as

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the number of arrangement of 6 numbers 1, 2, 3, 4, 5, 6 by taking all at a time.

So, the number of favourable cases = 6!

Hence, the required probability = $\frac{6!}{6^6}$

Example 4 Five persons enter a lift on the ground floor of an 8-floor apartment. Assuming that each one of them independently and with equal probability can leave the lift at any floor beginning with the first. What is the probability that all the five persons are leaving the lift at different floors?

Solution Apart from the ground floor, there are 7 floors.

A person can leave the lift at any of the seven floors. Hence, the total number of ways in which each of the five persons can leave the lift at any of the 7 floors = 7^5

So, the sample space = 7^5

Five persons can leave the lift at five different floor = ${}^7P_5$ ways.

So, the favourable number of ways = ${}^7P_5$

Hence, the required probability = $\frac{{}^7P_5}{7^5}$

Example 5 If you have 3 tickets of a lottery for which 10 tickets were sold and 5 prizes are to be given, then the probability that you will win at least one prize is:

(*JMET 2005*)

(a) $\frac{7}{12}$ (b) $\frac{9}{12}$ (c) $\frac{1}{12}$ (d) $\frac{11}{12}$

Solution Probability that you will win at least one prize = 1 – probability that you will not win any prize.

$$= 1 - \frac{5C_3}{10C_3} = \frac{11}{12}$$

Odds in Favour and Odds Against

$$\text{Odds in favour} = \frac{\text{Number of favourable cases}}{\text{Number of unfavourable cases}}$$

$$\text{Odds in against} = \frac{\text{Number of unfavourable cases}}{\text{Number of favourable cases}}$$

Understanding And/Or: To understand the role played by And/Or in our calculation, let us take the example of throwing an unbiased dice. Let A and B be the two events associated with it such that

A = getting an even number and B = getting a multiple of 3.

Then, A = {2, 4, 6} and B = {3, 6}

We now define a new event 'A or B' that occurs if A or B or both occur, that is, at least one of A or B occurs. The event 'A or B' occurs if the outcome is any one of the outcomes {2, 3, 4, 6}. Therefore, the event 'A or B' is represented by the subset A U B.

Similarly, 'A and B' means occurrence of both A and B, which is possible if the outcome is {6}.

Hence, it is represented by the subset $A \cap B$.

ADDITION THEOREM

If A and B are two events associated with a random experiment, then

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Corollary: If the events are mutually exclusive, then $P(A \cup B) = P(A) + P(B)$.

Addition theorem can be extended for any number of events.

Example 6 A basket contains 20 apples and 10 oranges out of which 5 apples and 3 oranges are defective. If a person takes out 2 at random, then what is the probability that either both are apples or both are good?

Solution Out of 30 items, two can be selected in ${}^{30}C_2$ ways. So, exhaustive number of cases = ${}^{30}C_2$

Consider the following events:

A = getting two apples; B = getting two good items

Required probability $= P(A \cup B) = P(A) + P(B) - P(A \cap B)$

There are 20 apples, out of which two can be drawn in ${}^{20}C_2$ ways.

$$\therefore P(A) = \frac{20C_2}{{}^{30}C_2}$$

There are 8 defective pieces and the remaining 22 are good. Out of 22 good pieces, two can be selected in ${}^{22}C_2$ ways.

$$\therefore P(B) = \frac{22C_2}{{}^{30}C_2}$$

Since there are 15 pieces that are good apples of which 2 can be selected in ${}^{15}C_2$ ways, therefore,

$P(A \cap B)$ = Probability of getting 2 pieces which are good apples $= \frac{15C_2}{{}^{30}C_2}$

Required probability = $P(A) + P(B) - P(A \cap B)$

$$\frac{{}^{20}C_2}{{}^{30}C_2} + \frac{{}^{22}C_2}{{}^{30}C_2} + \frac{{}^{15}C_2}{{}^{30}C_2} = \frac{316}{435}$$

CONDITIONAL PROBABILITY

Let A and B be the two events associated with a random experiment. Then, the probability of occurrence of A

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under the condition that B has already occurred and $P(B) \neq 0$ is called conditional probability and it is denoted by $P(A/B)$.

Therefore, $P(A/B)$ = probability of occurrence of A given that B has already happened.

Similarly, $P(B/A)$ = probability of occurrence of B given that A has already happened.

Sometimes, $P(A/B)$ is also used to denote the probability occurrence of A when B occurs. Similarly, $P(B/A)$ is used to denote the probability of occurrence of B when A occurs.

Following examples illustrate the various meanings of these notations:

Example 7 A bag contains 5 white and 4 red balls. Two balls are drawn form the bag one after the other without replacement. Consider the following events.

A = drawing a white ball in the first draw and B = drawing a red ball in the second draw.

Solution Now, $P(B/A)$ = probability of drawing a red ball in the second draw given that a white ball has already been drawn in the first draw.

Since 8 balls are left after drawing a white ball in the first draw and out of these 8 balls, 4 balls are red, therefore,

$$P(B/A) = \frac{4}{8} = \frac{1}{2}$$

Note that $P(A/B)$ is not meaningful in this experiment because A cannot occur after the occurrence of B.

Example 8 One card is drawn from a pack of 52 cards, each of the 52 cards being equally likely to be drawn. Find the probability that the card drawn is:

(i) A king
(ii) Either a red or a king
(iii) A red and a king

Solution Out of 52 cards, one card can be drawn in ${}^{52}C_1$ ways. Therefore, exhaustive number of cases = ${}^{52}C_1 = 52$

(i) There are 4 kings in a pack of cards, out of which one can be drawn in ${}^{4}C_1$ ways. Therefore, the favourable number of cases = ${}^{4}C_1 = 4$, so the required probability = $\frac{4}{52} = \frac{1}{13}$

(ii) There are 28 cards in a pack of cards which are either a red or a king. Therefore, one can be drawn in ${}^{28}C_1$ ways. Therefore, the favourable number of cases = ${}^{28}C_1 = 28$.

So, the required probability = $\frac{28}{52} = \frac{7}{13}$

(iii) There are 2 cards that are red and king, that is red kings. Therefore, the favourable number of cases = ${}^{2}C_1 = 2$. So, the required probability = $\frac{2}{52} = \frac{1}{6}$

Example 9 An urn contains 9 blue, 7 white, and 4 black balls. If 2 balls are drawn at random, then find the probability that

(i) Both the balls are blue
(ii) One ball is white

Solution There are 20 balls in the bag out of which 2 balls can be drawn in ${}^{20}C_2$ ways. So, the total number of cases(sample space) = ${}^{20}C_2 = 190$.

(i) There are 9 blue balls out of which 2 balls can be drawn in ${}^{9}C_2$ ways. Therefore, the favourable number of cases = ${}^{9}C_2 = 36$. So, the required probability = $\frac{36}{190} = \frac{18}{95}$

(ii) There are 7 white balls out of which 1 white ball can be drawn in ${}^{7}C_1$ ways. One ball from the remaining 13 balls can be drawn in ${}^{13}C_1$ ways. Therefore, 1 white and 1 other colour ball can be drawn in ${}^{7}C_1 \times {}^{13}C_1$ ways. So, the favourable number of cases = ${}^{7}C_1 \times {}^{13}C_1 = 91$

Hence, the required probability = $\frac{91}{190}$

Example 10 Three persons A, B, and C are to speak at a function along with five others. If they all speak in random order, the probability that A speaks before B and B speaks before C is:

(a) 3/8 (b) 1/6
(c) 3/5 (d) None of these

Solution The total number of ways in which 8 persons can speak is ${}^{8}P_8 = 8!$. The number of ways in which A, B, and C can be arranged in the specified speaking order is ${}^{8}C_3$. There are 5! ways in which the other five can speak. So, the favourable number of ways is ${}^{8}C_3 = 5!$

Hence, the required probability = $\frac{{}^{8}C_3 \times 5!}{8!} = \frac{1}{6}$

Example 11 There is a point inside a circle. What is the probability that this point is closer to the circumference than to the centre?

Solution

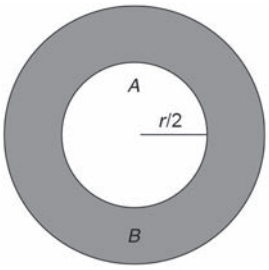
Assume that the radius of the bigger circle is r, and the radius of the inner circle is $r/2$. Point will be closer to the circumference than to the centre if the point is lying in the segment B.

Area of segment B = $3/4\ \pi r^2$

Hence, the probability of point being closer to the circumference = $3/4\ \pi r^2/\pi r^2 = 3/4$

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Example 12 Given that $P(A) = 1/3$ and $P(B) = 2/9$. Consider two cases: Case I—when the events are independent events and Case II—when the events are mutually exclusive events. What is the probability that

(a) Exactly one will happen?
(b) None will happen?
(c) At least one will happen?
(d) Both will happen?

Solution

Independent event	Mutually exclusive event
(a) Exactly One = $1/3 \times 7/9 + 2/3 \times 2/9 = 11/27$	(a) Exactly One = $1/3 + 2/9 = 5/9$
(b) None = $2/3 \times 7/9 = 14/27$	(b) None = $2/3 \times 7/9 = 14/27$
(c) At least one = 1 – none = $1 - 14/27 = 13/27$	(c) At least one = Exactly One = 5/9
(d) Both = $1/3 \times 2/9 = 2/27$	(d) Both = 0 (Zero)

Practice Exercises

Warm Up

Q.1 What is the chance of drawing an ace from a deck of cards?
(a) 8/52 (b) 16/52 (c) 2/52 (d) 4/52

Q.2 A three-digit number is to be formed by using the digits 1, 2, 3, …, 9. What is the probability that the number formed is greater than 500, if repetition is not allowed?
(a) 280/504 (b) 54/280 (c) 58/204 (d) 24/504

Q.3 Mayank and Amit are throwing an unbiased dice. If Mayank throws 1, then what is the Amit's chance of throwing a higher number?
(a) 1/5 (b) 5/6
(c) 2/12 (d) None of these

Q.4 A bag contains 4 red and 7 green balls. If three balls are drawn from the bag, replaced, and once again three balls are drawn from the bag, then what is the probability of obtaining 3 red balls on the first drawing and 3 green balls on the second drawing?
(a) 14/5445 (b) 14/27225
(c) 28/5445 (d) None of these

Q.5 One number is selected at random from the first 25 natural numbers. What is the probability that it is a multiple of either 5 or 7?
(a) 2/12 (b) 8/25
(c) 4/25 (d) None of these

Q.6 A bag contains 5 green apples and 7 red apples. If two apples are drawn from that bag, then what is the probability that one is red and the other is green?
(a) 12/66 (b) 35/66 (c) 2/12 (d) 2/35

Q.7 The probability of a problem being solved by A is 1/4 and B solving the same problem is 1/5. If they try independently, then what is the probability that the problem is not solved?
(a) 2/5 (b) 4/9 (c) 3/5 (d) 2/9

Q.8 Ten horses are running in a race; the chances that A will win are 30%, B will win are 20%, and C will win are 10%. What is the probability that exactly one of A or B or C will win?
(a) 0.689 (b) 0.598 (c) 0.498 (d) 0.398

Q.9 A's chance of winning a single game against B is 3/4. Find the chance that in a series of five games with B, A wins exactly 3 games?
(a) 135/1024 (b) 135/512
(c) 235/1024 (d) None of these

Q.10 It is known that at noon, the sun is hidden by clouds on an average of two days out of every three days. Find the probability that at noon on at least four out of five days the sun will be shining.
(a) 9/11 (b) 11/243
(c) 11/81 (d) None of these

Q.11 If on an average 1 vessel in every 10 is wrecked, find the chance that out of 5 vessels expected, at least 4 will arrive safely?
(a) 61,246/50,000 (b) 4592/5000
(c) 35,927/50,000 (d) None of these

Q.12 Two fair dice are thrown. What is the probability that the number of dots on the first dice exceeds 3 and that on the second exceeds 4?
(a) 2/6 (b) 3/6 (c) 1/6 (d) 5/6

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Q.13 What is the probability that a card drawn at random from a pack of 52 cards is either a King or a Spade?
(a) 2/52 (b) 4/13
(c) 4/26 (d) None of these

Q.14 Nazneen was dialing a telephone number of six digits. She forgot the last three digits of the six-digit telephone number but remembered that the number formed by the last three digits in the same order was a perfect square. What is the probability that the she dialed the right number?
(a) 1/32 (b) 2/32 (c) 2/16 (d) 1/23

Q.15 What is the probability that there are 53 Sundays and 53 Tuesdays in a leap year?
(a) 0 (b) 1
(c) 0.5 (d) None of these

Q.16 What is the probability that there are at least 52 Sundays in a leap year?
(a) 0 (b) 1
(c) 0.5 (d) None of these

Q.17 What is the probability that there are 53 Sundays and 53 Saturdays in a leap year?
(a) 1/7 (b) 2/7
(c) 0 (d) None of these

Q.18 What is the probability that there are 53 Sundays in a leap year?
(a) 2/7 (b) 1/7
(c) 0 (d) None of these

FOUNDATION

Q.1 Five boys and three girls are seated at random in a row. The probability that no boy sits between two girls is:
(a) $\frac{1}{56}$ (b) $\frac{1}{8}$
(c) 3/28 (d) None of these

Q.2 In a convex hexagon, two diagonals are drawn at random. The probability that the diagonals intersect at an interior point of the hexagon is:
(a) $\frac{5}{12}$ (b) $\frac{7}{12}$
(c) $\frac{2}{5}$ (d) None of these

Q.3 Two cards are drawn at random from a pack of 52 cards. Find the probability of getting at least a spade and an ace.
(a) $\frac{1}{34}$ (b) $\frac{8}{221}$ (c) $\frac{1}{26}$ (d) $\frac{2}{51}$

Q.4 If the letters of the word ATTEMPT are written down at random, then find the chance that all the Ts are consecutive.
(a) $\frac{1}{42}$ (b) $\frac{6}{71}$
(c) $\frac{1}{7}$ (d) None of these

Q.5 Seven white balls and 3 black balls are placed in a row at random. The probability that no two black balls are adjacent is:
(a) $\frac{1}{3}$ (b) $\frac{7}{15}$ (c) $\frac{2}{15}$ (d) $\frac{1}{3}$

Q.6 Four gentlemen and 4 ladies take seats at random on a round table. The probability that they are sitting alternately is:
(a) $\frac{4}{35}$ (b) $\frac{1}{70}$ (c) $\frac{2}{35}$ (d) $\frac{1}{35}$

Q.7 Let $x = 33^n$. The index n is given a positive integral value at random. The probability that the value of x will have 3 in the units place is:
(a) $\frac{1}{4}$ (b) $\frac{1}{2}$
(c) $\frac{1}{3}$ (d) None of these

Q.8 Three dice are thrown simultaneously. The probability of getting a sum of 15 is:
(a) $\frac{1}{72}$ (b) $\frac{5}{36}$
(c) $\frac{5}{72}$ (d) None of these

Q.9 Three dice are thrown. The probability of getting a sum which is a perfect square is:
(a) $\frac{2}{5}$ (b) $\frac{9}{20}$
(c) $\frac{1}{4}$ (d) None of these

Q.10 A box contains 6 red balls, 7 green balls, and 5 blue balls. Each ball is of a different size. The probability that the red ball being selected is:
(a) $\frac{1}{18}$ (b) $\frac{1}{3}$ (c) $\frac{1}{6}$ (d) $\frac{2}{3}$

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Q.11 Two distinct numbers are selected at random from the first twelve natural numbers. The probability that the sum will be divisible by 3 is:

(a) $\frac{1}{3}$ (b) $\frac{23}{66}$

(c) $\frac{1}{2}$ (d) None of these

Q.12 Three different numbers are selected at random from set $A = \{1, 2, 3, \ldots, 10\}$. The probability that the product of two of the numbers is equal to the third is:

(a) $\frac{3}{4}$ (b) $\frac{1}{40}$

(c) $\frac{1}{8}$ (d) None of these

Q.13 There are 7 seats in a row. Three persons take seats at random. The probability that the middle seat is always occupied and no two persons are consecutive is:

(a) $\frac{9}{70}$ (b) $\frac{9}{35}$

(c) $\frac{4}{35}$ (d) None of these

Q.14 From a group of 10 persons consisting of 5 lawyers, 3 doctors, and 2 engineers, four persons are selected at random. The probability that the selection contains at least one of each category is:

(a) $\frac{1}{2}$ (b) $\frac{1}{3}$

(c) $\frac{2}{3}$ (d) None of these

Direction for Questions 15 and 16: Read the passage below and solve the questions based on it.

A and B are two possible events of an experiment such that $P(A \cup B) = 0.7$ and $P(A) = 0.4$, then find $P(B)$ given that

Q.15 A and B are mutually exclusive events.

(a) 0.6 (b) 0.3 (c) 0.2 (d) 0.5

Q.16 A and B are independent events.

(a) 0.5 (b) 0.3 (c) 0.6 (d) 0.2

Q.17 The odds against an event are 3 to 4 and the odds in favour of another independent event are 2 to 5. Find the probability that exactly one of them occurs.

(a) $\frac{23}{49}$ (b) $\frac{26}{49}$ (c) $\frac{6}{49}$ (d) $\frac{8}{49}$

Q.18 In the previous question, what is the probability that at least one of them can occur?

(a) $\frac{34}{49}$ (b) $\frac{15}{49}$ (c) $\frac{23}{49}$ (d) $\frac{26}{49}$

Q.19 Chunmun picks a letter from the set of English alphabet and finds it to be a vowel. What is the probability that the letter is E?

(a) 1/26 (b) 5/6

(c) 1/5 (d) None of these

Q.20 There are 3 red, 3 blue, and 3 green balls. What is the probability that out of 3 selected balls 2 are red?

(a) 1/4 (b) 2/5

(c) 3/7 (d) None of these

Q.21 In the horse race, the horse Exaclibur's chances of winning are $\frac{1}{6}$, the horse Noorjehan's chances are $\frac{1}{10}$, and those of the horse Bombay are $\frac{1}{8}$. If the race starts with twelve horses and only one horse wins, then what are the chances that one of these three horses will win?

(a) $\frac{57}{120}$ (b) $\frac{47}{120}$ (c) $\frac{37}{120}$ (d) $\frac{27}{120}$

Q.22 If a year has 360 days and all the months with 30 days. What is the probability that your birthday falls on a Monday and that is an even day of an even month, given that January 1 is a Monday?

(a) $\frac{1}{10}$ (b) $\frac{13}{360}$ (c) $\frac{1}{28}$ (d) $\frac{11}{360}$

Q.23 If 4 whole numbers are taken at random and multiplied together, then the probabaility that the last digit in the product is 1, 3, 7, or 9 will be:

(a) 13/976 (b) 17/529

(c) 16/625 (d) 13/625

Q.24 There are 500 students in an examination. 150 students passed the first paper, 350 students passed the second paper, and 50 students passed both the papers. Find the probability that a student selected at random has failed in both the papers.

(a) 1/5 (b) 1/10 (c) 3/10 (d) 3/5

Q.25 A person is asked to randomly pick two balls from a bag that has 15 yellow and 5 red balls. The probability that the person picks two balls of the same colour is:

(a) 33/38 (b) 23/38

(c) 38/43 (d) 15/38

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MODERATE

Q.1 If the integers m and n are chosen at random between 1 and 100, then the probability that a number of the form $7^m + 7^n$ is divisible by 5 is:

(a) $\frac{1}{5}$ (b) $\frac{1}{7}$ (c) $\frac{1}{4}$ (d) $\frac{1}{49}$

Q.2 If the probability of A failing in an examination is $\frac{1}{5}$ and that of B failing the same examination is $\frac{3}{10}$, then the probability that either A or B fails is:

(a) $\frac{1}{2}$ (b) $\frac{11}{25}$

(c) $\frac{19}{50}$ (d) None of these

Q.3 India plays two matches each with West Indies and New Zealand. In any match, the probabilities of India getting 0,1, and 2 points are 0.45, 0.05, and 0.50, respectively. Assuming that the outcomes are independent, the probability of India getting at least 7 points is:

(a) 0.0875 (b) $\frac{1}{16}$

(c) 0.1125 (d) None of these

Q.4 Let A and B be two independent events such that $P(A) = \frac{1}{5}$, $P(A \cup B) = \frac{7}{10}$. Then, $P(\overline{B})$ is equal to:

(a) $\frac{3}{8}$ (b) $\frac{2}{7}$

(c) $\frac{7}{9}$ (d) None of these

Q.5 Let A and B be two independent events in such a way that their probabilities are $\frac{3}{10}$ and $\frac{2}{5}$, respectively. The probability of exactly one of the events happening is:

(a) $\frac{25}{30}$ (b) $\frac{1}{2}$

(c) $\frac{31}{50}$ (d) None of these

Q.6 A, B, and C are three events for which $P(A) = 0.6$, $P(B) = 0.4$, $P(C) = 0.5$, $P(A \cup B) = 0.8$, $P(A \cap B) = 0.3$, and $P(A \cap B \cap C) = 0.2$. If $P(A \cup B \cup C) \geq 0.85$, then find the interval of values of $P(B \cap C)$.

(a) [0.2, 0.35] (b) [0.55, 0.7]
(c) [0.2, 0.55] (d) None of these

Q.7 A dice is thrown $2n + 1$ times, $n \in N$. The probability that the faces with even numbers show odd number of times is:

(a) $\frac{2n+1}{4n+1}$ (b) Less than $\frac{1}{2}$

(c) Greater than $\frac{1}{2}$ (d) None of these

Q.8 A shooter firing at a target has 10% chance of hitting the target in one shot. The number of times he must fire at the target to have above 50% chance of hitting the target is:

(a) 11 (b) 9 (c) 7 (d) 5

Q.9 Let A = {2, 3, 4, …, 20, 21}. A number is chosen at random from the set A and it is found to be a prime number. The probability that it is more than 10 is:

(a) $\frac{9}{10}$ (b) $\frac{1}{10}$

(c) $\frac{1}{5}$ (d) None of these

Q.10 The probabilities that a student passes in Mathematics, Physics, and Chemistry are m, p, and c, respectively. Of these subjects, a student has a 75% chance of passing in at least one, a 50% chance of passing in at least two, and a 40% chance of passing in exactly two subjects. Which of the following relations are true?

(a) $p + m + c = \frac{19}{20}$ (b) $p + m + c = \frac{27}{20}$

(c) $pmc = \frac{1}{20}$ (d) $pmc = \frac{1}{4}$

Q.11 If A and B are two events such that $P(A \cup B) \geq \frac{3}{4}$ and $\frac{1}{8} \leq P(A \cap B) \leq \frac{3}{8}$, then:

(a) $P(A) + P(B) \leq \frac{11}{8}$

(b) $P(A) \cdot P(B) \leq \frac{3}{8}$

(c) $P(A) + P(B) \geq \frac{7}{8}$

(d) None of these

Q.12 Three natural numbers are taken at random from set A = $\{x | x, 1 \leq x \leq 100, x \in N\}$. The probability that the AM of the numbers taken is 25 is:

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(a) $\frac{^{77}C_2}{^{100}C_3}$ (b) $\frac{^{25}C_2}{^{100}C_3}$

(c) $\frac{^{74}C_2}{^{100}C_3}$ (d) None of these

Q.13 The probability of a number n showing in a throw of a dice marked 1 to 6 is proportional to n. Then, the probability of the number 3 showing in throw is:

(a) $\frac{1}{2}$ (b) $\frac{1}{6}$ (c) $\frac{1}{7}$ (d) $\frac{1}{21}$

Direction for Questions 14 to 16: Read the passage below and solve the questions based on it.

Three mountaineers Amit, Vinit, and Nishit are climbing up a mountain with their respective probabilities of reaching the summit being $\frac{1}{3}$, $\frac{1}{5}$, and $\frac{1}{4}$, respectively. What is the probability of the following:

Q.14 All of them reach the summit?

(a) $\frac{3}{60}$ (b) $\frac{1}{60}$ (c) 1 (d) 0

Q.15 At least one of them reaches the summit?

(a) $\frac{59}{60}$ (b) $\frac{36}{60}$ (c) $\frac{17}{60}$ (d) $\frac{1}{60}$

Q.16 Exactly one of them reaches the Summit?

(a) $\frac{13}{30}$ (b) $\frac{17}{30}$ (c) $\frac{19}{30}$ (d) $\frac{11}{30}$

Q.17 The coach of the football team says that the probability that player A plays and team wins is $\frac{1}{4}$, player B plays and team wins is $\frac{1}{2}$, and both play and team wins is $\frac{1}{4}$. He also added that the team has $\frac{3}{4}$ chances to win. What is the probability that when either player A or player B played, the team won?

(a) $\frac{2}{3}$ (b) $\frac{1}{2}$

(c) Cannot be determined (d) None of these

Direction for Questions 18 and 19: Read the passage below and solve the questions based on it.

Two teams Patel Nagar and Boring Road are playing a cricket tournament. If team Patel Nagar wins the toss, probability of it winning the match is 0.55. If Boring Road wins the toss, probability of its winning the match is 0.65. Probability of winning the toss is same for both the teams Patel Nagar and Boring Road.

Q.18 In a five-match tournament, what is the probability of Patel Nagar winning all the matches, if it wins at most two tosses?

(a) 0.45 (b) 0.41 (c) 0.39 (d) 0.51

Q.19 What is the probability of Boring Road winning all the matches, if Patel Nagar wins 3 tosses in a five-match tournament?

(a) 0.16 (b) 0.14

(c) 0.21 (d) None of these

Q.20 Rakesh went to take an exam. After the exam, while explaining the structure of the question paper, the teacher said that 40% of the total questions were difficult, 20% of those were on algebra, and 30% of those were on geometry. Also, 10% and 20% of easy questions were on algebra and geometry, respectively. All the questions that Rakesh attempted were either on algebra or on geometry. What is the probability that Rakesh found all the questions difficult (questions are either easy or difficult), if Rakesh attempted 15% questions?

(a) $\frac{14}{100}$ (b) $\frac{14}{27}$

(c) $\frac{20}{27}$ (d) None of these

Direction for Questions 21 and 22: Read the passage below and solve the questions based on it.

In a game, one can score as many points as he gets in every throw till he gets a score (in one throw) which is not a multiple of points scored in the previous throw.

Q.21 If a person scores 5 points in total, then what is the probability that the dice was thrown 2 times?

(a) $\frac{3}{4}$ (b) $\frac{3}{5}$

(c) $\frac{1}{2}$ (d) None of these

Q.22 If a person throws the dice 3 times, what is the probability that the total score is 11?

(a) 5/43 (b) 5/41

(c) 5/37 (d) None of these

Q.23 Triangles are formed by joining vertices of an octagon. Any one of those triangle is selected at random. What is the probability that the selected triangle has no side common with the octagon?

(a) 3/7 (b) 2/7 (c) 5/7 (d) 1/7

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Q.24 What is the probability that a 4-digit number of the type abcd (a, b, c, and d are all different) is a multiple of 11?
(a) 0.142 (b) 0.048 (c) 0.072 (d) 0.091

Q.25 A month is randomly selected from the months in a non-leap year and it is found that it has five Sundays. What is the probability that it has five Mondays?
(a) 1/7 (b) 2/7 (c) 2/3 (d) 20/33

ADVANCED

Q.1 Manoj throws a fair dice. He is promised an amount thrice the value of the number showing up if the number showing up is odd and an amount twice the value of the number showing up if it is even. What is the maximum amount Manoj is willing to pay each time to throw the dice, if in the long run he wants to make an average profit of ₹5 per throw?
(a) ₹3.50 (b) ₹8.5
(c) ₹5 (d) None of these

Direction for Questions 2 to 4: Read the passage below and solve the questions based on it.

A shopkeeper inspected some items in a box containing a, b, and c number of items of type P, Q, and R, respectively. He picked up an item randomly, and put it back into the box if he found that the item selected was regularly sold; he also added the equal number of items of the same type as those previously in the box; otherwise, he removed all the items of that type.

Q.2 If exactly one of the three items is regularly sold, then what is the probability that the item selected in the second draw is similar to that in the first draw? (Given that the item selected in the second draw is a regularly sold item.)

(a) $2\left(\frac{a}{a+b+c}+\frac{b}{a+b+c}+\frac{c}{a+b+c}\right)$

(b) $\left(\frac{2a^2}{2a+b+c}+\frac{2b^2}{a+2b+c}+\frac{2c^2}{a+b+2c}\right)\frac{1}{a+b+c}$

(c) $\left(\frac{a}{2a+b+c}+\frac{b}{1+2b+c}+\frac{c}{a+b+2c}\right)\frac{1}{a+b+c}$

(d) None of these

Q.3 In the above problem, what is the probability that the item selected in the second draw is not a regularly sold item?

(a) $\frac{a}{a+b+c}+\frac{b}{a+b+c}+\frac{c}{a+b+c}$

(b) $\frac{b+c}{2a+b+c}+\frac{a+c}{a+2b+c}+\frac{a+b}{a+b+2c}$

(c) $\frac{a}{b+c}+\frac{b}{a+c}+\frac{c}{a+b}$

(d) None of these

Q.4 What is the probability that the items selected in the second draw is different from that in the first draw, given that the item selected in the first draw is a regularly sold item? (Assume exactly that two of the three items are not regularly sold items.)

(a) $\left(\frac{a(a+c)}{2a+b+c}+\frac{b(a+c)}{a+2b+c}+\frac{c(a+b)}{a+b+2c}\right)\frac{1}{a+b+c}$

(b) $\left(\frac{a(b+c)}{2a+b+c}+\frac{b(a+c)}{a+2b+c}+\frac{c(a+b)}{a+b+2c}\right)\frac{1}{a+b+c}$

(c) $\left(\frac{a}{2a+b+c}+\frac{b}{a+2b+c}+\frac{c}{a+b+2c}\right)\frac{1}{a+b+c}$

(d) 0

Q.5 There is a frame of a cuboid of length 6 units, breadth 5 units, and height 7 units. The cuboid is only composed of skeleton of 210 cubes of side 1. An insect is on one corner of the cube and it wants to travel to the opposite end of the longest diagonal. It can only travel along the sides of the small cube and it always takes the shortest possible route. Find the probability that it passes through at least one of the corners.

(a) $\frac{1}{6}$ (b) $\frac{7}{12}$

(c) $\frac{5}{18}$ (d) None of these

Q.6 A coin is so biased that heads occur four times as frequently as tails. Another coin is biased such that heads occur 65% of the times. When the two coins are tossed simultaneously, what is the probability of at least one tail turning up?
(a) 35% (b) 87% (c) 48% (d) 73%

Q.7 Only one bomb is needed to destroy a bridge and there are three bombers engaged in an attempt to destroy the bridge. The probabilities of the bombers hitting the bridge are, respectively, 0.3, 0.4, and 0.7. What is the probability of the bridge being destroyed?
(a) 0.874 (b) 0.126 (c) 0.916 (d) 0.084

Q.8 A pack of playing cards was found to contain only 51 cards. If the first 13 cards, which are examined, all are red, then what is the probability that the missing card is black?
(a) 3/7 (b) 5/7 (c) 2/3 (d) 3/8

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Q.9 A speaks truth in 75% of the cases and B in 80% of the cases. In what percentage of cases are they likely to contradict each other in stating the same fact?
(a) 15% (b) 20% (c) 5% (d) 35%

Direction for Questions 10 to 12: Read the passage below and solve the questions based on it.

In a manufacturing unit, machines A, B, and C manufacture 15%, 25%, and 60% of the total production of bolts, respectively. Of all the bolts manufactured by the machines 4% by A, 2% by B, and 3% by C are defective, respectively. A bolt is drawn at random and is found to be defective.

Q.10 What is the probability that it was produced by machine A?
(a) 6/29 (b) 5/29 (c) 18/29 (d) 4/29

Q.11 What is the probability that it was produced by machine B?
(a) 6/29 (b) 5/29 (c) 18/29 (d) 4/29

Q.12 What is the probability that it was produced by machine C?
(a) 6/29 (b) 5/29 (c) 18/29 (d) 4/29

Q.13 There is a 30% chance that it rains on any particular day. What is the probability that there is at least one rainy day within a period of 3 days?
(a) $(0.7)^3$ (b) $1 - (0.7)^3$
(c) 0.3^3 (d) None of these

Q.14 The numbers are selected at random, one at a time, from the two-digit numbers 00, 01, 02, …, 99 with replacement. An event E occurs only if the product of the two digits of a selected numbers is 18. If four numbers are selected, then find the probability that the event E occurs at least 3 times.
(a) 0.0025 (b) 0.00025 (c) 0.025 (d) 0.25

Q.15 A and B are two independent events. The probability that both A and B occur is 1/6 and the probability that neither of them occurs in 1/3. Find the probability of the occurrence of A.
(a) 1/6 (b) 1/4
(c) 1/3 (d) Cannot be determined

Q.16 A biased dice is tossed and the respective probabilities for various faces to turn up are as follows:

Face	1	2	3	4	5	6
Probability	0.1	0.24	0.19	0.18	0.15	0.14

If an even face has turned up, then the probability of face 2 or face 4 is:
(a) 0.25 (b) 1.42 (c) 0.75 (d) 0.9

Q.17 A natural number x is chosen at random from the first one hundred natural numbers. What is the probability that $x + \frac{100}{x} > 50$?
(a) 13/20 (b) 3/5 (c) 9/20 (d) 11/20

Q.18 A dice is loaded in such a way that the probability of getting i on the dice is proportional to i. This dice is rolled three times. What is the probability of getting a product of 15?
(a) 10/1029 (b) 14/1029
(c) 16/1029 (d) 18/1029

Q.19 A target may be either at point I with probability $\frac{8}{9}$ or at point II with probability $\frac{1}{9}$. We have 21 shells, each of which can be fired either at point I or at point II. Each shell may hit the target, independently of the other shells, with probability $\frac{1}{2}$. How many shells must be fired at point I to hit the target with maximum probability?
(a) 10 (b) 11 (c) 9 (d) 12

Q.20 Let N cells be numbered 1, 2, 3, … N. We randomly throw balls into them. The process is continued until any one of the cells receives two balls. What is the probability that more than n throws will be necessary?
(a) ${}^{N}P_{N}/N^n$ (b) ${}^{N}/N^n$
(c) ${}^{N}C_{n}/N^n$ (d) None of these

Q.21 The probability that a man will live 10 more years is $\frac{1}{4}$ and the probability that his wife will live 10 more years is $\frac{1}{3}$. Then, the probability that neither will be alive in 10 years is:
(a) $\frac{5}{12}$ (b) $\frac{7}{12}$ (c) $\frac{1}{2}$ (d) $\frac{11}{12}$

Q.22 A locker can be opened by dialing a fixed three-digit code between 000 and 999. A stranger, who does not know the code, tries to open the locker by dialing three digits at random. Find the probability that the stranger succeeds at the sth trial. (Assume that the stranger does not repeat unsuccessful combinations.)
(a) 2s/1000 (b) 2s/999 (c) s/1000 (d) s/999

Q.23 The probability of a bomb hitting a bridge is $\frac{1}{2}$, and two direct hits are needed to destroy it. The least number of bombs required so that the probability of the bridge being destroyed is greater than 0.9 is:
(a) 8 (b) 9 (c) 10 (d) 11

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Q.24 From a bag, which is known to contain 4 balls each of which is just as likely to be black as white, a ball is drawn and is found to be white. What is the probability that the bag contained 3 white and 1 black balls?
(a) 3/7 (b) 5/8 (c) 3/8 (d) 4/7

Q.25 On a loaded dice, any even number occurs thrice as frequently as an odd number. If the dice is thrown twice, what is the probability that the sum is 10 or more?
(a) 55/72 (b) 17/72 (c) 39/72 (d) 23/72

Answers

Warm Up

1. (d)	**2.** (a)	**3.** (b)	**4.** (c)	**5.** (b)	**6.** (b)	**7.** (c)	**8.** (d)	**9.** (b)	**10.** (b)
11. (d)	**12.** (c)	**13.** (b)	**14.** (a)	**15.** (a)	**16.** (b)	**17.** (a)	**18.** (a)		

Foundation

1. (c)	**2.** (a)	**3.** (c)	**4.** (c)	**5.** (b)	**6.** (d)	**7.** (a)	**8.** (d)	**9.** (d)	**10.** (c)
11. (a)	**12.** (b)	**13.** (c)	**14.** (a)	**15.** (b)	**16.** (a)	**17.** (b)	**18.** (a)	**19.** (c)	**20.** (d)
21. (b)	**22.** (b)	**23.** (c)	**24.** (b)	**25.** (b)					

Moderate

1. (a)	**2.** (b)	**3.** (a)	**4.** (a)	**5.** (a)	**6.** (d)	**7.** (d)	**8.** (c)	**9.** (d)	**10.** (c)
11. (d)	**12.** (c)	**13.** (c)	**14.** (b)	**15.** (b)	**16.** (a)	**17.** (a)	**18.** (b)	**19.** (d)	**20.** (b)
21. (b)	**22.** (a)	**23.** (b)	**24.** (a)	**25.** (d)					

Advanced

1. (a)	**2.** (b)	**3.** (d)	**4.** (a)	**5.** (d)	**6.** (c)	**7.** (c)	**8.** (c)	**9.** (d)	**10.** (a)
11. (b)	**12.** (c)	**13.** (b)	**14.** (b)	**15.** (d)	**16.** (d)	**17.** (d)	**18.** (a)	**19.** (d)	**20.** (c)
21. (c)	**22.** (c)	**23.** (b)	**24.** (c)	**25.** (b)					

Hints and Solutions

Warm up

1. There are four aces out of total 52 cards.
Hence, probability of drawing an ace $= \frac{4}{52} = \frac{1}{13}$

2. Total number of three-digit numbers that can be formed without repetition $= 9 \times 9 \times 8$
Total number of three-digit numbers greater than 500 that can be formed without repetition $= 5 \times 9 \times 8$
Therefore, the required probability $= \frac{5 \times 9 \times 8}{9 \times 9 \times 8}$
Hence, option (a) is the answer.

3. If Mayank throws 1, then to throw a higher number, Amit can throw 2 or 3 or 4 or 5 or 6.
Hence, the required probability $= \frac{5}{6}$

4. The required probability $= \frac{{}^4C_3 \times {}^7C_3}{{}^{11}C_3 \times {}^{11}C_3} = \frac{140}{165 \times 165}$.
Hence, option (c) is the answer.

5. Total multiples of 5 = 5, 10, 15, 20, 25 = 5 numbers
Total multiples of 7 = 7, 14, 21 = 3 numbers

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Hence, total required numbers = 8 numbers

The required probability = $\frac{8}{25}$

6. The required probability = $\frac{{}^5C_1 \times {}^7C_1}{{}^{12}C_2} = \frac{35}{66}$

7. Probability that the problem is not solved = Probability that A does not solve the problem × probability that B does not solve the problem $= \frac{3}{4} \times \frac{4}{5} = \frac{3}{5}$

8. Exactly one of them wins = A wins but other two loses + B wins but other two loses + C wins but other two loses = 30% × 80% × 90% + 70% × 20% × 90% + 70% × 80% × 10% = 0.216 + 0.126 + 0.056 = 0.398

9. A will win three games and lose two games. We also need to select three games out of five games.

Hence, the required probability = ${}^5C_3 \times \left(\frac{3}{4}\right)^3 \times \left(\frac{1}{4}\right)^2 = 10 \times \frac{27}{1024} = \frac{135}{512}$

10. Probability that the sun is hidden = 2/3; so, the probability that the sun is not hidden by clouds = Sun will be shining = 1/3

At least four out of five days, sun will be shining = Probability of exactly four days + probability of exactly five days

$= {}^5C_4 \times (1/3)^4 \times (2/3)^1 + {}^5C_5 \times (1/3)^5 \times (2/3)^0 = 5 \times \frac{2}{243} + 1 \times \frac{1}{243} = \frac{11}{243}$

11. Probability that the vessel will be wrecked = 1/10, and probability that it will reach safely = 9/10

At least four out of five will reach safely = Probability of exactly four + probability of exactly five

$= {}^5C_4 \times (9/10)^4 \times (1/10)^1 + {}^5C_5 \times (9/10)^5 \times (1/3)^0$.

12. Ways in which number of dots on the first dice exceed 3 = 4, 5, 6 = 3 ways

Ways in which number of dots on the second dice exceed 4 = 5, 6 = 2 ways

Hence, the required probability = $\frac{3 \times 2}{6 \times 6} = 1/6$

13. Total number of cards which are either a king or a spade = 16

Hence, the required probability = $\frac{16}{52} = 4/13$

14. Total three digit numbers which are squares = 000 to 961 = 32 numbers

Hence, the required probability = 1/32

15. In a leap year, there are 366 days = 52 weeks + 2 days extra

If there are 53 Sundays, then the other extra day will be either a Saturday or a Monday. Hence, the required probability = 0.

16. All the days will occur atleast 52 times. Hence, the required probability = 1.

17. There are 7 different possibilities. Hence, the required probability = 1/7.

18. There are two extra days and seven different possibilities viz. (Sunday, Monday), (Monday, Tuesday), (Tuesday, Wednesday), (Wednesday, Thursday), (Thursday, Friday), (Friday, Saturday), and (Saturday, Sunday).

Hence, the required probability = 2/7.

MODERATE

1. For the sum to be divisible by 5, unit digit of $7^m + 7^n$ should be either 0 or 5.

Different unit digits possible for 7 are 7, 9, 3, and 1.

So, if 7^m is giving unit digit 7, then 7^n should give 3 or vice-versa.

Similarly, if 7^m is giving unit digit 9, then 7^n should give 1 or vice-versa.

Hence, probability = $\frac{25 \times 25 \times 2 \times 2}{100 \times 100} = \frac{1}{4}$

2. $P(A \cup B) = \frac{1}{5} + \frac{3}{10} - \frac{1}{5} \times \frac{3}{10} = \frac{10 + 15 - 3}{50} = \frac{22}{50}$

Hence, option (b) is the answer.

3. Earning at least 7 points means earning either 7 or 8 points. It can be done in 5 ways, which forms 5 mutually exclusive events.

1. India gets 2 points in all 4 matches and its probability is $(0.5)^4$.

2–5. India gets 2 points in 3 matches and 1 point in 1 match (it could be either for 1st, 2nd, 3rd, or 4th match).

So, its probability is $4 \times 0.05 \times (0.5)^3$.

So, total probability of getting at least 7 points is $= (0.5)^4 + [4 \times 0.05 \times (0.5)^3] = 0.0875$

Hence, the correct option is (a).

4. A and B are independent events, so $P(A \cap B) = P(A) \times P(B)$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

So, $7/10 = 1/5 + P(B) - [1/5 \times P(B)]$

$P(B) = 5/8$

Hence, $P(\bar{B}) = 1 - \frac{5}{8} = \frac{3}{8}$

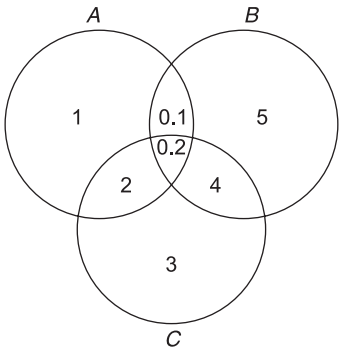
Hence, the correct option is (a).

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5. Do as given in Q.2.

6. Using the data given in the question, the following diagram can be drawn:



In the above diagram (please note that 1, 2, 3, 4, and 5 denotes 5 parts of $A \cup B \cup C$ and not the actual numbers), the $P(A \cup B \cup C)$ is shown in 7 parts, and we know the values of 2 parts $P(A \cap B \cap C) = 0.2$ and $P(A \cap B) = 0.3$, and hence, $P(A \cap B) - P(A \cap B \cap C) = 0.1$. Now, $P(B) = 0.4$ and as we can see $P(B) = 0.1 + 0.2 + 4 + 5$. So, $4 + 5 = 0.1$

Now, $P(B \cap C) = 0.2 + 4$. Here, we can notice that minimum and maximum values of 4 are 0 and 0.1. So, $P(B \cap C)$ must be in the interval [0.2, 0.3].

Hence, the correct option is (d).

7. Required probability is simply $\frac{1}{2}$.

Hence, the correct option is (d).

8. He must fire at target till the probability of missing the target becomes less than 50%. The probability of hitting target for any shot is 10%, and so, the probability of missing target for any shot is 90%, that is, 0.9. Let he fires the target n times, then $0.9^n < 0.5$. Minimum value of n for it is 7 (as $0.9^7 = 0.478$). So, he must fire 7 times at the target.

Hence, the correct option is (c).

9. Total numbers of primes = 8 and number of numbers more than 10 = 11

10. Let A, B, and C be events of passing subject Mathematics, Physics, and Chemistry, respectively. Now, P(A) = m, P(B) = p, and P(C) = c. Probability of at least two of A, B, and C is 0.5 and probability of exactly two of A, B, and C is 0.4. So, probability of all the three of A, B, and C is 0.1.

$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$

$= P(A) + P(B) + P(C) - P(\text{exactly 2 of } A, B, \text{ and } C) - 2P(A \cap B \cap C)$

or $0.75 = m + p + c - 0.4 - (2 \times 0.1)$. So, $m + p + c = 1.35 = 27/20$

Hence, the correct option is (b).

11. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$P(A) + P(B) = P(A \cup B) + P(A \cap B)$

Using the given information, $P(A) + P(B) \geq (3/4 + 1/8)$

$\Rightarrow P(A) + P(B) \geq 7/8$

Hence, the correct option is (c).

12. If AM of 3 numbers is 25, then their sum must be 75. So, we have to choose 3 numbers from first 100 natural numbers such that their sum is 75. It can be done in ${}^{75+3-1}C_{3-1} = {}^{77}C_2$ ways. Total no. of ways of selecting 3 numbers from first 100 natural numbers is ${}^{100}C_3$. So, the required probability is ${}^{77}C_2/{}^{100}C_3$.
Hence, the correct option is (a).

13. Probability of showing 3 is P(3) = 3/(1 + 2 + 3 + 4 + 5 + 6) = 3/21 = 1/7

14. All of them reach the summit = 1/3 × 1/5 × ¼ = 1/60

15. To find out 'At least one of them reaches the summit', we will find the probability of no body reaching the summit = 2/3 × 4/5 × 3/4 = 2/5. Hence, the probability of at least one of them reaching the summit = 1 – 2/5 = 3/5

16. This is same as: Amit reaches the summit, but Vinit and Nishit are not reaching the summit or Vinit reaches the summit, but Amit and Nishit are not reaching the summit or Nishit reaches the summit, but Vinit and Amit are not reaching the summit.
Amit reaches the summit, but Vinit and Nishit are not reaching the summit = 1/3 × 4/5 × 3/4 = 1/5, and similarly, we will find all the other possibilities.

17. There is no way to connect the probability of Player A playing and team winning the match and probability of Player B playing and team winning the match. So, it cannot be determined.
Hence, the correct option is (C).

18. If Patel Nagar win at most 2 tosses, then it means it might have won 0, 1, or 2 tosses. Its probability of winning the match if it wins toss is 0.55, and so, its probability of winning match if it loses the toss is 0.45.

Probability of winning 0 toss and winning all 5 matches is ${}^5C_0 \times (0.45)^5$.

Probability of winning 1 toss and winning all 5 matches is ${}^5C_1 \times (0.45)^4 \times 0.55$.

Probability of winning 2 toss and winning all 5 matches is ${}^5C_2 \times (0.45)^3 \times (0.55)^2$.

So, probability of Patel Nagar winning all the matches if it wins at most two tosses

$= [{}^5C_0 \times (0.45)^5] + [{}^5C_1 \times (0.45)^4 \times 0.55] + [{}^5C_2 \times (0.45)^3 \times (0.55)^2] = 0.41$

Hence, the correct option is (b).

19. If Patel Nagar wins 3 tosses, then Boring Road must have won 2 tosses. Now, the probability of Boring Road

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winning the match if it wins the toss is 0.65, and so, the probability of Boring Road winning the match if it loses the toss is 0.35.

Probability of boring road winning all the matches, if Patel Nagar wins 3 tosses = ${}^5C_2 \times (0.65)^2 \times (0.35)^3 = 0.18$

Hence, the correct option is (d).

20. Let there be total 100 questions. We have 40 difficult and 60 easy questions. There are 20 difficult and 18 easy questions from algebra and geometry (both algebra and geometry are taken together). Now, if the student solved only 15 questions, then the total selection may be ${}^{38}C_{15}$, and no. of selection of choosing questions from only difficult questions is ${}^{20}C_{15}$. So, the required probability is $\frac{{}^{20}C_{15}}{{}^{38}C_{15}}$.

Hence, the correct option is (d).

23. Total number of triangles formed = ${}^8C_3 = 56$

Triangles having two sides common = 8

Triangles having one side common = ${}^8C_1 \times {}^4C_1 = 32$

Triangles having three sides common = 0

Triangles having no side common = 56 – 40 = 16

So, probability = 16/56 = 2/7

25. In a non-leap year, February has 28 days, and so, it must have each day of week exactly 4 times. Now, we know 7 months have 31 days and 4 months have 30 days.

If a month has 31 days and it has 5 Sundays, then it is possible for 5 Fridays, Saturdays, and Sundays, or 5 Saturdays, Sundays, and Mondays, or 5 Sundays, Mondays, and Tuesdays. So, the probability of having 5 Tuesdays if it has 5 Sundays is 2/3.

If a month has 30 days and it has 5 Sundays, then it is possible for 5 Saturdays and Sundays or 5 Sundays and Mondays. So, the probability of having 5 Tuesdays if it has 5 Sundays is 1/2.

So, if a month is selected randomly, then probability of having 5 Mondays if it has 5 Sundays must be:

$(7/11 \times 2/3) + (4/11 \times 1/2) = 20/33$

Hence, the correct option is (d).

ADVANCED

1. The average earning per throw for Manoj can be calculated by summing the multiplication of probability of showing up of each number and the earning it will result into. As, die is fair, the probability of showing of each number is 1/6.

Average earning per throw is $(1/6 \times 3) + (1/6 \times 4) + (1/6 \times 9) + (1/6 \times 8) + (1/6 \times 15) + (1/6 \times 12) = 8.5$

So, to earn average profit of ₹5 per throw, he must be willing to pay ₹3.5 per throw.

Hence, the correct option is (a).

2. As item selected in second draw is regularly sold item, and so, in first draw too, there must be the regularly sold item and it is also given that there is exactly one of three items which is regularly sold item. So, it could be either of type P, Q, or R.

Let the regularly sold item be P. Then, the probability of getting it in first draw is $a/(a + b + c)$. Now, the shopkeeper keeps an item of P again, and now, there are $2a$ items of P. Therefore, the probability of getting P again in second draw is $2a/(2a + b + c)$. So, total probability is $a/(a + b + c) \times 2a/(2a + b + c)$. Similarly, it is possible that the regularly sold item is Q or R. So, the required probability is as given in option (b).

Hence, the correct option is (b).

3. If P is regularly sold item and Q is selected in first draw, then the probability of selecting Q in first draw is $b/(a + b + c)$. Now, the shopkeeper withdraws all the items of type Q, and now, in second draw, we must draw R type of item and its probability is $c/(a + c)$. Similarly, other cases can be considered.

So, the required probability is not the ones shown in options (a), (b), and (c).

Hence, the correct option is (d).

4. If P is regularly sold item and selected in 1st draw then, the probability of selecting it in 1st draw is $a/(a + b + c)$. Now, the shopkeeper keeps items of type P more, and now, in 2nd draw, we must draw either Q or R type of item, and its probability is $(b + c)/(2a + b + c)$. Similarly, the item selected in 1st draw may be Q or R.

So, required probability is as shown in option (b).

Hence, the correct option is (b).

6. Probability of at least one tail turning up = 1 – probability of head in both tosses = $1 - (4/5 \times 65\%) = (1 - 0.8 \times 0.65) = 1 - 0.52 = 0.48 = 48\%$

Hence, the correct option is (c).

7. It is nothing but [1 – probability of bridge not getting destroyed] = 1 – 0.084

8. This problem can be thought of calculating the probability of selecting the black card if we have to select 1 card from a pack of 13 red and 26 black cards. So, the required probability is $26/(26 + 13) = 2/3$

Hence, the correct option is (c).

9. Contradiction can occur if A speaks truth and B lies, and B speaks truth and A lies.

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So, the required probability = 0.75 × 0.2 + 0.8 × 0.25 = 0.35

10. Required probability = probability that bolt was manufactured by A and it was defective/probability of bolt being defective

Required probability = (15% × 4%)/[(15% × 4%) + (25% × 2%) + (60% × 3%)]

= (0.15 × 0.04)/[(0.15 × 0.04) + (0.25 × 0.02) + (0.60 × 0.03)] = 0.006/(0.006 + 0.005 + 0.018)

= 0.006/0.029 = 6/29

Hence, the correct option is (a).

11. Required probability = probability that bolt was manufactured by B and it was defective/probability of bolt being defective

Required probability = (25% × 2%)/[(15% × 4%) + (25% × 2%) + (60% × 3%)] = (0.25 × 0.02)/[(0.15 × 0.04) + (0.25 × 0.02) + (0.60 × 0.03)] = 0.005/(0.006 + 0.005 + 0.018) = 0.005/0.029 = 5/29

Hence, the correct option is (b).

12. Required probability = probability that bolt was manufactured by C and it was defective/probability of bolt being defective

Required probability = (60% × 3%)/[(15% × 4%) + (25% × 2%) + (60% × 3%)]

= (0.60 × 0.03)/[(0.15 × 0.04) + (0.25 × 0.02) + (0.60 × 0.03)] = 0.018/(0.006 + 0.005 + 0.018) = 0.018/0.029 = 18/29

Hence, the correct option is (c).

13. The probability that it rains = 0.3, so the probability that it does not rain = 0.7.

14. The possible numbers are 29, 36, 63, and 92, and there are total 100 numbers. So, the probability of selection of a number whose product of digits is 18 is 4/100 = 0.04. If we are to select 4 numbers with replacement, then event 'E occurs at least 3 times' means it occurs 3 or 4 times.

Probability of E occurring 3 times = ${}^4C_3 \times (0.04)^3 \times 0.96$

Probability of E occurring 4 times = $(0.04)^4$

So, the required probability = $({}^4C_3 \times (0.04)^3 \times 0.96) + (0.04)^4 = 0.00025$

Hence, the correct option is (b).

15. 1 – Neither of them occurs = at least one of them occurs. So, 2/3 = P(A) + P(B) – P(A) × P(B). In this equation, there are two unknowns, namely P(A) and P(B). So, it cannot be determined.

16. Required probability = probability of getting 2 or 4/probability of getting an even number = probability of getting 2 or 4/probability of getting 2 or 4 or 6 = (0.24 + 0.18)/(0.24 + 0.18 + 0.14) = 0.42/0.56 = 3/4 = 0.75

Hence, the correct option is (c).

17. The given condition is satisfied for all numbers from 51 to 100. It is also satisfied for 50, 49, 48, 1, and 2. So, there are total 55 numbers from first 100 natural numbers for which the given condition is satisfied. Therefore, the required probability = 55/100 = 11/20

Hence, the correct option is (d).

18. Here, the probability of getting 1, 2, 3, 4, 5, and 6 are 1/21, 2/21, 3/21, 4/21, 5/21, and 6/21, respectively. The only multiple of 15 which is possible to obtain by rolling a die thrice is 15. It is possible by the following ways:

(i) Getting two 6 and one 3 with probability ${}^3C_1 \times (6/21)^2 \times 3/21$

(ii) Getting three 5 with probability $(5/21)^3$

(iii) by getting one 4, one 5, and one 6 with probability $6 \times (4/21) \times (5/21) \times (6/21)$

19. As the probability of any shell hitting the target is 1/2, and so, the probability of any shell missing the target is also 1/2. Let we fire n shells to Point 1 and remaining $21 - n$ to Point 2. Then, the probability of target being missed is $[(8/9 \times (1/2)^n] + [(1/9 \times (1/2)^{21-n}]$. We have to choose the value of n for which it is minimum, so that the probability of target being hit will be maximum. It is minimum for value of $n = 12$ with value 0.000434. So, to hit the target with maximum probability, we must hit 12 shells at Point 1.

Hence, the correct option is (d).

21. The probability that a man will not live 10 more years $= \frac{3}{4}$, and the probability that his wife will not live 10 more years $= \frac{2}{3}$

Hence, the probability that neither of them will be alive after 10 years $= \frac{3}{4} \times \frac{2}{3} = \frac{1}{2}$

23. As the probability of bomb hitting the bridge is 1/2, the probability of bomb not hitting the bridge (i.e., missing the bridge) is also 1/2. We will use n bombs to destroy the bridge, so that the probability of bridge not being destroyed is less than 90% (so that the probability of bridge being destroyed will be more than 90%). The probability of 2 bombs hitting the bridge and it still not destroyed is ${}^nC_2 \times (1/2)^2 \times (1/2)^{n-2}$. For the value of $n = 9$, this value first goes below to 10%, and hence, we must use minimum of 9 bombs.

Hence, the correct option is (b).

25. Let the probability of occurrence of odd numbers be x. Then, the probability of occurrence of any even number is $3x$. So, the probability of getting 1, 3, or 5 is x, & of getting 2, 4, or 6 is $3x$. As total probability must be 1. So, solving it, we get $x = 1/12$. Hence, probability of getting 1, 3, or 5 is 1/12, whereas probability of getting 2, 4, or 6 is 1/4.

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Now, if we throw die and we get total 10 or more, then it means that we get total 10, 11, or 12.

(i) We get 10 in 3 ways (4, 6), (5, 5), or (6, 4) with probability $(1/4 \times 1/4 + 1/12 \times 1/12 + 1/4 \times 1/4)$.

(ii) We can get 11 as (5, 6) or (6, 5) with probability $(1/12 \times 1/4 + 1/4 \times 1/12)$.

(iii) We can get 12 as (6, 6) with probability $(1/4 \times 1/4)$.

So, the required probability is $(1/4 \times 1/4 + 1/12 \times 1/12 + 1/4 \times 1/4) + (1/12 \times 1/4 + 1/4 \times 1/12) + (1/4 \times 1/4)$

$= 1/16 + 1/144 + 1/16 + 1/48 + 1/48 + 1/16) = 34/144 = 17/72$

Hence, the correct option is (b).

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BENCHMARKING TEST 3

This test paper contains 30 questions of 3 marks each. One-third of the mark allotted to a particular question will be deducted in case of wrong answer.

Time Given: 60 Minutes **Total Marks: 90** **Qualifying Marks: 41**

Break-up of the Expected Percentile

(assuming this to be a CAT paper)

Marks	Expected percentile
66+	99+
62	98.5+
58.5	98
41.25	92
30	85
22.5	80

Q.1 The first term, of an AP = the common ratio of a GP and the first term of the GP = common difference of the AP. If the sum of the first two terms of the GP is equal to the sum of the first 2 terms of the AP, then the ratio of first term of GP to the first term of AP is:
(a) >1
(b) Independent of first term of AP
(c) Independent of first term of GP
(d) <1

Q.2 In a college of 300 students, every student reads 5 newspapers and every newspaper is read by 60 students. The number of newspapers is:
(a) At least 30 (b) At most 20
(c) Exactly 25 (d) None of these

Q.3 The condition that both the roots of a quadratic equation $ax^2 + bx + c = 0$ are positive is
(a) a and c have opposite sign that of b.
(b) b and c have opposite sign that of a.
(c) a and b have opposite sign that of c.
(d) None of these

Q.4 The 4th degree polynomial equation $p(x) = x^4 - 2x^3 + 3x^2 - 4x + 5 = 0$ has:
(a) At least one real root
(b) No real root
(c) Two real roots
(d) All real roots

Q.5 Let $g(x)$ be a function such that $g(x + 1) + g(x - 1) = g(x)$ for every real x. Then, for what value of p is the relation $g(x + p) = -g(x)$ necessarily true for every real x?
(a) 5 (b) 3 (c) 2 (d) 6

Q.6 If f is real-valued function and $f(x + y) = f(y)\ \forall\ x \in R$.
Given $f(1) = 3$ and $\sum_{r=1}^{r=n} f(r) = 1092$. Find the value of n.
(a) $n = 5$ (b) $n = 6$
(c) $n = 7$ (d) None of these

Q.7 Let $f(x) = 0$ be the polynomial equation of the least possible degree with rational co-efficients. One of the roots of this equation is $\sqrt{7} + \sqrt[3]{49}$. What is the product of all the roots of $f(x) = 0$?
(a) 21 (b) 35 (c) 42 (d) 56

Q.8 If $f(x)$ is a real-valued function and $f\{x + f(y)\} = x + f(y)$, then $f(100)$ is:
(a) 100 (b) 1000 (c) 10000 (d) 100000

Q.9 The quadratic equation $px^2 + qx + r = 0$ satisfies the following conditions
(i) p, q, and r are distinct.
(ii) $p, q, r \{1, 2, 3, 4, 5, 6\}$
(iii) $(x + 1)$ is a factor of quadratic equation $px^2 + qx + r = 0$.
How many such quadratic equations are possible?

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(a) 8 (b) 12
(c) 16 (d) None of these

Q.10 Given that $f(x) = 4^x + 4^{-x}$ and $g(x) = 5/4$. At how many points both the functions will intersect?
(a) 0 (b) 1 (c) 2 (d) Infinite

Q.11 How many values of p ($p > 1$) satisfy the equation $\log_2 p.\log_4 p.\log_6 p = \log_2 p.\log_4 p + \log_2 p.\log_6 p + \log_4 p.\log_6 p$?
(a) 0 (b) 1
(c) 2 (d) More than 2

Q.12 If $\sum_{r=0}^{\infty} \frac{1}{(2r-1)^2} = N$, then which of the following is equal to $\sum_{r=0}^{\infty} \frac{1}{r^2}$?
(a) (3/4 N) – 1 (b) (4/3N) + 1
(c) 3/4(N + 1) (d) 4/3 (N – 1)

Q.13 Let p, q, r, and s be real numbers. If N = maximum [minimum(p, q) and minimum(r, s)] and S = minimum[maximum (p, r) and maximum(q, s)], then which of the following is true?
(a) $N \le S$, for all values of p, q, r, and s
(b) $N \ge S$, for all values of p, q, r, and s
(c) $N = S$, for all values of p, q, r, and s
(d) Cannot be uniquely determined

Q.14 The maximum possible value of $x^2 + 4y^2 + 9z^2$ is N. What is the value of N if $x + 2y + 3z = 12$ (x, y, and z are real numbers)?
(a) 224 (b) 16
(c) 4096 (d) Infinite

Q.15 x, y, z, and t are positive numbers that satisfy the following conditions:
(i) If $x > y$, then $z > t$.
(ii) If $x > z$, then $y < t$.
Then, which of the following is true?
(a) If $x < y$, then $z < t$.
(b) If $x < z$, then $y > t$.
(c) If $x > y + z$, then $z > y$.
(d) If $x > y + z$, then $z < y$.

Q.16 If $h(x)$ is the greatest integer less than or equal to $h/2$, then how many of the following statements is/are true?
(i) $h(x)$ is half the least even integer greater than or equal to h.
(ii) $h(x)$ is the least integer greater than or equal to $\frac{h}{2} - 1$.
(iii) $h(x)$ is one less than half the least even integer greater than h.
(a) 0 (b) 1 (c) 2 (d) 3

Q.17 The number of ways in which three distinct numbers in AP and GP can be selected from 1, 2, 3, …, 24 is A and G, respectively. Which of the following is the value of A:G?
(a) 21 (b) 12
(c) 9 (d) None of these

Q.18 $N_1 + N_2 + N_3 + \ldots + N_N = 100$, where N_1, N_2, N_3 … and N_N are $n(n > 1)$ consecutive natural numbers such that $N_1 < N_2 < N_3 < N_4 \ldots < N_N$. If $N_1 > 8$, then how many values of n is/are possible?
(a) 4 (b) 8
(c) 2 (d) Infinite

Q.19 If $P = \frac{10^{999} + 9.10^{998} + 9^2.10^{997} + \ldots + 9^{999}}{10^{10^3} - 9^{10^3}}$, then find the value of $\frac{(P+1)(P+100)}{P(P-2)}$?
(a) 1 (b) 101
(c) 202 (d) None of these

Q.20 Let $f(x)$ be a function such that $f(x - 1) + (x + 1) = \sqrt{2}\, f(x)$. Then, for what value of y is the relation $f(x + y) = f(x)$ necessarily true for every real x?
(a) 4 (b) 6 (c) 8 (d) 12

Q.21 The sum of all the real roots of the equations $|x - 2|^2 + |x - 2| - 2 = 0$ is:
(a) 2 (b) –4 (c) –2 (d) 4

Q.22 Let p and q be roots of the equation $x^2 - 2x + A = 0$ and let r and s be the roots of the equations $x^2 - 18x + B = 0$. If $p < q < r < s$ are in an arithmetic progression, then what is the value of A + B?
(a) –77 (b) 74 (c) –74 (d) 77

Q.23 Let 't' be any root of the equation $x^n + x^{n-1} + x^{n-2} + \ldots + 1 = 0$. Then, $(t^{2n+2} + 3)(t^{3n+3} - 4)$ equals:
(a) 12 (b) –12 (c) 6 (d) –6

Q.24 An n-digit number is a positive number with exactly n digits. Nine hundred distinct n-digit numbers are to be formed using only the three digits 2, 5, and 7. The smallest value of n for which this is possible is:
(a) 6 (b) 7 (c) 8 (d) 9

Q.25 Let $a_1, a_2 \ldots, a_{10}$ be in AP and $h_1, h_2 \ldots, h_{10}$ be in HP. If $a_1 = h_1 = 2$ and $a_{10} = h_{10} = 3$, then $a_4 h_7$ is:
(a) 2 (b) 3 (c) 5 (d) 6

Q.26 If the integers m and n are chosen at random between 1 and 100, then the probability that a number of the form $7^m + 7^n$ is divisible by 5 equals:
(a) $\frac{1}{4}$ (b) $\frac{1}{7}$ (c) $\frac{1}{8}$ (d) $\frac{1}{49}$

Q.27 Let $S_1, S_2 \ldots$ be squares such that for each $n \ge 1$, the length of a side of S_n equals the length of a diagonal of

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S_{n+1}. If the length of a side of S_1 is 10 cm^2, then for how many values of n is the area of S_n less than 1 cm^2?

(i) 7 (ii) 8 (iii) 9 (iv) 10

(a) (i) (b) (ii) (c) (iii) (d) (iv)

Q.28 If a, b, c, and d are positive real numbers such that $a + b + c + d = 2$, then $M = (a + b)(c + d)$ satisfies the relation:

(a) $0 \le M \le 1$ (b) $1 \le M \le 2$

(c) $2 \le M \le 3$ (d) $3 \le M \le 4$

Q.29 How many different nine-digit numbers can be formed from the number 223,355,888 by rearranging its digits so that the odd digits occupy even positions?

(a) 16 (b) 36 (c) 60 (d) 180

Q.30 The area bounded by the curves $y = |x| - 1$ and $y = -|x| + 1$ is:

(a) 1 (b) 2 (c) $2\sqrt{2}$ (d) 4

Answers

Benchmarking Test 3

1. (c)	**2.** (c)	**3.** (a)	**4.** (b)	**5.** (b)	**6.** (b)	**7.** (d)	**8.** (a)	**9.** (b)	**10.** (a)
11. (b)	**12.** (d)	**13.** (a)	**14.** (d)	**15.** (c)	**16.** (d)	**17.** (b)	**18.** (c)	**19.** (d)	**20.** (c)
21. (d)	**22.** (b)	**23.** (b)	**24.** (b)	**25.** (d)	**26.** (a)	**27.** (c)	**28.** (a)	**29.** (c)	**30.** (b)

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MODULE

4 MEASUREMENT

- Geometry
- Mensuration
- Co-ordinate Geometry

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CHAPTER

18

Geometry

LEARNING OBJECTIVES

After completion of this chapter, the reader should be able to understand:

- Triangles, polygons, circles, and quadrilaterals
- Different theorems and their applications
- Similarity and conguency
- Kinds of questions asked in the CAT
- Methods of solving questions

INTRODUCTION

Over the years, geometry has become as important as the number system. In other words, it is one of those areas that can help students in getting good marks in QA on its own. In CAT, questions for close to 20% marks are asked from geometry alone. However, this is only the secondary reason of this topic being so significant; the primary reasons being (i) creation of level-playing field for everyone, as all the questions asked from this chapter pertain to close to Level 10 Mathematics only and (ii) most of the questions can be solved by the method of measurement of simple construction, without using any theorem of geometry.

Geometry deals with measures and properties of points, lines, surfaces, and solids. Here, we will discuss the following:

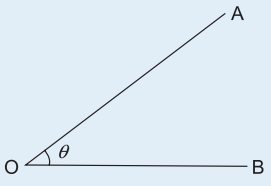
1. Lines and their properties
2. Polygons and their properties
3. Triangles and their properties
4. Quadrilaterals and their properties
5. Circles and their properties

LINES AND THEIR PROPERTIES

A line is a set of points placed together that extends into infinity in both directions.

Different Angles and Pairs of Angles

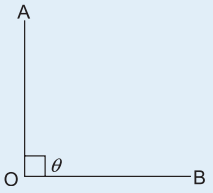
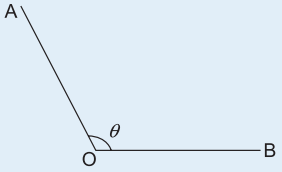
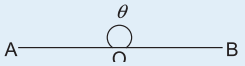
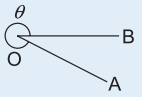
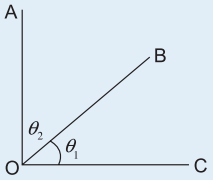
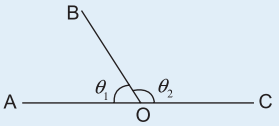
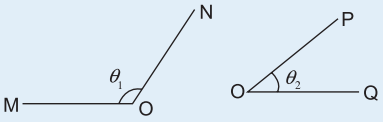
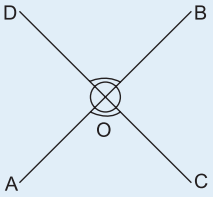
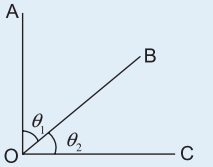
Measurement and nomenclature

Types of angles	Property	Diagram
Acute angle	$0° < \theta < 90°$ ($\angle AOB$ is an acute angle.)	

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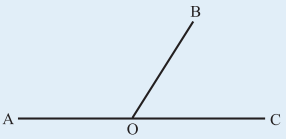
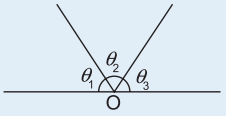
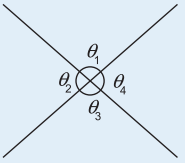
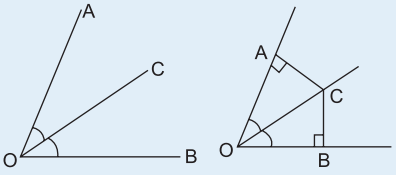
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Types of angles	Property	Diagram
Right angle	$\theta = 90°$	
Obtuse angle	$90° < \theta < 180°$	
Straight line	$\theta = 180°$	
Reflex angle	$180° < \theta < 360°$	
Complementary angle	$\theta_1 + \theta_2 = 90°$ Two angles, whose sum is 90°, are complementary to each other.	
Supplementary angle	$\theta_1 + \theta_2 = 180°$ Two angles, whose sum is 180°, are supplementary to each other.	 
Vertically opposite angle	∠DOA = ∠BOC and ∠DOB = ∠AOC	
Adjacent angles	∠AOB and ∠BOC are adjacent angles. Adjacent angles must have a common side (e.g., OB).	

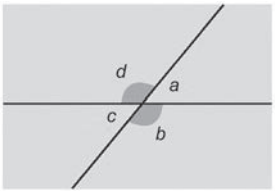
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Types of angles	Property	Diagram
Linear pair	∠AOB and ∠BOC are linear pair angles. One side must be common (e.g., OB), and these two angles must be supplementary.	
Angles on one side of a line	$\theta_1 + \theta_2 + \theta_3 = 180°$	
Angles round the point	$\theta_1 + \theta_2 + \theta_3 + \theta_4 = 360°$	
Angle bisector	OC is the angle bisector of ∠AOB, that is ∠AOC = ∠BOC $\frac{1}{2}$ > (∠AOB).	 Angle bisector is equidistant from the two sides of the angle, that is when a line segment divides an angle equally into two parts, it is said to be the angle bisector (OC).

Angles Associated with Two or More Straight Lines

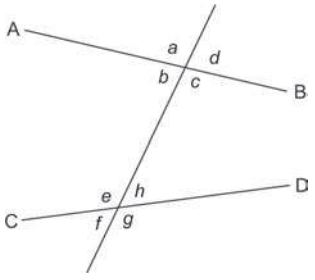


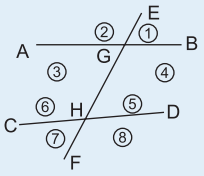
When two straight lines cross each other, ∠*d* and ∠*b* are pair of vertical angles, and ∠*a* and ∠*c* are pair of vertical angles.

Vertical angles are equal in value.

Alternate Angles and Corresponding Angles

In the figure given below, corresponding angles are ∠*a* and ∠*e*, ∠*b*, and ∠*f*, ∠*d*, and ∠*h*, and ∠*c*, and ∠*g*. The alternate angles are ∠*b* and ∠*h*, and ∠*c* and ∠*e*.



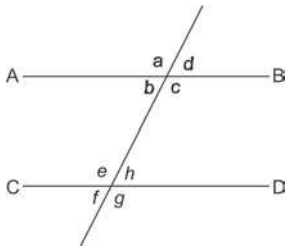
Corresponding angles	When two lines are intersected by a transversal, they form four pairs of corresponding angles: (a) ∠AGE, ∠CHG = (∠2, ∠6) (b) ∠AGH, ∠*CHF* = (∠3, ∠7) (c) ∠EGB, ∠GHD = (∠1, ∠5) (d) ∠BGH, ∠DHF = (∠4, ∠8)	

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Angles Associated with Parallel Lines

A line passing through two or more lines in a plane is called a transversal. When a transversal cuts two parallel lines, the set of all the corresponding angels will be equal, and similarly, the set of all the alternate angles will be equal.



In the figure given above, corresponding $\angle a = \angle e$ and corresponding $\angle b = \angle f$.

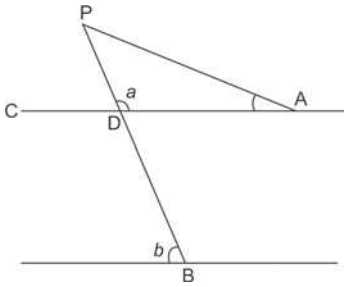
Similarly, alternate $\angle b = \angle h$ and alternate $\angle c = \angle e$.

Now, $\angle b + \angle c = 180°$; so, $\angle b + \angle e = \angle h + \angle c = 180°$

So, we can conclude that the sum of the angles on one side of the transversal and between the parallel lines will be equal to 180°.

Converse of the above theorem is also true. When a transversal cuts two lines, and if the corresponding angles are equal in size, or if alternate angles are equal in size, then the two lines are parallel.

Example 1 In the figure given below, find the value of $\angle b$ in terms of $\angle a$.



Solution In the given figure, $\angle b$ = corresponding $\angle PDC = 180° - \angle PDA = 180° - a$

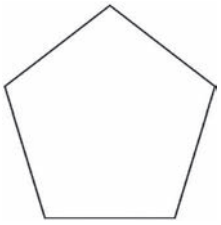
POLYGONS AND THEIR PROPERTIES

Any closed plane figure with n sides is known as polygon. If all the sides and the angles of this polygon are equivalent, then the polygon is called regular polygon. Polygons can be convex or concave. The word 'polygon' derives from the Greek, *poly* meaning 'many,' and *gonia* meaning 'angle.' The most familiar type of polygon is the regular polygon, which is a convex polygon with equal side lengths and angles.

The generalization of a polygon into three dimensions is called *polyhedron* and into four dimensions is called *polychoron*.

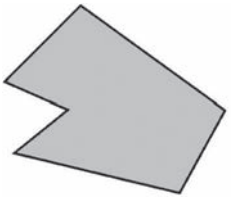
A **convex polygon** is a simple polygon that has the following features:

- Every internal angle is at most 180°.
- Every line segment between the two vertices of the polygon does not go outside the polygon (i.e., it remains inside or on the boundary of the polygon). In other words, all the diagonals of the polygon remain inside its boundary.
- Every triangle is strictly a convex polygon.



Convex polygon

If a simple polygon is not convex, it is called **concave**. At least one internal angle of a concave polygon is larger than 180°.



Concave polygon

As we can see in the above figure, one of the internal angles is more than 180°.

Here onwards, all the discussions about polygon refer to regular polygons only.

Polygons are named on the basis of the number of sides they have. A list of some of the polygons are given below:

Number of sides	Name of the polygon
3	Triangle
4	Quadrilateral
5	Pentagon

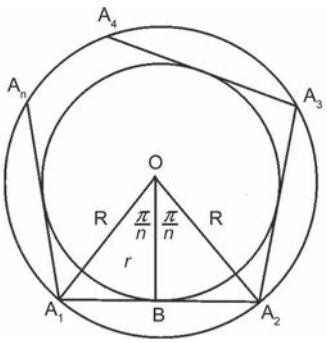
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Number of sides	Name of the polygon
6	Hexagon
7	Heptagon
8	Octagon
9	Nonagon
10	Decagon

Area and Perimeter of a Regular Polygon



Given $A_1, A_2, A_3, A_4, \ldots, A_n$ is a regular polygon with 'n' sides.

$A_1A_2 = A_2A_3 = A_3A_4 = \ldots = A_{(n-1)}A_n = a$ units

OB (in radius) $= r$ and $OA_1 = OA_2$ (Circumradius) $= R$

(i) Perimeter $(p) = na$

(ii) Area $= \frac{na}{2} \times r = \frac{p}{2} \times r$

(iii) Area $= \frac{na}{2}\sqrt{R^2 - \left(\frac{a}{2}\right)^2} = \frac{p}{2}\sqrt{R^2 - \left(\frac{a}{2}\right)^2}$

(iv) Area $= \frac{na^2}{4} \times \cot\left(\frac{\pi}{n}\right)^\circ$

(v) Area $= nr^2 \times \tan\left(\frac{\pi}{n}\right)^\circ$

(vi) r (in radius) $\frac{a}{2}\cot\left(\frac{\pi}{n}\right)^\circ$

Properties of a Polygon

1. Interior angle + Exterior angle = 180°
2. The number of diagonals in an n-sided polygon $= n(n-3)/2$
3. The sum of all the exterior angles of any polygon = 360°
4. The measure of each exterior angle of a regular polygon $= \frac{360^\circ}{n}$
5. The ratio of the sides of a polygon to the diagonals of a polygon is $2:(n-3)$
6. The ratio of the interior angle of a regular polygon to its exterior angle is $(n-2):2$
7. The sum total of all the interior angles of any polygon $= (2n-4) \times 90^\circ$

   So, each interior angle in a regular polygon $= \frac{(2n-4)90^\circ}{n}$

Some Frequently Used Polygons

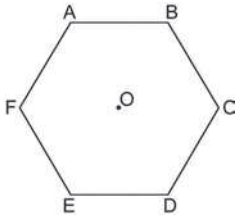
Apart from triangles and quadrilaterals, regular hexagon and regular octagon are also worth mentioning.

Regular Hexagon

In the figure given below, ABCDEF is a regular hexagon with each side measuring a unit. Point O inside the hexagon is the centre of the hexagon.

Sum of the interior angles = 720°
Each interior angle = 120°
Each exterior angle = 60°

Area $= \frac{3\sqrt{3}}{2}a^2$

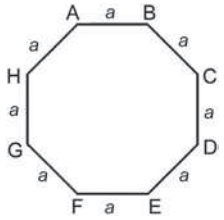


Regular Octagon

In the figure given below, ABCD-EFGH is a regular octagon with each side measuring 'a' unit.

Sum of the interior angles = 1080°
Each interior angle = 135°
Each exterior angle = 45°

Area $= 2a^2(1+\sqrt{2})$

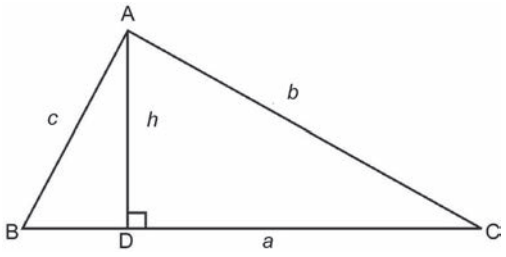


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TRIANGLES AND THEIR PROPERTIES

A triangle is a figure enclosed by three sides. In the figure given below, ABC is a triangle with sides AB, BC, and CA measuring c, a, and b units, respectively. Line AD represents the height of the triangle corresponding to the side BC and is denoted by h.



In any triangle ABC,

$$\text{Area} = \frac{1}{2} \times \text{BC} \times \text{AD} = \frac{1}{2}\ a \times h$$

Properties of a Triangle

- The sum of all the angles of a triangle = 180°
- The sum of lengths of the two sides > length of the third side
- The difference of any two sides of any triangle < length of the third side
- The area of any triangle can be found by several methods:

  (a) Area of any triangle $= \frac{1}{2} \times$ base $\times$ perpendicular to base from the opposite vertex.

  (b) Area of any triangle $= \sqrt{s(s-a)(s-b)(s-c)}$, where s is the semi-perimeter of the triangle and a, b, and c are the sides of a triangle.

  (c) Area of any triangle $= \left(\frac{1}{2}\right) \times$ bc Sin A

  Besides, there are some formulae that we use exclusively in some particular cases.

Example 2 What is the number of distinct triangles with integral valued sides and perimeter as 14?

(a) 6 (b) 5
(c) 4 (d) 3

Solution The sum of the lengths of the two sides > the length of the third side

So, the maximum length of any particular side can be 6 units.

Now, if $a = 6$, then $b + c = 8$, then the possible sets are (6, 6, 2), (6, 5, 3), and (6, 4, 4).

If $a = 5$, then $b + c = 9$, so the possible set is (5, 5, 4).

So, the number of distinct triangles = 4

CLASSIFICATION OF TRIANGLES

Based Upon Sides

1. *Scalene Triangle*

A triangle whose all sides are of different lengths is a scalene triangle.

$$\text{Area} = \sqrt{s-(s-a)(s-b)(s-c)}, \text{ where}$$

$$S \text{ (semi-perimeter)} = \frac{a+b+c}{2}$$

Example 3 What is the area of the triangle with side lengths 4 units, 5 units, and 10 units?

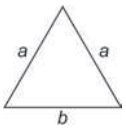
Solution This triangle is not possible, as the sum of lengths of the two sides > length of the third side.

2. *Isosceles Triangle*

A triangle whose two sides are of equal length is an isosceles triangle.

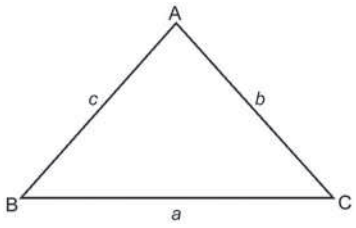
$$\text{Height} = \frac{\sqrt{4a^2 - b_2}}{2}$$

$$\text{Area} = \frac{b}{4}\sqrt{4a^2 - b^2}$$



3. *Equilateral Triangle*

A triangle whose all sides are of equal length is called an equilateral triangle.



In any equilateral triangle, all the three sides are of equal length, so $a = b = c$.

$$\text{Height} = (\text{side})\ \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{2}a$$

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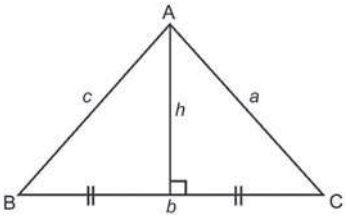
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$$\text{Area} = \frac{\sqrt{3}}{4}(\text{side})^2 = \frac{\sqrt{3}}{4}a^2$$

Based Upon Angles

1. *Right-angled Triangle*

A triangle whose one angle is of 90° is called a right-angled triangle. The side opposite to the right angle is called the hypotenuse.



$$\text{Area} = \frac{1}{2} \times \text{base} \times \text{perpendicular}$$

Pythagoras Theorem

Pythagoras theorem is applicable in case of right-angled triangle. It says that the square of the hypotenuse is equal to the sum of the squares of the other two sides.

$$(\text{Hypotenuse})^2 = (\text{Base})^2 + (\text{Perpendicular})^2$$

$$a^2 + b^2 = c^2$$

The smallest example is $a = 3$, $b = 4$, and $c = 5$. You can check that

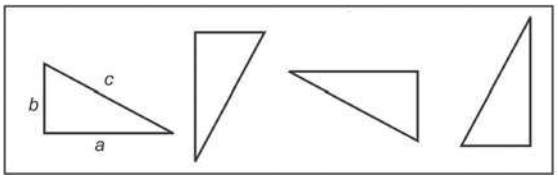
$$3^2 + 4^2 = 9 + 16 = 25 = 5^2$$

Sometimes, we use the notation (a, b, c) to denote such a triple.

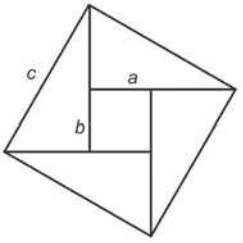
Notice that the greatest common divisor of the three numbers 3, 4, and 5 is 1. Pythagorean triples with this property are called *primitive*.

Proofs of Pythagoras theorem

Proof 1



Now, we start with four copies of the same triangle. Three of these have been rotated at 90°, 180°, and 270°, respectively. Each has the area $ab/2$. Let us put them together without additional rotations so that they form a square with side c.



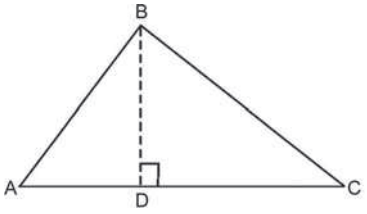
The square has a square hole with the side $(a - b)$. By summing up its area $(a - b)^2$ and $2ab$, the area of the four triangles $(4 \cdot ab/2)$, we get

$$C^2 = (a - b)^2 + 2ab = a^2 + b^2. \text{ QED}$$

Proof 2

ABC is a right-angled triangle at B

To Prove: $AC^2 = AB^2 + BC^2$



Construction: Draw BD ⊥ AC

Proof: ΔADB ~ΔABC (Property 8.5)

$$\therefore \frac{AD}{AB} = \frac{AB}{AC} \text{ (Sides are proportional)}$$

$$\text{Or } AB^2 = AD \times AC \quad \text{(i)}$$

Also, ΔCDB ~ΔCBA

$$\therefore \frac{CD}{BC} = \frac{BC}{CA}$$

$$\therefore \frac{BC^2}{AC^2} = \frac{BD \times BA}{AB \times AD} \text{ (Sides are proportional)}$$

$$\text{Or, } BC^2 = CD \times CA \quad \text{(ii)}$$

Adding (i) and (ii)

$$AB^2 + BC^2 = AD \times AC + CD \times CA$$
$$= AC\,[AD + CD]$$
$$= AC \times AC = AC^2$$

Pythagorean Triplets

A *Pythagorean triplet* is a set of three positive whole numbers a, b, and c that are the lengths of the sides of a right triangle.

$$a^2 + b^2 = c^2$$

It is noteworthy to see here that all of a, b, and c cannot be odd simultaneously. Either of a or b has to be even and c can be odd or even.

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The various possibilities for a, b and c are tabled below:

a	b	c
Odd	Odd	Even
Even	Odd	Odd
Odd	Even	Odd
Even	Even	Even

Some Pythagoras triplets are:

3	4	5	$(3^2 + 4^2 = 5^2)$
5	12	13	$(5^2 + 12^2 = 13^2)$
7	24	25	$(7^2 + 24^2 = 25^2)$
8	15	17	$(8^2 + 15^2 = 17^2)$
9	40	41	$(9^2 + 40^2 = 41^2)$
11	60	61	$(11^2 + 60^2 = 61^2)$
20	21	29	$(20^2 + 21^2 = 29^2)$

Note: If each term of any pythagorean triplet is multiplied or divided by a constant (say, P, $P > 0$), then the triplet so obtained will also be a Pythagorean triplet. This is because if $a^2 + b^2 = c^2$, then $(Pa)^2 + (Pb)^2 = (Pc)^2$, where $P > 0$.

For example, 3×2 4×2 5×2 gives
6 8 10 $(6^2 + 8^2 = 10^2)$

Using Pythagoras theorem to determine the nature of triangle If $c^2 = a^2 + b^2$, then the triangle is right-angled triangle.

If $c^2 > a^2 + b^2$, then the triangle is an obtuse-angled triangle.

If $c^2 < a^2 + b^2$, then the triangle is an acute-angled triangle.

Mechanism to derive a Pythagorean triplet If the length of the smallest side is odd, assume the length of the smallest side = 5

Step 1 Take the square of 5 (length of the smallest side) = 25

Step 2 Break 25 into two parts P and Q, where $P - Q = 1$. In this case, $P = 13$ and $Q = 12$. Now, these two parts P and Q along with the smallest side constitute pythagorean triplet.

However, there is another general formula for finding out all the primitive pythagorean triplets:

$a = r^2 - s^2$
$b = 2\,rs$
$c = r^2 + s^2$
$r > s > 0$ are whole numbers
$r - s$ is odd
The greatest common divisor of r and s is 1.

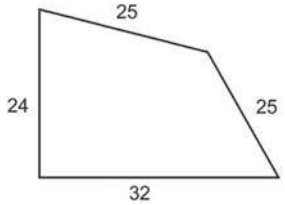
Table of small primitive Pythagorean triplets Here is a table of the first few primitive Pythagorean triplets:

r	s	a	b	c
2	1	3	4	5
3	2	5	12	13
4	1	15	8	17
4	3	7	24	25
5	2	21	20	29
5	4	9	40	41
6	1	35	12	37
6	5	11	60	61
7	2	45	28	53

Perimeter, area, inradius, and shortest side The perimeter P and area K of a Pythagorean triple triangle are given by:

$P = a + b + c = 2r(r + s)\ d$
$K = ab/2 = rs\ (r^2 - s^2)\ d^2$

Example 4 Two sides of a plot measure 32 m and 24 m and the angle between them is a right angle. The other two sides measure 25 m each and the other three angles are not right angles.

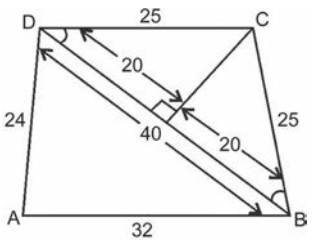


What is the area of the plot (in m^2)?
(a) 768 (b) 534 (c) 696 (d) 684

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Solution The figure given above can be seen as



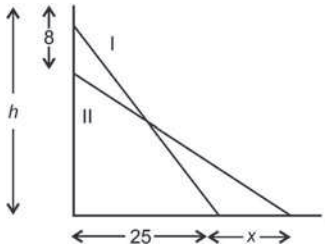
Since ABD is a right-angled triangle, so it will satisfy the Pythagoras theorem. And the triplet used here is –3(×8), 4(×8), and 5(×8). Similarly, the other part of the figure can also be bifurcated by drawing a perpendicular from C on BD.

So, the area of the plot is:

Area (ΔABD) + Area (ΔCBD) $= \frac{1}{2} \times 24 \times 32 + 2 \times (\frac{1}{2} \times 20 \times 15) = 684 \text{ m}^2$

Example 5 A ladder of length 65 m is resting against a wall. If it slips 8 m down the wall, then its bottom will move away from the wall by N m. If it was initially 25 m away from it, what is the value of x?

(a) 60 m (b) 39 m (c) 14 m (d) 52 m



Solution Using Pythagorean triplets, (5, 12, 13), $\Rightarrow h = 60$

After it has slipped by 8 m, the new height = 52 m, and the length of the ladder = 65 m.

So, $25 + x = 39$ (3, 4, 5 triplet)

$\Rightarrow x = 14$ m

2. Obtuse-angled Triangle

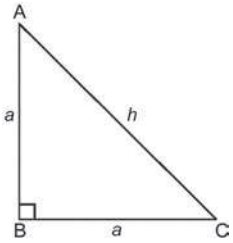
If one of the angles of the triangle is more than 90°, then the triangle is known as an obtuse angled triangle. Obviously, in this case, rest of the two angles will be less than 90°.

3. Acute-angled Triangle

If all the angles of the triangle are less than 90°, then the triangle is known as acute angled triangle.

4. Isosceles Right-angled Triangle

A right-angled triangle, whose two sides containing the right angle are equal in length, is an isosceles right triangle.



In this case, Hypotenuse $(h) = a\sqrt{2}$

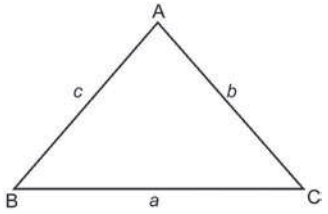
Perimeter $= 2a + h = 2a + a\sqrt{2}$

$= a\sqrt{2}(\sqrt{2} + a)$

$= h(1 + \sqrt{2})$

$=$ Hypotenuse $(1 + \sqrt{2})$

Trigonometric formulae In any Δ ABC,



Area of a triangle

Area of $\Delta = \frac{1}{2} bc \text{ Sin } A$, where $\angle A = \text{BAC}$

Area of $\Delta = \frac{1}{2} ac \text{ Sin } B$, where $\angle B = \text{ABC}$

Area of $\Delta = \frac{1}{2} ab \text{ Sin } C$, where $\angle C = \text{ACB}$

Cosine Rule and Sine Rule In any Δ, we have six quantities namely the three angles and the three sides. Using the following rules, we can find any of the three quantities if we are provided with the remaining three quantities.

$$\text{Cos } A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\text{Cos } B = \frac{a^2 + c^2 - b^2}{2ac}$$

$$\text{Cos } C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$\text{Sine Rule } \frac{a}{\text{Sin } A} = \frac{b}{\text{Sin } B} = \frac{c}{\text{Sin } C}$$

Sine and Cosine formulae are particularly more important in cases where we have one side and two angles out of the three angles of the triangle and we have to find out the value of all the sides and angles.

$$\frac{\text{AB}}{\sin \angle \text{ACB}} = \frac{\text{BC}}{\sin \angle \text{BAC}} = \frac{\text{AC}}{\sin \angle \text{ABC}} = 2R$$

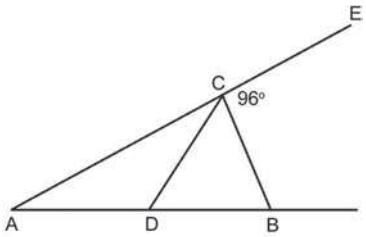
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Two important points

- The side opposite to the largest angle will be the largest.
- The side opposite to the smallest angle will be the smallest.

Example 6 In the figure given below, AD = CD = BC. What is the value of ∠CDB?

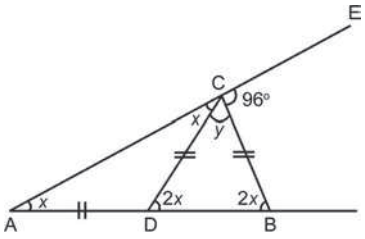


Solution $x + y = 180 - 96 = 84°$ (i)

In ΔCDB, $4x + y = 180$ (ii)

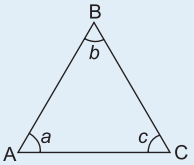
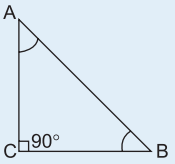
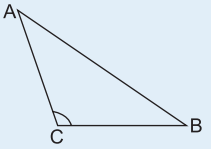
Solving (i) and (ii), $x = 32°$

So, $2x = 64°$

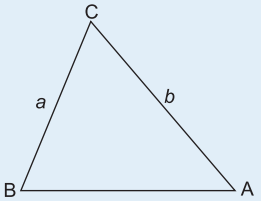


Summarizing the above Classification

(a) According to the measurement of angle

Types of triangles	Property/Definition	Diagram
Acute-angled triangle	Each angle of a triangle is less than 90°, that is $a < 90°$, $b < 90°$, $c < 90°$	 $\{\angle a, \angle b, \angle c\} < 90°$
Right-angled triangle	If one of the angles is equal to 90°, then it is called a right-angled triangle. The rest two angles are complementary to each other.	 $\angle C = 90°$
Obtuse-angled triangle	If one of the angles is obtuse (i.e., greater than 90°), then it is called an obtuse-angled triangle.	 $\angle C > 90°$

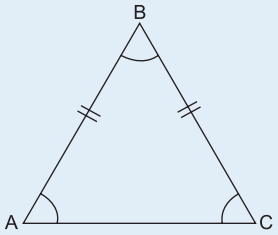
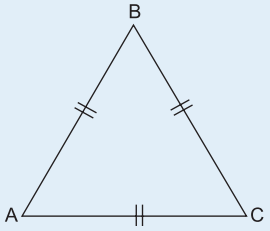
(b) According to the length of sides

Types of triangles	Property/Definition	Diagram
Scalene triangle	A triangle in which none of the three sides are equal is called a scalene triangle. In this triangle, all the three angles are also different.	 $a \neq b \neq c$

(*Continued*)

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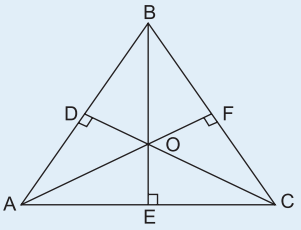
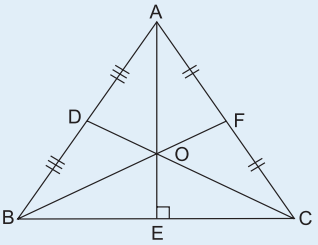
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Types of triangles	Property/Definition	Diagram
Isosceles triangle	A triangle in which two sides are equal is called an isosceles triangle. In this triangle, the angles opposite to the congruent sides are also equal.	 $AB = BC$ $\angle A = \angle C$
Equilateral triangle	A triangle in which all the three sides are equal is called an equilateral triangle. In this triangle, each angle is congruent and equal to 60°.	 $AB = BC = AC$, $\angle A = \angle B = \angle C = 60°$

Points of a Triangle

Before we move ahead to discuss different points inside a triangle, we need to be very clear about some of the basic definitions.

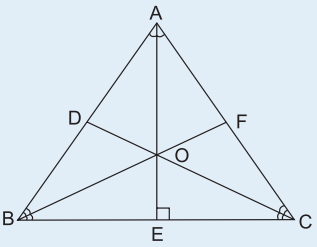
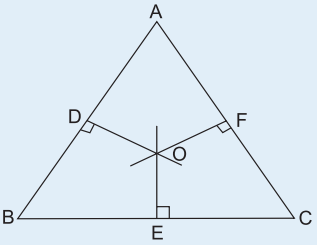
Basic Definitions

Nomenclature	Property/Definition	Diagram
Altitude (or height)	The perpendicular drawn from the opposite vertex of a side in a triangle is called an altitude of the triangle. There are three altitudes in a triangle.	 AF, CD, and BE are the altitudes.
Median	The line segment joining the mid-point of a side to the vertex opposite to the side is called a median. There are three medians in a triangle. A median bisects the area of the triangle. Area (ABE) = Area (AEC) = $\frac{1}{2}$ Area (ΔABC), etc.	 AE, CD, and BF are the medians. (BE = CE = AD = BD = AF = CF)

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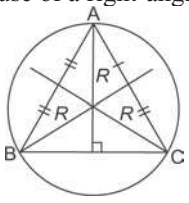
Nomenclature	Property/Definition	Diagram
Angle bisector	A line segment that originates from a vertex and bisects the same angle is called an angle bisector. ($\angle BAE = \angle CAE = \frac{1}{2}\angle BAC$), etc.	 AE, CD, and BF are the angle bisectors.
Perpendicular bisector	A line segment which bisects a side perpendicularly (i.e., at right angle) is called a perpendicular bisector of a side of triangle. All points on the perpendicular bisector of a line are equidistant from the ends of the line.	 DO, EO, and FO are the perpendicular bisectors.

Circumcentre

Circumcentre is the point of intersection of the three perpendicular bisectors of a triangle. The circumcentre of a triangle is equidistant from its vertices and the distance of the circumcentre from each of the three vertices is called circumradius (R) of the triangle. These perpendicular bisectors are different from altitudes, which are perpendiculars but not necessarily bisectors of the side.

The circle drawn with the circumcentre as the centre and circumradius as the radius is called the circumcircle of the triangle and it passes through all the three vertices of the triangle.

The circumcentre of a right-angled triangle is the mid-point of the hypotenuse of a right-angled triangle.



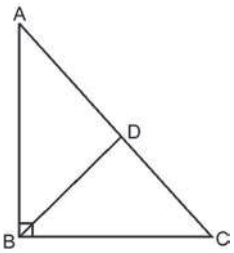
AB = c, BC = a, AC = b

The process to find the circumradius (R) For any triangle $R = \frac{abc}{4A}$, where a, b, and c are the three sides, and A = area of a triangle.

For equilateral triangle, $R = \frac{\text{Side}}{\sqrt{3}}$

Positioning of the Circumcentre

- If the triangle is acute-angled triangle, then the circumcentre will lie inside the triangle.
- If the triangle is obtuse-angled triangle, then the circumcentre will lie outside the triangle.
- If the triangle is a right-angled triangle, then the circumcentre will lie on the mid-point of the hypotenuse. This can be seen through the following diagram:



Here, D is the circumcentre. So, $AD = CD = BD$

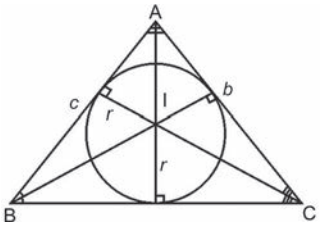
Incentre

Incentre is the point of intersection of the internal bisectors of the three angles of a triangle. The incentre is equidistant from the three sides of the triangle, that is the perpendiculars drawn from the incentre to the three sides are equal in length and are called the inradius of the triangle.

The circle drawn with incentre as the centre and inradius as the radius is called the incircle of the triangle and it touches all the three sides from the inside.

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AB = c, BC = a, CA = b

To find inradius (r)

For any triangle $r = \frac{A}{S}$, where

A = Area of triangle and

S = Semi-perimeter of the triangle $\frac{(a+b+c)}{2}$

For equilateral triangle, $r = \frac{\text{side}}{2\sqrt{3}}$

$\angle BIC = 90° + \angle A/2$

Important derivation In a right-angled triangle, Inradius = Semiperimeter – length of Hypotenuse.

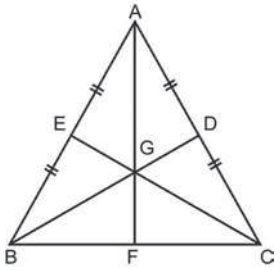
Euler's formula for inradius and circumradius of a triangle Let O and I be the circumcentre and incentre of a triangle with circumradius R and inradius r. Let d be the distance between O and I. Then

$d^2 = R(R - 2r)$

From this theorem, we obtain the inequality $r \geq 2r$. This is known as Euler's inequality.

Centroid

Centroid is the point of intersection of the three medians of a triangle. The centroid divides each of the medians in the ratio 2:1, the part of the median towards the vertex being twice in length to the part towards the side.



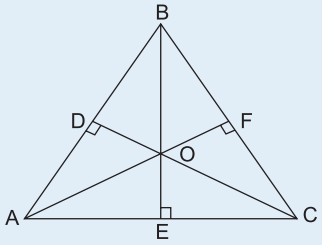
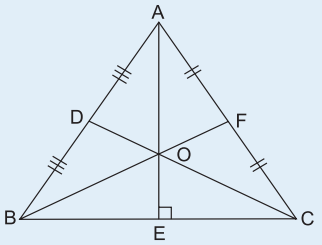
$$\frac{AG}{GF} = \frac{BG}{GD} = \frac{CD}{GE} = \frac{2}{1}$$

Median divides the triangle into two equal parts of the same area.

Orthocentre

The point of concurrency of the altitudes is known as the orthocentre.

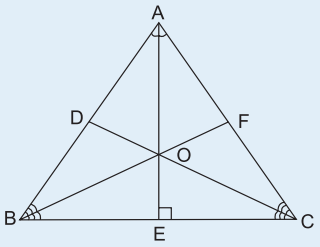
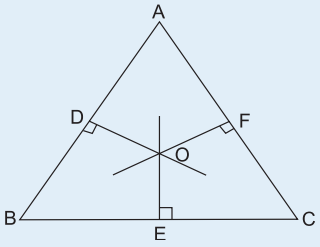
Summarizing the above discussion regarding the points of the triangle:

Orthocentre	The point of intersection of the three altitudes of the triangle is known as the orthocentre. $\angle BOC = 190 - \angle A$ $\angle COA = 190 - \angle B$ $\angle AOB = 180 - \angle B$	 'O' is the orthocentre
Centroid	The point of intersection of the three medians of a triangle is called the centroid. A centroid divides each median in the ratio 2:1 (vertex: base) $\frac{AO}{OE} = \frac{CO}{OD} = \frac{BO}{OF} = \frac{2}{1}$	 'O' is the centroid

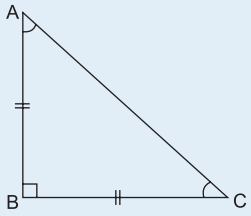
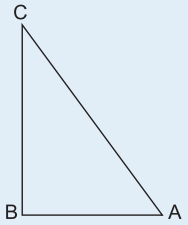
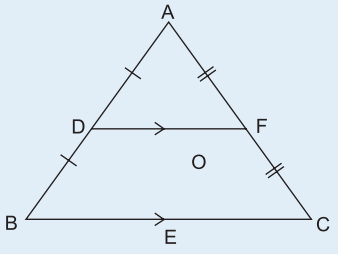
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Incentre	The point of intersection of the angle bisectors of a triangle is known as the incentre. Incentre O is the always equidistant from all three sides, that is the perpendicular distance between the sides.	 'O' is the incentre
Circumcentre	The point of intersection of the perpendicular bisectors of the sides of a triangle is called the circumcentre. OA = OB = OC = (circum radius). Circumcentre O is always equidistant from all the three vertices A, B, and C perpendicular bisectors need not be originating from the vertices.	 'O' is the circumcentre

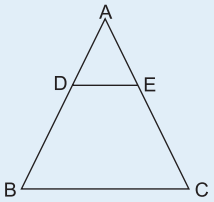
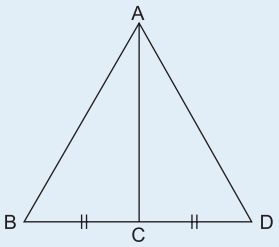
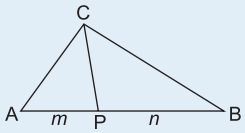
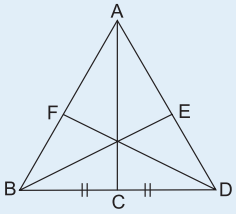
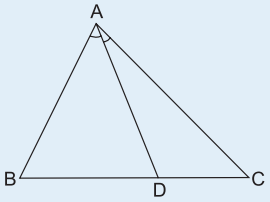
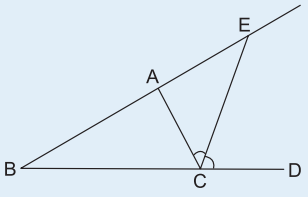
Important Theorems Related to Triangle

Theorem	Statement/Explanation	Diagram
45° – 45° – 90°	If the angles of a triangle are 45°, 45°, and 90°, then the hypotenuse (i.e., longest side) is $\sqrt{2}$ times of any smaller side. Excluding hypotenuse rest two sides are equal. That is, AB = BC and AC = $\sqrt{2}$ AB = $\sqrt{2}$ BC AB:BC:AC = 1:1:$\sqrt{2}$	 $\angle A = 45°$ $\angle B = 90°$ $\angle C = 45°$
30° – 60° – 90°	If the angles of a triangle are 30°, 60°, and 90°, then the sides opposite to 30° angle is half of the hypotenuse and the side opposite to 60° is $\frac{\sqrt{3}}{2}$ times the hypotenuse, e.g., AB – $\frac{AC}{2}$ and = $\frac{\sqrt{3}}{2}$ AC AB:BC:AC = 1: $\sqrt{3}$: 2	 $\angle C = 30°$, $\angle B = 90°$, $\angle A = 60°$
Basic proportionality theorem (BPT)	Any line parallel to one side of a triangle divides the other two sides proportionally. So, if DE is drawn parallel to BC, then it would divide sides AB and AC proportionally, i.e., $\frac{AD}{DB} = \frac{AF}{FC}$ or $\frac{AD}{AB} = \frac{AF}{AC}$ $\frac{AD}{DF} = \frac{AB}{BC}$	

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Theorem	Statement/Explanation	Diagram
Mid-point theorem	Any line joining the mid-points of two adjacent sides of a triangle are joined by a line segment, then this segment is parallel to the third side, that is if AD = BD and AE = CE, then DE||BC.	
Apollonius' theorem	In a triangle, the sum of the squares of any two sides of a triangle is equal to twice the sum of the square of the median to the third side and square of half the third side. That is, $AB^2 + AD^2 = 2(AC^2 + BC^2)$	
Stewarts theorem/generalization of Apollonius theorem	If the length of AP = m and PB = n, then $m \times CB^2 + n \times AC^2 = (m + n)\,PC^2 + mn\,(m + n)$ Here, it is also understood that m and n are length of segments, and not their ratio.	
Extension of Apollonius' theorem	In the given Δ ABC, AC, BE, and DF are medians. 3 (Sum of squares of sides) = 4 (Sum of squares of medians) $3(AB^2 + AD^2 + DB^2) = 4(AC^2 + EB^2 + FD^2)$	
Interior angle Bisector theorem	In a triangle, the angle bisector of an angle divides the opposite side to the angle in the ratio of the remaining two sides, that is $\frac{BD}{CD} = \frac{AB}{AC}$ and $BD \times AC = CD \times AB = AD^2$	
Exterior angle Bisector theorem	In a triangle, the angle bisector of any exterior angle of a triangle divides the side opposite to the external angle in the ratio of the remaining two sides, that is $\frac{BE}{AE} = \frac{BC}{AC}$	

Congruency of Triangles

Two figures are said to be congruent if, when placed one over the other, they completely overlap each other. They would have the same shape, the same area and will be identical in all respects.

So, we can say that all congruent triangles are similar triangles, but vice versa is not always true.

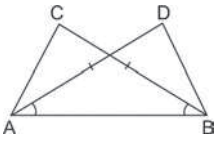
Rules for Two Triangles to be Congruent

1. $S - S - S$
   If in any two triangles, each side of one triangle is equal to a side of the other triangle, then the two triangles are congruent. This rule is $S - S - S$ rule.
2. $S - A - S$

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In Δ ABC and Δ ABD,
AB = AB (common side)
∠ABC = ∠ BAD (given)
BC = AD (given)

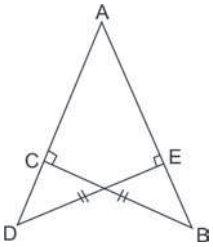


Therefore, by rule $S-A-S$, the two triangles are congruent.

This rule holds true, when the angles that are equal have to be included between the two equal sides (i.e., the angle should be formed between the two sides that are equal).

3. $A-S-A$

In Δ ABC and Δ ADE,
∠ACB = ∠AED (given)
∠BAC = ∠DAE (common angle)
BC = DE (given)
Therefore, by rule $A-S-A$ the two triangles are congruent.
For this rule, the side need not be the included side.



$A-S-A$ can be written as $A-A-S$ or $S-A-A$ also.

4. $R-H-S$

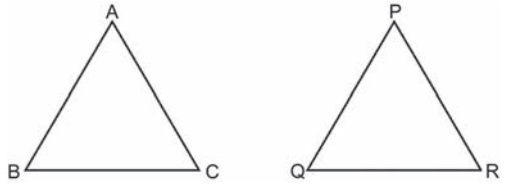
This rule is applicable only for right-angled triangles. If two right-angled triangles have their hypotenuse and one of the sides as same, then the triangles will be congruent.

Similarity of the Triangles

If we take two maps of India of different sizes (breadths and lengths), then the map of all the 29 states of India will cover proportionally the same percentage area in both the maps.

Lets see this in geometry:

Criteria for Similarity of Two Triangles



Two triangles are similar if (i) their corresponding angles are equal and/or (ii) their corresponding sides are in the same ratio. That is, if in two triangles, ABC and PQR,

(i) $\angle A = \angle P, \angle B = \angle Q, \angle C = \angle R$, and/or

(ii) $\frac{AB}{PQ} = \frac{BC}{QR} = \frac{CA}{RP}$, the two triangles are similar.

All regular polygons of the same number of sides such as equilateral triangles or squares, are similar. In particular, all circles are also similar.

Theorems for Similarity

1. If in two triangles, the corresponding angles are equal, then their corresponding sides will also be proportional (i.e., in the same ratio). Therefore, the two triangles are similar.
   This property is referred to as the *AAA* similarity criterion for two triangles.
   Corollary: If two angles of a triangle are, respectively, equal to two angles of another triangle, then the two triangles are similar. This is referred to as the *AA* similarity criterion for the two triangles. It is true due to the fact that if two angles of one triangle are equal to the two angles of another triangle, then the third angle of both the triangles will automatically be the same.
2. If the corresponding sides of two triangles are proportional (i.e., in the same ratio), their corresponding angles will also be equal and so the triangles are similar. This property is referred to as the *SSS* similarity criterion for the two triangles.
3. If one angle of a triangle is equal to one angle of the other and the sides including these angles are proportional, then the triangles are similar. This property is referred to as the *SAS* similarity criterion of the two triangles.
4. The ratio of the areas of the two similar triangles is equal to the ratio of the squares of their corresponding sides.
5. If a perpendicular is drawn from the vertex of the right angle of a right-angled triangle to the hypotenuse, the triangles on each side of the perpendicular are similar to the whole triangle and to each other.

Similar Polygons

Two polygons of the same number of sides are similar, if

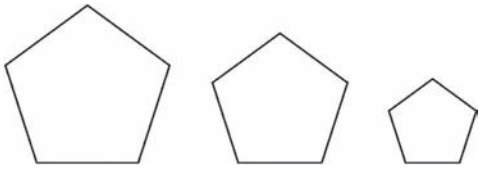
(i) their corresponding angles are equal (i.e., they are equiangular) and

(ii) their corresponding sides are in the same ratio (or proportional).

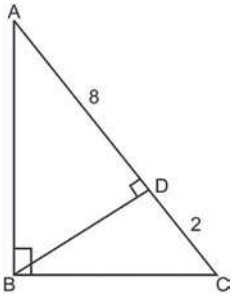
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This can be seen in the figures given below:



Example 7 ΔABC is a right-angled triangle BD ⊥ AC. If AD = 8 cm and DC = 2 cm, then BD = ?



(a) 4 cm (b) 4.5 cm
(c) 5 cm (d) Cannot be determined

Solution Δ ADB ~ ΔBDC

$\therefore \frac{AD}{BD} = \frac{BD}{DC}$

$\therefore BD^2 = AD \times DC = 8 \times 2$

$\therefore BD^2 = 16$

$\therefore BD = 4$ cm

Important Result of this question $BD^2 = AD \times DC$ can be used as a standard result also.

Example 8 Circles with radii 3, 4, and 5 units touch each other externally. If P is the point of intersection of the tangents to these circles at their point of contact, find the distance of P from the point of contacts of the circles.

Solution Let A, B, and C be the centres of the three circles. So, the point P will be the incentre of triangle ABC and distance of P from the point of contacts of the circles will be the inradius (r).

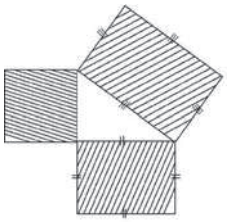
So, $r = \frac{A}{S}$

Sides of triangle ABC will be 7 units, 8 units and 9 units.

So, $r = \sqrt{5}$

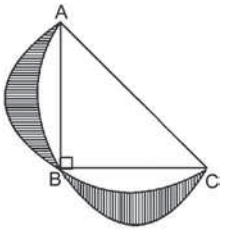
Some Interesting Facts

1. If we draw regular polygons on all the sides of a right angled triangle, taking the sides of the triangle as one of the sides of the figures, then the area of the shaded portion is equal to twice the area of the figure on the hypotenuse. This will also hold true for semi-circles whose diameters will be the side of the right-angled triangle.

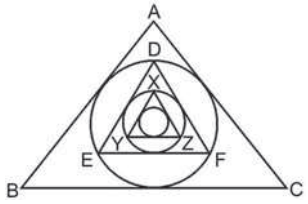


Area of the shaded region = 2 (area of figure on hypotenuse)

2. In the figure below, two semicircles are drawn with diameters equal to the sides of the right-angled triangle. The area of the shaded region (the crescents) is equal to the area of the right-angled triangle. Area of the shaded region (is equal to) the area of the right-angled triangle.

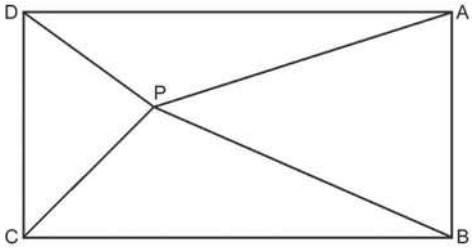


3. In the figure given below, all triangles are equilateral triangles and circles are inscribed in these triangles. If the side of triangle ABC = a, then the side of triangle DEF = $\frac{a}{2}$ and the side of triangle $xyz = \frac{a}{4}$.



In other words, we can say in order to obtain the side of the next inner triangle divide the side of the immediate outer triangle by 2. The same algorithm holds true for the inscribed circles.

4. In the figure given below, if P is any point inside rectangle ABCD, then $PA^2 + PC^2 = PB^2 + PD^2$

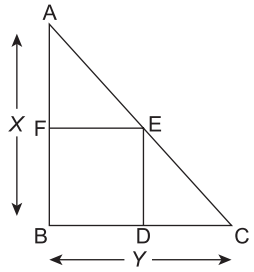


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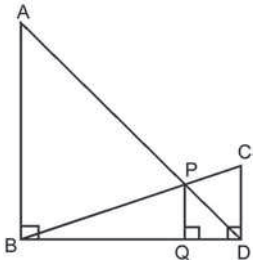
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Some Important Points

1. (a) In an acute-angled triangle, the circumcentre lies inside the triangle.
   (b) In a right-angled triangle, the circumcentre lies on the middle point of the hypotenuse.
   (c) In an obtuse-angled triangle, the circum centre lies outside the triangle.
2. (a) In an acute-angled triangle, the orthocentre lies inside the triangle.
   (b) In a right-angled triangle, the orthocentre lies on the vertex is where the right angle is formed, that is the vertex opposite to the hypotenuse.
   (c) In an obtuse-angled triangle, the orthocentre lies outside the triangle.
3. In a right-angled triangle, the length of the median drawn to the hypotenuse is equal to half the hypotenuse. This median is equal to the circumradius (R) of the right angled triangle.
4. In the figure given below, triangle ABC is a right-angled triangle, right angled at B. Side AB measures x units and BC measures y units. If a square (BDEF), the maximum area is inscribed in the triangle as shown, below then the side of the square is equal to $\frac{xy}{x+y}$.



Example 9 In the figure given below, ∠ABD = ∠CDB = ∠PQD = 90°. If AB:CD = 3:1, then what is the ratio of CD: PQ?

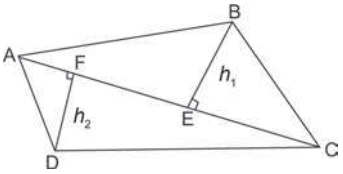


Solution Using the proportionality theorem, AB/PQ = BD/QD and PQ/CD = BQ/BD. Multiplying both these equations, we get AB/CD = BQ/QD = 3:1

Hence, CD/PQ = BD/BQ = 4:3.

QUADRILATERALS AND THEIR PROPERTIES

A quadrilateral is a figure bounded by four sides. In the figure given below, ABCD is a quadrilateral. Line AC is the diagonal of the quadrilateral (denoted by d) and BE and DF are the heights of the triangles ABC and ADC, respectively (denoted by h_1 and h_2).



AC = d, BE = h, and DE = h_2

(i) Area = $\frac{1}{2}$ × one diagonal × (sum of perpendiculars to the diagonal from the opposite vertexes) = $\frac{1}{2}$ d (h1 + h2)

(ii) Area = $\frac{1}{2}$ × product of diagonals × sine of the angle between them

(iii) Area of the cyclic quadrilateral

$= \sqrt{(s-a)(s-b)(s-c)(s-d)}$ where a, b, c, and d are the sides of quadrilateral and s = semiperimeter

$= \frac{a+b+c+d}{2}$

(iv) *Brahmagupta's formula*: For any quadrilateral with sides of length a, b, c, and d, the area A is given by

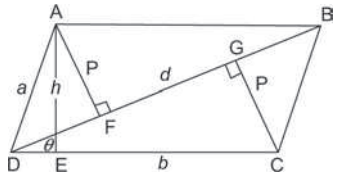
$$A = \sqrt{(s-a)(s-b)(s-c)(s-d) - abcd\cos^2\frac{1}{2}(A+B)}$$

Where $s = \frac{a+b+c+d}{2}$ is known as the semi-perimeter, A is the angle between sides a and d, and B is the angle between the sides b and c.

Different Types of Quadrilaterals

Parallelogram

A parallelogram is a quadrilateral when its opposite sides are equal and parallel. The diagonals of a parallelogram bisect each other.



Given: AD = BC = a and AB = DC = b

BD = d

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AF (height of ΔABD) = CG (height of ΔCBD) and AE = height of the parallelogram = h

$\angle ADC = \theta$

(i) Area = base × height

(ii) Area = (any diagonal) × (perpendicular distance to the diagonal from the opposite vertex)

(iii) Area = (product of adjacent sides) × (sine of the angle between them) Area = AB sin θ

(iv) Area = $2\sqrt{s(s-a)(s-b)(s-d)}$, where a and b are the adjacent sides and d is the diagonal.

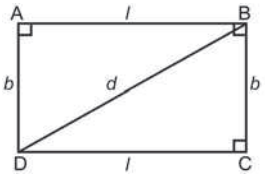
(v) $AC^2 + BD^2 = 2(AB^2 + BC^2)$

(vi) The parallelogram that is inscribed in a circle is a rectangle.

(vii) The parallelogram that is circumscribed about a circle is a rhombus.

(viii) A parallelogram is a rectangle if is diagonals are equal.

Rectangle



A rectangle is a quadrilateral when its opposite sides are equal and each internal angle equals 90°. The diagonals of a rectangle are equal and bisect each other.

Given: AD = BC = b and AB = DC = l, BD = d

(i) Area = length × breadth Area = lb

(ii) Perimeter = 2 (length + breadth) Perimeter = 2 $(l + b)$

(iii) Diagonal^2 = length^2 + breadth^2 (Pythagoras Theorem) $d^2 = l^2 + b^2$ $d = \sqrt{l^2 + b^2}$

(iv) Finding area using Brahmagupta' formula: In this case, we know that $a = c$ and $b = d$, and A + B = π. So, area of rectangle

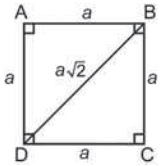
$$= \sqrt{(a+b-a)(a+b-b)(a+b-a)(a+b-b) - a\cdot b\cdot a\cdot b\cos^2 90^\circ}$$

$$= ab$$

(v) The quadrilateral formed by joining the mid-points of intersection of the angle bisectors of a parallelogram is a rectangle.

Square

A square is a quadrilateral when all its sides are equal and each internal angle is of 90°. The diagonals of a square bisect each other at right angles (90°)



Given: AB = BC = CD = DA = a

BD (diagonal) = $a\sqrt{2}$

(i) Area = $(\text{side})^2 = \dfrac{(\text{diagonal})^2}{2} = \dfrac{(\text{perimeter})^2}{16}$

Area = $a^2 = \dfrac{d^2}{2} = \dfrac{P^2}{16}$

(ii) Using Brahmagupta's formula to find out the area of a square: We know that $a = b = c = d$ and A + B = π So, area of square

$$= \sqrt{(s-a)(s-b)(s-c)(s-d) - abcd\cos^2\frac{1}{2}(A+B)}$$

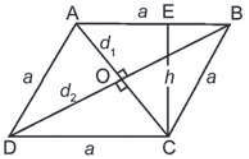
$$= \sqrt{(2a-a)(2a-a)(2a-a)(2a-a) - a\cdot a\cdot a\cdot a\cdot\cos^2 90^\circ}$$

$$= a^2$$

(iii) Perimeter = 4 (side) $\Rightarrow$ Perimeter = $4a$

Rhombus

A rhombus is a quadrilateral when all sides are equal. The diagonals of a rhombus bisects each other at right angles (90°)



Given = AB = BC = CD = DA = a

$\angle AOB = \angle BOC = \angle COD - \angle DOA = 90^\circ$

AC = d, (AO = OC) and BD = d_2 (BO = OD) CE (height) = h

(i) Area = $\dfrac{1}{2}$ × (product of the diagonals)

Area = $\dfrac{1}{2} d_1 d_2$

(ii) Area = base × height

Area = $a \times h$

(iii) A parallelogram is a rhombus if its diagonals are perpendicular to each other. Remember, the sum of the square of the diagonals is equal to four time the square of the side, that is $d_1^2 + d_2^2 = 4a^2$

Trapezium

A trapezium is a quadrilateral in which only one pair of the opposite sides is parallel

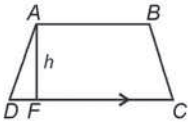


Figure 1

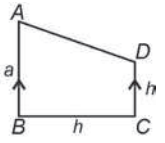


Figure 2

Given: AB = a and CD = b

In Fig. 1, AF (height) = h, and in Fig. 2, BC (height) = h

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(i) Area $= \frac{1}{2} \times$ (sum of the parallel sides) $\times$ (distance between the parallel sides)

Area $= \frac{1}{2}(a + b)\, h$

(ii) The line joining the mid-points of the non-parallel sides is half the sum of the parallel sides and is known as median.

(iii) If we make non-parallel sides equal, then the diagonals will also be equal to each other.

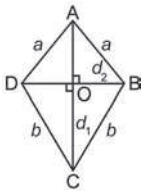
(iv) Diagonals intersect each other proportionally in the ratio of the lengths of the parallel sides.

(v) If a trapezium is inscribed inside a circle, then it is an isosceles trapezium with oblique sides being equal.

Kite

Kite is a quadrilateral when two pairs of adjacent sides are equal and the diagonals bisect each other at right angles (90°).

Given: AB = AD = a and BC = DC = b



AC = d_1 (AO = OC) and BD = d_2 (BO = OD)

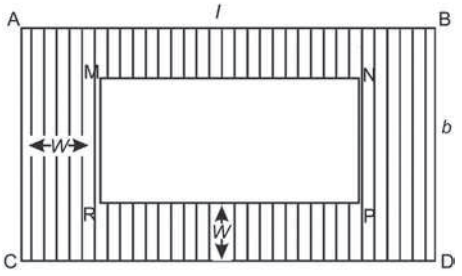
$\angle AOB = \angle BOC = \angle COD = \angle DOA = 90°$

(i) Area $= \frac{1}{2} \times$ (Product of the diagonals)

Area $= \frac{1}{2} d_1 d_2$

Area of Shaded Paths

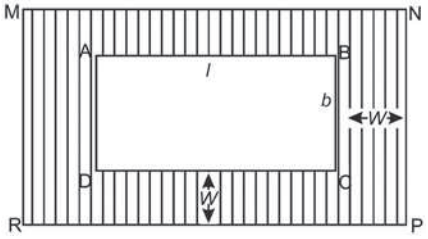
Case I When a pathway is made outside a rectangle having length = l and breadth = b



ABCD is a rectangle with length = l and breadth = b, the shaded region represents a pathway of uniform width = W

Area of the shaded region/pathway = $2w\ (1 + b - 2w)$

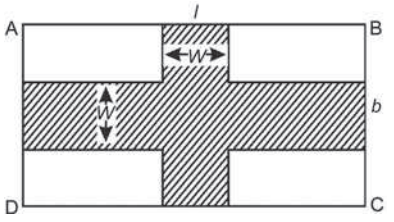
Case II When a pathway is made inside a rectangle having length = l and breadth = b



ABCD is a rectangle with length = l and breadth = b, the shaded region represents a pathway of uniform width = w

Area of the shaded region/pathway = $2w\ (l + b + 2w)$

Case III When two pathways are drawn parallel to the length and breadth of a rectangle having length = l and breadth = b



ABCD is a rectangle with length = l and breadth = b, the shaded region represents two pathways of a uniform width = w

Area of the shaded region/pathway = $W\ (l + b - w)$

From the above figure, we can observe that the area of the paths does not change on shifting their positions as long as they are perpendicular to each other.

We can conclude from here that:

1. Every rhombus is a parallelogram, but the converse is not true.
2. Every rectangle is a parallelogram, but the converse is not true.
3. Every square is a parallelogram, but the converse is not true.
4. Every square is a rhombus, but the converse is not true.
5. Every square is a rectangle, but the converse is not true.

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Construction of New Figures by Joining the Mid-points

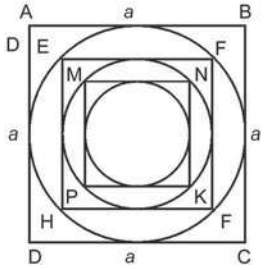
Lines joining the mid-points of adjacent sides of original figure	form	Resulting figure
Quadrilateral		Parallelogram
Parallelogram		Parallelogram
Rectangle		Rhombus
Rhombus		Rectangle
Trapezium		Four similar Δ

Properties of Diagonals

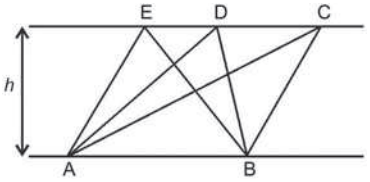
Properties		Types of quadrilaterals				
Sl. No.		Square	Rectangle	Parallelogram	Rhombus	Trapezium
1.	Diagonals are equal	Y	Y	N	N	N
2.	Diagonals bisect each other	Y	Y	Y	Y	N
3.	Diagonals bisect vertex angles	Y	N	N	Y	N
4.	Diagonals are at right angles	Y	N	N	Y	N
5.	Diagonals make congruent triangles	Y	N	N	Y	N

Some Important Points

1. In the figure given below, all the side quadrilaterals are squares and circles are inscribed in these squares. If the side of the square ABCD = a, then the side of square EFGH = $\frac{a}{\sqrt{2}}$ and the side of square MNKP = $\frac{a}{2}$. In other words, we can say that in order to obtain the side of the next inner square, divide the side of the immediate outer square by $\sqrt{2}$. The same procedure will be applied for the inscribed circles, that is we divide the radius of the immediate outer circle by $\sqrt{2}$ to obtain the radius of the next inner circle.



2. Triangles on the same base and between the same parallel lines are equal in area.



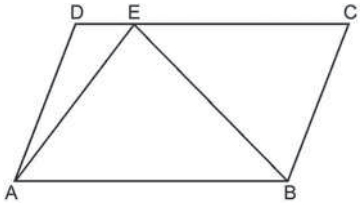
Area (∠ABC) = Area (ΔABD) = Area (ΔABE)

$= \frac{1}{2}(AB) \times h$

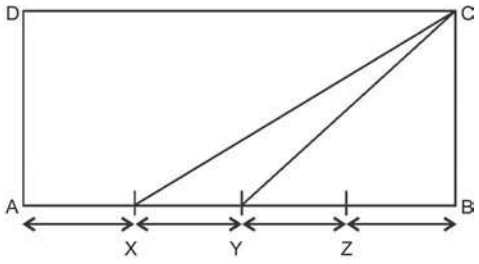
(Base AB is the same and the height 'h' is also the same)

3. If a parallelogram and a triangle are drawn on the same base and between the same parallel lines, then the area of the parallelogram is twice the area of the triangle.

Area of ABCD = 2 × (Area of Δ ABE)



Example 10 Side AB of a rectangle ABCD is divided into four (4) equal parts as shown in the figure. Find the ratio of the area (ΔXYC) and area (ABCD)?



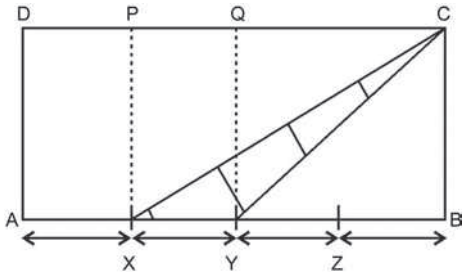
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Solution Let the area of the rectangle ABCD = A

Area of the rectangle XYQP = $A/4$

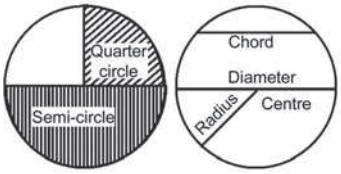
Rectangle XYQP and ΔXYC are on the same base and between the same parallel lines.



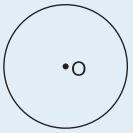
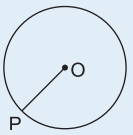
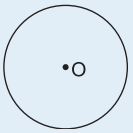
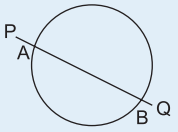
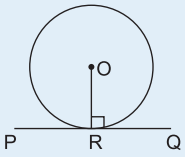
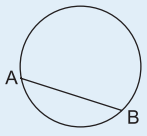
Therefore, area (XYC) = $A/8$

CIRCLES AND THEIR PROPERTIES

A circle is the path travelled by a point which moves in such a way that its distance from a fixed point remains constant. The fixed point is known as the centre and the fixed distance is called the radius.



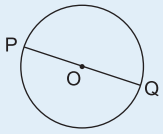
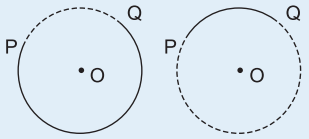
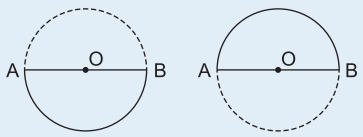
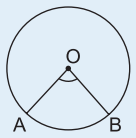
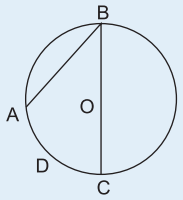
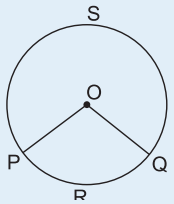
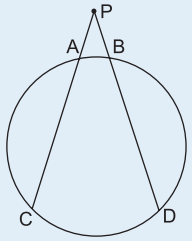
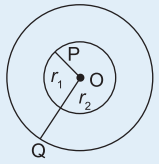
Before we move ahead, let us understand the basics definitions of circle.

Nomenclature	Definition	Diagram
Centre	The fixed point is called the centre. In the given diagram, 'O' is the centre of the circle.	
Radius	The fixed distance is called a radius. In the given diagram, OP is the radius of the circle. (point P lies on the circumference)	
Circumference	The circumference of a circle is the distance around a circle, which is equal to $2\pi r$. ($r \rightarrow$ radius of the circle)	
Secant	A line segment which intersects the circle in two distinct points is called as secant. In the given diagram, secant PQ intersects circle at two points at A and B.	
Tangent	A line segment which has one common point with the circumference of a circle, i.e., it touches only at only one point is called as tangent of circle. The common point is called as point of contact. In the given diagram, PQ is a tangent which touches the circle at a point R.	 (R is the point of contact) Note: Radius is always perpendicular to tangent.
Chord	A line segment whose end points lie on the circle. In the given diagram, AB is a chord.	

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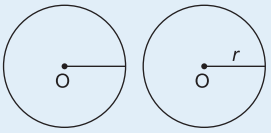
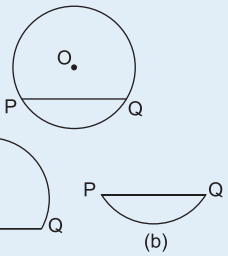
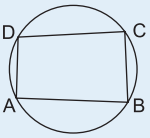
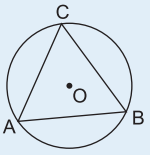
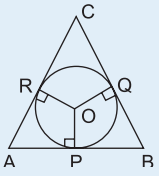
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Nomenclature	Definition	Diagram
Diameter	A chord which passes through the centre of the circle is called the diameter of the circle. The length of the diameter is twice the length of the radius. In the given diagram, PQ is the diameter of the circle. (O → is the centre of the circle)	
Arc	Any two points on the circle divides the circle into two parts, the smaller part is called as minor arc and the larger part is called as major arc. It is denoted as 'Arc'. In the given diagram PQ is arc.	
Semicircle	A diameter of the circle divides the circle into two equal parts. Each part is called a semicircle.	
Central angle	An angle formed at the centre of the circle is called the central angle. In the given diagram, ∠AOB in the central angle.	
Inscribed angle	When two chords have one common end point, then the angle included between these two chords at the common point is called the inscribed angle. ∠ABC is the inscribed angle by the arc ADC.	
Measure of an arc	Basically, it is the central angle formed by an arc. For example (a) measure of a circle = 360° (b) measure of a semicircle = 180° (c) measure of a minor arc = ∠POQ (d) measure of a major arc = 360 − ∠POQ	 m(arc PRQ) = m ∠POQ m(arc PSQ) = 360° − m (arc PRQ)
Intercepted arc	In the given diagram, AB and CD are the two intercepted arcs, intercepted by ∠CPD. The end points of the arc must touch the arms of ∠CPD, i.e., CP and DP.	
Concentric circles	Circles having the same centre at a plane are called the concentric circles. In the given diagram, there are two circles with radii r_1 and r_2 having the common (or same) centre. These are called as concentric circles.	

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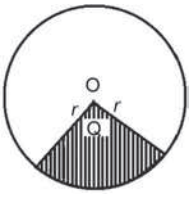
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Nomenclature	Definition	Diagram
Congruent circles	Circles with equal radii are called as congruent circles.	
Segment of a circle	A chord divides a circle into two regions. These two regions are called the segments of a circle: (a) major segment (b) minor segment.	
Cyclic quadrilateral	A quadrilateral whose all the four vertices lie on the circle.	
Circumcircle	A circle that passes through all the three vertices of a triangle. Therefore, the circumcentre is always equidistant from the vertices of the triangle. OA = OB = OC (circumradius)	
Incircle	A circle which touches all the three sides of a triangle, i.e., all the three sides of a triangle are tangents to the circle is called an incircle. Incircle is always equidistant from the sides of a triangle.	

Now come to different formula and theorems attached to circle:

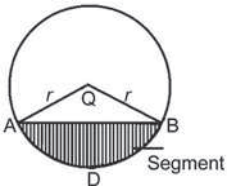
Circumference of a circle = $2\pi r$

Area of a circle = πr^2, where r is the radius.



$$\text{Area of a sector} = \pi r^2 \frac{\theta}{360^\circ}$$

$$\text{Perimeter of a sector} = 2r\left(\frac{\pi\theta}{360} + 1\right)$$



Area of a segment = Area of a sector OADB − Area of triangle OAB

$$\text{Area of a segment} = \pi r^2 \frac{\theta}{360} - \frac{1}{2} r^2 \text{Sin}\theta$$

Common Tangents and Secants of Circles

Depending upon the positioning of the circles, two or more than two circles can have a common tangent. Following is a list indicating the number of common tangents in case of two circles:

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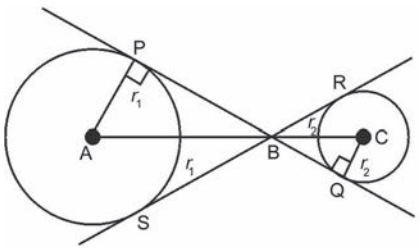
Sl. No.	Position of two circles	Number of common tangents
1.	One circle lies entirely inside the other circle	Zero
2.	Two circles touch internally	One
3.	Two circles intersect in two distinct points	Two
4.	Two circles touch externally	Three
5.	One circle lies entirely outside the other circle	Four

Direct Common Tangents and Transverse Common Tangents

1. *Transverse common tangent* In the figure given below, PQ and RS are the transverse common tangents. Transverse common tangents intersect the line joining the centre of the two circles. They divide the line in the ratio $r_1:r_2$.

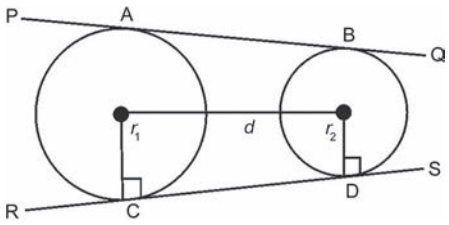
AB:BC = $r_1:r_2$

Assume AC = Distance between centres = d



$PQ^2 = RS^2 = d^2 - (r_1 + r_2)^2$

2. *Direct common tangent*

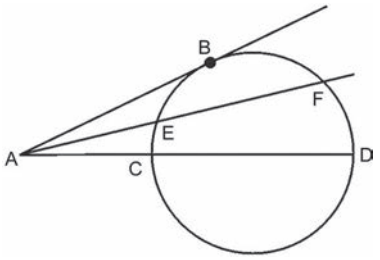


In the figure given above, PQ and RS are direct common tangents.

Points A and C are the point of tangency for the first circle and similarly, points B and D are the point of tangency for the second circle. AB and CD are known as lengths of the direct common tangents and they will be same.

$CD^2 = AB^2 = d^2 - (r_1 - r_2)^2$

Secants



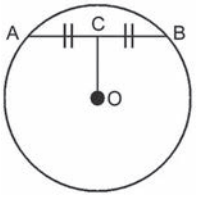
In the figure given above, AB is a tangent and ACD is a secants

(i) $AB^2 = AC \times AD$

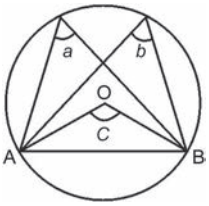
(ii) $AE \times AF = AC \times AD$

Important theorems related to circle

1. If C is the mid-point of AB, then OC is perpendicular to AB. And vice versa is also true.

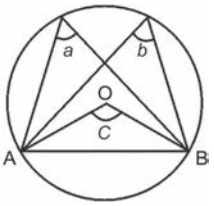


2. Angles in the same segment will be equal.



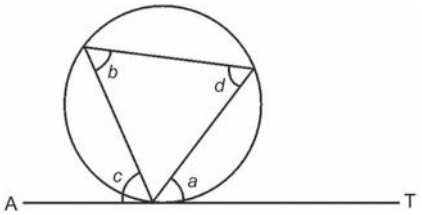
In the figure given above, $a = b$.

3. Angle subtended by a chord at the centre is two times the angle subtended on the circle on the same side. In the figure given below, $2a = 2b = c$.



4. Angle subtended by a diameter of the circle is a right angle.

5. Alternate segment theorem

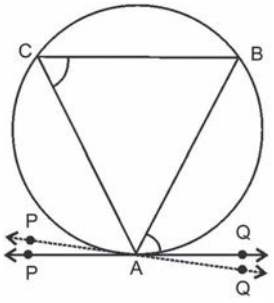


In the figure above, AT is the tangent. $\angle a$ = Alternate segment $\angle b$ $\angle c$ = alternate segment $\angle d$

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6. Converse of alternate segment theorem: If a line is drawn through an end point of a chord of a circle so that the angle formed by it with the chord is equal to the angle subtended by the chord in the alternate segment, then the line is a tangent to the circle.



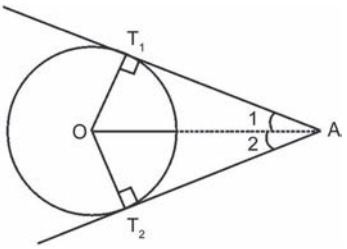
AB is a chord of a circle and a line PAQ such that ∠BAQ = ∠ACB, where C is any point in the alternate segment ACB, then PAQ is a tangent to the circle.

7. Tangent drawn to a circle from a point are same in length.

In the figure below, tangents are drawn to the circle from point A and AT_1 and AT_2 are the tangents.

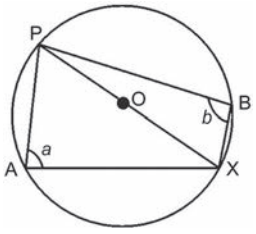
(i) $AT_1 = AT_2$ (ii) $\angle 1 = \angle 2$

(iii) $AT_1^2 + OT_1^2 = AT_2^2 + OT_2^2 = AO^2$



Cyclic Quadrilateral

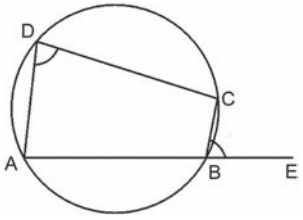
Consider the figure given below:



If we have $a + b = 180°$ and quadrilateral AXBP has all its vertices on a circle, then such a quadrilateral is called a cyclic quadrilateral.

For a cyclic quadrilateral, the sum of the opposite angles of a quadrilateral in a circle is 180°.

It can also be seen that exterior ∠CBE = internal ∠ADC = 180° − ∠ABC.



Using Brahmagupta's formula to find out the area of a cyclic quadrilateral We know that A + B = p. So, area of cyclic quadrilateral

$= \sqrt{(s-a)(s-b)(s-c)(s-d)}$

Where terms used are having their meaning.

[Cos 90° = 0]

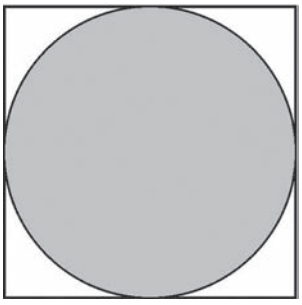
Circle Packing

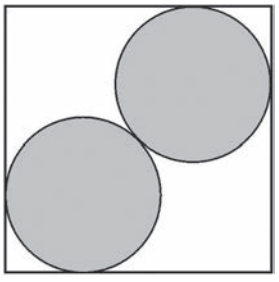
N circles have been packed inside a square of side length R unit and radius of the circle is to be calculated.

Case I When $n = 1$

Obviously, in this case, diameter of the circle = side of the square.

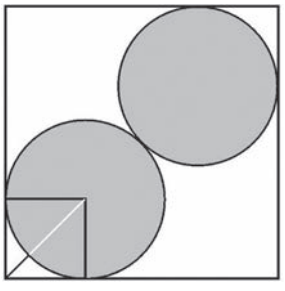
So, the radius of the circle $= \frac{1}{2} R$ unit





Case II When $n = 2$

Consider this figure



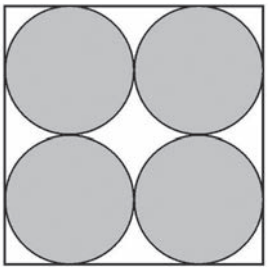
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This is a right-angled triangle with sides r, r, and $r\sqrt{2}$. Hence, the diagonal of the square $= 2r + 2\ r\sqrt{2} = \sqrt{2}R$

So, $r = \dfrac{\sqrt{2}}{2+2\sqrt{2}}R$

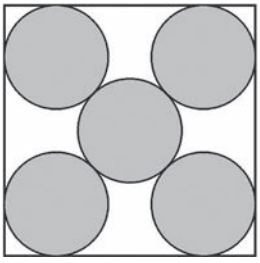
Case III When $n = 4$



Obviously, in this case, 2 × diameter of a circle = side of square.

So, the radius of a circle $= \dfrac{1}{4}R$

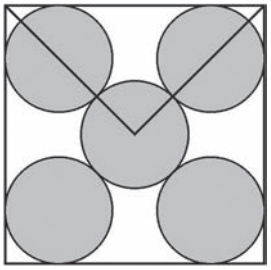
Case IV When $n = 5$



Consider this figure:



If the radius of the circle is r, then the distance between the centre of the circle and the vertex of a square = $r\sqrt{2}$

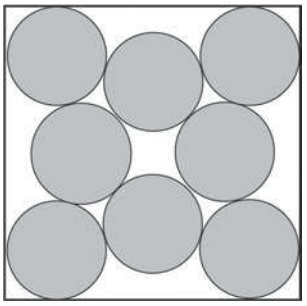


Now, consider this figure, the triangle formed here is a right-angled triangle:

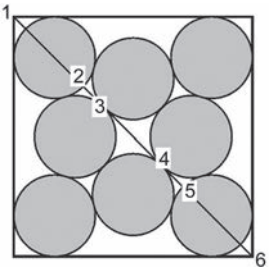
$1/\sqrt{2}$(1/2 of the diagonal of the square) $= r + r + r\sqrt{2}$

$= r(2 + \sqrt{2})$, so, $r = \dfrac{1}{\sqrt{2}\times(2+\sqrt{2})}R$

Case V When $n = 8$



We will find out the value of r in 3 steps here:



Step 1 Find 1 – 2 = 5 – 6

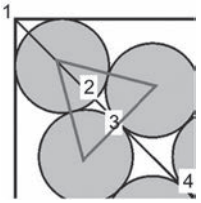
Step 2 Find 2 – 3 = 4 – 5

Step 3 Find 3 – 4

Step 1 $1 - 2 = 5 - 6 = r + r\sqrt{2}$

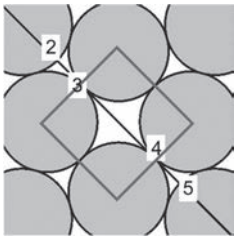
Step 2 For 2 – 3

This is an equilateral triangle with side length = $2r$. Height of this triangle $r\sqrt{3}$



So, $2 - 3 = 4 - 5 = r\sqrt{3} - r$

Step 3



This is a square with side length 2 r and diagonal = $2r\sqrt{2}$

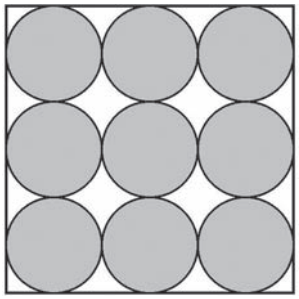
So, $3 - 4 = 2r$

Hence, $l - 6 = 2(r + r\sqrt{2}) + 2(r\sqrt{3} - r) + 2r = R\sqrt{2}$. Now, r can be calculated.

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Case VI When $n = 9$

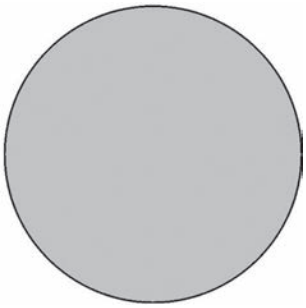


Radius of circle = r
So, the side of a square = 3 × diameter of a circle = $6r$
So, $6r = R$ unit, $r = R/6$ unit

Circles Inside Circles

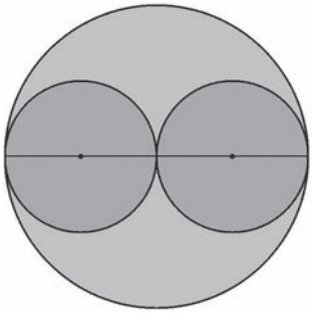
Radius of the outer circle $R = 1$ unit and n similar circles of 'r' radius have been inserted inside this outer circle.

$N = 1$



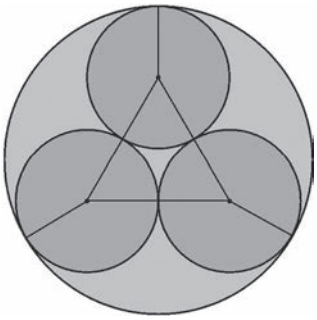
Obviously, $R = r$

$N = 2$



Obviously, $4r = 2R$, hence, $r = R/2$

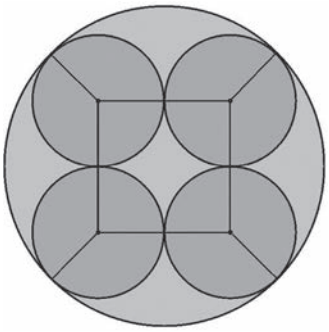
$N = 3$



The triangle formed inside will be an equilateral triangle of the side length $2r$. The centre of this circle will be incentre/centroid/circumcentre of this equilateral triangle. So, the distance between the centre of any smaller circle to the centre of the bigger circle = 2/3 median = $x = 2/3\ [\sqrt{3}/2 \times a]$, where $a = 2r$.

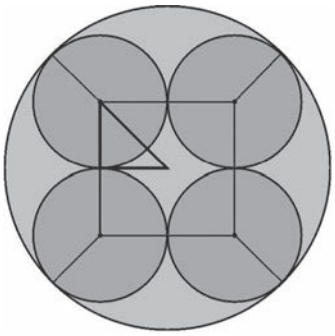
Now, $r + x = R$. Now put the values of x from the above condition to find the value of r.

$N = 4$



The square formed inside will be of the side length $2r$. Now, consider this figure,

$N = 5$

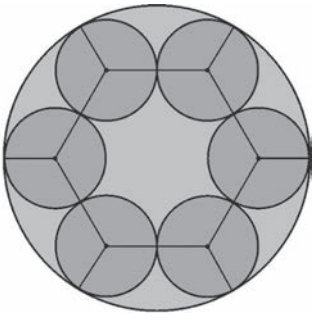


The figure inside is a right-angled triangle, with base and height being equal to r, and hypotenuse = $r\sqrt{2} = r + x$. Hence, $x = r(\sqrt{2} - 1)$

Now, $2R = 4r + 2x$. Hence, $r = 2(R - x)/4$

[Now put the value of x from the above condition.]

$N = 6$



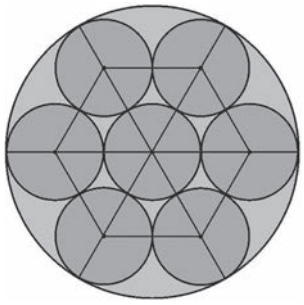
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Here, we can put a similar circle inside all the six circles making it a 7 circle figure. Now, the situation is the same as that of a circle packing with $N = 7$

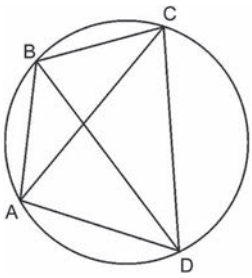
In this case, $6r = R$, so, $r = 1/6$ R

$N = 7$



In this case, $6r = R$, so, $r = 1/6\ R$

Ptolemy's Theorem of Cyclic Quadrilateral



For a cyclic quadrilateral, the sum of the products of the two pair of the opposite sides equal the product of the diagonals.

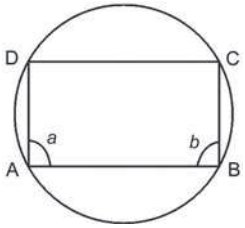
In the figure given above, AB × CD + BC × DA = AC × BD

In any cyclic quadrilateral of side lengths a, b, c, and d, the length of the diagonals p and q will be equal to:

$$p = \sqrt{\frac{(ab+cd)(ac+bd)}{ad+bc}} \qquad q = \sqrt{\frac{(ac+bd)(ad+bc)}{ab+dc}}$$

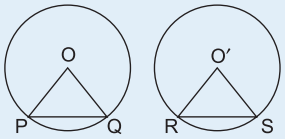
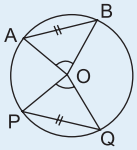
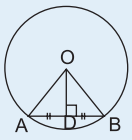
Example 11 Under some special conditions, it is given that a cyclic quadrilateral ABCD is a parallelogram. What kind of figure will ABCD be?

Solution ABCD is a parallelogram. Thus, $a + b = 180°$. And $\angle A + \angle C = 180°$



This means that a parallelogram drawn inside a circle is always a rectangle.

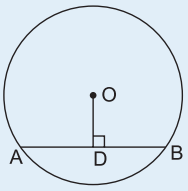
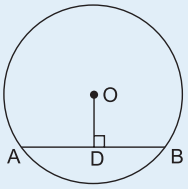
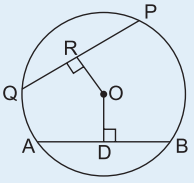
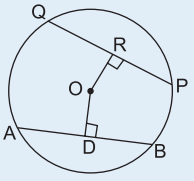
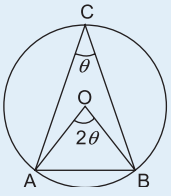
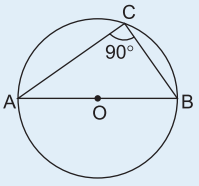
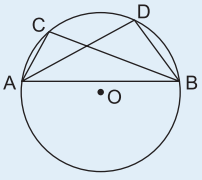
Summarizing the discussion regarding circle

Sl. No.	Theorem/Property	Diagram
1.	In a circle (or congruent circles) equal chords are made by equal arcs. (OP = OQ) = (O′R = O′S) PQ = RS and PQ = RS	
2.	Equal arcs (or chords) subtend equal angles at the centre PQ = AB (or PQ = AB) ∠POQ = ∠AOB	
3.	The perpendicular from the centre of a circle to a chord bisects the chord, i.e., if OD ⊥ AB (OD is perpendicular to AB).	

(*Continued*)

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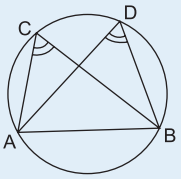
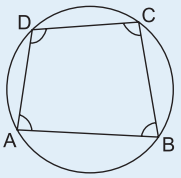
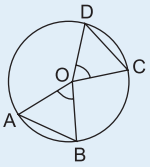
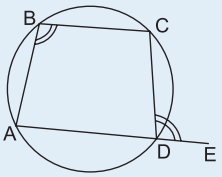
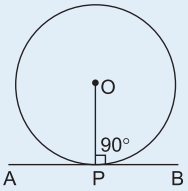
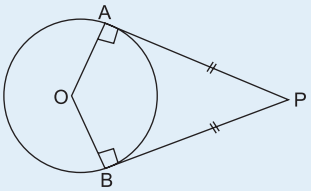
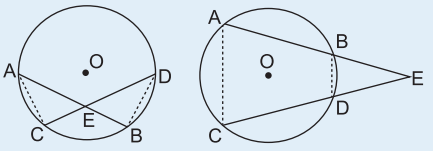
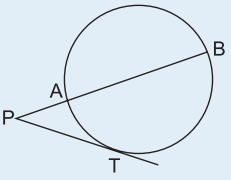
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Sl. No.	Theorem/Property	Diagram
4.	The line joining the centre of a circle to the mid-point of a chord is perpendicular to the chord. AD = DB OD ⊥ AB	
5.	Perpendicular bisector of a chord passes through the centre, i.e., OD ⊥ AB and AD = DB ∴ O is the centre of the circle.	
6.	Equal chords of a circle (or of congruent circles) are equidistant from the centre. ∴ AB = PQ ∴ OD = OR	
7.	Equidistant chords of a circle from the centre are of equal length. If OD = OR, then ∴ AB = PQ	
8.	The angle subtended by an arc (the degree measure of the arc) at the centre of a circle is twice the angle subtended by the arc at any point on the remaining part of the circle. $m\angle AOB = 2\, m\angle ACB$.	
9.	Angle in a semicircle is a right angle.	
10.	Angles in the same segment of a circle are equal, i.e., ∠ACB = ∠ADB.	

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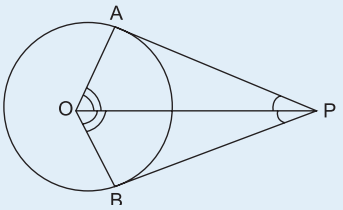
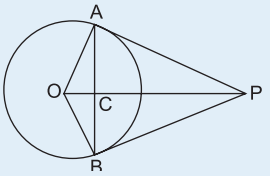
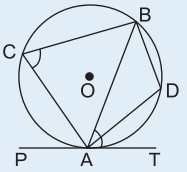
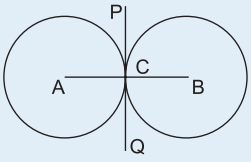
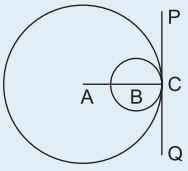
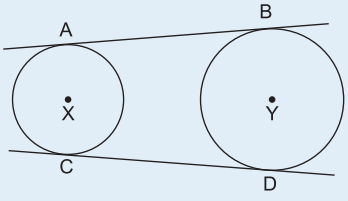
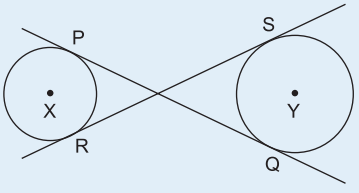
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Sl. No.	Theorem/Property	Diagram
11.	If a line segment joining two points subtends equal angle at two other points lying on the same side of the line containing the segment, then the four points lie on the same circle. $\angle ACB = \angle ADB$ $\therefore$ Points A, C, D, and B are concyclic, i.e., lie on the circle.	
12.	The sum of pair of opposite angles of a cyclic quadrilateral is 180°. $\angle DAB + \angle BCD = 180°$ $\angle ABC + \angle CDA = 180°$ (Inverse of this theorem is also true.)	
13.	Equal chords (or equal arcs) of a circle (or congruent circles) subtended equal angles at the centre. AB = CD (or AB = CD) $\angle AOB = \angle COD$ (Inverse of this theorem is also true.)	
14.	If a side of a cyclic quadrilateral is produced, then the exterior angle is equal to the interior opposite angle. $m\angle CDE = m\angle ABC$	
15.	A tangent at any point of a circle is perpendicular to the radius through the point of contact. (Inverse of this theorem is also true.)	
16.	The lengths of two tangents drawn from an external point to a circle are equal, that is AP = BP.	
17.	If two chords AB and CD of a circle, intersect inside a circle (outside the circle when produced at a point E), then AE × BE = CE × DE.	
18.	If PB be a secant which intersects the circle at A and B and PT be a tangent at T, then $PA \times PB = (PT)^2$.	

(*Continued*)

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Sl. No.	Theorem/Property	Diagram
19.	From an external point from which the tangents are drawn to the circle with centre O, then (a) they subtend equal angles at the centre (b) they are equally inclined to the line segment joining the centre of that point ∠AOP = ∠BOP and ∠APO = ∠BPO.	
20.	If P is an external point from which the tangents to the circle with centre O touch it at A and B then OP is the perpendicular bisector of AB. OP ⊥ AB and AC = BC	
21.	If from the point of contact of a tangent, a chord is drawn then the angles which the chord makes with the tangent line are equal respectively to the angles formed in the corresponding alternate segments. In the adjoining diagram, ∠BAT = ∠BCA and ∠BAP = ∠BDA	
22.	The point of contact of two tangents lies on the straight line joining the two centres. (a) When two circles touch externally then the distance between their centres is equal to sum of their radii, i.e., AB = AC + BC. (b) When two circles touch internally the distance between their centres is equal to the difference between their radii, i.e., AB = AC – BC.	 
23.	For the two circles with centre X and Y and radii r_1 and r_2. AB and CD are two Direct Common Tangents (DCT), then the length of DCT = $\sqrt{(\text{distance between centres})^2 - (r_1 - r_2)^2}$	
24.	For the two circles with centre X and Y and radii r_1 and r_2 PQ and RS are two transverse common tangent, then length of TCT = $\sqrt{(\text{distance between centres})^2 - (r_1 - r_2)^2}$	

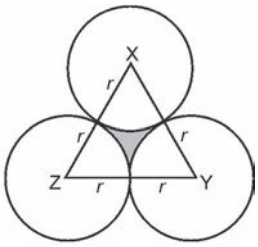
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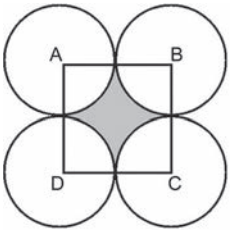
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Some Important Points

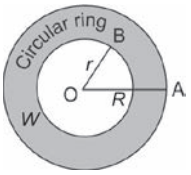
1. If three circles, each of radius r, are so kept that each circle touches the other two, then the area of the shaded region is $r^2\,[\sqrt{3} - (\pi/2)]$ which is approximately equal to $(4/25)\,r^2$.



2. If four circles, each of the radius r, are so kept that each circle touches the other two, then the area of the shaded region is $r^2\,(4 - \pi)$ which is equal to $(6/7)r^2$.



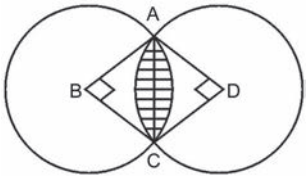
3. In the diagram given below, if the radius (OA) of the outer circle is R and the radius (OB) of the inner circle is r, then the width (w) of the ring is $(R - r)$ and the area of the shaded region is $\pi(R^2 - r^2)$ or $\pi(R - r)\,(R + r)$ or $\pi(w)\,(R + r)$



Example 12 Two identical circles intersect so that their centres, and the points at which they intersect, form a square of side 1 cm. The area in sq. cm of the portion that is common to the two circles, is:

(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{2} - 1$ (c) $\frac{\pi}{5}$ (d) $\sqrt{2} - 1$

Solution



Shaded area = 2 × (area of sector ADC − area of ΔADC)

$$= 2 \times \left(\frac{\pi}{4} \times 1^2 - \frac{1}{2} \times 1 \times 1 \right)$$

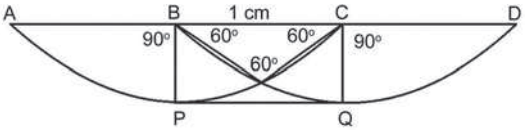
$$= \frac{\pi}{2} - 1$$

Hence, option (b) is the answer.

Example 13 Four points A, B, C, and D lie on a straight line in the X–Y plane, such that AB = BC = CD, and the length of AB is 1 cm. An ant at A wants to reach a sugar particle at D. But there are insect repellents kept at points B and C. The ant would not go within one metre of any insect repellent. The minimum distance the ant must traverse to reach the sugar particle (in m) is:

(a) $\frac{3}{\sqrt{2}}m$ (b) $1 + \pi$ (c) $\frac{4\pi}{3}$ (d) 5

Solution In the drawn figure, ant will go along APQD since it cannot be within a distance of 1 cm from the repellents kept at B and C.



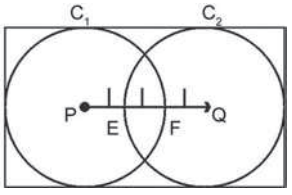
$$AP = \frac{90}{360} \times 2\pi \times 1 = \frac{\pi}{2}$$

Also, $AP = QD = \frac{\pi}{2}$

So, the minimum distance = AP + PQ + QD

$$= \frac{\pi}{2} + 1 + \frac{\pi}{2} = 1 + \pi$$

Example 14 Two circles C_1 and C_2, having the same radius of 2 cm and centres at P and Q, respectively, intersect each other such that the line of centres PQ intersects C_1 and C_2 at F and E, respectively. EF = 1 cm. The whole assembly is enclosed in a rectangle of the minimum area. The perimeter of the rectangle is:



Hence, breadth of the rectangle = 4 cm
And length = 7 cm
Perimeter = 2 × (7 + 4) = 22 cm

Example 15 Semicircle C_1 is drawn with a line segment PQ as its diameter with centre at R. Semicircles C_2 and C_3

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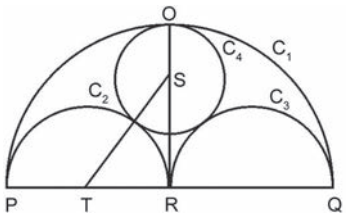
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are drawn with PR and QR as diameters, respectively, both C_2 and C_3 lying inside C_1. A full circle C_4 is drawn in such a way that it is tangent to all the three semicircles C_1, C_2, and C_3. C_4 lies inside C_1 and outside both C_2 and C_3. The radius of C_4 is:

(a) $\frac{1}{3}$PQ (b) $\frac{1}{6}$PQ

(c) $\frac{1}{\sqrt{2}}$PQ (d) $\frac{1}{4}$PQ

Solution



Assume that the radius of $C_4 = r$ and PQ = k.

Now, PR = $k/2$ = RQ = RO

$\Rightarrow$ RS = $(k/2) - r$

RT = $k/4$

ST = $(k/4) + r$

Applying Pythagoras theorem in triangle STR.

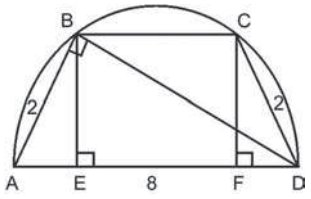
$$\left(\frac{k}{4}+r\right)^2 = \left(\frac{k}{4}\right)^2 + \left(\frac{k}{2}-r\right)^2$$

$\Rightarrow r = k/6 = PQ/6$

Example 16 On a semicircle with diameter AD, chord BC is parallel to AD. Further, each of the chords AB and CD has length 2 units, while AD = 8 units. What is the length of BC?

(a) 7.5 (b) 7 (c) 7.75 (d) 8

Solution



Finding area of DABD $\frac{1}{2}$ AB × BD = $\frac{1}{2}$ AD × BE

So, BE = $\frac{(\sqrt{15})}{2}$

Hence, AE = $\frac{1}{2}$

Now, BC = EF = $8 - \left(\frac{1}{2}+\frac{1}{2}\right) = 7$

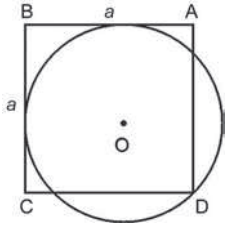
Alternatively, this question can be solved by using Ptolemy's theorem also.

Example 17 The adjacent sides AB, BC of a square of side 'a' units are tangent to a circle. The vertex D of the square lies on the circumference of the circle. The radius of the circle could be:

(a) $a(2 - \sqrt{2})$ (b) $a(\sqrt{2} - 1)$

(c) $a(\sqrt{2} + 1.5)$ (d) $a(\sqrt{2} + 1)$

Solution

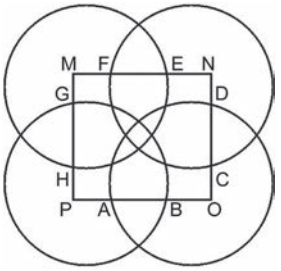


OD = r, OB = $r\sqrt{2}$

Hence, $r + r\sqrt{2} = \sqrt{2}\,a$

Hence, $r = a(2 - \sqrt{2})$

Example 18 M, N, O and P are centres of four intersecting circles each having a radius of 15 cm. If AB = 7 cm, CD = 5 cm, EF = 6 cm, GH = 8 cm, what is the perimeter of the quadrilateral MNOP?



(a) 84 cm (b) 45 cm (c) 94 cm (d) 124cm

Solution MN = 30 − 6 = 24 cm

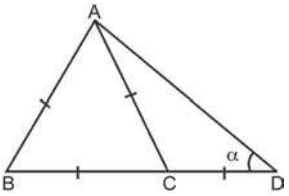
NO = 30 − 5 = 25 cm

OP = 30 − 7 = 23 cm

PM = 30 − 8 = 22 cm

So, the perimeter = 24 + 25 + 23 + 22 = 94 cm

Example 19



AB = BC = AC = CD. Find $\angle\alpha$.

(a) 30° (b) 60°

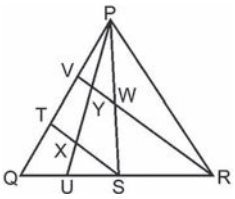
(c) 15° (d) None of these

Solution $\angle$ACB = 60° $\angle$ACD = 180 − 60 = 120° $\angle$CAD = $\angle$CDA = 30° $\Rightarrow \angle\alpha = 30°$

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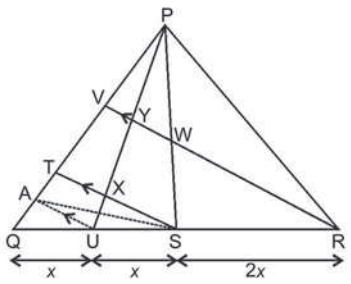
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Example 20 In the figure shown here QS = SR, QU = SU, PW = WS and ST||RV. What is the value of the $\frac{\text{Area of } \Delta PSX}{\text{Area of } \Delta PQR}$



(a) $\frac{1}{5}$ (b) $\frac{1}{7}$ (c) $\frac{1}{6}$ (d) $\frac{1}{9}$

Solution



Draw a line form U such that it is parallel to ST (and hence, RV, also) joining AS, to get ΔQSA.

Now, let us first find the area of ΔSUX.

PW = WS (W is the mid-point of PS)

Consider the ΔPXS, SX||YW and PX and PS are transversals to those parallel lines, we must have

$$\frac{PY}{YX} = \frac{PW}{WS} = \frac{1}{1} = 1$$

$\Rightarrow$ PY = YX

Similarly, we also get

PV = TV (in ΔPTS; VW||ST) and QA = AT (in ΔQTS: AU||ST) and QT = VT (in ΔVQR; ST||VR)

Combining all these $QA = AT = \frac{1}{2}(VT) + \frac{1}{2}(VP)$

Now, in ΔRUY

$$\frac{TS}{SR} = \frac{x}{2x} = \frac{1}{2} = \frac{UX}{XY} \Rightarrow UX = \frac{1}{2}(XY)$$

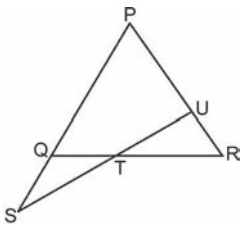
so, in ΔPUS, $UX = \frac{1}{2}XY = \frac{1}{2}PY \Rightarrow \frac{\text{Area}(\Delta SUX)}{\text{Area}(\Delta PUS)} = \frac{1}{5}$

$$\Rightarrow \text{Area}(\Delta SUX) = \frac{1}{5}\text{Area}(\Delta PUS)$$

$$= \frac{1}{5}\left(\frac{1}{4}\text{area}\Delta PQR\right) \Rightarrow \text{Area}(\Delta SUX) = \frac{1}{2}(\Delta SUX), \text{ then}$$

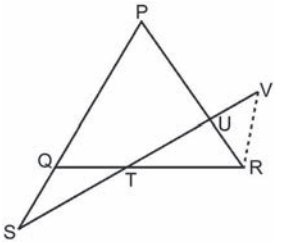
Area (ΔPSX) = Area (ΔPSU) − area (ΔSUX) $= \frac{1}{4} = \frac{1}{5}$ (ΔPQR)

Example 21 Through T, the mid-point of the side QR of a ΔPRQ a straight line is drawn to meet PQ produced to S and PR at U, so that PU = PS. If length of UR = 2 units, then the length of QS is:



(a) $2\sqrt{2}$ units (b) $\sqrt{2}$ units
(c) 2 units (d) Cannot be determined

Solution We have QT = TR and PU = PS and UR = 2 units



Draw RV||PS that meets SU extended at V.

Now, in ΔQST and ΔTVR

∠QST = ∠TVR (alternate angles as PS||VR) and ∠QTS = ∠VTR

QT = TR

∴ ΔQST and ΔTVR are congruent.

∴ QS = VR (i)

Now, ∠QST = ∠PUS = ∠VUR = ∠UVR

[∴ PS = PU]

In ΔUVR, ∠VUR = ∠RVU

or, RV = UR = 2 (ii)

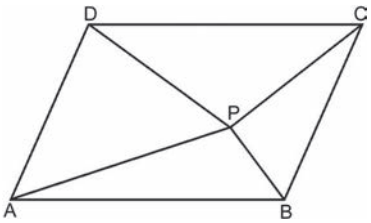
From (i) and (ii)

QS = VR = UR = 2 units

Example 22 ABCD is a parallelogram and P is any point within it. If the area of the parallelogram ABCD is 20 units, then what is the sum of the areas of the ΔPAB and ΔPCD?

(a) 5 units (b) 10 units
(c) 12 units (d) Cannot be determined

Solution



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Let AB = CD = a and x, y be the lengths of the perpendiculars from P on AB and CD, respectively, then

$$\text{Area of } (\Delta PAB + \Delta PCD) = \frac{1}{2}ax + \frac{1}{2}ay$$
$$= \frac{1}{2}a(x+y) = \frac{1}{2}ah$$

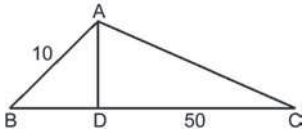
Given area of the parallelogram = 20 units

$$\text{Area of } (\Delta PAB + \Delta PCD) = \frac{1}{2} \times 20 = 10 \text{ units}$$

Example 23 Nayantara bought a triangular piece of land of area 150 m^2. He took a piece of rope and measured the two sides of the plot and found the largest side to be 50 m and another side to be 10 m. What is the exact length of the third side?

(a) $40\sqrt{3}$m (b) $30\sqrt{2}$m (c) $\sqrt{1560}$m (d) 32 m

Solution



Let the triangle be ABC.

Now, area of ΔABC = (1/2) (BC) (AD) (where D is a point on BC such that AD⊥BC)

Now, AD has to be equal to $\frac{\text{Area } \Delta 2}{BC}$

$$AD = \frac{(150)(2)}{50} \text{ m} = 6 \text{ m}$$

Now, ΔBDA is a right-angled triangle

$\therefore BD^2 = AB^2 - AD^2 = 10^2 - 6^2 = 8^2 \text{ m}^2$

$\Rightarrow$ BD = 8 m and DC = BC − BD = 42 m

In a right-angled ΔADC, $AC^2 = AD^2 + DC^2 = (6)^2 + (42)^2$

$\Rightarrow AC = \sqrt{1800} = 30\sqrt{2}$

Alternatively, going through the options, we get

$30\sqrt{2} = 42$

$40\sqrt{3} = 68$

$\sqrt{1560} = 39$

$\therefore$ Using the principle that the sum of any two sides of a triangle is greater than the third side. The two given sides are 50 and 10.

From the choices the above condition is satisfied only for choice (b).

In choice (a), 10 + 50 < 68

In choice (c), 39 + 10 < 50

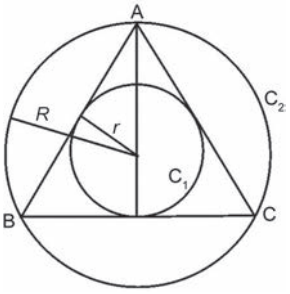
In choice (d), 32 + 10 < 50

Example 24 C_1 and C_2 are two concentric circles with radii 5 cm and 9 cm, respectively. If A, B, and C are points on C_2 such that AB and AC are tangent to C_1 at how many points does BC intersect C_1?

(a) 0 (b) 1

(c) 2 (d) Cannot be determined

Solution

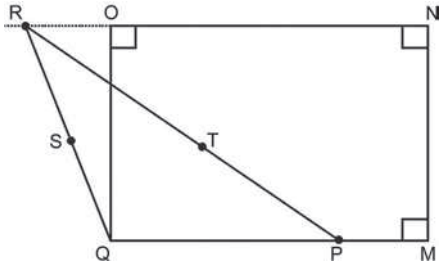


Consider two concentric circles with radii r and R ($r < R$). If $r = \frac{R}{2}$ and A, B, and C are points on the outer circle such that AB and AC are tangent to the inner circle, then BC is also a tangent.

(In this case, C_1 and C_2 become the incircle and circumcircle of ABC respectively, which will be an equilateral triangle)

If $r < \frac{R}{2}$, BC does not intersect or touch the inner circle and if $r > \frac{R}{2}$ then BC intersects the inner circle at two points.

Example 25 A rectangle MNOQ is drawn and length 'NO' is extended to point R and a triangle QPR is drawn with $QP = \frac{2}{3}$ QM. Angle ORP = 45° and side QR = $4\sqrt{17}$cm, S and T are the mid-points of sides QR and PR, respectively. If ST = 6 units, find the area (in sq. cm) of the rectangle.



(a) 112 (b) 144 (c) 288 (d) 256

Solution Since the line joining the mid-points of two sides of a triangle is parallel and equal to half the third side, we have PQ = 2(ST), $\Rightarrow$ PQ = 12 cm

$$\text{Since, } PQ = \frac{2}{3} QM$$

Now, $\angle ORP = 45°$

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Draw PV⊥ON

In ΔRVP, $\tan 45° = \frac{PV}{VR} = \frac{PV}{OR+OV} = \frac{\text{breadth}}{12+OR}$

(Since OV = PQ = 12 cm) also, $\text{breadth}^2 + OR^2 = (QR)^2$

In ΔROQ, ⇒ breadth = x + 12 where OR = x

$\Rightarrow (x+12)^2 = (4\sqrt{17})^2 - x^2$ (given that QR = $(4\sqrt{17})$

$\Rightarrow 2x^2 + 24x - 128 = 0$

$\Rightarrow x^2 + 12x - 24 = 0$

$\Rightarrow x = 4$ or -16

Since $x > 0$, $x = 4$

Breadth = x + 12, that is, 16 cm, and the area of MNOQ = 16 × 18 = 288 sq. cm.

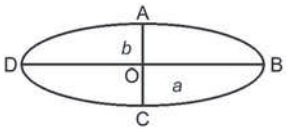
Example 26 ΔABC has sides AB and AC measuring 2001 and 1002 units, respectively. How many such triangles are possible with all integral sides?

(a) 2001 (b) 1002 (c) 2003 (d) 1004

Solution Value of BC will lie in between 999 and 3003. Hence, 999 < BC < 3003.

So, the total values possible for BC = 2003

Ellipse The path of a moving point which moves in such a way that its distance from a fixed point (focus) bears a constant ratio with its distance from a fixed line (directrix)



Given: OB = OD = a (semimajor axis)

OA = OC = b (semi = minor axis)

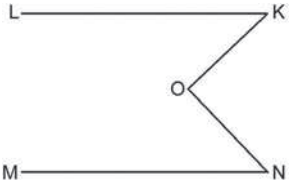
- Area = πab
- Perimeter = $\pi(a + b)$

Maths behind the formula (Area = πab) One way, to see why the formula is true, is to realize that the ellipse is just a unit circle that has been stretched by a factor 'a' in the x-direction and by a factor 'b' in the y-direction. Hence, the area of the ellipse is just $a \times b$ times the area of unit circle.

Practice Exercises

Warm Up

Q.1 In the given figure, MN and KL are parallel lines.

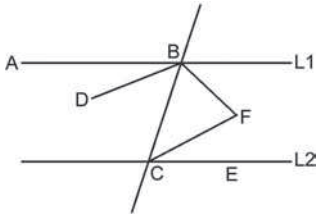


∠LKO = 70°, ∠KON = 100°

Find ∠MNO.

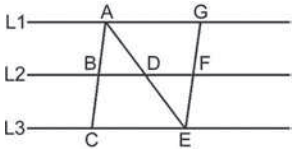
(a) 20° (b) 30° (c) 40° (d) 50°

Q.2 In the given figure, L1 and L2 are parallel lines. ∠ABC = 80°. If the lines BD and CF are parallel and ∠DBC = 30°, find ∠FCE.



(a) 30° (b) 45° (c) 50° (d) 60°

Q.3 The lines L1, L2, and L3 are parallel. If AB = 4 cm, BC = 6 cm, AD = 12 cm, and FG = DE, then find EF.



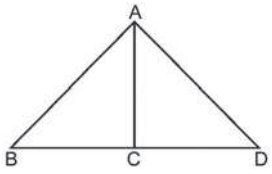
(a) 27 cm (b) 36 cm

(c) 48 cm (d) None of these

Q.4 The supplement of an angle is five times the angle. What is the measurement of the angle?

(a) 30° (b) 65° (c) 40° (d) 45°

Q.5 In the given figure, AB = AD. ∠ACB = 95° + ∠BAC and ∠BAD = 150°. Find ∠ACB.



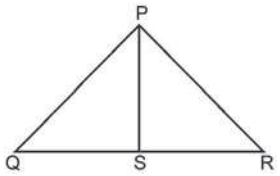
(a) 110° (b) 120° (c) 130° (d) 140°

Q.6 In the given figure, PS is the altitude drawn to the side QR of the triangle PQR. ∠PRS = ∠QPS − 20°,

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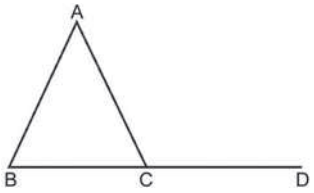
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∠PQS = 20°, ∠PRQ = 50°. What is the value of ∠QPR?



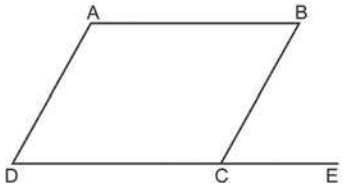
(a) 100° (b) 110° (c) 120° (d) 130°

Q.7 In ΔABC, ∠ACD = 100° and ∠ABC = 40°. Find ∠BAC.



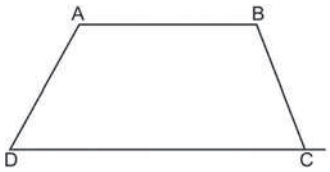
(a) 40° (b) 100° (c) 80° (d) 60°

Q.8 In the given figure, ABCD is a parallelogram. Find 2∠ABC − ADC if A = ∠135°.



(a) 50° (b) 60° (c) 45° (d) 75°

Q.9 ABCD is an isosceles trapezium with lines AB parallel to CD. If ∠DCB = 40°, ∠BAD equals:

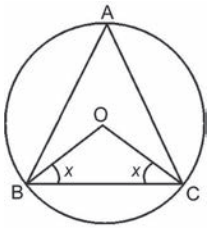


(a) 40° (b) 80° (c) 100° (d) 140°

Q.10 If each interior angle of a regular polygon is 135°, find the number of diagonals.

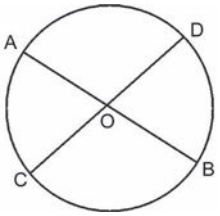
(a) 54 (b) 48
(c) 20 (d) None of these

Q.11 In the given figure, ΔABC is circumscribed by a circle having its centre at O. If OB and OC are the angle bisector of ∠B and ∠C, respectively, what is the value of ∠BAC?



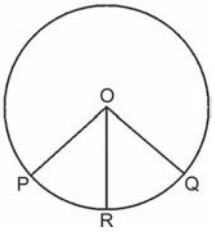
(a) 45° (b) 60° (c) 75° (d) 50°

Q.12 In the given figure, AB and CD are two chords of a circle intersecting at O. If AO = 4 cm, OB = 6 cm, and OC = 3 cm, then find OD.



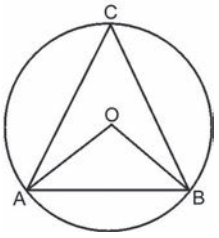
(a) 4 cm (b) 6 cm (c) 8 cm (d) 10 cm

Q.13 In the given figure, if 2 ∠POR = 3 ∠ROQ and ∠POQ = 100°, then find ∠POR.



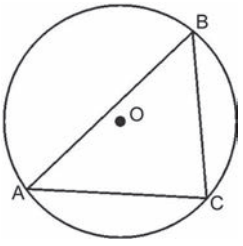
(a) 20° (b) 30° (c) 45° (d) 60°

Q.14 In the above figure, O is the centre of the circle, and ∠AOB = 120°. Find ∠ACB.



(a) 60° (b) 30° (c) 45° (d) 50°

Q.15 In the given figure, O is the centre of the circle, ∠AOC = 160°. If AB = AC, then find ∠BAC.



(a) 45° (b) 20° (c) 60° (d) 75°

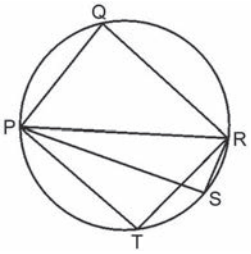
Q.16 PQRS is a cyclic quadrilateral. If PQR is an equilateral triangle, find ∠RSP.

(a) 120° (b) 60° (c) 30° (d) 150°

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Q.17 In the given figure, PT = TR, ∠PQR = 100°. Find ∠PRT.

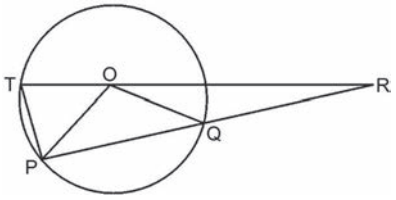


(a) 40° (b) 50° (c) 60° (d) 70°

Q.18 In a triangle ABC, the incentre is at O. If ∠BOC = 100°, find ∠BAC.

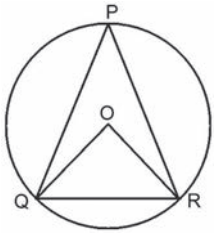
(a) 10° (b) 20° (c) 30° (d) 40°

Q.19 In the given figure, O is the centre of the circle, OQ = QR and ∠QRO = 15°. Find ∠QOP.



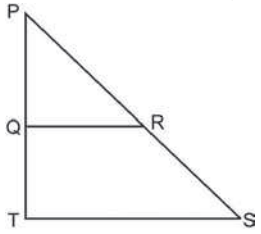
(a) 120° (b) 150° (c) 135° (d) 105°

Q.20 In the given figure, O is the centre of the circle and ∠OQP + ∠ORP = 70°. Find ∠ORQ.



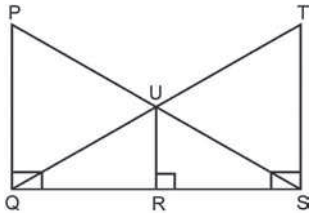
(a) 20° (b) 30° (c) 40° (d) 50°

Q.21 In the given figure PQ × PS = PT × PR. If ∠PQR = ∠PST + 30° and ∠PTS = 100°, then find ∠PRQ.



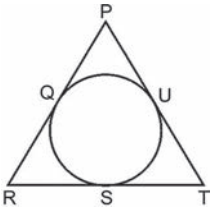
(a) 100° (b) 70° (c) 130° (d) 50°

Q.22 In the above figure, QR = 4 cm and RS = 12 cm. TS = 8 cm and QU is extended to T. Find PQ.



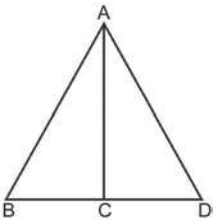
(a) 3 cm (b) 24 cm
(c) 21 cm (d) None of these

Q.23 In the given figure PR, PT and RT are tangents to the circle at Q, U and S, respectively. PR = (RT + 3) cm; PR = (PT + 1) cm. If the perimeter of triangle RPT is 26 cm, what is value of QR + PT?



(a) 12 cm (b) 3 cm
(c) 14 cm (d) 13 cm

Q.24 In the given figure, ∠CAB = 1/2 ∠ACD, ∠BAD = 85° and ∠ADC = 40°. Find ∠ACB.



(a) 45° (b) 55° (c) 70° (d) 60°

Q.25 How many different regular polygons can be formed with the interior angle exceeding the exterior angle and the sum of the interior angles not exceeding 180°?

(a) 7 (b) 8
(c) 9 (d) None of these

FOUNDATION

Q.1 The ratio of the sides of ΔABC is 1:2:4. What is the ratio of the altitudes drawn onto these sides?

(a) 4:2:1 (b) 1:2:4
(c) 1:4:16 (d) None of these

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Q.2 The largest angle of a triangle of sides 7 cm, 5 cm, and 3 cm is:
(a) 45° (b) 60° (c) 90° (d) 120°

Q.3 The three sides of a triangle measure 6 cm, 8 cm, and 10 cm, respectively. A rectangle equal in area to the triangle has a length of 8 cm. The perimeter of the rectangle is:
(a) 11 cm (b) 22 cm
(c) 16 cm (d) None of these

Q.4 X and Y are, respectively, two points on the sides DC and AD of parallelogram ABCD. The area of ΔABX is equal to:
(a) $\frac{1}{3}\times$ area of ΔBYC
(b) Area of ΔBYC
(c) $\frac{1}{2}\times$ area of ΔBYC
(d) $2\times$ area of ΔBYC

Q.5 A rectangular enclosure 40 m × 36 m has a horse tethered to a corner with a rope of 14 m in length. What is the ratio of the respective areas it can graze, if it is outside the enclosure and if it is inside the enclosure?
(a) 2 (b) 2:3 (c) 1:4 (d) 3:1

Q.6 What is the area of the triangle having the sides 4, 6, and 12 units?
(a) 14.28 (b) 12.6
(c) 16.3 (d) Not possible

Q.7 Identical spherical balls are spread on a table top so as to form an equilateral triangle. How many balls are needed so that a side of the equilateral triangle contains n balls?
(a) $\frac{n(n+1)}{2}$ (b) n^2-1 (c) $n(n-1)$ (d) $n!$

Direction for Questions 8 and 9: Read the passage below and solve the questions based on it.

The area of a square is equal to the area of a rectangle. Moreover, the perimeter of the square is also equal to the perimeter of the rectangle.

Q.8 The side of the square is equal to the:
(a) Length of the rectangle
(b) Breadth of the rectangle
(c) Cannot be determined
(d) Both (a) and (b)

Q.9 The length of the rectangle is equal to the:
(a) Breadth of the rectangle
(b) Side of the square
(c) Cannot be determined
(d) Both (a) and (b)

Q.10 A quadrilateral is inscribed in a circle. If an angle is inscribed in each of the segments outside the quadrilateral, then what is the sum of the four angles?
(a) 270° (b) 360° (c) 540° (d) 720°

Q.11 In a triangle PQR, PQ = 6 cm, QR = 8 cm, and PR = 10 cm. The length of the median bisecting the shortest side is (approx.):
(a) 10 cm (b) 8.5 cm
(c) 9 cm (d) None of these

Q.12 A square is inscribed in a circle which is inscribed in an equilateral triangle. If one side of the triangle is 'a', find the area of the square.
(a) $\frac{1^2}{3}$ (b) $\frac{a^2}{6}$ (c) $\frac{3a^2}{8}$ (d) $\frac{\pi a^2}{12}$

Q.13 A cyclic quadrilateral is such that two of its adjacent angles are divisible by 6 and 10, respectively. One of the remaining angles will necessarily be divisible by:
(a) 3 (b) 4
(c) 8 (d) None of these

Q.14 In the above question, if one of the remaining angles is necessarily divisible by 16, then its measure can be:
(a) 128° (b) 48° (c) 132° (d) 112°

Q.15 The volume of spheres are proportional to the cubes of their radii. Two spheres of the same material weigh 3.6 kg and 2.7 kg and the radius of the smaller one is 2 cm. If the two were melted down and formed into a single sphere, what would be its radius?
(a) 4 cm (b) 4.3 cm (c) 3 cm (d) 2.6 cm

Direction for Questions 16 and 17: Read the passage below and solve the questions based on it.

$f(x)$ is the area of a square where, x is the side of a square. $g(x)$ is the perimeter of square where, x is the side of a square. $h(x, y)$ is the area of a rectangle where x is the length and y is the breadth. $i(x, y)$ is the perimeter of a rectangle where x is the length and y is the breadth.

Q.16 The value of $g[h\{f\{i\,(2, 3)\}, 2\}$ is equal to:
(a) 450 (b) 600 (c) 800 (d) 1000

Q.17 The value of $\frac{[f(3)-g(2)]}{h(4,2)-i(6,2)}\times\frac{g(6)}{f(1)}$ is equal to:
(a) −3 (b) 2 (c) 1 (d) −4

Direction for Questions 18 and 19: Read the passage below and solve the questions based on it.

A 10-m long piece of wire is cut into two pieces, one of which is bent into a circle and the other into the square enclosing it.

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Q.18 The area of the square outside the circle is:
(a) 1200 cm² (b) 2400 cm²
(c) 3600 cm² (d) 4200 cm²

Q.19 The ratio of the radius of the circle to the perimeter of the square is:
(a) 8:1 (b) 1:8 (c) 4:1 (d) 1:4

Q.20 There are two regular polygons with the number of sides in the ratio of 4:5 and the interior angles in the ratio of 25:26. The number of sides in the first polygon are:
(a) 8 (b) 10 (c) 12 (d) 15

Q.21 A square is inscribed in a semi-circle of radius 10 cm. What is the area of the inscribed square? (Given that the side of the square is along the diameter of the semicircle.)
(a) 70 cm² (b) 50 cm² (c) 25 cm² (d) 80 cm²

Q.22 ABCD is a parallelogram in which $\angle B = 70°$. Find the number of points X in the plane of the parallelogram such that it is equidistant from its vertices.
(a) zero (b) one (c) two (d) three

Q.23 There is a fan with 3 blades at 120° to each other whose central circular disc has an area of 3π cm² and a blade is $(20-\sqrt{3})$ cm long. If the tips of the blades are joined so as to form an equilateral triangle, what will be its area?
(a) 900 cm² (b) $300\sqrt{3}$ cm²
(c) $(900+9\pi)$ cm² (d) $(3\pi+300)$ cm²

Q.24 Two circles of an equal radii are drawn, without any overlap, in a semicircle of radius 2 cm. If these are the largest possible circles that the semicircle can accommodate, what is the radius (in cm) of each of the circles?
(a) 0.414 (b) 0.828 (c) 0.172 (d) 0.586

Q.25 The sides of a regular octagon are extended to form a star. Find the measure of the internal angle at each point of the star.
(a) 45° (b) 90° (c) 135° (d) 60°

Q.26 PQRS is trapezium, in which PQ is parallel to RS, and PQ = 3 (RS). The diagonal of the trapezium intersect each other at X, then the ratio of ΔPXQ and ΔRXS is
(a) 6:1 (b) 3:1 (c) 9:1 (d) 7:1

Direction for Questions 27 and 28: Read the passage below and solve the questions based on it.

All the angles of a quadrilateral form a GP and its common ratio is a natural number. Exactly two of the angles are obtuse and none of the angles is a right angle.

Q.27 What is the measurement of the largest angle?
(a) 124° (b) 156°
(c) 192° (d) None of these

Q.28 What is the measurement of the smallest angle?
(a) 18° (b) 24°
(c) 36° (d) None of these

Q.29 Two circles with centres C1 and C2 and radii 6 cm and 8 cm, respectively, cut each other at right angles. Find the length of the common chord.
(a) 10 cm (b) 4.8 cm (c) 9.6 cm (d) 5 cm

Q.30 Let ABCDEF be a regular hexagon. What is the ratio of the area of the triangle ACE to that of the hexagon ABCDEF?
(a) 1/3 (b) 1/2 (c) 2/3 (d) 5/6

Q.31 The sides of a triangle are in the ratio of $\frac{1}{2}:\frac{1}{3}:\frac{1}{4}$.
If the perimeter is 52 cm, then the length of the smallest side is:
(a) 9 cm (b) 10 cm (c) 11 cm (d) 12 cm

Q.32 A pond 100 m in diameter is surrounded by a circular grass walk-way 2 m wide. How many square metres of grass is the on the walk-way?
(a) 98 π (b) 100 π (c) 204 π (d) 202 π

Q.33 A triangle and a parallelogram are constructed on the same base such that their areas are equal. If the altitude of the parallelogram is 100 m, then the altitude of the triangle is:
(a) 100 m (b) 200 m
(c) $100\sqrt{2}$m (d) $10\sqrt{2}$ m

Q.34 A rhombus OABC is drawn inside a circle, whose centre is at O, in such a way that the vertices A, B, and C of the rhombus are on the circle. If the area of the rhombus is $32\sqrt{3}$ m², then the radius of the circle is:
(a) 64 m (b) 8 m (c) 32 m (d) 46 m

Q.35 Let A be the area of a square inscribed in a circle of radius 'r', and let B be the area of a hexagon inscribed in the same circle. Then, B/A equals:
(a) $2\sqrt{3}$ (b) $\frac{3\sqrt{3}}{4}$
(c) $\frac{2\sqrt{3}}{4}$ (d) None of these

Q.36 The dimensions of a rectangular box are in the ratio of 1:2:4 and the difference between the costs of covering it with the cloth and a sheet at the rate of ₹20 and ₹20.5 per sq m, respectively, is ₹126. Find the dimensions of the box.
(a) 2 m, 6 m, 9 m (b) 6 m, 12 m, 24 m
(c) 1 m, 2 m, 4 m (d) None of these

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Q.37 If the sides 50 m and 130 m of the triangular field meet at an angle of 72°, then find the area in which wheat is cultivated. (sin 72° = 0.9510, cos 72° = 0.309)
(a) 100 πm² (b) 125 πm²
(c) 160 πm² (d) None of these

Q.38 Four horses are tied on the four corners of a square field of 14 m length so that each horse can touch just the other two horses. They were able to graze in the area accessible to them for 11 days. For how many days is the ungrazed area sufficient for them?
(a) 3 days (b) 4 days (c) 5 days (d) 2 days

Q.39 The ratio of the area of a square inscribed in a semi-circle to that of the area of a square inscribed in the circle of the same radius is:
(a) 2:1 (b) 2:3 (c) 2:5 (d) 2:7

Q.40 The area of a rectangle is given by $A = 8x^2 - 2x - 15$. If the length was larger than the breadth, then the length can be:
(a) $(3x + 5)$ (b) $(5x + 3)$ (c) $(4x + 5)$ (d) $(2x-3)$

Q.41 The ratio of the area of a square to that of the square drawn on the its the diagonal is:
(a) 1:4 (b) 2:1 (c) 1:2 (d) 1:3

Q.42 An equilateral triangle had 7 cm long sides. Equal circles were drawn about the three vertices of the triangle so that each circle touched the other two. Calculate the area of space enclosed between these circles.
(a) 3.934 cm² (b) 1.967 cm²
(c) 3.9 cm² (d) 1.95 cm²

Q.43 What is the area of the triangle in which two of its medians 9 cm and 12 cm long intersect at the right angles?
(a) 72 (b) 60 (c) 56 (d) 48

Q.44 If one leg of an isosceles right-angled triangle is increased by 6 cm and that of the other leg decreased by 4 cm, then the area of the triangle decreases by 24 sq cm. Find the length of the leg of the original triangle.
(a) 36 cm (b) 30 cm
(c) 24 cm (d) None of these

Q.45 ABCD is a parallelogram in which $\angle D = 120°$. The bisector of $\angle D$ bisects the side AB. If the length of the bisector is 5 cm, then what is the perimeter of the parallelogram?
(a) 30 cm (b) 25 cm (c) 40 cm (d) 15 cm

Q.46 Four horses are tethered at four corners of a square plot of side 14 m so that the adjacent horses can just reach one another. There is a small circular pond of area 20 m² at the centre. Find the ungrazed area.
(a) 22 m² (b) 42 m² (c) 84 m² (d) 168 m²

Q.47 Euclid has a triangle in mind. Its longest side has length 20 and another of its sides has length 10. Its area is 80. What is the exact length of its third side?
(a) $\sqrt{260}$ (b) $\sqrt{250}$ (c) $\sqrt{240}$ (d) $\sqrt{270}$

Q.48 AB is the hypotenuse in the right angled triangle ABC. N is the point inside the triangle which divides the triangle in three equal parts (ΔABN, ΔCAN, ΔBCN). What is the distance between the circumcentre of this triangle from this point N?
(a) $\frac{AB}{4}$ (b) $\frac{AB}{6}$ (c) $\frac{AB}{3}$ (d) $\frac{2}{1+\sqrt{3}}$

Q.49 Two sides of a triangle are 4 and 5. Then, for the area of the triangle, which one of the following bounds is the sharpest?
(a) <10 (b) ≤10 (c) ≤8 (d) >5

Q.50 ABC is a triangle and P is a point inside it such that $\angle BPC = \angle CPA = \angle APB$. Then, P is:
(a) The point of intersection of medians
(b) The incentre
(c) The circumcentre
(d) None of these

Moderate

Q.1 The angles of a triangle are in the ratio of 4:1:1. Then, the ratio of the largest side to the perimeter is:
(a) $\frac{2}{3}$ (b) $\frac{1}{2+\sqrt{3}}$ (c) $\frac{\sqrt{3}}{2+\sqrt{3}}$ (d) $\frac{2}{1+\sqrt{3}}$

Q.2 In ΔABC, P and Q are mid-points of sides AB and BC, respectively, right angled at B, then:
(a) $AQ^2 + CP^2 = AC^2$
(b) $AQ^2 + CP^2 = \frac{4}{5}AC^2$
(c) $AQ^2 + CP^2 = \frac{3}{4}AC^2$
(d) $AQ^2 + CP^2 = \frac{5}{4}AC^2$

Q.3 Three identical right angle cones with base radius r are placed on their bases so that each is touching the other two. The radius of the circle drawn through their vertices is:
(a) Smaller than r
(b) Equal to r

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(c) Larger than r
(d) Depends on the height of the cones

Q.4 The line AB is 6 m in length and is tangent to the inner of the two concentric circles at point C. It is known that the radii of the two circles are integers. The radius of the outer circle is....., where A and B are points on the outer circle.
(a) 5 m (b) 4 m (c) 6 m (d) 3 m

Q.5 Under the usual 2-dimensional coordinate system the equation $|x| + |y| = 1$, where x and y are real numbers, represents:
(a) A rhombus which is not a square.
(b) A parallelogram which is not a rhombus.
(c) A square whose sides are not parallel to the coordinate axis.
(d) A square with sides parallel to the coordinate axis.

Q.6 Two chords of lengths a and b of a circle subtend 60° and 90° angles at the centre, respectively. Which of the following is correct?
(a) $b = \sqrt{2a}$ (b) $b = \sqrt{2b}$ (c) $a = 2b$ (d) $b = 2a$

Q.7 PQR is a right-angled triangle with $\angle Q = 90°$, S is the mid-point of PR, and QS $= \sqrt{117}$ cm. The sum of the length of sides PQ and QR is 30 cm. Area of ∇ PQR is:
(a) 216 cm² (b) 108 cm²
(c) 54 cm² (d) 162 cm²

Q.8 In a trapezium, the diagonals intersect at point O. The ratio of the length of one of the diagonals from one vertex of the trapezium to the point O to its entire length is 2:5. Find the ratio of its parallel sides (smaller side:larger side).
(a) 2:5 (b) 2:3 (c) 2:7 (d) 5:7

Q.9 A chord of length 32 cm is placed inside a circle of radius 20 cm and a point whose distance from the centre of the circle is 13 cm, is marked on the chord. Calculate the lengths of the segment of the chord.
(a) 21 cm and 11 cm (b) 19 cm and 13 cm
(c) 16 cm each (d) 18 cm and 14 cm

Q.10 The sum of the lengths of the hypotenuse and one of the perpendicular sides of a right angled triangle is L. When the area of this triangle is maximum, the angle between these two sides is:
(a) 45° (b) 22.5°
(c) 60° (d) None of these

Q.11 In a triangle, the lengths of the two larger sides are 8 cm and 7 cm. The angles of the triangle are in an arithmetic progression. The length of the remaining side can be:
(a) 3 cm (b) 5 cm
(c) Either 3 cm or 5 cm (d) None of these

Q.12 The biggest possible regular hexagon H is cut out of an equilateral triangle X. The biggest possible equilateral triangle Y is cut out from the hexagon H. What is the ratio of the areas of the equilateral triangles X and Y?
(a) 5:1 (b) 6:1 (c) 8:1 (d) 3:1

Q.13 Nine parallel chords are drawn in a circle of diameter 10 cm. If the distance between any two of the adjacent chords is 1 cm, which of the following statements is always true?
(a) One of the chords is the diameter of the circle.
(b) At least two of the chords must be of an equal length.
(c) The difference between the lengths of any two adjacent chords on the same side of the diameter is greater than 1 cm.
(d) None of these

Direction for Questions 14 and 15: Read the passage below and solve the questions based on it.

There are three equal circles of unit radii touching each other.

Q.14 Find the area of the triangle circumscribing the three circles.
(a) $\sqrt{3}(\sqrt{3}+1)^2$ (b) $(\sqrt{3}+1)^2$
(c) $\sqrt{3}(\sqrt{3}+1)$ (d) None of these

Q.15 What would be the area of the remaining portion if the same three circles are circumscribed by another circle?
(a) $\pi\left(\frac{2}{\sqrt{3}}+1\right)^2$ (b) $\pi\left(\frac{2}{\sqrt{3}}+1\right)^2 - 3\pi$
(c) $\left(\frac{3}{\sqrt{2}}+1\right)^2 \pi - 3\pi$ (d) $\left(\frac{1}{\sqrt{3}}+1\right)^2 \pi - 3\pi$

Q.16 A circle passes through the vertex A of an equilateral triangle ABC and is tangent to BC at it mid-point. Find the ratio in which the circle divides each of the sides AB and AC.
(a) 1:1 (b) 3:2 (c) 3:1 (d) 2:1

Q.17 In any quadrilateral ABCD, the diagonal AC and BD intersect at a point X. If E, F, G, and H are the mid-points of AX, BX, CX, and DX, respectively, then what is the ratio of (EF + FG + GH + HE) to (AD + DC + CB + BA)?
(a) $\frac{1}{2}$ (b) $\frac{3}{2}$
(c) $\frac{3}{4}$ (d) Data insufficient

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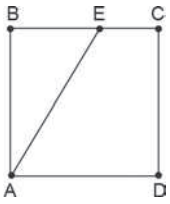
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Q.18 Points A and B are on a circle of radius 5 and AB = 6. Point C is the mid-point of the minor arc AB. What is the length of the line segment AC?

(a) $\sqrt{10}$ (b) $\frac{7}{2}$ (c) $\sqrt{14}$ (d) $\sqrt{15}$

Q.19 Square ABCD has side length 10. Point E is on $\overline{BC}$, and the area of ΔABE is 40. What is BE?

(a) 4 (b) 5 (c) 8 (d) 7



Q.20 A vertical lamp-post OP stands at the centre O of a square ABCD. Let h and b denote the length OP and AB, respectively. Suppose ∠APB = 60° then the relationship between h and b can be expressed as:

(a) $2b^2 = h^2$ (b) $2h^2 = b^2$
(c) $3b^2 = 2h^2$ (d) $3h^2 = 2b^2$

Q.21 Consider two different cloth-cutting processes. In the first one, *n* circular cloth pieces are cut from a square cloth piece of side a in the following steps: the original square of side a is divided into n smaller squares, not necessarily of the same size; then a circle of the maximum possible area is cut from each of the smaller squares. In the second process, only one circle of the maximum possible area is cut from the square of side a and the process ends there. The cloth pieces remaining after cutting the circles are scraped in both the process. The ratio of the total area of the scrap cloth generated in the former to that in the latter is:

(a) 1:1 (b) $\sqrt{2}:1$

(c) $\frac{n(4-x)}{4n-x}$ (d) $\frac{4n-x}{n(4-x)}$

Q.22 Let S_1 be a square of side a. Another square S_2 is formed by joining the mid-points of the sides of S_1. The same process is applied to S_2 to form yet another square S_3, and so on. If $A_1, A_2, A_3, \ldots$ be the areas and $P_1, P_2, P_3, \ldots$ be the perimeters of $S_1, S_2, S_3, \ldots$, respectively, then the ratio of $\frac{P_1+P_2+P_3+.....}{A_1+A_2+A_3+.....}$ equals.

(a) $\frac{(2+\sqrt{2})}{a}$ (b) $\frac{2(2-\sqrt{2})}{a}$

(c) $\frac{2(2+\sqrt{2})}{a}$ (d) $\frac{2(1+2\sqrt{2})}{a}$

Q.23 One of the angles of a parallelogram is of 150°. Altitudes are drawn from the vertex of this angle. If these altitudes measure 6 cm and 8 cm, then find the perimeter of the parallelogram.

(a) 28 cm (b) 42 cm (c) 56 cm (d) 64 cm

Q.24 A square, whose side is 2 m, has its corners cut away so as to form an octagon with all sides equal. Then, the length of the each side of the octagon, in metres is:

(a) $\frac{\sqrt{2}}{\sqrt{2}+1}$ (b) $\frac{\sqrt{2}}{\sqrt{2}-1}$ (c) $\frac{\sqrt{2}}{\sqrt{2}-1}$ (d) $\frac{\sqrt{2}}{\sqrt{2}+1}$

Q.25 A certain city has a circular wall around it and this wall has four gates pointing north, south, east, and west. A house stands outside the city 3 km north of the north gate, and it can be seen from a point 9 km east of the south gate. What is the diameter of the wall that surrounds the city?

(a) 6 km (b) 9 km
(c) 12 km (d) None of these

Q.26 The perimeter of a right-angled triangle is four times the shortest side. The ratio of the other two sides is:

(a) 5:6 (b) 3:4 (c) 4:5 (d) 2:3

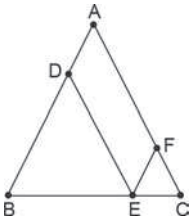
Q.27 A circle is inscribed in an equilateral triangle and a square is inscribed in the circle. The ratio of the area of the triangle to the area of the square is:

(a) $\sqrt{3}:\sqrt{2}$ (b) $3\sqrt{3}:2$ (c) $3:\sqrt{2}$ (d) $\sqrt{2}:1$

Q.28 The sides of a triangle are given to be x^2+x+1, $2x+1$ and x^2-1. Then, the largest of the three angles of the triangle is

(a) 75° (b) $\frac{x}{x+1}\pi$ (c) 120° (d) 135°

Q.29 In ΔABC, $\overline{AB} = \overline{AC} = 28$ and $\overline{BC} = 20$. Points D, E, and F are on sides $\overline{AB}$, $\overline{BC}$, and $\overline{AC}$, respectively, such that $\overline{DE}$ and $\overline{EF}$ are parallel to $\overline{AC}$ and $\overline{AB}$, respectively. What is the perimeter of parallelogram ADEF?



(a) 48 (b) 52 (c) 54 (d) 56

Q.30 In a triangle ABC, the medians AM and CN to the sides BC and AB, respectively, intersect at the point O. Let P be the mid-point of AC and let MP intersect CN at Q. If the area of the triangle OMQ is s square units, the area of ABC is:

(a) 16s (b) 18s (c) 12s (d) 24s

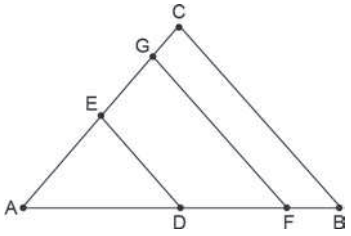
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Q.31 ABCD is a cyclic quadrilateral and the points A, B and C form an equilateral triangle. What is the sum of the lengths of line segments DA and DC?

(a) DB (b) DB/2 (c) $\sqrt{2}$ DB (d) DB/$\sqrt{2}$

Q.32 Triangle ABC is equilateral with AB = 1. Points E and G are on $\overline{AC}$ and points D and F are on $\overline{AB}$ such that both $\overline{EF}$ and $\overline{FG}$ are parallel to $\overline{BC}$. Furthermore, triangle ADE and trapezoids DFGE and FBCG all have the same perimeter. What is DE + FG?



(a) 1 (b) $\frac{3}{2}$ (c) $\frac{21}{13}$ (d) $\frac{13}{8}$ (e) $\frac{5}{3}$

Q.33 The side AB of a rectangle ABCD is a tangent to a circle which passes through the points C and D. The centre of the circle does not lie within the rectangle ABCD. If the length of the rectangle is twice the breadth, then what is the radius of the circle (in terms of breadth of the rectangle)?

(a) Breadth $(2 - \sqrt{2})$ (b) Breadth

(c) $2 \times$ Breadth (d) None of these

Q.34 Let C_1 and C_2 be the inscribed and circumscribed circles of a triangle with sides 3 cm, 4 cm and 5 cm. The $\frac{\text{area of } C_1}{\text{area of } C_2}$ equals.

(a) $\frac{16}{25}$ (b) $\frac{4}{25}$ (c) $\frac{9}{25}$ (d) $\frac{9}{16}$

Q.35 The sides of a triangle are given by:

$\sqrt{b^2+c^2}, \sqrt{c^2+a^2}$ and $\sqrt{a^2+b^2}$ where a, b, c are positive. Then, the area of the triangle equals.

(a) $\frac{1}{2}\sqrt{b^2c^2+c^2a^2+a^2b^2}$ (b) $\frac{1}{2}\sqrt{a^4+b^4+c^4}$

(c) $\frac{\sqrt{3}}{2}\sqrt{b^2c^2+c^2a^2+a^2b^2}$ (d) $\frac{\sqrt{3}}{2}(bc+ca+ab)$

Q.36 Let the bisector of the angle at C of a triangle ABC intersect the side AB in a point D. Then, the geometric mean of CA and CB

(a) is less than CD.

(b) is equal to CD.

(c) is greater than CD.

(d) does not always satisfy any one of the foregoing property.

Q.37 There is a pole at the centre of a stadium which is in the shape of an equilateral triangle. The angle subtended by any side of the stadium at the top of the tower is right-angled. What is the ratio of the height of the pole to the perimeter of the stadium?

(a) $\sqrt{6}$ (b) $\frac{\sqrt{6}}{18}$ (c) $\frac{\sqrt{6}}{3}$ (d) $\frac{\sqrt{6}}{4}$

Q.38 In ΔABC, AB = 86 and AC = 97. A circle with centre A and radius AB intersects $\overline{BC}$ at points B and X. Moreover, $\overline{BX}$ and $\overline{CX}$ have integer lengths. What is BC?

(a) 11 (b) 61 (c) 33 (d) 28

Q.39 The length of the common chord of two circles of radii 15 cm and 20 cm, whose centres are 25 cm apart, is (in cm):

(a) 24 (b) 25 (c) 15 (d) 20

Q.40 PQRS is a trapezium with PQ and RS parallel. PQ = 6 cm, QR = 5 cm, RS = 3 cm, PS = 4 cm. The area of PQRS is:

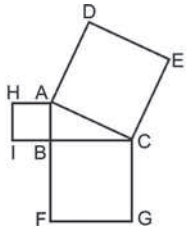
(a) 27 cm^2 (b) 12 cm^2

(c) 18 cm^2 (d) Cannot be determined

Q.41 If the number of square inches in the area of a square is equal to the number of inches in its circumference, then the diagonal of the square is equal to:

(a) 4 (b) $4\sqrt{2}$ (c) $3\sqrt{2}$ (d) $\sqrt{2}$

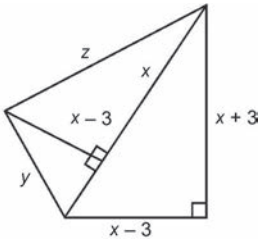
Q.42 All the three quadrilaterals ADEC, ABIH and BCGF are squares and ABC = 90°. If the area of ADEC = x^2 and the area of AHIB = y^2 ($x^2 > y^2$), then the area of BCGF is:



(a) $(x+y)(x-y)$ (b) $(x+y)^2$

(c) $(x-y)^2$ (d) None of these

Q.43 Based on the figure below, what is the value of x, if $y = 10$.



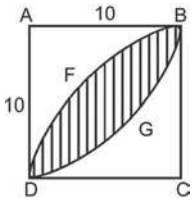
(a) 10 (b) 11

(c) 12 (d) None of these

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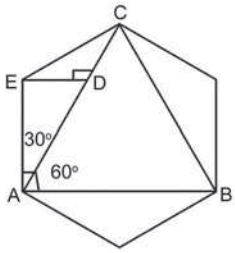
Q.44 In the figure, ABCD is a square with side 10. BFD is an arc of a circle with centre C. BGD is an arc of a circle with centre A. What is the area of the shaded region?



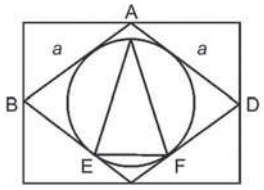
(a) $100\pi - 50$ (b) $100 - 25\pi$
(c) $50\pi - 100$ (d) $25\pi - 10$

Q.45 In the figure below, if the perimeter of ∇ABC is p, then the perimeter of the regular hexagon is:

(a) $\frac{3p}{\sqrt{2}}$ (b) $\frac{\sqrt{2}p}{3}$ (c) $\frac{\sqrt{3}p}{2}$ (d) $\frac{2p}{\sqrt{3}}$



Q.46 What is the area of the inner equilateral triangle if the side of the outermost square is 'a'? (ABCD is a square).

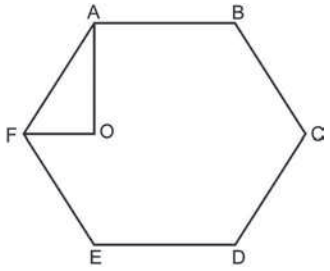


(a) $\frac{3\sqrt{3}a^2}{32}$ (b) $\frac{3\sqrt{3}a^2}{64}$ (c) $\frac{5\sqrt{3}a^2}{32}$ (d) $\frac{\sqrt{3}a^2}{16}$

Q.47 Each side of a given polygon is parallel to either the X or the Y-axis. A corner of such a polygon is said to be convex if the internal angle is 90° or concave if the internal angle is 270°. If the number of convex corners in such a polygon is 25, then the number of concaves must be:

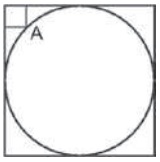
(a) 20 (b) 0 (c) 21 (d) 22

Q.48 In the figure below, ABCDEF is a regular hexagon and AOF = 90°. FO is parallel to ED. What is the ratio of the area of the triangle AOF to that of the hexagon ABCDEF?



(a) $\frac{1}{12}$ (b) $\frac{1}{6}$ (c) $\frac{1}{24}$ (d) $\frac{1}{18}$

Q.49 In the figure below, the rectangle at the corner measures 10 cm × 20 cm. The corner A of the rectangle is also a point on the circumference of the circle. What is the radius of the circle in cm?



(a) 10 cm (b) 40 cm
(c) 50 cm (d) None of these

Q.50 A piece of paper is in the shape of a right-angled triangle and is cut along a line that is parallel to the hypotenuse, leaving a smaller triangle. There was a 35% reduction in the length of the hypotenuse of the triangle. If the area of the original triangle was 34 square inches before the cut, what is the area (in square inches) of the smaller triangle?

(a) 16.665 (b) 16.565 (c) 15.465 (d) 14.365

ADVANCED

Q.1 A ball of diameter 15 cm is floating so that the top of the ball is 5 cm above the smooth surface (water) of the pond. What is the circumference in centimetres of the circle formed by the contact of the water surface with the ball?

(a) $10\sqrt{2}\pi$ (b) 50 p (c) 10 p (d) $5\sqrt{2}\pi$

Q.2 Let S be as an arbitrary point on the side PQ of an acute-angled ∇PQR. Let T be the point of intersection of QR extended with the straight line PT drawn parallel to SR through P. Let U be the point of intersection of PR extended with the straight line QU drawn parallel to SR through Q. If PT = a and QU = b, then the length of SR is:

(a) $\frac{a+b}{ab}$ (b) $\frac{a-b}{ab}$ (c) $\frac{ab}{a+b}$ (d) $\frac{ab}{a-b}$

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Q.3 The municipal authorities of three Cities A, B, and C jointly constructed three straight roads connecting these cities. The area enclosed by three roads is $150\sqrt{39}$ km^2. The authorities of City B found the length of the road connecting it to the City A is 40 km and that to the City C is 50 km. What is the length of the road connecting the Cities A and C, if it is the longest of the three roads?

(a) $20\sqrt{3}$ km (b) $30\sqrt{3}$ km
(c) $30\sqrt{6}$ km (d) $10\sqrt{55}$ km

Q.4 In an equilateral D, 3 coins of radii 1 unit each are kept in such a way that they touch each other and also the sides of the triangle. What is the area of the triangle (in sq. units)?

(a) $4 + 5\sqrt{2}$ (b) $6 + 4\sqrt{3}$ (c) $4 + 6\sqrt{3}$ (d) $3+8\sqrt{3}$

Q.5 In a scalene Δ, sum of all the sides can be at most 13 units. How many triangles are possible?

(a) 3 (b) 4 (c) 5 (d) 6

Q.6 Three circles of equal radii have been drawn inside an equilateral triangle, of side a, such that each circle touches the other two circles as well as two sides of the triangles. Then, the radius of each circle is:

(a) $\frac{a}{2(\sqrt{3}+1)}$ (b) $\frac{a}{2(\sqrt{3}-1)}$ (c) $\frac{a}{\sqrt{3}-1}$ (d) $\frac{a}{\sqrt{3}-1}$

Q.7 Consider a circle with unit radius. There are seven adjacent sectors, S_1, S_2, S_3, …, S_7, in the circle such that their total area is $\frac{1}{8}$ th of the area of the circle. Further, the area of the jth sector is twice that of the (j − 1)th sector, for j = 2, …, 7. What is the angle, in radians, subtended by the arc of S_1 at the centre of the circle?

(a) $\frac{\pi}{508}$ (b) $\frac{\pi}{2040}$ (c) $\frac{\pi}{1016}$ (d) $\frac{\pi}{1524}$

Q.8 In a triangle ABC, let C_1 be any point on the side AB other than A or B. Join CC_1. The line passing through A and parallel to CC_1 intersects the line BC extended at A_1. The line passing through B and parallel to CC_1 intersects the line AC extended at B_1. The lengths AA_1, BB_1, CC_1 are given to be p, q, r units, respectively. Then:

(a) $r = \frac{pq}{p+q}$ (b) $r = \frac{p+q}{4}$

(c) $r = \frac{\sqrt{pq}}{2}$ (d) None of these

Q.9 Two circles APQC and PBDQ intersect each other at the points P and Q, and APB and CQD are two parallel straight lines. Then, only one of the following statements is always true. Which one is it?

(a) ABDC is a cyclic quadrilateral.
(b) AC is parallel to BD.
(c) ABDC is a rectangle.
(d) ΔACQ is a right angle.

Q.10 Let s = $\{(x, y):|x| + |y| = 2\}$. Then, the diameter of S is:

(a) 2 (b) $4\sqrt{2}$ (c) 4 (d) $2\sqrt{2}$

Q.11 Through the centroid of an equilateral triangle, a line parallel to the base is drawn. On this line, an arbitrary point P is taken inside the triangle. Let h denote the distance of P from the base of the triangle. Let h_1 and h_2 be the distance of P from the other two sides of the triangle. Then:

(a) $h = \frac{h_1 + h_2}{2}$ (b) $h = \sqrt{h_1 h_2}$

(c) $h = \frac{2h_1h_2}{h_1 + h_2}$ (d) None of these

Q.12 A pyramid has a square base ABCD and vertex E. The area of square ABCD is 196, and the areas of ΔABE and ΔCDE are 105 and 91, respectively. What is the volume of the pyramid?

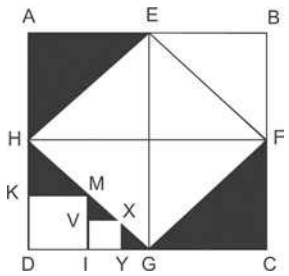
(a) 392 (b) $196\sqrt{6}$ (c) 784 (d) $392\sqrt{3}$

Q.13 Let A (x_1, y_1), B (x_2, y_2), C (x_3, y_3), D (x_4, y_4) be four points such that x_1, x_2, x_3, x_4 and y_1, y_2, y_3, y_4 are both in an AP. If Δ denotes the area of the quadrilateral ABCD, then:

(a) Δ = 0
(b) Δ = 1
(c) Δ < 1
(d) Δ depends on the coordinates A, B, C, and D.

Direction for Questions 14 to 16: Read the passage below and solve the questions based on it.

In the figure given below, EFGH is a square formed by joining the mid-points of the sides of the square ABCD.KMID and VXYI are the squares formed inside the right angled triangles HDG and MLG, respectively. The side of the square ABCD is equal to 'a' cm.



Q.14 What is the ratio of areas of ABCD and KMID?

(a) 4:1 (b) 8:1 (c) 16:1 (d) 12:1

Q.15 What is the ratio of areas of EFGH and VXYI?

(a) 32:1 (b) 64:1 (c) 16:1 (d) 8:1

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Q.16 What is the area of shaded region (in cm^2)?

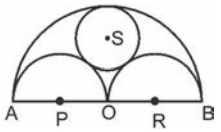
(a) $\frac{5}{16}a^2$ (b) $\frac{19}{64}a^2$ (c) $\frac{7}{16}a^2$ (d) $\frac{27}{64}a^2$

Q.17 Corners are sliced off a unit cube so that the six faces each become regular octagons. What is the total volume of the removed tetrahedron?

(a) $\frac{5\sqrt{2}-7}{3}$ (b) $\frac{10-4\sqrt{2}}{3}$

(c) $\frac{3-2\sqrt{2}}{3}$ (d) $\frac{8\sqrt{2}-11}{3}$

Q.18 Three horses are grazing within a semi-circular field. In the diagram given below, AB is the diameter of the semi-circular field with centre at O. Horses are tied up at P, R, and S such that PO and RO are the radii of semi-circles with centres at P and R, respectively, and S is the centre of the circle touching the two semi-circles with diameters AO and OB. The horses tied at P and R can graze within the respective semi-circle and the horse tied at S can graze within the circle centred at S. The percentage of the area of the semi-circles with diameter AB that cannot be grazed by the horses is nearest to:



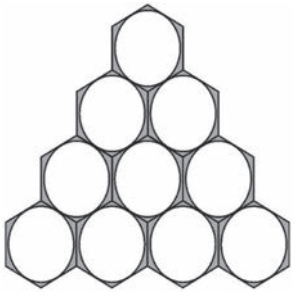
(a) 20 (b) 28 (c) 36 (d) 40

Q.19 Inradius of a right-angled ΔABC is 32 units and the sum of the length of its two sides other than the hypotenuse is 42 units. What is its circumradius?

(a) 26
(b) 52
(c) More than one value possible
(d) Triangle ABC is not possible

Direction for Questions 20 and 21: Read the passage below and solve the questions based on it.

In the figure given above, each hexagon is regular and has side measuring 'a' cm. it is also given that the last row has 'n' such regular hexagons and all the hexagons are inscribed by circles.



Q.20 Find the ratio of area of the shaded region to the area of unshaded region.

(a) $\frac{3a^2}{2}[n(n+1)]\left(\sqrt{3}-\frac{\pi}{2}\right)$

(b) $\frac{a^2}{2}[n(n+1)]\left(\frac{3\sqrt{3}}{2}-\pi\right)$

(c) $\frac{3a^2}{2}[n(n+1)]\left(\sqrt{3}-\frac{\pi}{2}\right)$

(d) None of these

Q.21 Suppose the above figure is a 3-D figure and each hexagon is inscribed by a sphere of maximum volume. The height of the hexagonal figure is just enough to contain the sphere, then find the volume of the shaded region (there are 'n' hexagonal figures of side 'a' in the last row).

(a) $\frac{\sqrt{3}}{2}a^3\left(\frac{3\sqrt{3}}{2}-\pi\right)[n(n+1)]a^3$

(b) $\frac{\sqrt{3}}{4}a^3\left(\frac{3\sqrt{3}}{4}-\pi\right)[n(n+1)]$

(c) $\frac{1}{2}\frac{\sqrt{3}}{2}a^3\left(\frac{3\sqrt{3}}{4}-\pi\right)[n(n+1)]$

(d) $\frac{\sqrt{3}}{2}a^3(3\sqrt{3}-\pi)[n(n+1)]$

Q.22 If a sphere of the maximum volume is placed inside a hollow right circular cone with radius 'r' and slant height 'λ' such that the base of the cone touches the sphere, then the volume of the sphere is:

(a) $\frac{4}{3}\pi\left(\frac{\ell+r}{\ell-r}\right)^3$ (b) $\frac{4}{3}\pi r^3\left(\frac{\ell-r}{\ell+r}\right)^{\frac{3}{2}}$

(c) $\frac{4}{3}\pi\left(\frac{\ell-r}{\ell+r}\right)^3$ (d) $\frac{4}{3}\pi r^3\left(\frac{\ell+r}{\ell-r}\right)^{\frac{3}{2}}$

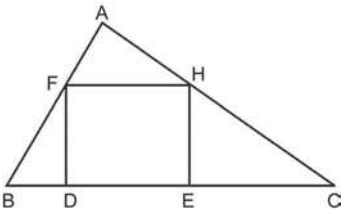
Q.23 A hollow right circular cylinder of radius r and height $4r$ is standing vertically on a plane. If a solid right circular cone of radius $2r$ and height $6r$ is placed with its vertex down in the cylinder, then the volume of the portion of the cone outside the cylinder is:

(a) $\frac{8}{3}\pi r^3$ (b) $2\pi r^3$ (c) $\frac{9}{8}\pi r^3$ (d) $7\pi r^3$

Q.24 Consider the following figure: AB = 10 cm, AC = 17 cm, BC = 21 cm and EHFD is a square. Find the length of the side of square (in cm).

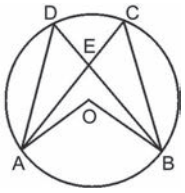
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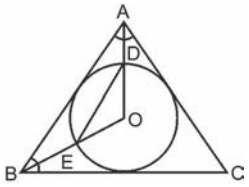
(a) 10.5 (b) 12
(c) 13.5 (d) None of these

Q.25 In the given figure, O is the centre of the circle and ∠AOB = 120°. What is the mesure of ∠AEB?



(a) 100° (b) 90°
(c) 110° (d) Cannot be determined

Q.26 For the equilateral triangle ABC, D and E are mid-points of AO and OB, and DE = 3 units. The circumference of the circle is equal to:



(a) 4π units (b) $\pi\sqrt{2}$ units
(c) $2\pi\sqrt{3}$ units (d) None of these

Q.27 The length of the sides CB and CA of a triangle ABC are given by a and b, and the angle C is $2\pi/3$. The line CD bisects the angle C and meets AB at D. Then, the length of CD is:

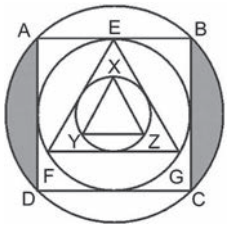
(a) $\frac{a^2+b^2}{2(a+b)}$ (b) $\frac{ab}{2(a+b)}$ (c) $\frac{1}{a+b}$ (d) $\frac{ab}{a+b}$

Q.28 For a regular octagon inscribed in a circle of radius 1 cm, the product of the distance from a fixed vertex to the other seven vertices is:

(a) 4 (b) 8 (c) 12 (d) 16

Direction for Questions 29 to 31: Read the passage below and solve the questions based on it.

In the figure given below, a square ABCD is inscribed in a circle of radius 5 cm. The square ABCD is inscribed by a circle which is inscribed by a triangle EFG. The triangle EFG is inscribed by a circle which is again inscribed by a triangle XYZ.



Q.29 Find the ratio of the radius of the outermost circle to the radius of the circle inscribed in triangle XYZ.

(a) 4:1 (b) $2\sqrt{2}:1$ (c) $4\sqrt{2}:1$ (d) 8:1

Q.30 Find the ratio of the area of square ABCD and the area of triangle XYZ.

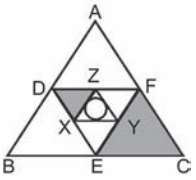
(a) $16:3\sqrt{3}$ (b) $64:3\sqrt{3}$ (c) 16:3 (d) $64:\sqrt{3}$

Q.31 Find the area of the shaded region (in cm^2).

(a) $\frac{500\pi-800-75\sqrt{3}}{16}$ (b) $\frac{900\pi-1600-75\sqrt{3}}{}$

(c) $\frac{400\pi-800+75\sqrt{3}}{32}$ (d) None of these

Q.32 D, E, and F are the mid-points of the sides AB, BC, and CA, respectively, and X, Y, and Z the are mid-points of DE, EF, and FD, respectively. It is given that the circumradius of triangle ABC is $8\sqrt{3}$ cm and triangle XYZ is inscribed by a circle. (in cm^2). What is the ratio of shaded area to that of area of ΔABC?



(a) $\frac{3}{16}$ (b) $\frac{5}{16}$ (c) $\frac{7}{16}$ (d) $\frac{9}{16}$

Q.33 The diameter of a right conical tent is 6 m. If a pole of length 2 m can be fixed in the tent at half the distance of the radius from the centre of the base, then the area of the canvas required is (in m^2):

(a) 10π (b) 12π (c) 15π (d) 16π

Q.34 A pyramid has its base as an equilateral triangle, of each side being one m. Its slant edge is 3 m. The whole surface area of the pyramid is equal to:

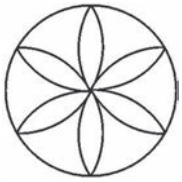
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(a) $\frac{\sqrt{3}+3\sqrt{13}}{4}m^2$ (b) $\frac{\sqrt{3}+2\sqrt{35}}{4}m^2$

(c) $\frac{\sqrt{3}+2\sqrt{13}}{4}m^2$ (d) $\frac{\sqrt{3}+3\sqrt{35}}{4}m^2$

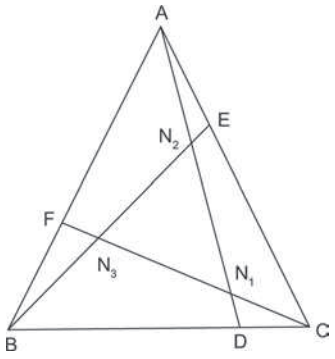
Q.35 In the figure given below, if a flower is inscribed in a circle of radius 1 cm, then find the area of the flower.



(a) $\left(\pi-\frac{3\sqrt{3}}{2}\right)\text{cm}^2$ (b) $(2\pi-3\sqrt{3})\text{cm}^2$

(c) $\left(\frac{5\pi}{3}-\frac{5\sqrt{3}}{2}\right)\text{cm}^2$ (d) $(4\pi-3\sqrt{3})\text{cm}^2$

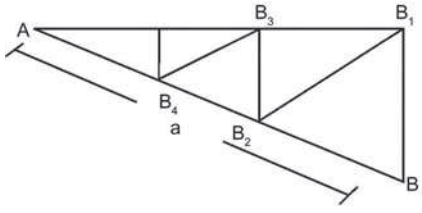
Q.36 In the figure given below, CD, AE, and BF are one third of their respective sides. It follows that $AN_2: N_2N_1:N_1D = 3:3:1$ and similarly, for lines BE and CF. Then, the area of triangle $N_1N_2N_3$ is:



(a) $\frac{1}{7}\Delta ABC$ (b) $\frac{1}{6}\Delta ABC$

(c) $\frac{1}{9}\Delta ABC$ (d) $\frac{1}{10}\Delta ABC$

Q.37 In the figure given below, two rays are drawn through a point A at an angle of 30°. A point B is taken on one of them at a distance of 'a' from the point A. A perpendicular is drawn from the point B to the other ray, and another perpendicular is drawn from its foot to meet AB at another point from where the similar process is repeated indefinitely. The length of all such lines will be:

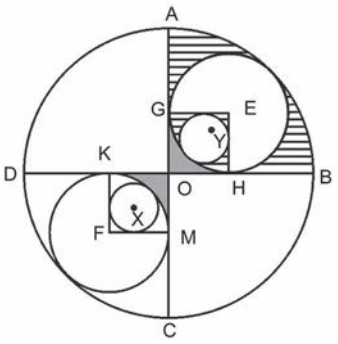


(a) $a(1-\sqrt{3})$ (b) $2a(2+\sqrt{3})$

(c) $\frac{3}{2a}$ (d) a

Direction for Questions 38 to 40: Read the passage below and solve the questions based on it.

In the figure given below, O is the centre of circles have radius (OB) + r. E and F are the centre of circles inscribed in the quarter circles AOB and COD, respectively, whereas X and Y are the centre of circles inscribed in the quarter circle KFM and GEH, respectively.



Q.38 Find the area of the shaded region (not including the lined region).

(a) $\frac{3}{7}r^2(3-2\sqrt{2})$ (b) $\frac{3}{14}r^2(3-2\sqrt{2})$

(c) $3\frac{3}{14}r^2(3+2\sqrt{2})$ (d) None of these

Q.39 Find the ratio of the radii of the circle with centre Y, E, and O.

(a) $1:(\sqrt{2}-1):(3+2\sqrt{2})$ (b) $1:(\sqrt{2}+1):(3-2\sqrt{2})$

(c) $1:(\sqrt{2}-1)(3-2\sqrt{2})$ (d) 1:2:4

Q.40 Find the area of the lined region.

(a) $\left(\frac{594\sqrt{2}-836}{14}\right)r^2$ (b) $\left(\frac{600\sqrt{2}-836}{14}\right)r^2$

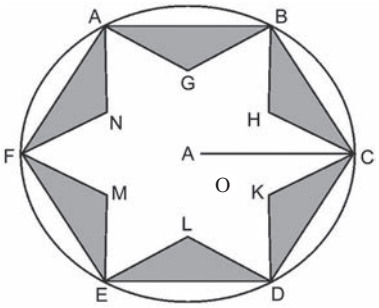
(c) $\left(\frac{594\sqrt{2}-845}{14}\right)r^2$ (d) $\left(\frac{600\sqrt{2}-845}{14}\right)r^2$

Direction for Questions 41 to 43: Read the passage below and solve the questions based on it.

O is the centre of a circle having radiua (OC) = r. ABCDEF is a regular hexagon and AGBHCKDLEMFNA is a regular six-pointed star.

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Q.41 Find the ratio of the circumference of a circle to the perimeter of the hexagon ABCDEF to the perimeter of the star AGBHCKD LEMFNA.

(a) $\pi : \sqrt{3}$ (b) $\pi : 3 : 2\sqrt{3}$

(c) $2 \cdot \pi : 6 : 4\sqrt{2}$ (d) $\pi : 2\sqrt{3}$

Q.42 Find the ratio of the area of the shaded region to the area of the star.

(a) 1:1 (b) 1:2 (c) 1:3 (d) 2:3

Q.43 If the triangles ANG, BGH, CHK, DKL, ELM, and FMN and the hexagon NGHKLM are inscribed by circles of the maximum area, then find the sum of the areas of all those circles.

(a) $\frac{\pi}{2}r^2$ (b) $\frac{\pi r^2}{3}$ (c) $\frac{5\pi r^2}{12}$ (d) $\frac{7\pi r^2}{12}$

Q.44 A triangle ABC with an obtuse angle B is inscribed in a circle. The altitude AD of the triangle is tangent to the circle. The side BC has length 12 cm and the segment BD has length 4 cm. Find the area of the triangle ABC (in cm^2).

(a) 48 (b) 60 (c) 72 (d) 36

Q.45 If the sum of the squares of the sides of a triangle is denoted by A and its perimeter by P, then which of the following is true about $\frac{A^2}{P}$?

(a) $\frac{A^2}{P} < 2$ (b) $1 \le \frac{A^2}{P} \le 4$

(c) $2 \le \frac{A^2}{P} \le 3$ (d) $4 \le \frac{A^2}{P}$

Q.46 A ring of diameter 10 cm is suspended from a point 12 cm vertically above the centre by six equal strings. The strings are attached to the circumference of the ring at equal intervals, thus keeping the ring in a horizontal plane. The value of cosine of the angle between the two adjacent strings lies in between:

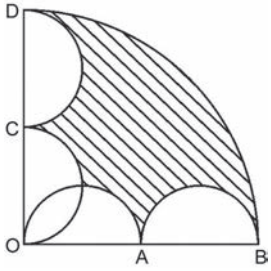
(a) 0 and 1/2 (b) 1/2 and 1

(c) −1/2 to 0 (d) −1 to −1/2

Q.47 A square of side 4 cm is drawn and a circle C_1 is inscribed in it. Now, four more circles C_2, C_3, C_4, and C_4, each of radius 2 cm, are drawn with centres as A, B, C, and D, respectively. In the resultant figure, what is the total area that is common to at least two of the five circles (in sq. cms)?

(a) $6\pi - 16$ (b) $7\pi - 16$ (c) $8\pi - 16$ (d) $9\pi - 16$

Q.48 In the figure, find the area of the shaded portion, that is, the portion of the quadrant DOB which is not included in the 4 semicircles. Given that OA = AB = OC = CD = 2 and ∠DOB = 90°.



(a) $5\pi - \frac{1}{2}$ (b) $5\pi - 1$ (c) $\frac{5\pi}{2} - \frac{1}{2}$ (d) $\frac{5\pi}{2} - 1$

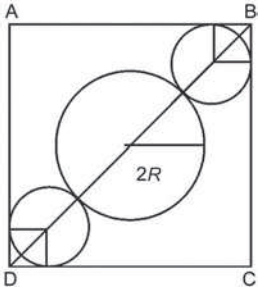
Q.49 Rajat cut out two identical triangular pieces of cardboard, each of area 300 sq. cm, and then placed them upon a table, one on top of the other such that the triangles completely coincide with each other. Now, if he rotated one of the two triangles by 180° about a vertical axis passing through its centroid, find the area that is common to both the triangles.

(a) 100 cm^2 (b) 150 cm^2

(c) 200 cm^2 (d) $133\frac{1}{3}$ cm^2

Q.50 See the figure given below:

Inside a square ABCD, three circles are drawn touching one another as shown in the figure. Radius of two smaller circles is R units and radius of the bigger circle is 2R units. Diagonal BD of the square passes through the centre of all the circles. What is the ratio of the radius of the smaller circles to the side of the square?



(a) 2:7 (b) 3:8

(c) 1:4 (d) None of these

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Answers

Warm Up

1. (b)	**2.** (c)	**3.** (a)	**4.** (a)	**5.** (c)	**6.** (b)	**7.** (d)	**8.** (c)	**9.** (d)	**10.** (c)
11. (b)	**12.** (c)	**13.** (d)	**14.** (a)	**15.** (b)	**16.** (a)	**17.** (b)	**18.** (b)	**19.** (a)	**20.** (a)
21. (b)	**22.** (d)	**23.** (d)	**24.** (c)	**25.** (b)					

Foundation

1. (d)	**2.** (d)	**3.** (b)	**4.** (b)	**5.** (d)	**6.** (d)	**7.** (d)	**8.** (d)	**9.** (d)	**10.** (c)
11. (b)	**12.** (b)	**13.** (a)	**14.** (b)	**15.** (d)	**16.** (c)	**17.** (a)	**18.** (d)	**19.** (b)	**20.** (c)
21. (d)	**22.** (a)	**23.** (b)	**24.** (b)	**25.** (b)	**26.** (c)	**27.** (c)	**28.** (b)	**29.** (c)	**30.** (b)
31. (d)	**32.** (c)	**33.** (b)	**34.** (b)	**35.** (b)	**36.** (d)	**37.** (d)	**38.** (a)	**39.** (c)	**40.** (c)
41. (c)	**42.** (b)	**43.** (a)	**44.** (d)	**45.** (a)	**46.** (b)	**47.** (a)	**48.** (b)	**49.** (b)	**50.** (d)

Moderate

1. (c)	**2.** (d)	**3.** (c)	**4.** (a)	**5.** (c)	**6.** (a)	**7.** (b)	**8.** (b)	**9.** (a)	**10.** (c)
11. (c)	**12.** (d)	**13.** (d)	**14.** (a)	**15.** (b)	**16.** (c)	**17.** (a)	**18.** (a)	**19.** (c)	**20.** (b)
21. (a)	**22.** (c)	**23.** (c)	**24.** (a)	**25.** (b)	**26.** (c)	**27.** (b)	**28.** (c)	**29.** (d)	**30.** (d)
31. (a)	**32.** (c)	**33.** (b)	**34.** (b)	**35.** (a)	**36.** (c)	**37.** (b)	**38.** (b)	**39.** (a)	**40.** (c)
41. (b)	**42.** (a)	**43.** (d)	**44.** (c)	**45.** (d)	**46.** (d)	**47.** (c)	**48.** (a)	**49.** (c)	**50.** (d)

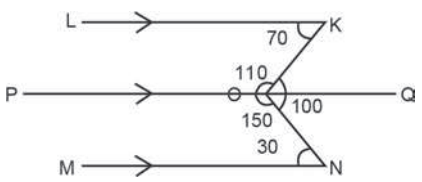
Advanced

1. (a)	**2.** (c)	**3.** (b)	**4.** (b)	**5.** (c)	**6.** (a)	**7.** (c)	**8.** (a)	**9.** (b)	**10.** (c)
11. (a)	**12.** (c)	**13.** (a)	**14.** (c)	**15.** (a)	**16.** (b)	**17.** (b)	**18.** (b)	**19.** (d)	**20.** (d)
21. (d)	**22.** (b)	**23.** (d)	**24.** (d)	**25.** (d)	**26.** (c)	**27.** (d)	**28.** (b)	**29.** (b)	**30.** (b)
31. (d)	**32.** (b)	**33.** (c)	**34.** (d)	**35.** (b)	**36.** (a)	**37.** (c)	**38.** (a)	**39.** (c)	**40.** (d)
41. (b)	**42.** (b)	**43.** (c)	**44.** (a)	**45.** (c)	**46.** (b)	**47.** (c)	**48.** (d)	**49.** (c)	**50.** (d)

Hints and Solutions

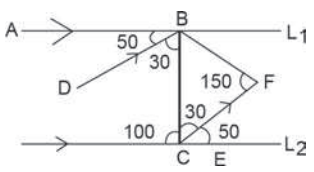
Warm up

1.



Hence, $\angle MNO = 30°$

2.

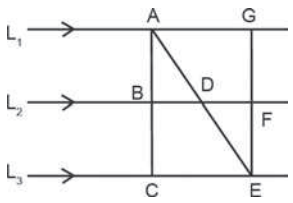


$\angle FCE = 50°$

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3.



$\frac{GF}{FE}\frac{AB}{BC}=\frac{AD}{DE};\frac{4}{6}=\frac{12}{DE}$

$DE=\frac{12\times6}{4}=18\ \therefore\ GF=18$

$\frac{GF}{FE}=\frac{4}{6}\Rightarrow\frac{18}{FE}=\frac{4}{6}$

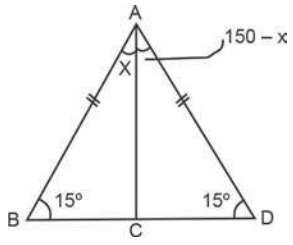
$FE=27$

4. $x+5x=180°$

$6x=180°$

$x=30°$

5.



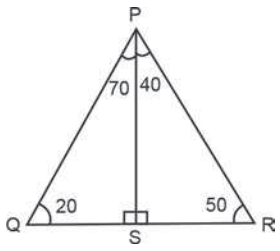
$150°-x+15°=95°+x$

$2x=70°$

$x=35°$

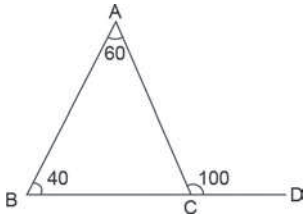
Therefore, $\angle ACB=95°+35°=130°$

6.

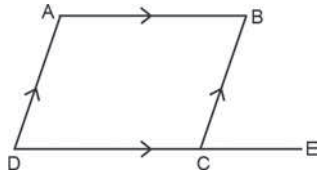


$\angle QPR=110°$

7.



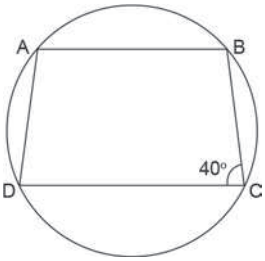
8.



Since A = 135°, So, B = D = 65°

We will find that option (c) is the answer.

9.



Isosceles trapezium is always cyclic.

The sum of opposite angles of a cyclic quadrilateral is 180°.

$\therefore\ \angle BCD+\angle BAD=180°$

$40°+\angle BAD=180°$

$\angle BAD=140°$

10. $\frac{(2n-4)\times90}{n}=135$

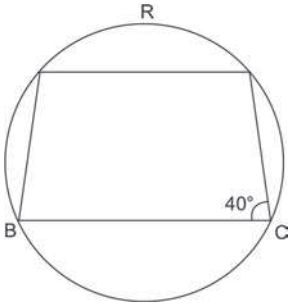
$(2n-4)\ 2=3\ n$

$4n-8=3n$

$n=8$

Number of diagonals $=\frac{1}{2}n(n-3)=20$

11.



$6x=180°$

$x=30°$

$\angle BAC=2x=2\times30=60°$

12. $AO\times OB=OC\times OD$

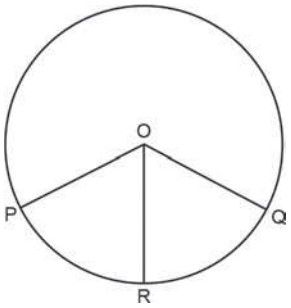
$4\times6=3\times OD$

$OD=8\ CM$

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13.



Let $\angle POR = x$

Therefore, $\angle ROQ = \frac{2}{3}x$

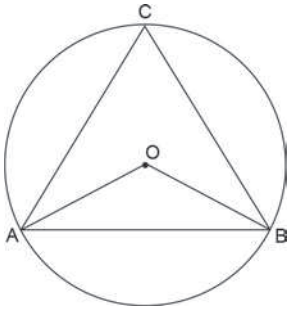
$x + \frac{2}{3}x = 100$

$\frac{5x}{3} = 100$

$x = 60°$

Therefore, $\angle POR = 60°$

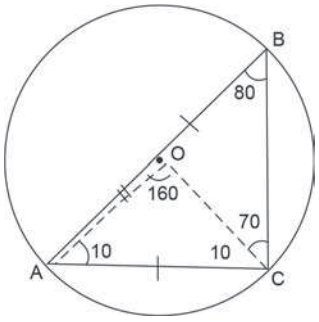
14.



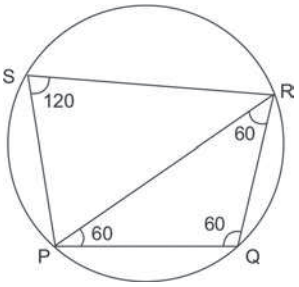
$\frac{1}{2}\angle AOB = \angle ACB$

$\angle ACB = 60°$

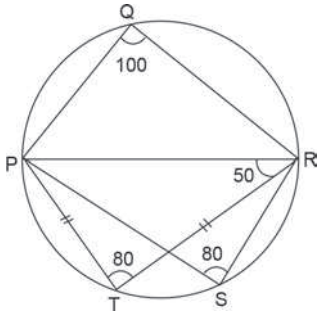
15.



16.

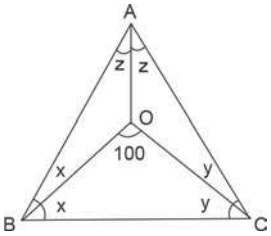


17.



$\angle PRT = 50$

18.



In $\angle BOC$, $x + y = 80$

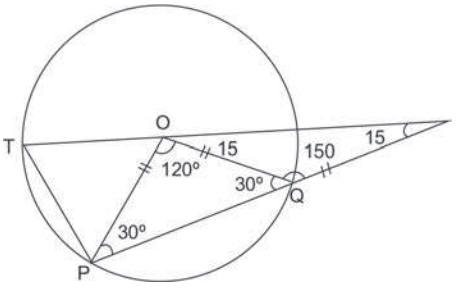
$2x + 2y = 160°$

Also, $2x + 2y + 2z = 180°$

$2z = 20°$

$\angle BAC = 2z = 20°$

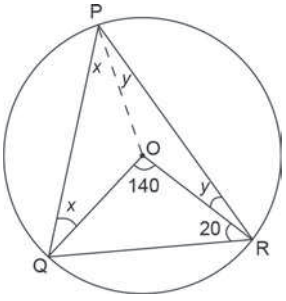
19.



$\angle OQP = 30°$

Therefore, $\angle QOP = 120°$

20.



$\angle OQP + \angle ORP = 70°$

$x + y = 70°$

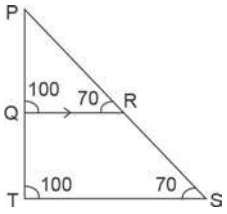
$\angle QPR = 70°$

$\angle ORQ = 20°$

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21.



$PQ \times PS = PT \times PR$

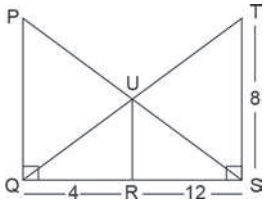
$QR \parallel TS$

$\angle PQR = \angle PST + 30°$

$\angle PST = 70°$

$\angle PRQ = 70°$

22.



In Δ QST and Δ QRU (Δ QST and Δ QRU are similar triangles).

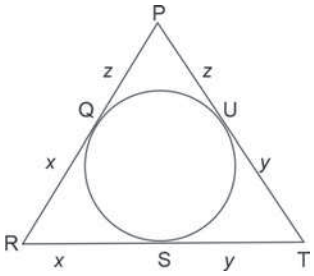
$$\frac{QR}{QS} = \frac{RU}{TS} \Rightarrow \frac{4}{16} = \frac{RU}{8} = RU = 2$$

Also, $\frac{QR}{QS} = \frac{RU}{TS} \Rightarrow \frac{4}{16} = \frac{RU}{8} = RU = 2$

$$\frac{1}{PQ} = \frac{1}{2} - \frac{1}{8} = \frac{4-1}{8} = \frac{3}{8}$$

$PQ = \frac{8}{3}$ cm

23.



$PR = RT + 3$

$x + z = x + y + 3$

$z - y = 3$ (i)

Similarly, $z + x = z + y + 1$

$x - y = 1$ (ii)

From (i) and (i)

$z - 3 = x - 1$

$z - x = 2$ (iii)

Also, $2x + 2y + 2z = 26$

$x + y + z = 13$ (iv)

$3x + 1 = 13$

$3x = 12$

$x = 4$

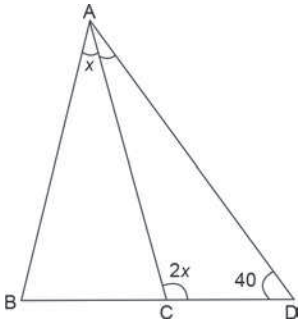
$z = 6$

$y = 3$

$QR + PT = x + y + z$

$= 4 + 6 + 3 = 13$

24.



$85° - x + 2x + 40° = 180°$

$x = 55°$

$\angle ACB = 70°$

25. Only 8 such regular polygons are possible.

FOUNDATION

1. The triangle is not possible. So, (d) is the correct option.

2. Clearly, triangle is obtuse. So, (d) is the correct option.

3. Clearly, triangle is a right-angled triangle.

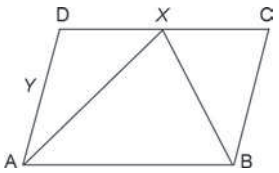
Its area $= \frac{1}{2} \times 6 \times 8 = 24$ cm^2

Now, $8 \times b = 24$

$b = 3$

Required perimeter $= 2\ (8 + 3) = 22$ cm

4.

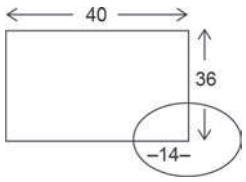


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We know that, between the same parallels and on the same base, the area of triangle is half the area of the parallelogram. Hence, option (d) is the answer.

5.



Required ratio = 3:1

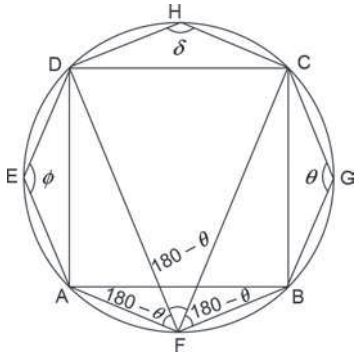
6. Triangle is not possible.

7. Putting the value of $n = 2, 3$, and so on, we get option (a) as answer.

8. This is possible only when both length and breadth of rectangle are equal to the side of the square.

9. Clearly (d)

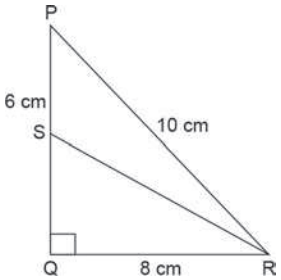
10.



Required sum

$= \theta + 180° - \theta + 180° - \theta + 180° - \theta + \theta + 8 = 540°$

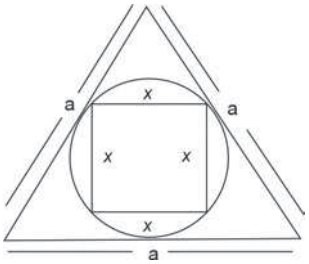
11.



Clearly, it is a right angle triangle at Q

$$RS = \sqrt{8^2 + 3^2} = \sqrt{64 + 9} = \sqrt{73} \approx 8.5\text{cm}$$

12.



Radius of circle $R = \dfrac{\frac{\sqrt{3}}{4}a^2}{\frac{3a}{2}} = \dfrac{\sqrt{3}}{6}a$

Diagonal of square = Diameter of circle

$= 2.\ \dfrac{\sqrt{3}}{6}a$

$\sqrt{2}x = \dfrac{\sqrt{3}}{3}a$

$x = \dfrac{\sqrt{3}}{3\sqrt{2}}a = \dfrac{\sqrt{6}}{6}a$

$\text{Area} = x^2 = \left[\dfrac{(\sqrt{6}a)}{6}\right]^2 = \dfrac{6}{36}a^2 = \dfrac{1}{6}a^2$

13. The two adjacent angles can be 30° and 60°; therefore, others angle can be 150° and 120°.

14. Going through the options, we find that only (d) satisfies the condition.

15. 1 kg = 1000 cm^3

$2700 = k.2^3$

$k = \dfrac{2700}{8}$

$6300 = k\ .\ r^3$

$r^3 = \dfrac{6300}{k} = \dfrac{6300}{2700} = \dfrac{8}{3}, r^3 = \dfrac{56}{3},\ r = 2.6\text{ cm}$

16. $i\ (2, 3) = 2\ (2 + 3) = 10$

$f(10) = 100$

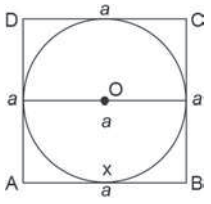
$h\ (100, 2) = 200$

$g\ (200) = 80$

17. $\dfrac{f(3) - g(2)}{h(4,2) - i(6,2)} \times \dfrac{g(6)}{f(1)} = \dfrac{9-8}{8-16} \times \dfrac{24}{1}$

$= \dfrac{1}{-8} \times 24 = -3$

18.



$4\ a + \pi a = 10$

$a = \dfrac{10}{4 + \pi} = 1.4$

Required Area $= (1.4)^2 - \pi(.7)^2$

$= 1.96 - \dfrac{22}{7} \times \dfrac{7}{10} \times \dfrac{7}{10} = 1.96 - 1.54$

$= 0.42\text{ m}^2 = 4200\text{ cm}^2$

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19. $\frac{.7}{4\times1.42}=1.8$

20. We can easily eliminate option (b) and option (d) as it is not divisible by 4.

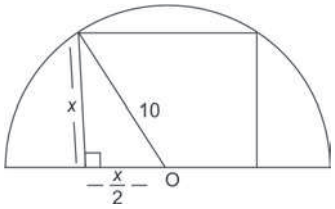
Checking option (a) Number of sides are 8 and 10 sides.

Ratio of one internal angle = 135:144 = 15:16. Hence, option (a) is not the answer.

Checking option (c) Number is sides are 12 and 15.

Ratio of one internal angle = 150:156 = 25:26. Hence, option (c) is not the answer.

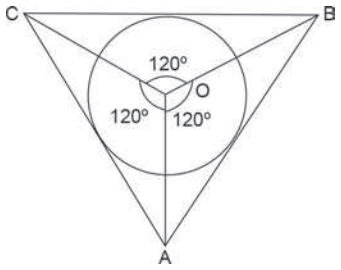
21.



$x^2+\frac{x^2}{4}=100,\ x^2=80$

22. No such point is possible.

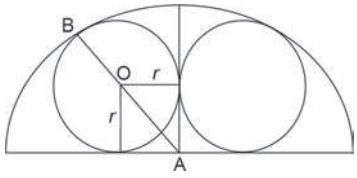
23.



Radius of disc $=\sqrt{3}$, OB = 20

Use Sine law to find the length of AB = BC = AC.

24.



Let radius of small circles

AD = DB = r

Then, AC = OD = r (Since AC = r (radius)

Since, it is given that $OO_1 = 2$

OB = OA = $2 - r$

Now, in ΔADO

$AD^2 + DO^2 = OA$

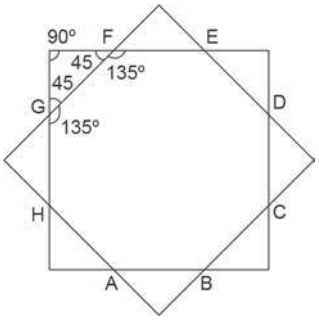
$r^2 + r^2 = (2-r)^2 = 4 + r^2 - 4r$

$r^2 + 4r - 4 = 0$

Then, $r=\frac{-b\pm\sqrt{b^2-4ac}}{2a}=\frac{+1.656}{2}$ or $\frac{-9.656}{2}$

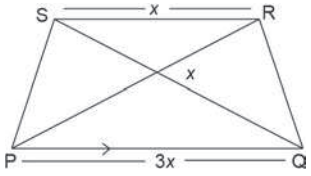
$\therefore r=\frac{1.656}{2}=0.828$ cm

25.



Required angle = 90°

26.



$\frac{ar\ (\Delta P\times Q)}{ar\ (\Delta R\times S)}=\frac{PQ^2}{RS^2}=\frac{3(x)^2}{x^2}=9:1$

27. Let the angles be x, xr, xr^2 and xr^3.

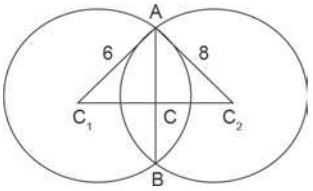
$x + xr + xr^2 + xr^3 = 360°$

$x\ (1 + r + r^2 + r^3) = 360°$

Therefore, angles are 24°, 48°, 96°, and 192°, which satisfies the given condition.

Hence, answer to 27 is (c) and answer to 28 is (b).

29.



It is given that

$AC_1 = 6,\ AC_2 = 8$

$\angle C_1AC_2 = 90°$

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Then, in ΔAC_1C_2

$AC_1^2 + AC_2^2 = C_1C_2^2 \rightarrow 36 + 84 = C_1C_2^2$

$C_1C_2^2 = 100 \rightarrow C_1C_2 = 10$

Let $AC = x$

Then, area of ΔAC_1C_2

$= \frac{1}{2} C_1C_2 \times AC = \frac{1}{2} \times 10 \times x = 5x$ (i)

$\angle C_1AC_2 = 90°$

Then, area of ΔAC_1C_2

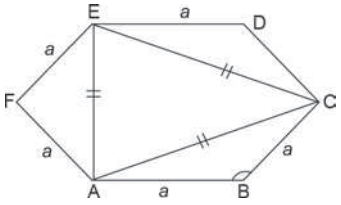
$= \frac{1}{2} \times AC_1 \times AC_2 = \frac{1}{2} \times 6 \times 8 = 24$ (ii)

From equation (i) and (ii)

$5x = 24$, so, $x = 4.8$

Since $AC = \frac{1}{2} AB$, so $AB = 2AC = 2 \times 4.8 = 9.6$.

30.



Clearly, Δ ACE is an equilateral triangle.

$$\frac{AC}{\text{Sin}120°} = \frac{a}{\text{Sin}30°}, \text{or}, \frac{AC}{\text{Sin}60°} = \frac{a}{\text{Sin}30°}$$

$$\rightarrow AC = \frac{a\text{Sin}60°}{\text{Sin}30°} = \frac{a \cdot \sqrt{3}/4}{1/2} = \sqrt{3}a$$

$$\text{Required ratio} = \frac{\frac{\sqrt{3}}{4}(\sqrt{3}a)^2}{6 \cdot \frac{\sqrt{3}}{4} a^2} = \frac{3a^2}{6a^2} = \frac{1}{2}$$

31. $\frac{1}{2} : \frac{1}{3} : \frac{1}{4} = 6 : 4 : 3$

$6x + 4x + 3x = 52$, or, $13x = 52$ and $x = 4$

Required length = 12 cm

32. Required Area $= \pi\left(\frac{104}{2}\right)^2 - \pi.50^2$

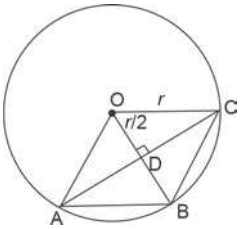
$= \pi.52^2 - \pi.50^2 = \pi(52 - 50)(52 + 50)$

$= \pi.2.102 = 204\pi$

33. Let h_1 = attitude of parallelogram and h_2 = altitude of triangle, then $\frac{1}{2} h_2 = h_1$

$h_2 = 2h_1 = 200m$

34.

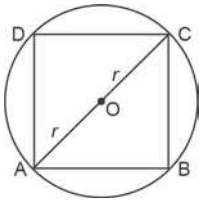


$$AD = \sqrt{r^2 - \frac{r^2}{4}} = \frac{\sqrt{3}}{2} r$$

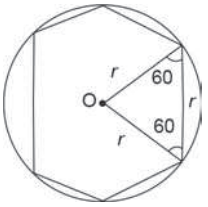
$$\frac{1}{2} \cdot r \cdot \sqrt{3}r = 32 \times \sqrt{3}, r^2 = 32 \times 2$$

$r^2 = 64$ and $r = 8$

35.



$A = (\sqrt{2}r)^2 = 2r^2$



$B = 6 \cdot \frac{\sqrt{3}}{4} r^2$

$$\frac{B}{A} = \frac{\frac{3}{2}\sqrt{3}r^2}{2r^2} = \frac{3\sqrt{3}}{4}$$

36. Let dimensions be x, $2x$, and $4x$.

A/Q

$20.5 \times 2[2x^2 + 8x^2 + 4x^2] - 20 \times 2[2x^2 + 8x^2 + 4x^2]$

$= 120$

$2 \times 14x^2 \times 0.5 = 126$

$x^2 = \frac{63 \times 2}{14} \Rightarrow x = 3$

Therefore, dimensions are 3, 6, and 12 m.

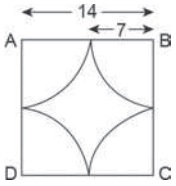
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37. Area $= \frac{1}{2} \cdot 50 \cdot 130 \text{Sin } 72°$

$= \frac{1}{2} \cdot 50 \cdot 130 \cdot 0.9510 = 3090.75 \text{m}^2$

38.



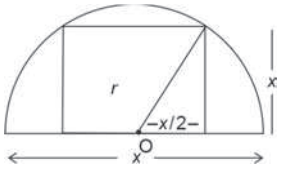
Area grazed by 4 horses $= \pi \cdot 7^2 = 154 \text{m}^2$

Area ungrazed $= (196 - 154)\ \text{m}^2 = 42\ \text{m}^2$

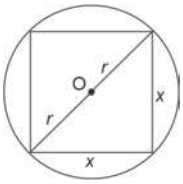
154 m^2 is grazed in 11 days.

42 m^2 is grazed in 3 days.

39.



Area of square $= \left(\frac{2}{\sqrt{5}} r\right)^2 = \frac{4}{5} r^2$



Area $= 2r^2$

Required ratio $= \frac{\frac{4}{5} r^2}{2r^2} = 2:5$

40. $A = 8x^2 - 2x - 15$

$= 8x^2 - 12x + 10x - 15$

$= 4x\ (2x - 3) + 5\ (2x - 3)$

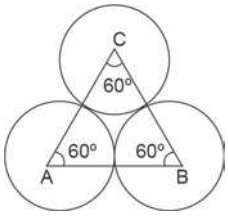
$A = (4x + 5)\ (2x - 3)$

Clearly, length $= (4x + 5)$

and breadth $= (2x - 3)$

41. Required ratio $= \frac{a^2}{(\sqrt{2}a)^2} = 1:2$

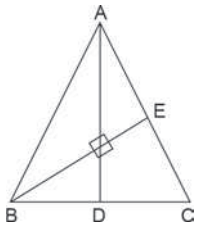
42.



Required area

$= \frac{\sqrt{3}}{4} \cdot 7^2 - 3 \cdot \frac{60}{360} \frac{22}{7} \times \left(\frac{7}{2}\right)^2 = 1.967 \text{ cm}^2$

43.



Area Δ AOB $= \frac{1}{2} \cdot 8.6 = 24 \text{ cm}^2$

Area of Δ BOD $= \frac{1}{2} \cdot 8.3 = 12 \text{ cm}^2$

Area of Δ ABD $= 36 \text{ cm}^2$

Area of ΔABC $= 72 \text{ cm}^2$

44. $\frac{1}{2}(x + 6)(y - 4) + 24 = \frac{1}{2}xy$

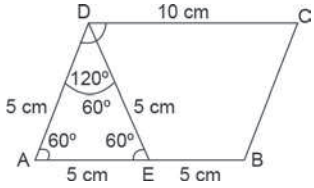
$\frac{1}{2}xy - \frac{1}{2}x \cdot 4 + \frac{1}{2}6 \cdot y - \frac{1}{2}6.4 + 24 = \frac{1}{2}xy$

$-2x + 3y - 12 + 24 = 0$

$2x - 3y = 12$

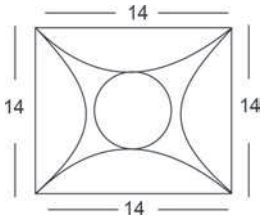
Now, putting the value of x or y from options, we get (d) as the answer.

45.



Perimeter = 30 cm

46.

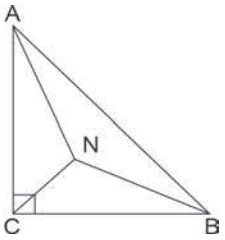


47. Going through the options, we get (a) as the answer.

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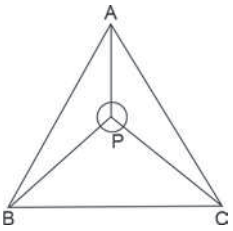
48.



49. Let AB = 4 and BC = 5 and AB is perpendicular to BC,

then area $= \frac{1}{2}\text{AB} \cdot \text{AC} = \frac{1}{2} \cdot 4 \cdot 5 = 10.$

50.



First three options do not have the properties as given in the question.

MODERATE

1. $\angle A : \angle B : \angle C = 4:1:1 = x$ (let)

$4x + x + x = 180°$

$x = 30°$

Now, $\frac{\sin A}{\sin B} = \frac{A}{B}$ is solved to get the answer $\frac{A}{B} = \frac{\sqrt{3}}{1}$

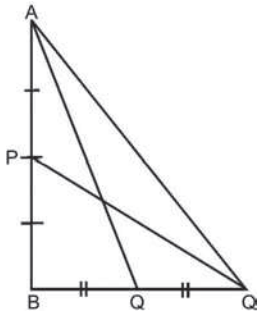
2. Use Pythagoras theorem, in right-angled ΔABQ

$$\text{AQ}^2 = \text{AB}^2 + \left(\frac{\text{BC}}{2}\right)^2 \quad \text{(i)}$$

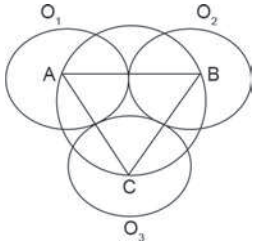
and in right-angled ΔCBP

$$\text{CP}^2 = \left(\frac{\text{AB}^2}{2}\right) + (\text{BC})^2 \quad \text{(ii)}$$

Add (i) and (ii) to get the answer.



3.



It is given that radius of circle O_1, O_2

AB = BC = CA = 2r

Then, the radius of circle ABC

$$= \frac{\text{ABC}}{4\text{A}} = \frac{2r \times 2r \times 2r}{4 \times \frac{1}{2} \times 2r \times 2r \times \text{Sin}60°}$$

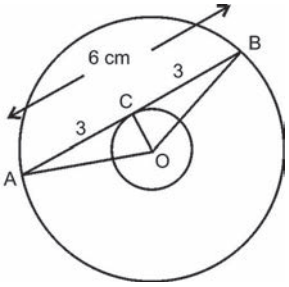
$$= \frac{8r^3}{4r^2\sqrt{3}} = \frac{2r}{\sqrt{3}}, \text{ which is greater than } r.$$

4. AC = 3 m

OC ⊥r AB

So, Δ AOI is a right-angled triangle.

CO = 4 and AO = 5 cm



5. It is given that $|x| + |y| = 1$.

This is the graph of a square with its sides not parallel to the coordinate axes.

6. In ΔCOD,

OC = OD = OB = OA = radius = r

$b^2 = 2r^2$

$r = \frac{b}{\sqrt{2}}$

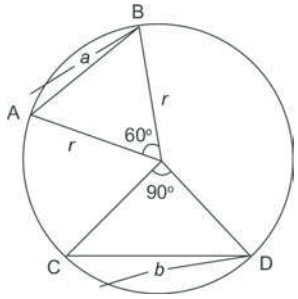
ΔAOB = equilateral triangle

Hence, $a = r$

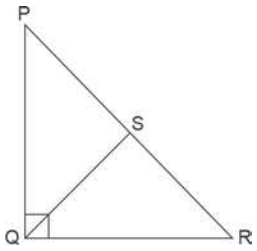
$r = \frac{b}{\sqrt{2}}$, hence, $b = \sqrt{2a}$

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7.



Since $\angle PQR = 90°$

S is the mid-point of PR.

It means QS = PS = SR = circum radius of the triangle.

Then, $PR = 2QS = 2\sqrt{117}$ (i)

Since PQ + QR = 30

Then, $(PQ + QR)^2 = (30)^2 \rightarrow PQ^2 + QR^2 + 2PQ \times QR = 900$

Then, $PQ^2 + QR^2 = 900 - 2\ PQ \times QR$ (ii)

Now, in ΔPQR, $PQ^2 + QR^2 = PR^2$

From equations (i) and (ii)

$900 - 2\ PQ \times QR = (2\sqrt{117})^2 = 468$

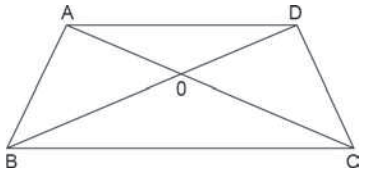
$2PQ \times QR = 432 \rightarrow PQ \times QR = 216$

Since it is given that PQ + QR = 30

Then, PQ = 18 and QR = 12

Then, area of $\Delta PQR = \frac{1}{2}\ PQ \times QR = \frac{1}{2} \times 18 \times 12 = 108\ cm^2$

8.



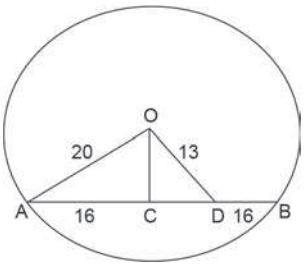
It is given that $\frac{OD}{DB} = \frac{2}{5}$

Then, $\frac{OD}{OB} = \frac{2}{3}$

We know that $\frac{AD}{BC} = \frac{OD}{OB} = \frac{OA}{OC}$

Then, $\frac{AD}{BC} = \frac{2}{3}$

9.



It is given that $OA = 20, AC = \frac{AB}{2} = 16$ and OD = 16

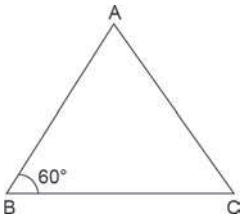
Then, OC = 12 (from Pythagoras)

Now, in ΔOCD

$OC^2 + CD^2 = OD^2 \rightarrow CD = 5$

Then, lengths of segment AD and DB are 21 and 11, respectively.

11.



It is given that all the angles are in AP. It means that angle B should be equal to 60°.

Let A > B > C

Then, A = 8, B = 7, and C = c

Then, $\text{Cos B} = \frac{a^2 + c^2 - b^2}{2ab}$

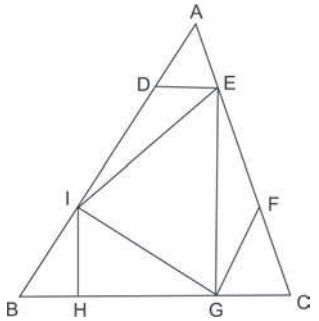
$\text{Cos } 60 = \frac{8^2 + c^2 - 7^2}{2 \times 8 \times a} \rightarrow c^2 - 8c + 15 = 0$

Then, c = 3 and 5

12. If AB = 3a

Then, DI = a = DE = EF = FG = GH = HI and IE = EG = IG = $\sqrt{3a}$

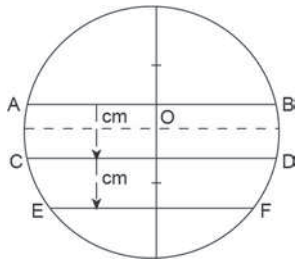
Now, find the area.



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13.



Let there are three parallel chords, which are drawn in a circle of diameter 4 cm.

Now, in this figure, the centre of the circle is O, and three chords are drawn, namely AB, CD, and EF.

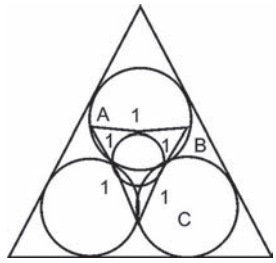
So, from this figure, option (d) is the answer.

14. ΔABC is equilateral.

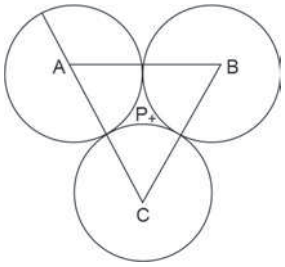
Tan 30 = $r/x = x = \sqrt{3}r$ (where $r = 1$)

$AB = 2 + 2x = 2 + 2\sqrt{3} = 2(1+\sqrt{3})$

Then, area = $\sqrt{3}(1+\sqrt{3})^2$



15. Following is the figure:



ΔABC will be an equilateral Δ.

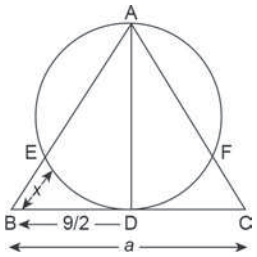
PA will be the circumradius of ΔABC = $\frac{a}{\sqrt{3}} = \frac{2}{\sqrt{3}}$

Hence, radius of outer circle = $\frac{2}{\sqrt{3}} + 1$

Hence, the required portion = $\pi\left(\frac{2}{\sqrt{3}}+1\right)^2 - 3 \times \pi \times 1^2$

$= \pi\left[\left(\frac{2}{\sqrt{3}}+1\right)^2 - 3\right]$ sq. units.

16.



Since BA is a secant which intersects the circle at E and A, and BD be a tangent at D. Then,

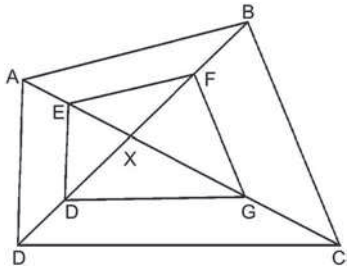
$BD^2 = BE \times BA$

$\left(\frac{a}{2}\right)^2 = x \times a \rightarrow x = \frac{a}{4}$

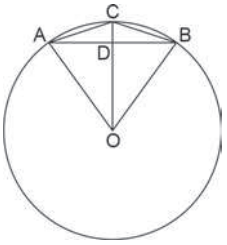
Since, $BE = x = \frac{a}{4}$, then $EA = a - x = a - \frac{a}{4} = \frac{a}{3}$

Then, $\frac{EA}{EB} = \frac{AF}{FC} = \frac{3}{1}$

17. This problem is based on mid-point theorem.



18. Method 1



Define D as the mid-point of line segment $\overline{AB}$, and O the centre of the circle. Then, O, C, and D are collinear, and since D is the mid-point of AB, m angle ODA = 90 degree, and so, $OD = \sqrt{5^2 - 3^2} = 4$. Since OD = 4, CD = 5 − 4 = 1, and so, $AC = \sqrt{3^2 + 1^2} = \sqrt{10}$. Hence, option (a) is the answer.

Method 2

Let α be the angle that subtends the arc AB. By the law of cosines, $6^2 = 5^2 + 5^2 - 2 \times 5 \times 5 \cos(\alpha)$ implies $\cos(\alpha) = 7/25$.

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Using Trigonometry:

$$\cos(\alpha/2) = \frac{\sqrt{1+\cos(\alpha)}}{2} = \sqrt{\frac{32/25}{2}} = \sqrt{\frac{16}{25}} = \frac{4}{5}$$

The law of cosines tells us AC

$$= \sqrt{5^2 + 5^2 - 2\times 5\times 5\times \frac{4}{5}} = \sqrt{50 - 50\frac{4}{5}} = \sqrt{10}$$

Hence, option (a) is the answer.

19. We are given that the area of ΔABE is 40, and that AB = 10.

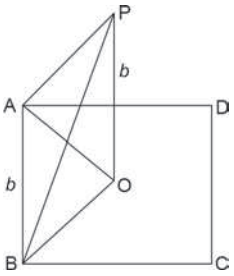
The area of a triangle:

$$A = \frac{bh}{2}$$

Using AB as the height of ΔABE,

$40 = \frac{10b}{2}$ and solving for b, $b = 8$. Hence, option (c) is the answer.

20.



In the ΔPAB, AP = PB and ∠APB = 60°. It means that ΔPAB is a equilateral triangle.

Then, AB = AP = BP = b (i)

Now, in ΔAOB,

∠AOB = 90°, AB = b, and OA = OB

Then, $OB = \frac{b}{\sqrt{2}}$ (ii)

Now, in ΔAPO

∠POA = 90°, $AO = OB = \frac{b}{\sqrt{2}}$ and $OP = h$

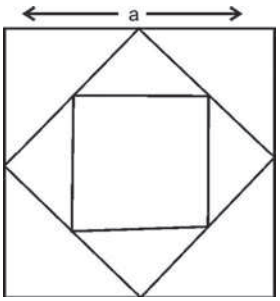
Then, $h^2 + \frac{b^2}{2} = b^2$ [Using Pythagoras theorem]

$2h^2 = b^2$

21. In this question, in first step, whatever we take the value of n and irrespective of the sizes of squares, the remaining cloth will be equal to the second process. So, the ratio is 1:1.

22. Find the sides of all the squares.

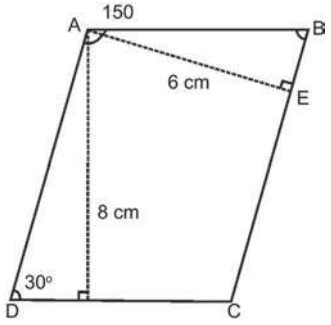
Then, use geometric progression concept and get the answer.



23. In ΔAEB, find AB and in ΔABF find AD.

Then, P = AB + BC + CD + AD

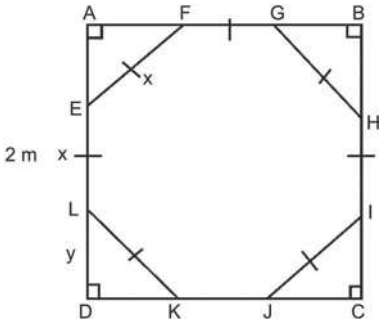
Use this to get the answer.



24. $x + 2y = 2$ (i)

In ΔAEF, $x^2 = 2y^2$ (ii)

Use (i) and (ii) to get the answer.



26. Let the sides be x, y, and z, and x is the smallest side.

Then, $x + y + z = 4x$ [It is given]

$y + z = 3x$

$x = \frac{y+2}{3}$ (i)

Since it is right angle triangle, $x^2 + y^2 = z^2$

Put $\frac{y+2}{3} = x$ from equation (i)

Then, $\left(\frac{y+z}{3}\right)^2 + y^2 = z^2$

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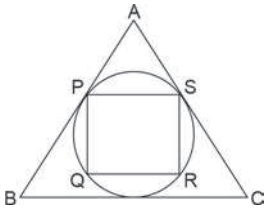
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$$\frac{y^2+z^2+2yz}{9}+y^2=z^2 \rightarrow 8z^2-2yz-10y^2=0$$

$8z^2+8zy-10zy+10y^2=0$, or, $(8z-10y)(z+y)=0$

$8z-10y=0 \rightarrow 8z=10y$, then $\frac{z}{y}=\frac{8}{10}=\frac{4}{5}$

27.



Let the side of the ΔABC is a

Then, the radius of the circle (In radius)

$r=\frac{a}{2\sqrt{3}}$ and area $=\frac{\sqrt{3}}{4}a^2$

Then, diameter of the circle (PR) $=2\times\frac{a}{2\sqrt{3}}=\frac{a}{\sqrt{3}}$

Since PR is the diagonals of the square PQRS.

Then, side of the square $=\frac{a}{\sqrt{6}}$

So, area of the square $\frac{a^2}{6}$

Then, ratio of $\frac{\text{Area of }\Delta ABC}{\text{Area of PQRS}}=\frac{(\sqrt{3}14)a^2}{(1/6)a^2}=3\sqrt{3}:2$

28. Cos A $=\frac{b^2+c^2-a^2}{2bc}$

Cos A $=\frac{(2x+1)^2+(x^2-1)^2-(x^2+x+1)^2}{2(2x+1)(x^2-1)}$

$=\frac{4x^2+4x+1+x^4-2x^2+1-x^4-x^2-1-2x^3-2x^2-2x}{4x^3-4x+2x^2-2}$

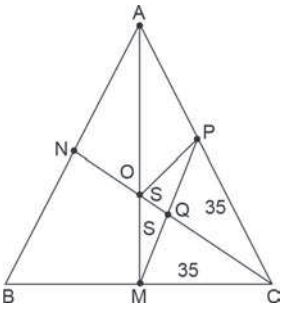
Cos A $=-\frac{1}{2}$, or, Cos A = Cos 120. Hence, ∠A = 120°

29. Note that because $\overline{DE}$ and $\overline{EF}$ are parallel to the sides of ΔABC, the internal triangles ΔBDE and ΔEFC are similar to ΔABC, and therefore, they are also isosceles triangles.

It follows that BD = DE. Thus, AD + DE = AD + DB = AB = 28.

Since opposite sides of parallelograms are equal, the perimeter is 2 × (AD + DE) = 56.

30.



We knows that medians intersect each other in the ratio of 2:1.

Then, $\frac{AO}{OM}=\frac{CO}{ON}=\frac{BO}{OP}=\frac{2}{1}$

Since P and M are the mid-points of AC and BC

$\therefore \frac{AP}{PC}=\frac{BM}{MC}=\frac{NQ}{QC}=\frac{1}{1}$

In the ΔABC $\frac{CO}{ON}=\frac{1}{2}$

$\frac{NQ}{QC}=\frac{1}{1}$

Then, $\frac{OQ}{QC}=\frac{1}{3}$

Then, in ΔOMQ and ΔQMC

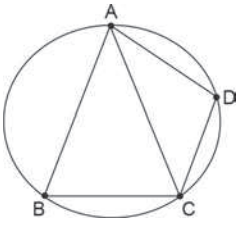
Heights are same and base are in the ratio of 1:3.

Then, ratio of area = s:3s

Then, area of ΔOMC = 4s

Then, area of ΔABC = 6 × 4s = 24s

31.



Since ΔABC is a equilateral triangle.

Let AB = BC = CA = x

Then, from the Ptolemy theorem

AB × CD + AD × BC = AC × BD

x CB + x AD = x BD. Hence, CB + AD = BD

32. Let AD = x and AG = y. We want to find DE + FG, which is nothing but $x + y$.

Based on the fact that ADE, DEFG, and BCFG have the same perimeters, we can say the following:

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$3x = x + 2(y - x) + y = y + 2(1 - y) + 1$

Simplifying, we can find that

$3x = 3y - x = 3 - y$

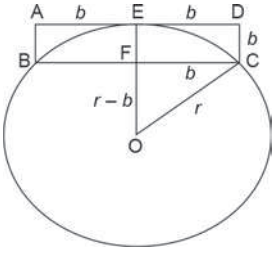
Since $3 - y = 3x$, $y = 3 - 3x$

After substitution, we find that $9 - 10x = 3x$, and $x = \frac{9}{13}$.

Again substituting, we find $y = \frac{12}{13}$

Therefore, $x + y = \frac{21}{13}$. Hence, option (c) is the answer.

33.



In the figure, EO = OC = radius = r and OF = OE – FE

$OF = r - b$ (i)

Then, in ΔFCO, $OC^2 = FC^2 + FO^2$

Or, $r^2 = b^2 + (r - b)^2$

Or, $r^2 = b^2 + r^2 + b^2 - 2br$

Or, $2b^2 = 2br$, or, $2b = r$

Hence, $r = b$

Then, radius of the circle = Breadth of the rectangle

34. Since sides of Δ are: 3, 4, and 5, so this Δ will be a right angle triangle.

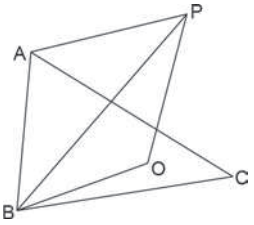
Then, radius of inscribed circle $C_1 = \frac{A}{S} = \frac{6}{6} = 1$

radius of circumscribed circle $C_2 = \frac{abc}{4A} = \frac{3 \times 4 \times 5}{4 \times 6} = \frac{5}{2}$

Then, $\frac{\text{Area of } C_1}{\text{Area of } C_2} = \frac{\pi r_1^2}{\pi r_2^2} \quad \frac{(r_1)^2}{(r_2)^2} = \frac{(1)^2}{(5/2)^2} = \frac{4}{25}$

36. Use similarity theorem

37.



Let the Δ is ABC and the centre of triangle is O.

Let the side of stadium = a

Then, in ΔABP, $\angle P = 90°$. So, $AP^2 + PB^2 = AB^2$

$2PB^2 = AB^2$ [Since AP = PB]

$PB^2 = \frac{a^2}{2} \rightarrow PB = \frac{a}{\sqrt{2}}$

Now, in ΔPBO, $\angle O = 90°$. So, $OB^2 + OP^2 = PB^2$

$OP^2 = \left(\frac{a}{\sqrt{2}}\right)^2 - OB^2$

$OP^2 = \frac{a^2}{2} - \frac{a^2}{3}$ $\left[OB = \frac{a}{\sqrt{3}}\right]$

$OP = \frac{a}{\sqrt{6}}$

Hence, $\frac{\text{Height of polo}}{\text{Perimetre of the stadium}} = \frac{a/\sqrt{6}}{3a} = \frac{\sqrt{6}}{18}$

38. Method 1

Let CX = x, BX = y. Let the circle intersect AC at D and the diameter including AD intersect the circle again at E. Use power of a point on point C to the circle centred at A.

So, $CX \times CB = CD \times CE$ $x(x + y) = (97 - 86)(97 + 86)$ $x(x + y) = 3 \times 11 \times 61$.

Obviously, $x + y > x$, so, we have three solution pairs for $(x, x + y) = (1, 2013)$, $(3, 671)$, $(11, 183)$, and $(33, 61)$. By the Triangle Inequality, only $x + y = 61$ yields a possible length of BX + CX = BC.

Therefore, the answer is 61.

Method 2

Let BX = q, CX = p, and AC meet the circle at Y and Z, with Y on AC. Then, AZ = AY = 86. Using the Power of a Point, we get that $p(p + q) = 11(183) = 11 \times 3 \times 61$. We know that $p + q > p$, and that $p > 13$ by the triangle inequality on Δ ACX. Thus, we get that BC = $p + q$ = 61.

Method 3

Let x represent CX, and let y represent BX. Since the circle goes through B and X, AB = AX = 86. Now, use Stewart's Theorem.

$xy(x + y) + 86^2(x + y) = 97^2y + 86^2x$

$x^2y + xy^2 + 86^2x + 86^2y = 97^2y + 86^2x$

$x^2 + xy + 86^2 = 97^2$

(Since y cannot be equal to 0, dividing both sides of the equation by y is allowed.)

$x(x + y) = (97 + 86)(97 - 86)$

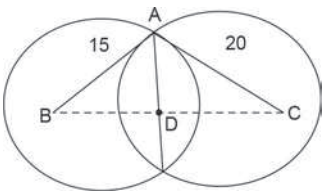
$x(x + y) = 2013$

The prime factors of 2013 are 3, 11, and 61. Obviously, $x < x + y$. In addition, by the Triangle Inequality, BC < AB + AC, so $x + y < 183$. Therefore, x must equal to 33, and $x + y$ must equal to 61.

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39.



In ΔABC, the sides are – 15, 20, and 25.

Then, from the Pythagoras ΔABC will be a right angle triangle at ∠A.

Then, area of $\Delta ABC = \frac{1}{2}$ AB × AC = $\frac{1}{2} \times 15 \times 20$ = 150 cm² (i)

Let height of ΔABC is AD

Then, area of $\Delta ABC = \frac{1}{2} \times BC \times AD = \frac{1}{2} \times 25 \times AD$ (ii)

From equations, (i) and (ii)

AD = 12 cm (iii)

Hence, length of the common chord = 2 AD = 24 cm

40. Area of trapezium $= \frac{1}{2} \times \text{height}(l_1 + l_2)$

SN⊥PQ and RM⊥PQ gives RS = MN = 3, So, PN + QM = 3

Assume PN = x

Then, in ΔPNS and ΔQMR (SN = RM)

$(SN)^2 = (RM)^2 = 4^2 - x^2 = 5^2 - (3 - x)^2$

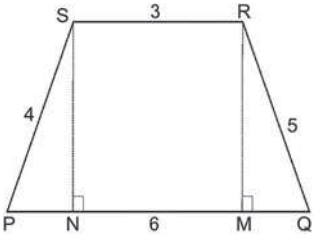
Finally, x = 0 = zero

That means,

$(SN)^2 = 4^2 - x^2$

SN = 4

So, area $= \frac{1}{2} \times 4(3+6) = 18 \text{ cm}^2$



Alternatively,

In trapezium PQRS, PN + MQ = 3. If we join ΔPSN and ΔMRQ through their altitude, it forms a triplet (3, 4, 5) of right-angled Δ making right angle at P, i.e.,

∠SPQ = 90°. Then, altitude = 4 cm.

Area of trapezium $= \frac{1}{2} \times (3+6) = 18 \text{ cm}^2$

41. Let side of square = a, then perimeter = $4a$

Area of square = a^2

According to the question, numerically $4a = a^2$

So, a = 4

Diagonal of square = $\sqrt{2}a = 4\sqrt{2}$

42. In square ADCE, area = x^2

So, side AC = x

In square AHIB, area = y^2

So, side AB = y

As $x^2 > y^2$, so, $x > y$

and in ΔABC (∠ABC = 90°)

$(AC)^2 - (AB)^2 = (BC)^2$

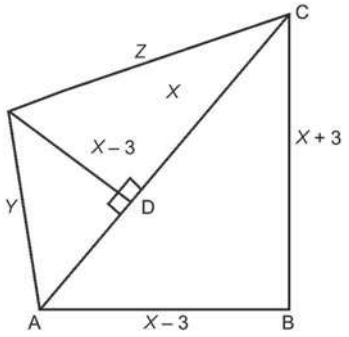
Hence, area BCGF = $(AC)^2 - (AB)^2 = x^2 - y^2 = (x - y)(x + y)$

43. $AD = (y^2 - (x-3)^2)1/2 = (100 - (x-3)^2)1/2$

DC = x

In ΔABC

$(x-3)^2 + (x+3)^2 = \left(x + \sqrt{100 - (x-3)^2}\right)^2$



Now, putting the values of the given options, we can calculate the answer. It is none of these.

Alternatively, this question can be done very easily using the options.

44. In square ABCD,

Area of shaded region = Area of sector BCD – Area of ΔBCD + Area of sector ABD – Area of ΔABD

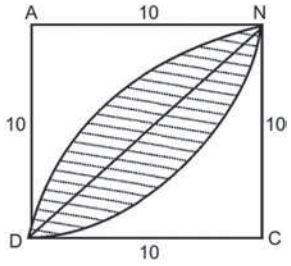
Since, all sides of a square are same,

Area of sectors and triangles will be the same having same radius (side of square) and angle 90°.

So, area of shaded region = 2 (Area of sector – Area of ΔABD)

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$$= 2\left(\frac{\pi}{4}(AB)^2 - \frac{1}{2} \times AD \times BA\right)$$

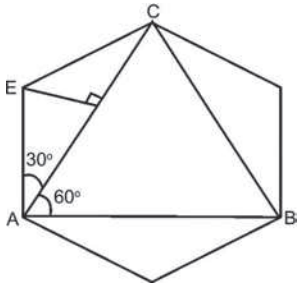
$$= (AB)^2\left(\frac{\pi}{2}\right) = 100\left(\frac{\pi}{2} - 1\right)$$

$$= 50\pi - 100$$

45. As hexagon is regular and AD = CD

So, ΔABC is equilateral triangle with AB = BC $AC = \frac{P}{3}$

$AD = \frac{P}{6}$



So,

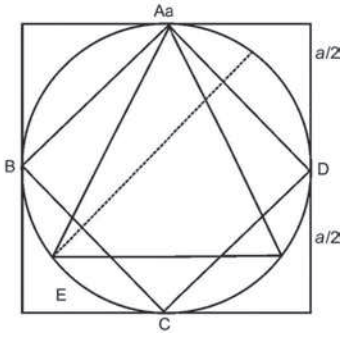
$\text{Cos } 30° = \frac{\sqrt{3}}{2} = \frac{AD}{AE}$

$AE = AD \times \frac{2}{\sqrt{3}} = \frac{2p}{3\sqrt{3}}$

So, perimeter of hexagon $= 6 \times \frac{2p}{6\sqrt{3}} = \frac{2p}{\sqrt{3}}$

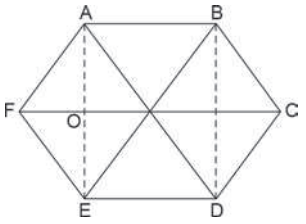
46. $AD = \left(\left(\frac{a}{2}\right)^2 + \left(\frac{a^2}{2}\right)\right)^{1/2}$

Then, radius of circle $= \frac{a}{2\sqrt{2}}$



47. For a figure to be enclosed with the given conditions, the number of concaves corners must be 4 less than convex corners.

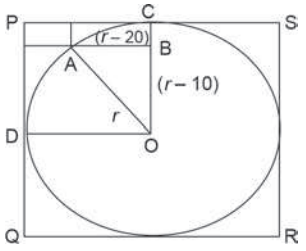
48.



For regular hexagon. The area of all small triangles will be equal, and there are 12 small equal triangles.

Then, $\frac{\text{Area of } \Delta AOF}{\text{Area of ABCDEF}} = \frac{1}{12}$

49.



Let the centre of circle is O.

Then, OA = OD = OC = r, or, OB = r − 10 and AB = r − 20

Then, in ΔABO,

$AO^2 = OB^2 + AB^2$, or, $r^2 = (r-10)^2 + (r-20)^2$

Or, $r^2 = r^2 - 20r + 100 + r^2 - 40r + 400$

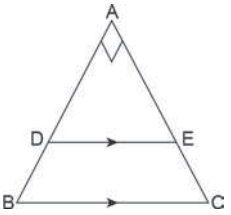
Or, $r^2 = 2r^2 - 60r + 500$

Or, $r^2 - 60r + 500 = 0 \Rightarrow r^2 - 50r - 10r + 500 = 0$

$\Rightarrow r(r - 50) - 10(r - 50) = 0$

$(r - 10)(r - 50) = 0$. Hence, R = 10 or 50. Since 10 is not possible, hence, r = 50

50.



Let length of hypotenuses

BC = 100 cm

Then, DE = 65 cm

Since ΔADE ~ ΔABC

$\frac{\text{Area of } \Delta ADE}{\text{Area of } \Delta ABC} = \frac{DE^2}{BC^2}$

$\frac{\text{Area of } \Delta ADE}{34} = \frac{6s^2}{100^2} = \frac{4225 \times 34}{100 \times 100} = = 14.365$ sq. unit

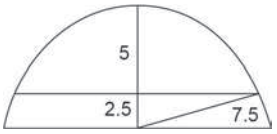
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ADVANCED

1. Radius of the circle cm $\sqrt{(7.5)^2-(2.5)^2}=5\sqrt{2}$

Hence, circumference $= 10\sqrt{2}\pi$ cm

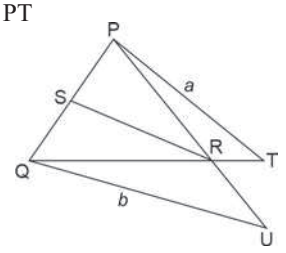


2. $\Delta PQU \sim PSR$

$$\Rightarrow \frac{PS}{PQ}=\frac{SR}{QU} \qquad \text{(i)}$$

$\Delta PQT \sim SQR$

$$\Rightarrow = \frac{QS}{PQ}=\frac{SR}{PT} \qquad \text{(ii)}$$



From (i) and (ii),

$$PQ \times SR = PS \times QU = SQ \times PT \Rightarrow = \frac{SQ}{PS}=\frac{b}{a}$$

Now, use componendo and equation (i) to obtain SR = $\frac{ab}{a+b}$

3. Go through the options.

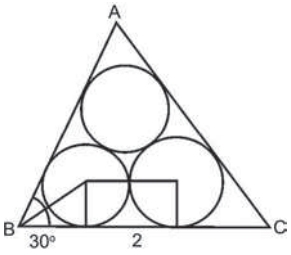
4. It can be seen that the side of the triangle

$= \sqrt{3}+2+\sqrt{3}=2+2\sqrt{3}$

Then, area $= \frac{\sqrt{3}}{4}a^2$

$= \frac{\sqrt{3}}{4}(2+2\sqrt{3})^2$

$= 4\sqrt{3}+6$



5. In a triangle, where a is the bigger side, and b, c are the smaller side

Then, $a < b + c$

Since $a + b + c = 13$

Then, a should not be bigger than 6.

When $a = 6$

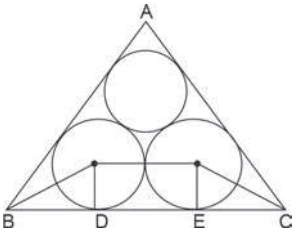
Then, b and $c = 6$ and 1, 5 and 2, 4 and 3,

when $a = 5$

Then, b and $c = 5$ and 3, 4 and 4.

So, only five triangles are possible.

6.



Let the radius of circles $= r$

Then, BD = EC = $r\sqrt{3}$

DE = $2r$

Then, BC = BD + DE + EC = $2(r + r\sqrt{3})$

BC = $2r(1 + \sqrt{3})$

$2r(1 + \sqrt{3}) = a$ [since BC = a]

$$r = \frac{a}{2+2\sqrt{3}} = \frac{a}{2(\sqrt{3}+1)}$$

7. $\frac{\pi}{8} = S, (2+4+8+16+32+64+128)$

Get S, from the above relation again $\frac{\pi\theta}{306^\circ} = S_1$

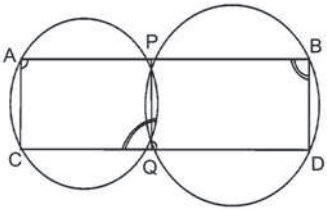
Now, θ can easily be calculated from here.

9. From alternate segment theorem

$\angle PAC = \angle PQD$ and $\angle PBD = \angle PQC$

But, $\angle PQD$ and $\angle PQC$ are supplementary.

Now, since $\angle PAC$ and $\angle PBD$ are supplementary, AC is parallel to BD.

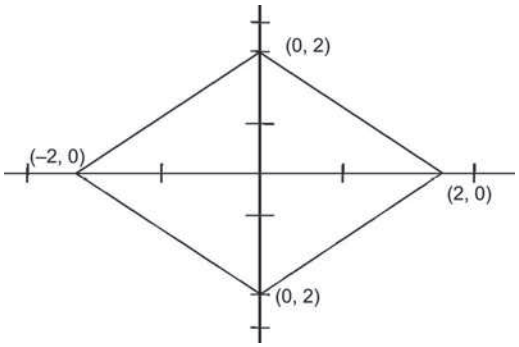


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10. $x + y = 2$ when $x \geq 0$ and $y \geq 0$

$x + y = -2$ when $x \leq 0$ and $y \leq 0$



12. Let h be the height of the pyramid and a be the distance from h to CD. The side length of the base is 14. The side lengths of ΔABE and ΔCDE are $2 \times 105 \div 14 = 15$ and $2 \times 91 \div 14 = 13$, respectively. We have a systems of equations through the Pythagorean Theorem:

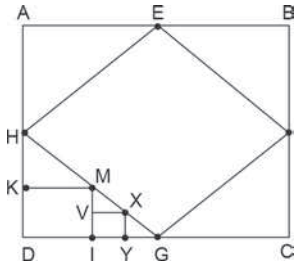
$13^2 - (14 - a)^2 = h^2$

$15^2 - a^2 = h^2$

Setting them equal to each other and simplifying gives $-27 + 28a = 225 \rightarrow a = 9$.

Therefore, $h = \sqrt{15^2 - 9^2} = 12$, and the volume of the pyramid is $\frac{bh}{3} = \frac{12 \times 196}{3} = 784$. Hence, option (c) is the answer.

14.



The side of square KMID $= \frac{1}{4}$ side of square ABCD

Let side of KMID $= b$

Then, $b = \frac{a}{4}$

Then, area of KMID $= b^2 = \left(\frac{a}{4}\right)^2 = \frac{a^2}{16}$ (i)

area of ABCD $= a^2$

So, ratio of area of ABCD and KMID

$= \frac{a^2}{a^2/16} = \frac{16}{1} = 16:1$

15. Side of square EFGH $= \frac{a}{\sqrt{2}}$

side of square VXYI $= \frac{a}{8}$

Then, ratio $= \frac{\text{Area of EFGH}}{\text{Area of VXYI}} = \frac{\left(a/\sqrt{2}\right)^2}{(a/8)^2} = 32:1$

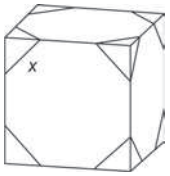
16. Area of shaded region

= Area of ABCD − (Area of EFGH + Area of KMID + Area of VXYI + Area of EBF)

$= a^2 - \left[\frac{a^2}{2} + \frac{a^2}{16} + \frac{a^2}{64} + \frac{a^2}{8}\right]$

$= a^2 - \left[\frac{45}{64}a^2\right] = \frac{19}{64}a^2$

17.



Assume sides of the regular polygon = x. Examine one edge of the unit cube: each contains two slanted diagonal edges of an octagon and one straight edge. The diagonal edges form 45-45-90Δ right triangles, making the distance on the edge of the cube $\frac{x}{\sqrt{2}}$. Thus, $2 \times \frac{x}{\sqrt{2}} + x = 1$, and $x = \frac{1}{\sqrt{2}+1} \times \left(\frac{\sqrt{2}-1}{\sqrt{2}-1}\right) = \sqrt{2} - 1$.

Each of the cut-off corners is a pyramid, whose volume can be calculated by $V = \frac{1}{3}Bh$. Use the base as one of the three congruent isosceles triangles, with the height being one of the edges of the pyramid that sits on the edges of the cube. The height is $\frac{x}{\sqrt{2}} = 1 - \frac{1}{\sqrt{2}}$. The base is a 45-45-90 triangle with leg of length $\left[1 - \frac{1}{\sqrt{2}}\right]$, making its area $\frac{1}{2}\left(1 - \frac{1}{\sqrt{2}}\right)^2 = \frac{3 - 2\sqrt{2}}{4}$. Using this, we get that the area of one of the tetrahedron is

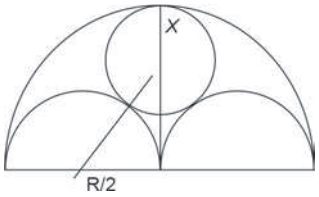
$\frac{1}{3}\left(1 - \frac{1}{\sqrt{2}}\right)\left(\frac{3 - 2\sqrt{2}}{4}\right) = \frac{10 - 7\sqrt{2}}{24}$. Since there are 8 removed corners, we get an answer of $\frac{10 - 7\sqrt{2}}{3}$. Hence, option (b) is the answer.

18. $\left(\frac{R}{2} + x\right)^2 = \left(\frac{R}{2}\right)^2 + (R - x)^2$

$x = \frac{R}{3}$

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19. We know that

In radius = Semi-perimeter – Hypotenuse

Let hypotenuse = x

Then, semi-perimeter = $\frac{42+x}{2} = 21+\frac{x}{2}$

Then, $32 = 21 + \frac{x}{2} - x$

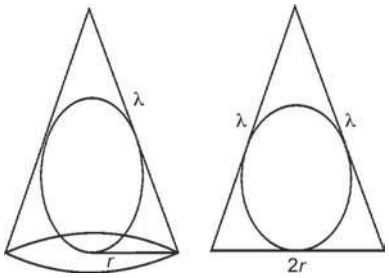
$32 = 21 - \frac{x}{2}$

$\frac{x}{2} = -11 \rightarrow x = -\frac{11}{2}$ which is not possible. So, triangle is not possible.

21. Height of the octagonal prism will be the diameter of the sphere inscribed in it. Hence, the answer will be:

$$\frac{\sqrt{3}}{4}a^3\left(3\sqrt{3}-r\right)[n(n+1)]$$

22. Consider the biggest cross-section of the cone as a isosceles triangle; therefore, the circle inscribed in the triangle will be the biggest cross-section of the sphere.

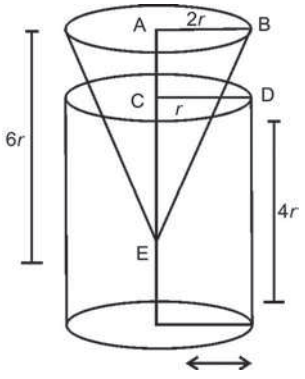


We know that in radius × semi-perimeter = Area of the triangle.

A little calculation will lead to the answer, i.e.,

$$\frac{4}{3}\pi r^3\left(\frac{\lambda-r}{\lambda+r}\right)^{3/2}$$

23. ΔECD is similar to ΔEAB. Hence, EC = 3r



So, the answer will be $7\pi r^3$.

24. Side of the square should always be less than the attitude of the triangle.

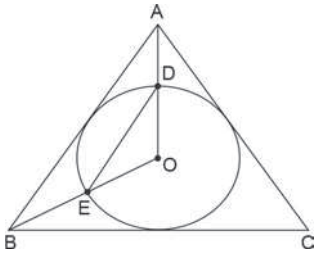
25. It cannot determined because ∠E is not dependent on ∠O.

Then, ratio of area of ABCD and area of ΔXYZ = x^2:

$\frac{3\sqrt{3}}{64}x^2$

$= 64:3\sqrt{3}$

26.



It is given that DE = 3 units

Let the radius of circle = r

Using Sine Rule,

$$\frac{\text{Sin O}}{\text{ED}} = \frac{\text{Sin E}}{\text{OD}}$$

$$\frac{\text{Sin}120^\circ}{3} = \frac{\text{Sin}30^\circ}{r} \rightarrow r = \sqrt{3}$$

Then, circumference of the circle = $2\pi r = 2\pi\sqrt{3}$

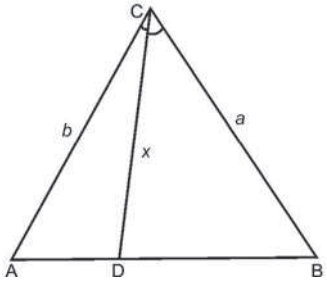
27. Let CD = x

Area ΔABC = ΔACD + ΔDCB

$$\Rightarrow \frac{1}{2}ab\sin 120^\circ = \frac{1}{2}bx\sin 60^\circ + \frac{1}{2}ax\sin 60^\circ$$

Now, it can be observed that

$x = \frac{ab}{a+b}$



28. We have to calculate

AH × AG × AF × AE × AD × AC × AB

AE = 2

∠AOB = 45°

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Use Cosine formula and get AB = AH. ∠AHG = 135°

Again by Cosine formula get AG (= AC) and AF (= AD).

29. Let the radius of outermost circle = R

Then, the radius of circle EFG = $\frac{R}{\sqrt{2}}$

Then, the radius of circle XYZ = $\frac{\left(R/\sqrt{2}\right)}{2} = \frac{R}{2\sqrt{2}}$

So, ratio of the radius of the outermost circle to the radius of circle inscribed in ΔXYZ = $1: \frac{1}{2\sqrt{2}} = 2\sqrt{2}:1$

30. Let the side of ABCD = x

Then, the radius of circle EFG = $\frac{x}{2}$

Then, side of ΔEFG = $\frac{\sqrt{3}}{2}x$

So, side of ΔXYZ = $\frac{\sqrt{3}}{4}x$

Then, area of ΔXYZ = $\frac{3\sqrt{3}}{64}x^2$

31. Area of the shaded region

$= \frac{\text{Area of circle ABCD} - \text{Area of square ABCD}}{2}$

Area of circle ABCD = $\pi r^2 = 25\pi$

Area of square ABCD = $(5\sqrt{2})^2 = 50$

Then, area of shaded region = $\frac{25\pi - 50}{2}$

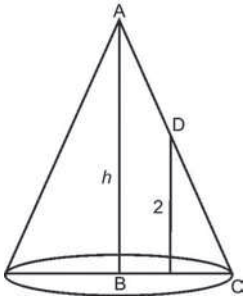
32. $\frac{\frac{1}{4}\text{ of ABC} + \frac{1}{4}\text{ of }\frac{1}{4}\text{ of ABC}}{a(\Delta ABC)} = \frac{5}{16}$

33. DE = 2

EC = 3/2

Find DC and then find l.

Thus, the answer will be 15 πm^2.

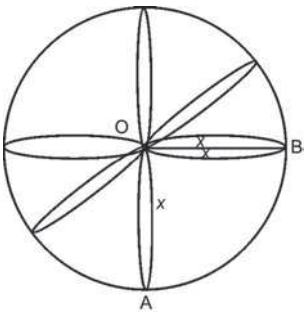


35. OA = OB = AB = 1 cm

∠OAB = 60°

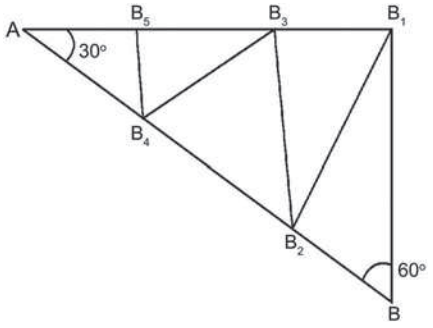
Area x = Area of sector OAB − DOAB

$12x$ = Area of the flower



37. AB = a BB = a/2

B_2B = a/4 $\Delta AB_2B_3 \sim \Delta ABB$

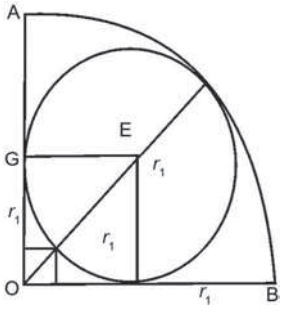


$\frac{B_2B_3}{B_dB_1} = \frac{3}{4} = \frac{AB_3}{AB_1}$

Now, we have two infinite GP series having common ratio 3/4. Calculate the sum of these series.

38. $r - r_1 = \sqrt{2}\, r_1$

Now, find r, and solve to get the answer.



41. Let OC = r

ABCDEF is a regular hexagon

ED = r

Let EC = x

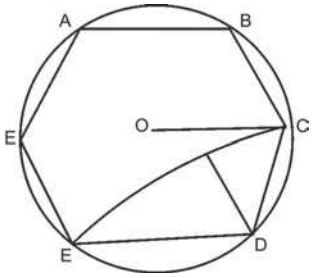
In ΔEDC, using cosine formula we can get, $x = \sqrt{3}\, r$

Side of the star will be $r/\sqrt{3}$.

Now, the desired ratio can be easily calculated.

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46. Consider the problem as a pyramid to the base of a regular hexagon.

Now, in Δ POR

OP = 12 cm

OR = 5 cm

PR = 13 cm

Now, cosine formula will give the answer.

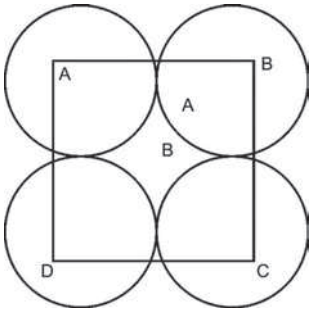
47. AB = 4 cm

$a = \frac{16 - 4\pi}{4} = 4 - \pi$ cm

Area of common region = $b = \frac{4\pi}{4} - 4 + \pi$

$b = 2\pi - 4$

Total area = $4b = 8\pi - 16$



49. $\frac{\sqrt{3}}{4}a^2 = 300 \text{ cm}^2$

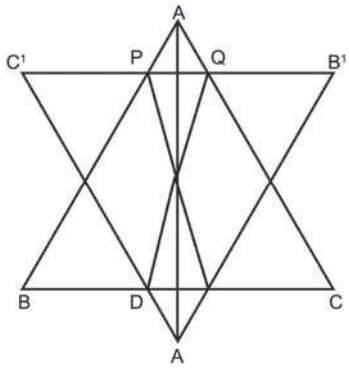
$a = \sqrt{\frac{1200}{\sqrt{3}}}$

So, AP will be equal to AB/3.

So, the area common to both triangles

$= 300 - 3 \times \frac{\sqrt{3}}{4} \times \frac{1}{9} \times \frac{1200}{\sqrt{3}}$

$= 200 \text{ cm}^2$



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CHAPTER

19

Mensuration

LEARNING OBJECTIVES

After completion of this chapter, the reader should be able to understand:

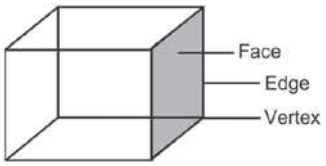
- Relationship between 2-D and 3-D figures
- Definitions and properties
- Different structures
- Process of finding out volume, lateral surface area, curved surface area and total surface area
- Methods of solving questions

INTRODUCTION

There is a direct connection between geometry and mensuration. It can be seen through analogy that the relationship shared by two dimension (2-D) and geometry is the same as the relationship shared by three dimension (3-D) and mensuration in most cases. Mensuration is another topic from which questions have been asked in the CAT, although not as much as from geometry. A fair number of questions from this topic are being asked in the CAT. The importance of this topic lies therein. Mensuration should be seen as a nice juxtaposition of 3-D figures, which conforms to the properties of both 2-D geometry and 3-D geometry.

SOLIDS

A solid has three dimensions, namely length, breadth or width, and height or thickness. The plane surfaces that bind it are called its faces and the solid so generated is known as polyhedron.



The volume of any solid figure is the amount of space enclosed within its bounding faces. A solid has edges, vertices, and faces, which are shown in the figure.

A solid has the following two types of surface areas:

Lateral Surface Area Lateral surface area (LSA) of a solid is the sum of the areas of all the surfaces it has except the top and the base.

Total Surface Area Total surface area (TSA) of a solid is the sum of the LSA and the areas of the base and the top.

Note: In case of solids, like the cube and cuboid, the LSA consists of plane surface areas (i.e., area of all surfaces except the top and base), whereas in case of solids, like cone and cylinder, it consists of curved surface areas (CSA). Therefore, for such solids, the LSA is also called CSA.

Euler's Rule

Euler's rule states that for any regular solid:

Number of faces (F) + Number of vertices (V) = Number of edges (E) + 2

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CUBOID

A cuboid is a rectangular solid having six rectangular faces. The opposite faces of a cuboid are equal rectangles. A cuboid has a length (l), breadth (b), and height (h).

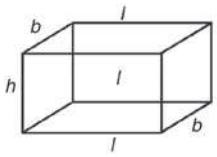


Figure 1

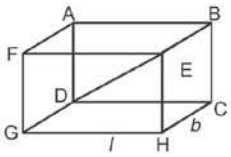


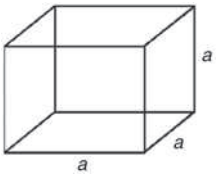
Figure 2

In Figure 2, ED is the diagonal of the cuboid. Moreover, the area of the surface GDCH is x, the area of the surface HEBC is y, and the area of the surface GFEH is z.

(i) Volume = Area of base × height = lbh
(ii) Volume = $\sqrt{xyz}$
(iii) Volume = $xh = yl = zb$
(iv) Lateral surface area (LSA) or area of the four walls = $2(l + b)h$
(v) Total surface area (TSA) = $2(x + y + z) = 2(lb + bh + lh)$
(vi) Diagonal = $\sqrt{l^2 + b^2 + h^2}$

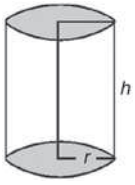
CUBE

A cube is a solid figure having six faces. All the faces of a cube are equal squares (let us say of the side 'a'). Therefore, the length, breadth, and height of a cube are equal.



(i) Volume = a^3
(ii) Lateral surface area (LSA) or area of the four walls = $4a^2$
(iii) Total surface area (TSA) = $6a^2$
(iv) Diagonal = $a\sqrt{3}$

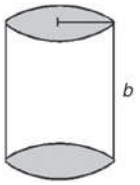
RIGHT CIRCULAR CYLINDER

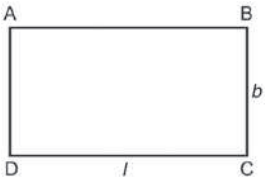


In the above figure, r is the radius of the base and h is the height of a right circular cylinder. A cylinder is generated by rotating a rectangle or a square by fixing one of its sides.

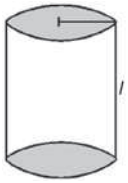
(i) Volume = area of base × height
(ii) Volume = $\pi r^2 h$
(iii) Curved surface area (CSA) = Perimeter of base × height
(iv) LSA = $2\pi rh$
(v) Total surface area (TSA) = LSA + area of the top + area of the base
(vi) TSA = $2\pi rh + \pi r^2 + \pi r^2$
(vii) TSA = $2\pi r(r + h)$

Some Important Deductions





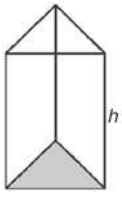
If the above rectangular sheet of paper (ABCD) is rolled along its length to form a cylinder, then the radius (r) of the cylinder will be ($L/2n$) and its height will be b and volume of this cylinder = $\frac{L^2 b}{4\pi}$, where l is the length of the rectangle.



If the above rectangular sheet of paper (ABCD) is rolled along its breadth to form a cylinder, then the radius (r) of the cylinder will be $\frac{b}{2\pi}$ and its height will be L. Volume of this cylinder = $\frac{b^2 L}{4\pi}$.

PRISM

A prism is a solid having identical and parallel top and bottom faces, that is, they will be identical polygons of any number of sides. The side faces of a prism are rectangular and are known as lateral faces. The distance between two bases is known as the height or the length of the prism.



(i) Volume = Area of base × Height

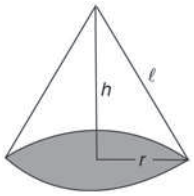
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(ii) Lateral surface area (LSA) = Perimeter of the base × Height

(iii) Total surface area (TSA) = LSA + (2 × Area of the base)

RIGHT CIRCULAR CONE



In the above figure, 'r' is the radius of the base, h is the height, and l is the slant height of the right circular cone.

(i) Volume $= \frac{1}{3} \times$ Area of the base height × height

Volume $= \frac{1}{3} \pi r^2 h$

(ii) Slant height $= l = \sqrt{r^2 + h^2}$

(iii) Curved surface area (CSA) $= \pi r l$

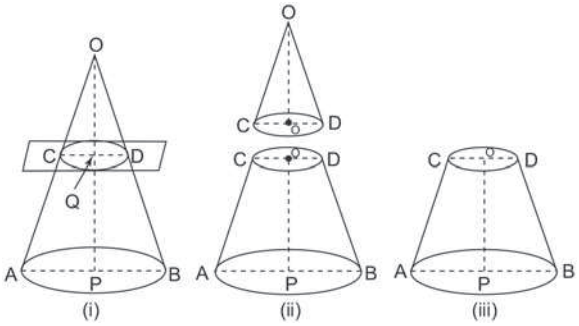
(iv) Total surface area (TSA) = (CSA + Area of the base)

$$TSA = \pi r l + \pi r^2$$

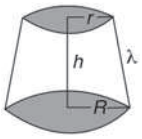
Frustum of Cone

A cone whose top portion is sliced off by a plane which is parallel to the base is called frustum of cone.

Formation of frustum:



However, for the sake of representing the formula, we will use another form of frustum right now as given below:



In the above figure, r is the radius of the base, h is the vertical height of the frustum, and λ is the slant height of the frustum.

(i) Volume $= \frac{\pi h}{3}(R^2 + r^2 + Rr)$

(ii) Slant height $= \lambda = \sqrt{(R-r)^2 + h^2}$

(iii) Curved surface area (CSA) $= \pi (R + r) \lambda$

(iv) Total surface area (TSA) = CSA + Area of the top + area of the base

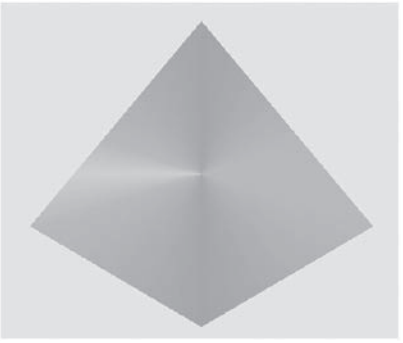
TSA $= \pi (R + r) \lambda + \pi r^2 + \pi R2$

TSA $= \pi (R\lambda + r\lambda + r^2 + R^2)$

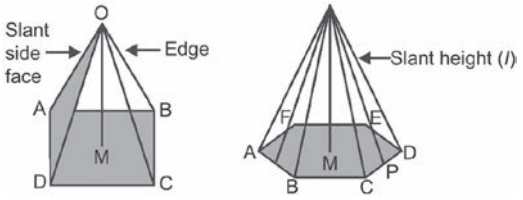
(v) To find the height (H) of original cone.

$$H = \frac{Rh}{R-r}$$

PYRAMID



A pyramid is a solid having an n-sided polygon at its base. The side faces of a pyramid are triangular with the top as a point.



In the above figures, OM is the height of the pyramid.

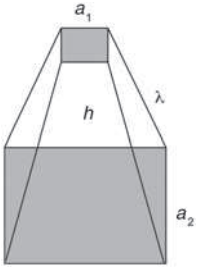
(i) Volume $= \frac{1}{3} \times$ Area of the base × Height

(ii) Lateral surface area (LSA) $= \frac{1}{2} \times$ (Perimeter of the base) × Slant Height

(iii) Total surface area (TSA) = LSA + Area of the base

Frustum of Pyramid

A pyramid whose top portion is sliced off by a plane that is parallel to the base is called the frustum of a pyramid.



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In the above figure, a_1 is the area of the top face of the frustum, a_2 is the area of the bottom face of the frustum, h is the height of the frustum, and l is the slant height of the frustum.

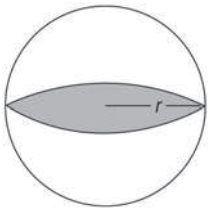
(i) Volume $= \frac{1}{3}h\left(a_1 + a_2 + \sqrt{a_1 a_2}\right)$

(ii) Lateral surface area (LSA) $= \frac{1}{2}(P_1 + P_2)l$

where P_1 and P_2 are perimeters of the top and the bottom faces.

(iii) Total surface area (TSA) = LSA + $a_1 + a_2$

SPHERE

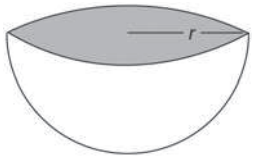


In the above figure, r is the radius of the sphere.

(i) Volume $= \frac{4}{3}\pi r^3$

(ii) Surface area $= 4\pi r^2$

HEMISPHERE



(i) Volume $= \frac{2}{3}\pi r^2$

(ii) Curved surface area (CSA) $= 2\pi r^2$

(iii) Total surface area (TSA) = LSA + Area of the top face (read circle)

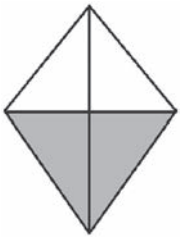
TSA $= 2\pi r^2 + \pi r^2$

TSA $= 3\pi r^2$

SOME MORE SOLIDS

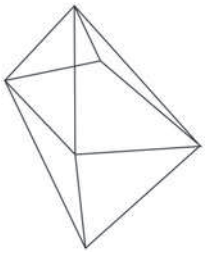
Tetrahedron

A tetrahedron is a solid with four faces. All the faces of a tetrahedron are equilateral triangles. A tetrahedron has four vertices and six edges.



Octahedron

An octahedron is a solid that has eight faces. All the faces of an octahedron are equilateral triangles. An octahedron has six vertices and 12 edges.



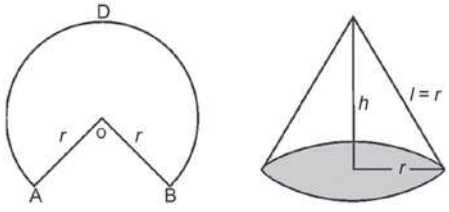
Inscribed and Circumscribed Solids

If a sphere of the maximum volume is inscribed in a cube of edge 'a', then the radius of the sphere $= \frac{a}{2}$.

If a cube of the maximum volume is inscribed in a sphere of radius 'r', then the edge of the cube $= \frac{2r}{\sqrt{3}}$.

If a cube of the maximum volume is inscribed in a hemisphere of radius 'r', then the edge of the cube $= \sqrt{\frac{2}{3}} \times r$.

Some Important Deductions



If a cone is made by a sector of a circle (AOBD), then the following two things must be remembered:

The area of the sector of a circle (AOBD) = The CSA of the cone

Radius of the circle (r) = Slant height (l) of the cone

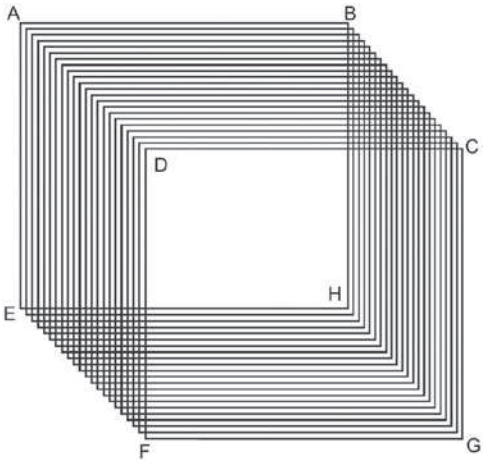
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VISUAL MENSURATION

In this section, we will be required to visualize some of the unknown dimensions of any structure with the help of the given dimension. Some of the typical examples of visual mensuration are given below:

Given is a cube ABCDEFGH of side length 'a' units. Its top face is ABCD and its bottom face is EFGH. Since the side length of this cube is 'a' units, AB = BC = CD = AD = AE = EF = FD = FG = GH = EH = BH = CG = 'a'



Minimum Length Between Vertex A and Vertex G

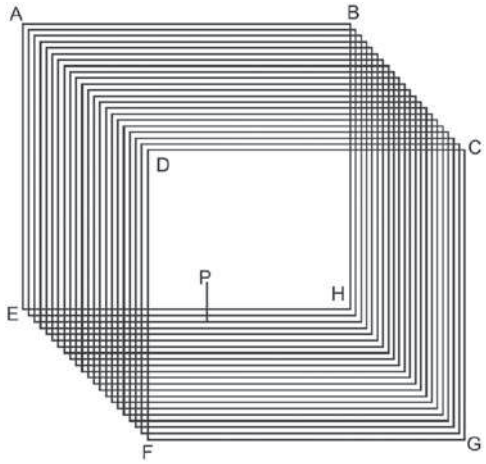
We can find the minimum distance between these two vertices in the following two ways:

- *Aerial distance* Aerial distance can be understood by assuming that there is a fly at vertex A and it has to reach vertex G through the minimum possible distance. This distance will be the diagonal distance between the vertices A and G

$$= \sqrt{a^2 + a^2 + a^2} = a\sqrt{3}$$

- *Physical distance* Physical distance can be understood by assuming that there is an ant at vertex A and it has to reach vertex G through the minimum possible distance. Since it cannot fly, so ant will cover this distance along the two faces viz., face ABEH and face EFGH or face ADFE and face CDFG or face ABCD and face CDFG or face ABCD and face BCGH.

The shortest possible distance can be the diagonal only. Let us assume that the ant is going via face ABEH and face EFGH. So, the ant will cover first the diagonal distance between A and P, where P is the mid-point of side EH, and then from P to vertex G.



Now, $AP^2 = EP^2 + GP^2 = \frac{a\sqrt{5}}{2}$

$$GP^2 = HP^2 + GH^2 = \frac{a\sqrt{5}}{2}$$

Hence, the minimum possible physical distance = $2\frac{a\sqrt{5}}{2} = a\sqrt{5}$

Let us find out why the distance AE + EG cannot be the shortest?

First calculate AE and EG.

$AE = a$

$EG = \sqrt{a^2 + a^2} = a\sqrt{2}$

So, $AE + EG = a + \sqrt{a^2 + a^2} = a\sqrt{2} + a = a(1 + \sqrt{2})$

Obviously, $a\sqrt{5} < a(1 + \sqrt{2})$ [For $a > 0$]

We can understand this phenomenon by having a bit of *mental mapping*. As we have seen earlier that the minimum possible distance can be the diagonal distance only. Now, let us cut open the face ABHE by making a cut mark at EH so that the faces ABHE and EFGH are in a plane, lying horizontal on the ground. Now, the minimum possible distance between A and G will be the diagonal of the newly formed rectangle AFGB. This diagonal will pass through the midpoint of EH.

So, $AG^2 = AF^2 + FG^2 = (2a)^2 + (a)^2 = 5a^2$

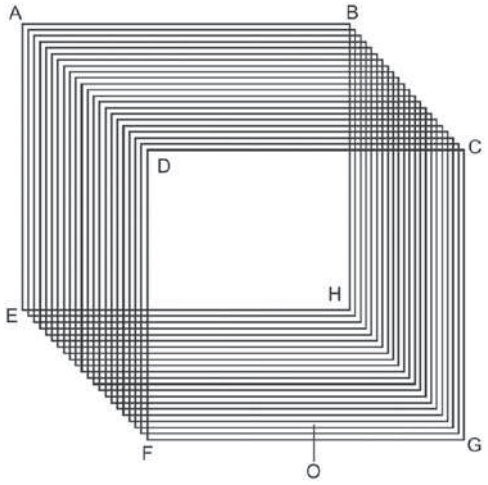
$AG = a\sqrt{5}$

Minimum Length Between Vertex A and Vertex O, where O is the Mid-point of FG

- *Aerial distance* We can see this situation vis-à-vis a cuboid of side lengths 'a' units, 'a' units and $\frac{a}{2}$ units.

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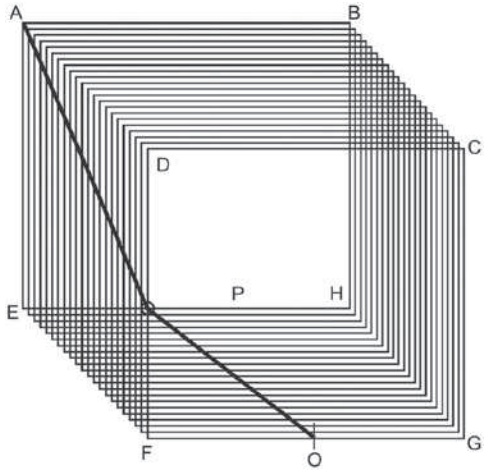
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Minimum aerial distance between A and O

$$= \sqrt{\frac{a^2}{4} + a^2 + a^2} = \frac{3a}{2}$$

- *Physical distance* Let us assume that the ant is moving through the faces ABHE and EFGH. Ant will first go the point Q, where Q is the mid-point of E and P. Then, the ant will cover QO.



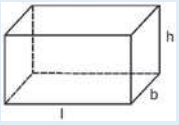
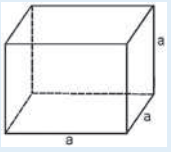
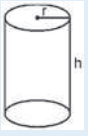
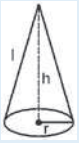
$$AQ^2 = AE^2 + EQ^2 = a^2 + \frac{a^2}{16} = \frac{17a^2}{16}$$

$$\text{So, } AQ = QO = \frac{a\sqrt{17}}{4}$$

So, the minimum distance between the vertex A and point

$$O = AQ + QO = \frac{a\sqrt{17}}{4}$$

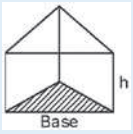
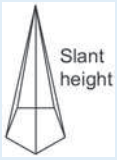
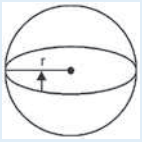
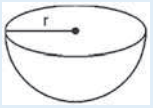
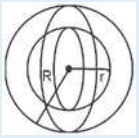
FORMULAE

S.No.	Name	Figure	Nomenclature	Volume	Curved/ lateral surface area	Total Surface area
1.	Cuboid		l = length b = breadth h = height	lbh	$2(l + b)$ h	$2(lb + bh + hl)$
2.	Cube		a = edge/side	a^3	$4a^2$	$6a^2$
3.	Right circular cylinder		R = radius of base h = height of the cylinder	$\pi r^2 h$	$2\pi rh$	$2\pi r\,(r + h)$
4.	Right circular cone		r = radius h = height l = slant height $l = \sqrt{r^2 + h^2}$	$\frac{\pi r^2 h}{3}$	πrl	$\pi r\,(l + r)$

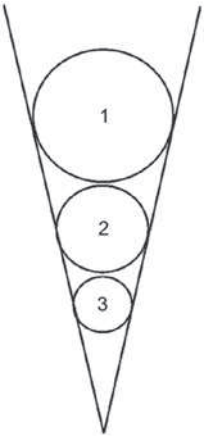
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S.No.	Name	Figure	Nomenclature	Volume	Curved/ lateral surface area	Total Surface area
5.	Right triangular prism		–	Area of the base × Height	Perimeter of the base × Height	Lateral surface area + 2(Area of base)
6.	Right pyramid		– height	1/3 area of the base × Height	1/2 × Perimeter of the base × Slant height	Lateral surface area + Area of base
7.	Sphere		r = radius	$\frac{4}{3}\pi r^3$	–	$4\pi r^2$
8.	Hemisphere		r = radius	$\frac{2}{3}\pi r^3$	$2\pi r^2$	$3\pi r^2$
9.	Spherical Shell		r = inner radius R = outer radius	$\frac{4}{3}\pi(R^3 - r^3)$	–	$4\pi(R^2 + r^2)$
10.	Frustum of a Cone		–	–	–	Lateral surface area + Area of top + Area of base

Example 1 Three spheres are kept inside a cone, as given in the figure. Spheres are touching both the slant sides of the cone and the adjacent spheres. If the radius of the first sphere and the third sphere are 5 units and 20 units, respectively, then find the radius of the second sphere.



Solution We can see in Figure 1, that the AOA′P, BOB′Q, and COC′R will be similar.

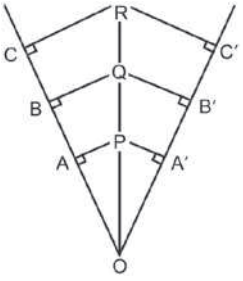


Figure 1

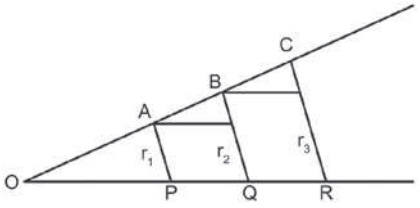


Figure 2

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In Figure 2, $(r_2 - r_1)/(r_2 + r_1) = (r_3 - r_2)/(r_3 + r_2) = K$
Using Componendo and Dividendo, $r_2/r_1 = r_3/r_2$

Hence, the three radii are in a GP.

So, $r_2/20 = 5/r_2$, and hence, $r_2 = 10$ units

Result of this question can be used as a formula also.

Example 2 The diameter of a metallic sphere is 6 cm. The sphere is melted and drawn into a wire of uniform circular cross-section. If the length of the wire is 36 m, find its radius.

Solution The diameter of the metallic sphere is 6 cm. Hence, radius of the sphere is 3 cm. Now, let the radius of the cross section of the wire be r cm. As we know, metallic sphere is converted into a cylindrical shaped wire, then their volumes will be equal to 4.

$$\text{So, } \frac{4}{3} \times \pi \times 3^3 = \pi \times r^2 \times 3600$$

$$\text{or, } 4 \times 9\ \pi \times 3^3 = 3600\ \pi r^2$$

$$\text{This gives } r^2 = 0.01$$

$$\text{So, } r = 0.1$$

Example 3 Sardar Sarovar Dam which is rectangular in shape can produce electricity only if the height of the water level in it is at least 7 cm. Now, the water is pumped in at the rate of 5 kmph through a pipe of diameter 14 cm into the dam area of dimensions 50 m × 44 m. In what time, the dam will be able to produce electricity?

Solution The volume of water flowing through the cylindrical pipe in 1 h at the rate of 5 km (5000 m) per hour

$$= \frac{22}{7} \times \frac{7}{100} \times \frac{7}{100} \times 5000 \text{ m}^3$$

So, radius = 7 cm = $\frac{7}{100}$ m = 77 m^3

Therefore, 77 m^3 of water will fall into the tank in 1 h. Since the level of the water is required to be raised by 7 cm, that is, $\frac{7}{100}$ m.

Volume of the required quantity of water = 50 × 44 × $\frac{7}{100}$ m^3 = 154 m^3

Since 77 m^3 of water falls into the tank in 1 h, therefore, 154 m^3 of water will fall into the dam in $\frac{154}{77}$ h, that is, 2 h.

Hence, the level of water will rise by 7 cm in 2 h.

Example 4 A right-angled triangle ABC whose two sides other than the hypotenuse are 15 cm and 20 cm. The triangle is made to revolve about its hypotenuse. Find the volume of the double cone so formed.

Solution Let ΔABC be the right angled triangle right angled at A, whose sides AB and AC measures 15 cm and 20 cm, respectively.

The length of the side BC (hypotenuse)

$$= \sqrt{15^2 + 20^2} \text{ cm} = 25$$

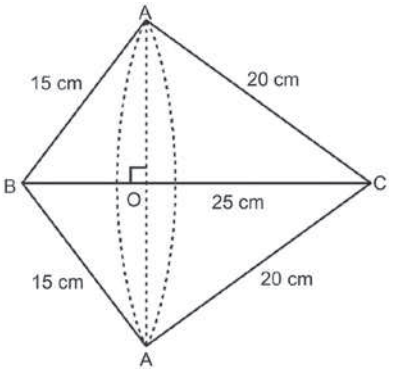
Here, AO (and A′O) is the radius of the common base of the double cone formed by revolving the ΔABC about BC.

Height of the cone BBA′ is BO and the slant height is 15 cm.

Height of the cone CAA′ is CO and the slant height is 20 cm

Using AA similarity,

Now, ΔAOB ~ ΔCAB



Therefore, $\frac{AO}{20} = \frac{15}{25}$

This gives AO = $\frac{20 \times 15}{25}$ cm = 12 cm

Also, $\frac{BO}{15} = \frac{15}{25}$ cm

Therefore, CO = 25 cm − 9 cm = 16 cm

Now, volume of the double cone

$$= \left(\frac{1}{3} \times 3.14 \times 12^2 \times 9 + \frac{1}{3} \times 3.14 \times 12^2 \times 16\right) \text{cm}^2$$

$$= \frac{3.14}{3} \times 12^2 \times (9 + 16) \text{cm}^2$$

$$= 3768 \text{ cm}^3$$

Example 5 When shopping in Big Bazar, I saw a peculiar solid toy in the form of a hemisphere surmounted by a right circular cone. Height of the cone was 2 cm and the diameter of the base was 4 cm. If a right circular cylinder circumscribed the solid, find out how much more space

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will it have, provided the height of the cone was 2 cm and diameter of the base was 4 cm, respectively?

Solution See the figure below.

Assume BPC be the hemisphere and ABC is the cone standing on the base of the hemisphere.

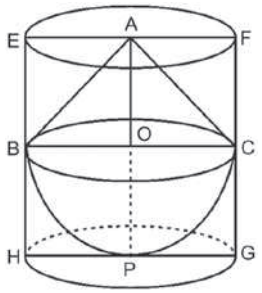
Radius BO of the hemisphere (as well as of the cone) $= \frac{1}{2} \times 4 \text{ cm} = 2 \text{ cm}$

Now, let the right circular cylinder EFGH circumscribe the given solid.

Radius of the base of the right circular cylinder = HP = BO = 2 cm

Height of the cylinder = AP = AO + OP = 2 cm + 2 cm = 4 cm

Now, volume of the right circular cylinder − volume of the solid



$$= \left[\pi \times 2^2 \times 4 - \left(\frac{2}{3} \times \pi 2^3 + \frac{1}{3} \times \pi \times 2^3\right)\right] \text{cm}^3$$

$= (16\pi - 8\pi) \text{ cm}^3$

$= 8\pi \text{ cm}^3$

Hence, the right circular cylinder is having 8π cm^3 more space than the solid.

Example 6 A toy consists of a base that is the section of a sphere and a conical top. The volume of the conical top is 30π sq. units and its height is 10 units. The total height of the toy is 19 units. The volume of the sphere (in cubic units), from which the base has been extracted, is:

(a) $\frac{256}{3}\pi$ (b) $\frac{64}{3}\pi$

(c) $\frac{108}{3}\pi$ (d) $\frac{500}{3}\pi$

Solution Height of the cone = 10 units

Volume of the cone = 30π cubic units = Diameter of the cross-section from where the sphere has been sectioned = 6 units

$= r + \sqrt{2^2 - 3^2} = 9$

$= r = 5$ units

= Volume of the original sphere from which the base has been sectioned $= \frac{4}{3}\pi r^3 = \frac{500}{3}\pi$

Practice Exercises

Warm Up

Q.1 If the side of a cube is increased by 100%, find by what percentage the surface area of the cube is increased?
(a) 150% (b) 200% (c) 300% (d) 350%

Q.2 A banquet hall has the dimensions 30 m × 12 m × 6 m. Each person should get 8 m^3 of space. Find the number of persons who can be accommodated in this hall.
(a) 240 (b) 250 (c) 270 (d) 300

Q.3. Find the height of the cylinder whose volume is 511 cm^3 and the area of the base is 36.5 cm^2.
(a) 3.5 cm (b) 10.5 cm
(c) 14 cm (d) None of these

Q.4 The ratio between the curved surface area and the total surface area is 2:3 and the total surface is 924 cm^2. What is the volume of the cylinder?
(a) 2156 cm^3 (b) 2183 cm^3
(c) 2492 cm^3 (d) None of these

Q.5 If the curved surface area of a cylinder is 1320 cm^2 and its base radius is 21 cm, then what is its total surface area?
(a) 4092 cm^2 (b) 2084 cm^2
(c) 5104 cm^2 (d) None of these

Q.6 The radius of an iron rod is decreased to one-fourth of it actual radius. If its volume remains constant, then the length will become:
(a) 2 times (b) 12 times
(c) 8 times (d) 16 times

Q.7 A reservoir is in the shape of a frustum of a right circular cone. It is 8 m wide at the top and 4 m wide at the bottom. If it is 6 m deep, then what is its volume?
(a) 224 m^3 (b) 176 m^3
(c) 204 m^3 (d) None of these

Q.8 A spherical metal ball of 6 cm radius is melted and recast into three spherical balls. The radii of two of

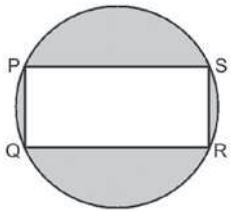
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these balls are 3 cm and 4 cm. What is the radius of the third ball?
(a) 4.5 cm (b) 5 cm (c) 6 cm (d) 7 cm

Q.9 If the diagonals of a rhombus are 18 cm and 24 cm, respectively, then find its perimeter.
(a) 15 cm (b) 42 cm (c) 60 cm (d) 68 cm

Q.10 In the figure shown below, PQRS is a rectangle of the dimension 8 cm × 6 cm and is inscribed in a circle. Find the area of the shaded portion.



(a) 44 cm^2 (b) 34.25 cm^2
(c) 32.50 cm^2 (d) None of these

Q.11 The radius of the base and the volume of the right circular cone are doubled. What is the ratio of the length of the larger cone to that of the smaller cone?
(a) 1:4 (b) 1:2 (c) 1:3 (d) 4:1

Q.12 A cone and a hemisphere have equal base radius and equal volumes. The ratio of their heights is:
(a) 3:1 (b) 2:1
(c) 4:1 (d) None of these

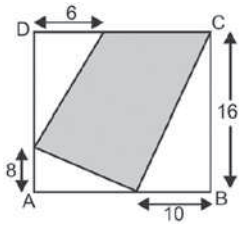
Q.13 If the right circular cone is cut into three solids of volumes V_1, V_2, and V_3 by two cuts which are parallel to the base and trisects the altitude, then $V_1:V_2:V_3$ is:
(a) 1:2:3 (b) 1:4:6
(c) 1:6:9 (d) None of these

Q.14 Water flows at the rate of 10 m per minute from a cylindrical pipe of radius 2.5 mm. A conical vessel whose diameter is 40 cm and depth 24 cm is filled with water flowing from this pipe. The time taken to fill the conical vessel is
(a) less than 30 min.
(b) less than 50 min but more than 30 min.
(c) more than 50 min but less than 75 min.
(d) more than 75 min.

Q.15 If a regular square pyramid has a base of side 8 cm and height of 30 cm, then what is its volume (in cc)?
(a) 360 (b) 480
(c) 640 (d) 800

Q.16 A cylinder circumscribes a sphere. The ratio of their volumes is:
(a) 2:1 (b) 3:2 (c) 4:3 (d) 6:5

Q.17 Find the area of the shaded region in the given figure of square ABCD.



(a) 128 cm^2 (b) 184 cm^2
(c) 154 cm^2 (d) 168 cm^2

Q.18 An open box is made of wood 2 cm thick. Its internal dimension is 86 cm × 46 cm × 38 cm. What is the cost of painting the outer surface of this box at ₹10 per m^2?
(a) ₹12.35 (b) ₹8.85 (c) ₹15.70 (d) ₹16.50

Q.19 A copper wire when bent in the form of a square encloses an area of 121 m^2. If the same wire is bent to form a circle, the area enclosed by it would be
(a) 124 m^2 (b) 136 m^2 (c) 154 m^2 (d) 184 m^2

Q.20 A spherical ball was painted black. After getting painted, it was cut into four similar pieces. What is ratio of the painted area to the non-painted area?
(a) 1:1 (b) 1:2 (c) 3:1 (d) 3:2

Q.21 Volumes of a sphere, a cube, a tetrahedron, and an octahedron are same. Find for which of the following structures, it will have the maximum surface area?
(a) Sphere (b) Cube
(c) Tetrahedron (d) Octahedron

Q.22 The ratio of diagonals of the two cubes is 3:2. What is the ratio of the surface areas of these two cubes, respectively?
(a) 27:8 (b) 3:2 (c) 9:4 (d) 16:9

FOUNDATION

Q.1 A rectangular classroom has to be painted covering all its surfaces excluding floor. The cost of painting a similar room is ₹1872. But, its length, breadth and height are 10% less, 10% more, and 10% less than the dimensions of the classroom, respectively. How much will it cost to paint the classroom if its length, breadth, and height are in the ratio of 1:2:3?
(a) ₹1956 (b) ₹2000
(c) ₹2100 (d) Cannot be determined

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Q.2 The top of a conical container has a circumference of 308 m. Water flows in at a rate of 12 m^3 every 2 seconds. When will the container be half-filled, if its depth is 12 m?
(a) 42 min (b) 68 min
(c) 54 min (d) 82 min

Q.3 What is the radius of a spherical ball in inches which is formed by melting a cylinder of base diameter 8 inches and height 160 inches, if the conversion wastage results in a 10% weight loss?
(a) 6 (b) 8 (c) 12 (d) 16

Q.4 A solid wooden toy is in the shape of a right circular cone mounted on the base of a hemisphere. If the radius of the hemisphere is 4.2 cm and the total height of the toy is 10.2 cm, find the volume of the wood needed to construct another such toy
(a) 104 cm^3 (b) 162 cm^3
(c) 427 cm^3 (d) 266 cm^3

Q.5 A well is dug 20 ft deep and the mud which came out is used to build a wall of width 1 ft around the well on the ground. If the height of the wall around the well is 5 ft, then what is the radius of the well?
(a) $\frac{\sqrt{5}+1}{4}$ (b) $\sqrt{5}$ (c) 1 (d) $\frac{1}{4}$

Q.6 The foot of a 15-m long ladder is 9 m from a house and its top reaches the upper part of a circular window. When the foot of the ladder is drawn to a distance of 3 m away from the house, the top reaches the lower edge of the window. The area of the window is:
(a) $5\frac{2}{3}$ (b) 6 m^2 (c) $7\frac{1}{14}$ (d) $8\frac{2}{3}$

Q.7 A brick wall is to be built 20 m long, 3 m high, and 40 cm thick. It has a door 3 m by 2 m. suppose each brick is to be 15 cm long, $7\frac{1}{2}$ cm broad, and 5 cm thick, how much will be the cost of the bricks at the rate of ₹800 per thousand bricks?
(a) ₹55,270 (b) ₹66,230
(c) ₹30,720 (d) ₹15,750

Q.8 A rectangular field is of the dimension 15.4 m × 12.1 m. A circular well of 0.7 m radius and 3 m depth is dug in the field. The mud, dug out from the well, is spread in the field. By how much would the level of the field rise?
(a) 1 cm (b) 2.5 cm (c) 3.5 cm (d) 4 cm

Q.9 The dimension of a room is thrice as long as it is height, but only twice as long as it is wide. The total cost of painting its walls at the rate of ₹2.50 m^2 is ₹360. What is the cost of laying a carpet on its floor at the rate of ₹3/m^2?
(a) ₹81 (b) ₹125 (c) ₹216 (d) ₹260

Q.10 A man, by walking diametrically across a circular grass plot, finds that it has taken 45 s less than if he had kept to the path around the outside. If he walks at the rate of 80 m per minute, the diameter of the grass plot is:
(a) 35 m (b) 65 m (c) 105 m (d) 145 m

Q.11 Within a rectangular courtyard of length 60 feet, a gravelled path, 3 feet wide, is laid down along all the four sides. The cost of gravelling the path is ₹2 per square feet. If the path had been twice as wide, the gravel would have cost ₹984 more. The width of the courtyard is:
(a) 24 feet (b) 40 feet
(c) 45 feet (d) 54 feet

Q.12 A vessel 2 m long, 1 m wide, and 1.5 m deep contains 2 m^3 water. How many bricks of 20 cm by 10 cm by 7.5 cm can be put in it so that water does not overflow provided that a brick is supposed to absorb 1/7 of its own volume of water?
(a) 666 (b) 111
(c) 555 (d) None of these

Q.13 In Snehans apartment, there is a playground of the dimension $7\frac{1}{2}$ by 7 m. In the centre of the ground, there is a flower-bed cut out of the dimension 3 m by $2\frac{1}{2}$ m. What fraction of the whole ground is occupied by the flower-bed?
(a) $\frac{3}{7}$ (b) $\frac{3}{5}$ (c) $\frac{1}{7}$ (d) $\frac{1}{11}$

Q.14 The height of a room is $\frac{1}{5}$ th of the sum of its length and breadth. The cost of preparing its wall at ₹4 per m^2 is ₹640. What is the height of the room?
(a) 4 m (b) 5 m (c) 6 m (d) 7 m

Q.15 A cylindrical container of 32 cm height and 18 cm radius is filled with sand. Now, all this sand is used to form a conical heap of sand. If the height of the conical heap is 24 cm, what is the radius of its base?
(a) 12 cm (b) 24 cm (c) 36 cm (d) 48 cm

Q.16 The total area of the four walls of a room is 150 m^2. If the area of the floor is 50 m^2 and the width of the floor is 3 m, then find the height of the room.
(a) 2.6 m (b) 3.8 m (c) 5.42 m (d) 7.32 m

Q.17 Anoop has a wooden box, each sides of which is an integer (in cm) and whose volume (neglecting the thickness of the box) is 216 cm^3. What is the minimum possible amount that Anoop will need to spend in order to get the outside of the box painted at the rate of ₹2 per cm^2?
(a) ₹320 (b) ₹225 (c) ₹236 (d) ₹432

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Direction for Questions 18 and 19: Read the passage below and solve the questions based on it.

There is a rectangular table of the dimension 2.20 m × 1.4 m. A rectangular tablecloth of the dimension 2.4 m × 0.8 m is spread unfolded on the table so that the longer sides of the cloth and the table are parallel.

Q.18 If the larger side of the tablecloth and the smaller side of the table are parallel, then what fraction of the area of the top of the table remains uncovered by the cloth?
(a) 3/7 (b) 4/9
(c) 7/11 (d) None of these

Q.19 In the previous question, what fraction of the cloth hangs perpendicularly over the edge of the table?
(a) 5/9 (b) 2/3 (c) 7/13 (d) 5/12

Q.20 A solid consists of a right circular cylinder; with the exact fitting, a right circular cone is placed on its top. The height of the cone is h. If the ratio of the total volume of the solid and volume of the cone is 3:1, then the height of the circular cylinder is:
(a) $h/2$ (b) $3h$
(c) $h/3$ (d) None of these

Direction for Questions 21 and 22: Read the passage below and solve the questions based on it.

There is an equilateral triangle of side 'a' units. Now, we join any two sides of the triangle to form a cone.

Q.21 What is the slant height of the cone formed?
(a) $\frac{a}{2\pi}$ (b) a (c) $\frac{a}{2}$ (d) $2\pi a$

Q.22 What is the radius of the cone formed?
(a) a (b) $\frac{a}{4}$
(c) $\frac{a}{\pi}$ (d) None of these

Q.23 The area of three adjacent faces of a cuboidal box are p, q, and r square cm. What is the volume of this box?
(a) $(p+q+r)$ cm^3 (b) $\sqrt{pqr}$ cm^3
(c) $\sqrt[3]{pqr}$ cm^2 (d) pqr cm^3

Q.24 Three equal cubes of unit side length are placed adjacent to each other in a row. Find the ratio of the total surface area of the new cuboid to that of the sum of the surface areas of all the three cubes.
(a) 3:5 (b) 4:5 (c) 6:7 (d) 7:9

Q.25 A classroom is to be built to accommodate 70 students. It should be done in such a way that for every student, there is 2.2 m^2 of floor and 11 m^3 of space. If the length of the room is 14 m, then find the breadth and height of the room.
(a) 12 m, 5.5 m (b) 11 m, 5 m
(c) 13 m, 6 m (d) 11 m, 4 m

Q.26 Length, breadth, and height of a rectangular parallelopiped are in the ratio of 6:5:4 and its total surface area is 33,300 m^2. Find length, breadth, and height of the parallelopiped (in cm).
(a) 42, 35, 28 (b) 60, 50, 40
(c) 102, 85, 68 (d) 90, 75, 60

Q.27 A rectangular tank is of dimension 30 m × 20 m. Water is being flown into it through a square pipe of side length 5 cm. Find the speed of the water if the level of water in the tank rises by 1 m in 8 h?
(a) 30 km/h (b) 36 km/h
(c) km/h (d) None of these

Q.28 A cube of side length 3 cm weighs 12 kg. What is the weight of the similar cube of same material whose side length is 12 cm?
(a) 768 kg (b) 678 kg (c) 964 kg (d) 864 kg

Q.29 A cube of side length 4 cm is cut into cubes of side 1 cm. Find the ratio of the sum of the surface area of all the small cubes to that of the large cube.
(a) 1:16 (b) 2:3 (c) 4:1 (d) 6:1

Q.30 A cylindrical well of depth 12 m with internal radius of 1.75 m is dug up. The mud, so obtained, is spread evenly to form a platform of dimension 10.5 m × 8.8 m. What is the height of the platform?
(a) 2.25 m (b) 3.25 m (c) 1.25 m (d) 4.25 m

Q.31 A hollow spherical ball of outer diameter 120 mm is cut into two equal hemispheres. One of the hemispheres is filled with honey and the total surface area of the other hemisphere is $389\frac{5}{7}$ cm^2. What is the volume of the honey put in the first hemisphere?
(a) 128 cm^3 (b) 134 cm^3
(c) 136 cm^3 (d) 138 cm^3

Q.32 The diameter of a cylindrical vessel is made twice as large. How should the height be changed so that it still contains the same quantity of water?
(a) $\frac{1}{2}$ × Height of the two cylinders
(b) $\frac{1}{4}$ × Height of the original cylinder
(c) Let it be the same
(d) None of these

Q.33 Two shopkeepers are offering Pepsi in two different types of cans. The first shopkeeper offers cylindrical cans with a circular base of radius 5 cm and the second shopkeeper offers the cans having a square base of length 10 cm. If height of both the cans is the same, then what is the difference in their capacities?

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(a) 350 cm^3 (b) 450 cm^3
(c) 250 cm^3 (d) Cannot be determined

Q.34 A bucket is a standard example of frustum of a cone. The circumference of such a bucket of one end is 48 cm and of the other end is 34 cm. If the height of the bucket is 10 cm, find its volume (in cm^3).
(a) 5400 (b) 1350 (c) 2700 (d) 4050

Q.35 The sum of the length, width, and depth of a cuboid is s and its diagonal is d. What is its surface area?
(a) s^2 (b) d^2 (c) $s^2 - d^2$ (d) $s^2 + d^2$

Q.36 A solid right circular cylinder with height 10 cm and radius of the base 6 cm is given. A right circular cone of the same height and base is removed from this cylinder. Find the volume (in cm^3) of the remaining solid.
(a) 855.5 (b) 754.3
(c) 1294 (d) None of these

Q.37 At the Amul co-operative society, right cylindrical vessels were used to store milk. But, after the uncere-monious exit of Mr Durien, it was decided that instead of right cylindrical vessels, right cones having the same diameter and height would be used. What would be the ratio of the number of the right cones to the number of right cylinders needed to hold the same quantity of milk?
(a) 2:1 (b) 3:1 (c) 4:1 (d) 5:1

Q.38 The radius of a right cylinder is doubled and its height is halved. What is the ratio between the new curved surface area and the previous curved surface area of the cylinder?
(a) 1:1 (b) 2:1 (c) 3:3 d) 2:3

Q.39 A solid cone is converted into a solid cylinder of the same radius. If the height of the cylinder is 5 m, what is the height of the cone?
(a) 25 m (b) 15 m (c) 20 m (d) 10 m

Q.40 The radii of a cylinder and a cone are equal and the height of the cylinder is equal to the slant height of the cone. Find the ratio of the curved surfaces of the cylinder and the cone.
(a) 1:1 (b) 2:1 (c) 3:1 (d) 4:1

Q.41 A right circular cone is placed inside a cube such that the edges of the base of the cone are touching the edges of one of the faces of the cube and the vertex is touching the opposite face of the cube. If the volume of the cube is 343 cm^3, what is the volume of the cone (approximately in cm^3)?
(a) 90 (b) 60 (c) 80 (d) 85

Q.42 When Natraj circus came to Patna for its show at Sonepur mela last year, I observed quite a few things. Its tent was cylindrical to a height of 3 m and conical above it and its diameter was 105 m and slant height of the conical portion was 53 m. But, I was not able to calculate the length of the canvas 5 m wide to make the tent. Can you please bail me out by telling this?
(a) 1857 m (b) 1647 m (c) 1947 m (d) 1847 m

Q.43 A cone and a cylinder have their height in the ratio of 3:2 and the radii of their bases are in the ratio of 4:3. Find the ratio of their volumes.
(a) 9:1 (b) 9:2 (c) 8:9 (d) 3:1

Q.44 A cylindrical structure standing on its base with radius 1.5 m and height 5 m is cut with a saw in such a way that the cutting planes goes through all the points at a distance of 0.625 m from the base. Find the volume of the remaining piece.
(a) 5.62π (b) 9.24π
(c) 9.04π (d) None of these

Q.45 The height of a right circular cylinder is 6 m. Three times the sum of the areas of its two circular faces is twice the area of its curved surface. The radius of the base is:
(a) 4 m (b) 2 m (c) 6 m (d) 1.5 m

Q.46 A solid cone kept on its base is cut at 2/3rd of its height along a plane parallel to its circular base. The base radius and the slant height are 14 cm and 50 cm, respectively. What is the ratio of the portion cut-out from the solid to the volume of the remaining solid?
(a) 1:20 (b) 1:25
(c) 1:36 (b) None of these

Q.47 A big cube of side 6 cm is formed by putting together 216 small identical cubes each of side 1 cm. Now, if the corner cubes in the topmost layer of the big cube are removed, then what will be the impact on the total surface area of the big cube?
(a) Will decrease
(b) Will increase
(c) Will remain unchanged
(d) Cannot be determined

Q.48 Mr Sinha and Mr Verma are two brothers. They inherited their paternal land in a rectangular form having the same area but different dimensions. For Mr Sinha, the length of the rectangular plot is increased by 10% and the breadth is decreased by 10%. For Mr Verma's rectangular plot, the length is decreased by 10% and the breadth is increased by 10%. Which of the following is true about the areas of the two rectangular plots now?
(a) The area of the first rectangle is 1% more than that of the second.
(b) The area of the second rectangle is 1% more than that of the first.
(c) The area of the two rectangles is the same.
(d) We cannot comment unless we know the dimensions of the individual rectangles.

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Q.49 A big solid sphere of diameter 10 cm is melted and made into solid spheres of diameters 2 cm. What is the percentage increase/decrease in the surface area of smaller spheres over that of the big sphere?
(a) 200% increase (b) 400% increase
(c) 400% decrease (d) None of these

Q.50 A cone is made of a sector of a circle of radius 21 cm and an angle of 90°. What is the total surface area of the cone (in cm^2)?
(a) 269.5 (b) 308
(c) 312.5 (d) 231

MODERATE

Q.1 There is a right circular cone with base radius 3 units and height 4 units. The surface of this right circular cone is painted. It is then cut into two parts by a plane parallel to the base so that the volume of the top part (the small cone) divided by the volume of the frustum equals the painted area of the top part divided by the painted area of the bottom part. The height of the small cone is:
(a) 7/3 (b) 5/4
(c) 5/2 (d) None of these

Direction for Questions 2 and 3: Read the passage below and solve the questions based on it.

My grandfather owned plenty of land, which easily encompassed an area of 1000×200 m^2. He wanted to give some part of it to his servant Ramu. But, he did not gift it directly. He supplied the material that could form a fence of length 100 m only. Then, he allowed Ramu to take any part with four sides that could be encased with the help of the given fencing material.

Q.2 What is the maximum possible land Ramu can take away from my grandfather? (All the four sides are fenced.)
(a) 100 m^2 (b) 2500 m^2 (c) 750 m^2 (d) 625 m^2

Q.3 Ramu had intimate knowledge of the land. Hence, he selected the site having a natural fencing of rocks on one side, because he could utilize the given material only on three sides of the plot. What is the maximum possible land that Ramu can claim now?
(a) 125 m^2 (b) 2500 m^2
(c) 625 m^2 (d) None of these

Q.4 The diameter of a road-roller is 42 cm and its length is 100 cm. It takes 400 complete revolutions moving once over to level the stretch of the road. If the cost of levelling is ₹100 per m^2, then the total cost of levelling works out to:
(a) ₹52,800 (b) ₹5280
(c) ₹5,28,000 (d) ₹528

Q.5 Inside a triangular garden, there is a flower-bed in the form of a similar triangle. Around the flower-bed runs a uniform path of such a width that the sides of the garden are double of the corresponding sides of the flower-bed. The areas of the path and the flower-bed are in the ratio
(a) 1:1 (b) 4:1 (c) 1:3 (d) 3:1

Q.6 ABCD is a square of side 10 cm. What is the area of the least-sized square that may be inscribed in ABCD with its vertices on the sides of ABCD?
(a) 0 cm^2 (b) 25 cm^2
(c) 50 cm^2 (d) 66.66 cm^2

Q.7 A cylinder of the maximum possible size is made out of a solid wooden cube. How much material is lost in this process (approximately)?
(a) 20% (b) 22%
(c) 24% (d) Cannot be determined

Q.8 Two friends, ND and SD, invited their friends for a party. A hemispherical bowl full of ice-cream of diameter 18 cm was served for dessert. Each of them had three scoops of the ice-cream and it was exactly sufficient for all of them. If the ice-cream was served in a hemispherical scoop of radius 1 cm, how many friends did they have for their party?
(a) 240 (b) 241 (c) 242 (d) 243

Q.9 The cost of white washing one m^2 is ₹50. What will be the maximum amount saved in painting the room in the most economical way, if the sum of the length, breadth, and height is 21 m and all the sides are integers (floor is not to be white washed)?
(a) ₹1,08,050 (b) ₹8400
(c) ₹9300 (d) ₹8540

Q.10 A small bird is taking rest at the centre of the base of a hemispherical cage. Suddenly it stands up, flies to the topmost point in the cage, then in a straight line to the cage door at the intersection of the curved surface and the base. In the process, it covers a total distance of 482 cm. What is the radius of the hemisphere?
(a) 100 cm (b) 150 cm
(c) 200 cm (d) 50 cm

Q.11 A cube is inscribed in a hemisphere of radius R, such that four of it vertices lie on the base of the hemisphere and the other four touch the hemispherical surface of the half-sphere. What is the volume of the cube?
(a) $0.25R^3$ (b) $0.67\sqrt{\frac{2}{3}}R^3$
(c) $0.5\sqrt{\frac{2}{3}}R^3$ (d) $0.67R^3$

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Q.12 The biggest possible cube is taken out of a right solid cylinder of radius 15 cm and height 20 cm, respectively. What will be the volume (in cm^3) of the cube?
(a) 375 (b) 800
(c) $375\sqrt{2}$ (d) None of these

Direction for Questions 13 and 14: Read the passage below and solve the questions based on it.

A larger cube is made up of $N = n^3$ smaller cubes.
Let A = number of smaller cubes with no exposed surfaces.
B = number of smaller cubes with one exposed surfaces.
C = number of smaller cubes with two exposed surfaces.
D = number of smaller cubes with three exposed surfaces.

Q.13 What is the ratio of the number of unexposed smaller cubes to the total number of cubes?
(a) $\frac{1}{27}$ (b) $\frac{(n-2)}{n}$ (c) $\left(\frac{n-2}{n}\right)^3$ (d) $\frac{n}{(n-2)}$

Q.14 What is the total number of smaller cubes in a bigger cube for which A = B?
(a) 512 (b) 256 (c) 1000 (d) 729

Q.15 A ball is fitted into the mouth of an open cone whose base radius is equal to that of the ball. How far can the ball definitely go inside the cone?
(a) Half the ball will fit into the cone.
(b) Less than half the ball will fit into the cone.
(c) More than half the ball will fit into the cone.
(d) One-fourth of the ball will fit into the cone.

Q.16 A well, 4 m in diameter and 35 m deep, is being dug out and the excavated soil is transported away in a truck trolley of size 5 m × 2 m × 0.5 m. Since the soil that is taken out is loose it occupies 20% more space. How many trips will the truck have to make to clear the excavated soil, if the truck can be filled only up to 80% of its height?
(a) 106 (b) 132 (c) 425 (d) 528

Q.17 The product of all the 12 edges of a rectangular box is equal to:
i. The fourth power of the volume of the box.
ii. The product of the areas of the six faces of the box.
iii. Sum of the squares of the areas of the six faces of the box.
(a) None of the above three values.
(b) The figure given by exactly one of the three statements.
(c) The figure given by exactly two of the three statements.
(d) The figure given by all the three statements.

Q.18 A right circular cone is given with a height 6 cm and its slant side is making an angle of 60° with its base. Now, a sphere of the maximum volume is placed inside this cone. If the radius of this sphere is 2.33 cm, then find the volume of the cone.
(a) 18π (b) 15π
(c) 24π (d) Cannot be determined

Q.19 A solid sphere of radius 7 cm is melted to form a number of small cones and cylinders. The requirement is such that the number of cones should be twice the number of cylinders. Also, the radius of the cone must be equal to its height, which should be equal to the radius of the cylinder and also should be half the height of the cylinder. If the height of one such cylinder is 4 cm, find the maximum number of cones which can be made out of the sphere.
(a) 18 (b) 21 (c) 36 (d) 42

Q.20 Let A and B be two solid spheres such that the surface area of B is 300% greater than the surface area of A. The volume of A is found to be K% lower than the volume of B. Then, the value of K must be
(a) 85.5 (b) 92.5 (c) 90.5 (d) 87.5

Q.21 A large cube is formed from the material obtained by melting three smaller cubes of 3, 4, and 5 cm side. What is the ratio of the total surface areas of the smaller cubes and the large cube?
(a) 2:1 (b) 3:2 (c) 25:18 (d) 27:20

Q.22 The volume of a cube is V. Then, the total length of its edges is:
(a) $6V^3$ (b) $8\sqrt{V}$ (c) $12V^{2/3}$ (d) $12V^{1/3}$

Q.23 The height of a cone is 30 cm. A small cone is cut off at the top by a plane parallel to the base. The volume of this smaller cone is 1/27 of the given cone. What is the height of the smaller cone?
(a) 13.5 cm (b) 11 cm (c) 10 cm (d) 12 cm

Q.24 The maximum distance between two points of the unit cube is:
(a) $\sqrt{2}+1$ (b) $\sqrt{2}$
(c) $\sqrt{3}$ (d) $\sqrt{2}+\sqrt{3}$

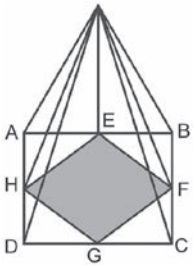
Q.25 A sealed cylindrical drum of radius r is 9% filled with paint. If the drum is tilted to rest on its side, find the fraction of its curved surface area (not counting the flat sides) that will be under the paint.
(a) Less than $\frac{1}{12}$ (b) Between $\frac{1}{12}$ and $\frac{1}{6}$
(c) Between $\frac{1}{6}$ and $\frac{1}{4}$ (d) Greater than $\frac{1}{4}$

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Direction for Questions 26 and 27: Read the passage below and solve the questions based on it.

In the figure given below, the base of the pyramid is a square ABCD of side 10 cm; another pyramid with base EFGH is formed where the points E, F, G, and H are obtained by joining the mid-points of the sides of the square ABCD. The height of the pyramid is 30 cm.



Q.26 Find the ratio of the volumes of pyramid with ABCD as base to the pyramid with EFGH as base.

(a) $\sqrt{2}:1$ (b) $2\sqrt{2}:1$
(c) 4:1 (d) 2:1

Q.27 Find the volume of the space left.

(a) 1500 cm^3 (b) 500 cm^3
(c) 1000 cm^3 (d) 800 cm^3

Q.28 All five faces of a regular pyramid with a square base are found to be of the same area. The height of the pyramid is 3 cm. Find the total area of all its surfaces (in cm^2).

(a) 8 (b) 10 (c) 12 (d) 16

Direction for Questions 29 and 30: Read the passage below and solve the questions based on it.

A right-angled triangle of sides 5 m, 12 m, and 13 m is made to spin about its hypotenuse.

Q.29 What kind of figure will be formed with this?

(a) Square (b) Double cone
(c) Sphere (d) None of these

Q.30 What is the volume of the figure thus formed?

(a) 270 m^3 (b) 290 m^3
(c) 320 m^3 (d) None of these

Q.31 A solid cube is cut into two halves by a plane passing through exactly two corners of the cube. What is the ratio of the total surface area of both the halves put together and the original total surface area of the cube?

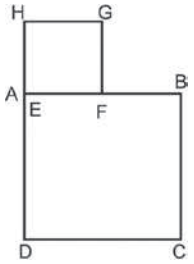
(a) $\sqrt{3}:\sqrt{3}+\sqrt{2}$ (b) $\sqrt{6}+1:\sqrt{6}$
(c) $6:6+\sqrt{6}$ (d) $3+\sqrt{2}:3$

Q.32 Sanjay wound a piece of string over a cylindrical block around its curved surface and found that the string could be wound for exactly half a round more than an integral number of rounds. He also figured that if the string had been five times as long, the string would have had wound for exactly 10 cm more than an integral number of the rounds. Find the radius of the cylindrical block. (Assume that all windings are done in the same plane and neglect any increase in the radius due to the winding of the string.)

(a) $2\frac{6}{22}$ cm (b) $3\frac{11}{22}$ cm
(c) $3\frac{4}{22}$ cm (d) $3\frac{9}{22}$ cm

Direction for Questions 33 and 34: Go through the figure given below and solve the questions based on it.

Consider the figure given below:



A cube with surface EFGH is put on a cube with a surface ABCD in such a way that vertices E and A of the two blocks coincide and side EF coincides with side AB. EF = 1 unit and AB = 2 units. Now, keeping vertex F fixed, the smaller cube is rotated along the larger cube till G coincides with B once. Then, keeping G fixed, the smaller block is again rotated till GH coincides with BC.

Q.33 How much is the aerial distance traversed by point E?

(a) $\sqrt{2\pi}$ units (b) π units
(c) $\sqrt{2\pi}+\frac{1}{2}\pi$ units (d) None of these

Q.34 In the above question, what is the total distance covered by point E?

(a) π units (b) $(1+\pi)$ units
(c) $(1+\sqrt{\pi})$ units (d) None of these

Q.35 How many cuboids of different dimensions can be made assembling 100 identical cubes?

(a) 9 (b) 8 (c) 12 (d) 10

Q.36 A cuboid is of the dimension $2\times2\times3$ units. Of the six surfaces of this cuboid, there are exactly x squares and exactly y rectangles (rectangles which are not squares.) Find the product of $x \times y$.

(a) 8 (b) 12
(c) 6 (d) None of these

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Q.37 A box without a lid has a height of x cm, a square base of side y cm, and a volume of 500 cm^3. For what value of x (in cm), the least quantity of the material will be needed to construct the box?
(a) $\sqrt{5}$ (b) $2\sqrt{5}$ (c) 5 (d) $5\sqrt{2}$

Q.38 Under the Indian Posts and Telegraph Act 1885, any package in the form of a right circular cylinder will not be accepted if the sum of its height and the diameter of its base exceeds 10 inches. The height (in inches) of a package of maximum volume that would be accepted is:
(a) 10/3 (b) 20/3 (c) 10 (d) 20

Q.39 A convex pentagon IJKLM is inscribed in a circle. If the angles subtended by the sides IJ, JK, KL, and LM at the centre are 50°, 80°, 60°, and 60°, respectively, what is the measurement of the smallest interior angle of the pentagon?
(a) 120° (b) 95°
(c) 85° (d) None of these

Q.40 A right circular cone, with radius to height ratio as 12:5, is cut parallel to its base to get a smaller cone and a frustum. If the height of the smaller cone to the height of the frustum are in the ratio of 3:1, by what percentage is the combined total surface area of the smaller cone and frustum will be more with respect to the original cone?
(a) 22% (b) 32%
(c) 46% (d) None of these

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Q.1 A cuboid of length 20 m, breadth 15 m, and height 12 m is lying on a table. The cuboid is cut into two equal halves by a plane which is perpendicular to the base and passes through a pair of diagonally opposite points of that surface. Then, a second cut is made by a plane which is parallel to the surface of the table again dividing the cuboid into two equal halves. Now, this cuboid is divided into four pieces. Out of these four pieces, one piece is now removed from its place. What is the total surface area of the remaining portion of the cuboid?
(a) 1290 m^2 (b) 1380 m^2
(c) 1440 m^2 (d) Cannot be determined

Q.2 A metal is made up of a cylindrical base and a conical top with the base of radius 5 cm. The ratio of height of the cone and the cylinder is 2:3. A cylindrical hole is drilled through the metal solid with height two-third the height of the metal solid. What should be the radius of the hole, so that the volume of the hole is 1/3 the volume of the metal solid after drilling?
(a) $\sqrt{45}$ cm (b) $\sqrt{\frac{55}{8}}$ cm
(c) $\sqrt{35}$ cm (d) $\sqrt{65}$ cm

Q.3 Sixteen cylindrical cans of Coke, each with a radius of 1 cm, are placed inside a wooden carton, four in a row. If the cans touch the adjacent cans and/or the walls of the box, then which of the following could be the internal area of the bottom of the carton (in cm^2)?
(a) 16 (b) 32
(c) 64 (d) None of these

Q.4 The areas of three adjacent faces of a cuboid are a, b, and c. If the volume of the cuboid is N, then N^2 is equal to:
(a) abc (b) $(ab + bc + ac)$
(c) $1/c$ (d) None of these

Q.5 A square tin sheet of side 12 inches is converted into a box with open top in the following steps: The sheet is placed horizontally. Then, equal-sized squares, each of side x inches, are cut from the four corners of the sheet. Finally, the four resulting sides are bent vertically upwards in the shape of a box. If x is an integer, then what value of x maximizes the volume of the box?
(a) 3 (b) 4 (c) 1 (d) 2

Q.6 A rectangular piece of cardboard 18 cm × 24 cm is made into an open box by cutting a square of 5 cm side from each corner and building up the side. What is the volume of the box (in cm^3)?
(a) 560 (b) 432
(c) 216 (d) None of these

Q.7 Consider the following function:
$E = f(A) = \sqrt{\frac{A}{7}}$ where length of the edge E of a Pyramid and the surface area A of the pyramid are the different terms used. How much longer is the edge of the pyramid with a surface area 3087 square units than the edge of one with a surface area 2023 square units?
(a) 1064 (b) 152
(c) 4 (d) None of these

Q.8 The ratio of the volumes of the two cylinders are a:b, and their heights are in the ratio c:d. What is the ratio of their diameters?
(a) $\frac{ad}{bc}$ (b) $\frac{d^2}{c^2}$
(c) $\sqrt{\frac{ad}{bc}}$ (d) $\sqrt{\frac{a}{b}} \times \frac{c}{d}$

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Q.9 From a solid right circular cone made of iron with base of radius 2 cm and height 5 cm, a hemisphere of diameter 2 cm and its centre, coinciding with the centre of the base of the cone, is cut out. The structure so obtained is then dropped in a right circular cylinder whose inner radius is 3 cm and inner height is 10 cm. Water is then poured into the cylinder to fill it up to its brim. What is the volume of the water required to fill it?

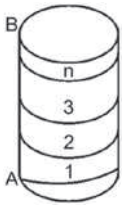
(a) 84π cm^3 (b) $\frac{250\pi}{3}$ cm^3

(c) $\frac{270\pi}{4}$ cm^3 (d) None of these

Direction for Questions 10 to 12: Read the passage below and solve the questions based on it.

Consider a cylinder of height h cm and radius $r = \frac{2}{\pi}$ cm as shown in the figure (not drawn to scale). A string of a certain length, when wound on its cylindrical surface, starting at a point A and ending at point B, gives a maximum of n turns (in other words, the string length is the minimum length to wind n turns.)

Q.10 What is the vertical spacing in cm between consecutive turns?



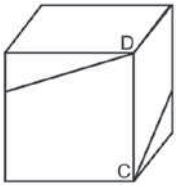
(a) $\frac{h}{n}$

(b) $\frac{h}{\sqrt{n}}$

(c) $\frac{h}{n^2}$

(d) Cannot be determined with the given information.

Q.11 The same string, when wound on the exterior four walls of a cube of side n cm, starting at point C and ending at point D, can give exactly one turn (see figure, not drawn to scale). The length of the string in cm is:



(a) $\sqrt{2}n$ (b) $\sqrt{17}n$

(c) n (d) $\sqrt{13}n$

Q.12 In the setup of the previous two questions, how is h related to n?

(a) $h = \sqrt{2}n$ (b) $h = \sqrt{17}n$

(c) $h = n$ (d) $h = \sqrt{13}n$

Q.13. An octagonal prism, of height 20 cm and a side of the base as $\sqrt{4\left(2-\sqrt{2}\right)}$ cm, is cut from its base to the top along the edges into two similar parts of equal volume. What is the total surface area of each part?

(a) $80 + 160\sqrt{2-\sqrt{2}} + 4\sqrt{2}$ cm^2

(b) $80 + 40\sqrt{2-\sqrt{2}}$ cm^2

(c) $80 + 160\sqrt{2-\sqrt{2}} + 8\sqrt{2}$ cm^2

(d) None of these

Direction for Questions 14 to 16: Read the passage below and solve the questions based on it.

There are 300 coins, each coin having radius 2 cm and height 1 cm. The coins are so kept that each coin touches the other two. The base has three coins and the figure is built upon this base.

Q.14 Find the volume of the region enclosed by the coins.

(a) $600\left(2\sqrt{3}-\pi\right)$cm^3 (b) $400\left(2\sqrt{3}-\pi\right)$cm^3

(c) $100\left(2\sqrt{3}-\pi\right)$cm^3 (d) $200\left(2\sqrt{3}-\pi\right)$cm^3

Q.15 Find the number of spheres of the maximum volume that can be accommodated in the above region.

(a) 324 (b) 323 (c) 162 (d) 161

Q.16 Find the volume of the prism circumscribing the whole structure of the coins.

(a) $100\sqrt{3\left(7+4\sqrt{3}\right)}$cm^3

(b) $2400\sqrt{3\left(2+\sqrt{3}\right)}$cm^3

(c) $800\sqrt{3\left(2+\sqrt{3}\right)}$cm^3

(d) None of these

Q.17. A cylinder of height 'h' cm and radius 6π cm is wound around with a string of width 'x' cm. The string covers the lateral surface area of the cylinder completely without keeping any space between two turns. What is the required length of the string.

(a) $\frac{12h}{x}$ cm (b) $\frac{12x}{h}$ cm

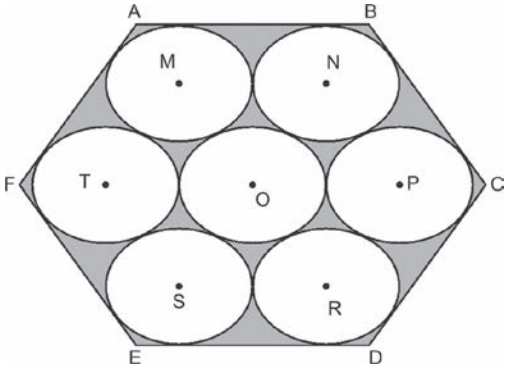
(c) 12 h cm (d) $\frac{6h}{x}$ cm

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Direction for Questions 18 to 20: Read the passage below and solve the questions based on it.

In the figure given below, a regular hexagon ABCDEF is inscribed by seven circles. The radius of each circle is equal to 6 cm. The centres of the circles touching the sides are M, N, P, R, S, and T, whereas the centre of the central circle is O.



Q.18 Find the ratio of the perimeter of the hexagon circumscribing the circle with centre O, the perimeter of hexagon formed by joining the points M, N, P, R, S, and T and the perimeter of the hexagon ABCDEF.
(a) $6:12:(12+2\sqrt{3})$
(b) $2:2\sqrt{3}:(2\sqrt{3}+1)$
(c) $1:\sqrt{3}:(\sqrt{3}+1)$
(d) 1:2:3

Q.19 Find the area of the shaded region (in cm^2).
(a) 136.83 (b) 129.63
(c) 139.83 (d) 138.83

Q.20 If the above figure is a 3-D figure and seven spherical balls of radius 6 cm each are tightly arranged in a hexagonal box in a single layer, then what will be the volume of the box unoccupied by the balls in cm^3?
(a) $3456\sqrt{3}+5184-252\pi$
(b) $3456\sqrt{3}+5184-288\pi$
(c) $576\sqrt{3}+864-288\pi$
(d) None of these

Q.21 Four spheres each of radius 10 cm lie on a horizontal table so that the centres of the spheres form a square of side 20 cm. A fifth sphere also of radius 10 cm is placed on them so that it touches each of these spheres without disturbing them. How many cm above the table is the centre of the fifth sphere?
(a) $10\sqrt{6}$ (b) $10(1+\sqrt{2})$
(c) $10(1+\sqrt{3})$ (d) $10(4-\sqrt{2})$

Q.22 A cube with an edge 'a' is cut from the angles by planes so that a regular octagon remains from each face, then the volume of the polygon so obtained will be:
(a) $a^3(7\sqrt{2}-9)$ (b) $\frac{7}{3}a^3(\sqrt{2}-1)$
(c) $7a^3(\sqrt{2}-1)$ (d) $\frac{a^3}{3}(7\sqrt{2}-9)$

Q.23 A square of side x cm is cut from each corner of a rectangular sheet of metal of 10 cm by 14 cm. The resulting projections are folded up and the seams welded to construct an open box. What is the volume of the box thus obtained?
(a) $4x^3-48x^2+140x$
(b) $4x^3+48x^2+140x$
(c) x^3+24x^2+140x
(d) None of these

Q.24 A sphere of radius 13 cm is cut by a plane whose distance from the centre of the sphere is 5 cm. What is the circumference of the plane circular section?
(a) 10π cm (b) 12π cm
(c) 24π cm (d) 26π cm

Q.25 The square of side 1 cm are cut from four corners of a sheet of tin (having length = l and breadth = b) in order to form an open box. If the whole sheet of tin was rolled along its length to form a cylinder, then the volume of the cylinder is equal to (343/4) cm^3. Find the volume of the box. (l and b are integers.)
(a) 154 cm^3 (b) 100 cm^3
(c) 126 cm^3 (c) Insufficient data

Q.26 Two spheres of radii 6 cm and 1 cm are inscribed in a right circular cone. The bigger sphere touches the smaller one and also the base of the cone. What is the height of the cone?
(a) 14 cm (b) 12 cm
(c) 85/6 cm (d) 72/5 cm

Q.27 If the total length of the diagonals of a cube is 12 cm, then what is the total length of the edges of the cube?
(a) $6\sqrt{3}$ cm (b) 12 cm
(c) $12\sqrt{3}$ cm (d) 15 cm

Q.28 A cylindrical vessel open at the top contains water up to 1/3rd of its height. A heavy sphere whose diameter is equal to the height of the cylinder is placed into the vessel touching its curved surface from all sides, then the water
(a) level will rise to half of its height.
(b) level will rise to 3/4th its height.
(c) level will rise to its height.
(d) will overflow.

Q.29 A sphere of maximum possible volume is to be completely immersed into a cylindrical container of radius 'a' containing water up to a height '$2a$'. What is the minimum height of the jar so that no water spills out of it?

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(a) $\frac{10a}{3}$ (b) $\frac{11a}{3}$

(c) $\frac{12a}{3}$ (d) $\frac{13a}{3}$

Q.30 In a rectangular parallelopiped A, a, b, and c are the lengths of the diagonals on three faces having a corner in common. Let d be the length of the largest diagonal that can be drawn in A, then:

(a) $2d^2 = a^2 + b^2 + c^2$
(b) $d^2 = a^2 + b^2 + c^2$
(c) $d^2 = 2(a^2 + b^2 + c^2)$
(d) None of these

Direction for Questions 31 to 33: Read the passage below and solve the questions based on it.

A right-angled triangle with base and height measuring 15 cm and 20 cm is rotated along its hypotenuse. The structure so formed is placed by two spheres such that they are just touching the sides of the cone.

Q.31 Find the volume of the structure formed before the double cone is placed (in cm^3)?

(a) 1200π (b) 3600π
(c) 1000π (d) None of these

Q.32 Find the sum of the volumes of both the sphere (in cm^3).

(a) $\frac{364}{3}\pi$ (b) $\frac{140}{3}\pi$
(c) 12π (d) 47π

Q.33 Find the curved surface area of the figure (in sq. units).

(a) 840π (b) 400π
(c) 420π (d) None of these

Q.34 A square hole of cross-sectional area 4 cm^2 is drilled across a cube with its length parallel to a side of the cube. If an edge of the cube measures 5 cm, what is the total surface area of the body so formed?

(a) 158 cm^2 (b) 190 cm^2
(c) 166 cm^2 (d) 182 cm^2

Q.35 The radius and the height of a right solid circular cone are r and h, respectively. A conical cavity of radius $r/2$ and height $h/2$ is cut out of the cone. What is the whole surface area of the rest of the portion?

(a) $\frac{\pi r}{4}\left(5\sqrt{(r^2+h^2)}\right)+3r$ (b) $\frac{5\pi r}{4}\sqrt{r^2+h^2}$

(c) $\frac{3\pi r}{4}\left(\sqrt{r^2+h^2}+r\right)$ (d) $\frac{3\pi r}{7}\left(\sqrt{r^3+h^2}+r\right)$

Q.36 John Nash, an avid mathematician, had his room constructed such that the floor of the room was an equilateral triangle in shape instead of the usual rectangular shape. One day he brought home a bird and tied it to one end of a string and then tied the other end of the string to one of the corners of his room. The next day, he untied the other end of the string from the corner of the room and tied it to a point exactly at the centre of the floor of the room. Assuming that the dimensions of the room are relatively large compared to the length of the string, find the number of times, by which the maximum possible space in which the bird can fly, increase.

(a) 4 (b) 5 (c) 6 (d) 7

Direction for Questions 37 to 40: Read the passage below and solve the questions based on it.

Homes of three of my friends—Bango, Mango, and Tatto—are located at the three vertices of the municipal park near my house. This park is triangular in shape. Given below are the directions to reach each of their homes from my home:

- Bango's house is100 m to the east and 150 m to the north.
- Mango's house is 200 m to the east and 50 m to the north.
- Tatto's house is 50 m to the west and 25 m to the north.
- Neglect dimensions of all the house while calculating the distances.

Q.37 What is the area of the municipal park?

(a) 27,500 m^2 (b) 11,500 m^2
(c) 13,750 m^2 (d) Cannot be determined

Q.38 There is a highway (a straight road) passing through the houses of Bango and Mango. Tatto starts from his house and takes the shortest possible route to reach the highway. What is the distance travelled by Tatto?

(a) $\frac{125}{\sqrt{3}}$ (b) $\frac{275}{\sqrt{2}}$

(c) $\frac{150}{\sqrt{2}}$ (d) None of these

Q.39 Travelling along one of the following routes will take me from my house to the point (on the highway) where Tatto reached in the previous question. Which one is it?

(a) 100 m to the east, 200 m to the north
(b) 87.5 m to the east and 62.5 m to the north
(c) 67.5 m to the east and 167.5 m to the north
(d) None of these

Q.40 What is the approximate distance between Mango's house and the point Tatto reached in Q.39?

(a) $\frac{225}{\sqrt{2}}$ m (b) $\frac{275}{\sqrt{2}}$ m

(c) $\frac{150}{\sqrt{2}}$ m (d) $\frac{325}{\sqrt{2}}$ m

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Answers

Warm Up

1. (c)	**2.** (c)	**3.** (c)	**4.** (a)	**5.** (a)	**6.** (d)	**7.** (b)	**8.** (b)	**9.** (c)	**10.** (d)
11. (b)	**12.** (b)	**13.** (d)	**14.** (c)	**15.** (c)	**16.** (b)	**17.** (a)	**18.** (c)	**19.** (c)	**20.** (a)
21. (c)	**22.** (c)								

Foundation

1. (d)	**2.** (a)	**3.** (c)	**4.** (d)	**5.** (a)	**6.** (c)	**7.** (c)	**8.** (b)	**9.** (c)	**10.** (c)
11. (b)	**12.** (d)	**13.** (c)	**14.** (a)	**15.** (c)	**16.** (b)	**17.** (d)	**18.** (c)	**19.** (d)	**20.** (d)
21. (b)	**22.** (d)	**23.** (b)	**24.** (d)	**25.** (b)	**26.** (d)	**27.** (a)	**28.** (a)	**29.** (c)	**30.** (c)
31. (b)	**32.** (b)	**33.** (d)	**34.** (d)	**35.** (c)	**36.** (b)	**37.** (b)	**38.** (a)	**39.** (b)	**40.** (b)
41. (a)	**42.** (c)	**43.** (c)	**44.** (d)	**45.** (a)	**46.** (d)	**47.** (c)	**48.** (c)	**49.** (b)	**50.** (a)

Moderate

1. (c)	**2.** (d)	**3.** (d)	**4.** (a)	**5.** (d)	**6.** (c)	**7.** (b)	**8.** (b)	**9.** (c)	**10.** (c)
11. (b)	**12.** (d)	**13.** (c)	**14.** (a)	**15.** (b)	**16.** (b)	**17.** (c)	**18.** (c)	**19.** (d)	**20.** (d)
21. (c)	**22.** (d)	**23.** (c)	**24.** (c)	**25.** (a)	**26.** (d)	**27.** (a)	**28.** (c)	**29.** (b)	**30.** (d)
31. (d)	**32.** (c)	**33.** (c)	**34.** (b)	**35.** (b)	**36.** (a)	**37.** (c)	**38.** (a)	**39.** (b)	**40.** (d)

Advanced

1. (d)	**2.** (b)	**3.** (c)	**4.** (a)	**5.** (d)	**6.** (a)	**7.** (c)	**8.** (c)	**9.** (a)	**10.** (a)
11. (b)	**12.** (c)	**13.** (d)	**14.** (d)	**15.** (d)	**16.** (c)	**17.** (a)	**18.** (c)	**19.** (d)	**20.** (b)
21. (b)	**22.** (b)	**23.** (a)	**24.** (c)	**25.** (b)	**26.** (d)	**27.** (c)	**28.** (c)	**29.** (a)	**30.** (a)
31. (a)	**32.** (b)	**33.** (c)	**34.** (d)	**35.** (a)	**36.** (b)	**37.** (c)	**38.** (b)	**39.** (b)	**40.** (a)

Hints and Solutions

Warm Up

1. Let the original side of the cube = a

Then, the surface area = $6a^2$ (i)

If side is increased by 100%, then new side = $2a$

and surface area = $6(2a)^2 = 24a^2$ (ii)

Hence, percentage change in surface area

$$= \frac{\text{change}}{\text{initial value}} \times 100 = \frac{18a^2}{6a^2} \times 100 = 300\%$$

2. Total volume of the hall = $30 \times 12 \times 6$ m^3

Given that volume of a man = 8 m^3

Then, the number of people = $\frac{30 \times 12 \times 6}{8} = 270$

3. Height of the cylinder = $\frac{\text{Volume}}{\text{Area of the base}} = \frac{511}{36.5}$ m = 14 m

4. Total surface area = 924 cm^2

Then, the curved surface area

$$= \frac{924}{3} \times 2 = 308 \times 2 = 616 \text{ cm}^2$$

Area of the base = $\frac{924 - 616}{2} = 154$ cm^2

Let the radius of base = r and $\pi r^2 = 154$ cm^2

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So, $r^2 = \frac{154 \times 7}{22} = 49$ or $r = 7$ cm

Then, $2\pi r = 2 \times \frac{22}{7} \times 7 = 44$ cm

Hence, surface area $= 2\pi rh = 616$ cm^2

Or, $2 \times \frac{22}{7} \times 7 \times h = 616$ or $h = \frac{616}{44}$

Volume $= \pi r^2 h = 154 \times \frac{616}{44}$ cm^3 $= 2156$ cm^3

5. Radius = 21 cm

Then, curved surface area $= 2\pi rh = 1320$ cm^2

Or, $2 \times \frac{22}{7} \times 21 \times h = 1320$ cm^2 or $6 \times 22 \times h = 1320$

Hence, $h = \frac{1320}{6 \times 22}$ cm; so, $h = 10$ cm

Then, total surface area $= 2\pi r^2 + 2\pi rh = 2\pi r\,(r + 2h)$

$= 2 \times \frac{22}{7} \times 21(21 + 10) = 132 \times 31 = 4092$ cm^2

6. Let the original radius $= r$ and length $= l$

Then, volume $= \pi r^2 l$ (i)

When radius is decreased to one-fourth, then let length be l_1.

Then, volume $= \pi\left(\frac{1}{4}r\right)^2 l_1$ (ii)

From equations (i) and (ii), we get

$\pi r^2 l = \frac{\pi r^2 l_1}{16} \rightarrow \frac{l}{l_1} = \frac{1}{16}$

7. Let the height $= 2a$

Then, length $= 6a$ and width $= 3a$

Then, area of walls $= 2[l \times h + b \times h] = 36a^2$ (i)

The price of painting the walls = 360, and the rate of painting = 2.50/m^2

Then, area of the walls $= \frac{360}{2.5} = 144$

$36a^2 = 144$. Hence, $a = 2$

Area of the floor $= l \times b = 18a^2 = 72$ and cost of carpeting = 3/m^2

So, total cost $= 72 \times 3 =$ ₹216

8. Volume of the original ball $= \frac{4}{3}\pi r^3 = \frac{4}{3}\pi \times 216$ cm^3 (i)

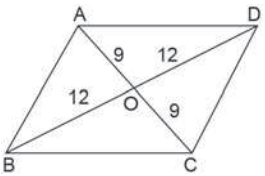
Let total volume of small balls $= \frac{4}{3}\pi\left[3^3 + 4^3 + a^3\right]$ (ii)

Then, from equations (i) and (ii), we get

$\frac{4}{3}\pi \times 216 = \frac{4}{3}\pi\left[27 + 64 + a^3\right]$

$a^3 = 216 - 91 = 125$ or $a = 5$ cm

9. From the figure:



$AD^2 = AO^2 + OD^2 = 9^2 + 12^2$

$AD = 15$

So, perimeter $= 15 \times 4 = 60$ cm

10. From the figure, $PR^2 = PS^2 + SR^2 = 8^2 + 6^2$

$PR = 10$ cm

Then, radius of the circle $= \frac{PR}{2} = 5$ cm

Then, area of shaded portion = Area of circle − Area of rectangle $= \pi(5)^2 - 8 \times 6 = \frac{22}{7} \times 25 - 48 = 29.50$ cm^2

11. Let the radius be r and height be h_1.

Then, radius of second cone $= 2r$

Let height $= h_2$

Then, from the question, $2\left[\frac{1}{3}\pi r^2 h_1\right] = \frac{1}{3}\pi(2r)^2 h_2$

$2h_1 = 4h_2$

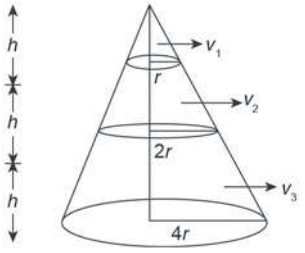
So, $\frac{h_2}{h_1} = \frac{1}{2}$

12. Let the radius be r and height of cone be h, respectively. Then, from the question, volume of hemisphere = volume of cone.

$\frac{2}{3}\pi r^3 = \frac{1}{3}\pi r^2 h$

$2r = h$; so, $\frac{h}{r} = \frac{2}{1}$

13.



From the question, the height would be same of all solids.

Let the radius of v_1 cone be r and height be h.

Then, volume of $v_1 = \frac{1}{3}\pi r^2 h$ (i)

Volume of v_2 solid $= \frac{\pi h}{3}\left[R^2 + r^2 + Rr\right]$

$= \frac{\pi h}{3}\left[4r^2 + r^2 + 2r^2\right] = \frac{7}{3}\pi r^2 h$ (ii)

Volume of v_3 solid $= \frac{\pi h}{3}[R^2 + r^2 + Rr]$

$= \frac{\pi h}{3}\left[16r^2 + 4r^2 + 8r^2\right] = \frac{28}{3}\pi r^2 h$ (iii)

Then, ratio of $v_1 : v_2 : v_3 =$

$\frac{1}{3}\pi r^2 h : \frac{7}{3}\pi r^2 h : \frac{7}{3}\pi r^2 h : \frac{28}{3}\pi r^2 h = 1 : 7 : 28$

14. The radius of pipe = 2.5 mm = 0.0025 m

Then, the volume of water that comes in 1 min

$= 0.0025 \times 10$ m^2 $= 0.025$ m^2 (i)

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The volume of conical vessel $= \frac{1}{3}\pi r^2 h$

$= \frac{1}{3} \times \frac{22}{7} \times \frac{(20)^2}{(100)^2} \times \frac{24}{100}$ (ii)

Then, time taken $= \frac{\frac{1}{3} \times \frac{22}{7} \times \frac{400}{10,000} \times \frac{24}{100}}{0.25\,m^2}$

15. The volume of pyramid $= \frac{1}{3}a^2h = \frac{1}{3}(8)^2 \times 30$

$= \frac{1}{3} \times 64 \times 30 = 640$

16. Let the radius of sphere be r.

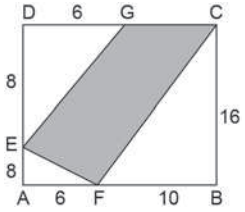
Then, volume $= \frac{4}{3}\pi r^3$ (i)

Length of cylinder be $2r$.

Then, volume $= \pi r^2 \times 2r = 2\pi r^3$ (ii)

The ratio $= \frac{2\pi r^3}{\frac{4}{3}\pi r^3} = \frac{3}{2}$

17.



Area of shaded region = Area of square − Area of [ΔDEG + ΔEAF + ΔCFB] = 128 cm²

18. Outer dimensions will be as follows:

Length = 86 + 4 = 90 cm, breadth = 46 + 4 = 50 cm, and height h = 38 + 2 = 40 cm

So, the area of outer surface $= l \times b + 2 \times l \times h + 2 \times b \times h =$ 4500 + 7200 + 4000 = 15,700 cm² = 1.57 m²

So, the cost of painting = 1.57 × 10 = ₹15.7

19. Area of square = 121 m² = a^2 = 121, and hence, side of the square (a) = 11 m

So, the perimeter of the square = 44 m

Given that perimeter of the square = Circumference of the circle $= 2\pi r^2 = 44$ m

$2 \times \frac{22}{7} \times r = 44$ or $r = 7$ m

So, the area of circle $= \pi r^2 = \frac{22}{7} \times 7 \times 7 = 154$ m²

20. Method 1 Visualization

Let the radius of ball = r; so, total surface area = $4\pi r^2$. When it is cut, there will be four parts, and each part will have two surfaces exposed with the area of each face = $\frac{1}{2}\pi r^2$. Hence, inside surface area of one part is πr^2.

So, total surface area = $4\pi r^2$

Therefore, ratio = 1:1

Method 2 Application of the formula

Let the radius of spherical ball be r, then surface area $= 4\pi r^2$ m² (i)

When it is cut into four smaller pieces, then total surface area of one piece $= \pi r^2 + \frac{4\pi r^2}{4} = \pi r^2 + \pi r^2 = 2\pi$

So, the total surface area of all smaller pieces = $\frac{4 \times 2\pi r^2}{2} = 8\pi r^2$ (ii)

In this area, $4\pi r^2$ area would be painted black.

So, non-painted area $= 8\pi r^2 - 4\pi r^2 = 4\pi r^2$

Hence, ratio = 1:1

21. If volumes are same, then which one of them has maximum side and has maximum surface area. Therefore, the answer is tetrahedron.

22. Let the side of first cube = a_1, then diagonal $= a_1\sqrt{3} = 3x$

Hence, $a_1 = \sqrt{3}x$

Then, surface area $= 6 \times (\sqrt{3}x)^2 = 18x^2$ (i)

Let the side of second cube = a_2, then the diagonal = $a_2\sqrt{3} = 2x$

Hence, $a_2 = \frac{2}{\sqrt{3}}x$

Then, surface area $= 6 \times \left(\frac{2}{\sqrt{3}}x\right)^2$

$= 6 \times \frac{4}{3}x^2 = 8x^3$ (ii)

Then, from equations (i) and (ii), we get

Ratio $= \frac{18x^2}{8x^2} = 9:4$

MODERATE

1. Take the total surface area of the initial cone into consideration and then proceed.

2. We know that when the four sides are equal, then the area will be maximum.

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So, perimeter $(4a) = 100 \rightarrow a = 25$

Then, area $(a^2) = (25)^2 = 625$ m^2

3. Here also, all sides should be equal. Let length of one side be a.

Then, $3a = 100$ or $a = \frac{100}{3}$

So, area $a^2 = \left(\frac{100}{3}\right)^2 = 1111.11$ m^2

4. Lateral surface area of roller $= 2\pi rh$

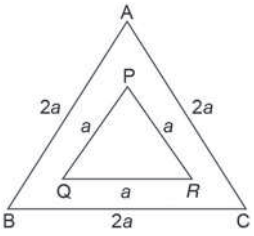
$= 2 \times \frac{22}{7} \times 21 \times 100 = 13{,}200$ cm^2

But, roller takes 400 complete revolutions to complete the work.

So, area covered by roller $= 400 \times 13{,}200 = 52{,}80{,}000$ cm^2 $= 528$ m^2

Therefore, total cost = Area (in m^2 × Rate/m^2) $= 528 \times 100$ = ₹52,800

5.



Let the flower bed be PQR, which has a side length of a unit.

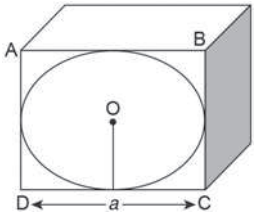
Park is ABC.

Then, $AB = BC = CA = 2a$

Then, $\frac{\text{Area of path}}{\text{Area of bed}} = \frac{\text{Area of ABC} - \text{Area of PQR}}{\text{Area of PQR}} = \frac{3}{1}$

6. The vertices of the smaller square will be on the mid-points of the sides of the larger square.

7.



Let the side of cube be a.

Then, from the figure, radius of cylinder

$$= \frac{\text{side of cube}}{2} = \frac{a}{2}$$

Then, percentage of material lost $= \frac{\text{Lost material}}{\text{Volume of cube}} \times 100$

$$= \frac{\text{Volume of cube} - \text{Volume of cylinder}}{\text{Volume of cube}} \times 100$$

$= 21.3\% = 22\%$ (approximately)

8. Let the friends be x.

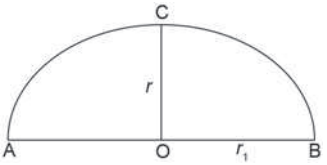
Then, from the question,

volume of hemisphere $= x$[3 × volume of scoop]

$$\frac{2}{3}\pi(9)^3 = 3x\left[\frac{2}{3}\pi(1)^3\right] x = 243$$

9. The amount saved would be maximum only when the difference of maximum area and minimum area to be painted is maximum.

10.



Let the radius of the cage $= r$.

Using the figure, $CB = r\sqrt{2}$.

Then, $OC + CB = 482$ or $r + r\sqrt{2} = 482$

Hence, $r = 199.66 = 200$ cm (approximately)

11. Let ABCDEFGH be the cube of side a and O be the centre of the hemisphere.

$AC = \sqrt{2}a$

$OD = OC = R$

Let P be the mid-point of AC

$OP = a$

Now, in ΔAOC,

$$R^2 = a^2 + \frac{a^2}{2}$$

$$\therefore a = \sqrt{\frac{2}{3}}R$$

12. For the biggest cube, the side would be 20 cm. So, volume of cube $= a^3 = (20)^3 = 8000$ cm^3

13. Take a value of n, say 3, and then, check the options.

14. We know that,

No. of small cubes with no exposed surface $= (n-2)^3$

No. of small cubes with one exposed surface $= 6(n-2)^2$

According to the question, A = B or $6(n-2)^2 = (n-2)^3$

$6 = n - 2$ or $n = 8$

So, total number of small cubes (for $n = 8$) $= n^3 = 8^3 = 512$

17. Let the edges be l, b, and h.

Then, product of all the 12 edges $= l^4 \times b^4 \times h^4$ (i)

Now, let us verify the options:

(i) The forth power of the volume of the box

$= (l \times b \times h)^4 = l^4 \times b^4 \times h^4$ (ii)

(ii) Product of areas of the six faces

$= (lb)^2 \times (bh)^2 \times (hl)^2 = l^4 \times b^4 \times h^4$ (iii)

(iii) Sum of the square of the six faces = $2[(lb)^2 + (bh)^2 + (hl)^2]$

Hence, option (c) is the answer.

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19. It is given that height of the cylinder = 4 cm.

Therefore, radius of the cylinder = 2 cm, and height and radius of the cone = 2 cm.

Now, let the number of cylinder = x. So, number of cones = $2x$.

Then, from the question,

x[volume of cylinder] + $2x$[volume of cone] = volume of sphere

20. Let the radius of A and B be r_1 and r_2, respectively.

Then, from the question,

$$\frac{\text{Surface area of B} - \text{Surface area of A}}{\text{Surface area of A}} \times 100 = 300$$

From here, we will find that $2r_1 = r_2$.

Now, solve the question.

21. Let the radius of new cone = r.

Then, $\frac{4}{3}\pi r^3 = \frac{4}{3}\pi(3)^3 + \frac{4}{3}\pi(4)^3 + \frac{4}{3}\pi(5)^3$

$r^3 = 27 + 64 + 125 = 216$ or $r = 6$

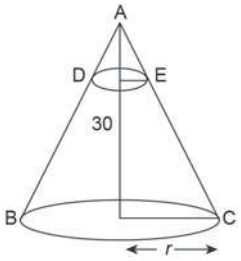
So, ratio of surface areas = 25:18

22. Let the side of cube be a.

Then, $a^3 = v \rightarrow a = \sqrt[3]{v}$

Then, total length of its edges = $12a = 12v^{1/3}$

23.



Let the radius of original cube be r.

We know that radius and height of the small cone will be in the ratio of radius and height of the original cube.

Then, radius and height of the small cube are rx and $30x$, respectively.

Now, from the question,

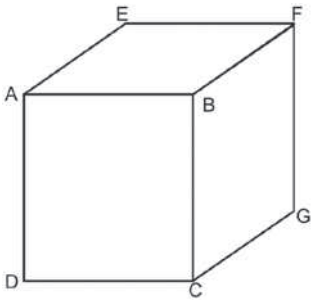
$$\frac{\frac{1}{3}\pi r^2(30)}{\frac{1}{3}\pi(rx)^2(30x)} = \frac{27}{1}$$

$$\frac{30r^2}{30r^2x^3} = \frac{27}{1} \rightarrow x^3 = \frac{1}{27} \rightarrow x = \frac{1}{3}$$

Then, height of the cone = $30x = 30 \times \frac{1}{3} = 10$ cm

24. The distance from any vertex at the base of the cube to the vertex that is perpendicular along height to the diametrically opposite vertex is required.

We have to calculate DF.



26. Side of square EFGH,

$$EF^2 = \left(\frac{AB}{2}\right)^2 + \left(\frac{AD}{2}\right)^2 = 25 + 25$$

$EF = 5\sqrt{2}$

$$\text{Now, } \frac{\text{Volume of ABCD}}{\text{Volume of EFGH}} = \frac{\frac{1}{3} \times (10)^2 \times 30}{\frac{1}{3}(5\sqrt{2})^2 \times 30} = \frac{2}{1} = 2:1$$

27. Volume of space left = Volume of ABCD − Volume of EFGH

$$= \frac{1}{3}(10)^2 \times 30 - \frac{1}{3}(5\sqrt{2})^2 \times 30$$

$= 1000 - 500 = 500$ cm^3

28. Equate the area of the square ABCD and triangle PDC and find a relation between the slant height and the length of the base of the pyramid.

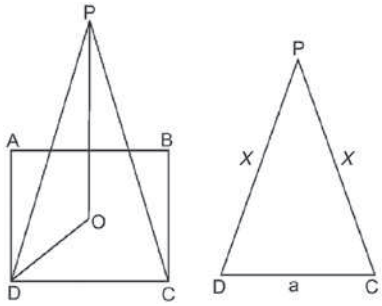
$$a^2 = \frac{a}{4}\sqrt{4x^2 - x^2}$$

$16ax^2 = 4x^2 = a^2$

$\Rightarrow 17a^2 = 4x^2$

$$\frac{a}{x} = \frac{2}{\sqrt{17}}$$

Let $a = 2\pi$; $x = \sqrt{17}$ P



Now, in Δ POD,

OD = $\sqrt{2}P$ PD = $\sqrt{17}$ P

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$AO = \sqrt{15}\ P$

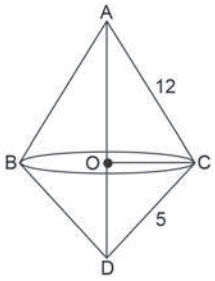
But, $AO = 3 \Rightarrow P = \dfrac{3}{\sqrt{15}}$

$2P = a = \dfrac{6}{\sqrt{15}}$

Area of the base $= \dfrac{36}{15}$

Total surface area $= \dfrac{36}{15} \times 5 = 12$

29.



From the figure, we can say that the figure will be double cone.

30. In ΔACD, let $OC = a$

Then, Area of $\Delta ABC = \dfrac{1}{2} \times 12 \times 5 = \dfrac{1}{2} \times 13 \times a = \dfrac{60}{30}$ m

Now, volume of the figure

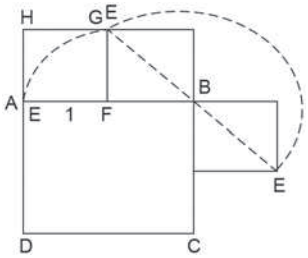
= Volume of cone ABC + Volume of cone BCD

$$= \frac{1}{3} \times \pi \left(\frac{60}{13}\right)^2 \times AO + \frac{1}{3}\pi \left(\frac{60}{13}\right)^2 \times OD$$

$$= \frac{1}{3} \times \frac{22}{7} \times \left(\frac{60}{13}\right)^2 [AO + OD] \frac{1}{3} \times \frac{22}{7} \times \left(\frac{60}{13}\right)^2 \times 13 = 290$$

32. In the first operation, the string that is extra after winding an integral number of times will be equal to 10 cm.

33.



From the figure EF = FG = 1 unit

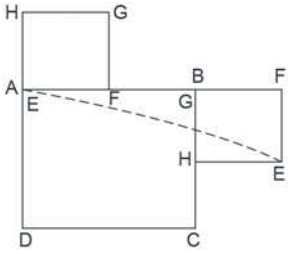
Then, $BG = \sqrt{2}$

Then, distance traversed by E

$$= 2\pi r_1 \times \frac{90}{300} + 2\pi r_2 \times \frac{180}{360}$$

$$= 2\pi \left[\frac{r_1}{4} + \frac{r_2}{2}\right] = 2\pi \left[\frac{1}{4} + \frac{\sqrt{2}}{2}\right] = \frac{\pi}{2} + \sqrt{2}\pi$$

34.



In the figure,

AF = AB + BF = 3 unit

And FE = 1 unit

Then, $AE_2 = AF^2 + FE^2$

$AE = \sqrt{3+1} = \sqrt{10}$

35. Use factors of 100.

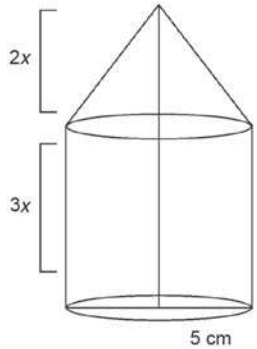
36. There are four faces of dimension 2×3, which are rectangle and two faces of dimension 2×2, which are squares. So, number of rectangles × Number of squares $= 4 \times 2 = 8$

38. Volume is maximum when radius is equal to height.

ADVANCED

1. Since we do not know that the cut is made parallel to which face, we cannot determine the surface area.

2. Suppose the height of the cone is 6 cm and that of cylinder is 9 cm. Therefore, the height of the hole is 10 cm, and now, a straight forward calculation leads to the answer.



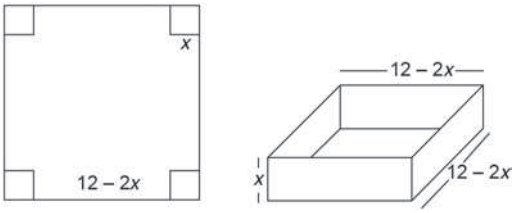
3. There are four cans of diameter 2 cm.
Then, the side of the carton = 2 × 4 = 8 cm, and so, the area $= a^2 \times 8 \times 8 = 64\ cm^2$

4. Let the length, breadth, and height of the cuboid be l, b, and h, respectively.
Then, $l \times b = a$, $b \times h = b$ and $h \times l = c$
Then, $a \times b \times c = l^2\ b^2\ h^2$
$(lbh)^2 = abc$
$N^2 = abc$ [since volume $l \times b \times h$]

5. Following are the figures:
Once we have removed the square of side x inches from the four corners, dimension of the box will be as follows:
Length of the box will be $(12 - 2x)$ inches, breadth will be $(12 - 2x)$ inches, and height will be x inches. Volume (V) of this open box will be $= (12 - 2x)\ (12 - 2x)(x)$ cubic inches

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Now, we have two methods to solve this question:

Finding dv/dx using calculus

Using the relation between sum of numbers and its maximum product

Method 1

Finding dv/dx is subject to the condition $0 < x < 6$. $V = (144 + 4x^2 - 48x)\ x = (144x + 4x^3 - 48x^2)$ $dv/dx = 144 + 12x^2 - 96x$

To find out the value of x, we will put $dv/dx = 0$ $dv/dx = 144 + 12x^2 - 96x = 0 \Rightarrow 12 + x^2 - 8x = 0$

Or, $x^2 + 8x + 12 = 0$ or $(x - 2)(x - 6) = 0$

Since x cannot be equal to 6 (as the length = breadth = 12 inches), $x = 2$ inches.

Method 2

We will go through the second one.

To maximize the volume, sum should be constant.

To make the sum constant, third dimension should be $4x$ instead of x.

It can be seen that the summation of dimensions = $(12 - 2x) + (12 - 2x) + 4x = 24$ cm

Product will be maximum when length = breadth = height

$= \frac{24}{3} = 8$ inches

Hence, $12 - 2x = 8$ or $x = 2$ inches

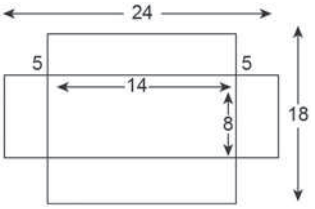
So, volume will be maximum when the square of size 2×2 inches is removed from the rectangular sheet.

Method 3

Going through the options.

From the option, at $x = 2$, the volume of box will be maximum.

6.



From the figure, the dimension of the base is 8×14 and height of the box is 5.

Then, volume of the box = $l \times b \times h = 14 \times 8 \times 5 = 560$ cm^3

7. $E = f(A) = \sqrt{\frac{A}{7}}$

When $A = 3087$, then $E = \sqrt{\frac{3087}{7}} = 21$ (i)

When $A = 2023$, then $E = \sqrt{\frac{2023}{7}} = 17$

So, answer = 21 – 17 = 4 sq. units

8. it is given that $\frac{\pi r_1^2 h_1}{\pi r_2^2 h_2} = \frac{a}{b} \rightarrow \frac{r_1^2 c}{r_2^2 d} = \frac{a}{b}$

$$\frac{r_1}{r_2} = \sqrt{\frac{ad}{bc}} \rightarrow \frac{2r_1}{2r_2} = \sqrt{\frac{ad}{bc}}$$

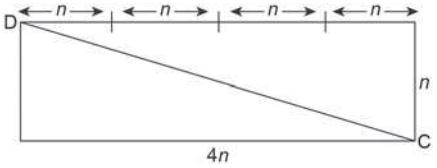
9. Volume of water required

= Volume of cylinder – (Volume of cone – Volume of hemisphere)

$$= \pi(3)^2 \times 10 - \left[\frac{1}{3}\pi(2)^2 \times 5 - \frac{2}{3}\pi(2)^3\right] = 90\pi - \left[\frac{4}{3}\pi\right]$$

10. If the cylinder is cut along its length, it becomes a rectangle.

11.



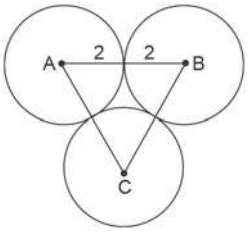
From the figure

$DC^2 = (4n)^2 + n^2 = 17n^2$

Hence, $DC = \sqrt{17}\,n$

12. Equate the lengths of strings obtained from the above two questions.

14.



Area of the hollow space = Area of $\Delta ABC - 3 \times$ Area of circle $\times \frac{60}{360}$

$$= \frac{1}{2}(4)^2 \times \sin 60 - \frac{1}{2}\pi(2)^2$$

$= 2(2\sqrt{3} - \pi)$ cm^2

Then, volume of hollow space = Area of base × Height

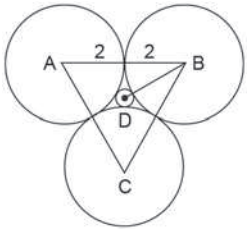
$= 2(2\sqrt{3} - \pi) \times 100$

$= 200(2\sqrt{3} - \pi)$ cm^3

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15.



In the figure, let the centre of ΔABC and sphere be D.

Then, DB = circumcentre of ΔABC

$DB = \frac{4}{\sqrt{3}}$

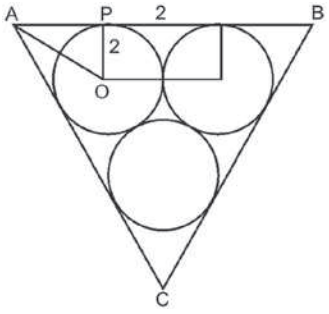
Then, radius of sphere = DB − Radius of circle

$$= \frac{4}{\sqrt{3}} - 2 = \frac{4 - 2\sqrt{3}}{\sqrt{3}}$$

Now, number of sphere = $\frac{\text{Height}}{\text{Diameter of sphere}}$

$$= \frac{100}{\frac{8 - 4\sqrt{3}}{3}} = 323$$

16.



$\angle AOP = 30°$

OP = 2 cm

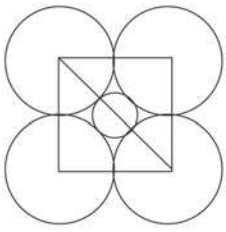
Calculate AP

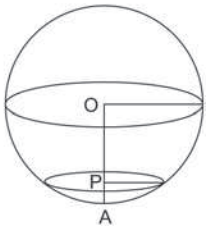
17. Lateral surface area = $2\pi r \cdot h = 12h$

Then, length of the string = $\frac{12h}{x}$ cm

19. The 3D shape of the hexagon will be a hexagonal prism.

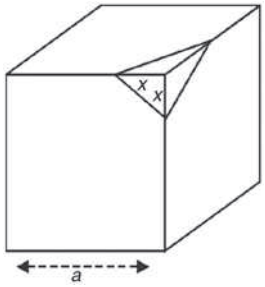
21. OA = 10





To find the value of PA, go through the options now.

22. In this question, we need to cut eight identical pyramids having a base of equilateral triangle of side equal to the side of the newly formed octagon.



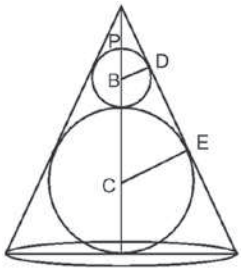
$A = 2x + \sqrt{2}x$

25. The length of the rectangle will be equal to the circumference of the base of the cylinder.

26. Let AP = x

ΔABD ~ ΔACE

Now, x can be calculated.



36. Consider the length of the string less than or equal to the in radius of the floor. At the corner of the floor, you will find a sixth part of a hemisphere, and at the centre, it will be a hemisphere.

Answers to Q.37 to 40:

Plot these points and direction on a graphical plane for a clear interpretation.

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CHAPTER

20

Co-ordinate Geometry

LEARNING OBJECTIVES

After completion of this chapter, the reader should be able to understand:

- Introduction to the topic
- Presenting the coordinates
- Equation of straight line—conditions for parallel and perpendicular lines
- Equation of circle
- Different types of questions and method of solving these questions

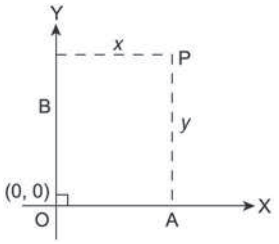
INTRODUCTION

Remember the famous movie 'Shashwank Redemption'? In that movie, the protagonist Andy Dufresne (Tim Robbins) tells his prison mate Ellis Boyd (Morgan Freeman) about a letter kept underneath the ground near a tree at a place which was quite far. Andy explains the way to reach the letter, and Boyd was successfully able to reach the place. Now think of the same situation in a desert—how would you be able to tell the other person the whereabouts of something that you want him/her to find? We have longitudes and latitudes for that. What if I have to show the same on my notebook? Here comes the role of coordinates.

Therefore, if I have to denote a point on my notebook, we use two restricting factors—distance from X-axis and distance from Y-axis. What if I have to show a point in the air in my room? Can I show it using only distances from X-axis and Y-axis? The answer is no. My room is a 3-dimensional space and to show the point in a 3-D space, I need to have three restricting factors—distance from X-axis, distance from Y-axis, and distance from Z-axis (or distance from the ceiling), or maybe the angle formed by that point from any of the corners of the room.

The method of denoting points with the help of coordinates was proposed by French Mathematician Descartes. Moreover, he also proposed that lines and curves are nothing but the collection of points and hence, can be represented by equations derived out of co-ordinate geometry. To honour his work, the coordinates of a point are often referred to as Cartesian coordinates.

Consider the Following Case



Suppose there is an ant at point O in the figure and it wants to go to point P?

One way to reach P is that the ant travels along OX, reaches A and then travels along AP and reaches P.

Hence, it first covers a distance 'X' horizontally and then covers a distance 'Y' vertically.

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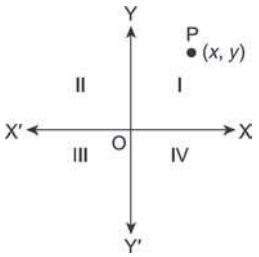
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First of all, we will assume a reference point, which is at a distance of 0 unit from both the X-axis and Y-axis and will call this origin.

If we assume 'O' as the origin, then the distances X and Y are the distances of this point P from Y-axis and X-axis, respectively. These are known as coordinates of point P and are written as P(X, Y).

Coordinate Axes and Representation of a Point

The figure given along with is called the X-Y Cartesian plane. The line XOX′ is called the X-axis and YOY′ is called the Y-axis.



If P(x, y) is a point in this plane, then x is the X-coordinate of P or abscissa of P and y is called the Y-coordinate of P or the ordinate of P.

Remember that X-coordinate of the point is the distance of the point from Y-axis and Y-coordinate of the point is the distance of the point from X-axis.

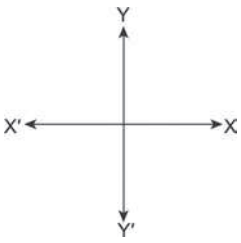
The X-Y Cartesian plane is divided into four equal parts called quadrants (I, II, III, and IV).

Sign convention

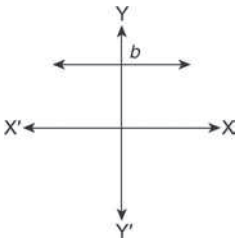
	1st Quadrant	2nd Quadrant	3rd Quadrant	4th Quadrant
X-axis	+ve	–ve	–ve	+ve
Y-axis	+ve	+ve	–ve	–ve

Equation and Graph of Coordinate Axis

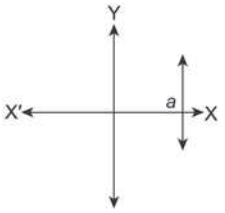
1. Equations of X- and Y-axes are $y = 0$ and $x = 0$, respectively.



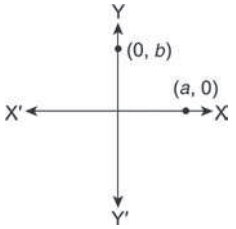
2. Equation of a line parallel to X-axis is $Y = b$ (b is constant).



3. Equation of a line parallel to Y-axis is $X = a$ (a is constant).



4. Any point on the X-axis can be taken as (a, 0) and any point on the Y-axis can be taken as (0, b).



5. To find out X and Y intercepts of a line, we will put $Y = 0$ and $X = 0$, respectively, in the equation of the line.

If you know the coordinates of two points:

- Find out the distance between them.
- Find the mid-point, slope, and equation of the line segment formed by these two points.

Some Standard Formula

1. **Distance between two points**
If there are two points A(x_1, y_1) and B(x_2, y_2) on the XY plane, then the distance between them is given by:

$$AB = d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Example 1 What is the distance between the points (3, 2) and (6, 6)?

Solution Distance $= \sqrt{(6-3)^2 + (6-2)^2} = \sqrt{25} = 5$ units

Example 2 Coordinates A and C of a square ABCD (points in order) are (4, 2) and (1, 4), respectively. What is the area of the square?

Solution For square ABCD, line segment AC will be its diagonal.

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$AC = \sqrt{(4-1)^2 + (2-4)^2} = \sqrt{13}$ units

Diagonal of square $= \sqrt{2}$ side $= \sqrt{13}$ units

So, side of square $= \dfrac{\sqrt{13}}{\sqrt{2}}$ units

Hence, area = (Side of square)$^2 = \left(\dfrac{13}{2}\right) = 6.5$ sq. units

Methods to identify if three points A, B, and C are in the same straight line:
If there are three points A, B, and C, they may be in the same straight line or form a triangle.

Method 1
Area formed by the three points = 0 [formula to find out the area of triangle given ahead]

Method 2
Slopes of any two line segments AB or BC or AC are equal. For example, slope of line AB = Slope of line AC

Method 3
Sum of any two line segments is equal to the third line segment. For example, AB + BC = AC

2. **Division of a line segment (if three points A, B, and C are in a straight line)**
   (i) **Internal**
   The coordinates of a point P that divides the line joining A(x_1, y_1) and B(x_2, y_2) internally in a ratio l:m are given by:

$$x = \frac{lx_2 + mx_1}{l+m}, y = \frac{ly_2 + my_1}{l+m}$$

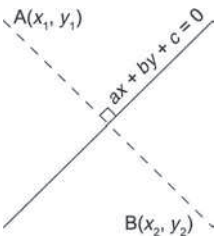
   (ii) **External**
   The coordinates of a point P that divides the line joining the point A(x_1, y_1) and B(x_2, y_2) externally in the ratio l:m are given by:

$$x = \frac{lx_2 - mx_1}{l-m}, y = \frac{ly_2 - my_1}{l-m}$$

The Image of a Point along the Mirror Placed on a Straight Line

The image of A(x_1, y_1) with respect to the line mirror $ax + by + c = 0$ be B(x_2, y_2) is given by:

$$\frac{x_2 - x_1}{a} = \frac{y_2 - y_1}{b} = \frac{-2(ax_1 + by_1 + c)}{(a^2 + b^2)}$$



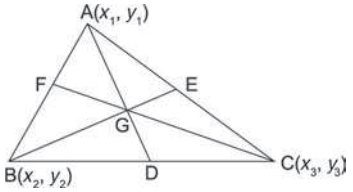
Foot of the Perpendicular

If the foot of the perpendicular from (x_1, y_1) to the line $lx + my + n = 0$ is (h, k), then

$$\frac{h - x_1}{l} = \frac{k - y_1}{m} = \frac{-(lx_1 + my_1 - n)}{l^2 + m^2}$$

Centroid of a Triangle

The point at which the medians of a triangle intersect is called centroid of the triangle.



Let ABC be a given triangle with vertices A(x_1, y_1), B(x_2, y_2), and C(x_3, y_3).

Since D is the mid-point of BC, its coordinates are $[(x_2 + x_3)/2, (y_2 + y_3)/2]$.

Let G(x, y) be a point dividing AD in the ratio 2:1.

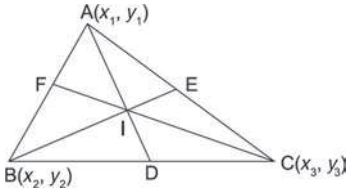
Then, $x = \dfrac{2\left(\dfrac{x_2 + x_3}{2}\right) + 1.x_1}{(2+1)} = \left(\dfrac{x_1 + x_2 + x_3}{3}\right)$

and $y = \dfrac{\left(\dfrac{y_2 + y_3}{2}\right) + 1.y_1}{(2+1)} = \left(\dfrac{y_1 + y_2 + y_3}{3}\right)$

Similarly, the coordinates of a point that divides BE in the ratio 2:1 as well as those of the point that divides CF in the ratio 2:1 are:

$$\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}\right)$$

Incentre of a Triangle



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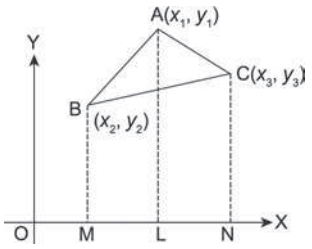
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The point at which the bisectors of the angles of a triangle intersect is called the incentre of the triangle. From geometry, we know that the bisector of an angle of a triangle divides the opposite side in the ratio of length of remaining sides. Hence, the bisectors of the angle of ΔABC are concurrent and meet at a point called incentre.

$$\left(\frac{ax_1+bx_2+cx_3}{a+b+c}, \frac{ay_1+by_2+cy_3}{a+b+c}\right)$$

Area of Triangle

Let ABC be a given triangle whose vertices are $A(x_1, y_1)$, $B(x_2, y_2)$, and $C(x_3, y_3)$.



Area of the triangle

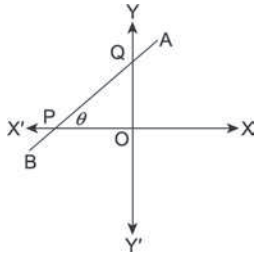
$$= \frac{1}{2}\left[x_1(y_1-y_2)+x_2(y_2-y_3)+x_3(y_3-y_1)\right]$$

If we interchange the order of any two vertices of the ΔABC, we obtain a negative value of the area. However, the area shall always be taken to be positive.

Equation of a Curve

An equation in two variables X and Y with the degree of the equation being equal to or more than two is called the equation of a curve. If the graph of that equation plotted on the X-Y Cartesian plane, it will give a shape of a curve and not a straight line.

For example, $x^2 + y^2 = 16$, $y = x^2$. Equation of circle has been discussed ahead in this chapter.



Straight Line

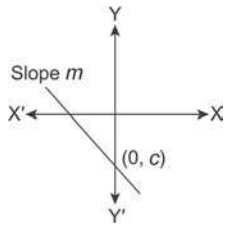
Any equation with the degree of equation being one is known as the equation of a straight line. General equation of straight line is given by $aX + bY + c = 0$, where X and Y are variables and a, b, and c are constants.

Any point lying on this line will satisfy the equation of the line.

If AB is a straight line on the XY plane, then the angle θ which the line makes with the X-axis in the anti-clockwise direction is called the inclination of the line and tangent of this angle θ ($\tan\theta$) is called the slope of the line AB. It is denoted by 'm'. The lengths OP and OQ are, respectively, known as the intercepts on X-axis and Y-axis made by the line.

DIFFERENT FORMS OF REPRESENTING A STRAIGHT LINE

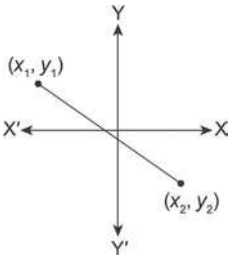
1. **Slope-intercept form**
   $y = mx + c$
   If 'm' is the slope of the line and 'c' the intercept made by the line on Y-axis, then the equation is $y = mx + c$.



2. **Point-slope form**
   If 'm' is the slope of the line and it passes through the point (x_1, y_1), then the equation of the line is given by:
   $y - y_1 = m(x - x_1)$

3. **Two-point form**
   If the line passes through two points (X_1, Y_1) and (X_2, Y_2), then the equation is:

$$(Y - Y_1) = \frac{Y_2 - Y_1}{X_2 - X_1}(X - X_1)$$



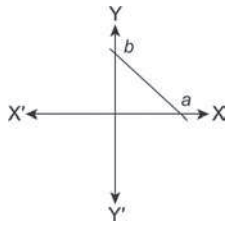
Using point-slope form and two-point form, we can find out the formula for slope also. Comparing the two equations, we get $m = \frac{Y_2 - Y_1}{X_2 - X_1}$.

4. **Slope-intercept form**
   If the line makes an intercept of a units on X-axis and b units on Y-axis, then the equation is:

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$$\frac{X}{a}+\frac{Y}{b}=1$$



Finding Slope of a Line:

(a) If equation of the line is $ax + by = c$, then slope of the line $= \frac{-a}{b}$.

For example, slope of the line $2x + 3y = 5$ is $\frac{-2}{3}$.

(b) If two points (x_1, y_1) and (x_2, y_2) are given, then slope of the line $= \frac{Y_2 - Y_1}{X_2 - X_1}$.

Angle between Two Intersecting Lines

The angle between two lines whose slopes are m_1 and m_2 is given by a formula such that $\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2}$ (where θ is the angle between the lines).

Condition for Two Straight Lines to be Parallel

It can be visualized that two straight lines can be parallel only if they make an equal inclination with the X-axis. This will, in turn, ensure that their slopes are equal.

The lines $y = m_1x + c_1$ and $y = m_2x + c_2$ are parallel, if and only if $m_1 = m_2$.

Condition for Two Straight Lines to be Perpendicular

The **General Equation** of a line parallel to a given line $ax + by + c = 0$ will be $ax + by + k = 0$, where k is any constant that can be found by additional information given in the question.

The **General Equation** of a line perpendicular to a given line $ax + by + c = 0$ will be $(bx - ay + k = 0)$ or $(-bx + ay + k = 0)$, where k is any constant that can be found by additional information given in the question.

The lines $y = m_1x + c_1$ and $y = m_2x + c_2$ are perpendicular if and only if $m_1 m_2 = -1$.

Example 3 Which of the following cannot be the equation of the straight line parallel to the straight line $4x - 6y = 10$?

(a) $2x - 3y = 8$ (b) $x - 1.5y = 2$

(c) $8x + 12y = 12$ (d) $2x - 3y = 4$

Solution Except option (c), all the other options can be written as $4x - 6y = K$ by multiplying the LHS by a suitable number. Hence, option (c) is the answer.

Point of Intersection of Two Lines

The coordinates of the point of intersection of the two intersecting lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ are:

$$\left(\frac{b_1c_2 - b_2c_1}{a_1b_2 - a_2b_1}, \frac{a_2c_1 - a_1c_2}{a_1b_2 - a_2b_1}\right)$$

However, to obtain the point of intersection, we are required to just solve the equations of the straight lines given as we do in the case of simultaneous equations.

Condition of Concurrency of Three Lines

Three lines are said to be concurrent if they pass through a common point, that is, if they meet at a point. The condition for three lines $a_1x + b_1y + c_1 = 0$, $a_2x + b_2y + c_2 = 0$, and $a_3x + b_3y + c_3 = 0$ is:

$$a_1(b_2c_3 - b_3c_2) + b_1(c_2a_3 - c_3a_2) + c_1(a_2b_3 - a_3b_2) = 0$$

Length of Perpendicular

The length of perpendicular (p) from (X_1, Y_1) on the line $AX + BY + C = 0$ is:

$$P = \frac{AX_1 + BY_1 + C}{\sqrt{A^2 + B^2}}$$

Distance between Two Parallel Lines

The distance between two parallel lines $AX + BY + C_1 = 0$ and $AX + BY + C_2 = 0$ is given by:

$$\left|\frac{C_1 - C_2}{\sqrt{A^2 + B^2}}\right|$$

Conditions for Points to be Collinear

If three points A, B, and C are co-linear, then any one of the following conditions should be true:

1. Area of triangle ABC = 0
2. Slope of AB = slope of BC = slope of AC
3. AB + BC = AC

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Depending upon the points given, we can use any one of the three to check if the points are collinear. It should also be mentioned that if one of these conditions is true, then other two will be definitely true.

Worked Problems

Example 4 Find the coordinates of the point that divides the line segment joining the points (5, –2) and (9, 6) in the ratio 3:1.

Solution Let the required point be (x, y). Then,

$$x = \left(\frac{3\times 9+1\times 5}{3+1}\right) = 8 \text{ and } y = \left(\frac{3\times 6+1\times(-2)}{3+1}\right) = 4$$

The coordinates of the required point are (8, 4).

Example 5 The ratio in which the point $(2, y)$ divides the line joining (–4, 3) and (6, 3) is 3:2. What is the value of y?

Solution Let the required ratio be k:1.

Then, $2 = \dfrac{6k-4\times 1}{k+1} \Rightarrow k = \dfrac{3}{2}$

$\therefore$ The required ratio is $\dfrac{3}{2}:1$ i.e., 3:2

Also, $y = \dfrac{3\times 3+2\times 3}{3+2} = 3$

When asked for ratio *m*:*n*, for convenience, we take ratio as *m*/*n*:1 or *k*:1.

Example 6 Two vertices of a triangle are (–1, 4) and (5, 2) and its centroid is (0, –3). Find the third vertex.

Solution Let the third vertex be (x, y). Then,

$$\frac{x+(-1)+5}{3} = 0 \text{ and } \frac{y+4+2}{3} = -3$$

$\therefore$ $x = -4$ and $y = -15$

Hence, the third vertex of the triangle is (–4, –15).

Example 7 Find the coordinates of incentre of the triangle whose vertices are (4, –2), (–2, 4), and (5, 5).

Solution

$$a = \text{BC} = \sqrt{(5+2)^2+(5-4)^2} = \sqrt{50} = 5\sqrt{2}$$
$$b = \text{AC} = \sqrt{(5-4)^2+(5+2)^2} = \sqrt{50} = 5\sqrt{2}$$
$$c = \text{AB} = \sqrt{(-2-4)^2+(4+2)^2} = \sqrt{72} = 6\sqrt{2}$$

Let (x, y) be the coordinates of incentre of ΔABC. Then,

$$x = \frac{ax_1+bx_2+cx_3}{a+b+c} = \left(\frac{20\sqrt{2}-10\sqrt{2}+30\sqrt{2}}{5\sqrt{2}+5\sqrt{2}+6\sqrt{2}}\right) = \frac{40\sqrt{2}}{16\sqrt{2}} = \frac{5}{2}$$

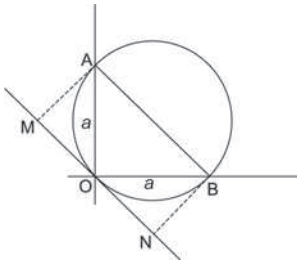
Similarly, y coordinate can be calculated.

$$y' = \frac{ay_1+by_2+cy_3}{a+b+c} = \left(\frac{20\sqrt{2}-10\sqrt{2}+30\sqrt{2}}{5\sqrt{2}+5\sqrt{2}+6\sqrt{2}}\right)$$
$$= \frac{40\sqrt{2}}{16\sqrt{2}} = \frac{5}{2}$$

So, the coordinates of the incentre are $\left(\frac{5}{2}, \frac{5}{2}\right)$.

Example 8 A line makes equal intercepts of length 'a' on the coordinate axes, intersecting the X-axis and Y-axis at A and B, respectively. A circle is circumscribed about the triangle OAB, where O is the origin of the coordinate system. A tangent is drawn to this circle at the point O. What is the sum of the perpendicular distances of the vertices A, B and O from this tangent?

Solution



$$\text{AM} + \text{BN} + \text{OO} = \frac{a}{\sqrt{2}} + \frac{a}{\sqrt{2}} + 0 = \sqrt{2}a$$

Example 9 What is the area of the triangle whose vertices are (4, 4), (3, –16), and (3, –2)?

Solution Let $x_1 = 4$, $x_2 = 3$, $x_3 = 3$ and $y_1 = 4$, $y_2 = -16$, $y_3 = -2$

Then, the area of the given triangle

$$= \frac{1}{2}\left[x_1(y_2-y_3)+x_2(y_3-y_1)+x_3(y_1-y_2)\right]$$
$$= \frac{1}{2}\left[4\cdot\{-16-(-2)\}+3\cdot(-2-4)+3\cdot\{4-(-16)\}\right]$$
$$= \frac{1}{2}\left[4\cdot\{-14+3(-6)\}+3\times 20\right] = \frac{1}{20}\left[-56-18+60\right] = -7$$

Since area of a triangle cannot be negative, the area of the triangle = 7 sq. units.

Example 10 What is the area of the triangle formed by the points (–5, 7), (–4, 5), and (1, –5)?

Solution Let $x_1 = -5$, $x_2 = -4$, $x_3 = 1$ and $y_1 = 7$, $y_2 = 5$, $y_3 = -5$

Area of the triangle formed by the given points

$$= \frac{1}{2}\left[x_1(y_2-y_3)+x_2(y_3-y_1)+x_3(y_1-y_2)\right]$$
$$= \frac{1}{2}\left[(-5)\{5-(-5)\}+(-4)(-5-7)+(7-5)\right] = 0$$

Hence, the given points are not forming any triangle, rather they are collinear.

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Example 11 Find the equation of the straight line that passes through (3, 4) and the sum of whose X and Y intercepts is 14.

Solution Let the intercepts made by the line X-axis and Y-axis be α and $(14 - \alpha)$, respectively.

Then, its equation is $\frac{X}{\alpha} + \frac{Y}{14-\alpha} = 1$

Since it passes through (3, 4), we have:

$$\frac{3}{\alpha} + \frac{4}{14-\alpha} = 1 \Rightarrow \alpha^2 - 13\alpha + 42 = 0$$

$$\Rightarrow (\alpha - 6)(\alpha - 7) = 0$$

$\alpha = 6$ and $\alpha = 7$

So, the required equation is $\frac{x}{6} + \frac{y}{8} = 1$ or $\frac{x}{7} + \frac{y}{7} = 1$, i.e., $4x + 3y = 24$ or $x + y = 7$

Example 12 What is the equation of a line that passes through the point (–1, 3) and is perpendicular to the straight-line $5x + 3y + 1 = 0$?

Solution The equation of any line perpendicular to the line $5x + 3y + 1 = 0$ is:

$$3x - 5y + K = 0$$

Since the required line passes through the point (–1, 3), we have:

$$3 \times (-1) - 5 \times 3 + K = 0, \text{ or, } K = 18$$

Hence, the required equation is $3x - 4y + 18 = 0$.

Example 13 What is the point of intersection of the lines $2x + 3y = 5$ and $3x - 4y = 10$?

Solution To find out the point of intersection, we just need to solve the simultaneous equations.

$$2x + 3y = 5 \quad \text{(i)}$$
$$3x - 4y = 10 \quad \text{(ii)}$$

Multiplying equation (i) by 3 and equation (ii) by 2,

$2x + 3y = 5$ becomes $6x + 9y = 15$ (equation 3) and $3x - 4y = 10$ becomes $6x - 8y = 20$ (equation 4).

Subtracting equation (3) – equation (4) gives:

$$\begin{array}{r} 6x + 9y = 15 \\ 6x - 8y = 20 \\ - \quad + \\ \hline 17y = 35 \end{array}$$

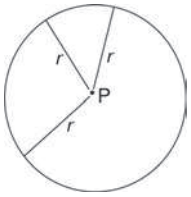
Hence, $y = 35/17$.

This gives $x = -10/17$.

So, the point of intersection $= \frac{-10}{17}, \frac{35}{17}$.

CIRCLE

Locus of points at a fixed distance r from point P is known as a circle. In this case, distance r is known as radius and point P is centre.



Equation of a Circle

If the coordinate of the centre is (a, b) and length of radius $= r$, then equation of circle is:

$$(x - a)^2 + (y - b)^2 = r^2$$

Equation of a circle may not be always given in the above format. Consider the following equation:

$$x^2 + y^2 - 2x - 4y - 4 = 0$$

It should be written as a summation of $(x - a)^2$ and $(y - b)^2$.

First of all, collect terms of x in one bracket and terms of y in other bracket.

$$(x^2 - 2x) + (y^2 - 4y) - 4 = 0$$

Then, start writing the terms in the bracket as squares, keeping the constant terms adjusted.

$$(x^2 - 2x + 1) + (y^2 - 4y + 4) - 4 - 5 = 0$$
$$\Rightarrow \quad (x - 1)^2 + (y - 2)^2 = 3^2$$

Hence, this circle has its centre at (1, 2) and its radius = 3 units.

General Equation of the Circle:

1. The general equation of a circle is $x^2 + y^2 + 2gx + 2fy + c = 0$.

   Centre of this circle is $(-g, -f)$ and its radius $= \sqrt{g^2 + f^2 - c}$.
2. If centre of the circle is origin (0, 0), then equation of the circle is $x^2 + y^2 = r^2$.

Example 14 Find the radius and centre of the circle $x^2 + y^2 - 6x + 4y - 12 = 0$.

Solution Centre of the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is $= (-g, -f)$.

Comparing this with the equation given in the question:

$$x^2 + y^2 - 6x + 4y - 12 = 0$$
$$\Downarrow \quad \Downarrow \quad \Downarrow$$
$$x^2 + y^2 + 2gx + 2fy + c = 0$$

$2g = -6 \Rightarrow -g = 3$

$2f = 4 \Rightarrow -f = -2$

Hence, centre $= (-g, -f) = (3, -2)$

Radius $= \sqrt{(-3)^2 + 2^2 - (-12)} = \sqrt{25} = 5$ units

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Example 15 Find the radius and centre of the circle $2x^2 + 2y^2 - 8x - 7y = 0$.

Solution We are first required to write it down in the format such that co-efficient of each of x^2 and $y^2 = 1$.

$2x^2 + 2y^2 - 8x - 7y = 0$ can be written as $x^2 + y^2 - 4x - \frac{7}{2}y = 0$

In this case, $g = -2$ and $f = -\frac{7}{4}$

Hence, centre $= (-g, -f) = \left(2, \frac{7}{4}\right)$

Radius $= \sqrt{g^2 + f^2 - c} = \sqrt{(-2)^2 + \left(\frac{7}{4}\right)^2 - 0} = \frac{1}{4}\sqrt{113}$

Practice Exercises

FOUNDATION

Q.1 The coordinates of the diagonals of a square are (2, 0) and (0, 5). What is the area of the square?

(a) 29 sq.units (b) $\frac{\sqrt{29}}{\sqrt{2}}$ sq.units

(c) $\frac{29}{4}$ sq.units (d) 14.5 sq.units

Q.2 Let P and Q be points on the line joining A(−2, 5) and B(3, 1) such that AP = PQ = QB. Then, the mid-point of PQ is:

(a) $\left(\frac{1}{2}, 3\right)$ (b) $\left(-\frac{1}{2}, 4\right)$

(c) (2, 3) (d) (−1, 4)

Q.3 If each of the point $(x_1, 4)$, $(-2, y_1)$ lies on the line joining the points (2, −1), (5, −3), then the point $p(x_1, y_1)$ lies on the line:

(a) $6(x + y) - 25 = 0$ (b) $2x + 6y + 1 = 0$

(c) $2x + 3y - 6 = 0$ (d) $6(x + y) + 25 = 0$

Q.4 Find the area enclosed by the x-axis, y-axis, and the graph $y = |x| - 4$ in the first quadrant (in sq. units).

(a) 8 (b) 16

(c) 14 (d) None of these

Q.5 The points $(-a, -b)$, (0, 0), (a, b), and (a^2, ab) are:

(a) Collinear

(b) Vertices of a parallelogram

(c) Vertices of a rectangle

(d) None of these

Q.6 The points $\left(0, \frac{8}{3}\right)$, (1, 3), and (82, 30) are vertices of:

(a) An obtuse-angled triangle

(b) An acute-angled triangle

(c) A right-angled triangle

(d) None of these

Q.7 The points (0, −1), (−2, 3), (6, 7), and (8, 3) are:

(a) Collinear

(b) Vertices of a parallelogram which is not a rectangle

(c) Vertices of a rectangle, which is not a square

(d) None of these

Q.8 Find the area enclosed by the graph $y = |X + 3|$ with the coordinate axes in square units.

(a) 9 (b) 4.5 (c) 0 (d) 12

Q.9 The coordinates of the middle points of the sides of a triangle are (4, 2) (3, 3), and (2, 2), then the coordinates of its centroid are:

(a) $\left(3, \frac{7}{3}\right)$ (b) (3, 3)

(c) (4, 3) (d) None of these

Q.10 The incentre of the triangle whose vertices are (−36, 7), (20, 7), and (0, −8) is:

(a) (0, −1) (b) (−1, 0)

(c) (1/2, 1) (d) None of these

Q.11 The triangle with vertices at (2, 4), (2, 6), and $(2 + \sqrt{3}, 5)$ is:

(a) Right angled

(b) Right angled and isosceles

(c) Equilateral

(d) Obtuse angled

Q.12 The area of the triangle with vertices at the point $(a, b + c)$, $(b, c + a)$, and $(c, a + b)$ is:

(a) 0 (b) $a + b + c$

(c) $ab + bc + ca$ (d) None of these

Q.13 The nearest point on the line $3x - 4y = 25$ from the origin is:

(a) (−4, 5) (b) (3, −4)

(c) (3, 4) (d) (3, 5)

Q.14 If three vertices of a rhombus taken in order are (2, −1), (3, 4), and (−2, 3), then the fourth vertex is:

(a) (−3, −2) (b) (3, 2)

(c) (2, 3) (d) (1, 2)

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Q.15 If a, b, c be in AP, then $ax + by + c = 0$ represents
(a) a single line.
(b) a family of concurrent lines.
(c) a family of parallel lines.
(d) None of these

Q.16 S_1 and S_2 are two light sources. Both the sources have originated from a single point (0, 1) and travel along a linear path. S_1 passes through (1, –2) and S_2 passes through (–1, 2). The path of S_1 and S_2 are:
(a) $x + 3y = 1, x + y = 1$
(b) $3x + y = 1, x - y = 1$
(c) $3x + y = 1, x + y = 1$
(d) $x - 3y = 1, x + 3y = 1$

MODERATE

Q.1 There is a straight line S perpendicular to the line $5x - y = 1$. The area of the triangle formed by the line S and the coordinate axis is 5 square units. Which of the following is the equation of the straight line S?
(a) $x + 5y = 5\sqrt{2}$ (b) $x + 5y + 5\sqrt{2} = 0$
(c) Both (a) and (b) (d) None of these

Q.2 Find the distance of a point (3, 5) from the line $x + y = 2$, measured along a line making an angle 45° with the positive x-axis. (The anti-clockwise direction is positive angle.)
(a) $\sqrt{2}$ unit (b) $2\sqrt{2}$ unit
(c) $3\sqrt{2}$ units (d) $4\sqrt{2}$ units

Q.3 On an axis expressed in kilometres, the coordinates of 2 points are A(17, 11) and B(22, 23). In how much time does a man travelling at 91 kmph cover the shortest possible distance between the two points A and B?
(a) $6\frac{2}{7}$ min (b) $8\frac{3}{7}$ min
(c) $8\frac{4}{7}$ min (d) $8\frac{6}{7}$ min

Q.4 Find the distance between the parallel lines $3x + 4y - 7 = 0$ and $6x + 8y + 3 = 0$.
(a) 4 units (b) $\frac{11}{10}$ units
(c) $\frac{17}{10}$ units (d) $\frac{13}{10}$ units

Q.5 The two vertices of the base of an isosceles triangle are $(2p, 0)$ and $(0, p)$. We also know that the equation of one of the sides of this triangle is $X = 2p$. What is the area of this isosceles triangle?
(a) $3.5\ p^2$ (b) $6.7\ p^2$ (c) $2.5\ p^2$ (d) $2\ p^2$

Q.6 Given four lines whose equations are $x + 2y - 3 = 0$, $2x + 3y - 4 = 0$, $3x + 4y - 7 = 0$, and $4x + 5y - 6 = 0$, then the lines are:
(a) Concurrent (b) Sides of a square
(c) Sides of a rhombus (d) None of these

Q.7 Points on the line $x + y = 4$ that lie at a unit distance from the line $4x + 3y - 10 = 0$ are:
(a) (3, 1) and (–7, 11) (b) (–3, 7) and (2, 2)
(c) (–3, 7) and (–7, 11) (d) None of these

Q.8 The number of points on the lines $4x - 3y + 7 = 0$ or $x - y + 3 = 0$, which are at a distance 5 units from the point (2, 5) are:
(a) 1 (b) 2 (c) 3 (d) 4

Q.9 If one vertex of an equilateral triangle is at (2, –1) and the base is $x + y - 2 = 0$, then find the length of each side.
(a) $\sqrt{3/2}$ (b) $\sqrt{2/3}$ (c) 2/3 (d) 3/2

Q.10 If the points (1, 3) and (5, 1) are two opposite vertices of a rectangle and the other two vertices lie on the line $y = 2x + c$, then find the value of c.
(a) 4 (b) –4
(c) 2 (d) None of these

Q.11 In the above question, the coordinates of the other two vertices are:
(a) (2, 0), (4, –4) (b) (2, 4), (4, 0)
(c) (–2, 0), (–5, 1) (d) None of these

Q.12 If the vertices of diagonal of a square are (–2, 4) and (–2, 2), then its other two vertices are at:
(a) (1, –1), (5, 1) (b) (1, 1), (5, –1)
(c) (1, 1), (–5, 1) (d) None of these

Q.13 If the sum of the distances of a point from two perpendicular lines in a plane is 1, then it will form a _____ structure.
(a) square (b) circle
(c) straight line (d) two intersecting lines

Q.14 If a, b, and c are in AP, then $ax + by + c = 0$ will always pass through a fixed point whose coordinates are:
(a) (1, –2) (b) (–1, 2)
(c) (1, 2) (d) (–1, –2)

Q.15 The points (2, 1), (8, 5), and $(x, 7)$ lie on a straight line. The value of x is:
(a) 10 (b) 11 (c) 12 (d) $11\frac{2}{3}$

Q.16 Find the radius of the circle $x^2 + y^2 - 2x - 4y - 20 = 0$.
(a) 10 (b) 5
(c) 2.5 (d) None of these

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Q.17 The area of the figure formed by the lines $ax \pm by \pm c = 0$ is:

(a) $\frac{c^2}{ab}$ (b) $\frac{2c^3}{ab}$

(c) $\frac{c^2}{2ab}$ (d) None of these

Q.18 The algebraic sum of the perpendicular distances from A(x_1, y_1), B(x_2, y_2), and A(x_3, y_3) to a variable line is zero, then the line passes through

(a) the orthocentre of ∇ABC.
(b) the centroid of ∇ABC.
(c) the circumcentre of ∇ABC.
(d) None of these

Q.19 Length of the median from B on AC where A(−1, 3), B(1, −1), and C(5, 1) is:

(a) $\sqrt{18}$ (b) $\sqrt{10}$ (c) $2\sqrt{3}$ (d) 4

Q.20 If A and B are two fixed points, then the locus of a point which moves in such a way that the angle APB is a right angle is:

(a) A circle (b) An ellipse
(c) A parabola (d) None of these

Q.21 The point (4, 1) undergoes the following three transformations successively:

I Reflection about the line $y = x$
II Translation through a distance 2 units along the positive direction of x–axis
III Rotation through an angle $\pi/4$ about the origin in the clockwise direction

The final position of the point is given by the coordinates.

(a) $\frac{1}{\sqrt{2}}, \frac{7}{\sqrt{2}}$ (b) $(-2, 7\sqrt{2})$

(c) $\left(-\frac{1}{\sqrt{2}}, \frac{7}{\sqrt{2}}\right)$ (d) $\left(\sqrt{2}, 7\sqrt{2}\right)$

Q.22 Find the centre of the circle $x^2 + y^2 - 2x - 4y - 20 = 0$.

(a) (1, 2) (b) (−1, −2)
(c) (2, 1) (d) (−2, −1)

Answers

Foundation

1. (d)	**2.** (a)	**3.** (b)	**4.** (a)	**5.** (a)	**6.** (d)	**7.** (c)	**8.** (b)	**9.** (a)	**10.** (b)
11. (a)	**12.** (a)	**13.** (b)	**14.** (a)	**15.** (b)	**16.** (c)				

Moderate

1. (c)	**2.** (c)	**3.** (c)	**4.** (c)	**5.** (c)	**6.** (d)	**7.** (a)	**8.** (d)	**9.** (a)	**10.** (b)
11. (a)	**12.** (c)	**13.** (a)	**14.** (a)	**15.** (b)	**16.** (b)	**17.** (b)	**18.** (b)	**19.** (b)	**20.** (a)
21. (a)	**22.** (a)								

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BENCHMARKING TEST 4

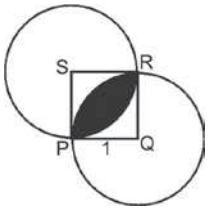
This test paper contains 30 questions of 4 marks each. One-third of the mark allotted to a particular question will be deducted in case of wrong answer.

Time Given: 70 Minutes | **Total Marks: 120** | **Qualifying Marks: 11**

Break-up of the Expected Percentile
(assuming this to be a CAT paper)

Marks	Expected percentile
82+	99+
77	98.5+
71	98
51	92
37	85
27	80

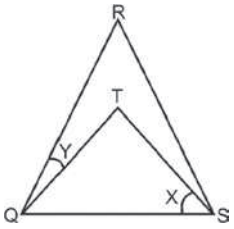
Q.1 Consider the figure given below:



PQRS is a square of side 1 unit and Q and S are the centres of the two circles. The area of the shaded portion is:

(a) $\frac{\pi}{2}$ (b) $\frac{1}{2}$ (c) $\frac{\pi}{4}-\frac{1}{2}$ (d) $\frac{\pi}{2}-1$

Q.2 In the figure given below, if QRS is an equilateral triangle and TRS is an isosceles triangle and $x = 47°$, then the value of y (in degrees) will be:

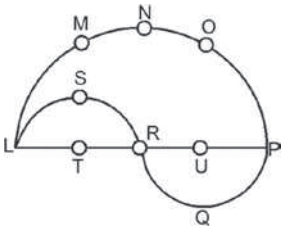


(a) 13° (b) 23° (c) 33° (d) 43°

Q.3 If the number representing the volume and the surface area of a cube are equal, then find the length of the edge of the cube in terms of the unit of measurement.

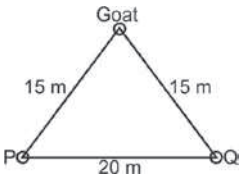
(a) 3 (b) 4 (c) 5 (d) 6

Q.4 LMNOP is semicircle with the centre R and diameter LP. LSR and RQP are also semicircles with centres at T and U, respectively, and diameters LR = RP = 1/2 LP. The ratio of the perimeter of LMNOP and LSRQP is:

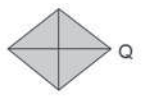


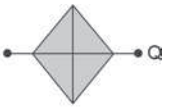
(a) 0.75:1 (b) 1:1 (c) 1:0.75 (d) 1.25:1

Q.5 A goat is tied to two poles P and Q with ropes that are 15 m long. P and Q are 20 m apart as shown in the given diagram:



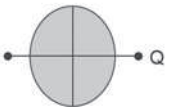
Which one of the following shaded portions indicates the total area over which the goat can graze?

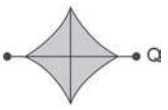
(a) 

(b) 

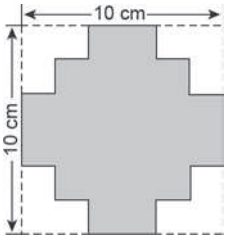
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(c) 

(d) 

Q.6 In the given figure, all line segments of the shaded portion are of the same length and at right angles to each other. The same can be cut out on a board of side 10 cm. What is the area of the shaded portion?



(a) 36 cm² (b) 48 cm² (c) 52 cm² (d) 64 cm²

Q.7 An equilateral triangle plate is to be cut into n number of identical small equilateral triangle plates. Which of the following can be a possible value of n?

(a) 196 (b) 216 (c) 256 (d) 296

Q.8 In an isosceles trapezium ABCD, P and Q are the mid-points of diagonals BD and AC, respectively, and PQ = 4 cm. The perpendicular drawn from A on BD is 5 cm and that from C on BD is 6 cm and each of the diagonals measure 8 cm. Find the length of the sides of AB and CD, given that the height of the trapezium is 4 cm.

(a) 12, 7 (b) 7, 15
(c) 5, 6 (d) None of these

Q.9 Consider a cylinder of radius 7 cm and height 14 cm. There is an intelligent ant that needs to travel on the surface of the cylinder. In order to travel to any point on the surface of the cylinder, the ant always chooses the shortest path. What is the distance, along the outer surface of the cylinder, of the point on the top of the cylinder for which the ant requires maximum time to reach?

(a) $\sqrt{340}$ (b) $\sqrt{680}$
(c) $\sqrt{240}$ (d) None of these

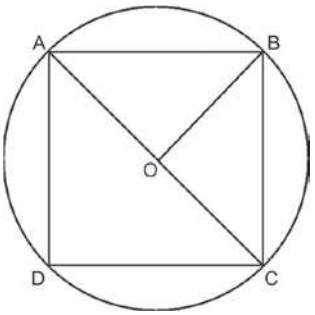
Q.10 There are 2 identical hollow cubical vessels A and B. A contains 64 identical solid spherical metallic balls. B contains 27 identical spherical balls of the same metal. In each cube, the balls are packed in such a manner that no more balls can be packed (of course balls in A will not be the same as balls in B). Which cube is heavier?

(a) A (b) B
(c) A = B (d) Cannot be determined

Q.11 There is a fan with 3 blades at 120° to each other. Its central circular disc has an area of 3π cm² and length of the blade is $\left(20-\sqrt{3}\right)$ cm. If the tips of the blades are joined so as to form an equilateral triangle, then what will be its area?

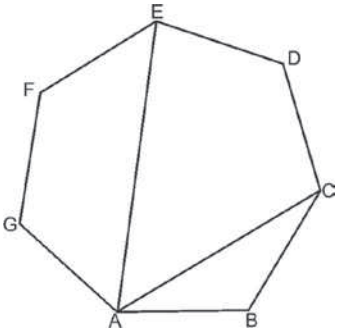
(a) 900 cm² (b) $300\sqrt{3}$ m²
(c) 900 + 9 cm² (d) None of these

Q.12 In the given figure, radius of the circle is 2.5 cm. O is the centre of the circle and OB bisects angle ABC. What can be the maximum area of ABCD?



(a) 2.5 (b) 12.5
(c) 12 (d) None of these

Q.13 In the given regular heptagon ABCDEFG, AB = a, AC = b, and AE = c. Which of the following statements are true?



(a) $c = b - a$ (b) $\frac{1}{b} = \frac{1}{c} + \frac{1}{a}$

(c) $\frac{1}{a} = \frac{1}{b} - \frac{1}{c}$ (d) None of these

Q.14 An insect wants to travel from one bottom corner to the opposite top corner inside a room of dimension $6 \times 8 \times 10$ units. What is the minimum distance that the insect needs to travel?

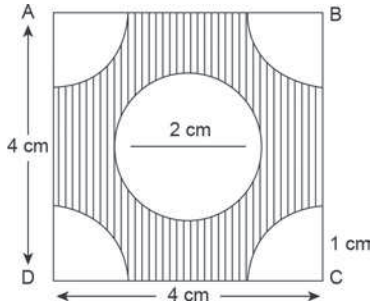
[Common sense hint: Insect can fly]

(a) $10\sqrt{2}$ units (b) $20\sqrt{2}$ units
(c) $15\sqrt{2}$ units (d) $25\sqrt{2}$ units

Q.15 In the figure given below, ABCD is a square of side length 4 units which has four symmetric cuts at all its corners. Find the area of the shaded portion.

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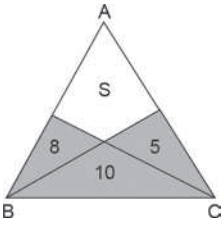
(a) 0.6 sq. units (b) 0.3 sq. units
(c) 0.9 sq. units (d) None of these

Q.16 If $(9p^2 + 4)$, $(2p + 7)$, and $(8p^2 - 1)$ are the lengths of three sides of a triangle ($p > 1$). Which of the following cannot be a value of p?
(a) 2 (b) $\sqrt{2}$
(c) $2\sqrt{2}$ (d) None of these

Q.17 AB is a line segment and P is its mid-point. Semi-circles are drawn with AP, PB, and AB as diameters on the same side of the line AB. A circle is drawn to touch all the three semicircles. Find the radius of this circle.
(a) 1/4 AB (b) 1/6 AB
(c) 1/7 AB (d) 1/8 AB

Q.18 In two circles C_1 and C_2, arcs of the same length subtend angles of 70° and 80° at their respective centres. What is the ratio of their radii (in the same order)?
(a) 7:8 (b) 8:7
(c) 8:15 (d) Cannot be determined

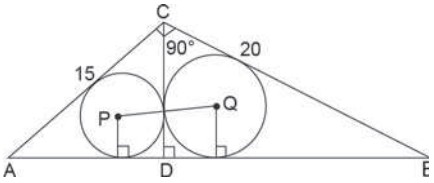
Q.19 Consider the figure given below:



Δ ABC is divided into four parts by straight lines from two of its vertices. The areas of the three triangular parts are 8 sq. units, 5 sq. units, and 10 sq. units. What is the area of the remaining part S (in sq. units)?
(a) 32 (b) 40
(c) 54 (d) None of these

Q.20 In a convex quadrilateral ABCD, the diagonal BD bisects neither the angle ABC nor the angle CDA. A point P lies inside ABCD and satisfies ∠PBC = ∠DBA and ∠PDC = ∠BDA. ABCD is cyclic quadrilateral if and only if:
(a) AP = CP (b) AB = BC
(c) BP = AP (d) None of these

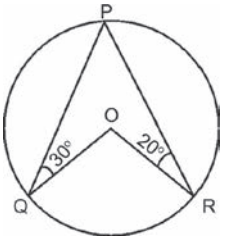
Q.21



In the above figure, ACB is a right-angled triangle. CD is the altitude. Circles are inscribed within the ΔACD and ΔBCD. P and Q are the centres of the circles. Find the distance of PQ.
(a) 5 (b) $\sqrt{50}$ (c) 7 (d) 8

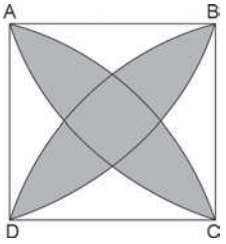
Q.22 If a, b, and c are the sides of a triangle, and $a^2 + b^2 + c^2 = bc + ca + ab$, then the triangle is:
(a) Equilateral (b) Isosceles
(c) Right-angled (d) Obtuse-angled

Q.23



In the figure given above, if ∠OQP = 30° and ∠ORP = 20°, then ∠QOR is equal to:
(a) 100° (b) 120° (c) 130° (d) 140°

Q.24 In the figure given below, ABCD is a square of side length $4\sqrt{2}$ units. Shaded parts are the sectors of the different circles with A, B, C, and D as the centres. What is the area (in sq. units) of the shaded region?

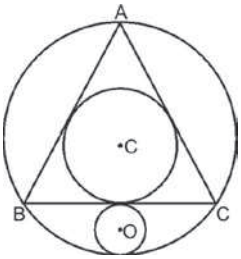


(a) $4\pi + 16$ (b) $8(\sqrt{2} + 1)\pi$
(c) $16\sqrt{2}\pi$ (d) None of these

Q.25 In the figure given below, ABC is an equilateral Δ of side length 3 units and circle with centre C is the in-circle. Another circle with centre O is drawn in such a way that side BC of the triangle ABC is tangent to the circle and circumcircle is having a point of tangency with this circle. What will be the length of the radius of the circle with centre O?

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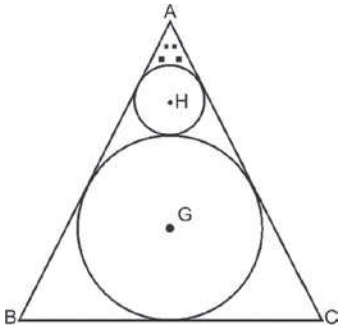
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(a) $\sqrt{3}$ (b) $\frac{\sqrt{3}+1}{2}$

(c) $\frac{\sqrt{3}-1}{2}$ (d) None of these

Q.26 In the figure given below, ΔABC is an equilateral triangle with side length 3 units. Triangles are arranged inside this triangle in such a way that the circle with centre G is the incircle of the triangle ABC and the circle with centre H is formed on the circle with centre G and so on. The next circle will be created on this circle with centre H and so on, and the infinite circles are created one above the other.



What is the sum of the radius of all such circles formed?

(a) 4.5 units (b) 2.25 units

(c) 3 units (d) 1.5 units

Q.27 An octagon is inscribed in a circle. One set of alternate vertices forms a square of area 5 units. The other set forms a rectangle area of 4 units. What is the maximum possible area for the octagon (in sq. units)?

(a) $3\sqrt{5}$ (b) $\frac{3}{\sqrt{5}}$

(c) $5\sqrt{3}$ (d) None of these

Q.28 What is the largest number of the quadrilaterals formed by four adjacent vertices of an convex polygon of n sides that can have an inscribed circle?

(a) n (b) $n/2$

(c) $n/3$ (d) None of these

Q.29 There are two circles with centres at A and B, respectively. The circle with centre A has a radius of 8 units and the circle with centre B has a radius of 6 units and the distance of AB is 12 units. Both the circles meet at points P and S. A line through P meets the circles again at Q and R (with Q on the larger circle) in such a way that QP = PR. Find the length of QP.

(a) $\sqrt{160}$ units

(b) $\sqrt{140}$ units

(c) $\sqrt{80}$ units

(d) None of these

Q.30 A circle of 4 units is taken. Now, n circles of the same radii are inserted in this circle ($1 \le n \le 10$, where n is a natural number) in such a way that they are encompassing the maximum possible areas of the circle and are inside the bigger circle along its circumference. (Obviously, for $n = 1$, radius of the inside circle will be same as the radius of the outside circle. Similarly, for $n = 2$, radius of the inside circle will be half of the outside bigger circle and so on.) For how many values of n, radius of the circle will be an integer?

(a) 2 (b) 3 (c) 4 (d) 5

Answers

Benchmarking Test 4

1. (d)	**2.** (a)	**3.** (d)	**4.** (b)	**5.** (b)	**6.** (c)	**7.** (c)	**8.** (b)	**9.** (b)	**10.** (a)
11. (b)	**12.** (b)	**13.** (d)	**14.** (a)	**15.** (b)	**16.** (c)	**17.** (b)	**18.** (b)	**19.** (d)	**20.** (a)
21. (b)	**22.** (a)	**23.** (a)	**24.** (d)	**25.** (d)	**26.** (b)	**27.** (a)	**28.** (b)	**29.** (d)	**30.** (b)

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MODULE

5 MISCELLANEOUS

- Set Theory
- Statistics
- Miscellaneous

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CHAPTER

21

Set Theory

LEARNING OBJECTIVES

After completion of this chapter, the reader should be able to understand:

- Types of sets and methods of their formation
- Definitions and properties
- Maxima and minima in set theory
- Kinds of questions asked in CAT
- Methods of solving the questions

INTRODUCTION

The concept of set theory has always been an important topic for the CAT, both in terms of direct and indirect applications of this concept and as a principle of counting. Questions from this topic are asked very regularly in the CAT and other major B-school exams. However, this concept is still in an evolutionary phase, as can be seen from the last few years' QA paper for this examination. The questions from this topic are not conventional. Maxima and minima can emerge as an important concept of set theory.

SET THEORY

A set is a collection of well-defined objects. The members of a set can be literally anything like letters of English alphabet, or different types of alphabets, or name of the countries, or numbers, or marks obtained by a student.

Given, capital letters are representing a standard set.

A = (a, b, c, d, e, f)—The first six letters of English alphabet.

B = (US, China, Japan, India)—The names of the top four countries in terms of their GDP.

C = (2, 4, 6, 8, 10)—The first five even natural numbers.

Here, A, B, and C are different sets that are representing different groups of objects.

In this chapter, we will confine ourselves with the following:

1. Type of sets
2. Solving techniques
3. Maxima and minima

Different Ways of Representing a Set

1. **Roster method:** With the help of this method, a set is represented by all the elements of it written under the brackets separated by commas.
   For example, A = {1, 2, 3, 4, 5}
2. **Set builder method:** With the help of this method, a set is represented by the common property of all its elements. It is written as:
   A = {x| x, $p(x)$ holds}
   A = (x:x, $p(x)$ holds}; where $p(x)$ is the common property shared by all the elements of set A.
   For example, A = {x e N |$x \leq 5$), which can be written in the roster method as A = {1, 2, 3, 4, 5}.

Types of Sets

1. **Empty or null set:** A set having no element is known as a empty or a null set and it is denoted as ϕ or { }.

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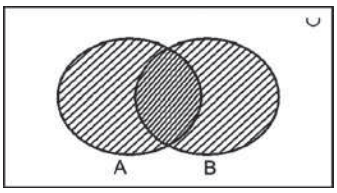
For example, A = set of even prime numbers excluding 2

2. **Singleton set:** A set having only one element is known as a singleton set.
   For example, A = set of even prime number(s)
3. **Finite set:** A set having a countable number of elements is known as the finite set.
   For example, A = set of odd numbers from 100 to 890
4. **Infinite set:** A set whose elements cannot be counted is known as infinite set.
   For example, A = set of all irrational numbers between 2 and 3
5. **Equal sets:** Two sets are said to be equal sets if all the elements of set A are included in set B and all the elements of set B are included in set A. If two sets A and B are equal, then it is represented by A = B, and if A and B are not equal, then it is written as A ≠ B, that is, all the elements of set A are not included in set B and all the elements of set B are not included in set A.
   For example, A = {a, b, c} and B = {c, b, a} are equal sets. Hence, in this case, we can write set A = set B or simply A = B.
6. **Subsets set:** Set A is said to be the subset of another set B if all the elements of set A are included in set B. 'Set A is subset of set B' is shown by A ⊆ B. We can now say that every element of set A is a member of set B.
   For example, if A = {a, b, c} and B = {a, b, c, d, e}, then A ⊆ B or A is a subset of B.

Some Important Results on Subsets

1. Every set is a subset of itself.
2. Every set has an empty set as its subset.
3. Total number of subsets of a set having n elements in 2^n.

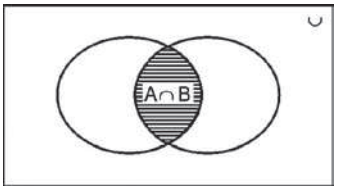
7. **Universal set:** A set which contains all the sets in a given context is a universal set.
   For example, when we are using sets containing natural numbers, N is the universal set.
   If A = {a, b, c}, B = {b, c, d}, and C = {c, d},
   then we can take ∪ = {a, b, c, d} as universal set.



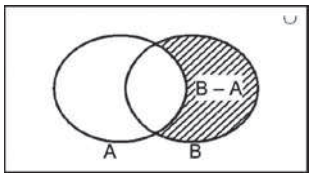
8. **Power set:** The collection of all the subsets of a set is known as the power set of that set. If A is the set, then a set containing all the subsets of A is known as the power set of A. It is denoted by P(A).
   Let A = {1, 2}, then the number of subsets of this set will be 2^2 and the subsets are {}, {1}, {2}, and {1, 2}, and the set containing all these four sets is known as the power set represented as P(A).
9. **Venn Diagram:** Swiss mathematician Euler first gave the idea of representing sets by a diagram. Later, British mathematician Venn brought this into practice. So, it is known as Euler–Venn diagram or simply Venn diagram. In this way of representing sets, we use a closed curve, generally a circle, to denote sets and their operations.

OPERATIONS ON SETS

1. **Union of sets:** If there are two sets A and B, then union of A and B is defined as the set that have all the elements which belong to either A or B or both. It is represented by A ∪ B. In other words, all the elements which are present in atleast one of the sets will be counted as the elements of union of the sets.
2. **Intersection of sets:** If there are two sets A and B, then the intersection of A and B is defined as the set that have all the elements which belong to both A and B.
   It is represented by A ∩ B.
   For example, find A ∪ B and A ∩ B if A = {1, 2, 3, 4, 5} and B = {2, 4, 6}.
   A ∪ B = {1, 2, 3, 4, 5, 6}
   A ∩ B = {2, 4}



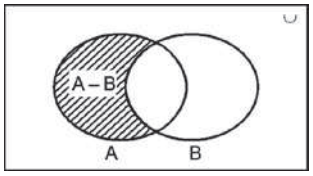
3. **Disjoint sets:** Two sets are said to be disjoint if A ∩ B = 0, that is, not a single element is common to both of these two sets.
   For example, If A = {Set of all odd numbers} and B = {Set of all even numbers}, then set A and set B are disjoint sets.



4. **Difference of sets:** For two sets A and B, A – B is the set of all those elements of A that do not belong to B. Similarly, B – A is the set of all those elements of B that do not belong to A.
   For example, A = {1, 2, 3, 4, 5, 6} and B = {2, 4, 8}, then A – B = {1, 3, 5, 6,} and B – A = {8}

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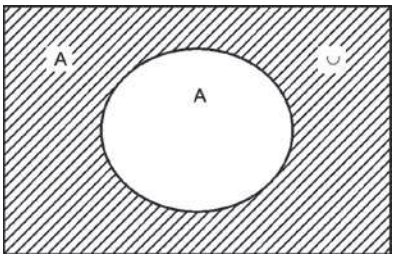
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5. **Complement of a set:** If U is the universal set and set A is such that $A \subseteq U$, then complement of the set A is defined as $U - A$ and represented as A' or Ac
For example, U = {Set of all prime numbers} and A = {Set of all even prime numbers}, then U – A = {Set of all odd prime numbers}.

ALGEBRAIC LAWS IN SET THEORY

1. **Commutative laws:** For any two sets A and B,
$A \cup B = B \cup A$ and $A \cap B = B \cap A$



2. **Associative laws:** For any three sets A, B, and C,
$(A \cup B) \cup C = A \cup (B \cup C)$ and
$A \cap (B \cap C) = (A \cap B) \cap C$
3. **Distributive laws:** For any three sets A, B, and C,
$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ and
$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
4. **Identity laws:**
$A \cup \phi = A$, where ϕ is the null set
$A \cap U = A$, where U is the universal set
5. **De–Morgan's law:** If A and B are two sets, then
$(A \cup B)' = A' \cap B'$
$(A \cap B)' = A' \cup B'$
6. $A \cup A = A$ and $A \cap A = A$

Some Important Results

If A and B are any two sets, then

1. $A - B = A \cap B'$
2. $B - A = B \cap A'$
3. $(A - B) \cup B = A \cup B$
4. $(A - B) \cap B = f$
5. $(A - B) \cup (B - A) = (A \cup B) - (A \cap B)$

Some Important Results on the Number of Elements in Sets

If A, B, and C are three finite sets, then

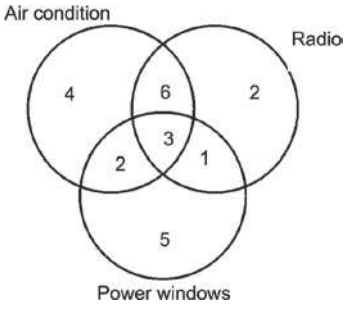
1. $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
2. $n(A \cup B) = n(A) + n(B)$, if set A and set B are two disjoint sets.
3. $n(A - B) = n(A) - n(A \cap B)$
4. $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$
5. $n(A' \cup B') = n((A \cap B)') = n(U) - n(A \cap B)$
6. $n(A' \cap B') = n((A \cup B)') = n(U) - n(A \cup B)$

Example 1 A survey on a sample of 25 new cars being sold at a local auto dealer was conducted to see which of the three popular options—air conditioning, radio, and power windows—were already installed. The following information were found in the survey:

(a) Fifteen cars had air conditioning.
(b) Two cars had air conditioning and power windows but no radios.
(c) Twelve cars had radio.
(d) Six cars had air conditioning and radio but no power windows.
(e) Eleven cars had power windows.
(f) Four cars had radio and power windows.
(g) Three cars had all the three options.

What is the number of cars that had none of the options?
(a) 4 (b) 3 (c) 1 (d) 2

Solution From the given conditions, we have



When we add all the values, we get a total of 23 cars.

So, 2 cars do not have the air conditioning or the radio, or the power windows.

Example 2 If A = {1, 3, 5, 7} and B = {1, 2, 3, 4}, then what is the value of (A U B) and (A ∩ B)?

Solution (A U B) = {1, 2, 3, 4, 5, 7}
(A ∩ B) = {1, 3}

Example 3 How many numbers from 1 to 100 are not divisible by either 2 or 4 or 5?

Solution Let us first understand the meaning of the statement given in the question.

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It is given that the numbers are from 1 to 100; so, while counting, we will include both the limits, that is 1 and 100. Had the question been, 'How many numbers in between 1 to 100 are …', then we would not have included either 1 or 100.

Now to solve this question, we will first find out the number of numbers from 1 to 100 that are divisible by either 2 or 5 (since all the numbers that are not divisible by 2 will not be divisible by 4 also, then we do not need to find the numbers divisible by 4). Then, we will subtract this from the total number of numbers, that is 100. It can be seen below.

Total number of numbers = numbers that are divisible + numbers that are not divisible

So, $n(2 \cup 5) = n(2) + n(5) - n(2 \cap 5)$

Now, $n(2) = 50$

$n(5) = 20$

$n(2 \cap 5) = 10$

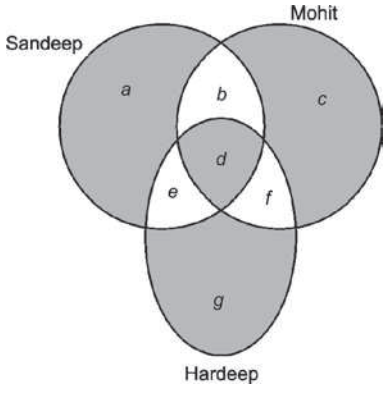
$n(2 \cup 5) = 50 + 20 - 10 = 60$

Numbers that are not divisible = total number of numbers – numbers that are divisible

$= 100 - 60 = 40$

Example 4 Sandeep, Mohit, and Hardeep are three financial analysts working at Due North Consultants. The sum of the number of projects handled by Sandeep and Hardeep individually is equal to the number of projects in which Mohit is involved. All three consultants are involved together in 6 projects. Sandeep works with Mohit in 14 projects. Hardeep has 2 projects with Mohit but without Sandeep, and 3 projects with Sandeep but without Mohit. The total number of projects is one less than twice the number of projects in which more than one consultant is involved. In how many projects is Mohit involved alone?

Solution Let us see the situation given above through the Venn–diagram.



$a + g = b + d + f + c$ (i)

$d = 6$

$b + d = 14$; so, $b = 8$

$f = 2$ and $e = 3$

Finally, $a + b + c + d + e + f + g$

$= 2(b + d + e + f) - 1$ (ii)

Using equations (i) and (ii),

c = Number of projects handled by Mohit alone = 1

MAXIMA AND MINIMA IN SET THEORY

Possibility of maxima and minima occurs in a set theory due to the fact that intersections of two or more than two values, in most of the cases, have a finite range. In other words, they can lie in between x and y, where x and y can be any number depending upon the given conditions.

Let us understand this with the following example:

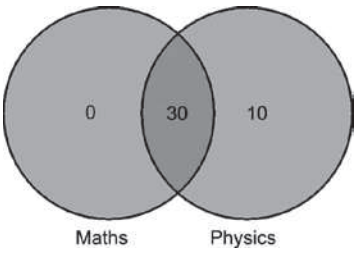
Example 5 In a class of 50 students, a test for 2 subjects Maths and Physics was conducted. Thirty students passed in Maths and 40 students passed in Physics.

- What is the maximum number of students who passed in both the subjects?
- What is the minimum number of students who passed in both the subjects?

Solution

- Since 30 students passed in Maths, so we can assume that these are the same 30 students out of 40 students who passed Physics also. So, the maximum number of students who passed in both the subjects = 30. (Remember that in this case, there was no restriction regarding the number of students who passed any of the subject. Had it been given that all these 50 students are passing at least one subject, then we would not have been in a position to do this.)

  The Venn- diagram representation of the same is given below.



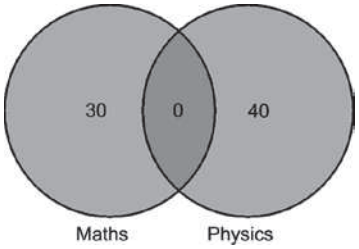
- To find out the minimum number of students who passed in both the subjects, intersection (Maths ∩ Physics) should be minimum, that is individually Maths and Physics each should be maximum.

Let us do a bit of hit-and-trial before we finally move on to do it with its proper solution.

If we take (Maths ∩ Physics) = 0, then the situation would be like:

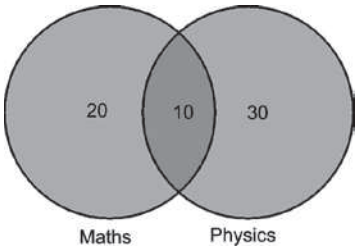
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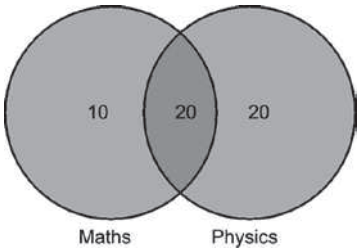
It can be seen that total number of students in this situation = 30 + 40 = 70, which is not true.

Similarly, if we take (Maths ∩ Physics) = 10, then the situation would be like:



It can be seen that total number of students in this situation = 20 + 10 + 30 = 60, which is again not true.

Hence, now we can conclude that since the number of students = 50, so (Maths ∩ Physics) = 20, and this is the minimum value of (Maths ∩ Physics).



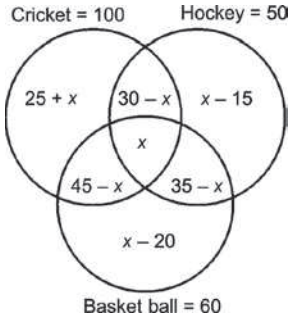
Alternatively, question (ii) can be done by the following set theory formula also: n(Maths ∪ Physics) = n(Maths) + n(Physics) − n(Maths ∩ Physics).

Remember In case of two sets, by using this formula of set theory, we get the minimum value, but if we use the same formula in case of three sets, then we would obtain the maximum value.

Example 6 There are 200 students in a school. Out of these, 100 students play cricket, 50 students play hockey, and 60 students play basketball. Thirty students play both cricket and hockey, 35 students play both hockey and basketball, and 45 students play both basketball and cricket.

- What is the maximum number of students who play at least one game?
- What is the maximum number of students who play all the 3 games?
- What is the minimum number of students playing at least one game?
- What is the minimum number of students playing all the 3 games?

Solution



Consider the Venn diagram given above.

First, we will convert all the values in terms of x, which can be seen above.

Since the number of students cannot be negative,

$x - 15 \geq 0$

$\therefore x - 20 \geq 0$

For the minimum number of students playing all three games, i.e., $x = 20$.

For the maximum value of x, again none of the categories should have –ve number of students.

$\therefore 30 - x \geq 0$

$x \leq 30$

If x is more than 30, then $30 - x$ would be –ve which is not possible.

Total number of students playing at least one game,

$= 100 + x - 15 + 35 - x + x - 20$

$= 100 + x$

So, the minimum number of students playing at least one game = 100 + 20 = 120

Hence, the maximum number of students playing at least one game = 100 + 30 = 130

Example 7 In a class of 50 students, 70% students pass in QA and 60% pass in RC. What is the minimum percentage of students who pass in both the papers?

Solution We know that $n(A \cup B) = n(A) + n(B) - n(A \cap B)$.

Assuming that $n(Q)$ = number of students passing QA and $n(R)$ = number of students passing RC.

Hence, $n(Q \cup R) = n(Q) + n(R) - n(Q \cap R)$

or $n(Q \cap R) = n(Q) + n(R) - n(Q \cup R) = 70\% + 60\% - n(Q \cup R)$

Now, the minimum value of $n(Q \cap R)$ will occur for the maximum value of $n(Q \cup R)$. The maximum possible value of $n(Q \cup R) = 100\%$

So, the minimum value of $n(Q \cap R) = 30\%$

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Example 8 In a class of 50 students, 70% students pass in QA, 90% students pass in EU, and 60% students pass in RC. What is the minimum percentage of students who pass in all the papers?

Solution We can do this question by using either the formula for three sets or we can simply keep on applying the formula for two sets required, for a number of times.

We know that $n(A \cup B) = n(A) + n(B) - n(A \cap B)$.

Assuming that $n(Q)$ = number of students passing QA, $n(E)$ = number of students passing EU, and $n(R)$ = number of students passing RC.

Hence, $n(Q \cup R) = n(Q) + n(R) - n(Q \cap R)$

or $n(Q \cap R) = n(Q) + n(R) - n(Q \cup R) = 70\% + 60\% - n(Q \cup R)$

Now, the minimum value of $n(Q \cap R)$ will occur for the maximum value of $n(Q \cup R)$. The maximum possible value of $n(Q \cup R) = 100\%$

So, the minimum value of $n(Q \cap R) = 30\%$

Now, the minimum value of EU, QA, and RC will be obtained by finding the minimum of EU and (QA $\cap$ RC).

The minimum value of EU and (QA $\cap$ RC) = 90% + 30% − 100% = 20%

Example 9 In a class of 50 students, 70% students pass in QA, 90% pass in EU, 85% pass in DI. and 60% pass in RC. What is the minimum percentage of students who pass in all the papers?

Solution This question is just the extension of the earlier question.

The minimum value of EU and (QA $\cap$ RC) = 90% + 30% − 100% = 20%

The minimum value of DI and EU and (QA $\cap$ RC) = 20% + 85% − 100% = 5%

Practice Exercises

Warm Up

Direction for Questions 1 to 5: Read the passage below and solve the questions based on it.

A newspaper agent delivers the morning papers to 240 families. Ninety buy only *Times of India*, 15 buy *Times of India* and *The Telegraph*, 100 buy *The Telegraph*, 30 buy *The Telegraph* and *The Hindu*, and 30 buy only *The Hindu*. No family buys all the three papers.

Q.1 How many buy the *Times of India*?
(a) 120 (b) 110 (c) 125 (d) 150

Q.2 How many buy *The Telegraph* only?
(a) 50 (b) 55 (c) 45 (d) 65

Q.3 How many people read only one paper?
(a) 165 (b) 175 (c) 145 (d) 195

Q.4 Which newspaper is the most popular?
(a) *The Hindu* (b) *The Telegraph*
(c) *Times of India* (d) None of these

Q.5 Which is the least popular newspaper?
(a) *The Hindu* (b) *The Telegraph*
(c) *Times of India* (d) None of these

Direction for Questions 6 and 7: Read the passage below and solve the questions based on it.

Of the 400 members of a sport club, 144 play squash (S), 135 play badminton (B), and 156 play cricket (C). Each member plays at least one game. The members who play badminton and cricket both must play squash also. For every three members who play at least two games, there are two members who play all the three games.

Q.6 Find the number of members who play all the three games.
(a) 14 (b) 23
(c) 11 (d) None of these

Q.7 Find the number of members who play only squash.
(a) 125 (b) 122 (c) 101 (d) 123

Direction for Questions 8 and 9: Read the passage below and solve the questions based on it.

The following are the statistics of 22 students of Career Zone who appeared for CAT. Fifteen students got calls from IIM-A, 12 students got calls from IIM-B, and 8 students got calls from IIM-C. Six students got calls from IIM-A and IIM-B, 7 students got calls from IIM-B and IIM-C, and 4 students got calls from IIM-A and IIM-C, and 4 students got calls from all the three IIMs.

Q.8 How many students of Career Zone got a call from IIM-A only?
(a) 12 (b) 9
(c) 7 (d) None of these

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Q.9 How many students of Career Zone got a call from IIM-C only?
(a) 3 (b) 4 (c) 1 (d) 2

Direction for Questions 10 to 12: Read the passage below and solve the questions based on it.

In order to study the breakfast habits of the people in the city of Longkong, a survey of 400 families was conducted. Of these 400 families, 156 had cereals, 288 had eggs, and 300 had toast. It was found that 212 families had egg and toast, and 104 had cereals and toast, and 128 had egg and cereals, and 84 had all the three.

Q.10 How many families had neither toast nor cereals nor egg?
(a) 9 (b) 16 (c) 32 (d) 64

Q.11 How many had egg or toast but not cereals?
(a) 196 (b) 252 (c) 228 (d) 238

Q.12 How many families had only two items?
(a) 192 (b) 182 (c) 212 (d) 312

Q.13 In a survey of 100 people, it was found that 60 people read *India Today*, 55 read *Outlook*, and 10 read neither of the two magazines. How many people read both the magazines?
(a) 45 (b) 35
(c) 25 (d) None of these

Q.14 In a class of 50 students, 28 like pop music and 20 like classical music. If 10 students like one of the two kinds of music, then find the number of the students who like both kinds of music?
(a) 8 (b) 5 (c) 7 (d) 6

FOUNDATION

Direction for Questions 1 to 5: Read the passage below and solve the questions based on it.

During 2001 census, 8000 people were surveyed in an extremely backward village, where most of the people were suffering from a certain number of diseases. In that village, 1264 people were suffering from Japanese fever and dehydration; 2976 people were suffering from dehydration; 3472 people from Japanese fever and cholera; 4720 people from cholera, and 1008 people from cholera, and dehydration. Three hundred people suffered from all the three given diseases.

Q.1 Find the number of persons who suffered from dehydration only.
(a) 586 (b) 368
(c) 492 (d) None of these

Q.2 Find the number of persons who suffered from dehydration or cholera but not Japanese fever.
(a) 2552 (b) 2252 (c) 3516 (d) 3216

Q.3 Find the number of persons who suffered from Japanese fever but neither cholera nor dehydration.
(a) 1258 (b) 1256
(c) 1254 (d) Cannot be determined

Q.4 Find the number of persons who suffered from dehydration but not cholera.
(a) 2268 (b) 2976 (c) 1968 (d) 2676

Q.5 Find the number of persons who suffered from cholera but not dehydration.
(a) 3712 (b) 4012
(c) 4920 (d) Cannot be determined

Q.6 In a class of 120 students, students are participating in the Sudoku or crossword classes. The ratio of students who participate in Sudoku to the number of students who participate in crossword is 2:1. Forty-five students do not participate in any of these two. What is the number of students who participate in Sudoku?
(a) 50 (b) 80 (c) 70 (d) 84

Q.7 In a group of 120 persons, there are 80 Biharis and 40 Punjabis. Further, 70 persons in the group are Muslims and the remaining are Hindus. Then, the number of Biharis who are also Muslims in the group is:
(a) Between 30 and 70
(b) Exactly 20
(c) Between 15 and 25
(d) Between 20 and 25

Q.8 Thirty monkeys went to a picnic. Twenty-five monkeys chose to irritate buffaloes, and 20 monkeys chose to irritate cows. How many monkeys chose to irritate both buffaloes and cows?
(a) 10 (b) 15 (c) 5 (d) 20

Q.9 A survey shows that 89% of the Australians like butter, whereas 79% like chocolate. It is known that they like at least one of the two. What is the percentage of Australians who like both butter and chocolate?
(a) 13% (b) 15%
(c) 11% (d) None of these

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Q.10 The market research department of a certain breakfast cereal firm interviewing 100 persons found that on a certain morning for breakfast

(i) 72 had cereals, 39 had fruits, and 75 had toast.

(ii) 32 had fruits and cereals, 53 had cereals and toast, and 26 had toast and fruits.

(iii) 21 had all the three.

How many of those interviewed had neither cereals nor toast nor fruits?

(a) 1 (b) 2

(c) 4 (d) None of these

Q.11 If S and T are two sets such that S has 21 elements, T has 32 elements, and S ∩ T has 11 elements, then how many elements does S ∩ T have?

(a) 36 (b) 38 (c) 42 (d) 40

Q.12 In a group of 52 persons, 16 drink tea but not coffee and 33 drink tea. Then, what is the number of persons who take coffee but not tea?

(a) 11 (b) 19

(c) 23 (d) Cannot be determined

Direction for Questions 13 to 16: Read the passage below and solve the questions based on it.

Out of 10,000 persons surveyed, 3700 liked City A, 4000 liked City B, and 5000 liked City C. Seven hundred persons liked Cities A and B, 1200 liked Cities A and C, and 1000 liked Cities B and C. Each person liked at least one city.

Q.13 Find the number of persons who liked all the three cities.

(a) 100 (b) 200 (c) 300 (d) 400

Q.14 Find the number of persons who liked at least two cities as a percentage of number of persons who liked exactly one city.

(a) 32.4% (b) 29.8% (c) 20.83% (d) 33.33%

Q.15 Find the number of persons who liked exactly two cities as a percentage of the number of persons who liked at least one city.

(a) 21.5% (b) 19.16%

(c) 25.5% (d) None of these

Q.16 Find the number of persons who liked City A and City B but not City C.

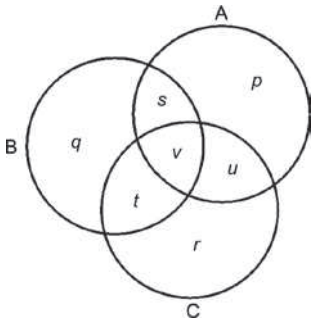
(a) 400 (b) 5500

(c) 600 (d) None of these

Q.17 A group of 50 students appeared for Physics and Mathematics examinations. Thirty-eight students passed in Physics and 37 students passed in Mathematics. If 30 students passed in both the subjects, then what is the number of students who failed in both the subjects?

(a) 2 (b) 3 (c) 4 (d) 5

Q.18 In the given diagram, Circle A represents teachers who can teach Physics, Circle B represents teachers who can teach Chemistry, and Circle C represents those who can teach Mathematics. Among the regions marked p, q, r, …, the one which represents teachers who can teach Physics and Mathematics but not Chemistry is:



(a) v (b) u (c) s (d) t

Q.19 In a group of 70 people, 37 like coffee, 52 like tea, and each person likes at least one of the two drinks. How many people like both coffee and tea?

(a) 19 (b) 21 (c) 17 (d) 15

Direction for Questions 20 and 21: Read the passage below and solve the questions based on it.

In a school with 727 students, 600 students take Mathematics and 173 students take both Mathematics and Physics. Each student is taking at least one of the two subjects.

Q.20 How many students took Physics?

(a) 250 (b) 300 (c) 280 (d) 290

Q.21 How many students took Physics only?

(a) 300 (b) 280 (c) 127 (d) 173

Q.22 Of the members of three athletic teams in a certain school, 21 are on the basketball team, 26 on the hockey team, and 29 on the football team. Fourteen play hockey and football, 12 play football and basketball, 15 play hockey and football, and 8 are on all the three teams. How many members are there altogether?

(a) 38 (b) 47 (c) 51 (d) 43

Direction for Questions 23 and 24: Read the passage below and solve the questions based on it.

In a survey of 500 students, it was found that 255 play table tennis, 195 play lawn tennis, 115 play tennis, 45 play table

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tennis and tennis, 70 play table tennis and lawn tennis, and 50 play lawn tennis and tennis. There are 80 students who do not play any of the three games.

Q.23 What is the number of students who play all the three games?
(a) 30 (b) 20
(c) 50 (d) None of these

Q.24 What is the number of students who play exactly one game?
(a) 295 (b) 300
(c) 325 (d) None of these

Q.25 In a party, 70% of the people drank coffee, 65% drank soft drinks, 27% did not drink anything, whereas 248 people drank both. What is the number of people who attended the party?
(a) 400 (b) 600 (c) 800 (d) 900

Direction for Questions 26 to 29: Read the passage below and solve the questions based on it.

In a class of 106 students, each student studies at least one of the three subjects Maths, Political Science, and Chemistry. Fourty-eight of them study Maths, 51 study Political Science, and 53 study Chemistry. Sixteen study both Maths and Political Science, 17 study both Maths and Chemistry, and 18 study both Political Science and Chemistry.

Q.26 What is the number of students who study exactly two subjects?
(a) 31 (b) 32 (c) 33 (d) 36

Q.27 What is the number of students who study more than one subject?
(a) 39 (b) 41 (c) 40 (d) 42

Q.28 What is the number of students who study all the three subjects?
(a) 5 (b) 6 (c) 7 (d) 4

Q.29 What is the number of students who study exactly one subject?
(a) 45 (b) 55 (c) 65 (d) 70

Q.30 In a survey conducted in Patna, it was found that $\frac{3}{4}$ th of the town owns colour TV, 85% of the people own refrigerator, and every 4 in 5 in the town owns music system. What is the minimum percentage of people who have all the three?
(a) 30% (b) 40%
(c) 55% (d) None of these

Moderate

Q.1 In a referendum about three proposals, 78% of the people were against at least one of the proposals, 50% of the people were against 1st proposal, 30% of the people were against 2nd proposal, and 20% of the people were against 3rd proposal. If 5% of the people were against all the three proposals, what percentage of people were against more than one of the three proposals?
(a) 10 (b) 12 (c) 17 (d) 22

Direction for Questions 2 to 4: Read the passage below and solve the questions based on it.

In a recent survey conducted by Cable TV Association among the people who watch DD, Zee, and Star TV, it is found that 80% of the people watched DD, 22% watched Star TV, and 15% watched Zee.

Q.2 What is the maximum percentage of people who can watch all the three channels?
(a) 12.5% (b) 8.5%
(c) 15% (d) Data insufficient

Q.3 If 5% of the people watched DD and Zee, and 10% watched Star TV and DD, then what percentage of the people surveyed watched Star TV and Zee only?
(a) 2% (b) 5%
(c) 8.5% (d) Cannot be determined

Q.4 In the previous question, what percentage of people watched all the three channels?
(a) 3.5% (b) 0%
(c) 8.5% (d) Cannot be determined

Q.5 Eighty thousand students appeared in the last year's CAT consisting of three sections QA, DI, and EU. Twenty-four thousand students passed in QA; 31,000 students passed in DI; 35,000 students passed in EU. Further, 49,000 students passed in at least one of QA and DI; 59,000 students passed in at least one of DI and EU; 54,000 students passed in at least one of QA and EU. Calls for IIMs are sent to only those students who passed in all the three sections in CAT. If only 2000 students got the IIM calls, then what is the number of students who failed in all the three papers (either you can pass or fail)?
(a) 6000 (b) 8000 (c) 10,000 (d) 4000

Direction for Questions 6 to 9: Read the passage below and solve the questions based on it.

CAT 2010 was having four sections, namely QA, LR, EU, and RC. Following is the data pertaining to CAT 2010:

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Answer the following questions based on the following information:

- The number of students who appeared = 1867
- The number of students who passed in exactly two sections = 232
- The number of students who passed in exactly one section = 132
- The number of students who failed in all the four sections = 292
- The number of students who failed in QA only = 200
- The number of students who failed in LR only = 177
- The number of students who failed in EU only = 108
- The number of students who failed in RC only = 211

A student is assumed to be pass in CAT Examination if he is passed in all the four sections. Further, a student can either pass or fail in the exam.

Q.6 What is the number of students who passed CAT 2010?
(a) 508 (b) 1011
(c) 812 (d) None of these

Q.7 What is the number of students who failed because of failing in more than two sections?
(a) 108 (b) 324
(c) 672 (d) None of these

Q.8 What is the number of students who failed in QA?
(a) 200 (b) 492
(c) 711 (d) Cannot be determined

Q.9 What is the number of students who failed in EU?
(a) 400 (b) 108
(c) 732 (d) Cannot be determined

Direction for Questions 10 to 13: Read the passage below and solve the questions based on it.

Five per cent of the passengers who boarded Delhi–Lucknow Shatabdi express on 27th April 2005 did not like coffee, tea, and lassi, and 10% like all the three, 20% like coffee and tea, 25% like lassi and coffee, and 25% like lassi and tea. Fifty-five per cent like coffee, 50% like tea, and 50% like lassi.

Q.10 The passengers who like only coffee is greater than the passengers who like only lassi by:
(a) 25% (b) 100% (c) 75% (d) 0 %

Q.11 The percentage of passengers who like both tea and lassi but not coffee is:
(a) 15 (b) 25 (c) 40 (d) 75

Q.12 The percentage of passengers who like at least 2 drinks out of coffee, tea, and lassi is:
(a) 30 (b) 45 (c) 50 (d) 60

Q.13 If the number of passengers is 180, then the number of passengers who like lassi only is:
(a) 10 (b) 18 (c) 27 (d) 36

Q.14 Sandeep, Mohit, and Hardeep are three financial analysts working at Due North Consultants. The sum of the number of projects handled by Sandeep and Hardeep individually is equal to the number of projects in which Mohit is involved. All three consultants are involved together in 6 projects. Sandeep works with Mohit in 14 projects. Hardeep has 2 projects with Mohit but without Sandeep, and 3 projects with Sandeep but without Mohit. The total number of projects is one less than twice the number of projects in which more than one consultant is involved. In how many projects Sandeep alone is involved?
(a) 1 (b) 0
(c) 3 (d) Cannot be determined

Q.15 There are a total of 78 persons in a big joint family. Each one of them watch at least one out of Ramayana, Mahabharat, and Chanakya. Of these, 36 watch Ramayana, 48 watch Mahabharat, and 32 watch Chanakya. If 14 people watch both Ramayana and Mahabharat, 20 people watch both Mahabharat and Chanakya, and 12 people watch both Ramayana and Chanakya, then find the ratio of the number of people who watch only Ramayana to the number of people who watch only Chanakya?
(a) 9:7 (b) 5:2 (c) 5:3 (d) 7:3

Q.16 In a class of 200 students, 70 played chess, 60 played hockey, and 80 played football. Thirty played chess and football, 30 played hockey and football, and 40 played chess and hockey. If 130 people played at least one game, then what is the number of people who played all the three games?
(a) 40 (b) 20
(c) 10 (d) None of these

Q.17 In a club, all the members are free to vote for one, two, or three of the candidates. Twenty per cent of the members did not vote, and 38% of the total members voted for at least two candidates. What percentage of the members voted for either one or three candidates if 10% of the total members who voted, voted for all the three candidates?
(a) 40% (b) 44%
(c) 36% (d) None of these

Q.18 In a class of 25 students, at least one of Mathematics or Statistics is taken by everybody. Twelve have taken Mathematics, and 8 have taken Mathematics but not Statistics. Find the difference in the number of students

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who have taken Mathematics and Statistics and those who have taken Statistics but not Mathematics?

(a) 13 (b) 9
(c) 5 (d) Cannot be determined

Q.19 In a recent survey (conducted by HLL) of 1000 houses, washing machine, vacuum cleaners, and refrigerators were counted. Each house had at least one of these products. 400 had no refrigerators, 380 had no vacuum cleaners, and 542 had no washing machines. Two hundred and ninety-four had both a vacuum cleaner and washing machine, 277 had both a vacuum cleaner and a refrigerator, and 120 had both a refrigerator and a washing machine. How many had only a vacuum cleaner?

(a) 132 (b) 234
(c) 342 (d) None of these

Q.20 A group of tourists (104 in all) speak Bengali or Spanish or Chinese. Out of these, 66 speak Bengali, 74 speak Spanish, and 64 speak Chinese. Forty-four speak Bengali and Spanish, 40 speak Bengali and Chinese, and 50 speak Chinese and Spanish. What is the number of tourists speaking all the three languages?

(a) 24 (b) 34 (c) 44 (d) 54

Q.21 In the above question, the number of people speaking neither Spanish nor Bengali is:

(a) 6 (b) 8
(c) 22 (d) Cannot be determined

ADVANCED

Q.1 We define, for two sets A and B, the following operations:

(i) A – B is the set of all elements of set A that are not the elements of set B.
(ii) A ∪ B is the set of all elements that appear in A or B or both A and B.
(iii) A ∩ B is the set of all elements that are common to both sets A and B.

If P, Q, and R are three sets, then P – (Q – R) is equivalent to:

(a) (P – Q) ∪ (P ∩ R) (b) P – (Q ∪ R)
(c) P – (Q ∩ R) (d) (P – Q) ∪ (P – R)

Q.2 We define, for two sets A and B, the following operations:

(i) A – B is the set of all elements of set A that are not the elements of set B.
(ii) A ∪ B is the set of all elements that appear in A or B or both A and B.
(iii) A ∩ B is the set of all elements that are common to both sets A and B.

If P, Q, and R are three sets, then (P – Q) – (Q – R) is equivalent to:

(a) P ∩ (Q ∩ R) (b) P – (Q ∪ R)
(c) (P ∩ Q) ∪ R (d) None of these

Q.3 A survey was conducted regarding the hobbies of a class having 200 students. The survey revealed that 85% of the students have played cricket, 80% have played basketball, 75% have played hockey, and 70% have played tennis. What is the minimum number of students who must have played all the four games?

(a) 15 (b) 25
(c) 40 (d) None of these

Q.4 There are three groups A = {all men}, B = {all women}, and C = {men and women}, who have four members in common, and each pair of groups has 5 members in common. If each of the group has only eight members and if the common members are men only, then what is the minimum number of men among the three groups taken together?

(a) 13 (b) 14
(c) 15 (d) None of these

Direction for Questions 5 and 6: Read the passage below and solve these questions based on it.

Following is the result of an examination.

Out of 1000 students who appeared

(a) 658 failed in Physics.
(b) 166 failed in Physics and Chemistry.
(c) 372 failed in Chemistry, and 434 failed in Physics and Biology.
(d) 590 failed in Biology, and 126 failed in Biology and Chemistry.

Assuming that none of them passed in all the subjects.

Q.5 Find the number of students who failed in Chemistry but not Physics.

(a) 312 (b) 226 (c) 266 (d) 206

Q.6 Find the number of students who failed in Physics or Biology but not in Chemistry.

(a) 560 (b) 710
(c) 620 (d) None of these

Direction for Questions 7 and 8: Read the passage below and solve these questions based on it.

During break, 123 students go to the school canteen that sells pastries, ice cream, and patties. Thirty-six students buy patties and 10 students buy only pastries. Fifteen students buy ice cream and patties, 10 students buy ice cream and

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pastries, 4 students buy pastries and patties but not ice cream, and 11 students buy ice cream and patties but not pastries. Forty-two students buy ice cream.

Q.7 How many students buy none of the items?
(a) 40 (b) 50 (c) 55 (d) 60

Q.8 How many students buy at least two items?
(a) 27 (b) 30 (c) 25 (d) 35

Direction for Questions 9 to 12: Read the passage below and solve these questions based on it.

A and B are two sets (e.g., A = mothers, B = women). The elements that could belong to both the sets (e.g., women who are mothers) is given by the set $C = A \times B$. The elements which could belong to either set A or B or both is indicated by the set $D = A \cup B$. A set that does not contain any elements is known as a null set, represented by ϕ (e.g., if none of the women in the set B is a mother, then $C = A \times B$ is a null set or $C = \phi$).

Let N signify the set of all the vertebrates; M the set of all the mammals; D the set of all the dogs; F the set of all the fish; A the set of all the alsatian and P is a dog named Party.

Q.9 Given that $X = M \times D$ such that $X = D$, then which of the following is true?
(a) All dogs are mammals.
(b) Some dogs are mammals.
(c) $X = \phi$
(d) All mammals are dog.

Q.10 Given that $Y = F \times (D \times N) = \phi$. Which of the following is true?
(a) All fish are vertebrates.
(b) All dogs are vertebrates.
(c) Some fish are dogs.
(d) None of these

Q.11 If $Z = (P \times D) \cup M$, then which of the following is true?
(a) The elements of Z consist of Party, the dog, or any other mammal.
(b) Z implies any dog or animal.
(c) Z implies Party or any dog that is a mammal.
(d) Z is a null set.

Q.12 If $P \times A = \phi$ and $P \cup A = D$, then which of the following is true?
(a) Party and alsatians are dogs.
(b) Party is an alsatian.
(c) Party is not an alsatian.
(d) D is a null set.

Q.13 In a class of 200 students, 70 played cricket, 60 played hockey, and 80 played football. Thirty played cricket and football, 30 played hockey and football, 40 played cricket and hockey.
Find the maximum number of people playing all the three games and also the minimum number of people playing at least one game?
(a) 200, 100 (b) 30, 110
(c) 30, 120 (d) None of these

Q.14 If class with n students is organized into four groups keeping the following conditions:
- Each student belongs to exactly two groups and
- Each pair of groups has exactly one student in common.

What is the value of n?
(a) $n = 11$ (b) $n = 7$
(c) $n = 9$ (d) None of these

Q.15 A survey show that 63% of the Americans like cheese, whereas 76% like apples. If x% of the Americans like both cheese and apples, then find the range of x.
(a) $0 \le x \le 23\%$ (b) $0 \le x \le 39\%$
(c) $4 \le x \le 35\%$ (d) $6 \le x \le 33\%$

Q.16 In Ranchi, only two newspapers *The Hindustan* and *The Indian Express* are published. It is known that 25% of the city population reads *The Hindustan* and 20% reads *The Indian Express*, whereas 8% read both newspapers. It is also known that 30% of those who read *The Hindustan* but not *The Indian Express* look into advertisements, whereas 50% of those who read both *The Hindustan* and *The Indian Express* look into advertisement. What percentage of the population looks into advertisement?
(a) 8.25% (b) 6.5%
(c) 11.75% (d) Cannot be determined

Q.17 In a certain examination of 3 subjects, 35 students failed in Maths, 40 in Physics, and 40 in Chemistry. Twenty failed in Maths and Physics, 17 in Physics and Chemistry, and 15 in Maths and Chemistry. If 132 students did not fail in any subject, then what is the maximum number of students who have appeared for the examination?
(a) 95 (b) 200 (c) 210 (d) 246

Direction for Questions 18 to 20: Read the passage below and solve these questions based on it.

There are 120 players, having a chest number 1 through 120, seated in a circular pattern in the same order. Since the selector is involved in game fixing, he tries to ensure that none of the right candidate for right game should go. For this to happen, he starts selecting the players in a very arbitrary way for four games, namely shot-put, javelin throw, discus throw, and sprint: Player numbered 1 is selected for sprint and after that, every 11th player is selected for sprint. The counting continues around the circle repeatedly until it ends at player number 1.

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Similarly, starting with player number 1, every 9th player is selected for shot-put, every 7th player for javelin throw, and every 8th player for discus throw.

Q.18 How many players are not selected for any of the four events?
(a) 45 (b) 65
(c) 33 (d) None of these

Q.19 How many players are selected for all the four events?
(a) 1 (b) 3
(c) 5 (d) None of these

Q.20 How many players were selected neither for discus throw nor shot-put?
(a) 12 (b) 14
(c) 16 (d) None of these

Answers

Warm Up

1. (c)	**2.** (b)	**3.** (b)	**4.** (c)	**5.** (b)	**6.** (a)	**7.** (d)	**8.** (b)	**9.** (c)	**10.** (b)
11. (c)	**12.** (a)	**13.** (c)	**14.** (a)	**15.** (d)					

Foundation

1. (d)	**2.** (b)	**3.** (d)	**4.** (c)	**5.** (a)	**6.** (a)	**7.** (a)	**8.** (b)	**9.** (d)	**10.** (c)
11. (c)	**12.** (d)	**13.** (b)	**14.** (d)	**15.** (b)	**16.** (a)	**17.** (d)	**18.** (b)	**19.** (a)	**20.** (b)
21. (c)	**22.** (d)	**23.** (a)	**24.** (c)	**25.** (a)	**26.** (d)	**27.** (b)	**28.** (a)	**29.** (c)	**30.** (b)

Moderate

1. (c)	**2.** (c)	**3.** (d)	**4.** (d)	**5.** (a)	**6.** (d)	**7.** (d)	**8.** (d)	**9.** (d)	**10.** (b)
11. (a)	**12.** (c)	**13.** (b)	**14.** (d)	**15.** (c)	**16.** (b)	**17.** (d)	**18.** (b)	**19.** (d)	**20.** (b)
21. (b)									

Advanced

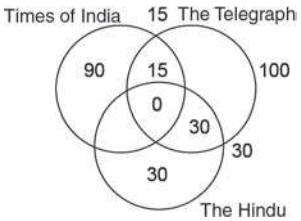
1. (a)	**2.** (d)	**3.** (d)	**4.** (d)	**5.** (d)	**6.** (d)	**7.** (b)	**8.** (c)	**9.** (a)	**10.** (c)
11. (a)	**12.** (c)	**13.** (b)	**14.** (d)	**15.** (b)	**16.** (d)	**17.** (c)	**18.** (a)	**19.** (b)	**20.** (d)

Hints and Solutions

Warm Up

Answers to Q.1 to 5:

Following is the Venn diagram:



Families who buy *The Telegraph* only = 100 – 30 – 15 = 55

Similarly, families who buy *Times of India* and *The Hindu* both = 240 – (100 + 30 + 90) = 20

1. Hence, option (c) is the answer.

2. Hence, option (b) is the answer.

3. Families who read only one newspaper = 90 + 30 + 55 = 175

Hence, option (b) is the answer.

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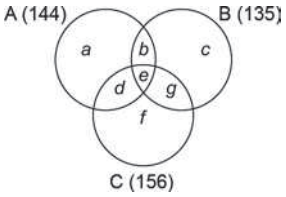
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4. Clearly, *Times of India* is the most popular newspaper as it is read by 125 families. Hence, option (c) is the answer.

5. *The Hindu* is the least popular newspaper as it is read by 80 families only. Hence, option (*a*) is the answer.

Answers to Q.6 and 7:

Following is the diagram:



Set A represents squash playing people, set B represents badminton playing people, and set C represents cricket playing people.

In the diagram given below, a = people who play only squash, b = people who play squash and badminton but not cricket, c = people who play only cricket, and so on.

6. It is given that $g = 0$ and $\frac{b+d+e+g}{e} = \frac{3}{2}$

or $2[b+d+e] = 3e$

$$b+d = \frac{e}{2} \quad \text{(i)}$$

According to the question, $A + B + C = 144 + 135 + 156$

$a + 2b + c + 2d + 3e + f = 435$ (ii)

It is also given that $a + b + c + d + e + f = 400$ (iii)

Subtracting equation (iii) from equation (ii) gives us, $b + d + 2e = 35$ (iv)

Substituting the value of $(b + d)$ from equation (i) in equation (iv) gives us

$$\frac{e}{2} + 2e = 35$$

Hence, $e = 14$

So, number of people who play all the three games = 14.

Hence, option (a) is the answer.

7. Number of players who play only squash = a

Using the data that we have calculated in previous question, $b + d + 2e = 35$ and $e = 14$

So, $b + d = 7$

Given that Set A = people who play squash = 144

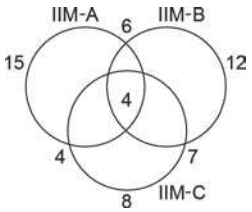
So, $a + b + d + e = 144$

$a + 7 + 14 = 144$

Hence, $a = 123$

Hence, option (d) is the answer.

Answers to Q.8 and 9:



Since 6 students got calls from both IIM-A and IIM-B and 4 students got calls from all the three IIMs, number of students who got calls from IIM-A and IIM-*B* only = $6 - 4 = 2$

Similarly, number of students who got calls from IIM-B and IIM-C only = $7 - 4 = 3$

Similarly, number of students who got calls from IIM-A and IIM-C only = $4 - 4 = 0$

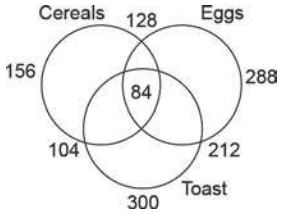
8. Students who got call from IIM-A only = $15 - [6 + 0] = 9$

Hence, option (b) is the answer.

9. Students who got call from IIM-C only = $8 - 7 = 1$

Hence, option (c) is the answer.

Answers to Q.10 to 12:



Families who have cereals and eggs only = $128 - 84 = 44$

Families who have eggs and toast only = $212 - 84 = 128$

Families who have cereals and toast only = $104 - 84 = 20$

Families who have cereals only = $156 - (128 + 20) = 8$

Families who have eggs only = $288 - (128 + 128) = 32$

Families who have toast only = $300 - (212 + 20) = 68$

10. Families who have neither of the three = $400 - [156 + 68 + 128 + 32] = 16$

Hence, option (b) is the answer.

11. Families who have eggs or toast but not cereals → $32 + 128 + 68 = 228$

Hence, option (c) is the answer.

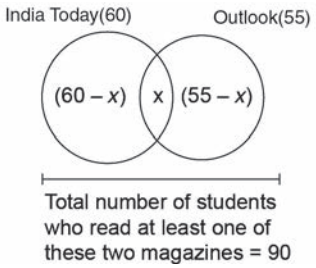
12. Families who had only 2 items = $44 + 128 + 20 = 192$

Hence, option (a) is the answer.

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13. From the following figure:



Method 1

Number of people who read at least one of the magazines = $n(India\ Today \cup Outlook) = 100 - 10 = 90$

$90 = 60 + 55 - n(India\ Today \cap Outlook)$

$n(India\ Today \cap Outlook) = 25$

Hence, option (c) is the answer.

Method 2

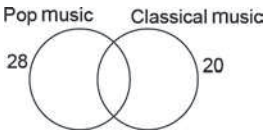
Assume that the number of people who read both magazines = x.

So, the number of people who read only *India Today* = $(60 - x)$ and the number of people who read only *Outlook* = $(55 - x)$

So, $60 - x + x + 55 - x = 90$ or $x = 25$

Hence, option (c) is the answer.

14.



Let the number of people who like both pop and classical music be x.

$\therefore$ People who like only pop music = $28 - x$

People who like only classical music = $20 - x$

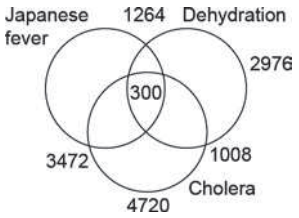
It is given that $28 - x + 20 - x = 10$

$48 - 2x = 10$ or $2x = 39$

Hence, $x = 19$

FOUNDATION

Answers to Q.1 to 5:



Dehydration and Japanese fever only → $1264 - 300 = 964$
Dehydration and cholera only → $1008 - 300 = 708$
Japanese fever and cholera only → $3472 - 300 = 3172$
Only dehydration → $2976 - (1264 + 708) = 1004$

1. Option (d) is the answer.

2. Dehydration or cholera but not Japanese fever = $1004 + 708 + 540 = 2252$

Hence, option (b) is the answer.

3. Only Japanese fever → cannot be determined, since we do not know the people suffering from Japanese fever.

4. Dehydration but not cholera = $1004 + 964 = 1968$

Hence, option (c) is the answer.

5. Cholera but not dehydration = $540 + 3172 = 3712$

Hence, option (a) is the answer.

6. $n(\text{Sudoku} \cup \text{Crossword}) = 120 - 45 = 75$

Let the total number of people in Sudoku be $2x$ and Crossword be x.

$2x + x = 75$

$x = 25$

So, the number of students participating in Sudoku = $2x = 50$ and number of students participating in Crossword = $x = 25$

Hence, option (a) is the answer.

8. $n(\text{buffaloes}) = 25$ and $n(\text{cows}) = 20$

$n(\text{buffaloes} \cup \text{cows}) = 30$

$n(\text{buffaloes} \cup \text{cows}) = n(\text{buffaloes}) + n(\text{cow}) - n(\text{buffaloes} \cap \text{cows})$

$30 = 25 + 20 - n(\text{buffaloes} \cap \text{cows})$ or $n(\text{buffaloes} \cap \text{cows}) = 15$

Hence, option (b) is the answer.

9. $n(\text{butter} \cup \text{chocolate}) = n(\text{butter}) + n(\text{chocolates}) - n(\text{butter} \cap \text{chocolate})$

Given that every Australian likes at least one of the two, and hence, $n(\text{butter} \cup \text{chocolate}) = 100\%$

or $100\% = 89\% + 79\% - n(\text{butter} \cap \text{chocolate})$

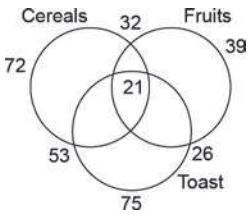
Therefore, $n(\text{butter} \cap \text{chocolate}) = 68\%$

Hence, option (d) is the answer.

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10.



Only cereals and fruits = 32 − 21 = 11

Only fruits and toast = 26 − 21 = 5

Only cereals and toast = 53 − 21 = 32

Only cereals = 72 − 32 − 32 = 8

Only fruits = 39 − 32 − 5 = 2

Neither of the three → only toast = 75 − 53 − 5 = 17

100 − [72 + 2 + 5 + 17] = 4

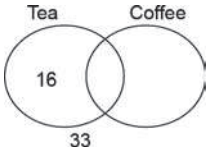
Hence, option (c) is the answer.

11. $n(S \cup T) = n(S) + n(T) - n(S \cap T)$

$n(S \cup T) = 21 + 32 - 11 = 42$

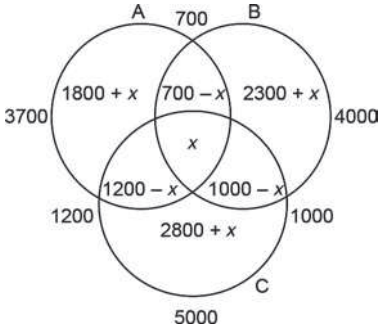
So, option (c) is the answer.

12.



Clearly, we cannot find out the number of people who drink only coffee because it is not known how many people people drink at least one of the two.

Answers to Q.13 to 16:



Let the number of people who liked all the three cities be x.

$3700 + 2300 + x + 1000 - x + 2800 + x = 10{,}000$

$x = 200$

Hence, option (b) is the answer.

14. Number of persons who liked at least 2 cities = 500 + 800 + 1000 + 200 = 2500

Number of persons who liked exactly 1 city = 2000 + 2500 + 3000 = 7500

$\therefore$ Required percentage $= \frac{2500}{7500} \times 100 = 33.33\%$

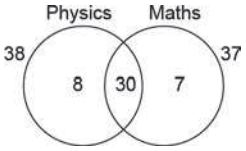
Hence, option(d) is the answer.

15. Number of persons who liked exactly 2 cities = 500 + 800 + 1000 = 2300

$\therefore$ Required percentage $= \frac{2300}{1000} \times 100 = 23\%$

16. Number of persons who liked Cities A and B but not City C = 2000 + 500 + 2500 = 5000

17.



Number of students who failed in both the subjects = 50 − 45 = 5

Hence, option (d) is the answer.

18. Clearly, we can see that (b) is the answer.

19. $n(\text{tea} \cup \text{coffee}) = n(\text{tea}) + n(\text{coffee}) - n(\text{tea} \cap \text{coffee})$

20. $n(\text{Maths} \cup \text{Physics}) = n(\text{Maths}) + n(\text{Physics}) - n(\text{Maths} \cap \text{Physics})$

$727 = 600 + n(\text{Physics}) - 173$

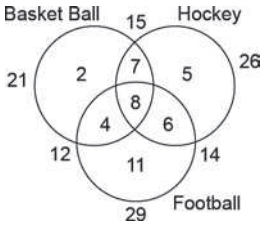
$n(\text{Physics}) = 300$

Hence, option (b) is the answer.

21. Physics only = 300 − 173 = 127

Hence, option (c) is the answer.

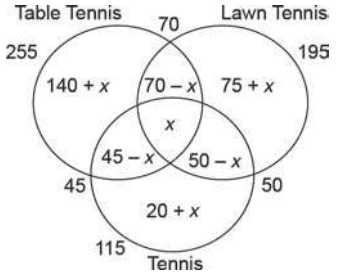
22.



Total number of members = 21 + 11 + 11 = 43

Hence, option (d) is the answer.

23.



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Let the number of people who play all the three be x.

$255 + 75 + x + 50 - x + 20 + x = 420$

$400 + x = 420$

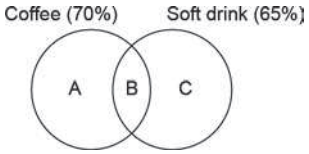
$x = 20$

Hence, option (b) is the answer.

24. Number of people who play exactly one game = 160 + 40 + 95 = 295

Hence, option (a) is the answer.

25. Method 1



From the diagram A + 2B + C = 70 + 65

$A + 2B + C = 135\%$ (i)

It is given that $A + B + C = 100 - 27 = 73\%$ (ii)

Then, from equation (i) and equation (ii)

B = 62% of the people and B = 248

Since 62% of the people = 248

Then, 100% of the people $= \frac{248}{62} \times 100 = 400$

Method 2

n(coffee ∪ soft drink) = n(coffee) + n(soft drink) − n(coffee ∩ soft drink)

$73 = 70 + 65 - n(\text{coffee} \cap \text{soft drink})$

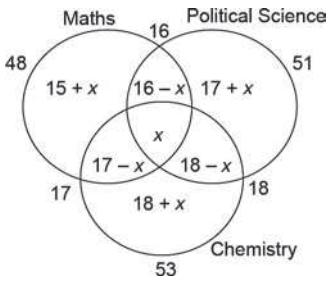
n(coffee ∩ soft drink) = 62%

62% = 248

$\therefore 100\% = \frac{248}{62} \times 100 = 400$

Hence, option (a) is the answer.

26.



Let the number of people who study all the three subjects be x.

$48 + 17 + x + 18 - x + 18 + x = 106$

$101 + x = 106$

$x = 5$

Number of people who study exactly 2 subjects = 11 + 13 + 12 = 36

Hence, option (d) is the answer.

27. Number of people who study more than one subject = 36 + 5 = 41

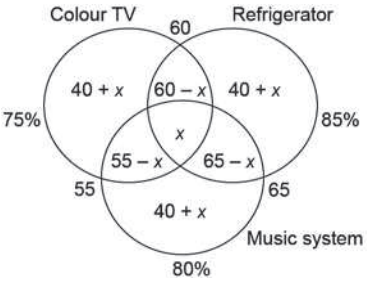
Hence, option (b) is the answer.

28. Number of students who study all the three subjects = **5**

29. Number of students who study exactly one subject = 20 + 22 + 23 = 65

Hence, option (c) is the answer.

30.



MODERATE

1. $n(1 \cup 2 \cup 3) = n(1) + n(2) + n(3) - n(1 \cap 2) - n(2 \cap 3) - n(1 \cap 3) + n(1 \cap 2 \cap 3)$

or $78\% = 50\% + 30\% + 20\% - n(1 \cap 2) - n(2 \cap 3) - n(1 \cap 3) + 5\%$

$n(1 \cap 2) + n(2 \cap 3) + n(1 \cap 3) = 27\%$

Only $[n(1 \cap 2) + n(2 \cap 3) + n(1 \cap 3)] = 27\% - 15\% = 12\%$, since $n(1 \cap 2 \cap 3)$ is present in all the three.

Percentage of people against more than one of the three proposals = only $[n(1 \cap 2) + n(2 \cap 3) + n(1 \cap 3)] + n(1 \cap 2 \cap 3) = 12\% + 5\% = 17\%$

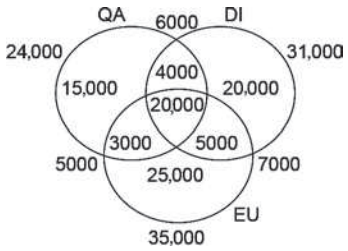
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2. To find out the maximum percentage of people watching all the three channels, we can assume that the people who watch ZEE, which is minimum among all the three channels, are watching all the three channels. Any value of percentage of people more than this 15% will make the percentage of people watching ZEE negative. Hence, maximum percentage of people watching all the three channels is 15%.

4. In this question, we can talk about a range of percentages who watch all the three channels. We cannot answer definitely what percentage of people watched all the three channels.

5.



$n(QA \cup DI) = 49{,}000$

$n(QA \cup DI) = n(QA) + n(DI) - n(QA \cap DI)$

$49{,}000 = 24{,}000 + 31{,}000 - n(QA \cap DI)$

$n(QA \cap DI) = 6000$

$n(DI \cup EU) = n(DI) + n(EU) - n(DI \cap EU)$

$59{,}000 = 31{,}000 + 35{,}000 - n(DI \cap EU)$

$n(DI \cap EU) = 7000$

$n(QA \cup EU) = n(QA) + n(EU) - n(QA \cap EU)$

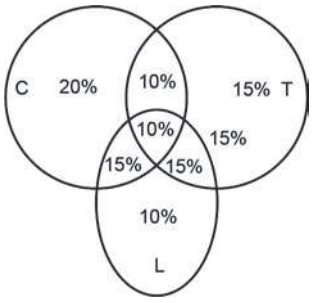
$54{,}000 = 24{,}000 + 35{,}000 - (QA \cap EU)$

$n(QA \cap EU) = 5000$

$\therefore$ Number of students failed = 80,000 − (24,000 + 50,000) = 6000

$\therefore$ Option (a) is the answer.

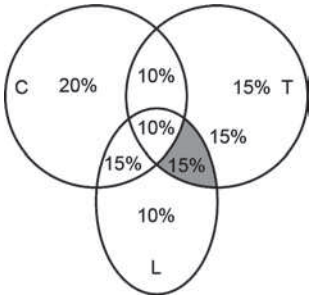
Answers to Q.10 to 13:



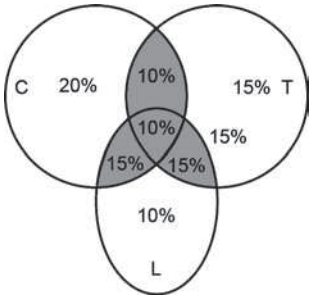
Where C = Coffee, T = Tea, and L = Lassi

10. The passengers who like only coffee = 20% and the passengers who like only lassi = 10%

11. It can be seen that the percentage of passengers who like both tea and lassi but not coffee = 15%. This is represented in the figure given below.



12. The percentage of passengers who like at least 2 drinks out of coffee, tea, and lassi can be seen in the below figure.

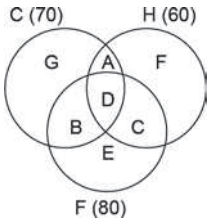


13. 10% of the people likes only lassi. So, the number of persons = 18

14. Go through the worked-out examples section.

15. Use the formula: $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$

16.



From the diagram, it is given that

A + D = 40, C + D = 30 and B + D = 30

It is also given that,

A + B + C + D + E + F + G = 130 and G + B + A + D = 70

F + A + C + D = 60

B + C + D + E = 80

Then from these equations, we can find D, which is equal to 20.

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17. The statement '12 have taken Mathematics, and 8 have taken Mathematics but not Statistics' means that 4 students have taken Mathematics and Statistics both.

Total number of students = 25. Out of this, 8 have taken only Mathematics, 4 have taken both. Hence, 13 have taken only Statistics.

So, the difference = 13 – 4 = 9

18. Solution to this question requires only the application of: $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$.

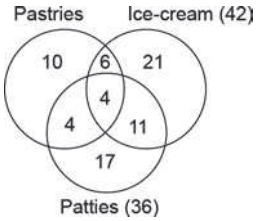
ADVANCED

Answers to Q.5 and 6:

It is a simple application of the formula:

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - (A \cap C) + (A \cap B \cap C)$$

Answers to Q.7 and 8:



7. From the diagram, A ∪ B ∪ C = 10 + 6 + 21 + 4 + 4 + 11 + 17 = 73

Total students = 123

Then, number of students who buy nothing = 123 – 73 = 50

8. Number of students who buy at least two item = 6 + 4 + 4 + 11 = 25

9. X = M × D → Mammals who are dogs. Now, since X = D, so all the dogs are mammals.

15. $n(C) = 63\%$

$n(A) = 76\%$

$n(C \cup A) = n(C) + n(A) - n(C \cap A)$

$100\% = 63\% + 76\% - X\%$

$X\% = 39\%$

18. n(sprint) = 19

n(shot-put) = 23

n(javelin) = 28

n(discus) = 25

n(sprint ∩ shot-put) = 2

n(shot-put ∩ javelin) = 3

n(javelin ∩ discus) = 3

n(discus ∩ sprint) = 0

n(sprint ∩ shot-put ∩ javelin ∩ discus) = 0

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CHAPTER

22

Statistics

LEARNING OBJECTIVES

After completion of this chapter, the reader should be able to understand:

- What is statistics?
- Different tools of statistics: mean, median, mode, standard deviation, and quartile
- Types and questions asked and methods to solve those

INTRODUCTION

In competitive examinations, questions are asked related to the interpretation of numerical data. Although a part of this has already been discussed in data interpretation, in this chapter, we are going to learn statistical terms and their related rules.

Questions based upon the below mentioned terms have been a regular feature at GMAT. Even Indian B-school exams like XAT, IIFT, JMET (discontinued now), etc., are asking questions based on the concepts discussed in this chapter.

WHAT IS STATISTICS

Statistics is defined as the mathematics of the collection, organization, and interpretation of numerical data, especially the analysis of population characteristics by inference from the sampling.

We are going to learn (a) mean, (b) median, (c) mode, (d) standard deviation, and (e) quartiles.

Mean, median, and mode are the measurement of central tendency for a set of data.

Calculation of Mean

Consider the following data related to the placement of seven students in a B-school:

Student's roll number	CTC offered (in ₹Lakhs)
Roll Number 1	42
Roll Number 2	7
Roll Number 3	9
Roll Number 4	10
Roll Number 5	24
Roll Number 6	7
Roll Number 7	6

$$\text{Mean} = \text{Mean} = \frac{\text{Sum of numbers}}{\text{Number of numbers}}$$

$$= \frac{42+7+9+10+24+7+6}{7} = \frac{105}{7} = 15$$

It should be noted that mean or arithmetic mean or average conveys exactly the same meaning for a set of data.

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Calculation of Median

Consider the following data related to the placement of seven students in a B-school:

Student's roll number	CTC offered (in ₹Lakhs)
Roll Number 1	42
Roll Number 2	7
Roll Number 3	9
Roll Number 4	10
Roll Number 5	24
Roll Number 6	7
Roll Number 7	6

To calculate median, arrange the data given either in ascending order or descending order. In case the number of items is odd, then the middle number is the median. If the number of items is even, then the average of middle two terms is the median.

In this case, since the number of data items is odd (7 data items are there), the middle term is the median. The following table gives the same values in ascending order:

Student's roll number	CTC offered (in ₹Lakhs)
Roll Number 7	6
Roll Number 2	7
Roll Number 6	7
Roll Number 3	9
Roll Number 4	10
Roll Number 5	24
Roll Number 1	42

Middle value = 9 = Roll Number 3. Hence, Median = 9

Calculation of Mode

Consider the following data related to the placement of seven students in a B-school:

Student's roll number	CTC offered (in ₹Lakhs)
Roll Number 1	42
Roll Number 2	7
Roll Number 3	9
Roll Number 4	10
Roll Number 5	24
Roll Number 6	7
Roll Number 7	6

Mode is defined as "the most frequently occurring data". In the above table, CTC offered ₹7 lakhs appears maximum number of times (twice). Hence, the mode for the given table = ₹7 lakhs.

Calculation of Range

Consider the following data related to the placement of seven students in a B-school:

Student's roll number	CTC offered (in ₹Lakhs)
Roll Number 1	42
Roll Number 2	7
Roll Number 3	9
Roll Number 4	10
Roll Number 5	24
Roll Number 6	7
Roll Number 7	6

Range is the difference between the largest and the smallest values.

In this table, the largest value = ₹42 lakhs and the smallest value = ₹6 lakhs

Hence, range = ₹42 lakhs – ₹6 lakhs = ₹36 lakhs

SOME SPECIAL CASES

Case I If there is no data repeating

Consider the set of data: 1, 2, 3, 4, 5

$$\text{Mean} = \frac{1+2+3+4+5}{4} = 3$$

Median = 3 is the middle value. Hence, median = 3

Mode = there is no data repeating, and each of the data occurs only once. In this case, we say that mode is not defined.

Range = Largest value – smallest value = 5 – 1 = 4

Case II If there are even number of terms

Consider the set of data: 1, 2, 3, 4, 5, 6

$$\text{Mean} = \frac{1+2+3+4+5+6}{4} - 3.5$$

Median = in case if there are even number of terms, then we take two middle values and calculate the average of these two values. For the given set of data, two middle values are 3 and 4. Average of 3 and 4 $= \frac{3+4}{2} = 3.5$. So, median for the given set of data = 3.5.

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Mode = since none of the data is repeating, the mode is not defined.
Range = largest value – smallest value = 6 – 1 = 5

Case III If there are even number of terms and more than one data occurs equal number of times
Consider the set of data: 3, 4, 6, 6, 8, 9, 9, 11

$$\text{Mean} = \frac{3+4+6+6+8+9+9+11}{8} = \frac{56}{8} = 7$$

Median = middle two values = 6 and 8. Median will be the average of 6 and 8 = 7. Hence, median = 7
Mode = both 6 and 9 occurs twice. Hence, there will be two modes: 6 and 9.
Range = largest value – smallest value = 11 – 3 = 8

TEST YOUR LEARNING

1. For the following set of data, find the mean, median, mode, and range:
   (a) 2, 3, 5, 6, 1, 8, 3
   (b) 3, 8, 1, 14, 7, 9
   (c) 14, 9, 40, 8, 1, 17, 8, 7
   (d) 3, 8, 12, 4, 7, 4, 20, 3

Answers:

(a) Mean = 4, median = 3, mode = 3, and range = 7
(b) Mean = 7, median = 7.5, mode = not defined, and range = 13
(c) Mean = 13, median = 8.5, mode = 8, and range = 39
(d) Mean = 7.625, median = 5.5, mode = 3 and 4, and range = 17

Standard Deviation

Standard deviation is the measurement of the dispersion of the data. In other words, standard deviation tells us that how spread the data are.

How Do We Calculate Standard Deviation?

Step 1 Calculate the mean of the data given.
Symbol:
We use μ (the Greek letter "mu") to denote the mean of all the values.
Step 2 Find out the deviation for each of the data from the mean.
Symbol:
We write this as $(x_i - \mu)$ [x_i denotes the individual data]
Step 3 Take the square of each of the deviations obtained in Step 2.
Symbol:
$(x_i - \mu)^2$ [x_i denotes the individual data]
Step 4 Calculate the mean of the squared deviations obtained in Step 3.
Symbol:

$$\frac{1}{N}\sum_{i=1}^{N}(x_1 - \mu)^2$$

[∑ notation is used to calculate the sum of similar values.]

Important Note: When we consider the *whole population* of a data, we divide the sum of squared deviation by *n* (number of elements in the population). Instead, if we are considering a *sample* of data (and not the population), then we divide the sum of squared deviation by *n* – 1 (one less than the number of elements in the sample).

Step 5 Take the square root of the mean obtained in Step 4. This is Standard Deviation.
Symbol:
Standard deviation is denoted by σ (the Greek letter sigma).

$$\sigma = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2}$$

Example: Consider the individual heights of 10 students in a class (in cm):
142, 161, 136, 164, 156, 178, 172, 153, 167, 147
Step 1 Calculate the mean of the data given.

$$\text{Mean} = \frac{142+161+136+164+156+178+172+153+167+147}{10} = 157.6 \text{ cm}$$

Step 2 Find out the deviation for each of the data (it does not matter you do 'individual data – mean' or 'mean – individual data' because in the next step, we are going to square it).

Data	Mean	Deviation
142	157.6	15.6
161	157.6	–3.4
136	157.6	21.6
164	157.6	–6.4
156	157.6	1.6
178	157.6	–20.4
172	157.6	–14.4
153	157.6	4.6
167	157.6	–9.4
147	157.6	10.6

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Step 3 Take the square of the deviations calculated in the last step.

Data	Mean	Deviation	Squared deviation
142	157.6	15.6	243.36
161	157.6	–3.4	11.56
136	157.6	21.6	466.56
164	157.6	–6.4	40.96
156	157.6	1.6	2.56
178	157.6	–20.4	416.16
172	157.6	–14.4	207.36
153	157.6	4.6	21.16
167	157.6	–9.4	88.36
147	157.6	10.6	112.36

Step 4 Mean = 161.04

Squared Deviation
243.36
11.56
466.56
40.96
2.56
416.16
207.36
21.16
88.36
112.36
Mean = 161.04

This is also known as 'Variance'.

Step 5 Take the square root of the value (mean) obtained in the previous step. This is the standard deviation of the whole data.

$$\text{Standard deviation} = \sqrt{161.04} = 12.69$$

Relationship among Mean, Median, and Mode

If the data is symmetrically distributed, then the following relationship among mean, median, and mode holds true:

$$\text{Mode} = 3\ \text{Median} - 2\ \text{Mean}$$

[Although a word of caution is required to be mentioned here, that unless it is specified that data is symmetrically distributed, we cannot assume otherwise, and if the data is not symmetrically distributed, then this formula cannot be used.]

Quartiles

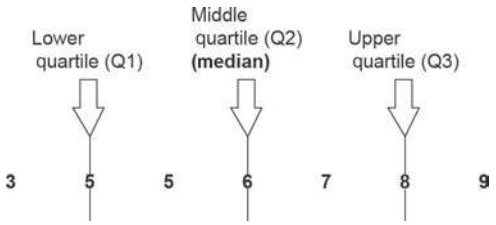
Quartiles are separators that divide the list of numbers into parts.

Example 1 Consider the set of values = 6, 8, 9, 3, 5, 5, 7

Solution First of all, we arrange the numbers in ascending order (exactly the same way as we did while finding median):

3, 5, 5, 6, 7, 8, 9

Now, cut the list into four equal parts.

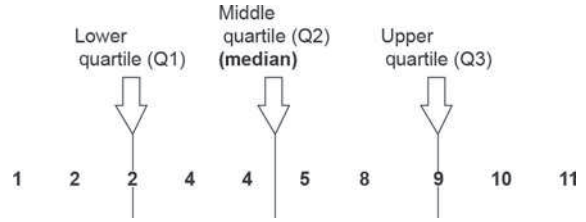


Here, lower quartile (Q1) = 5, middle quartile (Q2 or median) = 6, and upper quartile (Q3) = 8

Although sometimes we do not get a cut as clean as given above, a cut mark may be put in between two numbers, so as to have four equal parts.

Example 2 Consider the set of values = 2, 9, 4, 5, 8, 4, 10, 2, 11, 1

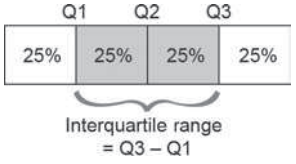
Solution Arrange these numbers in ascending order = 1, 2, 2, 4, 4, 5, 8, 9, 10, 11



Here, lower quartile (Q1) = 2, middle quartile (Q2 or median) = 4.5 (average of 4 and 5), and upper quartile (Q3) = 9

Interquartile Range

To calculate interquartile range, simply subtract Q1 from Q3.



Interquartile range = Q3 – Q1

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Practice Exercises

Warm Up

Q.1 Twenty students are asked how many hours they spend on facebook every month. Following are the results:

Student 1	12	Student 11	20
Student 2	8	Student 12	5
Student 3	10	Student 13	3
Student 4	9	Student 14	10
Student 5	2	Student 15	8
Student 6	22	Student 16	5
Student 7	12	Student 17	20
Student 8	8	Student 18	15
Student 9	8	Student 19	6
Student 10	10	Student 20	8

(a) Calculate mean, median, and mode.
(b) Find standard deviation, range, and interquartile range.
(c) Assuming that the data given above is a sample taken from the school log book containing data of 1000 students, what is the standard deviation now?

Q.2 A quality control manager is interested in analyzing the number of complaints received by the showroom about the quality of products sold over the year. Records over a 10-week period yield the data shown in the following table.

Week number	Number of complaints
Week 1	13
Week 2	15
Week 3	8
Week 4	16
Week 5	8
Week 6	4
Week 7	21
Week 8	11
Week 9	3
Week 10	15

(a) Find the mean, median, and mode.
(b) Find the variance,standard deviation, range, and interquartile range.

Q.3 First 45 natural numbers are placed in 5 groups of 9 each. Now median of each of these groups has been calculated. What is the maximum possible mean of the medians of these five groups?
(a) 25 (b) 31 (c) 15 (d) 26

Q.4 In a list of 7 integers, one integer, denoted as x, is unknown. The other six integers are 20, 4, 10, 4, 8, and 4. If the mean, median, and mode of these seven integers are arranged in ascending order, then they form an arithmetic progression. The sum of all such values of x is:
(a) 26 (b) 32 (c) 34 (d) 40
(*XAT 2011*)

Q.5 In the data set $\{2, 5, 7, 8, X\}$, the arithmetic mean is same as the median. Determine the value of X. Assume $X \geq 8$.
(a) 8 (b) 10 (c) 13 (d) 15
(*JMET 2009*)

Q.6 At the fag end of his career, Indian cricket team captain Dahendra Singh Moni was getting down the memory lane. He calculated that, during his captaincy, till 2013, Indian cricket team's winning percentage was below X%, and in 2014, Indian cricket team's winning percentage was above X%. Although he was surprised to know that during 2013–14, Indian cricket team's winning percentage was never equal to X%. Which of the following can be equal to X [assume that Dahendra Singh Moni was the captain of Indian team from 2007–2015, and all the matches resulted in either a win or a loss]?
(i) 50% (ii) 60% (iii) 80%
(a) (i), (ii), and (iii)
(b) Only (i) and (ii) but not (iii)
(c) Only (ii) and (iii) but not (i)
(d) Only (ii)

Q.7 A list of five natural numbers has mean = 12 and range = 18. Mode = Median = 8. How many different values are possible for the second largest number in the list?
(a) 3 (b) 4 (c) 5 (d) 6

Q.8 In the recent Business and Economy Conclave, random sample of 22 CEOs were asked to predict the percentage growth in the GDP over the previous year. The forecasts were (in percentage):
3.6 3.1 3.9 3.7 3.5 3.7 3.4 3.0 3.6 3.4 3.1
2.9 3.0 4.0 2.8 3.8 4.2 2.5 3.1 3.9 2.9 2.6

Find the mean, median, mode, and standard deviation of the sample given above.

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Q.9 Consider the box diagram given below:

8			26	x

Each of the boxes in the grey cell is the average of the cell on the right and the cell on the left. What is the value of x?

(a) 14 (b) 24 (c) 36 (d) 32

Answers

Warm Up

3. (b) **4.** (d) **5.** (c) **6.** (d) **7.** (d) **9.** (d)

Hints and Solutions

Warm Up

1. (a) Mean = x = 10.05

Median = 8.5

Mode = 8 (repeated 5 times)

(b) Standard deviation = 5.37

Range = 20

IQR = 12 − 7 = 5

(c) When we consider population of data, while calculating mean of squared deviation, we divide it by 'n'. Whereas when we calculate the standard deviation of a sample of data, then while calculating mean of squared deviation, we divide it by $(n - 1)$.

Standard deviation = 5.51

2. It can be seen that to analyse the data over the year, showroom has collected the data of 10 weeks. So, the data given in the question is NOT population, but sample.

(a) Mean = x = 11.4

Median = 12

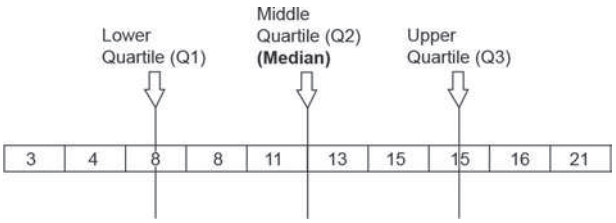
Mode = 8 and 15 (repeated 2 times)

(b) Variance = s^2 = 32.267

Standard Deviation = s = 5.680

Range = 18

IQR = Q3 − Q1 = 15 − 8 = 7



3. Since we have to maximize the mean (or average) of the medians, we should try to find the value of median in such a way that every group gives maximum median. Consider any group with 9 numbers ⇒ maximum possible median for this group is 41. [Assume the following groups a, b, c, d, 41, 42, 43, 44, 45, the median will be 41.]

We should try to assume minimum possible value of a, b, c, and d, as median will not be affected and will be still 41. For example, we can assume that for the 9 numbers 1, 2, 3, 4, 41, 42, 43, 44, 45, the median will be 41. Next group will be 5, 6, 7, 8, 36, 37, 38, 39, 40. This will have median = 36. Similarly, next medians will be = 31, 26, 21.

So, list of medians = 41, 36, 31, 26, 21. Therefore, maximum possible mean of the medians = 31.

Hence, option (b) is the answer.

4. Option (d) is the answer.

5. The best way of solving this question is going through the options.

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Evaluating option (a)—8

Data set = 2, 5, 7, 8, 8

Median = 7 and arithmetic mean = 6. Hence, option (a) is not the answer.

Evaluating option (b)—10

Data set = 2, 5, 7, 8, 10

Median = 7 and arithmetic mean = 7.4. Hence, option (b) is not the answer.

Evaluating option (c)—13

Data set = 2, 5, 7, 8, 13

Median = 7 and arithmetic mean = 7. Hence, option (c) is the answer.

6. Let us write down the values in fraction:

$50\% = \frac{1}{2}$ $60\% = \frac{3}{5}$ $80\% = \frac{4}{5}$

If the difference between numerator and denominator is 1, then it is not possible to bypass the ratio. In other words, for 50% and 80%, it is not possible to move from less than 50% (or less than 80%) to more than 50% (or more than 80%) without being equal to 50% (or 80%). Hence, option (d) is the answer.

7. Since mean of five natural numbers = 12 ⇒ Summation = 60

Median = 8 ⇒ 8 will be the middle number.

_ _ 8 _ _

Mode = 8 ⇒ 8 will occurs at least twice. We also understand that 8 cannot occur five times (in that case mean will not be equal to 12). At the same time, if 8 occurs four times, fifth number = 60 – 32 = 28. In that case, range = 28 – 8 = 20, which is not possible as per the question. So, 8 cannot occur four times.

Therefore, 8 occurs either twice (in that case, no other number occurs twice or thrice) or thrice (in that case, other number can occur twice, but not necessarily).

Hence, we have the following possibilities:

Case 1 _ 8 8 8 _

Case 2 8 8 8 _ _

Case 3 _ 8 8 _ _

Case 4 _ _ 8 8 _

We have to take care of two more facts that: Mean = 12 and Range = 18.

We get the following final possibilities:

Possibility 1	8	8	8	10	26
Possibility 2	3	8	8	20	21
Possibility 3	4	8	8	18	22
Possibility 4	5	8	8	16	23
Possibility 5	6	8	8	14	24
Possibility 6	7	8	8	12	25

So, the second largest number can be = 10, 12, 14, 16, 18, 20.

Hence, six values are possible. Hence, option (d) is the answer.

8. Mean = $x = 3.35$

Median = 3.4

Mode = 3.1 (repeated 3 times)

Since the sample data has been used, we will divide the sum of squared deviation by $(n - 1) = (22 - 1) = 21$

Standard deviation = 0.4768.

9. Consider the rightmost three boxes: Since 26 is the average of the box on the left-hand side and right-hand side, box on the right hand side = $52 - x$.

8		**(52 – *x*)**	**26**	*X*

Further, assume that the second box from left = y.

8	*y*	**(52 – *x*)**	**26**	*X*

So, $2y = 8 + 52 - x = 60 - x$

$\Rightarrow x + 2y = 6$ (i)

Next, $(52 - x)$ is the average of y and 26.

Hence, $2(52 - x) = 26 + y$

$\Rightarrow 104 - 2x = 26 + y$

$\Rightarrow 2x + y = 78$ (ii)

Solving both the equation, we get $x = 32$ and $y = 14$.

Hence, option (d) is the answer.

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CHAPTER

23

Miscellaneous

LEARNING OBJECTIVES

After completion of this chapter, the reader should be able to understand:

- Concept of calendars, inequality, and height and distance
- Definitions and properties
- Types of questions asked in the CAT
- Method of solving questions

INTRODUCTION

If we go through the history of CAT, then we will conclude that CAT has never spared the above collection of miscellaneous concepts while asking questions. Usually, 1 to 3 questions are asked every year from these topics.

Let us see these concepts one-by-one.

CALENDAR

We know that any non-leap year contains 365 days = 52 weeks + 1 day.

Leap year contains 366 days = 52 weeks + 2 days.

This 1 day and 2 days extra added to any year create changes in the calendar, and this is the reason why calendar of Nth year will not be same as $N + 1$th year.

Before we proceed ahead, we should be very clear about two things.

Which Years are Leap Years?

It takes the earth about 365.2422 days to go around the sun, but a normal calendar year is only 365 days. The extra fraction of a day added up four times makes four years (or four revolution of earth around sun), that is, 1460.9688 days, but four calendar years would only be 1460 days. The fraction 0.9688 is almost a whole day, so every four years, we add an extra day to our calendar, that is, February 29. We call that year a leap year. To make things easier, leap years are always divisible by four: 2004 and 2008 will both be leap years.

For hundreds of years, people used a calendar called the Julian calendar, which followed this rule, adding a leap year every four years. However, because 0.9688 is not *exactly* a whole day, the Julian calendar slowly began to disagree with the real seasons. In 1582, Pope Gregory fixed this problem by ordering everyone to use a new set of rules. These rules are known as the Gregorian calendar, named after him. The rules are given in the below table.

The Gregorian Calendar

Rule	Examples
Every fourth year is a leap year.	2004, 2008, and 2012 are leap years.
However, every centenary year is *not* a leap year.	1900 and 2100 are *not* leap years.
In case of centuries, every four hundred years, there's a leap year after all.	2000 and 2400 are leap years.

In layman terms, all the century years divisible by 400 will be leap years and all the non-century years divisible by 4 will be leap years. So, leap year next to 2096 AD is 2104 AD and not 2100 AD.

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Because 2000, 4000, 6000, etc., are leap years and 1000, 3000, 5000, etc., are not, the number of leap days in each millennium alternates between 242 and 243; with the first, third, etc., millennia (i.e., 1–1000, 2001–3000, etc.) having 242 leap days, and the second, fourth, etc. (i.e., 1001–2000, 3001– 4000, etc.) having 243 leap days.

How the Days of Consecutive Years Change?

Due to any non-leap year, calendar of next year go ahead by 1 day, and due to any leap year, calendar of next year goes ahead by 2 days, but this change in calendar will be there only before 29th February.

It can be seen through the example given below.

	1991	1992	1993
1st January	Sunday	Monday	Wednesday
28th February	Tuesday	Wednesday	Friday
1st March	Wednesday	Friday	Saturday

In the above example, we have assumed that 1st January of 1991 is Sunday. 1991 and 1993 are non-leap years and 1992 is a leap year.

Now, if we try to find out the symmetricity of calendars, then we can see this in the following way:

(i) For any leap year

Let us see, for example, the case of 1972.

Year	1972	1973	1974	1975	1976
Excess days	2	1	1	1	2

Since number of excess days are 7, the days of the year 1972 and year 1977 will be same from 1st of January and 28th of February.

(ii) For any (leap year + 1) year

Year	1973	1974	1975	1976	1977	1978
Excess days	1	1	1	2	1	1

Since number of excess days are 7, the calendar of the years 1973 and 1979 will be same for whole year.

(iii) For any (leap year + 2) year

Year	1974	1975	1976	1977	1978	1979
Excess days	1	1	2	1	1	1

Since excess days are 7, the calendar of the years 1974 and 1980 will be same till 28th of February.

(iv) For any (leap year + 3) year

Year	1975	1976	1977	1978	1979	1980	1981	1982	1983	1984	1985
Excess days	1	2	1	1	1	2	1	1	1	2	1

Since the number of excess days are 14, the calendar of the years 1975 and 1986 will be same for whole year.

This whole mechanism can be summed up in following way:

Nature of year	Number of years after which 1st January will be same
Leap year	5
Leap year + 1	6
Leap year + 2	6
Leap year + 3	11

Hence, 1st January of 1972 and 1st January of 1977 will be on same day.

Similarly, 1st January of 1973 and 1st January of 1979 will be on same day and so on.

Exception

No century year, which is not a leap year, should be included in this calculation.

Example 1 Sum of dates of last Monday of previous month and 1st Thursday of next month is 38. If both the dates are of the same year, then which month is the current month?

Solution Sum of dates of last Monday of previous month and 1st Thursday of next month is 38 is possible only if last Monday is 31st and 1st Thursday is 7th. (Since if we take 30 + 8 = 38, then 30 can be last Monday of any month but 8th cannot be the 1st Thursday of any month.)

So, 31st of last month is a Monday. Hence, 7th of current month, 14th of current month, 21st of current month, and 28th of current month will be a Monday.

Now, if current month is a month with 30 days, then 5th of next month will be a Monday, so 7th of next month cannot be a Thursday.

If current month is a month with 31 days, then 4th of next month will be a Monday, so 7th of next month will be a Thursday.

Finally, we can conclude that previous month and current month, both are having 31 days. Since both the dates are of the same year, so the current month is August.

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Finding Day of a Date by Using a Reference Date

Let us see this with the help of an example: If 9th Dec of 1972 is Sunday, then which day it will be on 14th December 1998?

Process: There are several processes to do this calculation: 1. Year method, 2. Days method, and 3. Actual calculation method.

1. **Year method:** We use the above given table to find out about any of the years.
   9.12.1972—Sunday
   1.1.1973—Tuesday (It is a Leap year + 1 year)
   So, 1.1.1979—Tuesday (It is a Leap year + 3 year)
   So, 1.1.1990—Tuesday (It is a Leap year + 2 year)
   So, 1.1.1996—Tuesday
   Now, we can find out all the next years one-by-one.
   1.1.1997—Thursday
   1.1.1998—Friday
   31.12.1998, 24.12.1998, 17.12.1998
   So, 14.12.1998—Tuesday

2. **Days method:** We use the number of excess days every year to find out the number of days the calendar will move ahead by.
   1.1.1973—Tuesday
   Due to 1973, calendar will go ahead by 1 day, similarly, due to 1974—1 day, due to 1975—1 day, and due to 1976—2 days.
   So, in four years, calendar will go ahead by 5 days.
   Using unitary method, in four years, the calendar will move ahead by 5 days.
   So, in 24 years, the calendar will move ahead by 30 days. Hence, the calendar will move ahead by 2 days.
   So, 1.1.1997 will be two days ahead of Tuesday, that is, Thursday.
   Now, it is calculated as given in year method.

3. **Actual calculation method:** With the help of this method, we can find the actual day of any date of 20th century. To use this method effectively, we need to remember the Month Codes of all the months.
   Let us learn this method by finding the date of 15th August 1947:
   At 1st, add the Date + Month code of August + Last two digits of year $+\left[\frac{\text{Last two digits of year}}{4}\right]$
   (where $[x]$ represents the greatest integer value of x)
   So, corresponding to 15th August 1947—15 + 3 + 47 + 11 = 76
   Now, divide this value by 7 to find out the remainder.
   If the remainder is 0 → then it is a Saturday.
   If the remainder is 1 → then it is a Sunday.
   If the remainder is 2 → then it is a Monday.
   If the remainder is 3 → then it is a Tuesday.
   If the remainder is 4 → then it is a Wednesday.
   If the remainder is 5 → then it is a Thursday.
   If the remainder is 6 → then it is a Friday.
   Here, remainder is 6, so 15th August 1947 was a Friday.(It should have been a 'Free'day.)

List of month code:

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Leap Year	0	3	4	0	2	5	0	3	6	1	4	6
Non-leap year	1	4	4	0	2	5	0	3	6	1	4	6

INEQUALITIES

While discussing equations and inequations in polynomials, we have seen what inequations are. However, while discussing inequalities here, our focus will be to discuss this concept in isolation with equations for real numbers only.

Tools of Inequality

'>' means 'greater than'
'<' means 'less than'
'≥' means 'greater than or equal to'
'≤' means 'less than or equal to'

If 'N' is any real number, then value of 'N' will be either positive or negative or zero.

When 'N' is positive, we write $N > 0$; which is read as 'N is greater than zero'.

Similarly, When 'N' is negative, we write $N < 0$; which is read as 'N is less than zero'. If 'N' is zero, we write $N = 0$, and in this case, 'N' is neither positive nor negative.

The two signs '>' and '<' are called the signs of inequalities.

Standard Definitions

For any set of real numbers M and N,

(i) M is said to be greater than N when $M - N$ is positive.
⇒ $M > N$ when $M - N > 0$
As we can see, 10 is greater than 5 because 10 – 5 = 5, and 5 is greater than zero.

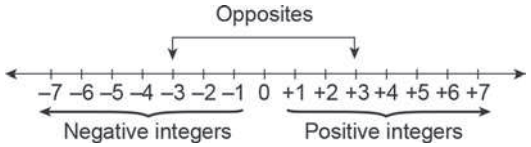
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(ii) M is said to be less than N when $M - N$ is negative.
$\Rightarrow$ $M < N$ when $M - N < 0$
As we can see, -10 is less than -5 because $-10 - (-5) = -5$, and -5 is less than zero.
However, in case of numbers, inequalities can be understood through number line also.

Number Line

The number line is used to represent the set of real numbers. Below is the brief representation of the number line:



Basics of Inequalities

It is quite pertinent here to understand some of the very basic properties related to inequalities. These properties should be seen as the building blocks of the concepts of inequalities.

Assume all the numbers used here are real numbers.
For any two real numbers M and N, either $M > N$ or $M < N$ or $M = N$.

If $M > N$, then $N < M$
If $M > N$ and $N > P$, then $M > P$
If $M < N$ and $N < P$, then $M < P$
If $M > N$, then $M \pm c > b \pm c$
If $M > N$ and $P > 0$ then $MP > NP$
If $M > N$ and $P < 0$, then $MP < NP$
If $M < N$ and $P > 0$, then $MP < NP$
If $M < N$ and $P < 0$, then $MP > NP$
If $M > N$ and $P > Q$, then $M + P > N + Q$
If $M < N$ and $P < Q$, then $M + P < N + Q$
However, if $M > N$ and $P < Q$ or $M < N$ and $P > Q$, then we cannot comment about the inequality between $(M + P)$ and $(N + Q)$.
If $M > N$ and $P > Q$, then we cannot infer the inequality sign between $(M - N)$ and $(P - Q)$. Depending on the values of M, N, P, and Q, it is possible to have $(M - N) > (P - Q)$, $(M - N) = (P - Q)$ or $(M - N) < (P - Q)$.
The square of any real number is always greater than or equal to 0.
The square of any non-zero real number is always greater than 0.
If $N > 0$, then $-N < 0$, and if $M > N$, then $-M < -N$.
If M and N are positive numbers and $M > N$, then
(i) $1/M < 1/N$
(ii) $M/P > N/P$ if $P > 0$
(iii) $M/P < N/P$ if $P < 0$

For any two positive numbers *M* and *N*
If $M > N$, then $M^2 > N^2$
If $M^2 > N^2$, then $M > N$
If $M > N$, then for any positive value of n, $M^n > N^n$

For two positive numbers *M* and *N*
If $M/N < 1$, then $M < N$
If $M/N = 1$, then $M = N$
If $M/N > 1$, then $M > N$

Relationship between a number and its square root
Let N be a natural number.
$\sqrt{N} = N$, for $N = 0$ or $N = 1$
$\sqrt{N} > N$, for $0 < N < 1$
$\sqrt{N} < N$, for $N > 1$
Let A, G, and H be the arithmetic mean, geometric mean, and harmonic mean of n positive real numbers, then $A \geq G \geq H$. Equality occurs only when all the numbers are equal.
If the sum of two positive quantities is given, then their product is greatest when they are equal; and if the product of two positive quantities is given, then their sum is least when they are equal.
For any positive number, the sum of the number and its reciprocal is always greater than or equal to 2.
i.e., $x + \frac{1}{x} \geq 2$ where $x > 0$.
The equality in this relationship will occur only when $x = 1$.

Cauchy–Schwarz Inequality

If $a_1, a_2, \ldots a_n$ and $b_1, b_2, \ldots b_n$ are $2n$ real numbers, then

$$(a_1b_1 + a_2b_2 + \ldots + a_nb_n)^2$$
$$\leq (a_1^2 + a_2^2 + \ldots + a_n^2)(b_1^2 + b_2^2 + \ldots + b_n^2)$$

With the equality holding if and only if

$$\frac{a_1}{b_1} = \frac{a_2}{b_2} = \ldots = \frac{a_n}{b_n}$$

Despite all the points given above, however, we should not let the LOGIC die. Most of the questions asked in CAT can be solved by using options and we will not be in need of using any concept of inequalities. However, this should not be seen as a case in support of not-going through the concepts.

Example 2 If $R = \frac{30^{65} - 29^{65}}{30^{64} + 29^{64}}$, then **(*CAT 2005*)**

(a) $0 < R \leq 0.1$ (b) $0.1 < R \leq 0.5$
(c) $0.5 < R \leq 1.0$ (d) $R > 10$

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Solution $R = \dfrac{30^{65} - 29^{65}}{30^{64} + 29^{64}} > 1$

As $30^{65} - 29^{65} > 30^{64} + 29^{64}$
or $30^{64}(30 - 1) > 29^{64}(29 + 1)$
or $30^{64} \times 29 > 29^{64} \times 30$
or $30^{63} > 29^{63}$
Hence, option (d) is the answer.

Example 3 If $13x + 1 < 2z$ and $z + 3 = 5y^2$, then **(*CAT 2003*)**

(a) x is necessarily less than y.
(b) x is necessarily greater than y.
(c) x is necessarily equal to y.
(d) None of the above is necessarily true.

Solution We have
$13x + 1 < 2z$ (i)
and $z + 3 = 5y^2$ (ii)
From equations (i) and (ii) we get,
$13x + 1 < 2(5y^2 - 3)$
$\Rightarrow \quad 13x + 1 < 10y^2 - 6$
$\Rightarrow \quad 13x < 10y^2 - 7$
$\Rightarrow \quad x < \dfrac{10y^2 - 7}{13}$
If $y = 1$, then we get $x < 0.230$
i.e., $y > x$
If $y = 2$, then we get $x < 2.538$.
i.e., $x > y$. This is not possible. Hence, the correct answer is option (d).

Example 4 If $|b| \geq 1$ and $x = -|a|\, b$, then which one of the following is necessarily true? **(*CAT 2003*)**

(a) $a - xb < 0$ (b) $a - xb \geq 0$
(c) $a - xb > 0$ (d) $a - xb \leq 0$

Solution Lets start assuming the values of a and b.
Assume $b = 2$ and $a = 1/2$. Then, $x = -1$
Therefore, options (a) and (d) have been eliminated.
Assuming $a = -1$ and $b = -1$, we get $x = 1$. Putting the values in options (a) and (b), we get the answer option (b).

HEIGHTS AND DISTANCES

General Theory on Heights and Distances

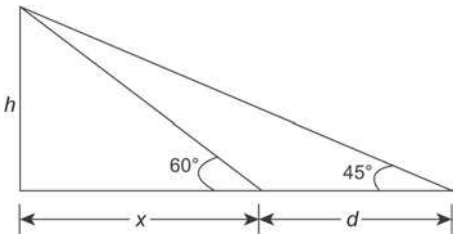
Angle of elevation If the object is above the horizontal level of the eyes (i.e., if it is above the eye level), then we have to turn our head upwards to view the object. In this process, our eyes move through an angle. Such an angle is called the angle of elevation of the object from our eyes.

Angle of depression Suppose a boy standing on the roof of a building observes a ball lying on the ground at some distance from the building. In this case, he has to move his head downwards to view the ball. In this process, his eyes again move through an angle. Such an angle is called the angle of depression of the object from the location of his eyes (figure).

SOME TRIGONOMETRIC RATIOS AND FORMULAE

	0°	30°	45°	60°	90°
Sin	0	1/2	$1/\sqrt{2}$	$\sqrt{3}/2$	1
Cos	1	$\sqrt{3}/2$	$1/\sqrt{2}$	1/2	0
Tan	0	$1/\sqrt{3}$	1	$\sqrt{3}$	∞

$\sin(-\theta) = -\sin\theta$; $\cos(-\theta) = \cos\theta$; $\tan(-\theta) = -\tan\theta$;
$\cot(-\theta) = -\cot\theta$; $\sec(-\theta) = \sec\theta$; $\operatorname{cosec}(-\theta) = -\operatorname{cosec}\theta$;
$\sin(90° - \theta) = \cos\theta$; $\cos(90° - \theta) = \sin\theta$
$\sin 2\theta = 2\sin\theta\cos\theta$
$\cos 2\theta = 2\cos^2\theta - 1 = 1 - 2\sin^2\theta$



Example 5 A car is being driven, in a straight line and at a uniform speed, towards the base of a vertical tower. The top of the tower is observed from the car and, in the process, it takes 10 min for the angle of elevation to change from 45° to 60°. After how much more time will this car reach the base of the tower?

(a) $5(\sqrt{3}+1)$ (b) $6(\sqrt{3}+\sqrt{2})$
(c) $7(\sqrt{3}-1)$ (d) $8(\sqrt{3}-2)$

Solution

$\tan 45° = \dfrac{h}{x+d}$ or
$h = x + d$

Similarly, $\tan 60° = \dfrac{h}{x}$ or
$h = x\sqrt{3}$

Hence, $x + d = x\sqrt{3}$
So, $x(\sqrt{3} - 1) = d$
The car takes 10 min in covering 'd' distance, so the car will take $10/(\sqrt{3} - 1)$ min in covering 'x' $[x = d/(\sqrt{3} - 1)]$ distance.
Now, $t = 10/(\sqrt{3} - 1)$

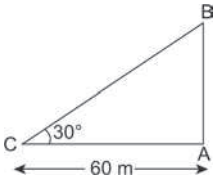
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Multiplying both numerator and denominator by $(\sqrt{3}+1)$, we get $t = 5(\sqrt{3}+1)$.

Example 6 The angle of elevation of the top of a tower from a point 60 m from its foot is 30°. What is the height of the tower?

Solution Let AB be the tower with its foot at A.
Let C be the point of observation.
Given $\angle ACB = 30°$ and AC = 60 m



From right triangle, BAC:

$$\frac{AB}{AC} = \tan 30°$$

$$\Rightarrow \quad AB = 60 \tan 30° = \left(\frac{1}{\sqrt{3}}\right) = 20\sqrt{3}$$

Example 7 Two pillars of equal height stand on either side of a road. At a point on the road between the pillars, the elevation of the tops of the pillars are 60° and 30°. Find the height of the pillars if it is given that the width of the road is 150 m.

Solution $\tan 60° = \sqrt{3} = h/x$

$\tan 30° = 1/\sqrt{3} = h/(150 - X)$
Solving we get, $x = 37.5$ m, $h = 64.95$ m

Practice Exercises

Foundation

Q.1 The total number of terms in the expansion of $(x+a)^{100} + (x-a)^{100}$ after simplification is:
(a) 202 (b) 51
(c) 50 (d) None of these

Q.2 The number of terms in the expansion of $(x+y+z)^{10}$ is:
(a) 11 (b) 33
(c) 66 (d) None of these

Q.3 In the expansion of $(1+x)^{50}$, the sum of the co-efficients of odd powers of x is:
(a) 0 (b) 2^{49} (c) 2^{50} (d) 2^{51}

Q.4 What is the number of leap years in between any 100 consecutive years?
(a) 24 (b) 25
(c) 26 (d) Cannot be determined

Q.5 Year 200X is a leap year. Which of the following year will definitely be a leap year again?
(a) 200X + 4 (b) 200X + 5
(c) 200X + 6 (d) Cannot be determined

Q.6 June 10, 1979, was a Sunday. Then, May 10, 1972, was a:
(a) Wednesday (b) Thursday
(c) Tuesday (d) Friday

Q.7 In a certain year, the month of January had exactly 4 Wednesdays and 4 Sundays. Then, January 1 of that year was a:
(a) Monday (b) Thursday
(c) Friday (d) Saturday

Q.8 A. I was born in a century in which the last year of the previous century was a leap year.
B. I was born in the first year after the first leap year of the century.
C. I was born after 1200 A.D. and before 2000 A.D.
What is the year of my birth?
(a) 1405 (b) 1505 (c) 1605 (d) 1705

Q.9 If $a \le 22$ and $b \ge 12$, then which of the following is always true?
(a) $a > b$ (b) $a - b > 10$
(c) $a - b \le 10$ (d) $a + b \ge 34$

Q.10 If x, y, and z are such that $x < y$ and $z > 0$, then:
(a) $\frac{x}{y} > 1$ (b) $\frac{z}{x} < \frac{z}{y}$
(c) $x > \frac{y}{z}$ (d) $xz < yz$

Q.11 Which of the following satisfies the condition $\frac{1}{x} > -10$?
(a) $x + 2 = 0$ (b) $x + 3 = 0$
(c) $x + 5 = 0$ (d) None of these

Q.12 Which of the following satisfies the condition $\frac{1}{x} < -10$?
(a) $x - 2 = 0$ (b) $x - 3 = 0$
(c) $x - 5 = 20$ (d) None of these

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Direction for Questions 13 to 15: Find the values of *x* for which the inequality holds true.

Q.13 $x+2<\sqrt{x+14}$

(a) $-2 \le x < 14$ (b) $-14 \le x < 2$
(c) $x \le -2$ or $x > 14$ (d) $x < -14$ or $x > 2$

Q.14 $(x-1)(3-x)(x-2)^2 > 0$

(a) $1 < x < 3$ (b) $1 < x < 3, x \ne 2$
(c) $-1 < x < 3, x \ne 2$ (d) $-1 < x < 3$

Q.15 $\frac{x}{x-5} > \frac{1}{2}$

(a) $-5 < x < 5$ (b) $x < -5$ or $x > 5$
(c) $-5 < x \le 0$ (d) $0 < x \le 5$

Q.16 $\frac{3}{x-2} < 1$

(a) $2 < x < 5$ (b) $-2 \le x < 5$
(c) $-5 \le x \le 2$ (d) $x < 2$ or $x > 5$

Q.17 For how many values of x is the following equation true $|2x+3|+x=10$?

(a) 1 (b) 2
(c) 0 (d) None of these

Q.18 From the top of a lighthouse of 60 m high with its base on the sea level, the angle of depression of a boat is 15°. Find the distance of the boat from the foot of the lighthouse.

(a) $60(2+\sqrt{3})$ m (b) $60(1+\sqrt{3})$ m
(c) $60\sqrt{3}$ m (d) $100(2+\sqrt{3})$ m

Q.19 From the top of a cliff of 200 m high, the angles of depression of the top and bottom of a tower are observed to be 30°and 60°, respectively. Find the height of the tower.

(a) $400\sqrt{3}$ m (b) $200\sqrt{3}$ m
(c) $100\sqrt{3}$ m (d) None of these

Q.20 The angle of elevation of a moon when the length of the shadow is equal to its height is:

(a) 30° (b) 60°
(c) 45° (d) None of these

Q.21 The ratio of the length of a rod and its shadow is $1:\sqrt{3}$ The angle of elevation of the sun is:

(a) 30° (b) 45° (c) 60° (d) 90°

Q.22 A tower stands on a horizontal plane. A man on the ground 100 m from the base of the tower finds the angle of elevation of the top of the tower to be 30°. What is the height of the tower?

(a) 100 m (b) $100\sqrt{3}$
(c) $100/\sqrt{3}$ (d) None of these

Q.23 When the sun is 30° above the horizontal, the length of shadow cast by a building 50 m high is:

(a) $\frac{50}{\sqrt{3}}$ m (b) $50\sqrt{3}$ m
(c) 25 m (d) $25\sqrt{3}$ m

Q.24 A pole being broken by the wind, the top struck the ground at an angle of 30° and at a distance of the 21 m from the foot of the pole. Find the total height of the pole.

(a) 21 m (b) $2.21/\sqrt{3}$
(c) $21\sqrt{3}$m (d) None of these

Q.25 The shadow of a tower is found to be 50 m longer when the altitude of the sun is 30° than when it is 60°. Find the height of the tower.

(a) $20\sqrt{3}$ (b) $25\sqrt{3}$
(c) $25/\sqrt{3}$ m (d) None of these

Q.26 From the top of a cliff 25 m high, the angle of elevation of a tower is found to be equal to the angle of depression of the foot of the tower. Find the height of the tower.

(a) 40 m (b) 48 m (c) 50 m (d) 52 m

Q.27 When the length of the shadow of a pole is equal to the height of the pole, then the elevation of source of light is:

(a) 30° (b) 45° (c) 60° (d) 75°

Q.28 If I set out from my office at quarter past six and reach the station at half past seven, then I will be 10 min earlier than the person who is 15 min late to the train. At what time does the train start?

(a) 7:25 pm (b) 7:35 pm
(c) 7:40 pm (d) 7:20 pm

Q.29 If 8th June 2006, is Thursday, then which day would be 15th August 1947?

(a) Sunday (b) Saturday
(c) Friday (d) Thursday

Q.30 Fifty kids are standing in a circle and one of them is having a ball. The ball is passed counter clockwise, skipping 0, 1, 2, 3, 4, and so on, respectively, that is, the 1st kid gives it to the 2nd kid, then 2nd kid gives it to 4th kid, then 4th kid gives it to 7th kid, and so on. After how many hands change will the ball be back to the 1st kid?

(a) 12 (b) 20
(c) 22 (d) None of these

Moderate

Q.1 Jappi and Diksha are playing a card game involving green and red cards. They get 3 points every time they draw a green card and 5 points every time they draw a red card. The winner is the first one to reach 50 points.

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At one stage, Diksha has 41 points. Which of the following could not be the total number of cards drawn by Diksha at that stage?
(a) 7 (b) 12
(c) 10 (d) None of these

Q.2 If the rate of change of the length of a rectangle is 5 m/s and the rate of change of breadth of the same rectangle is 3 m/s, then the rate of change of area of this rectangle, when length = 10 m and breadth = 2 m, is:
(a) 30 m^2 /s (b) 20 m^2 /s
(c) 50 m^2 /s (d) 40 m^2 /s

Q.3 A cube of side of 4 cm is painted on all 6 faces with the same colour. It is then broken up into smaller cubes, each of side 1 cm. What is the proportion of the number cubes with no paint to the number of cubes with exactly one side painted to those with exactly two sides painted?
(a) 1:2:3 (b) 3:6:8 (c) 1:3:4 (d) 1:3:3

Q.4 The following data are available for the monsoon season of the Hyderabad racing club. The data is for a total of 'y' days.
- There were races on 11 days – morning or evening.
- Whenever there was a race in the morning, there was no race in the evening.
- There were 8 mornings without any race.
- There were 5 evenings without any race.

What is the value of y?
(a) 10 (b) 12
(c) 14 (d) Cannot be determined

Q.5 A gathering of a certain number of families consists of people of two generations only. It is known that the number of families is less than the number of girls, the number of girls is less than the number of boys, and that the number of boys is less than the number of parents. If the minimum number of single parent families is two, then what is the minimum number of families in the gathering, given that no family has more than three children?
(a) 4 (b) 5
(c) 3 (d) None of these

Q.6 1st January of Year 19XY and 1st January of year 19PQ are same day. It is also given that no other year before 19PQ has its 1st January on the same day as that of 1st January of 19XY. What is the minimum possible value of | PQ-XY|, where PQ and XY are the last two digits of the year?
(a) 5 (b) 6 (c) 11 (d) 13

Q.7 In the above question, what is maximum possible value of | PQ-XY|, where PQ and XY are the last two digits of the year?
(a) 5 (b) 6 (c) 11 (d) 13

Direction for Questions 8 and 9: Read the passage below and solve the questions based on it.

January 1 of the year 1 of a century falls on a Sunday. Let Year X be the earliest year by which January 1 has fallen on all days of the week.

Q.8 What is year X?
(a) 7 (b) 10 (c) 8 (d) 9

Q.9 Which of the following occurs the most number of times by year X (including year X)?
(a) Mondays
(b) Sundays, Mondays, and Wednesdays
(c) Mondays and Wednesdays
(d) None of these

Q.10 Given that $-3 \le a \le 2$, $0.5 \le b \le 3$, and $-2.5 \le 0.5 - 0.5$.
If $x = \frac{a^2}{bc}$, then which of the following is necessarily true?
(a) $-16 \le x \le 0$ (b) $-6 \le x \le 16$
(c) $-18 \le x \le 35$ (d) $-36 \le x \le 0$

Q.11 Given that $-3 \le x \le 4$ and $1 \le y \le 6$. Which of the following expressions can have the least possible value?
(a) $x^2y + y^2x$ (b) $xy^3 - yx^2$
(c) $x^2y^2 - xy^3$ (d) $xy^2 + y^2x^3$

Q.12 n is a real number. It is given that $n!^2 > n^n$. For what values of n relationship holds good?
(a) All even numbers (b) All natural numbers
(c) All whole numbers (d) None of these

Q.13 If $|x^2 - 7x + 12| > x^2 - 7x + 12$, then:
(a) $x \le 3$ or $x \ge 4$
(b) $3 \le x \le 4$
(c) $3 < x < 4$
(d) x can take any value except $x = 3$ and $x = 4$

Q.14 The inequality
$$\frac{|x|^2 - |x| - 2}{2|x| - |x|^2 - 2} > 2$$
holds if and only if:
(a) $-1 < x < -2/3$ or $2/3 < x < 1$
(b) $-1 < x < 1$
(c) $\frac{2}{3} < x < 1$
(d) $x > 1$ or $x < -1$ or $-\frac{2}{3} < x < \frac{2}{3}$

Q.15 For natural numbers N, the inequality $2^N > N^2$ holds when:
(a) $N > 2$ (b) $N = 1$ and $N > 4$
(c) $N > 1$ (d) None of these

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Q.16 What are the values of x satisfying $|x-2| \le 2$ and $|x+3| \ge 4$?
(a) $1 < x < 4$ (b) $1 \le x \le 4$
(c) $1 \le x$ or $x \ge 4$ (d) $1 < x$ or $x > 4$

Direction for Questions 17 to 20: Find the values of x for which the inequality holds true.

Q.17 $\dfrac{x^2-5x+6}{|x|+7} < 0$
(a) $2 \le x < 5$ (b) $-2 \le x \le 3$
(c) $2 < x < 3$ (d) $2 \le x \le 3$

Q.18 $|x^2+3x|+x^2-2 \ge 0$
(a) $-1/2 \le x < 1/2$
(b) $x \le -1/2$ or $x \ge 1/2$
(c) $x \ge 1/2$ and $x \le 1/2$
(d) None of these

Q.19 $\dfrac{x^2-5x+6}{x^2+x+1} < 0$
(a) $2 < x < 3$ (b) $x < 2$ or $x > 3$
(c) $2 \le x < 3$ (d) $2 \le x \le 3$

Q.20 In a group of 120 persons, there are 80 Bengalis and 40 Gujaratis. Further, 70 persons in the group are Muslims and the remaining are Hindus. Then, the number of Bengali Muslims in the group is:
(a) Between 10 and 14
(b) Between 15 and 19
(c) Exactly 20
(d) 25 or more

Q.21 300 persons are participating in a meeting in India, out of which 120 are foreigners, and the rest are Indians. Out of the Indians, there are 110 men who are not judges; 160 are men judges, and 35 are women judges. There are no foreign judges. How many Indian women attended the meeting?
(a) 60 (b) 45 (c) 55 (d) 40

Q.22 There are six persons: A, B, C, D, E, and F.
A has 3 items more than C.
D has 4 items less than B.
E has 6 items less than F.
C has 2 items more than E.
F has 3 items more than D.
Which one of the following figures cannot be equal to the total number of items possessed by all the six persons?
(a) 41 (b) 53 (c) 47 (d) 58

Q.23 The angle of elevation of the top of a tower from a point 20 m away from its base is 45°. The height of the tower is:
(a) 10 m (b) 20 m
(c) 40 m (d) $200\sqrt{3}$ m

Q.24 The angle of elevation of the sun when the length of the shadow of a pole is $\sqrt{3}$ times the height of the pole will be:
(a) 30° (b) 60° (c) 90° (d) 45°

Q.25 A person walking along a straight road towards a hill observes at two points, distance $\sqrt{3}$ km, the angles of elevation of the hill to be 30° and 60°. The height of the hill is:
(a) $\frac{3}{2}$km (b) $\sqrt{\frac{2}{3}}$km
(c) $\frac{\sqrt{3}+1}{2}$km (d) $\sqrt{3}$km

Q.26 The tops of two poles of height 20 m and 14 m are connected by a wire. If the wire makes an angle 30° with the horizontal, then the length of the wire is:
(a) 12 m (b) 10 m
(c) 8 m (d) None of these

Q.27 The angles of elevation of the top of a tower from points at distance A and B from the base and in the same line with it are complementary. If $A > B$, then height of the tower is:
(a) $\sqrt{B-A}$ (b) $\sqrt{B/A}$ (c) $\sqrt{B+A}$ (d) $\sqrt{BA}$

Q.28 A man walks 10 m towards a lamp post and notices that the angle of elevation of the top of the post increases from 30° to 45°. Find the height of the post.
(a) 10 m (b) $5(\sqrt{3}+1)$ m
(c) $5(\sqrt{3}-1)$ m (d) None of these

Q.29 A 6-ft-tall man finds that the angle of elevation of the top of a 24-ft-high tower and the angle of depression of its base are complementary angles. The distance of the man from the pillar is:
(a) $2\sqrt{3}$ft (b) $8\sqrt{3}$ft
(c) $6\sqrt{3}$ft (d) None of these

Q.30 A vertical pole PO is standing at the centre O of a square ABCD. If AC subtends an angle 90° at the top P of the pole, then the angle subtended by a side of the square at P is:
(a) 45° (b) 30°
(c) 60° (d) None of these

Q.31 A vertical lamp post of height 9 m stands at the corner of rectangular field. The angle of evaluation of its top from the far farthest corner is 30°, whereas from another corner it is 45°. The area of the field is:
(a) $81\sqrt{2}$ m (b) 9 m (c) 81 m (d) 9 m

Q.32 A vertical lamp post, 6 m high, stands at a distance of 2 m from a wall of 4 m high. A 1.5 m-tall man starts to walk away from the wall to the other side of the wall,

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in line with the lamp post. The maximum distance to which the man can walk remaining in the shadow is:

(a) $\frac{5}{2}$ m (b) $\frac{3}{2}$ m
(c) 4 m (d) None of these

Direction for Questions 33 and 34: Read the passage below and solve the questions based on it.

Mayank had left his home at the age of 14. He remembers that the day was Monday. Since then, he has been fasting on every Tuesday. Today he is celebrating his 60th birth anniversary in a five-star hotel with his friends. As today is Tuesday again, so he is not taking anything except wine. At the end of the party, he discloses that it is his 2400th Tuesday of fasting.

Q.33 Today is October 9, 2001. On which date had he left his home?
(a) 10.10.1956 (b) 8.8.1955
(c) 10.10.1955 (d) 8.8.1955

Q.34 He was born on?
(a) Sunday (b) Monday
(c) Wednesday (d) Thursday

Q.35 All the possible two-character codes are formed using the digits 0 to 9 and the capital letters of the English alphabet A to Z. How many such codes are there, which upon writing the code on a piece of paper and then turning the paper around to read the code upside down will be mistaken for another code? There is no confusion between zero and the letter O or the number 1 and the letter I.
(a) 72 (b) 90 (c) 94 (d) 100

ADVANCED

Q.1 Let us express date in the DDMMYYYY format, where DD represents the day of the month, MM represents the month, and YYYY represents the year. If the last possible date in the 20th century with all the eight digits (in the date as expressed in the above format) being odd is a Sunday, then what day of the week will the first date of the 21st century with all the eight digits even will be?
(a) Monday (b) Saturday
(c) Friday (d) Tuesday

Q.2 A Computer Science student was using the following algorithm on his computer:
Step 1 : Read A
Step 2 : B = 1
Step 3 : If A > 1, then B = B × A
Step 4 : If A ≤ 1, then print B and stop
Step 5 : A = A − 1
Step 6 : go to step 3
If A is a non-negative integer, then what is the student trying to find using the algorithm?
(a) Square root of A
(b) Factorial of A
(c) Sum of first A natural numbers
(d) None of these

Direction for Questions 3 to 5: Read the passage below and solve the questions based on it.

Pankaj has 6 friends, and during a certain vacation, he met them at several dinners. He found that he dined with all 6 friends exactly one day; with every 5 of them of 2 days; with every 4 of them of 3 days; with every 3 of them of 4 days; with every 2 of them of 5 days. Further, every friend was present at 8 dinners and every friend was absent at 8 dinners.

Q.3 How many dinners did Pankaj have during the vacation?
(a) 220 (b) 173 (c) 16 (d) 12

Q.4 How many dinners did Pankaj have alone during the vacation?
(a) 6 (b) 1 (c) 7 (d) 2

Q.5 When Pankaj had dinner with every 3 of them of 4 days and every 2 of them of 5 days, what was the minimum number of friends who had dinner twice in these 9 days?
(a) 2 (b) 4 (c) 6 (d) 8

Q.6 A club with x members is organized into 4 committees according to the following rules:
1. Each member belongs to exactly two committees.
2. Each pair of committees has exactly one member in common.

Then, which of the following is true regarding the value of x?
(a) $x = 4$ (b) $x = 6$
(c) $x = 8$ (d) Cannot be determined

Q.7 In a football league, a particular team played 60 games in a season. The team never lost 3 games consecutively and never won five games consecutively in that season. If N is the number of games the team won in that season, then N satisfies:
(a) $24 < N < 50$ (b) $20 < N < 48$
(c) $12 < N < 40$ (d) $18 < N < 42$

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Q.8 From a group of 7 persons, 7 committee are formed. Any two committees have exactly one member in common. Each person is in exactly three committees. Then,
(a) at least one committee must have more than three members.
(b) each committee must have exactly three members.
(c) each committee must have more than three members.
(d) nothing can be said about the sizes of the committees.

Q.9 How many integers satisfy the inequality
$\frac{(x-1)(x+4)(x+2)}{(x-3)(x+8)(x+3)} < 0$?
(a) 2 (b) 4 (c) 6 (d) Infinite

Q.10 If x satisfies $|x-1|+|x-2|+|x-3| \geq 6$, then:
(a) $0 \leq x \leq 4$ (b) $x \leq -2$ or $x \geq 4$
(c) $x \leq 0$ or $x \geq 4$ (d) None of these

Q.11 For real values of x, the expression $\frac{(x-c)(x-b)}{(x-a)}$ will assume all real values provided.
(a) $a \leq c \leq b$ (b) $b \leq a \leq c$
(c) $b \leq c \leq a$ (d) $a \leq b \leq c$

Q.12 Let a_n = product of the first n natural numbers. Then, for all $n \geq N$:
(a) $n^n \geq a_n$ (b) $\left(\frac{n+1}{2}\right)^n \geq n!$
(c) $n^n \geq a_{n+1}$ (d) None of these

Q.13 Solve for the values of y:
$\frac{z}{y} < 4, z \times y > 3, -3 < z < 3$
(a) $y < -\frac{\sqrt{3}}{2}$ (b) $y > \frac{\sqrt{3}}{2}$
(c) All the real values (d) None of these

Q.14 If $a^2 + b^2 + c^2 = 1$, then $ab + bc + ca$ lies in between (including):
(a) -2 and 1 (b) -1 and -2
(c) 0 and 1 (d) None of these

Q.15 If a, b, c, and d are real numbers such that $b > 0$, $d > 0$ and $\frac{a}{b} < \frac{c}{d}$, then which of the following is true?
(a) $\frac{a}{b} < \frac{a-c}{b+d} < \frac{c}{d}$ (b) $\frac{a}{b} < \frac{a+c}{b+d} < \frac{c}{d}$
(c) $\frac{a}{b} < \frac{a-c}{b-d} < \frac{c}{d}$ (d) $\frac{a}{b} < \frac{a+c}{b-d} < \frac{c}{d}$

Q.16 Let x, y, z, and w be positive real numbers, which satisfy the following conditions:
(i) If $x > y$, then $z > w$
(ii) If $x > z$, then $y < w$
One of the following statements given below is a valid conclusion. Which one is it?
(a) If $x < y$, then $z < w$.
(b) If $x < z$, then $y > w$.
(c) If $x > y + z$, then $z < y$.
(d) If $x > y + z$, then $z > y$.

Q.17 If $xy + x + y = 2$, then which of the following is true?
(a) $\left(\frac{x+1}{y+1}\right) > 0$ (b) $\left(\frac{x+2}{y+2}\right) > 0$
(c) $\frac{x-1}{y-1} > 0$ (d) $\frac{x-2}{y-2} > 0$

Q.18 Let $N = \frac{3 \times 5 \times 7 \times \ldots \times 99}{2 \times 4 \times 6 \times \ldots \times 100}$
Then, which of the following is true regarding the value of N?
(a) $1/3 < N < 1/2$ (b) $1/5 < N < 1/4$
(c) $1/15 < N < 1/10$ (d) $1/10 < N < 1/5$

Q.19 The number of points (x, y) satisfying (i) $3y - 4x = 20$ and (ii) $x^2 + y^2 \leq 16$ is:
(a) 0 (b) 1
(c) 2 (d) infinite

Q.20 A sequence of terms, a_1, a_2, a_3, … a_i, is generated as follows. The first term (a_1) is chosen as (a). To obtain every subsequent term, a coin is tossed. If a heads (H) appears, then the next term is obtained by adding 2 to the previous term. If a tails (T) appears, only 1 is added to obtain the next term. For example, the sequence corresponding to HTH is 1, 3, 4, and 6.
If a_{11} =15, then what is the maximum number of consecutive tails that could have appeared up to that term?
(a) 6 (b) 9
(c) 7 (d) Cannot be determined

Q.21 The angle of elevation of a jet plane from a point A on the ground is 60°. After a flight of 15 seconds, the angle of elevation changes to 30°. If the jet plane is flying at a constant height of $1500\sqrt{3}$, then find the speed of jet plane.
(a) 720 kmph (b) 600 kmph
(c) 540 kmph (d) None of these

Q.22 A club has 108 members. Two-third of them are men and the rest are women. All members are married except for 9 women members. How many married women are there is the club?
(a) 20 (b) 24 (c) 27 (d) 30

Q.23 I see a bird flying away at a constant speed of 1.7568 kmph in the sky. The angle of elevation is 60°. After $\frac{1}{2}$ min,

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I see the bird again and this time the angle of elevation is 45°. What is the perpendicular (horizontal) distance of the bird from me?

(a) 10 m (b) $10\sqrt{3}$m

(c) 20 m (d) $20\sqrt{3}$m

Q.24 Amitabh makes a road trip to the Hooghly beach. If he averaged his speed 5 kmph more, he could have decreased his time by 10%. On the other hand, if he averaged his speed 5 kmph less, the journey to the beach would have taken an extra hour. What is the distance from Amitabh's home to the Hooghly beach?

(a) 380 km (b) 420 km (c) 275 km (d) 360km

Direction for Questions 25 and 26: Read the passage below and solve the questions based on it.

Two supersonic planes are flying in a straight line horizontally on different altitudes such that the distance between them is 200 km and the pilot of upper plane observes the lower plane at an angle of depression of 30°. After half-an-hour, the angle of depression becomes 60°.

Q.25 Find the distance between them after half-an-hour with respect to the initial point of reference.

(a) $\frac{100}{\sqrt{3}}$km (b) $200\sqrt{3}$km

(c) $\frac{200}{\sqrt{3}}$km (d) Cannot be determined

Q.26 If they are travelling in opposite directions and the ratio of speed of upper plane to speed of lower plane is 2:1, then find the speed of lower plane.

(a) $\frac{400}{\sqrt{3}}$kmph (b) $\frac{200}{\sqrt{3}}$kmph

(c) $\frac{800}{\sqrt{3}}$kmph (d) Cannot be determined

Q.27 The length of a ladder exactly equals the height of a wall. If the ladder is placed on a 2 ft tall stool placed 10 ft away from the wall, then its tip can just touch the top of the wall. The height of the wall is:

(a) 15 ft (b) 26 ft (c) 28 ft (d) 32 ft

Q.28 A tree of height 21 m on a roadside broke at a certain height and fell in such a way that its top touched the other edge of the road. If the breadth of the road is 12 m, then the height above the ground at which the tree broke was:

(a) 12 m (b) 6 m

(c) 9 m (d) None of these

Q.29 If a flagstaff of 6 metres high placed on the top of a tower throws a shadow of $2\sqrt{3}$ metres along the ground, then the angle (in degrees) that the sun makes with the ground is:

(a) 30° (b) 60°

(c) 45° (d) None of these

Q.30 In a chess tournament, each of the 5 players plays against every other player. No game results in a draw and the winner of each game gets one point and the loser gets zero. Then, which one of the following sequences cannot represent the scores of the five players?

(a) 3, 3, 2, 1, 1 (b) 3, 2, 2, 2, 1

(c) 2, 2, 2, 2, 2 (d) 4, 4, 1, 1, 0

Q.31 We consider the relation 'a person x shakes hand with a person y'. Obviously, if x shakes hand y, then y shakes hand with x. In a gathering of 99 persons, one of the following statements is always true, considering 0 to be an even number. Which one is it?

(a) There is at least one person who shakes hand exactly with an odd number of persons.

(b) There is at least one person who shakes hand exactly with an even number of persons.

(c) There is even number of persons who shake hand exactly with an even number of persons.

(d) None of these

Q.32 Let P, Q, R, S, and T be statements such that if P is true, then both Q and S are true, and if both R and S are true, then T is false. We then have:

(a) If T is true, then both P and R must be true.

(b) If T is true, then both P and R must be false.

(c) If T is true, then at least one of P and R must be true.

(d) If T is true, then at least one of P and R must be false.

Q.33 If A, B, C, and D are statements such that if at least one of A and B is true, then at least one of C and D must be true. Further more, both A and C are false. We then have:

(a) If D is false, then B is false.

(b) Both B and D are false.

(c) Both B and D are true.

(d) If D is true, then B is true.

Q.34 If x, y, and z are eliminated from the equations $\frac{x}{y}=\frac{a+1}{a-1}, \frac{y}{z}=\frac{b+1}{b-1}$, and, $\frac{z}{x}=\frac{c+1}{c-1}$, then which of the following is true regarding the relationship between a, b, and c?

(a) $a+b+c=-1$

(b) $abc=-1$

(c) $ab+bc+ca=-1$

(d) $a^2+b^2+c^2=-1$

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Answers

Foundation

1. (b)	**2.** (c)	**3.** (b)	**4.** (d)	**5.** (d)	**6.** (a)	**7.** (b)	**8.** (c)	**9.** (c)	**10.** (d)
11. (d)	**12.** (d)	**13.** (b)	**14.** (b)	**15.** (b)	**16.** (d)	**17.** (b)	**18.** (a)	**19.** (a)	**20.** (c)
21. (a)	**22.** (c)	**23.** (b)	**24.** (c)	**25.** (b)	**26.** (c)	**27.** (b)	**28.** (a)	**29.** (c)	**30.** (d)

Moderate

1. (d)	**2.** (d)	**3.** (d)	**4.** (b)	**5.** (b)	**6.** (a)	**7.** (c)	**8.** (b)	**9.** (b)	**10.** (a)
11. (a)	**12.** (d)	**13.** (c)	**14.** (a)	**15.** (b)	**16.** (b)	**17.** (c)	**18.** (d)	**19.** (a)	**20.** (d)
21. (c)	**22.** (b)	**23.** (b)	**24.** (a)	**25.** (a)	**26.** (a)	**27.** (d)	**28.** (b)	**29.** (c)	**30.** (c)
31. (a)	**32.** (a)	**33.** (c)	**34.** (d)	**35.** (c)					

Advanced

1. (c)	**2.** (b)	**3.** (b)	**4.** (a)	**5.** (d)	**6.** (d)	**7.** (b)	**8.** (b)	**9.** (b)	**10.** (c)
11. (b)	**12.** (a)	**13.** (d)	**14.** (d)	**15.** (b)	**16.** (d)	**17.** (a)	**18.** (c)	**19.** (b)	**20.** (b)
21. (a)	**22.** (c)	**23.** (d)	**24.** (d)	**25.** (d)	**26.** (b)	**27.** (b)	**28.** (d)	**29.** (b)	**30.** (d)
31. (b)	**32.** (d)	**33.** (a)	**34.** (c)						

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PART

2

CAT PAPERS

CAT 2002

CAT 2003

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CAT

CAT 2002

SECTION III

There are 50 questions in this paper. All questions carry equal mark.

There is negative marking for wrong answers.

Q.51 If there are 10 positive real numbers $n1 < n2 < n3... < n10$, how many triplets of these numbers ($n1$, $n2$, $n3$), ($n2$, $n3$, $n4$),…can be generated such that in each triplet, the first number is always less than the second number, and the second number is always less than the third number?

(a) 45 (b) 90 (c) 120 (d) 180

Q.52 In ΔABC, the internal bisector of ΔA meets BC at D. If AB = 4, AC = 3, and DA = 60°, then the length of AD is:

(a) $2\sqrt{3}$ (b) $\frac{12\sqrt{3}}{7}$ (c) $\frac{15\sqrt{3}}{8}$ (d) $\frac{6\sqrt{3}}{7}$

Q.53 Find the length of the common chord of two circles of radii 15 cm and 20 cm, whose centres are 25 cm apart.

(a) 24 cm (b) 25 cm (c) 15 cm (d) 20 cm

Q.54 If $f(x) = \log\left\{\frac{(1+x)}{1-x}\right\}$, find $f(x) + f(y)$.

(a) $f(x+y)$ (b) $f\left\{\frac{(x+y)}{(1+xy)}\right\}$

(c) $(x+y)f\left\{\frac{1}{(1+xy)}\right\}$ (d) $\frac{f(x)+f(y)}{(1+xy)}$

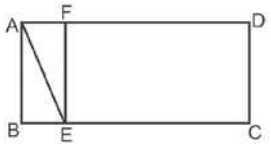
Q.55 Four horses are tethered at four corners of a square plot of side 14 m so that the adjacent horses can just reach one another. There is a small circular pond of area 20 m^2 at the centre. Find the ungrazed area.

(a) 22 m^2 (b) 42 m^2
(c) 84 m^2 (d) 168 m^2

Q.56 On a straight road XY, 100 m long, five heavy stones are placed 2 m apart beginning at the end X. A worker, starting at X, has to transport all the stones to Y, by carrying only one stone at a time. Find the minimum distance he has to travel.

(a) 472 m (b) 422 m (c) 744 m (d) 860 m

Q.57 In the figure given below, ABCD is a rectangle. The area of the isosceles right triangle ABE = 7 cm^2, EC = 3(BE). The area of ABCD (in cm^2) is:



(a) 21 cm^2 (b) 28 cm^2
(c) 42 cm^2 (d) 56 cm^2

Q.58 Find the area of the triangle whose vertices are (a, a), $(a + 1, a + 1)$, and $(a + 2, a)$.

(a) a^3 (b) 1 (c) $2a$ (d) $2^{1/2}$

Q.59 Instead of walking along two adjacent sides of a rectangular field, a boy took a short cut along the diagonal and saved a distance equal to half the longer side. Then, the ratio of the shorter side to the longer side is:

(a) $\frac{1}{2}$ (b) $\frac{2}{3}$ (c) $\frac{1}{4}$ (d) $\frac{3}{4}$

Q.60 Only a single rail track exists between Stations A and B on a railway line. One hour after the north-bound super fast train N leaves Station A for Station B, a south-bound passenger train S reaches

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Station A from Station B. The speed of the super fast train is twice that of a normal express train E, whereas the speed of a passenger train S is half that of E. On a particular day, N leaves for B from A, 20 min behind the normal schedule. In order to maintain the schedule, both N and S increased their speeds. If the super fast train doubles its speed, what should be the ratio (approximately) of the speeds of passenger train to that of the super fast train so that the passenger train S reaches exactly at the scheduled time at A on that day?

(a) 1:3 (b) 1:4 (c) 1:5 (d) 1:6

Q.61 On a 20-km tunnel, connecting two Cities A and B, there are three gutters (1, 2, and 3). The distance between gutters 1 and 2 is half the distance between gutters 2 and 3. The distance from City A to its nearest gutter, gutter 1, is equal to the distance of City B from gutter 3. On a particular day, the hospital in City A receives information that an accident has happened at gutter 3. The victim can be saved only if an operation is started within 40 min. An ambulance started from City A at 30 kmph and crossed gutter 1 after 5 min. If the driver had doubled the speed after that, what is the maximum amount of time would the doctor get to attend the patient at the hospital. Assume 1 min is elapsed for taking the patient into and out of the ambulance?

(a) 4 min
(b) 2.5 min
(c) 1.5 min
(d) The patient died before reaching the hospital

Q.62 Number S is obtained by squaring the sum of the digits of a two-digit number D. If difference between S and D is 27, then the two-digit number D is:

(a) 24 (b) 54 (c) 34 (d) 45

Q.63 The nth element of a series is represented as $X_n = (-1)^n X_{n-1}$

If $X_0 = x$ and $x > 0$, then which of the following is always true?

(a) X_n is positive if n is even.
(b) X_n is positive if n is odd.
(c) X_n is negative if n is even.
(d) None of these.

Q.64 If x, y, and z are real numbers such that $x + y + z = 5$ and $xy + yz + zx = 3$, what is the largest value that x can have?

(a) $\frac{5}{3}$ (b) $\sqrt{9}$
(c) $\frac{13}{3}$ (d) None of these

Q.65 Neeraj has agreed to mow a lawn, which is a 20 m × 40 m rectangle. He mows it with 1-m wide strip. If Neeraj starts at one corner and mows around the lawn towards the centre, then about how many times would he go round before he has mowed half the lawn?

(a) 2.5 (b) 3.5
(c) 3.5 (d) 4

Q.66 The owner of a local jewellery store hired three watchmen to guard his diamonds, but a thief still got in and stole some diamonds. On the way out, the thief met each watchman, one at a time. To each, he gave half of the diamonds he had then, besides 2 more. He escaped with one diamond. How many did he steal originally?

(a) 40 (b) 36
(c) 25 (d) None of these

Q.67 Mayank, Mirza, Little, and Jaspal bought a motorbike for $ 60. Mayank paid one-half of the sum of the amounts paid by the other boys. Mirza paid one-third of the sum of the amounts paid by the other boys. Little paid one-fourth of the sum of the amounts paid by the other boys. How much did Jaspal have to pay?

(a) $ 15 (b) $ 13
(c) $ 17 (d) None of these

Q.68 A rich merchant had collected many gold coins. He did not want anybody to know about his collection. One day, his wife asked, "How many gold coins do we have?" After a brief pause, he replied, 'Well! if I divide the coins into two unequal numbers, then 48 times the difference between the two numbers equals the difference between the squares of the two numbers'. The wife looked puzzled. Can you help the merchant's wife by finding out how many gold coins the merchant has?

(a) 96 (b) 53
(c) 43 (d) None of these

Q.69 Shyam visited Ram during his brief vacation. In the mornings, they both would go for yoga. In the evenings they would play tennis. To have more fun, they indulge in only one activity per day, i.e., either they went for yoga or played tennis each day. There were days when they were lazy and stayed home all day long. There were 24 mornings when they did nothing, 14 evenings when they stayed at home, and a total of 22 days when they did yoga or played tennis. For how many days Shyam stayed with Ram?

(a) 32 (b) 24
(c) 30 (d (d) None of these

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Q.70 Let S denotes the infinite sum $2 + 5x + 9x^2 + 14x^3 + 20x^4 + \ldots$, where $|x| < 1$ and the coefficient of x^{n-1} is $\frac{1}{2}n(n+3)$, $(n = 1, 2, \ldots)$. Then, S equals:

(a) $\frac{2-x}{(1-x)^3}$ (b) $\frac{2-x}{(1+x)^3}$

(c) $\frac{2+x}{(1-x)^3}$ (d) $\frac{2+x}{(1+x)^3}$

Q.71 If $x^2 + 5y^2 + z^2 = 2y\,(2x + z)$, then which of the following statements is/are necessarily true?

(A) $x = 2y$ (B) $x = 2z$ (C) $2x = z$

(a) Only A (b) B and C

(c) A and B (d) None of these

Q.72 Amol was asked to calculate the arithmetic mean of 10 positive integers, each of which had 2 digits. By mistake, he interchanged the 2 digits, say a and b, in one of these 10 integers. As a result, his answer for the arithmetic mean was 1.8 more than what it should have been. Then, $b - a$ equals:

(a) 1 (b) 2

(c) 3 (d) None of these

Q.73 A car rental agency has the following terms. If a car is rented for 5 h or less, then, the charge is ₹60 per hour or ₹12 per km whichever is more. On the other hand, if the car is rented for more than 5 h, the charge is ₹50 per hour or ₹7.50 per km whichever is more. Saurabh Gandhi rented a car from this agency, drove it for 30 km and ended up paying ₹300. For how many hours did he rent the car?

(a) 4 h (b) 5 h

(c) 6 h (d) None of these

Q.74 A child was asked to add first few natural numbers (i.e., 1 + 2 + 3 +…) so long as his patience permitted. As he stopped, he gave the sum as 575. When the teacher declared the result as wrong, the child discovered he had missed one number in the sequence during addition. The number he missed was:

(a) Less than 10 (b) 10

(c) 15 (d) More than 15

Q.75 Suppose for any real number x, $[x]$ denotes the greatest integer less than or equal to x. Let $L(x, y) = [x] + [y] + [x + y]$ and $R(x, y) = [2x] + [2y]$. Then, it is impossible to find any two positive real numbers x and y for which:

(a) $L(x, y) = R(x, y)$ (b) $L(x, y) \neq R(x, y)$

(c) $L(x, y) < R(x, y)$ (d) $L(x, y) > R(x, y)$

Q.76 Ten straight lines, no two of which are parallel and no three of which pass through any common point, are drawn on a plane. Find the total number of regions (including finite and infinite regions) into which the plane could be divided by the lines.

(a) 56 (b) 255

(c) 1024 (d) Not unique

Q.77 When 2^{256} is divided by 17, then the remainder would be:

(a) 1 (b) 16

(c) 14 (d) None of these

Q.78 The number of real roots of the equation $\frac{A^2}{x} + \frac{B^2}{x-1} = 1$, where A and B are real numbers and simultaneously not equal to zero is:

(a) None (b) 1 (c) 2 (d) 1 or 2

Q.79 At a bookstore, 'Modern Book Store' is a flashed using neon lights. The words as individually flashed at the intervals of $2\frac{1}{2}$s, $4\frac{1}{4}$s, and $5\frac{1}{8}$s, respectively, and each word is put off after a second. The least time after which the full name of the bookstore can be read again is:

(a) 49.5 seconds (b) 73.5 seconds

(c) 1744.5 seconds (d) 855 seconds

Q.80 Three pieces of cakes of weight $4\frac{1}{2}$ lb, $6\frac{3}{4}$ lb, and $7\frac{1}{5}$ lb, respectively, are to be divided into parts of equal weight. Each part must be as heavy as possible. If one such part is served to each guest, then what is the maximum number of guests who could be entertained?

(a) 54 (b) 72

(c) 20 (d) None of these

Q.81 After the division of a number successively by 3, 4, and 7, the remainders obtained are 2, 1, and 4, respectively. What will be the remainder if 84 divides the same number?

(a) 80 (b) 75 (c) 41 (d) 53

Q.82 Six persons are playing a card game. Suresh is facing Raghubir who is to the left of Ajay and to the right of Pramod. Ajay is to the left of Dhiraj. Yogendra is to the left of Pramod. If Dhiraj exchanges his seat with Yogendra and Pramod exchanges with Raghubir, who will be sitting to the left of Dhiraj?

(a) Yogendra (b) Raghubir

(c) Suresh (d) Ajay

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Direction for Questions 83 and 84: Answer the questions based on the following information:

A boy is asked to put one mango in a basket when ordered 'one', one orange when ordered 'two', one apple when ordered three, and is asked to take out from the basket one mango and an orange when ordered four. The sequence of orders is given as 1 2 3 3 2 1 4 2 3 1 4 2 2 3 3 1 4 1 1 3 2 3 4.

Q.83 How many total oranges were in the basket at the end of the above sequence?
(a) 1 (b) 4 (c) 3 (d) 2

Q.84 How many total fruits will be in the basket at the end of the above order sequence?
(a) 9 (b) 8 (c) 11 (d) 10

Direction for Questions 85 and 86: Answer the questions based on the following information:

Each of the 11 letters A, H, I, M, O, T, U, V, W, X, and Z appears same when looked at in a mirror. They are called symmetric letters. Other letters in the alphabet are asymmetric letters.

Q.85 How many four-letter computer passwords can be formed using only the symmetric letters (no repetition allowed)?
(a) 7920 (b) 330
(c) 14,640 (d) 419,430

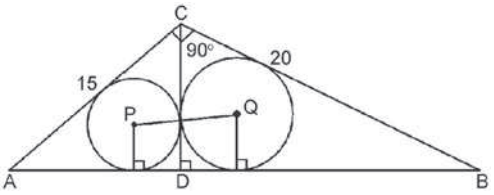
Q.86 How many three-letter computer passwords can be formed (no repetition allowed) with at least one symmetric letter?
(a) 990 (b) 2730
(c) 12,870 (d) 15,600

Q.87 A train approaches a tunnel AB. Inside the tunnel is a cat located at a point that is $\frac{3}{8}$ of the distance AB measured from the entrance A. When the train whistles, the cat runs away. If the cat moves to the entrance of the tunnel A, the train catches the cat exactly at the entrance. If the cat moves to the exit B, then the train catches the cat at exactly the exit. The speed of the train is greater than the speed of the cat by what order?
(a) 3:1 (b) 4:1
(c) 5:1 (d) None of these

Q.88 A piece of string is 40 cm long. It is cut into three pieces. The longest piece is three times as long as the middle-sized and the shortest piece is 23 cm shorter than the longest piece. Find the length of the shortest piece.
(a) 27 cm (b) 5 cm (c) 4 cm (d) 9 cm

Q.89 Three travellers are sitting around a fire and are about to eat a meal. One of them has 5 small loaves of bread, the second has 3 small loaves of bread. The third has no food, but has 8 coins. He offers to pay for some bread. They agree to share the 8 loaves equally among the three travellers, and the third traveller will pay 8 coins for his share of the 8 loaves. All loaves were of the same size. The second traveller (who had 3 loaves) suggests that he will be paid 3 coins, and that the first traveller be paid 5 coins. The first traveller says that he should get more than 5 coins. How much should the first traveller get?
(a) 5 (b) 7
(c) 1 (d) None of these

Q.90.



In the above figure, ACB is a right-angled triangle. CD is the altitude. Circles are inscribed within the ΔACD and ΔBCD. P and Q are the centres of the circles. The distance PQ is:
(a) 5 (b) 50 (c) 7 (d) 8

Q.91 If u, v is the weighted average and m is the natural number such that $u^m + v^m = w^m$, then which one of the following is true?
(a) $m \geq \min(u, v, w)$ (b) $m \geq \max(u, v, w)$
(c) $m < \min(u, v, w)$ (d) None of these

Q.92 In how many ways is it possible to choose a white square and a black square on a chessboard so that the squares must not lie in the same row or column?
(a) 56 (b) 896 (c) 60 (d) 768

Q.93 $7^{6n} - 6^{6n}$, where n is an integer > 0, is divisible by:
(a) 13 (b) 127
(c) 559 (d) All of these

Q.94 If $pqr = 1$, the value of the expression $\frac{1}{1+p+q^{-1}} + \frac{1}{1+q+r^{-1}} + \frac{1}{1+r+p^{-1}}$ is equal to:
(a) $p+q+r$ (b) $\frac{1}{p+q+r}$
(c) 1 (d) $p^{-1}+q^{-1}+r^{-1}$

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Q.95 It takes six technicians a total of 10 h to build a new server from a direct computer, with each working at the same rate. If six technicians start to build the server at 11 am, and one technician per hour is added beginning at 5 pm, at what time will the server be completed?

(a) 6:40 pm (b) 7 pm
(c) 7:20 pm (d) 8 pm

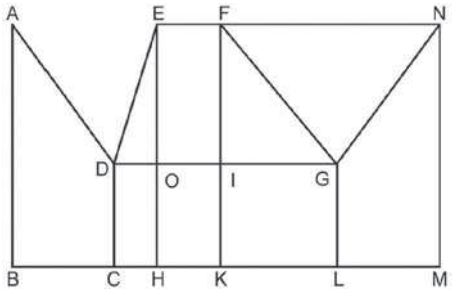
Q.96 Davji Shop sells samosas in boxes of different sizes. The samosas are priced at ₹2 per samosa up to 200 samosas. For every additional 20 samosas, the price of the whole lot goes down by 10 paise per samosa. What should be the maximum size of the box that would maximize the revenue?

(a) 240 (b) 300
(c) 400 (d) None of these

Q.97 Three small pumps and a large pump are filling a tank. Each of the three small pump works at $\frac{2}{3}$ rd the rate of the large pump. If all four pumps work at the same time, they should fill the tank in what fraction of the time that it would have taken the large pump alone?

(a) $\frac{4}{7}$ (b) $\frac{1}{3}$ (c) $\frac{2}{3}$ (d) $\frac{3}{4}$

Direction for Questions 98 to 100: Answer the questions based on the following diagram:



In the above diagram, ∠ABC = 90° = ∠DCH = ∠DOE = ∠EHK = ∠FKL = ∠GLM = ∠LMN
AB = BC = 2CH = 2CD = EH = FK = 2HK = 4KL = 2LM = MN

Q.98 The magnitude of ∠FGO is equal to:

(a) 30° (b) 45°
(c) 60° (d) None of these

Q.99 What is the ratio of the areas of the two quadrilaterals ABCD to DEFG?

(a) 1:2 (b) 2:1
(c) 12:7 (d) None of these

Q.100 How many numbers greater than 0 and less than a million can be formed with the digits 0, 7, and 8?

(a) 486 (b) 1084
(c) 728 (d) None of these

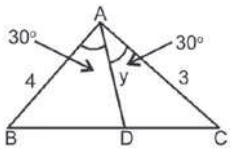
Answers

51. (c)	**52.** (b)	**53.** (a)	**54.** (b)	**55.** (a)	**56.** (d)	**57.** (d)	**58.** (b)	**59.** (d)	**60.** (d)
61. (c)	**62.** (b)	**63.** (d)	**64.** (d)	**65.** (c)	**66.** (b)	**67.** (b)	**68.** (d)	**69.** (c)	**70.** (a)
71. (c)	**72.** (b)	**73.** (c)	**74.** (d)	**75.** (d)	**76.** (a)	**77.** (a)	**78.** (d)	**79.** (b)	**80.** (d)
81. (d)	**82.** (c)	**83.** (d)	**84.** (c)	**85.** (a)	**86.** (c)	**87.** (a)	**88.** (c)	**89.** (b)	**90.** (b)
91. (d)	**92.** (d)	**93.** (d)	**94.** (c)	**95.** (d)	**96.** (b)	**97.** (b)	**98.** (d)	**99.** (c)	**100.** (c)

Hints and Solutions

51. Total possible arrangements = $10 \times 9 \times 8$.
Now, 3 number can be arranged among themselves in 3! way = 6 ways
Given condition is satisfied by only 1 out of 6 ways; hence, the required number of arrange-ments = $\frac{10 \times 9 \times 8}{6} = 120$
Hence, option (c) is the answer.

52.



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Let BC = x and AD = y

As per bisector theorem, $\frac{BD}{DC} = \frac{AB}{AC} = \frac{4}{3}$

Hence, BD = $\frac{4x}{7}$; DC = $\frac{3x}{7}$

In ΔABD, $\cos 30° = \frac{(4)^2 + y^2 - \frac{16x^2 49}{49}}{2 \times 4 \times y}$

$2 \times 4 \times y \times \frac{\sqrt{3}}{2} = 16 + y^2 - \frac{16x^2}{49}$

$4\sqrt{3}y = 16 + y^2 - \frac{16x^2}{49}$ (i)

Similarly, from DADC, $\text{Cos } 30° = \frac{9 + y^2 - \frac{9x^2}{49}}{2 \times 3 \times y}$

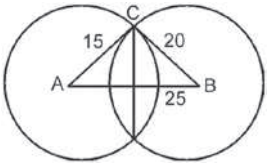
$\Rightarrow 3\sqrt{3}y = 9 + y^2 - \frac{9x^2}{49}$ (ii)

Now, (i) x 9 − 16 x (ii), we get

$36\sqrt{3}y - 48\sqrt{3}y = 9y^2 - 16y^2 \Rightarrow y = \frac{12\sqrt{3}}{7}$

Hence, option (b) is the answer.

53.



Let the chord be x cm.

$\therefore \frac{1}{2}(15 \times 20) = \frac{1}{2} \times 25 \times \frac{x}{2} \Rightarrow x = 24$ cm

Hence, option (a) is the answer.

54. $f(x) + f(y) = \log\left(\frac{1+x}{1-x}\right) + \log\left(\frac{1+y}{1-y}\right)$

$= \log\left(\frac{(1+x)(1+y)}{(1-x)(1-y)}\right)$

$= \log\left(\frac{1+x+y+xy}{1+xy-(x+y)}\right)$

$= \log\left(\frac{1+\left(\frac{x+y}{1-xy}\right)}{1-\left(\frac{x+y}{1-xy}\right)}\right)$

$= f\left(\frac{x+y}{1+xy}\right)$

Hence, option (b) is the answer.

55. Total area = $14 \times 14 = 196$ m^2

Grazed area = $\left(\frac{\pi \times r^2}{4}\right) \times 4 = \pi r^2 = 22 \times 7$

= 154 m^2

The ungrazed area is less than (196 - 154) = 42 m^2 for which there is only one option.

Hence, option (a) is the answer.

56. Every trip will need more than 180 m and there are $4\frac{1}{2}$ trips. Hence, the distance covered will be greater than 750 m for which there is only one option = 860.

Alternative Method:

For the first stone, he will cover 100 m.

For second, 200 – 4 = 196

For third, 200 – 8 = 192

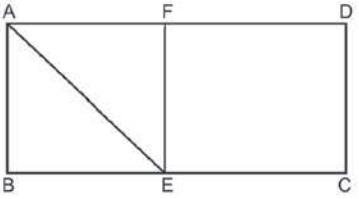
For fourth, 200 – 12 = 188

For fifth, 200 – 16 = 184

Hence, the total distance = 860 m

Hence, option (d) is the answer.

57.



Area of ΔABE = 7 cm^2

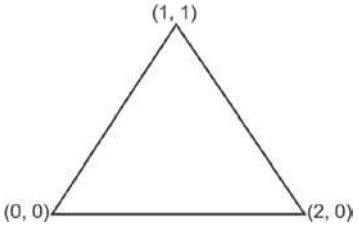
Area of ABEF = 14 cm^2

Area of ABCD = 14×4

= 56 cm^2

Hence, option (d) is the answer.

58.



Hence, area = $\frac{1}{2}$ (2) (1) = 1

Note: Answer should be independent of a and area of the triangle does not have a square root.

Hence, option (b) is the answer.

59. Check the options, e.g., $\frac{1}{2}$ diagonal = $\sqrt{5}$

Distance saved = 3 − $\sqrt{5}$ = 0.75 ≠ half the larger side; hence, incorrect.

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$\frac{3}{4} \Rightarrow$ Diagonal = 5

Distance saved = (4 + 3) − 5 = 2 = Half the larger side.

Hence, option (d) is the answer.

60. If speed of $N = 4$, speed of $g = 1$,

Average speed = $\frac{2 \times 4 \times 1}{4+1} = 1.6$

Because time available is $\frac{2}{3}$, speed = $\frac{3}{2}$

Now, average speed = 2, 4

Now, speed of $N = 8$

Now, speed of $S = y$

$\frac{2 \times 8 \times y}{8+y} = 2.4 \Rightarrow y = 1.3$

Required ratio = 1.3:8 = 1:6

Hence, option (d) is the answer.

61. $AG_1 = 5$ min at 30 kmph = 2.5 km

$G_1G_3 = 15$ km

Time for $AG_1 = 5$ min

Time for $G_1G_3 + G_3A = 32.5$ min = Total of 37.5 min 1 min is taken for transferring the patient into and out of the ambulance.

Hence, (40 − 37.5 − 1) = 1.5 min is remaining.

Hence, option (c) is the answer.

62. Check the options.

Option (b): 54 S = (5 + 4)2 = 81

$\Rightarrow$ D − S = 81 − 54 = 27.

Hence, the number = 54

Hence, option (b) is the answer.

63. $x_0 = x$

$x_1 = -x$

$x_3 = x$

$x_4 = x$

$x_5 = -x$

$x_6 = -x$

$\Rightarrow$ Choices (a), (b), and (c) are incorrect.

64. $(x + y + z)^2 = x^2 + y^2 + z^2 + 2\,(xy + zy + zx)$

$\Rightarrow x^2 + y^2 + z^2 = 19$

$\Rightarrow y + z$ cannot be simultaneously = 0

else, $xy + zy + zx = 0 \Rightarrow x^2 < 19$

$\Rightarrow x < \sqrt{19} \cong 4.4$

Hence, option (d) is the answer.

65. Area = 40 × 20 = 800

If 3 rounds are done, area = 34 × 14 = 476

Area > 3 rounds

If 4 rounds $\Rightarrow$ area left = 32 × 12 = 347

Hence, the area should be slightly less than 4 rounds.

Hence, option (c) is the answer.

66. Since thief escaped with 1 diamond,

Before 3rd watchman, he had (1 + 2) × 2 = 6

Before 2nd watchman, he had (6 + 2) × 2 = 16

Before 1st watchman, he had (16 + 2) × 2 = 36

Alternative Method:

Go through the options.

Hence, option (b) is the answer.

67. Mayank paid $\frac{1}{2}$ of what others paid.

$\Rightarrow$ Mayank paid $\frac{1}{3}$ rd of the total amount = $ 20.

Similarly, Mirza paid $15 and Little paid $ 12.

Remaining amount of $60 − $20 − $15 − $12 = $13 is paid by Jaspal.

Hence, option (b) is the answer.

68. Let the number of gold coins = $x + y$

$48(x - y) = x^2 - y$?

$48(x - y) = (x - y)\,(x + y) = -x + y = 48$

Hence, the correct option would be none of these.

Hence, option (d) is the answer.

69. Let us assume that

p days : They played tennis.

y days : They went for yoga.

T days : Total duration for which Ram and Shyam stayed together.

$\Rightarrow p + y = 22$

$T - y = 24$

$T - p = 14$

Adding all of them,

$2T = 22 + 24 + 14 - T = 30$ days

Hence, option (c) is the answer.

70. Co-efficient of $x = -(n + 1)(n + 4)$

$S = 2 + 5x + 9x^2 + 14x^3 +$

$xS = 2x + 5x^2 + \ldots$

$S(1 - x) = 23 + 3x + 4x^2 + 5x^3 + \ldots$

Let $S_1 = S(1 - x) \Rightarrow S_1 = 2 + 3x + 4x^2 + \ldots$

$x\,S_1 = 2x + 3x^2 + \ldots$

$S_1(1 - x) = 2 + x + x^2 + \ldots$

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$S_1(1-x^2) = 2 + \frac{x}{1-x}$

$S(1-x^2) = 2 + \frac{x}{1-x} \Rightarrow S = \frac{2-x}{(1-x)^3}$

Hence, option (a) is the answer.

71. $x^2 + 5y^2 + z^2 - 4yx + 2yz$

$(x^2 + 4y^2 - 4yx) + z^2 + y^2 - 2yz = 0$

$(x-2y)^2 + (z-y)^2 = 0$

It can be true only if $x = 2y$ and $z = y$.

Hence, option (c) is the answer.

72. Arithmetic mean is more by 18 means, sum is more by 18. So, $ba - ab = 18$

$b > a$ because sum has gone up, e.g., $31 - 13 = 18$

Hence, $b - a = 2$

Hence, option (b) is the answer.

73. By trial and error:

$30 \times 12 = 360 > 300$

$30 \times 7.5 = 225 < 300$

$50 \times 6 = 300$. Hence, he rented the car for 6 h.

Hence, option (c) is the answer.

74. $575 = -x$

$1150 = n^2 + n - 2x$

$n(n+1) \geq 1150$

The smallest value of n that will satisfy this will be $n = 34$.

For $n = 34$,

$40 = 2x \Rightarrow x = 20$

Hence, option (d) is the answer.

75. $x - 1 \leq [x] \leq x$

$2x + 2y - 3 \leq L(x, y) \leq 2x + 2y \Rightarrow a - 3 \leq L \leq a$

$2x + 2y - 2 \leq R(x, y) \leq 2x + 2y \Rightarrow a - 2 \leq R \leq a$

Therefore, $L \leq R$

Note: Option (b) is wrong, otherwise options (a) and (c) are also not correct. Choose the numbers to check.

Hence, option (d) is the answer.

76. Number of regions $= \frac{n(n+1)}{2} + 1$

where n = Number of lines, i.e., for 0 line, we have region = 1.

For 1 line, we have region = 2.

It can be shown as:

Number of lines	0	1	2	3	4	5	...	10
Number of regions	1	2	4	7	11	16	...	56

Therefore, for $n = 10$, it is $2 + 1 = 56$

Hence, option (a) is the answer.

77. $(2^4)^{64} = (17-1)^{64} = 17n + (-1)^{54} = 17n + 1$

Hence, remainder = 1

Hence, option (a) is the answer.

78. $\frac{A^2}{x} + \frac{B^2 - 1}{x-1} \Rightarrow A^2(x-1)\,B^2x = x^2 - x$

This is a quadratic equation. Hence, the number of roots = 2 or 1.

(1 in case when both roots are equal.)

Hence, option (d) is the answer.

79. Because each word is flashed for a second,

$$\text{LCM}\left(\frac{5}{2}+1, \frac{17}{4}+1, \frac{41}{8}+1\right) = \text{LCM}\left(\frac{7}{2}, \frac{21}{4}, \frac{49}{8}\right)$$

$$\frac{\text{LCM}(7,21,49)}{\text{HCM}(2,4,8)} = \frac{49 \times 3}{2} = 73.5\text{S}$$

Hence, option (b) is the answer.

80. $\text{HCF}\left(\frac{9}{2}, \frac{27}{4}, \frac{36}{5}\right) = \frac{\text{HCF}(9, 27, 36)}{\text{LCM}(2, 4, 5)} = \frac{9}{20}\text{lb}$

= weight of each piece total = 18.45 lb weight

Maximum number of guests $= \frac{18.45 \times 20}{9} = 41$

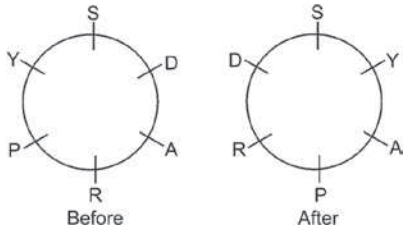
Hence, option (d) is the answer.

81. $3(4(7x+4)+1)+2 = 84x + 53$

Therefore, the remainder is 53.

Hence, option (d) is the answer.

82.



Suresh is sitting to the left of Dhiraj.

Hence, option (c) is the answer.

83. Number of oranges at the end of the sequence =

Number of $2s$ − Number of $4s = 6 - 4 = 2$

Hence, option (d) is the answer.

84. Number of $(1s + 2s + 3s) - 2 =$

Number of $4s = 19 - 8$

Hence, option (c) is the answer.

85. $11 \times 10 \times 9 \times 8 = 7920$

Hence, option (a) is the answer.

86. Total number of passwords using all letters − Total number of passwords using no symmetric letters

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$= (26 \times 25 \times 24) - (15 \times 14 \times 13) = 12870$

Hence, option (d) is the answer.

87. Let tunnel be 8 km and speed of cat be 1 kmph.

Time taken to reach the entrance of the tunnel by the cat = 3 h

Time taken to reach the exit of the tunnel by the cat = 5 h

Train will cover the sum (length of tunnel) = 2 h

Therefore, the ratio of the speed of the train and the cat = 4:1

Speed of the train is greater by 3:1 than that of the cat.

Hence, option (a) is the answer.

88. Let the largest piece = $3x$; middle = x; and shortest = $3x - 23$

$3x + x + (3x - 23) = 40$

$x = 9$

The shortest piece = $3(9) - 23 = 4$

Check the options.

The shortest piece has to be < 20 cm.

27 is a wrong choice.

The largest piece is a multiple of 3.

Or, (23 + shortest) should be a multiple of 3.

Answer = 4 cm (Among other choices)

Hence, option (c) is the answer.

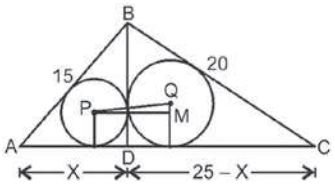
89. Each traveller had $\frac{8}{3}$ loaves.

First traveller has given $5 - \frac{8}{3}$ loaves to the third.

Second traveller sacrificed only $3 - \frac{8}{3} = \frac{1}{3}$ rd of a loaf. So, first should get 7 coins.

Hence, option (b) is the answer.

90.



$(15)^2 - x^2 = (20)^2 - (25 - x)^2$

$\Rightarrow x = 9$

$\Rightarrow$ BD = 12

Area of $\Delta ABD = \frac{1}{2} \times 12 \times 9 = 54$

$s = \frac{1}{2}(15 + 12 + 9) = 18$

$r_1 = \frac{\text{Area}}{s} \Rightarrow r_1 = 3$

Area of $\Delta BCD = \frac{1}{2} = 96$

$s = \frac{1}{2}(16 + 20 + 12) = 24$

$r_2 = \frac{\text{Area}}{s} \Rightarrow r_2 = 4$

In ΔPQM, $PM = r_1 + r_2 = 7$ cm

$QM = r_2 - r_1 = 1$ cm

Hence, $PQ = \sqrt{50}$ cm

Hence, option (b) is the answer.

91. $u^m + v^m = w^m$

$u^2 + v^2 = w^2$

Taking Pythagorean triplet 3, 4, and 5, we see in $< mm$ (u, v, w)

Also, $1' + 2' = 3'$, and hence, in $\leq mm$ (u, v, w)

Hence, option (d) is the answer.

92. A black square can be chosen in 32 ways. Once a black square is there, you cannot choose the 8 white squares in its row or column. So, the number of white squares available = 24.

Number of ways = $32 \times 24 = 768$

Hence, option (d) is the answer.

93. $7^{6n} - 6^{6n}$

Put $n = 1$

$7^6 - 6^6 = (7^3 - 6^3)(7^3 + 6^3)$

This is a multiple of $7^3 + 6^3 = 559$ and $7 + 6 = 13$

Hence, option (d) is the answer.

94. If $p = q = r = 1$, then expression = 1

Check the options only, one choice gives the value of expression = 1.

Hence, option (c) is the answer.

95. Total amount of work = 60 man hours

From 11 am to 5 pm, 6 technicians = 36 man hours

From 5 pm to 6 pm, 7 technicians = 7 man hours

From 6 pm to 7 pm, 8 technicians = 8 man hours

From 7 am to 8 pm, 9 technicians = 9 man hours

Total = 60 man hours

Hence, option (d) is the answer.

96. Number of samosas = $200 + 20n$

n is a natural number.

Price per samosa = ₹$(2 - 0.1n)$

Revenue = $(200 + 20n)(2 - 0.1n) = 400 + 20n - 2n^2$

For maxima $20 - 4n = 0$; by differentiation $n = 5$

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$\Rightarrow$ Maximum revenue will be at $(200 + 20 \times 5) = 300$ samosas

Hence, option (b) is the answer.

97. Three small pumps = Two large pumps

Three small + One large pump

= Three large pumps $\Rightarrow \frac{1}{3}$

Hence, option (b) is the answer.

98. If KL = 1, then IG = 1 and FI = 2

Hence, tan G $= \frac{2}{1} = 2$

Thus, q is none of 30°, 40°, and 60°.

Hence, option (d) is the answer.

99. Area of quadrilateral ABCD $= \frac{1}{2}(2x + 4x) \times 4x = 12x^2$

Area of quadrilateral DEFG $= \frac{1}{2}(5x + 2x) \times 2x = 7x^2$

Hence, option (c) is the answer.

100. Number of ways for single digit = 2

2 digits = $2 \times 3 = 6$

3 digits = $2 \times 3 \times 3 = 18$

4 digits = $2 \times 3 \times 3 \times 3 = 54$

5 digits = $2 \times 3 \times 3 \times 3 \times 3 = 162$

6 digits = $2 \times 3 \times 3 \times 3 \times 3 \times 3 = 486$

Total = 728

Hence, option (c) is the answer.

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CAT 2003

LEAKED PAPER

Direction for Questions 101 and 102: Answer the questions on the basis of the information given below.

A certain perfume is available at a duty–free shop at the Bangkok International Airport. It is priced in the Thai currency Baht but other currencies are also acceptable. In particular, the shop accepts Euro and US Dollar at the following rates of exchange:

1 US Dollar = 41 Bahts and 1 Euro = 46 Bahts

The perfume is priced at 520 Bahts per bottle. After one bottle is purchased, subsequent bottles are available at a discount of 30%. Three friends S, R, and M together purchase three bottles of the perfume, agreeing to share the cost equally. R pays 2 Euros, M pays 4 Euros and 27 Thai Bahts, and S pays the remaining amount in US Dollars.

Q.101 How much does R owe to S in Thai Baht?
(a) 428 (b) 416 (c) 334 (d) 324

Q.102 How much does M owe to S in US Dollars?
(a) 3 (b) 4 (c) 5 (d) 6

Direction for Questions 103 and 104: Answer the questions on the basis of the information given below:

New Age Consultants have three consultants Gyani, Medha, and Buddhi. The sum of the number of projects handled by Gyani and Buddhi individually is equal to the number of projects in which Medha is involved. All three consultants are involved together in 6 projects. Gyani works with Medha in 14 projects. Buddhi has 2 projects with Medha but without Gyani, and 3 projects with Gyani but without Medha. The total number of projects for the New Age Consultants is one less than twice the number of projects in which more than one consultant is involved.

Q.103 What is the number of projects in which Gyani alone is involved?
(a) Uniquely equal to zero
(b) Uniquely equal to 1
(c) Uniquely equal to 4
(d) Cannot be determined uniquely

Q.104 What is the number of projects in which Medha alone is involved?
(a) Uniquely equal to zero
(b) Uniquely equal to 1
(c) Uniquely equal to 4
(d) Cannot be determined uniquely

Direction for Questions 105 to 110: Answer the questions independently of each other:

Q.105 The number of non–negative real roots of $2^x - x - 1 = 0$ equals:
(a) 0 (b) 1 (c) 2 (d) 3

Q.106 When the curves $y = \log_{10} x$ and $y = x^{-1}$ are drawn in the $x - y$ plane, how many times do they intersect for values $x \geq 1$?
(a) Never (b) Once
(c) Twice (d) More than twice

Q.107 Let A and B be the two solid spheres such that the surface area of B is 300% higher than the surface area of A. The volume of A is found to be k% lower than the volume of B. The value of k must be:
(a) 85.5 (b) 92.5 (c) 90.5 (d) 87.5

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Q.108 Which one of the following conditions must p, q, and r satisfy so that the following system of linear simultaneous equations has at least one solution, such that
$p + q + r \neq 0$?
$x + 2y - 3z = p$
$2x + 6y - 11z = q$
$x - 2y + 7z = r$
(a) $5p - 2q - r = 0$ (b) $5p + 2q + r = 0$
(c) $5p + 2q - r = 0$ (d) $5p - 2q + r = 0$

Q.109 A leather factory produces two kinds of bags, standard and deluxe. The profit margin is ₹20 on a standard bag and ₹30 on a deluxe bag. Every bag must be processed on machine A and on Machine B. The processing times per bag on the two machines are as follows:

	Time required (h/bag)	
	Machine A	Machine B
Standard bag	4	6
Delux bag	5	10

The total time available on machine A is 700 h and on machine B is 1250 h. Among the following production plans, which one meets the machine availability constraints and maximizes the profit?
(a) Standard 75 bags, deluxe 80 bags
(b) Standard 100 bags, deluxe 60 bags
(c) Standard 50 bags, deluxe 100 bags
(d) Standard 60 bags, deluxe 90 bags

Q.110 The sum of the 3rd and the 15th elements of an arithmetic progression is equal to the sum of 6th, 11th, and 13th elements of the same progression. Then, which element of the series should necessarily be equal to zero?
(a) 1st (b) 9th
(c) 12th (d) None of the above

Direction for Questions 111 to 113: Answer the questions on the basis of the information given below:

A city has two perfectly circular and concentric ring roads, the outer ring road (OR) being twice as long as the inner ring road (IR). There are also four (straight line) chord roads from E1, the east end point of OR to N2, the north end point of IR; from N1, the north end point of OR to W2, the west end point of IR; from W1, the west end point of OR to S2, the south end point of IR; and from S1, the south end point of OR to E2, the east end point of IR. Traffic moves at a constant speed of 30 p kmph on the OR road, 20p kmph on the IR road, and $15\sqrt{5}$ kmph on all the chord roads.

Q.111 The ratio of the sum of the lengths of all the chord roads to the length of the outer ring road is:
(a) $\sqrt{5}$:2 (b) $\sqrt{5}$:2p
(c) $\sqrt{5}$:p (d) None of these

Q.112 Amit wants to reach N2 from S1. It would take him 90 min if he goes on minor arc S1 – E1 on OR, and then on the chord road E1 – N2. What is the radius of the outer ring road in kilometre?
(a) 60 (b) 40 (c) 30 (d) 20

Q.113 Amit wants to reach E2 from N1 using first chord N1 – W2 and then the inner ring road. What will be his travel time in min on the basis of information given in the above question?
(a) 60 (b) 45 (c) 90 (d) 105

Direction for Questions 114 to 120: Answer the questions independently of each other:

Q.114 A test has 50 questions. A student scores 1 mark for a correct answer, –1/3 for a wrong answer, and –1/6 for not attempting a question. If the net score of a student is 32, the number of questions answered wrongly by that student cannot be less than:
(a) 6 (b) 12 (c) 3 (d) 9

Q.115 Twenty–seven persons attend a party. Which one of the following statements can never be true?
(a) There is a person in the party who is acquainted with all the twenty–six others.
(b) Each person in the party has a different number of acquaintances.
(c) There is a person in the party who has an odd number of acquaintances.
(d) In the party there is no set of three mutual acquaintances.

Q.116 Let $g(x) = \max(5 - x, x + 2)$. The smallest possible value of $g(x)$ is:
(a) 4.0 (b) 4.5
(c) 1.5 (d) None of these

Q.117 The function $f(x) = |x - 2| + |2.5 - x| + |3.6 - x|$, where x is a real number, attains a minimum at:
(a) $x = 2.3$ (b) $x = 2.5$
(c) $x = 2.7$ (d) None of these

Q.118 How many even integers n, where $100 \leq n \leq 200$, are divisible neither by seven nor by nine?
(a) 40 (b) 37 (c) 39 (d) 38

Q.119 A positive whole number M less than 100 is represented in base 2 notation, base 3 notation, and base 5 notation. It is found that in all three cases, the last digit is 1, whereas in exactly two out of the three cases, the leading digit is 1. Then, M equals:
(a) 31 (b) 63 (c) 75 (d) 91

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Q.120 In a 4000–m race around a circular stadium having a circumference of 1000 m, the fastest runner and the slowest runner reach the same point at the end of the 5th min for the first time after the start of the race. All the runners have the same starting point and each runner maintains a uniform speed throughout the race. If the fastest runner runs at twice the speed of the slowest runner, what is the time taken by the fastest runner to finish the race?

(a) 20 min (b) 15 min
(c) 10 min (d) 5 min

Direction for Questions 121 to 125: Each question is followed by two Statements A and B. Answer each question using the following instructions:

Choose (a): If the question can be answered by one of the statements alone but not by the other.

Choose (b): If the question can be answered by using either statement alone.

Choose (c): If the question can be answered by using both the statements together but cannot be answered by using either statement alone.

Choose (d): If the question cannot be answered even by using both the statements together.

Q.121 Is $a^{44} < b^{11}$, given that $a = 2$ and b is an integer?

(A) b is even.
(B) b is greater than 16.

Q.122 What are the unique values of b and c in the equation $4x^2 + bx + c = 0$ if one of the roots of the equation is $(-1/2)$?

(A) The second root is 1/2.
(B) The ratio of c and b is 1.

Q.123 AB is a chord of a circle. AB = 5 cm. A tangent parallel to AB touches the minor arc AB at E. What is the radius of the circle?

(A) AB is not a diameter of the circle.
(B) The distance between AB and the tangent at E is 5 cm.

Q.124 Is $\left(\frac{1}{a^2}+\frac{1}{a^4}+\frac{1}{a^6}+\ldots\right) > \left(\frac{1}{a}+\frac{1}{a^3}+\frac{1}{a^5}+\ldots\right)$?

(A) $-3 \le a \le 3$
(B) One of the roots of the equation $4x^2 - 4x + 1 = 0$ is a.

Q.125 D, E, and F are the mid–points of the sides AB, BC, and CA of triangle ABC, respectively. What is the area of DEF in square centimeters?

(A) AD = 1 cm, DF = 1 cm, and perimeter of DEF = 3 cm
(B) Perimeter of ABC = 6 cm, AB = 2 cm, and AC = 2 cm

Direction for Questions 126 to 150: Answer the questions independently of each other:

Q.126 At the end of year 1998, Shepard bought nine dozen goats. Henceforth, every year he added $p\%$ of the goats at the beginning of the year and sold $q\%$ of the goats at the end of the year where $p > 0$ and $q > 0$. If Shepard had nine dozen goats at the end of year 2002 after making the sales for that year, which of the following is true?

(a) $p = q$ (b) $p < q$
(c) $p > q$ (d) $p = q/2$

Q.127 Each side of a given polygon is parallel to either the X or the Y axis. A corner of such a polygon is said to be convex if the internal angle is 90° or concave if the internal angle is 270°. If the number of convex corners in such a polygon is 25, the number of concave corners must be:

(a) 20 (b) 0 (c) 21 (d) 22

Q.128 The 288th term of the series a, b, b, c, c, c, d, d, d, d, e, e, e, e, e, f, f, f, f, f, f... is:

(a) u (b) v (c) w (d) x

Q.129 Let p and q be the roots of the quadratic equation $x^2 - (a - 2)x - a - 1 = 0$. What is the minimum possible value of $p^2 + q^2$?

(a) 0 (b) 3 (c) 4 (d) 5

Q.130 There are two concentric circles such that the area of the outer circle is four times the area of the inner circle. Let A, B, and C be three distinct points on the perimeter of the outer circle such that AB and AC are tangents to the inner circle. If the area of the outer circle is 12 cm^2, then the area (in sq. cm) of the triangle ABC would be:

(a) $\pi\sqrt{12}$ (b) $\frac{9}{\pi}$
(c) $\frac{9\sqrt{3}}{\pi}$ (d) $\frac{6\sqrt{3}}{\pi}$

Q.131 Let a, b, c, and d be four integers such that $a + b + c + d = 4m + 1$, where m is a positive integer. Given m, which one of the following is necessarily true?

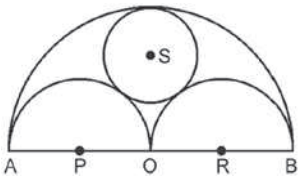
(a) The minimum possible value of $a^2 + b^2 + c^2 + d^2$ is $4m^2 - 2m + 1$
(b) The minimum possible value of $a^2 + b^2 + c^2 + d^2$ is $4m^2 + 2m + 1$
(c) The maximum possible value of $a^2 + b^2 + c^2 + d^2$ is $4m^2 - 2m + 1$
(d) The maximum possible value of $a^2 + b^2 + c^2 + d^2$ is $4m^2 + 2m + 1$

Q.132 Three horses are grazing within a semi–circular field. In the diagram given below, AB is the

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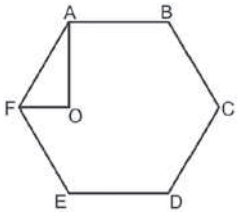
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diameter of the semicircular field with center at O. Horses are tied up at P, R, and S such that PO and RO are the radii of semicircles with centres at P and R, respectively, and S is the center of the circle touching the two semicircles with diameters AO and OB. The horses tied at P and R can graze within the respective semicircles and the horse tied at S can graze within the circle centred at S. The percentage of the area of the semicircle with diameter AB that cannot be grazed by the horses is nearest to:



(a) 20 (b) 28 (c) 36 (d) 40

Q.133 In the figure given below, ABCDEF is a regular hexagon and ∠AOF = 90°. FO is parallel to ED. What is the ratio of the area of the triangle AOF to that of the hexagon ABCDEF?



(a) $\frac{1}{12}$ (b) $\frac{1}{6}$ (c) $\frac{1}{24}$ (d) $\frac{1}{18}$

Q.134 How many three-digit positive integers, with digits x, y, and z in the hundreds, tens, and unit's place, respectively, exist such that $x < y$, $z < y$, and $x \neq 0$?

(a) 245 (b) 285 (c) 240 (d) 320

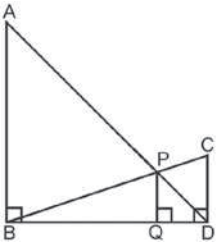
Q.135 A vertical tower OP stands at the centre O of a square ABCD. Let h and b denote the length OP and AB, respectively. Suppose ∠APB = 60°, then the relationship between h and b can be expressed as:

(a) $2b^2 = h^2$ (b) $2h^2 = b^2$
(c) $3b^2 = 2h^2$ (d) $3h^2 = 2b^2$

Q.136 In the triangle ABC, AB = 6, BC = 8, and AC = 10. A perpendicular dropped from B, meets the side AC at D. A circle of radius BD (with center B) is drawn. If the circle cuts AB and BC at P and Q, respectively, the AP:QC is equal to:

(a) 1:1 (b) 3:2 (c) 4:1 (d) 3:8

Q.137 In the diagram given below, ∠ABD = ∠CDB = ∠PQD = 90°. If AB:CD = 3:1, the ratio of CD:PQ is:



(a) 1:0.69 (b) 1:0.75
(c) 1:0.72 (d) None of the above

Q.138 There are 8436 steel balls, each with a radius of 1 cm, stacked in a pile, with 1 ball on top, 3 balls in the second layer, 6 in the third layer, 10 in the fourth, and so on. The number of horizontal layers in the pile is:

(a) 34 (b) 38 (c) 36 (d) 32

Q.139 If the product of n positive real numbers is unity, then their sum is necessarily:

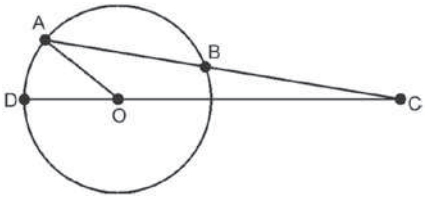
(a) A multiple of n (b) Equal to $n + \frac{1}{n}$
(c) Never less than n (d) A positive integer

Q.140 If $\log_3 2$, $\log_3 (2^x - 5)$, and $\log_3 (2^x - 7/2)$ are in an arithmetic progression, then the value of x is equal to:

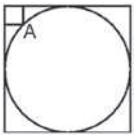
(a) 5 (b) 4 (c) 2 (d) 3

Q.141 In the figure given below, AB is the chord of a circle with center O. AB is extended to C such that BC = OB. The straight line CO is produced to meet the circle at D. If ∠ACD = y degrees and ∠AOD = x degrees such that $x = ky$, then find the value of k.

(a) 3 (b) 2
(c) 1 (d) None of the above



Q.142 In the figure given below, the rectangle at the corner measures 10 cm × 20 cm. The corner A of the rectangle is also a point on the circumference of the circle. What is the radius of the circle in cm?



(a) 10 cm (b) 40 cm
(c) 50 cm (d) None of these

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Q.143 Given that $-1 \le v \le 1$, $-2 \le u \le -0.5$, and $-2 \le z \le -0.5$ and $w = vz/u$, then which of the following is necessarily true?
(a) $-0.5 \le w \le 2$ (b) $-4 \le w \le 4$
(c) $-4 \le w \le 2$ (d) $-2 \le w \le -0.5$

Q.144 There are 6 boxes numbered 1, 2, …, 6. Each box is to be filled up either with a red or a green ball in such a way that at least 1 box contains a green ball and the boxes containing green balls are consecutively numbered. The total number of ways in which this can be done is:
(a) 5 (b) 21 (c) 33 (d) 60

Q.145 Consider the following two curves in the $x - y$ plane:
$y = x^3 + x^2 + 5$
$y = x^2 + x + 5$
Which of the following statements is true for $-2 \le x \le 2$?
(a) The two curves intersect once.
(b) The two curves intersect twice.
(c) The two curves do not intersect.
(d) The two curves intersect thrice.

Q.146 In a certain examination paper, there are n questions. For $j = 1, 2 \ldots n$, there are 2^{n-j} students who answered j or more questions wrongly. If the total number of wrong answers is 4095, then the value of n is:
(a) 12 (b) 11 (c) 10 (d) 9

Q.147 If x, y, and z are distinct positive real numbers, the $\dfrac{x^2(y+2)+y^2(x+z)+z^2+(x+y)}{xyz}$ would be:
(a) Greater than 4 (b) Greater than 5
(c) Greater than 6 (d) None of the above

Q.148 A graph may be defined as a set of points connected by lines called edges. Every edge connects a pair of points. Thus, a triangle is a graph with 3 edges and 3 points. The degree of a point is the number of edges connected to it. For example, a triangle is a graph with three points of degree 2 each. Consider a graph with 12 points. It is possible to reach any point from any point through a sequence of edges. The number of edges, e, in the graph must satisfy the condition:
(a) $11 \le e \le 66$ (b) $10 \le e \le 66$
(c) $11 \le e \le 65$ (d) $0 \le e \le 11$

Q.149 The number of positive integers n in the range $12 \le n \le 40$ such that the product $(n-1)(n-2) \ldots 3.2.1$ is not divisible by n is:
(a) 5 (b) 7 (c) 13 (d) 14

Q.150 Let T be the set of integers {3, 11 19, 27, 451, 459, 467} and S be a subset of T such that the sum of no two elements of S is 470. The maximum possible number of elements in S is:
(a) 32 (b) 28 (c) 29 (d) 30

Answers

101. (d)	**102.** (c)	**103.** (d)	**104.** (b)	**105.** (c)	**106.** (b)	**107.** (d)	**108.** (a)	**109.** (a)	**110.** (c)
111. (c)	**112.** (c)	**113.** (d)	**114.** (c)	**115.** (b)	**116.** (d)	**117.** (b)	**118.** (c)	**119.** (d)	**120.** (c)
121. (a)	**122.** (b)	**123.** (a)	**124.** (a)	**125.** (b)	**126.** (c)	**127.** (c)	**128.** (d)	**129.** (d)	**130.** (c)
131. (b)	**132.** (b)	**133.** (a)	**134.** (c)	**135.** (b)	**136.** (d)	**137.** (b)	**138.** (c)	**139.** (c)	**140.** (c)
141. (a)	**142.** (c)	**143.** (b)	**144.** (b)	**145.** (d)	**146.** (a)	**147.** (c)	**148.** (a)	**149.** (b)	**150.** (d)

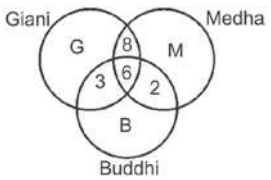
Hints and Solutions

103. Putting the value of M in either equation, we get G + B = 17.

Hence, neither of two can be uniquely determined.

Hence, option (d) is the answer.

104. As per the given data we get the following:



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G + B = M + 16

Also, M + B + G + 19 = (2 × 19) – 1

i.e., (G + B) = 18 – M

Thus, M + 16 = 18 – M, i.e., M = 1

Hence, option (b) is the answer.

105. $2^x - x - 1 = 0$

$\Rightarrow 2^x - 1 = x$

If we put $x = 0$, then this is satisfied, and if we put $x = 1$, then also this is satisfied.

Now, we put $x = 2$, then this is not valid.

Hence, option (c) is the answer.

106. For the curves to intersect, $\log_{10} x = x^{-1}$

Thus, $\log_{10} x = \frac{1}{x}$ or $x^x = 10$

This is possible for only one value of x $(2 < x < 3)$.

Hence, option (b) is the answer.

107. The surface area of a sphere is proportional to the square of the radius.

Thus, $\frac{S_B}{S_A} = \frac{4}{1}$ (SA of B is 300% higher than A)

$\therefore \frac{r_B}{r_A} = \frac{2}{1}$

The volume of a sphere is proportional to the cube of the radius.

Thus, $\frac{V_B}{V_A} = \frac{8}{1}$

V_A is $\frac{7}{8}$ th less than B, i.e., $\left(\frac{7}{8} \times 100\right)$ 87.5%

Hence, option (d) is the answer.

108. It is given that $p + q + r \neq 0$, if we consider the first option, and multiply the first equation by 5, second by –2 and third by –1, we see that the coefficients of x, y, and z all add up to zero.

Thus, $5p - 2q - r = 0$

No other option satisfies this.

Hence, option (a) is the answer.

109. Let 'x' be the number of standard bags and y be the number of deluxe bags.

Thus, $4x + 5y \leq 700$ and $6x + by \leq 1250$

Among the options, (c) and (d) do not satisfy the second equation.

Option (b) is eliminated, in order to maximize profits; the number of deluxe bags should be higher than the number of standard bags because the profit margin is higher in a deluxe bag.

Hence, option (a) is the answer.

110. Let the 1st term be 'a' and common difference be 'd', then we have 3rd term = $a + 2d$.

15th term = $a + 14d$

6th term = $a + 5d$

11th term = $a + 10d$

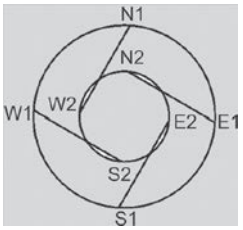
13th term = $a + 12d$

Since sum of 3rd and 15th term – sum of 6th, 11th, and 13th terms, therefore, we have $2a + 16d = 3a + 27d$

$\Rightarrow a + 11d = 0$, which is the 12th term.

Hence, option (c) is the answer.

Answers to Q.111 to 113:



If the radius of the inner ring road is r, then the radius of the outer ring road will be $2r$ (since the circumference is double).

The length of IR = $2\pi r$, that of OR = $4\pi r$ and that of the chord roads are $r\sqrt{5}$ (Pythagoras theorem)

The corresponding speeds are

20π, 30π, and $15\sqrt{5}$ kmph.

Thus, time taken to travel one circumference of IR = $\frac{r}{10}$ h, one circumference of OR = $\frac{r}{7.5}$ h

111. Sum of the length of the chord roads = $4r\sqrt{5}$ and the length of OR = $4\pi r$.

Thus, the required ratio = $\sqrt{5} : \pi$

Hence, option (c) is the answer.

112. The total time taken by the route given $= \frac{r}{30} + \frac{r}{15} = \frac{3}{2}$ (i.e., 90 min)

Thus, $r = 15$ km

The radius of OR = $2r = 30$ km

Hence, option (c) is the answer.

113. The total time taken $= \frac{r}{20} + \frac{r}{15} = \frac{7r}{60}$

Since $r = 15$, the total time taken = $\frac{7}{4}$ h

= 105 min

Hence, option (d) is the answer.

114. Let the number of correct answers be x, number of wrong answers be y, and number of questions not attempted be 'z'.

Thus, $x + y + z = 50$ (i)

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$$x - \frac{y}{3} - \frac{z}{6} = 32$$

The second equation can be written as,

$$6x - 2y - z = 192 \qquad \text{(ii)}$$

Adding equations (i) and (ii), we get

$$7x - y = 242 \text{ or } x = \frac{242}{7} + y$$

Since x and y are both integers, y cannot be 1 or 2. The minimum value that y can have is 3.

Hence, option (c) is the answer.

115. The number 27 has no significance here.

Option (b), will never be true for any number of people.

Let us take the case of 2 people.

If A knows B and B only knows A, both of them have 1 acquaintance each. Thus, B should be knowing at least one other person.

Let us say he knows 'C' as well. So, now 'B' has two acquaintances (A and C), but C has only one acquaintance (B), which is equal to that of A.

To close this loop, C will have to know A as well. In which case he will have two acquaintances, which is the same as that of C.

Thus, the loop will never be completed unless at least two of them have the same number of acquaintances. Besides, options (a), (c), and (d) can be true.

Note: If we consider the situation otherwise, to satisfy condition 2, the first person must have 26 acquaintances, the second 25, third 24, and so on. If we continue, the last one should have 0 acquaintance, which is not possible.

Hence, option (b) is the answer.

116. We can see that $x + 2$ is an increasing function and $5 - x$ is a decreasing function. This system of equation will have smallest value at the point of intersection of the two, i.e., $5 - x = x + 2$ or $x = 1.5$. Thus, the smallest value of $g(x) = 3.5$

Hence, option (d) is the answer.

117. **Case I** If $x < 2$, then $y = 2 - x + 2.5 - x + 3.6 - x = 8.1 - 3x$

This will be the least if x is the highest, i.e., just less than 2.

In this case, y will be just more than 2.1.

Case II If $2 \le x < 2.5$, then $y = x - 2 + 2.5 - x + 3.6 - x = 4.1 - x$

Again, this will be the least if x is the highest in this case, i.e., y will be just more than 1.6.

Case III If $2.5 \le x < 3.6$, then $y = x - 2 + x - 2.5 + 3.6 - x = x - 0.9$

This will be the least if x is least, i.e., the $x = 2.5$.

Case IV If $x \ge 3.6$, then

$$y = x - 2 + x - 2.5 + x - 3.6 = 3x - 8.1$$

The minimum value of this will be at $x = 3.6$ and $y = 2.7$.

Hence, the minimum value of y is attained at $x = 2.5$.

Hence, option (b) is the answer.

118. There are 101 integers in all, of which 51 are even.

From 100 to 200, there are 14 multiples of 7, of which 7 are even.

There are 11 multiples of 9, of which 6 are even.

But, there is one integer (i.e., 126) that is a multiple of both 7 and 9 and also even.

Hence, the answer is $(51 - 7 - 6 + 1) = 39$

Hence, option (c) is the answer.

119. Since the last digit in base 2, 3, and 5 is 1, the number should be such that on dividing by either 2, 3, or 5, we should get a remainder 1. The smallest such number is 31. The next set of numbers are 61 and 91. Among these, only 31 and 91 are a part of the answer choices.

Among these, $(31)_{10} = (11111)_2 = (1011)_3 = (111)_5$ Thus, all three forms have the leading digit 1.

Hence, the answer is 91.

Hence, option (d) is the answer.

120. The ratio of the speeds of the fastest and the slowest runners is 21. Hence, they should meet at only one point on the circumference, i.e., the starting point (as the difference in the ratio in the reduced form is 1). For the two of them to meet for the first time, the faster should have completed one complete round over the slower one. Since the two of them meet for the first time after 5 min, the faster one should have completed 2 rounds (i.e., 2000 m) and the slower one should have completed 1 round (i.e., 1000 m) in this time. Thus, the faster one would complete the race (i.e., 4000 m) in 10 min.

Hence, option (c) is the answer.

121. Solution cannot be found by using only Statement A since b can take any even number 2, 4, 6, …

But, we can arrive at solution by using Statement B alone.

If $b > 16$, say $b = 17$

Hence, $2^{44} < (16 + 1)^{11}$

$$2^{44} < (2^4 + 1)^{11}$$

Hence, option (a) is the answer.

122. Solution can be found using Statement A as we know both the roots for the equation $\left(\text{viz.}, \frac{1}{2} \text{and} -\frac{1}{2}\right)$

Also, Statement B is sufficient.

Since ratio of c and $b = 1$, $c = b$.

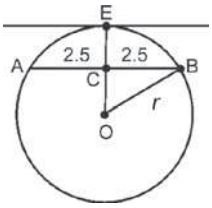
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Thus, the equation = $4x^2 + bx + b = 0$. Since $x = -\frac{1}{2}$ is one of the roots, substituting x, we get $1 - \frac{b}{2} + b = 0$ or $b = -2$. Thus, $c = -2$

Hence, option (b) is the answer.

123.



We can get the answer by using the second statement only. Let the radius be r. AC = CB = 2.5 and using Statement B, CE = 5, thus OC = $(r - 5)$.

Using Pythagoras theorem, $(r-5)^2 + (2.5)^2 = r^2$,

we get $r = 3.125$

Note: You will realize that such a circle is not possible (if $r = 3.125$ how can CE be 5). However, we need to check for data sufficiency and not for data consistency. Since we are able to find the value of r uniquely using second statement, the answer is option (a).

124. Both the series are infinitely diminishing series.

For the first series: First term $\frac{1}{a^2}$ and $r = \frac{1}{a^2}$

For the second series: First term $= \frac{1}{a}$ and $r = \frac{1}{a^2}$

The sum of the first series $= \frac{\frac{1}{a^2}}{1-\frac{1}{a^2}} = \frac{1}{a^2-1}$

The sum of the second series $= \frac{\frac{1}{a^2}}{1-\frac{1}{a^2}} = \frac{a}{a^2-1}$

Now, from the first statement, the relation can be anything (depending on whether 'a' is positive or negative).

But, the second statement tells us, $4a^2 - 4a + 1 = 0$ or $a = \frac{1}{2}$.

For this value of a, the sum of seond series will always be greater than that of the first.

Hence, option (a) is the answer.

125. The question tells us that the area of a triangle DEF will be the area of a triangle ABC. Thus, by knowing either of the statements, we can get the area of the triangle DEF.

Hence, option (b) is the answer.

126. The number of goats remains the same.

If the percentage that is added every time is equal to the percentage that is sold, then there should be a net decrease. The same will be the case if the percentage added is less than the percentage sold.

The only way, the number of goats will remain the same is if $p > q$.

Hence, option (c) is the answer.

127. In this kind of polygon, the number of convex angles will always be exactly 4 more than the number of concave angles.

Note: The number of vertices have to be even. Hence, the number of concave and convex corners should add up to an even number. This is true only for option (c).

128. The number of terms of the series forms the sum of first n natural numbers, i.e., $n(n + 1)$.

Thus, the first 23 letters will account for the first $23 \times 24 = 276$ terms of the series.

The 288 m term will be the 24 m letter, which is x.

Hence, option (d) is the answer.

129. $p + q = a - 2$ and $pq = -a - 1$

$(p + q)^2 = p^2 + q^2 + 2pq$

Thus, $(a-2)^2 = p^2 + q^2 + 2(-a - 1)$

$p^2 + q^2 = a^2 - 4a + 4 + 2a + 2$

$p^2 + q^2 = a^2 - 2a + 6$

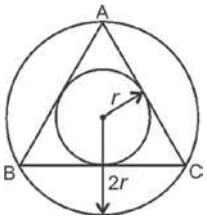
$p^2 + q^2 = -2a + 1 + 5$

$p^2 + q^2 - (a - 1)^2 + 5$

Thus, the minimum value of $p^2 + q^2$ is 5.

Hence, option (d) is the answer.

130.



Since the area of the outer circle is 4 times the area of the inner circle, the radius of the outer circle should be 2 times that of the inner circle.

Since AB and AC are the tangents to the inner circle, they should be equal. Also, BC should be a tangent to inner circle. In other words, triangle ABC should be equilateral.

The area of the outer circle is 12. Hence, the area of the inner circle is 3 or the radius is $\sqrt{\frac{3}{\pi}}$. The area of the equilateral triangle = $3\sqrt{3}\ r^2$, where r is the radius. Hence, option (c) is the answer.

131. $(a + b + c + d)^2 - (4m + 1)^2$

Thus, $a^2 + b^2 + c^2 + d^2 + 2(ab + ac + ad + bc + bd + cd) - 16m^2 + 8m + 1$

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$a^2 + b^2 + c^2 + d^2$ will have the minimum value if $(ab + ac + ad + bc + bd + cd)$ is the maximum.

This is possible if $a = b = c = d = (m + 0.25)$ … since $a + b + c + d = 4m + 1$

In that case, $2(ab + ac + ad + bc + bd + cd)$

$= 12 (m + 0.25)^2 - 12 m^2 + 6 m + 0.75$

Thus, the minimum value of $a^2 + b^2 + c^2 + d^2$

$= (16 m^2 + 8 m + 1) - 2 (ab + ac + ad + bc + bd + cd)$

$= (16 m^2 + 8 m + 1) - (12 m^2 + 6 m + 0.75)$

$= 4 m^2 + 2 m + 0.25$

Since it is an integer, the actual minimum value $= 4 m^2 + 2 m + 1$

Hence, option (b) is the answer.

132. If the radius of the field is r, then the total area of the field $= \frac{\pi r^2}{2}$

The radius of the semi-circles with centres P and $R = \frac{r}{2}$

Hence, their total area $= \frac{\pi r^2}{4}$

Let the radius of the circle with centre S be x.

Thus, OS $= (r - x)$, OR = and RS $= \left(\frac{r}{2} + x\right)$.

Applying Pythagoras theorem, we get

$$x = \frac{r}{3}$$

Thus, the area of the circle with centre S $= \frac{\pi r^2}{9}$

The total area that can be grazed $= \pi r^2\left(\frac{1}{4} + \frac{1}{9}\right) = \frac{13\pi r^2}{36}$

Thus, the fraction of the field that can be grazed

$$= \frac{26}{36}\left(\frac{\text{Area that can be grazed}}{\text{Area of the field}}\right)$$

The fraction that cannot be grazed $= \frac{10}{36} = 28\%$

Hence, option (b) is the answer.

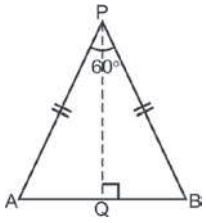
133. It is very clear that a regular hexagon can be divided into six equilateral triangles, and triangle AOF is half of an equilateral triangle. Hence, the required ratio = 1 : 12.

Hence, option (a) is the answer.

134. If $y = 2$ (it can not be 0 or 1), then x can take 1 value and z can take 2 values. Thus, with $y = 2$, a total of $1 \times 2 = 2$ numbers can be formed. With $y = 3$, $2 \times 3 = 6$ numbers can be formed. Similarly, checking for all values of y from 2 to 9 and adding up, we get the answer as 240.

Hence, option (c) is the answer.

135.



Given ∠APB = 60° and AB = b

$$\therefore \quad PQ = \frac{b}{2} \times \sqrt{3}$$

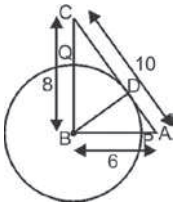
Next, $\frac{b}{2}$, h and PQ from a right-angled triangle.

$$\therefore \quad \frac{b^2}{4} + h^2 = \frac{3b^2}{4}$$

$$\therefore \quad 2h^2 = b^2$$

Hence, option (b) is the answer.

136.



Triangle ABC is a right-angled triangle.

Thus, $\frac{1}{2} \times CB \times AB = \frac{1}{2} \times BD \times AC$

$6 \times 8 = BD \times 10$. Thus, BD = 4.83

Therefore, BP = BQ

So, AP = AB − BP = 6 − 4.8 = 1.2 and CQ = BC − BQ = 8 − 4.8 = 3.2

Thus, AP:CQ = 1.2:3.2 = 3:8

Hence, option (d) is the answer.

137. Using the Basic Proportionality Theorem,

$$\frac{AB}{PQ} = \frac{BQ}{QD} \text{ and } \frac{PQ}{CD} = \frac{BQ}{BD}$$

Multipying the two, we get $\frac{AB}{CD} = \frac{BQ}{QD} = 3:1$

Thus, CD:PQ = BD:BQ = 4:3 = 1:0.75

Hence, option (b) is the answer.

138. Assume the number of horizontal layers in the pile be n.

So, $\sum \frac{n(n+1)}{2} = 8436$

$\Rightarrow \frac{1}{2}\left[\sum n^2 + \sum n\right] = 8436$

$n(n+1)\left[\frac{2n+4}{12}\right] = 8436$

$\Rightarrow n(n + 1)(n + 2) = 36 \times 37 \times 38$

So, $n = 36$

Hence, option (c) is the answer.

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139. The best way to do this is to take some value and verify.

e.g., 2, $\frac{1}{2}$, and 1

Thus, $n = 3$ and the sum of the three numbers = 3.5.

Thus, options (a), (b), and (d) get eliminated.

Alternative Method:

Let the n positive numbers be $a_1, a_2, a_3 \ldots a_n$.

We know that AM ≥ GM.

Hence, $\frac{1}{n}(a_1 + a_2 + a_3 + \ldots a_n) \geq (a_1\, a_2 \ldots a_n)^{1/n}$

or $(a_1 + a_2 + a_3 + \ldots a_n) \geq n$

Hence, option (c) is the answer.

140. Using $\log a - \log b = \log\left(\frac{a}{b}\right)$, $\frac{2}{y-5} = \frac{y-5}{y-3.5}$ where $y = 2^x$

Solving we get $y = 4$ or 8, i.e., $x = 2$ or 3. It cannot be 2 as log of negative number is not defined (see the second expression).

Hence, option (c) is the answer.

141. ∠BOC = 10° (Opposite equal sides)

∠OBA = 20° (External angle of ABOC)

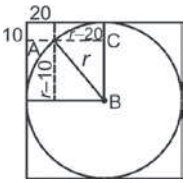
∠OAB = 20° (Opposite equal sides)

∠AOD = 30° (External angle of AAOC)

Thus, $k = 3$

Hence, option (a) is the answer.

142.



Let the radius be r. Thus, by Pythagoras' theorem for ΔABC, we have

$$(r-10)^2 + (r-20)^2$$

i.e., $r^2 - 60r + 500 = 0$. Thus, $r = 10$ or 50

It would be 10, if the corner of the rectangle had been lying on the inner circumference. But, as per the given diagram, the radius of the circle should be 50 cm.

Hence, option (c) is the answer.

143. u is always negative. Hence, to have a minimum vz value of $\frac{vz}{u}$, vz should be positive. Also, for the least u value, the numerator has to be the maximum positive value and the denominator has to be the smallest negative value. In other words, vz has to be 2 and u has to be −0.5.

Hence, the minimum value of $\frac{vz}{u} = \frac{2}{-0.5} = 4$

For us to get the maximum value, vz has to be the smallest negative value and u has to be the highest negative value. Thus, vz has to be −2 and u has to be −0.5.

Hence, the maximum value of $\frac{vz}{u} = \frac{2}{-0.5} = 4$

Hence, option (b) is the answer

144. GRRRRR, RGRRRR, RRGRRR, RRRGRR, RRRRGR, RRRRRG

GGRRRR, RGGRRR, RRGGRR, RRRGGR, RRRRGG, GGGRRR, RGGGRR, RRGGGR, RRRGGG, GGGGRR, RGGGGR, RRGGGG

GGGGGR, RGGGGG, GGGGGG

Hence, 21 ways

Hence, option (b) is the answer.

145. When we substitute two values of x in the above curves, at $x = -2$, we get

$$y = -8 + 4 + 5 = 1$$

$$y = 4 - 2 + 5 = 7$$

Hence, at $x = -2$, the curves do not intersect.

At $x = 2$, $y_1 = 17$ and $y_2 = 11$

At $x = -1$, $y_1 = 5$ and 2 and $y_2 = 5$

When $x = 0$, $y_1 = 5$ and $y_2 = 5$

At $x = 1$, $y_1 = 7$, and $y_2 = 7$

Therefore, the two curves meet thrice when $x = -1$, 0, and 1.

Hence, option (d) is the answer.

146. Let us say there are only 3 questions. Thus, there are $2^{3-1} = 4$ students who have done 1 or more questions wrongly, $2^{3-2} = 2$ students who have done 2 or more questions wrongly, and $2^{3-3} = 1$ student who must have done all 3 wrongly. Thus, the total number of wrong answers = 4 + 2 + 1 = $7 = 2^3 - 1 = 2^n - 1$

In our question, the total number of wrong answers = $4095 = 2^{12} - 1$

Thus, $n = 12$

Hence, option (a) is the answer.

147. Here x, y, and z are distinct positive real numbers.

So, $\frac{x^2(y+z) + y^2(x+2) + z^2(x+y)}{xyz}$

$= \frac{x}{y} + \frac{x}{z} + \frac{y}{x} + \frac{y}{z} + \frac{z}{x} + \frac{z}{y}$ (we know that

$= \left(\frac{x}{y} + \frac{y}{x}\right) + \left(\frac{y}{z} + \frac{z}{y}\right) + \left(\frac{z}{x} + \frac{x}{z}\right)$ that $\frac{a}{b} + \frac{b}{a} > 2$ if a and b are distinct numbers

$> 2 + 2 + 2$

> 6

Hence, option (c) is the answer.

148. The least number of edges will be when one point is connected to each of the other 11 points, giving a total

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of 11 lines. One can move from any point to any other point via the common point. The maximum edges will be when a line exists between any two points. Two points can be selected from 12 points in ${}^{12}C_2$, i.e., 66 lines.

Hence, option (a) is the answer.

149. From 12 to 40, there are 7 prime number, i.e., 13, 17, 19, 23, 29, 31, and 37, which is not divisible by $(n-1)!$

Hence, option (b) is the answer.

150. $T_n = a + (n-1)d$

$467 = 3 + (n-1)\,8$

$n = 59$

Half of $n = 29$ terms

29th term is 227 and 30th term is 235, and when these two terms are added, the sum is less than 470.

Hence, the maximum possible values the set S can have is 30.

Hence, option (d) is the answer.

RETEST PAPER

Direction for Questions 51 to 53: Answer the questions on the basis of the information given below:

The seven basic symbols in a certain numeral system and their respective values are as follows:

I = 1, V = 5, X = 10, L = 50, C = 100, D = 500, and M = 1000

In general, the symbols in the numeral system are (read from left to right) starting with the symbol representing the largest value; the same symbol cannot occur continuously more than three times: the value of the numeral is the sum of the values of the symbols. For example, XXVII = 10 + 10 + 5 + 1 + 1 = 27. An exception is that left-to-right reading occurs when a symbol is followed immediately by a symbol of greater value; then, the smaller value is subtracted from the larger.

For example, XLVI (50 − 10) + 5 + 1 = 46

Q.51 The value of the numeral MDCCLXXXVII is:

(a) 1687 (b) 1787

(c) 1887 (d) 1987

Q.52 The value of the numeral MCMXC IX is:

(a) 1999 (b) 1899

(c) 1989 (d) 1889

Q.53 Which of the following represent the numeral for 1995?

I. MCMLXXV II. MCMXCV

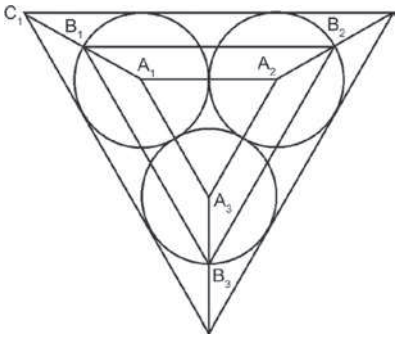
III. MVD IV. MVM

(a) Only I and II (b) Only Ill and IV

(c) Only II and IV (d) Only IV

Direction for Questions 54 to 56: Answer the questions on the basis of the information given below:

Consider three circular parks of equal size with centres at A_1, A_2, and A_3, respectively. The parks touch each other at the edge as shown in the figure (not drawn to scale). There are three paths formed by the triangles $A_1A_2A_3$, $B_1B_2B_3$, and $C_1C_2C_3$ as shown. Three sprinters A, B, and C begin running from points A_1, B_1, and C_1, respectively. Each sprinter traverses her respective triangular path clockwise and returns to her starting point.



Q.54 Let the radius of each circular park be r, and the distances to be traversed by the sprinters A, B, and C be a, b, and c, respectively. Which of the following is true?

(a) $b - c = c - b = 3\sqrt{3r}$

(b) $b - c = c - b = \sqrt{3r}$

(c) $b = \dfrac{a+c}{2} = 2\left(1+\sqrt{3}\right)r$

(d) $c = 2b - a = \left(2+\sqrt{3}\right)r$

Q.55 Sprinter A traverses distance A_1A_2, A_2A_3, and A_3A_1 at average speeds of 20, 30, and 15, respectively. B traverses her entire path at a uniform speed of $\left(10+\sqrt{3}+20\right)$. C traverses distances C_1C_2, C_2C_3, and C_3C_1 at an average speed of $\dfrac{40}{3}\left(\sqrt{3}+1\right), \dfrac{40}{3}\left(\sqrt{3}+1\right)$, and 120, respectively. All speeds are in the same unit. Where would B and C be, respectively, when A finishes her sprint?

(a) B_1, C_1

(b) B_3, C_3

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(c) B_1, C_3
(d) B_1, somewhere between C_3 and C_1

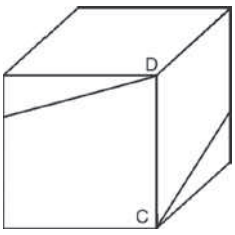
Q.56 Sprinters A, B, and C traverse their respective paths at uniform speeds of u, v, and w, respectively. It is known that $u^2:v^2:w^2$ is equal to Area A:Area B: Area C, where Area A, Area B, and Area C are the areas of triangles $A_1A_2A_3$, $B_1B_2B_3$, and $C_1C_2C_3$, respectively. Where would A and C be when B reaches point B_3?
(a) A_2, C_3
(b) A_3, C_3
(c) A_3, C_2
(d) Some where between A_2 and A_3, somewhere between C_3 and C_1

Direction for Questions 57 to 59: Answer the questions on the basis of the information given below:

Consider a cylinder of height h cm and radius $r = \frac{2}{\pi}$ cm as shown in the figure (not drawn to scale). A string of a certain length, when wound on its cylindrical surface, starting at point A and ending at point B, gives a maximum of n turns (in other words, the string's length is the minimum length required to wind n turns).

Q.57 What is the vertical spacing between the two consecutive turns?
(a) $\frac{h}{n}$ cm
(b) $\frac{h}{\sqrt{n}}$ cm
(c) $\frac{h}{n^2}$ cm
(d) Cannot be determined

Q.58 The same string, when wound on the exterior four walls of a cube of side n cm, starting at point C and ending at point D, can give exactly one turn (see figure, not drawn to scale). Find the length of the string.



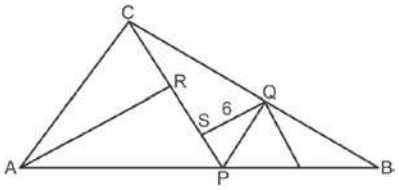
(a) $\sqrt{2}\ n$ cm
(b) $\sqrt{17}\ n$ cm
(c) n cm
(d) $\sqrt{13}\ n$ cm

Q.59 In the set–up of the previous two questions, how is h related to n?
(a) $h = \sqrt{2}\ n$
(b) $h = \sqrt{17}\ n$
(c) $h = n$
(d) $h = \sqrt{13}\ n$

Direction for Questions 60 to 93: Answer the following questions independently:

Q.60 There are 12 towns grouped into four zones with three towns per zone. It is intended to connect the towns with telephone lines such that every two towns are connected with three direct lines if they belong to the same zone, and with only one direct line otherwise. How many direct telephone lines are required?
(a) 72 (b) 90 (c) 96 (d) 144

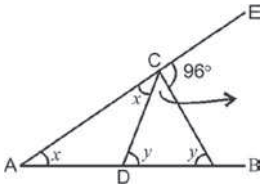
Q.61 In the figure (not drawn to scale) given below, P is a point on AB such that AP:PB = 4:3. PQ is parallel to AC and QD is parallel to CP. In ΔARC, ∠ARC = 90° and in ΔPQS, ∠PSQ = 90°. The length of QS is 6 cm. What is the ratio of AP:PD?



(a) 10:3 (b) 2:1 (c) 7:3 (d) 8:3

Q.62 A car is being driven, in a straight line and at a uniform speed, towards the base of a vertical tower. The top of the tower is observed from the car and, in the process, it takes 10 min for the angle of elevation to change from 45° to 60°. After how much more time will this car reach the base of the tower?
(a) $5\left(\sqrt{3}+1\right)$
(b) $6\left(\sqrt{3}+\sqrt{2}\right)$
(c) $7\left(\sqrt{3}-1\right)$
(d) $8\left(\sqrt{3}-2\right)$

Q.63 In the figure (not drawn to scale) given below, if AD = CD = BC and ∠BCE = 96°, how much is the value of ∠DBC?



(a) 32°
(b) 84°
(c) 64°
(d) Cannot be determined

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Q.64 If both a and b belong to the set $\{1, 2, 3, 4\}$, then the number of equations of the of $ax^2 + bx + 1 = 0$ having real roots is:
(a) 10 (b) 7 (c) 6 (d) 12

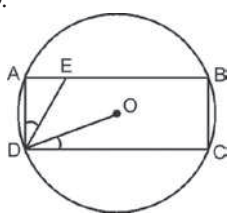
Q.65 If $\log_{10}x - \log_{10}\sqrt{x} = 2\log_x 10$, then the possible value of x is given by:
(a) 10 (b) $\frac{1}{100}$
(c) $\frac{1}{1000}$ (d) None of these

Q.66 What is the sum of all two–digit numbers that give a remainder of 3 when they are divided by 7?
(a) 666 (b) 676 (c) 683 (d) 777

Q.67 An intelligence agency forms a code of two distinct digits selected from 0, 1, 2,…, 9 such that the first digit of the code is non-zero. The code, handwritten on a slip, can, however, potentially create confusion when read upside down; for example, the code 91 may appear as 16. How many code are there for which no such confusion can arise?
(a) 80 (b) 78 (c) 71 (d) 69

Q.68 Consider two different cloth–cutting processes. In the first one, n circular cloth pieces are cut from square cloth piece of side a in the following steps: the original square of side a is divided into smaller squares, not necessarily of the same size, then a circle of maximum possible area is from each of the smaller squares. In the second process, only one circle of maximum possible area is cut from the square of side a and the process ends there. The cloth pieces remaining after cutting the circles are scrapped in both the processes. The ratio of the total area of scrap cloth generated in the former to that in the latter is:
(a) 1:1 (b) $\sqrt{2}:1$
(c) $\frac{n(4-\pi)}{4n-\pi}$ (d) $\frac{4n-\pi}{n(4-\pi)}$

Q.69 In the figure below (not drawn to scale), rectangle ABCD is inscribed in the circle with centre at O. The length of side AB is greater than side BC. The ratio of the area of the circle to the area of the rectangle ABCD is $\pi:\sqrt{3}$. The line segment DE intersects AB at E such that $\angle ODC = \angle ADE$. Find the ratio of AE:AD.



(a) $1:\sqrt{3}$ (b) $1:\sqrt{2}$
(c) $1:2\sqrt{3}$ (d) 1:2

Q.70 If $\frac{1}{3}\log_3 M + 3\log_3 N = 1 + \log_{0.008} 5$, then:
(a) $M^9 = \frac{9}{N}$ (b) $N^9 = \frac{9}{M}$
(c) $M^3 = \frac{3}{N}$ (d) $N^9 = \frac{3}{M}$

Q.71 Using only 2, 5, 10, 25, and 50 paisa coins, what will be the minimum number of coins required to pay exactly 78 paise, 69 paise, and ₹1.01 to three different persons?
(a) 19 (b) 20 (c) 17 (d) 18

Q.72 The length of the circumference of a circle equals the perimeter of a triangle of equal sides and also the perimeter of a square. The areas covered by the circle, triangle, and square are c, t, and s, respectively. Then,
(a) $s > t > c$ (b) $c > t > s$
(c) $c > s > t$ (d) $s > c > t$

Q.73 What is the remainder when 4^{96} is divided by 6?
(a) 0 (b) 2 (c) 3 (d) 4

Q.74 If x and y are integers, then the equation $5x + 1\,9y = 64$ has:
(a) No solution for $x < 300$ and $y < 0$
(b) No solution for $x > 250$ and $y > -100$
(c) A solution for $250 < x < 300$
(d) A solution for $-59 < y < -56$

Q.75 What is the sum of 'n' terms in the series
$\log m + \log\left(\frac{m^2}{n}\right) + \log\left(\frac{m^3}{n^2}\right) + \log\left(\frac{m^4}{n^3}\right) + \ldots$?
(a) $\log\left[\frac{n^{(n-1)}}{m^{(n+1)}}\right]^{\frac{n}{2}}$ (b) $\log\left[\frac{m^m}{n^n}\right]^{\frac{n}{2}}$
(c) $\log\left[\frac{m^{(1-n)}}{n^{(1-m)}}\right]^{\frac{n}{2}}$ (d) $\log\left[\frac{m^{(n+1)}}{n^{(n-1)}}\right]^{\frac{n}{2}}$

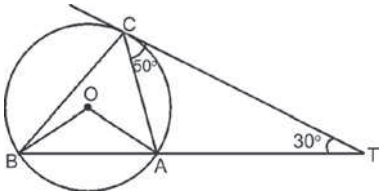
Q.76 Let S_1 be a square of side a. Another square S_2 is formed by joining the mid-points of the sides of S_1. The same process is applied to S_2 to form yet another square S_3 and so on. If $A_1, A_2, A_3\ldots$ be the areas and $P_1, P_2, P_3,\ldots$ be the perimeters of S_1, S_2, S_3, respectively, then the ratio $\frac{P_1+P_2+P_3+\ldots}{A_1+A_2+A_3+\ldots}$ equals:
(a) $\frac{2\left(1+2\sqrt{2}\right)}{a}$ (b) $\frac{2\left(2-2\sqrt{2}\right)}{a}$
(c) $\frac{2\left(2+2\sqrt{2}\right)}{a}$ (d) $\frac{2\left(1+2\sqrt{2}\right)}{a}$

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Q.77 If three positive real numbers x, y, and z satisfy $y - x = z - y$ and $x\,y\,z = 4$, then what is the minimum possible value of y?
(a) $21^{1/3}$ (b) $2^{2/3}$ (c) $2^{1/4}$ (d) $2^{3/4}$

Q.78 In the figure given below (not drawn to scale), A, B, and C are three points on a circle with centre O. The chord BA is extended to a point T such that CT becomes a tangent to the circle at point C. If ∠ATC = 30° and ∠ACT = 50°, then the angle ∠BOA is:



(a) 100°
(b) 150°
(c) 80°
(d) Not possible to determine

Q.79 The infinite sum $1+\frac{4}{7}+\frac{9}{7^2}+\frac{16}{7^3}+\frac{25}{7^4}+\ldots$ equals
(a) $\frac{27}{14}$ (b) $\frac{21}{13}$ (c) $\frac{49}{27}$ (d) $\frac{256}{147}$

Q.80 Consider the sets $T_n = \{n, n+1, n+2, n+3, n+4\}$, where n = 1, 2, 3, …96. How many of these sets contain 6 or any integral multiple thereof (i.e., any one of the numbers 6, 12, 18, ...)?
(a) 80 (b) 81 (c) 82 (d) 83

Q.81 Let ABCDEF be a regular hexagon. What is the ratio of the area of the DACE to that of the hexagon ABCDEF?
(a) $\frac{1}{3}$ (b) $\frac{1}{2}$ (c) $\frac{2}{3}$ (d) $\frac{5}{6}$

Q.82 The number of roots common between the two equations $x^3 + 3x^2 + 4x + 5 = 0$ and $x^3 + 2x^2 + 7x + 3 = 0$ is:
(a) 0 (b) 1 (c) 2 (d) 3

Q.83 A real number x satisfying $1 - \frac{1}{n} < x \le 3 + \frac{1}{n}$, for every positive integer n, is best described by:
(a) $1 < x < 4$ (b) $1 < x \le 3$
(c) $0 < x \le 4$ (d) $1 \le x \le 3$

Q.84 If n is such that $36 \le n \le 72$, then $x = \frac{n^2 + 2\sqrt{n}(n+4) + 16}{n + 4\sqrt{n} + 4}$ satisfies:
(a) $20 < x < 54$ (b) $23 < x < 58$
(c) $25 < x < 64$ (d) $28 < x < 60$

Q.85 If $13x + 1 < 2z$ and $z + 3 = 5y^2$, then:
(a) x is necessarily less than y.
(b) x is necessarily greater than y.
(c) x is necessarily equal to y.
(d) None of these are necessarily true.

Q.86 Let n (>1) be a composite integer such that $\sqrt{n}$ is not an integer. Consider the following statements:
(A) n has a perfect integer–valued divisor, which is greater than 1 and less than $\sqrt{n}$.
(B) n has a perfect integer–valued divisor, which is greater than $\sqrt{n}$ but less than n.
(a) Both A and B are false.
(b) A is true but B is false.
(c) A is false but B is true.
(d) Both A and B are true.

Q.87 If $|b| \ge 1$ and $x = -|a|b$, then which one of the following is necessarily true?
(a) $a - xb < 0$ (b) $a - xb \ge 0$
(c) $a - xb > 0$ (d) a $- xb \le 0$

Q.88 A piece of paper is in the shape of a right-angled triangle and is cut along a line that is parallel to the hypotonuse, leaving a smaller triangle. There was 35% reduction in the length of the hypotonuse of the triangle. If the area of the original triangle was 34 square inches before the cut, what is the area (in square inches) of the smaller triangle?
(a) 16.665 (b) 16.565
(c) 15.465 (d) 14.365

Q.89 Two straight roads R_1 and R_2 diverge from a point A at an angle of 120°. Ram starts walking from point A along R_1 at a uniform speed of 3 kmph. Shyam starts walking at the same time from A along R_2 at a uniform speed of 2 kmph. They continue walking for 4 h along their respective roads and reach points B and C on R_1 and R_2, respectively. There is a straight line path connecting B and C. Then, Ram returns to point A after walking along the line segments BC and CA. Shyam also returns to A after walking along line segments BC and CA. Their speeds remain unchanged. The time interval (in h) between Ram's and Shyam's return to the point A is:
(a) $\frac{10\sqrt{19}+26}{3}$ (b) $\frac{2\sqrt{19}+10}{3}$
(c) $\frac{\sqrt{19}+26}{3}$ (d) $\frac{\sqrt{19}+10}{3}$

Q.90 A square tin sheet of side 12 inches is converted into a box with open top in the following steps. The sheet is placed horizontally. Then, equal-sized

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squares, each of side x inches, are cut from the four corners of the sheet. Finally, the four resulting sides are bent vertically upwards in the shape of a box. If x is an integer, then what value of x maximizes the volume of the box?

(a) 3 (b) 4
(c) 1 (d) 2

Q.91 If a, $a + 2$, and $a + 4$ are prime numbers, then the number of possible solutions for a is:

(a) One (b) Two
(c) Third (d) More than three

Q.92 Let a, b, c, d, and e be integers such that $a = 6b = 12c$, and $2b = 9d = 12e$. Then, which of the following pairs contains a number that is not an integer?

(a) $\left(\frac{a}{27}, \frac{b}{e}\right)$ (b) $\left(\frac{a}{36}, \frac{c}{e}\right)$

(c) $\left(\frac{a}{12}, \frac{bd}{18}\right)$ (d) $\left(\frac{a}{6}, \frac{c}{d}\right)$

Q.93 In a coastal village, every year floods destroy exactly half of the huts. After the flood water recedes, twice the number of huts destroyed are rebuilt. The floods occurred consecutively in the last three years—2001, 2002, and 2003. If the floods are expected again in 2004, then find the number of huts expected to be destroyed.

(a) Less than the number of huts existing at the beginning of 2001.
(b) Less than the total number of huts destroyed by the floods in 2001 and 2003.
(c) Less than the total number of huts destroyed by the floods in 2002 and 2003.
(d) More than the total number of huts built in 2001 and 2002.

Direction for Questions 94 to 96: Answer the questions on the basis of the tables given below:

Two binary operations ⊕ and * are defined over the set {a, e, f, g, h} as per the following tables:

⊕	a	e	f	g	h
a	a	e	f	g	h
e	e	f	g	h	a
f	f	g	h	a	e
g	g	h	a	e	f
h	h	a	e	f	g

⊕	a	e	f	g	h
a	a	a	a	a	a
e	a	e	f	g	h
f	a	f	h	e	g
g	a	g	e	h	f
h	a	h	g	f	e

Thus, according to the first table $f \oplus g = a$, whereas according to the second table, $g * h = f$, and so on. Also, let $f^2 = f * f$, $g^3 = g*g*g$, and so on.

Q.94 What is the smallest positive integer n such that $g^n = e$?

(a) 4 (b) 5 (c) 2 (d) 3

Q.95 Upon simplification, $f \oplus [f * \{f \oplus (f * f)\}]$ equals:

(a) e (b) f (c) g (d) h

Q.96 Upon simplification, $\{a^{10} * (f^{10} \oplus g^g)\} \oplus e^8$ equals:

(a) e (b) f (c) g (d) h

Direction for Questions 97 and 98: Answer the questions on the basis of the information given below:

A string of three English letters is formed as per the following rules:

I. The first letter is any vowel.
II. The second letter is m, n, or p.
III. If the second letter is m, then the third letter is any vowel which is different from the first letter.
IV. If the second letter is n, then the third letter is e or u.
V. If the second letter is p, then the third letter is the same as the first letter.

Q.97 How many strings of letters can possibly be formed using the above rules?

(a) 40 (b) 45 (c) 30 (d) 35

Q.98 How many strings of letters can possibly be formed using the above rules such that the third letter of the string is e?

(a) 8 (b) 9 (c) 10 (d) 11

Direction for Questions 99 and 100: Answer the following questions independently:

Q.99 Let x and y be positive integers such that x is a prime and y is a composite. Then,

(a) $y - x$ cannot be an even integer.
(b) xy cannot be an even integer.

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(c) $\frac{(x+y)}{x}$ cannot be an even integer.

(d) None of these

Q.100 A survey on a sample of 25 new cars being sold at a local auto dealer was conducted to see which of the three popular options—air conditioning, radio and power windows—were already installed.

Following were the observation of the survey:

I. 15 had air conditioning.

II. 2 had air conditioning and power windows but no radios.

III. 12 had radio.

IV. 6 had air conditioning and radio but no power windows.

V. 11 had power windows.

VI. 4 had radio and power windows.

VII. 3 had all the three options.

What is the number of cars that had none of the options?

(a) 4 (b) 3 (c) 1 (d) 2

Answers

51. (b)	**52.** (a)	**53.** (c)	**54.** (a)	**55.** (c)	**56.** (b)	**57.** (a)	**58.** (b)	**59.** (c)	**60.** (b)
61. (c)	**62.** (a)	**63.** (c)	**64.** (b)	**65.** (b)	**66.** (b)	**67.** (c)	**68.** (a)	**69.** (a)	**70.** (b)
71. (a)	**72.** (c)	**73.** (d)	**74.** (c)	**75.** (d)	**76.** (c)	**77.** (b)	**78.** (a)	**79.** (c)	**80.** (a)
81. (b)	**82.** (a)	**83.** (c)	**84.** (d)	**85.** (d)	**86.** (d)	**87.** (b)	**88.** (d)	**89.** (b)	**90.** (d)
91. (a)	**92.** (d)	**93.** (c)	**94.** (a)	**95.** (d)	**96.** (a)	**97.** (d)	**98.** (c)	**99.** (d)	**100.** (d)

Hints and Solutions

51. MDCCLXXXVII = 1000 + 500 + 100 + 100 + 50 + 10 + 10 + 10 + 5 + 1 + 1 = 1787

Hence, option (b) is the answer.

52. MCMXCIX = 1000 + (1000 − 100) + (100 − 10) + (10 − 1) = 1000 + 900 + 90 + 9 = 1999

Hence, option (a) is the answer.

53. (I) MCMLXXV = 1000 + (1000 − 100) + 50 + 10 + 10 + 5 = 1975

(II) MCMXCV = 1000 + (1000 − 100) + (100 − 10) + 5 = 1995

(III) MVD = 1000 + (500 − 5) = 1495

(IV) MVM = 1000 + (1000 − 5) = 1995

Therefore, the answer is (II) and (IV), i.e., option (c) is the answer.

Answers to Q.54 to 56:

$A_1A_2 = 2r$, $B_1B_2 = 2r + r\sqrt{3}$, $C_1C_2 = 2r + 2r\sqrt{3}\,r$

Hence, $a = 3 \times 2r$

$b = 3 \times (2r + r\sqrt{3})$

$c = 3 \times (2r + 2r\sqrt{3})$

54. Difference between options (*a*) and (*b*) is $3\sqrt{3}\,r$ and that between options (b) and (c) is $3\sqrt{3}\,r$.

Hence, option (a) is the answer.

55. Time taken by $A = \frac{2r}{20} + \frac{2r}{30} + \frac{2r}{15} = \left(\frac{2r \times 9}{60}\right) = \frac{3}{10}r$

Therefore, B and C will also travel for time $\frac{3}{10}r$

Now, the speed of $B = \left(10\sqrt{3} + 20\right)$

Therefore, the distance covered $= \left(10\sqrt{3} + 20\right)$

$\times \frac{3}{10}r = \left(\sqrt{3} + 2\right) \times 10 \times \frac{3}{10}r$

$= \left(2r + \sqrt{3}r\right) \times 3 = B_1B_2 + B_2B_3 + B_3B_1$

$\therefore$ B will be at B_1

Now, the time taken by C for each distance is

$\frac{C_1C_2}{\frac{40}{3}\left(\sqrt{3}+1\right)}, \frac{C_2C_3}{\frac{40}{3}\left(\sqrt{3}+1\right)}, \frac{C_3C_1}{120}$

$\frac{\left(2+2\sqrt{3}\right)r}{\left(\sqrt{3}+1\right)}, \frac{3}{40} \times \frac{\left(2+2\sqrt{2}\right)r}{\left(\sqrt{3}+1\right)}, \frac{\left(2+2\sqrt{3}\right)}{120}$

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i.e., $\frac{3}{40}\times 2r, \frac{3}{40}\times 2r, \frac{\left(1+\sqrt{3}\right)}{60}r$

i.e., $\frac{3}{20}r, \frac{3}{20}r, \frac{\left(1+\sqrt{3}\right)}{60}r$

We can observe that time taken for C_1C_2 and C_2C_3 combined is $\frac{3}{20}r+\frac{3}{20}r=\frac{3}{10}r$, which is same as the time taken by A. Therefore, C will be at C_3.

Hence, option (c) is the answer.

56. In similar triangles, the ratio of the area = ratio of the squares of the corresponding sides.

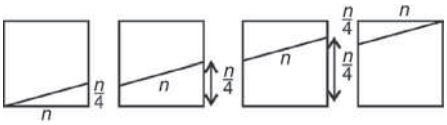
Thus, A and C reach A_3 and C_3, respectively.

Hence, option (b) is the answer.

57. The whole height h will be divided into n equal parts. Therefore, the spacing between two consecutive turns is $\frac{h}{n}$.

Hence, option (a) is the answer.

58. The four faces through which a string is passing can be shown as:



Therefore, the length of the string in each face

$$=\sqrt{n^2+\left(\frac{n}{4}\right)^2}$$

$$=\sqrt{n^2+\frac{n^2}{16}}=\frac{\sqrt{17}n}{4}$$

Therefore, the length of the string through four faces =

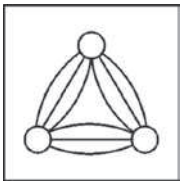
$$\frac{\sqrt{17}n}{4}\times 4=\sqrt{17}n$$

Hence, option (b) is the answer.

59. As h/n = Number of turns = 1 (as given). Thus, $h = n$.

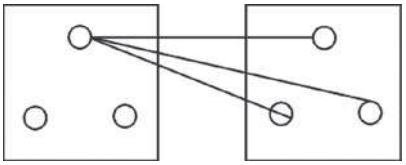
Hence, option (c) is the answer.

60. Consider the first zone. The number of telephone lines can be shown a follows.



Therefore, the total number of lines required for internal connections in each zone = $9 \times 4 = 36$ lines.

Now, consider the connection between any two zones.



Each town in the first zone can be connected to the three towns in the second zone.

Therefore, the lines required = $3 \times 3 = 9$

Therefore, the total number of lines required for connecting the towns of different zones = ${}^4C_2 \times 9 = 6 \times 9 = 54$

Therefore, the total number of lines in all = 54 + 36 = 90

Hence, option (b) is the answer.

61. PQ || AC

$$\therefore \frac{CD}{QB}=\frac{AB}{PB}=\frac{4}{3}$$

QD||PC

$$\therefore \frac{PD}{DB}=\frac{CQ}{QB}=\frac{4}{3}$$

As $\frac{PD}{DB}=\frac{4}{3}$

$$\therefore PD=\frac{4}{7}PB$$

$$\therefore \frac{AP}{PD}=\frac{AR}{\frac{4}{7}PB}$$

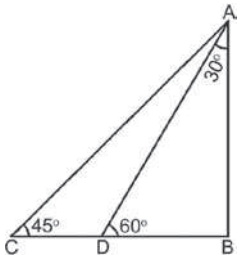
$$=\frac{7}{4}\times\frac{AP}{PB}$$

$$=\frac{7}{4}\times\frac{4}{3}$$

$$=7:3$$

Hence, option (c) is the answer.

62.



Let AB = 1

Therefore, BC = 1

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$\therefore \tan 60 = \frac{AB}{BD} \therefore \sqrt{3} = \frac{1}{BD}$

$\therefore \quad BD = \frac{1}{\sqrt{3}}$

$\therefore CD = BC - BD$

$= 1 - \frac{1}{\sqrt{3}}$

As time for travelling CD, i.e., $1 - \frac{1}{\sqrt{3}}$ is 10 min.

$\therefore$ Time required for travelling BD

$$= \frac{\frac{1}{\sqrt{3}}}{1 - \frac{1}{\sqrt{3}}} \times 10 \times 10$$

$$= \frac{1}{\sqrt{3} - 1} \times 10$$

$$= \frac{10}{\sqrt{3} - 1}$$

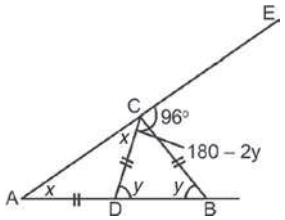
$$= \frac{10}{\sqrt{3} - 1} \times \frac{\sqrt{3} + 1}{\sqrt{3} + 1}$$

$$= \frac{10\left(\sqrt{3} + 1\right)}{2}$$

$$= 5\left(\sqrt{3} + 1\right)$$

Hence, option (a) is the answer.

63.



Using exterior angle theorem

$\angle A + \angle B = 96$

i.e., $x + y = 96$ (i)

Also, $x + (108 - 2y) + 96 = 180°$

$\therefore \quad x - 2y + 96 = 0$

$\therefore \quad x - 2y = -96$ (ii)

Solving (i) and (ii), we get

$y = 64°$ and $x = 32°$

$\therefore \angle DBC = y = 64°$

Hence, option (c) is the answer.

64. $ax^2 + bx + 1 = 0$

For real roots

$b^2 - 4ac \geq 0$

$\therefore b^2 - 4a(1) \geq 0$

$\therefore b^2 \geq 4a$

For $a = 1$, $4a = 4$, $\therefore b = 2, 3, 4$

$a = 2$, $4a = 8$, $\therefore b = 3, 4$

$a = 3$, $4a = 12$, $\therefore b = 4$

$a = 4$, $4a = 16$, $\therefore b = 4$

Hence, option (b) is the answer.

65. $\log_{10} x - \log_{10} \sqrt{x} = 2\log_x 10$

$$\log_{10}\left[\frac{x}{\sqrt{n}}\right] = \log_x 100$$

$\therefore \quad \log_{10} \sqrt{x} = \frac{\log_{10} 100}{\log_{10} x}$

$\therefore \quad \frac{1}{2}\log_{10} x = \frac{2}{\log_{10} x}$

$\therefore \quad (\log_{10} x)^2 = 4$

$\therefore \quad \log_{10} x = \pm 2$

$\therefore \quad \log_{10} x = 2$ or $\log_{10} x = -2$

$\therefore \quad 10^2 = x$ or $10^{-2} = x$

$\therefore \quad x = 100$ or $x = \frac{1}{100}$

Hence, option (b) is the answer.

66. Such numbers are 10, 17,…, 94.

These numbers are in an AP. There are 13 numbers.

$\therefore \quad \text{Sum} = \frac{10 + 94}{2} \times 13$

$= 52 \times 13 = 676$

Hence, option (b) is the answer.

67. The total codes which can be formed = $9 \times 9 = 81$. (Distinct digit codes)

The digits that can be confusing are 1, 6, 8, and 9. From these digits, we can form the codes = $4 \times 3 = 12$.

Out of these 12 codes, two numbers 69 and 96 will not create confusion.

Therefore, $(12 - 2) = 10$ codes will create a confusion.

Therefore, the total codes without confusion = $81 - 10 = 71$.

Hence, option (c) is the answer.

68. Consider a square of side x.

Therefore, its area = x^2

Therefore, the area of the largest circle = $\pi\left(\frac{x}{2}\right)^2$

Which can be cut from the square = $\frac{\pi x^2}{4}$

Therefore, the area scrapped

$= x^2 - \frac{\pi}{4}x^2 = x^2\left(1 - \frac{\pi}{4}\right)$

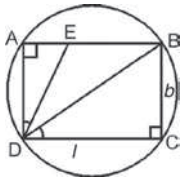
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$$\therefore \quad \frac{\text{Area scrapped}}{\text{Area scrapped}} = \frac{x^2\left(1-\frac{\pi}{4}\right)}{x^2} = 1-\frac{\pi}{4} = \text{Constant}$$

As this ratio is constant whether we cut a circle from the smaller square or the larger square, the scrapped area will be a fixed percentage of the square. Therefore, in this problem, as the two squares are of the same size, the ratio will be 1:1. Hence, option (a) is the answer.

69.



$BD = 2r$

$$\frac{\text{Area of circle}}{\text{Area rectangle}} = \frac{\pi r^r}{lb} = \frac{\pi}{\sqrt{3}}$$

$$\frac{\frac{d^2}{4}}{lb} = \frac{1}{\sqrt{3}}$$

$$\therefore \quad \frac{d^2}{4lb} = \frac{1}{\sqrt{3}}$$

$$\therefore \quad \frac{l^2+b^2}{lb} = \frac{1}{\sqrt{3}}$$

$$\therefore \quad \frac{l^2+b^2}{lb} = \frac{4}{\sqrt{3}}$$

$$\therefore \quad \frac{l}{b}+\frac{b}{l} = \frac{4}{\sqrt{3}} \qquad \text{(i)}$$

Now, $\Delta AEB \sim \Delta CBD$

$$\therefore \quad \frac{AE}{CB} = \frac{AD}{DC}$$

$$\therefore \quad \frac{AE}{AD} = \frac{BC}{DC}$$

$\therefore$ We have to find $\frac{AE}{AD}$, i.e., $\frac{b}{l}$

Let $\frac{b}{l} = x$

Therefore, from (i), we get

$$\frac{1}{x}+x = \frac{4}{\sqrt{3}}$$

$$\frac{1+x^2}{x} = \frac{4}{\sqrt{3}}$$

$$\sqrt{3}+\sqrt{3}x^2 = 4$$

$$\therefore \quad \sqrt{3}x^2-4x+\sqrt{3} = 0$$

$$\therefore \quad x = \frac{-(-4)\pm\sqrt{16-4\left(\sqrt{3}\right)\sqrt{3}}}{2\sqrt{3}}$$

$$= \frac{4\pm\sqrt{16-12}}{2\sqrt{3}}$$

$$= \frac{4\pm 2}{2\sqrt{3}}$$

$$= \frac{6}{2\sqrt{3}} \text{ or } \frac{2}{2\sqrt{3}}$$

$$= \frac{\sqrt{3}}{1} \text{ or } \frac{1}{\sqrt{3}}$$

From options, the answer is $\frac{1}{\sqrt{3}}$, i.e., $1:\sqrt{3}$.

Hence, option (a) is the answer.

70. $\frac{1}{3}\log_3 M + 3\log_3 N = 1+\log_{0.008} 5$

$$\log_3\left(M^{1/3}N^3\right) = 1+\frac{(\log 10+\log 2)}{\log 8-\log 1000}$$

$$\log_3\left(M^{1/3}N^3\right) = 1-\frac{(1-\log 2)}{3(1-\log 2)}$$

$$\Rightarrow \log_3\left(M^{1/3}N^3\right) = 1-\frac{1}{3} = \frac{2}{3}$$

$M^{1/3}N^3 = 3^{2/3}$

$MN^9 = 3^2$

$N^9 = 9/M$

Hence, option (b) is the answer.

71. Let us make the given sum by using minimum number of coins as

Value of coin	Number of coins	Number of coins	Number of coins	Total number of coins
50	1	1	1	3
25	–	–	1	1
10	1	2	2	5
5	1	–	–	1
2	2	4	3	9
Total Amount	69	78	101	19

Hence, option (a) is the answer.

72. It is the standard property among circle, square, and triangle. For a given parameter, area of a circle is the highest and area of the triangle is the least, whereas area of the square is in–between, i.e., $c > s > t$.

Hence, option (c) is the answer.

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73. Remainder when $\frac{4^{96}}{6}$

Let us come down to basic property of dividing the power of 4 by 6, i.e.,

$$\frac{4^1}{6} = 4$$
$$\frac{4^2}{6} = 4$$
$$\frac{4^3}{6} = 4$$
$$\frac{4^4}{6} = 4$$

Therefore, any power of 4 when divided by 6 leaves a remainder of 4. Hence, option (d) is the answer.

74. $5x + 19y = 64$

We see that if $y = 1$, we get an integer solution for $x = 9$, and if y changes (increases or decreases by $5x$, will the change decrease or increase) by 19. Looking at the options, if $x = 256$, we get $y = 64$. Using these values, we see options (a), (b), and (d) are eliminated and also that there exists a solution for $250 < x \leq 300$.

Hence, option (c) is the answer.

75. Sum of $\log m + \log\left(\frac{m^2}{n}\right) + \log\left(\frac{m^3}{n^2}\right) + \ldots$ + …n terms, such a problem must be solved by taking the value of number of the terms. Let us say 2 and check the given option. If we look at the sum of the 2 terms of the given series, it comes out to be $\log m + \log \frac{m^2}{n} \Rightarrow \log \frac{m \times m^2}{n} = \log\left(\frac{m^3}{n}\right)$. Now, look at the option and put number of terms as 2, only option (d) validates the abovementioned answer.

As $\log\left[\frac{m^{(n+1)}}{n^{(n-1)}}\right]^{\frac{n}{2}} \Rightarrow \log\left[\frac{m^3}{n}\right]^1 \Rightarrow \log\left(\frac{m^3}{n}\right)$

Hence, option (d) is the answer.

76. $$\frac{P + \frac{P}{\sqrt{2}} + \ldots\infty}{A + \frac{A}{2} + \ldots\infty} = \frac{\frac{P}{1 - \frac{1}{\sqrt{2}}}}{2A} = \frac{P\sqrt{2}}{\left(\sqrt{2}-1\right)} \times \frac{1}{2A}$$

$$= \frac{\sqrt{2}P\left(\sqrt{2}+1\right)}{2A} = \frac{\sqrt{2} \times 4a\left(\sqrt{2}+1\right)}{2 \times a^2}$$

$$= \frac{\sqrt{2} \times 2\left(\sqrt{2}+1\right)}{a} = \frac{2\left(2+\sqrt{2}\right)}{a}$$

Hence, option (c) is the answer.

77. $xyz = 4$

$y - x = z - y$

$2y = x + z$

y is the AM of x, y, z.

Also, $\sqrt[3]{xyz} = 4^{\frac{2}{3}} \Rightarrow \sqrt[3]{xyz} = 4^{\frac{1}{3}}$

AM ≥ GM

$Y \geq 2^{\frac{2}{3}}$

Therefore, the minimum value of y is $2^{\frac{2}{3}}$.

Hence, option (b) is the answer.

78. $\angle BAC = \angle ACT + \angle ATC = 50 + 30 = 80°$

$\angle ACT = \angle ABC$ (Angle in alternate segment)

So, $\angle ABC = 50°$

$\angle BCA = 1870 - (\angle ABC + \angle BAC)$

$= 180 - (50 + 80) = 50°$

Since $\angle BOA = 2\angle BCA = 2 \times 50 = 100°$

Alternative Method:

Join OC

$\angle OCT = 90°$ (TC is tangent to OC.)

$\angle OCA = 90° - 50° = 40°$

$\angle OAC = 40°$ (OA = OC being the radius)

$\angle BAC = 50° + 30° = 80°$

$\angle OAB = 80° - 40° = 40° = \angle OBA$ (OA = OB being the radius)

$\angle BOA = 180° - (\angle OBA + \angle OAB) = 100°$

Hence, option (a) is the answer.

79. Let $S = 1 + \frac{4}{7} + \frac{9}{7^2} + \frac{16}{7^3} + \frac{25}{7^4}$ (i)

$\therefore \quad \frac{1}{7}S = \frac{1}{7} + \frac{4}{7^2} + \frac{9}{7^3} + \frac{16}{7^4} + \frac{25}{7^5}$ (ii)

(i) – (ii) gives

$$S\left(1 - \frac{1}{7}\right) = 1 + \frac{3}{7} + \frac{5}{7^2} + \frac{7}{7^3} + \frac{9}{7^4} \quad \text{(iii)}$$

$$\frac{1}{7} \times S\left(1 - \frac{1}{7}\right) = \frac{1}{7} + \frac{3}{7^2} + \frac{5}{7^3} + \frac{7}{7^4} \quad \text{(iv)}$$

(iii) – (iv) gives

$$S\left(1 - \frac{1}{7}\right) - \frac{1}{7}S\left(1 - \frac{1}{7}\right) = 1 + \frac{2}{7} + \frac{2}{7^2} + \frac{2}{7^3} + \frac{2}{7^4}$$

$$\therefore \quad S\left(1 - \frac{1}{7}\right)\left(1 - \frac{1}{7}\right) = 1 + \frac{2}{7}\left[1 + \frac{1}{7} + \frac{1}{7^2} + \ldots\infty\right]$$

$$\therefore \quad S\left(1 - \frac{1}{7}\right)^2 = 1 + \frac{2}{7} \times \frac{1}{1 - \frac{1}{7}}$$

$$\therefore \quad S\left(\frac{6}{7}\right)^2 = 1 + \frac{2}{7} \times \frac{7}{6}$$

$$\therefore \quad S \times \frac{36}{49} = 1 + \frac{1}{3}$$

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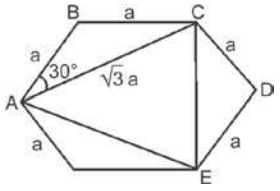
$S = \frac{49}{27}$

Hence, option (c) is the answer.

80. By observing, we see 6 will appear in 5 sets T_2, T_3, T_4, T_5, and T_6. Similarly, 12 will also appear in 5 sets and these sets will be distinct from the sets in which 6 appears, i.e., T_8, T_9, T_{10}, T_{11}, and T_{12}. Thus, each multiple of 6 will be appear in 5 distinct sets. Till T_{96}, there will be 16 multiples, which will appear in $16 \times 5 = 80$ sets.

Hence, option (a) is the answer.

81.



ΔACE is an equilateral triangle with side $\sqrt{3}a$.

Area of hexagon $= \frac{\sqrt{3}}{4}a^2 \times 6$

Area as $\Delta ACE = \frac{\sqrt{3}}{4}a^2 \times \left(\sqrt{3}a\right)^2$

Therefore, the ratio $= \frac{1}{2}$

Hence, option (b) is the answer.

82. Let a be the common root.

$\alpha^3 + 3\alpha^2 + 4\alpha + 5 = 0$

$\alpha^3 + 2\alpha^2 + 7\alpha + 3 = 0$

$\alpha^2 - 3\alpha + 2 = 0$

$\alpha = 2, \alpha = 1$

Hence, option (a) is the answer.

83. $1 - \frac{1}{n} < x \le 3 + \frac{1}{n}$

Put $n = 1$

$\therefore 0 < x \le 4$

Hence, option (c) is the answer.

84. $36 \le n \le 72$

$x = \frac{n^2 + 2\sqrt{2}(n+4) + 16}{n + 4\sqrt{n} + 4}$

Put $x = 36$

$\therefore \quad x = \frac{(36)^2 + 2 \times 6 \times 40 + 16}{36 + 24 + 4}$

Hence, option (d) is the answer.

85. $13x + 1 < 2z$ and $z + 3 = 5y^2$

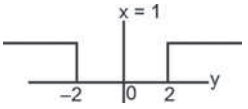
$13x + 1 < 2(5y^2 - 3)$

$13x + 1 < 10y^2 - 6$

$13x + 7 < 10y^2$ put $x = 1$

$20 < 10y^2$ $y^2 > 2$

$y^2 > \sqrt{2}$ $(y^2 - 2) > 0$



Hence, option (d) is the answer.

86. Let $n = 6$

Therefore, $\sqrt{n} = \sqrt{6} = 2.4$

Therefore, divisors of 6 are 1, 2, 3.

If we take 2 as divisor, then $\sqrt{n} > 2 > 1$

Statement A is true.

If we take 3 as divisor, then $6 > 3 > 2.4$, i.e., $n > 3\sqrt{n}$

Therefore, Statement B is true.

Hence, option (d) is the answer.

87. $x = -|a|b$

Now, $a - xb = a - (-|a|b)b$

$= a + |a|b^2$

$\therefore a - xb = a + ab^2 \ldots a \ge 0$ or $a - xb$

$= a - ab^2 \ldots a < 0$

$= a(1 + b^2) = a(1 - b^2)$

Consider the first case:

As $a \ge 0$ and $|b| \ge 1$, therefore, $(1 + b^2)$ is positive.

$\therefore a(1 + b^2) \ge 0$

$\therefore a - xb \ge 0$

Consider the second case:

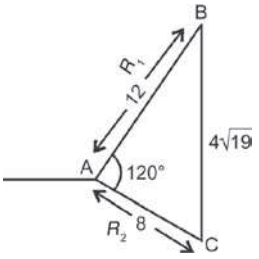
As $a < 0$ and $\}b\} \ge 1$, therefore, $(1 - b^2) \le 0$

$\therefore a(1 - b^2) \ge 0$ (since –ve ax –ve = +ve and $1 - b^2$ can be zero also), i.e., $a - xb \ge 0$

Therefore, in both cases $a - xb \ge 0$

Hence, option (b) is the answer.

88. The required answer is $34 \times 0.65 \times 0.65 = 14.365$. Because we get two similar triangles and the area is proportional to the square of its side.



$BC^2 = (12)^2 + 8^2 - 2 \times 12 \times \cos 120°$

$\therefore BC = 4\sqrt{19}$

$t_1 = \frac{4\sqrt{19} + 8}{3}$

$t_2 = \frac{4\sqrt{19} + 12}{2}$

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(Where t_1 and t_2 are the time taken by Ram and Shyam to reach the starting point.)

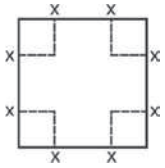
$$t_2 - t_1 = \frac{4\sqrt{19}+12}{2} - \frac{4\sqrt{19}+8}{3}$$

$$t_2 - t_1 = \frac{12\sqrt{19}+36-8\sqrt{19}+10}{6}$$

$$= \frac{4\sqrt{19}+20}{6} = \frac{2\sqrt{19}+10}{3}$$

Hence, option (d) is the answer.

90.



Volume = $l \times b \times h$

$V = (12 - 2x)(12 - 2x) \times x$

$V' = (12 - 2x)(12 - 2x)\,4x$

Where $V' = 4V$

Now, sum = $12 - 2x + 12 - 2x + 4x = 24$ (constant)

As sum is constant for the maximum product $12 - 2x = 12 - 2x = 4x$

Therefore, $x = 2$

Hence, option (d) is the answer.

91. As any prime number greater than 3 can be expressed in the form $6n \pm 1$, the minimum difference between the three consecutive prime numbers will be 2 and 4. The values that satisfy the given conditions are only 3, 5, and 7, i.e., only one set is possible.

Hence, option (a) is the answer.

92. $a = 6b = 12c$ and $2b = 9d = 12e$

Dividing the first equations by 12 and second by 36.

We get, $\frac{a}{12} = \frac{b}{2} = \frac{c}{1}$ and $\frac{b}{18} = \frac{d}{4} = \frac{e}{3}$

$\therefore \frac{a}{108} = \frac{b}{18} = \frac{c}{9} = \frac{d}{4} = \frac{e}{3}$

$\therefore a:b:c:d:e = 108:18:19:4:3$

$\therefore \frac{c}{d} = \frac{9}{4}$ is not an integer.

Hence, option (d) is the answer.

93. Let the number of huts before the floods in 2001 be 32.

$\therefore$ Huts destroyed $= \frac{32}{2} = 16$ and rebuilt = 32

$\therefore$ Total huts after floods in 2001 = 16 + 32 = 48

Similarly, in 2002, the destroyed huts = 24

Rebuilt = 48

Total = 24 + 48 = 72

Similarly, in 2003, the destroyed huts = 36

Rebuilt = 72

Total = 36 + 72 = 108

Hence, option (c) is the answer.

94. $g^2 = g * g = h$

$g^3 = g^2 * g = h * g = f$

$g^4 = g^3 * g = f * g = e$

$\therefore n = 4$

Hence, option (a) is the answer.

95. $f \oplus [f * \{f \oplus (f * F)\}]$

$= f \oplus [f * \{f \oplus h\}]$

$= f \oplus [f * e]$

$= f \oplus [f] = h$

Hence, option (d) is the answer.

96. $e^8 = e^2 * e^2 * e^2$

$= e * e * e$

$= e$

If we observe a * anything = a

$\therefore a^{10} = a$

$\therefore \{a^{10} * (f^{10} \oplus g^9)\} \oplus e^8$

$= a \oplus e$

$= e$

Hence, option (a) is the answer.

97. **Case I** ___m___

First place can be selected in five ways, and hence, the third in four ways.

$\therefore 5 \times 4 = 20$ ways

Case II ___n___

First place can be selected in 5 ways and the third in 2 ways.

$\therefore 5 \times 2 = 10$ ways

Case III ___p___

First place can be selected in 5 ways and the last letter will be the same, i.e., one way.

$\therefore 5 \times 1 = 5$ ways

Total ways = 20 + 10 + 5 = 35 ways

Hence, option (d) is the answer.

98. As the third letter is e, it can be selected in only one way.

Case I 4 m e = 4 ways

Case II 5 n e = 5 ways

Case III 1 p e = 1 way

= 10 ways

Hence, option (c) is the answer.

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99. It will go by elimination.

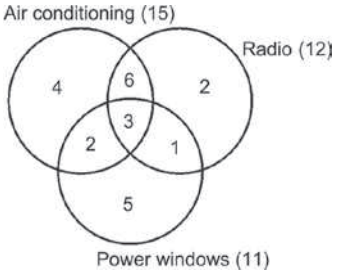
$9 - 7 = 2$ is even; therefore, option (a) is not possible.

$2 \times 9 = 18$ is even; therefore, option (c) is not possible.

$\frac{3+9}{3} = \frac{12}{3} = 4$ is even; therefore, option (c) is not possible.

Hence, option (d) is the answer.

100.



Total = 4 + 6 + 2 + 2 + 3 + 1 + 5 = 23

∴ Cars having none of the option = 25 − 23 = 2

Hence, option (d) is the answer.

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CAT 2004

This paper contains a total of 35 questions divided into two parts: Part A and Part B. Part A contains 20 questions and each question carry 1 mark each. Part B contains 15 questions and each question carry 2 marks each.

There is negative marking for wrong answers.

PART—A

Direction for Questions 39 to 52: Answer the questions independently of each other:

Q.39 Two boats, travelling at 5 and 10 kmph, head directly towards each other. They begin at a distance of 20 km from each other. How far apart are they (in km) 1 min before they collide?

(a) $\frac{1}{12}$ (b) $\frac{1}{6}$ (c) $\frac{1}{4}$ (d) $\frac{1}{3}$

Q.40 A rectangular sheet of paper, when halved by folding it at the mid-point of its longer side, results in a rectangle, whose longer and shorter sides are in the same proportion as the longer and shorter sides of the original rectangle. If the shorter side of the original rectangle is 2, what is the area of the smaller rectangle?

(a) $4\sqrt{2}$ (b) $2\sqrt{2}$

(c) $\sqrt{2}$ (d) None of these

Q.41 If the sum of the first 11 terms of an arithmetic progression equals that of the first 19 terms, then what is the sum of the first 30 terms?

(a) 0 (b) -1

(c) 1 (d) Not unique

Q.42 If a man cycles at 10 kmph, then he arrives at a certain place at 1 pm. If he cycles at 15 kmph, he will arrive at the same place at 11 am. At what speed must he cycle to get there at noon?

(a) 11 kmph (b) 12 kmph

(c) 13 kmph (d) 14 kmph

Q.43 On 1 January 2004, two new societies S_1 and S_2 are formed, each having n members. On the first day of each subsequent month, S_1 adds b members, whereas S_2 multiples its current numbers by a constant factor r. Both the societies have the same number of members on 2 July 2004. If $b = 10.5n$, what is the value of r?

(a) 2.0 (b) 1.9 (c) 1.8 (d) 1.7

Q.44 If $f(x) = x^3 - 4x + p$, and $f(0)$ and $f(1)$ are of opposite sign, then which of the following is necessarily true?

(a) $-1 < p < 2$ (b) $0 < p < 3$

(c) $-2 < p < 1$ (d) $-3 < p < 0$

Q.45 Suppose n is an integer such that the sum of digits on n is 2, and $10^{10} < n < 10^{11}$. The number of different values of n is:

(a) 11 (b) 10 (c) 9 (d) 8

Q.46 A milkman mixes 20 litres of water with 80 litres of milk. After selling one-fourth of this mixture, he adds water to replenish the quantity that he had sold. What is the current proportion of water to milk?

(a) 2:3 (b) 1:2 (c) 1:3 (d) 3:4

Q.47 If $\frac{a}{b+c} = \frac{b}{c+a} = \frac{c}{a+b} = r$, then r cannot take any value except:

(a) 1/2

(b) -1

(c) 1/2 or -1

(d) Cannot be determined

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Q.48 Let $y = \cfrac{1}{2+\cfrac{1}{3+\cfrac{1}{2+\cfrac{1}{3+\ldots}}}}$

What is the value of y?

(a) $\frac{\sqrt{11}+3}{2}$ (b) $\frac{\sqrt{11}-3}{2}$

(c) $\frac{\sqrt{15}+3}{2}$ (d) $\frac{\sqrt{15}-3}{2}$

Q.49 Karan and Arjun run a 100-m race, where Karan beats Arjun by 10 m. To do a favour to Arjun, Karan starts 10 m behind the starting line in a second 100 m race. They both run at their earlier speeds. Which of the following is true in connection with the second race?

(a) Karan and Arjun reach the finishing line simultaneously.
(b) Arjun beats Karan by 1 m.
(c) Arjun beats Karan by 2 m.
(d) Karan beats Arjun by 1 m.

Q.50 N persons stand on the circumference of a circle at distinct points. Each possible pair of persons, not standing next to each other, sings a 2-min song, one pair after the other. If the total time taken for singing is 28 min, what is N?

(a) 5 (b) 7
(c) 9 (d) None of the above

Q.51 In Nuts and Bolts factory, one machine produces only nuts at the rate of 100 nuts per minute and needs to be cleaned for 5 min after production of every 1000 nuts. Another machine produces only bolts at the rate of 75 bolts per minute and needs to be cleaned for 10 min after production of every 1500 bolts. If both the machines start production at the same time, what is the minimum duration required for producing 9000 pairs of nuts and bolts?

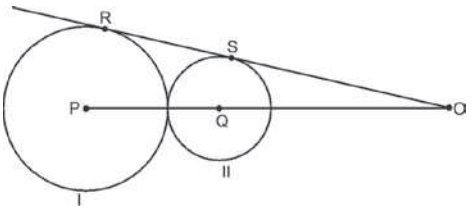
(a) 130 min (b) 135 min
(c) 170 min (d) 180 min

Q.52 A father and his son are waiting at a bus stop in the evening. There is a lamp post behind them. The lamp post, the father, and his son stand on the same straight line. The father observes that the shadows of his head and his son's head are incident at the same point on the ground. If the heights of the lamp post, the father, and his son are 6 m, 1.8 m, and 0.9 m, respectively, and the father is standing 2.1 m away from the post, then how far (in m) is the son standing from his father?

(a) 0.9 (b) 0.75 (c) 0.6 (d) 0.45

Direction for Questions 53 to 55: Answer the questions on the basis of the information given below:

In the adjoining figure, I and II are circles with P and Q as centres, respectively, The two circles touch each other and have a common tangent that touches them at points R and S, respectively. This common tangent meets the line joining P and Q at O. The diameters of I and II are in the ratio of 4:3. It is also known that the length of PQ is 28 cm.



Q.53 What is the ratio of the length of PQ to that of QO?

(a) 1:4 (b) 1:3 (c) 3:8 (d) 3:4

Q.54 What is the radius of the Circle II?

(a) 2 cm (b) 3 cm (c) 4 cm (d) cm

Q.55 The length of SO is:

(a) $8\sqrt{3}$cm (b) $10\sqrt{3}$ cm

(c) $12\sqrt{3}$ cm (d) $14\sqrt{3}$ cm

Direction for Questions 56 to 58: Answer the questions independently of each other:

Q.56 Let $f(x) = ax^2 - b|x|$, where a and b are constants. Then, at $x = 0$, $f(x)$ is

(a) maximized whenever $a > 0$, $b > 0$.
(b) maximized whenever $a > 0$, $b < 0$.
(c) minimized whenever $a > 0$, $b > 0$.
(d) minimized whenever $a > 0$, $b < 0$.

Q.57 Each family in a locality has at most two adults, and no family has fewer than 3 children. Considering all the families together, there are more adults than boys, more boys than girls, and more girls than families, Then, the minimum possible number of families in the locality is:

(a) 4 (b) 5 (c) 2 (d) 3

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Q.58 The total number of integers pairs (x, y) satisfying the equation $x + y = xy$ is:
(a) 0 (b) 1
(c) 2 (d) None of the above

PART – B

Number of Questions = 15

Note: Questions 59 to 73 carry 2 marks each.

Direction for Questions 59 to 62: Answer the questions independently of each other:

Q.59 Let C be a circle with centre P_0 and AR be a diameter of C. Suppose P_1 is the mid-point of the line segment P_0B, P_2 is the mid-point of the line segment P_1B, and so on. Let C_1, C_2, and C_3 be circles with diameters P_0P_1, P_1P_2, and P_2P_3, respectively. Suppose the circles C_1, C_2, and C_3 are all shaded. The ratio of the area of the unshaded portion of C to that of the original circle is:
(a) 8:9 (b) 9:10 (c) 10:11 (d) 11:12

Q.60 Consider the sequence of numbers $a_1, a_2, a_3, \ldots$ to infinity where $a_1 = 81.33$ and $a_2 = -19$ and $a_j = a_{j-1} - a_{j-2}$ for $j \geq 3$. What is the sum of the first 6002 terms of this sequence?
(a) −100.33 (b) −30.00
(c) 0 62.33 (d) 119.33

Q.61 A sprinter starts running on a circular path of radius r metres. Her average speed (in m/minute) is πr during the first 30 s, $\frac{\pi r}{2}$ during 1 min, $\frac{\pi r}{4}$ during next 2 min, $\frac{\pi r}{8}$ during next 4 min, and so on. What is the ratio of the time taken for the nth round to that for the previous round?
(a) 4 (b) 8 (c) 16 (d) 32

Q.62 Let $u = (\log_2 x)^2 - 6\log_2 x + 12$, where x is a real number. Then, the equation $x^u = 256$, has
(a) no solution for x.
(b) exactly one solution for x.
(c) exactly two distinct solutions for x.
(d) exactly three distinct solutions for x.

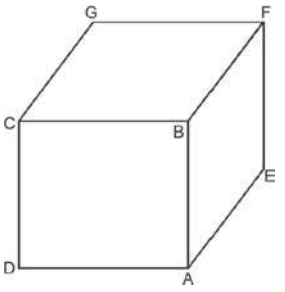
Direction for Questions 63 and 64: Answer the questions on the basis of the information given below:

$f_1(x) = x \quad 0 \leq x \leq 1$
$= 1 \quad x \geq 1$
$= 0$ Otherwise
$f_2(x) = f_1(-x)$ for all x
$f_3(x) = -f_2(x)$ for all x
$f_3(x) = f_3(-x)$ for all x

Q.63 How many of the following products are necessarily zero for every x?
$f_1(x)f_2(x)$, $f_2(x)f_3(x)$, $f_2(x)f_4(x)$?
(a) 0 (b) 1 (c) 2 (d) 3

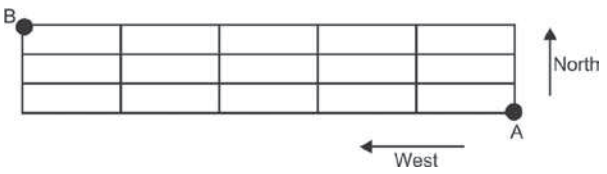
Q.64 Which of the following is necessarily true?
(a) $f_4(x) = f_1(x)$ for all x
(b) $f_1(x) = -f_3(-x)$ for all x
(c) $f_2(-x) = f_4(x)$ for all x
(d) $f_1(x) = f_3(x) = 0$ for all x

Q.65 If the lengths of diagonals DF, AG, and CE of the cube shown in the adjoining figure are equal to the three sides of a triangle, then the radius of the circle circumscribing that triangle will be



(a) equal to the side of cube.
(b) times the side of the cube.
(c) times the side of the cube.
(d) impossible to find from the given information.

Q.66 In the adjoining figure shown below, the lines represent one-way roads allowing travel only northwards or only westwards. Along how many distinct routes can a car reach point B from point A?

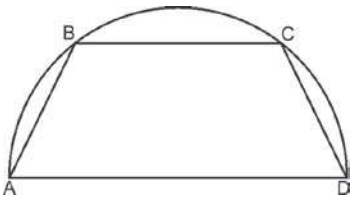


(a) 15 (b) 56 (c) 120 (d) 336

Q.67 On a semicircle with diameter AD, chord BC is parallel to the diameter. Further, each of the chords AB and CD have length 2, whereas AD has length 8. What is the length of BC?

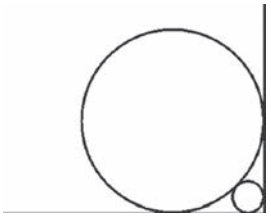
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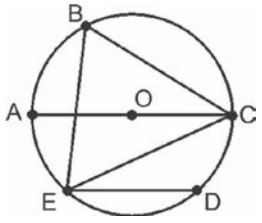
(a) 7.5 (b) 7
(c) 7.75 (d) None of these

Q.68 A circle with radius 2 is placed against a right angle. Another smaller circle is also placed as shown in the adjoining figure. What is the radius of the smaller circle?



(a) $3-2\sqrt{2}$ (b) $4-2\sqrt{2}$
(c) $7-4\sqrt{2}$ (d) $6-4\sqrt{2}$

Q.69 In the figure shown below, chord ED is parallel to the diameter AC of the circle. If $\angle CBE = 65^{o}$, then what is the value of $\angle DEC$?



(a) 35° (b) 55° (c) 45° (d) 25°

Direction for Questions 70 and 71: Answer the questions on the basis of the information given below:

In an examination, there are 100 questions divided into three Groups A, B, and C, such that each group contains at least one question. Each question in Group A carries 1 mark, each question in Group B carries 2 marks, and each question in Group C carries 3 marks. It is known that the questions in Group A together carry at least 60% of the total marks.

Q.70 If Group B contains 23 questions, then how many questions are there in Group C?
(a) 1
(b) 2
(c) 3
(d) Cannot be determined

Q.71 If Group C contains 8 questions and Group B carries at least 20% of the total marks, which of the following best describes the number of questions in Group B?
(a) 11 or 12 (b) 12 or 13
(c) 13 or 14 (d) 14 or 15

Direction for Questions 72 and 73: Answer the questions independently of each other:

Q.72 The reminder, when $(15^{23}+23^{23})$ is divided by 19, is:
(a) 4 (b) 15 (c) 0 (d) 18

Q.73 A new flag is to be designed with six vertical stripes using some or all of the colours yellow, green, blue, and red. Then, the number of ways this can be done so that no two adjacent stripes have the same colour, is:
(a) 12×81 (b) 16×192
(c) 20×125 (d) 24×216

Answers

39. (c)	**40.** (b)	**41.** (a)	**42.** (b)	**43.** (a)	**44.** (b)	**45.** (a)	**46.** (a)	**47.** (c)	**48.** (d)
49. (d)	**50.** (b)	**51.** (c)	**52.** (d)	**53.** (b)	**54.** (b)	**55.** (c)	**56.** (d)	**57.** (d)	**58.** (c)
59. (d)	**60.** (c)	**61.** (c)	**62.** (b)	**63.** (c)	**64.** (b)	**65.** (a)	**66.** (b)	**67.** (b)	**68.** (d)
69. (d)	**70.** (a)	**71.** (c)	**72.** (c)	**73.** (a)					

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Hints and Solutions

39. The boats will be colliding after a time which is given by:

After this time of 80 min, boat (1) has covered $80 \times \frac{5}{60}$ km

$= \frac{20}{3}$ km, whereas boat (2) has covered $80 \times \frac{10}{60}$ km $= \frac{40}{3}$

After 79 min, distance covered by the first boat

$= d_1 = \left(\frac{20}{3} - \frac{5}{60}\right)$

After 79 min, distance covered by the second boat

$= d_2 = \left(\frac{40}{3} - \frac{10}{60}\right)$

So, the separation between the two boats = 20 – $(d_1 + d_2)$

$= \frac{1}{4}$ km

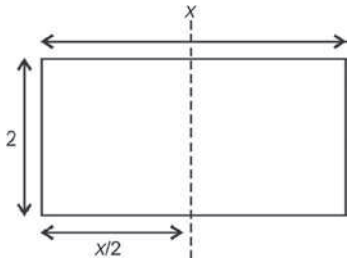
Alternative Method:

Relative speed of two boats = 5 + 10 = 15 kmph, i.e., in 60 min, they cover (together) = 15 km

$\therefore$ In 1 min, they will cover (together) $\frac{15}{60} = \frac{1}{4}$ km

Hence, option (c) is the answer.

40.



In original rectangle ratio $= \frac{x}{2}$

In smaller rectangle ratio $= \frac{2}{\left(\frac{x}{2}\right)}$

Given $\frac{x}{2} = \frac{2}{\frac{x}{2}} \Rightarrow x2\sqrt{2}$

Area of smaller rectangle $= \frac{x}{2} \times 2 = x = 2\sqrt{2}$

Hence, option (b) is the answer.

41. Given

$t_1 + t_2 + \ldots + t_{11} = t_1 + t_2 + \ldots + t_{19}$ (for an AP)

$\Rightarrow \frac{11}{2}[2a + (11-1)d] = \frac{19}{2}[2a + (19-1)d]$

$22a + 110d = 28a + 342d$

$16a + 232d = 0$

$\Rightarrow \frac{30}{2}[2a + (30-1)d] = 0$

$\Rightarrow S_{30 \text{ terms}} = 0$

Hence, option (a) is the answer.

42. When speed of the man = 10 kmph $= \frac{d}{t}$

When speed of the man = 15 kmph $= \frac{d}{t-2}$

Equating the value of d: $10 \times t = 15 \times (t-2) \Rightarrow t = 6$ h

Finally, the desired speed $\frac{d}{t-1} = \frac{10t}{t-1} = \frac{10 \times 6}{5} = 12$ kmph.

Hence, option (b) is the answer.

43. There will be an increase of 6 times. Number of members S will be in AP. On 2 July 2004, S_1 will have n + 6 b members $= n + 6 \times 10.5n = 64n$

Number of members in S_2 will be in GP.

On 2 July 2004, number of members in $S_2 = nr\,6$

They are equal, hence, $64n = nr^6$

$\Rightarrow 64 = r^6 \Rightarrow r = 2$

Hence, option (a) is the answer.

44. We have

$f(0) = 0^3 - 4(0) + p = p$

$f(1) = 1^3 - 4(1) + p = p - 3$

If P and $p - 3$ are of opposite signs, then $p\,(p-3) < 0$

Therefore, $0 < p < 3$

Hence, option (b) is the answer.

45. (a) We have

(1) $10^{10} < n < 10^{11}$

(2) Sum of the digits for 'n' = 2 clearly

(n) min = 10,00,00,00,001 (1 followed by 9 zeros and finally 1)

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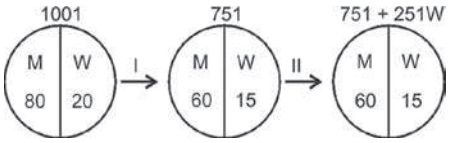
Obviously, we can form 10 such numbers by shifting '1' by one place from right to left again and again. Again, there is another possibility for n, i.e.,

$n = 20,00,00,00,000$

So, finally, number of different values for $n = 10 + 1 = 11$ answer

Hence, option (a) is the answer.

46.



The diagram is self-explanatory. Removal of 25 litres at stage I will result in the volume of milk being reduced by 80% of 25 litres, i.e., 20 litres and the volume of water being reduced by the remaining 5 litres. So, M = 60 litres and W = 15 litres. Addition of 25 litres water will finally given M = 60 litres and W = 40 m. Therefore, the ratio of W and M = 40:60 = 2:3

Hence, option (a) is the answer.

47. If $\frac{a}{b+c} = \frac{b}{c+a} = \frac{c}{a+b} = r$

Then, there are only two possibilities.

(1) If $a + b + c \neq 0$, then

$$\frac{a}{b+c} = \frac{b}{c+a} = \frac{c}{a+b}$$

$$= \frac{a+b+c}{(b+c)+(c+a)+(a+b)}$$

$$= \frac{a+b+c}{2(a+b+c)} = \frac{1}{2}$$

(2) If $a + b + c = 0$, then

$b + c = -a;\ c + a = -b;\ a + b = -c$

Hence, $\frac{a}{b+c} = \frac{a}{(-a)} = -1$

Similarly, $\frac{b}{c+a} = \frac{c}{a+b} = -1$

Hence, option (c) is the correct one $\frac{1}{2}$ or -1.

48. $y = \dfrac{1}{2 + \dfrac{1}{3+y}}$

$$\Rightarrow y = \frac{3+y}{7+2y}$$

$$\Rightarrow 2y^2 + 6y - 3 = 0$$

$$\Rightarrow y = \frac{-6 \pm \sqrt{36+24}}{4}$$

$$= \frac{-6 \pm \sqrt{60}}{4} = \frac{-3 \pm \sqrt{15}}{2}$$

Since 'y' is a +ve number, therefore,

$$y = \frac{\sqrt{15} - 3}{2}$$

Hence, option (d) is the answer.

49. **Situation I**

In whatever time Karan covers a distance of 100 m, Arjun covers 90 m in the same time.

Situation II

Now, Karan is 10 m behind the starting point. Once again to cover 100 m from this new point, Karan will be taking the same time as before. In this time, Ariun will cover 90 m only. This means that now both of them will be at the same point, which will be 10 m away from the finish point. Since both of them are required to cover the same distance of 10 m now and Karan has a higher speed, he will beat Ariun. No need for calculations as option (d) is the only such option available.

50. Each person will form a pair with all other persons except the two beside him. Hence, he will form $(n - 3)$ pairs.

If we consider each person, total pairs = $n(n - 3)$, but here, each pair is counted twice.

Hence, the actual number of pairs = $\frac{n(n-3)}{2}$

They will sing for $\frac{n(n-3)}{2} \times 2 = n(n-3)$ min

Hence, $n\ (n - 3) = 28$

$\Rightarrow n^2 - 3n - 28 = 0$

$\Rightarrow n = 7$ or -4

Discarding the –ve value: $n = 7$

Hence, option (b) is the answer.

51. **Machine I**

Number of nuts produced in 1 min = 100

To produce 1000 nuts, the time required = 10 min

Cleaning time for nuts = 5 min

Over all time to produce 1000 nuts = 15 min

Over all time to produce 9000 nuts = 138 min – 5 min = 133 min (i)

Machine II

To produce 75 bolts, time required = 1 min

To produce 1500 bolts, time required = 20 min

Cleaning time for bolts = 10 min

Effective time to produce 1500 bolts = 30 min

Effective time to produce 9000 bolts = $30 \times 6 - 10 = 170$ min (ii)

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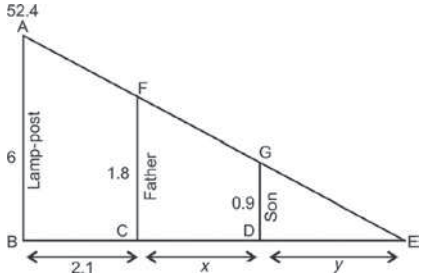
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From (i) and (ii)

Minimum time = 170 min

Hence, option (c) is the answer.

52.



$\Delta ABE \sim \Delta FCE$

$$\therefore \frac{6}{1.8} = \frac{2.1 + x + y}{x + y} \quad \text{(i)}$$

Also, $\Delta ABE \sim \Delta GDE$

$$\therefore \frac{6}{0.9} = \frac{2.1 + x + y}{x + y} \quad \text{(ii)}$$

From (i) and (ii), $x = 0.45$

Hence, option (d) is the answer.

53. $\frac{OP}{OQ} = \frac{PR}{QS} = \frac{4}{3}$

$OP = 28$

$OQ = 21$

$PQ = OP - OQ = 7$

$\frac{PQ}{OQ} = \frac{7}{21} = \frac{1}{3}$

Hence, option (b) is the answer.

54. $PR + QS = PQ = 7$

$= \frac{RP}{QS} = \frac{4}{3}$

$\Rightarrow QS = 3$

Hence, option (b) is the answer.

55. $\sqrt{OQ^2 - QS^2} = \sqrt{21^2 - 3^2}$

$= \sqrt{24 \times 18} = 12\sqrt{3}$

Hence, option (c) is the answer.

56. When $a > 0$, $b < 0$,

ax^2 and $-b|x|$ are non-negative for all x,

i.e., $ax^2 - b|x| \geq 0$ $ax^2 - b|x|$ is minimum at $x = 0$ when $a > 0$, $b < 0$.

Hence, option (d) is the answer.

57.

Family	Adults	Children
I	0, 1, 2	3, 4, 5...
II	0, 1, 2	3, 4, 5...
III	0, 1, 2	3, 4, 5...

As per the question, we need to satisfy three conditions, namely:

1. Adults (A) > Boys (B)
2. Boys (B) > Girls (G)
3. Girls (G) > Families (F)

Clearly, if the number of families is 2, maximum number of adults can only be 4. Now, for the second condition to be satisfied, every family should have at least two boys and one girl each. This will result in noncompliance with the first condition because adults will be equal to boys. If we consider the same conditions for 3 families, then all three conditions will be satisfied.

Hence, option (d) is the answer.

58. Given equation is $x + y = xy$.

$\Rightarrow xy - x - y + 1 = 1$

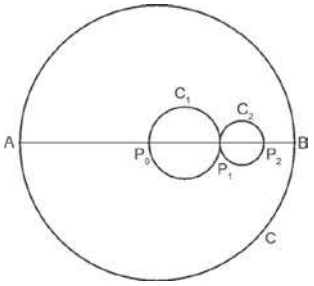
$\Rightarrow (x - 1)(y - 1) = 1$

$x - 1 = 1$ and $y - 1 = 1$ or $x - 1 = -1$ and $y - 1 = -1$

Clearly (0, 0) and (2, 2) are the only pairs that will satisfy the equation.

Hence, option (c) is the answer.

59.



Circle C	Radius r
C_1	$\frac{r}{4}$
C_2	$\frac{r}{8}$
C_3	$\frac{r}{16}$
:	:

$$= 1 - \frac{\text{Area of unshaded portion of C}}{\text{Area of C}}$$

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$$= 1 - \frac{\pi\left(\left(\frac{r}{4}\right)^2 + \left(\frac{r}{8}\right)^2 + \ldots\right)}{\pi r^2}$$

$$= 1 - 1 - \left(\frac{1}{4^2} + \frac{1}{8^2} + \ldots\right) = 1 - \frac{\frac{1}{16}}{1 - \frac{1}{4}}$$

$$= \frac{11}{12}$$

Hence, option (d) is the answer.

60. Given $a_1 = 81.33$; $a_2 = -19$

Also,

$aj = a_{j-1} - a_{j-2}$, for $j \geq 3$

$\Rightarrow a_3 = a_2 - a_1 = -100.33$

$a_4 = a_3 - a_2 = -81.33$

$a_5 = a_4 - a_3 = 19$

$a_6 = a_5 - a_4 = +100.33$

$a_7 = a_6 - a_5 = +81.33$

$a_8 = a_7 - a_6 = -19$

Clearly there is a cycle of 6 and the sum of terms in every such cycle = 0. Therefore, when we add a_1, a_2, a_3... upto '2', we will eventually be left with $a_1 + a_2$ only, i.e., 81.33 – 19 = 62.33.

Hence, option (c) is the answer.

61. As options are independent of n, let $n = 2$

Time taken for the first round $\frac{1}{2} + 1 + 2 + 4 = 7.5$ min

Time taken for the second round = 8 + 16 + 32 + 64 = 120 min

$$\text{Ratio} = \frac{120}{7.5} = 16 = 16$$

Hence, option (c) is the answer.

62. $u = (\log_2 x)^2 - 6\log_2 x + 12$

$x^u = 256$

Let $\log_2 x = y \Rightarrow x = 2^y$

$x^u = 2^8 \Rightarrow uy = 8 \Rightarrow u = \frac{8}{y}$

$\frac{8}{y} = t^2 - 6y + 12 \Rightarrow y^3 - 6y^2 - = 0$

$\Rightarrow (y-2)^3 = 0 \Rightarrow y = 2$

$\Rightarrow x = 4, u = 4$

Hence, option (b) is the answer.

63. $f_1 f_2 = f_1(x) f_1(-x)$

$$f_1(-x) = \begin{cases} -x & -1 \leq -x - 1 \\ 1 & -x \geq 1 \\ 0 & \text{otherwise} \end{cases}$$

$$= \begin{cases} -x & -1 \leq -x \leq 1 \\ 1 & -x \geq 1 \\ 0 & \text{otherwise} \end{cases}$$

$f_1 f_1(-x) = 0 \ \forall x$

Hence, option (c) is the answer.

64. Check with options:

Option (2)

$f_3(-x) = -f_2(-x) = -f_1(x)$

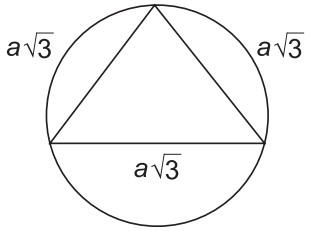
$= f_1(x) = -f_3(-x) \ \forall x$

Hence, option (b) is the answer.

65. DF, AG, and ~ E are body diagonals of cube.

Let the side of cube = a

Therefore, body diagonal is $a\sqrt{3}$.



Circum radius for equilateral triangle side = $\frac{\text{side}}{\sqrt{3}}$

Therefore, $\frac{a\sqrt{3}}{\sqrt{3}} = a$

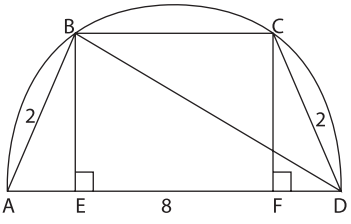
Hence, option (a) is the answer.

66. From A to B, there are 8 on-way roads out of which 3 roads are in northwards and 5 roads are westwards.

Therefore, number of distinct routes is = $\frac{8!}{513!} = 56$

Hence, option (b) is the answer.

67.



$$\frac{1}{2} \times AB \times BD = \frac{1}{2} \times AD \times BE$$

$$2\sqrt{8^2 - 2^2} = 8 \times BE$$

$$BE = \sqrt{\frac{60}{4}} = \sqrt{\frac{15}{2}}$$

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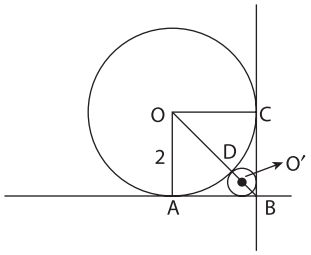
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$$AE = AE = \sqrt{2^2 - \left(\frac{\sqrt{15}}{2}\right)^2} = \sqrt{\frac{4-15}{4}} = \frac{1}{2}$$

$$BC = EF = 8 - \left(\frac{1}{2} + \frac{1}{2}\right) = 7$$

Hence, option (b) is the answer.

68.



Let the radius of smaller circle = r

$\therefore O'B = r\sqrt{2}$

$\therefore OB = O'B + O'D + OD$

$= r\sqrt{2} + r + 2$

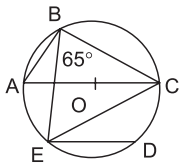
Also, $OB = 2\sqrt{2}$

$= r\sqrt{2} + r + 2 = 2\sqrt{2}$

$\Rightarrow r = 6 - 4\sqrt{2}$

Hence, option (d) is the answer.

69.



In ΔABC,

$\angle B = 90°$ (angles in semicircle)

Therefore, $\angle ABE = 90 - 65 = 25°$

Also, $\angle ABE = \angle ACE$ (angel subtended by same arc AE)

Also, $\angle ACE = \angle CED$ [AC ||ED]

Therefore, $\angle CED = 25°$

Hence, option (d) is the answer.

70. Since Group (B) contains 23 questions, the marks associated with this group are 46. Now check for option (a). If Group (C) has one question, then marks associated with this group will be 3. This means that the cumulative marks for these two groups taken together will be 49. Since total number of questions are 100, Group (A) will have 76 questions, the corresponding weightage being 76 marks. This satisfies all conditions, and hence, (a) is the correct option. It can be easily observed that no other option will fit the bill.

71. Since Group (C) contains 8 questions, the corresponding weightage will be 24 marks. This figure should be less than or equal to 20% of the total marks. Check from the options . Option (c) provides 13 or 14 questions in Group (B), with, a corresponding weightage of 26 or 28 marks. This means that number of questions in Group (A) will either be 79 or 78 and will satisfy the desired requirement.

Hence, option (c) is the answer.

72. $15^{23} = (19-4)^{23} - 19x + (-4)^{23}$ where x is a natural number.

$23^{23} - (19+4)^{23} = 19y + (4)^{23}$ where y is a natural number.

$15^{23} + 23^{23} - 19(x+y) + 4^{23} + (-4)^{23}$

$= 19\,(x+y)$

Hence, option (c) is the answer.

73. The first strip can be of any of the four colours. The 2nd can be of any colour except that of the first (i.e., 3). Similarly, each subsequent strip can be of any colour except that of the preceding strip (= 3).

Hence, the number of ways = $4 \times 35 = 12 \times 81$.

Hence, option (a) is the answer.

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CAT 2005

This paper contains a total of 30 questions divided into two parts: Part A and Part B. Part A contains 10 questions and each question carry 1 mark each. Part B contains 10 questions and each question carry 2 marks each.

There is a negative marking of 1/3 of the marks associated with the particular question.

PART—A

Direction for Questions 1 to 5: Answer the questions independently of each other.

Q.1 If $x = (16^3 + 17^{13} + 18^3 + 19^3)$, then x divided by 70 leaves a remainder of:
(a) 0 (b) 1 (c) 69 (a) 35

Q.2 A chemical plant has four tanks (A, B, C, and D), each containing 1000 litre of a chemical. The chemical is being pumped from one tank to another as follows:
From A to B at 20 *l*/min
From C to A at 90 *l*/min
From A to D at 10 *l*/min
From C to D at 50 *l*/min
From B to C at 100 *l*/min
From D to B at 110 *l*/min
Which tank gets emptied first, and how long does it take (in minute) to get empty, after the pumping starts?
(a) A, 16.66 (b) C, 20
(c) D, 20 (d) D, 25

Q.3 Two identical circles intersect so that their centres, and the points at which they intersect form a square of side 1 cm. The area (in cm^2) of the portion that is common to the two circles is:
(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{2}-1$
(c) $\frac{\pi}{5}$ (d) $\sqrt{2}-1$

Q.4 A jogging park has two identical circular tracks touching each other and a rectangular track enclosing the two circles. The edges of the rectangles are tangential to the circles. Two friends, A and B, start jogging simultaneously from the point where one of the circular tracks touches the smaller side of the rectangular track. A jogs along the rectangular track, whereas B jogs along the two circular tracks in a figure of eight. Approximately, how much faster B has to run than A, so that they take the same time to return to their starting point?
(a) 3.88% (b) 4.22%
(c) 4.44% (d) 4.72%

Q.5 In a chess competition involving some boys and girls of a school, every student had to play exactly one game with every other student. It was found that in 45 games, both the players were girls, and in 190 games, both were boys. The number of games in which one player was a boy and the other was a girl is:
(a) 200 (b) 216 (c) 235 (d) 256

Direction for Questions 6 and 7: Answer the questions on the basis of the information given below.

Ram and Shyam run a race between Points A and B, 5 km apart. Ram starts at 9 am from A at a speed of 5 kmph, reaches B, and returns to A at the same speed. Shyam starts at 9:45 am from A at a speed of 10 kmph, reaches B, and comes back to A at the same speed.

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Q.6 At what time do Ram and Shyam first meet each other?
(a) 10 am (b) 10:10 am
(c) 10:20 am (d) 10:30 am

Q.7 At what time does Shyam over take Ram?
(a) 10:20 am (b) 10:30 am
(c) 10:40 am (d) 10:30 am

Direction for Questions 8 to 10: Answer the questions independently of each other.

Q.8 If $R = \frac{30^{65} - 29^{65}}{30^{64} + 29^{64}}$, then:
(a) $0 < R \le 0.1$ (b) $0.1 < R \le 0.5$
(c) $0.5 < R \le 1.0$ (d) $R > 1$

Q.9 What is the distance (in cm) between two parallel chords of lengths 32 cm and 24 cm in a circle of radius 20 cm?
(a) 1 or 7 (b) 2 or 14
(c) 3 or 21 (d) 4 or 28

Q.10 For which value of k does the following pair of equations yield a unique solution of x such that the solution is positive?
$x^2 - y^2 = 0$ and $(x - k)^2 + y^2 = 1$
(a) 2 (b) 0
(c) $\sqrt{2}$ (d) -2

PART – B

Note: Questions 11 to 30 carry two marks each.

Q.11. Let $n! = 1 \times 2 \times 3 \times ... \times n$ for integer $n \ge 1$. If $p = 1! + (2 \times 2!) + (3 \times 3!) + ... + (10 \times 10!)$, then $p + 2$ when divided by 11! leaves a remainder of:
(a) 10 (b) 0 (c) 7 (d) 1

Q.12. Consider a triangle drawn on the X–Y plane with its three vertices of (41, 0), (0, 41), and (0, 0), each vertex being represented by its (X, Y) coordinates. The number of points with integer coordinates inside the triangle (excluding all the points on the boundary) is:
(a) 780 (b) 800 (c) 820 (d) 741

Q.13. The digits of a three-digit number A are written in the reverse order to form another three-digit number B. If $B > A$ and $B–A$ is perfectly divisible by 7, then which of the following is necessarily true?
(a) $100 < A < 299$ (b) $106 < A < 305$
(c) $112 < A < 311$ (d) $118 < A < 317$

Q.14. If $a_1 = 1$ and $a_{n+1} - 3a_n + 2 = 4n$ for every positive integer n, then a_{100} equals:
(a) $3^{99} - 200$ (b) $3^{99} + 200$
(c) $3^{100} - 200$ (d) $3^{100} + 200$

Q.15. Let S be the set of five-digit numbers formed by digits 1, 2, 3, 4, and 5, using each digit exactly once such that exactly two odd position are occupied by odd digits. What is the sum of the digits in the rightmost position of the numbers in S?
(a) 228 (b) 216 (c) 294 (d) 192

Q.16. The rightmost non-zero digits of the number 30^{2720} is:
(a) 1 (b) 3 (c) 7 (d) 9

Q.17. Four points A, B, C, and D lie on a straight line in the X–Y plane, such that AB = BC = CD, and the length of AB is 1 m. An ant at A wants to reach a sugar particle at D. But, there are insect repellents kept at points B and C. The ant would not go within 1 m of any insect repellent. The minimum distance (in m) the ant must traverse to reach the sugar particle is:
(a) $3\sqrt{2}$ (b) $1 + \pi$
(c) $\frac{4\pi}{3}$ (d) 5

Q.18. If $x \ge y$ and $y > 1$, then the value of the expression $\log_x\left(\frac{x}{y}\right) + \log_y\left(\frac{y}{x}\right)$ can never be:
(a) -1 (b) -0.5
(c) 0 (d) 1

Q.19. For a positive integer n, let p_n denote the product of the digits of n and S_n denote the sum of the digits of n. The number of integers between 10 and 1000 for which $p_n + s_n = n$ is:
(a) 81 (b) 16 (c) 18 (d) 9

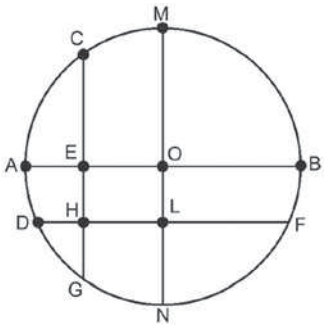
Q.20. Rectangular tiles each of the size 70 cm by 30 cm must be laid horizontally on a rectangular floor of size 110 cm by 130 cm such that the tiles do not overlap. A tile can be placed in any orientation as long as its edges are parallel to the edges of the floor. No tile should overshoot any edge of the floor. Find the maximum number of tiles that can be accommodated on the floor.
(a) 4 (b) 5 (c) 6 (d) 7

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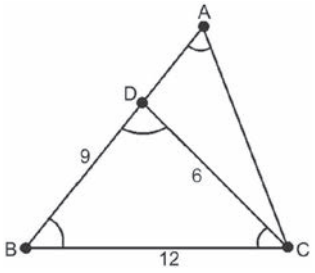
Q.21. In the X–Y plane, the area of the region bounded by the graph $|x+y|+|x-y|=4$ is:
(a) 8 (b) 12 (c) 16 (d) 20

Q.22. In the following figure, the diameter of the circle is 3 cm. AB and MN are two diameters such that MN is perpendicular to AB. In addition, CG is perpendicular to AB such that AE:EB = 1:2, and DF is perpendicular to MN such that NL:LM = 1:2. The length of DH (in cm) is:



(a) $2\sqrt{2}-1$ (b) $\dfrac{\left(2\sqrt{2}-1\right)}{2}$
(c) $\dfrac{\left(3\sqrt{2}-1\right)}{2}$ (d) $\dfrac{\left(2\sqrt{2}-1\right)}{2}$

Q.23. Consider the triangle ABC shown in the following figure where BC = 12 cm, DB = 9 cm, CD = 6 cm, and $\angle BCD = \angle BAC$.



What is the ratio of the perimeter of ΔADC to that of the ΔBDC?

Q.24. P, Q, S, and R are points on the circumference of a circle of radius r, such that PQR is an equilateral triangle and PS is a diameter of the circle. What is the perimeter of the quadrilateral PQSR?
(a) $2r\left(1+\sqrt{3}\right)$ (b) $2r\left(2+\sqrt{3}\right)$
(c) $r\left(1+\sqrt{5}\right)$ (d) $2r+\sqrt{3}$

Q.25. Let S be a set of positive integers such that every element n of S satisfies the conditions:
I. $1000 \le n < 1200$
II. Every digit in n is odd
Then, how many elements of S are divisible by 3?
(a) 9 (b) 10 (c) 11 (d) 12

Q.26. Let $x=\sqrt{4+\sqrt{4-\sqrt{4+\sqrt{\ldots\infty}}}}$, then x equals:
(a) 3 (b) $\left(\dfrac{\sqrt{3}-1}{2}\right)$
(c) $\left(\dfrac{\sqrt{3}+1}{2}\right)$ (d) $\sqrt{13}$

Q.27. Let $g(x)$ be a function such that $g(x+1)+g(x-1) = g(x)$ for every real x. Then, for what value of p is the relation $g(x+p) = g(x)$ necessarily true for every real x?
(a) 3 (b) 3 (c) 2 (d) 6

Q.28. A telecom service provider engages male and female operators for answering 1000 calls per day. A male operator can handle 40 calls per day, whereas a female operator can handle 50 calls per day. The male and the female operators get a fixed wage of ₹250 and ₹300 per day, respectively. In addition, a male operator gets ₹15 per call he answers and female operator gets ₹10 per call she answers. To minimize the total cost, how many male operators should the service provider employ assuming he has to employ more than 7 of the 12 female operators available for the job?
(a) 15 (b) 14 (c) 12 (d) 10

Q.29. Three Englishmen and three Frenchmen work for the same company. Each of them knows a secret not known to others. They need to exchange these secrets over person-to-person phone calls so that eventually each person knows all six secrets. None of the Frenchmen knows English, and only one Englishman knows French. What is the minimum number of phone calls needed for the above purpose?
(a) 5 (b) 10 (c) 9 (d) 15

Q.30. A rectangular floor is fully covered with square tiles of identical size. The tiles on the edges are white, and the tiles in the interior are red. The number of white tiles is same as the number of red tiles. A possible value of the number of tiles along one edge of the floor is:
(a) 10 (b) 12 (c) 14 (d) 16

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Answers

1. (a)	**2.** (c)	**3.** (b)	**4.** (d)	**5.** (a)	**6.** (b)	**7.** (b)	**8.** (d)	**9.** (d)	**10.** (c)
11. (d)	**12.** (a)	**13.** (b)	**14.** (c)	**15.** (b)	**16.** (a)	**17.** (b)	**18.** (d)	**19.** (d)	**20.** (c)
21. (c)	**22.** (b)	**23.** (a)	**24.** (a)	**25.** (a)	**26.** (c)	**27.** (d)	**28.** (d)	**29.** (c)	**30.** (b)

Hints and Solutions

1. $\frac{a^n + b^n + c^n + \ldots}{a + b + c + \ldots}$ is divisible for all the odd values of n.

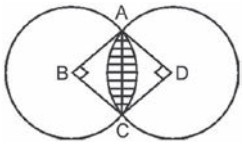
Hence, option (a) is the answer.

2.

A	B	C	D
–20	20		
90		–90	
–10			10
		–50	50
	–100	100	
	110		–110
Total + 60	30	–40	–50

Hence, option (c) is the answer.

3.



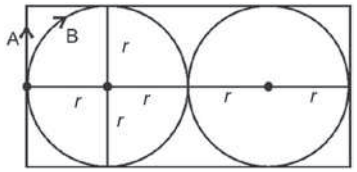
Shaded area = 2 × (area of sector ADC – area of ΔADC)

$= 2 \times \left(\frac{\pi}{4} \times 1^2 - \frac{1}{2} \times 1 \times 1 \right)$

$= \left(\frac{\pi}{2} - 1 \right)$

Hence, option (b) is the answer.

4.



A covers $2r + 2r + 4r + 4r = 12r$

B Covers $2\pi r + 2\pi r = 4\ \pi r$ distance

$\frac{4\pi r}{S_B} = \frac{12r}{S_A} \Rightarrow S_B = \frac{\pi}{3} S_A$

$\frac{S_B - S_A}{S_A} \times 100 = \frac{\pi - 3}{3} \times 100 = 4.72\%$

Hence, option (d) is the answer.

5. Let there be m boys and n girls.

${}^nC_2 = 45 = \Rightarrow n\,(n - 1) = 90 \Rightarrow n = 10$

${}^mC_2 = 190 \Rightarrow \frac{m(m-1)}{2} = 190 \Rightarrow m\,(m-1) = 380 \Rightarrow m = 20$

The number of games between one boy and one girl

$= {}^{10}C_1 \times {}^{20}C_1 = 10 \times 20 = 200$

Hence, option (a) is the answer.

6. Obviously, Ram and Shyam shall meet each other between C and B, sometime after 10 am. At 10 am, they are moving as shown below:

From now, the time taken to meet

$= \frac{2.5}{(10+5)} \times 60 \text{ min} = 10 \text{ min}$

So, they meet each other at 10:10 am.

Hence, option (b) is the answer.

7. It is obvious from the diagram that at 10:30 am, Shyam overtakes Ram.

Alternate: At 10:15 am, the situation is as shown:

Time taken for Shyam to overtake Ram $= \frac{2.5}{(10+5)} \times 60 \text{ min} = 15 \text{ min}$

Shyam overtakes Ram at 10:30 am.

Hence, option (b) is the answer.

8. $\frac{30^{65} - (30-1)^{65}}{30^{64} + (30-1)^{64}} = \frac{30^{65} - 30^{65}\left(1 - \frac{1}{30}\right)^{65}}{30^{64} - 30^{64}\left(1 - \frac{1}{30}\right)^{64}}$

$R = 30 \left\{ \frac{1 - \left(1 - \frac{1}{30}\right)^{65}}{1 + \left(1 - \frac{1}{30}\right)^{64}} \right\}$

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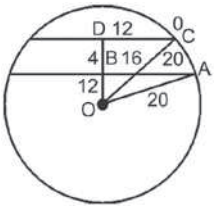
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$$R = 30\left\{\frac{1-(0.96)^{65}}{1+(0.96)^{64}}\right\}$$

The numerator is only slightly less then 1, and the denominator is only slightly more than 1. R is slightly less than 1.

Hence, option (d) is the answer.

9. **Case I** Chords on same side of the centre



$OB^2 = OA^2 - AB^2 = 20^2 - 16^2 = 144$

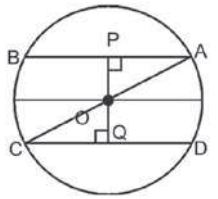
$OB = 12$

$OD^2 = 20^2 - 12^2 = 400 - 144 = 256$

$OD = 16$

$BD = 4$ cm

Case II Chords on opposite side of the centre



$AB = 32$ cm

$CD = 24$ cm

$OP = \sqrt{AO^2 - AP^2}$

$= \sqrt{(20)^2 - (16)^2}$

$OP = 12$ cm

$OQ = \sqrt{(OC)^2 - (CQ)^2}$

$= \sqrt{(20)^2 - (12)^2}$

$OQ = 16$ cm

Distance = PQ = 12 + 16 = 28 cm

Hence, option (d) is the answer.

10. $y^2 = x^2$

$2x^2 - 2kx + k^2 - 1 = 0$

$D = 0$

$\Rightarrow 4k^2 = 8k^2 - 6$

$\Rightarrow 4k^2 = 8$

$k^2 = 2 \Rightarrow 8$

$k^2 = 2 \Rightarrow k = \pm\sqrt{2}$ with $k = \pm\sqrt{2}$

The equation is $2x^2 + 2\sqrt{2}x + 1 = 0$

Root is $\frac{-b}{\sqrt{2a}} = +\frac{1}{\sqrt{2}}$

But with $k = -\sqrt{2}$, the equation is $-\frac{1}{\sqrt{2}}$

As this root is a –ve, it will reject $k = -\sqrt{2}$, the answer is $\Rightarrow k = +\sqrt{2}$ only.

Hence, option (c) is the answer.

11. If $p = 1! = 1$

Then, $p + 2 = 3$; when divided by 2!, the remainder will be 1.

If $p = 1! + 2 \times 2! = 5$

Then, $p + 2 = 7$; when divided by 3!, the remainder is still 1.

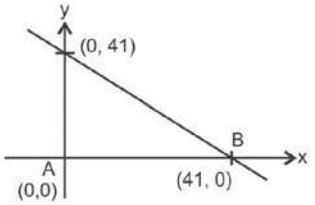
Hence, $p = 1! + (2 \times 2!) + (3 \times 3!) + \ldots + (10 \times 10!)$

When divided by 11!, the remainder is 1.

Thus, the remainder is 1.

Hence, option (d) is the answer.

12.



Equation of the line $= x + y = 41$. If the (x, y) coordinates of the points are integer, then their sum shall also be integers so that $x + y = k$ (l, a variable) as we have to exclude points lying on the boundary of the triangle; k can take all values from 1 to 40 only. $k = 0$ is also rejected because $k = 0$ will give the point A, which cannot be taken. Now, $x + y = k$ $(k = 1, 2, 3, \ldots, 40)$

With $k = 40$, $x + y = 40$ by taking integral solutions.We get points (1, 39), (2, 38), (3, 37),…, (39, 1), i.e., 39 points.

$x + y = 40$ will be satisfied by 39 points; similarly, $x + y = 39$ is satisfied by 38 points.

$x + y = 38$ is satisfied by 37 points.

$x + y = 3$ is satisfied by 2 points.

$x + y = 2$ is satisfied by 1 point.

$x + y = 1$ is satisfied by no points.

So, the total number of all such points is $39 + 38 + 37 + 36 + \ldots 3 + 2 + 1$

$= \frac{39 \times 40}{2} = 780$ points

Hence, option (a) is the answer.

13. Let $A = abc$, then $B = cba$

Given $B > A$, which implies $c > a$ (i)

$B - A = (100c + 10b + a) - (100a + 10b + 1)$

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$B - A = 100\,(c - a) + (a - c)$

$B - A = 99\,(c - a)$; $(B - A)$ is divisible by 7, and 99 is not divisible by 7 (no factor like 7 or 7^2). Therefore, $(c - a)$ must be divisible by 7 [i.e., $(c - a)$ must be 7, 7^2, etc.,] as c and a are single digits. $(c - a)$ must be 7 only, the possible values of (c, a) [with $c > a$] are (2) and (8,1).

With this we can write A as $A{:}abc = 1b8$ or $2b9$

As b can take values from 0 to 9, the smallest and the largest possible values are:

$A_{min} = 108$ and

$A_{max} = 299$

Only option (b) satisfies.

Hence, option (b) is the answer.

14. $a_1 = 1$, $a_{n+1} - 3a_n + 2 = 4n$

$a_{n+1} = 3a_n + 4n - 2$

When $n = 2$, then $a_2 = 3 + 4 - 2 = 5$

When $n = 3$, then $a_3 = 3 \times 5 + 4 \times 2 - 2 = 21$

From the options, we get an idea that a_n can be expressed in a combination of some power of 3 and some multiple of 100.

(a) $3^{99} - 200$ tells us that a_n could be: $3^{n-1} - 2 \times n$; but, it does not fit a_1 or a_2 or a_3.

(b) $3^{99} + 200$ tells us that a_n could be: $3^{n-1} + 2 \times n$; again not valid for a_1, a_2, etc.

(c) $3^{100} + 200$ tells $3^n + 2n$; again not valid.

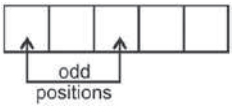
Hence, option (c) is the answer.

15.

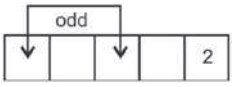


Odd positions can be counted in 2 ways.

(i) Counting from the LMD –till end:



We have 1, 2, 3, 4, and 5 to be filled in these blocks. Odd numbers (1, 3, 5) to be be filled at odd positions. Other places are to be filled by even numbers (2 or 4). Let us count, how many such numbers are there with 2 at the unit's digit.



Odd numbers can be filled ${}^3P_2 = 6$ ways

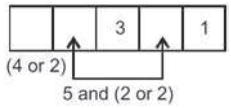
The remaining two places are to be filled by 2 numbers (one odd number left out of 1, 3, 5, and one even, i.e., 4) in = 2 ways

So, there are $6 \times 2 = 12$ number with 2 at the right most place. Similarly, there are 12 such numbers with 4 at the rightmost digits.

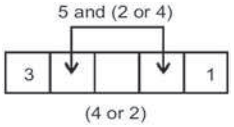
The sum of the rightmost digits in all such number = $12\,(2 + 4) = 72$

(ii) Now, counting from the RMD–till end:

Let us place 1 at the unit's place and check how many numbers are possible with (1, 3) at the odd positions.



Number of such cases = $2 \times 2 = 4$ ways



Here, again number of ways = $2 \times 2 = 4$ ways

So, there are $4 + 4 = 8$ numbers in which (1, 3) are at odd positions. Similarly, there are 8 numbers in which (1, 5) are at odd positions. So, in all, there are 16 numbers, where 1 is at the unit's place. Similarly, there are 16 numbers with 3 at the unit's place and 16 more with 5 at the unit's place.

Summing up all the odd unit's digits = $16\,(1 + 3 + 5) = 144$

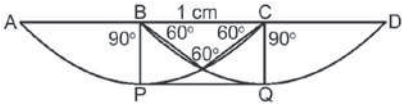
From (i) and (ii), we can now sum all (even or odd) numbers at unit place = $72 + 144 = 216$

Hence, option (b) is the answer.

16. $((30)^4)^{680} = (8100)^{680}$. Therefore, the rightmost non-zero digit is 1.

Hence, option (a) is the answer.

17.



Draw figure since it could not go within distance of 1 m, so it will go along APQD, which is the path of minimum distance.

$AP = \frac{90}{360} \times 2\pi \times 1 = \frac{\pi}{2}$

Also, $AP = QD = \frac{\pi}{2}$

So, the minimum distance $= AP + PQ + QD = \frac{\pi}{2} + 1 + \frac{\pi}{2}$
$= 1 + p$

Hence, option (b) is the answer.

18. $P = \log_x\left(\frac{x}{y}\right) + \log_y\left(\frac{y}{x}\right)$

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$= \log_x x - \log = x_y + \log_y y - \log_y x$

$= 2 - \log_x y - \log_y x$

Let, $t = \log_x y$

$\Rightarrow p = 2 - \frac{1}{t} - t = -\left[\sqrt{t} - \frac{1}{\sqrt{t}}\right]^2$

which can never be positive.

Out of the given option, it cannot assume a value of +1.

Hence, option (d) is the answer.

19. $10 < n < 1000$

Let n be two-digit number.

$n = 10a + b \Rightarrow p_n = ab, s_n = a + b$

Then, $ab + a + b = 10a + b$

$\Rightarrow ab = 9a \Rightarrow b = 9$

There are 9 such numbers 19, 29, 33, …, 99.

Then, let n be three-digit number.

$\Rightarrow n = 100a + 10b + c \Rightarrow P_n = abc, s_n = a + b + c$

Then, $ab + a + b + c = 100a + 10b + c$

$\Rightarrow abc = 99a + 9b$

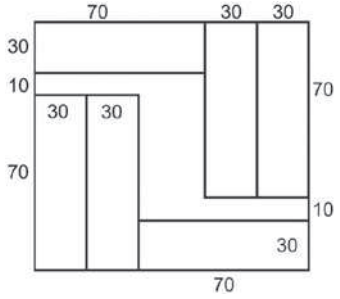
$\Rightarrow bc = 99 + 9\frac{b}{a}$

But, the minimum value for $bc = 81$ (i.e., when both b and c are 9).

RHS is more than 99. Therefore, no such number is possible.

Hence, option (d) is the answer.

20.



21. $|x + y| + |x - y| = 4$

Replace '$+x$' by '$-x$' and '$+y$' by 'y' everywhere in the curve; we again get the same equation.

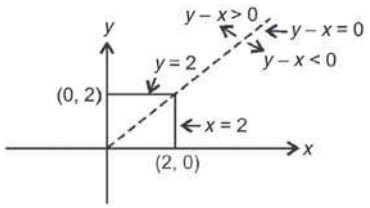
$\Rightarrow$ Curve is symmetric in the 4-quadrants of X–Y plane.

In I-quadrant $(x, y > 0)$

$|x + y| + |x - y| = 4$

$= \begin{cases} (x + y) + (y - x) = 4; y > x \\ (x + y) - (y - x) = 4; y < x \end{cases}$

Or, $\begin{cases} y = 2; y > x \\ x = 2; y < x \end{cases}$



We can now plot the graph:

Area in L-quadrant = $(2)^2 = 4$ unit^2

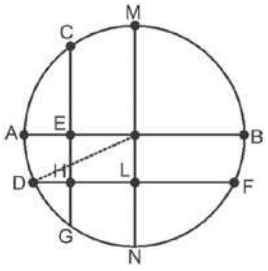
Total area of $|x + y| + |x - y| = 4$ is 4 x (area of I-quadrant)

$= 4 \times 4$

= 16 sq. unit

Hence, option (c) is the answer.

22.



AE = 1 cm

BE = 2 cm and NL = 1 cm

Ml = 2 cm

$HL = OE = \frac{1}{2}$

$DL = DH + HL$

$DL = DH + \frac{1}{2}$

OB = AO = radius = 1.5

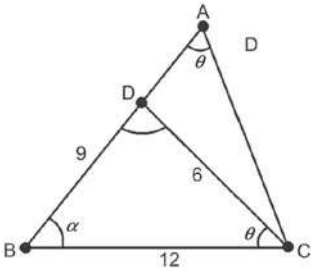
$DO^2 = OL^2 + Dl^2$

$\left(\frac{3}{2}\right)^2 = \left(\frac{1}{2}\right)^2 + \left(DH + \frac{1}{2}\right)^2$

$= \left(DH + \frac{1}{2}\right)^2 = 2 \Rightarrow DH\sqrt{2} - \frac{1}{2}$

Hence, option (b) is the answer.

23.



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Hence, $\angle ACB = \theta + [180 - (20 + a] = 180 - (\theta + \alpha)$

So, here we can say that the ΔBCD and ΔABC will be similar, ΔBCD ~ ΔBAC.

Hence, from the property of similarity,

$\frac{AB}{12} = \frac{12}{9}$ Hence, AB = 16

$\frac{AC}{6} = \frac{12}{9}$ Hence, AC = 8

Hence, AD = 7, AC = 8

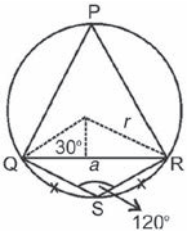
$S_{ADC} = 8 + 7 + 6 = 21$

$S_{BDC} = 27$

Therefore, $r = \frac{21}{27} = \frac{7}{9}$

Hence, option (a) is the answer.

24.



Here, $\cos 30° = \frac{a}{2r}$

$a = r\sqrt{3}$

From the diagram, $\cos 120° = \frac{x^2 + x^2 - a^2}{2x^2}$

$a^2 = 3x^2$

$x = r$

Therefore, the circumference will be $2r\left(1 + \sqrt{3}\right)$.

Hence, option (a) is the answer.

25. The 100th and 1000th position value will be only 1.

Now, the possibility of unit and tens digits are (1, 3), (1, 9), (3, 1), (5, 5), (7, 3), (7, 9), (9, 1), and (9, 7).

Hence, option (a) is the answer.

26. $x = \sqrt{4 + \sqrt{4 - x}} \Rightarrow x^2 = 4 + \sqrt{4 - x}$

$(X^2 - 4) = \sqrt{4 - x}$

Now put the values from the options. Only option (c) satisfies the condition.

27. $g(x + 1) + g(x - 1) = g(x)$

$g(x + 2) + g(x) = g(x + 1)$

Adding these two equations, we get

$g(x + 2) + g(x - 1) = 0$

$\Rightarrow g(x + 3) + g(x) = 0$

$\Rightarrow g(x + 4) + g(x + 1) = 0$

$\Rightarrow g(x + 5) + g(x + 2) = 0$

$\Rightarrow g(x + 6) + g(x + 3) = 0$

$\Rightarrow g(x + 6) - g(x) = 0$

Hence, option (d) is the answer.

28. Let us form both the equations first:

$40m + 50f = 1000$

$250m + 300f + 40 \times 15\text{ m} + 50 \times 10 \times f = A$

$850m + 8000f = A$

Where m and f are the number of males and females and A is the amount paid by the service provider.

Then, the possible values for f are 8, 9, 10, 11, and 12.

If $f = 8$, then $m = 15$

If $f = 9$, 10, and 11, then m will not be an integer, whereas if $f = 12$, then m will be 10.

By putting $f = 8$ and $m = 15$, and $A = 18800$. When $f = 12$ and $m = 10$, then $A = 18100$.

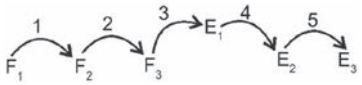
Thus, the number of males will be 10.

Hence, option (d) is the answer.

29. Frenchmen: F_1, F_2, F_3

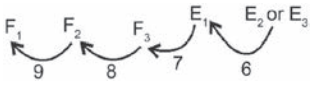
Englishmen: E_1, E_2, E_3

Let E_1 knows fresh 1 round of calls:



Persons	Secrets known after I-round
F_1	$F_1, F_2,$
$F_2,$	F_1, F_2, F_3
$F_3,$	F_1, F_2, F_3, F_4
$E_1,$	F_1, F_2, F_3, F_2
$E_2,$	$F_1, F_2, F_3, E_1, E_2, E_3$ all known
$E_3,$	$F_1, F_2, F_3, E_1, E_2, E_3$ all known

In the 6th call, E_1 knows all the secrets. Similarly, after the 9th call, everybody knows all the secrets.



Hence, option (c) is the answer.

30. Let the rectangle has m and n tiles along its length and breadth, respectively.

The number of white tiles

$W = 2m + 2(n - 2) = 2(m + n - 2)$

The number of red tiles = $R = mn - 2(m + n - 2)$

Given $W = R \Rightarrow 4(m + n - 2) = mn$

$\Rightarrow mn - 4m - 4n = -8$

$\Rightarrow (m - 4)(n - 4) = 8$

As m and n are integers so $(m - 4)$ and $(n - 4)$ are both integers. The possibilities are $(m - 4, n - 4) = (1, 8)$ or $(2, 4)$ giving (m, n) as (5, 12) or (6, 8); so the edges can have 5, 12, 6, or 8 tiles.

Hence, option (b) is the answer.

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CAT 2006

SECTION III

Each question carries 4 marks with a negative marking of 25%.

Q.51 If $a/b = 1/3$, $b/c = 2$, $c/d = 1/2$, $d/e = 3$, and $e/f = 1/4$, then what is the value of abc/def?
(a) 3/8 (b) 27/8 (c) 3/4 (d) 27/4
(e) 1/4

Q.52 If $x = -0.5$, then which of the following has the smallest value?
(a) $21/x$ (b) $1/x$ (c) $1/x2$ (d) $2X$
(e) $1/\sqrt{-x}$

Q.53 Consider a sequence where the nth term,
$t_n = n/(n+2)$, $n = 1, 2, \ldots$.
The value of $t_3 \times t_4 \times t_5 \times \ldots \times t_{53}$ equals:
(a) 2/495 (b) 2/477 (c) 12/55
(d) 1/1485 (e) 1/2970

Q.54 Which among 21/2, 31/3, 41/4, 61/6, and 121/12 is the largest?
(a) 21/2 (b) 31/3 (c) 41/4 (4) 61/6
(e) 21/12

Q.55 The length, breadth, and height of a room are in the ratio 3:2:1. If the breadth and height are halved and the length is doubled, then the total area of the four walls of the room will
(a) remain the same. (b) decrease by 13.64%.
(c) decrease by 15%. (d) decrease by 18.75%.
(e) decrease by 30%.

Q.56 A survey of a hundred people was conducted to find out whether they had read the recent issues of *Golmal*, a monthly magazine. A summary of the information regarding the readership in 3 months is given below:

Only September—18; September but not August —23; September and July—28
September—28; July—48; July and August—10
None of the three months—24
What is the number of surveyed people who have read exactly two consecutive issues (out of the three)?
(a) 7 (b) 9 (c) 12 (d) 14
(e) 17

Q.57 A semicircle is drawn with AB as its diameter. From C, a point on AB, a line perpendicular to AB is drawn meeting the circumference of the semicircle at D. Given that AC = 2 cm and CD = 6 cm, the area of the semicircle (in cm^2) will be:
(a) 32π (b) 50π (c) 40.5π (d) 81π
(e) Undeterminable

Direction for Questions 58 and 59: Answer the questions on the basis of the information given below.

An airline has a certain free luggage allowance and it charges for excess luggage at a fixed rate per kg. Two passengers, Raja and Praja have 60 kg of luggage between them, and are charged ₹1200 and ₹2400, respectively, for excess luggage. If the entire luggage belonged to one of them, then the excess luggage charge would have been ₹5400.

Q.58 What is the weight of Praja's luggage?
(a) 20 kg (b) 25 kg (c) 30 kg (d) 35 kg
(e) 40 kg

Q.59 What is the free luggage allowance?
(a) 10 kg (b) 5 kg (d) 20 kg (d) 25 kg
(e) 30 kg

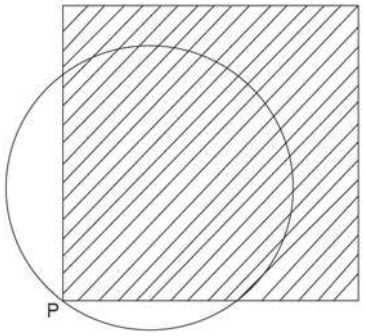
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Q.60 A group of 630 children is arranged in rows for a group photograph session. Each row contains three fewer children than the row in front of it. Which of the following number of rows is not possible?
(a) 3 (b) 4 (c) 5 (d) 6
(e) 7

Direction for Questions 61 to 62: Answer the questions on the basis of the information given below.

A punching machine is used to punch a circular hole of diameter two units from a square sheet of aluminium of width 2 units, as shown below. The hole is punched such that the circular hole touches one corner P of the square sheet and the diameter of the hole originating at P is in line with a diagonal of the square.



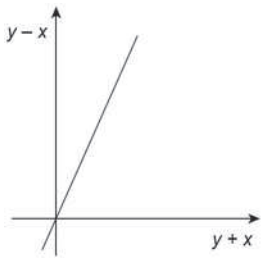
Q.61 The proportion of the sheet area that remains after punching is:
(a) $(\pi+2)/8$ (b) $(6-\pi)/8$
(c) $(4-\pi)/4$ (d) $(\pi-2)/4$
(e) $(14-3\pi)/6$

Q.62 Find the area of the part of the circle (round punch) falling outside the square sheet.
(a) $\pi/4$ (b) $(\pi-1)/2$
(c) $(\pi-1)/4$ (d) $(\pi-2)/2$
(e) $(\pi-2)/4$

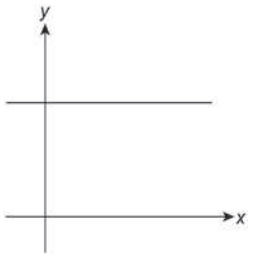
Q.63 What values of x satisfy $x^{2/3} + x^{1/3} - 2 \le 0$?
(a) $-8 \le x \le 1$ (b) $-1 \le x \le 8$
(c) $1 < x < 8$ (d) $1 \le x \le 8$
(e) $-8 \le x \le 8$

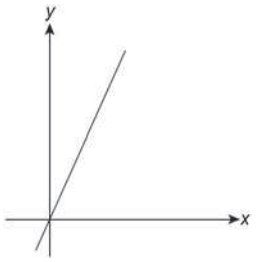
Q.64 Consider the set S = {1, 2, 3, . . ., 1000}. How many arithmetic progressions can be formed from the elements of S that start with 1 and end with 1000 and have at least 3 elements?
(a) 3 (b) 4 (c) 6 (d) 7
(e) 8

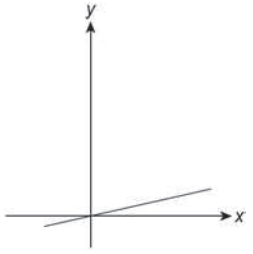
Q.65 The graph of $y - x$ against $y + x$ is as shown below. (All graphs in this question are drawn to scale and the same scale has been used on each axis.)

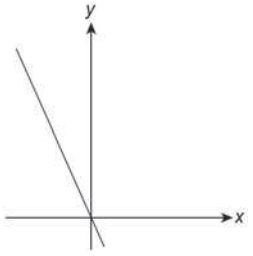


Which of the following shows the graph of y against x?

(a)


(b)


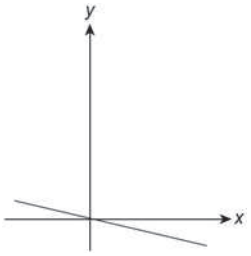
(c)


(d)


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(e)



Q.66 The sum of four consecutive two-digit odd numbers, when divided by 10, becomes a perfect square. Which of the following can possibly be one of these four numbers?
(a) 21 (b) 25 (c) 41 (d) 67
(e) 73

Q.67 The number of solutions of the equation $2x + y = 40$ where both x and y are positive integers and $x \le y$ is:
(a) 7 (b) 13 (c) 14 (d) 18
(e) 20

Q.68 The number of employees in Obelix Menhir Co. is a prime number and is less than 300. The ratio of the number of employees who are graduates and above, to that of employees who are not, can possibly be:
(a) 101:88 (b) 87:100
(c) 110:111 (d) 85:98
(e) 97:84

Q.69 There are 6 tasks and 6 persons. Task 1 cannot be assigned either to person 1 or to person 2, task 2 must be assigned to either person 3 or person 4. Every person is to be assigned one task. In how many ways can the assignment be done?
(a) 144 (b) 180 (c) 192 (d) 360
(e) 716

Q.70 If $\log_y x = (a \cdot \log_z y) = (b \cdot \log_x z) = ab$, then which of the following pairs of values for (a, b) is not possible?
(a) $(-2, 1/2)$ (b) $(1, 1)$
(c) $(0.4, 2.5)$ (d) $(\pi, 1/\pi)$
(e) $(2, 2)$

Q.71 What are the values of x and y that satisfy both the equations?

$2^{0.7x}.3^{-1.25y} = 8\sqrt{6}/27$

$4^{0.3x}.9^{0.2y} = 8.(81)^{1/5}$

(a) $x = 2, y = 5$ (b) $x = 2.5, y = 6$
(c) $x = 3, y = 5$ (d) $x = 3, y = 4$
(e) $x = 5, y = 2$

Q.72 Let $f(x) = \max(2x + 1, 3 - 4x)$, where x is any real number. Then, the minimum possible value of $f(x)$is:
(a) 1/3 (b) 1/2 (c) 2/3 (d) 4/3
(e) 5/3

Q.73 When you reverse the digits of the number 13, the number increases by 18. How many other two-digit numbers increase by 18 when their digits are reversed?
(a) 5 (b) 6 (c) 7 (d) 8
(e) 10

Q.74 An equilateral triangle BPC is drawn inside a square ABCD. What is the value of the angle APD in degrees?
(a) 75 (b) 90 (c) 120 (d) 135
(e) 150

Q.75 Arun, Barun, and Kiranmala start from the same place and travel in the same direction at speeds of 30, 40, and 60 kmph, respectively. Barun starts 2 h after Arun. If Barun and Kiranmala overtake Arun at the same instant, how many hours after Arun did Kiranmala start?
(a) 3 (b) 3.5 (c) 4 (d) 4.5
(e) 5

Answers

51. (a)	**52.** (b)	**53.** (a)	**54.** (b)	**55.** (e)	**56.** (b)	**57.** (b)	**58.** (d)	**59.** (*)	**60.** (d)
61. (b)	**62.** (d)	**63.** (a)	**64.** (d)	**65.** (d)	**66.** (c)	**67.** (b)	**68.** (e)	**69.** (a)	**70.** (e)
71. (e)	**72.** (e)	**73.** (b)	**74.** (e)	**75.** (c)					

* None of the options was correct.

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Hints and Solutions

51. $-a:b:c = 2:6:3$

$a:b:c:d:e:f = 6:18:9:18:6:24$

$abc/def = 3/8$

Hence, option (a) is the answer.

52. Go through the options.

Obviously options (a) and (d) are not possible (Since any power positive or negative raised to any positive number will be positive.)

Options (c) and (d) will be positive. Only option (b) is negative.

53. $-t_3 = 3/5$, $t_4 = 4/6$, t_5

$= 5/7, \ldots, t_{53} = 53/55$

$t_3 \times t_4 \times t_5 \times \ldots \times t_{53}$

$= 3/5 \times 4/6 \times 5/7 \times \ldots \times 53/55$

$= (3 \times 4)/(54 \times 55) = 2/495$

Hence, option (a) is the answer.

54. Take LCM of 2, 3, 4, 6, 12 = 12

$2^{1/2} = (2^6)^{1/12} = (64)^{1/12}$

$3^{1/3} = (3^4)^{1/12} = (81)^{1/12}$

$4^{1/4} = (4^3)^{1/12} = (64)^{1/12}$

$6^{1/6} = (6^2)^{1/12} = (36)^{1/12}$

$12^{1/12} = (12)^{1/12} = (12)^{1/12}$

Hence, option (b) is the answer.

55. Area of 4 walls = 2 H(L + B)

Assume that the length, breadth, and height of the room is 30 units, 20 units, and 10 units, respectively.

Initial total area of the four walls = 2×10 (50)

= 1000 sq. units.

After the changes, new length = 60 units,

new breadth = 10 units,

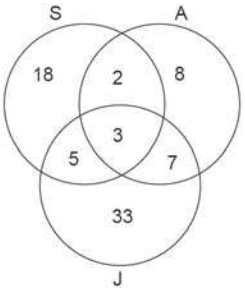
and height = 5 units

So, the new total area of the four walls = 700 sq. units

Thus, the area decreases by 30%.

Hence, option (e) is the answer.

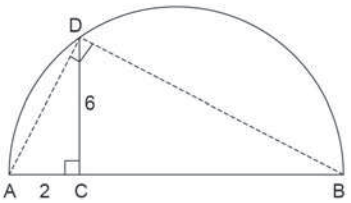
56.



As given in the above diagram, total persons reading in consecutive months, i.e., July and August and August and September is 2 + 7 = 9 people.

Hence, option (b) is the answer.

57.



$\angle\angle ADC = 90^o$ (Angle in a semicircle)

$CD^2 = AC \times CB$

$(6)^2 = 2 \times CB$; So, CB = 18

Hence, AB = AC + CB = 20

Area of semicircle = 50 π sq. units

Hence, option (b) is the answer.

58 and 59.

Let us assume that the total allowed luggage for Raja is A and excess luggage is E.

Hence, total luggage for Praja should be A + 2E.

E corresponds to ₹1200.

Hence, A must correspond to (5400 – 3600) = ₹1800

If E = 2N; A = 3N

So, total weight = 2(A) + 3E = 12N

N = 5

Hence, Praja's luggage weight = 7N = 35 kg

It should be noted here that none of the options given in Question number 59 was correct.

60. Let the number of students in front row be x.

So, the number of students in next rows be $x - 3$, $x - 6$, $x - 9$, and so on.

If n, i.e., number of rows be 3, then number of students $= x + (x - 3) + (x - 6) = 630$ or $3x = 639$

So, $x = 213$

Hence, we can conclude that total number of rows can be equal to 3.

Similarly, when $N = 4$, then

$x + (x - 3) + (x - 6) + (x - 9) = 630$ or $4x = 648$
So, number of rows = 4 is also possible

Similarly, when $n = 5$, then

$x + (x - 3) + (x - 6) + (x - 9) + (x - 12) = 630$ or $5x = 660$.
So, number of rows = 5 is also possible

Similarly, when $n = 6$, then

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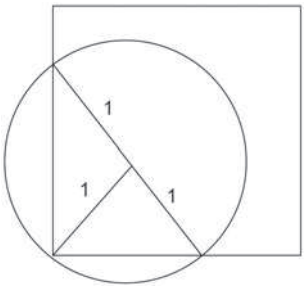
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$x + (x - 3) + (x - 6) + (x - 9) + (x - 12) + (x - 15) = 630$ or $6x = 675$

So, number of rows = 6 is not possible

Hence, option (d) is the answer.

61.

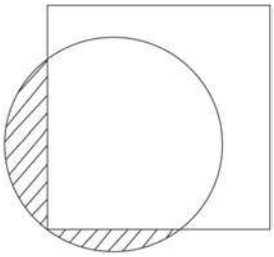


Remaining Area = $(6 - \pi)/2$

So, remaining proportion = $(6 - \pi)/8$

Hence, option (b) is the answer.

62.



Area of the required area = $\pi(1)^2 - [(\pi/2) + 1] = (\pi - 2)/2$

Hence, option (d) is the answer.

63. $x^{2/3} + x^{1/3} - 2 \le 0$

$(x^{1/3} - 1)(x^{1/3} + 2) \le 0$

$-8 \le x \le 1$

Hence, option (a) is the answer.

64. Let number of terms in progression be n, then using $t_n = a + (n - 1)d$

$1000 = 1 + (n - 1)d$ or $(n - 1)d = 999$

$= 3^3 \times 37$

Now, making a table of values of n and d gives us the following values of $(n - 1)$:

$(n - 1) = 3$ or 9 or 27 or 37 or 111 or 333 or 999

Thus, 7 progressions are possible.

Hence, option (d) is the answer.

65. From the graph of $(y - x)$ Vs. $(y + x)$, we can conclude the following:

Slope of line = $(y - x)/(y + x) = \pi$, $0 < \pi < 1$

Or, $(y - x)\pi = (y + x)$

Or, $y/x = (1 + \pi)/(1 - \pi)$ = slope of the line that shows the graph of y Vs. x

We know $0 < \pi < 1$

$(1 + \pi)/(1 - \pi)$ is –ve and is also greater than 1.

The slope of the graph y Vs. x must be negative and greater than 1. Accordingly, only option (d) satisfies.

Option (e) is negative slope but absolute value is less than 1.

Hence, option (d) is the answer.

66. Odd digits are 1, 3, 5, 7, and 9. If sum of four consecutive odd numbers is divisible by 10, then the digits should be 7, 9, 1, and 3.

Now, using the option (c), the numbers are 37, 39, 41, and 43.

Hence, option (c) is the answer.

67. $-2x + y = 40$, $x \le y$

$y = 40 - 2x$

Values of x and y that satisfy the equation can be calculated as:

x	y
1	38
2	36
..	
. ..	.
. ..	.
. ..	.
13 ..	14

There are 13 values.

Hence, option (b) is the answer.

68. The sum of numerator and denominator of the ratio should be a prime number.

Note that the number of employees is less then 300.

Consider the options.

(a) 101:88

101 + 88 = 189

189 × 2 > 300 ⇒ Number of employees who are graduates and above = 101 and the others are 88 in number, and 189 is not a prime number. Option (a) is eliminated.

(b) 87:100

87 + 100 = 187, 187 × 2 > 300

187 is not a prime number.

(c) 110:111

110 + 111 = 221, 221 × 2 > 300

221 is not a prime number.

(d) 85:98

85 + 98 = 183, 183 × 2 > 300

183 is not a prime number.

Hence, option (e) is the answer.

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69. Task 2 can be given to two persons only, viz. (3 and 4); so, number of ways = 2 ways.

First task can be done in 3 ways by 3 persons.

Finally, third task can be done by 4 persons.

So, 4 ways; similarly, for fourth, fifth, and sixth tasks, total number of ways of doing is 3, 2, and 1, respectively.

So, total number of ways = $2 \times 3 \times 4 \times 3 \times 2 \times 1$

$= 144$ ways

Hence, option (a) is the answer.

70. $\log_y x = a.\log_z y = a \times b$

$$\Rightarrow a \times b = \frac{\log_y x}{\log_z y} \times \left(\frac{\log_y x}{\log_z y}\right)$$

$$\left(\frac{\log_y x}{\log_z y}\right)^3 = (\log_y x)^3 = (ab)^3$$

$ab - a^3b^3 = 0$; hence, $ab = \pm 1$

Only option (e) does not satisfy.

71. Equation (ii) can be written as

$4^{0.3} \times 9^{0.2y} = 8 \times (81)^{1/5}$

$0.6x = 3$, so, $x = 5$ and $0.4y = 4/5$, hence, $y = 2$

If we put the values of x and y in first equation, these values satisfy the first equation also.

So, the answer is $x = 5$, $y = 2$

Hence, option (e) is the answer.

72. To find out minimum of $f(x)$ = maximum $(2x + 1, 3 - 4x)$, we should be taking the point of intersection of $(2x + 1)$ and $(3 - 4x)$. [Since one of these equations is increasing and other one is decreasing]

$$2x + 1 = 3 - 4x \text{ or } x = 1/3$$

$$2x + 1 = 3 - 4x = 5/3$$

Hence, option (e) is the answer.

73. We know that $xy - yx = 9(x - y)$, where xy is a two-digit number and $xy \geq yx$.

So, $x - y = 2$

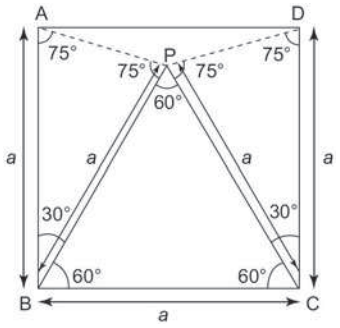
The possible pairs of (x, y) is (3, 1), (4, 2), (5, 3), (6, 4), (7, 5), (8, 6), and (9, 7)

But, we want the number other than 13, so there are 6 possible numbers, i.e., 24, 35, 46, 57, 68, 79.

So, total possible numbers are 6.

Hence, option (b) is the answer.

74.



$\angle PBC = \angle CPB = \angle BPC$ (equilateral triangle) and $PC = CD = a$

$\angle CPD = \angle PDC = 75°[(180° - 30°)/2]$

Similarly, $\angle BAP = \angle BPA = 75°$

Therefore, $\angle APD = 360° - (75° + 75° + 60°) = 150°$

Hence, option (e) is the answer.

75. Generate the equations to solve this question. Alternatively, options can also be used to solve this question.

Hence, option (c) is the answer.

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CAT

CAT 2007

There are three sections in this paper. Each section has 25 questions. Each section carries 100 marks. All questions carry 4 marks each. Each wrong answer will attract a penalty of 1 mark.

PART—A

Q.1 How many pairs of positive integers m, n satisfy:

$$\frac{1}{m}+\frac{4}{n}=\frac{1}{12}$$

Where n is an odd integer less than 60?

(a) 7 (b) 5 (c) 3 (d) 6
(e) 4

Q.2 A confused bank teller transposed the rupees and paise when he cashed a cheque for Shailaja, giving her rupees instead of paise and paise instead of rupees. After buying a toffee for 50 paise, Shailaja noticed that she was left with exactly three times as much as the amount on the cheque. Which of the following is a valid statement about the cheque amount?

(a) Over rupees 22 but less than rupees 23
(b) Over rupees 18 but less than rupees 19
(c) Over rupees 4 but less than rupees 5
(d) Over rupees 13 but less than rupees 14
(e) Over rupees 7 but less than rupees 8

Q.3 Suppose you have a currency, named Miso, in three denominations: 1 Miso, 10 Misos, and 50 Misos. In how many ways can you pay a bill of 107 Misos?

(a) 18 (b) 15 (c) 19 (d) 17
(e) 16

Direction for Questions 4 and 5: Answer the questions on the basis of the information given below:

Mr David manufactures and sells a single product at a fixed price in a niche market. The selling price of each unit is ₹30. On the other hand, the cost, in rupees, of producing x units is $240 + bx + cx^2$, where b and c are some constants. Mr David noticed that doubling the daily production from 20 to 40 units increases the daily production cost by $66\frac{2}{3}\%$. However, an increase in daily production from 40 to 60 units results in an increase of only 50% in the daily production cost. Assume that demand is unlimited and that Mr David can sell as much as he can produce. His objective is to maximize the profit.

Q.4 How many units should Mr David produce daily?

(a) 70 (b) 150 (c) 130 (d) 100
(e) Cannot be determined

Q.5 What is the maximum daily profit, in rupees, that Mr David can realize from his business?

(a) 840 (b) 760 (c) 620 (d) 920
(e) Cannot be determined

Direction for Questions 6 and 7: Answer the questions on the basis of the information given below:

Let $a_1 = p$ and $b_1 = q$, where p and q are positive quantities. Define

$a_n = pb_{n-1}$, $b_n = qb_{n-1}$, for even $n > 1$ and
$a_n = pa_{n-1}$, $b_n = qa_{n-1}$, for even $n > 1$

Q.6 Which of the following best describes $a_n + b_n$ for even n?

(a) $q^{\frac{1}{2}}(p+q)$ (b) $q^{\frac{1}{2}n}(p+q)^{\frac{1}{2}n}$
(c) $q(qp)^{\frac{1}{2}n-1}(p+q)^{\frac{1}{2}n}$ (d) $q(oq)^{\frac{1}{2}n-1}(p+q)$
(e) $qp^{\frac{1}{2}n-1}(p+q)$

Q.7 If $p=\frac{1}{3}$ and $q=\frac{2}{3}$, then what is the smallest odd n such that $a_n + b_n < 0.01$?

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(a) 11 (b) 9 (c) 15 (d) 7
(e) 13

Q.8 In a tournament, there are n teams T_1, T_2, ..., T_n, with $n > 5$. Each team consists of k players, $k > 3$. The following pairs of teams have one player in common:

T_1 and T_2, T_2 and T_3, ..., T_{n-1} and T_n, and T_n and T_1. No other pair of teams has any player in common. How many players are participating in the tournament, considering all the n teams together?

(a) $n(k-2)$ (b) $k(n-2)$
(c) $(n-1)(k-1)$ (d) $n(k-1)$
(e) $k(n-1)$

Q.9 Consider four-digit numbers for which the first two digits are equal and also equal. How many such numbers are perfect squares?

(a) 4 (b) 0 (c) 1 (d) 3
(e) 2

Direction for Questions 10 and 11: Answer the questions on the basis of the information given below:

Shabnam is considering three alternatives to invest her surplus cash for a week. She wishes to guarantee maximum returns on her investment. She has three options, each of which can be utilized fully or partially in conjunction with others.

Option A: Invest in a public sector bank. It promises a return of +0.10%.

Option B: Invest in mutual funds of ABC Ltd. A rise in the stock market will result in a return of +5%, whereas a fall will entail a return of –3%.

Option C: Invest in mutual funds of CBA Ltd. A rise in the stock market will result in a return of –2.5%, whereas a fall will entail of +2%.

Q.10 The maximum guaranteed return to Shabnam is:

(a) 0.20% (b) 0.15%
(c) 0.30% (d) 0.25%
(e) 0.10%

Q.11 What strategy will maximize the guaranteed return to Shabnam?

(a) 64% in option B and 36% in option C
(b) 1/3 in each of the three options
(c) 30% in option A, 32% in option B, and 38% in option C
(d) 100% in option A
(e) 36% in option B and 64% in option C

Direction for Questions 12 and 13: Answer the questions on the basis of the information given below:

Let S be the set of all pairs (i, j), where $1 \le i < j \le n$, and $n \ge 4$. Any two distinct members of S are called 'friends' if they have one constituent of the pairs in common and 'enemies' otherwise. For example, if $n = 4$, then S = {(1, 2), (1, 3), (1, 4), (2, 3), (2, 4), (3, 4)}. Here, (1, 2) and (1, 3) are friends, (1, 2) and (2, 3) are also friends, but (1, 4) and (2, 3) are enemies.

Q.12 For general n, how many enemies will each members of S have?

(a) $2n-7$ (b) $5-\frac{5k}{2}$ or $2k-3$
(c) $\frac{1}{2}(n^2-7n+14)$ (d) $n-3$
(e) $\frac{1}{2}(n^2-3n-2)$

Q.13 For general n, consider any two members of S that are friends. How many other members of S will be common friends of both these members?

(a) $(n-1)$ (b) $n-2$
(c) $\frac{1}{2}(n^2-7n+16)$ (d) $\frac{1}{2}(n^2-5n+8)$
(e) $2n-6$

Q.14 Two circles with centres P and Q cut each other at two distinct points A and B. The circles have the same radii and neither P nor Q falls within the intersection of the circles. What is the smallest range that includes all possible values of the angle AQP in degrees?

(a) Between 0 and 60 (b) Between 0 and 75
(c) Between 0 and 45 (d) Between 0 and 90
(e) Between 0 and 30

Q.15 The price of Darjeeling tea (in rupees per kilogram) is 100 + 0.10n, on the nth day of 2007 (n = 1, 2, …, 100), and then remains constant. On the other hand, the price of Ooty tea (in rupees per kilogram) is 89 + 0.15n, on the nth day of 2007 (n = 1, 2, …, 365). On which date in 2007 will the prices of these two varieties of tea be equal?

(a) May 20 (b) April 10
(c) June 30 (d) May 21
(d) April 11

Q.16 A quadratic function $f(x)$ attains a maximum of 3 at $x = 1$. The value of the function at $x = 0$ is 1. What is the value of $f(x) = 10$?

(a) –110 (b) –180 (c) –105 (d) –119
(e) –159

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Direction for Questions 17 and 18: Answer the questions on the basis of the information given below:

Cities A and B are in different time zones. A is located 3000 km east of B. The table below describes the schedules of an airline operating non-stop flights between A and B. All the times indicated are local and on the same day.

Departure		Arrival	
City	Time	City	Time
B	8:00 am	A	3:00 pm
A	4:00 pm	B	8:00 pm

Assume that planes cruise at the same speed in both directions. However, the effective speed is influenced by a steady wind blowing from east to west at 50 kmph.

Q.17 What is the time difference between A and B?
(a) 2 h and 30 min
(b) 1 h
(c) 1 h and 30 min
(d) 2 h
(e) Cannot be determined

Q.18 What is the plane's cruising speed in km per hour?
(a) 600 (b) 500 (c) 700 (d) 550
(e) Cannot be determined

Q.19 A function $f(x)$ satisfies $f(1) = 3600$, and $f(1) + f(2) + f(3) + \ldots\ldots + f(n) = n^2 f(n)$, for all positive integers $n > 1$. What is the value of $f(9)$?
(a) 200 (b) 100 (d) 120 (d) 80
(e) 240

Q.20 Consider the set S = {2, 3, 4, ..., $2n + 1$}, where n is a positive integer larger than 2007. Define X as the average of the odd integers in S and Y as the average of the even integers in S. What is the value of $X - Y$?
(a) $\frac{1}{2}n$ (b) $\frac{n+1}{2n}$ (c) 2008 (d) 0
(e) 1

Q.21 Ten years ago, the ages of the members of a joint family of eight people added up to 231 years. Three years later, one member died at the age of 60 years and a child was born during the same year. After another three years, one more member died, again at 60, and a child was born during the same year. The current average age of this eight-member joint family is nearest to:
(a) 21 years (b) 25 years
(c) 24 years (d) 23 years
(e) 22 years

Direction for Questions 22 and 25: Answer the questions on the basis of the information given below:

Each question is followed by two Statements A and B. Indicate your responses based on the following directives:

Mark (a): If the question can be answered using A alone but not using B alone.

Mark (b): If the question can be answered using B alone but not using A alone.

Mark (c): If the question can be answered using A and B together, but not using either A or B alone.

Mark (d): If the question cannot be answered even using A and B together.

Q.22 ABC Corporation is required to maintain at least 400 kilolitres of water at all times in its factory, in order to meet safety and regulatory requirements. ABC is considering the suitability of a spherical tank with uniform wall thickness for the purpose. The outer diameter of the tank is 10 meters. Is the tank capacity adequate to meet ABC's requirements?
A: The inner diameter of the tank is at least 8 meters.
B: The tank weighs 30,000 kg when empty, and is made of a material with density of 3 g/cc.

Q.23 Consider integers x, y, and z. What is the minimum possible value of $x^2 + y^2 + z^2$?
A: $x + y + z = 89$
B: Among x, y, and z, two are equal.

Q.24 The average weight of a class of 100 students is 45 kg. The class consists of two sections, I and II, each with 50 students. The average weight, W_I, of Section I is smaller than the average weight, W_{II}, of Section II. If the heaviest student, say Deepak, of Section II is moved to Section I, and the lightest student, say Poonam, of Section I is moved to Section II, then the average weights of the two sections are switched, i.e., the average weight of Section I becomes W_{II} and that of Section II becomes W_I. What is the weight of Poonam?
A: $W_{II} - W_I = 1.0$
B: Moving Deepak from Section II to I (without any move from I to II) makes the average weights of the two sections equal.

Q.25 Rahim plans to draw a square JKLM with a point O on the side JK but is not successful. Why is Rahim unable to draw the square?
A: The length of OM is twice that of OL.
B: The length of OM is 4 cm.

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Answers

1. (c)	**2.** (b)	**3.** (a)	**4.** (d)	**5.** (d)	**6.** (d)	**7.** (b)	**8.** (d)	**9.** (c)	**10.** (a)
11. (a)	**12.** (b)	**13.** (b)	**14.** (a)	**15.** (a)	**16.** (e)	**17.** (b)	**18.** (d)	**19.** (d)	**20.** (e)
21. (c)	**22.** (b)	**23.** (a)	**24.** (c)	**25.** (a)					

Hints and Solutions

1. $\frac{1}{m}=\frac{1}{13}-\frac{1}{n} \Rightarrow m=\frac{12n}{n-48}$

Since m is positive, n must be greater than 48. Possible odd values of n such that $48 < n < 60$ are 49, 51, 53, 55, 57 and 59, and 49, 51, and 57 give integral values of m. Hence, option (c) is the answer.

2. Let the original amount be ₹x and y paise. Then, interchanged amount = ₹y and x paise. From the given condition:

$3(100x + y) = 100y + x - 50$

$\Rightarrow 300x + 3y = 10y + x - 50$

$\Rightarrow 299x = 97y - 50 \Rightarrow y = \frac{299x+50}{97}$

Considering the options, only for $x = 18$, we get an integral value for y, i.e., $y = 56$.

Hence, option (b) is the answer.

3. Possible combinations

Case I Bill is paid with only two 50s misos

$2 \times 50 + 1 \times 10 + 7 \times 1 = 1$ way

Case II Bill is paid with only one 50 miso

$1\times50+5\times10+7\times1$
$1\times50+4\times10+17\times1$
. . .
. . .
. . .
$1\times50+0\times10+57\times1$
} i.e., 6 ways

Case III Bill is paid with only 50 misos

$10\times10\times7\times1$
$9\times10+17\times1$
. .
. .
. .
$1\times10+97\times1$
} i.e., 10 ways

Case IV Bill is paid with only 10 misos and only 50 misos

$1 \times 107 = 1$ way

Therefore, total number of ways = 1 + 6 + 10 + 1 = 18 ways

Hence, option (a) is the answer.

4.

Quantity produced	CP	SP	Profit
x	$240 + bx + cx^2$	$30x$	$30x - 240 - bx - cx^2$
20	$240 + 20b + 400c$	600	$600 - 240 - 20b - 400c$
40	$240 + 40b + 1600c$	1200	$1200 - 240 - 40b - 600c$
60	$240 + 60b + 3600c$	1800	$1800 - 240 - b + 3600c$

From the given conditions,

$$(240+40b+1600c)=\frac{5}{3}(240+20b+400c)\left(\frac{\pi}{2}-\theta\right) \quad \text{(i)}$$

Also,

$$240+60b+3600c=\frac{3}{2}(240+40b+1600b)$$
$$=\frac{5}{2}(240+20b+400c) \quad \text{(ii)}$$

From (i)

$$2800c + 20b - 480 = 0 \quad \text{(iii)}$$
$$200c + 20b - 720 = 0 \quad \text{(iv)}$$
$$2400c = 240$$
$$c = \frac{1}{10} \Rightarrow b = 10$$

Profit on x units $f(x) = 30x - 240 - 10x - \frac{x^2}{10}$

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i.e., $f(x) = -\frac{x^2}{10} + 20x - 240$

$f(x)$ is maximum at x if $f'(x) = 0$

i.e., $-\frac{2x}{10} + 20 = 0$

$2x = 200$ or $x = 100$

Hence, option (d) is the answer.

5. Maximum daily profit $= f(100)$

$= -1000 + 2000 - 240$

$=$ ₹760

Hence, option (d) is the answer.

Answers to Q.6 to 7:

$a_1 = p, b_1 = q$

$n = 2 : a_2 = pb_1 = pq ; b_2 = q \quad b_1 = q^2$

$n = 3 : a_3 = pq_2 = p^2q ; b_3 = qa_2 = pq^2$

$n = 4 : a_4 = pb_3 = p^2q^2 ; b_4 = qb_3 = pq^3$

$n = 5 : a_5 = pa_4 = p^3q^2 ; b_5 = qa_4 = p^2q^3$

$n = 6 : a_6 = pb_5 = p^3q^3 ; b_6 = qb_2 = p^2q^4$

$n = 7 : a_7 = pa_6 = p^4q^3 ; b_7 = qa_6 = p^3q^4$

6. $a_2 + b_2 = pq + q^2 = q(p + q)$

$a_4 + b_4 = p^2q^2 + pq^3 = pq^2(p + q) = q(pq)(p + q)$

$a_6 + b_6 = p^3q^3 + p^2q^4 = p^2q^3 (p + q) = q(pq)^2(p + q)$

Therefore, in general, $a_n + b_n = q(pq)^{\frac{1}{2}n-1}(p + q)$

Hence, option (d) is the answer.

7. $p = \frac{1}{3}, q = \frac{2}{3} \Rightarrow p + q = 1$ and $pq = \frac{2}{9}$

i.e., $a_1 + b_1 = 1$

Now, $a_3 + b_3 = pq(p + q) = pq$

$a_5 + b_5 = (pq)^2(p + q) = (pq)^2$

$a_7 + b_7 = (pq)^3(p + q) = (pq)^3$

In general, for odd 'n' and $p = \frac{1}{3}, q = \frac{2}{3}$

$$a_n + b_n = (pq)^{\frac{(n-1)}{2}} = \left(\frac{2}{9}\right)^{\left(\frac{n-1}{2}\right)}$$

Starting from the smallest option,

$$a_7 + b_7 = \left(\frac{2}{7}\right)^3 = 0.01$$

$$a_9 + b_9 = \left(\frac{2}{9}\right)^4 = 0.002 < 0.01$$

Hence, option (b) is the answer.

8. Total number of teams $= n$

Number of players in each team $= k$

Number of players common to two teams = number of teams $= n$

Thus, total number of players participating in the tournament

$$= nk - n = n(k - 1)$$

Hence, option (d) is the answer.

9. Let the four-digit number be

$1000a + 100a + 10b + b = 1100a + 11b$

This number will be a perfect square if

$1100a + 11b = k^2$; where k is an integer.

$\Rightarrow 11(100a + b) = k^2$

$\Rightarrow 100a + b = \frac{k^2}{11}$

Therefore, k should be a multiple of 11 such that $100a + b$ is a 3-digit number $k = 44, 55, 66, \ldots, 99$. Corresponding values of $100a + b$ will be 176, 275, 396, 539, 704, 891.

Therefore, $a, b < 9$ only '704' satisfies this.

Therefore, $1100a + 11b = 7700 + 44 = 7744 = 88^2$

Alternatively, we can solve this question by using elimination and fundamentals of number system.

Unit digit of a perfect square can be 0, 1, 4, 5, 6, 9.

Only these digits will get repeated at the last two places of the said square in this question.

So, the last two digits can only be 00, 11, 44, 55, 66, 99.

A perfect square cannot have 11, 55, 66, 99 as its last two digits.

(Reason: A perfect square, when divided by 4, should give either 0 or 1 as the remainder).

So, now we are left with = 00 and 44.

Perfect square lies in between 1000 and 9999 (both including). If the last two digits are 00, then the unit digit should be '0' in the number. Then, it is not possible to have other two digits (at the thousands and hundreds place) same.

So, the only option left, if possible = 44

Taking a cue from $12^2 = 144$, $88^2 = 7744$

This is the only number satisfying the condition given in the question.

Hence, option (c) is the answer.

Note:

1. The same question with a different format was asked in CAT 1999. The question was—Given $(aa)^2 = bbcc$, where a, b, and c are non-zero digits. Find the value of b.
2. $38^2 = 1444$ is another such interesting number. '444' occurring at the end of this square is the maximum number of times a non-zero digit is occurring at the end of any perfect square.

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10. Let the amount invested in option B and C be in the ratio 1:K.

So, depending on whether there is a rise or fall in the stock market, The amount earned will be $5-\frac{5k}{2}$ or $2k-3$

Therefore, guaranteed return

$= \text{minimum}\left\{\frac{5-5k}{2},\ 2k-3\right\}$

Therefore, the maximum guaranteed will be earned when

$\frac{5-5k}{2} = 2k-3$, i.e., $9k = 16$

i.e., $k = \frac{16}{9}$

Therefore, the maximum guaranteed return is when, the amounts invested are in the ratio 9:16, i.e., 36% and 64%. respectively. Now, the guaranteed return for this distribution is 0.2% (see Q.11). Since option A gives a return of 0.1%. which is lesser than this, no amount should be invested in option A.

Therefore, maximum guaranteed return = 0.20%

Hence, option (a) is the answer.

11. Let us assume that Shabnam has ₹100.

We calculate her guaranteed return in case of each of the given options:

Option (d): 0.1% of 100 = 0.1

Option (e): If there is a rise in the stock market, earning = 5% of 36 – 2.5% of 64

$\frac{-108+128}{100} = 0.2$

If there is a fall in the stock market, earning = –3% of 36 + 2% of + 64 = $\frac{-108+128}{100} = 0.2$

Therefore, guaranteed return = 0.2

Option (a): Rise in market gives earning = 5% if 64 + 2.5% of 36 = 2.4

Fall in market gives earning = –3% of 64 + 2% of 36 = –1.2, which is negative. Similarly, the guaranteed returns for options (b) and (c) are also negative.

Therefore, option (e) offers the highest guaranteed return.

12. A member (a, b) will have an enemy of the form (c, d), where c, d, a, and b are all distinct. Therefore, c and d can be chosen in ${}^{n-2}C_2$ ways.

So, number of enemies $= \frac{(n-2)(n-3)}{2}$

$= \frac{1}{2}(n^2-5n+6)$

Hence, option (b) is the answer.

13. Consider friends (a, b) and (a, c).

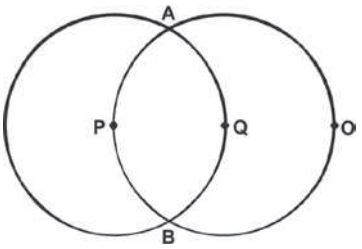
Their common friend can be either (b, c) or a member of the form (a, d) or (d, a), where d is different from a, b, c.

d can be chosen in (n – 3) ways.

So, number of common friends = (n – 3) + 1 = n – 2

Hence, option (b) is the answer.

14. P and Q do not lie within intersection of the circles. The extreme case can be that they lie on the circumference of the other circle as shown in the figure.



In this case, ΔAPQ will be an equilateral triangle.

Therefore, ∠AQP is more than 60º, then P and Q will lie within the intersection of the circles.

Hence, option (a) is the answer.

15. From 100 + 0.10n = 89 + 0.15n, n will be definitely greater than 100.

Price of Darjeeling tea on 100th day and onwards = 100 + 0.10 (100) = ₹110

Now, price of Ooty tea will be ₹110 When 89 + 0.15n = 110

$\Rightarrow 0.15n = 21 \Rightarrow n = 140$

Therefore, prices will be equal on 140th day, i.e., May 20.

Hence, option (a) is the answer.

16. Let $f(x) = ax^2 + bx + c$

f attains a maximum at $x = 1$.

$f(x) = 0$

$\Rightarrow \quad 2ax + b = 0$

$\Rightarrow \quad x = \frac{-b}{2a} = 1$

$\Rightarrow \quad -b = 2a$

Also, max $f(x) = 3$

$\Rightarrow \quad a + b + c = 3$

$\Rightarrow \quad a - 2a + c = 3$

$\Rightarrow \quad c - a = 3$

$f(0) = 1, \Rightarrow c = 1$ and $a = -2$

Therefore, $f(x) = -2x^2 + 4x + 1$

$f(10) = -2(100) + 4(10) + 1 = -200 + 41 = -159$

Hence, option (e) is the answer.

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Answers to Q.17 to 18:

Let the speed of the plane be x kmph.
Then, its speed from B to A = $(x + 50)$ kmph
Its speed from A to B = $(x - 50)$ kmph
The flight starts from City B (8:00 am) and arrives at city B (8:00 am) after taking a halt for 1 h in City A.
Therefore, total time taken = 11 h

i.e., $\frac{3000}{x+50} + \frac{3000}{x-50} = 11$

$\Rightarrow \frac{2x}{x^2 - 2500} = \frac{11}{3000}$

$\Rightarrow 11x^2 - 6000x - 27500 = 0$

Solving the quadratic equation, $x = 550$

Therefore, time taken from B to A = $\frac{3000}{500} = 6$ h

Flight reaches A when local time at B is 2:00 pm which is same as local time 3:00 pm at A. Therefore, required time difference = 1 h

19. $f(n) = \frac{f(1) + f(2) + \ldots + f(n-1)}{n^2 - 1}$

$f(1) = 3600$

$f(2) = \frac{f(1)}{3} = \frac{3600}{3} = 1200$

$f(3) = \frac{f(1) + f(2)}{8} = \frac{4800}{8} = 600$

$f(4) = \frac{f(1) + f(2) + f(3)}{15} = \frac{5400}{15} = 360$

$f(5) = \frac{f(1) + \ldots + f(4)}{24} = \frac{5760}{24} = 240$

$f(6) = \frac{f(1) + \ldots + f(5)}{35} = \frac{6000}{35} = \frac{1200}{7}$

$f(7) = \frac{f(1) + \ldots + f(6)}{48} = \frac{6000}{48} + \frac{1200}{7 \times 48} = 125 + \frac{25}{7}$

$f(8) = \frac{f(1) + \ldots + f(7)}{63} = \frac{6125}{63} + \frac{1225}{7 \times 63} = 100$

Similarly, $f(9) = 80$

Hence, option (d) is the answer.

20. $S = \{2, 3, 4, \ldots 2n + 1\}$

Total number of elements in S = $2n$

$$X = \frac{3 + 5 + \ldots + (2n+1)}{n}$$

$$Y = \frac{2 + 4 + 6 + \ldots + 2n}{n}$$

Therefore, $X - Y = \frac{1 + 1 + \ldots + 1(n \text{ times})}{n} = \frac{n}{n} = 1$

Hence, option (e) is the answer.

21. 10 years ago, total age of 8 members = 231

Three years later, sum of the ages = $231 + 8 \times 3 - 60$

$= 195$

Three more years later, sum of the ages

$= 195 + 8 \times 3 - 60 = 159$

Sum of current ages = $159 + 8 \times 4 = 191$

Now, current average age = $\frac{191}{8} \approx 24$ years

Hence, option (c) is the answer.

22. Using Statement A:

For minimum diameter, i.e., 8 m, the capacity of the tank

$= \frac{4}{3} \times \frac{22}{7} \times 4^3 \text{ m}^3 = 268.19 < 400$

For diameter greater than 8 m, for example, radius = 4.95 m, capacity of the tank = 508.25.71 > 400

Hence, Statement A alone cannot be used to answer the question.

Using Statement B:

Volume of material used = $\frac{\text{Mass}}{\text{Density}}$ = V (say), which is given

Therefore, Outer volume – Inner Volume = V

Therefore, we can find the inner volume and the question can be answered.

Hence, option (b) is the answer.

23. Using Statement A:

$$x + y + z = 89$$

For $x^2 + y^2 + z^2$ to be minimum, each of x, y, z must take integral value nearest to $\frac{89}{3}$.

Let $x = 30$, $y = 30$, and $z = 29$

Therefore, minimum value of

$$x^2 + y^2 + z^2 = (30)^2 + (30)^2 + (29)^2 = 2641$$

Thus, Statement A alone is sufficient to answer the question.

Hence, option (a) is the answer.

24. W_I = Average weight of Section I

W_{II} = Average weight of section II

$W_I + W_{II} = 90$ where $W_I < W_{II}$

Let weights of Deepak and Poonam be D and P kg, respectively.

$\frac{50 \times W_{II} - D + P}{50} = W_I$ and $\frac{50 \times W_{II} - D + P}{50} = W_I$

$\Rightarrow 50 (W_{II} - W_I) = D - P$

Using Statement A alone:

$$50 \times 1 = D - P$$

Thus, D and P can take various values.

So, Statement A alone

$$50 \times 1 = D - P \qquad \text{(i)}$$

Thus, D and P can take various values.

Hence, Statement A alone is not sufficient.

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Using Statement B alone:

$$\frac{50 \times W_I + D}{51} = \frac{50W_{II} - D}{49} \quad \text{(ii)}$$

Since values of W_I and W_{II} are not known, we cannot find the value of D.

Combining both the statements, values of W_I and W_{II} can be found, and therefore, value of D and P can be found, using (i) and (ii).

Hence, option (c) is the answer.

25. The farthest point from point M which is on the square is the diagonally opposite point.

Length OM $\leq \sqrt{2} \times$ side of the square $\Rightarrow$ Length OM $\leq$ $\sqrt{2} \times$ length of LK (i)

Length of OL $\geq$ length of LK (ii)

From (i) and (ii), we can conclude that such a point cannot be drawn.

Hence, option (a) is the answer.

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CAT

CAT 2008

SECTION I

This section contains 25 questions.

Q.1 The integers 1, 2,..., 40 are written on a blackboard. The following operation is then repeated 39 times: In each repetition, any two numbers, say a and b, currently on the blackboard are erased, and a new number $a + b - 1$ is written. What will be the number left on the board at the end?
(a) 820 (b) 821 (c) 781 (d) 819
(e) 780

Q.2 What are the last two digits of 7^{2008}?
(a) 21 (b) 61 (c) 01 (d) 41
(e) 81

Q.3 If the roots of the equation $x^3 - ax^2 + bx - c = 0$ are three consecutive integers, then what is the smallest possible value of b?
(a) $-\frac{1}{\sqrt{3}}$ (b) -1 (c) 0 (d) 1
(e) $\frac{1}{\sqrt{3}}$

Q.4 A shop stores x kg of rice. The first customer buys half this amount plus half a kg of rice. The second customer buys half the remaining amount plus half a kg of rice. Then, the third customer also buys half the remaining amount plus half a kg of rice. Thereafter, no rice is left in the shop. Which of the following best describes the value of x?
(a) $2 \le x \le 6$ (b) $5 \le x \le 8$
(c) $9 \le x \le 12$ (d) $11 \le x \le 14$
(e) $13 \le x \le 18$

Direction for Questions 5 and 6: Answer the questions on the basis of the information given below.

Let $f(x) = ax^2 + bx + c$, where a, b, and c are certain constants and $a \neq 0$. It is known that $f(5) = -3f(2)$ and that 3 is a root of $f(x) = 0$.

Q.5 What is the other root of $f(x) = 0$?
(a) -7 (b) -4 (c) 2 (d) 6
(e) Cannot be determined

Q.6 What is the value of $a + b + c$?
(a) 9 (b) 14 (c) 13 (d) 37
(e) Cannot be determined

Q.7 The number of common terms in the two sequences 17, 21, 25, ..., 417 and 16, 21, 26, ..., 466 is:
(a) 78 (b) 19 (c) 20 (d) 77
(e) 22

Q.8 How many integers, greater than 999 but not greater than 4000, can be formed with the digits 0, 1, 2, 3, and 4, if repetition of digits is allowed?
(a) 499 (b) 500 (c) 375 (d) 376
(e) 501

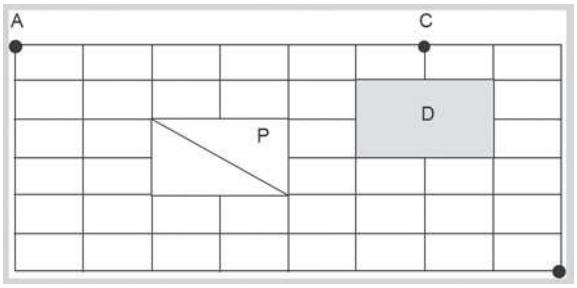
Direction for Questions 9 and 10: Answer the questions on the basis of the information given below.

The figure below shows the plan of a town. The streets are at right angles to each other. A rectangular park (P) is situated

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inside the town with a diagonal road running through it. There is also a prohibited region (D) in the town.



Q.9 Neelam rides her bicycle from her house at A to her office at B, taking the shortest path. Then, the number of possible shortest paths that she can choose is:
(a) 60 (b) 75 (c) 45 (d) 90
(e) 72

Q.10 Neelam rides her bicycle from her house at A to her club at C, via B taking the shortest path. Then, the number of possible shortest paths that she can choose is:
(a) 1170 (b) 630 (c) 792 (d) 1200
(e) 936

Q.11 Let $f(x)$ be a function satisfying $f(x)f(y) = f(xy)$ for all real x, y. If $f(2) = 4$, then what is the value of $f(1/2)$?
(a) 0 (b) $\frac{1}{4}$ (c) $\frac{1}{2}$ (d) 1
(e) Cannot be determined

Q.12 Suppose, the seed of any positive integer is defined as follows:

seed(n) = n, if $n < 10$
= seed(s(n)), otherwise

where $s(n)$ indicates the sum of digits of n. For example, seed (7) = 7, seed (248) = seed (2 + 4 + 8) = seed (14) = (1 + 4) = seed = 5, etc. How many positive integers n, such that $n < 500$, will have seed(n) = 9?
(a) 39 (b) 72 (c) 81 (d) 108
(e) 55

Q.13 In a triangle ABC, the lengths of the sides AB and AC equal 17.5 cm and 9 cm, respectively. Let D be a point on the line segment BC such that AD is perpendicular to BC. If AD = 3 cm, then what is the radius (in cm) of the circle circumscribing the triangle ABC?
(a) 17.05 (b) 27.85 (c) 22.45 (d) 32.25
(e) 26.25

Q.14 Consider obtuse–angled triangles with sides 8 cm, 15 cm, and x cm. If x is an integer, then how many such triangles exist?
(a) 5 (b) 21 (c) 10 (d) 15
(e) 14

Q.15 Consider a square ABCD with mid–points E, F, G, H of AB, BC, CD, and DA respectively. Let L denote the line passing through F and H. Consider points P and Q, on L and inside ABCD, such that the angles APD and BQC both equal 120°. What is the ratio of the area of ABQCDP to the remaining area inside ABCD?
(a) $\frac{4\sqrt{2}}{3}$ (b) $2+\sqrt{3}$
(c) $\frac{10-3\sqrt{3}}{9}$ (d) $1+\frac{1}{\sqrt{3}}$
(e) $2\sqrt{3}-1$

Q.16 What is the number of distinct terms in the expansion of $(a + b + c)^{20}$?
(a) 231 (b) 253 (c) 242 (d) 210
(e) 228

Direction for Questions 17 and 18: Answer the questions on the basis of the information given below.

Five horses, Red, White, Grey, Black, and Spotted participated in a race. As per the rules of the race, the persons betting on the winning horse get four times the bet amount and those betting on the horse that came in second get thrice the bet amount. Moreover, the bet amount is returned to those betting on the horse that came in third, and the rest lose the bet amount. Raju bets ₹3000, ₹2000, and ₹1000 on Red, White, and Black horses, respectively and ends up with no profit and no loss.

Q.17 Which of the following cannot be true?
(a) At least two horses finished before Spotted.
(b) Red finished last.
(c) There were three horses between Black and Spotted.
(d) There were three horses between White and Red.
(e) Grey came in second.

Q.18 Suppose, in addition, it is known that Grey came in fourth. Then, which of the following cannot be true?
(a) Spotted came in first.
(b) Red finished last.
(c) White came in second.
(d) Black came in second.
(e) There was one horse between Black and White.

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Direction for Questions 19 and 20: Answer the questions on the basis of the information given below.

Mark (a): If question can be answered from A alone but not from B alone.

Mark (b): If question can be answered from B alone but not from A alone.

Mark (c): If question can be answered from either of A or B alone.

Mark (d): If question can be answered from A and B together but not from any of them alone.

Mark (e): If question cannot be answered even from A and B together.

In a single elimination tournament, any player is eliminated with a single loss. The tournament is played in multiple rounds subject to the following rules:

(a) If the number of players, say n, in any round is even, then the players are grouped in to $n/2$ pairs. The players in each pair play a match against each other and the winner moves on to the next round.

(b) If the number of players, say it, in any round is odd, then one of them is given a bye, that is, he automatically moves on to the next round. The remaining $(n - 1)$ players are grouped into $(n - 1)/2$ pairs. The players in each pair play a match against each other and the winner moves on to the next round. No player gets more than one bye in the entire tournament.

Thus, if n is even, then $n/2$ players move on to the next round, whereas if n is odd, then $(n + 1)/2$ players move on to the next round. The process is continued till the final round, which obviously is played between two players. The winner in the final round is the champion of the tournament.

Q.19 Question: What is the number of matches played by the champion?

A: The entry list for the tournament consists of 83 players.

B: The champion received one bye.

Q.20 Question: If the number of players, say n, in the first round was between 65 and 128, then what is the exact value of n?

A: Exactly one player received a bye in the entire tournament.

B: One player received a bye while moving on to the fourth round from the third round.

Q.21 Two circles, both of radii 1 cm, intersect such that the circumference of each one passes through the centre of the other. What is the area (in sq cm) of the intersecting region?

(a) $\frac{\pi}{3} - \frac{\sqrt{3}}{4}$ (b) $\frac{2\pi}{3} + \frac{\sqrt{3}}{2}$

(c) $\frac{4\pi}{3} - \frac{\sqrt{3}}{2}$ (d) $\frac{4\pi}{3} + \frac{\sqrt{3}}{2}$

(e) $\frac{2\pi}{3} - \frac{\sqrt{3}}{2}$

Q.22 Rahim plans to drive from City A to Station C, at the speed of 70 kmph, to catch a train arriving there from B. He must reach C at least 15 min before the arrival of the train. The train leaves B, located 500 km south of A, at 8:00 am and travels at a speed of 50 kmph. It is known that C is located between west and northwest of B, with BC at 60° to AB. Also, C is located between south and southwest of A with AC at 30° to AB. The latest time by which Rahim must leave A and still catch the train is closest to:

(a) 6:15 am (b) 6:30 am

(c) 6:45 am (d) 7:00 am

(e) 7:15 am

Q.23 Three consecutive positive integers are raised to the first, second, and third powers, respectively, and then added. The sum so obtained is a perfect square whose square root equals the total of the three original integers. Which of the following best describes the minimum, say m, of these three integers?

(a) $1 \le m \le 3$ (b) $4 \le m \le 6$

(c) $7 \le m \le 9$ (d) $10 \le m \le 12$

(e) $13 \le m \le 15$

Q.24 Find the

$$\sqrt{1+\frac{1}{1^2}+\frac{1}{2^2}}+\sqrt{1+\frac{1}{2^2}+\frac{1}{3^2}}+\ldots+\sqrt{1+\frac{1}{2007^2}+\frac{1}{2008^2}}$$

(a) $2008 - \frac{1}{2008}$ (b) $2007 - \frac{1}{2007}$

(c) $2007 - \frac{1}{2007}$ (d) $2008 - \frac{1}{2007}$

(e) $2008 - \frac{1}{2009}$

Q.25 Consider a right circular cone of base radius 4 cm and height 10 cm. A cylinder is to be placed inside the cone with one of the flat surfaces resting on the base of the cone. Find the largest possible total surface area of the cylinder (in sq. cm).

(a) $100\frac{\pi}{3}$ (b) $80\frac{\pi}{3}$

(c) $120\frac{\pi}{7}$ (d) $130\frac{\pi}{9}$

(e) $110\frac{\pi}{7}$

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Answers

1. (c)	**2.** (c)	**3.** (b)	**4.** (b)	**5.** (b)	**6.** (e)	**7.** (c)	**8.** (d)	**9.** (d)	**10.** (a)
11. (b)	**12.** (e)	**13.** (e)	**14.** (c)	**15.** (e)	**16.** (a)	**17.** (d)	**18.** (c)	**19.** (d)	**20.** (d)
21. (e)	**22.** (b)	**23.** (a)	**24.** (a)	**25.** (a)					

Hints and Solutions

1. Let S = 1 + 2 + ... + 40. So, 1 + 2 + a + b + ... + 40 = S

After 1st step:

1 + 2 + ... + (a + b − 1) + ... + 40 = S − 1

So, after each step, the total reduces by 1. Continuing this way after 39 steps, numbers left $S - 39 = \frac{40 \times 41}{2} - 39$ = 781

Hence, option (c) is the answer.

2. $7^{2008} = (7^4)502 = (2401)^{502}$

To find tens digit, we need to take care only of last two digits of this number.

Last two digits of $(2401)^{502}$ = Last two digits of $(01)^{502}$

Thus, 7^{2008} end with 01.

Hence, option (c) is the answer.

3. Assume that the consecutive roots are (x − 1), x, and (x + 1)

Now, $b = x(x-1) + x(x+1) + (x-1)(x+1) = 3x^2 - 1$

Now, as $3x^2 \geq 0$ minimum value of b = $3 \times 0^2 - 1 = -1$

Hence, option (b) is the answer.

4. Let us make a table from the given information:

Quantity of rice in the shop	Quantity of rice bought	Quantity left
x	$\left(\frac{x}{2}+\frac{1}{2}\right)$	$\frac{x}{2}-\frac{1}{2}$
$\frac{x}{2}-\frac{1}{2}$	$\left(\frac{x}{2}-\frac{1}{2}\right)-\frac{1}{2}$	$\frac{x}{4}-\frac{3}{4}$
$\frac{x}{4}-\frac{3}{4}$	$\left(\frac{x}{8}-\frac{3}{8}\right)+\frac{1}{2}$	$\frac{x}{8}-\frac{7}{8}$

$\frac{x}{8}-\frac{7}{8}=0 \Rightarrow x=7$

Now, $5 < x < 8$

Hence, option (b) is the answer.

5. Given $f(x) = ax^2 + bx + c$ ($a \neq 0$)

Since 3 is a root of $f(x)$, so, $9a + 3b + c = 0$ (i)

Also, $f(5) = 3f(2)$

So, $25a + 5b + c = -3(4a + 2b + c) = -12a - 6b - 3c$

$\Rightarrow 37a + 11b + 4c = 0$ (ii)

From (i) and (ii), $a - b = 0 \Rightarrow a = b$

Thus, we get $f(x) = ax^2 + ax + c$

Dividing $f(x)$ by $x - 3$, we get $c = -12a$

$\Rightarrow f(x) = ax^2 + ax - 12a \Rightarrow f(x) = 0 \Rightarrow -4$ is second root

Hence, option (b) is the answer.

6. $a + b + c = a + a - 12a = -10a$

Value of a is not given, so cannot be determined.

Hence, option (e) is the answer.

7. In 1st series, terms are in the form of $4a + 1$ ($4 < a < 104$).

In 2nd series, terms are in the form of $5b + 1$ ($3 < b < 93$).

In order to have same terms, we should get $4a = 5b$.

This happens only 20 times.

Thus, we get 21, 41, 61, …, i.e., 20 common terms.

Hence, option (c) is the answer.

8. Numbers are of 4 digits, and less than or equal to 4000.

For numbers less than 4000, following is the box diagram:

$\underline{3} \times \underline{5} \times \underline{5} \times \underline{5} = 375$

Including 4000, total number of numbers = 376

Hence, option (d) is the answer.

9. Neelam has to take path XY

A to X = ${}^4C_2 = 6$ possibilities

Y to B = ${}^6C_2 = 15$ possibilities

So, total possibilities = 6 × 15 = 90 possibilities

Hence, option (d) is the answer.

10. From A to B = 90 paths

From B to C via N = 6 (and not via M)

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From B to C via M = 7 paths

In all, 90 V (6 + 7) = 1170 paths

Hence, option (a) is the answer.

11. We have $f(x)\,f(y) = f(xy)$

So, $f(1)f(1) = f(1 \times 1) = f(1)$

$\Rightarrow f(1)^2 = f(1) \Rightarrow f(1)^2 - f(1) = 0 \Rightarrow f(1)[f(1) - 1] = 0$

Hence, $f(1) = 0$ or $f(1) = 1$

If $f(1) = 0$, then $f(x) = 0$ for any x since $x = x \times 1$

$\Rightarrow f(1) = 1$

Now, $f(2) = 4$

$$\text{So, } 1 = f(1) = f\left(\frac{1}{2} \times 2\right) = f\left(\frac{1}{2}\right) f(2) = f\left(\frac{1}{2}\right) \times 4$$

$$\Rightarrow \quad f\left(\frac{1}{2}\right) = \frac{1}{4}$$

Hence, option (b) is the answer.

12. Our answer would be the number of integers between 1 and 500, which are divisible by 9.

The smallest is 9 and the largest is 495.

In the first 499 natural numbers, we have 495 as the last multiple of 9, and this is 55th multiple of 9.

Hence, option (e) is the answer.

13. Let the sides be a = AB, b = AC, and c = BC

$$A(\Delta ABC) = \frac{abc}{4R} \Rightarrow A\,(\Delta ABC) = \frac{1}{2}$$

$$\text{So,} \quad \frac{abc}{4R} = \frac{1}{2} \times c \times 3$$

$$R = \frac{a \times b}{2 \times 3} = \frac{17.5 \times 9}{2 \times 3} = 26.25 \text{ cm}^2$$

Hence, option (e) is the answer.

14. Let the three sides be $a = 8$, $b = 15$, and $c = x$

Case I c is the longest side.

We know that $15 < c < 23$. $(8 + 15 = 23)$. For $c = 17$, we get a right-angled triangle. So, $c > 17$ for an obtuse angled triangle.

Therefore, c is 18, 19, 20, 21, 22.

Case II b is the longest side. In this case, $7 < c < 15$ $(15 - 8 = 7)$

For a right-angled triangle, $c < \sqrt{15^2 - 8^2}$

$\Rightarrow c < \sqrt{161} \cong 12.7$

So, c can be 8, 9, 10, 11, or 12.

Therefore, in total, there are 10 possibilities.

Hence, option (c) is the answer.

16. $(a + b + c)^{20}$ has each term of the form $a^x b^y c^z$ where $x + y + z = 20$. This has ${}^{20+3-1}C_{3-1} = 231$ solutions.

(This question is same as distributing 20 apples among three children where any child can get any number of apples. Formula for this is ${}^{-n+r-1}C_{r-1}$).

Hence, option (a) is the answer.

17. There are 3 possibilities.

Horse	Rank		
	Case 1	Case 2	Case 3
Red	4/5	4/5	3
White	2	3	4/5
Black	5/4	1	2
Grey	1/3	2/4/5	1/4/5
Spotted	3/1	2/4/5	1/4/5

Hence, option (d) is the answer.

19. Using Statement A alone, if there are 83 players, the number of players in each subsequent round will be as follows:

Round	Players	
1	83	
2	42	
3	21	
4	11	
5	6	
6	3	
7	2	→ Final

The champion plays in the final and so can play either 6 or 7 matches (depending on whether he gets a bye or not). Hence, no unique answer.

Using Statement B alone, we can draw no conclusion without knowing the number of rounds. With both the statements together, we can say that the champion plays 6 matches.

Hence, option (d) is the answer.

20. Using Statement A alone, we get various possible cases. We can see that 7 rounds have to be played. The bye could have occurred in any of the first 6. Let us look at the table below:

Round	Bye in					
	Round 1	Round 2	Round 3	Round 4	Round 5	Round 6
1	127	126	124	120	112	96
2	64	63	62	60	56	47
3	32	32	31	30	28	24
4	16	16	16	15	14	12
5	8	8	8	8	7	6
6	4	4	4	4	4	3
7	2	2	2	2	2	2

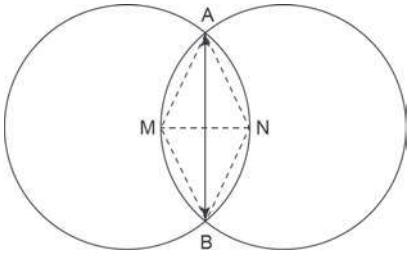
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Using Statement B alone, we cannot say anything as other players too might have received byes. Using both the statements together, we can say that the number of players is 124.

Hence, option (d) is the answer.

21. Area required region = 2 × [(Area of sector – AMB) – A (ΔANB)]



$$= 2 \times \frac{120^\circ}{360^\circ} \times \pi r^2 - 2 \times \text{Area}(\Delta ANB)$$

$$= \frac{2}{3}\pi - 2 \times \frac{\sqrt{3}}{4} \times 1 = \frac{2}{3}\pi - \frac{\sqrt{3}}{2} \text{ sq. cm}$$

Hence, option (e) is the answer.

23. $3^1 + 4^2 + 5^3 = 144$

$\sqrt{144} = 3 + 4 + 5$. So, the numbers are 3, 4, and 5.

Hence, option (a) is the answer.

Alternatively, assume the numbers to be n, $n + 1$, and $n + 2$.

So, $n^1 + (n + 1)^2 + (n + 2)^3 = (n + n + 1 + n + 2)^2 = (3n + 3)^2$

Expanding it, we get $n(n^2 - 2n - 3) = 0$

Since $n \neq 0$, so, $(n^2 - 2n - 3) = 0 \Rightarrow (n - 3)(n + 1) = 0$

Hence, $n = 3$

24. Go through the method of induction,

$$\sqrt{1 + \frac{1}{1^2} + \frac{1}{2^2}} + \sqrt{\frac{9}{4}} = \frac{3}{2} = 2 - \frac{1}{2}$$

$$\sqrt{1 + \frac{1}{1^2} + \frac{1}{2^2}} + \sqrt{1 + \frac{1}{2^2} + \frac{1}{3^3}}$$

$$= \frac{3}{2} + \frac{7}{6} = \frac{8}{3} = 3 - \frac{1}{3}$$

Similarly,

$$\sqrt{1 = \frac{1}{2^2} + \frac{1}{2^2}} + \ldots + \sqrt{1 + \frac{1}{200^7} + \frac{1}{2008^2}}$$

$$= 2008 - \frac{1}{2008}$$

Hence, option (a) is the answer.

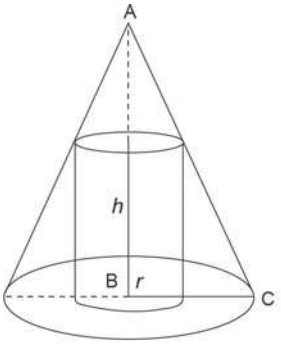
25. Consider the figures given below:

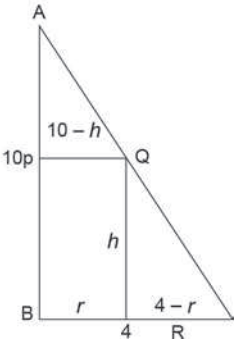
$$\frac{AP}{PB} = \frac{AQ}{QC} = \frac{10 - h}{h}$$

$$\frac{BR}{RC} = \frac{AQ}{QC} = \frac{R}{4 - r}$$

$$\Rightarrow \frac{10 - h}{h} = \frac{r}{4 - r} \Rightarrow 4h = 40 - 10r \quad \Rightarrow 2h = 20 - 5r$$

Surface area of cylinder = $2\pi r^2 + 2\pi rh = 2\pi r^2 + \pi r$ (20 – 5r)





$= 20\ \pi r - 2\ pr^2 = \pi\ (20r - 3r^2)$

So, we have to maximize $f(r) = 20r - 3r^2$

We know that the maximum of quadratic equation

$ax^2 + bx + c = 0$ arises for $x = \frac{-b}{2a}$ and the maximum value $= \frac{-d}{4a}$

In this case, $\frac{-d}{4a} = \frac{100}{3}$

So, the maximum surface area = $\frac{100\pi}{3}$

Hence, option (a) is the answer.

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PART

3

OTHER MBA ENTRANCE PAPERS

Model SNAP Paper 1

Model SNAP Paper 2

Model XAT Paper 1

Model XAT Paper 2

Model IIFT Paper

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SNAP

Model SNAP Paper 1

(Based upon Previous Years' SNAP Paper)

1. A dice is rolled three times and sum of three numbers appearing on the uppermost face is 15. The chance that the first roll was a four is:
(a) 2/5 (b) 1/5
(c) 1/6 (d) None of these

2. A boat covers a distance of 30 km downstream in 2 h, whereas it takes 6 h to cover the same distance upstream. What is the speed of the boat in kmph?
(a) 5 (b) 7.5 (c) 13 (d) 18

3. A five digit number is formed by using the digits 1, 2, 3, 4, and 5 without repetitions. What is the probability that the number is divisible by 4?
(a) 1/5 (b) 5/6
(c) 4/5 (d) None of these

4. If the algebraic sum of deviations of 20 observations measured from 23 is 70, then find the mean of these observations.
(a) 24 (b) 25
(c) 26 (d) None of these

5. An alloy of gold and silver weighs 50 g. It contains 80% gold. How much gold should be added to the alloy so that percentage of gold is increased to 90%?
(a) 50 g (b) 60 g (c) 30 g (d) 40 g

6. Weekly incomes of two persons are in the ratio of 7:3, and their weekly expenses are in the ratio of 5:2. If each of them saves ₹300 per week, then find the weekly income of the first person.
(a) ₹7500 (b) ₹4500
(c) ₹6300 (d) ₹5400

7. Wheat is now being sold at ₹27 per kg. During last month, its cost was ₹24 per kg. Find by how much per cent a family reduces its consumption so as to keep the expenditure fixed.
(a) 10.2% (b) 12.1% (c) 12.3% (d) 11.1%

8. There are 10 stations on a railway line. The number of different journey tickets that are required by the authorities is:
(a) 92 (b) 90
(c) 91 (d) None of these

9. The radius of circle is so increased that its circumference increased by 5%. The area of the circle then increases by:
(a) 12.5% (b) 10.25% (c) 10.5% (d) 11.25%

10. In how many ways can the letters of the word ABACUS be rearranged such that the vowels always appear together?
(a) 6!/2! (b) 3!*3!
(c) $(3! \times 3!)/2!$ (d) $(4! \times 3!)/2!$

11. In 4 years, ₹6000 amounts to ₹8000. In what time at the same rate will ₹525 amount to ₹700?
(a) 2 years (b) 3 years (c) 4 years (d) 5years

Direction for Questions 12–14: Answer the questions based on the information given below.

At the start of a game of cards, J and B together had four times as much money as T, whereas T and B together had three times as much as J. At the end of the evening, J and B together had three times as much money as T, whereas T and B together had twice as much as J, and B lost ₹200.

12. What fraction of the total money did T have at the beginning of the game?
(a) 1/3 (b) 1/8 (c) 2/9 (d) 1/5

13. What fraction of the total money did J win or lose?
(a) Won 1/12 (b) Lost 1/6
(c) Lost 1/3 (d) Won 1/5

14. What amount did B start with?
(a) ₹575 (b) ₹375 (c) ₹825 (d) ₹275

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15. If a and b are negative, and c is positive, then which of the following statement(s) is/are true?

I $a - b < a - c$

II If $a < b$, then $a/c < b/c$

III $a/b < a/c$

(a) I only (b) II only

(c) III only (d) II and III only

16. The diagonal of a square is $4\sqrt{2}$ cm. The diagonal of another square whose area is double that of the first square is:

(a) 8 cm (b) $8\sqrt{2}$ cm (c) $4\sqrt{2}$ cm (d) 16 cm

17. The maximum length of a pencil that can be kept in a rectangular box of dimensions 8 cm × 6 cm × 2 cm is:

(a) $2\sqrt{13}$ cm (b) $2\sqrt{14}$ cm

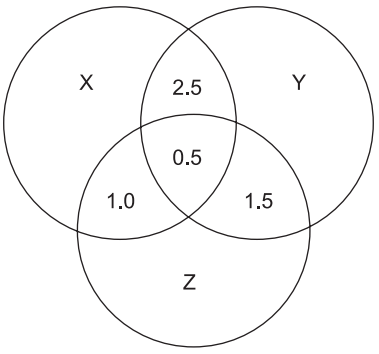
(c) $2\sqrt{26}$ cm (d) $10\sqrt{2}$ cm

Direction for Questions 18 to 19: Answer the questions based on the information given below.

The Venn diagram given below shows the estimated readership of 3 daily newspapers (X, Y, and Z) in a city. The total readership and advertising cost for each of these papers are as given below.

Newspapers	Readership (lakhs)	Advertisement (₹Per sq. cm)
X	8.7	6000
Y	9.1	6500
Z	5.6	5000

The total population of the city is estimated to be 14 million. The common readership (in lakhs) is indicated in the given Venn diagram.



18. The number of people (in lakhs) who read at least one newspaper is:

(a) 4.7 (b) 11.9 (c) 17.4 (d) 23.4

19. The number of people (in lakhs) who read only one newspaper is:

(a) 4.7 (b) 11.9 (c) 17.4 (d) 23.4

20. Sonali invests 15% of her monthly salary in insurance policies. She spends 55% of her monthly salary in shopping and on household expenses. She saves the remaining amount of ₹12,750. What is Sonali's monthly income?

(a) ₹42,500 (b) ₹38,800

(c) ₹40,000 (d) ₹35,500

21. How many kilograms of tea worth ₹25 per kg must be blended with 30 kg of tea worth ₹30 per kg so that by selling the blended variety at ₹30 per kg there should be a gain of 10%?

(a) 36 kg (b) 40 kg (c) 32 kg (d) 42 kg

22. In an examination, out of 80 students, 85% of the girls and 70% of the boys passed. How many boys appeared in the examination if total pass percentage was 75%?

(a) 370 (b) 340 (c) 320 (d) 360

23. Three hundred grams of salt solution has 40% salt in it. How much salt should be added to make it 50% in the solution?

(a) 40 g (b) 60 g (c) 70 g (d) 80 g

24. A five-digit number divisible by 3 is to be formed using numerical 0, 1, 2, 3, 4, and 5 without repetition. The total number of ways this can be done is:

(a) 122 (b) 210 (c) 216 (d) 217

25. A contract is to be completed in 50 days and 105 men were set to work, each working 8 hours a day. After 25 days, 2/5th of the work is finished. How many additional men be employed so that the work may be completed on time, each man now working 9 hours a day?

(a) 34 (b) 36 (c) 35 (d) 37

26. A can built up a structure in 8 days and B can break it in 3 days. A has worked for 4 days and then B joined towork with A for another 2 days only. In how many days will A alone build up the remaining part of the structure?

(a) 10 days (b) 9 days

(c) 12 days (d) None of these

27. The first two terms of a geometric progression add up to 12. The sum of the third and the fourth terms is 48. If the terms of the geometric progression are alternately positive and negative, then find the first term.

(a) −2 (b) −4 (c) −12 (d) 8

28. The mean of the numbers a, b, 8, 5, and 10 is 6 and the variance is 6.80. Then which one of the following gives possible values of a and b?

(a) $a = 0, b = 7$ (b) $a = 5, b = 2$

(c) $a = 3, b = 4$ (d) $a = 2, b = 4$

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29. An agent sells goods of value of ₹15,000. The commission which he receives at the ratio of 12½ % is:
(a) ₹1875 (b) ₹2125
(c) ₹2000 (d) ₹2700

30. $\sqrt{110.5} \times \sqrt{0.01} \div \sqrt{0.00025} - \sqrt{420.25}$ equal to:
(a) 0.75 (b) 0.50 (c) 0.64 (d) 0.73

Answers

1. (b)	**2.** (*)	**3.** (a)	**4.** (d)	**5.** (a)	**6.** (c)	**7.** (d)	**8.** (b)	**9.** (b)	**10.** (d)
11. (c)	**12.** (d)	**13.** (a)	**14.** (c)	**15.** (d)	**16.** (a)	**17.** (c)	**18.** (c)	**19.** (b)	**20.** (a)
21. (a)	**22.** (*)	**23.** (b)	**24.** (c)	**25.** (c)	**26.** (d)	**27.** (c)	**28.** (c)	**29.** (a)	**30.** (b)

* None of the options was correct.

Hints and Solutions

1. The sum of numbers can be 15 in the following three ways.

Case I $15 = 3 + 6 + 6$

The first, second, and third throws can be (3, 6, 6), (6, 3, 6), and (6, 6, 3), respectively.

∴ Total number of ways in which 3, 6, and 6 can be obtained is 3.

Case II $15 = 4 + 5 + 6$

The first, second, and third throws can be either of 4, 5, and 6.

∴ Total number of ways in which 4, 5, and 6 can be obtained is 6.

Case III $15 = 5 + 5 + 5$

The first, second, and third throws can be 5, 5, and 5.

∴ Total number of ways in which 5, 5, and 5 can be obtained is 1.

∴ The total number of ways in which the sum of throws can be 15 is $3 + 6 + 1 = 10$

The total number of ways in which the first row will be a 4 is 2.

$$\therefore \text{Required chance} = \frac{2}{10} = \frac{1}{5}$$

Hence, option (b) is the correct answer.

2. Let b and s be the speed of the boat and stream, respectively.

As per the given conditions,

$$\frac{30}{b+s} = 2 \quad \text{(i)}$$

$$\frac{30}{b-s} = 6 \quad \text{(ii)}$$

Dividing (i) by (ii), we get

$$\frac{b-s}{b+s} = \frac{1}{3}$$

$$\therefore b = 25$$

Substituting in (i), we get

$$\frac{30}{2s-s} = 2$$

$$\therefore s = 5$$

$\therefore b = 10$ kmph

None of the options are correct.

3. A number divisible by 4 formed using the digits 1, 2, 3, 4, and 5 have the last two digits 12 or 24 or 32 or 52.

In each of these cases, the five digit number can be formed using the remaining 3 digits in $3 \times 2 \times 1 = 6$ ways.

∴ A number divisible by 4 can be formed in $6 \times 4 = 24$ ways.

Total numbers that can be formed using the digits 1, 2, 3, 4, and 5 without repetitions = $5! = 120$

$$\therefore \text{Required probability} = \frac{24}{120} = \frac{1}{5}$$

Hence, option (a) is the correct answer.

4. Let a, b, c, d, e, and f be six numbers, such that

$a = e + m$

$b = e - k$

$c = e + l$

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$d = e - j$

$f = e + p$

Then, the sum of deviations of a, b, c, d, e, and f from e is $(m - k + l - j + p)$. The means of a, b, c, d, e, and f is given by

$$\frac{a+b+c+d+e+f}{6} = \frac{e+m+e-k+e-l+e-j+e+p}{6}$$

$$= \frac{6e + (m-k-l-j+p)}{6}$$

$$= \frac{6e + \text{Sum of deviations}}{6}$$

Using the similar logic, the required mean $= \frac{20 \times 23 + 70}{20} = 26.5$

Hence, option (d) is the correct answer.

5. Let x g of gold be added to the alloy to increase the percentage of gold in it to 90%.

$$\therefore \frac{80\% \text{ of } 50 + x}{50 + x} = \frac{90}{100}$$

$$\therefore \frac{40 + x}{50 + x} = \frac{90}{100}$$

$\therefore x = 50$ g

Hence, option (a) is the correct answer.

6. Let the incomes be $7x$ and $3x$ and let the expenses be $5y$ and $2y$.

$\therefore 7x - 5y = 300$ (i)

$3x = -2y = 300$ (ii)

Solving (i) and (ii) simultaneously, we get

$\therefore$ A's income $= 7x = 7 \times 900 =$ ₹6300

Hence, option (c) is the correct answer.

7. Assume the family consumes 1 kg usually.

To keep expenditure at ₹24, its new consumption should be $\frac{24}{27} = \frac{8}{9}$ kg

$\therefore$ Percentage decrease in consumption $= \frac{\left(1 - \frac{8}{9}\right)}{1} \times 100$

$= 11.1\%$

Hence, option (d) is the correct answer.

8. From a certain station, there will be a ticket for each of the other 9 stations, and there are 10 stations on the railway line.

$\therefore$ The number of different journey tickets $= 9 \times 10 = 90$

Hence, option (b) is the correct answer.

9. Circumference of a circle$= 2\pi r$, where r is the radius.

Circumference $\propto r$

As the circumference increases by 5%, the radius also increases by 5%.

$\therefore$ New radius $= 1.05r$

$\therefore$ As area $\propto$ (radius)^2

$\therefore$ New area $= (1.05)^2 \times$ old area

$= 1.1025 \times$ old area

$\therefore$ Percentage increase in area $= 10.25\%$

Hence, option (b) is the correct answer.

10. The three vowels in ABACUS are A, A, and U. These three can be arranged among themselves in 3!/2! = 3 ways.

As the three vowels are to appear together, we consider them as one entity. Thus, we have four letters; (AAU), B, C, and S to be arranged.

This can be done in 4! ways.

$\therefore$ The required number of ways $= \frac{4! \times 3!}{2}$

Hence, option (d) is the correct answer.

11. We can see that $\frac{8000}{6000} = \frac{700}{525} = \frac{4}{3}$

$\therefore$ The time that 525 requires to grow to 700 will be the same as the time that 6000 requires to grow to 8000.

$\therefore$ The required number of years = 4

Hence, option (c) is the correct answer.

12. At the start of the game J and B together had 4 times the money that T had.

Let the total money be m. Then, $j + b + t = 4t + t = 5t = m$

$\therefore$ The total money is divisible by 5.

Similarly, from the other three statements in the data, we find that the total money is divisible by 4 and 3 as well.

$\therefore$ The total money is divisible by $4 \times 3 \times 5 = 60$

Let the total money be $60y$.

At the start of the game,

$j + b = 4t$

$\therefore j + b + t = 5t = 60y$

$\therefore t = 12y$

$\therefore j + b = 48y$ (i)

Also,

$t + b = 3j$

$\therefore 4j = 60y$

$\therefore j = 15y$

Substituting in (i)

$15y + b = 48y$

$\therefore b = 33y$

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$\therefore$ The amounts with J, B, and T at the start were $15y$, $33y$, and $12y$, respectively.

Now, at the end of the game,

$j + b = 3t$

$\therefore 4t = 60y$

$\therefore t = 15y$

$\therefore j + b = 45y$ (ii)

Also, $t + b = 2j$

$\therefore 3j = 60y$

$\therefore j = 20y$

Substituting in (ii), we get

$b = 25y$

$\therefore$ The amounts with J, B, and T at the end were $20y$, $25y$, and $15y$, respectively.

$\therefore$ The fraction of total money that T had at the beginning was

$$\frac{12y}{60y} = \frac{1}{5}$$

Hence, option (d) is the correct answer.

13. J won $20y - 15y = 5y$

$\therefore$ The required fraction $= \frac{5y}{60} = \frac{1}{12}$

Hence, option (a) is the correct answer.

14. As B lost ₹200,

$\therefore 33y - 25y = 200$

$\therefore 8y = 200$

$\therefore y = 25$

$\therefore 33y = 825$

$\therefore$ B started with ₹825

Hence, option (c) is the correct answer.

15. Here, a and b are negative and c is positive.

Let $a = -4$, $b = -3$ and $c = 2$

I $a - b = -4 + 3 = -1$

$a - c = -4 - 2 = -6$

So, $a - b > a - c$

So, Statement I is not true.

II Taking the same values of a, b, and c, we have $a/c = -4/2 = -2$

$b/c = -3/2 = -1.5$

Since $-2 < -1.5$, $a/c < b/c$

So, Statement II is true.

III For any value of b and c,

$1/b < 1/c$ since b is negative and c is positive.

So, III is true.

So, Statements II and III are true.

Hence, option (d) is the correct answer.

16. Let the length of the side of the original square be x.

The length of the diagonal $= \sqrt{x^2 + x^2} = 4\sqrt{2}$

$\therefore 2x^2 = 32$

$\therefore x^2 = 16$

$\therefore x = 4$

Now, the area of the other square is twice that of the first square.

So, the area of the second square $= 2x^2 = 32$

If the length of the second square is y, then $y^2 = 32$

$\therefore$ Length of the diagonal of this square $= \sqrt{2y^2} = \sqrt{64}$

$= 8$ cm

Hence, option (a) is the correct answer.

17. Let the length, breadth, and height of the rectangular box be l, b, and h, respectively. Then, the maximum length of the pencil that can be accommodated in the box will be equal to the length of its greatest diagonal, which is the body diagonal.

This length is given as $\sqrt{l^2 + b^2 + h^2}$.

Dimensions of the box are given as $8 \times 6 \times 2$.

So, the maximum length of the pencil that can be kept in the box

$\sqrt{8^2 + 6^2 + 2^2} = \sqrt{104} = 2\sqrt{26}$ cm

Hence, option (c) is the correct answer.

18. Total readership of X is 8.7 lakhs.

$\therefore X + 2.5 + 0.5 + 1 = 8.7$

$\therefore X = 4.7$

Also, total readership of Y is 9.1 lakhs.

$\therefore 2.5 + 0.5 + 1.5 + Y = 9.1$

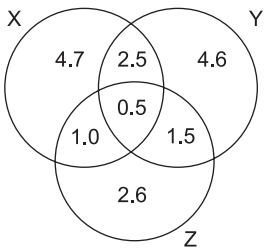
$\therefore Y = 4.6$

Also, total readership of Z is 5.6 lakhs.

$\therefore Z + 1 + 0.5 + 1.5 = 5.6$

$\therefore Z = 2.6$

Therefore, we have



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$\therefore$ Number of people who read at least one paper = 4.7 + 4.6 + 2.6 + 1 + 1.5 + 0.5 + 2.5 = 17.4

Hence, option (c) is the correct answer.

19. From the solution of the previous question,

Number of people who read only one newspaper = 4.6 + 4.7 + 2.6 = 11.9

Hence, option (b) is the correct answer.

20. Percentage of Sonali's salary invested in insurance policies = 15%

Percentage of salary spent in shopping and household expenses = 55%

$\therefore$ Percentage of salary saved = 100 − (15 + 55) = 30%

Let Sonali's monthly salary be x.

$\therefore$ 30% of x =12,750

$$x = \frac{12,750 \times 100}{30}$$

₹42,500

Hence, option (a) is the correct answer.

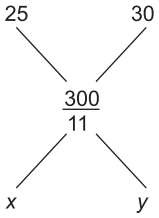
21. There are two varieties of tea: one worth ₹25 per kg and the other worth ₹30 per kg.

Now, by selling the blended variety at ₹30 per kg, profit should be 10%.

So, the cost price of the blended tea should be

$$\frac{30}{1.1} = ₹\frac{300}{11}\text{kg}$$

This problem can be depicted through alligation as below.



We get $\frac{x}{y} = \frac{30}{25}$

Amount of tea of ₹30 per kg to be used is 30 kg. If the amount of tea worth ₹25 per kg is a, then

$$\therefore \frac{30}{25} = \frac{a}{30}$$

$$\therefore a = \frac{30 \times 30}{25} = 36\text{kg}$$

Hence, option (a) is the correct answer.

22. Of the total 80 students, let there be b boys and g girls.

$\therefore b + g = 80$ (i)

Total pass percentage was 75%.

So, number of students who passed = 0.75 × 80 = 60

Now, it is given that 85% of girls and 70% of boys passed.

$\therefore 0.7b + 0.85g = 60$ (ii)

Solving (i) and (ii), we get

$g = 26.67$ and $b = 53.33$

None of the options are correct.

23. 300 g solution has 40% salt in it.

So, amount of salt in it $= \frac{40 \times 300}{100} = 120\,\text{g}$

For the salt percentage to be 50%,

$$\frac{120 + x}{300 + x} = \frac{1}{2}$$

$$\therefore 240 + 2x = 300 + x$$

$$\therefore x = 60\,\text{g}$$

Hence, option (b) is the correct answer.

24. Using the digits 0, 1, 2, 3, 4, and 5, five-digit numbers divisible by 3, can be formed using the following combinations.

Case I 1, 2, 3, 4, 5

Total number of numbers formed using these digits = 5! = 120

Case II 0, 1, 2, 4, 5

Total number of numbers formed using these digits = 4 × 4 × 3 × 2 = 96

Thus, total numbers = 120 + 96 = 216

Hence, option (c) is the correct answer.

25. Since $\frac{2}{5}$th of the work is completed in 25 days, remaining $\frac{3}{5}$th of the work is completed in 25 days.

Let x men work for 25 days to complete $\frac{3}{5}$th of the work.

$$\frac{M_1 D_1 H_1}{W_1} = \frac{M_2 D_2 H_2}{W_2}$$

$$\therefore \frac{25 \times 105 \times 8 \times 5}{2} = \frac{x \times 25 \times 9 \times 5}{3}$$

$$\therefore x = \frac{105 \times 8}{2 \times 3} = 140$$

$\therefore$ Additional men employed = 140 − 105 = 35

Hence, option (c) is the correct answer.

26. A can build the structure in 8 days.

$\therefore$ Fraction of structure built in a day by A = 1/8

Similarly, fraction of structure broken by B in a day = 1/3

Amount of work done by A in 4 days = 4/8 = 1/2

Now, both A and B work together for 2 days.

So, the fraction of structure built in 2 days

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$$= 2\left(\frac{1}{8} - \frac{1}{3}\right) = \frac{2\times(-5)}{24} = \frac{-5}{12}$$

Fraction of structure still to be built

$$= \frac{1}{2} + \frac{5}{12} = \frac{11}{12}$$

If A takes x days, to build up the remaining structure, then

$$\frac{x}{8} = \frac{11}{12}$$

$$\therefore x = \frac{22}{3}$$

Hence, option (d) is the correct answer.

27. Let the first term and the ratio of the geometric progression be a and r, respectively.

$\therefore a + ar = 12$

$\therefore a(1 + r) = 12$ (i)

Also, $ar^2 + ar^3 = 48$

$\therefore ar^2(1 + r) = 48$ (ii)

Dividing (ii) by (i), we get

$$\frac{ar^2(1+r)}{a(1+r)} = \frac{48}{12} = 4$$

$$\therefore r^2 = 4$$

$$\therefore r = \pm\sqrt{4} = \pm 2$$

Since the terms of the geometric progression are alternately positive and negative, $r = -2$

$\therefore$ From (i), $a(1 - 2) = 12$

$\therefore a = -12$

Hence, option (c) is the correct answer.

28. Mean of the given numbers $= \dfrac{a+b+23}{5}$

Mean is given as 6.

$$\therefore \frac{a+b+23}{5} = 6$$

$$\therefore a + b + = 30 - 23 = 7$$

So, option (d) can be eliminated.

Now, variance

$$= \frac{1}{5}[(a-6)^2 + (b-6)^2 + (5-6)^2 + (10-6)^2 + (8-6)^2]$$

$$\therefore \frac{1}{5}[(a-6)^2 + (b-6)^2 + 21] = 6.8$$

$$\therefore (a-6)^2 + (b-6)^2 + 21 = 34$$

$$\therefore (a-6)^2 + (b-6)^2 = 13$$

Only option (c) fits into the above equation.

Hence, option (c) is the correct answer.

29. Value of goods = ₹15,000

Commission the receives $= 12\frac{1}{2}\%$

$$\therefore \text{Commission received} = \frac{12.5\times 15{,}000}{100}$$

$$= 1875$$

Hence, option (a) is the correct answer.

30. $\sqrt{110.25} = 10.5$

$\sqrt{0.01} = 0.1$

$\sqrt{0.0025} = 0.05$

$\sqrt{420.25} = 20.5$

$$\therefore \sqrt{110.25}\times\sqrt{0.01} \div \sqrt{0.0025}\times\sqrt{420.25}$$

$$= 105 \times \frac{0.1}{0.05} - 20.5$$

$$= 1.05/0.05 - 20.5$$

$$= 21 - 20.5$$

$$= 0.5$$

Hence, option (b) is the correct answer.

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SNAP

Model SNAP Paper 2

(Based upon Previous Years' SNAP Paper)

1. In a retail outlet, the average revenue was ₹10,000 per day over a 30-day period. During this period, the average daily revenue on weekends (total 8 days) was ₹20,000 per day. What was the average daily revenue on weekends?
(a) 6364 (b) 5250 (c) 6570 (d) 8060

2. Two different prime numbers X and Y, both are greater than 2, then which of the following must be true?
(a) $X - Y = 23$ (b) $X + Y \neq 87$
(c) Both (a) and (b) (d) None of these

3. It takes 6 h for pump A, used alone, to fill a tank of water. Pump B used alone takes 8 h to fill the same tank. A, B, and another pump C all together fill the tank in 2 h. How long would pump C take, used alone, to fill the tank?
(a) 4.8 h (b) 6 h (c) 5.6 h (d) 3 h

4. A swimming pool can be filled by pipe A in 3 h and by pipe B in 6 h, each pump working on its own. At 9 am, pump A is started. At what time will the swimming pool be filled if pump B is started at 10 am?
(a) 11:20 am (b) 11:05 am
(c) 11:10 am (d) 10:50 am

5. The sum of prime numbers that are greater than 60 but less than 70 is:
(a) 128 (b) 191 (c) 197 (d) 260

6. Find out the appropriate next number in the series from the options given below:
0, 2, 6, 12, 20, 30, 42, ?
(a) 56 (b) 62 (c) 49 (d) 5

7. A bakery opened with its daily supply of 40 dozen rolls. Half of the rolls were sold by noon, and 60% of the remaining rolls were sold between noon and closing time. How many dozen rolls were left unsold?
(a) 6 (b) 8 (c) 10 (d) 12

8. Stuart, Jack, and Leo are colleagues working in a plant. Stuart and Jack can do a work in 10 days, Jack and Leo can do the same work in 15 days, whereas Stuart and Leo can do it in 12 days. All of them started the work together. After two days, Leo was shifted to some other work. How many days will Stuart and Jack take to finish the rest of the work?
(a) 9 (b) 12 (c) 8 (d) 7.5

9. Find the missing numbers in the series 1:1, 8:4, 9:27, 64:16, 25:125, ?:?, 49:343.
(a) 36:316 (b) 216:36
(c) 316:l6 (d) 32:316

10. The difference between the value of a number increased by 25% and the value of the original number decreased by 30% is 22. What is the original number?
(a) 70 (b) 65 (c) 40 (d) 90

11. Running at the same constant rate, 6 identical machines can produce a total of 180 bottles per hour. How many bottles could 15 such machines produce in 30 min?
(a) 225 (b) 300 (c) 250 (d) 350

12. Find the number whose fifth part increased by 4 is equal to its fourth part diminished by 10.
(a) 240 (b) 260 (c) 270 (d) 280

13. Which one of the following numbers will completely divide (461 + 462 + 463 + 464)?
(a) 3 (b) 10 (c) 11 (d) 13

Direction for Questions 14–20: Each of the following questions is followed by two statements:

Mark

(a) If Statement I alone is sufficient to answer the question.

(b) If Statement II alone is sufficient to answer the question.

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(c) If both Statements I and II together are necessary to answer the question.

(d) If both Statements I and II together are not sufficient to answer the question.

14. Is 'b' positive?
(I) $a + b$ is positive.
(II) $a - b$ is positive.

15. In a general body election, 3 candidates, P, Q, and R were contesting for a membership of the board. How many votes did each receive?
(I) P received 17 votes more than Q and 103 votes more than R.
(II) Total votes cast were 1703.

16. If C1 and C2 are the circumferences of the outer and inner circles, respectively. What is C1:C2?
(I) The two circles are concentric.
(II) The area of the ring is 2/3 the area of greater circle.

17. What is the middle number of 7 consecutive whole numbers?
(I) Product of number is 702,800.
(II) Sum of the number is 105.

18. Total marks obtained by P, Q, R, and S in Mathematics are 360. How many marks did P secure in Mathematics?
(I) P secured one-third marks of the total of Q, R, and S.
(II) Average marks obtained by Q and R are 20 more than that secured by S.

19. How many ice cubes can be accommodated in a container?
(I) The length and breadth of the container is 20 cm and 15 cm, respectively.
(II) The edge of the ice cube is 2 cm.

20. Ram got ₹1500 as dividend from a company. What is the rate of interest given by the company?
(I) The dividend paid last year was 10%.
(II) Ram has 350 shares of ₹10 denomination.

21. What is the number that is one-half of one-quarter of one-tenth of 400?
(a) 2 (b) 5 (c) 8 (d) 10

22. Consider a square ABCD with mid-points E, F, G, and H of sides AB, BC, CD, and DA. Let L denote the line passing through F and H. Consider points P and Q on the line L inside the square such that the angles APD and BQC are both equal to 120°. What is the ratio ABCQPD to the remaining area of ABCD?
(a) $4+\frac{\sqrt{2}}{3}$ (b) $2\sqrt{3}-1$
(c) $2+\sqrt{3}$ (d) $\left((10-3\sqrt{3})/9\right.$

23. The price of Darjeeling Tea (in rupees per kilogram) is $100 + 0.1n$, on the nth day of a non-leap year ($n = 1, 2, 3, ..., 100$) and then remains constant. On the other hand, the price of Ooty Tea (in rupees per kilogram) is $85 + 0.15n$, on the nth day ($n = 1, 2, ..., 365$). On which date of that year will the prices of these two varieties of the tea be equal?
(a) 27th October (b) 16th June
(c) 15th June (d) 28th October

Direction for Questions 24–27: Each of questions consist of a question followed by two statements numbered I and II:

Answer (a) if data in Statement I alone is sufficient to answer the question but the data in Statement II alone is not sufficient to answer the question.

Answer (b) if data in Statement II alone is sufficient to answer the question but the data in Statement I alone is not sufficient to answer the question.

Answer (c) if data in Statements I and II together are necessary to answer the question.

Answer (d) if data in Statements I and II together are not sufficient to answer the question.

24. ΔABC and ΔPQR are congruent.
(I) Area of ΔABC and ΔPQR are same.
(II) ΔABC and ΔPQR are right-angled triangles.

25. Salary of A and B is in the ratio 3:4 and expenditure is in ratio 4:5. What is the ratio of their saving?
(I) B's saving is 25% of his salary.
(II) B's salary is ₹2500.

26. What is the average height of the class?
(I) Average height of the class decreases by 1 cm if we exclude the tallest person of the class whose height is 56 cm.
(II) Average height of the class increases by 1 cm if we exclude the shortest person of the class whose height is 42 cm.

27. Ram is taller than Shyam and Jay is shorter than Vikram. Who is the shortest among them?
(I) Ram is the tallest.
(II) Shyam is taller than Vikram.

28. In September 2009, the sales of a product were two-third of that in July 2009. In November 2009, the sales of the product were higher by 5% as compared to September 2009. How much is the percentage of increase in sales in November 2009 with respect to the base figure in July 2009?
(a) +40% (b) −20% (c) −30% (d) +25%

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29. For what range of values of 'x' will be the inequality $15x - (2/x) > 1$?
(a) $x > 0.4$
(b) $x < 1/3$
(c) $-1/3 < x < 0.4, x > 15/2$
(d) $-1/3 < x < 0, x > 2/5$

30. How many litres of a 30% alcohol solution should be added to 40 litres of a 60% alcohol solution to prepare a 50% solution?
(a) 30 (b) 20 (c) 24 (d) 32

31. 66 cubic centimetres of silver is drawn into a wire of 1 mm diameter. The length of the wire in metres will be:
(a) 84 (b) 90 (c) 168 (d) 336

32. A train 108 m long moving at a speed of 50 kmph crosses a train 112 m long coming from opposite direction in 6 s. Find the speed of the second train.
(a) 48 kmph (b) 54 kmph
(c) 66 kmph (d) 82 kmph

33. R is a positive number. It is multiplied by 8 and then squared. The square is now divided by 4 and the square root is taken. The result of the square root is Q. What is the value of Q?
(a) $3R$ (b) $4R$ (c) $7R$ (d) $9R$

34. If the length, breadth, and height of the room are in the ratio 3:2:1. The breadth and height of the room are halved and length of the room is doubled. Then area of the four walls of the room will:
(a) Decrease by 13.64% (b) Decrease by 15%
(c) Decrease by 18.75% (d) Decrease by 30%

35. A survey was conducted of 100 people whether they have read recent issues of 'Golmal', a monthly magazine. Summarized information is presented below:
Only September: 18
September but not August: 23
September and July: 8
September: 28
July: 48
July and August: 10
None of the three months: 24
What is the number of surveyed people who have read exactly for two consecutive months?
(a) 7 (b) 9 (c) 12 (d) 14

Answers

1. (a)	**2.** (b)	**3.** (a)	**4.** (a)	**5.** (a)	**6.** (a)	**7.** (b)	**8.** (d)	**9.** (b)	**10.** (c)
11. (a)	**12.** (d)	**13.** (b)	**14.** (d)	**15.** (c)	**16.** (b)	**17.**	**18.** (a)	**19.** (d)	**20.** (b)
21. (b)	**22.** (b)	**23.** (b)	**24.** (d)	**25.** (a)	**26.** (c)	**27.** (b)	**28.** (c)	**29.**	**30.** (b)
31. (a)	**32.** (d)	**33.** (b)	**34.** (d)	**35.** (b)					

(For Q17 and Q29, none of the options given were correct).

Hints and Solutions

1. Total revenue of the month = $30 \times 10{,}000 = 300{,}000$
Revenue on weekend = $8 \times 20{,}000 = 160{,}000$
$\therefore$ Revenue on other 22 days = $300{,}000 - 160{,}000$
$= 140{,}000$
$\therefore$ Average daily revenue on weekdays = $\dfrac{140{,}000}{22}$
≈ 6364
Hence, option (a) is the correct answer.

2. Prime numbers X and Y are greater than 2; hence, they must be odd.
Difference of two odd numbers is even. Therefore, option (a) is definitely false.
Sum of two odd numbers is even. Therefore, option (b) must be true.
Hence, option (b) is the correct answer.

3. In 1 h, pump A and B can fill 1/6 and 1/8 of the tank, respectively.
If C fills $1/x$ of the tank in 1 h,
$$\frac{1}{6}+\frac{1}{8}+\frac{1}{x}=\frac{1}{2}$$
$\therefore$ $$x=\frac{24}{5}=4.8\text{h}$$

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4. Pipe A can fill the pool in 3 h and pipe B can fill the pool in 6 h.

Pump A starts at 9 am and B starts at 10 am.

A alone fills for 1 h. Let A and B together fill for n hours after 10 am. Then

$$\frac{1}{3}+n\left(\frac{1}{3}+\frac{1}{6}\right)=1$$

$$\therefore \quad n=\frac{4}{3}\text{h}=1\text{ h and 20 min}$$

Therefore, pool will be filled by 11:20 am.

Hence, option (a) is the correct answer.

5. 61 and 67 are the only prime numbers that are greater than 60 and less than 70.

$\therefore$ Sum of these numbers = 61 + 67 = 128

Hence, option (a) is the correct answer.

6. The difference between two consecutive numbers of the series is 2, 4, 6, 8, 10,12, …

Note that these differences are in A.P. with the common difference being 2.

$\therefore$ Next term = 42 + 14 = 56

Hence, option (a) is the correct answer.

7. Number of rolls initially =40 dozens.

After selling half the rolls, the number of rolls left = 20 dozens.

Now, between noon and closing time, 60% of the remaining rolls were sold.

Therefore, 40% of 20 dozen rolls = 8 dozen rolls were left unsold.

Hence, option (b) is the correct answer.

8. Let Stuart finish the work in s days, Jack in j days, and Leo in l days.

$$\frac{1}{s}+\frac{1}{j}=\frac{1}{10}$$

$$\frac{1}{j}+\frac{1}{l}=\frac{1}{15}$$

$$\frac{1}{s}+\frac{1}{l}=\frac{1}{12}$$

$$\therefore 2\left(\frac{1}{s}+\frac{1}{j}+\frac{1}{l}\right)=\frac{1}{10}+\frac{1}{15}+\frac{1}{12}=\frac{(6+4+5)}{60}=\frac{1}{4}$$

$$\frac{1}{s}+\frac{1}{j}+\frac{1}{l}=\frac{1}{8}$$

$\therefore$ Together they can finish $\frac{1}{8}$ th of the work in one day.

For two days, they worked together, after which Leo was shifted to some other work.

Let Stuart and Jack take n days to finish the remaining work.

$$\therefore \quad 2\times\frac{1}{8}+n\left(\frac{1}{10}\right)=1$$

$$\therefore \quad n=7.5\text{ days}$$

Hence, option (d) is the correct answer.

9. Consider the given series, 1:1, 8:4, 9:27, 64:16, 25:125, ?:?, 49:343

Note that the nth term of the series, if n is odd = n^2:n^3 and if n is even, n^{th} term = n^3:n^2

Therefore, sixth pair will be, 6^3:6^2 = 216:36

Hence, option (b) is the correct answer.

10. Let x be the required number. Now, increasing x by 25% gives $1.25x$ and 30% decrease in x gives $0.7x$.

Now, $1.25x - 0.7x = 0.55x = 22$

$$\therefore \quad x=\frac{22}{0.55}=40$$

Hence, option (c) is the correct answer.

11. Six machines can produce 180 bottles per hour.

Hence, 1 machine can produce 180/6 = 30 bottles per hour and 15 bottles in 30 min.

$\therefore$ 15 machines can produce 15 × 15 = 225 bottles per hour.

Hence, option (a) is the correct answer.

12. Let the number be x.

$$\therefore \quad \frac{x}{5}+4=\frac{x}{4}-10$$

$$\therefore \quad 14=\left(\frac{1}{4}-\frac{1}{5}\right)x$$

$$\therefore \quad x=20\times14=280$$

Hence, option (d) is the correct answer.

13. 461 + 462 + 463 + 464 = 1850

Among the given options, only 10 divides 1850.

Hence, option (b) is the correct answer.

14. If $a = 5$ and $b = 3$, then 5 + 3 = 8 is positive and 5 − 3 = 2 is positive.

If $a = 5$ and $b = -3$, then 5 − 3 = 2 is positive and 5 − (−3) = 8 is positive.

Thus, b can be positive or negative irrespective of what sign $(a + b)$ and $(a - b)$ take.

Hence, the question cannot be answered using Statement I alone or Statement II alone.

Using both the statements together,

$a + b + a - b$ is positive.

i.e., $2a$ is positive.

$\therefore$ a is positive.

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Using the same argument as above, we still cannot say anything about the sign of b.

Hence, option (d) is the correct answer.

15. Let p, q, and r be the number of votes received by P, Q, and R, respectively.

Using Statement I alone,

$P = q + 17$ and $p = r + 103$

But, this is not sufficient to find the number of votes that each received.

Hence, Statement I alone is not sufficient to answer the question.

Using Statement II alone,

Total votes = 1703

But, the breakup of these votes is not given.

$\therefore$ Statement II is not sufficient to answer the question.

Using both the statements together,

$3p - 120 = 1703$

Thus, p can be calculated, and hence, p, q, and r can be calculated.

Therefore, the question can be answered using both the statements together.

Hence, option (c) is the correct answer.

16. Let R and r be the radii of the outer and inner circles, respectively.

Using Statement I alone,

Statement I mentions nothing about the radii of the circles. Therefore, C1:C2 cannot be calculated.

Hence, Statement I alone is not sufficient to answer the question.

Using Statement II alone,

$$\frac{(\text{Area of ring})}{(\text{Area of greater circle})} = \frac{2}{3}$$

$$\therefore \quad \frac{\pi(R^2 - r^2)}{\pi R^2} = \frac{2}{3}$$

$$\therefore \quad \frac{R}{r} = \frac{\sqrt{3}}{1}$$

Now, $$\frac{C_1}{C_2} = \frac{2\pi R}{2\pi r} = \frac{R}{r}$$

Therefore, C_1:C_2 can be calculated using Statement II alone.

Hence, option (b) is the correct answer.

17. Let the numbers be $(n - 3)$, $(n - 2)$, $(n - 1)$, n, $(n + 1)$, $(n + 2)$, and $(n + 3)$.

Using Statement I alone,

$(n - 3)(n - 2)(n - 1)\, n\, (n + 1)(n + 2)(n + 3) = 702{,}800$

Note that the product of seven consecutive numbers will always be unique. Hence, the data is sufficient to find the value of n.

Taking $n = 7$, we get the product as 604,800, whereas taking $n = 8$, we get the product as 16,63,200.

Hence, there exists no whole number such that the given condition holds true.

Therefore, the question can be solved using Statement I alone.

Using Statement II alone, we get

$(n - 3) + (n - 2) + (n - 1) + n + (n + 1) + (n + 2) + (n + 3)$
$= 105$

$$\therefore \quad n = 15$$

Hence, the question can be solved using Statement II alone.

There is no such option for questions which can be solved using either of the statements.

18. Let p, q, r, and s be the marks obtained by P, Q, R, and S.

$$\therefore \quad p + q + r + s = 360$$

Using Statement I alone,

$$p = \frac{1}{3}(q + r + s)$$

$$\therefore \quad (q + r + s) = 3p$$

$$\therefore \quad p + (q + r + s) = 360$$

$$\therefore \quad p + 3p = 360$$

$$\therefore \quad p = 90$$

Hence, Statement I alone is sufficient to answer the question.

Using Statement II alone,

$$\frac{q + r}{2} = s + 20$$

$$\therefore \quad q + r = 2s + 40$$

$$\therefore \quad p + q + r + s = 360$$

$$\therefore \quad p + 2s + 40 + s = 360$$

$$\therefore \quad p + 3s + 40 = 360$$

Now, we are left with two variables and one equation.

Thus, the question cannot be answered using Statement II alone.

Hence, option (a) is the correct answer.

19. Using Statement I alone,

The height of the container is not known. Thus, neither the volume of the container nor the volume of the cube is known. Hence, the question cannot be answered using Statement I alone.

Using Statement II alone,

Volume of ice cube is given, but the volume of container is not known.

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Hence, the question cannot be answered using Statement II alone.

Using both the statements together,

Volume of the ice cube is known but the volume of container is not known. Therefore, the question cannot be answered.

Hence, option (d) is the correct answer.

20. Using Statement I alone,

The dividend paid last year does not tell us anything about the dividend paid in the current year.

Hence, the question cannot be answered using Statement I alone.

Using Statement II alone,

We know the number of shares Ram has and the denomination of shares.

$\therefore$ Dividend that Ram got = 1500 = (350 × 10 × Rate)/100

Thus, the rate can be calculated.

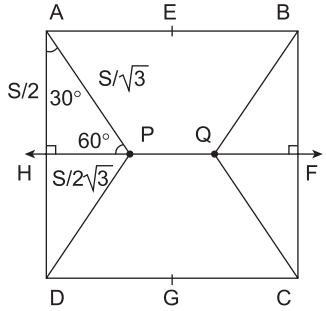
Therefore, the question can be answered using Statement II alone.

Hence, option (b) is the correct answer.

21. $x = \frac{1}{2} \times \frac{1}{4} \times \frac{1}{10} \times 400 = 5$

Hence, option (b) is the correct answer.

22. **Note:** If we solve for the ratio of the area of ABCQPD to the remaining area of ABCD, none of the options match. We assume that the question asks for the ratio of area of polygon ABQCDP to the remaining area of ABCD.



Let s be the side of the square.

Also, $\angle APH = 60°$, $\angle AHP = 90°$, and $\angle PAH = 30°$

Also, $AH = \frac{s}{2}$

$\therefore$ $HP = \frac{s}{2\sqrt{3}}$

$\therefore$ $A(\Delta APD) = \frac{1}{2} \times s \times \frac{s}{2\sqrt{3}} = \frac{s^2}{4\sqrt{3}}$

Similarly, $A(\Delta BQC) = \frac{s^2}{4\sqrt{3}}$

$A(ABQCDP) = A(\Delta ABCD) - A(\Delta APD) - A(\Delta BQC)$

$$= s^2 - \frac{s^2}{2\sqrt{3}} = \frac{s^2\left(2\sqrt{3}-1\right)}{2\sqrt{3}}$$

$\therefore$ Required ratio $= \frac{A(ABQCDP)}{2A(\Delta BQC)}$

$$= \frac{\left(2\sqrt{3}-1\right)}{2\sqrt{3}} \times \frac{2\sqrt{3}}{1} = 2\sqrt{3}-1$$

Hence, option (b) is the correct answer.

23. $100 + 0.1n = 85 + 0.15n$

$\therefore$ $n = 300$

But $n \neq 300$, as the price of Darjeeling Tea remains constant after the 100th day.

On the 100th day and later, the price of Darjeeling Tea is ₹110 per kg.

The price of Ooty Tea becomes 110 on the 167th day, which is 16th June.

Hence, option (b) is the correct answer.

24. Using Statement I alone,

Two triangles can have the same area even they are not congruent.

$\therefore$ Statement I is not sufficient to answer the question.

Using statement II alone,

Just by having the information that two triangles are right-angled triangle, one cannot conclude that the two triangles are congruent.

$\therefore$ Statement II is not sufficient to answer the question.

Using both the statements together,

Let b_1 and h_1 be the base and height of ΔABC and let b_2 and h_2 be the base and height of ΔPQR.

$\therefore$ $\frac{1}{2} b_1 h_1 = \frac{1}{2} b_2 h_2$

$\therefore$ $b_1 h_1 = b_2 h_2$

This does not necessarily mean that the base and height of the two triangles are correspondingly equal. Thus, the question cannot be answered using both the statements together.

Hence, option (d) is the correct answer.

25. Let A's salary and expenditure be $3x$ and $4y$, respectively.

Let B's salary and expenditure be $4x$ and $5y$, respectively.

$\therefore$ A's and B's savings are $(3x - 4y)$ and $(4x - 5y)$, respectively.

Using Statement I alone,

$4x - 5y = 0.25\ (4x)$

$\therefore$ $3x = 5y$

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We can find the ratio of savings using this.

∴ Statement I alone is sufficient to answer the question.

Hence, option (a) is the correct answer.

26. Let x be the average height of the class and let n be the number of students.

Using Statement I alone,

$$\frac{nx-56}{n-1}=x-1$$

$\therefore nx-56=nx-n-x+1$

$$\therefore \quad n+x=57 \qquad \text{(i)}$$

But, this is not sufficient to answer the question.

Hence, the question cannot be answered using Statement I alone.

Using Statement II alone,

$$\frac{nx-42}{n-1}=x+1$$

$\therefore nx-42=nx+n-x-1$

$$\therefore \quad x-n=41 \qquad \text{(ii)}$$

This is also not sufficient to answer the question.

Hence, the question cannot be answered using Statement II alone.

Using both the statements together,

From Statement I and II, we get

$n+x=57$ and

$x-n=41$

$x=49$

Hence, the question can be answered using both the statements together.

Hence, option (c) is the correct answer.

27. If '>' denotes the taller than relation, then we have Ram > Shyam and Vikram > Jay.

Using Statement I alone,

It is given that Ram is tallest, but still we cannot comment that whether Shyam or Jay is the shortest one.

Hence, the question cannot be answered using Statement I alone.

Using Statement II alone,

It is given that Shyam > Vikram

∴ We have the following relation: Ram > Shyam > Vikram > Jay.

∴ Jay is the shortest.

Thus, Statement II is sufficient to answer the question.

Hence, option (b) is the correct answer.

28. Let the sales in July be 300.

∴ Sales in September = 200

Sales in November $= 200 \times 1.05 = 210$

$$\therefore \text{ Required percentage increase} = \frac{210-300}{300}\times 100$$

$$= -30\%$$

Hence, option (c) is the correct answer.

29.
$$15x-\frac{2}{x}>1$$

$$\therefore \quad 15x-\frac{2}{x}-1>0$$

$$\therefore \quad 15x^2-2-x>0$$

$$\therefore \left(x+\frac{1}{3}\right)\left(x-\frac{2}{5}\right)>0$$

$$\therefore \quad x<-\frac{1}{3} \text{ or } x>\frac{2}{5}$$

The correct answer is not there in the options.

30. Let there be x litres of 30% alcohol solution.

$\therefore x(0.3)+40(0.6)=(40+x)0.5$

$\therefore x=20$

Hence, option (b) is the correct answer.

31. Radius of wire = 0.05 cm

As the volume of silver in the wire is 66 cc,

$66\pi\times(0.05)^2\times l$

$\therefore l=8400$ cm

$\therefore l=84$ m

Hence, option (a) is the correct answer.

32. Length of first train = 0.108 km

Length of second train = 0.112 km

Let the speed of second train be x.

As the trains are moving towards each other,

$$\frac{0.108+0.112}{50+x}=\frac{6}{60\times 60}$$

$$\therefore \quad x=82 \text{ kmph}$$

Hence, option (d) is the correct answer.

33.
$$\sqrt{\frac{(8R)^2}{4}}=Q$$

$$\therefore \quad Q=\sqrt{\frac{64R^2}{4}}$$

$$Q=4R$$

Hence, option (b) is the correct answer.

34. Let the original length, breadth, and height of the room be $3x$, $2x$, and x, respectively.

The area of the four walls = 2[(length × height) + (breadth × height)]

$= 2(3x^2+2x^2)=10x^2$

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The new length, breadth, and height are $6x$, x, and $x/2$, respectively.

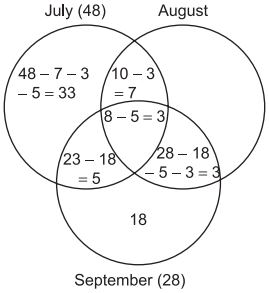
$\therefore$ The new area of walls $= 2\left[3x^2 + \frac{x^2}{2}\right] = 7x^2$

$\therefore$ Percentage decrease in area $= \frac{10x^2 - 7x^2}{10x^2} \times 100$

$= 30\%$

Hence, option (d) is the correct answer.

35. The following figure gives the distribution of the people who read 'Glomal' magazine in the given three months.



As 24 people did not read the magazine in any of the three months, 76 people read it in at least one of the three months.

$\therefore$ $76 - 48 - 18 - 2 = 8$ people read it only in August

$\therefore$ Number of people who read it exactly for two consecutive months (i.e. July and August but not September, and August and September but not July)

$= 7 + 2 = 9$

Hence, option (b) is the correct answer.

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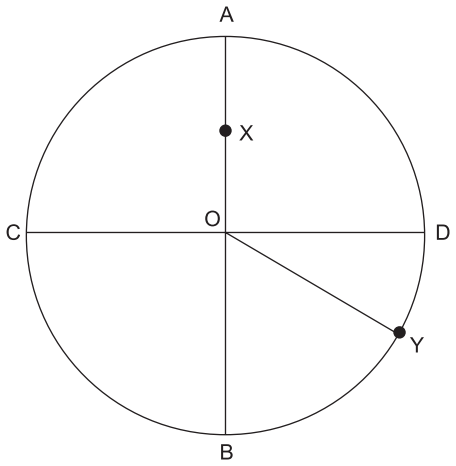
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XAT

Model XAT Paper 1

(Based upon Previous Years' XAT Paper)

1. In a circular field, AOB and COD are two mutually perpendicular diameters having length of 4 m. X is the mid-point of OA. Y is the point on the circumference such that ∠YOD = 30°. Which of the following correctly gives the relation among the three alternate paths from X to Y?



(a) XOBY : XODY : XADY :: 5.15 : 4.50 : 5.06
(b) XADY : XODY : XOBY :: 6.25 : 5.34 : 4.24
(c) XODY : XOBY : XADY :: 4.04 : 5.35 : 5.25
(d) ADY : XOBY : XODY :: 5.19 : 5.09 : 4.04
(e) XOBY : XADY : XODY :: 5.06 : 5.15 : 4.50

2. If x and y are real numbers, then find the minimum value of $x^2 + 4xy + 6y^2 - 4y + 4$.
(a) −4 (b) 0 (c) 2 (d) 4
(e) None of these

3. Let X be a four-digit positive integer such that the unit digit of X is prime and the product of all digits of X is also prime. How many such integers are possible?
(a) 4 (b) 8 (c) 12 (d) 24
(e) None of these

4. There are two types of employees in Sun Metals, namely general graduates and engineers. In which 40% of the employees are general graduates, and 75% of the engineers earn more than ₹5 lakh/year. If 50% of the organization's employees earn more than ₹5 lakh/year, then at what proportion of the general graduates employed by the organisation earn ₹5 lakh or less?
(a) 3/5 (b) 3/4 (c) 1/2 (d) 2/5
(e) None of these

5. In an equilateral triangle ABC, whose length of each side is 3 cm, D is the point on BC such that BD = 1/2 CD. What is the length of AD?
(a) $\sqrt{5}$ cm (b) $\sqrt{6}$ cm (c) $\sqrt{7}$ cm (d) $\sqrt{8}$ cm
(e) None of these

6. Two poles of height 2 m and 3 m are 5 m apart. The height of the point of intersection of the lines joining the top of each pole to the foot of the opposite pole is:
(a) 1.2 m (b) 1.0 m (c) 5.0 m (d) 3.0 m
(e) None of these

7. A manufacturer has 200 L of acid solution which has 15% acid content. How many litres of acid solution with 30% acid content may be added so that acid content in the resulting mixture will be more than 20% but less than 25%?
(a) More than 100 L but less than 300 L
(b) More than 120 L but less than 400 L
(c) More than 100 L but less than 400 L
(d) More than 120 L but less than 300 L
(e) None of these

Answer Questions 8–10 based on the following information:

An automobile company's annual sales of its small cars depend on the state of the economy as well as on whether the company uses some high profile individual as its brand ambassador in advertisements of its product. The state of the

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economy is 'good', 'okay', and 'bad' with probabilities 0.3, 0.4, and 0.3, respectively. The company may choose a high profile individual as its brand ambassador in TV ads or may go for the TV ads without a high profile brand ambassador.

If the company fixes price at ₹3.5 lakh, the annual sales of its small cars for different states of the economy and for different kinds of TV ads are summarized in Table 1. The figures in the first row are annual sales of the small cars when the company uses a high profile individual as its brand ambassador in its TV ads and the ones in the second row are that when the company does not use any brand ambassador in TV ads, for different states of the economy.

Table 1

	Good	Okay	Bad
With brand ambassador	100,000	80,000	50,000
Without brand ambassador	80,000	50,000	30,000

Without knowing what exactly will be the state of the company in the coming one year, the company will either have to sign a TV ad contract with some high profile individual, who will be the company's brand ambassador for its small car segment for the next one year, or go for a TV ad without featuring any high profile individual. It incurs a cost of ₹3.45 lakh (excluding the payment to the brand ambassador) to put a car on the road.

When the company's profit is ascertain, the company makes decisions on basis of its expected profit. If the company can earn a profit x_i with probability p_i (the probability depends on the state of economy), then the expected profit of the company is $\Sigma_i x_i p_i$.

8. The maximum that the company can afford to pay its brand ambassador is:
(a) ₹10.0 crores (b) ₹10.6 crores
(c) ₹10.8 crores (d) ₹12.0 crores
(e) ₹16.4 crores

9. Mr. Khan, a popular film actor, agrees to sign the contract to become the company's brand ambassador for ₹9 crores. The cost to the company of putting a car on the road also got escalated. The maximum escalation in cost of putting a car on the road, for which the company can afford to sign the contract with Mr. Khan, is:
(a) ₹900 (b) ₹967
(c) ₹1250 (d) ₹1267
(e) ₹1333

10. Mr. Khan, a popular film actor, agrees to sign the contract to become the company's brand ambassador for ₹9 crores. The cost to the company of putting a car on the road also got escalated by ₹1000. If the company signs the contract with Mr. Khan, then its profit will
(a) increase by ₹40 lakh. (b) increase by ₹60 lakh.
(c) decrease by ₹20 lakh. (d) decrease by ₹40 lakh.
(e) decrease by ₹50 lakh.

11. Determine the value(s) of 'a' for which the point (a, a^2) lies inside the triangle formed by the lines $2x + 3y = 1$, $x + 2y = 3$, and $5x - 6y = 1$.
(a) $(-3, -1) \cup (1/2, 1)$ (b) $(-\infty, 1/3) \cup (1/2, \infty)$
(c) $(-3/2, -1) \cup (1/2, 1)$ (d) $(-\infty, 1) \cup (1/3, 6)$
(e) None of these

12. The supervisor of a packaging unit of a milk plant is being pressurised to finish the job closer to the distribution time, thus giving the production staff more leeway to cater to last minute demand. He has the option of running the unit at normal speed or at 110% of normal, that is, 'fast speed'. He estimates that he will be able to run at a higher speed 60% of time. The packet is twice as likely to be damaged at the higher speed, which would mean temporarily stopping the process. If a packet on a randomly selected packaging runs has probability of 0.112 of damage, then what is the probability that the packet will not be damaged at normal speed?
(a) 0.81 (b) 0.93
(c) 0.75 (d) 0.60
(e) None of these

Direction for Questions 13 to 14:

Let $A_1, A_2, \ldots A_n$ be n points on the straight line $y = px + q$. The coordinates of A_k is (x_k, y_k), where k = 1, 2, ..., n such that $x_1, x_2, \ldots x_n$ are in arithmetic progression. The coordinates of A_2 is $(2, -2)$ and A_{24} is $(68, 31)$.

13. The y-ordinates of A_8 is:
(a) 13 (b) 10 (c) 7 (d) 5.5
(e) None of these

14. The number of point(s) satisfying the abovementioned characteristics and not in the first quadrant is/are:
(a) 1 (b) 2 (c) 3 (d) 7
(e) None of these

15. The operation (x) is defined by:
(i) $(1) = 2$
(ii) $(x + y) = (x) \cdot (y)$
for all positive integers x and y.
If $\sum_{x=1}^{n} (x) = 1022$, then find n.
(a) 8 (b) 9 (c) 10 (d) 11
(e) None of these

16. Amarendra and Dharmendra are brothers. One day they start at the same time from their home for Tatanagar railway station in their respective cars. Amarendra took 25 min to reach the station. After reaching the station, Amarendra found that Dharmendra

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is 2500 m away from the station. The distance of Tatanagar station from their home is 15 km. Next day, Dharmendra decided to start 7 min early. If they drive at the speed same as the previous day, then Amarendra will reach the station

(a) 120 s earlier than Dharmendra.
(b) 120 s later than Dharmendra.
(c) 300 s earlier than Dharmendra.
(d) 300 s later than Dharmendra.
(e) at the same time with Dharmendra.

17. Let S_1, S_2,, ... be the squares such that for each $n \geq 1$, the length of the diagonal of S_n is equal to the length of the side of S_{n+1}. If the length of the side of S_3 is 4 cm, then what is the length of the side of S_n?

(a) $2^{\left(\frac{2n+1}{2}\right)}$ (b) $2(n-1)$ (c) $2^{(n-1)}$ (d) $2^{\left(\frac{n+1}{2}\right)}$
(e) None of these

18. In a clock having a circular scale of 12 h, when time changes from 7:45 A.M. to 7:47 A.M., by how many degrees the angle formed by the hour hand and minute hand changes?

(a) 10 (b) 11 (c) 12 (d) 15
(e) None of these

Questions 19 and 20 are followed by two statements labelled as I and II. Decide if these statements are sufficient to conclusively answer the question. Choose the appropriate answer from the options given below:

A. Statement I alone is sufficient to answer the question.
B. Statement II alone is sufficient to answer the question.
C. Statement I and Statement II together are sufficient, but neither of the two alone is sufficient to answer the question.
D. Either Statement I or Statement II alone is sufficient to answer the question.
E. Both Statements I and II are insufficient to answer the question.

19. In the trapezoid PQRS, PS is parallel to QR. PQ and SR are extended to meet at A. What is the value of ∠PAS?

I. PQ = 3, RS = 4, and ∠QPS = 60°
II. PS = 10, QR = 5

20. A sequence of positive integer is defined as $A_{n+1} = A_n^2 + 1$ for each $n \geq 0$. What is the value of greatest common divisor of A_{900} and A_{1000}?

I. $A_0 = 1$ II. $A_1 = 2$

21. a, b, c, d, and e are integers such that $1 \leq a < b < c < d < e$. If a, b, c, d, and e are in geometric progression and lcm (m, n) is the least common multiple of m and n, then find the maximum value of

$$\frac{1}{\text{lcm}(a,b)} + \frac{1}{\text{lcm}(b,c)} + \frac{1}{\text{lcm}(c,d)} + \frac{1}{\text{lcm}(d,e)}.$$

(a) 1 (b) $\frac{15}{16}$ (c) $\frac{79}{81}$ (d) $\frac{7}{8}$
(e) None of these

Direction for Questions 22 and 23:

Books and More sells books, music CDs, and film DVDs. In December 2009, they earned 40% profit in music CDs and 25% profit in books. Music CDs contributed 35% towards their total sales in rupees. At the same time, total sales in rupees from books is 50% more than that of music CDs.

22. If *Books and More* have earned 20% profit overall, then in film DVDs they made

(a) 15.2% profit (b) 10.0% profit
(c) 10.0% loss (d) 16.3% loss
(e) 23.4% loss

23. If *Books and More* made 50% loss in film DVDs, then overall they made

(a) 12.3% profit (b) 8.7% profit
(c) 0.4% loss (d) 6.25% loss
(e) 20% loss

24. ABCD is a parallelogram with ∠ABC = 60°. If the longer diagonal is of length 7 cm and the area of the parallelogram ABCD is $15\frac{\sqrt{3}}{2}$ sq. cm, then find the perimeter of the parallelogram (in cm).

(a) 15 (b) $15\sqrt{3}$ (c) 16 (d) $16\sqrt{3}$
(e) None of these

25. If all letters of the word 'CHCJL' be arranged in an English dictionary, then what will be the 50th word?

(a) HCCLJ (b) LCCHJ
(c) LCCJH (d) JHCLC
(e) None of these

26. A manufacturer produces two types of products A and B, which are subjected to two types of operations, namely grinding and polishing. Each unit of product A takes 2 h of grinding and 3 h of polishing, whereas product B takes 3 h of grinding and 2 h of polishing. The manufacturer has 10 grinders and 15 polishers. Each grinder operates for 12 h/day and each polisher operates 10 h/day. The profit margin per unit of A and B are ₹5 and ₹7, respectively. If the manufacturer utilises all his resources for producing these two types of items, then what is the maximum profit that the manufacturer can earn?

(a) ₹280 (b) ₹294 (c) ₹515 (d) ₹550
(e) None of these

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27. A tank internally measuring 150 cm × 120 cm × 100 cm has 12,81,600 cm^3 water in it. Porous bricks are placed in the water until the tank is full up to its brim. Each brick absorbs one-tenth of its volume of water. How many bricks, of 20 cm × 6 cm × 4 cm, can be put in the tank without spilling over the water?
(a) 1100 (b) 1200 (c) 1150 (d) 1250
(e) None of these

28. The chance of India winning a cricket match against Australia is 1/6. What is the minimum number of matches India should play against Australia so that there is a fair chance of winning at least one match?
(a) 3 (b) 4 (c) 5 (d) 6
(e) None of these

29. A chocolate dealer has to send chocolates of three brands to a shopkeeper. All the brands are packed in boxes of same size. The number of boxes to be sent is 96 of Brand A, 240 of Brand B, and 336 of Brand C. These boxes are to be packed in cartons of same size containing equal number of boxes. Each carton should contain boxes of same brand of chocolates. What could be the minimum number of cartons that the dealer has to send?
(a) 20 (b) 14 (c) 42 (d) 38
(e) 16

Answers

1. (d)	**2.** (c)	**3.** (a)	**4.** (e)	**5.** (c)	**6.** (a)	**7.** (c)	**8.** (d)	**9.** (c)	**10.** (b)
11. (c)	**12.** (b)	**13.** (c)	**14.** (c)	**15.** (b)	**16.** (b)	**17.** (d)	**18.** (b)	**19.** (a)	**20.** (d)
21. (b)	**22.** (e)	**23.** (b)	**24.** (c)	**25.** (c)	**26.** (b)	**27.** (b)	**28.** (b)	**29.** (b)	

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XAT

Model XAT Paper 2

(Based upon Previous Years' XAT Paper)

1. In a locality, there are 10 houses in a row. On a particular night, a thief planned to steal from three houses of the locality. In how many ways can he plan such that no two of them are next to each other? *(1 mark)*
(a) 56 (b) 73 (c) 80 (d) 120
(e) None of these

2. If $= x\left(9+4\sqrt{5}\right)^{48} = [x] + f$, where $[x]$ is defined as integral part of x and f is a faction, then $x\,(1-f)$ equals: **(***5 marks***)**
(a) 1 (b) Less than 1
(c) More than 1 (d) Between 1 and 2
(e) None of these

3. Let a_n = 1 1 1 1 1 1 1… 1, where 1 occurs n number of times. Then,
(i) a_{741} is not a prime.
(ii) a_{534} is not a prime.
(iii) a_{123} is not a prime.
(iv) a_{77} is not a prime. *(5 marks)*
(a) (i) is correct.
(b) (i) and (ii) are correct.
(c) (ii) and (iii) are correct.
(d) All of them are correct.
(e) None of them is correct.

4. A straight line through point P of a triangle PQR intersects the side QR at the point S and the circumcircle of the triangle PQR at the point T. lf S is not the centre of the circumcircle, then which of the following is true? **(***3 marks***)**
(a) (1/PS) + (1/ST) < 2/√(QS)(QR)
(b) (1/PS) + (1/ST) < 4/QR
(c) (1/PS) + (1/ST) > 1√(QS)(QR)
(d) (1/PS) + (1/ST) > 4/QR
(e) None of these

5. What is the maximum possible value of (21 sin X + 72 cos X)? *(3 marks)*
(a) 21 (b) 57 (c) 63 (d) 75
(e) None of these

6. The scheduling officer for a local police department is trying to schedule additional patrol units in each of two neighbourhoods—southern and northern. She knows that on any given day, the probabilities of major crimes and minor crimes being committed in the northern neighbourhood were 0.418 and 0.612, respectively, and that the corresponding probabilities in the southern neighbourhood were 0.355 and 0.520. Assuming that all crimes occur independent of each other and likewise that crime in the two neighbourhoods are independent of each other, what is the probability that no crime of either type is committed in either neighbourhood on any given day? *(2 marks)*
(a) 0.069 (b) 0.225 (c) 0.69 (d) 0.775
(e) None of these

Answer Questions 7 and 8 based on the following information.

A man standing on a boat south of a lighthouse observes his shadow to be 24 m long, as measured at the sea level. On sailing 300 m eastwards, he finds his shadow as 30 m long, measured in a similar manner. The height of the man is 6 m above sea level.

7. The height of the lighthouse above the sea level is: *(3 marks)*
(a) 90 m (b) 94 m
(c) 96 m (d) 100 m
(e) 106 m

8. What is the horizontal distance of the man from the lighthouse in the second position? *(2 marks)*
(a) 300 m (b) 400 m
(c) 500 m (d) 600 m
(e) None of these

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9. A 25-ft long ladder is placed against the wall with its base 7 ft from the wall. The base of the ladder is drawn out so that the top comes down by half the distance that the base is drawn out. This distance is in the range: (*2 marks*)

(a) (2, 7) (b) (5, 8) (c) (9, 10) (d) (3, 7)
(e) None of these

10. The domain of the function $f(x) = \log_7\{\log_3(\log_5(20x - x^2 - 91))\}$ is: (*3 marks*)

(a) (7, 13) (b) (8, 12)
(c) (7, 12) (d) (12, 13)
(e) None of these

11. There are four machines in a factory. At exactly 8 pm, when the mechanic is about to leave the factory, he is informed that two of the four machines are not working properly. The mechanic is in a hurry and decides that he will identify the two faulty machines before going home and repair them next morning. It takes him 20 min to walk to the bus stop. The last bus leaves at 8:32 pm. If it takes 6 min to identify whether a machine is defective or not, and if he decides to check the machines at random, then what is the probability that the mechanic will be able to catch the last bus? **(***2 marks***)**

(a) 0 (b) 1/6
(c) 1/4 (d) 1/3
(e) 1

12. In a city, there is a circular park. There are four points of entry into the park, namely P, Q, R, and S. Three paths were constructed which connected the points PQ, RS, and PS. The length of the path PQ is 10 units, and the length of the path RS is 7 units. Later, the municipal corporation extended the paths PQ and RS past Q and R, respectively, and they meet at a point T on the main road outside the park. The path from Q to T measures 8 units, and it was found that the angle PTS is 60°. Find the area (in square units) enclosed by the paths PT, TS, and PS. (*2 marks*)

(a) $36\sqrt{3}$ (b) $54\sqrt{3}$ (c) $72\sqrt{3}$ (d) $90\sqrt{3}$
(e) None of these

Answer question numbers 13 and 14 based on the following information.

From a group of 545 contenders, a party has to select a leader. Even after holding a series of meetings, the politicians and the general body failed to reach a consensus. It was then proposed that all 545 contenders be given a number from 1 to 545. Then they will be asked to stand on a podium in a circular arrangement, and counting would start from the contender numbered 1. The counting would be done in a clockwise fashion. The rule is that every alternate contender would be asked to step down as the counting continued, with the circle getting smaller and smaller, till only one person remains standing. Therefore, the first person to be eliminated would be the contender numbered 2.

13. Which position should a contender choose if he has to be the leader? (*5 marks*)

(a) 3 (b) 67 (c) 195 (d) 323
(e) 451

14. One of the contending politicians, Mr Chanaya, was quite proficient in calculations and could correctly figure out the exact position. He was the last person remaining in the circle. Sensing foul play, the politicians decided to repeat the game. However, this time, instead of removing every alternate person, they agreed on removing every 300th person from the circle. All other rules were kept intact. Mr Chanaya did some quick calculations and found that for a group of 542 people the right position to become a leader would be 437. What is the right position for the whole group of 545 as per the modified rule? (*5 marks*)

(a) 3 (b) 194 (c) 249 (d) 437
(e) 543

15. Little Pika who is five-and-half years old has just learnt addition. However, he does not know how to carry. For example, he can add 14 and 5, but he does not know how to add 14 and 7. How many pairs of consecutive integers between 1000 and 2000 (both 1000 and 2000 included) can Little Pika add? (*2 marks*)

(a) 150 (b) 155 (c) 156 (d) 258
(e) None of these

16. The micromanometer in a certain factory can measure the pressure inside the gas chamber from 1 unit to 999,999 units. Lately this instrument has not been working properly. The problem with the instrument is that it always skips the digit 5 and moves directly from 4 to 6. What is the actual pressure inside the gas chamber if the micromanometer displays 003016?

(*3 marks***)**

(a) 2201 (b) 2202
(c) 2600 (d) 2960
(e) None of these

17. Consider a square ABCD of side 60 cm. lt contains arcs BD and AC drawn with centres at A and D, respectively. A circle is drawn such that it is tangent to side AB and the arcs BD and AC. What is the radius of the circle? (*5 marks*)

(a) 9 cm (b) 10 cm (c) 12 cm (d) 15 cm
(e) None of these

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18. In a plane rectangular coordinate system, points L, M, N, and O are represented by the coordinates (–5, 0), (1, –1), (0, 5), and (–1, 5), respectively. Consider a variable point P in the same plane. Find the minimum value of PL + PM + PN + PO. (*2 marks*)

(a) $1+\sqrt{37}$ (b) $5\sqrt{2}+2\sqrt{10}$

(c) $\sqrt{41}+\sqrt{37}$ (d) $\sqrt{41}+1$

(e) None of these

19. In a bank the account numbers are all 8-digit numbers, and they all start with the digit 2. So, an account number can be represented as $2x_1x_2x_3x_4x_5x_6x_7$. An account number is considered to be a 'magic' number if $x_1x_2x_3$ is exactly the same as $x_4x_5x_6$ or $x_5x_6x_7$ or both. x_i can take values from 0 to 9, but 2 followed by seven 0s is not a valid account number. What is the maximum possible number of customers having a 'magic' account number? (*3 marks*)

(a) 9989 (b) 19,980

(c) 19,989 (d) 19,999

(e) 19,990

20. In a list of 7 integers, one integer denoted as x is unknown. The other six integers are 20, 4, 10, 4, 8, and 4. If the mean, median, and mode of these seven integers are arranged in increasing order, then they form an arithmetic progression. Find the sum of all possible values of x. (*5 marks*)

(a) 26 (b) 32 (c) 34 (d) 38

(e) 40

21. Prof. Bee noticed something peculiar while entering the quiz marks of his five students into a spread sheet. The spread sheet was programmed to calculate the average after each score was entered. Prof. Bee entered the marks in a random order and noticed that after each mark was entered, the average was always an integer. In ascending order, the marks of the students were 71, 76, 80, 82, and 91. What were the fourth and fifth marks that Prof. Bee entered? (*3 marks*)

(a) 71 and 82 (b) 71 and 76

(c) 71 and 80 (d) 76 and 80

(e) 91 and 80

22. Rakhal is looking for a field where he can graze his cow. He finds a local farmer, Gopal, who agrees to rent his field to Rakhal for ₹1000 a year. Rakhal finds a post in the field and ties his cow to the post with a 25-ft rope. After some months, Gopal tells Rakhal that he will build a shed with four walls on the field with the post as one of the corner posts. The shed would be 15 ft by 10 ft. Rakhal agrees but he realizes that this arrangement would reduce the available area for grazing. What should be the modified rent to compensate for this loss of grazing area if Rakhal has to keep the cow tied to the same post with the same rope? (*2 marks*)

(a) ₹800 (b) ₹880

(c) ₹888 (d) ₹930

(e) None of these

Answers

1. (a)	**2.** (a)	**3.** (d)	**4.** (d)	**5.** (d)	**6.** (a)	**7.** (e)	**8.** (c)	**9.** (e)	**10.** (b)
11. (d)	**12.** (c)	**13.** (b)	**14.** (c)	**15.** (c)	**16.** (a)	**17.** (b)	**18.** (b)	**19.** (c)	**20.** (e)
21. (c)	**22.** (b)								

Hints and Solutions

1. Out of the 10 houses, we first place the 7 houses in which the thief will not steal. There are 8 spaces in between and at either ends of these 7 houses, where we can place the three houses in which he plans to steal. Thus, the three houses can be selected in ${}^8C_3 = 56$ ways.

Hence, option (a) is the answer.

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2. $x = \left(9+4\sqrt{5}\right)^{48}$

Let $y = \left(9-4\sqrt{5}\right)^{48}$

Now, $\left(9+4\sqrt{5}\right)^{48} \times \left(9-4\sqrt{5}\right)^{48} = (81-80)^{48} = 1$ (i)

Also, $\left(9+4\sqrt{5}\right)^{48} + \left(9-4\sqrt{5}\right)^{48}$

$$= \left[{}^{48}C_0 9^{48} + {}^{48}C_1 9^{47}\left(4\sqrt{5}\right) + {}^{48}C_2 9^{46}\left(4\sqrt{5}\right)^2 + \cdots + {}^{48}C_{47}(9)\left(4\sqrt{5}\right)^{47} + {}^{48}C_{48}\left(4\sqrt{5}\right)^{48}\right]$$

$$+ \left[{}^{48}C_0 9^{48} - {}^{48}C_1 9^{47}\left(4\sqrt{5}\right) + {}^{48}C_2 9^{46}\left(4\sqrt{5}\right)^2 - \cdots - {}^{48}C_{47}(9)\left(4\sqrt{5}\right)^{47} + {}^{48}C_{48}\left(4\sqrt{5}\right)^{48}\right]$$

$$= 2\left[{}^{48}C_0 9^{48} + {}^{48}C_2 9^{46}\left(4\sqrt{5}\right)^2 + \cdots {}^{48}C_{48}\left(4\sqrt{5}\right)^{48}\right]$$

$\therefore \quad x + y = 2(k) = \text{even}$

Now,

$0 < 9 - 4\sqrt{5} < 1$

$\therefore \quad 0 < \left(9-4\sqrt{5}\right)^{48} < 1$

$\therefore \quad 0 < y < 1$ (ii)

Also, $x = [x] + f,\ 0 < f < 1$ (iii)

$\therefore \quad [x] + f + y$ is even

As $[x]$ is an integer, $f + y$ is an integer.

From (ii) and (iii), we get

$0 < f + y < 2$

$\therefore \quad f + y = 1$

Now, $x(1-f) = xy$

But from (i), $xy = 1$

$\therefore \quad x(1-f) = 1$

Hence, option (a) is the answer.

3. a_{741} has 1 written 741 times.

$\therefore$ The sum of digits of a_{741} is divisible by 3.

$\therefore$ a_{741} is divisible by 3.

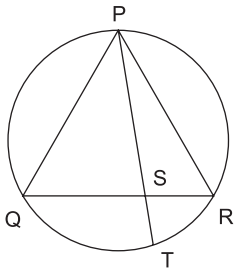
$\therefore$ a_{534} and a_{123} are also not prime by the same logic.

$\therefore$ (i), (ii), and (iii) are correct.

Option (d) can be safely assumed to be the correct option.

Hence, option (d) is the answer.

4.



As S is not the circumcentre, PS ≠ ST and QS ≠ SR

$\because$ PT and QR are chords of the circle intersecting at S.

$PS \times ST = QS \times SR$ (i)

We know that arithmetic mean ≥ geometric mean

$$\therefore \quad \frac{PS+ST}{2} \geq \sqrt{PS \times ST}$$

But, as PS ≠ ST,

$$\frac{PS+ST}{2} > \sqrt{PS \times ST}$$

$$\therefore \quad \frac{PS+ST}{2} > \sqrt{QS \times SR}$$

$$\therefore \quad \frac{PS+ST}{2} > 2\sqrt{QS \times SR}$$

$$\therefore \quad \frac{PS+ST}{PS \times ST} > \frac{2\sqrt{QS \times SR}}{QS \times SR}$$

$$\frac{1}{PS} + \frac{1}{ST} > \frac{2}{\sqrt{QS \times SR}} \qquad \text{(i)}$$

$\therefore$ Option (a) is false.

Also, $\dfrac{QS+SR}{2} > \sqrt{QS \times SR}$

$$\therefore \quad \frac{2}{QR} < \frac{1}{\sqrt{QS \times SR}}$$

$$\therefore \quad \frac{4}{QR} < \frac{2}{\sqrt{QS \times SR}}$$

$$\frac{1}{PS} + \frac{1}{ST} > \frac{2}{\sqrt{QS \times SR}} > \frac{4}{QR} \quad \text{from (i)}$$

Hence, option (d) is the answer.

Note: As the result is a general one, we can, without loss of generality, consider an equilateral triangle PQR with point S being the mid-point of QR and verify all the options using numbers.

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5. Let $f(X) = 21 \sin X + 72 \cos X$

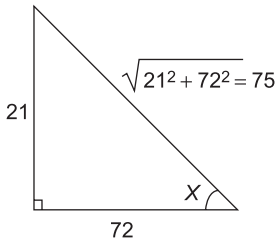
$\therefore \quad f'(X) = 21 \cos X - 72 \sin X$

At $f'(X) = 0$

$21 \cos X = 72 \sin X$

$\therefore \quad \tan x = \dfrac{21}{72}$

Drawing the corresponding right triangle, we have the following:



$\therefore \quad f''(X) = -21 \sin X - 72 \cos X$

$= -21 \times \dfrac{7}{25} - 72 \times \dfrac{24}{25} < 0$

$\therefore$ $f(X)$ has a maximum at $f'(X) = 0$

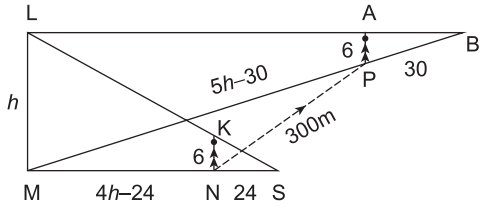
$\therefore$ Maximum value of $(X) = 21 \times \dfrac{21}{75} + 72 \times \dfrac{21}{75} = 75$

Hence, option (d) is the answer.

6. Required probability = $(1 - 0.418)(1 - 0.612)(1 - 0.355)(1 - 0.520) \approx 0.069$

Hence, option (a) is the answer.

7.



Let LM denote the light house of height h above the sea level.

Let KN denote the man and MN denote the south direction.

NS is the shadow of the man.

Then, KN = 6 and NS = 24

Also, $\angle$ KNS = 90° and $\angle$ LMS = 90°

By similarity of ΔLMS and ΔKNS,

$\dfrac{LM}{MS} = \dfrac{6}{24} = \dfrac{1}{4}$

$\therefore$ If LM = h, then MS = $4h$ and MN = $4h - 24$

The boat moves from N to P along the east.

$\therefore$ NP = 300

The man's new position is AP.

$\therefore$ AP = 6, PB = 30

$\Delta APB \sim \Delta LMB$

$\therefore \dfrac{LM}{MB} = \dfrac{6}{30} = \dfrac{1}{5}$

$\therefore$ MP = $5h - 30$

But, $MN^2 + 300^2 = MP^2$

$\therefore \quad 16(h-6)^2 + 300^2 = 25(h-6)^2$

$\therefore \quad 300^2 = 25(h-6)^2 - 16(h-6)^2$

$\therefore \quad 90{,}000 = 9(h-6)^2$

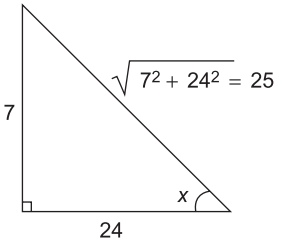
$\therefore \quad h = 106$

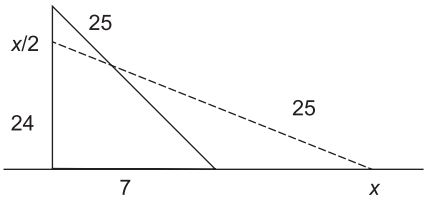
Hence, option (e) is the answer.

8. Refer first question's answer of this set. The horizontal distance of the man from the light house in the second position = $5h - 30 = 500$ m

Hence, option (c) is the answer.

9.





Let the base is drawn by x unit.

$\therefore$ Height of top of the ladder will decrease by = $\dfrac{x}{2}$

$\therefore$ By Pythagoras theorem,

$\therefore \quad \left(24 - \dfrac{x}{2}\right)^2 + (7 + x)^2 = 625$

$\therefore \quad \dfrac{x^2}{4} + x^2 - 24x + 14x = 0$

$\therefore \quad \dfrac{5x^2}{4} = 10x$

$x = 8$

Hence, option (e) is the answer.

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10. $f(x) = \log_7\{\log_3(\log_5(20x - x^2 - 91))\}$

As log of negative numbers is not defined,

$\log_3(\log_5(20x - x^2 - 91)) > 0$

$\therefore \log_5(20x - x^2 - 91) > 3^0$

$\therefore \log_5(20x - x^2 - 91) > 1$

$\therefore 20x - x^2 - 91 > 5$

$\therefore 20x - x^2 - 96 > 0$

$\therefore x^2 - 20x + 96 < 0$

$\therefore (x - 12)(x - 8) < 0$

$\therefore 8 < x < 12$

Hence, option (b) is the answer.

11. If the mechanic wants to catch the bus, then he will have 12 min to inspect the machines. As inspecting one machine takes 6 min, he will be able to identify the faulty machines if the first two machines he inspects are both faulty or both working properly.

Suppose A, B, C, and D are the machines, and A and B are faulty. He can inspect these machines in ${}^4P_4 = 24$ ways. If he inspects A and B first, he will be able to catch the bus. If he inspects C and D first, he will know that A and B are faulty and still he will be able to catch the bus.

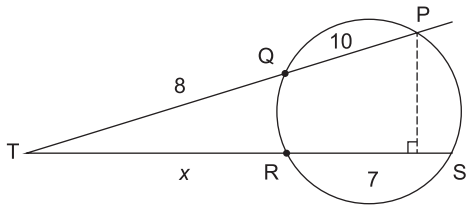
There are 4 ways in which he can inspect A and B first and 4 ways in which he can inspect C and D first.

$\therefore$ Probability that he will be able to catch the bus

$$= \frac{8}{24} = \frac{1}{3}$$

Hence, option (d) is the answer.

12. The following diagram represents the given scenario.



Let, $TR = x$

$\therefore 8 \times 18 = x(7 + x)$

$\therefore x^2 + 7x - 144 = 0$

$\therefore x = 9$ or $x = -16$

But, x cannot be negative.

Hence, $x = 9$

$$A(\Delta TPS) = \frac{1}{2} \times 18 \times 16 \times \sin 60°$$

$$= 72\sqrt{3} \text{ sq. units}$$

Hence, option (c) is the answer.

13. Let $f(n)$ represent the position of winner when n persons are standing in a circle.

$f(n) = 2l + 1$

Where, $n = 2^m + l$ and $0 \le l < 2^m$

Now, $n = 545$

$\therefore n = 512 + 33$

$\therefore n = 2^9 + 33$

$\therefore l = 33$

$\therefore f(545) = 2 \times 33 + 1 = 67$

Hence, option (b) is the answer.

Note: This question is based on the Josephus Flavius problem.

14. Let $f(n, k)$ represent the position of a winner when there are n people out of which every k^{th} person is eliminated.

We have,

$f(n, k) = (f(n - 1, k) + k) \bmod n$

Now, $f(542, 300) = 437$

Hence,

$f(543, 300) = (437 + 300) \bmod 543 = 194$

$f(544, 300) = (194 + 300) \bmod 544 = 494$

$f(545, 300) = (494 + 300) \bmod 545 = 249$

$\therefore$ A contender at 249th position will win the election.

Hence, option (c) is the answer.

Note: This question is based on the Josephus Flavius problem.

15. Let a, b, and c be three digits belonging to the set $\{0, 1, 2, 3, 4\}$.

The first number of the two consecutive numbers can be of following types:

Type	Count
1abc	$5 \times 5 \times 5 = 125$
1ab9	$5 \times 5 = 25$
1a99	5
1999	1

Therefore, total number of pairs $= 125 + 25 + 5 + 1 = 156$

Hence, option (c) is the answer.

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16. Skipping the digit 5 converts the counting base system to 9 from 10.

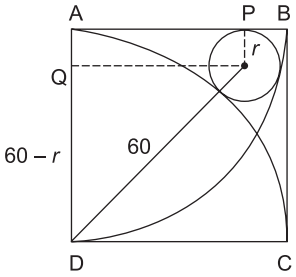
Actual values of digits greater than 5 will change, that is, face value of 6 will be 5, face value of 7 will be 6, and so on.

Hence, 3016 displayed by micromanometer is actually 3015 in base 9.

Now, $(3015)_9 = 3 \times 9^3 + 0 \times 9^2 + 1 \times 9 + 5 = 2201$

Hence, option (a) is the answer.

17.



Let O be the centre of the smaller circle.

Let the small circle touch AB at P. OQ is perpendicular to AD.

Now, AO = 60 – r and OP = r

Now,

$QO^2 = AP^2 = (60 - r)^2 - r^2$ (i)

Now, in ΔDQO,

$QO^2 = DO^2 - DQ^2$

Now, DO = 60 + r and DQ = 60 – r

DQ = 60 – r

$\therefore OQ^2 = (60 + r)^2 - (60 - r)^2$ (ii)

$\therefore (60 - r)^2 + 3600 - 120r = (60 + r)^2$

From equations (i) and (ii), we get

$\therefore r = 10$

Hence, option (b) is the answer.

18. PL + PM will be minimum if P lies on LN. PM + PO will be minimum if P lies on OM.

$\therefore$ PL + PM + PN + PO will be minimum, only if P is the point of intersection of diagonals of the quadrilateral LMNO.

Now, LN = $\sqrt{(-5-0)^2 + (0-5)^2} = 5\sqrt{2}$

MO = $\sqrt{1-(-1)^2 + (-1-5)^2} = 2\sqrt{10}$

$\therefore$ LN + MO = $5\sqrt{2} + 2\sqrt{10}$

$\therefore$ PL + PM + PN + PO = $5\sqrt{2} + 2\sqrt{10}$

Hence, option (b) is the answer.

19. $x_1x_2x_3x_4x_5x_6x_7$ can be of the form *abcdabc* or *abcabcd*.

abc can be chosen in $10 \times 10 \times 10 = 1000$ ways.

d can be chosen in 10 ways.

$\therefore x_1x_2x_3x_4x_5x_6x_7$ can be chosen in $2 \times 1000 \times 10 = 20000$ ways.

However, this includes the ways in which all of *a*, *b*, *c*, and *d* are 0, 1, 2, 3, …, 9, twice.

We subtract these numbers to get 20,000 – 10 = 19,990

We also do not want 20,00,000.

$\therefore$ The total number of ways is 19,989.

Hence, option (c) is the answer.

20. It can be observed that, irrespective of the value of x, mode of these numbers will be 4.

Now, the median of these numbers will depend on the value of x.

If $x < 4$, then the median of these seven numbers will be 4.

Now, as the mode is 4, the median cannot be 4.

(The question states that mean, median, and mode are arranged in ascending order.)

Hence, x cannot be less than 4.

Now,

If $4 < x \leq 8$, then the median will be x and the mean will be $\frac{50+x}{7}$.

Now, $\frac{50+x}{7} > 7$ and $x \leq 8$

$\therefore$ 4, x, and $\frac{50+x}{7}$ will form an AP only if $\frac{50+x}{7} > x$

$\therefore \frac{50+x}{7} - x = x - 4$

$\therefore \frac{50+x}{7} + 4 = 2x$

$\therefore x = 6$

Hence, $x = 6$ is a possible answer.

Now, if $x > 8$, then median will be 8 and mean will be $\frac{50+x}{7}$.

Now, if $x > 8$, then $\frac{50+x}{7}$ is greater than 8.

$\therefore$ Increasing order of mean, median, and mode will be 4, 8, and $\frac{50+x}{7}$, respectively.

Now, they are in arithmetic progression.

$\therefore 8 - 4 = \frac{50+x}{7} - 8$

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$\therefore 12 = \frac{50 + x}{7}$

$\therefore x = 12 \times 7 - 50$

$\therefore x = 34$

Therefore, sum of all possible values of $x = 6 + 34 = 40$

Hence, option (e) is the answer.

21. It is convenient to solve this question by evaluating the options.

Option (a):

If 71 and 82 are fourth and fifth numbers, then sum of first three numbers is 76 + 80 + 91 = 247

But, 247 is not divisible by 3.

Hence, they cannot be fourth and fifth numbers.

$\therefore$ Option (a) is not possible.

Option (b):

Fourth and fifth numbers are 71 and 76.

Sum of first three numbers will be:

80 + 91 + 82 = 253

It is also not divisible by 3.

Hence, option (b) is also not possible.

Option (c):

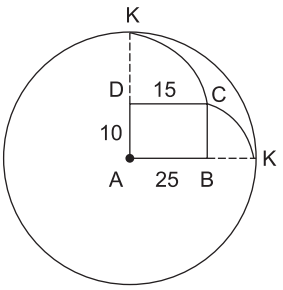
Fourth and fifth numbers are 71 and 80.

Therefore, sum of first three numbers will be 76 + 82 + 91 = 249, which is divisible by 3.

So, they can be fourth and fifth numbers.

Hence, option (c) is the answer.

22.



Area available after building the shed = area of major sector AKP + area of sector BPC + area of sector KDC

$\frac{3}{4} \times \pi \times 25^2 + \frac{1}{4} \times \pi \times 10^2 + \frac{1}{4} \times \pi \times 15^2 = 550\pi$

Initial area = 625π

Now, area of 625π costs ₹1000.

$\therefore$ Area of 550π will cost $\frac{1000}{625} \times 550 =$ ₹880

Hence, option (b) is the answer.

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IIFT

Model IIFT Paper

(Based upon Previous Years' IIFT Paper)

1. Find the sum of the following series:

$$\frac{2}{1!}+\frac{3}{2!}+\frac{6}{3!}+\frac{11}{4!}+\frac{18}{5!}+\ldots.$$

(a) $3e-1$ (b) $3(e-1)$ (c) $3(e+1)$ (d) $3e+1$

2. How many positive integers 'n' can be formed using the digits 3, 4, 4, 5, 6, 6, 7 if we want 'n' to exceed 60,00,000?

(a) 320 (b) 360 (c) 540 (d) 720

3. A techno company has 14 machines of equal efficiency in its factory. The annual manufacturing costs are ₹42,000 and establishment charges are ₹70,000. The annual output and manufacturing costs are directly proportional to the number of machines. The shareholders get 12.5% profit, which is directly proportional to the annual output of the company. If 7.14% machines remain closed throughout the year, then the percentage decrease in the amount of profit of the shareholders would be:

(a) 12% (b) 12.5%
(c) 13.0% (d) None of these

4. Sun Life Insurance Company issues standard, preferred, and ultra-preferred policies. Among the company's policy holders of a certain age, 50% are standard with a probability of 0.01 of dying in the next year, 30% are preferred with a probability 0.008 of dying in the next year, and 20% are ultra-preferred with a probability of 0.007 of dying in the next year. If a policy holder of that age dies in the next year, what is the probability of the deceased being a preferred policy holder?

(a) 0.1591 (b) 0.2727
(c) 0.375 (d) None of these

5. A metro train from Mehrauli to Gurgaon has capacity to board 900 people. The fare charged (in ₹) is defined by the function

$$f=\left(54-\frac{x}{32}\right)^2$$

where 'x' is the number of people per trip. How many people per trip will make the marginal revenue equal to zero?

(a) 1728 (b) 576 (c) 484 (d) 364

6. If each α, β, and γ is a positive acute angle such that $\sin(\alpha+\beta-\gamma)=\frac{1}{\sqrt{2}}$, $\operatorname{cosec}(\beta+\gamma-\alpha)=\frac{2}{\sqrt{3}}$, and $\tan(\gamma+\alpha-\beta)=\frac{1}{\sqrt{3}}$. What are the values of α, β, and γ?

(a) $\left(37\frac{1}{2},52\frac{1}{2},45\right)$ (b) (37, 53, 45)

(c) $9\frac{2}{7}\text{Hrs}\left(45,37\frac{1}{2},52\frac{1}{2}\right)$ (d) $\left(34\frac{1}{2},55\frac{1}{2},45\right)$

7. Shyam, Gopal, and Madhur are three partners in a business. Their capitals are, respectively, ₹4000, ₹8000, and ₹6000. Shyam gets 20% of total profit for managing the business. The remaining profit is divided among the three in the ratio of their capitals. At the end of the year, the profit of Shyam is ₹2200 less than the sum of the profit of Gopal and Madhur. How much profit, Madhur will get?

(a) ₹1600 (b) ₹2400
(c) ₹3000 (d) ₹5000

8. In how many ways can four letters of the word 'SERIES' be arranged?

(a) 24 (b) 42 (c) 84 (d) 102

9. The area of a triangle is 6; two of its vertices are (1, 1) and (4, –1), and the third vertex lies on $y=x+5$. Find the third vertex.

(a) $\left(\frac{2}{5},\frac{27}{5}\right)$ (b) $\left(-\frac{3}{5},\frac{22}{5}\right)$

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(c) $\left(\frac{3}{5}, \frac{28}{3}\right)$ (d) None of these

10. A small confectioner bought a certain number of pastries flavoured pineapple, mango, and black-forest from a bakery, giving for each pastry as many rupees as there were pastry of that kind; altogether, he bought 23 pastries and spent ₹211. Find the number of each kind of pastry that he bought, if mango pastry are cheaper than pineapple pastry and dearer than black-forest pastry.
(a) (10, 9, 4) (b) (11, 9, 3)
(c) (10, 8, 5) (d) (11, 8, 4)

11. Find the root of the quadratic equation $bx^2 - 2ax + a = 0$.

(a) $\frac{\sqrt{b}}{\sqrt{b \pm \sqrt{a-b}}}$ (b) $\frac{\sqrt{a}}{\sqrt{b \pm \sqrt{a-b}}}$

(c) $\frac{\sqrt{a}}{\sqrt{a \pm \sqrt{a-b}}}$ (d) $\frac{\sqrt{a}}{\sqrt{a \pm \sqrt{a+b}}}$

12. Three Professors Dr Gupta, Dr Sharma, and Dr Singh are evaluating answer scripts of a subject. Dr Gupta is 40% more efficient than Dr Sharma, who is 20% more efficient than Dr Singh. Dr Gupta takes 10 days less than Dr Sharma to complete the evaluation work. Dr Gupta starts the evaluation work and works for 10 days and then Dr Sharma takes over. Dr Sharma evaluates for next 15 days and then stops. In how many days, Dr Singh can complete the remaining evaluation work?
(a) 7.2 days (b) 9.5 days
(c) 11.5 days (d) None of these

13. If $[x]$ is the greater integer less than or equal to 'x', then find the value of the following series:

$$\left[\sqrt{1}\right]+\left[\sqrt{2}\right]+\left[\sqrt{3}\right]+\left[\sqrt{4}\right]+\ldots+\left[\sqrt{361}\right]$$

(a) 4389 (b) 4839 (c) 3498 (d) 3489

14. What is the value of $\sqrt{\frac{a}{b}}$, if

$\log_4 \log_4 4^{a-b} = 2\log_4\left(\sqrt{a}-\sqrt{b}\right)+1$?

(a) −5/3 (b) 2 (c) 5/3 (d) 1

15. Three pipes A, B, and C are connected to a tank. These pipes can fill the tank separately in 5 h, 10 h, and 15 h, respectively. When all the three pipes were opened simultaneously, it was observed that pipes A and B were supplying water at 3/4th of their normal rates for the first hour after which they supplied water at the normal rate. Pipe C supplied water at 2/3rd of its normal rate for first 2 h, after which it supplied at its normal rate. In how much time, the tank would be filled?
(a) 1.05 h (b) 2.05 h
(c) 3.05 h (d) None of these

16. The minimum value of $3^{\sin x} + 3^{\cos x}$ is:

(a) 2 (b) $2\left(3^{-\frac{1}{\sqrt{2}}}\right)$

(c) $3^{1-\frac{1}{\sqrt{2}}}$ (d) None of these

17. In a B-School, there are three levels of faculty positions, that is, Professor, Associate Professor, and Assistant Professor. It is found that the sum of the ages of all faculty present is 2160, their average age is 36; the average age of the Professor and Associate Professor is 39; of the Associate Professor and Assistant Professor is $32\frac{8}{11}$; of the Professor and Assistant Professor is $36\frac{2}{3}$; had each Professor been 1 year older, each Associate Professor 6 years older, and each Assistant Professor 7 years older, then their average age would increase by 5 years. What will be the number of faculty at each level and their average ages?
(a) (16, 24, 20 : 45, 35, 30 years)
(b) (18, 24, 20 : 42, 38, 30 years)
(c) (16, 20, 24 : 50, 30, 30 years)
(d) None of these

18. $\text{Log}_5 2$ is:
(a) An integer (b) A rational number
(c) A prime number (d) An irrational number

19. In a square of side 2 meters, isosceles triangles of equal area are cut from the corners to form a regular octagon. Find the perimeter and area of the regular octagon.

(a) $\frac{16}{2+\sqrt{2}}; \frac{4(1+\sqrt{2})}{3+2\sqrt{2}}$

(b) $\frac{8}{2+\sqrt{2}}; \frac{2(1+\sqrt{2})}{3+2\sqrt{2}}$

(c) $\frac{16}{1+\sqrt{2}}; \frac{3(1+\sqrt{2})}{3+2\sqrt{2}}$

(d) None of these

20. The smallest perfect square that is divisible by 7!
(a) 44,100 (b) 176,400
(c) 705,600 (d) 19,600

21. A survey shows that 61%, 46%, and 29% of the people watched '3 idiots', 'Rajneeti', and 'Avatar', respectively. Twenty-five percentage people watched exactly

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two of the three movies, and 3% watched none. What percentage of people watched all the three movies?
(a) 39% (b) 11% (c) 14% (d) 7%

22. In a triangle ABC, the length of side BC is 295. If the length of side AB is a perfect square, then the length of side AC is a power of 2, and the length of side AC is twice the length of side AB. Determine the perimeter of the triangle.
(a) 343 (b) 487
(c) 1063 (d) None of these

23. In Green View Apartment, the houses of a row are numbered consecutively from 1 to 49. Assuming that there is a value of 'x' such that the sum of the numbers of the houses preceding the house numbered 'x' is equal to the sum of the numbers of the houses following it. Then what will be the value of 'x'?
(a) 21 (b) 30 (c) 35 (d) 42

24. To start a new enterprise, Mr Yogesh has borrowed a total of ₹60,000 from two money lenders with the interest being compounded annually, to be repaid at the end of two years. Mr Yogesh repaid ₹38,800 more to the first money lender compared to the second money lender at the end of two years. The first money lender charged an interest rate, which was 10% more than what was charged by the second money lender. If Mr Yogesh had instead borrowed ₹30,000 from each at the respective initial rates for two years, he would have paid ₹7,500 more to the first money lender compared to the second. Find the money borrowed by Mr Yogesh from first money lender.
(a) 20,000 (b) 35,000
(c) 40,000 (d) 42,000

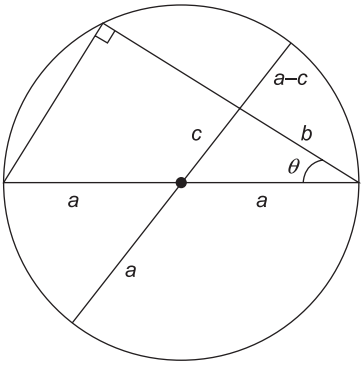
25. Find the coefficient of x^{12} in the expansion of $(1-x^6)^4(1-x)^{-4}$.
(a) 113 (b) 119 (c) 125 (d) 132

26. Mukesh, Suresh, and Dinesh travel from Delhi to Mathura to attend Janmashtami Utsav. They have a bike that can carry only two riders at a time as per traffic rules. Bike can be driven only by Mukesh. Mathura is 300 km from Delhi. All of them can walk at 15 kmph. All of them start their journey from Delhi simultaneously and are required to reach Mathura at the same time. If the speed of bike is 60 kmph, then what is the shortest possible time in which all three can reach Mathura at the same time?
(a) $8\frac{2}{7}$ h (b) $9\frac{2}{7}$ h
(c) 10 h (d) None of these

27. In a rocket-shape firecracker, explosive powder is to be filled up inside the metallic enclosure. The metallic enclosure is made up of a cylindrical base and conical top with the base of radius 8 centimetre. The ratio of height of cylinder and cone is 5:3. A cylindrical hole is drilled through the metal solid with height one third the height of the metal solid. What should be the radius of the hole, so that volume of the hole (in which gun powder is to be filled up) is half of the volume of metal solid after drilling?
(a) $4\sqrt{3}$ cm (b) 4.0 cm
(c) 3.0 cm (d) None of these

28. A small and medium enterprise imports two components A and B from Taiwan and China, respectively, and assembles them with other components to form a toy. Component A contributes to 10% of production cost. Component B contributes to 20% of the production cost. Usually the company sells this toy at 20% above the production cost. Due to increase in the raw material and labour cost in both the countries, component A became 20% costlier and component B became 40% costlier. Owing to these reasons, the company increased its selling price by 15%. Considering that cost of other components does not change, what will be the profit percentage, if the toy is sold at the new price?
(a) 15.5% (b) 25.5% (c) 35.5% (d) 40%

29. What is the value of c^2 in the given figure, where the radius of the circle is 'a' unit?



(a) $c^2 = a^2 + b^2 - 2ab\cos\theta$
(b) $c^2 = a^2 + b^2 - 2ab\sin\theta$
(c) $c^2 = a^2 - b^2 + 2ab\cos\theta$
(d) None of these

30. How many subsets of {1, 2, 3, …, 11} contain at least one even integer?
(a) 1900 (b) 1964 (c) 1984 (d) 2048

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Answers

1. (b)	**2.** (c)	**3.** (d)	**4.** (b)	**5.** (b)	**6.** (a)	**7.** (b)	**8.** (d)	**9.** (a)	**10.** (b)
11. (c)	**12.** (a)	**13.** (a)	**14.** (c)	**15.** (c)	**16.** (b)	**17.** (a)	**18.** (d)	**19.** (d)	**20.** (b)
21. (d)	**22.** (c)	**23.** (c)	**24.** (c)	**25.** (c)	**26.** (b)	**27.** (a)	**28.** (b)	**29.** (a)	**30.** (c)

Hints and Solutions

1. Let $S=\frac{2}{1!}+\frac{3}{2!}+\frac{6}{3!}+\frac{11}{4!}+\frac{18}{5!}+\ldots$

$\therefore \quad S=\sum_{n=1}^{\infty}\frac{n^2-2n+3}{n!}$

$\therefore \quad S=\sum_{n=1}^{\infty}\frac{n}{n-1!}-2\sum_{n=1}^{\infty}\frac{1}{(n-1)!}+\sum_{n=1}^{\infty}\frac{3}{n!}$

$\sum_{n=1}^{\infty}\frac{1}{n!}=\sum_{n=1}^{\infty}\frac{1}{(n-1)!}=e-1$

… (by definition of exponential series)

$\therefore S=\sum_{n=1}^{\infty}\frac{n}{(n-1)!}-2(e-1)+3(e-1)$

$=(e-1)+\sum_{n=1}^{\infty}\frac{n-1+1}{(n-1)!}$

$=(e-1)+\sum_{n=1}^{\infty}\frac{1}{(n-2)!}+\sum_{n=1}^{\infty}\frac{1}{(n-1)!}$

$=(e-1)+(e-1)+(e-1)=3(e-1)$

Hence, option (b) is the answer.

2. As n has to exceed 60,00,000, the first digit of n can be 6 or 7.

Case I The first digit of n is 6. Then, the other six digits are to be chosen from 3, 4, 4, 5, 6, and 7.

They can be arranged in $\frac{6!}{2!}=360$ ways

Case II The first digit of n is 7.

Then, the other six digits are to be chosen from 3, 4, 4, 5, 6, and 6.

They can be arranged in $\frac{6!}{2!\times 2!}=180$ ways

$\therefore$ Total number of arrangements = 360 + 180 = 540

Hence, option (c) is the answer.

3. Profit is directly proportional to the annual output, and the annual output is directly proportional to the number of machines.

$\therefore$ We can say that the profit is directly proportional to the number of machines.

$\therefore$ If 7.14% machines remain closed, the percentage decrease in profit is also 7.14%.

Hence, option (d) is the answer.

4. The required probability

$=\frac{0.3\times 0.008}{0.5\times 0.01+0.3\times 0.008+0.2\times 0.007}$

$=\frac{0.0024}{0.005+0.0024+0.0014}$

$=\frac{0.0024}{0.0088}\approx 0.2727$

Hence, option (b) is the answer.

5. Fare per person $f=\left(54-\frac{x}{32}\right)^2$

$\therefore$ Total fare $f_1=x\left(54-\frac{x}{32}\right)^2$

Marginal revenue is the change in total fare due to change in number of people by one unit. It is given by

$\frac{df_1}{dx}=x\left[2\left(54-\frac{x}{32}\right)\left(-\frac{1}{32}\right)\right]+\left(54-\frac{x}{32}\right)^2$

$=\frac{-x}{16}\left(54-\frac{x}{32}\right)+\left(54-\frac{x}{32}\right)^2$

$=\left(54-\frac{x}{32}\right)\left(54-\frac{3x}{32}\right)$

Now, $\frac{df_1}{dx}=0$

$\therefore\left(54-\frac{x}{32}\right)\left(54-\frac{3x}{32}\right)=0$

$\therefore x=1728$ or $x=576$

But, $x\le 900$

$\therefore x=576$

Hence, option (b) is the answer.

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6. We evaluate the options.
Option (a): Let $\alpha = 37.5$, $\beta = 52.5$, and $\gamma = 45$
$\therefore \alpha + \beta - \gamma = 45$

$$\therefore \sin(\alpha + \beta - \gamma) = \frac{1}{\sqrt{2}}$$

$$\beta + \gamma - \alpha = 60$$

$$\operatorname{cosec}(\beta + \gamma - \alpha) = \frac{2}{\sqrt{3}}$$

$$\gamma + \alpha - \beta = 30$$

$$\tan(\gamma + \alpha - \beta) = \frac{1}{3}$$

Hence, option (a) is the answer.

7. Capitals of Shyam, Gopal, and Madhur are in the ratio 2:4:3, respectively.

Let the total profit be x.

$\therefore$ By conditions,

$$0.8x \times \frac{7}{9} - \left[0.8x \times \frac{2}{9} + 0.2x\right] = 2200$$

$\therefore x = 9000$

$\therefore$ Madhur's share in the profit

$$= \frac{3}{9} \times 9000 \times 0.8 = ₹2400$$

Hence, option (b) is the answer.

8. The word SERIES has one R, one I, two E's, and two S's.

Four letters can be selected and arranged in the following ways:

Selection	Arrangements
2S, 2E	$\frac{4!}{2! \times 2!} = 6$ ways
2S, 1R, 1I	$\frac{4!}{2!} = 12$ ways
2E, 1R, 1I	12 ways
2S, 1R, 1E	12 ways
2S, 1I, 1E	12 ways
2E, 1R, 1S	12 ways
2E, 1I, 1S	12 ways
1S, 1E, 1R, 1I	24 ways

$\therefore$ Total number of arrangements = 6 + 12 + 12 + 12 + 12 + 12 + 12 + 24 = 102

Hence, option (d) is the answer.

9. If the coordinates of the vertices of a triangle are (x_1, y_1), (x_2, y_2), and (x_3, y_3), then its area is given by

$$A = \frac{1}{2}\begin{vmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ 1 & 1 & 1 \end{vmatrix}$$

$\therefore$ If the third vertex of the given triangle is (x, y).

$$6 = \frac{1}{2}\begin{vmatrix} 1 & 4 & x \\ 1 & -1 & y \\ 1 & 1 & 1 \end{vmatrix}$$

$$\therefore \quad 12 = 1(-1 - y) - 4(1 - y) + x(1 + 1)$$

$$\therefore \quad 12 = -1 - y - 4 + 4y + 2x$$

$$\therefore \quad 2x + 3y = 17 \qquad \text{(i)}$$

But, (x, y) also lies on $y = x + 5$ (ii)

Solving equations (i) and (ii)

$$x = \frac{2}{5}, y = \frac{27}{5}$$

Hence, option (a) is the answer.

10. Let p, m, and b be the number of pineapple, mango, and black-forest pastries, respectively.

$$p + m + b = 23 \qquad \text{(i)}$$

Each pastry cost as many rupees as there were pastries of that kind.

$$p^2 + m^2 + b^2 = 211 \qquad \text{(ii)}$$

Substituting options in equations (i) and (ii), we find that only option (b) satisfies both the equations.

Hence, option (b) is the answer.

11. $bx^2 - 2ax + a = 0$

Using the quadratic formula,

$$x = \frac{2a \pm \sqrt{4a^2 - 4ab}}{2b}$$

$$= \frac{2a \pm 2\sqrt{a^2 - ab}}{2b}$$

$$= \frac{a \pm \sqrt{a(a-b)}}{b}$$

$$= \frac{a \pm \sqrt{a}\sqrt{a-b}}{b}$$

$$= \frac{\left(a \pm \sqrt{a}\sqrt{a-b}\right)\left(a \pm \sqrt{a}\sqrt{a-b}\right)}{b\left(a \pm \sqrt{a}\sqrt{a-b}\right)}$$

$$= \frac{\left(a^2 - a(a-b)\right)}{b\left(a \pm \sqrt{a}\sqrt{a-b}\right)}$$

$$= \frac{\left(a^2 - a^2 + ab\right)}{b\left(a \pm \sqrt{a}\sqrt{a-b}\right)}$$

$$= \frac{a}{\sqrt{a}\left(\sqrt{a} \pm \sqrt{a-b}\right)}$$

$$= \frac{\sqrt{a}}{\left(\sqrt{a} \pm \sqrt{a-b}\right)}$$

The best answer is option (c).
Hence, option (c) is the answer.

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12. Let Dr Gupta take x days to complete the evaluation work.

$\therefore$ Dr Sharma takes $x + 10$ days

As Dr Gupta is 40% more efficient than Dr Sharma, we have

$$\frac{1}{x} = \frac{14}{x+10}$$

$$\therefore \quad x = 25$$

$$\therefore \quad x + 10 = 35$$

Also, Dr Sharma is 20% more efficient than Dr Singh.

$\therefore$ If Dr Singh takes y days to complete the evaluation work,

$$\frac{1}{35} = \frac{1.2}{y}$$

$\therefore\ y = 42$

Now, let Dr Singh complete the evaluation work in n days after Dr Gupta has worked for 10 days and Dr Sharma has worked for 15 days.

$$\therefore \frac{10}{25} + \frac{15}{35} + \frac{n}{42} = 1$$

$\therefore\ n = 7.2$

Hence, option (a) is the answer.

13.
$$\left[\sqrt{1}\right] = 1$$
$$\left[\sqrt{2}\right] = 1$$
$$\left[\sqrt{3}\right] = 1$$
$$\left[\sqrt{4}\right] = 2$$
$$\left[\sqrt{5}\right] = 2$$
$$\left[\sqrt{8}\right] = 2$$
$$\left[\sqrt{9}\right] = 3$$

and so on.

Thus, $\left[\sqrt{n}\right] = k$

where k^2 is the greatest perfect square less than or equal to n.

Also, the difference between two consecutive perfect squares $= (k+1)^2 - k^2 = 2k + 1$

$\therefore$ The required sum is

$$\left[\sum_{k=1}^{18} k(2k+1)\right] + \sqrt{361}$$

$$= \left[\sum_{k=1}^{18} 2k^2 + k\right] + 19$$

$$= 2\frac{(18)(18+1)(2\times 18+1)}{6} + \frac{(18)(18+1)}{2} + 19$$

$$= 4408$$

The closest answer is option (a).

Hence, option (a) is the answer.

14.
$$\log_4 \log_4 4^{a-b} = 2\log_4\left(\sqrt{a} - \sqrt{b}\right) + 1$$

$$\therefore \ \log_4(a-b)\log_4 4 = 2\log_4\left(\sqrt{a} - \sqrt{b}\right) + 1$$

$$\therefore \quad \log_4(a-b) = 2\log_4\left(\sqrt{a} - \sqrt{b}\right) + 1$$

$$\therefore \quad \log_4(a-b) = \log_4\left(\sqrt{a} - \sqrt{b}\right)^2 + 1$$

$$\therefore \log_4\left[\frac{a-b}{\left(\sqrt{a} - \sqrt{b}\right)^2}\right] = 1$$

$$\therefore \quad \frac{a-b}{a+b-2\sqrt{ab}} = 4$$

$$\therefore \quad a - b = 4a + 4b - 8\sqrt{ab}$$

$$\therefore \quad 8\sqrt{ab} = 3a + 5b$$

$$\therefore \quad 8 = 3\sqrt{\frac{a}{b}} + 5\sqrt{\frac{b}{a}}$$

Let $\sqrt{\frac{a}{b}} = x$

$$\therefore \quad 8 = 3x + \frac{5}{x}$$

$$\therefore \quad 8x = 3x^2 + 5$$

$$\therefore \quad 3x^2 - 8x + 5 = 0$$

$$\therefore \ 3x^2 - 3x - 5x + 5 = 0$$

$$\therefore 3x(x-1) - 5(x-1) = 0$$

$$\therefore (3x-5)(x-1) = 0$$

$$\therefore x = \frac{5}{3} \text{ or } x = 1z$$

But, $x \neq 1$ as $\sqrt{a} \neq \sqrt{b}$

$\therefore x \neq 5/3$

$$\therefore \sqrt{\frac{a}{b}} = \frac{5}{3}$$

Hence, option (c) is the answer.

15. Let the pipes work for n hours.

By the given conditions,

$$\frac{1}{5\times\frac{4}{3}} + \frac{1}{10\times\frac{4}{3}} + (n-1)\left[\frac{1}{5} + \frac{1}{10}\right] + \frac{2}{15\times\frac{3}{2}} + \frac{n-2}{15} = 1$$

$$\therefore \frac{3}{20} + \frac{3}{40} + (n-1)\frac{3}{10} + \frac{4}{45} + \frac{n-2}{15} = 1$$

$$\therefore \frac{9}{40} + \frac{3n-3}{10} + \frac{4}{45} + \frac{n-2}{15} = 1$$

$$\therefore \frac{9n-9}{30} + \frac{2n-4}{30} = 1 - \frac{9}{40} - \frac{4}{45}$$

$$\therefore \frac{11n-13}{30} = \frac{360-81-32}{360}$$

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$$\therefore 11n - 13 = \frac{247 \times 30}{360}$$

$$\therefore n \approx 3.05$$

Hence, option (c) is the answer.

16. $f(x) = 3^{\sin x} + 3^{\cos x}$

Using the inequality, AM ≥ GM

$f(x)$ will have a minimum value.

where $3^{\sin x} = 3^{\cos x}$

$\therefore \sin x = \cos x$

$$\therefore \sin x = \cos x = -\frac{1}{\sqrt{2}}$$

$$f(x)_{\min} = 3^{\frac{1}{\sqrt{2}}} + 3^{\frac{1}{\sqrt{2}}}$$

$$= 2\left(3^{\frac{1}{\sqrt{2}}}\right)$$

Hence, option (b) is the answer.

17. Let the number of professors, associate professors, and assistant professors be a, b, and c, respectively, and their average ages be p, q, and r, respectively.

$$\therefore a + b + c = \frac{2160}{36} = 60 \qquad \text{(i)}$$

$$\frac{ap + bq}{a + b} = 39$$

$$\frac{bq + cr}{b + c} = 32\frac{8}{11} \text{ and}$$

$$\frac{cr + ap}{a + c} = 36\frac{2}{3}$$

Also, $\frac{a(p+1) + b(q+6) + c(r+7)}{a + b + c} = 41$

As $\frac{ap + bq + cr}{a + b + c} = 36$

$$\therefore \frac{a + 6b + 7c}{a + b + c} = 5$$

$$a + 6b + 7c = 5a + 5b + 5c$$

$$b + 2c = 4a$$

$$b = 4a - c$$

Substituting in (i), we get

$$a + 4a - 2c + c = 60$$

$$\therefore \quad 5a - c = 60 \qquad \text{(ii)}$$

We find that option (a) satisfies equations (i) and (ii).

Further, the values of p, q, and r in option (a) also satisfy the other equations.

Hence, option (a) is the answer.

18. Let $\log_5 2$ be rational.

Then, $\log_5 2 = \frac{m}{n}$

$$\therefore 5^{\frac{m}{n}} = 2$$

$$\therefore \left(5^{m/n}\right)^n = 2^n$$

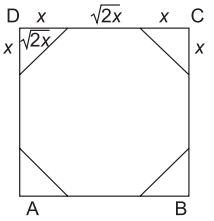
$$\therefore 5^m = 2^n$$

However, number 2 raised to any positive integer power must be even, but 5 raised to any positive integer power must be odd. Hence, we have a contradiction.

$\therefore \log_5 2$ is irrational.

Hence, option (d) is the answer.

19.



Let the side of isosceles triangle be x.

$\therefore$ The side of octagon is $x\sqrt{2}$.

$$\therefore x + x + x\sqrt{2} = 2$$

$$\therefore \quad 2x + x\sqrt{2} = 2$$

$$\therefore \qquad x = \frac{2}{2 + \sqrt{2}}$$

$$\therefore \text{Side of octagon} = x\sqrt{2} = \frac{2\sqrt{2}}{2 + \sqrt{2}} = \frac{2}{1 + \sqrt{2}}$$

$$\therefore \text{Perimeter of octagon} = 8 \times \frac{2}{1 + \sqrt{2}} = \frac{16}{1 + \sqrt{2}}$$

Area of octagon = Area of square − 4 × Area of isosceles triangle

$$= 2^2 - 4 \times \frac{1}{2}x^2$$

$$= 4 - 4 \times \frac{1}{2} \times \left(\frac{\sqrt{2}}{1 + \sqrt{2}}\right)^2$$

$$= 4 - \frac{4}{3 + 2\sqrt{2}} = \frac{8\left(1 + \sqrt{2}\right)}{3 + 2\sqrt{2}} \text{ sq. units}$$

Hence, option (d) is the answer.

20. $7! = 2^4 \times 3^2 \times 5 \times 7$

The required perfect square should be divisible by 16, 9, 5, and 7.

16 and 9 are already squares. Since 5 and 7 are not perfect squares, we multiply 7! by 5 and 7 to make it a perfect square.

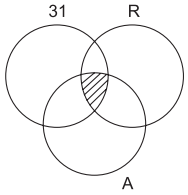
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So, the required perfect square = $2^4 \times 3^2 \times 5^2 \times 7^2 = 176,400$

Hence, option (b) is the answer.

21.



$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) - 2n(A \cap B \cap C)$

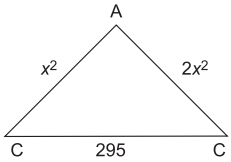
$\therefore 100 - 3 = 61 + 46 + 29 - 25 - 2(x)$

$\therefore 2x = 14$

$\therefore x = 7$

Hence, option (d) is the answer.

22.



$2x^2$ is a power of 2.

Therefore, x^2 is a power of 2, and also,

x is a power of 2.

Also, by triangle inequality, $3x^2 > 295$

$\therefore x^2 > 98.33$

$\therefore x \geq 10$

Hence, x is a power of 2, which is greater than 10.

We try with $x = 16$. Then we have,

AB = 256 and AC = 512

$\therefore$ Perimeter = 256 + 512 + 295 = 1063, which is there in the options.

Hence, option (c) is the answer.

23. Sum of numbers before x = sum of numbers after x

$\therefore 1 + 2 + 3 + \ldots + (x - 1) = (x + 1) + (x + 2) + \ldots + 49$

Adding $[1 + 2 + 3 + 4 + \ldots + (x - 1) + x]$ on both sides, we get

$2[1 + 2 + 3 + \ldots + (x - 1)] + x = (1 + 2 + 3 + \ldots + 49)$

$\therefore \frac{2 \times (x-1)x}{2} + x = \frac{49 \times 50}{2}$

$\therefore x^2 = 49 \times 25$

$\therefore x = 7 \times 5 = 35$

Hence, option (c) is the answer.

24. Let the rate of second lender be y%, then rate of first lender is $(y + 10)$%.

When they lend equal amounts,

$$30,000\left(1+\frac{y+10}{100}\right)^2 - 30,000\left(1+\frac{y}{100}\right)^2 = 7500$$

$$\therefore \quad \left(\frac{110+y}{100}\right)^2 - \left(\frac{100+y}{100}\right)^2 = \frac{1}{4}$$

$\therefore 110^2 - 100^2 + 220y - 200y = 2500 \; y = 20\%$

Now, we evaluate the options and option (c) satisfies the conditions.

$$40,000\left(1+\frac{30}{100}\right)^2 - 20,000\left(1+\frac{20}{100}\right)^2 = 38,800$$

Hence, option (c) is the answer.

25. $(1 - x^6)^4 (1 - x)^{-4}$

$= (1 - 4x^6 + 6x^{12} - 4x^{18} + x^{24})(1 - x)^{-4}$

To find the coefficient of x^{12} in the given expression, we need to find the coefficients of x^{12}, x^6, and x^0 terms in $(1 - x)^{-4}$.

We use the Binomial Theorem for negative coefficients.

Coefficient of x^{12}

$$= \frac{(-4)(-4-1)(-4-1-2)\ldots(-4-11)}{12!}$$

$$= \frac{4 \times 5 \times 6 \times \ldots \times 15}{12!} = \frac{13 \times 14 \times 15}{1 \times 2 \times 3}$$

$= 35 \times 13$

Coefficient of x^6

$$= \frac{(-4)(-4-1)\ldots(-4-5)}{6!}$$

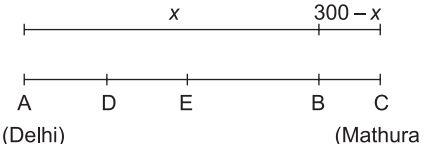
$$= \frac{7 \times 8 \times 9}{3!} = 84$$

The coefficient of x^0 is 1.

$\therefore$ The coefficient of x^{12} is $35 \times 13 + (-4) \times 84 + 6 = 125$

Hence, option (c) is the answer.

26.



Let Mukesh take Suresh on his bike till B and leave him there to walk till C (Mathura). In the meanwhile, Dinesh keeps walking to reach D, Mukesh comes back picks Dinesh and then both ride to Mathura. When Mukesh comes back, let us say he meets Dinesh at E.

Let AB = x, then BC = $300 - x$

Since Dinesh walks at 15 kmph and bike's speed is 60 kmph, we have AD = $\frac{x}{4}$.

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$$\therefore BD = \frac{3x}{4}$$

$$\therefore BE = \left(\frac{60}{75}\right)\left(\frac{3x}{4}\right) = \frac{3x}{5}$$

$$\text{Hence, } \frac{300-x}{15} = \frac{\frac{3x}{5}+\frac{3x}{5}+300-x}{60}$$

$$\therefore 4(300-x) = 300 + \frac{x}{5}$$

$$\therefore \quad 900 = 4x + \frac{x}{5}$$

$$\frac{4500}{21} = x$$

$$x = \frac{1500}{7}\text{km}$$

Hence, minimum time

$$= \frac{x}{60} + \frac{300-x}{15}$$

$$= \frac{1500}{60\times 7} + 20 - \frac{1500}{15\times 7}$$

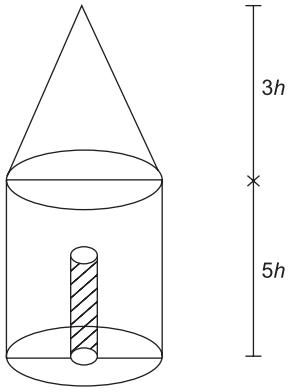
$$= \frac{25}{7} + 20 - \frac{100}{7}$$

$$= 20 - \frac{75}{7}$$

$$= \frac{65}{7} = 9\frac{2}{7}$$

Hence, option (b) is the answer.

27.



Let the height of the cylinder be $5h$ and that of the conical part be $3h$.

$$\therefore \text{Height of hole} = \frac{1}{3}(3h+5h)$$

$$= \frac{8h}{3}$$

Radius of cone = radius of cylinder = 8

Let radius of hole be r.

Now, $\frac{1}{2}$ (Total volume − Volume of hole) = Volume of hole

$$\therefore \text{Total volume} = 3 \text{ Volume of hole}$$

$$\therefore \pi\times 8^2\times 5h + \frac{1}{3}\times\pi\times 8^2\times 3h = 3\pi r^2\left(\frac{8h}{3}\right)$$

$$\therefore 8^2\times 6 = 8r^2$$

$$\therefore r^2 = 8\times 6$$

$$\therefore r = 4\sqrt{3}\,\text{cm}$$

Hence, option (a) is the answer.

28. Let initial production cost be 100.

Then, cost of A = 10 and cost of B = 20

Selling price = 120

Cost of rest = 100 − Cost of A − Cost of B = 100 − 10 − 20 = 70

$$\text{New cost of A} = 1 + \frac{20}{100}\times 10 = 12$$

$$\text{New cost of B} = 1 + \frac{40}{100}\times 20 = 28$$

$$\therefore \quad \text{New cost} = 28 + 12 + 70 = 110$$

$$\text{New S.P.} = \left(1 + \frac{15}{100}\right)\times \text{Original S.P.}$$

$$= \left(1 + \frac{15}{100}\right)\times 120$$

$$= 138$$

$$\text{New profit} = \frac{\text{New S.P.} - \text{New C.P.}}{\text{New C.P.}}\times 100$$

$$= \frac{138-110}{110}\times 100$$

$$\approx 25.5\%$$

Hence, option (b) is the answer.

29. By the cosine rule, in a ΔABC with $\angle ACB = \theta$ and $AB = c$, $BC = a$, and $AC = b$,

$$\cos\theta = \frac{a^2+b^2-c^2}{2ab}$$

$$\therefore 2ab\cos\theta = a^2 + b^2 - c^2$$

$$\therefore c^2 = a^2 + b^2 - 2ab\cos\theta$$

Hence, option (a) is the answer.

30. Total number of subsets = 2^{11}

Total number of subsets without any even integer = 2^6

Hence, the required number of subsets

$$= 2^{11} - 2^6$$

$$= 2048 - 64$$

$$= 1984$$

Hence, option (c) is the answer.

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PART

4

SECTION TESTS

Section Test 1

Section Test 2

Section Test 3

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1 Section Test 1

Instructions to the test takers

- There are 34 questions in this test.
- Each question carries 3 marks.
- There is 1/3rd NEGATIVE marking.
- Time allotted – 60 minutes

Q.1 How many even three-digit integers have the property that their digits, read from left to right, are strictly in increasing order?
(a) 21 (b) 34 (c) 51 (d) 72

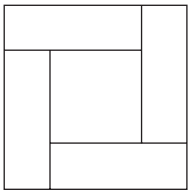
Q.2 Which of the following is equal to $1 + \dfrac{1}{1+\dfrac{1}{1+1}}$?

(a) $\frac{5}{4}$ (b) $\frac{3}{2}$
(c) $\frac{5}{3}$ (d) 2

Q.3 One dimension of a cube is increased by 1 unit, another is decreased by 1 unit, and the third is left unchanged. The volume of the new rectangular solid is 5 units less than that of the cube. What is the volume of the cube?
(a) 8 (b) 27 (c) 64 (d) 125

Q.4 Suppose that $P = 2^m$ and $Q = 3^n$. Which of the following is equal to 12^{mn} for every pair of integers (m, n)?
(a) P^2Q (b) P^nQ^m
(c) P^nQ^{2m} (d) $P^{2n}Q^m$

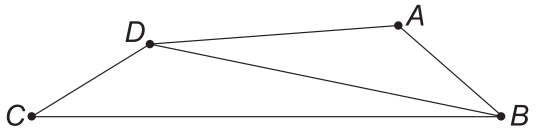
Q.5 Four congruent rectangles are placed as shown. The area of the outer square is 4 times that of the inner square. What is the ratio of the length of the longer side of each rectangle to the length of its shorter side?



(a) 3 (b) $\sqrt{10}$ (c) $2 + \sqrt{2}$ (d) $2\sqrt{3}$

Q.6 Given that $f(x + 3) = 3x^2 + 7x + 4$ and $f(x) = ax^2 + bx + c$. What is the value of $(a + b + c)$?
(a) −1 (b) 0 (c) 1 (d) 2

Q.7 In quadrilateral ABCD, AB = 5, BC = 17, CD = 5, DA = 9, and BD is an integer. What is BD?



(a) 11 (b) 12 (c) 13 (d) 14

Q.8 A class collects ₹50 to buy flowers for a classmate who is in the hospital. Roses cost ₹3 each, and carnations cost ₹2 each. No other flowers are to be used. How many different bouquets with at least one of each type could be purchased for exactly ₹50?
(a) 7 (b) 8 (c) 9 (d) 10

Q.9 At an apple orchard, apples are being packed in containers. Samar would take 9 h to pack one container full of apples and Aarna would take 10 h to pack one container full of apples when working alone.

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When they work together they talk a lot, and their combined packaging output is decreased by 10 apples per hour. Working together, they pack one container full of apples in 5 h. Then, find how many apples are packed in one container.
(a) 500 (b) 900 (c) 950 (d) 1000

Q.10 A cone-shaped mountain has its base on the ocean floor and has a height of 8000 feet. The top $\frac{1}{8}$ of the volume of the mountain is above water. What is the depth of the ocean at the base of the mountain (in feet)?
(a) 4000 (b) $200(4-\sqrt{2})$
(c) 6000 (d) 6400

Q.11 LCM of first 10 natural numbers is N. What is the LCM of first 15 natural numbers?
(a) 11 N
(b) $11 \times 12 \times 13 \times 14 \times 15$ N
(c) 143 N
(d) Cannot be determined

Q.12 What is the value of $(3x-2)(4x+1)-(3x-2)4x+1$ when $x=4$?
(a) 0 (b) 1 (c) 10 (d) 11

Q.13 If you have 3 tickets to a lottery for which 10 tickets were sold and 5 prices are to be given, then the probability that you will win at least one prize is:
(a) $\frac{7}{12}$ (b) $\frac{9}{12}$ (c) $\frac{1}{12}$ (d) $\frac{11}{12}$

Q.14 Let v, w, x, y, and z be the degree measures of the five angles of a pentagon. Suppose that $v < w < x < y < z$, and v, w, x, y, and z form an arithmetic sequence. Find the value of x.
(a) 72 (b) 84 (c) 90 (d) 108

Q.15 A square has sides of length 10 units, and a circle centred at one of its vertices has radius 10 units. What is the area of the union of the regions enclosed by the square and the circle?
(a) $200+25\pi$ (b) $100+75\pi$
(c) $75+100\pi$ (d) $100+100\pi$

Q.16 Point (–3, 2) is rotated 90° clockwise around the origin to point B. Point B is then reflected over the line $x=y$ to point C. What are the coordinates of C?
(a) (3, 2) (b) (–2, –3)
(c) (2, –3) (d) (2, 3)

Q.17 The two digits in Juhi's age are the same as the digits in Amir's age, but in reverse order. In five years, Juhi will be twice as old as Amir will be then. What is the difference in their current ages?
(a) 9 (b) 18 (c) 27 (d) 36

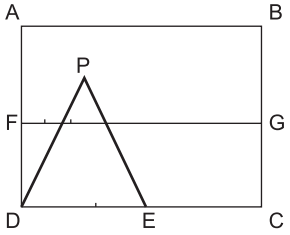
Q.18 Probability that A can solve a problem is $\frac{1}{3}$, probability that B can solve the same problem is $\frac{2}{3}$, and C can solve the same problem is $\frac{3}{5}$. If all three of them start solving the question, then what is the probability that problem will be solved by at least one of them?
(a) $\frac{4}{45}$ (b) $\frac{41}{45}$
(c) $\frac{1}{4}$ (d) None of these

Q.19 Sudhir is travelling from Patna to Dehradun. For half of the time, his speed is x kmph, and for other half of the time, his speed is y kmph. During return journey, for half of the distance, his speed is x kmph, and for other half of the distance, his speed is y kmph. If $x \neq y$, then which of the following is definitely correct?
(a) Average speed of onward journey > average speed of return journey
(b) Average speed of onward journey < average speed of return journey
(c) Average speed of onward journey = average speed of return journey
(d) Cannot be determined

Q.20 Mr and Mrs Lopez have two children. When they get into their family car, two people sit in the front, and the other two sit in the back. Either Mr Lopez or Mrs Lopez must sit in the driver's seat. How many seating arrangements are possible?
(a) 4 (b) 12 (c) 16 (d) 24

Q.21 The sequence $\log_{12} 162$, $\log_{12} x$, $\log_{12} y$, $\log_{12} z$, and $\log_{12} 1250$ is an arithmetic progression. What is x?
(a) $125\sqrt{3}$ (b) 270
(c) $162\sqrt{5}$ (d) 434

Q.22 Area of the square ABCD is 64 sq. units. Let E, F, and G be mid-points of DC, AD, and BC, respectively. If P is any point inside the rectangle ABGF and X is the area of the triangle DPE, then which one of the following is true?



(a) $8 < X < 16$ (b) $8 < X < 32$
(c) $16 < X 32$ (d) $16 < X < 64$

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Q.23 Parul Jaiswal has ₹11,200 in ₹100 and ₹500 notes. What is the ratio of the number of notes of the two types, if she has a total of 40 notes with her?
(a) 3:2 (b) 7:3 (c) 8:7 (d) 11:9

Q.24 A can do a piece of work in 36 days, B in 54 days, and C in 72 days. All three began the work together but A left 8 days and B left 12 days before the completion of the work. For how many days in all did C work?
(a) 24 days (b) 28 days
(c) 32 days (d) 36 days

Q.25 Five tables and eight chairs cost ₹7,350. Three tables and five chairs cost ₹4,475. What is the price of one table?
(a) ₹950 (b) ₹325 (c) ₹925 (d) ₹350

Q.26 Two boats, travelling at 5 kmph and 10 kmph, head directly towards each other. They begin at a distance of 20 km from each other. How far apart are they (in km) i min before they collide?
(a) $\frac{1}{12}$ (b) $\frac{1}{6}$ (c) $\frac{1}{4}$ (d) $\frac{1}{3}$

Q.27 Price of an article rose by 25% every odd year and fell by 20% every even year. What would be the percentage change after 40 years?
(a) 10% increase (b) 25% increase
(c) No change (d) 20% decrease

Q.28 Owing to heavy traffic on the route, a man finds that his speed is reduced by 25% today with respect to normal days. As a result, he reaches his office 20 min late than normal. How much time did he take today to reach the office?
(a) 40 min (b) 45 min
(c) 60 min (d) 75 min

Q.29 In a chess tournament, every person played one game with every other person in the group. The total number of games that men played between themselves exceeded those played by men with women by 18. If there were 4 women in the tournament, totally how many games were played in the tournament?
(a) 112 (b) 120 (c) 162 (d) 190

Q.30 For how many ordered pairs of positive integers (x, y) is $x + 2y = 100$?
(a) 33 (b) 49 (c) 50 (d) 99

Q.31 A basketball player made 5 baskets during a game. Each basket was worth either 2 or 3 points. How many different numbers could represent the total points scored by the player?
(a) 3 (b) 4 (c) 5 (d) 6

Q.32 Solve the inequality for x: $2\log_{10}(x - 4) < \log_{10}(x + 8)$.
(a) $x > 4$ (b) $-8 < x < 4$
(c) $4 < x < 8$ (d) $x < 8$

Q.33 Two roots of a quadratic equation $ax^2 + bx + c = 0$ are 2 and 3. What is the value of $(a + b + c)$?
(a) 2
(b) −2
(c) 4
(d) Cannot be determined

Q.34 A cube with 3-inch edges is to be constructed from 27 smaller cubes with 1-inch edges. Twenty-one of the cubes are coloured red and 6 are coloured white. If the 3-inch cube is constructed to have the smallest possible white surface area showing, then what fraction of the surface area is white?
(a) $\frac{5}{54}$ (b) $\frac{1}{2}$ (c) $\frac{5}{27}$ (d) $\frac{1}{3}$

Answers

1. (b)	**2.** (c)	**3.** (d)	**4.** (d)	**5.** (a)	**6.** (d)	**7.** (c)	**8.** (b)	**9.** (b)	**10.** (a)
11. (c)	**12.** (d)	**13.** (d)	**14.** (d)	**15.** (b)	**16.** (a)	**17.** (b)	**18.** (b)	**19.** (a)	**20.** (b)
21. (b)	**22.** (a)	**23.** (d)	**24.** (a)	**25.** (a)	**26.** (d)	**27.** (c)	**28.** (c)	**29.** (c)	**30.** (b)
31. (d)	**32.** (c)	**33.** (d)	**34.** (a)						

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Hints and Solutions

1. Let us set the middle (tens) digit first. The middle digit can be anything from 2 to 7 (if it was 1, then we would have the hundreds digit to be 0, and if it was more than 8, then the ones digit cannot be even).

If it was 2, then there is 1 possibility for the hundreds digit, and 3 for the ones digit. If it was 3, then there are 2 possibilities for the hundreds digit, and 3 for the ones digit. If it was 4, then there are 3 possibilities for the hundreds digit, and 2 for the ones digit, and so on.

So, the answer is $3(1 + 2) + 2(3 + 4) + 1(5 + 6) = (34)$. Hence, option (b) is the answer.

2. We have to find out the value of $1 + \dfrac{1}{1+\dfrac{1}{1+1}} = 1 + \dfrac{1}{1+\dfrac{1}{2}}$

$= 1 + \dfrac{1}{\dfrac{3}{2}} = 1 + \dfrac{2}{3} = \dfrac{5}{3}$

Hence, option (c) is the answer.

3. Let the original cube have edge length a. Then, its volume is a^3. The new box has dimensions $a - 1$, a, and $a + 1$, and hence, its volume is $(a - 1) \times a \times (a + 1) = a^3 - a$. The difference between the two volumes is a. As we are given that the difference is 5, we have $a = 5$, and the volume of the original cube was $5^3 = 125$.

Hence, option (d) is the answer.

4. We have $12^{mn} = (2 \times 2 \times 3)^{mn} = 2^{2mn} \times 3^{mn} = (2^m)^{2n} \times (3^n)^m = P^{2n}Q^m$

Hence, option (d) is the answer.

5. The area of the outer square is 4 times that of the inner square. Therefore, the side of the outer square is $\sqrt{4} = 2$ times that of the inner square.

Then, the shorter side of the rectangle is 1/4 of the side of the outer square, and the longer side of the rectangle is 3/4 of the side of the outer square, and therefore, their ratio is 3.

Hence, option (a) is the answer.

6. **Method 1**

As $f(x) = ax^2 + bx + c$, we have $f(1) = a \times 1^2 + b \times 1 + c = a + b + c$

To compute $f(1)$, set $x = -2$ in the first formula. We get $f(1) = f(-2 + 3) = 3(-2)^2 + 7(-2) + 4 = 12 - 14 + 4 = 2$

Method 2

Combining the two formulas, we know that $f(x + 3) = a(x + 3)^2 + b(x + 3) + c$

We can rearrange the right-hand side to $ax^2 + (6a + b)x + (9a + 3b + c)$

Comparing coefficients, we have $a = 3$, $6a + b = 7$, and $9a + 3b + c = 4$. From the second equation, we get $b = -11$, and then from the third, we get $c = 10$.

Hence, $a + b + c = 3 - 11 + 10 = (2)$

Hence, option (d) is the answer.

7. By the triangle inequality, we have BD < DA + AB = 9 + 5 = 14, and also BD + CD > BC; hence, BD > BC − CD = 17 − 5 = 12

We got that 12 < BD < 14, and as we know that BD is an integer, we must have BD = (13).

Hence, option (c) is the answer.

8. Assume number of roses are x and number of carnations are y.

According to the question, $3x + 2y = 50$

The class could send 22 carnations and 2 roses, 19 carnations and 4 roses, and so on, down to 1 carnation and 16 roses. There are 8 total possibilities (from 2 to 16 roses, increasing by 2 at each step).

Hence, option (b) is the answer.

9. Let h be the number of apples to be packed in one container.

Without talking, Samar and Aarna pack $\dfrac{h}{9}$ and $\dfrac{h}{10}$ containers per hour, respectively; so, together they pack $\dfrac{h}{9} + \dfrac{h}{10} - 10$ per hour together.

Since they pack one container in 5 hours, $\dfrac{h}{5} = \left(\dfrac{h}{9} + \dfrac{h}{10} - 10\right)$

Solving it, we get $h = 900$.

Hence, option (b) is the answer.

10. In a cone, radius and height each vary inversely with increasing height (i.e., the radius of the cone formed by cutting off the mountain at 4000 feet is half that of the original mountain). Therefore, volume varies as the inverse cube of increasing height (expressed as a percentage of the total height of cone):

$V_I \times \text{Height}^3 = V_N$

Plugging in our given condition, $1/8 = \text{Height}^3 \Rightarrow \text{Height} = 1/2$

$8000 \times 1/2 = 4000$

Hence, option (a) is the answer.

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11. There are two prime numbers from 10 to 15, namely 11 and 13. These two prime numbers have not appeared in the LCM of first 10 natural numbers, i.e., N.

Hence, LCM of first 15 natural numbers = $11 \times 13 \times$ LCM $(1 - 10) = 143$ N

Hence, option (c) is the answer.

12. Did you take out $(4x + 1)$ common? See the question carefully.

$(3x - 2)(4x + 1) - (3x - 2)\,4x + 1$

After minus sign, $(3x - 2)$ is multiplied by only $4x$ and not by $(4x + 1)$.

Solving it, we get

$(3x - 2)(4x + 1) - (3x - 2)\,4x + 1 = (12 - 2)(16 + 1) - (12 - 2)16 + 1 = 10 \times 17 - 10 \times 16 + 1 = 11$

Hence, option (d) is the answer.

13. Probability that you will win at least one prize = 1 – probability that you will not win any prize

$$= 1 - \frac{{}^{5}C_{3}}{{}^{10}C_{3}} = \frac{11}{12}$$

Hence, option (d) is the answer.

14. Sum of angles of any polygon $= (2n - 4) \times 90° = (2 \times 5 - 4) \times 90 = 540°$

If we let $v = x - 2d$, $w = x - d$, $y = x + d$, $z = x + 2d$, it follows that

$(x - 2d) + (x - d) + x + (x + d) + (x + 2d) = 5x = 540 \rightarrow x = 108$

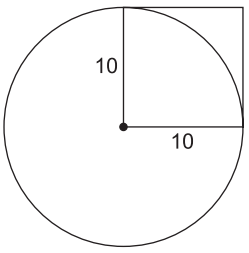
Hence, option (d) is the answer.

Note that since x is the middle term of an arithmetic sequence with an odd number of terms, it is simply the average of the sequence.

15. Area of the circle is $C = 100\pi$

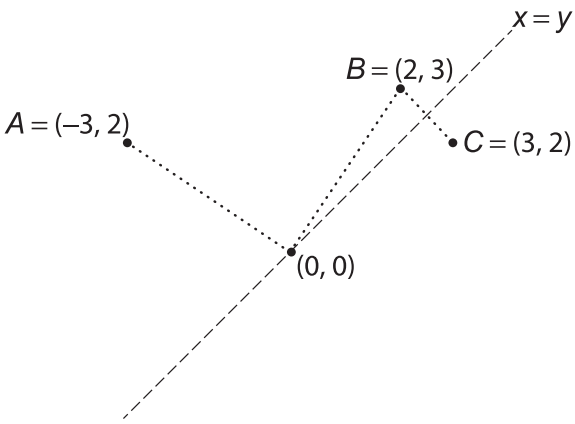
Area of the square is $S = 100$

Exactly $\frac{1}{4}$ of the circle lies inside the square. Thus, the total area is $\frac{3}{4}$ O + S $= (100 + 75\pi)$



Hence, option (b) is the answer.

16. The entire situation is given in the picture below. The correct answer is [(a) (3, 2)].



17. Age difference does not change in time. Thus, in 5 years, Amir's age will be equal to their age difference.

Age difference is $(10a + b) - (10b + a) = 9(a - b)$, and hence, it is a multiple of 9. Thus, Amir's current age should be of the format 9K + 4.

Possible values for Amir's age = {13, 22, 31, 40, 49, 58, 67, 76, 85, 94}

As Juhi is older, we only need to consider the cases where the tens digit of Amir's age is smaller than the ones digit. This leaves us with the options {13, 49, 58, 67}.

Checking each of them, we see that only 13 works and gives the solution 31 – 13 = (18).

Hence, option (b) is the answer.

18. Problem can be solved by at least one of them = 1 – (problem will not be solved by any one of them)

Probability that the problem will not be solved by A = 1 – (probability that the problem will be solved by A) =

$$1 - \frac{1}{3} = \frac{2}{3}$$

Probability that the problem will not be solved by B = 1 – (probability that the problem will be solved by B) =

$$1 - \frac{2}{3} = \frac{1}{3}$$

Probability that the problem will not be solved by C = 1 – (probability that the problem will be solved by C) =

$$1 - \frac{3}{5} = \frac{2}{5}$$

So, probability that problem will not be solved by any one of them $= \frac{2}{3} \times \frac{1}{3} \times \frac{2}{5} = \frac{4}{45}$

Therefore, the required probability $= 1 - \frac{4}{45} = \frac{41}{45}$

Hence, option (b) is the answer.

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19. We know that AM ≥ GM ≥ HM

In this case, average speed for onward journey = AM of x and $y = \frac{x+y}{2}$

Average speed for the return journey = HM of x and $y = \frac{2xy}{x+y}$

Since $x \neq y$, AM > HM.

Hence, option (a) is the answer.

20. First, we make the children seated.

First child can be seated in 3 spaces. Then, Second child can be seated in 2 spaces. Now, there are 2×1 ways to make the adults sit.

Therefore, total number of ways = $3 \times 2 \times 2 = 12$

Hence, option (b) is the answer.

21. **Method 1**

Since the sequence is in arithmetic progression

$\text{Log}_{12}\ 162 + 4d = \log_{12}\ 1250$, where d is the common difference.

Therefore, $4d = \log_{12}\ 1250 - \log_{12}\ 162 = \log_{12}\ (1250/162)$

and $d = \frac{1}{4}(\log_{12}\ (1250/162)) = \log_{12}\ (1250/162)^{1/4}$

Now, we found d, and we just add it to the first term to find x:

$\text{Log}_{12}\ 162 + \log_{12}\ (1250/162)^{1/4} = \log_{12}\ ((162) \times (1250/162)^{1/4})$

$x = (162) \times (1250/162)^{1/4} = (162) \times (625/81)^{1/4} = 162 \times \frac{5}{3} = 270$

Hence, option (b) is the answer.

Method 2

If $\log_x a$, $\log_x b$, and $\log_x c$ are in AP, then a, b, and c are in GP.

As the sequence $\log_{12}\ 162$, $\log_{12}\ x$, $\log_{12}\ y$, $\log_{12}\ z$, $\log_{12}\ 1250$ is an arithmetic progression, the sequence 162, x, y, z, 1250 must be a geometric progression.

If we factor the two known terms, then we get $162 = 2 \times 3^4$ and $1250 = 2 \times 5^4$; thus, the quotient is obviously $\frac{5}{3}$, and therefore, $x = 162 \times \frac{5}{3} = 270$

22. Area of square ABCD = 64 sq. units

Side length of square ABCD = 8 units

DE = FD = 4 units

Let PQ be the altitude of the triangle DPE. Since point P lies in the rectangle ABGF, value of PQ will be governed by:

$4 < \text{PQ} < 8 \Rightarrow \frac{1}{2} \times 4 \times 4 < \text{Area of triangle DPE} < \frac{1}{2} \times 4 \times 8 \Rightarrow 8 < \text{Area} < 16$

Hence, option (a) is the answer.

23. Assume total number of ₹100 notes is x and total number of ₹500 notes is y.

According to the question,

$x + y = 40$ and $100x + 500y = 11{,}200$

Solve these two equations to get the answer.

Hence, option (d) is the answer.

24. Let the total no. of days for completing the work be x.

According to the question,

$$(x-8) \times \frac{1}{36} + (x-12) \times \frac{1}{54} + x \times \frac{1}{72} = 1$$

On solving, we get $x = 24$.

Hence, option (a) is the answer.

25. Let the cost of one table is ₹x and cost of one chair is ₹y

$5x + 8y =$ ₹7350 (i)

$3x + 5y =$ ₹4475 (ii)

Solving equations (i) and (ii), we get

$x =$ ₹325 and $y =$ ₹950

Hence, option (a) is the answer.

26. Since boats are travelling towards each other, their relative speed will be the (sum of speed of boat 1 + speed of boat 2) = 5 kmph + 10 kmph = 15 kmph. In other words, they are reducing the gap by 15 kmph or $\frac{1}{4}$ km every minute. Therefore, the gap one minute before the collision will be $\frac{1}{4}$ km.

Hence, option (d) is the answer.

27. Let the price of article be ₹100.

Price after first year = ₹125

Next year, price is decreased by 20%. So, price after second year = 125 − 20% of 125 = ₹100

It can be seen that for every pair of years, net change in the price is zero. Hence, after 100 years, there will be no change in the price.

Hence, option(c) is the answer.

28. Speed is reduced by 25%. New speed becomes $\frac{3}{4}$ of original speed. Since speed and time are inversely proportional for a constant distance, new time taken will be 4/3 of the original time.

So, $\frac{4}{3}T - T = 20$ min

Therefore, $T = 60$ min

Hence, option (c) is the answer.

29. Assume that the number of men = m and number of women = n

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Using above concept note:

Number of games played by men among themselves = $\frac{m(m-1)}{2}$

Number of games played by women among themselves = $\frac{n(n-1)}{2} = \frac{4(4-1)}{2} = 6$

(Given that number of women = 4)

Total number of matches where one players is a male and other one is a female = $4m$

According to the question, total number of games men played between themselves exceeded those played by men with women by 18.

Hence, $\frac{m(m-1)}{2} - 4m = 8 \Rightarrow m(m-1) - 8m = 36$

Solving it,

$(m-12) \times (m+3) = 0 \Rightarrow m$ = number of men = 12

Hence, total number of people participating in the chess tournament = 12 + 4 = 16

So, total number of games played in the tournament = $\frac{16(16-1)}{2} = 120$

Hence, option (c) is the answer.

30. Every positive integer value of y leads to an integer solution for x.

Also, $y = \frac{100-x}{2}$; since x must be positive, $y < 50$.

This gives the following value of y.

$1 \le y < 50 \Rightarrow$ 49 values for y, which mean there are 49 solutions to the equation.

Hence, option (b) is the answer.

31. If the basketball player makes x three-point shots and $(5 - x)$ two-point shots, he scores $3x + 2(5 - x) = 10 + x$ points. Clearly, every value of x gives a different number of total points. Since he can make any number of three-point shots between 0 and 5 inclusive, the number of different point totals is 6.

Hence, option (d) is the answer.

32. For log to be defined:

$x - 4 > 0$ and $x + 8 > 0$ or $x > 4$ (i)

Further, according to the question,

$\log_{10}(x-4)^2 < \log_{10}(x+8)$

As the base is greater than 1, log is an increasing function.

$(x-4)^2 < x + 8$ or

$x^2 - 8x + 16 < x + 8$ or $x^2 - 9x + 8 < 0$ or $(x-1)(x-8) < 0$

Hence, $1 < x < 8$ (ii)

Combining (i) and (ii), we obtain the following range of value of x: $4 < x < 8$

Hence, option (c) is the answer.

33. At first glance, it appears that the equation is $(x - 2)(x - 3) = 0$ or

$x^2 - 5x + 6 = 0$

Hence, $(a + b + c) = 1 - 5 + 6 = 2$

There is a fallacy here. Look at the concept note.

In fact, $x^2 - 5x + 6 = 0$ will have 2 and 3 as roots. However, that is not the only quadratic equation that will have roots as 2 and 3.

$2(x^2 - 5x + 6 = 0) = 0$ or $2x^2 - 10x + 12 = 0$ will also have roots as 2 and 3 only, and in this case, value of $(a + b + c)$ will be twice the value of $(a + b + c)$ obtained in case of $x^2 - 5x + 6 = 0$

So, no unique values of $(a + b + c)$ can be obtained.

Hence, option (d) is the answer.

34. For the least possible surface area, we should have 1 cube in the centre and the other 5 with only 1 face exposed. This gives 5 square inches of white surface area. Since the cube has a surface area of 54 square inches, our answer is 5/54.

Hence, option (a) is the answer.

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2 Section Test 2

Instructions to the test takers

- There are 34 questions in this test.
- Each question carries 3 marks.
- There is 1/3rd NEGATIVE marking.
- Time allotted – 60 minutes

Q.1 A and B can complete a work in 15 days. A is 50% more efficient than B. How long would A take to complete the work working alone?
(a) 20 days (b) 22 days
(c) 24 days (d) None of these

Q.2 A circle has a radius of $\log_{10}(a^2)$ and a circumference of $\log_{10}(b^4)$. What is $\log_a b$?
(a) $\frac{1}{4\pi}$ (b) $\frac{1}{\pi}$ (c) π (d) 2π

Q.3 $f(x) = ax^2 + bx + c = 0$ has two real roots. If $ac < 0$, then how many of the roots of $f(x) = 0$ will be negative?
(a) 0
(b) 1
(c) 2
(d) Cannot be determined

Q.4 Sally has five red cards numbered 1 through 5 and four blue cards numbered 3 through 6. She stacks the cards so that the colours alternate and so that the number on each red card divides evenly into the number on each neighbouring blue card. What is the sum of the numbers on the middle three cards?
(a) 8 (b) 10 (c) 11 (d) 12

Q.5 How many two-digit natural numbers will have exactly three factors?
(a) 1 (b) 2 (c) 3 (d) 4

Q.6 A shopkeeper claims to sell his goods at his cost price. But, he gives only x grams at the place of 1000 g, and as a result, his profit percentage is 20%. What is the value of x (in grams)?
(a) 750 (b) 800 (c) 833.3 (d) 900

Q.7 Amar, Akbar, and Anthony purchased a gift worth ₹6000. Amar paid half of what others have paid, Akbar paid 1/3rd of what others have paid. What is the amount paid by Anthony?
(a) ₹2000 (b) ₹2500
(c) ₹2250 (d) ₹1500

Q.8 In a triangle with integer side lengths, one side is three times as long as the second side, and the length of the third side is 15. What is the greatest possible perimeter of the triangle?
(a) 43 (b) 44 (c) 45 (d) 46

Q.9 At ISL, a team will have 21 players each. League rules state that a player must be paid at least \$15,000, and that the total of all players' salaries for each team cannot exceed \$700,000. What is the maximum possible salary, in dollars, for a single player?
(a) 270,000 (b) 385,000
(c) 400,000 (d) 430,000

Q.10 There are two values of a for which the equation $4x^2 + ax + 8x + 9 = 0$ has only one distinct solution for x. What is the sum of these values of a?
(a) −16 (b) −8 (c) 0 (d) 8

Q.11 A line passes through A (1, 1) and B (100, 1000). How many other points with integer coordinates are on the line and strictly between A and B (not counting A and B)?
(a) 0 (b) 2 (c) 3 (d) 8

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Q.12 Three labelled boxes containing red and white cricket balls are mislabelled. It is known that one of the boxes contains only white balls and another one contains only red balls. The third contains a mixture of red and white balls. You are required to correctly label the boxes with the labels red, white, and red and white by picking a sample of one ball from only one box. What is the label on the box you should sample?
(a) White
(b) Red
(c) Red and white
(d) Not possible to determine from a sample of one ball

Q.13 A shipping clerk has five boxes of different but unknown weights each weighing less than 100 kg. The clerk weighs the boxes in pairs. The weights obtained are 110, 112, 113, 114, 115, 116, 117, 118, 120, and 121 kg. What is the weight of the heaviest box?
(a) 60 kg
(b) 62 kg
(c) 64 kg
(d) Cannot be determined

Q.14 Four persons A, B, C, and D need to cross a bridge at night. Unfortunately, they have only one torch and the bridge is too dangerous to cross without carrying torch. Further, bridge is strong enough only to support two people at a time. Not all people take the same time to cross the bridge. Time taken for each person: 1 min (for A), 2 min (for B), 7 min (for C), and 10 min (for D). What is the shortest time (minutes) in which all of them will cross the bridge?
(a) 19 (b) 17 (c) 15 (d) 20

Q.15 A, B, C, D, ..., X, Y, Z are 26 players who participated in a tournament. Everyone played with every other player exactly once. A win scores 2 points, a draw scores 1 point, and a loss scores 0 point. None of the matches ended in a draw. No two players scored the same score. At the end of the tournament, ranking list is published which is in accordance with the alphabetical order. Then, which of the following is correct?
(a) M wins over N.
(b) N wins over M.
(c) M does not play with N.
(d) None of these

Q.16 Given $A = 2^{65}$ and $B = (2^{64} + 2^{63} + 2^{62} + ... + 2^0)$, which of the following is true?
(a) $A = B$
(b) B is larger than A by 1.
(c) A is larger than B by 1.
(d) Cannot be determined

Q.17 Six X's have to be placed in the squares of the figure such that each row contains at least one X.



(a) 28 (b) 26 (c) 36 (d) 30

Q.18 A and B can do a work together in x days. A alone takes 9 days more than x days, and B alone takes 4 days more than x days. What is the value of x?
(a) 6 days (b) 8 days (c) 10 days (d) 4 days

Q.19 A square and an equilateral triangle have the same perimeter. Let A be the area of the circle circumscribed about the square and B the area of the circle circumscribed around the triangle. Find A/B.

(a) $\frac{9}{16}$ (b) $\frac{3}{4}$ (c) $\frac{27}{32}$ (d) $\frac{3\sqrt{6}}{8}$

Q.20 At compound interest, a sum of money amounts to ₹2880 at the end of 2 years and amount to ₹2400 at the end of first year. What will be the interest obtained in the third year if interest is compounded annually?
(a) ₹480
(b) ₹576
(c) ₹400
(d) Cannot be determined

Q.21 If $(x + y):(y + z):(z + x) = 6:7:8$ and $(x + y + z) = 14$, then find the value of c.
(a) 6 (b) 7 (c) 8 (d) 14

Q.22 Three jars have their volumes in the ratio 3:4:5. They are full of mixtures of milk and water. The mixtures contain milk and water in the ratio of (4:1), (3:1), and (5:2), respectively. The contents of all these three jars are poured into a fourth jar. Find the ratio of milk and water in the fourth jar.
(a) 4:1 (b) 151:48
(c) 157:53 (d) 5:2

Q.23 On an order of 5 dozen boxes of a consumer product, a retailer receives an extra dozen free. This is equivalent to allowing him a discount of:
(a) 15% (b) 16.66% (c) 16 2/3% (d) 20%

Q.24 Average of 10 positive numbers is x. If each number is increased by 30%, then x:
(a) Remains unchanged (b) Decreases by 30%
(c) Increases by 30% (d) Increases by 10%

Q.25 Ramesh goes from A to B at 40 kmph and comes back with a speed of N kmph. During his whole journey, his average speed = 80 kmph. What is the value of N?

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(a) 120 kmph (b) 140 kmph
(c) 250 kmph (d) None of these

Q.26 How many odd integers from 1000 to 8000 have none of its digits repeated?
(a) 1842 (b) 1736
(c) 1444 (d) None of these

Q.27 After striking the floor, a ball rebounds to $\frac{4}{5}$th of the height from which it has fallen. What is the total distance that it travels before coming to rest if it is gently dropped from a height of 120 m?
(a) 600 m (b) 480 m (c) 1080 m (d) 960 m

Q.28 A survey on a sample of 25 new cars being sold at a local auto dealer was conducted to see which of the three popular options—air conditioning, radio, and power windows—were already installed. The survey found 15 had air conditioning, 2 had air conditioning and power windows but no radios, 12 had radio, 6 had air conditioning and radio but no power windows, 11 had power windows, 4 had radio and power windows, and 3 had all three options. What is the number of cars that had none of the options?
(a) 0 (b) 1 (c) 2 (d) 3

Q.29 What is the value of

$$\frac{1}{1+\sqrt{2+\sqrt{3}}}+\frac{1}{1-\sqrt{2}+\sqrt{3}}+\frac{1}{1+\sqrt{2}-\sqrt{3}}+\frac{1}{1-\sqrt{2}-\sqrt{3}}$$

(a) 2 (b) −2 (c) √2 + √3 (d) √2−√3

Q.30 When the curves $y = \log_{10}x$ and $y = x^{-1}$ are drawn in the x–y plane, how many times do they intersect for values $x \geq 1$?
(a) Never (b) Once
(c) Twice (d) Thrice

Q.31 A GP consists of 1001 terms. Sum of the terms occupying the odd places is P_1, and the sum of the terms occupying the even places is P_2. Find the common ratio of this GP.
(a) P_2/P_1
(b) P_1/P_2
(c) $(P_2 - P_1)/P_1$
(d) Cannot be determined

Q.32 Let $N = \{x | x$ be a prime number and $x < 30\}$. The number of distinct rational numbers whose numerator and denominator belong to set N is:
(a) ${}^{10}C_2$ (b) ${}^{10}P_2$ (c) ${}^{10}C_2+1$ (d) ${}^{10}P_2+1$

Q.33 Consider the following data for a snacks shop "Santushti" located at Mukherjee Nagar, Delhi:

	2009	2010
CP/Unit	A	B
SP/Unit	C	D

It is given that $B - A = D - C$ (all the values A, B, C, and D are in natural numbers and rupees).
Which of the following is definitely true?
(a) Profit percentage per unit in 2010 is more than profit percentage per unit in 2009.
(b) Profit percentage per unit in 2009 is more than profit percentage per unit in 2010.
(c) Profit percentage per unit in 2010 is equal to the profit percentage per unit in 2009.
(d) Cannot be determined

Q.34 A telecom service provider engages male and female operators for answering 1000 calls per day. A male operator can handle 40 calls per day, whereas a female operator can handle 50 calls per day. The male and the female operators get a fixed wage of ₹250 and ₹300 per day, respectively. In addition, a male operator gets ₹15 per call he answers and female operator gets ₹10 per call she answers. To minimize the total cost, how many male operators should the service provider employ assuming he has to employ more than 7 of the 12 female operators available for the job?
(a) 15 (b) 14 (c) 12 (d) 10

Answers

1. (d)	**2.** (c)	**3.** (b)	**4.** (d)	**5.** (b)	**6.** (c)	**7.** (b)	**8.** (a)	**9.** (c)	**10.** (a)
11. (d)	**12.** (c)	**13.** (b)	**14.** (b)	**15.** (a)	**16.** (c)	**17.** (b)	**18.** (a)	**19.** (c)	**20.** (b)
21. (a)	**22.** (c)	**23.** (c)	**24.** (c)	**25.** (d)	**26.** (b)	**27.** (b)	**28.** (c)	**29.** (a)	**30.** (b)
31. (d)	**32.** (d)	**33.** (d)	**34.** (d)						

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Hints and Solutions

1. Assume that B does the whole job in x days. In that case, A does the whole job in $\frac{x}{1.5}$ days.

Since they do the whole job in 15 days working together, work done in one day $= \frac{1}{15}$

According to the question, $\frac{1}{x} + \frac{1.5}{x} = \frac{1}{15}$, hence $x = 37.5$ days

So, number of days taken by A $= \frac{37.5}{1.5}$ days $= 25$ days

Hence, option (d) is the answer.

2. Let C be the circumference of the circle, and let r be the radius of the circle.

Using log properties, $C = \log_{10}(b^4) = 4\log_{10}(b)$ and $r = \log_{10}(a^2) = 2\log_{10}(a)$

Since $C = 2\pi r$, $4\log_{10}(b) = 2\pi \times 2\log_{10}(a)$ or $\log_a b = \frac{\log_{10}(b)}{\log_{10}(a)} = \pi$

Hence, option (c) is the answer.

3. If $ac < 0$, then signs of a and c will be opposite (one will be positive and other will be negative).

Thus, product of the roots $= \alpha\beta = \frac{c}{a} < 0$

Therefore, α and β will have opposite signs. So, one of the roots will be positive and other one will be negative.

Hence, option (b) is the answer.

4. Let R_i and B_j designate the red card numbered i and the blue card numbered j, respectively.

B_5 is the only blue card that R_5 evenly divides, so R_5 must be at one end of the stack and B_5 must be the card next to it.

R_1 is the only other red card that evenly divides B_5, so R_1 must be the other card next to B_5.

B_4 is the only blue card that R_4 evenly divides, so R_4 must be at one end of the stack and B_4 must be the card next to it.

R_2 is the only other red card that evenly divides B_4, so R_2 must be the other card next to B_4.

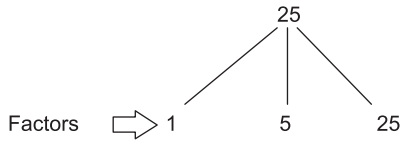
R_2 does not evenly divide B_3, so B_3 must be next to R_1, and so, B_6 must be next to R_2, and R_3 must be in the middle.

This yields the following arrangement from top to bottom: $\{R_5, B_5, R_1, B_3, R_3, B_6, R_2, B_4, R_4\}$

Therefore, the sum of the numbers on the middle three cards is $3 + 3 + 6 = 12$

Hence, option (d) is the answer.

5. If a number has three factors, then it should be a square of a prime number. We have following two-digit numbers as the square of prime number: 25 and 49. It can be seen that number 25 has following factors:



Therefore, there are two numbers.

Hence, option (b) is the answer.

6. **Method 1**

Shopkeeper sells his goods at his cost price.

If he sells x grams at the place of 1000 g, his profit percentage $= \frac{1000 - x}{x} \times 100$

According to the question, $20 = \frac{1000 - x}{x} \times 100 \Rightarrow x = \frac{1000}{1.2} = 833.33$ g

Method 2

If profit is 20%, then shopkeeper must have been charging the customer for 1.2 kg.

Therefore, $x = \frac{1000}{1.2} = 833.33$ g

Hence, option (c) is the answer.

7. Amar paid half of what others have paid and Amar paid 1/3rd of total $= 1/3 \times 6000 = ₹2000$

(To understand this, assume amount paid by Akbar + Anthony $= 2x$. Then, amount paid by Amar $= ₹x$. Therefore, total amount paid $= ₹3x$. Hence, amount paid by Amar ($₹x$) is $\frac{x}{3x} = \frac{1}{3}$ of total.)

Similarly, amount paid by Akbar = 1/4th of total $= 1/4 \times 6000 = ₹1500$

Thus, amount paid by Anthony $= ₹6000 - ₹2000 - ₹1500 = ₹2500$

Hence, option (b) is the answer.

8. If the second size has length x, then the first side has length $3x$, and we have the third side that has length 15 units.

Using triangle inequality, we have $x + 15 > 3x \rightarrow 2x < 15 \rightarrow x < 7.5$.

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Now, since we want the greatest perimeter, we want the greatest integer x, and if $x < 7.5$, then $x = 7$. Then, the first side has length $3 \times 7 = 21$, the second side has length 7, the third side has length 15, and so the perimeter is $21 + 7 + 15 = 43$

Hence, option (a) is the answer.

9. We want to find the maximum any player could make, so assume that everyone makes the minimum possible and that the combined salaries total the maximum of 700,000.

$700{,}000 = 20 \times 15{,}000 + x$

$x = 400{,}000$

The maximum any player could make is $400,000.

Hence, option (c) is the answer.

10. **Method 1**

A quadratic equation always has two roots, unless it has a repeated root. That means we can write the quadratic as a square, and the coefficients 4 and 9 suggest this. Completing the square, we get

$0 = (2x \pm 3)^2 = 4x^2 \pm 12x + 9$, so $\pm 12 = a + 8 \rightarrow a = 4, -20$

The sum of these is $-20 + 4 = -16$

Hence, option (a) is the answer.

Method 2

For a unique solution of x, the determinant must equal to 0.

Therefore, we can write $(a + 8)^2 - 144 = 0$, which factors to $a^2 + 16a - 80 = 0$

Sum of the solutions of this equation is $\frac{-b}{a} = -16$

Hence, option (a) is the answer.

11. For convenience's sake, we can transform A to the origin and B to (99,999) (this does not change the problem).

The line AB has the equation $y = \frac{999}{99}x = \frac{111}{11}x$

The coordinates are integers if x is divisible by 11. So, the values of x are 11, 22, …, 88, with a total of 8 coordinates.

Hence, option (d) is the answer.

12. Test the box labelled—Red and White. Now if the ball is Red, label the box Red. Now, the box which has the label white is either Red or Red and White. However, it cannot be Red. Hence, it is Red and White. The last box is White.

Hence, option (c) is the answer.

13. Sixty is wrong because then to arrive at a total of 121, the other box will have to weigh 61 kg, which will be obviously not the highest. Sixty-four is wrong too, because then to add up to 121, the other weight will have to be 57, and to make up to a total of 120, the next box shall have a weight 63 kg, which obviously makes the maximum possible total as 64 + 63 = 127. Sixty-two is the correct answer because the other boxes shall be 59, 54, and 56. These will give all the totals given above.

Hence, option (b) is the answer.

14. Order should be: First A and B will cross the bridge (in 2 min). Then, A will come back (1 min). Next C and D will go together (10 min). Then, B will come back (2 min). Then, A and B will cross the bridge (2 min).

Total time = 2 min + 1 min + 10 min + 2 min + 2 min = 17 min

15. Each one of the 26 players played 25 matches, and none of the matches ended in a draw. Hence, all the scores must be even. Also, each one of them scored different from the other. The maximum score possible is 50, and the minimum possible score is 0.

There are exactly 26 possible scores 50, 48, 46, 44, ..., 4, 2, 0. The ranking is in alphabetical order, which means A scored 50, B scored 48, and so on up to Z scored 0. This score is possible if A wins all the matches, B loses only to A and wins against all others, and so on. This implies that every player wins only against those players who are below him in the final ranking. Since M is above N in the alphabetical order and in the final ranking, M wins over N.

Hence, option (a) is the answer.

16. $B = (2^{64} + 2^{63} + 2^{62} + ... + 2^0) = (2^0 + 2^1 + 2^2 + ... + 2^{64})$

Using the summation formula of geometric progression:

$B = (2^0)\frac{2^{65} - 1}{2 - 1} = 2^{65} - 1$

Therefore, A is 1 more than B.

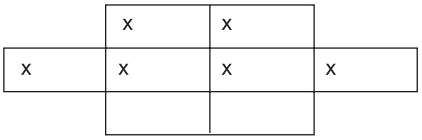
Hence, option (c) is the answer.

17. Total number of boxes = 8

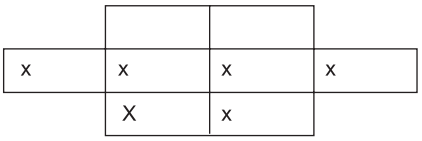
Without any restriction, number of ways boxes can be filled = ${}^8C_6 = 28$

Out of which, following two cases are not to be counted.

Case I



Case II



Therefore, total number of cases = 28 – 2 = 26.

Hence, option (b) is the answer.

18. Value of $x = \sqrt{9 \times 4} = 6$ days.

Hence, option (a) is the answer.

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19. Suppose that the common perimeter is P. Then, the side lengths of the square and triangle, respectively, are $\frac{P}{4}$ and $\frac{P}{3}$. The circle circumscribed about the square has a diameter equal to the diagonal of the square, which is $\frac{P\sqrt{2}}{4}$. Therefore, the radius is $\frac{P\sqrt{2}}{8}$, and the area of the circle is $\pi\times\left(\frac{P\sqrt{2}}{8}\right)^2=\pi\times\frac{2P^2}{64}=\frac{P^2\pi}{32}=(A)$

Now, consider the circle circumscribed around the equilateral triangle. Due to symmetry, the circle must share a centre with the equilateral triangle. The radius of the circle is simply the distance from the centre of the triangle to a vertex. This distance is $\frac{2}{3}$ of an altitude. By 30–60–90° right triangle properties, the altitude is $\frac{\sqrt{3}}{2}\times s$, where s is the side. So, the radius is $\frac{2}{3}\times\frac{\sqrt{3}}{2}\times\frac{P}{3}=\frac{P\sqrt{3}}{9}$

The area of the circle is $\pi\times\left(\frac{P\sqrt{3}}{9}\right)^2=\pi\times\frac{3P^2}{81}=\frac{P^2\pi}{27}=(B)$

So, $\frac{A}{B}=\frac{\frac{P^2x}{32}}{\frac{P^2x}{27}}=\frac{P^2\pi}{32}\times\frac{27}{P^2\pi}=\frac{27}{32}$

Hence, option (c) is the answer.

20. We know that CI for next year is calculated over the amount of previous year.

Interest obtained in second year = ₹2880 – ₹2400 = ₹480

Rate of interest = $\frac{480}{2400}\times 100=20\%$

Therefore, interest obtained in third year will be 20% of amount at the end of ₹2880 = 20% of 2880 = ₹576

Hence, option (b) is the answer.

21. Let $(x+y)=6k$, $(y+z)=7k$, and $(z+x)=8k$

Then, $2(x+y+z)=21k$ or $2\times 14=21k$

Thus, $k=\frac{28}{21}=\frac{4}{3}$

So, $(x+y)=\left(6\times\frac{4}{3}\right)=8$

Therefore, $z=(x+y+z)-(x+y)=(14-8)=6$

Hence, option (a) is the answer.

22. Let the three jars contain 30, 40, and 50 litres of mixtures, respectively.

Milk in first mixture = $\left(30\times\frac{4}{5}\right)$ L = 24 L

Water in first mixture = (30 – 24) L = 6 L

Milk in second mixture = $\left(40\times\frac{3}{4}\right)$ L = 30 L

Water in second mixture = (40 – 30) L = 10 L

Milk in third mixture = $\left(50\times\frac{5}{7}\right)$ L = $\frac{250}{7}$ L

Water in third mixture = $\left(50-\frac{250}{7}\right)$ L = $\frac{100}{7}$ L

Total milk in final mixture = $\left(24+30+\frac{250}{7}\right)$ L = $\frac{3140}{35}$ L

Total water in final mixture = $\left(6+10+\frac{100}{7}\right)$ L = $\frac{1060}{35}$ L

Ratio of milk and water in fourth jar = $\frac{3140}{35}:\frac{1060}{35}$ = 157:53

23. Clearly, the retailer gets 1 dozen out of 6 dozens free.

Therefore, equivalent discount = $(1/6\times 100)\%=16\ 2/3\%$

Hence, option (c) is the answer.

24. This question is a NO-brainer.

If all the numbers increase by 30% (or change by x%), then average will also increase by 30% (change by x%).

Hence, option (c) is the answer.

25. This is not possible. Average speed has to be less than twice the lower speed. In this case, lower speed = 40 kmph; therefore, average speed will be less than 80 kmph.

Hence, option (d) is the answer.

26. There are two restrictions in this question:

(i) For a number to be odd, unit digit should be either 1 or 3 or 5 or 7 or 9.

(ii) Thousand's place cannot be filled with 8 or 9. In other words, thousand's place should be filled with any digit from 1 to 7.

Ten's place digit and hundred's place digit can be filled with any available digit.

For unit's digit, when it is filled with 9, thousand's place can be filled in 7 ways, namely any digit from 1 to 7, and remaining two places can be filled in $8\times 7=56$ ways.

So, total number of numbers formed in this way = $56\times 7=392$

Now, if unit's place is filled with any of the four digits 1, 3, 5, or 7, then the thousand's place can be filled in 6 ways (0 will be excluded), and remaining two places can be filled in $8\times 7=56$ ways.

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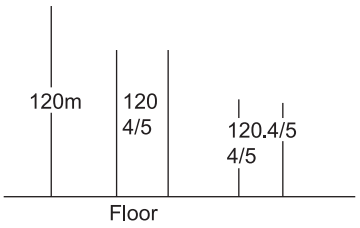
So, total number of numbers formed in this way = $56 \times 6 \times 4 = 1344$

So, total number of numbers = 392 + 1344 = 1736

Hence, option (b) is the answer.

27. Method 1

Distance covered before first rebound = 120 m



Then, the ball bounces to a height of $120 \times \frac{4}{5}$ m and then falls from the same height.

Next time ball will go up by $120 \times \frac{4}{5} \times \frac{4}{5}$ m and then will fall from the same height.

So, total distance covered = $120 + 2 \times 120 \times \frac{4}{5} + 2 \times 120 \times \frac{4}{5} \times \frac{4}{5} + \ldots \infty$

$= 120 + 2 \times 120\left(\frac{4}{5} + \frac{4}{5} \times \frac{4}{5} + \ldots \infty\right) = 120 + 240\left[\frac{\frac{4}{5}}{1 - \frac{4}{5}}\right]$

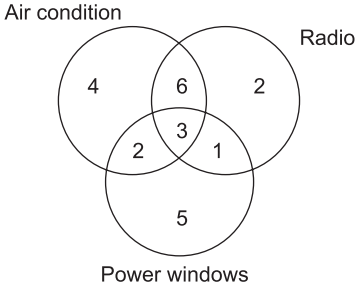
= 1080 m

Method 2

In these questions, if the ball rebounds to $\frac{a}{b}^{\text{th}}$ of original height H, then total distance covered before the ball comes to rest = $H \times \frac{b+a}{b-a}$

Using the above Concept Note, total distance covered = $120 \times \frac{5+4}{5-4} = 1080$ m

28. From the given conditions, we have



When we add all the values, we get a total of 23 cars.

So, 2 cars do not have any of air condition, radio, and power windows.

Hence, option (c) is the answer.

29.

$$\frac{1}{1+\sqrt{2}+\sqrt{3}} + \frac{1}{1+\sqrt{2}-\sqrt{3}} + \frac{1}{1-\sqrt{2}+\sqrt{3}} + \frac{1}{1-\sqrt{2}-\sqrt{3}} =$$

$$\frac{1+\sqrt{2}-\sqrt{3}+1+\sqrt{2}+\sqrt{3}}{\left((1+\sqrt{2})+\sqrt{3}\right)\left((1+\sqrt{2})-\sqrt{3}\right)} + \frac{1-\sqrt{2}-\sqrt{3}+1-\sqrt{2}+\sqrt{3}}{\left((1-\sqrt{2})+\sqrt{3}\right)\left((1-\sqrt{2})-\sqrt{3}\right)}$$

$$\frac{2+2\sqrt{2}}{2\sqrt{2}} - \frac{2-\sqrt{2}}{2\sqrt{2}} = \frac{2(2\sqrt{2})}{2\sqrt{2}} = 2$$

[Using $(a + b)(a - b) = a^2 - b^2$]

Hence, option (a) is the answer.

30. For the curves to intersect, $\log_{10} x = x^{-1}$

Thus, $\log_{10} x = \frac{1}{x}$ or $x^x = 10$

This is possible for only one value of x $(2 < x < 3)$.

Alternatively, it can be solved by drawing the graphs too.

31. We can calculate this for both equal number of odd terms and even terms.

For example, if the GP = 1, 2, 4, 8, 16, 32

Common ratio = Sum of all the odd numbered terms/sum of all even numbered terms = (1 + 4 + 16)/(2 + 8 + 32) = 21/42 = 1/2

Even if we take only first four terms, common ratio will remain same [(1 + 4)/(2 + 8)] = 1/2

However, if the GP contains only five terms or seven terms, we cannot have unique value of common ratio. Since in this question 1001 terms are given, common ratio cannot be determined.

Hence, option (d) is the answer.

32. $N = \{2, 3, 5, 7, 11, 13, 17, 19, 23, 29\}$

A rational number will be made by taking any two elements of this set in any order.

Total number of elements = 10

Total number of ways in which two elements can be taken = ${}^{10}P_2$

Total number of distinct rational numbers = ${}^{10}P_2$ + 1 (including 1)

Hence, option (d) is the answer.

33. Profit percentage = $\frac{\text{Profit}}{\text{CP}} \times 100$

It can be seen that the profit amount remains the same. The change in CP/unit = Change in SP/unit. However, we do not know that $A > B$ or $B > A$, so we do not have the information that if the base has increased or decreased. Therefore, it cannot be determined.

Hence, option (d) is the answer.

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34. Let us generate both the equations first:

$40m + 50f = 1000$

$250m + 300f + 40 \times 15m + 50 \times 10 \times f = A$

$850m + 8000f = A$

where m and f are the number of males and females and A is the amount paid by the service provider.

Then, the possible values for f are 8, 9, 10, 11, and 12

If $f = 8$, then $m = 15$

If $f = 9$, 10, 11, then m will not be an integer, while $f = 12$, then m will be 10

By putting $f = 8$ and $m = 15$, $A = 18{,}800$. When $f = 12$ and $m = 10$, then $A = 18{,}100$

Therefore, the number of males will be 10.

Hence, option (d) is the answer.

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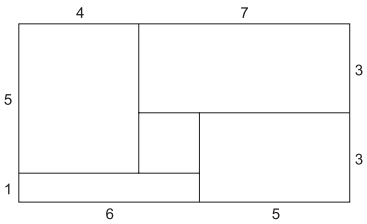
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3 Section Test 3

Instructions to the test takers

- There are 34 questions in this test.
- Each question carries 3 marks.
- There is 1/3rd NEGATIVE marking.
- Time allotted – 60 minutes

Q.1 Rose fills each of the rectangular regions of her rectangular flower bed with different types of flowers. The lengths, in feet, of the rectangular regions in her flower bed are as shown in the figure. She plants one flower per square foot in each region. Asters cost \$1 each, begonias cost \$1.50 each, cannas cost \$2 each, dahlias cost \$2.50 each, and Easter lilies cost \$3 each. What is the least possible cost, in dollars, for her garden?



(a) 108 (b) 115 (c) 132 (d) 144

Q.2 If a sum of money at simple interest doubles in 6 years, it will become 4 times in:

(a) 12 years (b) 14 years
(c) 16 years (d) 18 years

Q.3 The numbers x and y are three-digit positive integers, and $x + y$ is a four-digit integer. The tens digit of x equals 7 and the tens digit of y equals 5. If $x < y$, which of the following must be true?

I. The units digit of $x + y$ is greater than the units digit of either x or y.
II. The tens digit of $x + y$ equals 2.
III. The hundreds digit of y is at least 5.

(a) II only (b) III only
(c) I and II (d) I and III

Q.4 Let $a + ar_1 + ar_1^2 + ar_1^3 + \ldots$ and $a + ar_2 + ar_2^2 + ar_2^3 + \ldots$ be two different infinite geometric series of positive numbers with the same first term. The sum of the first series is r_1, and the sum of the second series is r_2. What is $r_1 + r_2$?

(a) 0 (b) $\frac{1}{2}$
(c) 1 (d) $\frac{1+\sqrt{5}}{2}$

Q.5 A bag contains 3 red, 4 black, and 2 white balls. What is the probability of drawing a red and a white ball in two successive draws, each ball being put back after it is drawn?

(a) 2/27 (b) 1/9 (c) 1/3 (d) 4/27

Q.6 There are only three kinds of books in the library, namely fiction, non-fiction, and biology. Ratio of fiction to non-fiction is 3:2; ratio of non-fiction to biology is 4:3. Total number of books is more than 1000. What is the minimum number of books in the library?

(a) 1014 (b) 1300 (c) 1001 (d) 1256

Q.7 If $a_1 = 1$ and $a_{n+1} - 3a_n + 2 = 4n$ for every positive integer n, then a_{100} equals:

(a) $3^{99} - 200$ (b) $3^{99} + 200$
(c) $3^{100} - 200$ (d) $3^{100} + 200$

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Q.8 A and B are two horses participating in a horse race with total six horses. How many different arrangements of finishes are there if A always finishes before B and there is no tie in ranking?
(a) 720 (b) 360 (c) 120 (d) 24

Q.9 In ΔABC, points P, Q, and R are the mid-points of side AB, BC, and CA, respectively. If area of ΔABC is 20 sq. units, then find the area of Δ PQR.
(a) 10 sq. units (b) $5\sqrt{3}$
(c) 5 sq. units (d) None of these

Q.10 In a hotel, P% of the total rooms are deluxe rooms and Q% of the total rooms are painted light blue. Which of the following best represents the minimum possible percentage of the light blue-painted deluxe rooms?
(a) $P - Q$ (b) $P - Q + 100$
(c) $P + Q - 100$ (d) None of these

Q.11 My name is Meeta. But, my son accidentally types the name by interchanging a pair of letters in my name. What is the probability that despite this interchange, the name remains unchanged?
(a) 0.5 (b) 0.1 (c) 0.2 (d) 0.25

Q.12 A boat takes 90 min less to travel 36 miles downstream than to travel the same distance upstream. If the speed of the boat in still water is 10 kmph, then the speed of the stream is:
(a) 2 kmph (b) 3 kmph
(c) 4 kmph (d) 5 kmph

Q.13 A rectangular courtyard 3.78 m long and 5.25 m broad is to be paved exactly with square tiles with same size. The minimum number of such tiles required is:
(a) 350 (b) 300 (c) 450 (d) 495

Q.14 If the number of subsets of a given set is 64, then find the number of ways of selecting any 2 elements from the set.
(a) 10 (b) 30 (c) 20 (d) 15

Q.15 A mixture of water and milk is sold at the same rate as that of the cost price of pure milk per litre. If total profit obtained is 20% on the whole mixture, then what is the ratio of milk to water in the mixture?
(a) 4:1 (b) 5:1 (c) 6:1 (d) 2:1

Q.16 A jet plane is at a vertical height of h metre above the ground. The angle of depression of two tanks on the horizontal ground are found to be $x°$ and $y°$ $(x > y)$. What is the distance between the tanks (in metre)?
(a) $\frac{h(\tan x - \tan y)}{xy}$ (b) $\frac{h(\tan x \times \tan y)}{\tan x - \tan y}$
(c) $\frac{h(\tan x - \tan y)}{\tan x + \tan y}$ (d) $\frac{h(\tan x - \tan y)}{\tan x \cdot \tan y}$

Q.17 Given that Sum of 1st n even consecutive natural numbers is 1560. What is the value of n?
(a) 38 (b) 39 (c) 76 (d) 78

Q.18 A certain deck of cards contains 2 blue cards, 2 red cards, 2 yellow cards, and 2 green cards. If two cards are randomly drawn from the deck one by one without replacement, what is the probability that none of them will be blue cards?
(a) $\frac{15}{28}$ (b) $\frac{1}{4}$
(c) $\frac{9}{28}$ (d) None of these

Q.19 How many 5-digit positive integers are there where no two consecutive digits are the same?
(a) $9 \times 9 \times 8 \times 7 \times 6$ (b) $9 \times 8 \times 7 \times 6 \times 5$
(c) 9^5 (d) 9×8^4

Q.20 ABC and DBC are right-angled triangles with common hypotenuse BC. The side AC and BD are extended to intersect at P. Then what is the value of $\frac{AP \times PC}{DP \times PB}$?
(a) 2 (b) $\frac{1}{3}$
(c) 1 (d) None of these

Q.21 A shopkeeper marks up his good by 10% and then gives a discount of 10%. What is the net effect on the shopkeeper's business?
(a) 1% profit (b) No profit, no loss
(c) 1% loss (d) 0.5% profit

Q.22 Rahul rows a certain distance downstream in 6 h and same distance upstream in 10 h. What is the ratio of speed of boat to the speed of river?
(a) 2:1 (b) 3:1 (c) 4:1 (d) 6:1

Q.23 12 men and 18 women can clean a building in 50 days. In how many days can 30 men and 45 women clean the same building?
(a) 25 days
(b) 30 days
(c) 20 days
(d) Cannot be determined

Q.24 A kid has 300 coins of different denominations—25 paise coins, 50 paise coins, and ₹1 coins. In all, he has ₹220. Ratio of number of 25 paise coins: Number of 50 paise coins: ₹1 coins = 2:5:8. How many 50 paise coins does he have?
(a) 100 (b) 50 (c) 200 (d) 20

Q.25 There are two flags to be hoisted in a yearlong function. Flag 1 is hoisted after every 3 days and Flag 2 is hoisted after every 4 days. Both the flags have

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been hoisted on 1st January. On which date both the flags will be hoisted again?
(a) 10th January (b) 12th January
(c) 13th January (d) 24th January

Q.26 What is the ratio of the area of a rectangle and the area of parallelogram if their height is same and their base length is equal?
(a) 2:1
(b) 1:2
(c) 1:1
(d) Cannot be determined

Q.27 There are 20 questions in a question paper. If no two students solve the same combination of questions but solve equal number of questions, then find the maximum number of students who appeared in the examination.
(a) $^{20}C_9$ (b) $^{20}C_{11}$
(c) $^{20}C_{10}$ (d) None of these

Q.28 Which of the following numbers will not divide $(4^{61} + 4^{62} + 4^{63} + 4^{64})$?
(a) 8 (b) 10 (c) 17 (d) 18

Q.29 If 5-digit number 653xy (where x is tens place digit and y is unit digit) is divisible by 80, then what is the value of $(x + y)$?
(a) 2 (b) 3 (c) 4 (d) 6

Q.30 P = Set of first N positive numbers. There are 16 numbers in Set P, which are divisible by both X and Y. There are 50 numbers in Set P, which are divisible by X but not by Y, and 34 numbers in Set P divisible by Y but not by X. How many numbers are divisible by X?
(a) 16 (b) 56 (c) 66 (d) 50

Q.31 In writing a quadratic equation of the form $x^2 + bx + c = 0$, a student writes the coefficient of x incorrectly and finds the roots as 7 and 8. Another student makes a mistake in writing the constant term and gets the roots as 8 and –3. Find the correct quadratic equation.
(a) $x^2 - 5x + 56 = 0$ (b) $x^2 - 5x + 21 = 0$
(c) $x^2 - 21x + 56 = 0$ (d) $x^2 - 7x + 21 = 0$

Q.32 Let A, B, and C be three distinct points on the graph of $y = x^2$ such that line AB is parallel to the x-axis, and ΔABC is a right triangle with area 2008 sq. units. What is the sum of the digits of the y-coordinate of C?
(a) 16 (b) 17 (c) 18 (d) 19

Q.33 Given that $4^{x1} = 5$, $5^{x2} = 6$, $6^{x3} = 7 \ldots 100^{x124} = 128$. What is the value of $x_1 \times x_2 \times x_3 \times \ldots \times x_{124}$?
(a) 1 (b) 2 (c) $\frac{7}{2}$ (d) $\frac{7}{3}$

Q.34 First term of a sequence is 2005. Each succeeding term is the sum of the cubes of the digits of the previous term. What is the 2005th term of the sequence?
(a) 29 (b) 55 (c) 85 (d) 250

Answers

1. (a)	**2.** (d)	**3.** (b)	**4.** (b)	**5.** (d)	**6.** (c)	**7.** (c)	**8.** (b)	**9.** (c)	**10.** (c)
11. (c)	**12.** (c)	**13.** (c)	**14.** (d)	**15.** (b)	**16.** (d)	**17.** (b)	**18.** (a)	**19.** (c)	**20.** (c)
21. (c)	**22.** (c)	**23.** (c)	**24.** (a)	**25.** (c)	**26.** (c)	**27.** (c)	**28.** (d)	**29.** (c)	**30.** (c)
31. (a)	**32.** (c)	**33.** (c)	**34.** (d)						

Hints and Solutions

1. The areas of the five regions from greatest to least are 21, 20, 15, 6, and 4 sq. units.

If we want to minimize the cost, then we want to maximize the area of the cheapest flower and minimize the area of the most expensive flower. Doing this, the cost is $1 \times 21 + 1.50 \times 20 + 2 \times 15 + 2.50 \times 6 + 3 \times 4$, which simplifies to \$108. Therefore, the answer is 108. Hence, option (a) is the answer.

2. Method 1

Let sum = x. then, SI = x

Hence, rate = $\frac{100x}{6x}\% = \frac{50}{3}\%$

Now, sum = x, SI = $3x$, rate = $\frac{50}{3}\%$

We know that SI = $\frac{PRT}{100}$

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Hence, time = $100 \times \frac{3x}{x} \times \frac{3}{50} = 18$ years

Method 2

Assume that sum invested initially = ₹100. It becomes ₹200 in 6 years. Hence, ₹100 interest is incurred in 6 years. For the sum to be 4 times, amount = ₹400. So, interest required to incurred = ₹300

Now, ₹100 interest is incurred in 6 years; hence, ₹300 interest would incur in 18 years.

Hence, option (d) is the answer.

3. Assume that $x = abc$ and $y = def$

According to the question, $x = a7c$ and $y = b5f$

$x < y$ and $x + y = wxyz$ (a 4-digit number)

I. The units digit of $x + y$ is greater than the units digit of either x or y. We really cannot be sure of this anyways and hence false.

II. The tens digit of $x + y$ equals 2. It can be 2 or 3 and hence false.

III. The hundreds digit of y is at least 5. Since $x < y$ and sum of x and y is a 4-digit number, y should be more than 500. Hence, hundreds digit of y should be at least 5. Hence, III is true.

Hence, option (b) is the answer.

4. Using the formula for the sum of a geometric series, we get that the sums of the given two sequences are $\frac{a}{1-r_1}$ and $\frac{a}{1-r_2}$.

Hence, we have $\frac{a}{1-r_1} = r_1$ and $\frac{a}{1-r_2} = r_2$. This can be rewritten as $r_1(1-r_1) = r_2(1-r_2) = a$

As r_1 and r_2 are distinct, these must be precisely two roots of the equation:

$x^2 - x + a = 0$

Using sum of roots of a quadratic equation, we get that the sum of these two roots is 1.

Hence, option (b) is the answer.

5. **Case I** First ball is red and second ball is white.

Probability = $\frac{3}{9} \times \frac{2}{9} = \frac{6}{81}$

Case II First ball is white and second ball is red

Probability = $\frac{2}{9} \times \frac{3}{9} = \frac{6}{81}$

Therefore, total probability = $\frac{6}{81} + \frac{6}{81} = \frac{12}{81} = \frac{4}{27}$

Hence, option (d) is the answer.

6. Given that:

Fiction:Non-fiction = 3:2 = 6:4

Non-fiction:Biology = 4:3

Combining both, we get Fiction:Non-fiction:Biology = 6:4:3

Or, $6x + 4x + 3x$ = Total number of books or $13x$ = Total number of books

Since total number of books is more than 1000 and this number should be divisible by 13, then the lowest number satisfying this is $1001 = 13 \times 11 \times 7$.

Hence, option (c) is the answer.

7. $a_1 = 1$, $a_{n+1} - 3a_n + 2 = 4n$

$a_{n+1} = 3a_n + 4n - 2$

When $n = 1$, then $a_2 = 3a_n + 4n - 2 = 3 + 4 - 2 = 5$

When $n = 2$, then $a_3 = 3 \times 5 + 4 \times 2 - 2 = 21$, and so on

Hence, next values can be obtained.

From the options, we get an idea that a_n can be expressed in a combination of some power of 3 and some multiple of 100. Let us verify the options one by one:

Option (a): $3^{99} - 200$; tells us that a_n could be: $3^{n-1} - 2 \times n$; but it does not fit a_1 or a_2 or a_3.

Option (b): $3^{99} + 200$; tells us that a_n could be: $3^{n-1} + 2 \times n$; again not valid for a_1, a_2, etc.

Option (c): $3^{100} - 200$; tells $3^n - 2n$, which is valid for the values of a_1 or a_2 or a_3 and so on.

Option (d): $3^{100} + 200$; tells $3^n + 2n$; again not valid.

Hence, option (c) is the answer.

8. **Method 1**

Out of six horses in a race, A has to finish before B, and remaining four can finish at any other positions. Understand that question is not saying that A has to finish immediately before B. There may be or may not be horses between A and B.

Let us see the cases:

Case I If A finishes at 1st rank, then B could be in any of other 5 positions in 5 ways and other horses can finish the race in 4! ways. Hence, total number of ways = $5 \times 4!$

Case II If A finishes at 2nd rank, then B could be in any of other 4 positions in 4 ways and other horses can finish the race in 4! ways. Hence, total number of ways = $4 \times 4!$

Case III If A finishes at 3rd rank, then B could be in any of other 3 positions in 3 ways and other horses can finish the race in 4! ways. Hence, total number of ways = $3 \times 4!$

Case IV If A finishes at 4th rank, then B could be in any of other 2 positions in 2 ways and other horses can finish the race in 4! ways. Hence, total number of ways = $2 \times 4!$

Case V If A finishes at 5th rank, then B could be in any of other 1 position in 1 way and other horses can finish the race in 4! ways. Hence, total number of ways = $1 \times 4!$

A cannot finish 6th, since A has to be ahead of B. Therefore, total number of ways = $5 \times 4! + 4 \times 4! + 3 \times 4! + 2 \times 4! + 4! = 120 + 96 + 72 + 48 + 24 = 360$

Hence, option (b) is the answer.

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Method 2

Total number of cases (without any constraint) = 6! Either A will finish before B or B will finish before A and both of these are equally likely. Hence, number of ways A will finish before B = Number of ways B will finish before A = $\frac{6!}{2} = 360$.

Hence, option (b) is the answer.

9. In case of triangle, the line joining the mid-points of any two sides is half the length of its third side. Hence, every side of ΔPQR would be half the length of respective side of ΔABC. Hence, area of ΔPQR would be $\frac{1}{4}$ of the area of ΔABC = $\frac{1}{4} \times 20 = 5$ sq. units.

Hence, option (c) is the answer.

10. We know that $n(P \cup Q) = n(P) + n(Q) - n(P \cap Q)$ or $n(P \cap Q) = n(P) + n(Q) - n(P \cup Q)$

Where P = % of total rooms that are deluxe rooms and Q = % of rooms painted light blue and $(P \cap Q)$ = intersection of the two sets = light blue-painted deluxe rooms.

According to the question, $n(P)$ and $n(Q)$ are fixed. To obtain the minimum value of $(P \cap Q)$, value of $(P \cup Q)$ should be maximum which is 100% (meaning that all the deluxe rooms are painted light blue).

Hence, minimum value of $n(P \cap Q)$ = Light blue-painted deluxe rooms = $P + Q - 100$.

Hence, option (c) is the answer.

11. There are actually 20 ways to interchange the letters, namely the first letter could be one of 5, and the other letter could be one of 4 left. So, total pairs by product rule = 20.

Now, there are two cases when it would not change the name. First, keeping them all the same. Second, interchanging the two EEs together. These two cases would leave the name intact.

Therefore, probability that name remains unchanged = $\frac{2}{20} = 0.1$.

Hence, option (b) is the answer.

12. Let the speed of the stream be x kmph. Then, speed downstream = $(10 + x)$ kmph and speed upstream = $(10 - x)$ kmph.

According to the question,

$$\frac{36}{(10-x)} - \frac{36}{(10+x)} = \frac{90}{60}$$

Or, $x^2 + 48x - 100 = 0$

Or, $(x + 50)(x - 2) = 0$

Therefore, $x = 2$ kmph

Hence, option (a) is the answer.

13. Length = 378 cm and breadth = 525 cm

Maximum length of a square tile = HCF of (378,525) = 21 cm

Number of tiles = $(378 \times 525)/(21 \times 21) = (18 \times 25) = 450$

Hence, option (c) is the answer.

14. The number of subsets of any set = 2^n, where n is the number of elements of the set.

Since the given set has 64 subsets, the number of elements will be 6 as value of $2^6 = 64$.

Number of ways of selecting 2 elements out of 6 elements = ${}^6C_2 = 15$ ways. Hence, option (d) is the answer.

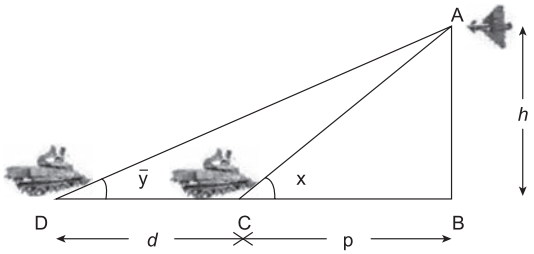
15. Understand that since cost price per litre is same as selling price per litre, profit % will be equal to the additional volume generated by mixing water into the pure milk.

In short, ratio of milk to water = 100%:20% = 5:1

Hence, option (b) is the answer.

Note: In such questions, answer will be = (100%:Profit percentage)

16. From the question, following diagram can be drawn:



Let A be the jet plane and C and D be the tanks.

In ∇ABC, $\tan x = \frac{AB}{BC}$ or $p = BC = \frac{h}{\tan x}$ (i)

In ∇ABD, $\tan y = \frac{AB}{BD} = \frac{h}{d+p}$

Hence, $d + p = \frac{h}{\tan y}$ (ii)

Putting the value of p from equation (i) in equation (ii), we get:

$$d + \frac{h}{\tan x} = \frac{h}{\tan y} \text{ or } d = \frac{h}{\tan y} - \frac{h}{\tan x} = \frac{h(\tan x - \tan y)}{\tan x \cdot \tan y}$$

Hence, option (d) is the answer.

17. Given that sum of 1st n even consecutive natural numbers = $1560 = 39 \times 40 = n \times (n + 1)$

Therefore, $n = 39$

Hence, option (b) is the answer.

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18. Chance of drawing a blue on the first draw $= \frac{2}{8}$, so chance of not drawing a blue in the first draw is $\frac{6}{8}$. Similarly, chance of not drawing blue on second draw $= \frac{5}{7}$.

Thus, probability of not drawing blue in two draws $= \frac{6}{8} \times \frac{5}{7} = \frac{15}{28}$

Hence, option (a) is the answer.

19. First place can be filled in 9 ways (any digit from 1 to 9 as 0 cannot come at the 1st place).

Number of ways 2nd place can be filled = 8 digits out of 1 to 9 (because the digit that came at the 1st place will not come at 2nd place) and zero = 9 digits.

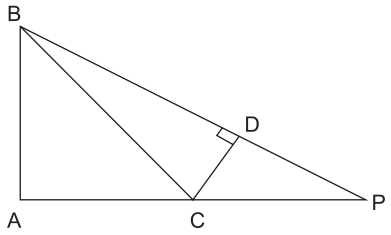
Number of ways 3rd place can be filled = 8 digits out of 1 to 9 (because the digit that came at the 2nd place will not come at 3rd place) and zero = 9 digits.

Similarly, number of ways 4th and 5th places each can be filled = 9 ways each.

Therefore, total number of ways $= 9 \times 9 \times 9 \times 9 \times 9 = 9^5$

Hence, option (c) is the answer.

20. From the question, following diagram can be drawn:



$\angle$ DPC = $\angle$ APB and $\angle$P is common in triangles ΔBAP and ΔCDP.

Hence, ΔBAP is similar to ΔCDP (angles taken in order).

So, ratio of sides will be proportionate.

$$\therefore \frac{DP}{AP} = \frac{CP}{BP} \Rightarrow \frac{DP \times BP}{AP \times PC} = 1$$

Hence, option (c) is the answer.

21. Assume that the cost price = ₹100.

So, mark-up price = ₹110.

From the fundamentals of profit and loss chapter, we know that discount is always calculated over the mark-up price (also known as tag price or list price).

Further, he gives a discount of 10% over his mark-up price. Hence, discount = 10% of ₹110 = ₹11.

So, net selling price = ₹110 – ₹11 = ₹99. Comparing it with the cost price (₹100), it can be clearly seen that loss percentage = 1%.

Hence, option (c) is the answer.

22. We know that $\frac{\text{Downstream speed}}{\text{Upstream speed}} = \frac{\text{Time taken upstream}}{\text{Time taken downstream}}$

Or, $\frac{B+R}{B-R} = \frac{10}{6}$ or $6(B+R) = 10(B-R)$. Solving it, we get B:R = 8:2 = 4:1

Hence, option (c) is the answer.

23. It can be observed that 30 men + 45 women = 2.5 × (12 men + 18 women)

Assume that 30 men + 45 women will take N days in completing the job.

Using the formula, $M_1D_1 = M_2D_2$

(12 men + 18 women) × 50 = (30 men + 45 women) × N

= 2.5 × (12 men + 18 women) × N

Therefore, $N = \frac{50}{2.5} = 20$ days

Hence, option (c) is the answer.

24. **Method 1**

Assume that the total number of 25 paise coins = x coins, total number of 50 paise coins = y coins, and total number of ₹1 coins = z coins

Value of total number of 25 paise coins = ₹x/4

Value of total number of 50 paise coins = ₹y/2

Value of total number of ₹1 coins = ₹z

Total value of all the coins = ₹x/4 + ₹y/2 + ₹z = ₹220 (i)

$x + y + z = 300$ (ii)

x:y:z = 2:5:8 (iii)

Solving equation (i), equation (ii), and equation (iii), we get $x = 40$, $y = 100$, and $z = 160$. Hence, option (a) is the answer.

Method 2

Note: In our opinion, in such questions, it is always better to go ahead with options directly instead of generating the equations.

Using option (a): Number of 50 paise coins = 100

Given that Number of 25 paise coins : Number of 50 paise coins : Number of ₹1 coins = 2:5:8 or Number of 25 paise coins: Number of 50 paise coins : Number of ₹1 coins = 40:100:160

Value of 40 coins of denomination 25 paise each = ₹10

Value of 100 coins of denomination 50 paise each = ₹50

Value of 160 coins of denomination ₹1 each = ₹160

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Sum of total value of all the coins = ₹10 + ₹50 + ₹160 = ₹220, which satisfies the condition given in the question.

Hence, option (a) is the answer.

25. Days after both the flags will be hoisted again = LCM (3 days, 4 days) = 12 days

Since the flags have been hoisted on 1st January, they will be hoisted again on 12 days after 1st January = 13th January.

Hence, option (c) is the answer.

26. If the height and base length are equal, then area of rectangle = Area of parallelogram = Length of base × Height.

Hence, option (c) is the answer.

27. This question is asking for maximum value of ${}^{n}C_{r}$ for $n = 20$. This is equal to $= {}^{20}C_{10}$.

Hence, option (c) is the answer.

28. $(4^{61} + 4^{62} + 4^{63} + 4^{64}) = 4^{61}(1 + 4 + 4^2 + 4^3) = 4^{61}(85) = 4^{61}(17 \times 5)$. It is divisible by 8, 10, and 17.

Hence, option (d) is the answer.

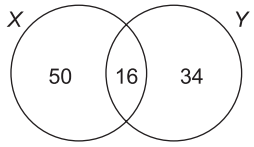
29. Since $653xy$ is divisible by 80, y has to be equal to 0.

Now, $653x$ should be divisible by 8 or $53x$ should be divisible by 8. Hence, $x = 4$.

So, $(x + y) = 4 + 0 = 4$.

Hence, option (c) is the answer.

30. Following Venn diagram presents the situation given in the question:



Numbers which are divisible by $X = 50 + 16 = 66$.

Hence, option (c) is the answer.

31. To obtain the equation from the roots, we have the following formula:

x^2 – (sum of roots)x + product of roots = 0

According to the question,

(i) When the coefficient of x is wrong, then product of roots = $7 \times 8 = 56$

(ii) When constant term is wrong, then sum of roots = $8 + (-3) = 5$

Thus, equation is $x^2 - 5x + 56 = 0$.

Hence, option (a) is the answer.

32. Suppose angle A = 90°, AC is perpendicular to AB. It means that on the x-axis, make AB a segment of the line $x = m$. However, that would mean that the coordinates of C are (m, m^2), contradicting the given points that A and C are distinct. So, angle A is not 90°. By a similar logic, angle B is also not 90°.

This means that angle C = 90° and AC is perpendicular to BC. Let C be the point (n, n^2). So, the slope of BC is the negative reciprocal of the slope of AC, yielding $m + n = \frac{1}{m-n} \rightarrow m^2 - n^2 = 1$.

Because $m^2 - n^2$ is the length of the altitude of triangle ABC from AB, and $2m$ is the length of AB, the area of $\Delta ABC = m(m^2 - n^2) = 2008$. Since $m^2 - n^2 = 1$, $m = 2008$. Substituting, $2008^2 - n^2 = 1 \rightarrow n^2 = 2008^2 - 1 = 4032063$, whose digits sum to 18.

Hence, option (c) is the answer.

33. $4^{x1} = 5$, $5^{x2} = 6$, $6^{x3} = 7$, … $100^{x124} = 128$ can be re-written as:

$(((4^{x1})^{x2})^{x3})\ldots)^{x124} = 128$ or $4^{x1 \times x2 \times x3} \ldots^{x124} = 128$

Or, $2^{2x1 \times x2 \times x3} \ldots {}^{x124} = 128 = 2^7$ or $2x_1 \times x_2 \times x_3 \times \ldots \times x_{124} = 7$

Therefore, $x_1 \times x_2 \times x_3 \times \ldots \times x_{124} = \frac{7}{2}$

Hence, option (c) is the answer.

34. $T_2 = 2^3 + 0^3 + 0^3 + 5^3 = 133$

$T_3 = 1^3 + 3^3 + 3^3 = 55$

$T_4 = 5^3 + 5^3 = 250$

$T_5 = 2^3 + 5^3 + 0^3 = 133$

And this completes the cycle. So, the cycle is 133, 55, 250, 133, 55, 250, and so on.

Therefore, 2005th term will be 250. Hence, option (d) is the answer.

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PART

5

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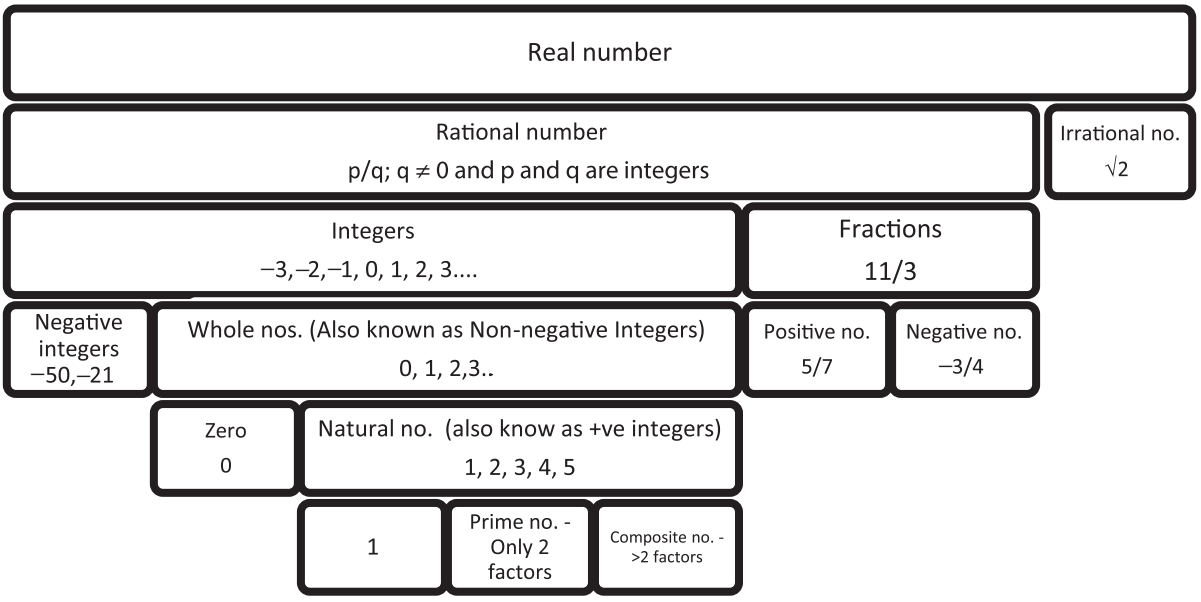
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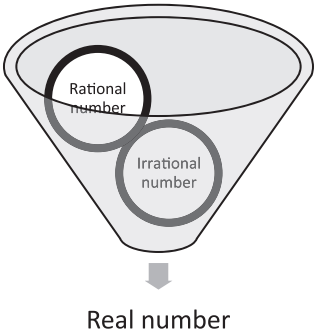
Special Section for Campus Recruitment Examinations

INTRODUCTION TO NUMBERS

Real Number Tree



Note: All the real numbers are either a rational number or an irrational number. No real number can be both a rational number and an irrational number.



Types of Rational Numbers

All the integers, terminating decimals, and recurring numbers are the types of rational numbers.

Note: π is an irrational number, though $\frac{22}{7}$ is a rational number. It is all because the fact that π is approximately equal to $\frac{22}{7}$.

All the real numbers other than rational numbers are irrational numbers.

Some important properties related to numbers are as follows:

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1. Odd ± Odd = Even Even ± Even = Even
   Odd ± Even = Odd Even ± Odd = Odd
   $(Odd)^{Even}$ = Odd $(Even)^{Odd}$ = Even
   Odd × Even = Even Odd × Odd = Odd
   Even × Even = Even
2. All the prime numbers greater than 3 are of the format $6N \pm 1$. It means that all the prime numbers will give either a remainder of +1 or −1 when divided by 6.
3. $(xy - yx) = 9 \times (x - y)$ [where xy is a two-digit number and yx is obtained by reversing it]
   For example, $51 - 15 = 9 \times (5 - 1) = 36$
4. Counting of numbers in any range:

If N is a natural number, then number of natural numbers in the range:

Both the ends included	
$A \le N \le B$	$(B-A)+1$
Only one end included	
$A \le N < B$	$B-A$
Only one of the ends included	
$A < N \le B$	$B-A$
None of the ends included	
$A < N < B$	$(B-A)-1$

Both the ends included	
$10 \le N \le 20$	$(20-10)+1=11$
Only one end included	
$10 \le N < 20$	$20-10=10$
Only one of the ends included	
$10 < N \le 20$	$20-10=10$
None of the ends included	
$10 < N < 20$	$(20-10)-1=9$

LCM / HCF

Standard Formula

1. LCM × HCF = Product of two numbers
   This formula can be applied only in case of two numbers.
2. LCM of fractions $= \dfrac{\text{LCM of numerators}}{\text{HCF of denominators}}$
3. HCF of fractions $= \dfrac{\text{HCF of numerators}}{\text{LCM of denominators}}$
4. For the same set of number, LCM ≥ each of the numbers ≥ HCF
5. LCM will be always divisible by HCF.
6. HCF of any two consecutive natural numbers = 1.
7. HCF of any two consecutive even natural numbers = 2.
8. HCF of any two consecutive odd natural numbers = 1.

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DIVISIBILITY RULES

Divisible by 2 – If unit digit of the number is 0, 2, 4, 6, or 8.

Divisible by 3 – If sum of digits of the number is divisible by 3.

Divisible by 4 – If last two digits of the number is divisible by 4.

Divisible by 5 – If last digit of the number is either 5 or 0.

Divisible by 6 – If last digit of the number is divisible by two and sum of all the digits of number is divisible by 3.

Divisible by 8 – If last three digits of the number are divisible by 8.

Divisible by 9 – If sum of digits of the number is divisible by 9.

Divisible by 10 – If the unit digit of the number is 0.

Divisible by 11 – If the difference between the sum of the digits at the even places and the sum of the digits at the odd places is divisible by 11 (zero is divisible by 11).

For example, 4553109 is divisible by 11 as the difference of $4 + 5 + 1 + 9 = 19$ and $5 + 3 + 0 = 8$ is 11.

Divisible by 12 – If the number is divisible by 3 and 4, then the number will be divisible by 12.

Divisible by 16 – if the number formed by the last 4 digits of the given number is divisible by 16.

PERCENTAGE

Basic statement 1: What is x% of y?

$$\Rightarrow \frac{x \otimes y}{100}$$

It can also be seen that x% of $y = y$ % of x.

Example 4.5% of 20,000 = 20,000% of 4.5

This simple fact can be used to divide or multiply any number by 50 or 25 or so. Let us explain this with the help of an example. We are trying to find out the value of 25×32, which is nothing but $32 \times 100/4 = 800$. Similarly, if we have to divide any number by 50, we should multiply the number by 100 and divide it by 2 finally.

Using this, we can observe that if we have to calculate 24% of 25 (or any other calculation of similar nature), it is better to find out 25% of $24 = (24 \times 100/4) = 600$

Example 1 What is 20% of 50% of 40% of 20?

Solution Percentage means 'per hundred'.

So, 20% of 50% of 40% of 20 = $(20/100) \times (50/100) \times (40/100) \times 20 = 0.8$

Here, we observed that even if we change the order of values, the final result will be same.

Basic statement 2: What percentage of x is y?

$$= \frac{y \times 100}{x}$$

Basic statement 3: Percentage change $= \frac{\text{Change}}{\text{Initial value}} \times 100$

Let us assume some values to understand the abovementioned concept:

Market share	2008–2009	2009–2010
Maruti	40%	48%
Honda	30%	26%

Percentage change in the market share of Maruti over the years $= \frac{48 - 40}{40} \times 100 = 20\%$

Percentage point change in the market share of Maruti over the years = 48 % – 40% = 8%

Similarly, if we have to increase any quantity N by S%, then it is equal to $N\left(1 + \frac{S}{100}\right)$ and when the same quantity N is to be decreased by S%, then final quantity = $N\left(1 - \frac{S}{100}\right)$

Observations

(i) An increase in 100% is equal to the final amount becoming 200% of initial value or twice the initial value.

(ii) An increase in 500% is equal to the final amount becoming 600% of initial value or six times the initial value.

(iii) A decrease in 100% is equal to the final amount becoming zero. Hence, 0% of initial value.

(iv) Concept of multiplier – Multiplier is the factor that provides the final value.

$$100 \rightarrow 20\%\uparrow \rightarrow 100 \times 1.2 = 120$$
$$100 \rightarrow 20\%\downarrow \rightarrow 100 \times 0.8 = 80$$

In the above examples, 1.2 and 0.8 are the multipliers obtained as a result of increasing by 20% and decreasing by 20%, respectively.

$$150 \rightarrow 30\%\uparrow \rightarrow 150 \times 1.3 = 195$$
$$150 \rightarrow 30\%\downarrow \rightarrow 150 \times 0.7 = 105$$

In the above examples, 1.3 and 0.7 are the multipliers obtained as a result of increasing by 30% and decreasing by 30%, respectively.

$$210 \rightarrow 27\%\uparrow \rightarrow 210 \times 1.27 = 266.7$$
$$210 \rightarrow 27\%\downarrow \rightarrow 210 \times 0.73 = 153.3$$

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In the above examples, 1.27 and 0.73 are the multipliers obtained as a result of increasing by 27% and decreasing by 27%, respectively.

So, if final value and percentage increase or percentage decrease is given and we have to find out the initial value, then it can be done in the similar way.

Using $S \rightarrow 30\%\uparrow \rightarrow S \times 1.3 = 195$

So, if final value 195 and 30%↑ is given, then initial value $S = \frac{195}{1.3} = 150$

Some More Types of Questions

Type 1 – Questions Based on Simple Calculations in Percentage

Example 2 Anoop got 273 marks in his graduation examination and got 5% more than the pass percentage. If Siddharth got 312 marks, then by what percentage above the pass marks did he pass the examination?

(a) 9% (b) 12.5% (c) 20% (d) 25%

Solution Let us assume that pass marks = N marks. Since Anoop got 5% more than the pass percentage, we can say the multiplier here is 1.05. So, $1.05 \times N = 273 \Rightarrow N = 260$. Siddharth got 312 marks ⇒ 52 marks more than the pass marks. So, he obtained $\frac{52}{260} \times 100 = 20\%$ more marks than the pass marks.

Hence, option (c) is the answer.

Example 3 Express 5:2 as percentage.

(a) 12.5% (b) 40% (c) 80% (d) 250%

Solution $5:2 = \frac{5}{2}$. To convert this in percentage, we need to multiple the ratio by $100 = \frac{5}{2} \times 100\ \% = 250\%$

Hence, option (d) is the answer.

Example 4 A's income is 60% of B's income, and A's expenditure is 70% of B's expenditure. If A's income is 75% of B's expenditure, find the ratio of A's saving to B's saving.

(a) 5:1 (b) 1:5 (c) 7:2 (d) 2:7

Solution Let us assume that income of B = ₹100, then income of A = ₹60. Now, using 'A's income is 75% of B's expenditure', B's expenditure = 60/0.75 = ₹80

So, B's savings = ₹20

Expenditure of B = ₹80 ⇒ Expenditure of A = 70% of 80 = ₹56

So, savings of A = Income of A – Expenditure of A = ₹60 – ₹56 = ₹4

So, savings of A: savings of B = ₹4:₹20 = 1:5

Hence, option (b) is the answer.

Example 5 Salary of Anil and Vinit is 20% and 30%, respectively, less than salary of Dheeraj. By what percentage is the salary of Anil more than the salary of Vinit?

(a) 33.33% (b) 50%

(c) 10% (d) 14.28%

Solution Assume that the salary of Dheeraj = ₹100 ⇒ salary of Anil = ₹80 and salary of Vinit = ₹70

To calculate 'By what percentage is the salary of Anil more than salary of Vinit', we would first calculate salary of Anil is how much more than the salary of Vinit = ₹10. Since 'than the salary of Vinit' is to be calculated, salary of Vinit will be the base (denominator). Percentage increase = $\frac{10}{70} \times 100 = 14.28\%$

Hence, option (d) is the answer.

Example 6 If 120 is 20% of a number, then 120% of that number will be:

(a) 20 (b) 120 (c) 360 (d) 720

Solution Let the number be x.

Then, 20% of x = 120 ↔ $(20/100 \times x) = 120 \leftrightarrow x = (120 \times 100/20) = 600$

Therefore, 120% of $x = (120/100 \times 600) = 720$

Hence, option (d) is the answer.

Alternatively, if 20% = 120, so 120 % will be six times of 120 = 720

Example 7 30% of 28% of 480 is the same as:

(a) 15% of 56% of 240 (b) 60% of 28% of 240

(c) 60% of 56% of 240 (d) None of these

Solution Clearly, 60% of 28% of 240 = $(60/100 \times 28/100 \times 240) = (30/100 \times 28/100 \times 2 \times 240) = (30/100 \times 28/100 \times 480)$ = 30% of 28% of 480

Hence, option (b) is the answer.

Example 8 When 35 is subtracted from a number, it reduces to its 80%. What is four-fifth of that number?

(a) 70 (b) 90 (c) 120 (d) 140

Solution Let the number be x.

Analyse the statement and look at the preposition 'to'—it reduces to its 80%—it means a loss of 20% = 35 subtracted from the number.

Hence, 100% = 35 × 5 = 175

Four-fifth of number = $175 \times \frac{4}{5} = 140$

Hence, option (d) is the answer.

PROFIT AND LOSS

1. Cost Price (CP)

This is the price that a person pays to purchase something or cost incurred while manufacturing something.

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Types of Cost

1. Fixed cost – As obvious from the name, it is that kind of cost which is fixed in all the cases.
2. Variable cost – Variable costs are those costs that vary according to the number of units produced.
3. Semi-variable cost – Semivariable costs are those costs that are fixed in one particular strata, but the costs varies among the different layers.

One good example of fixed cost, variable cost, and semi-variable cost is the bill we receive for the telephone connections at our home. A part of that bill, rental, is fixed cost; and the rest part of the bill is calculated on the basis of the number of calls made.

2. Selling Price (SP)

This is the price at which something is sold.

Now, there are three situations possible:

Case I Selling price > Cost price, then profit occurs.
Profit = SP − CP

$$\text{Profit percentage} = \frac{\text{Profit}}{\text{CP}} \times 100$$

Case II Cost price > Selling price, then loss occurs.
Loss = CP − SP

$$\text{Loss percentage} = \frac{\text{Loss}}{\text{CP}} \times 100$$

Case III Selling price = Cost price, then there is no profit no loss. We call it break-even case.

3. Marked Price Or Mark-up Price (MP)

This is the price which the shopkeeper fixes in the anticipation of some discount being asked by customer.

List price or tag price – As obvious from the name, this is the price which is printed on the tag of the article.

For our calculations related to the concept of PLD, till the moment nothing is stated in the questions, we would not see much difference between marked price and list price.

Types of Questions

Type 1:
CP and SP are given, and profit percentage or loss percentage is to be calculated.

Example 9 An article is bought for ₹600 and sold for ₹750. What is the profit percentage?
(a) 20% (b) 25%
(c) 30% (d) None of these

Solution CP = ₹600 and SP = ₹750
Since SP is more than CP, there will be profit.
Profit = SP − CP = ₹150

$$\text{Profit percentage} = \frac{\text{Profit}}{\text{CP}} \times 100 = \frac{150}{600} \times 100 = 25\%$$

Type 2:
CP and profit percentage/loss percentage are given, and SP is to be calculated.

If one of CP or SP is given along with profit percentage or loss percentage, then using the concept of multiplier makes the whole calculation simple. [To know about the concept of multiplier, read percentage chapter.]

In general,

(a) CP × Multiplier = SP

(b) $\text{CP} = \frac{\text{SP}}{\text{Multiplier}}$

(c) If there is a profit, multiplier will be more than 1, and if there is a loss, multiplier will be less than 1. [Irrespective of the fact that we have to find out CP or SP.]

Alternatively:

If there is a profit of R% and CP = C, then $\text{CP} = \frac{\text{SP}}{100 + R} \times 100$

If there is a loss of R%, $\text{CP} = \frac{\text{SP}}{100 - R} \times 100$

Example 10 Kitti buys a kinetic for ₹16,000. If he wants to gain 40%, how much should she charge for the kinetic?

Solution CP = ₹16,000
Profit percentage = 40%
SP = CP + 40% of CP = 1.4 × CP [Here the multiplier = 1.4] = 1.4 × 16,000 = ₹22,400

Example 11 By selling a VCD player for ₹1950, I got a profit of 30%. At what price should I have sold it in order to get a profit of 40%?
(a) ₹2000 (b) ₹2100
(c) ₹2500 (d) None of these

Solution SP = ₹1950 and profit percentage = 30%

$$\text{CP} = \frac{SP}{1 + \frac{30}{100}} = \frac{1950}{1.3} = ₹1500$$

Now, I want to earn a profit of 40%.
Hence, multiplier = 1.4
So, SP = CP × 1.4 = 1500 × 1.4 = ₹2100

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Example 12 By selling an article for ₹360, loss incurred is 10%. At what minimum price, should the article be solve to avoid loss?

(a) ₹320 (b) ₹324
(c) ₹396 (d) ₹400

Solution SP = ₹360
Loss percentage = 10%; hence, multiplier = 0.9

$$\text{Now, CP} = \frac{360}{0.9} = ₹400$$

Hence, to avoid any loss, SP should be atleast equal to the CP = ₹400.

Type 3:
Questions involving marked price/tag price

Example 13 A shopkeeper wants to earn a profit of 20%, and at the same time, the minimum discount that he wants to offer is 25%. What should be the minimum percentage mark-up over CP?

(a) 50% (b) 42.5% (c) 62.5% (d) 35%

Solution Assume that cost price = ₹100
To earn a profit of 20%, multiplier = 1.2
Hence, SP = CP × 1.2 = 100 × 1.2 = ₹120 (i)
Now, discount offered = 25%. Discount is always provided on tag price or mark-up price.
Multiplier related to 25% discount = 0.75. Assume that mark-up price = M
Hence, selling price = 25% discount on mark-up price = 0.75 M

$$\text{Using (i), } 0.75M = ₹120 \Rightarrow M = \frac{120}{0.75} = ₹150$$

$$\text{Hence, percentage mark-up} = \frac{\text{Mark Up}}{\text{CP}} \times 100 = \frac{(150-100)}{100} \times 100 = 50\%$$

Type 4:
Questions involving number of articles sold and number of articles bought

Example 14 Cost price of 40 apples is equal to selling price of 30 apples. What is the profit percentage?

Solution Best way to solve these questions is by assuming a value (ideally LCM of 30 and 40).
CP of 40 apples = SP of 30 apples = ₹120 (LCM of 30 and 40)
⇒ CP of one apple = ₹3 and SP of one apple = ₹4
Therefore, profit = SP − CP = ₹1

$$\text{Hence, profit percentage} = \frac{1}{3} \times 100 = 33.33\%$$

In general,

$$\text{Profit percentage/Loss percentage} = \frac{\text{Goods left/Goods added}}{\text{Number of articles sold}} \times 100$$

In this case, there are 10 apples left out (after selling 30 apples out of 40 apples).

$$\text{Hence, profit percentage} = \frac{10}{30} \times 100 = 33.33\% \text{ profit}$$

Example 15 Cost price of 40 apples is equal to selling price of 50 apples. What is the profit/loss percentage?

Solution Since number of articles sold is more than number of articles bought; hence, there is loss. [Shopkeeper is selling more number of articles than he has got.]

$$\text{Loss percentage} = \frac{10}{50} \times 100 = 20\%$$

Type 5:
Questions based upon faulty balance

In these questions, shopkeeper cheats his customers by selling less quantity than what he is professing.

If shopkeeper sells x g instead of 1000 grams (where $x < 1000$),

$$\text{Profit percentage} = \frac{1000-x}{x} \times 100\ \%$$

Example 16 A shopkeeper professes to sell his goods at cost price. However, he sells only 800 g at the place of 1000 g. What is his profit percentage?

Solution Profit will be obtained for 200 g (1000 − 800).

$$\text{Profit percentage} = \frac{200}{800} \times 100\% = 25\%$$

Example 17 A shopkeeper professes to sell his goods at cost price. However, he sells x g at the place of 1000 g and therefore earns a profit of 20%. What is the value of x?

Solution

$$\frac{1000-x}{x} \times 100 = 20 \Rightarrow \frac{1000-x}{x} = 1/5 \Rightarrow x = 833.33 \text{ g}$$

Some Important Results

When SPs of two articles are same, then:

(i) First article is sold at a profit of x% and second article is sold at a loss of x%.
In this case, there will be always a loss.

$$\text{Net loss} = \frac{x^2}{100}\% \text{ of CP}$$

(ii) First article is sold at a profit of x% and second article is sold at a profit of y%.
Ratio of CP1:CP2 = $(100 + y):(100 + x)$

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(iii) First article is sold at a profit of x% and second article is sold at a loss of y%.
Ratio of $CP_1:CP_2 = (100 - y):(100 + x)$
In this case, we cannot find if there will be net profit or loss.

(iv) First article is sold at a loss of x% and second article is sold at a loss of y%.
Ratio of $CP_1:CP_2 = (100 - y):(100 - x)$

(v) First article is sold at a loss of x% and second article is sold at a profit of y%.
Ratio of $CP_1:CP_2 = (100 + y):(100 - x)$
In this case, we cannot find if there will be net profit or loss.

SIMPLE INTEREST AND COMPOUND INTEREST

Interest is the cost of borrowing money. In other words, interest is also defined as 'time value of money'.

Two types of interest are as follows:

1. **Simple interest:** In case of simple interest, interest as well as principal remains fixed for every compounding period.
   Simple interest is calculated for original principal only. Accumulated interest from previous periods is not used in calculations for the next periods.

Example If the rate of interest = 10% and the principal = ₹1000, then:
Interest for first year = 10% of ₹1000 = ₹100
Interest for second year = 10% of ₹1000 = ₹100
Interest for third year = 10% of ₹1000 = ₹100
It can be seen that interest generated every year = ₹100

	Principal	Rate	Interest
First year	1000	10%	100
Second year	1000	10%	100
Third year	1000	10%	100

2. **Compound interest:** In case of compound interest, interest and principal keeps on changing for every compounding period. Interest keeps on increasing every compounding period because principal increases every year. Understand this in the following way:
   Principal of first year (Initially) = P
   Principal of second year = P + Interest of first year
   Principal of third year = P + Interest of first year + Interest of second year

In case of compound interest, interest gets added to the principal, and for next years, interest is accrued over (Principal + Interest accrued so far). Hence, compound interest is interest that is paid on both the principal and also on any interest from past years.

Example If the rate of interest = 10% and the principal = ₹1000, then:
Interest for first year = 10% of ₹1000 = ₹100

	Principal	Rate	Interest
First year	1000	10%	100
Second year	1000 + 100 = 1100	10%	110
Third year	1000 + 100 + 110 = 1210	10%	121

Expression for simple interest (SI)/compound interest (CI):

$$SI = \frac{\text{Principal} \times \text{Rate of Interest} \times \text{Time}}{100}$$

$$CI = \text{Principal} \times \left[1 + \frac{R}{100}\right]^N - \text{Principal}$$

Principal = Sum invested or lent
R = Rate of interest per annum
N = Number of years
It should be noted that the unit of rate of interest and time should be same. Therefore, if rate of interest is 'per year', then time should also be in 'Year'.
In case of compound interest, if the compounding is not done annually, then formula changes like the following:

(a) **Half yearly compounding:** It means interest is given after every six months. In this case, after every six months, interest will be added to the principal.

Rate of interest	Compounding period	Interest in 6 months (Half Yearly)	Number of compounding period in a year
R% per year	Half yearly	R%/2	2 (12 months/6 months)

$$CI = \text{Principal} \times \left[1 + \frac{R/2}{100}\right]^{2N} - \text{Principal}$$

(b) **Quarterly compounding:** It means interest is given after every three months. In this case, after every three months, interest will be added to the principal.

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Rate of interest	Compounding period	Interest in three months (Quarter year)	Number of compounding period in a year
R% per year	Quarterly	R%/4	4 (12 months / 3 months)

$$\text{CI} = \text{Principal} \times \left[1+\frac{R/4}{100}\right]^{4N} - \text{Principal}$$

Understand that the expression $\left\{\text{Principal} \times \left[1+\frac{R}{100}\right]^{N}\right\}$ in compound interest provides the amount = principal + interest. To calculate interest, we need to subtract principal from this.

Example 18 Find simple interest for the following data. Principal = ₹400, rate of interest = 20% per annum, and time = 4 months

Solution In this case, we can see that units of rate and time are not same. We can convert any one of the two to be in one single unit – either in months/year.

$$\text{Time} = 4 \text{ months} = \frac{4}{12} \text{year} = \frac{1}{3} \text{year}$$

$$\text{SI} = \frac{\text{Principal} \times \text{Rate of interest} \times \text{Time}}{100} = \frac{400 \times 20 \times \frac{1}{3}}{100}$$

$$= \frac{400 \times 20 \times 1}{100 \times 3} = ₹26.66$$

Example 19 Find compound interest for the following data: Principal = ₹400, rate of interest = 20% per annum, and time = 12 months. Interest is compounded half yearly.

Solution Since interest is compounded half yearly, in 12 months, interest will be added (or compounded) twice.

$$\text{Rate of interest for six months} = \frac{20}{2} = 10\%$$

$$\text{CI} = 400 \times \left[1+\frac{10}{100}\right]^{2} - 400 = 400\ (1.1)^2 - 400 =$$

₹484 – ₹400 = ₹84

Alternatively, it can be calculated through simple addition too:

Interest for first six months = 10% of ₹400 = ₹40

Interest for next six months = 10% of ₹40 (interest for the interest for first six months) + 10% of ₹400 = ₹44

Hence, total interest = ₹30 + ₹44 = ₹84

Example 20 A sum of money becomes three times in 5 years. Find in how many years will the same sum become six times at the same rate of SI?

Solution Sum of money gets three times, it means 200% is being added up to the original sum (Principal) in 5 years.

So, 500% will be added up in $12\frac{1}{2}$ years.

Example 21 Difference between two years of compound interest and simple interest at 10% over ₹X is ₹10. What is the value of *X*?

Solution

	SI	CI
At the end of first year	10%	10%
At the end of second year	10%	10% + 10% of 10% = 11%
	= 20%	= 21%

So, 1% = ₹10

⇒ 100% = ₹1000

Important Points

1. If the rate of interest = *R*% per annum for both CI and SI, then the difference between CI and SI for 2 years will be equal to (*R*% of *R*)% of principal = $\frac{R^2}{100}$% of principal
   In the above case, *R* = 10%, so the difference between CI and SI for 2 years = 1%
2. If a sum doubles itself in *n* years at simple interest, then rate of interest = $\frac{100}{n}$

Example 22 A sum of money doubles itself in 5 years at SI. What is the rate of interest?

Solution

$$\text{Rate of interest} = \frac{100}{5} = 20\%$$

Example 23 A sum of money amounts to ₹2600 in 3 years and to ₹3000 in next 2 years at simple interest. What is the rate of interest?

Solution Increase in interest in two years = ₹400

Increase in interest in 1 year = ₹200

Principal = 2600 – (3 × 200) = ₹2000

$$\text{Interest rate} = \frac{200}{2000} \times 100 = 10\%$$

Comparison between CI and SI

Assume two different sums are getting double at their respective rates of SI and CI in 5 years. Following table gives us the mechanism of getting money *n* times in above situation:

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	After 5 years	After 10 years	After 15 years	After 20 years
At SI	2 times	3 times	4 times	5 times
At CI	2 times	4 times	8 times	16 times

RATIO, PROPORTION AND VARIATION

Some Important Results

1. Continued proportion
   a, b, and c are said to be in continued proportion if $\frac{a}{b}=\frac{b}{c}$
   So, $b^2 = ac$. Here, b is known as the mean proportion.
   Similarly, if a, b, c, and d are in continued proportion, then we get
   $\frac{a}{b}=\frac{b}{c}=\frac{c}{d}$
2. Componendo
   If $\frac{a}{b}=\frac{c}{d}$, then $\frac{a+b}{b}=\frac{c+d}{d}$
3. Dividendo
   If $\frac{a}{b}=\frac{c}{d}$, then $\frac{a-b}{b}=\frac{c-d}{d}$
4. Componendo and dividendo
   If $\frac{a}{b}=\frac{c}{d}$, then $\frac{a+b}{a-b}=\frac{c+d}{c-d}$
5. Invertendo
   If $\frac{a}{b}=\frac{c}{d}$, then $\frac{b}{a}=\frac{d}{c}$
6. Alterando
   If $\frac{a}{b}=\frac{c}{d}$, then $\frac{a}{c}=\frac{b}{d}$
7. $\frac{a}{b}=\frac{c}{d}=\frac{a+c}{b+d}$
   In general, if $\frac{a}{b}=\frac{c}{d}=\frac{e}{f}=\ldots=K$
   Then, $\frac{a}{b}=\frac{c}{d}=\frac{e}{f}=\ldots=K=\frac{a+c+e+\ldots}{b+d+f+\ldots}$ = (any combination of numerator/any combination of corresponding denominator)
   Example $1/2 = 3/6 = 4/8 = \ldots = (1 + 3 + 4)/(2 + 6 + 8) = (3 + 4)/(6 + 8)$
8. If we multiply the numerator and denominator of a ratio by any no. $N(N \neq 0)$, then the ratio remains same.
   $\frac{A}{B}=\frac{nA}{nB}$
9. If we divide the numerator and the denominator of a ratio by any no. $N(N\neq0)$, then the ratio remains same.
   $A/B = (A/N)/(B/N)$
10. If $A:B = x:y$ and $B:C = p:q$
    Then, $A:B:C = xp:yp:yq$
    This can be seen below through the following diagram:

$$\begin{array}{l} A : B = x : y \\ \underline{B : C = p : q} \\ A : B : C = xp : yp : yq \end{array}$$

Note: In the above ratios, B is known as 'middle term'. Ratio among different quantities can be established only if there is a middle term present.

Example 24 Ratio of salary of A and salary of B = 2:3 and ratio of salary of B and salary of C = 4:5. What is the ratio of salary of A, B, and C?

Solution

$$\begin{array}{l} A : B = 2 : 3 \\ \quad\quad\ \diagup\!\diagdown \\ \underline{B : C = 4 : 5} \\ A : B : C = 2x4 : 3x4 : 3x5 \end{array}$$

So, $A:B:C$ = 8:12:15

Example 25 Ratio of salary of A and salary of B = 2:5 and ratio of salary of C and salary of D = 1:3. What is the ratio of salary of A, B, C, and D?

Solution There is no middle term here.
Hence, ratio among A, B, C, and D cannot be established.

Example 26 Three friends Aman, Parth, and Manoj purchased a music system for ₹6000 by pooling in money. Aman contributed half of what Parth and Manoj contributed combined. How much did Aman contribute?

Solution Since Aman contributed half of what Parth and Manoj contributed combined, Aman contributed one-third of total.

[An easier way of understanding this goes as follows:
Assume the sum contributed by Parth and Manoj combined = ₹$2x$
Since Aman contributed half of the sum contributed by Parth and Manoj combined, he contributed $1/2 \times 2x$ = ₹x. So, total sum = ₹$2x$ + ₹x = ₹$3x$. Out of this, Aman contributed ₹x, and hence, he contributed one-third of the total.]

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According to the question, ₹$3x$ = ₹6000
Hence, x = ₹2000

Example 27 A man distributed his wealth among his three children in the ratio of their ages. Ages of the children are 9 years, 12 years, and 18 years. What proportion of the whole wealth did the eldest son get?

Solution Age of eldest son = 18 years
Ratio of ages of sons = 9:12:18 = 3:4:6
According to the question, wealth distributed is in the ratio of their ages $3x$:$4x$:$6x$.
So, total wealth distributed = $3x + 4x + 6x = 13x$
Hence, proportion of the whole wealth did the eldest son get = $\frac{6x}{13x} = \frac{6}{13}$

Example 28 Given that $\frac{x}{y}$ = 3:2. What is the value of $(2x - y):(3x + 2y)$?

Solution Best way to solve such questions is 'through assuming the values'.

Given that $\frac{x}{y}$ = 3:2, so assume $x = 3$ and $y = 2$

Hence, $(2x - y):(3x + 2y) = (2 \times 3 - 2):(3 \times 3 + 2 \times 2)$ = 4:13

TIME AND WORK

1. Concept of Man-hour or Man-days

(a) If number of hours of work done per day is same but work done are different:

$$\frac{M_1D_1}{W_1} = \frac{M_2D_2}{W_2}$$

Where M_1 and M_2 are the number of persons involved in doing two works W_1 and W_2 in D_1 and D_2 days, respectively.

(b) If number of hours of work done per day are different but work done are different:

$$\frac{M_1D_1T_1}{W_1} = \frac{M_2D_2T_2}{W_2}$$

Where M_1 and M_2 are the number of persons involved in doing two works W_1 and W_2 in D_1 and D_2 days.

(c) If work done in two cases are same:

$$M_1D_1 = M_2D_2$$

Where M_1 and M_2 are the number of persons involved in doing the same work in D_1 and D_2 days.

2. Concept of Individual Work and Efficiency

(a) If A can do a job in 10 days, work done by A in one day = $\frac{1}{10}$ of the total work

In general, if A can finish a work in 'x' days, then in 'y' days, he would have completed $\frac{y}{x}$ part of the whole job.

(b) If ratio of efficiency of two persons is x:y, then the ratio of time taken by them in doing the same work = $\frac{1}{x} : \frac{1}{y} = y:x$

Vice versa of this is also true ⇒ if ratio of days taken by A and B in doing the same work is a:b, then ratio of efficiency of A and $B = \frac{1}{a} : \frac{1}{b} = b:a$

(c) Payment of wages is always done in ratio of efficiency.

3. Concept of Pipes and Cisterns

Concept of pipes and cisterns is just an extension of concept of individual work.

Basic statement: If tap A can fill a tank in 20 h, then part of tank filled by tap A in 1 h = $\frac{1}{20}$th of the total tank. Similarly, if leakage B can empty the whole tank in 25 h, then part of tank vacated by leakage in 1 h = $\frac{1}{25}$th of the total tank. It should be noted that sign of the work done by leakage is 'negative'.

WORKED OUT EXAMPLES

Example 29 20 men can plough 12 fields in 18 days. How many men are required to plough 24 fields in 10 days working same number of hours per day?

Solution Let us identify the terms involved here:
$M_1 = 20$ men, $W_1 = 12$ fields and $D_1 = 18$ days. Similarly, $M_2 = N$ men (assume), $W_2 = 2r$ fields and $D_2 = 10$ days
Using the formula:

$$\frac{M_1D_1}{W_1} = \frac{M_2D_2}{W_2}$$

$$\Rightarrow \frac{20 \times 18}{12} = \frac{N \times 10}{24}$$

Solving it, we get $N = 72$ men.

Example 30 A contractor estimated that 25 labours will be able to finish the whole project in 60 days. However, only 15 labours reported for the job. In how many days, will the project be finished?

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Solution Using the formula:

$$M_1D_1 = M_2D_2$$

$25 \times 60 = 15 \times D_2$
Hence, $D_2 = 100$ days

Example 31 Mohan can plough a field in 15 days. Ram can plough the same field in 30 days. Mohan and Ram start working together. They worked for initial few days together, but Ram quit the job 6 days before the whole field was ploughed. In how many days, the whole field is ploughed?

Solution Part of the field ploughed by Mohan in 1 day = $\frac{1}{15}$ of the whole field and part of the field ploughed by Ram in 1 day = $\frac{1}{30}$ of the whole field.

Part of the field ploughed by Ram and Mohan together in 1 day = $\frac{1}{15} + \frac{1}{30} = \frac{2+1}{30} = \frac{1}{10}$ of the whole field. (i)

According to the question:

Ram and Mohan together	Mohan alone
Initial few days	Last six days

Given that Ram quit the job 6 days before the whole field was ploughed. Hence, only Mohan has worked in the last six days. Part of the field ploughed by Mohan in last six days = $6 \times \frac{1}{15} = \frac{6}{15}$ of the whole field.

Remaining part of the field was ploughed by Ram and Mohan both in initial few days = $1 - \frac{6}{15} = \frac{9}{15}$ of the whole field. (ii)

Using (i) and (ii), number of days Ram and Mohan worked together = $\frac{\frac{9}{15}}{\frac{1}{10}}$ days = 6 days

So, total number of days to plough the whole field = 6 days + 6 days = 12 days

TIME, SPEED, AND DISTANCE

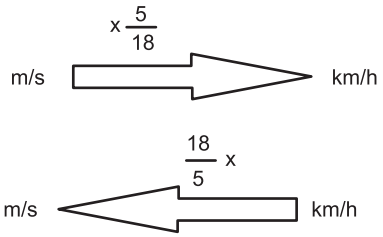
Speed is defined as the distance travelled in unit time. Mathematically, Speed = $\frac{\text{Distance}}{\text{Time}}$

Distance: If a person is moving at a speed of v m/s for t seconds, the distance covered by him will be = Speed × Time

Or, Distance = Speed × Time

Most commonly used units of speed are meter per second or km per hour.

To convert km/h unit into m/s, we multiply it by a factor of $\frac{5}{18}$ and if we have to convert m/s to km/h, it is multiplied by $\frac{18}{5}$.



Motion in a Straight Line

Important Formula

1. If the ratio of the speeds of two moving objects is in (x:y), then the ratio of the times taken by them to cover same distance is (y:x).
2. Average speed for the whole journey = $\frac{\text{Total distance}}{\text{Total time taken}}$

   Corollary to point (2):

   (a) However, if a moving object covers a certain distance with a speed of 'a' kmph and an equal distance at 'b' kmph, then the average speed for the whole journey = $\left(\frac{2ab}{a+b}\right)$ kmph.

   (b) If a moving object travels for 't' hours with a speed of 'a' kmph, and another 't' hours with a speed of 'b' kmph, then the average speed for the whole journey = $\left(\frac{a+b}{2}\right)$ kmph.
3. When two objects are travelling in the same direction, relative speed = difference of the speeds.
4. When two objects are travelling in the opposite direction, relative speed = sum of the speeds.

WORKED OUT EXAMPLES

Example 32 A man covers 70 m in 35 s. What is the speed of this man?

Solution

Speed = $\frac{\text{Distance}}{\text{Time}}$

Distance = 70 m and time = 35 s

So, speed = $\frac{70}{35} = 2$ m/s

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Example 33 A car is moving with a speed of 90 kmph for 2 h. What is the distance covered by car in that time?

Solution Distance = Speed × Time = 90 kmph × 2 h = 180 km

Example 34 A man covers a distance of 245 m while moving with a speed of 5 m/s. What is the time taken by the man?

Solution $\text{Time} = \frac{\text{Distance}}{\text{Speed}} = \frac{145}{5} = 49 \text{ s}$

Example 35 Samar travelled from point A to point B at the rate of 30 kmph and walked back to 'A' at the rate of 50 kmph. What was his average speed during the whole journey?

Solution As distance covered is same in both the cases,

$\text{average speed} = \left(\frac{2ab}{a+b}\right) = \left(\frac{2\times30\times50}{30+50}\right) = 37.5 \text{ kmph}$

Example 36 Tinki travels for 3 h at the speed of 25 kmph and another 3 h at the speed of 33 kmph. What is his average speed for the whole journey?

Solution As time invested is same in both the cases, average speed $= \frac{a+b}{2} = \frac{25+33}{2} = 29$ kmph

Example 37 A thief escaped from a jail at 11 am. After realizing the escape at 1 pm same day, a cop starts chasing the thief in his direction. If speed of thief is 30 kmph and speed of cop is 50 kmph, at what time will the thief be caught?

Solution Since the escape was unreported till 1 pm, the distance thief has covered from 11 am (since his escape) till cop starts chasing (1 pm) = 30 kmph × 2 h = 60 kms
Since cop and thief are travelling in the same direction, relative speed of cop over thief = 50 kmph – 30 kmph = 20 kmph

Time taken by cop to catch the thief $= \frac{\text{Distance}}{\text{Relative Speed}} = \frac{60 \text{ kms}}{20 \text{ kmph}} = 3 \text{ h}$

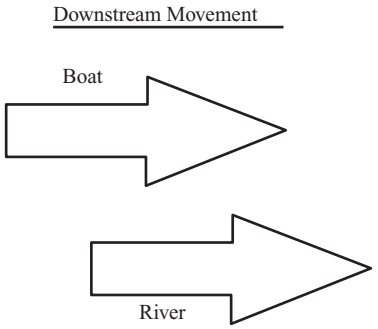
Since cop starts chasing the thief at 1 pm, he would catch thief 3 hours later at 4 pm.

Example 38 In previous question, how much distance cop has travelled before catching the thief?

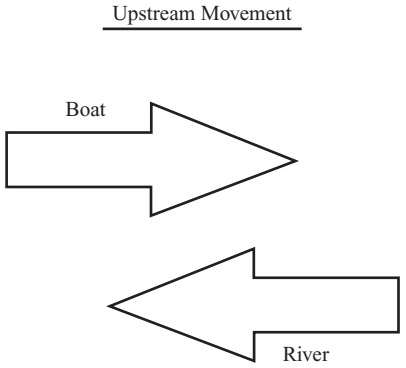
Solution Cop started chasing the thief at 1 pm and caught him at 4 pm. So, time taken by cop in catching the thief = 3 h.
Distance = Speed × Time = 50 × 3 = 150 km

BOATS AND STREAMS

Downstream movement – If a person rows a boat along the river, then it is called downstream movement.



Upstream movement – If a person rows a boat against the direction of flow of the river, then it is called upstream movement.



Important Formula

1. Let the speed of boat in still water be v_1 kmph and the speed of stream be v_2 kmph.
   Downstream speed = $(v_1 + v_2)$ kmph
   Upstream speed = $(v_1 - v_2)$ kmph
2. Speed of boat in still water = $\frac{1}{2}$ (Downstream speed + Upstream speed)
   Speed of river = $\frac{1}{2}$ (Downstream speed – Upstream speed)
3. If same distance is covered downstream and upstream:
   $\frac{\text{Downstream speed}}{\text{Upstream speed}} = \frac{\text{Time taken Upstream}}{\text{Time taken Downstream}}$

Example 39 A boat moves upstream at a speed of 8 kmph and downstream 16 kmph. What is the speed of boat in still water and what is the speed of stream?

Solution Speed of boat in still water = $\frac{1}{2}$ (8 + 16) kmph = 12 kmph
Speed of stream = $\frac{1}{2}$ (16 – 8) kmph = 4 kmph

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Example 40 Downstream speed of a boat is 15 kmph and upstream speed is 9 kmph. It takes 6 h in covering a certain distance downstream. How much time will it take in covering the same distance upstream?

Solution

$$\frac{\text{Downstream speed}}{\text{Upstream speed}} = \frac{\text{TIme taken upstream}}{\text{Time taken downstream}}$$

Assume time taken in covering the same distance Upstream = t hours

$$\text{Or, } \frac{15}{9} = \frac{t}{6} \Rightarrow t = 10 \text{ kmph}$$

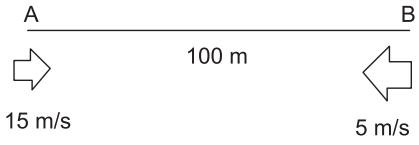
TRAINS

Let us first understand that why do we need to study concept of trains separately despite this concept is a part of time speed distance.

Consider the following case:

1a. A and B are two men running towards each other. Speed of A is 15 m/s and speed of B is 5 m/s. If the distance between them is 100 m, in how much time will they *meet* each other?

1b. A and B are two men running towards each other. Speed of A is 15 m/s and speed of B is 5 m/s. If the distance between them is 100 m, in how much time will they *cross* each other?

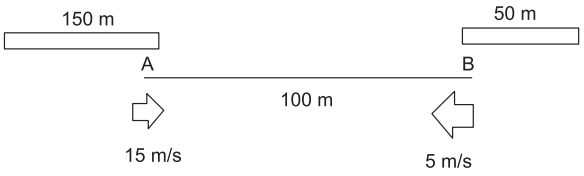


It can be seen that answer to both 1a and 1b will be same, that is, meeting and crossing each other are same in this case. Time = $\frac{\text{Distance}}{\text{Relative speed}} = \frac{100}{20} = 5$ s

Now, consider the following case:

2a. A and B are two trains running towards each other. Speed of train A is 15 m/s and speed of train B is 5 m/s. Length of train A is 150 m and length of train B is 50 m. If the distance between them is 100 m, in how much time will they *meet* each other?

2b. A and B are two trains running towards each other. Speed of train A is 15 m/s and speed of train B is 5 m/s. Length of train A is 150 m and length of train B is 50 m. If the distance between them is 100 m, in how much time will they *cross* each other?



Will the answers to questions 2a and 2b same?

It will not be same because in case of (2a), total distance travelled by trains = 100 m, Time = $\frac{\text{Distance}}{\text{Relative speed}} = \frac{100}{20} = 5$ s

In case of (2b), total distance travelled by trains = 150 m + 100 m + 50 m = 300 m

$$\text{Time} = \frac{\text{Distance}}{\text{Relative speed}} = \frac{300}{20} = 15 \text{ s}$$

(a) If a train of length 'L' is passing a pole (or any stationery object of negligible width) at a speed of v m/s, then the train covers a distance equal to its length = L m. Hence, time taken by train in crossing a pole = $\frac{\text{Length of train}}{\text{Speed of train}} = \frac{L}{v}$

(b) If a train of length is passing a platform at a speed of v m/s, then the train covers a distance equal to (its own length + length of platform). Hence, time taken by train in crossing a platform = $\frac{\text{Length of train + length of platform}}{\text{Speed of train}}$

(c) If two train or two bodies of length l_1 and l_2 m are moving in opposite direction on parallel path with v_1 m/s and v_2 m/s, then time taken by the trains or bodies to cross each other = $\left(\frac{l_1 + l_2}{v_1 + v_2}\right)$ second

(d) If two trains of length l_1 and l_2 are moving in same direction at v_1 m/s and v_2 m/s, then the time taken by the faster body to cross the slower body = $\left(\frac{l_1 + l_2}{v_1 - v_2}\right)$ second

Example 41 A train 260 m long is moving at a speed of 72 kmph. What is the time taken by this train to pass a pole near the railway line?

Solution

First, we will find out speed of the train, that is, $72 \times \frac{5}{18}$ m/s = 20 m/s

Distance covered in passing the standing pole = 260 m

So, required time taken = $\frac{260}{20} = 13$ s

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Example 42 Two trains, 200 m and 175 m in length, are moving in opposite direction on the parallel tracks at the speed of 32 kmph and 58 kmph, respectively. In how much time, they will cross each other?

Solution Since trains are moving in opposite direction, their relative speed = Sum of speeds of trains
Relative speed of the trains = 32 kmph + 58 kmph = 90 kmph
$= 90 \times \frac{5}{18} = 25$ m/s

Total distance required to be covered = Length of train 1 + length of train 2 = 200 m + 175 m = 375 m

Time taken by the trains to cross each other = $\frac{375}{25} = 15$ s

AVERAGE

Average is calculated by dividing the sum of all the numbers by the number of numbers.

$$\text{Average} = \frac{\text{Sum of numbers}}{\text{Number of numbers}}$$

Let us find the average of four numbers 214, 215, 219, and 224.

$$\text{Average} = \frac{214+215+219+224}{4} = 218$$

1. If the series is in Arithmetic Progression:

$$\text{Average} = \frac{\text{1st number} + \text{Last number}}{2}$$

[Arithmetic progression is a series in which difference between any two consecutive terms is same. To know more, go through the chapter – Sequence and Series in this book.]

Example 43 Average of first five multiples of 3 is:

Solution

Method 1
Using general definition of average = 3 (1 + 2 + 3 + 4 + 5)/5 = 45/5 = 9

Method 2
Since consecutive multiples will be in arithmetic progression,

$$\text{Average} = \frac{\text{1st number} + \text{Last number}}{2}$$

First number = 1st multiple of 3 = 3, and last number = 5th multiple of 3 = 15

Hence, average = $\frac{3+15}{2} = 9$

Example 44 Average of 5 observations p, $p + 2$, $p + 4$, $p + 6$, and $p + 8$ is 11. What is the average of last three observations?

Solution

Method 1
It can be seen that numbers are in AP.
Hence, Average = Middle number
Hence, Middle number = Average = $p + 4 = 11 \Rightarrow p = 7$
Average of last three observations = Middle number = $p + 6$
$= 7 + 6 = 13$

Method 2
Going through general approach:
$[p + (p + 2) + (p + 4) + (p + 6) + (p + 8)] /5 = 11$ or $5p + 20 = 55$ or $p = 7$
So, the numbers are 7, 9, 11, 13, and 15.
Hence, required mean = (11 + 13 + 15) /3 = 39/3 = 13

2. Average involving Time, Speed, and Distance:

$$\text{Average speed} = \frac{\text{Total distance}}{\text{Total time}}$$

Example 45 Lovely goes to Patna from New Delhi at a speed of 40 kmph and returns with a speed of 60 kmph. What is her average speed during the whole journey?

Solution First of all understand that, average speed in this case is NOT equal to $\frac{40+60}{2} = 50$ kmph.

To solve this question, we need to have the distance and time known. We can assume the distance either a variable (x) or a constant (number). In my perception, it is always better, in these cases, to assume number.

Let us assume that the total distance between Patna and New Delhi is 120 km (LCM of 40 and 60).

So, total time taken (Patna – New Delhi and New Delhi – Patna) = 3 + 2 = 5 h

So, average speed = $\frac{240}{5} = 48$ kmph

Alternatively:

(a) If a person covers the same distance with different speeds A and B,

Average speed for the whole journey = $\frac{2AB}{A+B}$ = Harmonic mean of A and B.

(b) If a person covers some distance with different speeds A and B for equal time,

Average speed for the whole journey = $\frac{A+B}{2}$ = Arithmetic mean of A and B.

Example 46 Lovely goes to Patna from New Delhi at a speed of 40 kmph and returns with a speed of N kmph. Her average speed during the whole journey is 60 kmph. What is the value of N?

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Solution Since the distance covered is same, average speed $= \frac{2AB}{A+B} = \frac{2\times 40\times N}{40+N} = 60$

$\Rightarrow (40+N)\times 60 = 80\times N \Rightarrow 2400N + 60N = 80N \Rightarrow N = 120$

Example 47 Munu Shing is going to Patna from New Delhi. She covers first half of the distance with a speed of 40 kmph and second half of the distance with a speed of 60 kmph. What is her average speed for the whole journey?

Solution This question is exactly same as Question 46. Hence, answer remains the same = 48 kmph

Example 48 Ram Prasad is going to Delhi from Dehradun. For half of the time, he travelled with a speed of 40 kmph and for other half of the time, he travelled with a speed of 60 kmph. What is his average speed during the whole journey?

Solution Since time invested is same, average speed $= \frac{A+B}{2} = \frac{40+60}{2} = 50$ kmph

Example 49 Average age of five members of a family is 20 years. The youngest member of the family is four years old. At the time of birth of this youngest member, average age of rest of the members of the family was N years. What is the average age of the family (in terms of N) excluding the youngest member?

Solution Sum of ages of all the members of the family = 100

Sum of ages of all the members of the family excluding the youngest number = 100 – 4 = 96

So, average age of all the members of the family excluding the youngest number = 96/4 = 24 = N

The average age of the family (in terms of N) excluding the youngest member = $N + 4$

3. Average involving number system

(a) Average of first n consecutive natural numbers $= \frac{n+1}{2}$

Example: Average of first five natural numbers = 3

(b) Average of first n consecutive even natural numbers = $n + 1$

Sum of first n consecutive even natural numbers = $n(n + 1)$

Example: Average of first five even natural numbers = 6

(c) Average of first n consecutive odd natural numbers = n

Sum of first n consecutive odd natural numbers = n^2

Example: Average of first five odd natural numbers = 5

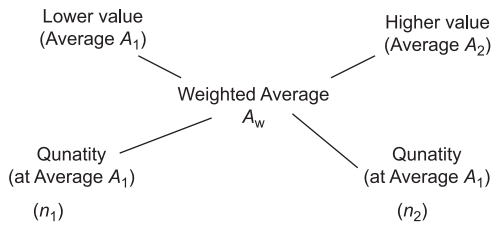
ALLIGATION (WEIGHTED AVERAGE)

Let us see a general scenario for two groups:

Group No.	No. of members	Average age of group
G_1	N_1	A_1
G_2	N_2	A_2

If we combine both the groups, then average age of all the members $= (N_1 \times A_1 + N_2 \times A_2)/(N_1 + N_2) = A_w$

We write this in the conventional 'Criss-Cross' method as given below:



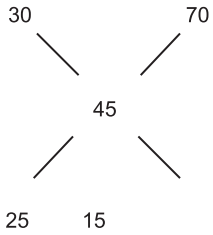
And we write this as: $\frac{n_1}{n_2} = \frac{A_2 - A_w}{A_w - A_1}$

i.e.,

$$\frac{\text{Quantity (Lower Priced)}}{\text{Quantity (Higher Priced)}} = \frac{\text{Higher Price} - \text{Average Price}}{\text{Average Price} - \text{Lower Price}}$$

Example 50 Two mixtures of 70% milk solution and 30% milk solution are mixed to from 45% milk solution. In what ratio they have been mixed?

Solution



Hence, quantity of 30% milk solution:70% milk solution = 25:15 = 5:3

PERMUTATION AND COMBINATION

1. Fundamental Principle of counting

(a) **Multiplication rule:** If a work is done only when all of a number of works are done, then number of ways of doing that work is equal to the product of number of ways of doing separate works.

(b) **Addition rule:** If a work is done only when any one of a number of works is done, then number of ways of doing that work is equal to the sum of number of ways of doing separate works. Therefore, if a work is done when exactly one of a member of works is done, then number of ways of doing this work = sum of number of ways of doing all the works.

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2. If ${}^{n}C_{x} = {}^{n}C_{y}$, then either $x = y$ or $x + y = n$
3. $\lfloor n = 1.2.3 \ldots n$; $\lfloor 0 = 1$
4. (a) Number of permutation of n different articles taken r at a time is denoted by ${}^{n}P_{r}$ and ${}^{n}P_{r} = \frac{\lfloor n}{\lfloor n-r}$
   (b) Number of permutations of n different articles taking all at a time = $\lfloor n$
   (c) Number of permutations of N articles, out of which P are alike and are of one type, Q are alike and are of second type and rest are all different = $\frac{N!}{P!Q!}$
5. Number of permutations (arrangements) of n different articles taking r at a time when articles can be repeated any number of times = $n \times n \times \ldots r$ times = n^{r}.
6. Circular permutations:
   (a) Number of circular permutations (arrangements) of n different articles = $\lfloor n-1$
   (b) Number of circular arrangements of n different articles when clockwise and anticlockwise arrangements are not different, that is, when observation can be made from both sides = $\frac{\lfloor n-1}{2}$
7. Number or combinations of n different articles taken r at a time is denoted by ${}^{n}C_{r}$ and ${}^{n}C_{r} = \frac{\lfloor n}{\lfloor r \lfloor n-r}$
8. Number of selections of r articles $(r \le n)$ out of n identical articles is 1
9. Total number of selections of zero or more articles from n distinct articles = ${}^{n}C_{0} + {}^{n}C_{1} + {}^{n}C_{2} + \ldots {}^{n}C_{n} = 2^{n}$
10. Total number of selections of zero or more articles from n identical articles = 1 + 1 + 1+… to $(n + 1)$ terms = $n + 1$
11. Number of ways of distributing n identical articles among r persons when each person may get any number of articles = ${}^{n+r-1}C_{r-1}$
12. Number of ways of dividing $m + n$ different articles in two groups containing m and n articles, respectively $(m \ne n) = {}^{m+n}C_{n} \times {}^{m}C_{m} = \frac{\lfloor m+n}{\lfloor m \lfloor n}$
13. Number of ways of dividing $2m$ different articles each containing m articles = $\frac{\lfloor 2m}{\lfloor m \lfloor m \lfloor 2}$
14. Number of ways of selecting n distinct articles taken r at a time when p particular articles are always included = ${}^{n-p}C_{r-p}$
15. ${}^{n}C_{r-1} + {}^{n}C_{r} = {}^{n+1}C_{r}$
16. Number of diagonals in an n-sided regular polygon = $nC_{2} - n - n\,(n-3)/2$
17. Maximum number of point of intersection among n lines = nC_{2}

Some Standard Formats of Questions

Word Formation

As we know that order of occurrence of letters decide the formation of words, so word formation is one standard example of permutation.

Let us understand with the help of some examples:

Example 51 How many words can be formed with the word 'LUCKNOW', which are having the following criteria?

(a) No restriction is there
(b) L is the first letter of the word
(c) L and W are the terminal letters
(d) All the vowels are together
(e) L always occurs before U
(f) L always occurs before U and U always occurs before W.

Solution

(a) Total number of distinct letters = 7 (L, U, C, K, N, O, W)
So, total number of words that can be formed = 7!

(b) Now, we can arrange only six letters (As place of L is restricted),
So, total number of words that can be formed = 6!

(c) Now, we can arrange only five letters (As place of L and W are restricted),
So, number of arrangements = 5!
But place of L and W can be interchanged between themselves.
So, total number of words that can be formed = 5! × 2!

(d) U and O should be together, so we will assume these two letters to be tied up with each other.
Now, we have six distinct things to be arranged - (L, UO, C, K, N, W)
So, number of arrangements = 6!
But place of U and O can be interchanged between themselves.
So, total number of words that can be formed = 5! × 2!

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(e) Occurrence of all the letters in the word are having equal likelihood; so, in half of the cases, L will occur before U and in the remaining half, U will occur before O.
So, total number of words that can be formed = 7!/2

(f) There are six possible arrangements (3!) corresponding to L, U and W. However, only one out of these six will be in the prescribed order: L always occurs before U and U always occurs before W.
So, corresponding to 7! arrangements, number of ways in which the condition will be satisfied = 7!/3! ways.

Example 52 How many new words can be formed with the word 'PATNA'?

Solution Total number of letters is 5. P, T, N are occurring once, whereas A is occurring twice.
So, total number of words that can be formed = 5!/2! = 60
Total number of new words = 60 –1 = 59

Example 53 How many words can be formed with the word 'ALLAHABAD'?

Solution Letters are: A – Four times
L – Twice
H, B, and D are occurring once.

So, total number of words = $\frac{9!}{4!2!}$

Number Formation

Number formation is another standard example of permutation. Here, we will introduce BOX diagram method of solving the questions.

If a three-digit number is to be constructed, then we will use

Hundred's Place	Ten's place	Unit's place

If a four-digit number is to be constructed, then we will use

Thousand's Place	Hundred's Place	Ten's place	Unit's place

And so on.

While solving questions related to **Number Formation**, we should have two things very clearly in our mind:

(i) While using box diagram, we should start with the digit which is having any restriction, that is, some condition is imposed on that digit.

(ii) When nothing about repetition of digits is mentioned in the question, we will have to assume that repetition is allowed.

Example 54 How many different three-digit numbers can be formed using the digits 1, 2, 3, 4, and 5.

(i) When repetition is not allowed.

(ii) When repetition is allowed.

Solution Below given box represents the respective positioning of digits in a three-digit number.

Hundred's Place	Ten's place	Unit's place

(i) Since repetition of the digits is not allowed, we can fill unit's place in 5 ways, ten's place in 4 ways, and hundred's place in 2 ways.

3	4	5

Using multiplication theorem, total number of numbers which can be formed = 5 × 4 × 3 = 60
<>Alternatively, 3 digits can be selected out of 5 digits in ${}^5P_3 = \mathbf{60}$

(ii) Since repetition of the digits is allowed here, we can fill each of the hundred's, ten's, and unit's place in 5 ways.

5	5	5

Using Multiplication Theorem, total number of numbers which can be formed = 5 × 5 × 5 = 125

For example, how many four-digit numbers divisible by 4 can be formed from the digits 0,1,2,3,4,5?

Solution Any number divisible by 4 will have number formed by its last two digits divisible by 4.
In this case, last two digits of the number can be 00, 04, 12, 20, 24, 32, 40, 44, 52.
Corresponding to any one of 00, 04, 12, 20, 24, 32, 40, 44, 52, we can have the following digits at its hundred's and thousand's place:

5	6	1	1

Thousand's place cannot be filled by 0, so it can be filled in 5 ways.

Hundred's place can be filled by any of the 0, 1, 2, 3, 4, 5; hence, 6 ways.

So, corresponding to any one of 00, 04, 12, 20, 24, 32, 40, 44, 52, total number of ways = 5 × 6 = 30

Hence, total number of numbers which can be formed = 30 × 9 = 270

PROBABILITY

Important notations and formula:

(a) Probability of an event = $\frac{\text{Number of desired outcomes}}{\text{Total number of possible outcomes}}$

(b) Probability of occurrence of an event is shown as P(E).

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(c) Probability of non-occurrence of an event is shown as $P'(E)$.

(d) $P(E) + P'(E) = 1$

(e) Certain event is an event that has probability = 1 or which are 100% certain. In other words, it is bound to happen. For example, if a ball is thrown up in the air, probability that it will fall on the earth (unless it has been thrown with escape velocity) = 1.

(f) Impossible event is an event which has probability = 0 or which is 0% probable. In other words, it will never happen. For example, probability that there will be 55 Sundays in a year = 0.

(g) Playing cards:
There are 52 playing cards in a pack: 13 each from heart, club, diamond, and spade. These 13 cards are Ace, no. 2, no. 3, no. 4, no.5, no. 6, no. 7, no. 8, no. 9, no. 10, Jack, Queen, and King.

(h) **ODDS in FAVOUR and ODDS in AGAINST**

$$\text{Odds in favour} = \frac{\text{Number of favourable cases}}{\text{Number of unfavourable cases}}$$

$$\text{Odds in against} = \frac{\text{Number of unfavourable cases}}{\text{Number of favourable cases}}$$

(i) **Understanding AND/OR**
To understand the role played by And/Or in our questions, let us take an example of throwing an unbiased die. Let A and B be two events associated with it such that Event A = getting an odd number, event B = getting a multiple of 3
Then, $A = \{1, 3, 5\}$ and $B = \{3, 6\}$
We now define a new event 'A or B' which occurs if event A or event B or both occur, that is, **at least one** of event A or event B occurs. Clearly, the event 'A or B' occurs if the outcome is any one of the outcomes {1, 3, 5, 6}. Therefore, the event 'A or B' is represented by the $A \cup B$.
Similarly, 'A and B' means occurrence of both A and B which is possible if the outcome is {3}.
Hence, it is represented by the subset $A \cap B$.

WORKED OUT EXAMPLES

Example 55 Using the data from the previous years, a school predicts that 6 out of total 300 students will get above 95% marks in 10th board. This year school has 400 students. How many of them are expected to get above 95% marks in 10th board?

Solution
Probability that a student will get above 95% marks $= \frac{6}{300} = \frac{1}{50}$
There are 400 students this year in the school.
Hence, expected number of students with above 95% marks $= \frac{1}{50} \times 400 = 8$ students

Example 56 In a single throw of a die, what is the probability that the number shown is either an even number or a prime number?

Solution Following numbers are even numbers = 2, 4, 6
Following numbers are prime numbers = 2, 3, 5
Hence, numbers that are either an even number or a prime number = 2, 3, 4, 5, 6. So, the number of favourable outcomes = 5 and number of total outcomes = 6
Hence, desired probability $= \frac{5}{6}$

Example 57 In a single throw of a die, what is the probability that the number shown is an even number and a prime number both?

Solution Following numbers are even numbers = 2, 4, 6
Following numbers are prime numbers = 2, 3, 5
Hence, numbers which is/are an even number and a prime number both = 2. So, number of favourable outcomes = 1, and number of total outcomes = 6
Hence, desired probability $= \frac{1}{6}$

Example 58 In a single throw of two die, what is the probability that the sum on the top face of both the die will be 8?

Solution When two die are thrown, sum of the numbers appearing on the faces can be anything from 2 to 12. To find out the number of favourable cases, we will be required to find out the cases in which sum = 8.
Following are the cases: (6, 2), (2, 6), (5, 3), (3, 5), (4, 4)
So, total number of favourable cases = 5
Total number of possible outcomes $= 6 \times 6 = 36$
Hence, probability $= \frac{5}{36}$

Example 59 Three coins are tossed simultaneously. What is the probability that exactly one of them shows a head?

Solution To get exactly one head, following ways are possible: (HTT), (THT), and (TTH).
Hence, total number of favourable cases = 3
Total number of possible outcomes $= 2 \times 2 \times 2 = 8$
Hence, probability $= \frac{3}{8}$

Example 60 Three coins are tossed simultaneously. What is the probability that at least one of them shows head?

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Solution At least one head = 1 – Probability that all three coins are showing tails = $1 - \frac{1}{8} = \frac{7}{8}$

Example 61 In an examination, probability that A passes the examination = 1/3, probability that B passes the examination = 1/4, and probability that C passes the examination = 2/5. What is the probability that

(a) none of them pass the examination?
(b) at least one of them passes the examination?
(c) exactly one of them passes the examination?
(d) all of them pass the examination?

Solution If three students are appearing for an examination, following EIGHT cases are possible:

A B C, ✗ B C, A ✗ C, A B ✗,
A ✗ ✗, ✗ B ✗, ✗ ✗ C, ✗ ✗ ✗,

✗ means student fails the exam.

(a) Probability that A fails the examination = 1 – 1/3 = 2/3
Probability that B fails the examination =1 – 1/4 = 3/4
Probability that C fails the examination = 1 – 2/5 = 3/5
None of them pass the examination = ✗ ✗ ✗,
Hence, probability that none of them pass the examination = 2/3 × 3/4 × 3/5 = 18/60 = 3/10

(b) At least one of them passes the exam = 1 – None of them pass the exam = 1 – 3/10 = 7/10

A B C, ✗ B C, A ✗ C, A B ✗,
A ✗ ✗, ✗ B ✗, ✗ ✗ C,

(c) Exactly one of them passes the examination = (1/3 × 3/4 × 3/5) + (2/3 × 1/4 × 3/5) + (2/3 × 3/4 × 2/5) = 27/60

A ✗ ✗, ✗ B ✗, ✗ ✗ C,

(d) All of them pass the examination = A B C, = 1/3 × 1/4 × 2/5 = 2/60 = 1/30

SET THEORY

Some Important Results on Numbers of Elements in Sets

If A, B, and C are three finite sets, then

1. $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
2. $n(A \cup B) = n(A) + n(B)$ if set A and set B are two disjoint sets.
3. $n(A - B) = n(A) - n(A \cap B)$
4. $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$
5. $n(A' \cup B') = n((A \cap B)') = n(U) - n(A \cap B)$
6. $n(A' \cap B') = n((A \cup B)') = n(U) - n(A \cup B)$

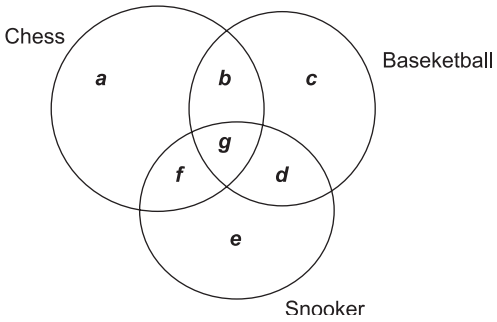
Approaching the Question

In case of set theory, the question setter tries to generate different situational questions using different languages. Let us understand this through the following example:

Example 62 In a class of 60 students, each of the students plays at least one of the three games—chess, basketball, or snooker. Twenty-four students play chess, 12 students play basketball only, and 16 students play snooker only. There are six students who play both chess and snooker, eight students play chess and basketball, and four students play snooker and basketball. How many students play basketball and snooker but not chess?

Solution Understand the language:
Twenty-four students play chess ⇒ It means that these 24 students play chess, and at the same time, they might be playing other games too.

Twelve students play basketball only ⇒ it means that these 12 students play only basketball, and they do not play any other game.



In the above diagram, a = chess only, c = basketball only, e = snooker only, b = chess and basketball but not snooker [(chess ∩ basketball) – (chess ∩ basketball ∩ snooker)], and so on g = (chess ∩ basketball ∩ snooker). We are required to calculate the value of d.

As per the question,

$$a + b + g + f = 24,\ c = 12,\ e = 16$$

Since each one of them plays at least one game, hence $a + b + c + d + e + f + g = 60$

Hence, $d = (a + b + c + d + e + f + g) - (a + b + g + f) - (c + e) = 60 - (24 + 12 + 16) = 8$

WORKED OUT EXAMPLES

Example 63 If $A = \{1, 3, 5, 7\}$, $B = \{1, 2, 3, 4\}$. What is the value of $(A \cup B)$ and $(A \cap B)$?

Solution
$(A \cup B) = \{1, 2, 3, 4, 5, 7\}$ and $(A \cap B) = \{1, 3\}$

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Example 64 How many numbers from 1 to 100 are not divisible by either of 2 or 4 or 5?

Solution Let us first understand the meaning of the statement given in the question. It is given that numbers are from 1 to 100, so while counting we will include both the limits, that is, 1 and 100. Had this been 'How many numbers in between 1 to 100 are ….', then we would not have included either 1 or 100.

Now, to solve this question, we will first find out the number of numbers from 1 to 100 which are divisible by either 2 or 5 (Since all the numbers which are not divisible by 2 will not be divisible by 4 also, so we are not needed to find out the numbers divisible by 4). And then, we will subtract this from the total number of numbers, that is, 100. It can be seen below:

Total number of numbers = Numbers which are divisible + Numbers which are not divisible

So, $n(2 \cup 5) = n(2) + n(5) - n(2 \cap 5)$

Now, $n(2) = 50$, $n(5) = 20$, $n(2 \cap 5) = 10$

$n(2 \cup 5) = 50 + 20 - 10 = 60$

Numbers which are not divisible = Total number of numbers – Numbers which are divisible

$= 100 - 60 = 40$

Example 65 There are 100 students in a particular class. 60% students play cricket, 30% students play football, and 10% students play both the games. What is the number of students who play neither cricket nor football?

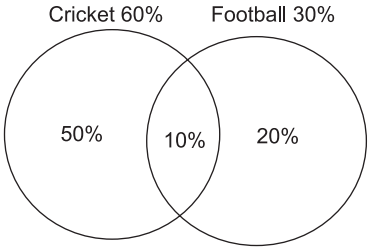
(a) 25% (b) 18% (c) 20% (d) 15%

Solution Of all the students, there are four categories of students possible:

	Cricket	Football
Category 1	Yes	Yes
Category 2	Yes	No
Category 3	No	Yes
Category 4	No	No

Question is asking us to find out Category 4 students. Obviously, summation of all the four categories = 100%.

Following is the Venn-Diagram presentation of the situation given in the question.



Since 60% students play cricket and 10% play both, hence, 50% students play only cricket, and since 30% students play football, and 10% play both ⇒ Hence, 20% students play only football.

Hence, percentage of students playing any one of the games = percentage of students who play only cricket + percentage of students who play only football + percentage of students who play both the games = 50% + 20% + 10% = 80%

Hence, percentage of students who do not play any games = 100 % – 80% = 20%

Hence, option (C) is the answer.

Example 66 In an examination, 80% students passed in Philosophy and 70% students passed in Maths. At the same time, 15% failed in both the subjects. If 390 students passed in both the subjects, then how many students appeared in the examination?

(a) 500 (b) 400 (c) 800 (d) 600

Solution First of all, understand all the possibilities and statements:

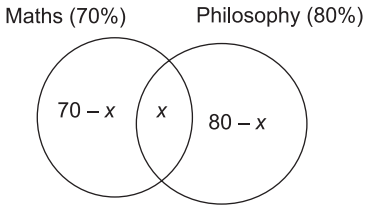
Philosophy	Pass	Fail	Pass	Fail
Maths	Pass	Pass	Fail	Fail

15% failed in both the subjects does not mean that 85% passed in both the subjects. It means summation of following three possibilities:

Philosophy	Pass	Fail	Pass
Maths	Pass	Pass	Fail

We have the information regarding the Philosophy pass percentage (and obviously fail percentage too can be calculated from this data) and Maths pass percentage (and obviously fail percentage too can be calculated from this data).

Using set theory will provide a better view of the whole scenario:



According to the question:

$85\% = (70\% - x) + x + (80\% - x) \Rightarrow x = 65\%$ = Students who passed in both the subjects = 390

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$$\Rightarrow 100\% = \frac{390}{65} \times 100 = 600$$

Hence, option (d) is the answer.

LOGARITHM

If any number N is expressed in the form a^x, then the index 'X' is called the logarithm of the number N to the base 'a'.

Therefore, if $N = a^x$ then, $X = \log_a N$. It is read as Log of N to the base a.

A simple way (though not exactly mathematical) to recollect the formula:



(i) Base of index will become the base of log.

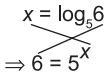
(ii) Left hand number will go on the right hand side and vice versa.

For example, $10^2 = 100$ will be written in logarithm format as follows:

$2 = \log_{10} 100$

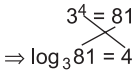
Example 67 **(a)** Write the following in the index format: $x = \log_5 6$

Solution Base of log will become the log of index and left-hand side number (x) will go the right-hand side and vice versa.



(b) Write the following in the logarithm format: $3^4 = 81$

Solution Base of index will become the base of logarithm and left-hand side number will go to the right-hand side number and vice versa.



Some Important Properties

In case of all the following properties given ahead, we will be using the following standard restrictions on logarithm:

1. $\log_a(XY) = \log_a X + \log_a Y$
2. $\log_a(X/Y) = \log_a X - \log_a Y$
3. (a) $\log_a(X^k) = k \log_a X$,
3. (b) $\log_a \sqrt[k]{X} = \frac{1}{k} \times \log_a x$
4. $\log_a 1 = 0$ (As $a^0 = 1$ provided $a \neq 1$)
5. $\log_x X = 1$
6. $\log_a X = \frac{1}{\log_x a}$
7. Base change rule
   $\log_a x = \log_a b \times \log_b x = \log_b x \times \log_a b$
8. $a^{(\log_a x)} = x$
9. If $a > 1$ and $x > a$, then $\log_a x > 0$
10. $\log_a 1 = 0$
11. $\log_a b = \frac{\log b}{\log a}$

Example 68 What is the value of $\log_{125} 625$?

Solution

$$\log_{125} 625 = \log_{5^3} 5^4 = 4 \times \frac{1}{3}(\log_5 5) = \frac{4}{3} \times 1 = \frac{4}{3}$$

Example 69 What is the value of $\log_3 2. \log_4 3. \log_5 4 \ldots \log_{16} 15$?

Solution

$$\log_3 2 = \frac{\log 2}{\log 3};\ \log_4 3 = \frac{\log 3}{\log 4}, \text{ and so on}$$

$$\log_3 2. \log_4 3. \log_5 4 \ldots \log_{16} 15 = \frac{\log 2}{\log 3} \cdot \frac{\log 3}{\log 4} \cdots \frac{\log 15}{\log 16} = \frac{\log 2}{\log 16} = \log_{16} 2$$

$$\Rightarrow \log_{16} 2 = \log_{2^4} 2 = \frac{1}{4}(\log_2 2) = \frac{1}{4} \times 1 = \frac{1}{4}$$

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Test 1

Q.1 In a dinner party there are 10 Indians, 5 Americans, and 5 Englishmen. In how many ways, can they be arranged in a row so that all persons of the same nationality sit together?
(a) 10!×5!×5! (b) 20!
(c) 3!×10!×5!×5! (d) 20! 3!

Q.2 Samar has 60 apples, 156 oranges, and 204 guavas. He distributes the fruits equally among a group of kids such that each kid gets maximum number of fruits, and all the kids get exactly one type of fruit. Find the number of fruits that each child will get.
(a) 12 (b) 24 (c) 18 (d) 6

Q.3 If $\log(xy^3) = 1$ and $\log(x^2y) = 1$, what is $\log(xy)$?
(a) $-\frac{1}{2}$ (b) 0 (c) $\frac{1}{2}$ (d) $\frac{3}{5}$

Q.4 Let $U = 2\times2004^{2005}$, $V = 2004^{2005}$, $W = 2003\times2004^{2004}$, $X = 2\times2004^{2004}$, $Y = 2004^{2004}$, and $Z = 2004^{2003}$. Which of the following is the largest?
(a) $U - V$ (b) $V - W$ (c) $W - X$ (d) $X - Y$

Q.5 There are 20 questions in a question paper. If no two students solve the same combination of questions but solve equal number of questions, then the maximum number of students who appeared in the examination is
(a) ${}^{20}C_9$ (b) ${}^{20}C_{11}$
(c) ${}^{20}C_{10}$ (d) None of these

Q.6 The sum of 49 consecutive integers is 7^5. What is their average?
(a) 7 (b) 7^2 (c) 7^3 (d) 7^4

Q.7 The greatest possible number of points of intersection of 8 straight lines and 4 circles is:
(a) 32 (b) 64 (c) 76 (d) 104

Q.8 If x and y are positive integers for which $2^x3^y = 1296$, what is the value of $x + y$?
(a) 8 (b) 9 (c) 10 (d) 11

Q.9 If four whole numbers are taken at random, and multiplied together, then the probability that the last digit in the product is 1, 3, 7, or 9 will be:
(a) 13/976 (b) 17/529 (c) 16/625 (d) 13/625

Q.10 There are 500 students sitting taking an examination. 150 students passed the first paper, 350 students passed the second paper, and 50 students passed both the papers. Find the probability that a student selected at random has failed in both the papers.
(a) 1/5 (b) 1/10 (c) 3/10 (d) 3/5

Q.11 Find the number of pairs of two natural numbers having product = 3600 and HCF = 30.
(a) 0 (b) 1 (c) 2 (d) 5

Q.12 Two digits in Jack's age are the same as the digits in Bill's age, but in reverse order. In five years, age of Jack will be twice as old as the age of Bill. What is the difference in their current ages?
(a) 9 (b) 18 (c) 27 (d) 36

Q.13 Two is 10% of x and 20% of y. What is $x - y$?
(a) 1 (b) 2 (c) 5 (d) 10

Q.14 In how many ways, three-digit numbers can be formed to satisfy the property that the middle digit is the average of the first and the last digits?
(a) 41 (b) 42 (c) 43 (d) 45

Q.15 Two distinct numbers a and b are chosen randomly from the set $\{2, 2^2, 2^3, \ldots, 2^{25}\}$. What is the probability that $\log_a b$ is an integer?
(a) $\frac{2}{25}$ (b) $\frac{31}{300}$ (c) $\frac{13}{100}$ (d) $\frac{7}{50}$

Q.16 A class collects ₹50 to buy flowers for a classmate who is in the hospital. Roses cost ₹3 each, and carnations cost ₹2 each. No other flowers are to be used. How many different bouquets could be purchased for exactly ₹50?
(a) 1 (b) 7 (c) 9 (d) 16

Q.17 Bricklayer NS would take 9 h to build a chimney alone, and bricklayer PA would take 10 h to build it alone. When they work together they talk a lot, and their combined output is decreased by 10 bricks per hour. Working together, they build the chimney in 5 h. How many bricks are in the chimney?
(a) 500 (b) 900 (c) 950 (d) 1000

Q.18 Which of the following is equal to $1 + \dfrac{1}{1+\dfrac{1}{1+1}}$?
(a) $\frac{5}{4}$ (b) $\frac{3}{2}$ (c) $\frac{5}{3}$ (d) 2

Q.19 A shopkeeper sells 25 articles at ₹45 per article after giving 10% discount on marked price and earns 50% profit. Had there been no discount, profit percentage would have been:
(a) 60% (b) 60⅔% (c) 66% (d) 66⅔%

Q.20 How many years would it take for a sum to grow from ₹1250 to ₹10,000 if it is invested at 12.5 pcpa at SI?
(a) 28 (b) 56 (c) 48 (d) 70

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Test 2

Q.1 Average of the nine numbers in the set {9,99,999,9999, …,999999999} is a 9-digit number M, all of whose digits are distinct. The number M does not contain the digit:
(a) 0 (b) 2 (c) 4 (d) 6

Q.2 What is the value of $\frac{2^{2014}+2^{2012}}{2^{2014}-2^{2012}}$?
(a) -1 (b) $\frac{5}{3}$ (c) 2013 (d) 2^{4024}

Q.3 In a bag of marbles, $\frac{3}{5}$ of the marbles are blue and the rest are red. If the number of red marbles is doubled and the number of blue marbles stays the same, what fraction of the marbles will be red?
(a) $\frac{2}{5}$ (b) $\frac{3}{7}$ (c) $\frac{4}{7}$ (d) $\frac{3}{5}$

Q.4 What is the product of all positive odd integers less than 10,000?
(a) $\frac{10,000!}{(5000!)^2}$ (b) $\frac{10,000!}{2^{5000}}$
(c) $\frac{9999!}{2^{5000}}$ (d) $\frac{10,000!}{2^{5000}.5000!}$

Q.5 It takes Mary 30 min to walk uphill 1 km from her home to school, but it takes her only 10 min to walk from school to her home along the same route. What is her average speed, in kmph, for the round trip?
(a) 3 (b) 3.125 (c) 3.5 (d) 4

Q.6 Suppose July of year N has five Mondays. Which of the following will definitely occur five times in the August of the same year?
(a) Monday (b) Thursday
(c) Wednesday (d) Tuesday

Q.7 What is the probability that a randomly drawn positive factor of 60 is less than 7?
(a) $\frac{1}{10}$ (b) $\frac{1}{4}$
(c) $\frac{1}{2}$ (d) None of these

Q.8 What is the value of $(3x - 2)(4x + 1) - (3x - 2)$ $4x + 1$ when $x = 4$?
(a) 0 (b) 1 (c) 10 (d) 11

Q.9 Sum of the squares of marbles the three children have is 600 and the sum of products taken two at a time of marbles is 150. Then, what is the sum of all the three numbers?
(a) 30 (b) 35 (c) 40 (d) 45

Q.10 Samar sets his watch to the correct time at noon. At the actual time of 1:00 pm, he notices that his watch reads 12:57 and 36 s. Assuming that his watch loses time at a constant rate, what will be the actual time when his watch first reads 10:00 pm?
(a) 10:22 pm and 24 s (b) 10:25 pm
(c) 10:25:24 pm (d) None of these

Q.11 A grocer makes a display of cans in which the top row has one can and each lower row has two more cans than the row above it. If the display contains 100 cans, how many rows does it contain?
(a) 8 (b) 9 (c) 10 (d) 11

Q.12 Josh and Mike live 13 km apart. Yesterday, Josh started to ride his bicycle towards Mike's house. A little later Mike started to ride his bicycle towards Josh's house. When they met, Josh had ridden for twice the length of time as Mike and at four-fifths of Mike's rate. How many kilometre had Mike ridden when they met?
(a) 4 (b) 8 (c) 6 (d) 5

Q.13 If $x < 0$, then which of the following must be positive?
(a) $-x^{-1}$ (b) $-x^2$ (c) -2^x (d) $\sqrt[3]{x}$

Q.14 Sum of the products of the ten numbers +1, +2, +3, +4, +5 taking two at a time is:
(a) 165 (b) –55
(c) 55 (d) None of these

Q.15 If $\log_k N = 6$, and $\log_{25k}(8N) = 3$, then k is:
(a) $(12.5)^3$ (b) $(12.5)^2$
(c) (12.5) (d) $(12.5)^{\frac{2}{3}}$

Q.16 Anil, Manoj, and Chandu play a game that has 6 rounds. In each round, there is a single winner, and the outcomes of the rounds are independent. For each round the probability that Anil wins is $\frac{1}{2}$, and Manoj is twice as likely to win as Chandu. What is the probability that Anil wins three rounds, Manoj wins two rounds, and Chandu wins one round?
(a) $\frac{5}{72}$ (b) $\frac{5}{36}$ (c) $\frac{1}{6}$ (d) $\frac{1}{3}$

Q.17 Ram runs 7/4 times as fast as Sham. If Ram gives Sham a start of 300 m, how far must the winning post be if both Ram and Sham have to end the race at the same time?
(a) 700 m (b) 1400 m (c) 350 m (d) 210 m

Q.18 A certain sum is lent out at a certain rate of interest for a certain period and the amount is 3.6 times as the principle. Had it been lent out on 60% higher rate of interest for 40% less time period the amount would have been how many times the principle?
(a) 2.75 times (b) 3.2 times
(c) 3.7 times (d) None of these

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Q.19 Radha can do 3/4th of a work in 12 days. After working for 8 days Mira joins her and both works together for three days after which Radha leaves. If Mira takes 12 more days to complete the work, then how many days Mira will take alone to do it?
(a) 16 (b) 12 (c) 48 (d) 24

Q.20 How many different integers can be expressed as the sum of three distinct members of the set {1, 4, 7, 10, 13, 16, 19}?
(a) 13 (b) 16 (c) 24 (d) 30

Test 3

Q.1 Chunmun picks a letter from the set of English alphabet and finds it to be a vowel. What is the probability that the letter is *E*?
(a) 1/26 (b) 5/6
(c) 1/5 (d) None of these

Q.2 Given $A = 2^{65}$ and $B = (2^{64} + 2^{63} + 2^{62} + ... + 2^0)$, which of the following is true?
(a) $A = B$ (b) B is larger than A by 1.
(c) A is larger than B by 1. (d) Cannot be determined

Q.3 Let S be the set of integers x such that
I. $100 \le x \le 200$
II. x is odd
III. x is divisible by 3 but not by 7.
How many elements does S contain?
(a) 16 (b) 12 (c) 11 (d) 13

Q.4 The integers 34,041 and 32,506, when divided by a three-digit integer n, leave the same remainder. What is the value of n?
(a) 289 (b) 367 (c) 453 (d) 307

Q.5 Eleven oranges are bought for ₹10, and 10 oranges are sold for ₹11. What is the profit or loss percentage?
(a) 21% loss (b) 21% profit
(c) 10% profit (d) No profit no loss

Q.6 In a race of 100 m, *A* beats *B* by 20 m or 4 s. What is the speed of *A*?
(a) 5 m/s (b) 10 m/s
(c) 6.25 m/s (d) None of these

Q.7 Of the 200 candidates who were interviewed for a position at a call centre, 100 had a two-wheeler, 70 had a credit card, and 140 had a mobile phone. 40 of them had both, a two-wheeler and a credit card; 30 had both, a credit card and a mobile phone, and 60 had both, a two wheeler and mobile phone, and 10 had all three. How many candidates had none of the three?
(a) 0 (b) 5 (c) 20 (d) 10

Q.8 In a chess tournament, every person played one game with every other person in the group. The total number of games that men played between themselves exceeded those played by men with women by 18. If there were four women in the tournament, totally how many games were played in the tournament?
(a) 112 (b) 120 (c) 162 (d) 190

Q.9 There are three inlet taps whose diameters are 1 cm, 2 cm, and 3 cm, respectively. Rate of flow of water is directly proportional to the square of the diameter. It takes 9 min for the smallest pipe to fill an empty tank. Find the time taken to fill an empty tank when all the three taps are opened.
(a) 9/14 (b) 14/9
(c) 3 (d) None of these

Q.10 A certain quantity of 40% solution is replaced with 25% solution such that the new concentration is 35%. What is the fraction of the solution that was replaced?
(a) 1/4 (b) 1/3 (c) 1/2 (d) 2/3

Q.11 If $-2 \le x \le 2$ and $3 \le y \le 8$, which of the following represents the range of all possible values of $y - x$?
(a) $5 \le y - x \le 6$ (b) $1 \le y - x \le 5$
(c) $1 \le y - x \le 6$ (d) $1 \le y - x \le 10$

Q.12 What is the difference between the sum of the first 2003 even natural numbers and the sum of the first 2003 odd natural numbers?
(a) 0 (b) 1 (c) 2 (d) 2003

Q.13 Let $a, b, c, d,$ and e be integers such that $a = 6b = 12c$, and $2b = 9d = 12e$. Then, which of the following pairs contains a number that is not an integer?
(a) $\left(\frac{a}{27}, \frac{b}{e}\right)$ (b) $\left(\frac{a}{36}, \frac{c}{e}\right)$
(c) $\left(\frac{a}{12}, \frac{bd}{18}\right)$ (d) $\left(\frac{a}{6}, \frac{c}{d}\right)$

Q.14 A new flag is to be designed with six vertical stripes using some or all of the colours yellow, green, blue, and red. Then, the number of ways this can be done so that no two adjacent stripes have the same colour is:
(a) 12×81 (b) 16×192
(c) 20×125 (d) 24×216

Q.15 LCM of three natural numbers 10, 20, and $N = 60$. How many values of N are possible?
(a) 2 (b) 5
(c) 6 (d) None of these

Q.16 Find the value of $3\log\frac{81}{80} + 5\log\frac{25}{24} + 7\log\frac{16}{15}$.
(a) log 2 (b) log 3
(c) 1 (d) None of these

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Q.17 Five persons entered the lift on the ground floor of an 8-floor apartment. Assuming that each one of them independently and with equal probability can leave the lift at any floor beginning with the first. What is the probability that all five persons are leaving the lift at different floors?

(a) $\frac{{}^{7}P_{5}}{7^{5}}$ (b) 22/49

(c) 21/343 (d) None of these

Q.18 Likhit earns x% on the first ₹2000 and y% on the rest of his income. If he earns ₹700 from ₹4000 income and ₹900 from ₹5000 income, find x%.

(a) 20% (b) 15%

(c) 25% (d) None of these

Q.19 A bag contains ₹600 in the form of one-rupee, 50 paise, and 25 paise coins in the ratio 3:4:12. Find the total number of 25 paise coins in the bag.

(a) 75 (b) 200 (c) 300 (d) 900

Q.20 An alloy of manganese, tin, and bronze contains 90% bronze, 7% manganese, and 3% tin. A second alloy of bronze and tin only is melted with the first and the mixture contains 85% of bronze, 5% of manganese, and 10% of tin. What is the percentage of bronze in the second alloy?

(a) 67.5% (b) 72.5% (c) 77.5% (d) 82.5%

Answers

Test 1

1. (c)	**2.** (a)	**3.** (d)	**4.** (a)	**5.** (c)	**6.** (c)	**7.** (d)	**8.** (a)	**9.** (c)	**10.** (b)
11. (b)	**12.** (b)	**13.** (d)	**14.** (d)	**15.** (b)	**16.** (c)	**17.** (b)	**18.** (c)	**19.** (d)	**20.** (b)

Test 2

1. (a)	**2.** (b)	**3.** (c)	**4.** (d)	**5.** (a)	**6.** (b)	**7.** (c)	**8.** (d)	**9.** (a)	**10.** (c)
11. (c)	**12.** (a)	**13.** (a)	**14.** (b)	**15.** (c)	**16.** (b)	**17.** (a)	**18.** (d)	**19.** (c)	**20.** (a)

Test 3

1. (c)	**2.** (c)	**3.** (d)	**4.** (d)	**5.** (b)	**6.** (c)	**7.** (d)	**8.** (b)	**9.** (a)	**10.** (b)
11. (d)	**12.** (d)	**13.** (d)	**14.** (a)	**15.** (c)	**16.** (a)	**17.** (a)	**18.** (b)	**19.** (d)	**20.** (b)

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